Polylogarithms and their Arithmetic Bridges

Special Values, Zeta Jets, Lattices and Experimental Discovery

ProveIt Contributors

Consolidated manuscript, October 2026

Preface

The purpose of this manuscript is to develop a common mathematical language for special values that first appear in rather different guises: nested sums, arctangent integrals, gamma products, lattice row sums and coefficients of Hurwitz zeta. Its central movement is from experimental discovery to exact structure. A well-chosen computation can suggest a transformation, reveal an unexpected character component or expose a missing generator. A proof then explains the phenomenon and determines its range of validity.

The material originates in thirty-nine articles, reports and literature surveys from the PolyLog programme in the Smithereens repository, together with successive continuation packages preserved in the accompanying source inventory. These supply exact reductions, unrestricted formal ranks, positive kernel geometry and certified asymptotics that supersede earlier gaps. The editorial ledger distinguishes integrated proofs from preserved reports whose analytic reconciliation remains active. The present text merges overlapping treatments, uses later corrected formulas, and revises claims whose original wording exceeded the evidence. It is organized by mathematical dependency rather than by the order of research sessions. The source inventory and editorial ledger accompany the manuscript.

There are three distinct meanings of "exact" in this work. An analytic identity may have a proof in ordinary mathematics. A row combination or a shuffle calculation may be checked by finite exact arithmetic, relative to its stated law vocabulary. A numerical identity may instead have a very small independently recomputed residual. These are valuable and different forms of evidence. In particular, an unsuccessful integer-relation search proves neither independence nor minimal depth, and a zero canary coefficient does not prove that a proposed relation is exact. No result in this project has yet been formalized in a proof assistant.

The chapter order follows this distinction. Nested sums and iterated integrals establish the algebra; cyclotomic Fourier transforms organize its arithmetic coordinates; algebraic-argument transformations and depth experiments identify candidate special values. Signed kernels and certified computation expose their analytic geometry. Gamma certificates and CM lattices connect these values to periods. Hurwitz-zeta jets unify integration and differentiation, and the Herglotz function carries these bridges into real-quadratic arithmetic. The final chapter returns to experimental design and the proof problems that this progression leaves open.

Historical residuals, performance measurements and search counts refer to the source snapshots, whose scripts and identity stores were not all imported into ProveIt. Fresh checks in the accompanying verification directory are labeled separately. Attribution is retained for classical identities and published theorems; an experimental rediscovery is not a claim of priority. The original drafts were prepared with agentic assistance from Anthropic Claude, with an external OpenAI review of the Stieltjes integration article; the present consolidation and additional audit were assisted by OpenAI Codex.

Contents

Preface **i**

1 From one polylogarithm to many **1**
1.1 Multiple polylogarithms: two faces . . . 1
1.2 Multiple zeta values and their relations . . . 4
1.3 Harmonic polylogarithms . . . 9
1.4 Nielsen's generalized polylogarithms . . . 12
1.5 The common algebra and the motivic boundary . . . 17

2 Cyclotomic values and the polygamma bridge **18**
2.1 The two-way exact DFT . . . 18
2.2 Clausen-type laws for polygamma values at rational arguments . . . 19
2.3 The boundary of the odd-weight Clausen collapse . . . 20
2.4 The fundamental constants, by conductor . . . 21
2.5 Reduction tables and the settled level-24 question . . . 21
2.6 A new closed form . . . 22
2.7 A weight parity swap . . . 22

3 Algebraic arguments, integrals and ladders **23**
3.1 The problem and the key geometric fact . . . 23
3.2 The weight-2 vertical-line theorem . . . 23
3.3 The Clausen form at $y = \tan\theta$ . . . 24
3.4 Weight 3: a half-lifted symmetry . . . 24
3.5 The integral and its provenance . . . 24
3.6 The master formula (Reshetnikov–Lee) and a differentiation proof . . . 25
3.7 Cleo's integral as the perfect-square instance . . . 26
3.8 Closed values and a bounded experimental survey . . . 26
3.9 New closed-form integrals from elementary χ_2 . . . 26
3.10 A sibling trichotomy . . . 27
3.11 An alternate form, and minimality . . . 27
3.12 Unit relations and the golden family . . . 28
3.13 Reshetnikov's formulas and their numerical evidence . . . 28
3.14 Reconstructing Tito's weight-6 and weight-7 formulas . . . 28
3.15 Explicit candidate formulas at weights 8 and 9 . . . 29
3.16 Reducing the tetralogarithm relation to a functional identity . . . 29
3.17 An exact proof of the rational tetralogarithm conjecture . . . 30
3.18 Canonical index-12/20/24 ladders and a ζ-elimination chain . . . 36

3.19 The minimal Pisot ladder base . . . 37
3.20 Hunting more ladder bases, and a methodological caveat . . . 37
3.21 Proof certificates for the cubic and quartic trilogarithm ladders . . . 37
3.22 Rigidity of complementary powers . . . 41
3.23 Exact proofs and log-free ladders at the plastic base . . . 46
3.24 The modified polylogarithm . . . 49
3.25 Ladders are Bloch-group elements . . . 49
3.26 The bases as number fields . . . 50
3.27 Regulator parity and the limits of ladder counting . . . 50
3.28 Attributed dilogarithm and inverse-binomial examples . . . 50
3.29 Weighted arctangent families in classical trilogarithms . . . 51

4 Gaussian and Eisenstein depth **56**
4.1 Depth conventions and Gaussian generators . . . 56
4.2 An infinite even-weight reduction . . . 56
4.3 Generators and their exact singularities . . . 57
4.4 Exact shuffle quotients and a two-product proof at weight five . . . 60
4.5 The mixed Gaussian weight-five identity and its exact proof . . . 65
4.6 An explicit binomial parity reduction for every double weight . . . 70
4.7 Classical reductions for every index with one two . . . 73
4.8 Classical reductions with one index equal to two . . . 75
4.9 From one-two reductions to general complementary depth . . . 81
4.10 An explicit complementary-depth theorem . . . 82
4.11 A double-polylogarithm formula for an all-weight family . . . 84
4.12 The colored stuffle identity . . . 85
4.13 Gaussian doubles: closed forms . . . 85
4.14 Gaussian doubles: the double-shuffle dimension . . . 87
4.15 Triple sums and the scope of formal quotient dimensions . . . 88
4.16 The Eisenstein analogue . . . 89
4.17 Why the standard evaluator degenerates at $w = 1$ . . . 90
4.18 A robust $w = 1$ evaluator . . . 90
4.19 Validation . . . 90
4.20 Closed forms for the Eisenstein mixed-point values . . . 91
4.21 Inverse colors: the exact reduction that the searches missed . . . 91
4.22 Mixed endpoints and all-weight cyclotomic product quotients . . . 95
4.23 Harmonic index, reflection and certified asymptotics . . . 101
4.24 The harmonic-index family and its integral representation . . . 101
4.25 An all-index reflection identity . . . 102
4.26 The first harmonic index revisited . . . 104
4.27 Complete monotonicity and globally enveloping truncations . . . 104
4.28 The joint depth–exponent transition . . . 106
4.29 Geometric evaluation with explicit error bounds . . . 110
4.30 Exponential cancellation at proportional harmonic depth . . . 111

5 Signed kernels and certified polylogarithm computation **116**
5.1 A signed-kernel representation . . . 116
5.2 The unique zero on the upper semicircle . . . 118
5.3 A four-term asymptotic for the zero . . . 118
5.4 Certified Euler evaluation . . . 120
5.5 An exact kernel and the sharp Euler error . . . 123
5.6 Real orders, the finite signed-measure threshold and its atoms . . . 130
5.7 Consequences for complex zeros and angular zeros . . . 134
5.8 A positive two-variable representation at all positive orders . . . 136
5.9 Below the measure threshold: positive differences settle the zero question . . . 137
5.10 Strict Gaussian envelope shape and a certified maximum . . . 145
5.11 Cartesian motion of the complete real-order zero arc . . . 147
5.12 Critical defects and the emergence of the atom . . . 148
5.13 Elementary critical-kernel identities and Hurwitz inequalities . . . 149
5.14 A proved local part of the normalized-radius conjecture . . . 150
5.15 Global normalized-radius motion at leading index one . . . 153
5.16 A proved failure of fractional global monotonicity . . . 156
5.17 The two atomic edges and the logarithmic corner . . . 160
5.18 Power-scale corrections and exponential mass concentration . . . 162
5.19 Euler enclosures and the failure of an unrestricted constant one . . . 165
5.20 A sharp Euler constant at every positive real order . . . 169
5.21 A uniform rational kernel inequality . . . 171
5.22 Strict depth, signed oscillation and minimal positive compensation . . . 174
5.23 Two bounded operators on signed moment densities . . . 175
5.24 Exact oscillation and two interlacing laws . . . 178
5.25 Canonical positive compensation and slit-plane nonvanishing . . . 179
5.26 Exactly $d-1$ angular zeros on every upper semicircle . . . 181
5.27 Small-radius and large-outer-index expansions . . . 185
5.28 Explicit logistic kernels and positive integral identities . . . 187
5.29 A divided-difference identity across depths and colors . . . 190
5.30 Exact rational evaluation and finite certificates . . . 193
5.31 Independent rational logarithm and Gaussian controls . . . 195
5.32 A strict radial theorem for every positive Stieltjes transform . . . 196
5.33 A geometric evaluation formula that survives $xy = 1$ . . . 198
5.34 Every moment reduces to classical polylogarithms . . . 200
5.35 Optimal geometric rate and the conductor dependence . . . 202
5.36 Outward-rounded evaluation and the treatment of cancellation . . . 203
5.37 Exact rational certificates for Gaussian values . . . 205

6 Gamma grids and exact certificates **210**
6.1 The lattice model . . . 210
6.2 Trig-root Γ identities: mechanism and census . . . 211
6.3 The trigamma analogue . . . 212
6.4 The harvest: legacy notebooks, verified and taxonomized . . . 213
6.5 Multiplier certificates . . . 213

7 Polygamma row sums at complex multiplication points **216**
7.1 Lattice normalization and row sums . . . 216
7.2 The elementary layer: reflection composed with conjugation . . . 216
7.3 Maximal symmetry: vanishing identities and Γ-periods . . . 218
7.4 Beyond maximal symmetry: $2i$ and $i\sqrt{2}$ . . . 219
7.5 Class number 2: discriminants -20 and -15 . . . 219
7.6 Class number 3: discriminant -23 . . . 220
7.7 Class number 4: discriminant -39 and genus structure . . . 220
7.8 Class number 5: discriminant -47 . . . 221
7.9 Exact CM row products from class polynomials . . . 222
7.10 Genus characters prove the recorded CM ratios . . . 232
7.11 Norms, exact quadratic degree and oscillating genus ratios . . . 238
7.12 The opaque components . . . 240
7.13 Summation order, differentiation and certified tails . . . 241
7.14 Regularized modular rows and all derivative evaluations . . . 241

8 Zeta jets, Stieltjes antiderivatives and negative polygamma **248**
8.1 Below the digamma: $\psi^{(-2)}$ at rationals . . . 248
8.2 The general log-gamma integral and the fifth-point obstruction . . . 249
8.3 Binet's second formula and Malmsten's integrals: the Lambert-kernel face . . . 250
8.4 The log-gamma moment ladder . . . 250
8.5 Log-gamma moments: poles, late coefficients, and effective remainders . . . 251
8.6 Sharp truncation of the log-gamma moment expansion . . . 260
8.7 Reflected log-gamma moments and their late coefficients . . . 265
8.8 The Stieltjes layer . . . 272
8.9 The common Hurwitz-jet calculus . . . 273
8.10 Underived Stieltjes grids and character coordinates . . . 274
8.11 The ladder . . . 274
8.12 Zeta jets and the conversion dictionary . . . 275
8.13 Reflection formulas . . . 276
8.14 Moments . . . 277
8.15 Explicit values of Γ_n . . . 278
8.16 The negative-order polygamma floor . . . 281
8.17 Fourier conventions and the Hurwitz semigroup . . . 287
8.18 Canonical finite parts of the Stieltjes functions . . . 288
8.19 All-index convolution closure and exact normal forms . . . 289
8.20 Ordinary integral realizations and explicit Stieltjes identities . . . 291
8.21 Conventions and endpoint normalization . . . 292
8.22 The shifted Hurwitz correlation . . . 293
8.23 An all-order linear Stieltjes closure theorem . . . 295
8.24 Coincident endpoints and canonical collision subtraction . . . 298
8.25 Ordinary integrals of all antiderivative orders . . . 301
8.26 Explicit log-gamma and mixed identities . . . 303
8.27 Rational shifts and exact arithmetic coordinates . . . 305
8.28 All circular convolution powers of log Gamma . . . 306
8.29 Nonintegral twists: Lerch jets, contact terms, and the half-twist . . . 308
8.30 Unequal twists through beta mixtures and finite resolvent identities . . . 317
8.31 Gamma coefficients prove every harmonic-order identity . . . 319

8.32 Mixed Gamma derivatives and exact harmonic tail identities . . . 320
8.33 All centered trigamma-square moments and a polylogarithmic generator . . . 324
8.34 Digamma powers, harmonic spectral data and exact finite parts . . . 328
8.35 The universal contour polynomial . . . 330
8.36 A finite-part formula for every positive integer power . . . 331
8.37 A finite algorithm in strict harmonic words . . . 335
8.38 The fifth power with every finite correction . . . 337
8.39 Normalized primitives and polygamma identities . . . 343

9 The parameter-derivative tower 348
9.1 The uniform master identity . . . 348
9.2 Differentiation and the contact-term correction . . . 349
9.3 A two-jet collapse for every polygamma convolution . . . 351
9.4 Contact terms in translated derivative products . . . 352
9.5 Dilation, distribution traces and exact logarithmic contacts . . . 353
9.6 The uniform rational-argument landscape . . . 355
9.7 Weighted distribution modules and exact rational-grid ranks . . . 355
9.8 Integral distribution, reflection and modular specialization . . . 362
9.9 Primitive-grid determinants, local defects and finite descent . . . 383
9.10 Explicit primitive-grid reduction coefficients . . . 385
9.11 All-orders Hurwitz–Stieltjes identities . . . 388
9.12 Finite conductor descent for all Stieltjes layers . . . 389
9.13 Cyclotomic polylogarithm traces . . . 391
9.14 Explicit tables for the second constant . . . 395
9.15 The derivative $\leftrightarrow$ negapolygamma duality . . . 396
9.16 One normalized functional equation for all derivative bridges . . . 397
9.17 Experimental independence tests . . . 401
9.18 Elementary grids and the sixth-point completion . . . 401
9.19 Fixed-coordinate periodic collision contacts . . . 402
9.20 The off-collision kernel and its second finite part . . . 404
9.21 The all-index collision-contact theorem . . . 405
9.22 Exact convergent integral identities from finite Fourier data . . . 408
9.23 A Hurwitz kernel for all positive-integer spectral jets . . . 409
9.24 Polylogarithm boundary values and invisible local terms . . . 411
9.25 Exact primitives and the log-gamma bridge . . . 412

10 Positive kernels and the geometry of Stieltjes zeros 414
10.1 A unified Stieltjes–Lerch derivative family . . . 414
10.2 Real-rootedness and a global zero bound . . . 416
10.3 An all-order concentration expansion . . . 417
10.4 Exactly n eventual zeros and their locations . . . 419
10.5 A recurrence for the next zero coefficient . . . 422
10.6 The classical half-unit bracket . . . 422
10.7 Exact low-index zero counts . . . 423
10.8 High parameter derivatives of Stieltjes constants . . . 425
10.9 The first classical failure of saturation: the fifth index . . . 432
10.10 Exact sixth- and seventh-index saturation thresholds . . . 436

11 Herglotz arithmetic and quadratic regulators **439**
11.1 The function and the exact dilogarithm sum . . . 439
11.2 Quadratic-unit values: Tables 1 & 2, and two errata . . . 439
11.3 Rational arguments: a family of theorems . . . 440
11.4 From rational examples to an exact formal criterion . . . 442
11.5 Cyclotomic symbols and formal reduction . . . 443
11.6 The all-conductor relation theorem . . . 444
11.7 Exact reduction criteria and constructive rational cores . . . 446
11.8 Complete classification and evaluations of $J(2/q)$ . . . 450
11.9 Bridges to the golden ladders and the Clausen lattice . . . 451
11.10 Positive measures and sharp approximation . . . 452
11.11 A positive measure and certified remainder bounds . . . 452
11.12 Uniform sharp asymptotics at the optimal scale . . . 455
11.13 The exact best cutoff and its transitions . . . 457
11.14 An arithmetic displacement beyond all algebraic orders . . . 459
11.15 Rational Herglotz derivatives and an exponentially small oscillation . . . 460
11.16 Quadratic units and Kronecker-limit coordinates . . . 468
11.17 Regulator candidates and field degrees . . . 468
11.18 Exact class-number-one certificates . . . 469

12 Experimental discovery and the route to proof **471**
12.1 Constructing a meaningful integer-relation experiment . . . 471
12.2 Controls and independent recomputation . . . 471
12.3 What exact linear algebra proves . . . 472
12.4 Computational conventions . . . 472
12.5 The surviving research programme . . . 472
12.6 Questions sharpened by the exact continuations . . . 473
12.7 From decimal fits to short executable proofs . . . 474
12.8 What the new reports change in the research programme . . . 475
12.9 A certified rejection beyond the discovery precision . . . 475
12.10 The complete Cayley ideal does not close the bounded weight-seven search . . . 476

A Literature problems and source provenance **482**
A.1 Polylogarithm questions . . . 482
A.2 Polygamma and Hurwitz questions . . . 485
A.3 The provenance boundary . . . 486
A.4 A version-specific external consistency check . . . 487
A.5 Two Stieltjes index and coefficient corrections . . . 487

Chapter 1

From one polylogarithm to many

An ordinary polylogarithm has two descriptions: a series and an iterated integral. Nesting the first and extending the alphabet of the second produce the same larger class of functions. Their two multiplication laws then provide exact relations. This chapter fixes conventions before those relations are used to organize special values.

1.1 Multiple polylogarithms: two faces

The classical polylogarithm $\mathrm{Li}_s(z) = \sum_{n\geq 1} z^n/n^s$ has, for the reader of Lewin, two complementary descriptions: a power series and an iterated integral $\mathrm{Li}_s(z) = \int_0^z \mathrm{Li}_{s-1}(t)\,dt/t$. Each survives the passage to many indices, and the resulting two descriptions of one and the same function are the source of essentially every nontrivial identity in this circle of ideas. We set up both here; the products they generate occupy the next section.

1.1.1 The nested sum

Definition 1.1. *For $s_1, \dots, s_k \in \mathbb{C}$ and $z_1, \dots, z_k \in \mathbb{C}$ the* multiple polylogarithm *is the strictly nested sum*

$$\mathrm{Li}_{s_1,\dots,s_k}(z_1,\dots,z_k) \;=\; \sum_{n_1>n_2>\cdots>n_k\geq 1} \prod_{i=1}^{k} \frac{z_i^{n_i}}{n_i^{s_i}}.$$

Its weight *is $w = s_1 + \cdots + s_k$ and its* depth *is k. The single-index case $k = 1$ recovers $\mathrm{Li}_s(z)$.*

Two features of this definition demand comment before anything is proved.

Remark 1.2 (Conventions differ; ours is fixed). The inequality is *strict*, $n_1 > n_2 > \cdots > n_k$. A sizable part of the literature (and several computer-algebra systems) instead uses a non-strict summation $n_1 \geq n_2 \geq \cdots \geq n_k$, or reverses the role of the indices so that the *innermost* summation variable carries the first parameter. These choices are genuinely different functions (they differ by lower-depth terms, exactly the diagonal contributions $n_i = n_{i+1}$), and the discrepancy is the single most common source of sign and indexing confusion when comparing sources. We adhere throughout to the strict, "outermost variable carries s_1" convention above. The arguments are likewise ordered with z_1 paired to the largest index n_1. When checking against `MultiplePolyLog`, confirm the convention numerically on a small case rather than assuming agreement.

Remark 1.3 (Multiple zeta values). Setting every $z_i = 1$ gives the *multiple zeta value*

$$\zeta(s_1,\dots,s_k) = \mathrm{Li}_{s_1,\dots,s_k}(1,\dots,1) = \sum_{n_1>\cdots>n_k\geq 1} \prod_i n_i^{-s_i},$$

the central special values of the theory (Zagier 1994; Hoffman 1992). The strict nesting is what makes $\zeta(s_1, \ldots, s_k)$ finite for $s_1 \geq 2$.

Convergence. The single-variable series $\mathrm{Li}_s(z)$ converges absolutely for $|z| < 1$ (any s), on $|z| = 1$ for $\Re s > 1$, and conditionally on $|z| = 1$, $z \neq 1$, for $\Re s > 0$. For the nested sum the corresponding statement is clean.

Proposition 1.4 (A sufficient absolute-convergence domain)**.** *For positive integer indices, the nested series converges absolutely when $|z_i| \leq 1$ for all i and $s_1 \geq 2$. It also converges absolutely for $|z_1| < 1$, $|z_i| \leq 1$ for $i > 1$, with $s_1 \geq 1$.*

Proof. The absolute value of the inner nested sum is at most $H^{k-1}_{n_1-1} = O((1 + \log n_1)^{k-1})$. The outer series is therefore bounded by $\sum |z_1|^{n_1}(1 + \log n_1)^{k-1}/n_1^{s_1}$, which converges in either stated case. □

These are sufficient conditions, not a complete classification of every boundary point. Oscillation can give conditional convergence: $\mathrm{Li}_1(-1)$ converges, but $\sum |(-1)^n/n|$ diverges. More generally, away from the polydisc an interior domain is given by $|z_1 \cdots z_j| < 1$ for each $1 \leq j \leq k$. At unity, for positive integer indices, admissibility is exactly $s_1 \geq 2$.

1.1.2 The iterated integral

The second face is Chen's iterated integral. Fix the two differential one-forms

$$\omega_0 = \frac{dt}{t}, \qquad \omega_1 = \frac{dt}{1-t},$$

and, for a word $u = a_1 a_2 \cdots a_w$ in the letters $\{0, 1\}$, write the iterated integral over the standard simplex

$$\int_0^z u \;=\; \int_{0<t_1<t_2<\cdots<t_w<z} \omega_{a_w}(t_1)\, \omega_{a_{w-1}}(t_2) \cdots \omega_{a_1}(t_w),$$

i.e. the leftmost letter a_1 is attached to the *outermost* (largest) variable t_w and one integrates from the top down. This is the orientation under which d/dz strips the outermost form. With this normalization the one-variable multiple polylogarithm is an iterated integral over a single explicit word.

Theorem 1.5 (Iterated-integral representation)**.** *For $s_1 \geq 2$ and the remaining $s_i \geq 1$,*

$$\mathrm{Li}_{s_1,\ldots,s_k}(z, 1, \ldots, 1) \;=\; \int_0^z \underbrace{0^{s_1-1}1}_{\text{block } 1}\, \underbrace{0^{s_2-1}1}_{\text{block } 2} \cdots \underbrace{0^{s_k-1}1}_{\text{block } k},$$

the iterated integral of the binary word

$$u(\mathbf{s}) = 0^{s_1-1}1\, 0^{s_2-1}1 \cdots 0^{s_k-1}1$$

of length w. Equivalently, with the alternative kernel $\widetilde{\omega}_1 = dt/(t-1) = -\omega_1$ one carries one sign per 1-letter, giving the textbook normalization

$$\mathrm{Li}_{s_1,\ldots,s_k}(z, 1, \ldots, 1) = (-1)^k \int_{0<t_1<\cdots<t_w<z} \prod_{j=1}^{w} \eta_{a_{w+1-j}}(t_j), \qquad \eta_0 = \tfrac{dt}{t},\ \eta_1 = \tfrac{dt}{t-1},$$

the factor $(-1)^k$ being exactly the number of 1-letters in $u(\mathbf{s})$, one per block.

The composition ↔ word dictionary. The map $\mathbf{s} = (s_1, \dots, s_k) \mapsto u(\mathbf{s})$ is a bijection between compositions (ordered tuples with each $s_i \geq 1$) and binary words that *end in a* 1. Reading u from left to right: a 1 *closes a block* and contributes a new summation index; the run of 0's immediately preceding that 1 sets the corresponding exponent, $s_i - 1$ zeros for exponent s_i. Thus depth $k =$ (number of 1's) and weight $w =$ (length of the word). Admissibility $s_1 \geq 2$ is the requirement that u *begin with a* 0: a leading 1 would attach ω_1 to the top variable and produce the logarithmic divergence at $t_w \to 1$ when $z = 1$. The endpoints play dual roles — ω_0 is singular at 0, ω_1 at 1 — so a convergent word neither starts with 1 nor ends with 0.

Origin. Expand the innermost form geometrically. The depth-one case is Lewin's $\mathrm{Li}_s(z) = \int_0^z 0^{s-1}1$, obtained by iterating $\mathrm{Li}_s(z) = \int_0^z \mathrm{Li}_{s-1}(t)\,\omega_0$ down to $\mathrm{Li}_1(z) = -\log(1-z) = \int_0^z \omega_1$. For general depth, $\int_0^z \omega_1$ on the innermost block introduces $\sum_{n_k \geq 1} t^{n_k}/n_k$, the subsequent ω_0's raise the power of n_k, and the next ω_1 couples a strictly larger index $n_{k-1} > n_k$; the strict inequality is forced by the nesting of the simplex $t_1 < \cdots < t_w$. Carrying this out reproduces the nested sum, and Chen's theory (1977) guarantees convergence and homotopy invariance of the integral. The general framework is due to Goncharov; the systematic exploitation for these functions is Goncharov's and Borwein–Bradley–Broadhurst–Lisoněk's. □

1.1.3 General alphabets and a fixed integral orientation

For a path from 0 to z avoiding its letters, use the outer-letter-first recursion

$$G(a_1, \dots, a_w; z) = \int_0^z \frac{G(a_2, \dots, a_w; t)}{t - a_1}\, dt, \qquad G(;z) = 1.$$

The all-zero word is regularized as $G(0^w; z) = \log^w z/w!$. For positive integer indices the dictionary in our decreasing-index convention is

$$\mathrm{Li}_{s_1,\dots,s_k}(z_1, \dots, z_k) = (-1)^k G\left(0^{s_1-1}, z_1^{-1}, 0^{s_2-1}, (z_1 z_2)^{-1}, \dots, 0^{s_k-1}, (z_1 \cdots z_k)^{-1}; 1\right).$$

Thus the letters are inverse *prefix* products. Reversing the integration order also reverses the letter word; mixing these conventions changes the function. The formula follows by expanding each nonzero kernel geometrically near the origin and integrating recursively, then continuing along a common path. Wolfram's generalized-polylogarithm interface must be calibrated against this orientation before numerical comparisons.

1.1.4 Two facts to carry forward

(1) **The differential strips the first letter.** In the integral orientation above,

$$z\frac{d}{dz}\,\mathrm{Li}_{s_1,\dots}(z, 1, \dots) = \begin{cases} \mathrm{Li}_{s_1-1,s_2,\dots}(z, 1, \dots), & s_1 \geq 2, \\ \dfrac{z}{1-z}\,(\text{remove the leading block}), & s_1 = 1, \end{cases}$$

i.e. differentiation removes one outer letter — a 0 lowers s_1, the leading 1 deletes the first index. This recursion (equivalently the integral equation defining the word) drives every analytic continuation and every series-rearrangement proof. It is also the recursion behind the appearance of `HarmonicPolyLog` in `DSolve`/`AsymptoticDSolveValue` output for Fuchsian systems with rational coefficients (Remiddi–Vermaseren 1999 for the $\{0,1\}$-alphabet case).

(2) **One function, two presentations.** The *same* $\mathrm{Li}_{s_1,\dots,s_k}$ is simultaneously the nested sum and the iterated integral. Multiplying two nested sums and re-sorting the indices yields the *stuffle* (harmonic, quasi-shuffle) product $*$; multiplying two iterated integrals over a common path and shuffling the letters yields the *shuffle* product Π. Both express a product of polylogarithms as a $\mathbb{Q}$-linear combination of polylogarithms, but through different combinatorics; equating them is the engine of the next section.

1.1.5 The smallest nontrivial example, in full

Take depth $k=2$, $\mathbf{s}=(2,1)$, the lightest admissible non-classical case. The nested sum is

$$\mathrm{Li}_{2,1}(z,1)=\sum_{m>n\geq 1}\frac{z^m}{m^2\,n}=\sum_{m=2}^{\infty}\frac{z^m}{m^2}\,H_{m-1},\qquad H_{m-1}=\sum_{n=1}^{m-1}\tfrac{1}{n},$$

exhibiting the characteristic harmonic-number coefficient. The associated word is

$$u(2,1)=\underbrace{0^{2-1}1}_{\text{block }(s_1=2)}\ \underbrace{0^{1-1}1}_{\text{block }(s_2=1)}=011,$$

a word of length $w=3$ (weight 3) with two 1's (depth 2), beginning in 0 (admissible) and ending in 1 (convergent). By Theorem 1.5,

$$\mathrm{Li}_{2,1}(z,1)=\int_0^z 011=\int_{0<t_1<t_2<t_3<z}\frac{dt_3}{t_3}\,\frac{dt_2}{1-t_2}\,\frac{dt_1}{1-t_1},$$

the leading 0 sitting on the outermost variable t_3 and the two 1's on t_2,t_1. (Equivalently $(-1)^2=+1$ in the $\eta_1=dt/(t-1)$ normalization, consistent with the unsigned $\omega_1=dt/(1-t)$ form.) A numerical check at $z=0.7$ gives both sides equal to $0.2384965\ldots$, and at $z=1$ the integral $\int_0^1 011$ recovers Euler's celebrated

$$\zeta(2,1)=\sum_{m>n\geq 1}\frac{1}{m^2 n}=\zeta(3),$$

the depth-two value whose collapse to a depth-one zeta was Euler's first multiple-zeta identity — and, in Wolfram 15, exactly the symbolic output of `MultipleZeta` on $\{2,1\}$.

1.2 Multiple zeta values and their relations

Specializing the multiple polylogarithm at $z_1=\cdots=z_k=1$ produces the central objects of this chapter, the *multiple zeta values* (MZVs):

$$\zeta(s_1,\dots,s_k)\;=\;\mathrm{Li}_{s_1,\dots,s_k}(1,\dots,1)\;=\sum_{n_1>n_2>\cdots>n_k\geq 1}\prod_{i=1}^{k}\frac{1}{n_i^{s_i}}. \tag{1.1}$$

The outermost summation index n_1 carries the exponent s_1, and the strict inequalities make the sum a genuine iterated, rather than ordinary, multiple series. The series (1.1) converges if and only if $s_1\geq 2$; such an index tuple is called *admissible*. (Any $s_i\geq 1$ for $i\geq 2$ is allowed; only the leading entry is constrained.) We call $w=s_1+\cdots+s_k$ the *weight* and k the *depth*. In Wolfram Language these are `MultipleZeta`, with $\zeta(2,1)$ written `MultipleZeta[{2,1}]`.

The depth-one values are the classical zeta values. Euler's formula is

$$\zeta(2n) = \frac{(-1)^{n+1} B_{2n} (2\pi)^{2n}}{2(2n)!} \in \pi^{2n}\mathbb{Q}.$$

The first genuinely new phenomenon is that distinct MZVs can collapse onto a single lower-depth value. The prototype is Euler's identity

$$\zeta(2,1) \;=\; \sum_{n>m\geq 1} \frac{1}{n^2\, m} \;=\; \zeta(3), \tag{1.2}$$

already known to Euler in the 1740s. The conjecture is that the $\mathbb{Q}$-vector space spanned by all MZVs is *weight-graded*: products and relations never mix weights, so one studies each weight w separately.

The sum theorem

Euler's $\zeta(2,1) = \zeta(3)$ is the depth-two case of the first structural theorem. Summing all admissible MZVs of a fixed weight w and fixed depth k recovers the single zeta value $\zeta(w)$.

Theorem 1.6 (Sum theorem; Euler for $k = 2$, Granville and Zagier in general, 1990s)**.** *For all* $w > k \geq 1$,

$$\sum_{\substack{s_1+\cdots+s_k=w \\ s_1\geq 2,\ s_i\geq 1}} \zeta(s_1,\ldots,s_k) \;=\; \zeta(w). \tag{1.3}$$

At $w = 3, k = 2$ the only admissible tuple is $(2,1)$, recovering (1.2). At $w = 4, k = 2$ the admissible tuples are $(3,1)$ and $(2,2)$, so

$$\zeta(3,1) + \zeta(2,2) = \zeta(4) = \tfrac{\pi^4}{90}. \tag{1.4}$$

This one equation does not separate $\zeta(3,1)$ from $\zeta(2,2)$; for that we need a *second* multiplication law, to which we now turn.

The two products

MZVs admit two structurally different multiplication rules, because $\zeta(s_1,\ldots,s_k)$ has two faithful representations: as a nested sum (1.1), and as an iterated integral. Each representation makes the product of two MZVs expand into a sum of MZVs, but the two expansions differ — and that discrepancy is the engine of the theory.

(a) The stuffle (harmonic / quasi-shuffle) product. Multiply two nested sums directly. For depth one,

$$\zeta(a)\,\zeta(b) = \Big(\sum_{m\geq 1} m^{-a}\Big)\Big(\sum_{n\geq 1} n^{-b}\Big) = \sum_{m>n} + \sum_{n>m} + \sum_{m=n},$$

splitting the lattice $\{(m,n)\}$ by the trichotomy $m > n$, $m < n$, $m = n$. The three regions are exactly two depth-two MZVs and one depth-one MZV:

$$\zeta(a)\,\zeta(b) = \zeta(a,b) + \zeta(b,a) + \zeta(a+b). \tag{1.5}$$

The general rule is the same combinatorial device — the *quasi-shuffle* of Hoffman — applied to the index words $(s_1,\ldots,s_k)$ and $(t_1,\ldots,t_\ell)$: interleave the two words in all order-preserving ways, and

additionally allow any aligned pair s_i, t_j to *collide* into a single slot $s_i + t_j$. (Setting $a = b = 2$ in (1.5) gives $\zeta(2)^2 = 2\,\zeta(2,2) + \zeta(4)$, used below.) A depth-two example of the general rule:

$$\zeta(a)\,\zeta(b,c) = \zeta(a,b,c) + \zeta(b,a,c) + \zeta(b,c,a) + \zeta(a+b,c) + \zeta(b,a+c),$$

the first three terms inserting a into the three gaps of (b,c), the last two colliding a with b and with c.

(b) The shuffle product. The integral representation reads each MZV as a Chen iterated integral over the simplex $1 > t_1 > \cdots > t_w > 0$ against the two one-forms $\omega_0 = \frac{dt}{t}$ and $\omega_1 = \frac{dt}{1-t}$. Encode an admissible index as a binary word: each s_i becomes $\underbrace{0\cdots0}_{s_i-1}1$, so that, e.g., $\zeta(2) \leftrightarrow 01$, $\zeta(3) \leftrightarrow 001$, $\zeta(2,1) \leftrightarrow 011$, $\zeta(3,1) \leftrightarrow 0011$, $\zeta(2,2) \leftrightarrow 0101$. Admissibility ($s_1 \geq 2$) is the condition that the word *begins with* 0; convergence at the inner end requires the word to *end with* 1. Multiplying two iterated integrals over the same simplex produces the integral over the product region, which decomposes into the $\binom{w_1+w_2}{w_1}$ order-preserving interleavings — the *shuffle product* Π — of the two letter words, *without* any collision term. Thus

$$\zeta(a)\,\zeta(b) = \sum_{\text{words } u \in (\text{word } a)\,\Pi\,(\text{word } b)} \zeta(u). \tag{1.6}$$

The double-shuffle relation: $\zeta(3,1)$ in closed form

The canonical demonstration computes $\zeta(2)^2$ by both products and equates the results. Stuffle, from (1.5) with $a = b = 2$:

$$\zeta(2)^2 = 2\,\zeta(2,2) + \zeta(4). \tag{1.7}$$

Shuffle: $\zeta(2)$ is the word 01, and we shuffle $01 \,\Pi\, 01$. Writing the two factors as 0_a1_a and 0_b1_b and listing the $\binom{4}{2} = 6$ interleavings that preserve $0_a \to 1_a$ and $0_b \to 1_b$:

$$0_a0_b1_a1_b,\;\; 0_a0_b1_b1_a,\;\; 0_b0_a1_a1_b,\;\; 0_b0_a1_b1_a,\;\; 0_a1_a0_b1_b,\;\; 0_b1_b0_a1_a.$$

Forgetting the colour labels, these are the words 0011 (four times) and 0101 (twice). Since $0011 \leftrightarrow \zeta(3,1)$ and $0101 \leftrightarrow \zeta(2,2)$,

$$\zeta(2)^2 = 4\,\zeta(3,1) + 2\,\zeta(2,2). \tag{1.8}$$

Now subtract (1.7) from (1.8): the $2\,\zeta(2,2)$ terms cancel and we obtain the *double-shuffle relation*

$$4\,\zeta(3,1) = \zeta(4) \qquad \Longrightarrow \qquad \zeta(3,1) = \tfrac{1}{4}\zeta(4) = \frac{\pi^4}{360}. \tag{1.9}$$

Feeding this back into the sum-theorem relation (1.4) also pins $\zeta(2,2) = \zeta(4) - \zeta(3,1) = \frac{3}{4}\zeta(4) = \pi^4/120$ — consistent with the all-2's family below.

Remark 1.7 (Regularized double shuffle). The naive double-shuffle relations above are produced only by admissible words. To capture the boundary case, one regularizes the divergent $\zeta(1)$ (the word 1). Comparing the two regularizations — shuffle versus stuffle — as polynomials in a formal variable T standing for the regularized $\zeta(1)$ yields the *extended* or *regularized double-shuffle relations* of Ihara, Kaneko, and Zagier (2006). These are conjectured to generate *all* $\mathbb{Q}$-linear relations among MZVs.

Other relation families

The double-shuffle relations sit inside a larger, heavily overlapping web. We list the principal families, one line each.

Duality. The integral substitution $t \mapsto 1-t$ swaps $\omega_0 \leftrightarrow \omega_1$ and reverses the order of integration, giving $\zeta(\mathbf{s}) = \zeta(\mathbf{s}^\dagger)$, where the *dual* word $\mathbf{s}^\dagger$ is obtained from the binary word by reading it backwards and exchanging $0 \leftrightarrow 1$. The simplest instance is $011 \mapsto 011$ read backwards is 110, complemented is 001: thus $\zeta(2,1) = \zeta(3)$ (1.2) is precisely a self/dual coincidence. Likewise $0011\,(\zeta(3,1))$ is *self*-dual—read backwards and complemented it is again 0011—so duality says nothing new there; the cleanest nontrivial higher instance is the depth-three identity $\zeta(2,1,1) = \zeta(4)$, from $0111 \mapsto 0001$.

Hoffman's relations. Relations obtained by comparing the regularized product with $\zeta(1)$ provide useful index identities. Their precise insertion and splitting terms matter; a schematic insertion of a single 1 in every position is not a valid general formula.

Derivation relations. An infinite family produced by certain Hopf-algebra derivations acting on the algebra of admissible words (Ihara–Kaneko–Zagier), equivalent over $\mathbb{Q}$ to large portions of the regularized double shuffle.

Ohno's relation. A single generating identity (Ohno 1999) that simultaneously generalizes the sum theorem (1.3) *and* duality, obtained by summing MZVs over all ways of distributing a fixed total increment among the indices of $\mathbf{s}$ and of its dual.

Cyclic sum formula. Hoffman–Ohno (and Ohno–Wakabayashi): a cyclic sum of MZVs of weight w equals a weighted sum of weight-w values, refining the sum theorem with a cyclic symmetry on the index word.

Closed-form families

Beyond individual reductions, several infinite families of MZVs evaluate to powers of π in closed form. Throughout, $\{s\}_n$ denotes the index s repeated n times.

All 2's. The generating function $\prod_{n\geq 1}(1 - x^2/n^2) = \sin(\pi x)/\pi x$ yields, after extracting the relevant power sums,

$$\zeta(\{2\}_n) = \zeta(\underbrace{2,2,\ldots,2}_{n}) = \frac{\pi^{2n}}{(2n+1)!}. \tag{1.10}$$

For $n=1$ this is $\zeta(2) = \pi^2/6$; for $n=2$, $\zeta(2,2) = \pi^4/120$, matching (1.9); for $n=3$, $\zeta(2,2,2) = \pi^6/5040$.

The $\{3,1\}$ family. A celebrated result of Borwein, Bradley, Broadhurst, and Lisoněk (1998), originally conjectured experimentally:

$$\zeta(\{3,1\}_n) = \zeta(\underbrace{3,1,3,1,\ldots,3,1}_{n \text{ blocks}}) = \frac{2\,\pi^{4n}}{(4n+2)!}. \tag{1.11}$$

For $n = 1$ this reproduces $\zeta(3,1) = 2\pi^4/6! = \pi^4/360$ of (1.9); for $n = 2$, $\zeta(3,1,3,1) = 2\pi^8/10!$. The proof is a striking generating-function argument identifying $\sum_n \zeta(\{3,1\}_n)x^{4n}$ with a ratio of cosh and cos.

The $\{2\}_a\,3\,\{2\}_b$ family. Zagier (2012) evaluated the two-parameter family $\zeta(\{2\}_a, 3, \{2\}_b)$ as an explicit $\mathbb{Q}$-linear combination of products $\pi^{2m}\zeta(2n+1)$, a result whose proof unexpectedly drew on the arithmetic of the modular-form-like generating series; it remains one of the deepest known closed-form families.

Dimensions and the weight grading

Let D_w be the $\mathbb{Q}$-vector space spanned by all MZVs of weight w (with $D_0 = \mathbb{Q}$, $D_1 = 0$ since $\zeta(1)$ diverges), and $d_w = \dim_{\mathbb{Q}} D_w$. Zagier (1994) conjectured the remarkably simple recursion

$$d_w = d_{w-2} + d_{w-3}, \qquad d_0 = 1,\ d_1 = 0,\ d_2 = 1, \tag{1.12}$$

giving the sequence

$$d_0, d_1, d_2, \cdots = 1, 0, 1, 1, 1, 2, 2, 3, 4, 5, 7, 9, \ldots.$$

The *upper* bound $d_w \le$ (the recursion value) is a theorem, proved independently by Goncharov and by Terasoma (around 2002), with the conceptual framework of mixed Tate motives completed by Deligne and Goncharov; it amounts to showing that the known relations (e.g. associator / motivic) suffice to cut the dimension down. Broadhurst and Kreimer (1997) refined the count by also grading by depth, conjecturing a two-variable generating function that brings in the dimensions of spaces of cusp forms (the first modular obstruction appearing at weight 12, depth 4). The matching *lower* bound — equivalently, that no further relations exist and the conjectured spanning sets are linearly independent — is wide open; even $\dim D_w \ge 2$ for some specific large w would already imply transcendence statements (such as the linear independence of π^6 and $\zeta(3)^2$ over $\mathbb{Q}$ at $w = 6$) that are presently out of reach.

A table of low-weight reductions

The following table records all admissible MZVs through weight 5 together with their reductions to the standard spanning sets $\{\zeta(2)\}$, $\{\zeta(3)\}$, $\{\zeta(4)\} \sim \{\pi^4\}$, $\{\zeta(5), \zeta(2)\zeta(3)\}$. Every entry has been verified symbolically.

w	spanning count	MZVs of weight w and their reductions
2	1	$\zeta(2) = \dfrac{\pi^2}{6}$
3	1	$\zeta(3)$; $\quad \zeta(2,1) = \zeta(3)$
4	1	$\zeta(4) = \dfrac{\pi^4}{90}$; $\quad \zeta(3,1) = \dfrac{\pi^4}{360}$, $\quad \zeta(2,2) = \dfrac{\pi^4}{120}$, $\quad \zeta(2,1,1) = \zeta(4)$
5	2	$\zeta(5)$, $\zeta(2)\zeta(3)$ span; $\quad \zeta(4,1) = 2\zeta(5) - \zeta(2)\zeta(3)$,
		$\zeta(3,2) = -\frac{11}{2}\zeta(5) + 3\,\zeta(2)\zeta(3)$, $\quad \zeta(2,3) = \frac{9}{2}\zeta(5) - 2\,\zeta(2)\zeta(3)$,
		$\zeta(3,1,1) = 2\zeta(5) - \zeta(2)\zeta(3)$, $\quad \zeta(2,1,1,1) = \zeta(5)$ (duality of $\zeta(5)$)
		$\zeta(2,2,1) = \zeta(3,2) = -\frac{11}{2}\zeta(5) + 3\zeta(2)\zeta(3)$,
		$\zeta(2,1,2) = \zeta(2,3) = \frac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3)$ (duality)

The table gives spanning sets of sizes $1, 1, 1, 2$ at weights $2, 3, 4, 5$. The two-element weight-five set is not known to be independent. Every depth-two value in that row reduces to products of single zeta values. At weight eight, values such as $\zeta(6, 2)$ are candidates for a primitive direction beyond the polynomial algebra of single zetas.

1.3 Harmonic polylogarithms

The multiple polylogarithms of the previous sections are nested *sums*. For the practitioner who must actually manipulate weight-w transcendentals—most insistently the particle physicist computing Feynman integrals—it is the *iterated-integral* avatar that is decisive, because integration variables, differential equations, and analytic continuation all speak the language of integrals. The harmonic polylogarithms (HPLs) of Remiddi and Vermaseren [42] are precisely this avatar, specialised to the three integration kernels that arise when a rational integrand has singularities only at $t = 0, 1, -1$. They have become the de facto standard basis for one-scale problems, and Wolfram 15 exposes them directly as `HarmonicPolyLog`.

1.3.1 Definition: kernels and iterated integrals

Fix the alphabet $\{0, 1, -1\}$ and the three rational *kernels*

$$f_0(t) = \frac{1}{t}, \qquad f_1(t) = \frac{1}{1-t}, \qquad f_{-1}(t) = \frac{1}{1+t}.$$

A harmonic polylogarithm $H(a_1, \dots, a_w; z)$ is indexed by a word $\vec{a} = (a_1, \dots, a_w)$ over this alphabet; the length w is the *weight*. The definition is recursive in the leftmost letter. When the leading letter is nonzero,

$$H(a_1, a_2, \dots, a_w; z) = \int_0^z f_{a_1}(t)\, H(a_2, \dots, a_w; t)\, dt, \qquad a_1 \neq 0, \tag{1.13}$$

with the depth-zero seed $H(; z) = 1$ (the empty word). The recursion also applies when $a_1 = 0$, provided the remaining word is not all zeros. Such an integrand is integrable at the origin. Only the all-zero words need the logarithmic regularization above; zeros at the end of a general word can be removed by shuffle identities. For example $H(0, 1; z) = \mathrm{Li}_2(z)$ is a convergent integral with a leading zero. The leftmost letter always labels the outermost integration.

1.3.2 The weight-one and weight-two cases

At weight one (1.13)–the all-zero regularization give the three building blocks

$$H(0; z) = \ln z, \qquad H(1; z) = -\ln(1-z), \qquad H(-1; z) = \ln(1+z), \tag{1.14}$$

since $\int_0^z dt/(1-t) = -\ln(1-z)$ and $\int_0^z dt/(1+t) = \ln(1+z)$. At weight two, integrating these against another kernel reproduces the classical dilogarithm and its companions:

$$H(0, 1; z) = \int_0^z \frac{dt}{t}\big(-\ln(1-t)\big) = \mathrm{Li}_2(z), \qquad H(0, -1; z) = \int_0^z \frac{dt}{t} \ln(1+t) = -\mathrm{Li}_2(-z), \tag{1.15}$$

$$H(1, 0; z) = \int_0^z \frac{\ln t}{1-t}\, dt = -\mathrm{Li}_2(z) - \ln z\, \ln(1-z), \qquad H(1, 1; z) = \tfrac{1}{2} \ln^2(1-z), \tag{1.16}$$

the last by (1.13) with $H(1;t) = -\ln(1-t)$ and $\int_0^z (-\ln(1-t))/(1-t)\,dt = \frac{1}{2}\ln^2(1-z)$. The pair $H(0,1;z) = \mathrm{Li}_2(z)$ and $H(0,-1;z) = -\mathrm{Li}_2(-z)$ shows the general pattern: the words $H(\underbrace{0,\dots,0}_{s-1},1;z) = \mathrm{Li}_s(z)$ and $H(\underbrace{0,\dots,0}_{s-1},-1;z) = -\mathrm{Li}_s(-z)$ recover the entire ladder of classical polylogarithms, so the HPLs are a strict superset of the functions of Lewin's treatise. In Wolfram 15 these read, e.g., `HarmonicPolyLog[{0,1},x]` $= \mathrm{Li}_2(x)$ and `HarmonicPolyLog[{0,-1},x]` $= -\mathrm{Li}_2(-x)$.

1.3.3 The condensed m-notation

Writing out long strings of 0's is wasteful, and Remiddi and Vermaseren introduced a condensed index that absorbs them. To each maximal block "$\underbrace{0,\dots,0}_{|m|-1}\,\sigma$" consisting of $|m|-1$ zeros followed by a sign letter $\sigma \in \{1,-1\}$ one assigns a single integer

$$m = \sigma \cdot (\text{number of zeros} + 1),$$

so the sign of m records σ and the magnitude $|m|$ records how many kernels (one $f_{\pm 1}$ preceded by $|m|-1$ copies of f_0) the block contributes. The resulting tuple is written as a subscript,

$$H_{m_1,\dots,m_k}(z) := H(\vec{a};z),$$

and the *depth* of the HPL is k, the number of nonzero letters, while the weight is $w = \sum_i |m_i|$. Thus (1.15) becomes $H_2(z) = \mathrm{Li}_2(z)$ and $H_{-2}(z) = -\mathrm{Li}_2(-z)$, the polylogarithm ladder is $H_s(z) = \mathrm{Li}_s(z)$, $H_{-s}(z) = -\mathrm{Li}_s(-z)$, and a mixed word such as $H(0,1,-1;z)$ is $H_{2,-1}(z)$. The m-notation is the input most reduction tables are organised by; the explicit word notation $H(\vec{a};z)$ is the one that makes the integral representation (1.13) transparent. (Some authors reverse the global sign or the letter ordering; we follow the convention compatible with (1.14)–(1.15) and with `HarmonicPolyLog`, where indices read outermost-integration-first.)

1.3.4 The shuffle algebra

Theorem 1.8 (Shuffle product)**.** *For a fixed argument z, the product of two harmonic polylogarithms is a $\mathbb{Q}$-linear combination of harmonic polylogarithms of the same argument:*

$$H(\vec{a};z)\,H(\vec{b};z) = \sum_{\vec{c}\in \vec{a}\,\Pi\,\vec{b}} H(\vec{c};z),$$

the sum running over all words $\vec{c}$ in the shuffle $\vec{a}\,\Pi\,\vec{b}$, i.e. all interleavings of $\vec{a}$ and $\vec{b}$ that preserve the internal order of each.

This is the iterated-integral analogue of integration by parts: a product of two iterated integrals over $[0,z]$ is the integral over the union of the two simplices, which decomposes into the $\binom{|\vec{a}|+|\vec{b}|}{|\vec{a}|}$ ordered shuffles. Concretely, at weight $1+1$,

$$H(0;z)\,H(1;z) = H(0,1;z) + H(1,0;z), \tag{1.17}$$

which is the statement $\ln z \cdot (-\ln(1-z)) = \mathrm{Li}_2(z) + \big(-\mathrm{Li}_2(z) - \ln z\,\ln(1-z)\big) = -\ln z\,\ln(1-z)$, i.e. the integration-by-parts consistency that re-expresses the product of the two weight-one HPLs through the two weight-two ones. (The dilogarithm *reflection* formula is a different relation—the

$z \mapsto 1-z$ image of $H(0,1;z)$, derived in (1.19) below.) The shuffle algebra is graded by weight, and crucially it lets one strip leading zeros: since $H(0;z)=\ln z$, repeated use of (1.8) expresses any $H(0,\vec{a};z)$ through $\ln z$ and HPLs whose words begin with a nonzero letter, which is the regularisation alluded to after the all-zero regularization.

1.3.5 Values at $z=1$: Euler sums and colored MZVs

Setting $z=1$ turns admissible HPLs (those whose word does not begin with 1, so that the boundary integral converges) into constants. Because the kernels live on the level-two alphabet $\{0,\pm1\}$, these constants are exactly the level-2, or *alternating*, multiple zeta values—the Euler sums. Words over $\{0,1\}$ give the ordinary MZVs,

$$H(\underbrace{0,\dots,0}_{s_1-1},1,\underbrace{0,\dots,0}_{s_2-1},1,\dots;1)=\zeta(s_1,s_2,\dots), \tag{1.18}$$

so for instance $H(0,1;1)=\mathrm{Li}_2(1)=\zeta(2)$ and $H(0,0,1;1)=\zeta(3)$. Admitting the letter -1 produces the alternating Euler sums: with $\bar{s}$ marking a sign,

$$H(0,-1;1)=-\mathrm{Li}_2(-1)=\tfrac{\pi^2}{12}=-\zeta(\bar{2}),\qquad H(0,0,-1;1)=-\mathrm{Li}_3(-1)=\tfrac{3}{4}\zeta(3),$$

in the alternating-MZV notation $\zeta(\bar{s}_1,\dots)=\sum_{n_1>\cdots}\prod(\mathrm{sgn})^{n_i}/n_i^{s_i}$. Thus the $\mathbb{Q}$-algebra of HPL values at 1 is the algebra of alternating MZVs; its dimensions and the structure of its relations were mapped by Borwein–Bradley–Broadhurst–Lisoněk and, on the motivic side, Deligne and Goncharov. This is the practical reason HPLs are the right basis: the boundary data of a Feynman integral land automatically in a well-understood number-theoretic ring.

1.3.6 Argument transformations and the alphabet

Changing variables transports the singular set of the differential forms. Negation and inversion, and the map $(1-z)/(1+z)$, preserve the extended three-letter singular set $\{0,1,-1,\infty\}$, with appropriate endpoint and branch constants. Reflection $1-z$ preserves the two-letter subalphabet $\{0,1\}$ but sends the letter -1 to 2. Thus $H(-1;1-z)=\log(2-z)$ cannot in general be expressed using only the original three-letter alphabet at z. Squaring likewise introduces letters $\pm i$ when applied to a word containing -1.

The useful algorithm is to differentiate, partially fraction the transported kernels, integrate in the resulting alphabet, and fix the endpoint constant. For the two-letter dilogarithm this gives

$$\mathrm{Li}_2(z)+\mathrm{Li}_2(1-z)=\frac{\pi^2}{6}-\log z\log(1-z),\qquad 0<z<1. \tag{1.19}$$

Analytic continuation requires a common choice of branches. Duplication is the special closed identity

$$H(0,1;z^2)=2\big(H(0,1;z)-H(0,-1;z)\big).$$

These examples illustrate transport of alphabets, rather than universal closure under every fractional-linear or quadratic substitution.

1.3.7 Relation to Nielsen polylogarithms and to Li

The harmonic polylogarithms contain the Nielsen generalised polylogarithms $S_{n,p}(z)$ of Nielsen (1909), studied analytically by Kölbig (1986). Indeed $S_{n,p}(z)$ is, exactly the single HPL whose word

is n zeros followed by p ones,

$$S_{n,p}(z) = H(\underbrace{0,\dots,0}_{n},\underbrace{1,\dots,1}_{p};z), \tag{1.20}$$

of weight $n+p$ and depth p; the classical case is $S_{n,1}(z) = \mathrm{Li}_{n+1}(z)$, consistent with (1.15) for $n=1, p=1$. Thus the chain of inclusions reads: classical Li_s (depth-1 HPLs over $\{0,1\}$) $\subset$ Nielsen $S_{n,p}$ (two-letter words 0^n1^p) $\subset$ harmonic polylogarithms (arbitrary words over $\{0,1,-1\}$) $\subset$ multiple polylogarithms (the $z=1$ specialisations being MZVs and Euler sums). Each enlargement adds exactly the words the previous class could not name.

1.3.8 Finite truncations and Wolfram 15

Behind every iterated integral stands a nested *sum*, and behind every value at $z=1$ stands the partial sum of that series. These partial sums are the multiple (finite) harmonic numbers

$$H_N^{(s_1,\dots,s_k)} = \sum_{N\geq n_1>n_2>\cdots>n_k\geq 1} \prod_{i=1}^{k} \frac{1}{n_i^{s_i}},$$

generalising the ordinary $H_N = H_N^{(1)} = \sum_{n\leq N} 1/n$; their $N\to\infty$ limits are the admissible MZVs, and they are the natural objects in which the series expansions of ε-dependent Feynman integrals are stored. Wolfram 15 provides all of these as first-class functions: `HarmonicPolyLog` for $H(\vec{a};z)$ (with the index/sign conventions used above; recall `HarmonicPolyLog[{1},x]` $= -\ln(1-x)$), `MultiplePolyLog` and `MultipleZeta` for the limiting transcendentals, and `MultipleHarmonicNumber` for the finite sums $H_N^{(\vec{s})}$. With `DSolve` and `AsymptoticDSolveValue` now returning harmonic polylogarithms for Fuchsian systems with rational coefficients, the entire Remiddi–Vermaseren workflow—solve the differential equation, expand at the boundary into Euler sums, continue by the transformations the transformations just discussed—is available without leaving the kernel.

Remark 1.9. The choice of alphabet $\{0,1,-1\}$ is what fixes the singular set at $\{0,1,-1,\infty\}$ and hence ties the boundary values to *alternating* MZVs. Allowing the additional kernels $f_{1/2}(t) = 1/(t-\frac{1}{2})$ or roots of unity of higher order enlarges the class to the two-dimensional harmonic polylogarithms and to Goncharov's multiple polylogarithms with general singular points (the `GeneralizedPolyLog` of the previous section), at the price of boundary constants drawn from level-N colored MZVs rather than the level-2 Euler sums. For one-scale problems the three-letter alphabet is almost always enough, which is the empirical reason HPLs, not their more general cousins, became the working basis of the field.

1.4 Nielsen's generalized polylogarithms

Long before multiple zeta values acquired their modern algebraic name, Nielsen (1909) introduced a two-parameter family of higher polylogarithms that already pervades Lewin's *Polylogarithms and Associated Functions* (1981, Chapter 7). They are the simplest objects that go beyond the ordinary $\mathrm{Li}_s(z)$ while remaining genuinely "one-variable": a single argument z, two integer indices $n,p\geq 1$, and a closed integral representation. The reader who is comfortable with the dilogarithm reflection and Landen's transformation will find here their direct higher-weight descendants, and – once we translate into the language of the previous sections – their place as one distinguished corner of the multiple-polylogarithm world.

1.4.1 Definition and elementary reductions

Definition 1.10. *For integers $n, p \geq 1$ the* Nielsen generalized polylogarithm *is*

$$S_{n,p}(z) \;=\; \frac{(-1)^{n+p-1}}{(n-1)!\,p!} \int_0^1 \frac{(\ln t)^{n-1}\,\ln^p(1-zt)}{t}\,dt, \tag{1.21}$$

analytically continued from $|z| < 1$ to $z \in \mathbb{C} \setminus [1, \infty)$ along the cut of $\ln(1-zt)$. *Its* weight *is $n+p$ and its* depth *is p.*

The normalizing factorials are chosen exactly so that two boundary specializations collapse onto familiar functions. Setting $p = 1$ and integrating $\ln(1-zt)/t$ termwise against $(\ln t)^{n-1}$ recovers the classical Euler integral for the polylogarithm:

$$S_{n,1}(z) \;=\; \mathrm{Li}_{n+1}(z), \qquad n \geq 1. \tag{1.22}$$

Thus the entire first column of the Nielsen table is nothing but the ordinary polylogarithm shifted by one weight; in particular the lowest member is the dilogarithm,

$$S_{1,1}(z) \;=\; \mathrm{Li}_2(z). \tag{1.23}$$

At the opposite corner $n = 1$ the inner factor $\ln t$ disappears and (1.21) reduces to a pure power of $\ln(1-zt)$, giving the elementary $S_{0,p}(z) = \frac{(-1)^p}{p!}\ln^p(1-z)$ in the degenerate $n = 0$ row (we keep $n \geq 1$ in the sequel, but this boundary is the seed for the recursions below).

1.4.2 Series and the multiple-polylogarithm identification

Expanding $\ln^p(1-zt)$ as a power of the series $-\sum_{m\geq 1}(zt)^m/m$ and integrating against $(\ln t)^{n-1}/t$ turns (1.21) into a single power series in z whose coefficients are nested harmonic sums:

$$S_{n,p}(z) \;=\; \sum_{k\geq 1} \frac{z^k}{k^{\,n+1}}\, c_k^{(p)}, \qquad c_k^{(p)} = \sum_{k > m_1 > \cdots > m_{p-1} \geq 1} \frac{1}{m_1 \cdots m_{p-1}}, \tag{1.24}$$

so $c_k^{(1)} \equiv 1$ (reproducing (1.22)) and in general $c_k^{(p)}$ is a depth-$(p-1)$ harmonic sum, an instance of `MultipleHarmonicNumber`. Comparing (1.24) with the strictly nested sum of the article preamble identifies $S_{n,p}$ as a multiple polylogarithm with a single repeated argument:

$$\boxed{S_{n,p}(z) \;=\; \mathrm{Li}_{n+1,\underbrace{1,\ldots,1}_{p-1}}\!\left(z, \underbrace{1,\ldots,1}_{p-1}\right)} \tag{1.25}$$

of weight $n+p$ and depth p. Its iterated-integral word over the alphabet $\{0,1\}$ (with $0 \leftrightarrow dt/t$, $1 \leftrightarrow dt/(1-t)$) is

$$0^{\,n}\,1^{\,p} \;=\; \underbrace{0\cdots 0}_{n}\,\underbrace{1\cdots 1}_{p},$$

a single block of n zeros followed by a single block of p ones. This is the defining structural fact: among all multiple polylogarithms, the Nielsen family is exactly the "one initial zero-block, one trailing one-block" subclass. Everything peculiar about $S_{n,p}$ – the two-index labelling, the clean ladders, the duality below – is a shadow of that two-block word.

Remark 1.11 (Sign and normalization). With the positive kernel $dt/(1-t)$ the word 0^n1^p equals $S_{n,p}(z)$ with no extra sign. Replacing each 1-kernel by $dt/(t-1)$ gives the factor $(-1)^p$. No factor $(-1)^{p-1}$ belongs to either convention.

1.4.3 Integral and differential relations

Differentiating (1.24) (or (1.21)) in z lowers the leading index by one and threads the function back into the ladder:

$$\frac{d}{dz}\,S_{n,p}(z) \;=\; \frac{1}{z}\,S_{n-1,p}(z) \quad (n\geq 1), \qquad \frac{d}{dz}\,S_{1,p}(z) \;=\; \frac{1}{z}\,S_{0,p}(z) \;=\; \frac{(-1)^p}{p\,!}\,\frac{\ln^p(1-z)}{z}, \tag{1.26}$$

the second relation being the $n=1$ boundary, where the dt/t block is exhausted and the degenerate row $S_{0,p}(z)=\frac{(-1)^p}{p!}\ln^p(1-z)$ is the pure power of $\ln(1-z)$. The two clean ladder moves are the integral forms of these derivatives,

$$\text{zero-block ladder:}\quad S_{n,p}(z)=\int_0^z S_{n-1,p}(u)\,\frac{du}{u} \qquad (n\geq 1), \tag{1.27}$$

$$\text{one-block ladder:}\quad S_{0,p}(z)=\int_0^z S_{0,p-1}(u)\,\frac{du}{1-u} \qquad (p\geq 1), \tag{1.28}$$

with seed $S_{0,0}(z)=1$ (the empty word): (1.27) peels the outermost 0 from the word 0^n1^p, and (1.28) peels the outermost 1 from the pure one-block 1^p; in iterated-integral terms both are simply "integrate against the next letter of 0^n1^p." The practical content is that the whole Nielsen table is generated from the empty word by first applying the 1-kernel integral $\int_0^z(\cdot)\,du/(1-u)$ a total of p times to build $S_{0,p}$ (so $S_{0,1}(z)=-\ln(1-z)$, $S_{0,2}(z)=\frac{1}{2}\ln^2(1-z)$, and so on), then the 0-kernel integral $\int_0^z(\cdot)\,du/u$ a total of n times to prepend the zeros; this is exactly the Fuchsian recursion that `HarmonicPolyLog` encodes, and it is why `DSolve` now returns Nielsen polylogs for the relevant rational-coefficient ODEs.

1.4.4 Special values

The point $z=1$ and Hoffman's duality

Setting $z=1$ in (1.25) turns the multiple polylogarithm into a multiple zeta value with the same two-block word. The first column reproduces Euler,

$$S_{n,1}(1)=\zeta(n+1), \qquad S_{1,2}(1)=\zeta(3), \tag{1.29}$$

and in general

$$S_{n,p}(1) \;=\; \zeta\big(n+1,\{1\}^{p-1}\big), \tag{1.30}$$

the MZV whose argument string is $n+1$ followed by $p-1$ ones (admissible because the leading entry $n+1\geq 2$). Now the iterated-integral word 0^n1^p of $S_{n,p}$ has a manifest symmetry: reversing it and exchanging $0\leftrightarrow 1$ sends $0^n1^p\mapsto 0^p1^n$, the word of $S_{p,n}$. This reversal-complement is exactly Hoffman's MZV *duality* (the Kontsevich integral symmetry $\omega\mapsto\overline{\omega}^{\mathrm{rev}}$), so

$$\boxed{\zeta\big(n+1,\{1\}^{p-1}\big) \;=\; \zeta\big(p+1,\{1\}^{n-1}\big), \qquad \text{equivalently} \qquad S_{n,p}(1)=S_{p,n}(1).} \tag{1.31}$$

This is a striking statement: the value of a Nielsen polylogarithm at 1 is invariant under swapping its two indices, even though $S_{n,p}(z)$ and $S_{p,n}(z)$ are wholly different functions of z. For $p=1$ it gives $\zeta(n+1)=\zeta(2,\{1\}^{n-1})$; the first nontrivial case is $S_{1,2}(1)=S_{2,1}(1)$, i.e. $\zeta(2,1)=\zeta(3)$, Euler's reflection. We confirmed (1.30) and (1.31) symbolically against `MultipleZeta` (e.g. $S_{1,2}(1)-\zeta(2,1)$ and $S_{n,p}(1)-S_{p,n}(1)$ vanish to >70 digits for all $n,p\leq 4$).

The points $z=-1$ and $z=\frac{1}{2}$ (Kölbig 1986)

At $z=-1$ the column reduction (1.22) gives $S_{n,1}(-1)=\mathrm{Li}_{n+1}(-1)=-(1-2^{-n})\,\zeta(n+1)$, and the first off-column value is the elegant

$$S_{1,2}(-1) \;=\; \tfrac{1}{8}\,\zeta(3). \tag{1.32}$$

Higher entries $S_{n,p}(-1)$ reduce (Kölbig 1986) to the alternating-MZV basis at their weight; for example $S_{2,2}(-1)$ and $S_{1,3}(-1)$ already require $\mathrm{Li}_4(\frac{1}{2})$ alongside π^4, $\ln^4 2$, $\pi^2\ln^2 2$ and $\zeta(3)\ln 2$, the unmistakable level-2 weight-4 signature (we recovered both reductions by integer-relation search). At the binary point $z=\frac{1}{2}$ the cleanest closed form is

$$\boxed{S_{1,2}\big(\tfrac{1}{2}\big) \;=\; \tfrac{1}{8}\,\zeta(3) \;-\; \tfrac{1}{6}\,\ln^3 2.} \tag{1.33}$$

We obtained (1.33) by PSLQ over $\{\zeta(3),\pi^2\ln 2,\ln^3 2\}$ – the relation $24\,S_{1,2}(\frac{1}{2})-3\,\zeta(3)+4\ln^3 2=0$ – and verified it to 40 digits. (Compare Lewin's dilogarithm value $\mathrm{Li}_2(\frac{1}{2})=\frac{\pi^2}{12}-\frac{1}{2}\ln^2 2$: the same flavour, one weight up.)

1.4.5 Functional equations

The reader who knows Lewin's dilogarithm chapter expects three staples – reflection $z\leftrightarrow 1-z$, inversion $z\leftrightarrow 1/z$, and the Landen ladder – to survive at higher weight. They do, as specializations of the harmonic-polylogarithm argument transformations of the previous section, but they grow extra lower-weight "tail" terms.

Reflection $z\mapsto 1-z$

At weight 2 the only reflection is the dilogarithm law itself, here in Nielsen dress ($n=p=1$):

$$S_{1,1}(z)+S_{1,1}(1-z) \;=\; \frac{\pi^2}{6}-\ln z\,\ln(1-z), \tag{1.34}$$

which is exactly Euler's reflection for Li_2. At weight 3 the genuinely new instance is the $S_{1,2}$ reflection; carrying out the integer-relation search over the natural weight-3 basis yields the clean

$$\boxed{S_{1,2}(z) \;=\; \zeta(3) \;-\; \mathrm{Li}_3(1-z) \;+\; \ln(1-z)\,\mathrm{Li}_2(1-z) \;+\; \tfrac{1}{2}\,\ln z\,\ln^2(1-z).} \tag{1.35}$$

We verified (1.35) to 40 digits at several rational and one negative argument. Two consistency checks are worth noting. Putting $z=1$ collapses the right side to $\zeta(3)-\mathrm{Li}_3(0)+0+0=\zeta(3)$, reproducing (1.29); and adding (1.35) to its $z\mapsto 1-z$ image recovers the symmetric Euler relation $S_{1,2}(z)+S_{1,2}(1-z)=2\zeta(3)-\mathrm{Li}_3(z)-\mathrm{Li}_3(1-z)+\cdots$ with the logarithmic tails arranging themselves into the $\zeta(3)+\mathrm{Li}_3$ shuffle. Identity (1.35) is precisely the statement that the iterated integral of 011 from 0 to z and from 0 to $1-z$ differ by an explicit lower-weight boundary correction – the weight-3 avatar of the dilogarithm reflection.

A finite inversion formula

The earlier draft inferred failure of closure from a search in the basket $\{S_{1,2}(z),\zeta(3),\pi^2\log z,\log^3 z\}$. That basket omits polylogarithms and mixed log products needed by the actual transformation. Inversion does close in the Nielsen family after those products and endpoint constants are included. The following direct coefficient construction makes the distinction precise.

Proposition 1.12 (Nielsen inversion with explicit endpoint data)**.** *For integers $n, p \geq 1$, let $u = \log(-z)$ on $\mathbb{C} \setminus [0, \infty)$ and define*

$$\mathcal{N}_{n,p}(z) = \frac{u^{n+p}}{(n+p)!} + \sum_{r=1}^{p} \sum_{\ell=0}^{p-r} (-1)^{\ell} \binom{n+\ell-1}{\ell} \frac{u^{p-r-\ell}}{(p-r-\ell)!} S_{n+\ell,r}(z).$$

Then

$$S_{n,p}(1/z) = (-1)^n \mathcal{N}_{n,p}(z) + C_{n,p}(u),$$

where $C_{n,p}$ is a polynomial of degree at most $n-1$. Its coefficients are determined entirely by Nielsen values at -1: for $0 \leq j < n$,

$$C_{n,p}^{(j)}(0) = (-1)^j S_{n-j,p}(-1) - (-1)^n \sum_{r=1}^{p} (-1)^{p-r} \binom{n-j+p-r-1}{p-r} S_{n-j+p-r,r}(-1).$$

All logarithms and polylogarithms use the common principal continuation.

Proof. First take $z < 0$. Extend the notation by $S_{0,r}(z) = (-1)^r \log^r(1-z)/r!$ and $S_{0,0} = 1$. Since $\log(1 - 1/z) = \log(1-z) - u$, the binomial theorem gives

$$S_{0,p}(1/z) = \sum_{r=0}^{p} \frac{u^{p-r}}{(p-r)!} S_{0,r}(z).$$

The ladder $dS_{m,r}/du = S_{m-1,r}$ implies, by Pascal's identity, that $d\mathcal{N}_{n,p}/du = \mathcal{N}_{n-1,p}$, with $\mathcal{N}_{0,p} = S_{0,p}(1/z)$. On the other hand $dS_{n,p}(1/z)/du = -S_{n-1,p}(1/z)$. Consequently the difference has vanishing nth derivative in u and is a polynomial of degree at most $n-1$. Evaluate its first n derivatives at $u = 0$, namely $z = -1$, to obtain the displayed endpoint coefficients. Both sides are holomorphic on the stated slit plane, so the identity theorem supplies the continuation. □

At weight three this gives the particularly useful exact identity

$$S_{1,2}(1/z) = -S_{1,2}(z) + \mathrm{Li}_3(z) - \log(-z)\,\mathrm{Li}_2(z) - \frac{1}{6}\log^3(-z) + \zeta(3).$$

Closure of a function representation does not establish independence of its values. Other argument changes should likewise be derived by transporting the kernels in the appropriate alphabet, rather than by an unsuccessful search in an underspecified basket.

1.4.6 Placement inside the multiple-polylogarithm world

Equation (1.25) identifies $S_{n,p}$ as the "one trailing block" corner of the depth-p multiple polylogarithms: the words $0^n 1^p$ form a single two-parameter slice of the full $\{0,1\}^*$ alphabet, the slice closest to the ordinary polylogarithm (recovered at $p = 1$) and stable under the two operations – evaluation at 1 to MZVs, and the duality swap $n \leftrightarrow p$ – that organize the modern theory. This is exactly why the Nielsen polylogarithms are the natural bridge between Lewin's classical apparatus and the MZV algebra: every Lewin-style identity (reflection, Landen, the special values at $\pm 1, \frac{1}{2}$) is the $p \leq 2$ or $n \leq 2$ shadow of a multiple-polylogarithm identity, and conversely the duality $S_{n,p}(1) = S_{p,n}(1)$ is the first place a classical reader meets a genuinely MZV-theoretic symmetry with no single-variable analogue. In the Wolfram Language the family is a first-class function: `PolyLog[n,p,z]` is exactly $S_{n,p}(z)$, with `PolyLog[n,1,z]` agreeing with `PolyLog[n+1,z]` per (1.22), and it sits one specialization step from `MultiplePolyLog` via (1.25) and one from `MultipleZeta` via (1.30).

1.5 The common algebra and the motivic boundary

The nested series and outer-letter-first integral describe one function. Choosing the alphabet distinguishes general multiple polylogarithms, cyclotomic multiple polylogarithms, HPLs and the Nielsen subfamily. General Goncharov letters can be arbitrary complex numbers; restricting them to roots of unity gives only the cyclotomic class.

One must distinguish the argument patterns of the two products. Shuffle at a common endpoint gives $\mathrm{Li}_1(z)^2 = 2\,\mathrm{Li}_{1,1}(z,1)$, whereas stuffle gives $\mathrm{Li}_1(z)^2 = 2\,\mathrm{Li}_{1,1}(z,z)+\mathrm{Li}_2(z^2)$. Both follow directly from their respective representations and therefore give a nontrivial identity when compared.

Iterated integrals also carry a coproduct organized by subsequences of marked points. Its reduced components and the associated symbol organize functional equations by weight. They lose some period constants: vanishing symbol or reduced coproduct alone is not a proof of a zero function. Endpoint constants and branches must be fixed separately. For the dilogarithm the differential $d\,\mathrm{Li}_2(z) = -\log(1-z)d\log z$ makes this elementary; differentiating Euler reflection and evaluating an endpoint proves the identity exactly.

Motivic MZVs [43] provide an algebraic setting in which dimension and spanning theorems can be proved. The numerical period map transports identities and upper bounds. Its injectivity, and thus the corresponding lower bounds for numerical periods, is a separate conjectural issue. The same caution will apply to the Gaussian and Eisenstein depth experiments.

Chapter 2

Cyclotomic values and the polygamma bridge

Roots of unity turn additive Fourier coordinates into arithmetic character coordinates. The same finite transform carries Hurwitz zeta, positive-order polygamma and Clausen values. It explains which half of a polylogarithm is elementary at a given weight and which character values remain to be studied.

2.1 The two-way exact DFT

We use the Clausen convention

$$\mathrm{Cl}_{2m}(\theta) = \Im\,\mathrm{Li}_{2m}(e^{i\theta}), \qquad \mathrm{Cl}_{2m+1}(\theta) = \Re\,\mathrm{Li}_{2m+1}(e^{i\theta}).$$

The sine series at odd weight is a different function; its rational-angle values reduce to powers of π through Bernoulli polynomials.

Law 2.1 (forward direction; classical)**.** *For every $n \geq 2$, every $q \geq 1$, and $\gcd(p, q) = 1$,*

$$\mathrm{Li}_n\big(e^{2\pi i p/q}\big) = \frac{(-1)^n}{q^n\,(n-1)!} \sum_{k=1}^{q-1} e^{2\pi i k p/q}\,\psi^{(n-1)}\Big(\frac{k}{q}\Big) \;+\; \frac{\zeta(n)}{q^n}.$$

This is identity 10.08.03.0007.01 of the `functions.wolfram.com` *PolyLog* reference [2]; it is provable by rearranging the absolutely convergent Hurwitz-zeta series, and it is the workhorse of the project's cyclotomic reducer (`cycloExact`): a finite polygamma sum, exact and fast, resolving Li_n at *every* root of unity with no PSLQ, no squarefree-radical basis, and no nested-radical limitation.

Result 2.2 (inverse direction). Inverting the discrete Fourier transform over $\mathbb{Z}/q$ gives, with $\zeta_q = e^{2\pi i/q}$,

$$\psi^{(n-1)}\Big(\frac{k}{q}\Big) = (-1)^n\,(n-1)!\;q^{\,n-1} \sum_{p=0}^{q-1} \zeta_q^{-pk}\;\mathrm{Li}_n\big(\zeta_q^{\,p}\big).$$

[historical experiment: verified to 40 digits at weights $2, 3$ across $q \leq 21$; re-verified by mpmath at 70 dps for $n - 1 = 1, 2, 3$ over the full store grid]

The bridge is therefore an *exact finite Fourier transform, in both directions.* Three non-obvious consequences are recorded.

1. **DFT-conjugacy of the relation lattices.** The distribution relation $\sum_j \mathrm{Li}_n(z\,\zeta_m^j) = m^{1-n}\,\mathrm{Li}_n(z^m)$ Fourier-transforms into the polygamma multiplication theorem, and polylogarithm inversion/reflection corresponds to the symmetry $k \leftrightarrow q-k$ of the ψ-grid. This is *why* the two relation lattices — computed independently by the project's two reducers — carry identical dimension counts $\varphi(q)/2$ per denominator grid.

2. **The parity split, and Catalan as the $q=4$ odd component.** The imaginary parts are the odd Fourier sector under $k \mapsto q-k$. The usual even-weight Clausen values belong to this sector, while odd-weight Clausen values are real parts in the even sector. In particular

$$G = \mathrm{Im}\,\mathrm{Li}_2(i) = \frac{\psi^{(1)}(1/4) - \psi^{(1)}(3/4)}{16},$$

   which identifies the well-known "arguments 1/4 and 3/4" pairing (Math.SE 893486) as the $q=4$ odd component of the parity split.

3. **Certificate transport.** Cyclotomic-polylogarithm proof certificates can in principle be transported to and from ψ-side lattice certificates through the exact $\mathbb{Q}(\zeta_q)$ DFT matrix. The ψ-side elementary atoms ($\pi^2\csc^2$ values) are far lighter than the sine-product algebraic objects that constrain the Γ-certificate sweeps on large grids, so the transport is a proposed escape route from that bottleneck (not yet implemented).

2.2 Clausen-type laws for polygamma values at rational arguments

MathWorld's *Polygamma Function* page lists fifteen closed forms in its special-cases block, eqs. (22)–(36) ($\psi^{(1)}$, $\psi^{(2)}$, $\psi^{(3)}$ at p/q with $q \in \{2,3,4,6\}$) in Catalan/Clausen/β atoms, several of them traditionally presented as isolated curiosities. All of them — and their extension to *every* rational argument — are instances of three sine/cosine-weighted Clausen-DFT laws, obtained by splitting the inverse DFT of Result 2.2 into its elementary (reflection) half and its Clausen half.

Law 2.3 (trigamma; Lewin 1991)**.** *For $0 < p < q$, $\gcd(p,q) = 1$,*

$$\psi^{(1)}\Big(\frac{p}{q}\Big) = \frac{\pi^2}{2\sin^2(\pi p/q)} + 2q \sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \sin\Big(\frac{2\pi jp}{q}\Big)\,\mathrm{Cl}_2\Big(\frac{2\pi j}{q}\Big).$$

[historical experiment: verified at $q = 3,4,5,8,12$ to 10^{-46} or better, then over the full store grid to 10^{-50} in-kernel and at 70 dps in mpmath]

This law is classical — MathWorld's *Trigamma Function* eq. (14) cites Lewin (1991) [1] — and was independently re-derived here from the DFT bridge before the attribution was spotted; the project's records explicitly re-frame the finding as *placement*, not novelty. MathWorld's scattered coefficients are its small-q shadows: the $3\sqrt{3}$ of $q=3$ is $2q\sin(2\pi/3)$, the $8G$ of $q=4$ is $2q\ \mathrm{Cl}_2(\pi/2)$.

Law 2.4 (tetragamma)**.**

$$\psi^{(2)}\Big(\frac{p}{q}\Big) = -\pi^3\cot\Big(\frac{\pi p}{q}\Big)\csc^2\Big(\frac{\pi p}{q}\Big) - 2q^2\left(\zeta(3) + 2\sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \cos\Big(\frac{2\pi jp}{q}\Big)\,\mathrm{Cl}_3\Big(\frac{2\pi j}{q}\Big) + \varepsilon_q\right),$$

where $\varepsilon_q = (-1)^p\,\mathrm{Cl}_3(\pi)$ for even q, $\varepsilon_q = 0$ for odd q, and $\mathrm{Cl}_3(\pi) = -\frac{3}{4}\zeta(3)$. [historical experiment: verified over the full store grid to 10^{-50} in-kernel and at 70 dps in mpmath]

Law 2.5 (pentagamma)**.**

$$\psi^{(3)}\Big(\frac{p}{q}\Big) = \pi^4\Big(2\cot^2\Big(\frac{\pi p}{q}\Big)\csc^2\Big(\frac{\pi p}{q}\Big) + \csc^4\Big(\frac{\pi p}{q}\Big)\Big) + 12\,q^3\sum_{j=1}^{\lfloor (q-1)/2\rfloor}\sin\Big(\frac{2\pi jp}{q}\Big)\ \mathrm{Cl}_4\Big(\frac{2\pi j}{q}\Big),$$

equivalently, in difference form, $\psi^{(3)}(p/q) - \psi^{(3)}(1-p/q) = 24\,q^3\sum_j \sin(2\pi jp/q)\ \mathrm{Cl}_4(2\pi j/q)$*.* *[historical experiment: same verification battery]*

The $\psi^{(3)}$ law is the Lewin law one weight up, by the same derivation; no literature citation has been located, and the project records it as "presumably in Lewin's orbit." MathWorld's $162\sqrt{3}$ and $768\beta(4)$ coefficients are $12q^3\sin(2\pi j/q)$-instances of the one-sided law ($24q^3$ in the difference form).

Result 2.6 (new instances beyond the MathWorld table). Among the instances at $q \in \{5, 8, 12\}$:

$$\psi^{(1)}\big(\tfrac{1}{5}\big) = \frac{\pi^2}{2\sin^2(\pi/5)} + 10\Big[\sin\tfrac{2\pi}{5}\ \mathrm{Cl}_2\big(\tfrac{2\pi}{5}\big) + \sin\tfrac{4\pi}{5}\ \mathrm{Cl}_2\big(\tfrac{4\pi}{5}\big)\Big],$$

$$\psi^{(1)}\big(\tfrac{1}{12}\big) = 40\,G + \frac{4\pi^2}{(\sqrt{3}-1)^2} + 30\sqrt{3}\ \mathrm{Cl}_2\big(\tfrac{2\pi}{3}\big),$$

the latter after duplication/triplication canonicalization collapses the raw five-atom DFT form onto the canonical atom set. Also, MathWorld's eq. (27) simplifies: from $\mathrm{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$,

$$\psi^{(2)}\big(\tfrac{1}{3}\big) = -\frac{4\pi^3}{3\sqrt{3}} - 26\,\zeta(3).$$

Result 2.7 (the polygamma–Clausen store). The corpus now carries `identities/polygamma-clausen-values.wl`: 54 entries $\psi^{(m)}(p/q)$ for $m \in \{1, 2, 3\}$, $q \in \{3, 4, 5, 6, 8, 12\}$, all p coprime to q (18 arguments $\times$ 3 orders), in canonicalized Clausen atoms. The residual atom basis across the store is

$$\{G,\ \beta(4),\ \zeta(3)\}\ \cup\ \mathrm{Cl}_2\big\{\tfrac{2\pi}{3}, \tfrac{2\pi}{5}, \tfrac{4\pi}{5}, \tfrac{\pi}{4}\big\}$$
$$\cup\ \mathrm{Cl}_3\big\{\tfrac{2\pi}{5}, \tfrac{\pi}{4}, \tfrac{\pi}{6}\big\}\ \cup\ \mathrm{Cl}_4\big\{\tfrac{2\pi}{3}, \tfrac{2\pi}{5}, \tfrac{4\pi}{5}, \tfrac{\pi}{4}\big\}.$$

Every entry is verified in-kernel to $|\mathrm{lhs} - \mathrm{rhs}| < 10^{-50}$ at 60 digits (including a disk round-trip) and independently by an mpmath script at 70 dps (tolerance 10^{-55}) that re-derives the raw DFT bridge, the three laws, all 17 Clausen canonicalization relations, and the L-function bridges of Section 2.3 from mpmath primitives, with positive and negative PSLQ controls.

2.3 The boundary of the odd-weight Clausen collapse

A folklore rule says that odd-weight Clausen values at rational angles collapse to rational multiples of zeta values ($\mathrm{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$ being the standard example). The store emission *sharpened the rule's scope*:

Proposition 2.8 (Crystallographic-angle reductions)**.** *At denominators* $q \in \{1, 2, 3, 4, 6\}$*, the odd Clausen value* $\mathrm{Cl}_3(2\pi j/q) = \Re\,\mathrm{Li}_3(e^{2\pi ij/q})$ *reduces to a rational multiple of* $\zeta(3)$*. At* $q = 5, 8, 12$ *the*

character decomposition instead gives

$$\mathrm{Cl}_3(2\pi/5) = -\frac{6}{25}\zeta(3) + \frac{\sqrt{5}}{4}L(3,\chi_5),$$
$$\mathrm{Cl}_3(\pi/4) = -\frac{3}{256}\zeta(3) + \frac{1}{\sqrt{2}}L(3,\chi_8),$$
$$\mathrm{Cl}_3(\pi/6) = \frac{1}{24}\zeta(3) + \frac{\sqrt{3}}{2}L(3,\chi_{12}).$$

Here χ_D is the primitive even quadratic character of conductor D.

These are exact coordinate identities from the residue-class Fourier sum. No theorem asserting that the three L-values are independent of $\zeta(3)$ or π^3 is supplied. The source searches at height 10^{10} are numerical evidence in a stated basis, not an "if and only if" classification of all possible reductions.

2.4 The fundamental constants, by conductor

The residue classes of a cyclotomic series can be organized either by additive Fourier modes or by Dirichlet characters. Sine, or imaginary, components use odd characters, while cosine, or real, components use even characters. With the Clausen convention used here, even-index Clausen functions are sine components and odd-index ones are cosine components. Distribution relations first remove non-coprime residues; the remaining finite character transform has $\varphi(n)/2$ formal odd coordinates. Its coefficients are generally algebraic, so this coordinate count is not a proof of the dimension of the numerical $\mathbb{Q}$-span of the periods.

For the levels $8, 12, 24$, convenient generators are

$$V = \mathrm{Cl}_2(\pi/3), \quad G = \mathrm{Cl}_2(\pi/2), \quad W_8 = \mathrm{Cl}_2(\pi/4), \quad W_{24} = \mathrm{Cl}_2(\pi/12).$$

There are no primitive odd characters of conductor 12; its odd characters are induced from conductors 3 and 4. This explains the distribution reductions below. Conductors 3 and 4 each have one primitive odd character, while conductor 5 has two conjugate primitive odd characters. The source's printed sequence of primitive-character counts started with two incorrect zeros; it is not used here.

Their canonical L-value (trigamma) forms are

$$G = \frac{\psi^{(1)}(\frac{1}{4}) - \pi^2}{8}, \qquad \sqrt{3}\,V = \frac{3\psi^{(1)}(\frac{1}{3}) - 2\pi^2}{6} \quad \Big(\text{i.e. } \psi^{(1)}(\tfrac{1}{3}) = \tfrac{2\pi^2}{3} + 2\sqrt{3}\,V\Big). \tag{2.1}$$

2.5 Reduction tables and the settled level-24 question

Every $\mathrm{Cl}_2(2\pi a/n)$ at $n \in \{8, 12, 24\}$ reduces to the basis above (PSLQ, ≥ 110 digits, Champernowne canary; all confirmed in Wolfram):

$$\begin{aligned}
\text{level } 12: &\quad \mathrm{Cl}_2(\tfrac{\pi}{6}) = \tfrac{2}{3}G + \tfrac{1}{4}V, \quad \mathrm{Cl}_2(\tfrac{5\pi}{6}) = \tfrac{2}{3}G - \tfrac{1}{4}V, \quad \mathrm{Cl}_2(\tfrac{2\pi}{3}) = \tfrac{2}{3}V;\\
\text{level } 8: &\quad \mathrm{Cl}_2(\tfrac{3\pi}{4}) = W_8 - \tfrac{1}{2}G;\\
\text{level } 24: &\quad \mathrm{Cl}_2(\tfrac{5\pi}{12}) = \tfrac{16W_8+24W_{24}-4G-3V}{24}, \quad \mathrm{Cl}_2(\tfrac{7\pi}{12}) = \tfrac{4W_8+6W_{24}-3G}{6},\\
&\quad \mathrm{Cl}_2(\tfrac{11\pi}{12}) = \tfrac{24W_{24}-8G-3V}{24}.
\end{aligned}$$

The displayed formulas prove that the level-24 values are generated by G, V, W_8, W_{24}. They give an upper bound of four on their rational span. The source did not prove that W_{24} lies outside the span of the other three, so the minimal number of numerical generators remains open. This distinction is central when a symbolic conductor count is compared with a PSLQ experiment.

2.6 A new closed form

Since $\mathrm{Cl}_2(\pi/12) = W_{24}$ is itself fundamental, the cleanest carrier is the dilogarithm at the primitive 24th root $e^{\mathrm{i}\pi/12}$:

$$\mathrm{Li}_2(e^{\mathrm{i}\pi/12}) = \frac{73\pi^2}{576} + \mathrm{i}\, W_{24}, \tag{2.2}$$

the real part being $\frac{\pi^2}{6} - \frac{\pi}{2}\cdot\frac{\pi}{12} + \frac{1}{4}(\frac{\pi}{12})^2$. Pulling this back through $z = 1/(1-e^{\mathrm{i}\theta})$ gives Reshetnikov-style closed forms at the corresponding $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{6})$ points; the cleanest stays the root-of-unity form (2.2).

2.7 A weight parity swap

The same character parity that makes Cl_2 (weight 2, odd characters) carry deep constants reverses one weight up. For primitive odd χ, the positive odd integer values are algebraic multiples of π^s; since $\operatorname{Im}\mathrm{Li}_3(e^{\mathrm{i}\theta}) = \sum \sin(k\theta)/k^3$ pairs with odd characters at $s = 3$,

$$\operatorname{Im}\mathrm{Li}_3(e^{2\pi\mathrm{i}a/n}) \in \mathbb{Q}\cdot\pi^3 \quad \text{for all } a, n, \tag{2.3}$$

e.g. $\operatorname{Im}\mathrm{Li}_3(e^{2\pi\mathrm{i}/3}) = \frac{2\pi^3}{81}$, $\operatorname{Im}\mathrm{Li}_3(\mathrm{i}) = \frac{\pi^3}{32}$, $\operatorname{Im}\mathrm{Li}_3(e^{\mathrm{i}\pi/3}) = \frac{5\pi^3}{162}$, $\operatorname{Im}\mathrm{Li}_3(e^{2\pi\mathrm{i}/5}) = \frac{4\pi^3}{125}$. Thus at weight 3 the *imaginary* part at roots of unity is elementary, and the deep constants ($\zeta(3)$ and the even-character $L(3,\chi)$) live in the *real* part—the exact reverse of weight 2. (On the line $\operatorname{Re} z = 1$ the picture is different again: there $\operatorname{Im}\mathrm{Li}_3(1+\mathrm{i})$ is the central-binomial Gaussian tower constant, because $1+\mathrm{i}$ is not a root of unity; see `polylog-vertical-line`.)

Chapter 3

Algebraic arguments, integrals and ladders

Away from roots of unity, functional equations still move values onto structured sets of arguments. Vertical lines and half-angle substitutions give tractable examples; unit relations in number fields lead to longer ladders. The organizing principle is to understand the map of arguments before choosing the constants in an integer-relation search.

3.1 The problem and the key geometric fact

MSE 1410276 records

$$\operatorname{Re}\operatorname{Li}_2(1\pm \mathrm{i}\sqrt{3}) = \frac{\pi^2}{24} - \tfrac{1}{2}\ln^2 2 - \tfrac{1}{4}\operatorname{Li}_2(\tfrac{1}{4}), \tag{3.1}$$

with $\operatorname{Im}\operatorname{Li}_2(1+\mathrm{i}\sqrt{3}) = \frac{\pi\ln 2}{2} + \frac{5\sqrt{3}}{72}\big[\psi_1(\frac{1}{3}) - \psi_1(\frac{2}{3})\big]$ (a/1410829, a/1410705). The point $1+\mathrm{i}\sqrt{3} = 2e^{\mathrm{i}\pi/3}$, but the structurally decisive fact is that for any $z = 1+\mathrm{i}y$ the reflection point $1-z = -\mathrm{i}y$ is *purely imaginary*. This is exactly what makes the dilogarithm functional equations collapse.

3.2 The weight-2 vertical-line theorem

Theorem 3.1. *For $y > 0$,*

$$\operatorname{Re}\operatorname{Li}_2(1+\mathrm{i}y) = \frac{\pi^2}{6} - \tfrac{1}{2}\ln(1+y^2)\ln y - \frac{\pi}{2}\arctan y - \tfrac{1}{4}\operatorname{Li}_2(-y^2), \tag{3.2}$$

$$\operatorname{Im}\operatorname{Li}_2(1+\mathrm{i}y) = \frac{\pi}{4}\ln(1+y^2) - \arctan y\,\ln y + \operatorname{Ti}_2(y), \qquad \operatorname{Ti}_2(y) = \operatorname{Im}\operatorname{Li}_2(\mathrm{i}y). \tag{3.3}$$

Proof. The reflection law $\operatorname{Li}_2(z)+\operatorname{Li}_2(1-z) = \zeta(2) - \ln z\ln(1-z)$ at $z = 1+\mathrm{i}y$, $1-z = -\mathrm{i}y$, gives $\operatorname{Li}_2(1+\mathrm{i}y) = \zeta(2) - \ln(1+\mathrm{i}y)\ln(-\mathrm{i}y) - \operatorname{Li}_2(-\mathrm{i}y)$. The duplication law $\operatorname{Li}_2(\mathrm{i}y)+\operatorname{Li}_2(-\mathrm{i}y) = \frac{1}{2}\operatorname{Li}_2(-y^2)$ together with $\overline{\operatorname{Li}_2(w)} = \operatorname{Li}_2(\bar{w})$ yields $\operatorname{Re}\operatorname{Li}_2(-\mathrm{i}y) = \frac{1}{2}\big(\operatorname{Li}_2(\mathrm{i}y) + \operatorname{Li}_2(-\mathrm{i}y)\big) = \frac{1}{4}\operatorname{Li}_2(-y^2)$ and $\operatorname{Im}\operatorname{Li}_2(-\mathrm{i}y) = -\operatorname{Ti}_2(y)$. Expanding $\ln(1+\mathrm{i}y) = \frac{1}{2}\ln(1+y^2) + \mathrm{i}\arctan y$ and $\ln(-\mathrm{i}y) = \ln y - \frac{\mathrm{i}\pi}{2}$ and taking real and imaginary parts gives (3.2)–(3.3). □

On the line $\operatorname{Re} z = 1$ the dilogarithm thus splits into a *real* part carried by the even real dilogarithm $\operatorname{Li}_2(-y^2)$ and an *imaginary* part carried by the odd inverse-tangent integral $\operatorname{Ti}_2(y)$. Equation (3.1) is the case $y = \sqrt{3}$ (using $\operatorname{Li}_2(-3) = \operatorname{Li}_2(\frac{1}{4}) - \zeta(2) + 2\ln 2\ln\frac{2}{3}$), and the trigamma form of the imaginary part is just $\operatorname{Ti}_2(\sqrt{3}) = \frac{\pi}{6}\ln 3 + \frac{5\sqrt{3}}{72}[\psi_1(\frac{1}{3}) - \psi_1(\frac{2}{3})]$ substituted into (3.3).

3.3 The Clausen form at $y = \tan\theta$

Since $1 + \mathrm{i}\tan\theta = e^{\mathrm{i}\theta}/\cos\theta$, Theorem 3.1 combines with Ramanujan's $\mathrm{Ti}_2(\tan\theta) = \theta\ln\tan\theta + \frac{1}{2}\big[\mathrm{Cl}_2(2\theta) + \mathrm{Cl}_2(\pi - 2\theta)\big]$ so that the $\theta\ln\tan\theta$ terms cancel, leaving a pure-Clausen imaginary part:

$$\operatorname{Im}\mathrm{Li}_2(1 + \mathrm{i}\tan\theta) = -\frac{\pi}{2}\ln\cos\theta + \tfrac{1}{2}\big[\mathrm{Cl}_2(2\theta) + \mathrm{Cl}_2(\pi - 2\theta)\big]. \tag{3.4}$$

Corollary 3.2. $\operatorname{Re}\mathrm{Li}_2(1+\mathrm{i}) = \dfrac{\pi^2}{16}$ *and* $\operatorname{Im}\mathrm{Li}_2(1+\mathrm{i}) = \dfrac{\pi\ln 2}{4} + G$ *(with G Catalan's constant), from $y = 1$ in Theorem 3.1 (using $\mathrm{Li}_2(-1) = -\pi^2/12$, $\mathrm{Ti}_2(1) = G$).*

3.4 Weight 3: a half-lifted symmetry

At weight 3 the real/imaginary split survives only in part. The *real* part still closes at special y:

$$\operatorname{Re}\mathrm{Li}_3(1+\mathrm{i}) = \frac{35}{64}\zeta(3) + \frac{\pi^2\ln 2}{32}, \qquad \operatorname{Re}\mathrm{Li}_3(1+\mathrm{i}\sqrt{3}) = \frac{35\zeta(3) + 18\,\mathrm{Li}_3(\frac{1}{4}) + 6\pi^2\ln 2 - 24\ln^3 2}{144}, \tag{3.5}$$

the second being the exact weight-3 analogue of the $\mathrm{Li}_2(\frac{1}{4})$ in (3.1). For the general real part, the displayed calculation introduces a further integral: differentiating, $\partial_y \operatorname{Re}\mathrm{Li}_3(1+\mathrm{i}y) = -\big(\operatorname{Im}\mathrm{Li}_2(1+\mathrm{i}y) - y\,\operatorname{Re}\mathrm{Li}_2(1+\mathrm{i}y)\big)/(1+y^2)$, and integrating the Ti_2 contribution produces $\int_0^y \arctan^2(t)/t\,dt$, a weight-three inverse-tangent-square integral. The displayed special cases at $y = 1, \sqrt{3}$ simplify, without giving an elementary-value classification for every rational angle. The *imaginary* part at $1+\mathrm{i}$ is represented here by the central-binomial Gaussian tower constant

$$\operatorname{Im}\mathrm{Li}_3(1+\mathrm{i}) = \mathcal{F}_3 - \frac{\pi^3}{192}, \qquad \mathcal{F}_3 = \sqrt{2}\,{}_4F_3\big(\{\tfrac{1}{2}\}^4; \{\tfrac{3}{2}\}^3; \tfrac{1}{2}\big), \tag{3.6}$$

the $n = 3$ entry of the $\operatorname{Im}\mathrm{Li}_n(1+\mathrm{i})$ tower (companion report `gaussian-1plusi-tower`). So on $\operatorname{Re} z = 1$ the imaginary part is precisely the deep object that the central-binomial ladder measures, while the real part remains tractable—the two halves of the polylogarithm part ways at weight 3.

Remark 3.3 (An unsupported nonreduction claim). Earlier prose asserted that this imaginary part was not elementary, without specifying a comparison algebra or supplying an obstruction. That assertion is an open reducibility question, rather than a consequence of (3.6). The displayed hypergeometric representation remains valid; any proposed reduction or nonreduction must name its algebra of constants and establish a separate proof.

3.5 The integral and its provenance

The object (Math.StackExchange question 564816) is

$$I = \int_0^{\pi/2} \arctan^2\!\left(\frac{6\sin x}{3 + \cos 2x}\right) dx = 1.33097039684256666612532 7\ldots \tag{3.7}$$

Cleo gave the dilogarithmic form

$$I = \frac{\pi}{2}\Big(2\,\mathrm{Li}_2\big((1-\sqrt{2})(2-\sqrt{5})\big) - 2\,\mathrm{Li}_2\big((1-\sqrt{2})(\sqrt{5}-2)\big) + \mathrm{Li}_2(3-\sqrt{8}) - \mathrm{Li}_2(\sqrt{8}-3) + \mathrm{Li}_2(9-\sqrt{80}) - \mathrm{Li}_2(\sqrt{80}-9)\Big), \tag{3.8}$$

while Sangchul Lee gave the Legendre-chi form

$$I = \pi\Big(\chi_2(3-2\sqrt{2}) + \chi_2(9-4\sqrt{5}) + 2\chi_2\big((\sqrt{2}-1)(\sqrt{5}-2)\big)\Big), \tag{3.9}$$

where $\chi_2(x) = \frac{1}{2}\big(\mathrm{Li}_2(x) - \mathrm{Li}_2(-x)\big) = \sum_{k\ge 0} \frac{x^{2k+1}}{(2k+1)^2}$. These are identical: $\mathrm{Li}_2(x) - \mathrm{Li}_2(-x) = 2\chi_2(x)$ turns each Li_2 pair in (3.8) into a χ_2, and $3-\sqrt{8} = (\sqrt{2}-1)^2$, $9-\sqrt{80} = (\sqrt{5}-2)^2$, $(1-\sqrt{2})(2-\sqrt{5}) = (\sqrt{2}-1)(\sqrt{5}-2)$, so both equal

$$I = \pi\Big[\chi_2\big((\sqrt{2}-1)^2\big) + \chi_2\big((\sqrt{5}-2)^2\big) + 2\,\chi_2\big((\sqrt{2}-1)(\sqrt{5}-2)\big)\Big]. \tag{3.10}$$

The arguments are a silver $\otimes$ golden pair: $\sqrt{2}-1 = \tan\frac{\pi}{8} = \delta_{\mathrm{S}}^{-1}$ (reciprocal silver ratio $\delta_{\mathrm{S}} = 1+\sqrt{2}$) and $\sqrt{5}-2 = \phi^{-3}$ (golden $\phi = \frac{1+\sqrt{5}}{2}$, $\phi^3 = \sqrt{5}+2$), so $I = \pi[\chi_2(\delta_{\mathrm{S}}^{-2}) + \chi_2(\phi^{-6}) + 2\chi_2(\delta_{\mathrm{S}}^{-1}\phi^{-3})]$.

3.6 The master formula (Reshetnikov–Lee) and a differentiation proof

The closed forms (3.8)–(3.9) are instances of a two-parameter master formula. Its seed is Reshetnikov's single-arctangent ladder $\int_0^{\pi/2} \arctan(t\sin x)\,dx = 2\chi_2(u_t)$; inspired by it, Lee (a/562492, 2013) conjectured and proved

Theorem 3.4 (Lee 2013)**.** *For real p, q,*

$$J(p,q) := \int_0^{\pi/2} \arctan(p\sin x)\,\arctan(q\sin x)\,dx = \pi\,\chi_2\big(u_p u_q\big), \qquad \text{where} \quad u_p = \frac{\sqrt{1+p^2}-1}{p} = \tan\big(\tfrac{1}{2}\arctan p\big). \tag{3.11}$$

Lee gives two proofs (a triple integral and a Fourier-orthogonality argument). We record a short third proof by differentiation, which is the most direct route and is machine-verifiable.

Proof. Both sides vanish on the axes, and $\partial_p J(p,0) = \partial_p[\pi\chi_2(u_p u_0)] = 0$ (since $u_0 = 0$), symmetrically in q; hence each side equals the double integral of its mixed partial over $[0,p]\times[0,q]$, and it suffices to match mixed partials. Differentiating under the integral and using the partial fraction $\frac{s^2}{(1+p^2s^2)(1+q^2s^2)} = \frac{1}{p^2-q^2}\big(\frac{1}{1+q^2s^2} - \frac{1}{1+p^2s^2}\big)$ together with $\int_0^{\pi/2} \frac{dx}{1+c\sin^2 x} = \frac{\pi}{2\sqrt{1+c}}$,

$$\partial_p\partial_q J = \frac{\pi}{2(p^2-q^2)}\Big(\frac{1}{\sqrt{1+q^2}} - \frac{1}{\sqrt{1+p^2}}\Big). \tag{3.12}$$

On the right, $\chi_2'(w) = \operatorname{arctanh}(w)/w$ gives $\partial_p\partial_q[\pi\chi_2(u_p u_q)] = \pi\,u_p' u_q'/(1-u_p^2u_q^2)$. The substitution $p = \frac{2u_p}{1-u_p^2}$ yields $\sqrt{1+p^2} = \frac{1+u_p^2}{1-u_p^2}$ and $u_p' = \frac{(1-u_p^2)^2}{2(1+u_p^2)}$, whence both sides reduce to the same rational function $\frac{\pi(1-u_p^2)^2(1-u_q^2)^2}{4(1+u_p^2)(1+u_q^2)(1-u_p^2u_q^2)}$. (Wolfram confirms this reconciliation symbolically: `FullSimplify` of the difference returns 0.) □

3.7 Cleo's integral as the perfect-square instance

The argument in (3.7) collapses to a sum of arctangents. With $\cos 2x = 1 - 2\sin^2 x$,

$$\frac{6\sin x}{3+\cos 2x} = \frac{3\sin x}{2-\sin^2 x} = \frac{(1+\frac{1}{2})\sin x}{1-(1\cdot\frac{1}{2})\sin^2 x},$$

the tangent-addition shape with roots $1, \frac{1}{2}$ of $t^2 - \frac{3}{2}t + \frac{1}{2}$, so for $0 \le x \le \frac{\pi}{2}$

$$\arctan\left(\frac{6\sin x}{3+\cos 2x}\right) = \arctan(\sin x) + \arctan(\tfrac{1}{2}\sin x), \tag{3.13}$$

so that $I = \int_0^{\pi/2} \big(\arctan(\sin x) + \arctan(\frac{1}{2}\sin x)\big)^2 dx$. Applying Theorem 3.4 to the expanded square, with $u_1 = \sqrt{2}-1$, $u_{1/2} = \sqrt{5}-2$, reproduces (3.10). The coefficient pattern $1, 2, 1$ is a perfect square: writing $a = \sqrt{2}-1$, $b = \sqrt{5}-2$,

$$\frac{I}{\pi} = \sum_{k\ge 0} \frac{\big((\sqrt{2}-1)^{2k+1} + (\sqrt{5}-2)^{2k+1}\big)^2}{(2k+1)^2}. \tag{3.14}$$

The coefficients $a^{2k+1} + b^{2k+1}$ are the sin-power Fourier coefficients of $\arctan(\sin x) + \arctan(\frac{1}{2}\sin x)$; (3.14) is the Parseval form underlying Lee's Fourier proof.

3.8 Closed values and a bounded experimental survey

Three classical values supply useful exact specializations. With $\phi = (1+\sqrt{5})/2$,

$$\begin{aligned} \chi_2(\phi^{-1}) &= \frac{\pi^2}{12} - \frac{3}{4}\log^2\phi, \\ \chi_2(\phi^{-3}) &= \frac{\pi^2}{24} - \frac{3}{4}\log^2\phi, \\ \chi_2(\sqrt{2}-1) &= \frac{\pi^2}{16} - \frac{1}{4}\log^2(1+\sqrt{2}). \end{aligned} \tag{3.15}$$

The source surveyed inverse metallic means m_n^{-k} with $1 \le n, k \le 6$, selected rational points, and $\tan(\pi/k)$ for $k = 5, 8, 10, 12$ and their squares. Searches at 130–140 digits found no further accepted reduction in their selected logarithmic and π^2 baskets. This does not classify all elementary values. Nor would independence of the individual terms alone prove that their sum cannot simplify.

3.9 New closed-form integrals from elementary χ_2

Conversely, (3.15) feeds the master formula to produce integrals that *do* close. The cleanest cross integral uses $u_2 = \phi^{-1}$, $u_{2/\sqrt{5}} = \phi^{-2}$, so $u_2 u_{2/\sqrt{5}} = \phi^{-3}$:

$$\boxed{\int_0^{\pi/2} \arctan(2\sin x)\,\arctan\big(\tfrac{2}{\sqrt{5}}\sin x\big)\,dx = \frac{\pi^3}{24} - \frac{3\pi}{4}\ln^2\phi} \qquad (= 0.7463164407\ldots). \tag{3.16}$$

The diagonal $\int_0^{\pi/2} \arctan^2(\lambda \sin x)\, dx = \pi\, \chi_2(u_\lambda^2)$ closes at three clean λ, since $u^2_{\sqrt{\phi}} = \phi^{-3}$, $u^2_{\sqrt{2+2\sqrt{2}}} = \sqrt{2}-1$, and $u^2_{2\phi^{3/2}} = \phi^{-1}$:

$$\int_0^{\pi/2} \arctan^2(\sqrt{\phi}\, \sin x)\, dx = \frac{\pi^3}{24} - \frac{3\pi}{4} \ln^2 \phi, \quad \int_0^{\pi/2} \arctan^2\big(2\phi^{3/2} \sin x\big)\, dx = \frac{\pi^3}{12} - \frac{3\pi}{4} \ln^2 \phi. \tag{3.17}$$

The $(1,1)$ diagonal is $\int_0^{\pi/2} \arctan^2(\sin x)\, dx = \pi\, \chi_2(3-2\sqrt{2})$. No elementary reduction or nonreduction theorem for this value is established here.

3.10 A sibling trichotomy

The master formula has two siblings under the companion half-angle substitution $v_p = \frac{1-\sqrt{1-p^2}}{p} = \tanh(\frac{1}{2} \operatorname{arctanh} p)$ (for $|p| < 1$). Replacing one or both arctangents by arctanh produces:

$$\int_0^{\pi/2} \begin{cases} \arctan(p \sin x)\ \arctan(q \sin x) \\ \operatorname{arctanh}(p \sin x)\ \operatorname{arctanh}(q \sin x) \\ \arctan(p \sin x)\ \operatorname{arctanh}(q \sin x) \end{cases} dx = \begin{cases} \pi\, \chi_2(u_p u_q), \\ \pi\, \chi_2(v_p v_q), \\ \pi\ \mathrm{Ti}_2(u_p v_q), \end{cases} \tag{3.18}$$

where $\mathrm{Ti}_2(z) = \Im\, \mathrm{Li}_2(\mathrm{i}z) = \int_0^z \frac{\arctan t}{t}\, dt$ is the inverse-tangent integral: like pairs give the Legendre chi, the mixed pair gives the inverse-tangent integral. The linear shadows are $\int_0^{\pi/2} \arctan(\lambda \sin x)\, dx = 2\chi_2(u_\lambda)$ (Reshetnikov) and, for $-1 < \lambda < 1$ with $\theta = \arcsin \lambda$,

$$\int_0^{\pi/2} \operatorname{arctanh}(\lambda \sin x)\, dx = \theta\, \ln \left|\tan \tfrac{\theta}{2}\right| + 2\, \mathrm{Cl}_2(\theta) - \tfrac{1}{2}\, \mathrm{Cl}_2(2\theta), \tag{3.19}$$

with the value at zero defined by continuity. Indeed, differentiation of the integral with respect to θ and the substitution $t = \cos x$ give $\cos\theta \int_0^1 (\cos^2\theta + \sin^2\theta\, t^2)^{-1} dt = \theta / \sin\theta$. The derivative of the right side is the same, using $\mathrm{Cl}_2'(t) = -\log|2\sin(t/2)|$, and both sides tend to zero at $\theta = 0$. This proves the real branch on both sides of zero. Monotone convergence and oddness extend the formula to $\lambda = \pm 1$, with values $\pm 2\, \mathrm{Cl}_2(\pi/2)$. The latter shows that the arctanh branch is *not* a clean $2\chi_2(v_\lambda)$ mirror—that coincidence is special to arctan. By $x \mapsto \frac{\pi}{2} - x$ the cos versions of all of these equal the sin versions. The weighted analogue $\int_0^{\pi/2} \arctan(p \sin x) \arctan(q \sin x)/\sin x\, dx$ has a different parameter structure. Theorem 3.26 evaluates it by two real trilogarithmic kernels at the hyperbolic sum and difference. Proposition 3.28 proves that no function of $u_p u_q$ alone can represent this family near the origin, replacing the earlier bounded-search observation with an exact obstruction.

3.11 An alternate form, and minimality

Landen-mirror form. Applying the χ_2 Landen law

$$\chi_2(x) + \chi_2\left(\frac{1-x}{1+x}\right) = \frac{\pi^2}{8} - \frac{1}{2} \ln x\, \ln \frac{1-x}{1+x} \qquad (0 < x < 1) \tag{3.20}$$

to the three arguments of (3.10) sends $(\sqrt{2}-1)^2 \mapsto \frac{1}{\sqrt{2}}$, $(\sqrt{2}-1)(\sqrt{5}-2) \mapsto \sqrt{5}-\sqrt{2}$, $(\sqrt{5}-2)^2 \mapsto \frac{2}{\sqrt{5}}$, giving a second three-term representation

$$\frac{I}{\pi} = \mathcal{E} - \Big[\chi_2(\tfrac{1}{\sqrt{2}}) + 2\, \chi_2(\sqrt{5}-\sqrt{2}) + \chi_2(\tfrac{2}{\sqrt{5}})\Big], \tag{3.21}$$

with the elementary constant $\mathcal{E} = \frac{\pi^2}{2} - \frac{1}{2}\ln 2\,\ln(1+\sqrt{2}) + 3\ln 2\,\ln\phi - \frac{3}{2}\ln 5\,\ln\phi + \big(\ln(1+\sqrt{2}) + 3\ln\phi\big)\ln(\sqrt{5}-\sqrt{2})$. This representation trades the small original arguments for larger ones and explicit logarithms. It follows directly by applying the Landen law three times.

Search for shorter forms. The source tested 209 algebraic arguments in $\mathbb{Q}(\sqrt{2},\sqrt{5},\sqrt{10})$ at at least 120 digits, with residual threshold 10^{-105}. It found no accepted one-term or two-term representation in the tested baskets. The three-term formula is an exact evaluation, while its minimality and non-elementarity remain unproved.

3.12 Unit relations and the golden family

With $\phi = \frac{1+\sqrt{5}}{2}$, $\rho = \phi^{-1}$, the defining relation $1-\rho^2 = \rho$ is the seed of the golden ladders. Reshetnikov's conjectures live among the values $\mathrm{Li}_n(\phi^{-2k})$. The source used PSLQ to recover coefficient vectors and tested the resulting relations numerically (`mpmath`, 130–420 digits, Champernowne canary), with Wolfram comparisons where the cancellation depth allowed. Exact functional-equation certificates, where supplied, are identified separately. This description concerns the evidence presented here; it does not assert that the corresponding identities lack proofs in the literature.

Remark 3.5. The source reports cancellation of $\sim 10^8$-sized terms with more than 100 matching decimal digits. Its historical Wolfram evaluation under-resolved some cancellations, so independently controlled precision in mpmath supplied a comparator. Agreement between numerical engines is a useful check, but does not establish exact equality.

3.13 Reshetnikov's formulas and their numerical evidence

The source recovered the following tetralogarithm relation (MSE 1373123, MO 261408) numerically. The exact proof below now establishes it. With $\alpha = \ln 2, \beta = \ln 3$, its form is

$$\begin{aligned} 720\,\mathrm{Li}_4(\tfrac{1}{2}) - 2160\,\mathrm{Li}_4(\tfrac{1}{3}) + 2160\,\mathrm{Li}_4(\tfrac{2}{3}) \\ + 270\,\mathrm{Li}_4(\tfrac{1}{4}) + 540\,\mathrm{Li}_4(\tfrac{3}{4}) + 135\,\mathrm{Li}_4(\tfrac{1}{9}) \\ = 19\pi^4 + 30(\pi^2-\beta^2)(10\alpha^2 - 12\alpha\beta + 3\beta^2) - 30\alpha^2(19\alpha^2 - 24\alpha\beta + 8\beta^2), \end{aligned} \tag{3.22}$$

The source recovered this relation in its weight-four basket; Theorem 3.7 proves it without a numerical-recognition premise. Uniqueness of all relations among the individual values remains a separate question. For the three Li_5 golden formulas below, the source reports residuals of approximately 10^{-257}. Their connection with the ladders of Abouzahra and Lewin [36] is described below:

$$\begin{aligned} 4455\,\mathrm{Li}_5(\phi^{-2}) - 1215\,\mathrm{Li}_5(\phi^{-4}) - 360\,\mathrm{Li}_5(\phi^{-6}) + 15\,\mathrm{Li}_5(\phi^{-12}) &= 3015\zeta(5) - 702\ln\phi^5 + 180\pi^2\ln\phi^3 - 38\pi^4\ln\phi, \\ 45000\,\mathrm{Li}_5(\phi^{-2}) - 16875\,\mathrm{Li}_5(\phi^{-4}) - 144\,\mathrm{Li}_5(\phi^{-10}) + 9\,\mathrm{Li}_5(\phi^{-20}) &= 28944\zeta(5) - 6000\ln\phi^5 + 1600\pi^2\ln\phi^3 - 356\pi^4\ln\phi, \\ 15660\,\mathrm{Li}_5(\phi^{-2}) - 19440\,\mathrm{Li}_5(\phi^{-4}) + 7680\,\mathrm{Li}_5(\phi^{-6}) & \\ +2430\,\mathrm{Li}_5(\phi^{-8}) - 15\,\mathrm{Li}_5(\phi^{-24}) &= 5025\zeta(5) + 2088\ln\phi^5 - 240\pi^2\ln\phi^3 - 28\pi^4\ln\phi. \end{aligned}$$

3.14 Reconstructing Tito's weight-6 and weight-7 formulas

Tito's "old answer" lists the golden family at weights 2–5 (numerically checked in the source), then states the weight-6 and weight-7 ladders only schematically, the integer coefficients "too

tedious to write." PSLQ recovers the following candidate coefficient vectors on the six-argument set $\{\phi^{-2},\phi^{-4},\phi^{-6},\phi^{-8},\phi^{-12},\phi^{-24}\}$:

$$\begin{aligned}
&98820\,\mathrm{Li}_6(\phi^{-2}) - 104490\,\mathrm{Li}_6(\phi^{-4}) + 33120\,\mathrm{Li}_6(\phi^{-6}) + 7290\,\mathrm{Li}_6(\phi^{-8}) - 50\,\mathrm{Li}_6(\phi^{-12}) - 15\,\mathrm{Li}_6(\phi^{-24})\\
&\quad = -13032\ln\phi^6 + 3240\pi^2\ln\phi^4 - 424\pi^4\ln\phi^2 + \mathbf{32155}\,\zeta(6), \qquad (3.23)\\
&74707920\,\mathrm{Li}_7(\phi^{-2}) - 39497220\,\mathrm{Li}_7(\phi^{-4}) + 8346240\,\mathrm{Li}_7(\phi^{-6}) + 1377810\,\mathrm{Li}_7(\phi^{-8}) - 6300\,\mathrm{Li}_7(\phi^{-12}) - 945\,\mathrm{Li}_7(\phi^{-24})\\
&\quad = 2814912\ln\phi^7 - 979776\pi^2\ln\phi^5 + 213696\pi^4\ln\phi^3 - 51448\pi^6\ln\phi + \mathbf{44689995}\,\zeta(7). \qquad (3.24)
\end{aligned}$$

The $\zeta(6)$ and $\zeta(7)$ coefficients reproduce Tito's stated 32155 and 44689995 exactly. The source also reports direct-substitution residuals 10^{-117} and 10^{-114} for (3.23)–(3.24), with primitive coefficient vectors. These are consistent numerical checks; an exact derivation of these displayed reconstructions is not supplied in this section.

3.15 Explicit candidate formulas at weights 8 and 9

Tito remarks that the index-24 ladder "can be extended even to $n = 9$" but gives no formula. The source recovers the following candidates. Its searches on the six-argument set of (3.23)–(3.24) detected numerical reductions at weights 2–7 and none at 8 or 9 within the tested bounds. The accepted higher-weight candidates use the *eight*-argument union of the index-12, 20, 24 sets, $\mathcal{A} = \{\phi^{-2},\phi^{-4},\phi^{-6},\phi^{-8},\phi^{-10},\phi^{-12},\phi^{-20},\phi^{-24}\}$:

$$\begin{aligned}
&-490795200\,\mathrm{Li}_8(\phi^{-2}) + 636131475\,\mathrm{Li}_8(\phi^{-4}) - 388684800\,\mathrm{Li}_8(\phi^{-6}) + 82668600\,\mathrm{Li}_8(\phi^{-8})\\
&\quad + 9507456\,\mathrm{Li}_8(\phi^{-10}) + 5250000\,\mathrm{Li}_8(\phi^{-12}) - 74277\,\mathrm{Li}_8(\phi^{-20}) - 18900\,\mathrm{Li}_8(\phi^{-24})\\
&\qquad = -23641920\ln\phi^8 + 7956480\pi^2\ln\phi^6 - 1434720\pi^4\ln\phi^4 + 184432\pi^6\ln\phi^2 - 148565886\,\zeta(8), \qquad (3.25)\\
&220857840000\,\mathrm{Li}_9(\phi^{-2}) - 143129581875\,\mathrm{Li}_9(\phi^{-4}) + 58302720000\,\mathrm{Li}_9(\phi^{-6}) - 9300217500\,\mathrm{Li}_9(\phi^{-8})\\
&\quad - 855671040\,\mathrm{Li}_9(\phi^{-10}) - 393750000\,\mathrm{Li}_9(\phi^{-12}) + 3342465\,\mathrm{Li}_9(\phi^{-20}) + 708750\,\mathrm{Li}_9(\phi^{-24})\\
&\qquad = -2364192000\ln\phi^9 + 1022976000\pi^2\ln\phi^7 - 258249600\pi^4\ln\phi^5\\
&\qquad\quad + 55329600\pi^6\ln\phi^3 - 14149132\pi^8\ln\phi + 125596001790\,\zeta(9). \qquad (3.26)
\end{aligned}$$

The source reports a residual printed as 0 at 150 working digits for (3.25), and approximately 10^{-190} at 200 working digits for (3.26); both coefficient vectors are primitive. The weight-9 vector was recovered in three searches with different padding arguments, all of which received coefficient 0. These observations support the formulas numerically. A residual printed as zero at finite precision does not establish exact equality.

Remark 3.6. These experiments support golden formulas through weight 9, using four or five arguments per index at weight 5, six arguments at weights 6–7, and eight arguments at weights 8–9. Higher-order golden extrapolations and their numerical construction appear in Abouzahra–Lewin [36] and Bailey–Broadhurst [38]. The present numerical evidence does not prove a maximal ladder weight. Completeness of a multiplicative seed list, additional higher-weight compatibility conditions, and numerical period independence are separate questions.

3.16 Reducing the tetralogarithm relation to a functional identity

Duplication reduces the proved tetralogarithm relation (3.22) to the following equivalent cross-base relation. The complete certificate proof that follows establishes both; no independence claim about the six individual values is needed.

Two duplications $\mathrm{Li}_4(\frac14) = 8\big(\mathrm{Li}_4(\frac12) + \mathrm{Li}_4(-\frac12)\big)$ and $\mathrm{Li}_4(\frac19) = 8\big(\mathrm{Li}_4(\frac13) + \mathrm{Li}_4(-\frac13)\big)$ reduce the left side to the reflected arguments, giving the following *cross-base Kummer specialization*:

$$\begin{aligned}
\mathrm{Li}_4(\tfrac34) + 4\,\mathrm{Li}_4(\tfrac23) = {}& -\tfrac{16}{3}\,\mathrm{Li}_4(\tfrac12) - 4\,\mathrm{Li}_4(-\tfrac12) + 2\,\mathrm{Li}_4(\tfrac13) - 2\,\mathrm{Li}_4(-\tfrac13) + \tfrac{19}{540}\pi^4 + \tfrac59\pi^2\ln^2 2 + \tfrac16\pi^2\ln^2 3\\
&- \tfrac23\pi^2\ln 2\ln 3 - \tfrac{19}{18}\ln^4 2 - \tfrac16\ln^4 3 + \tfrac43\ln^3 2\ln 3 - \ln^2 2\ln^2 3 + \tfrac23\ln 2\ln^3 3. \qquad (3.27)
\end{aligned}$$

Substituting the two duplication identities into (3.27) cancels the tetralogarithms in (3.22). The next proof supplies rational-function cancellation, real branch conventions, and endpoint descent directly, closing the earlier unproved-specialization gap.

3.17 An exact proof of the rational tetralogarithm conjecture

The six-term relation (3.22) was proposed in 2015 and 2017 [100, 101]. Its numerical discovery and the duplication reduction suggest a rational-function certificate. The continuation [102] supplies the forty-two rows below, together with the analytic descent and endpoint calculations. These jointly prove the relation; tensor cancellation alone would not fix the constants or branches. The proof uses elementary differential identities for classical polylogarithms and finite rational arithmetic, and also yields a one-parameter functional identity. No worldwide priority or period independence claim is needed.

Theorem 3.7 (The six rational arguments)**.** *Put $A = \log 2$ and $B = \log 3$. Then*

$$\begin{aligned} &720\,\mathrm{Li}_4(\tfrac{1}{2}) - 2160\,\mathrm{Li}_4(\tfrac{1}{3}) + 2160\,\mathrm{Li}_4(\tfrac{2}{3}) + 270\,\mathrm{Li}_4(\tfrac{1}{4}) \\ &\qquad + 540\,\mathrm{Li}_4(\tfrac{3}{4}) + 135\,\mathrm{Li}_4(\tfrac{1}{9}) \\ &= 19\pi^4 + 30(\pi^2 - B^2)(10A^2 - 12AB + 3B^2) - 30A^2(19A^2 - 24AB + 8B^2). \end{aligned} \tag{3.28}$$

In particular, the intermediate relation (3.27) above is also proved, by the two elementary duplications already recorded there.

3.17.1 A real generalized Rogers function and its exact descent

Write $\ell_n(x) = \mathrm{Re}\,\mathrm{Li}_n(x)$ on the principal branch. For $x \in \mathbb{R} \setminus \{0, 1\}$ set $L = \log|x|$, $M = \log|1 - x|$, so $\ell_1(x) = -M$. Define

$$\begin{aligned} R_2(x) &= \ell_2(x) - \tfrac{1}{2}L\ell_1(x), \\ R_3(x) &= \ell_3(x) - L\ell_2(x) + \tfrac{1}{3}L^2\ell_1(x), \\ R_4(x) &= \ell_4(x) - L\ell_3(x) + \tfrac{1}{2}L^2\ell_2(x) - \tfrac{1}{8}L^3\ell_1(x). \end{aligned}$$

These functions extend continuously by $R_n(0) = 0$ and $R_n(1) = \zeta(n)$. Direct telescoping differentiation gives

$$dR_n(x) = \frac{(-1)^n(n-1)}{n!}L^{n-2}(L\,dM - M\,dL), \qquad n = 2, 3, 4. \tag{3.29}$$

At a crossing of $x = 1$, all logarithmically singular terms have positive powers of L, which tend to zero; this proves the stated continuity and allows identities obtained on regular subintervals to be joined.

Let $V = \mathbb{Q}(t)^\times \otimes_{\mathbb{Z}} \mathbb{Q}$, written additively, and put

$$\beta_n[f] = f^{\otimes(n-2)} \otimes (f \wedge (1 - f)).$$

The sign of a rational function is torsion and vanishes in V. Rational prime factors do not vanish and must be retained.

Lemma 3.8 (Descent with several independent logarithms)**.** *Let $f_i(t)$ be nonconstant rational functions with no zero or pole on an interval I, and let $v_i \in \mathbb{Q}$ satisfy $\sum_i v_i\beta_4[f_i] = 0$. For every linear functional $\eta : V \to \mathbb{R}$ and each $n = 2, 3, 4$,*

$$\sum_i v_i\eta(f_i)^{4-n}R_n(f_i(t))$$

is constant on I, with the continuous interpretation at $f_i(t) = 1$.

Proof. Contract $4 - n$ of the first two tensor slots with η. The result is $\sum_i v_i \eta(f_i)^{4-n} \beta_n[f_i] = 0$. Evaluate the remaining ordinary tensor slots by $g \mapsto \log|g(t)|$ and the wedge by $g \wedge h \mapsto \log|g|\, d\log|h| - \log|h|\, d\log|g|$. Equation (3.29) gives a zero derivative on every regular subinterval. Continuity joins the constants across the finitely many crossings of $f_i = 1$. □

The point of the lemma is that $\eta(f_i)$ need not be a rational multiple of a single logarithm. At the final specialization $t = 1/2$, the needed values are independent linear forms in $A = \log 2$ and $B = \log 3$. Thus the one-logarithm golden descent does not have to be applied outside its stated hypotheses.

3.17.2 The complete finite certificate

For the following forty-two rows define

$$f_i(t) = \varepsilon_i 2^{d_i} t^{a_i} (1-t)^{b_i} (1+t)^{c_i}. \tag{3.30}$$

All $a_i \geq 0$, and all factors except the displayed sign are positive for $0 < t < 1$. The coefficients below are primitive integers.

i	v_i	ε_i	d_i	a_i	b_i	c_i	i	v_i	ε_i	d_i	a_i	b_i	c_i
1	-1	+	-1	0	0	1	22	3	+	-1	1	0	-1
2	-11	+	0	0	0	1	23	-4	+	0	1	0	-1
3	-6	-	0	0	0	1	24	6	-	0	1	0	-1
4	-3	+	1	0	0	1	25	1	+	1	1	0	-1
5	-4	+	0	0	1	-1	26	-3	+	-1	1	0	0
6	4	-	0	0	1	-1	27	-3	-	-1	1	0	0
7	-1	+	-1	0	1	0	28	6	+	0	1	0	0
8	22	+	0	0	1	0	29	-6	-	0	1	0	0
9	21	-	0	0	1	0	30	3	+	1	1	0	0
10	-3	+	1	0	1	0	31	3	-	1	1	0	0
11	4	+	0	0	1	1	32	1	+	-1	1	0	1
12	-1	+	0	0	2	1	33	1	-	-1	1	1	0
13	-1	-	0	0	3	0	34	3	+	0	1	1	0
14	-1	+	-1	1	-2	0	35	3	-	0	1	1	0
15	-6	-	0	1	-2	0	36	-4	-	0	2	-1	-1
16	-6	-	0	1	-1	-1	37	-3	-	0	2	-1	0
17	3	-	-1	1	-1	0	38	-1	+	1	2	-1	0
18	-21	+	0	1	-1	0	39	-3	+	0	2	0	-1
19	-7	-	0	1	-1	0	40	-1	+	1	2	0	-1
20	1	-	1	1	-1	0	41	1	+	0	3	-3	0
21	-1	-	-1	1	0	-2	42	1	-	0	3	-1	-2

Lemma 3.9 (Exact tensor cancellation)**.** *The displayed rational functions satisfy*

$$\sum_{i=1}^{42} v_i \beta_4[f_i] = 0. \tag{3.31}$$

Proof. The complete factor list needed for all f_i and $1 - f_i$, modulo sign, is

$$2, \quad t, \quad t-1, \quad t+1, \quad t-2, \quad t+2, \quad 2t-1, \quad 2t+1, \quad t^2-t-1, \quad t^2-t+1.$$

Use their exponent vectors as coordinates in V. For example, the constant 2 occurs in

$$1-\frac{1+t}{2}=\frac{1-t}{2},\qquad 1-\frac{t}{2(1-t)^2}=\frac{(2t-1)(t-2)}{2(1-t)^2}.$$

The first two slots of β_4 are symmetric. For each of the ten unordered pairs of coordinates of $2, t, 1-t, 1+t$, multiply the corresponding exponents and the two-by-two alternating minors of the vectors of $f_i, 1-f_i$, and sum with coefficient v_i. Every resulting coordinate is zero. There are 230 coordinates touched by the finite sum.

The accompanying `tetralogarithm_certificate.json` gives exactly the forty-two displayed rows. The self-contained verifier independently factors every rational function, checks the polynomial factorizations by multiplication, retains the prime-2 coordinate, and performs the indicated sums over rational integers. It does not search for a relation or use numerical polylogarithms in this check. This is a finite exact verification of (3.31). □

3.17.3 The functional identity proved by the certificate

For $0<t<1$, put

$$A=\log 2,\qquad X=\log t,\qquad Y=\log(1-t),\qquad Z=\log(1+t),$$

and define

$$\begin{aligned}P_4=&\frac{A^4}{4}-\frac{A^3(Y+Z)}{6}+\frac{A^2(Y^2+Z^2)}{4}+\frac{5A(Y^3+Z^3)}{6}-XY(3Y+Z)(Y+Z)\\&+\frac{17Y^4+8Y^3Z+9Y^2Z^2+2YZ^3+2Z^4}{6},\end{aligned}\tag{3.32}$$

$$P_2=-6A^2-4A(Y+Z)+\frac{5}{2}Y^2+8YZ-8Z^2.\tag{3.33}$$

Theorem 3.10 (A rational-function identity)**.** *With the forty-two rows of the table,*

$$\sum_{i=1}^{42}v_i\,\mathrm{Re}\,\mathrm{Li}_4(f_i(t))-4\,\mathrm{Li}_4(\tfrac12)=P_4+\zeta(2)P_2-\frac{71}{4}\zeta(4),\qquad 0<t<1.\tag{3.34}$$

The principal real parts and continuous values are used when an argument is greater than one or equals one.

Proof. Fix $t\in(0,1)$ and write $L_i=\log|f_i(t)|$. Apply Lemma 3.8 with the functional sending a prime or irreducible factor $2, s, 1-s, 1+s$ respectively to A, X, Y, Z, and extend it linearly to the factor space, assigning arbitrary values to its other basis elements. Unique factorization shows that such a functional exists, and its value on every $f_i(s)$ is L_i. This definition remains valid when one of the additional factors of $1-f_i(t)$ vanishes; no logarithm of zero is used to define the functional. The table functions have limits

$$e_i=\lim_{s\to0^+}f_i(s)=\begin{cases}0, & a_i>0,\\ \varepsilon_i 2^{d_i}, & a_i=0.\end{cases}$$

Thus for $n=2,3,4$,

$$\sum_i v_iL_i^{4-n}R_n(f_i(t))=\sum_{a_i=0}v_iL_i^{4-n}R_n(e_i).$$

The elementary triangular identity

$$\ell_4(x) = R_4(x) + LR_3(x) + \tfrac{1}{2}L^2R_2(x) + \tfrac{1}{24}L^3\ell_1(x)$$

then supplies an exact ordinary-polylogarithm identity. More explicitly, put $M_i = \log|e_i| = d_iA$ for $a_i = 0$, and set $Q_i = -M_i^3/8 + L_iM_i^2/3 - L_i^2M_i/4$. The result is

$$\begin{aligned}\sum_i v_i\ell_4(f_i(t)) = \sum_{a_i=0} v_i\big\{\ell_4(e_i) + (L_i - M_i)\ell_3(e_i) + \tfrac{1}{2}(L_i - M_i)^2\ell_2(e_i) + Q_i\ell_1(e_i)\big\} \\ + \frac{1}{24}\sum_i v_iL_i^3\ell_1(f_i(t)).\end{aligned} \tag{3.35}$$

When $e_i = 1$, $Q_i = 0$ and the corresponding last term in braces is zero; no divergent value is assigned to $\ell_1(1)$. Likewise $L_i^3\ell_1(f_i(t))$ is assigned its limiting value zero at $f_i(t) = 1$.

Only $e_i \in \{1, -1, 1/2, 2\}$ occur. The required lower values are

e	$\ell_1(e)$	$\ell_2(e)$	$\ell_3(e)$
-1	$-A$	$-\zeta(2)/2$	$-3\zeta(3)/4$
$1/2$	A	$\zeta(2)/2 - A^2/2$	$7\zeta(3)/8 - \zeta(2)A/2 + A^3/6$
2	0	$3\zeta(2)/2$	$7\zeta(3)/8 + 3\zeta(2)A/2$

and $\ell_2(1) = \zeta(2)$, $\ell_3(1) = \zeta(3)$. For completeness, the lower reflection and Landen identities are

$$\begin{aligned}\operatorname{Li}_2(x) + \operatorname{Li}_2(1-x) &= \zeta(2) - \log x\log(1-x), \\ \operatorname{Li}_2\left(-\frac{x}{1-x}\right) &= -\operatorname{Li}_2(x) - \frac{1}{2}\log^2(1-x), \\ \operatorname{Li}_3(x) + \operatorname{Li}_3(1-x) + \operatorname{Li}_3\left(-\frac{x}{1-x}\right) &= \zeta(3) + \zeta(2)\log(1-x) \\ &\quad - \frac{1}{2}\log x\log^2(1-x) + \frac{1}{6}\log^3(1-x),\end{aligned}$$

for $0 < x < 1$. Differentiating the first two identities gives zero differences, with constants fixed at $x \to 0^+$. Using them in the derivative of the third gives zero as well, and the same endpoint fixes its constant. Substitution $x = 1/2$ proves the displayed half values. The values at 2 follow by real inversion. Thus these endpoint reductions require no integer-relation recognition.

The endpoint weight-four sum is exactly

$$\sum_{a_i=0} v_i\ell_4(e_i) = 4\operatorname{Li}_4(\tfrac{1}{2}) + \frac{A^4}{4} - 6\zeta(2)A^2 - \frac{71}{4}\zeta(4).$$

Substitute $L_i = d_iA + a_iX + b_iY + c_iZ$ and the factorizations of $1 - f_i$ in (3.35). All logarithms of the six additional polynomial factors cancel, as do the $\zeta(3)$ terms. The remaining polynomial is precisely $P_4 + \zeta(2)P_2 - 71\zeta(4)/4$, which proves the theorem. The verifier expands both the master formula and the displayed P_4, P_2 independently and checks their exact polynomial difference is zero. □

3.17.4 The rational specialization, including all inversion terms

For $r > 1$, the real inversion formulas at weight four are

$$\begin{aligned}\ell_4(r) + \operatorname{Li}_4(r^{-1}) &= -\frac{\log^4 r}{24} + \zeta(2)\log^2 r + 2\zeta(4), \\ \operatorname{Li}_4(-r) + \operatorname{Li}_4(-r^{-1}) &= -\frac{\log^4 r}{24} - \frac{1}{2}\zeta(2)\log^2 r - \frac{7}{4}\zeta(4).\end{aligned}$$

They follow by repeated differentiation from the real logarithm and dilogarithm inversion laws, fixing the constants at $r = 1$. Negative arguments inside the unit interval are removed by

$$\mathrm{Li}_4(-r) = \tfrac{1}{8}\,\mathrm{Li}_4(r^2) - \mathrm{Li}_4(r), \qquad 0 \le r \le 1.$$

At $t = 1/2$ one has $X = Y = -A$, $Z = B - A$ and $f_i(1/2) = \varepsilon_i 2^{d_i - a_i - b_i - c_i} 3^{c_i}$. Applying the three displayed rules to the left side of (3.35), after subtracting its endpoint weight-four sum and multiplying by 180, gives exactly the six tetralogarithms in (3.28). The remaining polynomial from those inversions is

$$\begin{aligned} I = {} & 120A^4 - 510A^3B + 765A^2B^2 - 510AB^3 + 150B^4 \\ & + \zeta(2)(90A^2 + 2880AB - 1980B^2) - 1710\zeta(4). \end{aligned}$$

The non-weight-four part of the specialized descent master formula is

$$\begin{aligned} D = {} & -450A^4 + 210A^3B + 225A^2B^2 - 150AB^3 + 60B^4 \\ & + \zeta(2)(1890A^2 + 720AB - 1440B^2). \end{aligned}$$

Consequently the six-term sum equals $D - I$, namely

$$\begin{aligned} & -570A^4 + 720A^3B - 540A^2B^2 + 360AB^3 - 90B^4 \\ & \quad + \zeta(2)(1800A^2 - 2160AB + 540B^2) + 1710\zeta(4). \end{aligned}$$

Using $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$ gives exactly (3.28), proving Theorem 3.7.

3.17.5 A further proved rational specialization

The functional identity also produces explicit identities beyond the six-term conjecture. The specialization $t = 1/3$ is convenient because all logarithms reduce to $A = \log 2$ and $B = \log 3$. The following corollary is a new entry in this report, with no claim of priority among classical tetralogarithm identities.

Corollary 3.11 (Twenty-five positive rational arguments)**.** *For the argument–coefficient pairs (r, c_r) in the table,*

$$\begin{aligned} \sum_r c_r\,\mathrm{Li}_4(r) = {} & \frac{109}{3}A^4 - 116A^3B + 54A^2B^2 + 36AB^3 - 21B^4 \\ & + \zeta(2)(316A^2 - 408AB + 132B^2) + 53\zeta(4). \end{aligned} \tag{3.36}$$

All arguments lie in $(0, 1)$, and the integer coefficient list is primitive: its greatest common divisor is one.

r	c_r	r	c_r
$1/1024$	1	$9/64$	-6
$9/1024$	-1	$1/6$	16
$1/81$	1	$2/9$	8
$1/64$	-4	$1/4$	-107
$1/36$	-6	$8/27$	8
$1/32$	-8	$1/3$	80
$4/81$	3	$3/8$	64
$1/16$	9	$4/9$	24
$1/12$	-24	$1/2$	-200
$64/729$	-1	$16/27$	-8
$3/32$	8	$3/4$	112
$1/9$	-14	$8/9$	32
$1/8$	64		

Proof. Set $t = 1/3$ in Theorem 3.10. The logarithms are $X = -B$, $Y = A - B$, and $Z = 2A - B$. Apply the real inversion formulas to arguments whose absolute value exceeds one, then the duplication formula $\mathrm{Li}_4(-r) = \mathrm{Li}_4(r^2)/8 - \mathrm{Li}_4(r)$ to negative arguments in $[-1, 0]$. Multiplication by eight clears the resulting coefficient denominators. Collecting equal positive arguments gives exactly the table, and collecting all inversion terms gives the displayed polynomial. The same certificate verifier performs these reductions over $\mathbb{Q}$ and checks that the ordinary-polynomial residual is zero. As an independent audit only, a direct evaluation of the reduced identity at ninety working decimal digits has absolute residual below 8×10^{-90}. □

The argument list and polynomial are supplied separately in `tetralogarithm_one_third.json`, and are also included in the shared certificate data. The replay checks the original conjecture, the functional identity, and this corollary in a single run.

3.17.6 Verification, provenance, and further questions

The exact verifier finishes with a zero tensor residual, a zero ordinary-polynomial residual, and the six specified rational arguments with coefficients $(720, -2160, 2160, 270, 540, 135)$. Separately, it evaluates the functional identity at $t = 1/5, 37/100, 1/2, 4/5$ using 90 working decimal digits. The largest numerical discrepancy is approximately 7.9×10^{-90}. These calculations check the branch conventions; the proof is the finite tensor cancellation and the analytic descent.

The original questions are https://math.stackexchange.com/questions/1373123/ and https://mathoverflow.net/questions/261408/. The rational weight-four question and the separate golden weight-five ladders have different proof obligations: the certificate here settles the rational weight-four formula directly. The method is compatible with the higher-Bloch-condition approach to functional equations, but no unproved period conjecture or transcendence assertion is used.

Three useful next problems are to decompose the forty-two rows into a small set of named classical Kummer specializations; to specialize the proved functional identity at other rational or quadratic arguments and systematically reduce the resulting baskets; and to search for similarly

complete certificates for the unresolved golden weights six and seven. The Gaussian S_6 conjecture remains a distinct problem and is not promoted by this result. No minimality or arithmetic independence of the six rational tetralogarithms is asserted.

The complete delivered certificate and verifier are preserved under `reports/uniform-transition-continuation/continuations/tetralogarithm-euler-polynomial-zeros/code/`. The fresh isolated replay is recorded under `verification/seventh-replay/tetra/`. It checks all forty-two rows and 230 touched tensor coordinates, the exact lower-weight polynomial descent, both rational specializations, and separately labeled numerical branch diagnostics. The other claims in that package have independent proofs and retain their editorial-ledger audit status.

3.18 Canonical index-12/20/24 ladders and a ζ-elimination chain

Reshetnikov's Li_5 conjectures are the weight-5 entries of three first-degree golden ladders (Abouzahra–Lewin). For positive integer n, use the following canonical (monic, a^{n-1}-normalized) notation. Put $\rho = \phi^{-1}$, $\ell = \ln \rho$, and

$$\mathrm{tw}(n) = D_1 \frac{\ell^n}{n!} + D_2 \zeta(2) \frac{\ell^{n-2}}{(n-2)!} + D_3 \zeta(4) \frac{\ell^{n-4}}{(n-4)!},$$

where a summand with a negative factorial index is omitted. The previously implicit L_{12} is reconstructed from the first displayed weight-5 formula using the same normalization:

$$\begin{aligned}
L_{12}(n) &= \frac{\mathrm{Li}_n(\rho^{12})}{12^{n-1}} - \frac{3}{2}\frac{\mathrm{Li}_n(\rho^6)}{6^{n-1}} - \frac{\mathrm{Li}_n(\rho^4)}{4^{n-1}} + \frac{11}{48}\frac{\mathrm{Li}_n(\rho^2)}{2^{n-1}} + \mathrm{tw}(n),\\
L_{20}(n) &= \frac{\mathrm{Li}_n(\rho^{20})}{20^{n-1}} - \frac{\mathrm{Li}_n(\rho^{10})}{10^{n-1}} - 3\frac{\mathrm{Li}_n(\rho^4)}{4^{n-1}} + \frac{1}{2}\frac{\mathrm{Li}_n(\rho^2)}{2^{n-1}} + \mathrm{tw}(n),\\
L_{24}(n) &= \frac{\mathrm{Li}_n(\rho^{24})}{24^{n-1}} - \frac{4}{3}\frac{\mathrm{Li}_n(\rho^{12})}{12^{n-1}} - 2\frac{\mathrm{Li}_n(\rho^8)}{8^{n-1}}\\
&\quad + \frac{7}{3}\frac{\mathrm{Li}_n(\rho^4)}{4^{n-1}} - \frac{205}{576}\frac{\mathrm{Li}_n(\rho^2)}{2^{n-1}} + \mathrm{tw}(n).
\end{aligned}$$

For L_{12}, L_{20}, L_{24} respectively, the triples (D_1, D_2, D_3) are

$$(-13/48, 1/48, -19/1728), \quad (-1/2, 1/25, -89/4000), \quad (179/576, -5/192, 629/41472).$$

The displayed weight-5 formulas are equivalent, by exact rational recombination and normalization, to

$$\begin{aligned}
L_{12}(5) &= \frac{67}{6912}\zeta(5), \qquad L_{20}(5) = \frac{201}{10000}\zeta(5),\\
L_{24}(5) &= -\frac{1541}{110592}\zeta(5).
\end{aligned}$$

For the last formula, first add 64/3 times the first displayed weight-5 formula to the third, eliminating $\mathrm{Li}_5(\rho^6)$, then divide by $-15 \cdot 24^4$. This supplies the missing notation; it does not provide a new proof of those evaluations. The source's low-order vanishing statements, with the factorial convention above, concern $1 \le n \le 4$. The rational combinations $M_{20} = L_{20} - \frac{1296}{625} L_{12}$ and $M_{24} = L_{24} + \frac{23}{16} L_{12}$ cancel the displayed $\zeta(5)$ coefficients exactly. The source's subsequent numerical extrapolation reaches weight 7 and forms $T = M_{24} + \frac{409375}{373248} M_{20}$ to cancel its $\zeta(7)$ term and reach weight 9, proposing

$$T(9) = \frac{66452911}{41278242816000}\zeta(9) + \frac{3537283}{2063912140800}\zeta(8)\ell - \frac{11527}{2866544640}\zeta(6)\frac{\ell^3}{3!}.$$

The source reports direct substitution and an independent PSLQ check at 460 digits for this expression. Those checks record numerical evidence for the higher-order extrapolation, rather than a functional-equation certificate. This presentation follows the Abouzahra–Lewin ladder construction [36]; the single eight-argument formulas (3.25)–(3.26) are written separately.

3.19 The minimal Pisot ladder base

Beyond the golden ratio, the reciprocal of the minimal Pisot number $\omega = 0.7548776\ldots$, the real root of $x^3 + x^2 - 1$, carries its own ladders. Abouzahra, Lewin, and Xiao Hongnian studied this field in their published 1987 sequel [37]. Its seeds include the pure-unit relations $1-\omega = \omega^5$, $1-\omega^2 = \omega^3$, $1-\omega^3 = \omega^2$, and $1-\omega^5 = \omega$. The source records the following low-weight formulas, with $\lambda = \ln\omega$:

$$\tfrac{1}{2}\operatorname{Li}_2(\omega^2) - \operatorname{Li}_2(\omega^5) - \lambda^2 = 0, \qquad -\operatorname{Li}_3(\omega) + \tfrac{6}{4}\operatorname{Li}_3(\omega^2) + \tfrac{3}{9}\operatorname{Li}_3(\omega^3) - \operatorname{Li}_3(\omega^5) - \tfrac{6\lambda^3}{3!} = 0,$$
$$-71\operatorname{Li}_4(\omega) - \tfrac{120}{8}\operatorname{Li}_4(\omega^2) + \tfrac{1053}{27}\operatorname{Li}_4(\omega^3) - 17\operatorname{Li}_4(\omega^5) + \tfrac{5488}{343}\operatorname{Li}_4(\omega^7) - \operatorname{Li}_4(\omega^{14}) + 51\zeta(4) + \tfrac{2010\lambda^4}{4!} - \tfrac{84\zeta(2)\lambda^2}{2!} = 0.$$

The weight-four formula uses the extra legs ω^7, ω^{14}. The later Bloch-framework investigation identifies higher ladders for this field; the earlier assertion that the tower stops at weight four is discarded. A fixed argument set can stop yielding detected relations while larger sets continue to produce them.

3.20 Hunting more ladder bases, and a methodological caveat

Pure-unit seeds $1 - \rho^a = \rho^b$ offer a useful way to discover ladders. They do not exhaust general multiplicative relations, and a bounded search over polynomial coefficients does not cover all monic polynomials of a given degree. Root disguises are easy to miss: the root of $x^4 + x^2 - 1$ becomes a golden reciprocal after squaring. The following two cubic and quartic relations illustrate why negative searches require this care.

Two bases the earlier survey had dismissed as "weight-2 only" do, in fact, carry weight-3 ladders. With ρ the root of $x^3 + x - 1$ (the supergolden reciprocal) and of $x^4 + x - 1$ (the reciprocal of the second-smallest Pisot number $x^4 - x^3 - 1$),

$$-60\operatorname{Li}_3(\rho) + 54\operatorname{Li}_3(\rho^2) - 32\operatorname{Li}_3(\rho^3) - 3\operatorname{Li}_3(\rho^6) = 24\ln^3\rho - 4\pi^2\ln\rho - 36\zeta(3),$$
$$-12\operatorname{Li}_3(\rho) + 12\operatorname{Li}_3(\rho^3) - 12\operatorname{Li}_3(\rho^4) - 3\operatorname{Li}_3(\rho^6) = 22\ln^3\rho - 2\pi^2\ln\rho - 12\zeta(3),$$

each verified in two engines (and by direct substitution to 400+ digits). The lesson is methodological: **a PSLQ "no relation" is not a proof of non-existence.** The first relation above was *missed* by a pure-tower PSLQ search at 300 digits, yet passed direct substitution at 400 digits. Its exact proof below establishes that this was a false negative. Recomputing a frozen candidate at precision well above its discovery precision is a useful diagnostic; the analytic proofs below establish the two identities. Whether these two bases climb past weight 3 remains open in the present investigation; the question is delicate precisely because PSLQ negatives cannot settle it.

3.21 Proof certificates for the cubic and quartic trilogarithm ladders

The two displayed formulas record two trilogarithm combinations at the positive roots of $r^3 + r = 1$ and $r^4 + r = 1$, with substantial experimental evidence. Their exact functional-equation certificates now explain the coefficients. We give such derivations. The quartic identity belongs to a uniform family; the cubic identity has a short certificate using two specializations of Kummer's nine-term

relation and four Rogers pentagons. These are exact consequences of classical functional equations, not claims that the underlying equations are new.

3.21.1 A uniform ladder from a complementary pair of powers

For $0 < x < 1$ the real Landen relation for the trilogarithm is

$$\operatorname{Li}_3(x) + \operatorname{Li}_3(1-x) + \operatorname{Li}_3(1-x^{-1}) = \zeta(3) + \frac{\pi^2}{6}\log x - \frac{1}{2}\log^2 x \log(1-x) + \frac{1}{6}\log^3 x. \tag{3.37}$$

For a modern account and its arithmetic analogues see Nakamura and Shiraishi, *Landen's trilogarithm functional equation and ℓ-adic Galois multiple polylogarithms* [63]. For completeness, a real-variable proof is especially short. Differentiation of the left side, followed by

$$\operatorname{Li}_2(x) + \operatorname{Li}_2(1-x) = \frac{\pi^2}{6} - \log x \log(1-x), \qquad \operatorname{Li}_2(1-x^{-1}) = -\operatorname{Li}_2(1-x) - \frac{1}{2}\log^2 x,$$

gives

$$\frac{\pi^2/6 - \log x \log(1-x)}{x} + \frac{\log^2 x}{2x(1-x)}.$$

This is the derivative of the right side. The difference tends to zero as $x \to 1^-$, proving (3.37). The two dilogarithm relations used here follow themselves by one differentiation and an endpoint value. All logarithms in the calculation are real, so there is no branch ambiguity.

Theorem 3.12 (Complementary-power trilogarithm ladder)**.** *Let a, b be positive integers with $a \le b$, and let $0 < \rho < 1$ be the unique positive solution of $\rho^a + \rho^b = 1$. Set $\lambda = \log \rho$ and $d = b - a$. Then*

$$\begin{aligned} &\operatorname{Li}_3(\rho^a) + \operatorname{Li}_3(\rho^b) - \operatorname{Li}_3(\rho^d) + \frac{1}{4}\operatorname{Li}_3(\rho^{2d}) \\ &\qquad = \zeta(3) + \frac{a\pi^2}{6}\lambda + \frac{a^2(a-3b)}{6}\lambda^3. \end{aligned} \tag{3.38}$$

When $a = b$, the powers ρ^0 are interpreted as 1.

Proof. The polynomial $t^a + t^b$ is strictly increasing from 0 to 2 on $[0, 1]$, which proves existence and uniqueness. Substitute $x = \rho^a$ in (3.37); its other arguments are $1 - x = \rho^b$ and $1 - x^{-1} = -\rho^{b-a}$. The absolutely convergent series identity

$$\operatorname{Li}_3(z) + \operatorname{Li}_3(-z) = \frac{1}{4}\operatorname{Li}_3(z^2), \qquad 0 \le z \le 1,$$

eliminates the negative argument. The logarithmic terms become exactly the right side of (3.38). □

In particular, if $r_m^m + r_m = 1$ and $\lambda_m = \log r_m$, then

$$\operatorname{Li}_3(r_m) + \operatorname{Li}_3(r_m^m) - \operatorname{Li}_3(r_m^{m-1}) + \frac{1}{4}\operatorname{Li}_3(r_m^{2m-2}) = \zeta(3) + \frac{\pi^2}{6}\lambda_m - \frac{3m-1}{6}\lambda_m^3. \tag{3.39}$$

This supplies an infinite family of finite exact reductions, although different pairs (a, b) can describe the same base after a power substitution. It does not assert that these exhaust all ladders for any one number field.

Corollary 3.13 (The quartic identity)**.** *If $r^4 + r = 1$ and $\lambda = \log r$, then*

$$-12\operatorname{Li}_3(r) + 12\operatorname{Li}_3(r^3) - 12\operatorname{Li}_3(r^4) - 3\operatorname{Li}_3(r^6) = 22\lambda^3 - 2\pi^2\lambda - 12\zeta(3). \tag{3.40}$$

Proof. Take $(a,b)=(1,4)$ in Theorem 3.12 and multiply by -12. □

For $m=2$, (3.39) reduces to the familiar golden value

$$\mathrm{Li}_3(r_2^2)=\frac{4}{5}\zeta(3)+\frac{2\pi^2}{15}\log r_2-\frac{2}{3}\log^3 r_2.$$

For $m=3$ it gives a useful additional relation on the arguments r,r^2,r^3,r^4; the manuscript's cubic identity instead uses r,r^2,r^3,r^6. The change of the last argument requires more than the one-variable Landen relation.

3.21.2 An elementary dilogarithm certificate in the cubic field

Throughout this subsection let $r^3+r=1$, $0<r<1$, and put $\lambda=\log r$, $A=\log(1+r)$. Direct algebra in $\mathbb{Q}[r]/(r^3+r-1)$ gives

$$1-r=r^3, \qquad 1-r^2=r^3(1+r), \qquad 1-r^3=r, \tag{3.41}$$

$$1-r^4=r^2(1+r), \qquad 1+r^3=r^2(1+r)^2, \qquad 1-r^6=r^3(1+r)^2. \tag{3.42}$$

Define the Rogers dilogarithm on $(0,1)$ by

$$R(x)=\mathrm{Li}_2(x)+\frac{1}{2}\log x\log(1-x).$$

It satisfies $R(x)+R(1-x)=\zeta(2)$ and the pentagon

$$R(x)+R(y)=R(xy)+R\left(\frac{x(1-y)}{1-xy}\right)+R\left(\frac{y(1-x)}{1-xy}\right), \qquad 0<x,y<1. \tag{3.43}$$

The latter is a real-domain form of the classical five-term relation. It may also be verified by differentiating in x with fixed y and using the limit at $x=0$.

Lemma 3.14 (Cubic dilogarithm identity)**.**

$$\mathrm{Li}_2(r^6)=6\,\mathrm{Li}_2(r^2)-2\,\mathrm{Li}_2(r)-4\,\mathrm{Li}_2(r^3). \tag{3.44}$$

Proof. Set $s=r/(1+r)$ and $t=r^4/(1+r)$, so every argument below lies in $(0,1)$. Write $R_k=R(r^k)$ and $R_s=R(s)$, $R_t=R(t)$. Applying (3.43) at $(x,y)=(r,r)$ gives

$$2R_1=R_2+2R_s.$$

At $(x,y)=(r,r^3)$ its two remaining arguments are $1-s$ and t. Since $R_1+R_3=\zeta(2)$, reflection gives

$$R_t=R_s-R_4.$$

At $(x,y)=(r^2,r^4)$ its two remaining arguments are s and t, and therefore

$$R_6=R_2+2R_4-2R_s.$$

Finally, at $(x,y)=(r,r^2)$ these arguments are $1-r^2$ and r^4. Using reflection twice gives

$$R_4=2R_2-2R_3.$$

Eliminating R_s,R_t,R_4 now yields $R_6=6R_2-2R_1-4R_3$. The difference between $R(r^k)$ and $\mathrm{Li}_2(r^k)$ is $(k\lambda/2)\log(1-r^k)$. In the indicated combination these terms cancel: by (3.42), the required check is

$$3\lambda(3\lambda+2A)-6\lambda(3\lambda+A)+3\lambda^2+6\lambda^2=0.$$

This proves the claimed ordinary dilogarithm identity. □

3.21.3 Two Kummer substitutions prove the cubic trilogarithm

The single-valued trilogarithm used in the manuscript is

$$P_3(z) = \operatorname{Re}\operatorname{Li}_3(z) - \log|z|\operatorname{Re}\operatorname{Li}_2(z) - \frac{1}{3}\log^2|z|\log|1-z|.$$

The value at $z=1$ is defined by continuity. It obeys

$$P_3(z^{-1}) = P_3(z), \qquad P_3(z) + P_3(-z) = \frac{1}{4}P_3(z^2),$$

and the single-valued Landen relation $P_3(x) + P_3(1-x) + P_3(1-x^{-1}) = \zeta(3)$ for real $0 < x < 1$. The latter follows from (3.37) and the two dilogarithm identities in its proof.

We use Kummer's classical equation in the form stated and justified by Zagier, *Special values and functional equations of polylogarithms*, §7, Example 3, equation (7) [62]:

$$\begin{aligned} &2P_3(x) + 2P_3(y) + 2P_3\left(\frac{x(1-y)}{x-1}\right) + 2P_3\left(\frac{y(1-x)}{y-1}\right) \\ &\quad + 2P_3\left(\frac{1-x}{1-y}\right) + 2P_3\left(\frac{x(1-y)}{y(1-x)}\right) - P_3(xy) - P_3(x/y) \\ &\quad - P_3\left(\frac{x(1-y)^2}{y(1-x)^2}\right) = 2\zeta(3). \end{aligned} \tag{3.45}$$

We only use specializations with nonzero denominators and finite arguments. Since P_3 is single-valued, some real arguments exceeding 1 cause no branch correction in this equation; after inversion and duplication, all remaining arguments are $r^k \in (0,1)$.

Theorem 3.15 (The cubic identity)**.** *Let $r^3 + r = 1$ with $0 < r < 1$, and let $\lambda = \log r$. Then*

$$\begin{aligned} &-60\operatorname{Li}_3(r) + 54\operatorname{Li}_3(r^2) - 32\operatorname{Li}_3(r^3) - 3\operatorname{Li}_3(r^6) \\ &\qquad = 24\lambda^3 - 4\pi^2\lambda - 36\zeta(3). \end{aligned} \tag{3.46}$$

Proof. Write $P_k = P_3(r^k)$. Landen at r and duplication give

$$\mathsf{A} := P_1 - P_2 + P_3 + \tfrac{1}{4}P_4 = \zeta(3).$$

The nine arguments of (3.45) at $(x,y) = (r, r^3)$ and $(-r, r^2)$, in their displayed order, are

	1	2	3	4	5	6	7	8	9
(r, r^3)	r	r^3	$-r^{-1}$	$-r^5$	r^2	r^{-4}	r^4	r^{-2}	r^{-6}
$(-r, r^2)$	$-r$	r^2	r^4	$-r^{-1}$	r^{-3}	$-r^2$	$-r^3$	$-r^{-1}$	$-r^5$

respectively. Using inversion and $P_3(-r^k) = \frac{1}{4}P_{2|k|} - P_{|k|}$, these two equations reduce to

$$\begin{aligned} \mathsf{B} &:= \tfrac{3}{2}P_2 + 2P_3 + P_4 - 2P_5 - P_6 + \tfrac{1}{2}P_{10} = 2\zeta(3), \\ \mathsf{C} &:= -3P_1 + \tfrac{3}{4}P_2 + 3P_3 + \tfrac{5}{2}P_4 + P_5 - \tfrac{1}{4}P_6 - \tfrac{1}{4}P_{10} = 2\zeta(3). \end{aligned}$$

The exact rational row operation $4\mathsf{C} + 2\mathsf{B} - 48\mathsf{A}$ therefore proves

$$-60P_1 + 54P_2 - 32P_3 - 3P_6 = -36\zeta(3). \tag{3.47}$$

To recover ordinary trilogarithms, put $c_1 = -60, c_2 = 54, c_3 = -32, c_6 = -3$ and

$$D = \sum_{k\in\{1,2,3,6\}} c_k \operatorname{Li}_3(r^k), \quad S = \sum_{k\in\{1,2,3,6\}} c_k k \operatorname{Li}_2(r^k), \quad T = \sum_{k\in\{1,2,3,6\}} c_k k^2 \log(1-r^k).$$

Expansion of the definition gives $\sum c_k P_k = D - \lambda S - \lambda^2 T/3$. Lemma 3.14, together with $\operatorname{Li}_2(r) + \operatorname{Li}_2(r^3) = \pi^2/6 - 3\lambda^2$, gives

$$S = 72\lambda^2 - 4\pi^2.$$

The factorizations (3.42) give

$$T = -60(3\lambda) + 216(3\lambda + A) - 288\lambda - 108(3\lambda + 2A) = -144\lambda.$$

Consequently (3.47) becomes $D - 24\lambda^3 + 4\pi^2\lambda = -36\zeta(3)$, which is (3.46). □

3.21.4 What the certificate establishes and what remains open

The proofs settle the two displayed cubic and quartic trilogarithm identities without period-independence assumptions, numerical recognition, or unproved higher Bloch-group assertions. The universal family (3.38) also gives a principled generator of additional exact identities. It does not justify pseudointegration to weight four or above: differentiating with respect to an algebraic base does not preserve its isolated defining relation.

Natural next problems are to derive comparable local certificates for the weight-five through weight-nine golden formulas, find the smallest useful functional-equation closure for a chosen base, and extend the exact argument/substitution checker to arbitrary algebraic number fields. Any claim of minimality of the surviving period basket remains distinct from the existence of the displayed reductions.

3.22 Rigidity of complementary powers

The earlier complementary-power ladder starts from a relation $r^a + r^b = 1$, where $0 < r < 1$ and a, b are positive integers. That construction raises a concrete completeness question: can one base have many different pairs of complementary powers? We settle this question for pure powers, including arbitrary common divisors in the exponents. The decisive arithmetic input is a classical irreducibility theorem for trinomials. Our conclusions concern this precisely specified class of seeds; they do not classify general multiplicative relations among $1 - r^k$, or bound the maximum weight of polylogarithm ladders.

For $0 < r < 1$, define

$$\mathcal{C}(r) = \{(a, b) \in \mathbb{Z}^2_{>0} : a \le b,\ r^a + r^b = 1\}.$$

Let ω be the unique root in $(0, 1)$ of $X^3 + X^2 - 1$. Thus ω is the reciprocal of the plastic number.

Theorem 3.16 (Complete classification of collisions)**.** *For every real $0 < r < 1$, the set $\mathcal{C}(r)$ has at most two elements. It has two elements if and only if $r^d = \omega$ for some positive integer d, in which case*

$$\mathcal{C}(r) = \{(d, 5d), (2d, 3d)\}.$$

If $\mathcal{C}(r)$ contains (a, a), then $\mathcal{C}(r) = \{(a, a)\}$ and $r = 2^{-1/a}$.

The theorem supplies an exact answer to the seed-classification question posed after the earlier cubic and quartic ladder certificates. It is proved below as a consequence of the Ljunggren–Tverberg theorem, with all additional arguments included. We make no claim that this consequence is globally new in the literature. The related theorem of Aarts, Fokkink, and Kruijtzer classifies the two morphic numbers, by requiring both $p+1$ and $p-1$ to be powers of p; their proof also uses trinomial irreducibility [74]. Our formulation permits arbitrary positive exponent pairs and explicitly handles power substitutions.

3.22.1 The irreducibility input and the entire cyclotomic factor

For $1 \le a < b$, put $d = \gcd(a, b)$, $A = a/d$, $B = b/d$, and

$$F_{a,b}(X) = X^b + X^a - 1.$$

The following precise specialization is the arithmetic input.

Theorem 3.17 (Ljunggren–Tverberg, specialized)**.** *The polynomial $F_{a,b}$ has exactly one irreducible factor over $\mathbb{Q}$ whose roots are not roots of unity. Its other factor is*

$$C_{a,b}(X) = \begin{cases} X^{2d} - X^d + 1, & (A, B) \equiv (1, 5) \text{ or } (5, 1) \pmod 6, \\ 1, & \text{otherwise.} \end{cases}$$

In particular $H_{a,b} = F_{a,b}/C_{a,b}$ is irreducible, monic, and has constant term -1. Moreover $H_{a,b}(X) = H_{A,B}(X^d)$.

This is the trinomial result of Ljunggren and Tverberg [75, 76]; a modern explicit formulation, including the assertion that $H_{A,B}(X^d)$ remains irreducible, is Boyd–Martin–Thom, Lemma 4.1(a),(b) [77]. Only the trinomial theorem is used. Its separate quadrinomial analogue has a more delicate history and plays no role here.

For clarity, the displayed cyclotomic factor can be obtained directly, without relying on a table of sign cases. Suppose $|z| = 1$ and $z^a + z^b = 1$. The two unit vectors must be the two numbers $e^{i\pi/3}$ and $e^{-i\pi/3}$, in either order. If $w = z^d$, then w^A and w^B are these primitive sixth roots. A Bezout identity for A, B implies $w^6 = 1$, and w^A of order six then implies that w itself has order six. Consequently A, B have opposite residues $1, 5$ modulo six. Conversely those residues make every root of $X^{2d} - X^d + 1$ a root of $F_{a,b}$. These roots are simple: a common zero of $F_{a,b}$ and its derivative on the unit circle would give $a|z^a| = b|z^b|$, hence $a = b$, a contradiction. Thus the formula for $C_{a,b}$ accounts for the *entire* cyclotomic factor. The remaining assertion, irreducibility of $H_{a,b}$, is the cited theorem.

3.22.2 A conjugate-rotation invariant

Lemma 3.18 (The common divisor is intrinsic)**.** *Suppose $r^a + r^b = r^c + r^e = 1$, where $0 < r < 1$, $1 \le a < b$, and $1 \le c < e$. Then*

$$\gcd(a, b) = \gcd(c, e).$$

Proof. Set $d = \gcd(a, b)$. Since $0 < r < 1$, it is not a root of unity, so its minimal polynomial is $H_{a,b}$. Theorem 3.17 shows that this polynomial is a polynomial in X^d. If ζ is a primitive dth root of unity, $r\zeta$ is therefore a conjugate of r. The second equation also holds at this conjugate:

$$r^c\zeta^c + r^e\zeta^e = 1.$$

Taking absolute values gives equality in the triangle inequality, because $r^c + r^e = 1$. Both summands must have the argument of their positive real sum. Thus $\zeta^c = \zeta^e = 1$, proving $d \mid \gcd(c, e)$. Interchanging the pairs proves the reverse divisibility. □

This step prevents a gap that would arise from normalizing the two pairs independently. The cyclotomic factor can have degree $2d$, so a degree comparison before proving that the common divisors agree would not suffice.

3.22.3 The primitive coefficient comparison

Proof of Theorem 3.16. First suppose an equal pair occurs. Then $r^a = 1/2$, so r cannot be an algebraic integer: otherwise the rational number $r^a = 1/2$ would be an algebraic integer. Any unequal pair would make r a root of the monic integer polynomial $F_{c,e}$, a contradiction. There is only one equal pair because the powers of r strictly decrease.

Now suppose two distinct unequal pairs occur. By Lemma 3.18 they have a common gcd d. Replace r by r^d and divide all four exponents by d. Both pairs are now primitive. Their trinomials have the same irreducible noncyclotomic factor, and their cyclotomic factors belong to $\{1, \Phi_6\}$, where $\Phi_6 = X^2 - X + 1$. If the two cyclotomic factors agree, the trinomials agree, contrary to distinctness. After interchanging them if necessary, there must therefore be a polynomial identity

$$(X^2 - X + 1)(X^n + X^m - 1) = X^{n+2} + X^k - 1, \quad 1 \le m < n, \quad 1 \le k < n + 2. \tag{3.48}$$

The coefficient of X^{n+1} on the left is -1 unless $m = n - 1$, in which case it is zero. The right-hand coefficient is zero or one. Hence $m = n - 1$, and the identity reduces to

$$X^{n+2} + X^{n-1} - X^2 + X - 1 = X^{n+2} + X^k - 1.$$

For $n = 2$ the remaining polynomial is $2X - X^2$, which is not a monomial. For $n > 3$ its coefficient at X^2 is -1. Thus $n = 3$, which gives $m = 2$ and $k = 1$. The common factor is $X^3 + X^2 - 1$, and the two normalized pairs are $(1, 5)$ and $(2, 3)$. Conversely,

$$X^5 + X - 1 = (X^2 - X + 1)(X^3 + X^2 - 1)$$

shows that these two pairs do occur at ω. Finally, a third pair would have to be the same exceptional companion of either existing pair, so no third pair occurs. □

3.22.4 All signed pure-power unit equations

The functional equations for polylogarithms use signs and inverses as well as positive powers. The preceding classification automatically controls all such complementary arguments.

Corollary 3.19 (Sharp signed-unit count)**.** *Let* $\Gamma_r = \{\pm r^k : k \in \mathbb{Z}\}$ *and*

$$\mathcal{U}(r) = \{x \in \Gamma_r : 1 - x \in \Gamma_r\}.$$

Then $|\mathcal{U}(r)|$ *is one of* $0, 3, 6, 12$*. For every unequal pair* $(a, b) \in \mathcal{C}(r)$*, its contribution is the six-element set*

$$\left\{r^a, r^b, r^{-a}, r^{-b}, -r^{b-a}, -r^{a-b}\right\}. \tag{3.49}$$

These sets are disjoint for different unordered pairs. An equal pair contributes precisely $\{1/2, 2, -1\}$*. The case* $|\mathcal{U}(r)| = 12$ *occurs exactly when* $r^d = \omega$ *for some positive integer* d*.*

Proof. The transformations $x \mapsto 1-x$ and $x \mapsto 1/x$ preserve $\mathcal{U}(r)$. For example, $1 - 1/x = -(1-x)/x \in \Gamma_r$ whenever $x, 1-x \in \Gamma_r$. Their orbit consists of

$$x, \quad 1-x, \quad x^{-1}, \quad (1-x)^{-1}, \quad x/(x-1), \quad (x-1)/x.$$

Every real orbit has a representative in $(0,1)$: use $1/x$ if $x > 1$ and $1/(1-x)$ if $x < 0$. Such a representative and its complement are r^a, r^b with $a, b > 0$. The two members of a nondegenerate real orbit that lie in $(0,1)$ are exactly x and $1-x$, so its unordered positive pair is unique. Substituting $x = r^a$, $1-x = r^b$ gives (3.49). All six entries are distinct unless $a = b$, when they reduce to the stated three values. The cardinalities now follow from Theorem 3.16. □

All four possibilities occur: $r = 1/3$, $1/2$, the reciprocal golden ratio, and ω give respectively $0, 3, 6, 12$. The conclusion classifies *pure signed-power complements*. It does not say that there are at most twelve useful polylogarithm arguments: products of units $1 - r^k$ can introduce many more arguments.

Corollary 3.20 (Rational exponents)**.** *The same collision classification holds for positive rational exponents, with the exceptional common scale d allowed to be a positive rational number. In particular there are at most two unordered positive rational pairs, and*

$$\#\{x \in \{\pm r^q : q \in \mathbb{Q}\} : 1 - x \in \{\pm r^q : q \in \mathbb{Q}\}\} \le 12.$$

Proof. Given two pairs, choose a common denominator N and apply Theorem 3.16 to $r^{1/N}$. A third rational pair could be included in the same denominator and would contradict the integer theorem. The orbit proof is unchanged. □

3.22.5 A provably complete bounded-degree search

Corollary 3.21 (Degree and an exhaustive exponent bound)**.** *If $r^a + r^b = 1$ with $1 \le a < b$, then*

$$[\mathbb{Q}(r) : \mathbb{Q}] = \begin{cases} b - 2d, & (a/d, b/d) \equiv (1,5) \text{ or } (5,1) \pmod 6, \\ b, & \text{otherwise}, \end{cases} \qquad d = \gcd(a,b).$$

Consequently all such bases of degree at most D are found by enumerating

$$1 \le a < b \le \left\lfloor \frac{5D}{3} \right\rfloor$$

and retaining those whose displayed degree is at most D. For primitive pairs it suffices to enumerate $b \le D + 2$. The only repeated bases in either enumeration are the classified plastic pairs. Equal pairs add the separate bases $2^{-1/a}$, with $a \le D$.

Proof. The degree formula is Theorem 3.17. In the exceptional case $b/d \ge 5$, so $b - 2d \ge 3d$. If the degree is at most D, then $d \le D/3$ and $b = (b-2d) + 2d \le 5D/3$. The nonexceptional case is immediate. For primitive pairs the possible degree loss is only two. The equal-pair polynomial $2X^a - 1$ has degree a and is irreducible: its reciprocal is, up to a nonzero scalar, $X^a - 2$, which is Eisenstein at 2. □

This bound is sharp for infinitely many D: the pair $(d, 5d)$ has degree $3d$ and largest exponent $5d$. Unlike a search over bounded polynomial coefficients, the algorithm has an explicit completeness theorem for its stated class. No numerical equality test is required. A base is stored by the exact

irreducible polynomial $H_{a,b}$ together with its unique root in $(0,1)$, and duplicate detection is exact polynomial comparison.

For illustration, the complete unequal-pair list through degree four is

degree	complement pair(s)	minimal polynomial
2	$(1,2)$	X^2+X-1
3	$(1,3)$	X^3+X-1
3	$(1,5),(2,3)$	X^3+X^2-1
4	$(1,4)$	X^4+X-1
4	$(2,4)$	X^4+X^2-1
4	$(3,4)$	X^4+X^3-1

The degree-four entry with pair $(2,4)$ is exactly a square root of the reciprocal golden ratio. Its presence illustrates why normalization of exponents is necessary when counting bases rather than primitive seed types.

The exact enumeration in the accompanying certificate gives:

D	1	2	3	4	5	6	7	8	9	10	11	12
unequal-pair bases	0	1	3	6	10	15	20	27	37	46	56	67
primitive bases	0	1	3	5	9	11	16	20	28	32	42	46
all bases	1	3	6	10	15	21	27	35	46	56	67	79

Each row counts bases of degree at most D. A primitive base has an unequal complementary pair of gcd one; the intrinsic-gcd lemma makes this definition independent of the choice of pair. The last row also includes the D equal-pair bases $2^{-1/a}$, $1 \le a \le D$.

3.22.6 An exact counting formula

The classification also counts bases without constructing or factoring their minimal polynomials. For $n \ge 1$ define

$$e(n) = \#\{a : 1 \le a < n,\ \gcd(a,n) = 1,\ (a,n) \equiv (1,5) \text{ or } (5,1) \pmod 6\}.$$

In particular $e(n) = 0$ unless $n \equiv 1$ or $5 \pmod 6$, and $e(1) = 0$.

Corollary 3.22 (Counting all bases by algebraic degree)**.** *Let $N(D)$ be the number of real $0 < r < 1$ with $[\mathbb{Q}(r) : \mathbb{Q}] \le D$ and $\mathcal{C}(r) \ne \varnothing$. For every positive integer D,*

$$N(D) = \frac{D(D+1)}{2} - \left\lfloor \frac{D}{3} \right\rfloor + \sum_{d=1}^{\lfloor D/3 \rfloor} \left\{ e\left(\left\lfloor \frac{D}{d} \right\rfloor + 1 \right) + e\left(\left\lfloor \frac{D}{d} \right\rfloor + 2 \right) \right\}. \tag{3.50}$$

In particular,

$$N(D) = \frac{1}{2} D^2 + O(D \log D), \qquad D \longrightarrow \infty.$$

Proof. There are $D(D-1)/2$ unequal pairs with largest exponent at most D, and their associated bases all have degree at most D. A pair with largest exponent greater than D can still have degree

at most D only in the exceptional cyclotomic case. Write that pair (dA, dB), where $\gcd(A, B) = 1$ and the residues are opposite $1, 5$ modulo six. The two inequalities are

$$dB > D, \qquad d(B-2) \le D.$$

Equivalently, $B \in \{\lfloor D/d \rfloor + 1, \lfloor D/d \rfloor + 2\}$. Since the exceptional normalized degree $B-2$ is at least three, only $d \le D/3$ can occur. This gives the sum in (3.50). Every base counted twice is $\omega^{1/d}$, of degree $3d$; there are exactly $\lfloor D/3 \rfloor$ such duplicates. Finally the D equal-pair bases are distinct from all unequal-pair bases and from each other. This proves the exact formula.

For the asymptotic statement, use $0 \le e(n) \le n$ to bound the sum by

$$\sum_{d \le D/3} \left(2 \left\lfloor \frac{D}{d} \right\rfloor + 3 \right) \le 2D \sum_{d \le D/3} \frac{1}{d} + D.$$

The harmonic sum is $O(\log D)$, proving the claim. □

For an alternative exact implementation, when $n > 1$ is coprime to six, Mobius inversion gives

$$e(n) = \sum_{d \mid n} \mu(d) \left\lfloor \frac{n/d + 1}{6} \right\rfloor. \tag{3.51}$$

Indeed, after writing $a = dj$, the condition $a \equiv -n \pmod 6$ becomes $j \equiv -(n/d) \pmod 6$ because every unit modulo six is its own inverse. For t coprime to six, the number of integers $1 \le j < t$ in the residue class $-t$ is $\lfloor (t+1)/6 \rfloor$. Summing with $\mu(d)$ imposes $\gcd(a, n) = 1$.

3.23 Exact proofs and log-free ladders at the plastic base

The exceptional base in Theorem 3.16 has two complementary pairs. This additional structure produces short proofs of two previously numerically supported formulas in the manuscript, as well as log-free identities for $\zeta(2)$ and $\zeta(3)$.

Throughout this section, $0 < q < 1$, $q^2 + q^3 = 1$, and $\lambda = \log q$. The factorization in the preceding proof gives

$$1 - q = q^5, \qquad 1 - q^2 = q^3, \qquad 1 - q^3 = q^2, \qquad 1 - q^5 = q. \tag{3.52}$$

All logarithms used in the final identities are real.

3.23.1 The dilogarithm formula: one Rogers pentagon

Write

$$R(x) = \mathrm{Li}_2(x) + \frac{1}{2} \log x \log(1-x), \qquad 0 < x < 1.$$

The Rogers reflection and pentagon equations are

$$R(x) + R(1-x) = \zeta(2), \tag{3.53}$$

$$R(x) + R(y) = R(xy) + R\left(\frac{x(1-y)}{1-xy} \right) + R\left(\frac{y(1-x)}{1-xy} \right). \tag{3.54}$$

They hold for $0 < x, y < 1$ and follow from the classical dilogarithm functional equations; alternatively one can differentiate in x and evaluate at $x = 0$. This real-domain formulation avoids branch choices [62].

Theorem 3.23 (The manuscript's plastic dilogarithm formula)**.**

$$\frac{1}{2}\operatorname{Li}_2(q^2) - \operatorname{Li}_2(q^5) = \lambda^2. \tag{3.55}$$

Proof. Set $R_k = R(q^k)$. At $(x, y) = (q, q^2)$ the three pentagon arguments are

$$xy = q^3, \qquad \frac{x(1-y)}{1-xy} = q^2, \qquad \frac{y(1-x)}{1-xy} = q^5.$$

Hence $R_1 + R_2 = R_3 + R_2 + R_5$, or $R_1 = R_3 + R_5$. Reflection gives $R_1 + R_5 = R_2 + R_3 = \zeta(2)$, so $R_2 = 2R_5$. Expanding the logarithmic corrections by (3.52) gives

$$\operatorname{Li}_2(q^2) + 3\lambda^2 = 2\operatorname{Li}_2(q^5) + 5\lambda^2,$$

as required. □

3.23.2 The trilogarithm formula: one Kummer specialization

For real nonzero $z \neq 1$, define the single-valued trilogarithm

$$P(z) = \operatorname{Re}\operatorname{Li}_3(z) - \log|z|\operatorname{Re}\operatorname{Li}_2(z) - \frac{1}{3}\log^2|z|\log|1-z|,$$

with the continuous value $P(1) = \zeta(3)$. The inversion and duplication equations are

$$P(z^{-1}) = P(z), \qquad P(z) + P(-z) = \frac{1}{4}P(z^2).$$

Kummer's equation is (3.45), with $P = P_3$. We use the form recorded by Zagier, Section 7, Example 3, equation (7) [62].

The single-valued Landen equation is

$$P(x) + P(1-x) + P(1-x^{-1}) = \zeta(3), \qquad 0 < x < 1.$$

Its ordinary real counterpart is

$$\operatorname{Li}_3(x) + \operatorname{Li}_3(1-x) + \operatorname{Li}_3(1-x^{-1}) = \zeta(3) + \zeta(2)\log x - \frac{1}{2}\log^2 x \log(1-x) + \frac{1}{6}\log^3 x. \tag{3.56}$$

One obtains it by differentiation using dilogarithm reflection and Landen, with the integration constant determined at $x = 1$. See also Nakamura–Shiraishi [63].

Theorem 3.24 (The manuscript's plastic trilogarithm formula)**.**

$$-\operatorname{Li}_3(q) + \frac{3}{2}\operatorname{Li}_3(q^2) + \frac{1}{3}\operatorname{Li}_3(q^3) - \operatorname{Li}_3(q^5) = \lambda^3. \tag{3.57}$$

Proof. Put $P_k = P(q^k)$. At $(x, y) = (q, q^2)$, the nine arguments in (3.45), in displayed order, are

$$q, \quad q^2, \quad -q^{-1}, \quad -q^4, \quad q^2, \quad q^{-3}, \quad q^3, \quad q^{-1}, \quad q^{-5}.$$

Using inversion and duplication, the equation becomes

$$\mathsf{K} = -P_1 + \frac{9}{2}P_2 + P_3 - 2P_4 - P_5 + \frac{1}{2}P_8 = 2\zeta(3).$$

Landen at $x = q$, with $1 - q = q^5$, similarly gives

$$\mathsf{A} = P_1 + P_5 - P_4 + \frac{1}{4}P_8 = \zeta(3).$$

Consequently the exact row operation $(\mathsf{K} - 2\mathsf{A})/3$ proves

$$-P_1 + \frac{3}{2}P_2 + \frac{1}{3}P_3 - P_5 = 0. \tag{3.58}$$

It remains to convert this to ordinary trilogarithms. Denote the left side of (3.57) by D, and put

$$S = -\operatorname{Li}_2(q) + 3\operatorname{Li}_2(q^2) + \operatorname{Li}_2(q^3) - 5\operatorname{Li}_2(q^5).$$

The two ordinary reflection equations and Theorem 3.23 give

$$\begin{aligned} \operatorname{Li}_2(q) + \operatorname{Li}_2(q^5) &= \zeta(2) - 5\lambda^2, \\ \operatorname{Li}_2(q^2) + \operatorname{Li}_2(q^3) &= \zeta(2) - 6\lambda^2, \\ \tfrac{1}{2}\operatorname{Li}_2(q^2) - \operatorname{Li}_2(q^5) &= \lambda^2. \end{aligned}$$

Taking minus the first row, plus the second, plus four times the third yields $S = 3\lambda^2$. The remaining logarithmic coefficient in the definition of P is

$$T = \{-1 \cdot 1^2 \cdot 5 + (3/2) \cdot 2^2 \cdot 3 + (1/3) \cdot 3^2 \cdot 2 - 1 \cdot 5^2 \cdot 1\}\lambda = -6\lambda.$$

Thus the left side of (3.58) is

$$D - \lambda S - \frac{\lambda^2}{3}T = D - 3\lambda^3 + 2\lambda^3 = D - \lambda^3.$$

This proves (3.57). □

The proof promotes the two displayed manuscript formulas from numerical support to exact consequences of classical functional equations. It does not rely on pseudointegration at an isolated algebraic base.

3.23.3 Two log-free identities

Theorem 3.25 (Log-free plastic ladders)**.** *At the same base q, the following identities hold:*

$$6\operatorname{Li}_2(q) - 5\operatorname{Li}_2(q^2) - 5\operatorname{Li}_2(q^3) + 6\operatorname{Li}_2(q^5) = \zeta(2), \tag{3.59}$$

$$12\operatorname{Li}_3(q) - 5\operatorname{Li}_3(q^2) - 4\operatorname{Li}_3(q^3) - 8\operatorname{Li}_3(q^4) + 8\operatorname{Li}_3(q^5) + 2\operatorname{Li}_3(q^8) = 4\zeta(3). \tag{3.60}$$

Proof. Six times the first ordinary reflection equation minus five times the second proves (3.59). For the trilogarithm, write $L_k = \operatorname{Li}_3(q^k)$. Equation (3.56) at $x = q$ and $x = q^2$, with duplication to eliminate negative arguments, gives

$$\begin{aligned} A &= L_1 + L_5 - L_4 + \frac{1}{4}L_8 = \zeta(3) + \zeta(2)\lambda - \frac{7}{3}\lambda^3, \\ B &= \frac{5}{4}L_2 + L_3 - L_1 = \zeta(3) + 2\zeta(2)\lambda - \frac{14}{3}\lambda^3. \end{aligned}$$

The logarithmic tails are proportional. Therefore $4(2A - B) = 4\zeta(3)$, exactly the claimed six-argument identity. □

These formulas are explicit additional identities for the package. Their derivations and the classical origins of the functional equations are given in full; no global priority claim is made. They also give exponentially convergent algebraic series, for example

$$\zeta(3) = \sum_{n=1}^{\infty} \frac{3q^n - \frac{5}{4}q^{2n} - q^{3n} - 2q^{4n} + 2q^{5n} + \frac{1}{2}q^{8n}}{n^3}. \tag{3.61}$$

Absolute convergence justifies expansion of the polylogarithms. If this series is truncated after N terms, its absolute tail is at most

$$\frac{39}{4}\,\frac{q^{N+1}}{(N+1)^3(1-q)},$$

obtained by summing the absolute coefficients and using $q^{kn} \le q^n$ for $k \ge 1$. The constant $39/4$ is deliberately simple; the sharper sum of the six individual geometric tails can also be used.

3.24 The modified polylogarithm

Zagier's one-valued modified polylogarithm (his eq. (33)) is

$$P_m(x) = \mathrm{Re}_m\Big(\sum_{j=0}^{m-1} \frac{2^j B_j}{j!}\,(\log|x|)^j\ \mathrm{Li}_{m-j}(x)\Big), \qquad \mathrm{Re}_m = \begin{cases} \mathrm{Re}, & m \text{ odd}, \\ \mathrm{Im}, & m \text{ even}, \end{cases} \tag{3.62}$$

with B_j the Bernoulli numbers and $\mathrm{Li}_1(x) = -\log(1-x)$. It is continuous on the Riemann sphere away from $\{0, 1, \infty\}$, satisfies $P_m(\bar{x}) = (-1)^{m-1}P_m(x)$, and $P_m(1) = \zeta(m)$ for odd m, 0 for even m; $P_2(\mathrm{i})$ is Catalan's constant. We verified our implementation against Zagier's worked examples $P_5(-\frac{1}{8}) - 126P_5(\frac{1}{2}) - 162P_5(-\frac{1}{2}) = \frac{403}{16}\zeta(5)$ (his eq. (825)) and the long thirteen-term relation (825)/(833), to 60 digits.

3.25 Ladders are Bloch-group elements

The Bloch-group viewpoint records the obstruction to simplifying a dilogarithm combination. At weight two the boundary sends $[x] \mapsto x \wedge (1-x)$; a vanishing boundary makes a combination a candidate for the Bloch group, modulo the five-term relation. Higher weights require further relations and compatibility conditions: a weight-one unit relation alone does not prove that its pseudointegrations are higher Bloch-group elements. Modified polylogarithms remove explicit logarithmic tails and make the following ladder combinations transparent:

$$\text{golden}: \quad 15\,P_3(\phi^{-2}) = 12\,\zeta(3), \tag{3.63}$$

$$\text{supergolden}: \quad -60\,P_3(\rho) + 54\,P_3(\rho^2) - 32\,P_3(\rho^3) - 3\,P_3(\rho^6) = -36\,\zeta(3), \tag{3.64}$$

both exact. Compare (3.63) with the "raw" form $15\,\mathrm{Li}_3(\phi^{-2}) = 10\ln^3\phi - 2\pi^2\ln\phi + 12\zeta(3)$: the modified trilogarithm removes the explicit logarithmic tail. At odd weight, comparable modified combinations require the appropriate lower-weight identities. At even weight the definition takes an imaginary part, and every term is real on $0 < x < 1$, so $P_{2j}(x) = 0$. The ordinary even-weight golden formulas retain their content; under this definition they cannot become nonzero even-zeta multiples. This distinction concerns the chosen modification, not a universal bound on regulator directions.

3.26 The bases as number fields

base ρ	minimal polynomial of $1/\rho$	field F	signature	disc
golden ϕ^{-1}	$x^2 - x - 1$	$\mathbb{Q}(\sqrt{5})$	$(2,0)$ totally real	5
minimal Pisot ω	$x^3 - x - 1$	$\mathbb{Q}(\theta)$	$(1,1)$	-23
supergolden recip.	$x^3 - x^2 - 1$	$\mathbb{Q}(\sigma)$	$(1,1)$	-31

The middle row is Zagier's own running example (§4–5, §9C): the field $\mathbb{Q}(\theta)$, $\theta^3 - \theta - 1 = 0$, whose ladder he calls "particularly spectacular," and whose defining unit relations $1 - \theta = -\theta^{-4}$, $1 + \theta = \theta^3, \ldots$ are precisely the seeds we rediscovered. Our supergolden base gives a *different* complex cubic field, discriminant -31.

3.27 Regulator parity and the limits of ladder counting

For $m \geq 2$, Borel's rank theorem gives rank $r_1 + r_2$ for $K_{2m-1}(F)$ when m is odd, and r_2 when m is even. The modified polylogarithm uses its real part at odd weight and imaginary part at even weight. In particular P_{2m} vanishes on real arguments, even when the number field has nonzero regulator directions at its complex embeddings. This explains why a real-argument search can miss arithmetic content visible at a complex embedding.

Cyclotomic unit searches compute $\operatorname{Norm} \Phi_k(\rho) = \operatorname{Res}(p, \Phi_k)$ and select the unit cases. For a fixed finite seed set, imposing successive regulator constraints is a finite linear-algebra problem. A dimension count can give a lower bound for a kernel or exhaust that chosen construction. It does not prove there are no additional seeds, no dependencies among constraints, and no other functional identities at higher weight.

The later project source identifies minimal-Pisot ladders through weight six. The published literature already includes the 1987 Abouzahra–Lewin–Xiao paper [37], which distinguishes analytic low-weight results from numerical higher ladders, and Gangl's proved weight-six and weight-seven relations for single-valued modified polylogarithms [39]. The modified and ordinary polylogarithms must be distinguished when comparing these statements, particularly at real arguments and even weight. Cohen–Lewin–Zagier's published numerical sixteenth-order Lehmer ladder [40] supplies another substantial example. These results supersede earlier bounded negative searches. The project source's stronger assertion that the supergolden field has no weight-five real ladder is not justified by exhausting its selected seed set, and is omitted. A maximal-weight theorem needs completeness for the chosen class of higher-weight relations; completeness of multiplicative seeds alone does not establish it.

3.28 Attributed dilogarithm and inverse-binomial examples

The following examples are attributed to Hakimoglu-Brown [41]. Their displayed numerical checks in the source are historical.

All to ≥ 38 digits (`tools/scratch/dilog-inverse-binomial/`); a representative selection:

- **Sun's open conjecture** (eq. (24)), proved in the paper via the $\mathbb{Q}(\sqrt{3})$ evaluation (25):

$$\sum_{n\geq 0} \frac{\binom{2n}{n}}{(2n+1)8^n} \Big(\sum_{0\leq k<n} \frac{(-1)^k}{2k+1} - \frac{(-1)^n}{2n+1} \Big) = -\frac{\sqrt{2}}{16}\pi^2.$$

- **Theorem 2** ($\mathbb{Q}(\sqrt{3})$): $8\,\mathrm{Li}_2(2-\sqrt{3}) - \mathrm{Li}_2(4\sqrt{3}-7) = \frac{5\pi^2}{12} + \ln(2-\sqrt{3})\ln(2+\sqrt{3})$, and $3\,\mathrm{Li}_2(7-4\sqrt{3}) = 6\,\mathrm{Li}_2(2-\sqrt{3}) + 6\,\mathrm{Li}_2(\sqrt{3}-2)$.
- **Lima's identity** ($\mathbb{Q}(\sqrt{2})$, eq. (26)): $\mathrm{Li}_2(\frac{1}{\sqrt{2}}) - \mathrm{Li}_2(1-\sqrt{2}) = \frac{\pi^2}{6} - \frac{\ln^2 2}{8} + \frac{1}{2}\ln 2\,\ln(\sqrt{2}-1)$.
- **Zagier's four-term** (eq. (27)): $3L(-\frac{1}{6}) - L(\frac{1}{8}) + L(\frac{1}{9}) + L(\frac{1}{28}) = -\frac{\pi^2}{12}$, with the (unnormalized) Rogers $L(x) = \mathrm{Li}_2(x) + \frac{1}{2}\ln|x|\ln|1-x|$; and its compressed three-term form.
- the rational-argument $\frac{\pi^2}{18} + 2\ln 2\ln 3 - 2\ln^2 2 - \frac{2}{3}\ln^2 3 = \mathrm{Li}_2(\frac{1}{4}) + \frac{1}{3}\mathrm{Li}_2(\frac{1}{9})$, the inverse-tangent/$\mathrm{Cl}_2(\frac{\pi}{3})$ relation (30), and $\mathrm{Li}_2(\frac{4}{3}) + \mathrm{Li}_2(4) = \frac{1}{2}(\pi^2 - \ln^2 3)$ (real parts).
- the **engine**: ${}_4F_3(\{1,1,1,\frac{3}{2}\};\{\frac{5}{2},\frac{5}{2},\frac{5}{2}\};1) = \frac{27}{2}\big(7\zeta(3) + (3-2G)\pi - 12\big)$ (D'Aurizio) and the cube-binomial ${}_4F_3$ of eq. (1).
- the paper's **claimed-original elliptic ${}_4F_3(1)$ evaluations** (eqs. 32, 33, 35):

$$\begin{aligned}
\mathrm{Cl}_2(\tfrac{\pi}{3}) &= \tfrac{\sqrt{3}}{2}\,{}_4F_3\big(\{1,\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}\};\{\tfrac{3}{2},\tfrac{5}{6},\tfrac{7}{6}\};1\big),\\
{}_4F_3\big(\{1,\tfrac{1}{2},\tfrac{2}{3},\tfrac{1}{3}\};\{\tfrac{3}{2},\tfrac{5}{4},\tfrac{3}{4}\};1\big) &= \tfrac{3\sqrt{3}}{4}\big(2\ln(4+3\sqrt{2}) + \ln(\tfrac{7}{2}-2\sqrt{3})\big),\\
\ln^2\!\Big(\tfrac{5+\sqrt{13}}{2\sqrt{3}}\Big) - \tfrac{\pi^2}{12} &= \tfrac{1}{180}\,{}_4F_3\big(\{1,1,1,\tfrac{3}{2}\};\{\tfrac{7}{6},\tfrac{11}{6},2\};-\tfrac{1}{324}\big).
\end{aligned}$$

All check out. One caution worth recording: equation (27) uses the *unnormalized* Rogers L (so the right side is $-\frac{\pi^2}{12}$, not a pure rational); with the $\frac{6}{\pi^2}$-normalized L the identity reads $3L(-\frac{1}{6}) - L(\frac{1}{8}) + L(\frac{1}{9}) + L(\frac{1}{28}) = -\frac{1}{2}$.

Connection. The paper evaluates inverse-binomial series with $\binom{2n}{n}^2$ and $\binom{3n}{n}$ via beta integrals; this is the reciprocal-binomial sibling of the central-binomial ladder $\sum\binom{2n}{n}4^{-n}(2n+1)^{-p}$ studied in `harvested-central-binomial-arcsin.wl`. The verified identities are stored (with attribution) in `src/PolyLog/identities/harvested-dilogarithm-ladders-2506.10206.wl`.

3.29 Weighted arctangent families in classical trilogarithms

The bilinear integral underlying Section 3.7 reduces to the product of its half-angle parameters. Inserting the weight $1/\sin x$ changes that structure, but it still leaves an exact evaluation in classical dilogarithms and trilogarithms. Hyperbolic sums and differences of the parameters replace their product. This provides a new arithmetic route at real algebraic parameters, alongside the unit-circle route of the inverse-hyperbolic companion below. The incoming continuation [120] supplies these two families; we give their complete proofs and strengthen its coefficient obstruction to arbitrary one-variable functions.

$$K(p,q) = \int_0^{\pi/2} \frac{\arctan(p\sin x)\arctan(q\sin x)}{\sin x}\,dx, \qquad p,q\in\mathbb{R}. \tag{3.65}$$

The integral in (3.65) is ordinary and convergent. Its integrand is $pq\,x + O(x^3)$ at zero. The principal real arctangent is used throughout. The polylogarithms below use convergent real series or absolutely convergent unit-circle boundary series, with the classical conventions of DLMF §25.12 [96].

3.29.1 The hyperbolic addition formula

Write

$$\chi_j(z) = \frac{\mathrm{Li}_j(z) - \mathrm{Li}_j(-z)}{2} = \sum_{k=0}^{\infty} \frac{z^{2k+1}}{(2k+1)^j} \qquad (|z| < 1)$$

for the Legendre chi function. The values used in the next theorem are real and lie in its ordinary convergent series domain, including the endpoints $\chi_2(1) = \pi^2/8$ and $\chi_3(1) = 7\zeta(3)/8$.

Theorem 3.26 (A real two-term weighted evaluation)**.** *Define the even function*

$$\mathfrak{F}(s) = \frac{\pi^2|s|}{8} - \frac{7}{4}\zeta(3) + |s|\chi_2(e^{-|s|}) + 2\chi_3(e^{-|s|}), \qquad s \in \mathbb{R}. \tag{3.66}$$

Its value at zero is $\mathfrak{F}(0) = 0$. The apparent lack of smoothness from the absolute values is removable: $\mathfrak{F}$ is real analytic on $\mathbb{R}$, and it is equivalently characterized by

$$\mathfrak{F}''(s) = \frac{s}{2\sinh s}, \qquad \mathfrak{F}''(0) = \frac{1}{2}, \qquad \mathfrak{F}(0) = \mathfrak{F}'(0) = 0. \tag{3.67}$$

For every $p, q \in \mathbb{R}$, put $\alpha = \operatorname{arcsinh} p$ and $\beta = \operatorname{arcsinh} q$. Then

$$\boxed{K(p,q) = \mathfrak{F}(\alpha+\beta) - \mathfrak{F}(\alpha-\beta).} \tag{3.68}$$

In particular the formula includes both diagonals $p = \pm q$, either zero parameter, and arbitrary signs without branch changes.

Proof. First, for $s > 0$ use $d\chi_j(e^{-s})/ds = -\chi_{j-1}(e^{-s})$ to obtain

$$\mathfrak{F}'(s) = \frac{\pi^2}{8} - \chi_2(e^{-s}) - s\chi_1(e^{-s}), \qquad \mathfrak{F}''(s) = s\chi_0(e^{-s}) = \frac{s}{2\sinh s}.$$

The limits at zero follow from the endpoint values of χ_2 and χ_3 and from $s\chi_1(e^{-s}) = \frac{s}{2}\log\coth(s/2) \to 0$. The unique twice-normalized primitive of the real analytic even function $s/(2\sinh s)$ is real analytic and even. Hence (3.66) agrees with this primitive on the whole real line; this proves all assertions in (3.67).

Differentiate (3.65) once in each parameter. On compact real parameter sets the differentiated integrand is bounded by an integrable function, and therefore

$$\begin{aligned} K_{pq}(p,q) &= \int_0^{\pi/2} \frac{\sin x\, dx}{(1+p^2\sin^2 x)(1+q^2\sin^2 x)} \\ &= \frac{p\operatorname{arcsinh} p/\sqrt{1+p^2} - q\operatorname{arcsinh} q/\sqrt{1+q^2}}{p^2-q^2} \qquad (p^2 \neq q^2). \end{aligned}$$

For the second equality use partial fractions and

$$\int_0^{\pi/2} \frac{\sin x}{1+p^2\sin^2 x}\, dx = \frac{\operatorname{arcsinh} p}{p\sqrt{1+p^2}},$$

whose zero-parameter value is 1 and which follows by $v = \cos x$.

Let $\widetilde{K}(\alpha,\beta)=K(\sinh\alpha,\sinh\beta)$. Multiplication by $\cosh\alpha\cosh\beta$ and the addition formulas for sinh now give

$$\begin{aligned}\widetilde{K}_{\alpha\beta}&=\frac{\alpha\sinh\alpha\cosh\beta-\beta\sinh\beta\cosh\alpha}{\sinh^2\alpha-\sinh^2\beta}\\&=\frac{1}{2}\left\{\frac{\alpha+\beta}{\sinh(\alpha+\beta)}+\frac{\alpha-\beta}{\sinh(\alpha-\beta)}\right\}.\end{aligned}\tag{3.69}$$

Both quotients on the last line have their removable value 1 at zero. Thus the equality holds at $\alpha=\pm\beta$ by continuity. By (3.67), this is the mixed derivative of $\mathfrak{F}(\alpha+\beta)-\mathfrak{F}(\alpha-\beta)$. Both this expression and $\widetilde{K}$ vanish on $\alpha=0$ and $\beta=0$. Integration of their equal mixed derivatives over the oriented rectangle from $(0,0)$ to (α,β) proves (3.68) for every real α,β. □

For algebraic real p,q, the two arguments $e^{-|\alpha+\beta|}$ and $e^{-|\alpha-\beta|}$ are algebraic. Indeed $e^{\alpha}=p+\sqrt{1+p^2}$ and $e^{\beta}=q+\sqrt{1+q^2}$ are positive algebraic numbers, and taking the absolute value in either exponent selects an algebraic number or its reciprocal. Thus (3.68) expresses every such integral by classical polylogarithms at real algebraic arguments and logarithms of positive algebraic numbers. It asserts no independence or minimality of the resulting constants.

The diagonal specializes particularly simply:

$$K(p,p)=\mathfrak{F}(2\operatorname{arcsinh}p).\tag{3.70}$$

For $\ell=\log(1+\sqrt{2})$ and $r=3-2\sqrt{2}$, this gives the exact identity

$$\boxed{\int_0^{\pi/2}\frac{\arctan^2(\sin x)}{\sin x}\,dx=\frac{\pi^2\ell}{4}-\frac{7}{4}\zeta(3)+2\ell\chi_2(r)+2\chi_3(r).}\tag{3.71}$$

Its numerical value is $0.71806431974110340703632871069 3\ldots$; the decimal only identifies the normalization, while the theorem proves the identity.

An exact weighted counterpart of the manuscript's Cleo integral also follows. Set $\alpha=\log(1+\sqrt{2})$ and $\beta=\log\bigl((1+\sqrt{5})/2\bigr)$. Then

$$\begin{aligned}&\int_0^{\pi/2}\frac{1}{\sin x}\arctan^2\!\left(\frac{6\sin x}{3+\cos 2x}\right)dx\\&\qquad=\mathfrak{F}(2\alpha)+2\mathfrak{F}(\alpha+\beta)-2\mathfrak{F}(\alpha-\beta)+\mathfrak{F}(2\beta).\end{aligned}\tag{3.72}$$

To verify the reduction put $s=\sin x\in[0,1]$. Since $\arctan s+\arctan(s/2)<\pi/2$, the tangent addition formula gives

$$\arctan\frac{6\sin x}{3+\cos 2x}=\arctan(\sin x)+\arctan(\tfrac{1}{2}\sin x).$$

Squaring and integrating yields $K(1,1)+2K(1,1/2)+K(1/2,1/2)$, proving (3.72) by (3.68).

3.29.2 The weighted inverse-hyperbolic companion

The same parameter method has a trigonometric form. It gives a second exact family, including convergent improper endpoint values.

Theorem 3.27 (An inverse-hyperbolic bilinear evaluation)**.** *For $|r|, |s| \le 1$ define*

$$L(r,s) = \int_0^{\pi/2} \frac{\operatorname{arctanh}(r \sin x)\operatorname{arctanh}(s \sin x)}{\sin x}\, dx,$$

using real inverse hyperbolic tangents and interpreting an endpoint with $|r| = 1$ or $|s| = 1$ as an improper integral. Put

$$\mathfrak{G}(t) = \frac{7}{4}\zeta(3) - t \operatorname{Im} \chi_2(e^{it}) - 2 \operatorname{Re} \chi_3(e^{it}), \qquad |t| \le \pi. \tag{3.73}$$

The chi values on the unit circle are defined by their absolutely convergent series. Then $\mathfrak{G}$ is even, continuous on $[-\pi, \pi]$, and real analytic on $(-\pi, \pi)$, with

$$\mathfrak{G}''(t) = \frac{t}{2\sin t}, \qquad \mathfrak{G}''(0) = \frac{1}{2}, \qquad \mathfrak{G}(0) = \mathfrak{G}'(0) = 0. \tag{3.74}$$

For $a = \arcsin r$, $b = \arcsin s$, including either sign of either parameter,

$$\boxed{L(r,s) = \mathfrak{G}(a+b) - \mathfrak{G}(a-b).} \tag{3.75}$$

In particular,

$$\boxed{\int_0^{\pi/2} \frac{\operatorname{arctanh}^2(\sin x)}{\sin x}\, dx = \frac{7}{2}\zeta(3).} \tag{3.76}$$

Proof. Conjugation of the chi series proves evenness and endpoint continuity of (3.73). For $0 < t < \pi$, polylogarithm differentiation gives

$$\mathfrak{G}'(t) = \operatorname{Im} \chi_2(e^{it}) - t \operatorname{Re} \chi_1(e^{it}), \qquad \mathfrak{G}''(t) = t \operatorname{Im} \chi_0(e^{it}) = \frac{t}{2\sin t}.$$

Here $\operatorname{Re} \chi_1(e^{it}) = \frac{1}{2}\log\cot(t/2)$, so the limit of the first derivative at zero is zero. Also $\mathfrak{G}(0) = 0$ by $\chi_3(1) = 7\zeta(3)/8$. The normalized even primitive of $t/(2\sin t)$ is real analytic on $(-\pi, \pi)$, proving (3.74).

For $|r|, |s| < 1$, differentiation and partial fractions give

$$L_{rs}(r,s) = \frac{r \arcsin r/\sqrt{1-r^2} - s \arcsin s/\sqrt{1-s^2}}{r^2 - s^2} \qquad (r^2 \ne s^2).$$

The underlying elementary integral is $\int_0^{\pi/2} \sin x/(1 - r^2 \sin^2 x)\, dx = \arcsin r/(r\sqrt{1-r^2})$, again obtained by $v = \cos x$. For $\widetilde{L}(a,b) = L(\sin a, \sin b)$ the sine addition formulas therefore yield

$$\widetilde{L}_{ab}(a,b) = \frac{1}{2}\left\{ \frac{a+b}{\sin(a+b)} + \frac{a-b}{\sin(a-b)} \right\}, \qquad |a|, |b| < \pi/2.$$

The zero quotients take their removable value 1. This is the mixed derivative of the right side of (3.75), and both sides vanish on the two axes. Rectangle integration proves the identity in the open parameter square.

For all $|r|, |s| \le 1$ the absolute value of the integrand is bounded by $\operatorname{arctanh}^2(\sin x)/\sin x$. This majorant is $O(x)$ at zero and $O(\log^2(\pi/2 - x))$ at the other endpoint, so it is integrable. Dominated convergence extends the equality to the closed square. Finally $\chi_3(-1) = -7\zeta(3)/8$ and $\operatorname{Im} \chi_2(-1) = 0$ give $\mathfrak{G}(\pi) = 7\zeta(3)/2$, proving (3.76). The endpoint value also follows directly from $u = \operatorname{arctanh}(\sin x)$:

$$\int_0^\infty \frac{u^2}{\sinh u}\, du = 4 \sum_{k=0}^{\infty} \frac{1}{(2k+1)^3} = \frac{7}{2}\zeta(3).$$

□

For algebraic $r, s \in [-1, 1]$, the chi arguments in this theorem are algebraic points on the unit circle, since $e^{ia} = \sqrt{1-r^2} + ir$ and $e^{ib} = \sqrt{1-s^2} + is$. The theorem has been proved directly over real parameters; it does not depend on an unstated continuation of the arctangent formula.

3.29.3 An exact obstruction to the one-product ansatz

The distinction from the unweighted factorization has a short exact proof. Set

$$u(p) = \frac{p}{1+\sqrt{1+p^2}}, \qquad u(0) = 0.$$

The unweighted integral is a function of $u(p)u(q)$. The weighted integral has no such factorization into any one-variable function on a real neighborhood of zero; no regularity assumption on that function is needed.

Proposition 3.28. *There is no function G defined on a real neighborhood of zero for which $K(p,q) = G(u(p)u(q))$ holds for all sufficiently small real p, q.*

Proof. Suppose such a function exists. Fix a sufficiently small $q_0 \neq 0$ and put $c = u(q_0) \neq 0$. The local inverse of u is $p = 2u/(1-u^2)$. The asserted identity therefore forces

$$G(t) = K\left(\frac{2(t/c)}{1-(t/c)^2}, q_0\right)$$

for every sufficiently small real t. The integral K is real analytic near the origin: on compact subdisks of $|p|, |q| < 1$, the product of the arctangent series divided by $\sin x$ converges normally and is dominated by $C \sin x$ for a parameter-independent constant C. Integration of this normally convergent series proves analyticity despite the apparent endpoint singularity. Hence the displayed expression forces G to be real analytic near zero.

For $|p|, |q| < 1$, integration of the absolutely convergent arctangent series gives

$$K(p,q) = \sum_{j,k\geq 0} \frac{(-1)^{j+k} p^{2j+1} q^{2k+1}}{(2j+1)(2k+1)} \frac{4^{j+k}}{(2j+2k+1)\binom{2j+2k}{j+k}}. \tag{3.77}$$

In particular,

$$K(p,q) = pq - \frac{2}{9}(p^3q + pq^3) + \text{terms of total degree at least six},$$

whereas

$$u(p)u(q) = \frac{pq}{4} - \frac{p^3q + pq^3}{16} + \text{terms of total degree at least six}.$$

The coefficient of pq would force $G'(0) = 4$, and hence the coefficient of p^3q would have to be $-1/4$, contradicting $-2/9$. Terms of order at least two in G cannot contain p^3q. □

The obstruction concerns this functional ansatz and implies no arithmetic independence. Theorem 3.26 gives a short alternative in hyperbolic sums and differences. Independent exact checks and signed, diagonal and endpoint quadratures audit the normalization; the analytic proofs establish the identities. The ordinary primitive and higher Euler ladder remain separate integration targets in the editorial ledger.

Chapter 4

Gaussian and Eisenstein depth

The classical transformations suggest a sharper question: how much nesting is necessary after known identities have been used? We first prove an infinite family of reductions. The shuffle and parity proofs below settle several of the source's numerical relations. The other families illustrate how a successful experimental identity gives an upper bound on depth, while a failed search leaves open both a large-coefficient reduction and a missing argument family.

4.1 Depth conventions and Gaussian generators

Write $G = \beta(2)$, $\beta(s) = \sum_{n\geq 0}(-1)^n/(2n+1)^s$, and

$$g_{a,b} = \Im \operatorname{Li}_{a,b}(i,1), \qquad h_{a,b} = \Re \operatorname{Li}_{a,b}(i,1), \qquad \lambda_n = \Im \operatorname{Li}_n((1+i)/2).$$

Subscripts such as g_{41} mean $g_{4,1}$. Series depth counts nested indices. In the experimental tables, a "depth-one reduction" means a polynomial in the stated single-value basket, which is a different convention from the additive depth filtration on a motivic algebra. Products of two depth-two factors have additive depth at most four; their membership in an algebra generated by doubles does not make them depth at most two in that additive filtration.

4.2 An infinite even-weight reduction

Define

$$S_p = \sum_{n\geq 0} \frac{(-1)^n H_n}{(2n+1)^p}, \qquad p \geq 1.$$

For $p = 1$ the sum is conditionally convergent; for $p > 1$ it is absolute.

Theorem 4.1 (Odd-denominator harmonic sums at even weight)**.** *For every integer $m \geq 0$,*

$$S_{2m+1} = (2m+1)\beta(2m+2) - [t^{2m}]\frac{\pi}{4\cos(\pi t/2)}\left(2\log 2 + \sum_{j\geq 1}(2-2^{-2j})\zeta(2j+1)t^{2j}\right). \tag{4.1}$$

In particular, each sum of even weight $2m+2$ belongs to the algebra generated by $\pi, \log 2$, odd zeta values and even beta values.

Proof. Put

$$I(t)=\int_0^1 \frac{x^t\log(1+x^2)}{1+x^2}\,dx,\quad J(t)=\int_0^1 \frac{x^t}{1+x^2}\,dx,\quad A(t)=\int_0^\infty \frac{x^t\log(1+x^2)}{1+x^2}\,dx.$$

These are analytic near $t=0$. Splitting A at one and substituting $x=1/u$ in the second part gives $A(t)=I(t)+I(-t)-2J'(-t)$. The generating function $\sum H_n z^n=-\log(1-z)/(1-z)$, used first with an Abel factor and then by dominated convergence, gives $I^{(2m)}(0)=-(2m)!S_{2m+1}$. Also $J^{(2m+1)}(0)=-(2m+1)!\beta(2m+2)$, by the geometric series for $1/(1+x^2)$; its integrated absolute majorant is summable. Consequently

$$S_{2m+1}=(2m+1)\beta(2m+2)-\frac{[t^{2m}]A(t)}{2}.$$

Differentiating the beta integral in its second parameter yields

$$A(t)=\frac{\pi}{2\cos(\pi t/2)}\left[\psi(1)-\psi\left(\frac{1-t}{2}\right)\right],\qquad |\Re t|<1.$$

Its even part follows from $\psi(1)-\psi(1/2)=2\log 2$ and $\psi^{(2j)}(1/2)=-(2j)!(2^{2j+1}-1)\zeta(2j+1)$. Extracting the coefficient proves the formula. □

The first cases are

$$S_1=G-\frac{\pi}{2}\log 2, \tag{4.2}$$

$$S_3=3\beta(4)-\frac{\pi^3}{16}\log 2-\frac{7\pi}{16}\zeta(3), \tag{4.3}$$

$$768S_5=3840\beta(6)-5\pi^5\log 2-42\pi^3\zeta(3)-372\pi\zeta(5), \tag{4.4}$$

$$92160S_7=645120\beta(8)-61\pi^7\log 2-525\pi^5\zeta(3)-5580\pi^3\zeta(5)-45720\pi\zeta(7). \tag{4.5}$$

The source's high-precision experiments suggested the last two identities; the generating integral proves them and every higher odd-index case at once. The proof gives a sufficient reduction. It does not prove that the even beta values are independent, nor that the odd-weight sums have minimal depth two. No such conclusion follows merely from the absence of a new odd beta value.

Literature and normalization. The odd-index formulas specialize the established Euler-sum theory of Xu and Wang [59]. In their notation

$$R^{1,-1}_{1,q}=\sum_{n\ge 1}\frac{(-1)^{n-1}H_{n-1}}{(n-1/2)^q}=2^qS_q.$$

The independent beta-integral and reflection proofs are retained because they make the normalization and the later certified generalizations transparent. No claim of worldwide priority is made for this evaluation.

4.3 Generators and their exact singularities

For this section, put $\mathfrak{q}(x)=\log(1+x^2)/(1+x^2)$. The beta-integral calculation above gives

$$\mathcal{M}(s):=\int_0^\infty x^{s-1}\mathfrak{q}(x)\,dx=\frac{\pi}{2\sin(\pi s/2)}\left[\psi(1)-\psi(1-s/2)\right],\qquad 0<\Re s<2. \tag{4.6}$$

The harmonic generating function also gives, for integer $p \geq 1$,

$$S_p = -\frac{1}{(p-1)!}\int_0^1 (-\log x)^{p-1}\mathfrak{q}(x)\,dx. \tag{4.7}$$

The first-omitted-term theorem below gives $|S_p| \leq 3^{-p}$. These facts make the actual poles visible even when a special-function expression contains removable singularities.

Theorem 4.2 (Even generator)**.** *The power series*

$$\mathcal{S}_e(t) = \sum_{m\geq 0} S_{2m+1}t^{2m}$$

has radius of convergence exactly 3*. It extends to an even meromorphic function on* $\mathbb{C}$*, with the closed form*

$$\boxed{\mathcal{S}_e(t) = \frac{\psi_1((1+t)/4) - \psi_1((3+t)/4)}{16} + \frac{\pi\,[\psi((1+t)/2) - \psi(1)]}{4\cos(\pi t/2)},} \tag{4.8}$$

where $\psi_1 = \psi'$*, and apparent singularities are interpreted by meromorphic continuation. Its only poles are simple and occur at* $t = \pm(2n+1)$*,* $n \geq 1$*, with*

$$\operatorname{Res}_{t=2n+1}\mathcal{S}_e(t) = \frac{(-1)^{n+1}H_n}{2}, \qquad \operatorname{Res}_{t=-(2n+1)}\mathcal{S}_e(t) = \frac{(-1)^n H_n}{2}. \tag{4.9}$$

In particular, the apparent singularities at $t = \pm 1$ *are removable, and*

$$\mathcal{S}_e(1) = \mathcal{S}_e(-1) = -\frac{\pi^2}{48}. \tag{4.10}$$

Proof. The Mellin representation (4.7) and the bound $|S_p| \leq 3^{-p}$ first define E on $|t| < 3$. Summing the even exponential series under the Mellin integral gives

$$\mathcal{S}_e(t) = -\frac{1}{2}\int_0^1 (x^t + x^{-t})\mathfrak{q}(x)\,dx. \tag{4.11}$$

For $|t| < 3$, absolute values of the summed integrand are bounded by $\mathfrak{q}(x)\cosh(|t|\log x)$, which is integrable at zero since $\mathfrak{q}(x) = O(x^2)$. The right-hand side of (4.11) is holomorphic on the larger vertical strip $|\operatorname{Re} t| < 3$: on each compact subset, the same argument applies with a fixed exponent strictly smaller than 3, and all parameter derivatives add only powers of $|\log x|$. It also proves evenness there.

To evaluate the integral, let

$$J(t) = \int_0^1 x^t\mathfrak{q}(x)\,dx, \qquad K(t) = \int_0^1 \frac{x^t\log x}{1+x^2}\,dx.$$

For $|\operatorname{Re} t| < 1$, splitting the full Mellin integral at one and inverting the second part gives

$$\mathcal{M}(1-t) = J(-t) + J(t) - 2K(t).$$

In this strip, expansion of $(1+x^2)^{-1}$ and absolute summability after integration give

$$\begin{aligned} K(t) &= -\sum_{n\geq 0}\frac{(-1)^n}{(2n+1+t)^2} \\ &= -\frac{1}{16}\big[\psi_1((1+t)/4) - \psi_1((3+t)/4)\big]. \end{aligned}$$

Since $\mathcal{S}_e(t) = -\mathcal{M}(1-t)/2 - K(t)$, substitution of (4.6) proves (4.8) in $|\operatorname{Re} t| < 1$.

The exact global singularity statement should not be inferred merely from the individual terms of that formula, because they have cancelling poles. Instead, retain the conditionally convergent coefficient S_1 separately and sum all the absolutely convergent higher coefficients. For $|t| < 3$, this gives

$$\mathcal{S}_e(t) = S_1 + \sum_{n\ge1} \frac{(-1)^n H_n t^2}{(2n+1)((2n+1)^2 - t^2)}. \tag{4.12}$$

The interchange is valid because

$$\sum_{n\ge1} \frac{H_n}{2n+1} \sum_{m\ge1} \left(\frac{|t|}{2n+1}\right)^{2m} = \sum_{n\ge1} \frac{H_n |t|^2}{(2n+1)((2n+1)^2 - |t|^2)} < \infty.$$

For any compact set avoiding the displayed poles in (4.12), the summands for sufficiently large n are uniformly $O((1+\log n)/n^3)$. Thus the series converges normally there and defines a meromorphic function on $\mathbb{C}$. At a point $2n+1$ or $-(2n+1)$, precisely one summand is singular; its elementary residue is (4.9). These residues are nonzero, and there are no further poles. The right-hand side of (4.8) is also meromorphic on $\mathbb{C}$ and agrees on an open strip, so the meromorphic identity theorem proves that the two representations agree everywhere. In particular, all additional apparent poles in the polygamma expression cancel. The nearest actual poles to zero are 3 and -3, proving that the Taylor radius is exactly 3.

Finally, (4.11) can be evaluated directly at $t = 1$, inside its holomorphy strip:

$$\mathcal{S}_e(1) = -\frac{1}{2}\int_0^1 \frac{\log(1+x^2)}{x}\,dx = \frac{1}{4}\operatorname{Li}_2(-1) = -\frac{\pi^2}{48}.$$

Evenness gives the same value at -1, proving (4.10) without a delicate cancellation of separate Laurent expansions. □

4.3.1 The full generator and the unresolved parity direction

The symmetry can be stated more precisely by keeping all indices.

Proposition 4.3. *The full generating function*

$$\mathcal{S}(t) = \sum_{p\ge1} S_p t^{p-1}$$

has Taylor radius 3*, integral representation*

$$\mathcal{S}(t) = -\int_0^1 x^{-t}\mathfrak{q}(x)\,dx, \qquad \operatorname{Re} t < 3, \tag{4.13}$$

and global meromorphic representation

$$\mathcal{S}(t) = S_1 + \sum_{n\ge1} \frac{(-1)^n H_n t}{(2n+1)(2n+1-t)}. \tag{4.14}$$

Its only poles are simple, at $t = 2n+1$, $n \ge 1$, with residue $(-1)^{n+1}H_n$. Moreover,

$$\frac{\mathcal{S}(t) + \mathcal{S}(-t)}{2} = \mathcal{S}_e(t), \qquad \frac{\mathcal{S}(t) - \mathcal{S}(-t)}{2} = \sum_{m\ge1} S_{2m} t^{2m-1} \quad (|t| < 3). \tag{4.15}$$

Proof. For $|t| < 3$, insert (4.7) and sum the exponential series $\sum_{p\ge1} t^{p-1}(-\log x)^{p-1}/(p-1)! = x^{-t}$. The absolute sum is bounded by $x^{-|t|}\mathfrak{q}(x)$, which is integrable. The resulting integral is holomorphic for $\operatorname{Re} t < 3$, by the same compact-set bounds as above. Separating $p = 1$ and summing the geometric series for $p \ge 2$ gives (4.14); the needed absolute bound is

$$\sum_{n\ge1} \frac{H_n|t|}{(2n+1)(2n+1-|t|)} < \infty, \qquad |t| < 3.$$

On a compact set avoiding positive odd integers at least three, the resulting rational summands are uniformly $O((1+\log n)/n^2)$. Normal convergence gives the stated global continuation. The single summand singular at $t = 2n+1$ has residue $(-1)^{n+1}H_n$, and that residue is nonzero. This proves both the complete pole list and the exact radius. Taking even and odd parts of the Taylor series proves (4.15). □

The inversion $x \mapsto 1/x$ determines the even part of A through the full Mellin transform. It leaves the odd part as a different analytic quantity. The absence of the same reduction for that part is not a proof that its coefficients lie outside the algebra generated by single polylogarithmic values. In particular, no result here proves nonmembership for S_{2m}, which have odd total weight. Such a claim requires an additional arithmetic or motivic argument for the specified constants and specified algebra. The presence of an even beta value likewise supplies no independence certificate: even the irrationality of $G = \beta(2)$ is not known. The correction to the draft is therefore precise: replace the proposed two-sided alternation theorem by (4.1) and its proved even-weight membership statement, and retain the opposite direction as a research question.

4.4 Exact shuffle quotients and a two-product proof at weight five

4.4.1 The quotient being computed

Let $\mathcal{H}^1$ be the rational vector space on the empty word and all binary words ending in 1, with shuffle multiplication. It is bigraded by word length w and number of ones d. Let $\mathcal{J}$ be its augmentation ideal, spanned by nonempty words. The indecomposable quotient

$$Q_{w,d} = (\mathcal{J}/\mathcal{J}^2)_{w,d}$$

sets products of two positive-weight elements equal to zero. This is a formal combinatorial construction. Shuffle equalities evaluate to analytic identities at z; the corresponding map on indecomposable quotients requires also quotienting the evaluated algebra by products. Evaluation can introduce further relations. In particular, the formal dimension is not the dimension of a space of numerical periods.

Theorem 4.4 (The exact formal shuffle count)**.** *For $w \ge 1$ and $1 \le d \le w$,*

$$\dim_{\mathbb{Q}} Q_{w,d} = \ell_{w,d} := \frac{1}{w} \sum_{k \mid \gcd(w,d)} \mu(k) \binom{w/k}{d/k}.$$

A basis is represented by the binary Lyndon words of length w with d ones. Every such word belongs to $\mathcal{H}^1$. Every other word has a finite, exact polynomial expression in the full family of Lyndon generators, each monomial having the original total bidegree. The number of independent product relations in this component is

$$\binom{w-1}{d-1} - \ell_{w,d}.$$

Proof. Order the alphabet by $0 < 1$ and words lexicographically. A Lyndon word is strictly smaller than every nontrivial rotation. Every word has a unique Chen–Fox–Lyndon factorization into a nonincreasing concatenation of Lyndon words. In characteristic zero, shuffle monomials in Lyndon words form a polynomial basis [60]. One can see the constructive point directly: if the factorization is $u_1^{m_1} \cdots u_r^{m_r}$ with distinct $u_1 > \cdots > u_r$, then the shuffle product $u_1^{\amalg m_1} \amalg \cdots \amalg u_r^{\amalg m_r}$ has this concatenation as its lexicographically largest word, with coefficient $m_1! \cdots m_r!$. All other words are smaller. Induction therefore gives a triangular polynomial expression, and the same triangularity proves linear independence of the monomials.

Every Lyndon word of length greater than one that contains both letters starts with 0 and ends with 1. The single-letter words are 0 and 1. The factorization of a word ending in 1 cannot contain the factor 0, because that smallest factor would be last, forcing the concatenation to end in 0. Thus $\mathcal{H}^1$ is exactly the polynomial algebra on all Lyndon words other than 0. Passing to its augmentation quotient leaves one class for each generator.

It remains to count. A primitive binary word of length w with d ones has w distinct rotations and exactly one Lyndon rotation. Every binary word is a power of a unique primitive word. Möbius inversion on this power decomposition gives $\sum_{k \mid \gcd(w,d)} \mu(k) \binom{w/k}{d/k}$ primitive words, proving the formula. Finally, there are $\binom{w-1}{d-1}$ words ending in 1 in the component, which gives the claimed product rank. □

Weight w	$d=1$	$d=2$	$d=3$	$d=4$	$d=5$
3	1	1	0	–	–
4	1	1	1	0	–
5	1	2	2	1	0
6	1	2	3	2	1
7	1	3	5	5	3
8	1	3	7	8	7
9	1	4	9	14	14

Table 4.1: The exact formal indecomposable counts $\ell_{w,d}$. They count Lyndon generators, not independent numerical constants.

For $w = 6, d = 3$, the ten words reduce modulo products to three Lyndon words:

$$000111, \qquad 001011, \qquad 001101,$$

corresponding respectively to $F_{4,1,1}$, $F_{3,2,1}$, and $F_{3,1,2}$. This explains the manuscript's finite rank-seven observation and gives its uniform extension. The companion code records the full polynomial normal forms of all ten triples. For a compact view of the quotient, put $X = [F_{4,1,1}], Y = [F_{3,2,1}], Z = [F_{3,1,2}]$, where brackets mean classes modulo products. Exact triangular elimination gives

	X	Y	Z		X	Y	Z
$[F_{4,1,1}]$	1	0	0	$[F_{1,4,1}]$	6	3	1
$[F_{3,2,1}]$	0	1	0	$[F_{1,3,2}]$	−12	−6	−2
$[F_{3,1,2}]$	0	0	1	$[F_{1,2,3}]$	12	6	2
$[F_{2,3,1}]$	−9	−4	−1	$[F_{1,1,4}]$	−6	−3	−1
$[F_{2,2,2}]$	12	4	0	$[F_{2,1,3}]$	−3	−1	0

For example, one full identity before taking the quotient is

$$F_{2,2,2}(z) = 12F_{4,1,1}(z) + 4F_{3,2,1}(z) - 2\,\mathrm{Li}_2(z)F_{3,1}(z) + \frac{\mathrm{Li}_2(z)^3}{6}. \tag{4.16}$$

Re-expanding its right side into words gives the single word 010101 with coefficient one. This supplies a finite rational certificate, including the product terms that a quotient table suppresses.

4.4.2 An elementary all-weight double elimination

At depth two the count simplifies to

$$\ell_{w,2} = \left\lfloor \frac{w-1}{2} \right\rfloor.$$

There is also a direct triangular proof that avoids general shuffle algebra. For $p+q=w$, shuffling the two single-polylogarithm words gives

$$\mathrm{Li}_p(z)\,\mathrm{Li}_q(z) = \sum_{j=1}^{p} \binom{w-j-1}{q-1} F_{w-j,j}(z) + \sum_{j=1}^{q} \binom{w-j-1}{p-1} F_{w-j,j}(z). \tag{4.17}$$

The binomial coefficient counts the interleavings of the zero blocks before the second nonzero letter.

Proposition 4.5 (Independent double product rows)**.** *For fixed w, the rows of (4.17) with $1 \le p \le \lfloor w/2 \rfloor$ have rank $\lfloor w/2 \rfloor$ on the $w-1$ double coordinates. They express the entire family through products of singles and the $\lfloor (w-1)/2 \rfloor$ coordinates*

$$\{F_{w-j,j} : 1 \le j \le \lfloor (w-1)/2 \rfloor\}.$$

This set is empty when $w = 2$.

Proof. In the row indexed by p, the largest inner index appearing is $q = w-p$. Its coefficient is 1 when $p < q$, and 2 when $p = q$. No larger inner index occurs. Order the rows with p decreasing and use the columns with $j = w-p$; the corresponding square block is triangular with nonzero diagonal. All products of two depth-one words have already been included, since swapping p, q repeats a row. This proves the rank and the elimination statement. □

At $z = i$, taking imaginary parts of these equations gives an unconditional spanning bound for the $g_{a,b}$, with explicit single-value terms. It can be sharpened in the parity sector, as the next section shows.

4.4.3 A normalized Pascal inverse for the shuffle rows

The triangular elimination can be made explicit in outer-index coordinates. Let $m = \lfloor w/2 \rfloor$ and let $A_{p,a} = \binom{a-1}{p-1} + \binom{a-1}{w-p-1}$, for $1 \le p \le m$, $1 \le a < w$. Divide row p by $c_p = 1 + \mathbf{1}_{2p=w}$. Its first m columns are $P_{p,a} = \binom{a-1}{p-1}$, with inverse

$$(P^{-1})_{i,j} = (-1)^{j-i}\binom{j-1}{i-1} \quad (j \ge i), \qquad (P^{-1})_{i,j} = 0 \quad (j < i).$$

Indeed the matrix product reduces to $\binom{j-1}{i-1}\sum_{k=i}^{j}(-1)^{j-k}\binom{j-i}{k-i} = \delta_{ij}$. Writing $R_p = \operatorname{Im}(\operatorname{Li}_p(\mathrm{i})\operatorname{Li}_{w-p}(\mathrm{i}))$, the constructive solution is

$$g_{a,w-a} = \sum_{p=a}^{m}(-1)^{p-a}\binom{p-1}{a-1}\left[\frac{R_p}{c_p} - \sum_{j=m+1}^{w-1}\frac{A_{p,j}}{c_p}g_{j,w-j}\right], \qquad 1 \le a \le m. \tag{4.18}$$

This explicitly leaves the same $\lfloor (w-1)/2 \rfloor$ formal coordinates as Proposition 4.5; evaluation can impose further relations.

4.4.4 The weight-five relation is two shuffles

The words for $\operatorname{Li}_1\operatorname{Li}_4$ and $\operatorname{Li}_2\operatorname{Li}_3$ give

$$\operatorname{Li}_1(z)\operatorname{Li}_4(z) = 2F_{4,1}(z) + F_{3,2}(z) + F_{2,3}(z) + F_{1,4}(z), \tag{4.19}$$
$$\operatorname{Li}_2(z)\operatorname{Li}_3(z) = 6F_{4,1}(z) + 3F_{3,2}(z) + F_{2,3}(z). \tag{4.20}$$

The exact single values are

$$\operatorname{Li}_1(i) = -\frac{L}{2} + \frac{\pi i}{4}, \quad \operatorname{Li}_2(i) = -\frac{\pi^2}{48} + iG, \quad \operatorname{Li}_3(i) = -\frac{3\zeta(3)}{32} + \frac{\pi^3 i}{32},$$

$$\operatorname{Li}_4(i) = -\frac{7\pi^4}{11520} + i\beta(4).$$

Consequently

$$2g_{41} + g_{32} + g_{23} + g_{14} = -\frac{1}{2}\beta(4)L - \frac{7\pi^5}{46080}, \tag{4.21}$$
$$6g_{41} + 3g_{32} + g_{23} = -\frac{\pi^5}{1536} - \frac{3G\zeta(3)}{32}. \tag{4.22}$$

Here g_{41} means $g_{4,1}$, and similarly for other subscripts.

Corollary 4.6 (Proof and strengthening of the recorded relation)**.** *The experimentally discovered identity*

$$576g_{41} + 288g_{32} + 736g_{23} + 960g_{14} = 21G\zeta(3) - 480\beta(4)L \tag{4.23}$$

holds. Moreover the four g-values are spanned by g_{41}, g_{32} and the single-value products $\pi^5, G\zeta(3), \beta(4)L$. Explicitly,

$$g_{23} = -6g_{41} - 3g_{32} - \frac{\pi^5}{1536} - \frac{3G\zeta(3)}{32}, \tag{4.24}$$
$$g_{14} = 4g_{41} + 2g_{32} + \frac{23\pi^5}{46080} + \frac{3G\zeta(3)}{32} - \frac{1}{2}\beta(4)L. \tag{4.25}$$

Proof. Multiply (4.21) by 960, multiply (4.22) by -224, and add. The π^5 coefficients cancel exactly. Solving the two independent equations instead gives (4.24) and (4.25). □

This proof explains why a search can return an apparently complicated relation while missing the simpler structure: eliminating a chosen single-value coordinate can enlarge coefficients. The mathematical content is visible before the elimination.

4.4.5 An infinite odd-weight shuffle family

Theorem 4.7 (An infinite odd-weight family with the pure π term removed)**.** *For $m \geq 2$, set $W = 2m + 1$ and*

$$A_W = 2g_{W-1,1} + \sum_{a=1}^{W-2} g_{a,W-a},$$

$$B_W = 2(W-2)g_{W-1,1} + (W-2)g_{W-2,2} + \sum_{a=2}^{W-3} (a-1)g_{a,W-a}, \tag{4.26}$$

$$\kappa_m = \frac{12(2^{2m-1}-1)B_{2m}}{(2m)(2m-1)E_{2m-2}}. \tag{4.27}$$

Then κ_m is a positive rational number and

$$A_W - \kappa_m B_W = -\frac{1}{2}\beta(2m)L + \kappa_m 2^{-(2m-1)}(1-2^{2-2m})G\zeta(2m-1). \tag{4.28}$$

Proof. Equation (4.17) identifies A_W and B_W with $\mathrm{Im}(\mathrm{Li}_1(\mathrm{i})\,\mathrm{Li}_{2m}(\mathrm{i}))$ and $\mathrm{Im}(\mathrm{Li}_2(\mathrm{i})\,\mathrm{Li}_{2m-1}(\mathrm{i}))$ respectively. Writing $c_k = 2^{-k}(1-2^{1-k})$, these products are

$$A_W = -\tfrac{1}{2}L\beta(2m) - \tfrac{\pi}{4}c_{2m}\zeta(2m),$$
$$B_W = -c_{2m-1}G\zeta(2m-1) - \tfrac{\pi^2}{48}\beta(2m-1).$$

Their pure-π^W terms have ratio

$$\frac{12c_{2m}\zeta(2m)}{\pi\beta(2m-1)}.$$

Euler's even-zeta formula and the classical odd-beta evaluation turn this ratio into (4.27). The signs of B_{2m} and E_{2m-2} agree, so the ratio is positive. Subtraction proves (4.28). □

The first three coefficients are

$$\kappa_2 = \frac{7}{30}, \qquad \kappa_3 = \frac{31}{525}, \qquad \kappa_4 = \frac{127}{8540}.$$

For example, at weight seven the identity can be written without fractions as

$$268800A_7 - 15872B_7 = 465G\zeta(5) - 134400\beta(6)L. \tag{4.29}$$

This theorem does not reduce every odd-weight $g_{a,b}$ individually. It supplies one explicit family of linear combinations in the parity component not individually reduced by Theorem 4.12.

At weight seven a second combination complements the pure-π-free family above:

$$430g_{6,1} + 215g_{5,2} + 79g_{4,3} + 11g_{3,4} - 7g_{2,5} = \frac{105}{512}G\zeta(5) - \frac{75}{32}\beta(4)\zeta(3). \tag{4.30}$$

Its exact certificate is -7 times the $(2,5)$ shuffle row plus 25 times the $(3,4)$ row, after evaluating their single values. The exact row calculation fixes the homogeneous weight-seven products.

4.5 The mixed Gaussian weight-five identity and its exact proof

Put $G = \beta(2)$ and

$$g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1), \qquad S_p = \sum_{n=0}^{\infty} \frac{(-1)^n H_n}{(2n+1)^p}.$$

The manuscript's equation `gauss:eq:S4-closed` was explicitly classified as a numerically supported identity. We prove precisely that identity. The contribution is a reduction of its mixed Gaussian term to three specified single-color doubles and elementary products. It is not a claim that the remaining doubles are independent or irreducible, nor a claim that S_4 previously lacked every representation in special functions. Indeed the 2021 discussion by Hintze and pisco already records the representation by two colored double values and relates it to hypergeometric evaluations [70, 71].

Theorem 4.8 (Certified mixed Gaussian reduction)**.** *With the series convention* $\operatorname{Li}_{a,b}(x, y) = \sum_{n>m\geq 1} x^n y^m/(n^a m^b)$*,*

$$S_4 = \frac{4g_{4,1} - 3g_{3,2} - 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27}{224} G\zeta(3) - 2\beta(4)\log 2. \tag{4.31}$$

Equivalently,

$$\operatorname{Im} \operatorname{Li}_{4,1}(i, -1) = -\frac{3g_{4,1} + 3g_{3,2} + 9g_{2,3}}{7} + \frac{\pi^5}{224} - \frac{27}{224} G\zeta(3) - 2\beta(4)\log 2. \tag{4.32}$$

The proof consists of convergent changes of variables, standard shuffle/stuffle identities with the explicitly described single-divergence regularization, and a finite rational linear combination supplied as an independently replayable certificate.

4.5.1 The arithmetic translation

For $n = 2k + 1$,

$$H_k = \sum_{m=1}^{n-1} \frac{1 + (-1)^m}{m}.$$

Taking imaginary parts removes even n, while $\operatorname{Im}(i^{2k+1}) = (-1)^k$. Consequently, for $p \geq 1$,

$$S_p = \operatorname{Im}\left(\operatorname{Li}_{p,1}(i, 1) + \operatorname{Li}_{p,1}(i, -1)\right). \tag{4.33}$$

For $p > 1$ the interchange is justified by $\sum_{n\geq 2} H_{n-1}/n^p < \infty$. At $p = 1$, use the common radial Abel limit and the ordinary convergence proved by summation by parts. Thus (4.31) and (4.32) are equivalent.

4.5.2 A convergent octahedral change of variables

Write $\omega_a = dt/(t - a)$ and

$$\mathcal{I}(a_1 \cdots a_w) = \int_{0<t_w<\cdots<t_1<1} \omega_{a_1}(t_1) \cdots \omega_{a_w}(t_w),$$

extended multilinearly in the letters. A convergent word has first letter different from 1 and last letter different from 0. For colors $c_j \in \mathbb{Z}/4\mathbb{Z}$, set

$$Z((s_1, c_1), \ldots, (s_d, c_d)) = \operatorname{Li}_{s_1,\ldots,s_d}(i^{c_1}, \ldots, i^{c_d}).$$

Its integral word is

$$Z(\boldsymbol{s};\boldsymbol{c}) = (-1)^d \mathcal{I}\left(0^{s_1-1} i^{-c_1}\, 0^{s_2-1} i^{-(c_1+c_2)} \cdots 0^{s_d-1} i^{-(c_1+\cdots+c_d)}\right). \tag{4.34}$$

This fixes both the color convention and every depth-dependent sign.

Consider the involution

$$\phi(t) = \frac{1-t}{1+t}.$$

It preserves the puncture set $\{0, \infty, 1, -1, i, -i\}$ and reverses the real interval. Direct differentiation gives

$$\phi^*\omega_a = \omega_{\phi(a)} - \omega_{-1} \quad (a \neq -1), \qquad \phi^*\omega_{-1} = -\omega_{-1},$$

where $\phi(0) = 1$, $\phi(1) = 0$, $\phi(i) = -i$, and $\phi(-i) = i$. Denote this linear map on letters by τ. Changing variables and reversing the order of the simplex proves

$$\mathcal{I}(a_1 \cdots a_w) = (-1)^w \mathcal{I}\big(\tau(a_w) \cdots \tau(a_1)\big). \tag{4.35}$$

No tangential-basepoint correction occurs here: an original last letter other than 0 transforms into first letters other than 1, and an original first letter other than 1 transforms into last letters other than 0. Every term of the expansion therefore remains convergent. The broader use of octahedral symmetries to supplement standard level-four relations goes back to Zhao [72, 69]; the elementary involution above is sufficient for the present certificate.

Only one instance of (4.35) is needed. The word of $\mathrm{Li}_{1,4}(i, 1)$ is $(-i)000(-i)$, and hence

$$D := \mathrm{Im}\left\{\mathcal{I}(e_{-i}e_0^3 e_{-i}) + \mathcal{I}\big((e_i - e_{-1})(e_1 - e_{-1})^3(e_i - e_{-1})\big)\right\} = 0. \tag{4.36}$$

Here e_a means the letter at a, multiplication is concatenation, and powers are powers in the noncommutative word algebra. The transformed product expands into exactly 32 integral words, with coefficients ± 1. The first integral is the single-color double; the second group is converted to colored polylogarithms using (4.34).

4.5.3 The standard relation rows

The certificate uses two schemas, interpreted over $\mathbb{Q}$ before imaginary parts are taken. First, for two admissible multiple-polylogarithm index/color lists u, v, let $\mathsf{H}(u, v)$ be the integral shuffle product and $\mathsf{S}(u, v)$ the series stuffle product. Then

$$R_{u,v} := \mathrm{Im}\big(\mathsf{H}(u, v) - \mathsf{S}(u, v)\big) = 0. \tag{4.37}$$

The same row is used when $u = ((1, 0))$ and v is admissible, with the standard single-divergence regularization. This point must not be suppressed: $\mathrm{Li}_1(1)$ itself is divergent. Its shuffle and stuffle regularizations are both denoted by a formal variable T. A single leading $(1, 0)$ gives a polynomial of degree at most one in T, and the regularization comparison is the identity in degrees zero and one. Explicitly, its comparison operator satisfies

$$\rho(e^{Tu}) = \exp\left(\sum_{n\geq 2} \frac{(-1)^n \zeta(n)}{n} u^n\right) e^{Tu},$$

so $\rho(1) = 1$ and $\rho(T) = T$ [69, Theorem 2.9]. Subtracting the two product expansions cancels their common divergent word and leaves the convergent identity (4.37). This is the single-divergence case

of regularized double shuffle [73, 69]. The verifier rejects all other divergent input patterns and checks that every surviving word is admissible.

Second, for a list $z = ((s_1, c_1), \dots, (s_d, c_d))$ with $c_j \in \{0, 1\}$ and with all terms below convergent, ordinary distribution is

$$Z(\boldsymbol{s}; 2\boldsymbol{c}) - 2^{w-d} \sum_{\boldsymbol{\epsilon} \in \{0,2\}^d} Z(\boldsymbol{s}; \boldsymbol{c} + \boldsymbol{\epsilon}) = 0, \qquad w = s_1 + \cdots + s_d. \tag{4.38}$$

Indeed summing signs forces every nested summation index to be even, so the factor is exactly 2^{w-d}. The certificate multiplies some such relations by an admissible value of complementary weight and expands that product by integral shuffle. These are *lifted convergent distribution* rows; no regularized distribution formula is assumed. At weight one this includes the elementary seed $\mathrm{Li}_1(-1) = \mathrm{Li}_1(i) + \mathrm{Li}_1(-i)$.

4.5.4 The exact certificate identity

In the formal real vector space whose generators are imaginary parts of convergent colored values of weight five, define

$$\begin{aligned} F = {} & 1120 \operatorname{Im} \mathrm{Li}_{4,1}(i, -1) + 480 g_{4,1} + 480 g_{3,2} + 1440 g_{2,3} - 1536 \operatorname{Im} \mathrm{Li}_5(i) \\ & + 135 \operatorname{Im}\big(\mathrm{Li}_2(i) \mathrm{Li}_3(1)\big) - 2240 \operatorname{Im}\big(\mathrm{Li}_4(i) \mathrm{Li}_1(-1)\big). \end{aligned} \tag{4.39}$$

The two products in this expression are expanded by integral shuffle. Complex conjugation is imposed only by $\operatorname{Im} Z(\boldsymbol{s}; -\boldsymbol{c}) = -\operatorname{Im} Z(\boldsymbol{s}; \boldsymbol{c})$.

Lemma 4.9 (Finite rational certificate)**.** *The accompanying file* `S4_certificate.json` *gives* 911 *rational numbers* q_r *and* 911 *standard-relation descriptors* R_r *such that the following is an exact equality of formal vectors:*

$$F = -1120 D + \sum_{r=1}^{911} q_r R_r. \tag{4.40}$$

The rows consist of 713 *convergent double-shuffle rows,* 181 *rows with one regularized weight-one factor, and* 17 *lifted convergent distribution rows.*

Proof. The supplied program `verify_s4.py` rebuilds every row from the indices and colors in its descriptor. Integral shuffles are generated by choosing interleaving positions, series stuffles by the three defining recursive cases, and distribution by sign enumeration. The program also reconstructs the single duality seed and the target independently of their stored expansions. All coefficients are integers or instances of Python's exact `Fraction` type. Summing the left side minus the right side of (4.40) yields the empty coefficient dictionary, so every formal-word coefficient is exactly zero. The verifier neither reruns a rank calculation nor uses numerical polylogarithms, integer-relation searches, cached matrices, or modular arithmetic. Thus the certificate is a finite rational proof of the stated formal equality. □

Proof of Theorem 4.8. Every R_r vanishes by (4.37) or (4.38), and $D = 0$ by (4.36). The certificate therefore gives $F = 0$. Now

$$\operatorname{Im} \mathrm{Li}_5(i) = \beta(5) = \frac{5\pi^5}{1536}, \qquad \mathrm{Li}_3(1) = \zeta(3), \qquad \mathrm{Li}_1(-1) = -\log 2.$$

Substituting these evaluations into (4.39) gives

$$1120 \operatorname{Im} \mathrm{Li}_{4,1}(i, -1) + 480 g_{4,1} + 480 g_{3,2} + 1440 g_{2,3} - 5\pi^5 + 135 G\zeta(3) + 2240 \beta(4) \log 2 = 0.$$

Division by 1120 proves (4.32), and (4.33) proves (4.31). □

Independent numerical audit. An independent real-quadrature evaluation uses

$$S_4 = \frac{1}{6}\int_0^1 \frac{\log^3 x\,\log(1+x^2)}{1+x^2}\,dx, \qquad g_{a,b} = \frac{1}{(a-1)!}\int_0^1 (-\log x)^{a-1}\,\mathrm{Im}\,\frac{i\,\mathrm{Li}_b(ix)}{1-ix}\,dx.$$

At 70 decimal working digits it gives

$$S_4 = -0.010493456297335043453356751006201751963510618877319761513129 4,$$

with discrepancy about 4.6×10^{-71} between the two sides. This numerical agreement is an audit of conventions, not a step in the proof; the proof is the exact certificate above.

Scope and further work. The existing two relations among $g_{4,1}, g_{3,2}, g_{2,3}, g_{1,4}$ may be used to shorten (4.31); they are prior results and are not counted as new here. The present proof establishes an upper bound on a specific constant's expression complexity. It says nothing about a lower bound, linear independence, or the impossibility of a classical-polylogarithm reduction. A useful next problem is to compress the 911 standard rows into a short conceptual derivation, and to apply the same convergent involution to the analogous odd-weight mixed sums $S_6, S_8, \ldots$.

Corollary 4.10 (The two-coordinate form)**.**

$$S_4 = \frac{58g_{4,1}+24g_{3,2}}{7} + \frac{19\pi^5}{3584} - 2\beta(4)\log 2. \tag{4.41}$$

Proof. Use the two weight-five shuffle eliminations (4.24)–(4.25) in Theorem 4.8. No independence assumption enters. □

4.5.5 A weight-three companion with its full small certificate

Theorem 4.11 (The first even-index harmonic sum)**.** *With $G=\beta(2)$ and $L=\log 2$,*

$$\begin{aligned} S_2 &= -2g_{2,1} + \frac{\pi^3}{32} - 2GL \\ &= 2\,\mathrm{Im}\,\mathrm{Li}_3\!\left(\frac{1+i}{2}\right) - GL - \frac{\pi L^2}{16} - \frac{\pi^3}{64}. \end{aligned}$$

This is the first even-index companion of the weight-five reduction. It is presented as a proved addition, without a claim of priority in the broader Euler-sum literature.

Proof. Use the convergent integral word $J(w)=\mathcal{I}(w)$ of the preceding S_4 proof, with forms $dt/(t-a)$. The short certificate makes the weight-three argument fully explicit. Code the letters by

$$a=-i, \qquad b=-1, \qquad c=1, \qquad d=i, \qquad o=0.$$

Let $\mathcal{D}(U,V)$ be signed stuffle minus shuffle in this integral-word convention. Let P be the letter pullback of the preceding octahedral involution, so $P(a)=d-b$, $P(b)=-b$, $P(c)=o-b$, $P(d)=a-b$ and $P(o)=c-b$. Put $\mathcal{O}(w)=w+\mathrm{rev}\,P(w)$ at weight three. Both evaluate to zero, with the degree-one regularization used when $U=c$, as justified by the comparison operator in the preceding S_4 proof. Every final word is convergent. Let $\mathfrak{I}$ denote projection to the formal imaginary word space: $\mathfrak{I}(\bar{w})=-\mathfrak{I}(w)$, and entirely real words are zero. Here the space has $(80-12)/2=34$ coordinates, with no assertion of numerical independence.

The target is

$$T_3 = 12oaa + 4oad + (d-a)^{\Pi_3} - 8(oa)\,\Pi\, b.$$

The complete certificate is

$$\mathfrak{I}(T_3) = \sum_{(U,V,k) \text{ in Table } 4.2} k\mathfrak{I}(\mathcal{D}(U,V)) + 8\mathfrak{I}(\mathcal{O}(aab)) - 8\mathfrak{I}(\mathcal{O}(aaa)). \tag{4.42}$$

This is a finite rational word identity: expand each shuffle and signed stuffle, apply the five-letter pullback, and identify conjugates. Every coefficient agrees exactly. Both provided implementations reconstruct and verify the identity using rational arithmetic; the second imports none of the first implementation's word code.

U	V	k	U	V	k
a	aa	-12	b	ba	24
a	ab	-6	c	ba	-48
a	ad	12	a	dc	-14
a	bb	-48	a	db	10
a	bc	32	c	ad	24
a	oa	6	c	aa	-24
a	da	12	b	ad	-4
a	dd	12	a	bd	-4
a	ac	6	a	od	-10
b	ab	16	a	oc	-26
a	ob	22	c	oa	48
b	ac	8			

Table 4.2: All 23 double-shuffle coefficients in the S_2 certificate. The four rows with first factor c use one degree of regularization.

The two additional octahedral rows are completely explicit:

w	$\mathcal{O}(w)$	coefficient
aab	$aab - b(d-b)^2$	8
aaa	$aaa + (d-b)^3$	-8

All products inside these two word expressions are concatenations, not shuffle products. For example, $P(a) = d - b$ and $P(b) = -b$; reversal therefore gives $\operatorname{rev} P(aab) = -b(d-b)^2$.

To translate the target, the elementary harmonic parity split yields

$$S_2 = g_{2,1} + u_{2,1}, \qquad u_{2,1} = \operatorname{Im} \operatorname{Li}_{2,1}(i,-1).$$

Furthermore,

$$J(d) - J(a) = \frac{i\pi}{2}, \qquad J(b) = L, \qquad J(oa) = -\operatorname{Li}_2(i).$$

Consequently

$$\operatorname{Im} J(T_3) = 4\left(S_2 + 2g_{2,1} - \frac{\pi^3}{32} + 2GL\right).$$

All rows on the right of (4.42) evaluate to zero, proving the first formula. The height-one reduction below, obtained from (4.58), gives

$$g_{2,1} = -\frac{GL}{2} + \frac{\pi L^2}{32} - \operatorname{Im} \operatorname{Li}_3\left(\frac{1+i}{2}\right) + \frac{3\pi^3}{128}$$

then gives the second formula by substitution. □

4.6 An explicit binomial parity reduction for every double weight

Throughout this section the convention is

$$F_{a,b}(z)=\mathrm{Li}_{a,b}(z,1)=\sum_{n>m>0}\frac{z^n}{n^a m^b},\qquad U_n(z)=\mathrm{Re}\,\mathrm{Li}_n(z),\quad V_n(z)=\mathrm{Im}\,\mathrm{Li}_n(z).$$

We take $|z|=1$, $z\neq 1$, and use radial boundary values. These are also the ordinary convergent series values: summation by parts applies to $H_{n-1}^{(b)}/n^a$, whose total variation is finite and which tends to zero. The result below is an elementary explicit specialization of the established multiple-polylogarithm parity theorem. It should not be presented as a new general parity theorem: Panzer proved the general theorem in 2017, including explicit formulas at depths two and three; Umezawa supplied a general-depth explicit formula in 2025 [33, 61]. Its contribution here is a short independently checkable formula in the manuscript's convention, including the boundary cases with an index equal to one.

For $w\geq 2$ define the parity component

$$E_{a,b}(z)=\begin{cases}\mathrm{Im}\,F_{a,b}(z), & w=a+b \text{ even},\\ \mathrm{Re}\,F_{a,b}(z), & w=a+b \text{ odd},\end{cases}\qquad W_n(z)=\begin{cases}V_n(z), & w \text{ even},\\ U_n(z), & w \text{ odd}.\end{cases}$$

The choice of W_n depends on the fixed total weight w, not on n.

Theorem 4.12 (Explicit double parity formula)**.** *Let $w\geq 2$, $1\leq p\leq w-1$, and $q=w-p$. Put*

$$A_p(z)=\sum_{j=1}^{p}\binom{w-j-1}{q-1}E_{w-j,j}(z),$$

$$P_p(z)=\begin{cases}U_q(z)V_p(z), & w \text{ even},\ p \text{ odd},\\ U_p(z)V_q(z), & w \text{ even},\ p \text{ even},\\ -V_p(z)V_q(z), & w \text{ odd},\ p \text{ odd},\\ U_p(z)U_q(z), & w \text{ odd},\ p \text{ even}.\end{cases}$$

Then

$$\begin{aligned}A_p(z)=P_p(z)&+\sum_{\substack{3\leq j\leq p\\ j \text{ odd}}}\binom{w-j-1}{q-1}\zeta(j)W_{w-j}(z)-\sum_{\substack{3\leq j\leq q\\ j \text{ odd}}}\binom{w-j-1}{p-1}\zeta(j)W_{w-j}(z)\\ &-\frac{1}{2}\left[\binom{w-1}{q}-\binom{w-1}{p}\right]W_w(z)-\mathbf{1}_{w \text{ odd}}\frac{(-1)^p}{2}\zeta(w).\end{aligned}\tag{4.43}$$

Each individual double is recovered by the explicit inverse transform

$$E_{w-b,b}(z)=\sum_{p=1}^{b}(-1)^{b-p}\binom{w-p-1}{b-p}A_p(z),\qquad 1\leq b<w.\tag{4.44}$$

Consequently every imaginary part of even total weight and every real part of odd total weight in the same-argument double family is an explicit polynomial in single polylogarithms and ordinary zeta values.

Proof. We supply the analytic identity responsible for the reduction, so that no rank computation or numerical recognition enters the proof. Define

$$T_{p,q}(k) = \sum_{m=1}^{\infty} \frac{1}{(m+k)^p m^q}, \qquad k \geq 1.$$

Partial fractions at $m = 0$ and $m = -k$ give

$$\begin{aligned} \frac{1}{(m+k)^p m^q} = &\sum_{j=1}^{q} (-1)^{q-j} \binom{w-j-1}{p-1} \frac{1}{k^{w-j} m^j} \\ &+ (-1)^q \sum_{j=1}^{p} \binom{w-j-1}{q-1} \frac{1}{k^{w-j}(m+k)^j}. \end{aligned} \tag{4.45}$$

The two $j = 1$ divergences cancel. Summing first to a finite cutoff and then letting it tend to infinity yields

$$\begin{aligned} T_{p,q}(k) = &\sum_{j=2}^{q} (-1)^{q-j} \binom{w-j-1}{p-1} \frac{\zeta(j)}{k^{w-j}} + (-1)^q \sum_{j=2}^{p} \binom{w-j-1}{q-1} \frac{\zeta(j)}{k^{w-j}} \\ &+ (-1)^{q+1} \sum_{j=1}^{p} \binom{w-j-1}{q-1} \frac{H_k^{(j)}}{k^{w-j}}. \end{aligned} \tag{4.46}$$

Write

$$\mathcal{A}_p(z) = \sum_{j=1}^{p} \binom{w-j-1}{q-1} F_{w-j,j}(z), \qquad C_p = \binom{w-1}{q},$$

and

$$\mathcal{U}_{p,q}(z) = \sum_{j=2}^{q} (-1)^{q-j} \binom{w-j-1}{p-1} \zeta(j) \operatorname{Li}_{w-j}(z) + (-1)^q \sum_{j=2}^{p} \binom{w-j-1}{q-1} \zeta(j) \operatorname{Li}_{w-j}(z).$$

Using $H_k^{(j)} = H_{k-1}^{(j)} + k^{-j}$ and the hockey-stick identity, equation (4.46) becomes

$$R_{p,q}(z) := \sum_{k \geq 1} z^k T_{p,q}(k) = \mathcal{U}_{p,q}(z) + (-1)^{q+1} \big(\mathcal{A}_p(z) + C_p \operatorname{Li}_w(z) \big). \tag{4.47}$$

There are two product identities. The familiar same-argument shuffle is

$$\operatorname{Li}_p(z) \operatorname{Li}_q(z) = \mathcal{A}_p(z) + \mathcal{A}_q(z). \tag{4.48}$$

The Fourier-convolution identity is

$$\operatorname{Li}_p(z) \operatorname{Li}_q(z^{-1}) = \zeta(w) + R_{p,q}(z) + R_{q,p}(z^{-1}). \tag{4.49}$$

For completeness, this remains valid when $p = 1$ or $q = 1$. Each $\operatorname{Li}_p(e^{i\theta})$ belongs to L^2 because $\sum n^{-2p} < \infty$. The Fourier coefficients of their product are therefore the absolutely convergent convolutions, equal to $\zeta(w)$ at index zero and to $T_{p,q}(k)$ and $T_{q,p}(k)$ at positive and negative indices. Both sequences $T_{p,q}(k)$ decrease to zero. Dirichlet's test gives convergence of the two Fourier series, uniform on closed subarcs away from $z = 1$. Abel–Poisson summation of the L^1 product converges

to the continuous product at every such point, proving (4.49) there. Thus the derivation has never multiplied divergent zeta series.

If w is even, take imaginary parts in (4.48)–(4.49). Since p and q have the same parity, these give the sum and difference of A_p, A_q. Moreover,

$$\begin{aligned}&\operatorname{Im}\bigl(\mathcal{U}_{p,q}(z)+\mathcal{U}_{q,p}(z^{-1})\bigr)\\&\quad=2(-1)^p\left[\sum_{\substack{3\le j\le p\\ j\text{ odd}}}\binom{w-j-1}{q-1}\zeta(j)V_{w-j}(z)-\sum_{\substack{3\le j\le q\\ j\text{ odd}}}\binom{w-j-1}{p-1}\zeta(j)V_{w-j}(z)\right].\end{aligned}$$

Solving the two elementary equations gives (4.43) for even w. If w is odd, take real parts. Now p, q have opposite parity and the same two equations again give the sum and difference. The analogous last display has $-2(-1)^p$ in place of $2(-1)^p$, with U in place of V. The constant term $\zeta(w)$ contributes $-(-1)^p\zeta(w)/2$. This proves (4.43) in the odd case too.

Finally the coefficient matrix in the definition of A_p is lower triangular with diagonal one. Its inverse follows from

$$\binom{w-j-1}{p-j}\binom{w-p-1}{b-p}=\binom{w-j-1}{b-j}\binom{b-j}{p-j}$$

and $\sum_{p=j}^{b}(-1)^{b-p}\binom{b-j}{p-j}=\delta_{b,j}$. This proves (4.44). □

4.6.1 Consequences for Gaussian and Eisenstein values

At $z=i$ we have $V_n(i)=\beta(n)$, $U_1(i)=-\log 2/2$, and

$$U_n(i)=-2^{-n}(1-2^{1-n})\zeta(n),\qquad n\ge 2.$$

The odd beta values are rational multiples of powers of π. Thus Theorem 4.12 supplies an explicit all-weight reduction to the Gaussian single-value algebra in the appropriate parity. It proves all five previously numerical weight-six equations:

$$2048\operatorname{Im}\operatorname{Li}_{5,1}(i,1)=-64\pi^3\zeta(3)-527\pi\zeta(5)+4096\beta(6), \tag{4.50}$$

$$1536\operatorname{Im}\operatorname{Li}_{4,2}(i,1)=96\pi^3\zeta(3)-32\pi^2\beta(4)+1581\pi\zeta(5)-8448\beta(6), \tag{4.51}$$

$$1024\operatorname{Im}\operatorname{Li}_{3,3}(i,1)=-3\pi^3\zeta(3)+64\pi^2\beta(4)-1581\pi\zeta(5)+4608\beta(6), \tag{4.52}$$

$$23040\operatorname{Im}\operatorname{Li}_{2,4}(i,1)=-14\pi^4\beta(2)+135\pi^3\zeta(3)-1440\pi^2\beta(4) \tag{4.53}$$

$$+23715\pi\zeta(5)-69120\beta(6), \tag{4.54}$$

$$92160\operatorname{Im}\operatorname{Li}_{1,5}(i,1)=-150\pi^5\log 2+56\pi^4\beta(2)-270\pi^3\zeta(3) \tag{4.55}$$

$$+1920\pi^2\beta(4)-675\pi\zeta(5). \tag{4.56}$$

An especially simple all-order subfamily is

$$\operatorname{Im}\operatorname{Li}_{2m-1,1}(z,1)=(m-1)V_{2m}(z)+U_{2m-1}(z)V_1(z)-\sum_{k=1}^{m-1}\zeta(2k+1)V_{2m-2k-1}(z),\qquad m\ge 1. \tag{4.57}$$

For example, this gives the further proved weight-eight identity

$$\operatorname{Im}\operatorname{Li}_{7,1}(i,1)=3\beta(8)-\frac{5\pi^5}{1536}\zeta(3)-\frac{\pi^3}{32}\zeta(5)-\frac{8255\pi}{32768}\zeta(7).$$

At $z=\rho=e^{2\pi i/3}$ write $C_n=\operatorname{Im}\operatorname{Li}_n(\rho)$. The same theorem applies unchanged; in particular

$$\operatorname{Im}\operatorname{Li}_{5,1}(\rho,1)=2C_6-\frac{2\pi^3}{81}\zeta(3)-\frac{121\pi}{486}\zeta(5),\qquad \operatorname{Im}\operatorname{Li}_{3,1}(\rho,1)=C_4-\frac{13\pi}{54}\zeta(3).$$

These statements assert reduction identities, not numerical or motivic independence of the remaining single values.

4.7 Classical reductions for every index with one two

This section uses the one-variable abbreviation $\operatorname{Li}_{k_1,\dots,k_d}(z)=\operatorname{Li}_{k_1,\dots,k_d}(z,1,\dots,1)$. Let $\{1\}^r$ denote r consecutive ones. Work first on $0<z<1$, and then continue analytically along paths avoiding the cuts of the displayed functions. In particular, the formulas hold at $z=i$ by continuation through the upper half-plane, with the principal values used below.

Theorem 4.13 (One two at arbitrary depth)**.** *For $d\ge 1$, putting $a=\log(1-z)$ gives*

$$\operatorname{Li}_{2,\{1\}^{d-1}}(z)=\zeta(d+1)+\frac{(-1)^d a^d\log z}{d!}+\sum_{k=0}^{d-1}\frac{(-1)^{k+1}a^k}{k!}\operatorname{Li}_{d+1-k}(1-z). \tag{4.58}$$

For all $r,s\ge 0$, put $\alpha=-\log(1-z)$. Then

$$\operatorname{Li}_{\{1\}^r,2,\{1\}^s}(z)=\sum_{j=0}^{r}(-1)^j\binom{s+j+1}{j}\frac{\alpha^{r-j}}{(r-j)!}\operatorname{Li}_{2,\{1\}^{s+j}}(z). \tag{4.59}$$

Consequently every value with exactly one index equal to two and all other indices equal to one reduces to classical polylogarithms and logarithms.

The height-one functional equation is classical; a modern treatment and arithmetic analogue appear in [63]. The proof below also records the shuffle reduction when the two occupies any position.

Proof. The iterated integral for d ones is $\operatorname{Li}_{\{1\}^d}(z)=\alpha^d/d!$. Hence

$$\operatorname{Li}_{2,\{1\}^{d-1}}(z)=\frac{(-1)^d}{d!}\int_0^z\frac{\log^d(1-t)}{t}\,dt.$$

Set $s=1-t$ and integrate by parts d times. The antiderivative

$$-\log^d s\log(1-s)+\sum_{j=1}^{d}(-1)^j\frac{d!}{(d-j)!}\log^{d-j}s\operatorname{Li}_{j+1}(s)$$

has derivative $\log^d(s)/(1-s)$. At $s=1$ its limiting value is $(-1)^d d!\zeta(d+1)$. This gives (4.58) including the constant term and its sign.

For the second identity, locate the unique zero letter in the iterated integral word $1^r0\,1^{s+1}$. Its variable is t. Integrating the r outer and $s+1$ inner identical letters gives

$$\operatorname{Li}_{\{1\}^r,2,\{1\}^s}(z)=\frac{1}{r!(s+1)!}\int_0^z\big(\alpha(z)-\alpha(t)\big)^r\alpha(t)^{s+1}\frac{dt}{t}.$$

Expanding the first factor and using the preceding integral formula proves (4.59). Analytic continuation then gives the asserted complex values. □

4.7.1 Proof of all three weight-four Gaussian triple formulas

Put $L = \log 2$, $G = \beta(2)$, $r = (1+i)/2$, and $\lambda_n = \operatorname{Im}\operatorname{Li}_n(r)$, $\mu_n = \operatorname{Re}\operatorname{Li}_n(r)$. The elementary dilogarithm reflection, Landen transformation, and trilogarithm reflection give

$$\begin{aligned} \mu_2 &= \frac{5\pi^2}{96} - \frac{L^2}{8}, & \lambda_2 &= G - \frac{\pi L}{8}, \\ \mu_3 &= \frac{35}{64}\zeta(3) - \frac{5\pi^2 L}{192} + \frac{L^3}{48}. \end{aligned} \tag{4.60}$$

For clarity, the trilogarithm identity used for the last formula is

$$\begin{aligned} &\operatorname{Li}_3(z) + \operatorname{Li}_3(1-z) + \operatorname{Li}_3\!\left(-\frac{z}{1-z}\right) \\ &\quad = \zeta(3) + \zeta(2)\log(1-z) - \frac{1}{2}\log z \log^2(1-z) + \frac{1}{6}\log^3(1-z). \end{aligned}$$

It follows by differentiation using the dilogarithm identities and fixing the constant as $z \to 0$; it is then continued to $z = r$, where $1-r = \bar{r}$ and $-r/(1-r) = -i$. Taking real parts proves the displayed μ_3 evaluation. The first two formulas follow respectively from $\operatorname{Li}_2(r) + \operatorname{Li}_2(\bar{r}) = \zeta(2) - \log r \log \bar{r}$ and $\operatorname{Li}_2(i) + \operatorname{Li}_2(\bar{r}) = -\frac{1}{2}\log^2(1-i)$.

To evaluate the height-one formulas at i, write

$$a = \log(1-i) = \frac{L}{2} - \frac{\pi i}{4}, \qquad c = \log(-(1-i)) = \frac{L}{2} + \frac{3\pi i}{4}.$$

Since $(1-i)^{-1} = r$, the needed principal-branch inversion formulas are

$$\begin{aligned} \operatorname{Li}_2(1-i) &= -\operatorname{Li}_2(r) - \frac{\pi^2}{6} - \frac{c^2}{2}, \\ \operatorname{Li}_3(1-i) &= \operatorname{Li}_3(r) - \frac{\pi^2 c}{6} - \frac{c^3}{6}, \\ \operatorname{Li}_4(1-i) &= -\operatorname{Li}_4(r) - \frac{7\pi^4}{360} - \frac{\pi^2 c^2}{12} - \frac{c^4}{24}. \end{aligned}$$

Substitution into (4.58) with $d = 2, 3$, using $\log i = \pi i/2$, yields

$$\begin{aligned} \operatorname{Re}\operatorname{Li}_{2,1}(i) &= \frac{29}{64}\zeta(3) - \frac{\pi G}{4}, \\ \operatorname{Im}\operatorname{Li}_{2,1}(i) &= -\frac{GL}{2} + \frac{\pi L^2}{32} - \lambda_3 + \frac{3\pi^3}{128}, \\ \operatorname{Im}\operatorname{Li}_{2,1,1}(i) &= \lambda_4 + \frac{L\lambda_3}{2} - \frac{G(\pi^2 - 4L^2)}{32} - \frac{\pi(8L^3 + 105\zeta(3))}{768}. \end{aligned}$$

The real-part evaluation of $\operatorname{Li}_{2,1}(i)$ is also printed in Panzer [33], after reversing his index convention. The remaining two identities follow from exact shuffle relations:

$$\begin{aligned} \operatorname{Li}_1(z)\operatorname{Li}_{2,1}(z) &= \operatorname{Li}_{1,2,1}(z) + 3\operatorname{Li}_{2,1,1}(z), \\ \operatorname{Li}_2(z)\operatorname{Li}_{1,1}(z) &= 3\operatorname{Li}_{2,1,1}(z) + 2\operatorname{Li}_{1,2,1}(z) + \operatorname{Li}_{1,1,2}(z). \end{aligned}$$

Using $\operatorname{Li}_1(i) = -a$, $\operatorname{Li}_{1,1}(i) = a^2/2$, and $\operatorname{Li}_2(i) = -\pi^2/48 + iG$, one obtains

$$\begin{aligned} \operatorname{Im}\operatorname{Li}_{1,2,1}(i) &= -3\lambda_4 - L\lambda_3 + \frac{G(\pi^2 - 4L^2)}{32} - \frac{3\pi^3 L}{256} + \frac{\pi(2L^3 + 67\zeta(3))}{128}, \\ \operatorname{Im}\operatorname{Li}_{1,1,2}(i) &= 3\lambda_4 + \frac{L\lambda_3}{2} + \frac{5\pi^3 L}{192} - \frac{163\pi\zeta(3)}{256}. \end{aligned}$$

These are exactly the three numerical formulas in the original experimental draft, now proved. The arbitrary-depth theorem proves additional such identities without a new integer-relation search. It does not address the independent status of the weight-six triples $(3,1,2),(3,2,1),(4,1,1)$, which contain more than one zero letter and lie outside this reduction mechanism.

4.8 Classical reductions with one index equal to two

This section settles the three Gaussian weight-four triple identities printed in Chapter 4 of the source manuscript. The proof gives a uniform formula for every multiple polylogarithm whose indices consist of one 2 and any number of 1's. The general phenomenon is classical: Borwein and Straub give the complementary-argument formula for $\mathrm{Li}_{2,\{1\}^{p-1}}$ in [66, Eq. (33)], including the weight-four instance in their Eq. (34). The results below reorganize that reduction at the reciprocal complementary argument, make the dependence on the position of the 2 explicit, and prove the specific Gaussian formulas in the manuscript. We do not claim that classical reducibility of this family is a new phenomenon.

Throughout this section, a single displayed argument abbreviates the remaining arguments 1:

$$\mathrm{Li}_{s_1,\dots,s_d}(z) := \mathrm{Li}_{s_1,\dots,s_d}(z,1,\dots,1).$$

The defining series has strictly decreasing summation indices. Set

$$U_{a,b}(z) = \mathrm{Li}_{\{1\}^a,2,\{1\}^b}(z), \qquad H_p(z) = \mathrm{Li}_{2,\{1\}^{p-1}}(z),$$

where $a,b \geq 0$ and $p \geq 1$. The weight of $U_{a,b}$ is $N = a+b+2$, and its series depth is $N-1$.

4.8.1 A logarithmic moment and its branch conventions

We use principal branches and first work on

$$\mathcal{D} = \mathbb{C} \setminus [0,\infty), \qquad w = \frac{1}{1-z}, \qquad L = \operatorname{Log} w = -\operatorname{Log}(1-z).$$

On this domain, w avoids the standard polylogarithm cut $[1,\infty)$ and the logarithm cut $(-\infty,0]$. All terms in the formulas below are therefore single-valued analytic functions. The formulas also extend to the usual slit plane by consistent continuation; no such extension is needed at $z = i$ or at a nonreal root of unity.

Lemma 4.14 (Logarithmic moment)**.** *Let $q(t) = -\operatorname{Log}(1-t)$. For an integration path from 0 to z avoiding $\{0,1\}$ apart from its initial point 0, with the branches continued along that path,*

$$U_{a,b}(z) = \frac{1}{a!(b+1)!} \int_0^z \frac{(q(z)-q(t))^a q(t)^{b+1}}{t}\, dt. \tag{4.61}$$

Consequently,

$$U_{a,b}(z) = \sum_{j=0}^{a} (-1)^j \binom{b+j+1}{j} \frac{q(z)^{a-j}}{(a-j)!}\, H_{b+j+1}(z). \tag{4.62}$$

Proof. The iterated-integral word of $U_{a,b}$ is $1^a 0\, 1^{b+1}$, with forms $dt/(1-t)$ and dt/t attached to 1 and 0, respectively. The innermost $b+1$ identical forms contribute $q(t)^{b+1}/(b+1)!$. The a identical forms between t and z contribute $(q(z)-q(t))^a/a!$. Integrating the single remaining 0 form proves

(4.61). Equivalently, one may verify the same formula by differentiating successively with respect to z and using the zero initial values at $z = 0$. Expanding the binomial and using

$$H_p(z) = \frac{1}{p!} \int_0^z \frac{q(t)^p}{t}\, dt$$

proves (4.62). □

Theorem 4.15 (Explicit reduction for every position of the 2)**.** *For $a, b \geq 0$, $N = a + b + 2$, and $z \in \mathcal{D}$,*

$$U_{a,b}(z) = \sum_{k=a+1}^{N} (-1)^{a+k} \binom{k-1}{a} \frac{L^{N-k}}{(N-k)!} \operatorname{Li}_k(w) - \frac{L^N}{N!} + (-1)^{b+1} \sum_{j=0}^{a} \binom{b+j+1}{j} \frac{L^{a-j}}{(a-j)!} \zeta(b+j+2). \tag{4.63}$$

In particular,

$$H_p(z) = (-1)^p \zeta(p+1) - \frac{L^{p+1}}{(p+1)!} + \sum_{k=1}^{p+1} (-1)^k \frac{L^{p+1-k}}{(p+1-k)!} \operatorname{Li}_k(w). \tag{4.64}$$

Proof. We first prove (4.64) for real $z < 0$, so that $0 < w < 1$ and every intermediate integral is real. Substitute $v = 1/(1-t)$ in the logarithmic moment. Then

$$\frac{dt}{t} = \frac{dv}{v(v-1)}, \qquad q(t) = \log v,$$

and

$$H_p(z) = \frac{1}{p!} \int_1^w \frac{\log^p v}{v(v-1)}\, dv.$$

The differential identity

$$\frac{d}{dv} \left[\sum_{k=1}^{p+1} \frac{(-1)^{k-1} p!}{(p+1-k)!} \log^{p+1-k} v \, \operatorname{Li}_k(v) \right] = \frac{\log^p v}{1-v} \tag{4.65}$$

follows by cancellation of adjacent terms, using $\operatorname{Li}_1(v) = -\log(1-v)$ and $\operatorname{Li}_k'(v) = \operatorname{Li}_{k-1}(v)/v$ for $k \geq 2$. Since $1/[v(v-1)] = -1/v - 1/(1-v)$, integration gives (4.64). At $v = 1$, every term in (4.65) with a positive power of $\log v$ tends to zero. This includes the apparently singular term $\log^p v \, \operatorname{Li}_1(v)$, which is $O((v-1)^p \log(1-v))$ and tends to zero because $p \geq 1$. The sole nonzero endpoint term is $(-1)^p p! \zeta(p+1)$. Thus the constant in (4.64) is fixed, not inferred from a differential identity alone. Analytic continuation on the connected domain $\mathcal{D}$ proves that identity throughout the stated domain.

Insert (4.64) in (4.62), using $q(z) = L$. The zeta terms immediately give the second line of (4.63). The coefficient of $(-1)^k L^{N-k} \operatorname{Li}_k(w)$ is

$$\sum_{j=0}^{a} \frac{(-1)^j \binom{b+j+1}{j}}{(a-j)!(b+j+2-k)!} = \frac{(-1)^a \binom{k-1}{a}}{(N-k)!}.$$

Terms containing a factorial of a negative integer are omitted. To verify this finite identity, write its left side as

$$\frac{(k-1)!}{(b+1)!a!}\sum_{j=0}^{a}(-1)^j\binom{a}{j}\binom{b+j+1}{k-1}$$

and extract the coefficient of x^{k-1} in

$$(1+x)^{b+1}\sum_{j=0}^{a}(-1)^j\binom{a}{j}(1+x)^j=(-x)^a(1+x)^{b+1}.$$

The coefficients vanish when $k\le a$, explaining the lower endpoint $a+1$ in the sum. Finally, the coefficient of $-L^N$ equals

$$\begin{aligned}\sum_{j=0}^{a}\frac{(-1)^j\binom{b+j+1}{j}}{(a-j)!(b+j+2)!}&=\frac{1}{a!(b+1)!}\int_0^1 t^{b+1}(1-t)^a\,dt\\&=\frac{1}{N!}.\end{aligned}$$

This proves the complete formula. □

For the Gaussian endpoint the path itself can be inspected explicitly:

$$z=ix\quad(0\le x\le 1),\qquad w(x)=\frac{1+ix}{1+x^2}.$$

For $x>0$ this path lies in the upper half-plane and does not meet the polylogarithm cut. Its initial point is $w(0)=1$, where the endpoint limits in the proof apply, and its final point is $(1+i)/2$. This verifies the branch choices directly, without using an inversion formula with an unspecified logarithmic correction.

The boundary series at $z=i$ agree with these analytic values. Indeed the finite nested harmonic sums in a coefficient satisfy a bound $O((1+\log n)^{d-1})$. After division by n^{s_1} their coefficients tend to zero and have summable successive differences. Summation by parts against the bounded partial sums of i^n proves convergence, even when $s_1=1$; Abel's theorem then identifies the series with the radial analytic limit.

Corollary 4.16 (A short formula when the 2 is last)**.** *For every $N\ge 2$,*

$$\begin{aligned}U_{N-2,0}(z)=(N-1)\operatorname{Li}_N(w)-L\operatorname{Li}_{N-1}(w)-\frac{L^N}{N!}\\-\sum_{m=0}^{N-2}(N-1-m)\zeta(N-m)\frac{L^m}{m!}.\end{aligned}\tag{4.66}$$

Thus only two nonconstant classical polylogarithms are needed in this subfamily, regardless of series depth.

Proof. Set $a=N-2$, $b=0$ in (4.63) and reindex the zeta sum. □

Corollary 4.17 (The top two series-depth layers)**.** *Every single-endpoint multiple polylogarithm of weight $N\ge 2$ and series depth at least $N-1$ has an explicit expression in classical polylogarithms at $(1-z)^{-1}$, zeta values, and logarithms, on the domain $\mathcal{D}$.*

Proof. Positive indices with sum N and length N are all 1, and $\mathrm{Li}_{\{1\}^N}(z) = L^N/N!$. Those with length $N-1$ have exactly one index 2 and all other indices 1, so Theorem 4.15 applies. These exhaust the possibilities. □

This statement concerns the literal depth of the defining series. It does not identify that depth with a motivic filtration or prove a minimal depth for the resulting numerical constants.

4.8.2 One possible new direction at each weight

Put

$$\ell = \log 2, \qquad w = \frac{1+i}{2}, \qquad \mu_k = \Re\,\mathrm{Li}_k(w), \qquad \lambda_k = \Im\,\mathrm{Li}_k(w).$$

For $N \geq 2$, let $\mathcal{P}_N$ be the rational span of products of at least two positive-weight atoms selected from

$$\pi, \ell, \quad \zeta(k), \mu_k, \lambda_k \quad (2 \leq k < N),$$

whose assigned weights sum to N; the assigned weights are 1 for π, ℓ and k for the atoms with subscript or argument k. This definition is a concrete span of numerical products. It does not assume that numerical periods form a direct sum by weight.

Corollary 4.18 (Signed binomial coefficients modulo products)**.** *For $a+b+2 = N$, set $c_{N,a} = (-1)^{N+a}\binom{N-1}{a}$. Then*

$$U_{a,b}(i) \equiv c_{N,a}\big(\mathrm{Li}_N(w) - \zeta(N)\big) \pmod{\mathcal{P}_N + i\mathcal{P}_N}, \tag{4.67}$$

$$\Re U_{a,b}(i) \equiv c_{N,a}\big(\mu_N - \zeta(N)\big) \pmod{\mathcal{P}_N}, \tag{4.68}$$

$$\Im U_{a,b}(i) \equiv c_{N,a}\lambda_N \pmod{\mathcal{P}_N}. \tag{4.69}$$

In particular, the imaginary parts of this entire family contribute at most one direction after products of the stated lower-weight atoms have been factored out.

Proof. At $z = i$, $L = -\ell/2 + i\pi/4$, and $\mathrm{Li}_1(w) = -\overline{L}$. In (4.63), every term is a product of lower-weight atoms except the term $k = N$ and the term $j = a$ in the zeta sum. Their respective coefficients are $c_{N,a}$ and $-c_{N,a}$. Taking real and imaginary parts proves the statements. □

The coefficients of the highest-weight half-point imaginary value are

N	$c_{N,0}, \ldots, c_{N,N-2}$
4	$1, -3, 3$
5	$-1, 4, -6, 4$
6	$1, -5, 10, -10, 5.$

The corollary is an upper bound, and proves neither that λ_N is nonzero in the quotient nor that it is independent of constants from other argument families. It also does not identify series depth with an additive motivic depth filtration.

4.8.3 A sixth-root specialization in every position

The special role of sixth roots in relations for Nielsen polylogarithms is studied by Borwein and Straub [66]. The preceding position formula gives a direct specialization in that established setting, including an explicit rational coefficient for one component at every weight. Throughout this subsection,

$$\omega = e^{i\pi/3}, \qquad N = a+b+2, \qquad a, b \geq 0,$$

and all logarithms and polylogarithms use their principal branches.

Corollary 4.19 (Sixth-root closure and a pure parity component)**.** *At $z=\omega$, formula (4.63) reduces $U_{a,b}(\omega)$ to classical values $\mathrm{Li}_k(\omega)$, zeta values, and powers of $i\pi$, since*

$$\frac{1}{1-\omega}=\omega, \qquad -\operatorname{Log}(1-\omega)=\operatorname{Log}\omega=\frac{i\pi}{3}. \tag{4.70}$$

Moreover,

$$U_{a,b}(\omega)+(-1)^N\overline{U_{a,b}(\omega)}=R_{N,a}(i\pi)^N, \qquad R_{N,a}\in\mathbb{Q}. \tag{4.71}$$

Thus $\Re U_{a,b}(\omega)$ is a rational multiple of π^N when N is even, and $\Im U_{a,b}(\omega)$ is a rational multiple of π^N when N is odd. Explicitly, with Bernoulli polynomials defined by

$$\frac{te^{xt}}{e^t-1}=\sum_{k\ge 0}B_k(x)\frac{t^k}{k!}, \qquad B_k=B_k(0),$$

one may take

$$\begin{aligned} R_{N,a}=&-\frac{2}{3^N N!}-\sum_{k=a+1}^{N}\frac{(-1)^{a+k}\binom{k-1}{a}\,2^kB_k(1/6)}{3^{N-k}(N-k)!k!}\\ &+(-1)^{N-a}\sum_{\substack{N-a\le m\le N\\ m\ \text{even}}}\frac{\binom{m-1}{N-a-1}\,2^mB_m}{3^{N-m}(N-m)!m!}. \end{aligned} \tag{4.72}$$

The same formulas at $\overline{\omega}$ follow by complex conjugation.

Proof. The identities in (4.70) follow from $1-\omega=\overline{\omega}$. Along the radial segment $z=r\omega$, $0\le r\le 1$, the real part of $1-z$ is at least $1/2$, so the indicated logarithm has no branch crossing. The transformed path lies in the upper half-plane for $r>0$, starts at 1, and ends at ω; the endpoint limits already used to prove (4.63) apply. Consequently that formula has exactly the stated principal values. The ordinary boundary multiple series agree with their radial limits by the same summation-by-parts argument used at $z=i$.

Put $T=i\pi$ and $L=T/3$. The classical Bernoulli Fourier identity is

$$\mathrm{Li}_k(e^{2\pi ix})+(-1)^k\,\mathrm{Li}_k(e^{-2\pi ix})=-\frac{(2\pi i)^k}{k!}B_k(x), \qquad 0<x<1, \quad k\ge 1. \tag{4.73}$$

For $k=1$, this follows directly from $\mathrm{Li}_1(z)=-\operatorname{Log}(1-z)$ on the stated arc. Successive integration, with the zero average of both sides fixing the constants, proves every higher case. Since $\overline{L}=-L$, the parity projection of a classical term in (4.63) is

$$\begin{aligned} &L^{N-k}\,\mathrm{Li}_k(\omega)+(-1)^N\overline{L^{N-k}\,\mathrm{Li}_k(\omega)}\\ &\qquad=L^{N-k}\big(\mathrm{Li}_k(\omega)+(-1)^k\,\mathrm{Li}_k(\overline{\omega})\big)=-\frac{2^kB_k(1/6)}{3^{N-k}k!}\,T^N. \end{aligned}$$

The polynomial term $-L^N/N!$ contributes $-2T^N/(3^NN!)$. A zeta term with zeta weight m contributes zero when m is odd. For even $m\ge 2$, use

$$\frac{2\zeta(m)}{(i\pi)^m}=-\frac{2^mB_m}{m!}.$$

In the last sum of (4.63), write $m=b+j+2=N-a+j$. Its binomial coefficient becomes $\binom{m-1}{N-a-1}$, while its sign after the even-zeta evaluation is $(-1)^b=(-1)^{N-a}$. Combining these three contributions proves (4.72) and hence (4.71). □

There is a precise geometric reason that this specialization closes on a unit-circle argument. If $|z| = 1$ and $z \neq 1$, then

$$\left|\frac{1}{1-z}\right| = 1 \quad \Longleftrightarrow \quad |1-z|^2 = 2 - 2\Re z = 1 \quad \Longleftrightarrow \quad z \in \{\omega, \overline{\omega}\}.$$

Both of these points are fixed by $z \mapsto (1-z)^{-1}$. This classifies the unit-circle endpoint geometry of this particular transformation.

Corollary 4.20 (The two weight-three values)**.** *Define the ordinary Clausen value by its absolutely convergent series*

$$C = \mathrm{Cl}_2(\pi/3) = \sum_{n \geq 1} \frac{\sin(n\pi/3)}{n^2}.$$

Then

$$U_{0,1}(\omega) = \mathrm{Li}_{2,1}(\omega) = \frac{2}{3}\zeta(3) - \frac{\pi}{3}C + \frac{i\pi^3}{324}, \tag{4.74}$$

$$U_{1,0}(\omega) = \mathrm{Li}_{1,2}(\omega) = -\frac{4}{3}\zeta(3) + \frac{\pi}{3}C + \frac{i\pi^3}{324}. \tag{4.75}$$

Proof. The classical values needed are

$$\mathrm{Li}_1(\omega) = \frac{i\pi}{3}, \qquad \mathrm{Li}_2(\omega) = \frac{\pi^2}{36} + iC, \qquad \mathrm{Li}_3(\omega) = \frac{\zeta(3)}{3} + \frac{5i\pi^3}{162}.$$

The real dilogarithm component and imaginary trilogarithm component follow from (4.73). The real trilogarithm component follows by grouping the defining series into residue classes, or from the distribution identity

$$\mathrm{Li}_s(\omega) + \mathrm{Li}_s(\overline{\omega}) = (1 - 2^{1-s})(1 - 3^{1-s})\zeta(s), \qquad s > 1.$$

For example, summing over the three cube roots gives $(3^{1-s} - 1)\zeta(s)$ for the two nontrivial cube roots; applying the two-term distribution formula to the pair of sixth roots multiplies that sum by $2^{1-s} - 1$. Now (4.64) at $p = 2$ gives, with $L = i\pi/3$,

$$U_{0,1}(\omega) = \zeta(3) - \mathrm{Li}_3(\omega) + L\,\mathrm{Li}_2(\omega) - \frac{2}{3}L^3.$$

Substitution yields (4.74). The shuffle identity

$$\mathrm{Li}_1(\omega)\,\mathrm{Li}_2(\omega) = U_{1,0}(\omega) + 2U_{0,1}(\omega)$$

then proves (4.75). In both positions, (4.72) gives $R_{3,a} = -1/162$. □

The independent script `code/independent/one_two/sixth_root.py` computes (4.72) with exact rational arithmetic and compares the specialized closed formula, its pure parity component, and the displayed weight-three examples with the logarithmic moment. It covers every position of the 2 at weights 2 through 8, at both ω and $\overline{\omega}$, using 100 working decimal digits. Across these 56 cases, the largest closed-form versus moment residual is below $6.59 \cdot 10^{-100}$; the largest parity-projection residual is below $7.15 \cdot 10^{-102}$. The four checks of the two explicit examples and their conjugates have residual below $2.16 \cdot 10^{-101}$. These are numerical checks of the proved identities, rather than rigorous quadrature error bounds.

4.8.4 Further exact identities at weights five and six

The same calculation gives formulas beyond the three displayed manuscript rows. They involve the unavoidable output of this particular reduction, namely higher classical values at w; no independence claim about those values is implied.

Corollary 4.21 (Two higher-weight Gaussian reductions)**.** *With* $H_4 = \mathrm{Li}_{2,1,1,1}$ *and* $H_5 = \mathrm{Li}_{2,1,1,1,1}$,

$$\begin{aligned}\Im H_4(i) &= -\lambda_5 - \frac{\ell\lambda_4}{2} + \frac{\pi\mu_4}{4} + \frac{\pi^2 - 4\ell^2}{32}\lambda_3 \\ &\quad + \frac{G\ell(3\pi^2 - 4\ell^2)}{192} + \frac{35\pi\ell\zeta(3)}{512} + \frac{\pi\ell^2(\ell^2 - \pi^2)}{384} - \frac{17\pi^5}{92160}, \qquad (4.76)\\ \Im H_5(i) &= \lambda_6 + \frac{\ell\lambda_5}{2} - \frac{\pi\mu_5}{4} + \frac{4\ell^2 - \pi^2}{32}\lambda_4 - \frac{\pi\ell\mu_4}{8} \\ &\quad + \frac{\ell(4\ell^2 - 3\pi^2)}{192}\lambda_3 + \frac{G(16\ell^4 - 24\ell^2\pi^2 + \pi^4)}{6144} \\ &\quad + \frac{35\pi(\pi^2 - 12\ell^2)\zeta(3)}{24576} + \frac{\pi\ell^3(5\pi^2 - 3\ell^2)}{5760}. \qquad (4.77)\end{aligned}$$

Proof. Use $p = 4$ and $p = 5$ in (4.64), substitute $L = -\ell/2 + i\pi/4$ and Lemma (4.60), and take imaginary parts. The accompanying coefficient generator performs these polynomial operations over $\mathbb{Q}$ and also emits every position of the 2 at weights four, five, and six. □

The real parts can be treated by the same formula. For example,

$$\Re\,\mathrm{Li}_{2,1,1}(i) = \mu_4 + \frac{\pi\lambda_3}{4} + \frac{\pi G\ell}{8} + \frac{35\ell\zeta(3)}{128} + \frac{\ell^4}{384} - \frac{11\pi^2\ell^2}{768} - \frac{1249\pi^4}{92160}. \qquad (4.78)$$

The machine-readable tables include both real and imaginary parts, so this companion family can be reproduced without manual transcription.

4.9 From one-two reductions to general complementary depth

For positive indices we abbreviate

$$\mathrm{Li}_{s_1,\dots,s_d}(z) := \mathrm{Li}_{s_1,\dots,s_d}(z,1,\dots,1) = \sum_{n_1>\cdots>n_d\geq 1} \frac{z^{n_1}}{n_1^{s_1}\cdots n_d^{s_d}}, \qquad |z| < 1. \qquad (4.79)$$

The indices decrease in the summation order, as in the manuscript. Weight is $w = s_1 + \cdots + s_d$ and series depth is d. An empty index denotes the multiplicative identity. We use convergent multiple zeta values $\zeta(\boldsymbol{s}) = \mathrm{Li}_{\boldsymbol{s}}(1)$ when $s_1 \geq 2$.

Let

$$\omega_0(t) = \frac{dt}{t}, \qquad \omega_1(t) = \frac{dt}{1-t}.$$

Words are written *outer letter first*. Thus

$$H_{0^{s_1-1}1\cdots 0^{s_d-1}1}(z) = \mathrm{Li}_{s_1,\dots,s_d}(z). \qquad (4.80)$$

In particular $H_{01} = \mathrm{Li}_2$, not its negative or a reversed word. The recursion is $H_{av}(z) = \int_0^z \omega_a(t)H_v(t)$ whenever this is an ordinary convergent integral. Pure-zero words are defined by $H_{0^r}(z) = \log^r z/r!$,

and trailing-zero words are obtained by shuffle regularization. Every word ending in 1 is convergent at its initial endpoint and has the ordinary interpretation in (4.80).

Throughout the functional identities we use

$$D = \mathbb{C} \setminus [0, \infty), \qquad q = \frac{1}{1-z}, \qquad L = \log q = -\log(1-z), \qquad z \in D, \tag{4.81}$$

with principal logarithms and principal polylogarithms. The image of D avoids both $(-\infty, 0]$ and $[1, \infty)$ in the q-plane, so the hyperlogarithms with possible trailing zeros have unambiguous branches. The origin is a limiting point, not an included point of D.

We first prove formulas on $z < 0$, where $0 < q < 1$ and all logarithms in q are real. Analytic continuation on the connected slit domain D then proves the stated complex identities. This method avoids an unstated choice of a value on the polylogarithm cut. It includes both $z = i$ and $z = \rho = e^{2\pi i/3}$:

$$z = i: \quad q = \frac{1+i}{2}, \qquad L = -\frac{1}{2}\log 2 + \frac{i\pi}{4}, \tag{4.82}$$

$$z = \rho: \quad q = \frac{1}{2} + \frac{i\sqrt{3}}{6}, \qquad L = -\frac{1}{2}\log 3 + \frac{i\pi}{6}. \tag{4.83}$$

Their transformed moduli are $2^{-1/2}$ and $3^{-1/2}$ respectively.

Definition 4.22 (Depth with logarithmic coefficients)**.** *A term $L^j \operatorname{Li}_{\boldsymbol{a}}(q)\zeta(\boldsymbol{b})$ has* reduced depth $\operatorname{dep}(\boldsymbol{a}) + \operatorname{dep}(\boldsymbol{b})$. *The factor L^j is a coefficient, not an additional nested sum.*

This is an explicit convention for the following reduction. It is not an identification with every motivic depth filtration, and does not turn arbitrary products of deep factors into depth-one objects.

4.10 An explicit complementary-depth theorem

4.10.1 Endpoint transport with convergent constants

For a word v ending in 0, write $J_v(q)$ for the iterated integral from 1 to q using the same forms and the same outer-first convention. These integrals converge at 1: the innermost form is ω_0, which is regular there; preceding ω_1 integrations do not destroy integrability. Set $J_\varnothing = 1$.

Lemma 4.23 (Connection formula)**.** *For every nonempty word v ending in 0 and $0 < q < 1$,*

$$J_v(q) = \sum_{v=uv'} (-1)^{|v'|} H_u(q) H_{\mathrm{rev}(v')}(1). \tag{4.84}$$

The sum includes empty prefixes and suffixes. Every nonempty endpoint word $\mathrm{rev}(v')$ starts in 0, so its value at 1 is convergent after the usual trailing-zero normalization at 0.

Proof. We give the endpoint justification rather than silently substituting into a divergent path identity. Fix a maximum word length N and work in the truncated noncommutative algebra on e_0, e_1. Let $F(q) = \sum_{|u| \le N} H_u(q) u$. It satisfies

$$F'(q) = \left(\frac{e_0}{q} + \frac{e_1}{1-q}\right) F(q).$$

Put $\ell = -\log(1-q)$. The function $e^{-\ell e_1} F(q)$ has a finite coefficientwise limit Z as $q \to 1^-$: its differential equation has coefficient $e^{-\ell e_1} e_0 e^{\ell e_1}/q$, a finite polynomial in ℓ at each truncated weight,

and this is integrable at 1. Consequently $F(q) = e^{\ell e_1} Z + O((1-q)\log^M(1-q))$ coefficientwise, for a sufficiently large finite M.

Shuffle identities say that F is group-like; the same holds for Z. The inverse coefficient in a group-like word series is $(Z^{-1})_{v'} = (-1)^{|v'|} Z_{\mathrm{rev}(v')}$, by the shuffle antipode, or equivalently by reversal of an iterated path. The coefficients of $R(q) = F(q)Z^{-1}$ solve the same differential recursion and tend to zero at 1 for every nonempty word ending in 0. Induction on word length therefore identifies them with the convergent integrals $J_v(q)$.

If v' is a nonempty suffix of v, its reverse begins in 0. The corresponding coefficient of Z is simply $H_{\mathrm{rev}(v')}(1)$: multiplication by $e^{-\ell e_1}$ only changes words beginning with 1. Taking the coefficient of v in $F(q)Z^{-1}$ proves (4.84). Since N was arbitrary, this proves every finite word identity. □

4.10.2 The word formula and the depth bound

For a word w ending in 1, let K be its number of zeros. Replace each original 1 by 0, and replace each original 0 independently by either 0 or 1. Denote the resulting 2^K words by $\tau_A(w)$, where A records the zero positions replaced by 1. Every $\tau_A(w)$ ends in 0.

Theorem 4.24 (Complementary depth)**.** *For $z \in D$ and every nonempty word w ending in 1,*

$$H_w(z) = (-1)^K \sum_A \sum_{\tau_A(w)=uv} (-1)^{|v|} H_u(q) H_{\mathrm{rev}(v)}(1). \tag{4.85}$$

After trailing-zero normalization, this is a finite rational linear combination of terms

$$L^j \operatorname{Li}_{\boldsymbol{a}}(q)\zeta(\boldsymbol{b}), \qquad j + \mathrm{wt}(\boldsymbol{a}) + \mathrm{wt}(\boldsymbol{b}) = |w|, \quad \mathrm{dep}(\boldsymbol{a}) + \mathrm{dep}(\boldsymbol{b}) \le K. \tag{4.86}$$

Nonempty zeta indices are convergent. Thus a one-variable polylogarithm of weight w and depth d has the stated reduction with $K = w - d$.

Proof. On $z < 0$, make the substitution $u = (1-t)^{-1}$ in every integration. Its pullbacks are

$$\frac{dt}{t} = -\frac{du}{u} - \frac{du}{1-u}, \qquad \frac{dt}{1-t} = \frac{du}{u}. \tag{4.87}$$

The initial endpoint $t = 0$ becomes $u = 1$. Multilinearity gives $H_w(z) = (-1)^K \sum_A J_{\tau_A(w)}(q)$. Lemma 4.23 proves (4.85).

It remains to check the claimed normal form. The identity $H_v H_0 = H_{v\Pi 0}$ removes a trailing zero. If a word ends in exactly r zeros, its coefficient in the shuffle of its prefix with 0 is r; all other words in that shuffle have fewer trailing zeros. Induction therefore expresses each $H_u(q)$ as a rational combination of $L^j H_{u'}(q)$ with u' empty or ending in 1. The operation preserves the number of 1 letters.

For an endpoint word beginning in 0, the same normalization preserves its initial 0 in every nonzero term with no logarithmic factor. Terms containing a positive power of $\log 1$ vanish. The surviving nonempty terms begin in 0 and end in 1, hence are convergent MZVs. In each split of $\tau_A(w)$ the total number of 1 letters in the two factors is $|A| \le K$. This proves the depth bound and the weight identity. Finally continue analytically from $z < 0$ to D. □

Corollary 4.25 (An argument-enlarged depth bound)**.** *For $w \ge 2$, every one-variable MPL of weight w can be represented using nonlogarithmic factors of depth at most $\lfloor w/2 \rfloor$, allowing both the original argument z and $q = (1-z)^{-1}$, MZVs, and powers of L as coefficients. More precisely, use the original expression when $d \le w/2$, and Theorem 4.24 otherwise. The all-one case is $L^w/w!$.*

This corollary is deliberately not phrased as a theorem about arbitrary multivariable MPLs, or as a reduction inside an unchanged root-of-unity basket. At $z = i$ the half-point q is not a root of unity. The theorem explains an argument-enlarged upper bound; it does not prove the absence or presence of new cyclotomic periods in a smaller comparison algebra.

For the next family put $F_r(z) = \mathrm{Li}_{2,\{1\}^{r-1}}(z)$, as in the one-two formulas above. Their reciprocal-complementary form is

$$F_r(z) = \sum_{k=1}^{r+1} \frac{(-1)^k L^{r+1-k}}{(r+1-k)!} \mathrm{Li}_k(q) + (-1)^r \zeta(r+1) - \frac{L^{r+1}}{(r+1)!}.$$

4.11 A double-polylogarithm formula for an all-weight family

The next formula addresses the first case beyond the classical ladder. It reduces the height-one family with initial index 3 to singles and doubles at q, irrespective of its original depth. Let

$$C_1(q) = 0, \qquad C_k(q) = \sum_{a=1}^{k-1} \mathrm{Li}_{a,k-a}(q) \quad (k \geq 2). \tag{4.88}$$

All MPLs on the right have the one-variable meaning (4.79).

Theorem 4.26 (Two-zero height-one identity)**.** *For every $w \geq 3$ and $z \in D$,*

$$\begin{aligned}\mathrm{Li}_{3,\{1\}^{w-3}}(z) = \frac{L^w}{w!} + \sum_{k=1}^{w} \frac{(-1)^k L^{w-k}}{(w-k)!} \big((k-2)\,\mathrm{Li}_k(q) + C_k(q)\big) \\ + (-1)^{w-1} \Big((L + \mathrm{Li}_1(q))\zeta(w-1) + (w-2)\zeta(w) - \zeta(w-1,1) \Big).\end{aligned} \tag{4.89}$$

The right side has reduced depth at most two.

Proof. Write $D_q = q\,d/dq = d/dL$. Direct differentiation of the nested series, followed by continuation, gives

$$D_q C_k = C_{k-1} + \frac{q}{1-q}\,\mathrm{Li}_{k-1} \quad (k \geq 2).$$

Put $V_k = (k-2)\,\mathrm{Li}_k + C_k$, with $V_1 = -\,\mathrm{Li}_1$. Then

$$D_q V_k = V_{k-1} + \frac{\mathrm{Li}_{k-1}}{1-q} \quad (k \geq 2), \qquad D_q V_1 = -\frac{q}{1-q}.$$

Differentiating the finite sum in (4.89) telescopes and yields

$$\frac{F_{w-2}(z) - (-1)^{w-2}\zeta(w-1)}{q-1}.$$

The derivative of the explicit $(L + \mathrm{Li}_1(q))\zeta(w-1)$ term supplies the missing constant term, because $D_q(L + \mathrm{Li}_1(q)) = 1/(1-q)$. The derivative of the entire right side is therefore $F_{w-2}(z)/(q-1)$. The left side has the same derivative, since $d\,\mathrm{Li}_{3,\{1\}^{w-3}}(z)/dz = F_{w-2}(z)/z$ and $dz/z = dL/(q-1)$.

For the endpoint constant, recall the convergent depth-two sum identity

$$\sum_{a=2}^{w-1} \zeta(a, w-a) = \zeta(w), \qquad w \geq 3. \tag{4.90}$$

It follows, for example, by subtracting the shuffle and stuffle expansions of the once-regularized product $\zeta(1)\zeta(w-1)$: its common divergent term $\zeta(1,w-1)$ cancels, leaving precisely (4.90). Only one logarithmic divergence is involved, so no higher regularization correction enters. Equivalently one may use finite cutoffs and subtract the common harmonic-logarithmic part before taking the limit.

The convergent colored stuffle identity, for $0<q<1$, gives

$$\mathrm{Li}_{1,w-1}(q)-\mathrm{Li}_1(q)\zeta(w-1)=-\mathrm{Li}_{w-1,1}(1,q)-\mathrm{Li}_w(q).$$

The right side tends to $-\zeta(w-1,1)-\zeta(w)$ by dominated convergence. Combining this with (4.90) yields

$$\lim_{q\to1^-}\big(C_w(q)-\mathrm{Li}_1(q)\zeta(w-1)\big)=-\zeta(w-1,1).$$

All positive powers of L multiplying at most logarithmically singular terms vanish. The remaining constants in (4.89) cancel, so its right side tends to zero as $z\to0^-$. The left side does too. Equality of derivatives and the endpoint limit proves the result; continue to D. □

Remark 4.27 (The sum identity and the proof dependency). Equation (4.90) is the classical depth-two Euler sum formula, not a new conjecture assumed here. The same two-zero formula can also be obtained directly from Theorem 4.24 by expanding its two zero positions and normalizing trailing zeros. The supplied code verifies the agreement of these two independently specified formulas through weight twelve.

At weight six, (4.89) is a complete depth-two representation of $\mathrm{Li}_{3,1,1,1}(i)$ and of $\mathrm{Li}_{3,1,1,1}(\rho)$. In particular a depth-four index does not by itself identify a new depth-four period in the enlarged comparison algebra. The endpoint double is harmless as a named convergent constant; standard Euler reductions can simplify it further. For example,

$$\zeta(4,1)=2\zeta(5)-\zeta(2)\zeta(3),\qquad \zeta(5,1)=\frac{3}{4}\zeta(6)-\frac{1}{2}\zeta(3)^2.$$

These last simplifications are not required for the theorem.

4.12 The colored stuffle identity

Whenever the defining sums are absolutely convergent, sorting the two indices in a product proves

$$\mathrm{Li}_a(x)\,\mathrm{Li}_b(y)=\mathrm{Li}_{a,b}(x,y)+\mathrm{Li}_{b,a}(y,x)+\mathrm{Li}_{a+b}(xy). \tag{4.91}$$

At conditionally convergent boundary points, Abel limits give the same identity when the limiting products and constituent series exist.

4.13 Gaussian doubles: closed forms

Write $\mathrm{Li}_{a,b}$ at the three Gaussian base points beyond $(\mathrm{i},1)$:

$$(\mathrm{i},\mathrm{i}),\qquad(-1,\mathrm{i}),\qquad(\mathrm{i},-1),$$

all generated to 700 digits by an accelerated evaluator (Section 12) and agreeing with Wolfram 15 to full precision. Under the level-4 complex conjugation $\mathrm{i}\mapsto-\mathrm{i}$ the imaginary parts are odd and the real parts even.

4.13.1 Family 1: the stuffle closed forms

Specializing (4.91) at the Gaussian points ($uv = -1$ at (i,i); $uv = -\mathrm{i}$ at the mixed pair) gives a complete set of exact closed forms for the symmetric combinations:

$$\mathrm{Li}_{a,b}(\mathrm{i},\mathrm{i}) + \mathrm{Li}_{b,a}(\mathrm{i},\mathrm{i}) = \mathrm{Li}_a(\mathrm{i})\,\mathrm{Li}_b(\mathrm{i}) - \mathrm{Li}_{a+b}(-1), \tag{4.92}$$

$$\mathrm{Li}_{a,b}(-1,\mathrm{i}) + \mathrm{Li}_{b,a}(\mathrm{i},-1) = \mathrm{Li}_a(-1)\,\mathrm{Li}_b(\mathrm{i}) - \mathrm{Li}_{a+b}(-\mathrm{i}), \tag{4.93}$$

$$\mathrm{Li}_{a,b}(\mathrm{i},-1) + \mathrm{Li}_{b,a}(-1,\mathrm{i}) = \mathrm{Li}_a(\mathrm{i})\,\mathrm{Li}_b(-1) - \mathrm{Li}_{a+b}(-\mathrm{i}),$$

with the clean diagonal corollary

$$\mathrm{Li}_{a,a}(\mathrm{i},\mathrm{i}) = \tfrac{1}{2}\big(\mathrm{Li}_a(\mathrm{i})^2 - \mathrm{Li}_{2a}(-1)\big)\,. \tag{4.94}$$

Every single polylogarithm on the right is elementary in the level-4 vocabulary: $\operatorname{Im}\mathrm{Li}_n(\mathrm{i}) = \beta(n)$ is the Dirichlet beta value, $\operatorname{Re}\mathrm{Li}_n(\mathrm{i})$ and $\mathrm{Li}_n(-1) = -\eta(n)$ are rational multiples of $\zeta(n)$ for $n \geq 2$ (or π^n at even n), and $\mathrm{Li}_n(-\mathrm{i}) = \overline{\mathrm{Li}_n(\mathrm{i})}$. At $n = 1$, the real values are $\operatorname{Re}\mathrm{Li}_1(\mathrm{i}) = -\log 2/2$ and $\mathrm{Li}_1(-1) = -\log 2$. So Family 1 reduces every symmetric and diagonal Gaussian double to single L-values in closed form.

Remark 4.28. The three stuffle identities (4.92)–(4.93) were checked against the stored 700-digit values to $\sim 10^{-692}$ for every admissible a, b, and the closed forms were confirmed symbolically and numerically by Wolfram 15 at low weight, including the weight-6 diagonal. Its exact homogeneous expression follows directly from stuffle:

$$\mathrm{Li}_{3,3}(\mathrm{i},\mathrm{i}) = \frac{9}{2048}\zeta(3)^2 + \frac{47\pi^6}{1935360} - \frac{3\mathrm{i}\pi^3\zeta(3)}{1024}.$$

Indeed $\mathrm{Li}_3(\mathrm{i}) = (-3\zeta(3) + \mathrm{i}\pi^3)/32$ and $\mathrm{Li}_6(-1) = -31\pi^6/30240$.

After the symmetric stuffle combinations have been removed, the remaining information is carried by $\mathrm{Li}_{a,b}(x,y) - \mathrm{Li}_{b,a}(y,x)$, exchanging both indices and colors. Equal indices alone do not force it to vanish at mixed points. The same-color diagonal is a special case.

Here is an exact counterexample entirely inside the manuscript's alphabet:

$$\Im\big(\mathrm{Li}_{1,1}(-1,i) - \mathrm{Li}_{1,1}(i,-1)\big) = \frac{\pi}{2}\log 2 - G = \int_0^1 \frac{\log(1+t^2)}{1+t^2}\,dt > 0. \tag{4.95}$$

To prove the first equality, differentiating the convergent defining series in the disk gives

$$\frac{\partial}{\partial x}\,\mathrm{Li}_{1,1}(x,y) = \frac{\mathrm{Li}_1(xy)}{1-x}.$$

Continuation along the segment to $x = -1$ is nonsingular, and hence

$$\Im\,\mathrm{Li}_{1,1}(-1,i) = \int_0^1 \frac{\arctan t}{1+t}\,dt.$$

Put $K = \int_0^1 \log(1+t)/(1+t^2)\,dt$. With $t = \tan\theta$,

$$K = \int_0^{\pi/4} \big(\tfrac{1}{2}\log 2 + \log\sin(\theta+\pi/4) - \log\cos\theta\big)\,d\theta = \frac{\pi}{8}\log 2.$$

The last two integrals cancel after reflection about $\pi/2$. Integration by parts now gives $\Im\,\mathrm{Li}_{1,1}(-1,i) = \pi\log 2/4 - K = \pi\log 2/8$. Stuffle and

$$\mathrm{Li}_1(-1) = -\log 2, \qquad \mathrm{Li}_1(i) = -\tfrac{1}{2}\log 2 + \tfrac{\pi i}{4}, \qquad \Im\,\mathrm{Li}_2(-i) = -G$$

give

$$\Im\big(\mathrm{Li}_{1,1}(-1,i)+\mathrm{Li}_{1,1}(i,-1)\big)=G-\frac{\pi}{4}\log 2.$$

Subtracting proves the first equality in (4.95); the second also follows directly by setting $t=\tan\theta$. Indeed,

$$\int_0^{\pi/4}\log\cos\theta\,d\theta=-\frac{\pi}{4}\log 2+\frac{G}{2},$$

which follows from the sum and difference of the corresponding log-sine and log-cosine integrals: their difference is G, and their sum is $-\pi\log 2/2$ by the double-angle formula. Thus the displayed integral is $-2\int_0^{\pi/4}\log\cos\theta\,d\theta$. Its integrand is positive away from zero. The difference is approximately 0.17282745097458205019574093418642286289514247590297.

4.13.2 Family 2: antisymmetric reductions to the $(\mathrm{i},1)$ algebra

Against the basis of single-polylog monomials together with the $(\mathrm{i},1)$ generators $g_{a,b}=\operatorname{Im}\mathrm{Li}_{a,b}(\mathrm{i},1)$, $h_{a,b}=\operatorname{Re}\mathrm{Li}_{a,b}(\mathrm{i},1)$, the *low-weight* antisymmetric parts reduce, each reduction a new closed-form identity found by integer-relation search and confirmed in Wolfram 15. A representative instance, with $\mu_n=\operatorname{Re}\mathrm{Li}_n(\frac{1+\mathrm{i}}{2})$ and $\lambda_n=\operatorname{Im}\mathrm{Li}_n(\frac{1+\mathrm{i}}{2})$:

$$\operatorname{Re}[\mathrm{Li}_{1,2}(\mathrm{i},\mathrm{i})-\mathrm{Li}_{2,1}(\mathrm{i},\mathrm{i})]=3\zeta(3)-4\log 2\,\mu_2-\pi\lambda_2-4h_{1,1}\log 2+3h_{1,2}. \tag{4.96}$$

Such reductions were found at weights 2–4 at the mixed points and at weight 3 at (i,i). At higher weights the reported searches did not find reductions against the specified $(\mathrm{i},1)$ basket. This records the outcome of those searches; it does not prove numerical nonreduction. Exact shuffle relations and a precisely specified formal quotient determine which directions survive the available relation vocabulary.

4.14 Gaussian doubles: the double-shuffle dimension

4.14.1 A cautionary integer-relation count

A first attempt to count the genuinely-new depth-2 directions proceeds by integer-relation search: test each antisymmetric part for membership in the span of the $(\mathrm{i},1)$ generators and single-polylog products, and tabulate those that fail. Carried out at 700 digits, this reports **30** independent new directions through weight 6. The number is wrong. It is an artifact of the spurious scale: the genuine reductions are of large height, and against a basis of a hundred-odd atoms the search cannot see them. (Three searches against bases of 46, 55, and the $(\mathrm{i},1)$-plus-single set returned "dimensions" of 3, 6, and 30 respectively—all different, all wrong, the count *decreasing* as the basis grew and the ceiling fell.)

4.14.2 The shuffle, and the correct count

The shuffle settles it. The key relation is that the shuffle of a (-1)-polylogarithm with an i-polylogarithm produces (i,i) and $\overline{(-1,\mathrm{i})}$ doubles, for instance

$$\mathrm{Li}_2(-1)\ \mathrm{Li}_2(\mathrm{i})=\mathrm{Li}_{2,2}(\mathrm{i},\mathrm{i})+2\,\mathrm{Li}_{3,1}(\mathrm{i},\mathrm{i})+\overline{\mathrm{Li}_{2,2}(-1,\mathrm{i})}+2\,\overline{\mathrm{Li}_{3,1}(-1,\mathrm{i})} \tag{4.97}$$

(verified to $\sim 10^{-694}$), whereas the *stuffle* of the same product produces the $(-1,\mathrm{i}),(\mathrm{i},-1)$ doubles. Equating ties all three Gaussian points together. Collecting all 240 stuffle and shuffle relations

from the products $\mathrm{Li}_p(s)\,\mathrm{Li}_q(t)$, $s,t \in \{\mathrm{i},-1,-\mathrm{i}\}$, $2 \le p+q \le 6$, verifying each to $\sim 10^{-600}$, and computing the exact rational rank of the linear system on the doubles, modulo single polylogarithms and the $(\mathrm{i},1)$ generators, gives:

weight	unknowns (Re+Im)	dim (stuffle only)	dim (double shuffle)
4	18	8	**1**
5	24	12	**2**
6	30	14	**2**
total		34	**5**

Observation 4.29. Modulo single polylogarithms and the $(\mathrm{i},1)$ generators, the Gaussian double points $(\mathrm{i},\mathrm{i}),(-1,\mathrm{i}),(\mathrm{i},-1)$ contribute exactly **5** candidate depth-two generators through weight 6—one at weight 4, two at weight 5, two at weight 6. A representative basis is

$$\mathrm{Re}\,\mathrm{Li}_{3,1}(\mathrm{i},-1); \quad \mathrm{Im}\,\mathrm{Li}_{3,2}(\mathrm{i},-1),\ \mathrm{Im}\,\mathrm{Li}_{4,1}(\mathrm{i},-1); \quad \mathrm{Re}\,\mathrm{Li}_{4,2}(\mathrm{i},-1),\ \mathrm{Re}\,\mathrm{Li}_{5,1}(\mathrm{i},-1).$$

Every other Gaussian double of weight ≤ 6 is an explicit combination of these five, the $(\mathrm{i},1)$ generators, and products of single L-values.

The rank is verified two ways: each relation reproduces its value to 10^{-600}, and the exact-rational rank is precision-independent. As a final gold-check at weight 4, fixing the single survivor $\mathrm{Re}\,\mathrm{Li}_{3,1}(\mathrm{i},-1)$ to its true value and solving the system predicts all other 17 weight-4 components to $\sim 10^{-692}$.

Remark 4.30 (the methodological moral). Observation 4.29 corrects the integer-relation count of 30 by a factor of six. The reductions are exact consequences of the small-height double-shuffle generators, but the *net* reductions are of large height—so a search at a specified precision and coefficient bound can miss them. This is the recurring trap of high-precision experimental mathematics, here biting in the direction of *over*-stating independence. The exact-rank calculation determines the formal quotient by the supplied valid relations. It does not make an arithmetic-independence claim.

4.15 Triple sums and the scope of formal quotient dimensions

The word for $\mathrm{Li}_{a,b,c}(i,1,1)$ is $0^{a-1}(-i)0^{b-1}(-i)0^{c-1}(-i)$ in the outer-letter-first convention. Shuffle relations express products of lower-weight functions as sums of such triples. At weight six the source reports rank seven for thirteen relations on ten triple coordinates, leaving three formal coordinates represented by $(3,1,2),(3,2,1),(4,1,1)$. Rank seven means that seven coordinates can be eliminated using the chosen relations and the three remaining coordinates. It does not mean that seven particular triples are individually products, or that the remaining three numerical values are independent.

The source also claimed binomial dimensions $\binom{n}{p}$ for the actual depth filtration from a total Hilbert series $1/(1-2t)$. A total Hilbert series alone cannot determine a depth grading. A grading by the length of a chosen motivic word basis must first be identified with the filtration under discussion; moreover a motivic lower bound does not give a numerical lower bound without injectivity of the period map. We therefore make no unconditional claim that genuine numerical depth three first occurs at weight six. A rigorous reduction or obstruction for these specific triples is an open problem. Panzer's parity theorem [33] supplies exact lower-depth conjugation combinations, but does not assert independence of the complementary components.

4.16 The Eisenstein analogue

Everything above has a level-3 counterpart at the cube roots of unity. Write $\rho = e^{2\pi i/3}$, so $\rho^2 = \bar{\rho}$.

4.16.1 Family 1, with an arithmetic twist

The stuffle closed forms read

$$\mathrm{Li}_{a,b}(\rho,\rho) + \mathrm{Li}_{b,a}(\rho,\rho) = \mathrm{Li}_a(\rho)\,\mathrm{Li}_b(\rho) - \mathrm{Li}_{a+b}(\rho^2), \tag{4.98}$$

$$\mathrm{Li}_{a,b}(\rho^2,\rho) + \mathrm{Li}_{b,a}(\rho,\rho^2) = \mathrm{Li}_a(\rho^2)\,\mathrm{Li}_b(\rho) - \zeta(a+b), \tag{4.99}$$

with the diagonal corollary $\mathrm{Li}_{a,a}(\rho,\rho) = \frac{1}{2}(\mathrm{Li}_a(\rho)^2 - \mathrm{Li}_{2a}(\rho^2))$. The arithmetic twist is in (4.99): because $\rho^2 \cdot \rho = \rho^3 = 1$, the colored tail degenerates to the *ordinary* zeta value $\zeta(a+b)$, not a level-3 value. The single cube-root L-values at $n \geq 2$ split cleanly under the distribution relation $\mathrm{Li}_n(\rho) + \mathrm{Li}_n(\rho^2) = (3^{1-n} - 1)\zeta(n)$:

$$\mathrm{Re}\,\mathrm{Li}_n(\rho) = \tfrac{1}{2}(3^{1-n} - 1)\,\zeta(n), \qquad \mathrm{Im}\,\mathrm{Li}_n(\rho) = \mathrm{Sl}_n(2\pi/3), \tag{4.100}$$

At weight one, $\mathrm{Li}_1(\rho) = -\frac{1}{2}\log 3 + i\pi/6$. For the displayed higher-weight formula, $\mathrm{Sl}_n(\theta) = \sum_{k\geq 1} \sin(k\theta)/k^n$. The real part is a rational multiple of $\zeta(n)$; the sine series agrees with Cl_n at even weight and is elementary at rational angles at odd weight. Also $\mathrm{Cl}_2(2\pi/3) = 2V/3$, where $V = \mathrm{Cl}_2(\pi/3)$ is the Gieseking constant. All of (4.98)–(4.100) are confirmed in Wolfram 15.

4.16.2 The (ρ,ρ) frontier: one constant

Running the level-3 double shuffle as in Section 4.14—90 stuffle and shuffle relations from products $\mathrm{Li}_p(s)\,\mathrm{Li}_q(t)$, $s,t \in \{\rho,\rho^2\}$, each verified to $\sim 10^{-600}$, exact rational rank—collapses the (ρ,ρ) doubles still more sharply than the Gaussian:

weight	unknowns (Re+Im)	dim (stuffle only)	dim (double shuffle)
2	2	0	0
3	4	2	0
4	6	2	0
5	8	4	**1**
6	10	4	0
total		12	**1**

Observation 4.31. Modulo single polylogarithms and the $(\rho,1)$ generators, the entire (ρ,ρ) family through weight 6 reduces to a *single* candidate depth-two generator—$\mathrm{Im}\,\mathrm{Li}_{3,2}(\rho,\rho)$, at weight 5—with weights $2,3,4$ and 6 collapsing completely.

A gold-check at weight 5 fixes the survivor and predicts all seven other weight-5 components to $\sim 10^{-696}$. The pattern mirrors the Gaussian: there too the weight-5 survivors are imaginary $\mathrm{Li}_{3,2}/\mathrm{Li}_{4,1}$ directions.

Remark 4.32 (a genuine cube-root obstacle). The decisive shuffle $\mathrm{Li}_p(\rho^2)\,\mathrm{Li}_q(\rho) \to (\rho,\rho) + (\rho^2,\rho^2)$ lands only on points with $w = z_1 z_2 \neq 1$, so the computation never needs the mixed points $(\rho^2,\rho),(\rho,\rho^2)$, where $w = \rho^3 = 1$. Those points presented a numerical obstacle in the earlier evaluator: both the Lerch-plus-Wynn evaluator and a direct cumulative-sum evaluator stall there at $\sim$8 digits—the period-three, non-oscillating tail defeats Wynn's ε-algorithm, unlike the even-period

i, −1 cases, which converge to full precision. Folding the mixed cube-root points in (for the *full* level-3 frontier) requires a period-three-aware accelerated evaluator, or Wolfram; the following mixed-point method addresses that computational gap.

4.17 Why the standard evaluator degenerates at $w = 1$

With the convention the nested-series definition, the accelerated evaluator of the article writes the inner tail as a Lerch transcendent and sums the *outer* series with an acceleration factor $w^n = (z_1 z_2)^n$. At the mixed points $w = 1$, that factor is identically 1: the oscillation Wynn's ε-algorithm relies on disappears, leaving a bare period-three power-law tail that the algorithm cannot resolve past a handful of digits. Summing over the *larger* index instead keeps an oscillating factor $z_1^{n_1}$, but for $z_1 = \rho^2$ this is a *period-three* oscillation, and a plain ε-algorithm (tuned for geometric tails) again stalls.

4.18 A robust $w = 1$ evaluator

Two ingredients fix it.

(a) A smooth correction identity. Writing $l = n_1$ (the larger index) and splitting the inner partial sum $P(l) = \sum_{n<l} z_2^n/n^b = \mathrm{Li}_b(z_2) - R(l)$ with tail $R(l) = \sum_{n\geq l} z_2^n/n^b = z_2^l\,\Phi(z_2, b, l)$ (Lerch Φ), one gets the exact identity

$$\boxed{\mathrm{Li}_{a,b}(z_1, z_2) = \mathrm{Li}_a(z_1)\ \mathrm{Li}_b(z_2)\ -\ \sum_{l\geq 1} \frac{w^l}{l^a}\,\Phi(z_2, b, l)} \tag{4.101}$$

At $w = 1$ the correction summand $\Phi(z_2, b, l)/l^a$ is *smooth*—every oscillating factor $z_1^l z_2^l = w^l$ collapses to 1—with the clean asymptotic $\Phi(z_2, b, l) \sim \sum_{j\geq 0} \binom{-b}{j} c_j(z_2)\, l^{-b-j}$, where $c_0 = \frac{1}{1-z_2}$ and $c_{j\geq 1} = \mathrm{Li}_{-j}(z_2)$ (rational in z_2). The subtle point—the source of a factor-of-one bug that cost five digits in a first draft—is that $c_0 = \sum_{k\geq 0} z_2^k = \frac{1}{1-z_2}$ is *not* $\mathrm{Li}_0(z_2) = \frac{z_2}{1-z_2}$; they differ by exactly 1. This gives an Euler–Maclaurin tail in Hurwitz zetas, $\sum_{l>M} \Phi/l^a = \sum_j \binom{-b}{j} c_j\, \zeta(a + b + j, M + 1)$, a fast and accurate cross-check at low to medium precision.

(b) Period-aware ρ-acceleration for production. For high precision the production evaluator sums the outer cumulative series directly, samples the partial sums at the common period $N = \mathrm{lcm}(\mathrm{ord}\, z_1, \mathrm{ord}\, z_2)$ (so every oscillating factor realigns to 1 at the sample points), and accelerates the resulting smooth, power-law-convergent sequence with **Wynn's ρ-algorithm** —the algebraic-convergence counterpart of the ε-algorithm. The ρ-algorithm yields empirical convergence on the tested families; precision inflation and independent comparison are needed to assess its conditioning (`mpl_w1.py`).

The general certified evaluators and their geometric bounds are developed in Chapter 5. The historical Lerch evaluator above remains a useful independent comparison.

4.19 Validation

The evaluator is checked by three independent, Wolfram-free routes plus one Wolfram cross-check.

1. **Exact symmetric closed form** (single polylogs only, an entirely separate code path):

$$\mathrm{Li}_{a,b}(\rho^2,\rho)+\mathrm{Li}_{b,a}(\rho,\rho^2)=\mathrm{Li}_a(\rho^2)\,\mathrm{Li}_b(\rho)-\zeta(a+b), \tag{4.102}$$

   and the Gaussian analogue $\mathrm{Li}_{a,b}(\mathrm{i},-\mathrm{i})+\mathrm{Li}_{b,a}(-\mathrm{i},\mathrm{i})=\mathrm{Li}_a(\mathrm{i})\,\mathrm{Li}_b(-\mathrm{i})-\zeta(a+b)$. (The collision term is $\mathrm{Li}_{a+b}(z_1z_2)=\mathrm{Li}_{a+b}(1)=\zeta(a+b)$.)

2. **Conjugation** $\mathrm{Li}_{a,b}(\rho,\rho^2)=\overline{\mathrm{Li}_{a,b}(\rho^2,\rho)}$.

3. **Two independent summation methods**—the ρ-accelerated cumulative sum and the Euler–Maclaurin smooth-correction series—agree to full working precision.

4. **Wolfram 15** `MultiplePolyLog` at moderate precision.

All four agree; the symmetric closed form (4.102) holds to the full generated precision ($\sim 10^{-500}$) for every (a,b).

4.20 Closed forms for the Eisenstein mixed-point values

With 500-digit values in hand, an integer-relation (PSLQ/LLL) search against the level-3 basis —monomials in the single L-values $\{\pi,\log 3,\zeta(3),\zeta(5),\mathrm{Cl}_2(\frac{2\pi}{3}),\mathrm{Cl}_4(\frac{2\pi}{3}),\mathrm{Cl}_6(\frac{2\pi}{3})\}$ times the $(\rho,1)$ generators $g_{a,b}=\mathrm{Im}\,\mathrm{Li}_{a,b}(\rho,1)$, $h_{a,b}=\mathrm{Re}\,\mathrm{Li}_{a,b}(\rho,1)$, and the lone (ρ,ρ) double-shuffle survivor $\mathrm{Im}\,\mathrm{Li}_{3,2}(\rho,\rho)$—**reduces every mixed value through weight 4, and every weight-5 imaginary part**, to explicit closed forms (each canary-clean, height ≤ 10, re-verified to 10^{-425} against the *complete* weight-graded basis). The two simplest:

$$\mathrm{Im}\,\mathrm{Li}_{1,1}(\rho^2,\rho)=-\big(\mathrm{Cl}_2(\tfrac{2\pi}{3})+\mathrm{Im}\,\mathrm{Li}_{1,1}(\rho,1)\big),\qquad \mathrm{Re}\,\mathrm{Li}_{1,1}(\rho^2,\rho)=5\,\mathrm{Re}\,\mathrm{Li}_{1,1}(\rho,1)-\tfrac{1}{2}\log^2 3. \tag{4.103}$$

(The full list of $\sim$16 reductions is in `harvest_eis_mixed.py`.) The original weight-five real and weight-six searches failed in their selected baskets. The exact inverse-color reduction below supersedes the interpretation of that failure as a new mixed-point obstruction.

4.21 Inverse colors: the exact reduction that the searches missed

The numerical evaluator above made the mixed points accessible, but its failed relation searches did not identify the correct quotient. An exact partial-fraction argument now proves that inverse-color doubles add no directions relative to the one-direction family and single-value products. This supersedes the earlier mixed-point candidate counts at weights five and six, in both the Gaussian and Eisenstein cases.

Put $M_{a,b}(z)=\mathrm{Li}_{a,b}(z,z^{-1})$ and $w=a+b$. We allow $0<|z|<1$, and $|z|=1, z\neq 1$, with positive integers a,b. Boundary sums use the stated outer-index order. They converge absolutely if $a\ge 2$. For $a=1$, the inner colored harmonic sum equals its limit plus $O(n^{-b})$; the outer sum is therefore a convergent oscillatory series plus an absolutely convergent remainder.

Theorem 4.33 (All-weight inverse-argument reduction)**.** *In this domain,*

$$\begin{aligned} M_{a,b}(z) = (-1)^{b+1}\sum_{j=1}^{a}\binom{w-j-1}{b-1}\operatorname{Li}_{w-j,j}(z,1) + (-1)^{b+1}\binom{w-1}{b}\operatorname{Li}_w(z) \\ + \sum_{j=2}^{b}(-1)^{b-j}\binom{w-j-1}{a-1}\zeta(j)\operatorname{Li}_{w-j}(z) \\ + (-1)^b\sum_{j=2}^{a}\binom{w-j-1}{b-1}\zeta(j)\operatorname{Li}_{w-j}(z). \end{aligned} \tag{4.104}$$

Empty sums are zero. In particular, $F_a = M_{a,1}$ satisfies

$$F_a(z) = a\operatorname{Li}_{a+1}(z) + \sum_{j=1}^{a}\operatorname{Li}_{a+1-j,j}(z,1) - \sum_{j=2}^{a}\zeta(j)\operatorname{Li}_{a+1-j}(z). \tag{4.105}$$

Proof. Set $h = n - m$. Principal parts at $m = 0, -h$ give

$$\begin{aligned} \frac{1}{m^b(m+h)^a} = \sum_{j=1}^{b}\frac{(-1)^{b-j}\binom{w-j-1}{a-1}}{h^{w-j}m^j} \\ + (-1)^b\sum_{j=1}^{a}\frac{\binom{w-j-1}{b-1}}{h^{w-j}(m+h)^j}. \end{aligned} \tag{4.106}$$

Indeed, the difference has no poles and tends to zero at infinity. The $j = 1$ coefficients are opposite; their sum over m is finite because $\sum_{m\geq 1}(1/m - 1/(m+h)) = H_h$. For $j \geq 2$, the shifted sum is $\zeta(j) - H_h^{(j)}$. Consequently

$$\begin{aligned} c_h := \sum_{m\geq 1}\frac{1}{m^b(m+h)^a} = (-1)^{b+1}\sum_{j=1}^{a}\binom{w-j-1}{b-1}\frac{H_h^{(j)}}{h^{w-j}} \\ + \sum_{j=2}^{b}(-1)^{b-j}\binom{w-j-1}{a-1}\frac{\zeta(j)}{h^{w-j}} + (-1)^b\sum_{j=2}^{a}\binom{w-j-1}{b-1}\frac{\zeta(j)}{h^{w-j}}. \end{aligned}$$

Multiply by z^h, sum, and use

$$\sum_{h\geq 1}\frac{z^h H_h^{(j)}}{h^{w-j}} = \operatorname{Li}_{w-j,j}(z,1) + \operatorname{Li}_w(z), \qquad \sum_{j=1}^{a}\binom{w-j-1}{b-1} = \binom{w-1}{b}.$$

For $|z| < 1$ these operations are absolute.

To justify the boundary even when $a = 1$, let O_N be the original sum over $n \leq N$ and $D_N = \sum_{h=1}^{N-1} z^h c_h$. Partial summation gives

$$\left|\sum_{h=h_0}^{h_1}\frac{z^h}{(m+h)^a}\right| \leq \frac{4}{|1-z|(m+h_0)^a}.$$

Splitting the discrepancy at $m = N$ now yields

$$|D_N - O_N| \leq \frac{4}{|1-z|}\left(N^{-a}H_N^{(b)} + \sum_{m>N} m^{-a-b}\right) \longrightarrow 0.$$

The coefficients c_h decrease to zero, the latter by dominated convergence with majorant m^{-a-b}. Dirichlet's test proves convergence of $\sum z^h c_h$. The one-direction doubles on the right also converge. This proves the identity in the original boundary convention. □

Corollary 4.34 (The mixed quotient is zero)**.** *Let $\mathcal{A}_w(z)$ be the $\mathbb{Q}$-span of*

$$\mathrm{Li}_{r,w-r}(z,1)\ (1 \le r < w), \quad \mathrm{Li}_w(z), \quad \zeta(j)\,\mathrm{Li}_{w-j}(z)\ (2 \le j < w).$$

Every $M_{a,b}(z)$ of weight w belongs to $\mathcal{A}_w(z)$. Adding the mixed values contributes zero dimension modulo this space.

At $z = \mathrm{i}$, this is the Gaussian family already used in the report. At $z = \omega^2$, the one-direction values are conjugates of its $(\omega, 1)$ values, whose real and imaginary parts it includes. Thus the proposed additional mixed directions disappear at every weight. The result concerns inverse-color pairs, and does not assert that all colored double polylogarithms reduce to one direction.

4.21.1 An independent integral representation

Proposition 4.35. *In the same domain,*

$$M_{a,b}(z) = \frac{z}{(a-1)!}\int_0^1 \frac{(-\log t)^{a-1}\,\mathrm{Li}_b(t)}{1-zt}\,dt, \tag{4.107}$$

$$\mathrm{Li}_{a,b}(z,1) = \frac{z}{(a-1)!}\int_0^1 \frac{(-\log t)^{a-1}\,\mathrm{Li}_b(zt)}{1-zt}\,dt. \tag{4.108}$$

Proof. Insert $n^{-a} = (a-1)!^{-1}\int_0^1 (-\log t)^{a-1} t^{n-1}\,dt$. The $h = n-m$ geometric sum gives $zt/(1-zt)$; the m-sum gives $\mathrm{Li}_b(t)$ or $\mathrm{Li}_b(zt)$. At a fixed boundary point $z \neq 1$, the denominator is bounded away from zero on $[0,1]$, and the possible logarithmic endpoint singularities are integrable. Dominated convergence and the preceding boundary argument complete the proof. □

These formulas supply independent numerical checks without the power-law acceleration problem at $z_1 z_2 = 1$. Quadrature itself is not a rigorous certificate unless its numerical error is enclosed.

4.21.2 The mixed parity component reduces to single values

Panzer's theorem predicts lower depth for the real component at odd weight and the imaginary component at even weight [33]. Here is a direct proof for the mixed F_a family.

Theorem 4.36 (Mixed parity formulas)**.** *For $|z| = 1, z \neq 1$ and $m \ge 1$,*

$$\begin{aligned}\operatorname{Re} F_{2m}(z) = {}& \frac{2m+1}{2}\operatorname{Re}\mathrm{Li}_{2m+1}(z) - \sum_{j=1}^{m}\zeta(2j)\operatorname{Re}\mathrm{Li}_{2m+1-2j}(z)\\ & - \frac{1}{2}\zeta(2m+1) + \operatorname{Re}\mathrm{Li}_{2m}(z)\operatorname{Re}\mathrm{Li}_1(z).\end{aligned} \tag{4.109}$$

For $m \ge 0$,

$$\begin{aligned}\operatorname{Im} F_{2m+1}(z) = {}& (m+1)\operatorname{Im}\mathrm{Li}_{2m+2}(z) - \sum_{j=1}^{m}\zeta(2j)\operatorname{Im}\mathrm{Li}_{2m+2-2j}(z)\\ & + \operatorname{Im}\mathrm{Li}_{2m+1}(z)\operatorname{Re}\mathrm{Li}_1(z).\end{aligned} \tag{4.110}$$

Proof. For $a \geq 2$, set $\mathcal{B}_a(z) = -\operatorname{Li}_a(z) - (-1)^a \operatorname{Li}_a(z^{-1})$. The Fourier series $-\mathcal{B}_a(z) = \sum_{n\neq 0} z^n/n^a$ is absolutely summable, while $\operatorname{Li}_1(z^{-1}) = \sum_{\ell\geq 1} z^{-\ell}/\ell$ belongs to L^2 on the circle. Their product has positive h-th coefficient $c_h = \sum_{\ell\geq 1}[\ell(\ell+h)^a]^{-1}$, constant coefficient $\zeta(a+1)$, and negative h-th coefficient

$$q_{-h} = \sum_{\ell\geq 1,\ \ell\neq h} \frac{1}{\ell(\ell-h)^a}.$$

The difference $q_{-h} - (-1)^a c_h$ is

$$\sum_{\ell\in\mathbb{Z}\setminus\{0,h\}} \frac{1}{\ell(\ell-h)^a} = (-1)^a \left[2 \sum_{j=1}^{\lfloor a/2\rfloor} \frac{\zeta(2j)}{h^{a+1-2j}} - \frac{a+1}{h^{a+1}} \right]. \tag{4.111}$$

To check this absolutely convergent bilateral sum, use

$$\frac{1}{\ell(\ell-h)^a} = \frac{1}{(-h)^a \ell} - \sum_{r=1}^{a} \frac{1}{(-h)^{a+1-r}(\ell-h)^r}.$$

Symmetric summation of the $r=1$ terms gives respectively $-1/h$ and $1/h$. For $r \geq 2$, the shifted sum is $(1+(-1)^r)\zeta(r) - (-h)^{-r}$. Substitution gives (4.111). Comparing all Fourier coefficients now gives

$$\begin{aligned} F_a(z) + (-1)^a F_a(z^{-1}) = (-1)^a &\left[(a+1)\operatorname{Li}_{a+1}(z^{-1}) - 2\sum_{j=1}^{\lfloor a/2\rfloor} \zeta(2j)\operatorname{Li}_{a+1-2j}(z^{-1}) \right] \\ &- \zeta(a+1) - \operatorname{Li}_1(z^{-1})\mathcal{B}_a(z). \end{aligned} \tag{4.112}$$

The equality holds first in L^2, then pointwise by continuity away from $z=1$. Real parts for even a, and imaginary parts for odd a, give the claims. The remaining case $a=1$ follows from $F_1(z) = \operatorname{Li}_2(z) + \frac{1}{2}\operatorname{Li}_1(z)^2$, itself a consequence of (4.105) and the elementary shuffle identity. □

4.21.3 Four explicit replacements for the claimed survivors

Corollary 4.37. *The four highlighted mixed-point constants satisfy*

$$\operatorname{Re}\operatorname{Li}_{4,1}(\mathrm{i},-\mathrm{i}) = -\frac{587}{1024}\zeta(5) + \frac{3}{32}\zeta(2)\zeta(3) + \frac{135}{256}\zeta(4)\log 2, \tag{4.113}$$

$$\operatorname{Im}\operatorname{Li}_{5,1}(\mathrm{i},-\mathrm{i}) = 3\beta(6) - \zeta(2)\beta(4) - \zeta(4)G - \frac{5\pi^5\log 2}{3072}, \tag{4.114}$$

$$\operatorname{Re}\operatorname{Li}_{4,1}(\omega^2,\omega) = -\frac{281}{162}\zeta(5) + \frac{4}{9}\zeta(2)\zeta(3) + \frac{20}{27}\zeta(4)\log 3, \tag{4.115}$$

$$\operatorname{Im}\operatorname{Li}_{5,1}(\omega^2,\omega) = -3C_6 + \zeta(2)C_4 + \zeta(4)C_2 + \frac{\pi^5\log 3}{729}. \tag{4.116}$$

Proof. Use Theorem 4.36 with $m=2$, together with

$$\operatorname{Re}\operatorname{Li}_s(\mathrm{i}) = -2^{-s}(1-2^{1-s})\zeta(s), \qquad \operatorname{Re}\operatorname{Li}_s(\omega) = \frac{3^{1-s}-1}{2}\zeta(s) \quad (s>1),$$

and

$$\operatorname{Re}\operatorname{Li}_1(\mathrm{i}) = -\tfrac{1}{2}\log 2, \quad \operatorname{Re}\operatorname{Li}_1(\omega) = -\tfrac{1}{2}\log 3, \quad \operatorname{Im}\operatorname{Li}_5(\mathrm{i}) = \frac{5\pi^5}{1536}, \quad \operatorname{Im}\operatorname{Li}_5(\omega) = \frac{2\pi^5}{729}.$$

The real identities follow by residue-class splitting; the imaginary fifth-order values follow from the odd-beta and Bernoulli Fourier formulas, with $B_5(1/3) = -5/243$. Conjugation changes the sign of imaginary parts at ω^2. □

The four values, in the displayed order, are approximately

$$\begin{aligned} &-0.01342026735728779, && 0.03269424445221378,\\ &-0.03903921845650667, && -0.02533938136807792. \end{aligned}$$

Their right sides already belong to the report's listed single-value bases. Its `thm:parallel` is therefore false in the printed convention. The evaluator scripts and value stores cited by that report were not imported into ProveIt, so their precise failure cannot be diagnosed from the available files. Replace the survivor claims by Theorem 4.33 and Corollary 4.37, and recompute any dimension conclusions that depended on the proposed new directions.

4.22 Mixed endpoints and all-weight cyclotomic product quotients

Put $A_{a,b} = \operatorname{Im}\operatorname{Li}_{a,b}(\mathrm{i},\mathrm{i})$ and $K_{a,b} = \operatorname{Im}\operatorname{Li}_{a,b}(\mathrm{i},-1)$; retain $g_{a,b}$ and S_p as above. The all-exponent endpoint identity relates the mixed family to the same-color family before any period-independence question is asked.

4.22.1 An all-exponent mixed endpoint identity

Theorem 4.38 (Gaussian mixed endpoint)**.** *For every integer $p \geq 1$,*

$$K_{p,1} + (-1)^{p-1}A_{1,p} = \beta(p+1) - 2\log 2\,\beta(p) + \sum_{j=2}^{p}(-1)^j\eta(j)\beta(p+1-j). \tag{4.117}$$

Moreover,

$$S_p = g_{p,1} + K_{p,1}. \tag{4.118}$$

Proof. All the boundary values used here converge. First-index-one values with first argument i can be obtained by Dirichlet's test or radial Abel limits. The shuffle and stuffle identities may therefore be justified by convergent iterated integrals and Abel limiting.

Fix $w = p+1$, and define homogeneous generating polynomials of degree $w-2$:

$$\mathcal{A}(X,Y) = \sum_{a+b=w} A_{a,b}X^{a-1}Y^{b-1}, \quad \mathcal{K}(X,Y) = \sum_{a+b=w} K_{a,b}X^{a-1}Y^{b-1}.$$

Let $\mathcal{V}$ denote the same polynomial for $\operatorname{Im}\operatorname{Li}_{a,b}(-1,\mathrm{i})$, and put

$$P(X,Y) = \sum_{a+b=w} \operatorname{Im}[\operatorname{Li}_a(-1)\operatorname{Li}_b(\mathrm{i})]X^{a-1}Y^{b-1}, \qquad H(X,Y) = \sum_{a+b=w} X^{a-1}Y^{b-1}.$$

Stuffle and shuffle respectively give

$$\mathcal{V}(X,Y) + \mathcal{K}(Y,X) = P(X,Y) + \beta(w)H(X,Y),$$

$$\mathcal{A}(X+Y,X) - \mathcal{V}(X+Y,Y) = P(X,Y).$$

The collision term in the first equation is $-\operatorname{Im}\operatorname{Li}_w(-\mathrm{i}) = \beta(w)$. Eliminating $\mathcal{V}$ yields

$$\mathcal{K}(U,V) = P(V-U,U) + P(V,U) + \beta(w)H(V,U) - \mathcal{A}(V,V-U).$$

Set $V = 0$ and compare coefficients of U^{w-2}. The evaluation $\operatorname{Li}_j(-1) = -\eta(j)$ gives (4.117), including the two occurrences of $-\log 2\,\beta(p)$.

For (4.118), taking imaginary parts of the outer index in $\operatorname{Li}_{p,1}(\mathrm{i},1) + \operatorname{Li}_{p,1}(\mathrm{i},-1)$ retains indices $2n+1$, with sign $(-1)^n$. Their inner sum is

$$\sum_{m=1}^{2n} \frac{1+(-1)^m}{m} = H_n.$$

Absolute convergence proves the result directly for $p > 1$. An Abel limit gives the endpoint $p = 1$. □

At $p = 4$ this specializes to

$$K_{4,1} = A_{1,4} + \frac{191\pi^5}{23040} - \frac{3}{4}G\zeta(3) - 2\beta(4)\log 2. \tag{4.119}$$

4.22.2 A precisely defined formal presentation

Fix a level $L = 3$ or 4 and a weight $w \geq 2$. Let $\omega = e^{2\pi\mathrm{i}/L}$. Introduce rational formal symbols

$$D_a(r,s) \quad (1 \leq a < w, \quad r,s \in \mathbb{Z}/L\mathbb{Z})$$

for imaginary parts of $\operatorname{Li}_{a,w-a}(\omega^r,\omega^s)$. Impose conjugation $D_a(-r,-s) = -D_a(r,s)$. Symbols fixed by this involution vanish. Put all single values and imaginary parts of products of two single values in the background, and also set $D_a(1,0) = 0$. The latter is precisely quotienting by the chosen g-family.

The only further rows imposed are, for $p+q = w$,

$$D_p(r,s) + D_q(s,r) = 0, \tag{4.120}$$

$$\sum_{j=0}^{p-1}\binom{q+j-1}{j} D_{q+j}(s,r-s) + \sum_{j=0}^{q-1}\binom{p+j-1}{j} D_{p+j}(r,s-r) = 0. \tag{4.121}$$

They are the stuffle and shuffle rows of two single values, with their one-leading-one regularizations. A regularized product involving the real parameter T belongs to the background. At $w = 2$, the only two-divergent-factor correction is real and disappears on taking imaginary parts. Call the resulting quotient $V_{L,w}$.

This is *not* the full cyclotomic double-shuffle algebra. Products involving a depth-two factor, relations lifted through higher depths, additional functional equations, and motivic period assertions have not been imposed. In particular, Zhao's standard relations [81] are substantially broader than this presentation. The theorem below computes $V_{L,w}$, not a dimension of actual numbers.

For a scalar assignment to the symbols, introduce

$$\mathcal{F}^{r,s}(X,Y) = \sum_{a=1}^{w-1} D_a(r,s)X^{a-1}Y^{w-a-1}.$$

The conditions that the assignment annihilate all rows are exactly

$$\mathcal{F}^{r,s}(X,Y) + \mathcal{F}^{s,r}(Y,X) = 0, \tag{4.122}$$

$$\mathcal{F}^{s,r-s}(X+Y,X) + \mathcal{F}^{r,s-r}(X+Y,Y) = 0. \tag{4.123}$$

These follow by comparing binomial coefficients after substitution. They describe the dual of $V_{L,w}$, so their solution dimension is the required quotient dimension.

4.22.3 All-weight dimension theorems

Theorem 4.39 (Level four)**.** *The formal quotient satisfies*

$$\dim_{\mathbb{Q}} V_{4,w} = \begin{cases} 0, & w \text{ even}, \\ (w-1)/2, & w \text{ odd}. \end{cases} \qquad \sum_{w\geq 2} \dim V_{4,w}\, t^w = \frac{t^3}{(1-t^2)^2}.$$

At odd weight, its dual is parameterized by arbitrary antisymmetric homogeneous polynomials of degree $w-2$.

Proof. Up to conjugation, the six color pairs are $(0,1),(1,0),(1,1),(1,2),(1,3),(2,1)$. Write their polynomials as U, G, A, K, M, V, respectively, and let $n = w-2$. The background condition gives $G = 0$; stuffle gives

$$U = 0, \quad A(Y,X) = -A(X,Y), \quad V(X,Y) = -K(Y,X), \quad M(Y,X) = M(X,Y).$$

The shuffle with colors $(1,0)$ forces $M = 0$. The shuffle with colors $(2,1)$ gives

$$A(X+Y,X) + K(Y,X+Y) = 0,$$

hence $K(X,Y) = -A(Y,Y-X)$. The shuffle with colors $(1,3)$ is $K(X,Y) = K(X,X-Y)$. After substitution and antisymmetry this condition is $A = (-1)^{n+1}A$.

For even n all polynomials therefore vanish. For odd n, any antisymmetric A, with K, V as above and $U = G = M = 0$, satisfies the rows. To check exhaustiveness, the unordered color classes up to conjugation are

$$(0,0),(0,1),(0,2),(1,1),(1,2),(1,3),(2,2).$$

The real classes vanish, and the remaining classes are precisely the conditions just used, together with their transposes. Thus there are no unexamined rows. A basis for A at odd w is

$$X^{a-1}Y^{w-a-1} - X^{w-a-1}Y^{a-1}, \qquad 1 \leq a \leq (w-1)/2.$$

Counting these basis vectors and summing over odd weights proves the formulas. □

Theorem 4.40 (Level three)**.** *The formal quotient satisfies*

$$\dim_{\mathbb{Q}} V_{3,w} = \begin{cases} 0, & w \text{ even}, \\ \lfloor (w+1)/6 \rfloor, & w \text{ odd}, \end{cases} \qquad \sum_{w\geq 2} \dim V_{3,w}\, t^w = \frac{t^5}{(1-t^2)(1-t^6)}.$$

Writing $\Delta = XY(X-Y)$, $Q = X^2 - XY + Y^2$, the dual is parameterized by the homogeneous polynomials

$$A(X,Y) = \Delta\, P(Q, \Delta^2)$$

of degree $w-2$.

Proof. Up to conjugation the four pairs are $(0,1),(1,0),(1,1),(1,2)$. As before the first two vanish and the shuffle with a zeta factor kills the inverse-argument polynomial. The remaining polynomial A satisfies exactly

$$A(Y,X) = -A(X,Y), \qquad A(X,X-Y) = A(X,Y).$$

Antisymmetry implies divisibility by $X-Y$; the second symmetry then forces divisibility by Y, and antisymmetry forces divisibility by X. Hence $A = \Delta B$. The polynomial Δ has the same two transformation signs as A, so B is invariant under swapping X, Y and under $(X,Y) \mapsto (X, X-Y)$.

On the plane $a+b+c=0$, with $(a,b,c)=(X,-Y,Y-X)$, the second transformation swaps b,c; the first is global negation followed by the transposition of a,b. The generated group contains global negation, obtained by cubing the product, and all permutations. The elementary symmetric-polynomial theorem therefore identifies its invariant ring with

$$\mathbb{Q}[e_2,e_3^2]=\mathbb{Q}[Q,\Delta^2].$$

Conversely every $\Delta P(Q,\Delta^2)$ has the required symmetries, and direct substitution verifies all the original color rows. Counting monomials of degrees $3+2a+6b=w-2$ gives the result. Explicitly, at odd $w\geq 5$ a basis is

$$\Delta^{2b+1}Q^{(w-5)/2-3b},\qquad 0\leq b\leq \lfloor (w-5)/6\rfloor.$$

□

For comparison, the exact dimensions at the first few odd weights are

w	3	5	7	9	11	13	15
$\dim V_{3,w}$	0	1	1	1	2	2	2
$\dim V_{4,w}$	1	2	3	4	5	6	7

and all even-weight dimensions are zero. The exact matrix script checks both formulas through weight 16, including that the displayed polynomial bases annihilate every raw row and have the stated rank.

4.22.4 A separating functional and the scope of the S4 proof

At every odd weight $w\geq 3$, choose $A(X,Y)=(Y-X)^{w-2}$ in the level-four theorem. Then $K(X,Y)=X^{w-2}$, $V(X,Y)=-Y^{w-2}$. This gives a rational linear functional ℓ that annihilates every defining row and every background value:

$$\ell(D_a(1,1))=(-1)^{a+1}\binom{w-2}{a-1},$$
$$\ell(D_{w-1}(1,2))=1,\qquad \ell(D_1(2,1))=-1. \tag{4.124}$$

All other entries vanish, apart from conjugates with opposite signs.

At weight five its nonzero entries are particularly transparent:

symbol	$A_{1,4}$	$A_{2,3}$	$A_{3,2}$	$A_{4,1}$	$\operatorname{Im}\operatorname{Li}_{1,4}(-1,\mathrm{i})$	$K_{4,1}$
ℓ	1	−3	3	−1	−1	1

The polynomial proof verifies all weights; the delivered certificate also checks this six-entry vector against every exact weight-five row.

The now-proved identity of Theorem 4.8 has an informative image in this restricted presentation.

The exact identity (4.119) makes this equivalent to

$$161280A_{1,4}+69120(g_{4,1}+g_{3,2}+3g_{2,3})+617\pi^5-101520G\zeta(3)=0. \tag{4.125}$$

The functional ℓ sends its left side to 161280. Consequently this relation does not follow from (4.120)–(4.121) and the stated background alone. This is an exact obstruction to a specific proof strategy. The convergent octahedral relation used in its proof supplies precisely information outside this presentation.

4.22.5 A new weight-seven candidate

Proposition 4.41 (Two weight-seven coordinate baskets)**.** *The Gaussian doubles satisfy the exact identity*

$$g_{2,5} = 30g_{6,1} + 15g_{5,2} + 5g_{4,3} - \frac{15}{512}G\zeta(5) + \frac{3}{16}\beta(4)\zeta(3) - \frac{11\pi^7}{368640}. \tag{4.126}$$

Proof. The convergent same-point shuffles give

$$g_{2,5} + 2g_{3,4} + 3g_{4,3} + 5g_{5,2} + 10g_{6,1} = \mathrm{Im}(\mathrm{Li}_2(i)\,\mathrm{Li}_5(i)),$$
$$g_{3,4} + 4g_{4,3} + 10g_{5,2} + 20g_{6,1} = \mathrm{Im}(\mathrm{Li}_3(i)\,\mathrm{Li}_4(i)).$$

Subtract twice the second row from the first and use

$$\mathrm{Im}(\mathrm{Li}_2(i)\,\mathrm{Li}_5(i)) = -\frac{15}{512}G\zeta(5) - \frac{5\pi^7}{73728},$$

$$\mathrm{Im}(\mathrm{Li}_3(i)\,\mathrm{Li}_4(i)) = -\frac{3}{32}\beta(4)\zeta(3) - \frac{7\pi^7}{368640}.$$

These product evaluations follow from the ordinary depth-one fourth-root formulas. The coefficient of $g_{3,4}$ cancels, giving the identity. □

The new Cayley report's proposed formula

$$S_6 = -\frac{12334}{527}g_{6,1} - \frac{384}{31}g_{5,2} - \frac{2136}{527}g_{4,3} + \frac{3179\pi^7}{5713920} - \frac{801}{2108}\beta(4)\zeta(3) - 2\beta(6)\log 2$$

is therefore exactly the same conjecture as the following earlier form, written in a different Gaussian basket. If the two residuals mean left side minus right side, their difference is $-128/155$ times the zero relation obtained by moving (4.126) to the left. The exact two-row check has zero rational residual. This proves equivalence of the candidates, not the conjectured vanishing of either one.

Conjecture 4.42 (A reduction for S_6)**.** *With the same definitions,*

$$S_6 = \frac{722}{527}g_{6,1} + \frac{40}{527}g_{4,3} - \frac{128}{155}g_{2,5} + \frac{15191}{28569600}\pi^7 - \frac{3}{124}G\zeta(5) - \frac{2373}{10540}\beta(4)\zeta(3) - 2\beta(6)\log 2. \tag{4.127}$$

The candidate was found in the ordered basket

$$(S_6, g_{6,1}, g_{4,3}, g_{2,5}, \pi^7, G\zeta(5), \beta(4)\zeta(3), \beta(6)\log 2).$$

Its primitive integer relation vector is

$$(485683200, -665395200, -36864000, 401080320, -258247, 11750400, 109347840, 971366400).$$

A 100-digit search used coefficient bound 10^{12} and tolerance 10^{-90}. After the coefficients were frozen, reevaluation by Mellin integrals at 220 working digits gave an integer-normalized residual of approximately -2.315×10^{-213}. An independent Euler transformation of the defining alternating series, using 900 terms and 300 working digits, gave approximately 2.854×10^{-263}. The script records the full values and the exact integer vector. These are numerical evidence, not a proof or an interval certificate.

For the second method, the defining series are

$$g_{a,b} = \sum_{n\geq 0} \frac{(-1)^n H_{2n}^{(b)}}{(2n+1)^a}, \qquad H_m^{(b)} = \sum_{j=1}^{m} j^{-b},$$

and the analogous series with H_n for S_6. Thus the second evaluation does not reuse the discovery integral. Last transformed terms are recorded as diagnostics; their small size alone is not asserted to bound the complete tail.

The weight-seven functional (4.124) sends the integer-normalized candidate to 485683200. Its proof therefore also requires information beyond the restricted single-product presentation. The S4 certificate suggests seeking a corresponding weight-seven functional equation for (4.127), with the nonzero functional serving as a check that the missing relation has actually been introduced.

4.22.6 A distinct frozen candidate at weight nine

The later signed/zero continuation supplies a new S_8 candidate in a Gaussian basket with odd inner indices. It is distinct from the vector rigorously rejected in Proposition 12.1.

Conjecture 4.43 (The next mixed Gaussian reduction)**.**

$$\begin{aligned} S_8 = {} & \frac{52334}{37973} g_{8,1} + \frac{2824}{37973} g_{6,3} - \frac{6832}{189865} g_{4,5} - \frac{139760}{265811} g_{2,7} \\ & + \frac{998183\pi^9}{17283022848} - \frac{78615}{19442176} G\zeta(7) - \frac{201021}{4252976} \beta(4)\zeta(5) \\ & - \frac{290505}{1063244} \beta(6)\zeta(3) - 2\beta(8)\log 2. \end{aligned} \tag{4.128}$$

In the order

$$(S_8, g_{8,1}, g_{6,3}, g_{4,5}, g_{2,7}, \pi^9, G\zeta(7), \beta(4)\zeta(5), \beta(6)\zeta(3), \beta(8)\log 2),$$

its primitive integer vector is

$$\begin{aligned} (& -10974719508480, 15125246115840, 816174858240, -394908401664, \\ & -5770366156800, 633846205, -44376595200, -518730670080, \\ & -2998569369600, -21949439016960). \end{aligned}$$

All entries have greatest common divisor one; each displayed right-side coefficient is minus its vector entry divided by the first entry.

The discovery used 720 Euler terms at 250 working digits and then PSLQ at 150 digits, with tolerance 10^{-140} and coefficient bound 10^{20}. After freezing the vector, separate Mellin integrals at 220 digits and a 1400-term Euler evaluation at 500 digits corroborated it. Their residuals are numerical diagnostics and are not used to prove equality.

The fresh standard-library replay reconstructs two rational proximity certificates: the 1100-term source-bound scheme and the 1200-term direct scheme. For R_8 equal to the left side minus the right side of (4.128), the latter proves

$$|R_8| < 10^{-355}.$$

The same scheme improves the normalized S_6 residual to that bound. Each residual interval contains zero. The exact endpoints and frozen vectors are preserved with the signed/zero continuation. These enclosures prove proximity and leave both identities conjectural.

4.23 Harmonic index, reflection and certified asymptotics

The first-index sum S_p is part of a larger family. Here the harmonic index measures the displayed nesting length, not the minimal arithmetic or motivic depth. The same beta kernel gives reflection identities, signed truncation certificates and a joint index–exponent transition. These are analytic consequences of a positive kernel, and therefore remain useful independently of unresolved period-independence questions.

4.24 The harmonic-index family and its integral representation

Definition 4.44. *For integers $n, r \geq 0$, put*

$$e_r(n) = [u^r] \prod_{j=1}^{n} \left(1 + \frac{u}{j}\right), \qquad e_0(n) = 1. \tag{4.129}$$

Thus $e_r(n) = 0$ for $n < r$, $e_1(n) = H_n$, and $e_2(n) = (H_n^2 - H_n^{(2)})/2$. For real $a, p > 0$, define

$$T_{p,r}(a) = \sum_{n \geq 0} \frac{(-1)^n e_r(n)}{(n+a)^p}, \qquad S_{p,r} = 2^{-p} T_{p,r}(1/2). \tag{4.130}$$

The original S_p is $S_{p,1}$, while $S_{p,0} = \beta(p)$.

Lemma 4.45 (Convergence)**.** *For fixed $r \geq 0$ and $a, p > 0$, the series (4.130) converges. It converges absolutely for $p > 1$.*

Proof. The bound $e_r(n) \leq H_n^r/r!$ proves absolute convergence for $p > 1$ and convergence of the summand magnitude to zero for every $p > 0$. For eventual monotonicity, when $r \geq 1$ and $n \geq r$, counting each r-element subset in r ways gives

$$r e_r(n) = \sum_{|A|=r-1} \left(\prod_{j \in A} \frac{1}{j}\right) \left(H_n - \sum_{j \in A} \frac{1}{j}\right) \geq (H_n - H_{r-1}) e_{r-1}(n).$$

Together with $e_r(n+1) = e_r(n) + e_{r-1}(n)/(n+1)$, this shows that the logarithm of the ratio of consecutive magnitudes is at most

$$\frac{r}{(n+1)(H_n - H_{r-1})} - p \log\left(1 + \frac{1}{n+a}\right) < 0$$

for sufficiently large n. The alternating-series test applies. The case $r = 0$ is immediate. □

Proposition 4.46 (Mellin integral)**.** *For the same parameters,*

$$T_{p,r}(a) = \frac{(-1)^r}{r!\Gamma(p)} \int_0^1 t^{a-1}(-\log t)^{p-1} \frac{\log^r(1+t)}{1+t}\,\mathrm{d}t. \tag{4.131}$$

In particular, $(-1)^r T_{p,r}(a) > 0$.

Proof. The binomial theorem and (4.129) give, for $|t| < 1$,

$$\sum_{n\geq 0}(-1)^n e_r(n)t^n = [u^r](1+t)^{-1-u} = \frac{(-1)^r \log^r(1+t)}{r!(1+t)}. \tag{4.132}$$

Insert an Abel factor ρ^n with $0 < \rho < 1$, integrate termwise using $\int_0^1 t^{n+a-1}(-\log t)^{p-1}\,\mathrm{d}t = \Gamma(p)/(n+a)^p$, and let $\rho \uparrow 1$. Dominated convergence applies on the integral side, and Lemma 4.45 identifies the Abel limit with the ordinary sum. The integrand in (4.131) is strictly positive almost everywhere after removing the sign. □

4.24.1 Exact cyclotomic realizations

For integer $p \geq 1$, the series connects directly to the repository's Gaussian multiple polylogarithms:

$$S_{p,r} = \sum_{\epsilon_1,\ldots,\epsilon_r \in \{1,-1\}} \Im\,\mathrm{Li}_{p,\underbrace{1,\ldots,1}_{r}}(\mathrm{i}, \epsilon_1, \ldots, \epsilon_r). \tag{4.133}$$

Indeed, write the outer index as $m_0 = 2n+1$ and the inner indices as $m_j = 2k_j$. The 2^r from $1/k_j = 2/m_j$ cancels the factors $1/2$ in the even-index filters. Finally, $\Im(\mathrm{i}^{m_0})$ is exactly $(-1)^n$ for odd m_0 and zero for even m_0. This proves (4.133), first with an Abel factor and then by convergence. For $r = 1$, both second arguments 1 and -1 occur; S_p is not simply one of the draft's $g_{a,b}$.

More generally, let $a = m/q$ with $0 < m < q$ and $\xi = e^{\pi\mathrm{i}/q}$. Residue-class filters give

$$\begin{aligned} T_{p,r}(m/q) = q^{p-1} \sum_{h=0}^{q-1} \sum_{b_1,\ldots,b_r=0}^{q-1} & \xi^{-(2h+1)m} \\ & \cdot \mathrm{Li}_{p,1,\ldots,1}(\xi^{2h+1}, \xi^{2b_1}, \ldots, \xi^{2b_r}). \end{aligned} \tag{4.134}$$

To check the normalization, the denominator rescaling contributes q^{p+r}, while the $r+1$ residue filters contribute q^{-r-1}. Thus rational parameters lead to level $2q$ values. This is a representation of nesting length at most $r+1$, not a minimal-depth assertion.

4.25 An all-index reflection identity

Define, initially for $\Re(a-z) > 0$,

$$\Phi_a(z,u) = \int_0^1 t^{a-z-1}(1+t)^{-1-u}\,\mathrm{d}t. \tag{4.135}$$

Near $(z,u) = (0,0)$, coefficient extraction in (4.131) yields

$$\Phi_a(z,u) = \sum_{p\geq 1}\sum_{r\geq 0} T_{p,r}(a) z^{p-1}u^r. \tag{4.136}$$

There are no factorials in this ordinary Taylor series. This convention matters for every coefficient below.

Theorem 4.47 (Reflection)**.** *If $0 < a < 1$, then as analytic germs at the origin,*

$$\boxed{\Phi_a(z,u) + \Phi_{1-a}(-z-u,u) = \frac{\Gamma(a-z)\Gamma(1-a+z+u)}{\Gamma(1+u)}.} \tag{4.137}$$

More generally the identity holds whenever $\Re(a-z) > 0$ and $\Re(1-a+z+u) > 0$.

Proof. The beta integral evaluates

$$\int_0^\infty t^{a-z-1}(1+t)^{-1-u}\,\mathrm{d}t = \frac{\Gamma(a-z)\Gamma(1-a+z+u)}{\Gamma(1+u)}.$$

The part over $(0,1)$ is (4.135). In the part over $(1,\infty)$, substitute $t = 1/x$. The transformed integrand is $x^{-a+z+u}(1+x)^{-1-u}$, which is exactly the second term of (4.137). The stated inequalities justify both integrals. □

Let

$$C_{p,r}(a) = [z^{p-1}u^r]\frac{\Gamma(a-z)\Gamma(1-a+z+u)}{\Gamma(1+u)}. \tag{4.138}$$

Corollary 4.48 (Finite triangular reduction)**.** *For integers $p \geq 1, r \geq 0$ and $0 < a < 1$,*

$$\begin{aligned} T_{p,r}(a) + (-1)^{p-1}T_{p,r}(1-a) = C_{p,r}(a) \\ - \sum_{j=1}^{r}(-1)^{p-1+j}\binom{p-1+j}{j}T_{p+j,r-j}(1-a). \end{aligned} \tag{4.139}$$

Every sum on the right has smaller harmonic index.

Proof. The coefficient of $z^{p-1}u^r$ in $\Phi_{1-a}(-z-u,u)$ is

$$\sum_{j=0}^{r}(-1)^{p-1+j}\binom{p-1+j}{j}T_{p+j,r-j}(1-a).$$

Separate $j = 0$ in Theorem 4.47. □

4.25.1 A finite compiler for the gamma coefficient

Put $R_a(z,u)$ equal to the gamma quotient in (4.138), and write $R_a = (\pi/\sin\pi a)\exp(\sum_{d\geq 1}\ell_d)$, with ℓ_d homogeneous of total degree d. Then

$$\ell_d(z,u) = \frac{\psi^{(d-1)}(a)(-z)^d + \psi^{(d-1)}(1-a)(z+u)^d - \psi^{(d-1)}(1)u^d}{d!}. \tag{4.140}$$

Here $\psi^{(0)} = \psi$. If E_n is the homogeneous degree-n part of the exponential, then

$$E_0 = 1, \qquad E_n = \frac{1}{n}\sum_{d=1}^{n} d\ell_d E_{n-d}. \tag{4.141}$$

Only degrees through $p+r-1$ are needed for $C_{p,r}$. These recurrences are a finite identity certificate, not a relation search.

For rational a, the coefficients reduce to ordinary cyclotomic values and their products. For derivatives of positive order, use $\psi^{(k)}(a) = (-1)^{k+1}k!\zeta(k+1,a)$. For $s \geq 2$,

$$\zeta(s, m/q) = q^{s-1} \sum_{h=0}^{q-1} e^{-2\pi \mathrm{i} hm/q} \operatorname{Li}_s(e^{2\pi \mathrm{i} h/q}).$$

The Euler constants cancel in the linear part of (4.140); the remaining rational digamma combinations are expressible in logarithms and trigonometric constants. Thus (4.139) reduces the reflection combination to lower-index sums and products of depth-one values. It does not prove that the complementary combination is irreducible.

4.25.2 The self-reflecting Gaussian case

Set

$$F(z,u) = \tfrac{1}{2}\Phi_{1/2}(z/2,u) = \sum_{p\geq 1, r\geq 0} S_{p,r} z^{p-1} u^r.$$

Then

$$F(z,u) + F(-z-2u,u) = K(z,u), \qquad K(z,u) = \frac{\Gamma((1-z)/2)\Gamma((1+z)/2+u)}{2\Gamma(1+u)}. \tag{4.142}$$

For odd p,

$$2S_{p,r} = [z^{p-1}u^r]K - \sum_{j=1}^{r} (-1)^j 2^j \binom{p-1+j}{j} S_{p+j,r-j}. \tag{4.143}$$

For example,

$$S_{p,2} = pS_{p+1,1} - p(p+1)\beta(p+2) + \tfrac{1}{2}[z^{p-1}u^2]K \qquad (p \text{ odd}).$$

At even p, the coefficient of $S_{p,r}$ cancels. The identity then supplies relations among lower-index quantities rather than an unsupported converse to the odd-p reduction.

4.26 The first harmonic index revisited

At $r = 1$, the preceding reflection identity recovers equation (4.1). Its explicit coefficient form is equivalent to the generating-integral proof at the beginning of this chapter. This is one theorem reached from two useful representations: the full beta integral and the incomplete-beta reflection germ.

4.27 Complete monotonicity and globally enveloping truncations

Define

$$h_r(t) = \frac{1}{1+t}\left(\frac{\log(1+t)}{t}\right)^r, \qquad h_r(0) = 1. \tag{4.144}$$

The elementary representation

$$h_r(t) = \int_{[0,1]^r} \frac{1}{1+t} \prod_{j=1}^{r} \frac{1}{1+x_j t}\, \mathrm{d}x_1 \cdots \mathrm{d}x_r \tag{4.145}$$

is the source of the signed bounds.

Lemma 4.49. *For $t \geq 0$, h_r is completely monotone: $(-1)^k h_r^{(k)}(t) \geq 0$ for all $k \geq 0$. Moreover,*

$$h_r(t) = r! \sum_{j \geq 0} (-1)^j e_r(r+j) t^j \quad (|t| < 1). \tag{4.146}$$

For every integer $M \geq 0$ and $0 \leq t \leq 1$,

$$0 \leq (-1)^M \left(h_r(t) - r! \sum_{j=0}^{M-1} (-1)^j e_r(r+j) t^j \right) \leq r! e_r(r+M) t^M. \tag{4.147}$$

Proof. Every factor in (4.145) has derivatives of alternating sign, and products preserve this property by Leibniz's rule. Integration preserves it as well. Formula (4.146) follows from (4.132) after removing the zero of order r at $t = 0$. Taylor's integral remainder has sign $(-1)^M$. Its magnitude is at most $|h_r^{(M)}(0)| t^M / M!$ because complete monotonicity makes $|h_r^{(M)}(t)|$ nonincreasing. This is (4.147); $M = 0$ means $0 \leq h_r(t) \leq 1$. □

Theorem 4.50 (First-omitted-term certificate)**.** *For every real $p, a > 0$, integer $r \geq 0$, and integer $M \geq 0$,*

$$0 \leq (-1)^{r+M} \left(T_{p,r}(a) - \sum_{n=r}^{r+M-1} \frac{(-1)^n e_r(n)}{(n+a)^p} \right) \leq \frac{e_r(r+M)}{(r+M+a)^p}. \tag{4.148}$$

In the Gaussian normalization, replace every $(n+a)^p$ by $(2n+1)^p$.

Proof. Multiply (4.147) by $t^{a+r-1}(-\log t)^{p-1}/(r!\Gamma(p))$ and integrate. Equation (4.131) gives the middle term, and each monomial integrates explicitly. □

Remark 4.51. The certificate holds even when the first few summand magnitudes increase. Its proof is not an application of the alternating-series remainder theorem at an unjustified starting index. For example r may be large and p small; the positivity of the integral, rather than early term monotonicity, controls the error.

Since $e_r(r) = 1/r!$ and $e_r(r+1) = (r+2)/(2r!)$, we obtain

$$0 \leq 1 - (-1)^r r!(r+a)^p T_{p,r}(a) \leq \frac{r+2}{2} \left(\frac{r+a}{r+a+1} \right)^p. \tag{4.149}$$

The lower expression furnished by this bound may be negative for small p; positivity from Proposition 4.46 remains available. For fixed r, a and $p \to \infty$,

$$T_{p,r}(a) = \frac{(-1)^r}{r!(r+a)^p} + \frac{(-1)^{r+1}(r+2)}{2r!(r+a+1)^p} + R_{p,r}, \quad |R_{p,r}| \leq \frac{e_r(r+2)}{(r+a+2)^p}. \tag{4.150}$$

More generally, every finite number of exponential terms has its own signed explicit remainder. The first-omitted-term constant is asymptotically sharp for fixed r, a, M, by applying the bound once more to the remainder.

4.27.1 An exact symmetric-function certificate

Expanding (4.145) at zero gives

$$r!e_r(r+j) = \sum_{k_0+\cdots+k_r=j} \prod_{i=1}^{r} \frac{1}{k_i+1}. \tag{4.151}$$

Equivalently, it is the integral of the complete homogeneous polynomial $h_j(1, x_1, \ldots, x_r)$ over $[0,1]^r$. This identity is checked using exact rational arithmetic by two different recurrences in the supplied code. In particular, for fixed j,

$$r!e_r(r+j) = \frac{r^j}{2^j j!} + O_j(r^{j-1}). \tag{4.152}$$

The leading contribution comes from j distinct variables among $x_1, \ldots, x_r$; repeated variables and use of the distinguished entry 1 contribute $O_j(r^{j-1})$.

4.27.2 Pole stripping and exact Taylor radius

Let $F_{a,r}(z) = [u^r]\Phi_a(z,u)$. The integral representation shows that it extends holomorphically to $\Re z < a + r$, because the numerator vanishes to order r at the origin. Subtracting finitely many terms of (4.132) then continues it meromorphically to the entire z-plane, with precisely the simple poles

$$z = a + n \quad (n \geq r), \qquad \operatorname{Res}_{z=a+n} F_{a,r}(z) = (-1)^{n+1} e_r(n). \tag{4.153}$$

For clarity, subtract through degree N inside the integral. The remainder is $O(t^{N+1})$ at zero and is holomorphic for $\Re z < a + N + 1$; the subtracted terms are $(-1)^n e_r(n)/(a+n-z)$. Every residue in (4.153) is nonzero. Therefore the Taylor series $F_{a,r}(z) = \sum_{p \geq 1} T_{p,r}(a) z^{p-1}$ has exact radius $a + r$. This explains the first exponential scale in (4.150).

4.28 The joint depth–exponent transition

For $r \geq 1$, set

$$\mathcal{R}_{p,r}(a) = (-1)^r r!(r+a)^p T_{p,r}(a), \qquad L = p/r, \quad c = L - \log r, \quad \lambda = \tfrac{1}{2}e^{-c}. \tag{4.154}$$

At $a = 1/2$, this is $(-1)^r r!(2r+1)^p S_{p,r}$. By the strictly positive integral and $0 < h_r(t) < 1$ for $t > 0$, we have $0 < \mathcal{R}_{p,r}(a) < 1$.

Proposition 4.52 (Gamma expectation)**.** *If Y has gamma distribution with shape p and rate $r + a$, then*

$$\mathcal{R}_{p,r}(a) = \mathbb{E}h_r(e^{-Y}). \tag{4.155}$$

Proof. Set $t = e^{-y}$ in (4.131) and factor e^{-ry} out of $\log^r(1+e^{-y})$. Multiplying by the normalization in (4.154) leaves precisely the gamma density $(r+a)^p y^{p-1} e^{-(r+a)y}/\Gamma(p)$. □

Theorem 4.53 (Uniform transition and first correction)**.** *Let a range over a fixed compact subset of $(0,\infty)$ and let c range over a fixed compact subset of $\mathbb{R}$. Put $p = r(\log r + c)$ and $L = \log r + c$. As the integer $r \to \infty$,*

$$\mathcal{R}_{p,r}(a) = e^{-\lambda}\left[1 + \frac{1}{r}\left\{\left(\frac{\lambda^2}{2} - (a+\tfrac{1}{2})\lambda\right)L + \frac{5\lambda^2}{6} - 2\lambda\right\}\right] + O\left(\frac{L^2}{r^2}\right), \qquad \lambda = \tfrac{1}{2}e^{-c}. \tag{4.156}$$

The implied constant is uniform on those compact sets. In particular,

$$\mathcal{R}_{p,r}(a) \longrightarrow \exp(-e^{-c}/2). \tag{4.157}$$

For integer p, use $c = p/r - \log r$ in the expansion; rounding $r(\log r + c_0)$ to an integer preserves the limit with c_0.

We prove a uniform kernel lemma first. It is also the basis of the all-orders result.

Lemma 4.54 (Uniform kernel expansion)**.** *Write*

$$\log\left(\frac{\log(1+t)}{t}\right) = \sum_{m\geq 1} b_m t^m, \qquad b_1 = -\tfrac{1}{2}, \quad b_2 = \tfrac{5}{24}. \tag{4.158}$$

Define polynomials $\mathcal{U}_j(v)$ by

$$\sum_{j\geq 0} \mathcal{U}_j(v)\varepsilon^j = \exp\left(\sum_{k\geq 1} \varepsilon^k \left(2^{k+1} b_{k+1} v^{k+1} + \frac{(-1)^k 2^k v^k}{k}\right)\right). \tag{4.159}$$

For each fixed $J \geq 0$,

$$\sup_{0\leq t\leq 1} \left| h_r(t) - e^{-rt/2} \sum_{j=0}^{J} \frac{\mathcal{U}_j(rt/2)}{r^j} \right| \leq C_J r^{-J-1}. \tag{4.160}$$

In particular $\mathcal{U}_0 = 1$ and $\mathcal{U}_1(v) = 5v^2/6 - 2v$.

Proof. Put $v = rt/2$. Expanding $r\log(\log(1+t)/t) - \log(1+t)$ at zero gives $-v$ plus the exponent in (4.159) with $\varepsilon = 1/r$. Fix a sufficiently small $\delta > 0$. On $0 \leq t \leq \delta$, Taylor remainders are bounded by r^{-J-1} times a fixed polynomial in v, after expansion through the required order. The perturbation of $-v$ is bounded by a small fixed multiple of v (enlarge the threshold for r if needed). Taylor's remainder for the exponential is consequently dominated by $r^{-J-1}\Theta_J(v)e^{-\eta v}$ for some $\eta > 0$. Its supremum is finite.

On $\delta \leq t \leq 1$, the ratio $\log(1+t)/t$ is bounded above by a number strictly less than one. Hence $h_r(t)$ decays exponentially in r, while every term of the proposed approximation is a polynomial in r times $e^{-r\delta/2}$. Both are $O_J(r^{-J-1})$. Combining the two ranges proves the claim. All constants here depend only on J and the chosen fixed δ. □

Proof of Theorem 4.53. Let $\Delta = Y - L$. Its mean and centered gamma moments give, uniformly in the stated parameter sets,

$$\begin{aligned} \mathbb{E}\Delta &= -\frac{aL}{r} + O(L/r^2), & \mathbb{E}\Delta^2 &= \frac{L}{r} + O(L^2/r^2), \\ \mathbb{E}\Delta^3 &= O(L^2/r^2), & \mathbb{E}\Delta^4 &= O(L^2/r^2). \end{aligned} \tag{4.161}$$

For example $\mathbb{E}Y = rL/(r+a)$, $\operatorname{Var} Y = rL/(r+a)^2$, and the centered third and fourth moments are $2rL/(r+a)^3$ and $3(rL)^2/(r+a)^4 + 6rL/(r+a)^4$. These formulas imply (4.161) by translation.

Define $v = \lambda e^{-\Delta}$, $f(\Delta) = e^{-\lambda e^{-\Delta}}$, and $g(\Delta) = e^{-v}(5v^2/6 - 2v)$. All derivatives of f and g of fixed order are bounded on the real line: each is a polynomial in $v \geq 0$ times e^{-v}. Lemma 4.54 with $J = 1$ gives

$$\mathcal{R}_{p,r}(a) = \mathbb{E}f(\Delta) + r^{-1}\mathbb{E}g(\Delta) + O(r^{-2}).$$

Taylor-expand f through degree three, retaining the fourth-order remainder. Since $f'(0) = \lambda e^{-\lambda}$ and $f''(0) = (\lambda^2 - \lambda)e^{-\lambda}$, (4.161) yields

$$\mathbb{E}f(\Delta) = e^{-\lambda}\left[1 + \frac{L}{r}\left(\frac{\lambda^2}{2} - (a + \tfrac{1}{2})\lambda\right)\right] + O(L^2/r^2).$$

A second-order Taylor bound gives $\mathbb{E}g(\Delta) = e^{-\lambda}(5\lambda^2/6 - 2\lambda) + O(L/r)$. Substitute the two estimates. Since $L \asymp \log r$, this is exactly (4.156). □

4.28.1 A second proof of the limiting profile

Theorem 4.50 and (4.152) provide a useful independent explanation. For fixed j and $p = r(\log r + c)$,

$$r!e_r(r + j)\left(\frac{r + a}{r + a + j}\right)^p \longrightarrow \frac{\lambda^j}{j!}.$$

Every even and odd truncation bounds $\mathcal{R}_{p,r}(a)$ from opposite sides. Taking $r \to \infty$ for a fixed truncation and then increasing the truncation squeezes the limit between the alternating Taylor bounds for $e^{-\lambda}$. This argument does not interchange an uncontrolled infinite sum with a limit. It also gives compact-c uniform convergence by making the finite exponential remainder uniformly small.

4.28.2 The threshold outside the compact transition window

For fixed $a > 0$ (also uniformly on compact a-sets), if $p/r - \log r \to +\infty$, then (4.149) implies $\mathcal{R}_{p,r}(a) \to 1$. One may split into $p/r \le 2\log r$ and $p/r > 2\log r$ to verify the exponential estimate without assuming p/r^2 is small.

If $p/r - \log r \to -\infty$, with $p > 0$, then eventually $L \le \log r$, and $Y - L \to 0$ in probability because its variance and mean shift tend to zero. Thus $re^{-Y} \to \infty$ in probability. The elementary inequality $\log(1 + t)/t \le 1 - t/6$ on $0 \le t \le 1$ gives $h_r(t) \le e^{-rt/6}$. Bounded convergence in probability in (4.155) now gives $\mathcal{R}_{p,r}(a) \to 0$. Therefore $p \sim r\log r$ is the actual threshold for first-pole dominance.

4.28.3 All-orders transition polynomials

Define the operator on polynomials in v

$$\mathcal{D}P(v) = vP(v) - vP'(v), \tag{4.162}$$

and the polynomials

$$d_j(t; a) = \frac{(-1)^j}{j + 1}\left(a^{j+1} - (a - t)^{j+1}\right) \quad (j \ge 1). \tag{4.163}$$

The operator convention follows from $\partial_\Delta(e^{-v}P(v)) = e^{-v}\mathcal{D}P(v)$ for $v = \lambda e^{-\Delta}$.

Theorem 4.55 (All-orders compiler). *For every fixed $J \ge 0$, in the parameter range of Theorem 4.53,*

$$\mathcal{R}_{p,r}(a) = e^{-\lambda}\sum_{j=0}^{J}\frac{\Theta_j(L, \lambda; a)}{r^j} + O_J\left(\frac{L^{J+1}}{r^{J+1}}\right), \tag{4.164}$$

uniformly on compact a, c-sets. The Θ_j have rational coefficients and degree at most j in L. They are determined by the finite formal prescription

$$\sum_{j\ge 0}\Theta_j(L,v;a)\varepsilon^j=\exp\left(L\sum_{k\ge 1}\varepsilon^k d_k(\mathcal{D};a)\right)\left(\sum_{k\ge 0}\varepsilon^k\mathcal{U}_k(v)\right). \tag{4.165}$$

In particular

$$\Theta_1=\left(\frac{v^2}{2}-(a+\tfrac{1}{2})v\right)L+\frac{5}{6}v^2-2v. \tag{4.166}$$

Proof. The exact moment-generating function of Δ is

$$\log\mathbb{E}e^{t\Delta}=-Lt+rL\{\log(1+a/r)-\log(1+(a-t)/r)\}=L\sum_{j\ge 1}r^{-j}d_j(t;a).$$

Consequently expectation, formally through any fixed order in r^{-1}, is the differential operator $\exp(L\sum r^{-j}d_j(\partial_\Delta;a))$ evaluated at $\Delta=0$. Conjugation by e^{-v} turns ∂_Δ into (4.162), proving the coefficient rule.

For the error justification, apply Lemma 4.54 through order J. For the term containing $r^{-k}\mathcal{U}_k$, Taylor-expand $e^{-v}\mathcal{U}_k(v)$ as a function of Δ through degree $2(J-k)+1$. Its fixed-order derivatives are bounded, and the gamma moments imply $\mathbb{E}|\Delta|^{2s}=O_s((L/r)^s)$ for each fixed s, since $L\asymp\log r$. The Taylor remainder therefore contributes $O(r^{-J-1}L^{J-k+1})$. The finitely many retained moments are explicit rational functions of $r+a$ with polynomial dependence on rL; expanding them through order J yields the stated formal operator and an error $O_J(L^{J+1}/r^{J+1})$. This also controls the uniform kernel error. Finally, each factor in the operator exponent carries one L and at least one power of ε, so the coefficient of ε^j has degree at most j in L. □

For reproducibility, write $\Theta_2=A_2L^2+B_2L+C_2$, where

$$\begin{aligned}
A_2&=\frac{v^4}{8}-\left(\frac{a}{2}+\frac{3}{4}\right)v^3+\left(\frac{a^2}{2}+\frac{3a}{2}+\frac{7}{8}\right)v^2-\left(\frac{a^2}{2}+\frac{a}{2}+\frac{1}{8}\right)v,\\
B_2&=\frac{5v^4}{12}-\left(\frac{5a}{6}+\frac{11}{4}\right)v^3+\left(\frac{8a}{3}+\frac{11}{3}\right)v^2+\left(a^2-a-\frac{2}{3}\right)v,\\
C_2&=\frac{25v^4}{72}-\frac{8}{3}v^3+4v^2.
\end{aligned}$$

The supplementary symbolic script produces $\Theta_0,\dots,\Theta_4$ from (4.165); it does not fit them to numerical data.

4.28.4 The complementary fixed-exponent regime

The transition theorem is not uniform down to fixed p. A separate endpoint expansion describes that regime.

Theorem 4.56 (Fixed p, growing harmonic index)**.** *Let $b=\log 2$. For p,a in compact subsets of $(0,\infty)$ and $r\to\infty$,*

$$T_{p,r}(a)=\frac{(-1)^r2^{p-1}b^{r+p}}{r!r^p}\left[1+\frac{(1-2a)bp+(b-1)p(p+1)/2}{r}+O(r^{-2})\right]. \tag{4.167}$$

Thus

$$S_{p,r}=\frac{(-1)^rb^{r+p}}{2r!r^p}\left[1+\frac{(b-1)p(p+1)}{2r}+O_p(r^{-2})\right]. \tag{4.168}$$

Proof. In the $y = -\log t$ form of (4.131), the factor $\log^r(1+e^{-y})$ localizes the integral near $y = 0$. Its logarithm expands as

$$\log\left(\frac{\log(1+e^{-y})}{b}\right) = -\frac{y}{2b} + \frac{b-1}{8b^2}y^2 + O(y^3),$$

while $e^{-ay}/(1+e^{-y}) = \frac{1}{2}(1+(1/2-a)y+O(y^2))$. Set $y = 2bv/r$. The integrand becomes $v^{p-1}e^{-v}$ times

$$1 + \frac{(1-2a)bv + (b-1)v^2/2}{r} + O\big(r^{-2}P(v)\big)$$

in a fixed small-y interval, with an integrable exponential majorant after the usual smaller-interval split. The complementary y-interval is exponentially small relative to $b^r r^{-p}$: the logarithmic ratio is uniformly below one there, and its large-y decay supplies integrability. Integrating the displayed terms uses $\Gamma(p+1)/\Gamma(p) = p$ and $\Gamma(p+2)/\Gamma(p) = p(p+1)$. Uniform Taylor bounds on compact a, p-sets give the stated remainder. Finally divide by 2^p at $a = 1/2$. □

4.29 Geometric evaluation with explicit error bounds

The signed partial-sum bounds are exact but may converge slowly at small p. An incomplete-beta expansion supplies geometric convergence. In (4.135), put $y = t/(1+t)$; then

$$\Phi_a(z,u) = \int_0^{1/2} y^{a-z-1}(1-y)^{u-a+z}\,\mathrm{d}y = 2^{-a+z}\sum_{n\ge 0}\frac{(a-z-u)_n}{n!\,2^n(n+a-z)}. \tag{4.169}$$

This is the usual binomial expansion of the incomplete-beta integral, with all variables and normalizations displayed.

Theorem 4.57 (Explicit Cauchy tail bound)**.** *Let $p \ge 1$ and $r \ge 0$ be integers, and let $a > 0$. Choose $0 < \rho < a$, $\sigma > 0$, and let $A = a + \rho + \sigma$. For an integer $N \ge 1$, put*

$$q_N = \frac{1}{2}\max\left(1, \frac{N+A}{N+1}\right), \qquad B_N = 2^{-a+\rho}\frac{(A)_N}{N!2^N(N+a-\rho)}.$$

If $q_N < 1$, the error after retaining $0 \le n < N$ in (4.169) has modulus at most $B_N/(1-q_N)$ on $|z| \le \rho, |u| \le \sigma$. Consequently the error in $T_{p,r}(a)$ is at most

$$\rho^{-(p-1)}\sigma^{-r}\frac{B_N}{1-q_N}. \tag{4.170}$$

Proof. For every factor of the rising factorial, $|a-z-u+j| \le A+j$. Also $|n+a-z| \ge n+a-\rho$ and $|2^{-a+z}| \le 2^{-a+\rho}$. The resulting positive majorant term has ratio

$$\frac{n+A}{2(n+1)}\frac{n+a-\rho}{n+1+a-\rho} \le q_N \qquad (n \ge N).$$

Sum the geometric majorant. The bivariate Cauchy coefficient estimate gives (4.170). □

Corollary 4.58 (A rational Gaussian error bound)**.** *For $a = 1/2$ take $\rho = \sigma = 1/4$. Let $S_{p,r}^{[N]}$ be the coefficient approximation from (4.169), divided by 2^p. Then*

$$\boxed{|S_{p,r} - S_{p,r}^{[N]}| \le \frac{2^{p+2r-1-N}}{N+1/4}.} \tag{4.171}$$

Proof. Here $A = 1$, $(A)_N/N! = 1$, $q_N = 1/2$, and $2^{-1/4} < 1$. Insert these values into (4.170) and multiply by 2^{-p}. □

4.29.1 How the supplied exact certificates are constructed

For rational $a = 1/2$, the finite approximation has the form

$$S_{p,r}^{[N]} = 2^{-p-1/2} \sum_{\ell=0}^{p-1} c_\ell (\log 2)^\ell, \qquad c_\ell \in \mathbb{Q}. \tag{4.172}$$

The coefficients are computed exactly. Maintain the truncated bivariate polynomial $\Theta_n(z,u) = (1/2 - z - u)_n/n!$ using

$$\Theta_{n+1} = \frac{n + 1/2 - z - u}{n+1} \Theta_n.$$

Multiply by the truncated geometric expansion of $(n + 1/2 - z)^{-1}$, accumulate the 2^{-n} terms, and finally multiply by the Taylor polynomial of $e^{z \log 2}$. All operations up to (4.172) are rational.

The code then encloses $\log 2$ using

$$\log 2 = 2 \sum_{k=0}^{K-1} \frac{1}{(2k+1)3^{2k+1}} + \eta_K, \qquad 0 < \eta_K \le \frac{9}{4(2K+1)3^{2K+1}},$$

and encloses $2^{-1/2}$ by an integer square-root computation on a dyadic scale. Rational interval Horner evaluation controls the error in (4.172), and the rational bound (4.171) controls the infinite tail. Thus the resulting interval endpoints are exact fractions. They do not depend on trusting a floating-point quadrature result or a special-function library.

These certificates establish enclosures for individual constants. Overlap of two intervals would not, by itself, prove an identity between the constants; the analytic proofs in earlier sections perform that task. Conversely, disjoint validated intervals can rigorously disprove a proposed equality. A finite-height exclusion program would require a further certified enumeration or lattice argument, which is not included here.

4.30 Exponential cancellation at proportional harmonic depth

The positive integral representation now leads to an analytic question left open by the earlier alternating-harmonic report Chapter 4: how rapidly does its normalized sum tend to zero below the first-term transition? The case of proportional outer exponent and harmonic depth has an explicit rate function. This is a continuation of that report's results, not another proof of its already established reflection identity.

For integers $h, n \ge 0$, define

$$e_h(n) = [u^h] \prod_{j=1}^{n} (1 + u/j), \qquad T_{p,h}(a) = \sum_{n \ge 0} \frac{(-1)^n e_h(n)}{(n+a)^p}, \quad a, p > 0.$$

Thus $e_0(n) = 1$, $e_1(n) = H_n$, and $e_2(n) = (H_n^2 - H_n^{(2)})/2$. Ordinary convergence holds for $p > 0$; absolute convergence holds for $p > 1$. The Gaussian normalization is $S_{p,h} = 2^{-p} T_{p,h}(1/2)$. The positive Mellin integral

$$T_{p,h}(a) = \frac{(-1)^h}{h!\,\Gamma(p)} \int_0^\infty \frac{x^{p-1} e^{-ax} \log^h(1 + e^{-x})}{1 + e^{-x}}\, dx \tag{4.173}$$

follows from

$$\sum_{n\ge 0} e_h(n) z^n = \frac{(-\log(1-z))^h}{h!(1-z)}$$

by inserting an Abel factor, using the gamma integral, and taking its limit. The exponential decay at infinity and the bound $\log(1+e^{-x}) \le \log 2$ near zero justify the integral limit. The convergence and Abel-limit argument are also proved in the integral representation above. In particular, $(-1)^h T_{p,h}(a) > 0$.

Put

$$R_{p,h}(a) = (-1)^h h!(h+a)^p T_{p,h}(a). \tag{4.174}$$

The earlier report proves $0 < R_{p,h}(a) < 1$ and identifies the threshold $p \sim h\log h$. When $p/h - \log h \to -\infty$, it proves that $R_{p,h}(a) \to 0$. The following theorem quantifies this statement when p/h remains in a fixed positive compact interval.

4.30.1 The unique saddle and the rate function

Write

$$L(x) = \log(1+e^{-x}), \qquad w(x) = \frac{1}{(1+e^x)L(x)} = -\frac{d}{dx}\log L(x), \quad x > 0. \tag{4.175}$$

Lemma 4.59. *The function w is strictly increasing, with*

$$\frac{1}{2\log 2} < w(x) < 1.$$

For every $\alpha > 0$ there is exactly one $x_\alpha > 0$ satisfying

$$x_\alpha w(x_\alpha) = \alpha. \tag{4.176}$$

Moreover $\alpha < x_\alpha < 2\alpha\log 2$.

Proof. Set $t = e^{-x} \in (0,1)$. Then $w = t/((1+t)\log(1+t))$ and

$$\frac{d}{dt}\log w = \frac{\log(1+t) - t}{t(1+t)\log(1+t)} < 0.$$

Consequently w increases with x. Its endpoint limits give the bounds. The function $xw(x)$ has derivative $w + xw' > 0$, tends to zero at zero, and tends to infinity at infinity. This proves existence, uniqueness, and the stated enclosure. □

For $x = x_\alpha$, set

$$\Phi_\alpha(y) = \alpha\log y + \log L(y), \tag{4.177}$$

$$A_\alpha = \frac{\alpha}{x^2} + w'(x) > 0, \tag{4.178}$$

$$I(\alpha) = -\alpha\log(x/\alpha) - \alpha - \log L(x), \tag{4.179}$$

$$C(\alpha, a) = \frac{\sqrt{\alpha/A_\alpha}}{x(1+e^{-x})}\exp\big(a(\alpha - x)\big). \tag{4.180}$$

All of these functions are real analytic for $\alpha > 0$.

Theorem 4.60 (Proportional-depth asymptotics)**.** *Let a and α range over fixed compact subsets of $(0,\infty)$, and put $p=\alpha h$. As the integer $h\to\infty$,*

$$R_{\alpha h,h}(a)=C(\alpha,a)e^{-hI(\alpha)}\big(1+O(h^{-1})\big), \tag{4.181}$$

uniformly on those compact sets. In fact

$$R_{\alpha h,h}(a)=C(\alpha,a)e^{-hI(\alpha)}\left(1+\frac{D(\alpha,a)}{h}+O(h^{-2})\right), \tag{4.182}$$

where, with $x=x_\alpha$, $A=A_\alpha$, and $g(y)=e^{-ay}/(y(1+e^{-y}))$,

$$D(\alpha,a)=\frac{g''(x)}{2Ag(x)}+\frac{\Phi_\alpha'''(x)g'(x)}{2A^2g(x)}+\frac{\Phi_\alpha''''(x)}{8A^2}+\frac{5\Phi_\alpha'''(x)^2}{24A^3}-\frac{\alpha a^2}{2}-\frac{1}{12\alpha}. \tag{4.183}$$

Every fixed further order in h^{-1} exists and is effectively obtained from derivatives of L and g at x_α and the Stirling series. The remainder at every fixed order is uniform on the same compact sets.

Proof. Equation (4.173) gives

$$R_{\alpha h,h}(a)=\frac{(h+a)^{\alpha h}}{\Gamma(\alpha h)}\int_0^\infty e^{h\Phi_\alpha(y)}g(y)\,dy.$$

We have $\Phi_\alpha'(y)=\alpha/y-w(y)$ and $\Phi_\alpha''(y)=-\alpha/y^2-w'(y)<0$. The sole critical point is therefore a strict global maximum. Its location stays in a fixed compact subinterval of $(0,\infty)$ as α varies in the prescribed compact set, and its curvature is bounded away from zero.

Here are explicit tail controls for the application of Laplace's method [78, §2.3(iii)] [78, Section 2.3(iii)]. Near zero, $\exp(h\Phi_\alpha(y))\le y^{h\alpha}(\log 2)^h$, while $g(y)=O(y^{-1})$ uniformly in a. Choosing the endpoint interval small enough makes its integral exponentially smaller than the maximum, uniformly in α. At infinity, $L(y)<e^{-y}$ implies $\Phi_\alpha(y)<\alpha\log y-y$; this supplies an integrable exponential bound uniform in the same compact parameters. On the remaining compact set outside a fixed neighborhood of x_α, strict concavity and compactness give a uniform negative gap from the maximum. These observations justify discarding both tails with an exponentially small relative error.

In the neighborhood, substitute $y=x+v/\sqrt{h}$. Taylor expansion, followed by Gaussian integration, gives

$$\int_0^\infty e^{h\Phi_\alpha(y)}g(y)\,dy=e^{h\Phi_\alpha(x)}g(x)\sqrt{\frac{2\pi}{hA}}\left(1+\frac{E}{h}+O(h^{-2})\right),$$
$$E=\frac{g''}{2Ag}+\frac{\Phi'''g'}{2A^2g}+\frac{\Phi''''}{8A^2}+\frac{5(\Phi''')^2}{24A^3},$$

with all derivatives evaluated at x. Odd Gaussian moments vanish; the even moments $1/A$, $3/A^2$, and $15/A^3$ produce precisely these four terms. The analytic functions have uniformly bounded derivatives on a common neighborhood of the saddle. Taylor remainders there, together with the tail bounds just given, prove the claimed uniform error and also permit expansion to any prescribed order.

Finally Stirling's formula, uniformly for α in its compact set, and $\alpha h\log(1+a/h)=\alpha a-\alpha a^2/(2h)+O(h^{-2})$ give

$$\frac{(h+a)^{\alpha h}}{\Gamma(\alpha h)}=\sqrt{\frac{\alpha h}{2\pi}}e^{h\alpha(1-\log\alpha)+a\alpha}\left(1-\frac{\alpha a^2/2+1/(12\alpha)}{h}+O(h^{-2})\right).$$

Multiplying the two expansions proves (4.181)–(4.183). □

Remark 4.61 (Integer exponents and rounding). For integer p, set $\alpha = p/h$ in the theorem. If instead one fixes α_0 and rounds $\alpha_0 h$ to p, the $O(1/h)$ change of α must remain inside $e^{-hI(\alpha)}$: replacing it by $e^{-hI(\alpha_0)}$ changes the leading constant by a generally nontrivial bounded factor. This is a practical distinction in numerical comparisons.

4.30.2 Monotonicity, convexity, and matching limits

Proposition 4.62. *The rate I is positive, strictly decreasing, and strictly convex on $(0,\infty)$. More explicitly,*

$$I'(\alpha) = \log\frac{\alpha}{x_\alpha} = \log w(x_\alpha) < 0, \qquad I''(\alpha) = \frac{w'(x_\alpha)}{w(x_\alpha)(w(x_\alpha)+x_\alpha w'(x_\alpha))} > 0. \tag{4.184}$$

Its endpoint limits are

$$I(0+) = -\log\log 2, \qquad I(+\infty) = 0.$$

As $\alpha \to \infty$,

$$x_\alpha = \alpha + \frac{\alpha}{2}e^{-\alpha} + O(\alpha^2 e^{-2\alpha}), \tag{4.185}$$

$$I(\alpha) = \frac{1}{2}e^{-\alpha} - \frac{3\alpha+5}{24}e^{-2\alpha} + O(\alpha^2 e^{-3\alpha}), \tag{4.186}$$

$$C(\alpha,a) = 1 - \left(\frac{(a+1/2)\alpha}{2} + 1\right)e^{-\alpha} + O_a(\alpha^2 e^{-2\alpha}). \tag{4.187}$$

Also $C(\alpha,a) \to 1/2$ as $\alpha \downarrow 0$.

Proof. Differentiating the optimized value in (4.179) cancels its x'_α terms by (4.176). This gives the first identity in (4.184). Differentiating $\alpha = xw(x)$ gives the second. Positivity also follows directly: $\log L(y) < -y$, so

$$\max_{y>0}\{\alpha\log(y/\alpha) + \alpha + \log L(y)\} < \max_{y>0}\{\alpha\log(y/\alpha) + \alpha - y\} = 0.$$

The strict inequality is valid because the first maximum is attained; at its maximizing point the second expression is at most zero.

The small-α limits follow from $x_\alpha/\alpha \to 2\log 2$ and $A_\alpha \sim \alpha/x_\alpha^2$. At infinity, put $t = e^{-y}$ and use

$$\log L(y) = -y - \frac{t}{2} + \frac{5t^2}{24} + O(t^3), \qquad w(y) = 1 - \frac{t}{2} + \frac{5t^2}{12} + O(t^3).$$

Substitution in (4.176) first gives (4.185). Taylor expansion of (4.179) around $y = \alpha$, retaining the quadratic displacement, gives (4.186). Expansion of (4.180) and $A_\alpha = \alpha/x_\alpha^2 + w'(x_\alpha)$ gives (4.187). The remainders stated above follow from analyticity of $\log(\log(1+t)/t)$ for t near zero, with $x_\alpha - \alpha = O(\alpha e^{-\alpha})$. □

The rate formula explains the scale of cancellation even when the terms of the original alternating sum are much larger than its value. For example, $I(1) = 0.15091647898554\ldots$ and $C(1,1/2) = 0.66325765726435\ldots$. Thus

$$R_{h,h}(1/2) \sim 0.66325765726435\ldots\, e^{-0.15091647898554\ldots h}.$$

This is a proved exponential rate, rather than just a limit of zero.

The large-α expansions also explain why the transition parameter in the previous report was $\lambda = he^{-\alpha}/2$. Formally putting $\alpha = \log h + c$ in $hI(\alpha)$ gives the leading value $e^{-c}/2$. This agrees with the independently proved transition $R \to \exp(-e^{-c}/2)$ in Chapter 4. Such matching is a consistency calculation: the compact-α theorem proved here does *not* by itself establish a uniform error for growing α. Obtaining a single effective formula across that expanding range remains a precise further problem.

4.30.3 A probability law at the saddle

There is a useful formulation independent of the alternating series. Normalize the positive integrand in (4.173), with $p = \alpha h$, to a probability density and call the resulting random variable X_h. Then

$$\sqrt{hA_\alpha}(X_h - x_\alpha) \implies N(0,1). \tag{4.188}$$

Indeed, the same change of variables used in the proof of Theorem 4.60, its uniform Gaussian domination on a shrinking neighborhood, and its exponentially negligible tails prove convergence against every bounded continuous test function. The associated rate for X_h is $\Phi_\alpha(x_\alpha) - \Phi_\alpha(y)$. This identifies which part of the positive integral carries the very small alternating sum and gives a stable center for numerical quadrature.

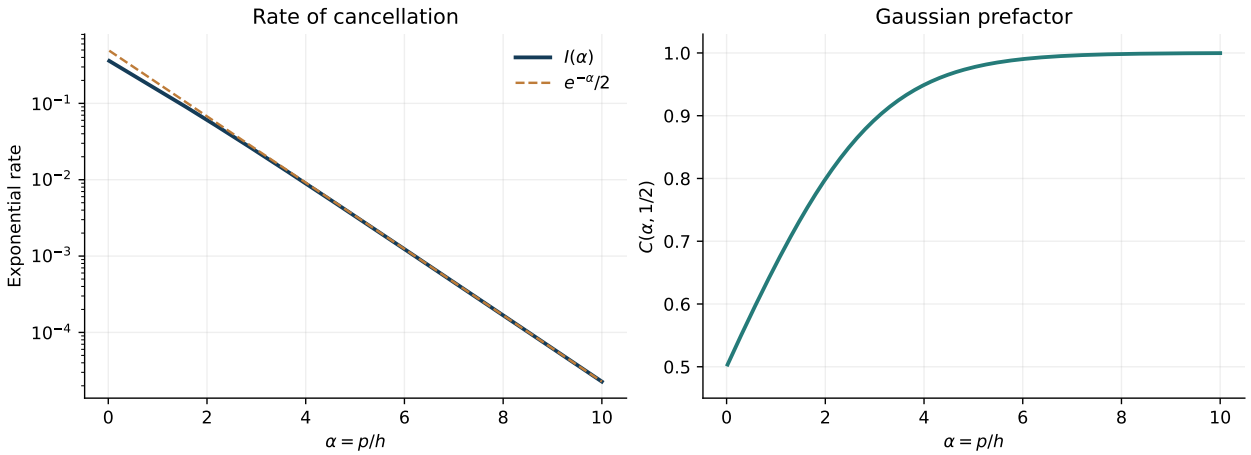


Figure 4.1: The proportional-depth rate and normalized asymptotic. The positive, decreasing and convex rate is proved in the text; sampled values illustrate it. The compact-parameter theorem does not by itself cover a growing $\alpha = \log h + c$ transition.

Chapter 5

Signed kernels and certified polylogarithm computation

The exact reductions suggest evaluation methods that expose more analytic structure. A signed density for $F_{a,b}(z) = \mathrm{Li}_{a,b}(z,1)$ proves a unique angular zero and makes Gaussian Euler errors one-sided. The general double integral supports Chebyshev and midpoint expansions, while Holder convolution provides uniform rational certificates for a wider alphabet. These methods connect experiments, geometry and finite arithmetic: their tail theorems and rounding contracts determine what a numerical comparison establishes. Write $D_{a,b}(x,y) = \mathrm{Li}_{a,b}(x,y)$. Throughout the signed-density discussion, $H_n^{(b)} = \sum_{j=1}^{n} j^{-b}$, $H_0^{(b)} = 0$, and $g_{a,b} = \operatorname{Im} F_{a,b}(\mathrm{i})$; real $b > 0$ is allowed where stated.

5.1 A signed-kernel representation

Definition 5.1. *For real $b > 0$ and $T > 0$, define*

$$\kappa_{1,b}(e^{-T}) = -\frac{T^b}{\Gamma(b+1)} + \frac{1}{\Gamma(b)} \int_T^\infty \frac{v^{b-1}}{e^v - 1}\, \mathrm{d}v. \tag{5.1}$$

For an integer $a \geq 2$, define successively

$$\kappa_{a,b}(u) = \int_u^1 \frac{\kappa_{a-1,b}(v)}{v}\, \mathrm{d}v, \qquad 0 < u < 1. \tag{5.2}$$

Theorem 5.2 (Moments and one sign change)**.** *For every integer $a \geq 1$ and real $b > 0$, the density $\kappa_{a,b}$ is integrable on $(0,1)$ and*

$$\int_0^1 u^{n-1} \kappa_{a,b}(u)\, \mathrm{d}u = \frac{H_{n-1}^{(b)}}{n^a}, \qquad n \geq 1. \tag{5.3}$$

In particular its total mass is zero. There is exactly one $r_{a,b} \in (0,1)$ such that

$$\kappa_{a,b}(u) < 0 \quad (0 < u < r_{a,b}), \qquad \kappa_{a,b}(u) > 0 \quad (r_{a,b} < u < 1). \tag{5.4}$$

The crossing is strict, and $r_{a,b} < r_{a-1,b}$ when $a \geq 2$.

Proof. First consider $a = 1$. The negative term in (5.1) has moment $-n^{-b-1}$. Substituting $u = e^{-T}$ and interchanging nonnegative integrands in the positive term gives

$$\frac{1}{\Gamma(b)} \int_0^\infty e^{-nT} \int_T^\infty \frac{v^{b-1}}{e^v - 1} \, \mathrm{d}v \, \mathrm{d}T$$
$$= \frac{1}{n\Gamma(b)} \int_0^\infty \frac{v^{b-1}(1 - e^{-nv})}{e^v - 1} \, \mathrm{d}v = \frac{1}{n} \sum_{k=1}^{n} k^{-b}.$$

The final equality uses the finite geometric identity $(1 - e^{-nv})/(e^v - 1) = \sum_{k=1}^{n} e^{-kv}$. Subtracting n^{-b-1} proves the base moment formula. At $n = 1$, both positive and negative terms have finite integral 1, so integrability and zero mass are established without subtracting divergent quantities.

As a function of T, the base density has derivative

$$\frac{\mathrm{d}}{\mathrm{d}T} \kappa_{1,b}(e^{-T}) = -\frac{T^{b-1}}{\Gamma(b)(1 - e^{-T})} < 0. \tag{5.5}$$

It is positive as $T \downarrow 0$: the limit is $\zeta(b) > 0$ if $b > 1$, and is $+\infty$ if $0 < b \leq 1$. It tends to $-\infty$ as $T \to \infty$. Thus it has exactly one crossing, with the sign pattern stated in the u coordinate.

For the induction, the integral operator in (5.2) is an L^1 contraction:

$$\int_0^1 \left| \int_u^1 \frac{f(v)}{v} \, \mathrm{d}v \right| \mathrm{d}u \leq \int_0^1 |f(v)| \, \mathrm{d}v.$$

Fubini's theorem gives the moment of the new density as $1/n$ times the previous moment; in particular it preserves zero mass. If the preceding crossing is r, the new density is positive on $[r, 1)$, while on $(0, r)$ its derivative is $-\kappa_{a-1,b}(u)/u > 0$. Zero mass forces it to be negative somewhere on that interval. Strict increase then gives exactly one crossing in $(0, r)$. This completes the induction. □

Corollary 5.3 (Integral continuation)**.** *For $z \in \mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = \int_0^1 \frac{z}{1 - zu} \kappa_{a,b}(u) \, \mathrm{d}u. \tag{5.6}$$

In particular, if $0 < \theta < \pi$,

$$\operatorname{Im} F_{a,b}(e^{\mathrm{i}\theta}) = \sin\theta \, J_{a,b}(\cos\theta), \qquad J_{a,b}(c) = \int_0^1 \frac{\kappa_{a,b}(u)}{1 - 2cu + u^2} \, \mathrm{d}u. \tag{5.7}$$

Proof. For $|z| < 1$, expand $z/(1 - zu)$ and use the moments. On compact subsets of the slit plane the denominator is bounded away from zero uniformly in $u \in [0, 1]$, so the integral is analytic there. This continues the disk identity. Taking the imaginary part of $e^{\mathrm{i}\theta}/(1 - ue^{\mathrm{i}\theta})$ gives (5.7). □

Lemma 5.4 (A sign test for zero-mass densities)**.** *Let κ have zero integral and the strict sign pattern (5.4), with crossing r. If q is strictly decreasing and the integral exists, then $\int q\kappa < 0$. If q is strictly increasing, the sign is positive.*

Proof. Subtract $q(r) \int \kappa = 0$. The product $(q(u) - q(r))\kappa(u)$ has the asserted strict sign on each side of r. □

5.2 The unique zero on the upper semicircle

Theorem 5.5 (Existence, uniqueness, and simplicity)**.** *Let $a \geq 1$ be an integer and let $b > 0$ be real. There is exactly one $\theta_{a,b} \in (0, \pi)$ such that $\operatorname{Im} F_{a,b}(e^{\mathrm{i}\theta_{a,b}}) = 0$. It satisfies*

$$0 < \theta_{a,b} < \frac{\pi}{2}. \tag{5.8}$$

The imaginary part is positive for $0 < \theta < \theta_{a,b}$ and negative for $\theta_{a,b} < \theta < \pi$. At its zero the angular derivative is strictly negative. Consequently,

$$g_{a,b} < 0. \tag{5.9}$$

Proof. Suppress the indices in J and κ. The kernel $u \mapsto (1+u^2)^{-1}$ is strictly decreasing, so Lemma 5.4 gives $J(0) < 0$.

We next show that the limit of $J(c)$ as $c \uparrow 1$ is positive, possibly infinite. The negative density is supported in $(0, r)$, away from the singular endpoint, and its contribution has a finite limit. On $(r, 1)$ monotone convergence applies. Expanding $(1-u)^{-2}$, treating these two parts separately, and using (5.3) yields

$$\lim_{c\uparrow 1} J(c) = \int_0^1 \frac{\kappa(u)}{(1-u)^2}\,\mathrm{d}u = \sum_{n\geq 0} \frac{H_n^{(b)}}{(n+1)^{a-1}} > 0. \tag{5.10}$$

Every summand from $n = 1$ onward is positive. Thus continuity produces at least one zero $c_0 \in (0, 1)$.

For uniqueness, write $K_c(u) = (1 - 2cu + u^2)^{-1}$. If $c' > c$, then

$$\frac{\mathrm{d}}{\mathrm{d}u}\frac{K_{c'}(u)}{K_c(u)} = \frac{2(c'-c)(1-u^2)}{(1-2c'u+u^2)^2} > 0. \tag{5.11}$$

At a zero $J(c) = 0$, the signed density $K_c\kappa$ still has one sign change and zero mass. Applying Lemma 5.4 to the ratio in (5.11) gives $J(c') > 0$ for every $c' > c$. The same argument with a smaller parameter gives $J(c') < 0$ for $c' < c$. There can be only one zero in $(-1, 1)$, with the indicated signs.

Finally,

$$J'(c_0) = \int_0^1 K_{c_0}(u)\kappa(u)\frac{2u}{1-2c_0u+u^2}\,\mathrm{d}u > 0,$$

because the last factor has derivative $2(1-u^2)/(1-2c_0u+u^2)^2 > 0$. Equation (5.7) therefore has derivative $-\sin^2\theta_{a,b}\,J'(c_0) < 0$ at $\theta_{a,b} = \arccos c_0$. Since $c_0 > 0$, this zero is below $\pi/2$. The Gaussian inequality is its value at $\theta = \pi/2$. □

Remark 5.6 (The first zero is elementary). The identity $F_{1,1}(z) = \frac{1}{2}\log^2(1-z)$ gives

$$\operatorname{Im} F_{1,1}(e^{\mathrm{i}\theta}) = \log\left(2\sin\frac{\theta}{2}\right)\frac{\theta-\pi}{2}.$$

Hence $\theta_{1,1} = \pi/3$ and $g_{1,1} = -\pi\log 2/8$. This example also checks the direction of the sign change in Theorem 5.5.

5.3 A four-term asymptotic for the zero

Put $\delta_{a,b} = \pi/2 - \theta_{a,b}$. The first surviving Fourier mode at $\pi/2$ is the $n = 3$ term, whereas the $n = 2$ term controls the derivative. Their ratio suggests $(2/3)^a$; the theorem below identifies both the next Fourier corrections and the first nonlinear correction.

Theorem 5.7 (Uniform zero asymptotic)**.** *For integer $a \to \infty$, uniformly for real $b \geq 1$, set $A = H_2^{(b)}$, $B = H_3^{(b)}$, and $C = H_4^{(b)}$. Then*

$$\delta_{a,b} = \frac{A}{2}\left(\frac{2}{3}\right)^a - \frac{C}{2}\left(\frac{2}{5}\right)^a + AB\left(\frac{1}{3}\right)^a - \frac{23A^3}{48}\left(\frac{8}{27}\right)^a + O\left(\left(\frac{2}{7}\right)^a\right). \tag{5.12}$$

The implied constant and the threshold for a can be chosen independently of $b \geq 1$.

Proof. For sufficiently large a, the boundary series and its derivative converge absolutely. Normalize the imaginary part by multiplying by 2^a, and write $\theta = \pi/2 - \delta$. The modes $n = 2, 3, 4, 5$ give

$$\begin{aligned} 0 = {} & \sin(2\delta) - Aq^a\cos(3\delta) - Bs^a\sin(4\delta) + Cr^a\cos(5\delta) \\ & + O\big(\delta\, 3^{-a} + (2/7)^a\big), \qquad q = \frac{2}{3}, \quad s = \frac{1}{2}, \quad r = \frac{2}{5}. \end{aligned} \tag{5.13}$$

The $n = 6$ mode is $O(\delta 3^{-a})$. For $n \geq 7$, the bound $H_{n-1}^{(b)} \leq 1 + \log n$ gives a uniform tail estimate: for $a \geq 3$,

$$\sum_{n\geq 7}(1+\log n)(2/n)^a \leq (2/7)^a \sum_{n\geq 7}(1+\log n)(7/n)^3.$$

The constant is finite and independent of b.

Since $1 \leq A \leq 3/2$ and B, C are bounded uniformly in b, the normalized expression is negative at $\delta = 0$ and positive at $\delta = q^a$ for all sufficiently large a. Theorem 5.5 identifies the root in that interval. Thus $\delta = O(q^a)$ uniformly.

Taylor expansion of (5.13) now gives

$$0 = 2\delta - Aq^a - 4Bs^a\delta + Cr^a + \frac{9}{2}Aq^a\delta^2 - \frac{4}{3}\delta^3 + O((2/7)^a). \tag{5.14}$$

For example, the omitted terms have exponential bases bounded by q^5, sq^3, rq^2, and $q/3$, all smaller than $2/7$. Insert first $\delta_0 = Aq^a/2$. The linear perturbation $-4Bs^a\delta_0$ requires the correction $AB(qs)^a$, while the $n = 5$ mode requires $-Cr^a/2$. The cubic contribution at δ_0 is

$$\frac{9}{2}Aq^a\delta_0^2 - \frac{4}{3}\delta_0^3 = \frac{23}{24}A^3q^{3a},$$

so its compensating correction is $-23A^3q^{3a}/48$. Substitution of the full displayed approximation in (5.14) leaves a residual $O((2/7)^a)$; all cross-products of the corrections have smaller exponential bases. The derivative of the normalized expression with respect to δ is $2 + o(1)$, uniformly on this interval. The mean-value theorem therefore converts the residual estimate into the error in (5.12). □

a	b	$\theta_{a,b}$, approximately	$\delta_{a,b}$, approximately
8	1	1.541868140889	0.02892818590626
8	2	1.546656477391	0.02413984940401
8	4	1.550268639150	0.02052768764464
12	1	1.565028587360	0.005767739435093
12	2	1.565988233290	0.004808093504888
12	4	1.566708834682	0.004087492112621
20	1	1.570570790993	0.0002255358018452
20	2	1.570608378737	0.0001879480579893
20	4	1.570636570306	0.0001597564888279

These entries are numerical illustrations from a 300-term Fourier sum, not certified root intervals. Existence, uniqueness, and the asymptotic estimate do not depend on the table. At $a = 20, 40, 80$ and $b = 1, 2, 8$, the computed next normalized remainder approaches $H_6^{(b)}/2$, as suggested by the $n = 7$ mode. A systematic expansion beyond the proved four terms is a separate research problem.

5.4 Certified Euler evaluation

5.4.1 A one-sided rational enclosure

Define the difference operator with the sign convention

$$(Df)(n) = f(n) - f(n+1).$$

This is the negative of the usual forward difference; it makes the Euler transformation particularly simple. Let

$$f_{a,b}(n) = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \qquad d_k = D^k f_{a,b}(0), \qquad E_N(a,b) = \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}}. \tag{5.15}$$

Theorem 5.8 (Signed Euler certificate)**.** *For positive integers a, b,*

$$d_0 = 0, \qquad d_k < 0 \quad (k \geq 1), \qquad g_{a,b} = \sum_{k\geq 0} \frac{d_k}{2^{k+1}}. \tag{5.16}$$

For every $N \geq 1$,

$$E_N(a,b) - C_b 2^{-N} \leq g_{a,b} \leq E_N(a,b), \qquad C_b = \begin{cases} 1, & b = 1, \\ b/(b-1), & b \geq 2. \end{cases} \tag{5.17}$$

The approximants decrease strictly from $E_1 = 0$ to $g_{a,b}$.

Proof. The moment representation gives

$$d_k = \int_0^1 (1-u^2)^k \kappa_{a,b}(u)\, \mathrm{d}u. \tag{5.18}$$

For $k = 0$ this is zero mass. For $k \geq 1$ the weight is strictly decreasing, so Lemma 5.4 gives $d_k < 0$.

We bound either signed mass of the density. When $b = 1$,

$$\kappa_{1,1}(u) = \log \frac{u}{1-u}, \qquad \int_0^1 (\kappa_{1,1})_+ \,\mathrm{d}u = \log 2 < 1.$$

When $b \geq 2$, the positive term of (5.1) is at most $\zeta(b)$, so the positive mass is at most $\zeta(b) < 1 + 1/(b-1) = C_b$. Zero mass makes the negative mass equal to the positive mass. The L^1 contraction used in Theorem 5.2, together with preservation of zero mass, gives the same bound for every a. Since $0 \leq (1-u^2)^k \leq 1$, we obtain $|d_k| \leq C_b$.

The series in (5.16) is now absolutely convergent. Summing its geometric kernel gives

$$\sum_{k\geq 0} \frac{d_k}{2^{k+1}} = \int_0^1 \frac{\kappa_{a,b}(u)}{1+u^2}\,\mathrm{d}u = g_{a,b},$$

where the last equality is (5.6) at $z = \mathrm{i}$. All tail terms are negative, and their total absolute value is at most $C_b \sum_{k\geq N} 2^{-k-1} = C_b 2^{-N}$. This proves the certificate and strict monotonicity. □

The theorem is stronger than applying a general-purpose extrapolation rule and observing stability. Every endpoint in (5.17) is rational. For $N = 400$, the interval width is at most $2 \cdot 2^{-400} < 8 \cdot 10^{-121}$. The accompanying files round the enclosures outward to convenient decimal strings; that display rounding is separate from the internal rational width.

5.4.2 An exact linear-work finite formula

Computing a full triangular difference table uses quadratic work. A finite rearrangement avoids it.

Proposition 5.9 (Binomial-tail weights)**.** *For $N \geq 1$,*

$$E_N(a,b) = 2^{-N} \sum_{n=0}^{N-1} (-1)^n \frac{H_{2n}^{(b)}}{(2n+1)^a} \sum_{j=n+1}^{N} \binom{N}{j}. \tag{5.19}$$

At fixed a, b, the right side can be formed with $O(N)$ exact rational or integer updates after choosing N. All denominators divide

$$2^N \operatorname{lcm}(1, 2, \ldots, 2N)^{a+b}. \tag{5.20}$$

In particular, the number of bit operations is polynomial in the requested precision at fixed indices.

Proof. Expand $D^k f(0) = \sum_{n=0}^{k} (-1)^n \binom{k}{n} f(n)$ and interchange the two finite sums. The required weight identity is

$$\sum_{k=n}^{N-1} 2^{N-k-1} \binom{k}{n} = \sum_{j=n+1}^{N} \binom{N}{j}.$$

It follows by induction on N using Pascal's recurrence, or by comparing binomial-tail generating functions. This proves (5.19).

The binomial coefficients are generated successively by $\binom{N}{n} = \binom{N}{n-1}(N-n+1)/n$, and the cumulative tail is updated by subtracting one coefficient. Each harmonic number is updated by two rational summands. Thus only linearly many updates are needed. Each summand's denominator divides the indicated least-common-multiple power, and the additional factor is 2^N. Finally, the elementary bound $\operatorname{lcm}(1, \ldots, 2N) \leq (2N)!$ gives bit length $O(N + (a+b)N \log N)$ for a common denominator; the numerator has a comparable polynomial bound. Since $N = O(d)$ suffices for d decimal places by (5.17), even elementary integer arithmetic proves polynomial bit complexity. No optimal multiplication or sharp least-common-multiple estimate is required. □

5.4.3 Independent depth-one interval arithmetic

The right sides of the reductions are evaluated using rational enclosures separate from the Gaussian evaluator. For $s \geq 1$, the sequences $(2n+1)^{-s}$ and $(n+1)^{-s}$ are moments of positive densities, so all their D-differences are nonnegative. Their Euler sums give lower bounds for $\beta(s)$ and $\eta(s)$, with tails at most 2^{-N}. For integer $s \geq 2$, divide the latter interval by $1 - 2^{1-s}$ to enclose $\zeta(s)$.

For π, the implementation uses Machin's formula

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and the alternating-series bounds for each rational arctangent series. The identity follows directly by tangent addition and the fact that the angle lies in the first quadrant. For $\log 2$, it uses

$$0 < \log 2 - 2 \sum_{n=0}^{N-1} \frac{1}{(2n+1)3^{2n+1}} \leq \frac{9}{4(2N+1)3^{2N+1}}. \tag{5.21}$$

The expansion is $2\operatorname{arctanh}(1/3)$; the bound replaces the remaining denominators by $2N+1$ and sums a geometric series.

Ordinary rational interval addition and multiplication then enclose every polynomial right side. An overlap check can detect an erroneous formula or implementation if the intervals separate. Overlap alone does not prove an identity; the proofs in Sections 4.6 and 4.4 do that.

5.4.4 Affine harmonic sequences as signed moments

The signed-measure formulation also certifies sequences whose harmonic and denominator steps differ. The Cayley continuation supplies the integer-order case; its Gamma-density argument extends without change to real outer order $a \geq 1$. It complements the sharper one-sided bounds above.

Theorem 5.10 (Affine signed-moment bound)**.** *Let r, d be positive integers, $a \geq 1$ real and $b > 0$ real. Set*

$$A_k = \frac{H_{rk}^{(b)}}{(dk+1)^a}, \qquad m_1 = 1, \qquad m_a = \frac{(a-1)^{a-1}e^{-(a-1)}}{\Gamma(a)} \quad (a > 1).$$

There is a finite signed measure ν on $[0,1]$, with no atom at one, such that $A_k = \int x^k \, d\nu(x)$ and

$$\|\nu\|_{\mathrm{TV}} \leq \frac{2r}{d} m_a b \zeta(b+1).$$

For $b > 1$ one also has $\|\nu\|_{\mathrm{TV}} \leq 2\zeta(b)$. Consequently the ordinary alternating series converges, and its N-term Euler transform E_N satisfies

$$\left| \sum_{k \geq 0} (-1)^k A_k - E_N \right| \leq C 2^{-N},$$

where C may be either applicable bound. For integer b, valid rational choices are $C = 10r/(3d)$ at $b = 1$ and $C = 10/3$ at $b \geq 2$.

Proof. Let Y have Gamma density $y^{a-1}e^{-y}/\Gamma(a)$, and let μ be the law of $X = e^{-dY}$. Then $\int x^k \, d\mu = (dk+1)^{-a}$. In the logarithmic coordinate $s = -\log x$, its density is

$$f(s) = \frac{s^{a-1}e^{-s/d}}{d^a \Gamma(a)} \mathbf{1}_{s>0}.$$

For every real $a \geq 1$ its maximum is m_a/d. Extended by zero to the negative axis, the total variation of its distributional derivative is $2m_a/d$: the rise and fall give this for $a > 1$, while at $a = 1$ the jump at zero and the decreasing exponential each contribute $1/d$. The L^1 distance from a bounded-variation density to its translate by $h \geq 0$ is at most h times this derivative variation. Indeed, integrate $f(s-h) - f(s) = -\int_{s-h}^{s} f'(v) \, dv$ and use Fubini; smooth convolution proves the same bound for the jump case. Thus, writing $D_\rho \mu$ for pushforward by $x \mapsto \rho x$,

$$\|\mu - D_{e^{-rt}}\mu\|_{\mathrm{TV}} \leq \min\left(2, \frac{2m_a r}{d} t\right).$$

The generalized harmonic integral gives the measure

$$\nu = \frac{1}{\Gamma(b)} \int_0^\infty \frac{t^{b-1}}{e^t - 1} (\mu - D_{e^{-rt}}\mu) \, dt.$$

This integral converges in total variation: the translation bound supplies an additional factor t at zero, and exponential decay controls infinity. Taking moments gives A_k, since the moments of the measure difference are $(dk+1)^{-a}(1-e^{-rkt})$. Its total-variation bounds follow from

$$\frac{1}{\Gamma(b)}\int_0^\infty \frac{t^b}{e^t-1}\,dt = b\zeta(b+1), \qquad \frac{1}{\Gamma(b)}\int_0^\infty \frac{t^{b-1}}{e^t-1}\,dt = \zeta(b) \quad (b>1).$$

Each constituent has no atom at one, and variation domination passes that property to ν. The finite alternating kernel is bounded by two and converges for $x<1$ to $(1+x)^{-1}$, so dominated convergence proves ordinary convergence. The exact remainder kernel in the Euler expansion is $2^{-N}(1-x)^N/(1+x)$, bounded by 2^{-N}; integrating against ν proves the stated error. Finally $m_a \le 1$, by bounding the maximum of the unit-scale Gamma density, and $\zeta(2) < 5/3$. For the displayed rational bound at $b=1$ one may also use only the integer a needed by the rational evaluator: then $m_a \le 1$ follows directly from $e^{a-1} \ge (a-1)^{a-1}/(a-1)!$. For the real-$a$ inequality, write the Gamma density as the convolution of an exponential density and a Gamma$(a-1,1)$ probability density when $a>1$; its supremum is at most one. For integer $b \ge 2$, $\zeta(b) \le \zeta(2)$ gives $10/3$. □

The real-order theorem does not make its centers rational at noninteger indices. The finite rational implementation uses integer a, b, r, d and binomial-tail Euler weights. No complete monotonicity of A_k is assumed: the representing measure is signed and has total mass zero, since $A_0 = 0$.

5.5 An exact kernel and the sharp Euler error

5.5.1 Separating the polynomial from the exponentially small part

For this section a, b are positive integers, $w = a+b$, and $T = -\log u$.

Theorem 5.11 (Explicit density)**.** *Define the polynomial*

$$P_{a,b}(T) = -\frac{T^{w-1}}{(w-1)!} + \sum_{\mu=b+1}^{w-1} (-1)^{\mu-b-1}\binom{\mu-1}{b-1}\zeta(\mu)\frac{T^{w-1-\mu}}{(w-1-\mu)!}. \tag{5.22}$$

Then, for $T>0$,

$$\kappa_{a,b}(e^{-T}) = P_{a,b}(T) + (-1)^{a-1}\sum_{j=0}^{b-1}\binom{w-j-2}{a-1}\frac{T^j}{j!}\operatorname{Li}_{w-1-j}(e^{-T}). \tag{5.23}$$

Consequently,

$$\kappa_{a,b}(u) = P_{a,b}(-\log u) + O\Big(u(1+|\log u|^{b-1})\Big) \quad (u \downarrow 0). \tag{5.24}$$

Proof. For $a=1$, expand $(e^v-1)^{-1} = \sum_{n\ge1} e^{-nv}$ in (5.1). The elementary incomplete-gamma integral for integer b gives

$$\frac{1}{(b-1)!}\int_T^\infty \frac{v^{b-1}}{e^v-1}\,\mathrm{d}v = \sum_{j=0}^{b-1}\frac{T^j}{j!}\operatorname{Li}_{b-j}(e^{-T}),$$

which is exactly the asserted base case.

Write $K_{a,b}(T) = \kappa_{a,b}(e^{-T})$. The recursion is $K_{a,b}(T) = \int_0^T K_{a-1,b}(s)\,\mathrm{d}s$. Differentiate the proposed right side of (5.23), using $\frac{\mathrm{d}}{\mathrm{d}T}\operatorname{Li}_s(e^{-T}) = -\operatorname{Li}_{s-1}(e^{-T})$ and Pascal's identity. The polylogarithmic coefficients reduce to those for $(a-1,b)$; the derivative of $P_{a,b}$ is $P_{a-1,b}$. When $a \geq 2$, the constant terms at $T = 0$ cancel because

$$\binom{w-2}{b-1} = \binom{w-2}{a-1}$$

and their signs are opposite. The asserted formula therefore has the correct derivative and initial value. This proves it by induction. Finally each $\operatorname{Li}_s(e^{-T})$ in the finite sum is $O(e^{-T})$ as $T \to \infty$, proving (5.24). □

5.5.2 The complete logarithmic error polynomial

The exact error is

$$E_N(a,b) - g_{a,b} = -2^{-N} \int_0^1 \kappa_{a,b}(u) \frac{(1-u^2)^N}{1+u^2}\,\mathrm{d}u. \tag{5.25}$$

This follows by summing only the first N terms of the geometric kernel in the proof of Theorem 5.8. Its localization near $u = 0$ explains why the polynomial in Theorem 5.11 determines the error.

Theorem 5.12 (Sharp error with all logarithmic terms)**.** *Define the polynomial of degree $w-1$*

$$\begin{aligned} Q_{a,b}(L) &= -\int_0^\infty e^{-t^2} P_{a,b}(L - \log t)\,\mathrm{d}t \\ &= -\sum_{j=0}^{w-1} \frac{(-1)^j P_{a,b}^{(j)}(L)}{j!\,2^{j+1}} \Gamma^{(j)}(1/2). \end{aligned} \tag{5.26}$$

For fixed a, b, as $N \to \infty$,

$$\begin{aligned} E_N(a,b) - g_{a,b} = 2^{-N}\Big\{ & N^{-1/2} Q_{a,b}\Big(\frac{1}{2}\log N\Big) \\ & + O\Big(N^{-1}(1+\log^{b-1} N) + N^{-3/2}(1+\log^{w-1} N)\Big)\Big\}. \end{aligned} \tag{5.27}$$

In particular,

$$E_N(a,b) - g_{a,b} \sim \frac{\sqrt{\pi}}{2^w (w-1)!} \frac{2^{-N}(\log N)^{w-1}}{\sqrt{N}}. \tag{5.28}$$

The leading coefficient depends only on the total weight, not on its split into a and b.

Proof. Choose a fixed $\varepsilon \in (0, 1/2)$. The part of (5.25) with $u \geq \varepsilon$ is bounded by $2^{-N}(1-\varepsilon^2)^N \|\kappa_{a,b}\|_1$ and is negligible at the displayed scales. In the remaining part substitute $u = t/\sqrt{N}$ and use (5.24). For $t \leq \varepsilon\sqrt{N}$,

$$\left| \frac{(1-t^2/N)^N}{1+t^2/N} - e^{-t^2} \right| \leq \frac{C}{N} e^{-t^2}(t^2 + t^4).$$

Indeed, $0 \leq -\log(1-x) - x \leq Cx^2$ for $0 \leq x \leq \varepsilon^2$, and the denominator changes the expression by at most t^2/N times the Gaussian bound.

The polynomial contribution, including the Jacobian $N^{-1/2}$, is therefore $N^{-1/2} Q_{a,b}(\frac{1}{2}\log N)$ up to $O(N^{-3/2}(1+\log^{w-1} N))$. Extending the t integral to infinity changes it only by an exponentially

small amount. The remainder in (5.24) contributes at most $O(N^{-1}(1+\log^{b-1} N))$; integrability of $te^{-t^2}|\log t|^j$ at both endpoints justifies this estimate.

To obtain the second line of (5.26), expand the polynomial and use

$$\int_0^\infty e^{-t^2}(\log t)^j \,\mathrm{d}t = 2^{-j-1}\Gamma^{(j)}(1/2),$$

which follows by differentiating $\frac{1}{2}\Gamma(s/2)$ at $s=1$. Finally the leading term of $-P_{a,b}$ is $T^{w-1}/(w-1)!$. Its Gaussian integral is $\sqrt{\pi}/2$, and $L=(\log N)/2$. This gives (5.28); both error terms in (5.27) are smaller than that leading term. □

For example,

$$Q_{1,1}(L) = \frac{\sqrt{\pi}}{2}\left(L + \frac{\gamma + 2\log 2}{2}\right), \tag{5.29}$$

where γ is Euler's constant. Keeping the lower logarithmic terms matters at practical truncation lengths. For $(a,b)=(1,1)$, the following ratios use the exact rational proxy $E_N - E_{2N}$; its distance from the true error is at most 2^{-2N}.

N	Ratio to the leading term	Ratio to the full $Q_{1,1}$ approximation
64	1.42496	0.967964
128	1.37730	0.980510
256	1.33791	0.988050
512	1.30501	0.992593

The asymptotic expansion is not substituted for the certified bound in the implementation. Its role is to explain the observed convergence and suggest more efficient future remainder enclosures.

Proposition 5.13 (Euler certificate for the odd-denominator harmonic family)**.** *For integer $p \geq 1$, let $f(n) = H_n/(2n+1)^p$, let $d_k = D^k f(0)$, and let $E_N^S = \sum_{k<N} d_k/2^{k+1}$. Then*

$$E_N^S - \frac{N+1}{3^p 2^N} \leq S_p \leq E_N^S, \qquad S_p = \sum_{n\geq 0} \frac{(-1)^n H_n}{(2n+1)^p}. \tag{5.30}$$

Proof. Use $H_n = \int_0^1 (1-t^n)/(1-t)\,\mathrm{d}t$ and the Mellin integral for $(2n+1)^{-p}$. Finite differencing gives

$$d_k = \frac{1}{\Gamma(p)}\int_0^1 (-\log u)^{p-1}\int_0^1 \frac{(1-u^2)^k - (1-u^2t)^k}{1-t}\,\mathrm{d}t\,\mathrm{d}u.$$

Thus $d_0 = 0$ and $d_k < 0$ for $k \geq 1$. The numerator has absolute value at most $ku^2(1-t)$, so $|d_k| \leq k/3^p$. Summing the Euler kernels yields

$$\sum_{k\geq 0}\frac{d_k}{2^{k+1}} = -\frac{1}{\Gamma(p)}\int_0^1 \frac{(-\log u)^{p-1}\log(1+u^2)}{1+u^2}\,\mathrm{d}u = S_p.$$

The final equality follows from the harmonic generating function, first with an Abel factor. The bound on d_k justifies summing and integrating, and $\sum_{k\geq N} k/2^{k+1} = (N+1)2^{-N}$ proves (5.30). □

At $N=400$, combine Proposition 5.13, Theorem 5.8, and the independent depth-one intervals. The exact rational computation encloses the difference between the two sides of (4.41) in

$$[-10^{-118},\, 10^{-118}]. \tag{5.31}$$

Both sides begin

$$-0.010493456297335043453356751006201751963510618877 32\ldots.$$

This is certified proximity, not proof of equality. The file `s4_interval_certificate.json` records the outward-rounded enclosures and the tail bound used.

5.5.3 The precise relation system being tested

We now specify a finite-dimensional linear question; it must not be confused with completeness of all polylogarithm relations. Let

$$X^{r,s}_{a,b} = \operatorname{Im} D_{a,b}(\mathrm{i}^r, \mathrm{i}^s), \qquad r,s \in \mathbb{Z}/4\mathbb{Z}, \qquad a+b=w.$$

Set $X^{r,s}=0$ when both r,s are even, and impose conjugation $X^{r,s}=-X^{-r,-s}$. Choose the lexicographically smaller of each conjugate pair. Remove the divergent imaginary coordinate $X^{0,1}_{1,w-1}$; its conjugate is already identified. This leaves $6w-7$ formal coordinates.

For every $p+q=w$ and colors r,s, take the imaginary parts of the finite product stuffle

$$D_{p,q}(x,y)+D_{q,p}(y,x)=\operatorname{Li}_p(x)\operatorname{Li}_q(y)-\operatorname{Li}_w(xy) \tag{5.32}$$

and product shuffle

$$\begin{aligned}&\sum_{j=0}^{p-1}\binom{q+j-1}{j}D_{q+j,p-j}(y,x/y)\\ &\quad+\sum_{j=0}^{q-1}\binom{p+j-1}{j}D_{p+j,q-j}(x,y/x)=\operatorname{Li}_p(x)\operatorname{Li}_q(y),\end{aligned} \tag{5.33}$$

where $x=\mathrm{i}^r$, $y=\mathrm{i}^s$. If exactly one factor is $\operatorname{Li}_1(1)$, retain the stuffle-minus-shuffle cancellation row: the common divergent double coordinate cancels, leaving the right side $-\operatorname{Im}\operatorname{Li}_w(xy)$. Such rows follow by subtracting before taking the regularized boundary limit. Zero rows are discarded; duplicate nonzero rows need not be discarded. This construction defines a rational coefficient matrix A_w and its depth-one right side. It is precisely the model implemented in `code/full_ds.py`.

These are only linear depth-two product rows in a fixed weight. They do not include every relation obtained from distribution, argument transformations, higher-depth elimination, or the full algebraic consequences of standard relations. Zhao's treatment [69] provides broader context for these distinctions.

5.5.4 An exact obstruction at weight five

Let $u_{4,1}=\operatorname{Im} D_{4,1}(\mathrm{i},-1)$. Splitting the inner harmonic sum into even and odd indices proves

$$S_4 = g_{4,1}+u_{4,1}. \tag{5.34}$$

Indeed, for an odd outer index $2n+1$, the sum of the two inner coefficients is $H_{2n}+\sum_{m=1}^{2n}(-1)^m/m = H_n$; even outer indices have zero imaginary part. Consequently, the left-hand coefficient vector needed for (4.31) is

$$T = 3g_{4,1}+3g_{3,2}+9g_{2,3}+7u_{4,1}. \tag{5.35}$$

Proposition 5.14 (A finite linear-module obstruction)**.** *The coefficient vector T in (5.35) is not in the rational row span of the explicitly defined weight-five matrix A_5. Therefore no rational linear combination of just those weight-five depth-two rows proves the S_4 identity.*

Proof. The matrix has 92 rows and 23 columns. In the canonical coordinate order above, define a vector v by the following nonzero entries, setting every other entry to zero:

(a,b,r,s)	v	(a,b,r,s)	v
$(1,4,1,0)$	$-1/12$	$(2,3,0,1)$	$1/24$
$(2,3,1,0)$	$1/8$	$(2,3,1,3)$	$-1/24$
$(3,2,0,1)$	$-1/8$	$(3,2,1,0)$	$-1/24$
$(3,2,1,3)$	$-1/24$	$(4,1,0,1)$	$1/12$

Substitution in the binomial rows (5.32)–(5.33) gives $A_5 v = 0$, while $T \cdot v = 1$. These are finite rational equalities; the complete integer matrix, coordinate order, target, and witness are included in `s4_linear_module_obstruction.json` and replayed by `verify.py`. Every row-span vector annihilates v, whereas T does not. This proves the assertion. □

The proposition concerns only the specified depth-two product rows. Theorem 4.8 adds a convergent octahedral duality and lifted relations through higher depths, outside this row vocabulary. The two statements are therefore compatible: the obstruction identifies the additional information an eventual successful proof had to supply.

5.5.5 Exact ranks from two polynomial reflections

Retain exactly the coefficient matrix A_w defined above, including the stuffle-minus-shuffle boundary rows and excluding the one divergent imaginary coordinate. No distribution or higher-depth relations are added. Let $A_w^{\neg g}$ denote the submatrix with the Gaussian columns deleted. The following theorem replaces the former rank conjecture.

Theorem 5.15 (All-weight ranks)**.** *For every odd $w \geq 3$,*

$$\operatorname{rank} A_w = 5w - 6, \qquad \operatorname{rank} A_w^{\neg g} = \frac{9w - 11}{2}. \tag{5.36}$$

For every even $w \geq 2$,

$$\operatorname{rank} A_w = 6w - 7, \qquad \operatorname{rank} A_w^{\neg g} = 5w - 6. \tag{5.37}$$

In particular, the Gaussian-supported row space has dimension $(w-1)/2$ in odd weight and $w-1$ in even weight. In odd weight it is exactly the span of the same-point shuffle rows.

Proof. Put $N = w-2$. Use six degree-N polynomials A, B, C, D, E, F to encode a homogeneous null vector. Their respective colors are $(0,1), (1,0), (1,1), (1,2), (1,3), (2,1)$. Coefficients of $X^{a-1}Y^{b-1}$ have $a + b = w$. Complete the missing coefficient of A by $v^{0,1}_{1,w-1} = -v^{1,0}_{w-1,1}$. This satisfies the missing stuffle row; the missing shuffle row follows from its retained difference with stuffle. The omitted coordinate occurs in no other row: outer index one in a shuffle sum forces one of its factors to be $\mathrm{Li}_1(1)$. Thus completion and restriction identify the matrix kernel with the full homogeneous polynomial solution space.

Writing $P^{\mathsf{T}}(X,Y) = P(Y,X)$, the stuffle equations are

$$A = -B^{\mathsf{T}}, \quad C = -C^{\mathsf{T}}, \quad D = -F^{\mathsf{T}}, \quad E = E^{\mathsf{T}}. \tag{5.38}$$

The shuffle equations are

$$\begin{aligned} E(X+Y,X)+A(X+Y,Y)&=0,\\ B(X+Y,X)+B(X+Y,Y)&=0,\\ C(X+Y,Y)-F(X+Y,X)&=0,\\ D(X+Y,Y)-D(X+Y,X)&=0. \end{aligned} \tag{5.39}$$

For general colors (r,s) the shuffle polynomial is $P_{s,r-s}(X+Y,X)+P_{r,s-r}(X+Y,Y)$; its coefficient of $X^{p-1}Y^{q-1}$ is the stated binomial row. The four equations above come from product colors $(0,1),(1,1),(1,2),(1,3)$. Swapping the factors, conjugating, and discarding even-even imaginary rows exhaust the other cases.

The first two equations give

$$E(X,Y)=B(X-Y,X), \qquad B(X,Y)+B(X,X-Y)=0.$$

With $\varepsilon=(-1)^N$, this implies $E(Y,X)=-\varepsilon E(X,Y)$. Since $E=E^{\mathsf{T}}$, the whole A,B,E block vanishes if N is even. If $N=2m-1$ is odd, its free polynomial B is precisely an anti-invariant under $Y\mapsto X-Y$, with basis

$$B_j=X^{N-2j-1}(2Y-X)^{2j+1}, \qquad 0\le j<m. \tag{5.40}$$

Indeed, $U=X, V=2Y-X$ changes this reflection into $V\mapsto -V$.

The remaining equations give

$$F(X,Y)=C(X,X-Y), \quad D(X,Y)=-C(Y,Y-X), \quad C=-C^{\mathsf{T}}.$$

They imply $D(X,X-Y)=-\varepsilon D(X,Y)$, while the last shuffle asks for equality. Hence the C,D,F block is zero when N is even. When $N=2m-1$, it has the basis determined by

$$C_j=X^jY^{N-j}-X^{N-j}Y^j, \qquad 0\le j<m. \tag{5.41}$$

Conversely, the displayed substitutions satisfy every homogeneous equation. The B_j and C_j are independent and are retained in full after deleting the missing coefficient of A. They therefore give an integral $\mathbb{Q}$-basis of the full nullspace.

For odd $w=2m+1$, the nullity is $2m=w-1$, proving $\operatorname{rank} A_w=(6w-7)-(w-1)=5w-6$. A null vector of $A_w^{\neg g}$ extends by zero on the Gaussian columns; this is exactly the C block, of dimension m. Its $5w-6$ columns thus have rank $5w-6-m=(9w-11)/2$. For even w all columns of A_w are independent, and so are all columns of its submatrix.

Projection of the row space onto the non-Gaussian columns gives the stated supported-row dimensions by taking the difference of the two ranks. In odd weight $w=2m+1$, the same-point shuffle coefficient on $g_{a,w-a}$ is $\binom{a-1}{p-1}+\binom{a-1}{w-p-1}$, $1\le p\le m$. Its first m columns form a triangular Pascal matrix with diagonal one. These m independent Gaussian-supported rows therefore exhaust that space. □

Exact membership and the harmonic-sum obstruction. Let K_w have the preceding integer null vectors as columns. A rational coefficient row t is in the specified row space exactly when $tK_w=0$. Otherwise any column with nonzero pairing is a separating certificate. The basis is constructible in $O(w^2)$ integer arithmetic operations and has $O(w)$-bit coefficients; coefficient recurrences and bit bounds are given in the accompanying article.

For odd $w = 2m + 1$, choose $B = 0$, $C = Y^{2m-1} - X^{2m-1}$. It vanishes on every Gaussian column and has $D(1, 0) = 1$. Since $S_{2m} = g_{2m,1} + \operatorname{Im} D_{2m,1}(\mathrm{i}, -1)$, it pairs to one with S_{2m}. Thus no rational linear combination of these fixed-weight product rows reduces S_{2m} to Gaussian doubles and the associated single-value right sides. In the formal quotient, adjoining it to the Gaussian span increases the dimension from m to $m + 1$. For the weight-five target above, this witness gives $T \cdot v = 7$. This is a uniform proof-system obstruction, not a numerical independence theorem and compatible with the already proved S_4 identity, whose certificate uses a larger vocabulary.

Restoring the right sides: an explicit compiler

Set $\ell_k(r) = \operatorname{Li}_k(\mathrm{i}^r)$, with the formal regularization $\ell_1(0) = 0$, and define known degree-N polynomials

$$\begin{aligned} P_{r,s}(X, Y) &= \sum_{p+q=w} \operatorname{Im}(\ell_p(r)\ell_q(s))X^{p-1}Y^{q-1}, \\ H_N(X, Y) &= \sum_{j=0}^{N} X^j Y^{N-j}, \\ S_{01} = P_{0,1} - \beta(w)H_N, &\qquad S_{12} = P_{1,2} + \beta(w)H_N. \end{aligned}$$

The same six letters now encode the actual imaginary double values, with the artificial missing coefficient $[X^0Y^N]A = -g_{w-1,1} - \beta(w)$. Put

$$\begin{aligned} e(X, Y) &= P_{0,1}(Y, X - Y) - S_{01}(X, X - Y), \\ d(X, Y) &= S_{12}(X, Y) + P_{1,2}(X, Y - X), \\ b(X, Y) &= P_{1,1}(Y, X - Y), \\ h(X, Y) &= P_{1,3}(Y, Y - X) - e(Y, Y - X) + e(Y - X, Y). \end{aligned}$$

Theorem 5.16 (Even-weight polynomial solution)**.** *For even $w \geq 2$ the full solution is*

$$\begin{aligned} B(X, Y) &= \tfrac{1}{2}\big(b(X, Y) + h(X, Y)\big), \\ C(X, Y) &= \tfrac{1}{2}\big(P_{1,3}(X, -Y) + P_{1,1}(Y, X) - d(X - Y, -Y) + d(X - Y, X)\big), \\ A(X, Y) &= S_{01}(X, Y) - B(Y, X), \\ E(X, Y) &= B(X - Y, X) + e(X, Y), \\ F(X, Y) &= C(X, X - Y) - P_{1,2}(Y, X - Y), \\ D(X, Y) &= -C(Y, Y - X) + d(X, Y). \end{aligned}$$

Discarding the artificial coefficient gives explicit single-value reductions for every convergent imaginary double at fourth roots of unity.

Proof. The affine stuffle equations are $A + B^{\mathsf{T}} = S_{01}$, $C + C^{\mathsf{T}} = P_{1,1}$, $D + F^{\mathsf{T}} = S_{12}$, $E - E^{\mathsf{T}} = P_{1,3}$. The four shuffle right sides in (5.39) are now $P_{0,1}, P_{1,1}, P_{1,2}, P_{1,3}$, respectively. Elimination gives the last four displayed formulas and $B(X, Y) + B(X, X - Y) = b(X, Y)$. Even homogeneity and the E stuffle give the difference $B(X, Y) - B(X, X - Y) = h(X, Y)$, proving the formula for B. For C, the last shuffle is $D(u, u - v) - D(u, v) = P_{1,3}(v, u - v)$. Use $C(u - v, -v) = C(v - u, v) = P_{1,1}(v - u, v) - C(v, v - u)$ and then $(X, Y) = (v, v - u)$ to get the asserted formula. Actual values satisfy these equations, and the zero homogeneous kernel makes the solution unique. □

Two mixed leading-index families. Let $c_p = \operatorname{Im} D_{p,1}(\mathrm{i},\mathrm{i})$ and $e_p = \operatorname{Im} D_{p,1}(\mathrm{i},-\mathrm{i})$. For all $m \geq 1$,

$$c_{2m-1} = (m-1)\beta(2m) - \tfrac{1}{2}\log 2\,\beta(2m-1) - \sum_{k=1}^{m-1}(1-2^{1-2k})\zeta(2k)\beta(2m-2k), \tag{5.42}$$

$$e_{2m-1} = m\beta(2m) - \tfrac{1}{2}\log 2\,\beta(2m-1) - \sum_{k=1}^{m-1}\zeta(2k)\beta(2m-2k). \tag{5.43}$$

For a direct coefficient proof, evaluate $C(1,0)$ and $E(1,0)$ in the compiler. The resulting equations are

$$\begin{aligned} 2C(1,0) &= P_{1,3}(1,0) + P_{1,1}(0,1) - d(1,0) + d(1,1), \\ 2E(1,0) &= P_{1,1}(1,0) + P_{1,3}(1,0) + e(1,0) + e(0,1). \end{aligned}$$

In both lines the first two terms are $-\log 2\,\beta(w-1)$, and

$$\begin{aligned} d(1,1) - d(1,0) &= P_{1,2}(1,1) - P_{1,2}(1,-1) + N\beta(w), \\ e(1,0) + e(0,1) &= -P_{0,1}(1,1) + P_{0,1}(1,-1) + w\beta(w). \end{aligned}$$

Only even inner, respectively even outer, indices survive the differences. Using $\ell_{2k}(2) = -(1-2^{1-2k})\zeta(2k)$ and $\operatorname{Im}\ell_j(1) = \beta(j)$ proves both formulas.

Status and evidence. The rank theorem proves both individual ranks, rather than inferring them from their difference. The previous word "Equivalently" should therefore be read as "In particular." Independent matrix construction, exact integer kernel checks and modular lower-bound certificates cover every weight 2–41; 210 even-weight formulas through weight 12 satisfy all 816 product rows, and both leading-index formulas are checked through weight 16. These finite checks support the uniform proofs; they are not their substitute. The known general parity theorem already gives the relevant reduction phenomenon. The new contribution here is the explicit solution of this specified coefficient module. This row-system theorem asserts no numerical period independence. The S_4 identity is proved separately in Theorem 4.8. Historical conjectural receipts are retained as provenance rather than silently rewritten.

5.6 Real orders, the finite signed-measure threshold and its atoms

The recursive integer-order density is part of a continuous family. Regularized Hurwitz kernels give the exact finite signed-measure domain, including its two atomic edges. The construction below supplies the supercritical part of the later all-positive-order zero theorem. Positive Hausdorff generating functions supply a classical context [98]; here the signed and hypersingular measures are constructed explicitly with their separate integrability conditions. Write

$$A_n(a,b) = \frac{H_{n-1}^{(b)}}{n^a}, \qquad F_{a,b}(z) = \sum_{n\geq 1} A_n(a,b)z^n \quad (|z|<1).$$

We permit $a = 0$ to state the sharp endpoint. Throughout $b > 0$; write $w = a + b$ and $c_n = A_n(a,b)$ in these real-order extensions.

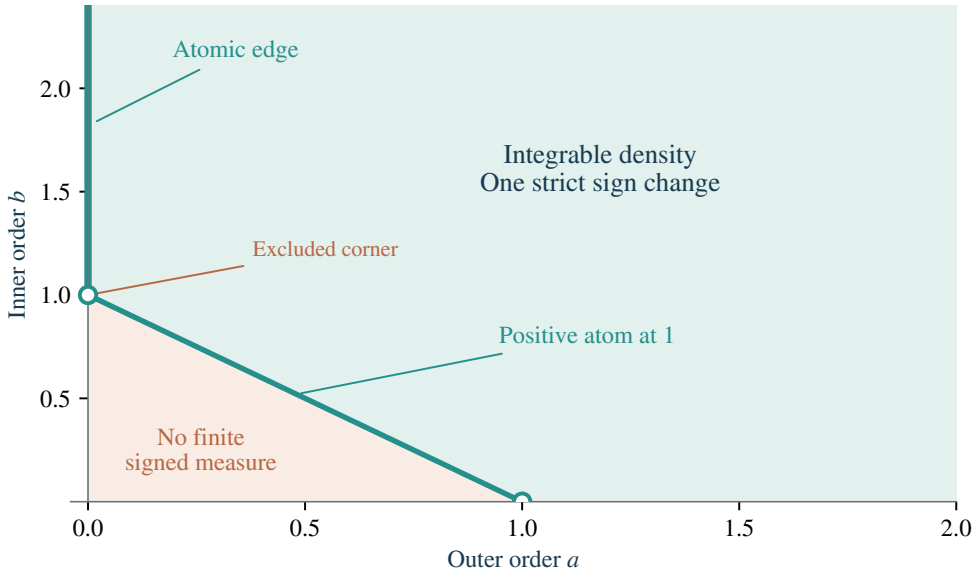


Figure 5.1: The exact finite signed-measure domain for $a \geq 0, b > 0$. The teal edges carry a positive atom at one. The point $(0, 1)$ is excluded, and the horizontal axis $b = 0$ is outside the stated domain. The entire positive quadrant has zero-free principal continuation by Theorem 5.26; finite signed-moment existence has the smaller domain shown here.

Theorem 5.17 (Exact finite-measure domain)**.** *For $a \geq 0$ and $b > 0$, a finite real signed measure $\nu_{a,b}$ on $[0, 1]$ satisfying*

$$\int_{[0,1]} u^{n-1}\, d\nu_{a,b}(u) = A_n(a, b) \qquad (n \geq 1) \tag{5.44}$$

exists if and only if

$$(a > 0 \text{ and } a + b \geq 1) \quad \text{or} \quad (a = 0 \text{ and } b > 1). \tag{5.45}$$

It is unique and has total mass zero. More precisely:

1. *If $a > 0$ and $a + b > 1$, it has an integrable density $\kappa_{a,b}(u)$ with exactly one strict sign change, from negative to positive as u increases, at a point $r_{a,b} \in (0, 1)$.*
2. *If $0 < b < 1$ and $a = 1 - b$, then $\nu_{a,b} = (1 - b)^{-1}\delta_1 + \kappa_{a,b}(u)\, du$, where $\kappa_{a,b}(u) < 0$ throughout $(0, 1)$.*
3. *If $a = 0$ and $b > 1$, then $\nu_{0,b} = \zeta(b)\delta_1 + \kappa_{0,b}(u)\, du$, where $\kappa_{0,b}(u) < 0$ throughout $(0, 1)$.*

In the two atomic cases all positive mass is at one and all negative mass is in $(0, 1)$. Outside (5.45), not even a finite signed measure with arbitrarily many sign changes can satisfy (5.44).

5.6.1 A regularized positive kernel

Set

$$q(t) = \frac{1}{1 - e^{-t}} - \frac{1}{t}, \qquad t > 0. \tag{5.46}$$

The elementary identities

$$q(t) = \frac{1}{2} + \frac{1}{2}\coth(t/2) - \frac{1}{t}, \qquad q'(t) = \frac{1}{t^2} - \frac{1}{4\sinh^2(t/2)} > 0$$

give $1/2 < q(t) < 1$, because $\sinh(t/2) > t/2$ and the endpoint limits are $1/2$ and 1. In particular q is positive, bounded and strictly increasing. Also $q(t) = 1/2 + O(t)$ as $t \downarrow 0$.

For later use put $k_v(T) = T^{v-1}/\Gamma(v)$ for $v > 0$. The positive convolution satisfies

$$C_{a,b}(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T-t)^{a-1} t^{b-1} q(t)\, dt = k_{a+b}(T)\mathbb{E}[q(TV)], \quad V \sim \mathrm{Beta}(b,a), \tag{5.47}$$

$$\tfrac{1}{2} k_{a+b}(T) < C_{a,b}(T) < k_{a+b}(T), \tag{5.48}$$

$$\int_0^\infty e^{-nT} C_{a,b}(T)\, dT = n^{-a}\left(\zeta(b,n) - \frac{n^{1-b}}{b-1}\right) \quad (b \neq 1). \tag{5.49}$$

At $b = 1$ the last expression is $n^{-a}(\log n - \psi(n))$. These identities follow by absolutely convergent convolution integrals.

The classical Hurwitz integral and one elementary subtraction give

$$\zeta(b,x) = \frac{x^{1-b}}{b-1} + \frac{1}{\Gamma(b)} \int_0^\infty e^{-xt} t^{b-1} q(t)\, dt, \qquad x > 0,\ b > 0,\ b \neq 1. \tag{5.50}$$

For $b > 1$ this is obtained by splitting $1/(1-e^{-t}) = 1/t + q(t)$ in the usual Gamma integral. The integral involving q is holomorphic for $\Re b > 0$, so the identity theorem extends the same formula to $0 < b < 1$. This is analytic continuation of an absolutely convergent regularized integral, not subtraction of two divergent real integrals. The standard starting integral and recurrence are DLMF 25.11.25 and 25.11.4 [93]. For $b = 1$ the corresponding formula is

$$\psi(x) = \log x - \int_0^\infty e^{-xt} q(t)\, dt, \tag{5.51}$$

which is DLMF 5.9.13 [95] and also follows by taking the finite part of (5.50) at $b = 1$.

For $0 < b < 1$, we shall use $\zeta(b) < 0$. Indeed the alternating Dirichlet series $\eta(b)$ is positive, while $\eta(b) = (1 - 2^{1-b})\zeta(b)$ and $1 - 2^{1-b} < 0$.

5.6.2 Explicit densities and their Laplace transforms

Put $T = -\log u$ and $K_{a,b}(T) = \kappa_{a,b}(e^{-T})$. For $a > 0$, $a + b > 1$ and $b \neq 1$, define

$$K_{a,b}(T) = \frac{\zeta(b) T^{a-1}}{\Gamma(a)} - \frac{T^{a+b-2}}{(b-1)\Gamma(a+b-1)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T-t)^{a-1} t^{b-1} q(t)\, dt. \tag{5.52}$$

When $b > 1$, an equivalent form is

$$K_{a,b}(T) = \frac{\zeta(b) T^{a-1}}{\Gamma(a)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T \frac{(T-t)^{a-1} t^{b-1}}{1 - e^{-t}}\, dt. \tag{5.53}$$

At $b = 1$, define

$$K_{a,1}(T) = \frac{T^{a-1}}{\Gamma(a)}\big(\gamma + \psi(a) - \log T\big) - \frac{1}{\Gamma(a)} \int_0^T (T-t)^{a-1} q(t)\, dt. \tag{5.54}$$

The Gamma factors and the term $\gamma + \psi(a)$ in this last formula are essential; in particular it reduces at $a = 1$ to $K_{1,1}(T) = -\log(e^T - 1)$, agreeing with the original signed kernel.

These functions are integrable against $e^{-T}\, dT$. At zero, (5.52) has only the integrable powers T^{a-1}, T^{a+b-2} and $O(T^{a+b-1})$; the logarithmic factor in (5.54) is likewise integrable for $a > 0$. At infinity all terms have at most polynomial times logarithmic growth. For $b > 1$, the integrability of the convolution in (5.53) at its lower endpoint uses $b > 1$ explicitly.

By the Gamma integral and the convolution rule, the Laplace transform of (5.52) is

$$\int_0^\infty e^{-nT} K_{a,b}(T)\, dT = n^{-a}\left[\zeta(b) - \frac{n^{1-b}}{b-1} - \frac{1}{\Gamma(b)} \int_0^\infty e^{-nt} t^{b-1} q(t)\, dt\right] = n^{-a}\big(\zeta(b) - \zeta(b,n)\big) = \frac{H_{n-1}^{(b)}}{n^a}.$$

Each convolution interchange is justified absolutely; q is bounded and the remaining factors have Gamma integrals. Formula (5.54) has Laplace transform

$$n^{-a}\left(\gamma + \log n - \int_0^\infty e^{-nt} q(t)\, dt\right) = n^{-a}(\gamma + \psi(n)) = \frac{H_{n-1}}{n^a}.$$

Here the Laplace transform of $T^{a-1}\log T$ is obtained by differentiating the Gamma integral, justified by absolute integrability. The substitution $u = e^{-T}$ changes $u^{n-1}\, du$ to $e^{-nT}\, dT$, proving the desired moment normalization. In particular $n = 1$ gives total mass zero.

5.6.3 The one-sign-change proof

For $b > 1$, divide (5.53) by T^{a-1} and rescale $t = Tv$:

$$T^{1-a} K_{a,b}(T) = \frac{\zeta(b)}{\Gamma(a)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 (1-v)^{a-1} v^{b-1} \frac{T^b}{1 - e^{-Tv}}\, dv.$$

For fixed $v > 0$, the logarithmic derivative of $T^b/(1 - e^{-Tv})$ is

$$\frac{b}{T} - \frac{v}{e^{Tv} - 1} > \frac{b-1}{T} > 0.$$

Thus the rescaled density is strictly decreasing. Its limit at zero is $\zeta(b)/\Gamma(a) > 0$, and its limit at infinity is $-\infty$. This proves one strict positive–negative crossing in the T coordinate.

For $b = 1$, the rescaled expression is

$$\Gamma(a) T^{1-a} K_{a,1}(T) = \gamma + \psi(a) - \log T - T \int_0^1 (1-v)^{a-1} q(Tv)\, dv.$$

It is strictly decreasing from $+\infty$ to $-\infty$, since q is positive and increasing. For $0 < b < 1$, instead divide by T^{a+b-2} to obtain

$$T^{2-a-b}K_{a,b}(T) = \frac{1}{(1-b)\Gamma(a+b-1)} + \frac{\zeta(b)}{\Gamma(a)}T^{1-b} - \frac{T}{\Gamma(a)\Gamma(b)}\int_0^1 (1-v)^{a-1}v^{b-1}q(Tv)\,dv. \tag{5.55}$$

Now the first term is positive, the second is strictly decreasing because $\zeta(b) < 0$, and the final term is strictly decreasing because q is positive and increasing. The endpoint limits are a positive constant and $-\infty$. Thus there is again exactly one strict crossing. The derivatives of these rescaled expressions are strictly negative; differentiation is dominated by the same integrable beta weights. At a zero of K, multiplication by the positive rescaling factor does not affect the sign of its derivative. Reversing $T = -\log u$ gives the asserted negative–positive sign change and a strictly positive u derivative at its unique crossing.

5.6.4 The critical atom and the exact boundary

Let $0 < b < 1$ and $a = 1 - b$. Replace the middle term of (5.52) by an atom of mass $(1-b)^{-1}$ at $u = 1$, and retain the density

$$K_{1-b,b}(T) = \frac{\zeta(b)T^{-b}}{\Gamma(1-b)} - \frac{1}{\Gamma(1-b)\Gamma(b)}\int_0^T (T-t)^{-b}t^{b-1}q(t)\,dt. \tag{5.56}$$

Both terms are strictly negative and integrable against $e^{-T}\,dT$. Its moments plus the atom are

$$\frac{1}{1-b} + n^{b-1}\left[\zeta(b) - \frac{1}{\Gamma(b)}\int_0^\infty e^{-nt}t^{b-1}q(t)\,dt\right] = \frac{H_{n-1}^{(b)}}{n^{1-b}},$$

by (5.50). Thus the measure has total mass zero, and its negative mass is exactly $(1-b)^{-1}$.

For $a = 0$, $b > 1$, the formula is especially simple:

$$\nu_{0,b} = \zeta(b)\delta_1 + K_{0,b}(-\log u)\,du, \qquad K_{0,b}(T) = -\frac{T^{b-1}}{\Gamma(b)(1-e^{-T})}. \tag{5.57}$$

The ordinary Hurwitz integral gives its moments immediately. Its negative mass is exactly $\zeta(b)$.

Every finite signed measure has bounded moments, since $|\int u^{n-1}\,d\nu| \le \|\nu\|_{\mathrm{TV}}$. If $0 < b < 1$, elementary integral comparison gives $H_{n-1}^{(b)} \sim n^{1-b}/(1-b)$, so the moments diverge to infinity when $a + b < 1$. If $a = 0, b = 1$, they diverge logarithmically. These are exactly the excluded cases for $a \ge 0, b > 0$. Finally, two finite signed measures with the same moments integrate every polynomial equally; polynomial density in $C([0,1])$ makes the measures identical. This proves Theorem 5.17.

5.7 Consequences for complex zeros and angular zeros

Theorem 5.18 (Nonvanishing throughout the finite-measure domain)**.** *For all parameters in* (5.45)*, $F_{a,b}$ has an analytic continuation to $\mathbb{C} \setminus [1,\infty)$ and has no zero there except for its double zero at the origin. Every upper open semicircle $z = \rho e^{i\theta}$, $0 < \rho \le 1$, $0 < \theta < \pi$, has exactly one simple zero of $\operatorname{Im} F_{a,b}(z)$, and it satisfies*

$$\arccos\rho < \theta_{a,b}(\rho) < \frac{\pi}{2}.$$

The imaginary part is positive below this angle and negative above it; its angular derivative at the zero is strictly negative. In the atomic cases, unit-circle values mean analytic continuation or Abel boundary values: the original Fourier summands do not tend to zero, so ordinary Fourier convergence is not asserted.

Proof. The signed transform

$$F_{a,b}(z) = z \int_{[0,1]} \frac{d\nu_{a,b}(u)}{1 - zu}$$

follows by geometric expansion in the disk and extends analytically to the slit plane by uniform domination on compact sets. Let r be the crossing from Theorem 5.17, or let $r = 1$ in the atomic cases. Zero mass gives

$$F_{a,b}(z) = \frac{z^2}{1 - rz} P_{a,b}(z), \qquad P_{a,b}(z) = \int_{[0,1]} \frac{u - r}{1 - zu}\, d\nu_{a,b}(u). \tag{5.58}$$

The measure $(u - r)\, d\nu_{a,b}(u)$ is positive and nonzero, including when $r = 1$: its atom disappears but its density is strictly positive on $(0, 1)$. Consequently $\operatorname{Im} P_{a,b}(z)$ has the strict sign of $\operatorname{Im} z$, while $P_{a,b}(x) > 0$ for $x < 1$. Apart from the explicit z^2, the factor $(1 - rz)^{-1}$ in (5.58) is analytic and nonzero on the slit plane. Since $P_{a,b}(0) = A_2 = 2^{-a}$, the zero at zero has multiplicity exactly two.

For angular zeros put

$$J_\rho(c) = \int_{[0,1]} \frac{d\nu(u)}{1 - 2\rho c u + \rho^2 u^2}, \qquad \operatorname{Im} F(\rho e^{i\theta}) = \rho \sin\theta\, J_\rho(\cos\theta).$$

The zero-mass one-crossing sign test holds for our atomic measures as well: subtract the value of the test function at r, and use the ordered supports of the two signs. It gives $J_\rho(0) < 0$. When $\rho < 1$, the kernel at $c = \rho$ is strictly increasing in u, so $J_\rho(\rho) > 0$. When $\rho = 1$ and the crossing is interior, separate the two signs before expanding $(1 - u)^{-2}$ to obtain

$$\lim_{c \uparrow 1} J_1(c) = \int_{[0,1]} \frac{d\nu(u)}{(1 - u)^2} = \sum_{n \geq 1} n A_n > 0,$$

allowing $+\infty$. The negative part is supported away from one, so this is the same justified monotone-limit argument as in the original signed-kernel theorem. In the atomic cases the limiting signed integral at $c = 1$ need not be defined, and we do not use that display. Instead the limit is $+\infty$ directly: the atom contributes $M/[2(1 - c)]$, and the negative density contributes $o((1 - c)^{-1})$. For the latter assertion, multiply its integral by $1 - c$ and use dominated convergence: the multiplier $(1 - c)/(1 - 2cu + u^2)$ is bounded by 1 for $0 \leq c < 1$ and tends to zero for every $u < 1$. The negative density is integrable and has no atom at one.

For $c' > c$, the quotient of the two kernels has derivative

$$\frac{d}{du} \frac{1 - 2\rho c u + \rho^2 u^2}{1 - 2\rho c' u + \rho^2 u^2} = \frac{2\rho(c' - c)(1 - \rho^2 u^2)}{(1 - 2\rho c' u + \rho^2 u^2)^2} > 0 \quad (0 < u < 1).$$

At any zero, apply the sign test to the signed measure obtained by weighting ν with the positive kernel. It follows that J_ρ is positive at every larger c and negative at every smaller c; this proves uniqueness and the signs. Finally its derivative at its zero is positive, because the additional factor $2\rho u/(1 - 2\rho c u + \rho^2 u^2)$ is strictly increasing in u. The angular derivative is therefore $-\rho \sin^2\theta\, J'_\rho(\cos\theta) < 0$. □

Corollary 5.19 (Ordinary and analytic boundary values)**.** *For $a, b > 0$ with $a + b > 1$, the defining series converges ordinarily at every $|z| = 1, z \neq 1$, and equals its analytic continuation. At $a + b \leq 1$, or on the axis $a = 0, b > 0$, its terms do not tend to zero at those boundary points. There the stated values are analytic or Abel values.*

Proof. In the absolutely continuous finite-measure case, expand the finite geometric sum and use domination of $z(1 - (zu)^N)/(1 - zu)$ against the total variation; there is no atom at one. Harmonic asymptotics give a positive nonzero limit or unbounded coefficients in the remaining cases, so the term test fails. The axis formula and the subsequent Stieltjes factorization still give well-defined continuation and radial Abel limits away from the cut. □

5.8 A positive two-variable representation at all positive orders

The threshold in Theorem 5.17 concerns a *one-variable finite signed measure*. It is not a threshold for analytic continuation or every positivity property.

Theorem 5.20 (Positive double kernel)**.** *For every $a, b > 0$ and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = z^2 \int_{0<y<x<1} \frac{W_{a,b}(x, y)}{(1 - xz)(1 - yz)} \, \mathrm{d}y \, \mathrm{d}x, \tag{5.59}$$

where

$$W_{a,b}(x, y) = \frac{(-\log x)^{a-1} (\log(x/y))^{b-1}}{\Gamma(a)\Gamma(b)}. \tag{5.60}$$

The weight is strictly positive and has total mass 2^{-a}.

Proof. For $|z| < 1$, expand the two denominators. Substitution $y = xv$ gives, for nonnegative integers j, k,

$$\int_{0<y<x<1} W_{a,b}(x, y) x^j y^k \, \mathrm{d}y \, \mathrm{d}x = \frac{1}{(j + k + 2)^a (k + 1)^b}.$$

Summing over $j + k = n - 2$ yields c_n. Absolute convergence justifies the rearrangement, and taking $j = k = 0$ gives the total mass. On each compact subset of $\mathbb{C} \setminus [1, \infty)$, both denominators are bounded away from zero uniformly in $x, y \in [0, 1]$. The integral is consequently holomorphic there and supplies the stated continuation. □

Corollary 5.21 (Disk nonvanishing below and above the threshold)**.** *For all $a, b > 0$, $F_{a,b}$ has no nonzero zeros in $\{z : |z| \leq 1,\ z \neq 1\}$. More precisely, when $\operatorname{Im} z > 0$ and $|z| \leq 1$,*

$$\operatorname{Im} \frac{F_{a,b}(z)}{z} > 0. \tag{5.61}$$

At any upper-half-disk zero of $\operatorname{Im} F_{a,b}(z)$, the value $F_{a,b}(z)$ itself is strictly negative real.

Proof. A direct calculation gives

$$\operatorname{Im} \frac{z}{(1 - xz)(1 - yz)} = \frac{(\operatorname{Im} z)(1 - xy|z|^2)}{|1 - xz|^2 |1 - yz|^2} > 0$$

for $0 < y < x < 1$, proving (5.61). The lower half follows by conjugation. On the real interval $z < 1$, the integral $F(z)/z^2$ is strictly positive. Finally, if $F(z) = R \in \mathbb{R}$ with $\operatorname{Im} z > 0$, then $\operatorname{Im}(R/z) = -R \operatorname{Im} z/|z|^2 > 0$, so $R < 0$. □

Corollary 5.22 (Gaussian negativity and strict log-convexity)**.** *For every $a, b > 0$,*

$$-g_{a,b} = \int_{0<y<x<1} W_{a,b}(x,y) \frac{x+y}{(1+x^2)(1+y^2)} \, \mathrm{d}y \, \mathrm{d}x > 0, \tag{5.62}$$

and

$$\frac{H_2^{(b)}}{4\,3^a} < -g_{a,b} < \frac{H_2^{(b)}}{3^a}. \tag{5.63}$$

The function $(a,b) \mapsto \Gamma(a)\Gamma(b)(-g_{a,b})$ is strictly log-convex on the positive quadrant.

Proof. Set $z = \mathrm{i}$ in (5.59) and rationalize the denominator. Since $1 < (1+x^2)(1+y^2) < 4$ and $\int W(x+y) = c_3 = H_2^{(b)}/3^a$, the bounds follow. For log-convexity apply Hölder's inequality to the positive integral with the gamma factors removed. Its parameter dependence is the exponential of $(a-1)\log(-\log x) + (b-1)\log\log(x/y)$. Equality in Hölder would require a nontrivial real linear combination of these two logarithms to be constant almost everywhere on the open triangle, which is impossible. This proves strictness for every nontrivial line segment of parameter pairs. □

5.8.1 All-order integral identities

Theorem 5.23 (Hurwitz–polylogarithm transport)**.** *For $a, b > 0$, $b \neq 1$, and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$\begin{aligned} F_{a,b}(z) = \zeta(b)\,\mathrm{Li}_a(z) - \frac{\mathrm{Li}_{a+b-1}(z)}{b-1} \\ - \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} q(t)\,\mathrm{Li}_a(ze^{-t})\,\mathrm{d}t. \end{aligned} \tag{5.64}$$

At $b = 1$ the pole-free identity is

$$F_{a,1}(z) = \gamma\,\mathrm{Li}_a(z) - \partial_a\,\mathrm{Li}_a(z) - \int_0^\infty q(t)\,\mathrm{Li}_a(ze^{-t})\,\mathrm{d}t. \tag{5.65}$$

In particular, the index $a+b-1$ in (5.64) is allowed to be zero or negative.

Proof. Inside the disk, multiply (5.50), with $x = n$, by $n^{-a}z^n$ and sum. Exponential decay at infinity, boundedness of q, and $b > 0$ justify termwise integration. The Hurwitz difference is $H_{n-1}^{(b)}$, so coefficient comparison gives (5.64). At $b = 1$, use $H_{n-1} = \gamma + \log n - \int_0^\infty e^{-nt} q(t)\,\mathrm{d}t$; the sum of $(\log n) n^{-a} z^n$ is $-\partial_a\,\mathrm{Li}_a(z)$. The integral expressions on the right are locally holomorphic on the slit plane. Near $t = 0$, $\mathrm{Li}_a(ze^{-t})$ stays bounded on compact slit sets, and at infinity it is $O(e^{-t})$. Analytic continuation therefore proves the identities throughout the claimed domain. Standard conventions for Li_s are those of DLMF 25.12 [96]. □

These identities are useful even when no signed Hausdorff measure exists. They are transformations to a depth-one polylogarithm integral, not assertions that a depth-two constant has been reduced to a finite basis of depth-one periods.

5.9 Below the measure threshold: positive differences settle the zero question

The failure of finite signed-moment representability below $a + b = 1$ does not imply a failure of positivity after a discrete derivative. Consecutive differences supply a finite positive measure and

resolve the slit-plane nonvanishing and angular-uniqueness questions left open by the real-order reports. They also extend the right-endpoint radial mechanism from the critical line to the entire subcritical region.

Write $c_n = H_{n-1}^{(b)}/n^a$ and $F_{a,b}(z) = \sum_{n\ge1} c_n z^n$. For $a > 0$, $b > 0$, $w = a + b < 1$, put $s = w - 1 \in (-1, 0)$ and

$$k_v(T) = T^{v-1}/\Gamma(v), \tag{5.66}$$

$$\mathfrak{h}(t) = (1 - e^{-t})^{-1} - t^{-1}, \tag{5.67}$$

$$\mathcal{C}_{a,b}(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T - t)^{a-1} t^{b-1} \mathfrak{h}(t)\, dt, \tag{5.68}$$

$$R_{a,b}(T) = -\zeta(b)k_a(T) - \frac{k_s(T)}{1 - b} + \mathcal{C}_{a,b}(T). \tag{5.69}$$

Here k_s is a signed function, not a positive Gamma density: $\Gamma(s) = \Gamma(s+1)/s < 0$. Since $0 < b < 1$, $\zeta(b) < 0$, and $1/2 < \mathfrak{h}(t) < 1$, all three terms of $R_{a,b}$ are strictly positive. The regularized Hurwitz identity used here is

$$\zeta(b, n) = \frac{n^{1-b}}{b - 1} + \frac{1}{\Gamma(b)} \int_0^\infty e^{-nt} t^{b-1} \mathfrak{h}(t)\, dt;$$

it follows from the convergent Hurwitz integral by subtraction and analytic continuation in b, as in the preceding construction [93].

Theorem 5.24 (Positive Hausdorff differences)**.** *For $a, b > 0$ with $a + b < 1$, define*

$$d\omega_{a,b}(u) = (1 - u)R_{a,b}(-\log u)\, du, \qquad 0 < u < 1.$$

This is a finite positive measure, with strictly positive density and mass 2^{-a}. For every $n \ge 1$,

$$c_n = \int_0^1 (1 - u^{n-1})R_{a,b}(-\log u)\, du, \tag{5.70}$$

$$c_{n+1} - c_n = \int_0^1 u^{n-1}\, d\omega_{a,b}(u). \tag{5.71}$$

The first measure has infinite mass, but every displayed integral is absolutely convergent. Thus the shifted sequence c_{n+1} is a discrete Bernstein sequence, and its differences are strictly completely monotone. Here $\Delta f_n = f_{n+1} - f_n$. In particular, for all $j \ge 0$ and $n \ge 1$,

$$(-1)^j \Delta^j (c_{n+1} - c_n) = \int_0^1 u^{n-1}(1 - u)^j\, d\omega_{a,b}(u) > 0.$$

The positive difference measure is uniquely determined by these moments.

Proof. At $T = 0$, the leading term of R is a positive constant times T^{w-2}; the other terms have orders T^{a-1} and T^{w-1}. Multiplication by $1 - e^{-T}$ therefore makes it integrable, since $w > 0$. At infinity all terms grow at most polynomially and the change of variables $u = e^{-T}$ supplies e^{-T}. This proves finiteness of ω, convergence of (5.70), and divergence of the unweighted mass at $u = 1$.

For $s \in (-1, 0)$, integration by parts gives the convergent identity

$$\int_0^\infty (e^{-T} - e^{-nT})k_s(T)\, dT = 1 - n^{-s}.$$

Both boundary terms vanish: near zero their order is T^{s+1}, and infinity has exponential decay. For k_a the same identity is the ordinary Gamma integral. The convolution has Laplace transform

$$\int_0^\infty e^{-nT}\mathcal{C}_{a,b}(T)\,dT = n^{-a}\left(\zeta(b,n)+\frac{n^{1-b}}{1-b}\right).$$

Every convolution interchange is absolute. Insert these three identities into (5.69), with the factor $e^{-T}-e^{-nT}$ retained. The constant and power terms cancel, leaving $n^{-a}(\zeta(b)-\zeta(b,n))=c_n$. No divergent Laplace transforms are subtracted. Subtract the two convergent formulas at $n+1$ and n to obtain (5.71). At $n=1$ this gives mass $c_2=2^{-a}$. The difference formula proves every strict sign, and polynomial density on $[0,1]$ proves uniqueness of the finite measure. □

On the critical line $a=1-b>0$, use instead $R_{a,b}=-\zeta(b)k_a+\mathcal{C}_{a,b}$. Its finite mass is $1/a$, and its Laplace transform is $1/a-c_n$, so the same formulas for $\omega=(1-u)R(-\log u)\,du$ hold. This is precisely the negative density of the critical signed measure; its positive atom at one disappears after multiplication by $1-u$. On the whole axis $a=0, b>0$, the difference measure is simply

$$d\omega_{0,b}(u)=\frac{(-\log u)^{b-1}}{\Gamma(b)}\,du, \qquad c_{n+1}-c_n=n^{-b},$$

and $F_{0,b}(z)=z\,\mathrm{Li}_b(z)/(1-z)$. These measures also have strictly positive densities, with mass 2^{-a} and no endpoint atoms.

Corollary 5.25 (A sharp complementary positivity classification)**.** *For $a\ge 0, b>0$, the consecutive differences $c_{n+1}-c_n$ are moments of a finite positive measure on $[0,1]$ if and only if $a=0$ or $a+b\le 1$. In the same parameter region the function*

$$\Phi_{a,b}(t)=(t+1)^{-a}\{\zeta(b)-\zeta(b,t+1)\}, \qquad t\ge 0,$$

with $\gamma+\psi(t+1)$ replacing the braces at $b=1$, is a Bernstein function: $\Phi(0)=0$, $\Phi(t)>0$ for $t>0$, and $(-1)^j\Phi^{(j+1)}(t)>0$ for all $j\ge 0$.

Proof. The preceding proof works for every real $n\ge 1$, since the Gamma and regularized Hurwitz identities used there do not require integer n. Thus $\Phi(t)=\int(1-u^t)\,d\rho(u)$ with $d\rho=d\omega/(1-u)$. Differentiation gives

$$(-1)^j\Phi^{(j+1)}(t)=\int_0^1(-\log u)^{j+1}u^t\,d\rho(u)>0.$$

All these integrals converge even at $t=0$: the endpoint at one is controlled by $\int(1-u)\,d\rho<\infty$, and the logarithmic-coordinate density at zero has exponential decay times polynomial growth. The same representation follows on the axis from the ordinary Hurwitz integral, or its convergent difference at $b\le 1$.

Conversely, for $a>0, a+b>1$, elementary harmonic asymptotics give $c_n\to 0$ while $c_2=2^{-a}>0$. The sequence cannot be nondecreasing, so its differences cannot all be positive-measure moments and its continuous interpolation cannot be a Bernstein function. This proves both sharp parameter statements. □

Theorem 5.26 (Global nonvanishing at all positive inner orders)**.** *For every $a\ge 0$, $b>0$, the principal continuation of $F_{a,b}$ has no zero on $\mathbb{C}\setminus[1,\infty)$ except its double zero at zero. For $a>0$, $a+b\le 1$, and also for $a=0, b>0$, the stronger factorization is*

$$F_{a,b}(z)=\frac{z^2}{1-z}P_{a,b}(z), \qquad P_{a,b}(z)=\int_0^1\frac{d\omega_{a,b}(u)}{1-zu}. \tag{5.72}$$

In particular $P(0)=2^{-a}$, $\operatorname{Im}P(z)$ has the strict sign of $\operatorname{Im}z$, and $P(x)>0$ for every real $x<1$.

Proof. The coefficient of z^m in $(1-z)F(z)/z^2$ is $c_{m+2}-c_{m+1}$, including $m=0$ because $c_1=0$. The difference moments give (5.72) in the disk. The finite positive measure gives holomorphic continuation to the slit plane, where its imaginary-part and real-axis signs prove nonvanishing. The mass proves multiplicity exactly two. For $a>0, a+b>1$, the one-crossing finite signed measure already gives $F=z^2P_r/(1-rz)$ with a nonzero positive Stieltjes transform and $0<r<1$. The same argument applies. This proves the complete parameter statement while retaining the sharp boundary for finite signed moments. □

5.9.1 Angular uniqueness and a full subcritical radial theorem

Theorem 5.27 (Right-endpoint Stieltjes geometry)**.** *For $a,b>0$ with $a+b\le 1$, and for $a=0, b>0$, every open upper semicircle of radius $0<\rho\le 1$ has exactly one simple zero of $\operatorname{Im} F_{a,b}$. Its angle satisfies*

$$\arccos\rho<\theta_{a,b}(\rho)<\arccos(\rho/2), \qquad \tfrac{1}{2}<\eta_{a,b}(\rho)=\cos\theta_{a,b}(\rho)/\rho<1.$$

The imaginary part is positive before that angle and negative after it. The normalized quantity $\eta_{a,b}$ is strictly increasing in ρ. Together with the supercritical angular theorem, this proves angular uniqueness for every $a\ge 0, b>0$, including the formerly conjectural subcritical region. Boundary values are analytic or Abel values where the ordinary Fourier series does not converge.

Proof. The following argument works for any finite positive measure ω with a strictly positive density on $(0,1)$. Set $x=\rho^2$, $m=2\cos\theta/\rho-1$, and

$$D(u)=1+x\{u^2-(1+m)u\}, \qquad G(m,x)=\int_0^1 \frac{m-u}{D(u)}\,d\omega(u).$$

From (5.72), direct rationalization gives

$$\operatorname{Im} F(\rho e^{i\theta})=\frac{\rho^3\sin\theta}{|1-\rho e^{i\theta}|^2}G(m,x).$$

For $0<m<1$ all denominators are positive, and

$$G_m=\int_0^1 \frac{1-xu}{D(u)^2}\,d\omega(u)>0.$$

We have $G(0,x)<0$. For $x<1$, $G(1,x)>0$. At $x=1$, its negative part for $u>m\ge 1/2$ is bounded in magnitude by $2\omega((m,1))$, which tends to zero. Its positive part tends pointwise to $d\omega(u)/(1-u)$ and has a strictly positive limit inferior by Fatou's lemma, with infinity allowed. Hence $G(m,1)>0$ for some $m<1$. This proves a unique root in $(0,1)$ even when $\omega/(1-u)$ has infinite mass. Outside that interval the numerator has a fixed sign, proving uniqueness on the full upper arc. At the zero, the angular derivative is $-2\rho^2\sin^2\theta\,G_m/|1-\rho e^{i\theta}|^2<0$.

At a root, the probability measure proportional to $d\omega/D$ has mean m. For $d(u)=u^2-(1+m)u$, the identities $d(m)=-m$ and $d(u)-d(m)=(u-m)(u-1)$ give

$$G_x=-\frac{1}{1-xm}\int_0^1 \frac{(u-m)^2(1-u)}{D(u)^2}\,d\omega(u)<0.$$

Consequently

$$\frac{dm}{dx}=\frac{\displaystyle\int (u-m)^2(1-u)D(u)^{-2}\,d\omega(u)}{\displaystyle(1-xm)\int (1-xu)D(u)^{-2}\,d\omega(u)}>0, \qquad \eta'(\rho)=\rho m'(\rho^2)>0.$$

All differentiations are dominated locally at the interior root, including at $x = 1$ since $m < 1$. The implicit-function theorem proves smooth dependence. This argument uses only the finite measure ω; it does not apply a finite-measure limit theorem to its infinite unweighted predecessor. □

5.9.2 One-sided Euler certificates even for divergent boundary series

Theorem 5.28 (Subcritical Euler continuation and a sharp fixed-pair constant)**.** *For the parameters of Theorem 5.27, put* $A_k = c_{2k+1}$, $g_{a,b} = \operatorname{Im} F_{a,b}(i)$ *and*

$$E_N = \sum_{j=0}^{N-1} 2^{-j-1} \sum_{k=0}^{j} (-1)^k \binom{j}{k} A_k.$$

Then E_N *decreases strictly to the analytic Gaussian value, and*

$$-g_{a,b} = \frac{1}{2} \int_0^1 \frac{1+u}{1+u^2}\, d\omega(u), \tag{5.73}$$

$$E_N - g_{a,b} = 2^{-N} \int_0^1 \frac{(1+u)(1-u^2)^{N-1}}{1+u^2}\, d\omega(u) > 0. \tag{5.74}$$

The least constant in $E_N - g \le C2^{-N}$ *for all* $N \ge 1$ *is* $C = -2g$, *and*

$$2^{-a} < -2g_{a,b} < \frac{1+\sqrt{2}}{2}\, 2^{-a} < \frac{5}{4}.$$

Thus $E_N - (5/4)2^{-N} < g_{a,b} < E_N$ *is a rational enclosure throughout this domain. It concerns analytic continuation when the ordinary alternating series diverges.*

Proof. For $j \ge 1$, the finite binomial difference of A_k is

$$-\int_0^1 (1+u)(1-u^2)^{j-1}\, d\omega(u),$$

while the order-zero difference is $A_0 = 0$. Sum the geometric tail against the finite measure to get (5.74). Evaluating (5.72) at i gives the same limiting value. The scaled errors decrease strictly to zero, and their largest value is $2(E_1 - g) = -2g$. Finally $(1+u)/(1+u^2)$ has minimum one at the endpoints and maximum $(1+\sqrt{2})/2$ at $u = \sqrt{2} - 1$. The positive density makes both bounds strict, and the measure has mass 2^{-a}. Since $\sqrt{2} < 3/2$, the rational bound follows. □

The global integer-outer-order normalized-radius conjecture remains open. The new radial theorem covers the full subcritical and critical region and the zero outer-order axis; it does not assert the same radial direction in the supercritical region, where fractional turning points are possible. Finite signed-moment existence, analytic nonvanishing, angular uniqueness and radial motion now have explicitly different scopes.

Corollary 5.29 (An exact Gaussian identity on the outer-order axis)**.** *For every* $b > 0$, *with analytic or Abel interpretation when necessary,*

$$g_{0,b} = -\frac{1}{2}\left(\beta(b) + 2^{-b}\eta(b)\right), \qquad \eta(b) = \sum_{n\ge1} \frac{(-1)^{n-1}}{n^b}.$$

In particular $g_{0,1} = -\pi/8 - \log 2/4$ *and* $g_{0,2} = -G/2 - \pi^2/96$.

Proof. The defining residue classes give $\mathrm{Li}_b(i) = -2^{-b}\eta(b) + i\beta(b)$. Since $F_{0,b}(i) = i\,\mathrm{Li}_b(i)/(1-i) = (-1+i)\,\mathrm{Li}_b(i)/2$, its imaginary part is the claimed expression. Both Dirichlet series converge for $b > 0$; the depth-two series retains its stated boundary interpretation. These formulas also identify the sharp fixed-pair Euler constant as $\beta(b) + 2^{-b}\eta(b)$ on this axis. □

Theorem 5.30 (Strict Gaussian decrease in the real outer order)**.** *For every fixed $b > 0$, the positive magnitude $-g_{a,b}$ decreases strictly as a runs through $[0,\infty)$. In particular, for $a > 0$,*

$$0 < -2g_{a,b} < \beta(b) + 2^{-b}\eta(b) < \frac{1+\sqrt{2}}{2}.$$

The supremum over nonnegative outer orders is the explicit axis value. This theorem is a magnitude comparison. The separate contraction proof in Section 5.20 establishes the Euler remainder interpretation of the same axis expression throughout the parameter domain.

Proof. The outer Gamma integral, first by coefficient comparison in the disk and then by locally dominated continuation to the slit plane, is

$$F_{a,b}(z) = \frac{1}{\Gamma(a)} \int_0^\infty t^{a-1} F_{0,b}(ze^{-t})\, dt \qquad (a > 0).$$

Near zero its boundary value at $z = i$ is bounded; at infinity the integrand without t^{a-1} is $O(e^{-2t})$. Let $h_b(t) = e^t[-\operatorname{Im} F_{0,b}(ie^{-t})]$. The positive axis measure gives, with $\rho = e^{-t}$,

$$h_b(t) = \int_0^1 \frac{(1+u)\rho^2}{(1+\rho^2)(1+u^2\rho^2)}\, d\omega_{0,b}(u).$$

The derivative in ρ of the displayed kernel is

$$\frac{2\rho(1+u)(1-u^2\rho^4)}{(1+\rho^2)^2(1+u^2\rho^2)^2} > 0 \qquad (0 < \rho \le 1,\ 0 < u < 1).$$

Thus h_b is positive, bounded and strictly decreasing in t. The Gamma integral is equivalently $-g_{a,b} = \mathbb{E}[h_b(T_a)]$, where $T_a \sim \mathrm{Gamma}(a,1)$. Independent Gamma variables satisfy $T_{a+d} \overset{d}{=} T_a + T_d$; strict decrease of h_b proves strict decrease of the expectation. At $a = 0$ take $T_0 = 0$, and bounded continuity gives the same strict comparison and the limit as $a \downarrow 0$. For $a > 0$, differentiation also gives the exact negative covariance $\partial_a(-g_{a,b}) = \mathrm{Cov}(\log T_a, h_b(T_a)) < 0$. The axis identity and its positive-measure bound complete the proof. □

Theorem 5.31 (Gaussian monotonicity at every fixed total order)**.** *For every $w > 0$, $a \mapsto -2g_{a,w-a}$ decreases strictly on $(0,w)$. Its endpoint limits give the exact strict inequalities*

$$2\{\beta(w) - \beta(w-1)\} < -2g_{a,w-a} < \beta(w) + 2^{-w}\eta(w).$$

Here $\beta(w-1)$ uses analytic continuation when necessary. In particular, this proves the previously conjectural strict decrease on the entire critical line $w = 1$.

Proof. In the positive double kernel put $x = e^{-s}$, $y = e^{-(s+t)}$, then $T = s + t$ and $R = s/T$. Its Gaussian value becomes

$$-g_{a,b} = \mathbb{E}[K(R,T)], \qquad K(R,T) = \frac{X(X+Y)}{(1+X^2)(1+Y^2)}, \quad X = e^{-RT}, \quad Y = e^{-T},$$

where $T \sim \text{Gamma}(w,1)$ and $R \sim \text{Beta}(a, w-a)$ are independent. The Jacobian gives $e^{-2s-t} = e^{-T}X$; the remaining power is T^{w-1}, so every normalization is explicit. For fixed $T > 0$ the derivative with respect to X is

$$\frac{2X + Y - YX^2}{(1+X^2)^2(1+Y^2)} > 0.$$

Thus K decreases strictly in R. The beta score along fixed w is an increasing function of R, namely $\log(R/(1-R))$ minus its mean. Differentiation therefore gives a strictly negative covariance. All logarithmic beta moments are finite for $0 < a < w$. At $a \downarrow 0$, $R \to 0$ in probability; at $a \uparrow w$, $R \to 1$. Bounded convergence gives the two endpoint values. The first is the axis formula. The second also follows from $F_{w,0}(z) = \text{Li}_{w-1}(z) - \text{Li}_w(z)$ and its analytic value at i. □

5.9.3 A unique global Gaussian maximum and the sharp critical Euler constant

Put $C(b) = \beta(b) + 2^{-b}\eta(b)$ and $\phi(T) = (1+e^{-T})/(1+e^{-2T})$. The axis measure gives $C(b) = \mathbb{E}[\phi(T_b)]$ for $T_b \sim \text{Gamma}(b,1)$. The function ϕ increases and then decreases, with its unique maximum at $T = \log(1+\sqrt{2})$.

Theorem 5.32 (A unique, nondegenerate axis maximum)**.** *The function C has exactly one stationary point b_* on $(0,\infty)$. It is a nondegenerate global maximum; $C' > 0$ below b_* and $C' < 0$ above it. Moreover,*

$$1 < b_* < 2, \qquad \frac{227}{200} < C(b_*) < \frac{1+\sqrt{2}}{2}.$$

The supremum of $-2g_{a,b}$ over $a \geq 0, b > 0$ is this maximum, attained only at $(a,b) = (0, b_*)$. Over $a, b > 0$ it is a supremum approached at the same axis point.

Proof. The gamma law tends to zero in probability as $b \downarrow 0$ and to infinity in probability as $b \to \infty$. Hence $C(b) \to 1$ at both ends. For every finite $b > 0$, $1 < C(b) < \max \phi$, since the density is positive throughout $(0,\infty)$. Thus an interior maximum exists.

We prove a strict two-intersection property. For $1 < \lambda < \max \phi$, the function $f(T) = \phi(T) - \lambda$ has sign pattern negative, positive, negative, with exactly two sign changes $T_1 < T_2$. If $C(b_i) = \lambda$ at three distinct orders $b_1 < b_2 < b_3$, each of the three integrals $\int f(T)T^{b_i-1}e^{-T}\,dT$ vanishes. Write $v = T^{b_2-b_1}$ and $p = (b_3-b_1)/(b_2-b_1) > 1$. Subtract from v^p its secant line through the values at T_1, T_2, and multiply by $-T^{b_1-1}$. Strict convexity gives a linear combination of the three powers with the same sign pattern as f. Its integral against $f(T)e^{-T}$ is strictly positive, contradicting the three vanishing integrals. Thus every horizontal level has at most two intersections.

At a stationary point b_0, take $\lambda = C(b_0)$. If also $C''(b_0) = 0$, the Mellin integral and its first two derivatives vanish there. Their linear combination with kernel $-T^{b_0-1}(\log T - \log T_1)(\log T - \log T_2)$ again has strictly positive integral, a contradiction. Therefore every stationary point is nondegenerate. It cannot be a local minimum: the end limits one, the minimum itself, and the two adjacent larger values would give three distinct intersections with its own horizontal level. Every stationary point is consequently a strict maximum, and there can be only one. This proves both derivative signs and nondegeneracy.

The exact axis certificate at $b = 3/2$ gives $C(3/2) > 227/200$. For comparison, $\pi < 22/7$ and $\log 2 < 25/36$ give $C(1) < 571/504 < 227/200$. The latter logarithmic bound follows by summing the first two terms of its artanh series and bounding the remaining positive geometric tail. Also the first five terms of the alternating Catalan series give $G < 37/40$, hence $C(2) = G + \pi^2/48 < 37/40 + 121/588 < 227/200$. Strict unimodality now forces $1 < b_* < 2$. The magnitude theorem proves the remaining supremum assertions. □

The numerical axis experiment finds $b_* \approx 1.30221658710124120959$ and $C(b_*) \approx 1.13656110333950956095$. This experiment supplies decimal diagnostics; the preceding proof gives the broad bracket $1 < b_* < 2$ and exact value bounds independently. The sharper rational enclosures in Section 5.10 subsequently certify the displayed location and height. Neither uniqueness proof relies on a numerical scan.

Corollary 5.33 (The sharp uniform constant on and below the critical line)**.** *For every $a, b > 0$ with $a + b \le 1$ and every integer $N \ge 1$,*

$$0 < E_N - g_{a,b} < C_{\text{critical}} 2^{-N}, \qquad C_{\text{critical}} = \frac{\pi}{4} + \frac{\log 2}{2}.$$

This is the least constant uniform over the critical line and is also the least constant uniform over the whole positive subcritical triangle. Thus the critical constant conjecture and its stronger strict-monotonicity claim are both settled. The rational choice $57/50$ is valid on this triangle.

Proof. On the critical line, Theorem 5.31 bounds $-2g$ strictly by its left endpoint $C(1) = C_{\text{critical}}$. For $a + b < 1$, the outer-order magnitude theorem gives $-2g_{a,b} < C(b)$, and $b < 1 < b_*$ gives $C(b) < C(1)$. Apply the fixed-pair Euler constant theorem. Sharpness follows from the critical endpoint limit $a \downarrow 0, b \uparrow 1$ and the error at $N = 1$, where $E_1 = 0$. Finally $C(1) < 571/504 < 57/50$. The conclusion does not extend the same bound to arbitrary supercritical orders or to the entire outer-order axis. □

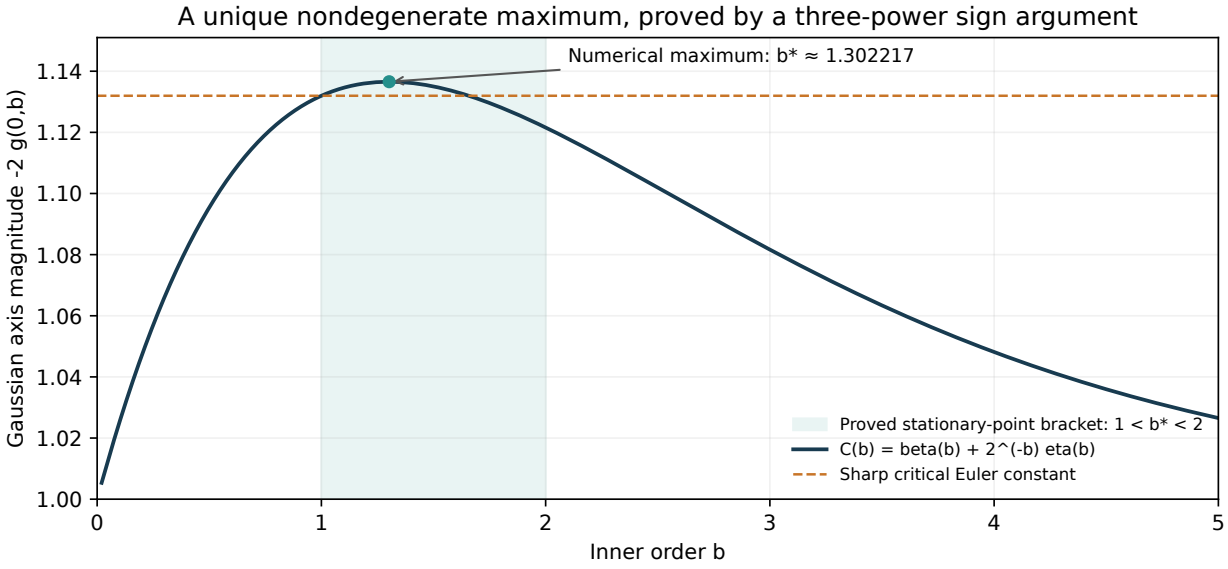


Figure 5.2: Numerical values of the exact Gaussian axis expression and its stationary-point marker. The shaded interval is the proved bracket for the unique maximum; the horizontal line is the proved sharp critical Euler constant. The marker was plotted numerically; its parameter and height are subsequently certified in Section 5.10. The universal Euler interpretation is established separately by the tail-kernel contraction in Theorem 5.64.

5.10 Strict Gaussian envelope shape and a certified maximum

The incoming extremal continuation [99] independently develops the axis maximum and strengthens its description. Its strict normalized Mellin comparison gives a second proof of unimodality, Turan inequalities and a rigorous enclosure of the maximum. We retain the earlier three-power proof as a distinct route and reconcile the common axis normalization before using the stronger certificate.

Lemma 5.34 (A strict normalized Mellin comparison)**.** *Suppose $f : (0,\infty) \to (0,\infty)$ is strictly log-concave, and that its Mellin integrals and their first two parameter derivatives are finite and admit differentiation under the integral. Then*

$$M(p) = \frac{1}{\Gamma(p)} \int_0^\infty T^{p-1} f(T)\, dT \quad (p > 0) \quad \textit{satisfies} \quad (\log M)''(p) < 0.$$

Proof. Normalize $T^{p-1}f(T)$ to a probability density P. Choose the gamma density Q of shape p and rate $\lambda > 0$ so that $\psi(p) - \log\lambda = \mathbb{E}_P \log T$. The two densities have matching mass and matching logarithmic mean. Their log likelihood ratio is $\log f(T) + \lambda T$ plus a constant, hence strictly concave. Its positive set is an interval, and matching mass forces both signs. A single crossing at T_0 is impossible: $\log(T/T_0)(P-Q)$ would have one strict sign while its integral is zero by the two matching moments. There must therefore be two crossings $T_1 < T_2$ and the sign pattern of $P - Q$ is negative, positive, negative.

It follows that

$$\int (\log T - \log T_1)(\log T - \log T_2)(P-Q)(T)\, dT < 0.$$

The constant and linear terms vanish by the matching moments. Thus $\mathrm{Var}_P(\log T) < \mathrm{Var}_Q(\log T) = \psi_1(p)$. Differentiation of $\log M$ gives exactly this variance difference. □

Theorem 5.35 (Strict envelope log-concavity)**.** *Put $D(b) = C(b) - 1$, with C defined in Theorem 5.32. For every $b > 0$,*

$$D(b) > 0, \qquad (\log D)''(b) < 0.$$

Consequently $D(b)^2 > D(b-h)D(b+h)$ for $0 < h < b$. Moreover $2^b D(b)$ increases strictly from zero to one.

Proof. Subtract the ordinary gamma integral for one from the axis integral:

$$D(b) = \frac{1}{\Gamma(b)} \int_0^\infty T^{b-1} f(T)\, dT, \qquad f(T) = \frac{e^{-T}(1-e^{-T})}{2\cosh T}.$$

Here $f(T) = T/2 + O(T^2)$ near zero, $f(T) < e^{-2T}$ at infinity, and

$$(\log f)''(T) = -\frac{e^T}{(e^T-1)^2} - \operatorname{sech}^2 T < 0.$$

All differentiated integrals are locally dominated on $b > 0$, so the lemma applies. Strict midpoint concavity gives the Turan inequality. The endpoint limits $D(b) \to 0$ at zero and infinity also give a second proof of a unique nondegenerate maximum: at its stationary point $D'' = D(\log D)'' < 0$.

For the final assertion let T_b have gamma shape b and rate two. Then

$$2^b D(b) = \mathbb{E}\frac{1 - e^{-T_b}}{1 + e^{-2T_b}}.$$

The displayed function increases strictly from zero to one. Gamma addition gives strict increase with b. The gamma laws tend to zero and infinity in probability at the respective parameter endpoints; bounded convergence proves the limits. □

For example, the Turan inequality at $b = 2, h = 1$ gives the explicit constant comparison

$$\left(G + \frac{\pi^2}{48} - 1\right)^2 > \left(\frac{\pi}{4} + \frac{\log 2}{2} - 1\right)\left(\frac{\pi^3}{32} + \frac{3\zeta(3)}{32} - 1\right).$$

5.10.1 Outward rational enclosure of the axis maximum

Theorem 5.36 (Certified location and height)**.** *The unique maximum satisfies the rational inequalities*

$$1.302216587101241209592371 70 < b_* < 1.302216587101241209592371 71,$$
$$1.136561103339509560952586375779942387681723540 < C(b_*)$$
$$C(b_*) < 1.136561103339509560952586375779942387681723541.$$

In particular $C(b_*) < 57/50$.

Proof. The incoming standard-library verifier `code/certify_gaussian_envelope.py` was replayed on an isolated copy. We give the analytic budgets used in its finite certificate, so its transcendental enclosures have explicit proof premises. For $d = 1, 2$, put

$$A_d(b) = \sum_{j\ge 0} \frac{(-1)^j}{(dj+1)^b}, \qquad T_{d,N}(b) = 2^{-N} \sum_{j=0}^{N-1} \frac{(-1)^j}{(dj+1)^b} \sum_{h=j+1}^{N} \binom{N}{h}.$$

These give $A_1 = \eta$ and $A_2 = \beta$. For T with gamma shape b and rate one, finite geometric summation gives

$$A_d - T_{d,N} = 2^{-N}\mathbb{E}\frac{(1 - e^{-dT})^N}{1 + e^{-dT}} \in (0, 2^{-N}).$$

Differentiating the gamma density introduces the centered score $\log T - \psi(b)$. Cauchy–Schwarz and $\sqrt{\psi_1(b)} \le \pi/\sqrt{6} < 2$ for $b \ge 1$ yield $|A'_d - T'_{d,N}| < 2^{1-N}$. For $C_N = T_{2,N} + 2^{-b}T_{1,N}$ this proves

$$0 < C - C_N \le (1 + 2^{-b})2^{-N}, \qquad |C' - C'_N| < 4\,2^{-N} \quad (1 \le b \le 2).$$

The finite terms are enclosed on an outward-rounded 10^{90} grid. After reducing an integer logarithm to $1 \le r \le 2$, its expansion

$$\log r = 2\sum_{j=0}^{M-1} \frac{v^{2j+1}}{2j+1} + R_M, \quad v = \frac{r-1}{r+1}, \quad 0 \le R_M \le \frac{2v^{2M+1}}{(2M+1)(1-v^2)}$$

uses $M = 160$. Exponentials are scaled to $[0, 1]$, bounded by their positive Taylor polynomial through degree 180 and twice the first omitted term, then recovered by four outward-rounded squarings and positive reciprocals. Multiplication by the logarithm gives the derivative enclosure with the same rational arithmetic.

With $N = 160$, the resulting certificate proves $C'(\ell) > 0$ and $C'(u) < 0$ at the displayed location endpoints. Its separate checks give $C'(1) > 0.03260$ and $C'(2) < -0.03561$. Strict log-concavity of D provides the tangent upper bound

$$D(b_*) \le D(\ell)\exp\left(\frac{C'(\ell)}{D(\ell)}(u - \ell)\right).$$

The outward upper endpoint of this expression and the lower bound $C(\ell)$ give the displayed value interval. The fresh exact receipt is `verification/fifth-replay/extremal/results/gaussian_envelope_certificate.json`. The final comparison with $57/50$ is rational. No numerical optimizer or floating-point library evaluation is a premise. □

The old plotted stationary-point marker remains a diagnostic. The enclosures above now certify its location and the axis height to the displayed widths; they do not certify every plotted value. The further inverse generalized-power expansion in the incoming report is retained as a separate analytic integration task.

5.11 Cartesian motion of the complete real-order zero arc

The normalized radial motion can have different signs above the critical line, but its Cartesian abscissa has one direction throughout the entire nonnegative-outer-order domain.

Theorem 5.37 (Strict outward abscissa motion)**.** *For every $a \geq 0, b > 0$, put $s = \rho^2$ and $X_{a,b}(s) = \rho\cos\theta_{a,b}(\rho)$. The unique upper-arc zero has a real-analytic abscissa on $(0,1]$, extending real analytically through zero, with*

$$0 < X_{a,b}(s) < s, \qquad X'_{a,b}(s) > 0 \quad (0 < s \leq 1).$$

This is an abscissa theorem; it does not assert one direction for $\cos\theta/\rho$ in every supercritical parameter pair.

Proof. For $a > 0, a + b > 1$, let ν be the finite signed measure of Theorem 5.17. Set

$$G(X,s) = \int_0^1 \frac{d\nu(u)}{D(u)}, \qquad D(u) = 1 - 2Xu + su^2.$$

The angular root satisfies $G = 0$ and $0 < X < s \leq 1$. At that root the measure $d\tau = d\nu/D$ has zero total mass and the same ordered signs. The sign test gives

$$G_X = 2\int \frac{u}{D}\,d\tau > 0, \qquad \left(\frac{u}{D}\right)' = \frac{1-su^2}{D^2} > 0,$$
$$G_s = -\int \frac{u^2}{D}\,d\tau < 0, \qquad \left(\frac{u^2}{D}\right)' = \frac{2u(1-Xu)}{D^2} > 0.$$

Thus $X' = -G_s/G_X > 0$. At $(X,s) = (0,0)$, $G = 0$ and $G_X = 2c_2 > 0$, proving analytic extension by the implicit-function theorem.

For $a + b \leq 1$ and on the full axis, the right-endpoint factorization gives $X = s\eta(\sqrt{s})$, with $\eta > 0$ and $\partial_s\eta > 0$ by Theorem 5.27. Its positive-measure equation has derivative $G_m = \omega([0,1]) > 0$ at $s = 0$, so m and $\eta = (1+m)/2$ are analytic there. This proves the same derivative conclusion. In every case the root is away from the cut at $s = 1$, and all denominators are locally uniformly nonzero; the analytic implicit argument also applies there. □

5.11.1 The opposite endpoint mechanism in the large-inner-order limit

For fixed $a > 0$, the disk coefficients tend as $b \to \infty$ to $c_1 = 0$, $c_n = n^{-a}$ for $n \geq 2$. The limit function is $F_{a,\infty}(z) = \mathrm{Li}_a(z) - z$. Its signed measure is the positive Gamma moment law minus a unit atom at zero. Thus its normalized-radius motion has the opposite direction from the right-endpoint case.

Theorem 5.38 (Strict decrease for the left-endpoint limit)**.** *For every $a > 0$, the unique upper-arc zero of $\operatorname{Im} F_{a,\infty}$ satisfies*

$$\arccos(\rho/2) < \theta_{a,\infty}(\rho) < \pi/2, \qquad 0 < \eta_{a,\infty}(\rho) < \tfrac{1}{2}, \qquad \eta'_{a,\infty}(\rho) < 0$$

for $0 < \rho \leq 1$.

Proof. Let $U = e^{-Y}$, $Y \sim \mathrm{Gamma}(a, 1)$, and let $d\omega(u) = u\, d\Pr(U \le u)$. The measure has strictly positive density on $(0, 1)$ and mass 2^{-a}. Moment expansion gives

$$F_{a,\infty}(z) = z^2 \int_0^1 \frac{d\omega(u)}{1 - zu}.$$

Set $x = \rho^2$, $m = 2\cos\theta/\rho$, and

$$D(u) = 1 + x(u^2 - mu), \qquad G(m, x) = \int_0^1 \frac{m - u}{D(u)}\, d\omega(u).$$

Direct rationalization shows that the imaginary part has the sign of G. We have $G(0, x) < 0 < G(1, x)$, and $G_m = \int D^{-2}\, d\omega > 0$. Outside $0 < m < 1$ the numerator has a fixed sign. Thus the zero is unique and simple on the full upper arc. At the root, differentiation gives

$$G_x = \int \frac{u(u - m)^2}{D(u)^2}\, d\omega(u) > 0, \qquad \frac{dm}{dx} = -\frac{\int u(u - m)^2 D(u)^{-2}\, d\omega(u)}{\int D(u)^{-2}\, d\omega(u)} < 0.$$

All denominators are uniformly positive at $0 < m < 1$, $0 \le x \le 1$. Hence $\eta'(\rho) = \rho m'(\rho^2) < 0$, proving the claims. The theorem concerns the exact limit family, not an unproved transfer of its radial sign to every finite inner order. □

5.12 Critical defects and the emergence of the atom

Theorem 5.39 (Completely monotone critical defects)**.** *Let $a = 1 - b \in (0, 1)$ and*

$$D_n = \frac{1}{a} - \frac{H^{(b)}_{n-1}}{n^a}, \qquad n \ge 1.$$

Then $D_n = \int_0^1 u^{n-1}\, \mathrm{d}\sigma_{a,b}(u)$ where $d\sigma_{a,b}(u) = -K_{1-b,b}(-\log u)\, du$ is the strictly positive critical negative part. If $\Delta D_n = D_{n+1} - D_n$, then $(-1)^k \Delta^k D_n > 0$ for every $k \ge 0, n \ge 1$. Moreover,

$$-\zeta(b)n^{-a} + \frac{1}{2n} < D_n < -\zeta(b)n^{-a} + \frac{1}{n}. \tag{5.75}$$

Every finite Hankel matrix $(D_{j+k+1})_{j,k=0}^{r}$ is positive definite, and $D_n D_{n+2} > D_{n+1}^2$.

Proof. Subtract the critical signed-measure moments from its atomic mass. Then

$$(-1)^k \Delta^k D_n = \int_0^1 u^{n-1}(1 - u)^k\, \mathrm{d}\sigma(u) > 0.$$

On $w = 1$, $k_w(T) = 1$, so (5.48) reads $1/2 < C_{a,b}(T) < 1$. Integrating these inequalities against $e^{-nT}\, \mathrm{d}T$ gives (5.75). For a nonzero real vector $(v_0, \dots, v_r)$,

$$\sum_{j,k=0}^{r} v_j v_k D_{j+k+1} = \int_0^1 \left(\sum_{j=0}^{r} v_j u^j \right)^2 \mathrm{d}\sigma(u) > 0,$$

since a nonzero polynomial cannot vanish on an interval and the density is positive everywhere inside $(0, 1)$. Strict log-convexity follows either from its 2×2 minors or from strict Cauchy–Schwarz with the weight $u^{n-1}\, \mathrm{d}\sigma$. □

The theorem also gives a continuous version. For $q > 0$,

$$D(q) = \frac{1}{a} - q^{-a}\big(\zeta(b) - \zeta(b,q)\big) = \int_0^\infty e^{-qT}\big[-\zeta(b)k_a(T) + C_{a,b}(T)\big]\,\mathrm{d}T. \tag{5.76}$$

Thus $(-1)^j D^{(j)}(q) > 0$ for all integers $j \geq 0$. All derivatives are justified by exponential domination. These are inequalities for specific harmonic/Hurwitz expressions, not claims about independence of their special values.

5.13 Elementary critical-kernel identities and Hurwitz inequalities

In this section $0 < b < 1$, $a = 1 - b$, and $C_b(T) = C_{1-b,b}(T) = \mathbb{E}[q(TV)]$ with $V \sim \mathrm{Beta}(b, 1-b)$. In particular $1/2 < C_b(T) < 1$ and C_b is strictly increasing.

Theorem 5.40 (An elementary convergent series)**.** *For every $T > 0$,*

$$C_b(T) = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{\sin\big(b\arctan(T/(2\pi k))\big)}{\pi k\big(1 + (T/(2\pi k))^2\big)^{b/2}}. \tag{5.77}$$

Every summand is positive. The tail after $K \geq 1$ terms is bounded by

$$0 < C_b(T) - \frac{1}{2} - \sum_{k=1}^{K}(\mathit{summand}) \leq \frac{bT}{2\pi^2 K}. \tag{5.78}$$

For $|T| < 2\pi$ it also has the convergent expansion

$$C_b(T) = \frac{1}{2} + \sum_{j=1}^{\infty} \frac{B_{2j}(b)_{2j-1}}{(2j)!(2j-1)!} T^{2j-1}, \tag{5.79}$$

where $(b)_m$ is the rising factorial and B_{2j} is a Bernoulli number, not a Bernoulli polynomial evaluated at b.

Proof. The classical partial fraction for coth gives

$$q(t) = \frac{1}{2} + 2t\sum_{k=1}^{\infty} \frac{1}{t^2 + (2\pi k)^2}. \tag{5.80}$$

This follows from DLMF 4.36.3 after rescaling [97]. The summands are positive for $t > 0$, so Tonelli permits averaging over V. For $|z| < 1$, the beta moments yield

$$\mathbb{E}\frac{1}{1 - zV} = \sum_{j \geq 0} z^j \frac{(b)_j}{j!} = (1-z)^{-b}.$$

Both sides are analytic for $z \in \mathbb{C} \setminus [1, \infty)$, so the same identity holds there. At $z = \mathrm{i}q$, its imaginary part is

$$\mathbb{E}\frac{qV}{1 + q^2V^2} = \frac{\sin(b\arctan q)}{(1+q^2)^{b/2}}.$$

Taking $q = T/(2\pi k)$ proves (5.77). Since $\sin(b\arctan q) \leq bq$ and the denominator factor is at least one, its tail is at most $bT(2\pi^2)^{-1}\sum_{k>K} k^{-2} \leq bT/(2\pi^2 K)$.

The usual Bernoulli expansion of (5.46) has radius 2π: $q(t) = 1/2 + \sum_{j\geq 1} B_{2j}t^{2j-1}/(2j)!$. For $|T| < 2\pi$ it is uniformly absolutely convergent for $0 \leq V \leq 1$. Average termwise and use $\mathbb{E}V^{2j-1} = (b)_{2j-1}/(2j-1)!$ to obtain (5.79). □

For instance,

$$C_{1/2}(T) = \frac{1}{2} + \frac{T}{24} - \frac{T^3}{2304} + \frac{T^5}{122880} + O(T^7). \tag{5.81}$$

The companion program regenerates exact coefficients through degree 13 at $b = 1/10, 1/2, 9/10$ by two algebraically equivalent constructions. The numerical tests of (5.77) use independent beta quadrature, with the analytic tail bound displayed separately.

Theorem 5.41 (A completely monotone normalized Hurwitz expression)**.** *For $0 < b < 1$ and $q > 0$, define*

$$L_b(q) = q^{b-1}\zeta(b, q) + \frac{1}{1-b}. \tag{5.82}$$

Then

$$L_b(q) = \int_0^\infty e^{-qT} C_b(T)\, \mathrm{d}T, \qquad \frac{1}{2q} < L_b(q) < \frac{1}{q}. \tag{5.83}$$

In particular, $(-1)^j L_b^{(j)}(q) > 0$ for all $j \geq 0$. The function

$$qL_b(q) = q^b\zeta(b, q) + \frac{q}{1-b} \tag{5.84}$$

decreases strictly from 1 to 1/2 as q increases from zero to infinity.

Proof. Formula (5.83) is (5.49) with $a = 1 - b$. Bounds and strict complete monotonicity follow from $1/2 < C_b < 1$ and differentiating the positive Laplace integral. Substitution $T = s/q$ gives

$$qL_b(q) = \int_0^\infty e^{-s} C_b(s/q)\, \mathrm{d}s.$$

Strict increase of C_b proves strict decrease with q. The endpoint limits follow by dominated convergence and the limits $C_b(0+) = 1/2$, $C_b(\infty) = 1$. □

The beta average is the common source of the elementary series and the Hurwitz inequalities. The inequalities, finite-difference signs, and Hankel determinants are analytic consequences of a positive integral, not numerical pattern extrapolations.

5.14 A proved local part of the normalized-radius conjecture

The earlier *Angular Zero Continuation* report [94] asks whether

$$\eta_{a,b}(\rho) = \frac{\cos\theta_{a,b}(\rho)}{\rho}$$

decreases on the whole interval $0 < \rho \leq 1$ for every integer $a \geq 2$, $b > 0$. At $a = 1$ it predicts decreasing behavior for $b > 1$, increasing behavior for $0 < b < 1$, and the constant $1/2$ for $b = 1$. That report identifies the sign of the first nonconstant Taylor coefficient as a smaller explicit open problem. We solve that problem for the full stated parameter range and describe the intervening fractional-order threshold.

Throughout this section put

$$U = 1 + 2^{-b}, \qquad V = 1 + 2^{-b} + 3^{-b}, \qquad W = 1 + 2^{-b} + 3^{-b} + 4^{-b}.$$

The parameter ρ in this section is a disk radius. It is unrelated to the Lerch deformation parameter used later.

Lemma 5.42 (The local radial coefficient)**.** *For fixed $a \geq 1$, $b > 0$, $\eta_{a,b}$ extends as a real analytic even function near $\rho = 0$, and*

$$\eta_{a,b}(\rho) = \frac{U}{2}\left(\frac{2}{3}\right)^a + K(a,b)\rho^2 + O_{a,b}(\rho^4), \tag{5.85}$$

where

$$K(a,b) = UV\,3^{-a} - \frac{W}{2}\left(\frac{2}{5}\right)^a - \frac{U^3}{2}\left(\frac{8}{27}\right)^a. \tag{5.86}$$

Proof. Set $\delta = \pi/2 - \theta$ and divide $\operatorname{Im} F_{a,b}(\rho e^{i\theta})$ by $2^{-a}\rho^2$. The resulting equation is analytic near $(\rho, \delta) = (0,0)$ and starts with $\sin(2\delta)$; its δ derivative there is 2. The implicit-function theorem gives a unique analytic solution, odd under $(\rho, \delta) \mapsto (-\rho, -\delta)$. Writing $\delta = \alpha\rho + \beta\rho^3 + O(\rho^5)$, the modes with outer indices $2, 3, 4, 5$ give

$$\alpha = \frac{U}{2}(2/3)^a, \qquad \beta = UV\,3^{-a} - \frac{W}{2}(2/5)^a - \frac{23U^3}{48}(8/27)^a.$$

All higher modes enter only at order ρ^5 or later in the normalized equation. Since $\eta = \sin\delta/\rho$, its quadratic coefficient is $\beta - \alpha^3/6$, which is (5.86). This also rederives the earlier report's local coefficient without assuming its sign. □

Theorem 5.43 (Complete sign in the conjectured local range)**.** *The coefficient* (5.86) *satisfies*

1. *$K(a,b) < 0$ for every real $a \geq 2$ and every $b > 0$;*
2. *$K(1,b) < 0$ if $b > 1$, $K(1,b) > 0$ if $0 < b < 1$, and $K(1,1) = 0$;*
3. *for $b \geq 1$, $K(a,b) < 0$ whenever $a > 1$;*
4. *for each $0 < b < 1$, there is exactly one $a_*(b) \in (1,2)$ with $K(a_*(b), b) = 0$. The coefficient is positive on $1 \leq a < a_*(b)$ and negative for $a > a_*(b)$.*

The function $a_*(b)$ is real analytic on $(0,1)$. Every strict coefficient sign gives the corresponding strict monotonicity of $\eta_{a,b}$ on some interval $0 < \rho < \rho_0(a,b)$. No all-radius monotonicity claim is made here.

5.14.1 Monotonicity in the outer order

Define

$$G_a = W + U^3\left(\frac{20}{27}\right)^a - 2UV\left(\frac{5}{6}\right)^a.$$

Then $K(a,b) = -\frac{1}{2}(2/5)^a G_a$, so it suffices to determine the sign of G_a. We first show

$$\frac{\partial G_a}{\partial a} > 0 \qquad (a \geq 1, b > 0). \tag{5.87}$$

Put $x = 2^{-b}$ and $y = 3^{-b}$. Since $y \geq x^2$ and $0 < x < 1$,

$$\frac{U^2}{V} \leq \frac{(1+x)^2}{1+x+x^2} \leq \frac{4}{3}.$$

The last inequality is equivalent to $(1-x)^2 \geq 0$. Elementary alternating-series bounds for $\log(1+t)$ give

$$\log(27/20) < \frac{7273}{24000}, \qquad \log(6/5) > \frac{9}{50}.$$

Consequently

$$\frac{U^2}{2V}\frac{\log(27/20)}{\log(6/5)} < \frac{7273}{6480} < \frac{9}{8} \leq \left(\frac{9}{8}\right)^a.$$

This is precisely the inequality obtained by differentiating G_a and moving its negative term to the other side. It proves (5.87).

5.14.2 A finite positive certificate at $a = 2$

Because $\log_2 3 > 3/2$, $y \leq x^{3/2}$. The coefficient of y in G_2 is $-(7+25x)/18 < 0$. With $x = t^2$, $0 < t < 1$, we obtain

$$G_2 \geq P_6(t) := \frac{800t^6 - 2025t^5 + 1833t^4 - 567t^3 - 192t^2 + 233}{1458}. \tag{5.88}$$

This polynomial has the degree-six Bernstein representation

$$P_6(t) = \sum_{j=0}^{6} b_j \binom{6}{j} t^j (1-t)^{6-j},$$

with the exact coefficient vector

$$(b_0, \ldots, b_6) = \left(\frac{233}{1458}, \frac{233}{1458}, \frac{367}{2430}, \frac{665}{5832}, \frac{55}{486}, \frac{95}{1458}, \frac{41}{729}\right). \tag{5.89}$$

Every coefficient is positive, and the Bernstein basis is nonnegative and sums to one on $[0,1]$. Hence $G_2 > 0$. Together with (5.87), this proves the first assertion of the theorem. The accompanying verifier expands (5.89) over $\mathbb{Q}$ and checks equality with (5.88); no floating-point polynomial sign test is used.

5.14.3 The sign change at $a = 1$, $b = 1$

A direct simplification gives

$$27G_1 = 20x^3 + 42x^2 - 3x + 2 - (45x + 18)y. \tag{5.90}$$

If $b > 1$, let $t = \sqrt{2x} \in (0,1)$. Since $y = (2x)^{\log_2 3}/3 \leq t^3/3$, substitution in (5.90) yields

$$G_1 \geq (1-t)P_5(t), \qquad P_5(t) = \frac{-5t^5 + 10t^4 - 11t^3 + t^2 + 4t + 4}{54}.$$

The degree-five Bernstein coefficients of P_5 are

$$\left(\frac{2}{27}, \frac{4}{45}, \frac{19}{180}, \frac{14}{135}, \frac{1}{10}, \frac{1}{18}\right). \tag{5.91}$$

They are all positive, so $G_1 > 0$ for $b > 1$.

Suppose instead $0 < b < 1$, so $1/2 < x < 1$. Put $p = \log_2 3$. The exact integer inequality $2^{11} < 3^7$ gives $p > 11/7$, while $p < 2$. For $y(x) = x^p$ this implies

$$y(1/2) = \frac{1}{3}, \quad y'(1/2) > \frac{22}{21}, \quad y''(x) = p(p-1)x^{p-2} > \frac{44}{49} \quad (1/2 < x < 1).$$

Taylor's formula with an integral remainder therefore gives

$$y \geq \frac{1}{3} + \frac{22}{21}(x - 1/2) + \frac{22}{49}(x - 1/2)^2.$$

Using the negative coefficient of y in (5.90), we find

$$G_1 \leq -\frac{(2x-1)(10x^2 - 337x + 334)}{2646} < 0.$$

Indeed the quadratic is decreasing on $[1/2, 1]$ and its value at one is $7 > 0$. At $b = 1$, direct substitution $x = 1/2$, $y = 1/3$ gives $G_1 = 0$. The exact identity $F_{1,1}(z) = \frac{1}{2}\log^2(1 - z)$ gives $\eta_{1,1}(\rho) = 1/2$ for every $0 < \rho \leq 1$.

Finally, G_a is strictly increasing in $a \geq 1$. The signs of G_1 and G_2 give all the remaining assertions and the unique threshold. Its analyticity follows from the analytic implicit-function theorem and (5.87). Differentiating (5.85) shows that a nonzero K controls the sign of η' for sufficiently small positive ρ.

Remark 5.44 (What remains of the conjecture). This proves the local sign statement explicitly requested in the earlier report, including all real $a \geq 2$. It does not show that a derivative cannot change sign farther from the origin. At the fractional threshold $a = a_*(b)$, the quadratic coefficient vanishes; its quartic successor is needed even for local monotonicity. The fractional-turning development below proves opposite quartic signs at two inner orders, producing both maxima and minima. The next section proves the complete $a = 1$ full-radius branch. The higher-integer global problem and the general quartic-sign classification require separate arguments; the low-order local coefficient does not exclude later changes of radial direction.

5.15 Global normalized-radius motion at leading index one

The local $a = 1$ sign extends to the full disk. Here the zero curve is defined by the imaginary part; unrestricted slit-plane nonvanishing was already proved above. The continuation [103] supplies a broader Green-kernel theorem, an outer-radius cutoff, moment inequalities, and exact sums over inner orders. The proof needed for the present assertion follows here.

Theorem 5.45 (The complete leading-index-one branch)**.** *Let $b > 0$, and let $\theta_{1,b}(\rho)$ be the unique upper-arc zero of $\operatorname{Im} F_{1,b}(\rho e^{i\theta})$ for $0 < \rho \leq 1$. Then $\eta_b(\rho) = \cos\theta_{1,b}(\rho)/\rho$ is strictly decreasing for $b > 1$, strictly increasing for $0 < b < 1$, and identically $1/2$ for $b = 1$. Its analytic even extension at zero has value $\eta_b(0) = (1 + 2^{-b})/3$.*

Define

$$w_b(u) = \frac{(-\log u)^{b-1}}{\Gamma(b)},$$
$$Q_b(u) = (1-u)\int_0^u \frac{w_b(t)}{1-t}\,dt + u\int_u^1 \frac{w_b(t)}{t}\,dt. \tag{5.92}$$

The ratio $\min(u,t)(1-\max(u,t))/(t(1-t))$ lies in $[0,1]$. Dominated convergence therefore extends Q_b continuously to zero at both endpoints. Its first derivative is integrable, since each positive term in

$$Q_b'(u) = \int_u^1 \frac{w_b(t)}{t}\,dt - \int_0^u \frac{w_b(t)}{1-t}\,dt$$

has integral 1. Thus Q_b is absolutely continuous, positive inside, and

$$Q_b''(u) = -\frac{w_b(u)}{u(1-u)} < 0. \tag{5.93}$$

It is strictly log-concave as well. Direct integration of the Green kernel gives

$$\int_0^1 u^j Q_b(u)\,du = \frac{H_{j+1}^{(b)}}{(j+1)(j+2)}.$$

Hence coefficient comparison and integration by parts establish

$$F_{1,b}(z) = z^2 \int_0^1 \frac{Q_b(u)}{(1-zu)^2}\,du = z\int_0^1 \frac{-Q_b'(u)}{1-zu}\,du \tag{5.94}$$

on the principal slit plane. The mean of the probability density $2Q_b$ is $\mu_b = (1+2^{-b})/3$.

Lemma 5.46 (A unique nondegenerate Cauchy mode)**.** *For every $y>0$, the function $A_y(x) = \int_0^1 Q_b(u)((x-u)^2+y^2)^{-1}\,du$ has one stationary point $m_b(y)\in(0,1)$, and $A_y''(m_b(y))<0$.*

Proof. With $K(v) = (v^2+y^2)^{-1}$, A_y' is positive on $x\le 0$ and negative on $x\ge 1$. At any stationary point x, the signed integrand $Q_b(x-v)K'(v)$ has zero integral, positive sign for $v<0$, and negative sign for $v>0$. The score $S(v) = Q_b'(x-v)/Q_b(x-v)$ is strictly increasing by strict log-concavity. Integration by parts, then subtraction of $S(0)$, gives

$$A_y''(x) = \int_{x-1}^{x} Q_b(x-v)K'(v)(S(v)-S(0))\,dv < 0.$$

The integrability of Q_b' and the zero endpoint values justify the calculation even if endpoint scores diverge. Every stationary point is consequently a strict downward crossing of A_y', so there is only one. The implicit-function theorem makes m_b real analytic. □

Lemma 5.47 (Reflected curvature controls the height derivative)**.** *Put*

$$I(x,t) = \int_0^1 \frac{(u-x)Q_b(u)}{((u-x)^2+t)^2}\,du.$$

At its zero $x = m_b(\sqrt{t})$, $I_t>0$ for $b>1$ and $I_t<0$ for $0<b<1$. For $b=1$ the zero is $x=1/2$ for all t.

Proof. Suppose $b>1$, so w_b is strictly decreasing. Extend Q_b by zero outside $[0,1]$, and let $\Delta_x(v) = Q_b(x+v) - Q_b(x-v)$ for $v\ge 0$. At $x=1/2$, it is strictly convex between two zero endpoint values, because

$$\Delta_{1/2}''(v) = \frac{w_b(1/2-v) - w_b(1/2+v)}{1/4-v^2} > 0.$$

Thus $I(1/2,t)<0$, and the unique mode lies below $1/2$. For $0<x<1/2$ and $0<v<x$,

$$\Delta_x''(v)=\frac{w_b(x-v)}{(x-v)(1-x+v)}-\frac{w_b(x+v)}{(x+v)(1-x-v)}>0.$$

The first numerator is larger and its denominator smaller. Therefore $\Delta_x(v)/v$ is strictly increasing on $(0,x)$. Also $\Delta_x(x)=Q_b(2x)>0$, and $\Delta_x(v)>0$ on $x<v<1-x$. At the mode, the paired equation

$$I(x,t)=\int_0^\infty \frac{v\Delta_x(v)}{(v^2+t)^2}\,dv=0$$

forces a negative part, so Δ_x has exactly one negative-to-positive crossing $v_0\in(0,x)$. Consequently

$$I_t=\int_0^\infty \frac{v\Delta_x(v)}{(v^2+t)^2}\left(-\frac{2}{v^2+t}+\frac{2}{v_0^2+t}\right)dv>0.$$

For $0<b<1$, reflect the increasing weight w_b under $u\mapsto 1-u$ and apply the same argument to the resulting decreasing weight; the sign reverses. At $b=1$ the density is symmetric, so the mode is $1/2$. □

Proof of Theorem 5.45. For $z=(x-\mathrm{i}y)^{-1}$, (5.94) gives

$$\operatorname{Im} F_{1,b}(z)=-2yI(x,y^2).$$

Thus the upper-half-plane imaginary zero has $x=m_b(y)$. To convert this to disk radius, set $s=\rho^2$ and $\mathcal{I}(x,s)=I(x,s^{-1}-x^2)$. At the angular zero $x=\eta_b(\sqrt{s})$.

The signed density $\kappa=-Q_b'$ has zero mass and one strict negative-to-positive crossing. Put

$$J_\rho(c)=\int_0^1 \frac{\kappa(u)}{1-2\rho cu+\rho^2u^2}\,du.$$

At its unique zero, $J_\rho'(c)>0$: the signed measure obtained by multiplying κ by the positive denominator reciprocal still has zero mass and one crossing, and the derivative score $2\rho u/(1-2\rho cu+\rho^2u^2)$ is strictly increasing in u for $0<\rho\le 1$. Subtraction of its value at the crossing proves the sign. The root lies in $0<c<\rho$, since $J_\rho(0)<0<J_\rho(\rho)$; the upper endpoint at $\rho=1$ is interpreted by monotone convergence of its positive part.

Since $\operatorname{Im} z=sy$, comparison of the two integral formulas gives

$$J_{\sqrt{s}}(\sqrt{s}\,x)=-\frac{2}{s}\mathcal{I}(x,s),\qquad \mathcal{I}_x<0 \text{ at the root.}$$

At fixed x, $\mathcal{I}_s=-s^{-2}I_t$. Hence

$$\frac{d}{ds}\eta_b(\sqrt{s})=\frac{s^{-2}I_t}{\mathcal{I}_x}. \tag{5.95}$$

Lemma 5.47 gives all strict signs. Symmetry gives $\eta_1=1/2$. The local analytic extension and its value at zero follow either from the existing local coefficient calculation or, after substituting $c=\rho\eta$, from $J/\rho^2=\eta-\mu_b+O(\rho^2)$. □

Corollary 5.48 (The local coefficient is a central moment)**.** *Let U_b have density $2Q_b$ and mean μ_b. Then*

$$K(1,b)=-2\mathbb{E}(U_b-\mu_b)^3.$$

More generally, every odd central moment of order $2m+1$, $m\ge 1$, has the sign of $b-1$.

Proof. At $x = \mu_b$, the mean identity gives $\int_0^\infty v\Delta_x(v)\,dv = 0$. For $b > 1$, the reflected-curvature argument gives exactly one negative-to-positive crossing v_0. Multiplication by the increasing factor $v^{2m} - v_0^{2m}$ shows $\int v^{2m+1}\Delta_x(v)\,dv > 0$. Reflection handles $b < 1$ and symmetry handles $b = 1$. The raw moments from (5.92) yield

$$\mathbb{E}(U_b - \mu_b)^3 = \frac{H_4^{(b)}}{10} - \frac{H_2^{(b)}H_3^{(b)}}{6} + \frac{2(H_2^{(b)})^3}{27},$$

which agrees with $-K(1, b)/2$. □

Remark 5.49 (Scope). The theorem resolves the full-radius branch $a = 1$, not the remaining higher-outer-order problem. Monotonicity should also not be replaced by a uniform convexity claim: the coefficient of ρ^4 in η_2 is $-263/777600$, whereas in η_4 it is $504047/3149280000$. The separate article proves these values and provides exact verification artifacts.

The isolated exact replay certifies 1,938 finite coefficient and moment checks and eighteen rational angular brackets, including fractional inner orders 1/4 and 1/2; the separate symbolic replay checks five identities. These catch finite algebra and transcription errors. The continuum monotonicity theorem rests on the Green-density and single-crossing proof above, not on the historical quadrature grid. Fresh receipts are in `verification/seventh-replay/leading-one/`.

5.16 A proved failure of fractional global monotonicity

The local sign threshold does not extend directly to a classification of radial monotonicity on the whole unit disk. There are fractional orders for which $\eta_{a,b}(\rho) = \cos\theta_{a,b}(\rho)/\rho$ first increases and then decreases. This is an analytic theorem, supported by a finite exact certificate for its hypotheses.

5.16.1 The next local coefficient

Put $p_n = H_{n-1}^{(b)}/n^a$ and

$$\mu = \frac{p_3}{2p_2}, \qquad K = -\frac{4p_3\mu^2 - 4p_4\mu + p_5}{2p_2}.$$

Thus $\mu = \eta_{a,b}(0)$ and K is the previously identified quadratic coefficient. The next coefficient is

$$Q = -\frac{(8p_3\mu - 4p_4)K + 8p_4\mu^3 - 12p_5\mu^2 + 6p_6\mu - p_7}{2p_2}, \tag{5.96}$$

and the convergent local expansion is

$$\eta_{a,b}(\rho) = \mu + K\rho^2 + Q\rho^4 + O(\rho^6). \tag{5.97}$$

The coefficients and remainder are real analytic in a, b locally within the parameter range under consideration.

To check (5.96), write $c = \rho\eta$ and use $\sin(n\theta) = \sin\theta\, U_{n-1}(c)$. Division of the imaginary-part equation by $\rho^3 \sin\theta$ gives

$$\begin{aligned} 0 = {} & 2p_2\eta - p_3 + \rho^2(4p_3\eta^2 - 4p_4\eta + p_5) \\ & + \rho^4(8p_4\eta^3 - 12p_5\eta^2 + 6p_6\eta - p_7) + O(\rho^6). \end{aligned}$$

Every omitted mode has order at least ρ^6. This equation is analytic in $a, b, t = \rho^2, \eta$ near $t = 0$; it follows either from the convergent disk series or from the parity of each Chebyshev term. Its derivative in η at zero is $2p_2 > 0$. The analytic implicit-function theorem therefore gives (5.97), and substitution proves the formula for Q.

Theorem 5.50 (Emergence of a strict radial maximum)**.** *Fix $b = 1/2$. There is a unique zero $a_\star$ of $K(a, 1/2)$ in the rational interval*

$$1.31298577512189 < a_\star < 1.31298577512190. \tag{5.98}$$

There exist $\delta > 0$ and $\rho_0 > 0$ such that, for every $a \in (a_\star - \delta, a_\star)$, the function $\eta_{a,1/2}(\rho)$ has exactly one critical point in $(0, \rho_0)$. It is a strict local maximum: the function increases before it and decreases after it within that interval. Writing this point as $\rho_{\max}(a)$, one has

$$\rho_{\max}(a) = \left(\frac{\partial_a K(a_\star, 1/2)}{2Q(a_\star, 1/2)}\right)^{1/2} (a_\star - a)^{1/2} + O((a_\star - a)^{3/2}). \tag{5.99}$$

The positive prefactor is rigorously enclosed by

$$44.0912731287 < \left(\frac{\partial_a K(a_\star, 1/2)}{2Q(a_\star, 1/2)}\right)^{1/2} < 44.0912738633.$$

In particular these fractional normalized-radius curves are not monotone on $(0, 1]$, despite having $K > 0$.

Proof. The accompanying standard-library verifier `certify_fractional_turning.py` uses only integer and rational arithmetic. Let I be the closed rational interval in (5.98). Its output proves

$$K(\inf I, 1/2) > 8.16 \cdot 10^{-17}, \qquad K(\sup I, 1/2) < -8.72 \cdot 10^{-17},$$

and, throughout I,

$$-0.016895831331 < \partial_a K(a, 1/2) < -0.016895831329,$$
$$-4.345546 \cdot 10^{-6} < Q(a, 1/2) < -4.345545 \cdot 10^{-6}.$$

These signs alone give existence and uniqueness of $a_\star$ in I, independently of the more general local-threshold theorem. That theorem also identifies it as the unique threshold in $(1, 2)$.

Here is the complete arithmetic basis of the certificate. For $2 \le n \le 7$, it encloses $\log n$ by the first 120 terms of

$$\log n = 2 \sum_{j \ge 0} \frac{z^{2j+1}}{2j+1}, \qquad z = \frac{n-1}{n+1},$$

with positive tail bounded by $2z^{241}/(241(1 - z^2))$. For a nonnegative rational x, the first 60 terms after the constant term of e^x have tail at most

$$\frac{x^{61}}{61!} \frac{1}{1 - x/62};$$

all arguments used here are less than three. Monotonicity encloses $e^{a \log n}$ at both ends of the interval arguments, and reciprocation encloses n^{-a}. The factors $j^{-1/2}$ in the harmonic numbers are bounded by exact integer-square-root operations. All arithmetic operations round outwards to multiples of 2^{-128}. Substitution in the formulas for K, Q and their elementary derivatives proves the displayed signs. Exact rational endpoints, including the tighter prefactor enclosure, are recorded in `fractional_turning_certificate.json`. The written tail bounds and integer inequalities make this a finite exact proof certificate; no floating-point sign decision is used.

Write $\eta_{a,1/2}(\rho) = \Phi(a,t)$, $t = \rho^2$, using the analytic function constructed above. Set $U(a,t) = \partial_t \Phi(a,t)$. Then

$$U(a_\star, 0) = 0, \qquad \partial_t U(a_\star, 0) = 2Q(a_\star, 1/2) < 0, \qquad \partial_a U(a_\star, 0) = \partial_a K(a_\star, 1/2) < 0.$$

The analytic implicit-function theorem gives a unique local solution $t = t(a)$ of $U(a,t) = 0$, with

$$t(a) = \frac{\partial_a K(a_\star, 1/2)}{2Q(a_\star, 1/2)}(a_\star - a) + O((a_\star - a)^2).$$

It is positive for $a < a_\star$ sufficiently close. On a sufficiently small fixed rectangle, $\partial_t U < 0$; at its upper t boundary, $U < 0$, while $U(a,0) = K(a,1/2) > 0$ for $a < a_\star$. Thus there is exactly one such positive critical point and the derivative $\partial_\rho \eta = 2\rho U(a,\rho^2)$ changes from positive to negative there. At the critical point its second derivative is $4t\partial_t U < 0$. Taking the positive square root of the expansion for $t(a)$ proves (5.99). □

5.16.2 Exploration beyond the existence theorem

A separate 70-digit diagnostic evaluates the defining disk series and its derivative at $b = 1/4, 1/2, 3/4$, at the local threshold and at offsets $-0.02, -10^{-4}, +0.02$, for radii $0.3, 0.7, 0.9, 0.98$. For each b, the offset -10^{-4} gives a positive derivative at 0.3 and a negative derivative at 0.98. Thus the same phenomenon appears numerically at all three sampled inner orders; the theorem above proves it at $b = 1/2$.

For the exact rational pair $a = 21/16$, $b = 1/2$, the diagnostic finds

$$\eta'(0.9) = 1.2937181281564777 \cdot 10^{-6}, \qquad \eta'(0.98) = -2.6349934635137804 \cdot 10^{-6}.$$

These two particular signs are numerical observations, not an extra certified instance of the theorem's unspecified interval in a. The disk-series truncation errors for F and zF' are bounded analytically by 10^{-50}, but rounding is not interval-enclosed. The existence theorem relies instead on the exact finite coefficient certificate above. Extending this argument to other b requires determining the sign of $Q(a_\star(b), b)$. The complementary minimum theorem below proves that this sign is not always negative.

5.16.3 A complementary minimum and a change of quartic sign

The sign of the quartic coefficient along the local threshold also changes. In particular, the negative sign at $b = 1/2$ cannot extend to every $0 < b < 1$. The following exact certificate exhibits the opposite local bifurcation at a rational inner order close to one.

Theorem 5.51 (Emergence of a strict radial minimum)**.** *Set $b_0 = 999/1000$. The unique local threshold $a_0 = a_\star(b_0)$ satisfies the rational enclosure*

$$1.00057049936559087278 < a_0 < 1.00057049936559087279. \tag{5.100}$$

Moreover,

$$Q(a_0, b_0) > 0, \qquad \partial_a K(a_0, b_0) < 0.$$

There exist $\delta > 0$ and $\rho_0 > 0$ such that, for every $a \in (a_0, a_0 + \delta)$, the function $\eta_{a,b_0}(\rho)$ has exactly one critical point in $(0, \rho_0)$. It is a strict local minimum: the function decreases before it and increases after it within this interval. Its location satisfies

$$\rho_{\min}(a) = \left(\frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)}\right)^{1/2} (a - a_0)^{1/2} + O((a - a_0)^{3/2}), \tag{5.101}$$

where the prefactor has the certified enclosure

$$17076.2396129683 < \left(\frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)}\right)^{1/2} < 17076.2396548521.$$

Thus fractional normalized-radius curves can also fail global monotonicity while their initial quadratic coefficient is negative. The theorem asserts a sufficiently small neighborhood; it gives no numerical lower bound on δ.

Proof. Let I be the closed rational interval in (5.100). The exact interval verifier `certify_fractional_minimum.py` proves

$$K(\inf I, b_0) > 4.08027 \cdot 10^{-23}, \qquad K(\sup I, b_0) < -1.29944 \cdot 10^{-22}.$$

Throughout I, it also proves

$$2.92779195 \cdot 10^{-11} < Q(a, b_0) < 2.92779198 \cdot 10^{-11},$$
$$-0.017074763285092 < \partial_a K(a, b_0) < -0.017074763285091.$$

The arithmetic uses the same outward dyadic intervals and rational logarithm/exponential remainder bounds as the maximum certificate. For this rational inner order, each j^{-b_0} is enclosed as the reciprocal of $e^{b_0 \log j}$, for $1 \le j \le 6$; each n^{-a} is enclosed in the same way, for $2 \le n \le 7$. All exponential arguments are less than three. Substitution into the finite formulas for K, Q and $\partial_a K$ establishes the signs. Exact rational endpoints and the sharper prefactor interval are in `fractional_minimum_certificate.json`. Thus both the small positive quartic coefficient and the much smaller endpoint values of K are certified without floating-point arithmetic.

Write $\eta_{a,b_0}(\rho) = \Phi(a, t)$ with $t = \rho^2$, and let $U = \partial_t \Phi$. At $(a_0, 0)$ one has

$$U = 0, \qquad U_t = 2Q(a_0, b_0) > 0, \qquad U_a = \partial_a K(a_0, b_0) < 0.$$

The analytic implicit-function theorem gives a unique local zero $t = t(a)$ with

$$t(a) = \frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)}(a - a_0) + O((a - a_0)^2).$$

It is positive for $a > a_0$ sufficiently close. On a sufficiently small fixed rectangle, $U_t > 0$; hence U has exactly one zero in the indicated positive t interval and changes from negative to positive there. Since $\eta_\rho = 2\rho U$, this is the asserted minimum, with $\eta_{\rho\rho} = 4tU_t > 0$ at the critical point. Taking its positive square root proves (5.101). □

Corollary 5.52 (A zero of the quartic coefficient along the threshold)**.** *There exists $b_\dagger \in (1/2, 999/1000)$ for which*

$$Q(a_\star(b_\dagger), b_\dagger) = 0.$$

Proof. The threshold equation has nonzero a derivative at every $a_\star(b)$, so $a_\star(b)$ is real analytic on $(0, 1)$. The finite formula for Q then makes $q(b) = Q(a_\star(b), b)$ real analytic there. The exact maximum and minimum certificates give $q(1/2) < 0 < q(999/1000)$, so the intermediate value theorem applies. □

The finite-coefficient diagnostic `locate_quartic_sign_transition.py`, run at 100 decimal digits, locates one apparent sign change at

$$b_\dagger \approx 0.99749378987342042107460559 31, \qquad a_\star(b_\dagger) \approx 1.00143018096149513812935172 94.$$

These displayed decimal locations are diagnostic, not certified enclosures. No uniqueness of $b_\dagger$ is claimed. The 11-point finite-coefficient sweep has negative $q(b)$ at every requested sample from 0.001 through 0.99, and positive $q(0.999)$; the latter sign is proved by the independent exact certificate. Determining the number of zeros of q and the higher-order radial behavior at its zeros remains a concrete research problem.

5.17 The two atomic edges and the logarithmic corner

Theorem 5.53 (Boundary-layer scale and weak convergence)**.** *Fix $0 < b < 1$, let $a = 1 - b + \varepsilon$ with $\varepsilon > 0$, and let $T_\varepsilon = -\log r_{a,b}$. Define*

$$\tau_\varepsilon = \left(\frac{\Gamma(1-b+\varepsilon)}{(1-b)(-\zeta(b))\Gamma(\varepsilon)} \right)^{1/(1-b)}.$$

Then, as $\varepsilon \downarrow 0$,

$$T_\varepsilon = \tau_\varepsilon \left(1 + O_b\left(\frac{\tau_\varepsilon}{\varepsilon} \right) \right), \qquad 0 < T_\varepsilon < \tau_\varepsilon, \tag{5.102}$$

and in particular

$$1 - r_{1-b+\varepsilon,b} \sim \left(\frac{\Gamma(1-b)}{(1-b)(-\zeta(b))} \varepsilon \right)^{1/(1-b)}. \tag{5.103}$$

The measures converge weakly to the critical measure in Theorem 5.17, and their positive masses converge to $(1-b)^{-1}$ while their positive supports shrink to $u = 1$.

Proof. In (5.55), put $A_\varepsilon = ((1-b)\Gamma(\varepsilon))^{-1}$ and $B_\varepsilon = (-\zeta(b))/\Gamma(1-b+\varepsilon)$. The zero equation is

$$A_\varepsilon - B_\varepsilon T^{1-b} - T C_\varepsilon(T) = 0, \quad C_\varepsilon(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 (1-v)^{a-1} v^{b-1} q(Tv)\, dv.$$

Since $1/2 < q < 1$ and $a + b = 1 + \varepsilon$,

$$\frac{1}{2\Gamma(1+\varepsilon)} < C_\varepsilon(T) < \frac{1}{\Gamma(1+\varepsilon)}.$$

By definition $B_\varepsilon \tau_\varepsilon^{1-b} = A_\varepsilon$, so the equation at $T = \tau_\varepsilon$ has strictly negative left side and $T_\varepsilon < \tau_\varepsilon$. Furthermore

$$0 \le 1 - (T_\varepsilon/\tau_\varepsilon)^{1-b} = \frac{T_\varepsilon C_\varepsilon(T_\varepsilon)}{A_\varepsilon} \le \frac{(1-b)\tau_\varepsilon}{\varepsilon}.$$

The final equality in the upper bound uses $\Gamma(1+\varepsilon) = \varepsilon\Gamma(\varepsilon)$. Now $\tau_\varepsilon = O_b(\varepsilon^{1/(1-b)})$, hence $\tau_\varepsilon/\varepsilon \to 0$. The last inequality proves (5.102). Expanding the Gamma ratio and using $1 - e^{-T} \sim T$ proves (5.103).

For weak convergence, the positive term in (5.52), regarded as a measure in the T coordinate, is

$$\frac{1}{1-b} \frac{e^{-T} T^{\varepsilon-1}}{\Gamma(\varepsilon)}\, dT.$$

It is $(1-b)^{-1}$ times a Gamma$(\varepsilon,1)$ law. Its mean is ε, so it converges weakly to mass $(1-b)^{-1}$ at $T=0$, equivalently at $u=1$. The two negative components in (5.52) converge in $L^1(e^{-T}\,dT)$ to those in (5.56). For the power term, dominated convergence holds with an exponent bounded above -1. For the convolution, either use the rescaled beta integral and the same domination, or use L^1 continuity of the Gamma densities and the integrable convolution factor $e^{-t}t^{b-1}q(t)$.

The unique positive support is $0<T<T_\varepsilon$, which shrinks to zero. The Gamma$(\varepsilon,1)$ mass in this interval tends to one: since T_ε is a fixed power of ε to leading order,

$$\frac{1}{\Gamma(\varepsilon)}\int_0^{T_\varepsilon} e^{-T}T^{\varepsilon-1}\,dT = \frac{T_\varepsilon^{\varepsilon}}{\Gamma(1+\varepsilon)}(1+o(1)) \longrightarrow 1.$$

The negative components have vanishing mass on the shrinking interval by their L^1 convergence to an integrable limit. Thus the actual positive mass of the signed density tends to $(1-b)^{-1}$, completing the proof. □

5.17.1 The other edge and the logarithmic corner

The second atomic edge has a complementary power law. The corner $b=1$ has an exponential scale and no finite limiting measure.

Proposition 5.54 (Two further endpoint scales)**.** *For fixed $b>1$, as $a\downarrow 0$,*

$$1-r_{a,b} \sim \big(\Gamma(b)\zeta(b)a\big)^{1/(b-1)}. \tag{5.104}$$

The measures converge weakly to $\nu_{0,b}$, and their positive masses converge to $\zeta(b)$. For $b=1$, set $\sigma_a=\exp(\gamma+\psi(a))$. Then

$$-\log r_{a,1} = \sigma_a\big(1+O(\sigma_a/a)\big), \qquad 0<-\log r_{a,1}<\sigma_a, \tag{5.105}$$

where

$$\sigma_a = \exp\big(-1/a+\zeta(2)a-\zeta(3)a^2+O(a^3)\big).$$

If $M_{a,1}$ denotes either signed mass of $\nu_{a,1}$, then

$$M_{a,1} \sim \frac{1}{ea}. \tag{5.106}$$

Thus the finite signed measures have unbounded total variation at this corner, consistently with the impossibility of a finite measure at $a=0,b=1$.

Proof. For $b>1$, the equation obtained from (5.52) after division by T^{a-1} is

$$0=\frac{\zeta(b)}{\Gamma(a)}-\frac{T^{b-1}}{(b-1)\Gamma(a+b-1)}-\frac{T^b}{\Gamma(a)\Gamma(b)}\int_0^1(1-v)^{a-1}v^{b-1}q(Tv)\,dv.$$

The last integral, after its Gamma normalization, lies between $1/(2\Gamma(a+b))$ and $1/\Gamma(a+b)$. The ratio of this last term to the preceding negative term is therefore $O_b(T)$, uniformly for small positive a. The balancing point of the first two terms is

$$\left(\frac{(b-1)\Gamma(a+b-1)\zeta(b)}{\Gamma(a)}\right)^{1/(b-1)} \sim \big(\Gamma(b)\zeta(b)a\big)^{1/(b-1)}.$$

The strict monotonicity of the rescaled density, or the same two-sided comparison used in Theorem 5.53, makes the actual zero asymptotic to this point. The positive term is $\zeta(b)$ times a Gamma$(a,1)$ law in the T coordinate. The negative term in (5.53), after multiplication by e^{-T}, is the convolution of that Gamma probability density with $e^{-t}t^{b-1}/(\Gamma(b)(1-e^{-t}))$, an integrable function of mass $\zeta(b)$. As $a \downarrow 0$, this convolution converges in L^1 to the latter function; Gamma$(a,1)$ is an approximate identity since its mean tends to zero. This proves weak convergence. As before, the Gamma mass below the crossing tends to one, since $a \log T \to 0$, and the limiting negative density is integrable. Hence the positive mass tends to $\zeta(b)$.

For $b = 1$, put $J_a(T) = \int_0^1 (1-v)^{a-1} q(Tv)\, dv$, so that $1/(2a) < J_a(T) < 1/a$. At the crossing T_a, formula (5.54) says

$$\log(\sigma_a/T_a) = T_a J_a(T_a), \qquad 0 < \log(\sigma_a/T_a) < \sigma_a/a.$$

This gives (5.105). The digamma expansion $\psi(a) = -1/a - \gamma + \zeta(2)a - \zeta(3)a^2 + O(a^3)$ gives the stated scale.

Finally the positive mass is the integral of $e^{-T}T^{a-1}[\log(\sigma_a/T) - TJ_a(T)]/\Gamma(a)$ over $0 < T < T_a$. Omitting the factor e^{-T}, the TJ_a term, and the vanishing interval between T_a and σ_a changes the result by a relative $o(1)$. To see this explicitly, the first omission has relative error at most σ_a; the TJ_a integral is at most $\sigma_a^{a+1}/(a(a+1)\Gamma(a))$, whereas the main integral is $\sigma_a^a/(a^2\Gamma(a))$. For the removed interval, write $T = \sigma_a v$ and use $1 - T_a/\sigma_a = O(\sigma_a/a)$; its relative contribution is $O(\sigma_a^2)$. Therefore

$$M_{a,1} \sim \frac{\sigma_a^a}{a^2\Gamma(a)} \sim \frac{1}{ea},$$

as asserted. □

5.18 Power-scale corrections and exponential mass concentration

The atom in Theorem 5.17 raises a quantitative question: how does an absolutely continuous positive part collapse into that atom? Two distinct scales occur. Its outermost point approaches the endpoint by a power of the distance to the critical line. Most of its mass lies much closer to the endpoint, on an exponential scale described by a limiting exponential distribution.

Fix $b \in (0,1)$, and write

$$p = 1 - b, \qquad a = p + \varepsilon, \qquad K_\varepsilon(T) = K_{p+\varepsilon,b}(T), \qquad 0 < \varepsilon \le \varepsilon_0 < b. \tag{5.107}$$

Let T_ε be the unique logarithmic crossing of K_ε, so $K_\varepsilon > 0$ on $(0, T_\varepsilon)$. All constants below may depend on the fixed b; the local expansion constants may also depend on its chosen radius T_0.

Lemma 5.55 (A uniform local kernel expansion)**.** *For any fixed $T_0 < 2\pi$, uniformly for $0 \le \varepsilon \le \varepsilon_0$ and $0 < T \le T_0$,*

$$\begin{aligned} K_\varepsilon(T) = T^{\varepsilon-1}\Bigg\{ &\frac{\varepsilon}{p\Gamma(1+\varepsilon)} + \frac{\zeta(b)}{\Gamma(p+\varepsilon)}T^p - \frac{T}{2\Gamma(1+\varepsilon)} \\ &- \frac{bT^2}{12\Gamma(2+\varepsilon)} + O_{b,T_0}(T^4)\Bigg\}. \end{aligned} \tag{5.108}$$

More generally the bracket has the convergent expansion

$$\begin{aligned} &\frac{\varepsilon}{p\Gamma(1+\varepsilon)} + \frac{\zeta(b)}{\Gamma(p+\varepsilon)}T^p - \frac{T}{2\Gamma(1+\varepsilon)} \\ &\quad - \sum_{\ell \ge 1} \frac{B_{2\ell}\Gamma(b+2\ell-1)}{(2\ell)!\,\Gamma(b)\Gamma(\varepsilon+2\ell)}T^{2\ell}, \qquad 0 < T < 2\pi. \end{aligned} \tag{5.109}$$

Here $B_{2\ell}$ are the Bernoulli numbers, with $B_2 = 1/6$.

Proof. In (5.52), put $a = p + \varepsilon$ and $a + b = 1 + \varepsilon$. The common regularizer has the convergent expansion

$$q(t) = \frac{1}{2} + \sum_{\ell \geq 1} \frac{B_{2\ell}}{(2\ell)!} t^{2\ell - 1}, \qquad |t| < 2\pi.$$

Its positive convolution therefore has expansion

$$C_{a,b}(T) = \frac{T^{\varepsilon}}{2\Gamma(1+\varepsilon)} + \sum_{\ell \geq 1} \frac{B_{2\ell}\Gamma(b + 2\ell - 1)}{(2\ell)!\Gamma(b)\Gamma(\varepsilon + 2\ell)} T^{\varepsilon + 2\ell - 1}.$$

The beta integral evaluates each coefficient. Substitute this into (5.52) and use $\Gamma(1+\varepsilon) = \varepsilon\Gamma(\varepsilon)$ to obtain (5.109). For uniform remainders choose $T_0 < T_1 < 2\pi$; Cauchy estimates bound the regularizer's Taylor remainder on $|t| \leq T_0$, while $v^{b-1}(1-v)^{p-1}$ dominates all beta weights. The Gamma factors are bounded on the fixed compact parameter interval. This proves the uniform truncation and the full locally convergent series. □

Theorem 5.56 (Crossing scale and its first correction)**.** *Define*

$$\tau_{\varepsilon} = \left\{ \frac{\Gamma(p+\varepsilon)\varepsilon}{p[-\zeta(b)]\Gamma(1+\varepsilon)} \right\}^{1/p}. \tag{5.110}$$

As $\varepsilon \downarrow 0$,

$$T_{\varepsilon} = \tau_{\varepsilon} - \frac{\tau_{\varepsilon}^2}{2\varepsilon} + O_b\left(\frac{\tau_{\varepsilon}^3}{\varepsilon^2} \right). \tag{5.111}$$

In particular, with

$$K_b = \left\{ \frac{\Gamma(1-b)}{(1-b)[-\zeta(b)]} \right\}^{1/(1-b)}, \tag{5.112}$$

one has

$$T_{\varepsilon} \sim K_b \varepsilon^{1/(1-b)}, \qquad 1 - r_{p+\varepsilon,b} \sim K_b \varepsilon^{1/(1-b)}. \tag{5.113}$$

Proof. For every fixed $T > 0$, the kernel depends continuously on ε and its limit $K_{p,b}(T)$ is strictly negative by (5.56). The unique crossing therefore satisfies $T_{\varepsilon} \to 0$. At that crossing, multiply the bracket in (5.108) by $\Gamma(1+\varepsilon)$ and write

$$C_{\varepsilon} = \frac{[-\zeta(b)]\Gamma(1+\varepsilon)}{\Gamma(p+\varepsilon)} > 0.$$

This gives

$$\frac{\varepsilon}{p} = C_{\varepsilon} T_{\varepsilon}^{p} + \frac{T_{\varepsilon}}{2} + \frac{bT_{\varepsilon}^2}{12(1+\varepsilon)} + O_b(T_{\varepsilon}^4). \tag{5.114}$$

Since $p < 1$, the last three terms are lower order than T_{ε}^{p}. Hence $T_{\varepsilon}/\tau_{\varepsilon} \to 1$.

Put $y = T_{\varepsilon}/\tau_{\varepsilon}$ and $q = \tau_{\varepsilon}/\varepsilon$. The leading equivalence implies $q = O_b(\varepsilon^{b/p}) \to 0$. Dividing (5.114) by ε/p gives, uniformly for y near one,

$$1 = y^p + \frac{p}{2} q y + O_b(\varepsilon q^2).$$

The mean value theorem first yields $y - 1 = O_b(q)$. Taylor expansion then yields $p(y-1) + (p/2)q = O_b(q^2)$, so $y = 1 - q/2 + O_b(q^2)$. This proves (5.111). Formula (5.113) follows from continuity of the gamma factors and $1 - e^{-T} \sim T$. □

At the symmetric boundary point $b = p = 1/2$, the analytic correction from the gamma quotient and the nonlinear root correction have the same relative order. Using $\psi(1/2) = -\gamma - 2\log 2$ gives the explicit consequence

$$T_\varepsilon = \frac{4\pi}{\zeta(1/2)^2}\varepsilon^2\left[1 - \left(4\log 2 + \frac{2\pi}{\zeta(1/2)^2}\right)\varepsilon + O(\varepsilon^2)\right]. \tag{5.115}$$

For other b, retaining the exact gamma quotient in τ_ε keeps the algebraic and fractional-power corrections in (5.111) unambiguous.

Theorem 5.57 (Condensation and the exponential positive-mass law)**.** *Let $\nu_\varepsilon = \nu_{p+\varepsilon,b}$, and denote its positive and negative parts by ν_ε^+ and ν_ε^-, so $\nu_\varepsilon = \nu_\varepsilon^+ - \nu_\varepsilon^-$. Then*

$$\nu_\varepsilon^- \longrightarrow -\kappa_{p,b}(u)\,du \quad \text{in total variation}, \qquad \nu_\varepsilon^+ \Longrightarrow \frac{1}{p}\delta_1. \tag{5.116}$$

In particular $\nu_\varepsilon \Longrightarrow \nu_{p,b}$.

Let λ_ε be the pushforward of the finite positive measure $p\,K_\varepsilon(T)^+ e^{-T}\,dT$ under $Y_\varepsilon = -\varepsilon\log T$. For all sufficiently small ε it is supported in $[0,\infty)$ and

$$\left\|\lambda_\varepsilon - e^{-y}\mathbf{1}_{y>0}\,dy\right\|_{\mathrm{TV}} = O_b(\varepsilon|\log\varepsilon|)\,. \tag{5.117}$$

The same convergence and rate hold after normalizing λ_ε to a probability measure.

Proof. For $T \le T_0 < 1$, Lemma 5.55 gives

$$K_\varepsilon(T) = \frac{\varepsilon}{p\Gamma(1+\varepsilon)}T^{\varepsilon-1} + R_\varepsilon(T), \qquad |R_\varepsilon(T)| \le C_b T^{p+\varepsilon-1}. \tag{5.118}$$

The first term is positive, so the negative part is bounded by $C_b T^{p-1}$, an integrable bound independent of ε. For $T \ge T_0$, formula (5.52) and $0 < C_{a,b}(T) < k_{a+b}(T)$ give a common polynomial majorant as a ranges over $[p, p+\varepsilon_0]$; hence $e^{-T}|K_\varepsilon(T)| \le C_b e^{-T}(1+T^{\varepsilon_0})$. Pointwise convergence and dominated convergence prove the negative-part assertion in (5.116). Its limiting mass is $1/p$ by Theorem 5.17. Each ν_ε has total mass zero, so the positive masses have the same limit. Their supports lie in $(e^{-T_\varepsilon}, 1)$, which shrinks to 1; this proves the positive weak limit.

For the quantitative distributional statement set $y_\varepsilon = -\varepsilon\log T_\varepsilon$. Theorem 5.56 gives $y_\varepsilon = O_b(\varepsilon|\log\varepsilon|)$ and $y_\varepsilon > 0$ for small ε. On the positive support $0 < T < T_\varepsilon$, replace the density in (5.118) by its leading term and replace e^{-T} by one. The total variation error is at most

$$C_b T_\varepsilon^{p+\varepsilon} + C_b\varepsilon T_\varepsilon^{1+\varepsilon} = O_b(\varepsilon),$$

because $T_\varepsilon^p = O_b(\varepsilon)$. Total variation cannot increase under pushforward. With $T = e^{-y/\varepsilon}$, the pushforward of the resulting leading measure is exactly

$$\frac{e^{-y}}{\Gamma(1+\varepsilon)}\mathbf{1}_{y>y_\varepsilon}\,dy.$$

Its total variation difference from the unit exponential density is at most $O(\varepsilon) + \int_0^{y_\varepsilon} e^{-y}\,dy$. This proves (5.117). Its mass consequently differs from one by the same order; normalizing changes the total variation by at most that mass discrepancy. □

Thus the power-law crossing width does not describe the typical positive-mass location. Under the normalized positive part, $-\varepsilon \log(-\log u)$ tends to an exponential random variable of mean one. A typical logarithmic distance $T = -\log u$ is therefore exponentially small in $1/\varepsilon$, even though the outer edge of the positive region is of order $\varepsilon^{1/(1-b)}$.

The accompanying boundary diagnostics compare the two-term root formula at $b = 1/4, 1/2, 3/4$ and decreasing ε, and compare positive-mass distribution functions with the exponential law. The convergent series (5.109) provides a stable evaluator with a recorded analytic truncation bound. Numerical rounding is still separate from that bound; these are diagnostics, not new proof assumptions.

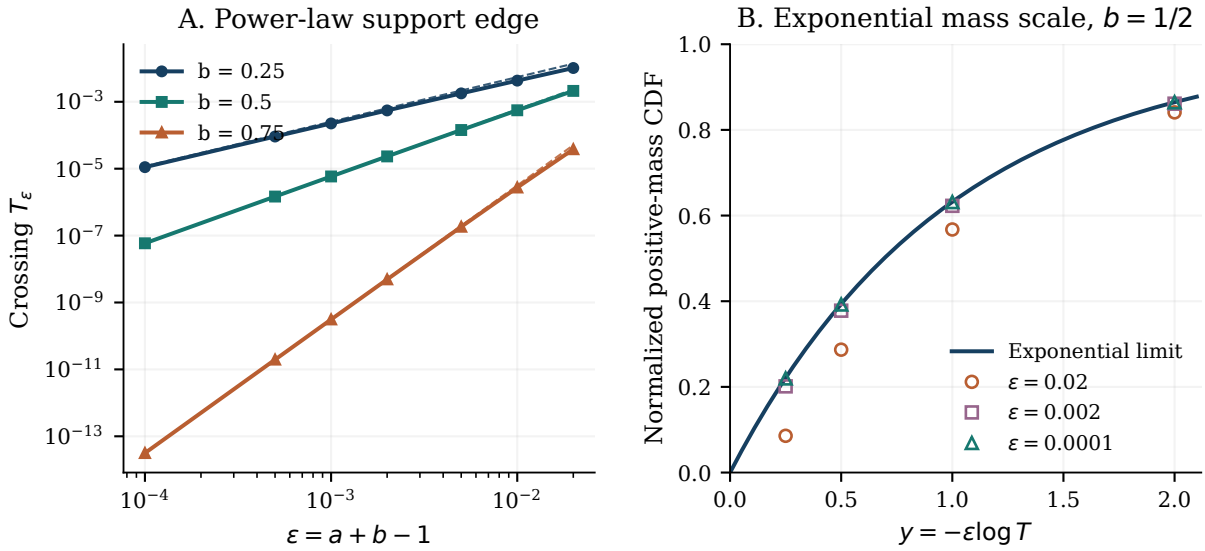


Figure 5.3: The two scales of endpoint concentration. Left: numerical crossings (markers) and the leading power laws (dashed curves) at three fixed inner orders. Right: four sampled distribution-function values for each indicated ε at $b = 1/2$, compared with the proved unit-exponential limit. These are diagnostics of the analytic results.

All limits in this section keep b fixed. As $b \downarrow 0$, the small parameter $\tau_\varepsilon/\varepsilon$ has exponent $b/(1-b) \downarrow 0$, so the stated hierarchy is not uniform at that endpoint. As $b \uparrow 1$, the exponent $1/(1-b)$ and the constants become singular. Joint limits involving either endpoint require additional analysis and are not consequences of these theorems.

5.19 Euler enclosures and the failure of an unrestricted constant one

Let $a, b > 0$ with $w \geq 1$. Define

$$f(n) = c_{2n+1} = \frac{H_{2n}^{(b)}}{(2n+1)^a}, \qquad d_k = \sum_{n=0}^{k} (-1)^n \binom{k}{n} f(n), \qquad E_N = \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}}. \tag{5.119}$$

The error is measured against $g_{a,b} = \operatorname{Im} F_{a,b}(\mathrm{i})$. On the critical line this is an Abel value; the ordinary alternating series with terms $(-1)^n f(n)$ diverges by the term test.

Theorem 5.58 (Extended one-sided Euler bounds)**.** *For every $k \geq 1$ one has $d_k < 0$, while $d_0 = 0$. The E_N decrease strictly to $g_{a,b}$, and*

$$E_N - g_{a,b} = -2^{-N} \int_{[0,1]} \frac{(1-u^2)^N}{1+u^2} \, \mathrm{d}\nu_{a,b}(u) > 0. \tag{5.120}$$

Consequently

$$E_N - M(a,b)2^{-N} < g_{a,b} < E_N \qquad (N \geq 1), \tag{5.121}$$

with the following elementary choices:

$$M(a,b) = \begin{cases} 1, & a \geq 1, \\ (1-b)^{-1}, & 0 < a < 1,\ 0 < b < 1,\ w \geq 1, \\ b/(b-1), & 0 < a < 1,\ b > 1, \\ 1 + 1/a, & 0 < a < 1,\ b = 1. \end{cases} \tag{5.122}$$

On the critical line the sharper uniform choice is proved in Corollary 5.33; its matching subcritical Euler theorem uses the positive difference measure. The choice one in the first line is the prior optimal universal constant on that restricted domain; it is not asserted to work in the other lines.

Proof. The binomial theorem and (5.44) give $d_k = \int (1-u^2)^k \, \mathrm{d}\nu(u)$. Thus $d_0 = 0$, while the strictly decreasing weight gives $d_k < 0$ for $k \geq 1$. With $m = m(\nu)$, we have $|d_k| \leq m$. Therefore the Euler series is absolutely convergent and may be summed under the measure integral:

$$\sum_{k \geq 0} \frac{d_k}{2^{k+1}} = \int \frac{\mathrm{d}\nu(u)}{1+u^2} = \operatorname{Im} F(\mathrm{i}).$$

Summing the geometric tail gives (5.120). Its weight is strictly decreasing, bounded by one, and strictly less than one for every $u > 0$. Since the negative Jordan part has no atom at zero, $0 < E_N - g < m2^{-N}$. It remains to bound m.

For $b < 1, w > 1$, rewrite (5.52) as

$$K_{a,b} = \frac{k_{w-1}}{1-b} - \big[-\zeta(b)k_a + C_{a,b}\big].$$

The two displayed densities are positive, and each has mass $1/(1-b)$ in the u measure, by zero total mass. Hence $m < 1/(1-b)$. On $w = 1$ the exact Jordan mass is $1/a = 1/(1-b)$. For $b > 1$, (5.52) is a difference of positive densities $\zeta(b)k_a$ and $k_{w-1}/(b-1) + C_{a,b}$, so $m < \zeta(b) < 1 + 1/(b-1) = b/(b-1)$.

For $b = 1$ and $0 < a < 1$, use a translation estimate rather than the singular limit of either of these bounds. Let σ_a be the probability law of $u = e^{-Y}$ with $Y \sim \mathrm{Gamma}(a,1)$. Its logarithmic density $p_a(s) = e^{-s}s^{a-1}/\Gamma(a)$ decreases strictly. If $D_{e^{-t}}$ denotes pushforward under $u \mapsto e^{-t}u$, then direct comparison of $p_a(s)$ and $p_a(s-t)$ gives

$$\|\sigma_a - D_{e^{-t}}\sigma_a\|_{\mathrm{TV}} = 2\int_0^t p_a(s) \, \mathrm{d}s \leq \frac{2t^a}{\Gamma(a+1)}.$$

The signed-measure integral

$$\widetilde{\nu} = \int_0^\infty \frac{\sigma_a - D_{e^{-t}}\sigma_a}{e^t - 1} \, \mathrm{d}t$$

converges in total variation. Its $(n-1)$st moment is $n^{-a}\int_0^\infty (1-e^{-(n-1)t})(e^t-1)^{-1}\,\mathrm{d}t = c_n$. By uniqueness it is $\nu_{a,1}$, and

$$m \le \frac{1}{\Gamma(a+1)}\int_0^\infty \frac{t^a}{e^t-1}\,\mathrm{d}t = \zeta(a+1) < 1 + 1/a.$$

Finally, for $a \ge 1$ the earlier sharper bound can be recovered without any fractional-order gap. At $a = 1$ the equivalent base density is

$$K_{1,b}(T) = -\frac{T^b}{\Gamma(b+1)} + \frac{1}{\Gamma(b)}\int_T^\infty \frac{t^{b-1}}{e^t-1}\,\mathrm{d}t.$$

Its nth Laplace value is $-n^{-b-1} + n^{-1}\sum_{j=1}^n j^{-b} = H_{n-1}^{(b)}/n$, by Tonelli and a finite geometric sum. Moment uniqueness therefore identifies this base density with (5.52) or (5.54). Both positive terms in this difference have $e^{-T}\,\mathrm{d}T$ integral one. For the second term, interchange the integrals and use $(1-e^{-t})/(e^t-1) = e^{-t}$; the first is a gamma integral. Their positive overlap makes the Jordan mass strictly less than one. For $a > 1$, convolution by k_{a-1} in the T coordinate preserves moments after multiplication by $n^{-(a-1)}$ and contracts the $L^1(e^{-T}\,\mathrm{d}T)$ norm, because $\int_0^\infty e^{-T}k_{a-1}(T)\,\mathrm{d}T = 1$. Moment uniqueness identifies the result with the present kernel, so $m < 1$ for every real $a \ge 1$. This proves all four choices. □

The same translation proof gives, when $0 < a \le 1$ and $a + b > 1$, the optional non-elementary bound

$$m \le \frac{\Gamma(a+b)\zeta(a+b)}{\Gamma(a+1)\Gamma(b)}.$$

It may improve a particular estimate, but diverges as $a + b \downarrow 1$. The critical atomic construction is therefore not replaced by this translation argument.

Proposition 5.59 (Exact finite Euler weights)**.** *For $N \ge 1$,*

$$E_N = 2^{-N}\sum_{n=0}^{N-1}(-1)^n f(n)\sum_{j=n+1}^{N}\binom{N}{j}. \tag{5.123}$$

Thus the finite sum can be formed without constructing a triangular difference table.

Proof. Substitute the definition of d_k and interchange the finite sums. The coefficient identity is

$$\sum_{k=n}^{N-1} 2^{N-k-1}\binom{k}{n} = \sum_{j=n+1}^{N}\binom{N}{j},$$

which follows by induction from Pascal's identity. Harmonic numbers and binomial tails can then each be updated successively. This rearrangement is already used in the integer-order manuscript; the new feature in the implementation here is certified rational interval arithmetic for fractional powers. □

Theorem 5.60 (A certified scope obstruction)**.** *For $a = 1/10$ and $b = 9/10$,*

$$\begin{aligned} -0.52976193228577126309462936 55 &< g_{1/10,9/10} \\ &< -0.52976193228577126309462936 53. \end{aligned} \tag{5.124}$$

In particular, $g_{1/10,9/10} < -13/25 < -1/2$. *Since* $E_1 = 0$, *the inequality* $E_N - 2^{-N} < g_{a,b}$ *fails at* $N = 1$ *for this pair. A uniform constant one cannot be extended to the entire region* $a + b \geq 1$. *The failure also occurs strictly above the threshold, with ordinary boundary convergence:*

$$-0.530844252966159326172071260 2 < g_{1/10,1} < -0.530844252966159326172071260 0 < -0.53. \tag{5.125}$$

Proof. The accompanying standard-library program `code/certify.py` evaluates (5.123) with $N = 96$ by outward rational interval arithmetic. Every fractional power is bounded by integer tenth-root inequalities, as detailed in Section 5.19.1. The critical mass bound is the exact rational $M = 10$. Subtracting $10\,2^{-96}$ from the lower endpoint of the finite Euler enclosure and leaving its upper endpoint unchanged gives the interval recorded in `data/exact_certificates.json`. Outward decimal rounding yields (5.124); exact rational comparisons verify its upper endpoint is less than $-13/25$. No numerical polylogarithm, floating-point root, or unproved identity enters this certificate. For $(a, b) = (1/10, 1)$ the same computation uses $M = 1 + 1/a = 11$ and proves (5.125). Its ordinary boundary convergence follows from $a + b = 11/10 > 1$ and Corollary 5.19. □

This is *not* a counterexample to the baseline theorem, which explicitly assumes $a \geq 1$. It is a counterexample to dropping that hypothesis while retaining the same universal error constant.

Proposition 5.61 (The exact critical constant for fixed parameters)**.** *On* $a + b = 1$, *put* $d\sigma = -K_{1-b,b}(-\log u)\,du$. *The scaled errors* $2^N(E_N - g_{a,b})$ *decrease strictly to zero for* $N \geq 1$. *The smallest constant in the non-strict estimate* $E_N - g_{a,b} \leq C2^{-N}$, *valid for all integers* $N \geq 1$ *at that fixed pair, is* $C = -2g_{a,b}$.

Proof. The endpoint atom contributes zero to (5.120) when $N \geq 1$. Hence

$$2^N(E_N - g) = \int_0^1 \frac{(1-u^2)^N}{1+u^2}\,\mathrm{d}\sigma(u).$$

Strict pointwise decrease and dominated convergence prove the first claim. Its largest value occurs at $N = 1$ and is $2(E_1 - g) = -2g$. For the corresponding strict inequality the infimum of admissible constants is the same, but equality at $N = 1$ prevents attaining it. □

5.19.1 Exact arithmetic and independent native evidence

The new standard-library verifier encloses every rational fractional power by integer root inequalities on a 10^{100} grid. Products, harmonic sums and finite Euler weights are then combined as exact rational intervals. It certifies 616 strictly positive Hausdorff-difference signs, 112 negative Euler increments and seven Gaussian analytic-value enclosures at 192 Euler terms. The six triangle cases use the rational budget 57/50; the remaining axis case uses 5/4. The axis case $b = 3/2$ also proves $-2g_{0,3/2} > 227/200$, supplying the finite arithmetic premise in the axis-maximum theorem.

A separate native Wolfram calculation at 100 working digits evaluates the outer Gamma integral of a depth-one polylogarithm, or the direct axis value. All seven values lie inside their rational Euler enclosures. The native quadrature is a numerical diagnostic, not interval quadrature. The sign and remainder theorems depend on the written positive-measure proofs. The original threshold report's tenth-root certificates are preserved unchanged with its delivered code and data.

5.20 A sharp Euler constant at every positive real order

The Gaussian maximum does not bound arbitrary Euler errors merely because it bounds the value of the function. A contraction inequality for the Euler kernel supplies the missing implication. It removes the divergent parameter-dependent budgets near the signed-measure threshold, while retaining the stronger constant one on the earlier domain $a \geq 1$.

Lemma 5.62 (A difference-of-powers bound)**.** *For $m \geq 2$, $0 \leq r \leq 1$ and $0 \leq v \leq 1$,*

$$0 \leq (1 - rv)^m - (1 - r)^m \leq \frac{1 - v}{1 + v}.$$

For $0 < v < 1$, equality on the right is possible only when $m = 2$, and then it holds precisely at $r = 1/(1 + v)$.

Proof. The endpoints $v = 0, 1$ are immediate. Fix $0 < v < 1$ and put $q = v^{1/(m-1)}$. Differentiation shows that the unique maximum over $r \in [0, 1]$ is at $r = (1 - q)/(1 - vq)$, where its value is

$$\frac{(1 - v)^m}{(1 - v^{m/(m-1)})^{m-1}}.$$

Let $p = m/(m - 1) \in (1, 2]$. The function $p \mapsto \log(1 - v^p)$ has strictly negative second derivative $-(\log v)^2 v^p/(1 - v^p)^2$. Its chord between $p = 1$ and $p = 2$ therefore gives

$$1 - v^p \geq (1 - v)(1 + v)^{p-1},$$

strictly when $1 < p < 2$. Substitution in the maximum proves the bound and both equality assertions. □

Lemma 5.63 (Contraction of the Euler tail kernel)**.** *For an integer $N \geq 1$, set $H_N(t) = (1 - t)^N/(1 + t)$ on $[0, 1]$. For $0 \leq r, v \leq 1$,*

$$0 \leq H_N(rv) - H_N(r) \leq \frac{1 - v}{1 + v}. \tag{5.126}$$

If $0 < v < 1$, equality on the right holds precisely when $N = 1$ and $r = 1$.

Proof. For $N = 1$, the difference between the right side and the middle expression factors exactly as

$$\frac{(1 - v)(1 - r)(1 - rv)}{(1 + v)(1 + rv)(1 + r)}.$$

For $N \geq 2$, the uniformly convergent positive expansion

$$H_N(t) = \sum_{k \geq 0} \frac{(1 - t)^{N+k}}{2^{k+1}}$$

expresses its difference as a probability-weighted sum of the preceding power differences. Each is bounded as claimed, and the terms with $N + k > 2$ make the upper inequality strict for $0 < v < 1$. Decrease of H_N proves the lower inequality. □

Let ω_b be the probability measure

$$d\omega_b(v) = \frac{(-\log v)^{b-1}}{\Gamma(b)}\,dv, \qquad 0 < v < 1.$$

Let π_a be the law of $X = e^{-T_a}$ with $T_a \sim \mathrm{Gamma}(a,1)$ when $a > 0$, and put $\pi_0 = \delta_1$. These measures are independent. They retain ordinary probability mass throughout the parameter domain, including the critical and subcritical regions where the unweighted one-variable moment measure is not finite.

Theorem 5.64 (All-order Euler representation and the sharp universal constant)**.** *For every $a \geq 0, b > 0$, define d_k and E_N from the coefficients c_{2n+1} as in (5.119), and use the analytic value $g_{a,b} = \operatorname{Im} F_{a,b}(i)$. Then $d_0 = 0$, $d_k < 0$ for $k \geq 1$, and E_N decreases strictly to $g_{a,b}$. Its exact error is*

$$2^N(E_N - g_{a,b}) = \iint \frac{H_N(x^2v^2) - H_N(x^2)}{1-v}\,d\pi_a(x)\,d\omega_b(v). \tag{5.127}$$

Consequently, for every $N \geq 1$,

$$0 < 2^N(E_N - g_{a,b}) \leq C(b), \qquad C(b) = \beta(b) + 2^{-b}\eta(b). \tag{5.128}$$

Equality in the upper bound holds exactly when $a = 0$ and $N = 1$. The sharp uniform constant over all nonnegative outer orders is

$$C_* = C(b_*),$$

where b_* is the unique nondegenerate maximum from Theorem 5.32. It satisfies

$$\frac{227}{200} < C_* < \frac{1+\sqrt{2}}{2} < \frac{5}{4}.$$

For positive a, b the strict inequality $0 < E_N - g_{a,b} < C_* 2^{-N}$ holds, and C_* is its least possible uniform constant. When $a = 0$ is included, the non-strict upper bound is attained only at $a = 0, b = b_*, N = 1$.

Proof. The elementary moment laws give, for $n \geq 0$,

$$c_{2n+1} = \iint x^{2n}\frac{1-v^{2n}}{1-v}\,d\pi_a(x)\,d\omega_b(v).$$

Indeed the two factors integrate to $(2n+1)^{-a}$ and $H_{2n}^{(b)}$. The quotient is a finite geometric sum, so this identity includes $n = 0$ without an endpoint singularity. Finite binomial expansion now gives

$$d_k = \iint \frac{(1-x^2)^k - (1-x^2v^2)^k}{1-v}\,d\pi_a(x)\,d\omega_b(v).$$

For $k \geq 1$ its integrand is strictly negative and has absolute value at most $2k$, by the mean-value theorem. Finite summation is therefore legitimate for every pair of parameters.

The Gaussian double kernel, or the axis formula when $a = 0$, gives

$$g_{a,b} = \iint \frac{(1+x^2)^{-1} - (1+x^2v^2)^{-1}}{1-v}\,d\pi_a(x)\,d\omega_b(v).$$

Its integrand is $-x^2(1+v)/((1+x^2)(1+x^2v^2))$, hence bounded. Subtract the finite geometric sum defining E_N from this identity to obtain (5.127); no divergent boundary series is used.

Apply Lemma 5.63 with $r = x^2$ and its variable v replaced by v^2. The resulting majorant is

$$\frac{1 - v^2}{(1 - v)(1 + v^2)} = \frac{1 + v}{1 + v^2}.$$

Its ω_b integral is $C(b)$ by the axis identity. It is bounded by $(1 + \sqrt{2})/2$, so the exact error tends to zero at least geometrically. The strict signs of d_k give the strict decrease of E_N. The contraction equality criterion, together with $\pi_0 = \delta_1$ and the interior densities for $a, b > 0$, proves the fixed-b equality claim.

The unique axis maximum supplies the uniform constant and its bounds. Sharpness over $a, b > 0$ follows by fixing $b = b_*$, taking $a \downarrow 0$ at $N = 1$, and using bounded convergence in the Gaussian probability integral. Thus the supremum is approached at the axis, while inclusion of the axis gives the stated unique attaining triple. □

This theorem closes the previously open universal real-order Euler-budget question. The separate contraction argument is essential: Gaussian magnitude alone would not control later scaled errors. In particular, this result does not say that their maximum at each positive parameter pair is the first error, nor that the scaled errors always decrease. The constant one remains sharper on $a \geq 1$, and the sharp critical constant $\pi/4 + \log 2/2$ remains sharper on $a + b \leq 1$. The decimal diagnostic $C_* \approx 1.13656110333950956095$ inherits the axis experiment's status in the first proof. The independent rational enclosure in Theorem 5.36 now certifies this constant to the displayed width and proves that 57/50 is a valid uniform rational Euler budget at all positive real orders. The exact definition and universal contraction proof do not rely on a numerical stationary-point solver.

5.21 A uniform rational kernel inequality

The independent continuations [104, 105] separate the first Euler step from every later one. Retain the residual function $H_N(t)$ above, and define the continuously extended kernel

$$K_N(p, y) = \frac{H_N(py^2) - H_N(p)}{1 - y}, \qquad K_N(p, 1) = -2pH_N'(p).$$

It is nonnegative. The following finite polynomial certificate and analytic mixture reduction prove a stronger uniform budget for $N \geq 2$.

Theorem 5.65 (Kernel bound for every later truncation)**.** *For every integer $N \geq 2$ and $0 \leq p, y \leq 1$,*

$$0 \leq K_N(p, y) < \frac{9}{8}. \tag{5.129}$$

The lower bound is strict when $p > 0$ and $y < 1$. The strict upper bound includes the continuous boundary values.

The proof has two finite base cases and an all-N envelope. We give enough information to reconstruct every rational certificate without a symbolic algebra system.

5.21.1 Two explicit polynomial base cases

Multiplication by the positive denominator shows that $K_N < 9/8$ is equivalent to positivity of

$$P_N(p, y) = 8(1 + p)(1 + py^2)\{9/8 - K_N(p, y)\}.$$

For $N = 2, 3$ these are the following polynomials:

$$\begin{aligned} P_2 &= 8y^2(1+y)p^3 + (8y^3+17y^2+8y+8)p^2 \\ &\quad + (9y^2-24y-15)p+9, \end{aligned} \tag{5.130}$$

$$\begin{aligned} P_3 &= -8y^2(y+1)(y^2+1)p^4 \\ &\quad + (-8y^5-8y^4+16y^3+16y^2-8y-8)p^3 \\ &\quad + (24y^3+33y^2+24y+24)p^2 + (9y^2-32y-23)p+9. \end{aligned} \tag{5.131}$$

An identity check can avoid rational-function division altogether:

$$\begin{aligned} (1-y)P_N &= 9(1-y)(1+p)(1+py^2) \\ &\quad - 8\{(1-py^2)^N(1+p) - (1-p)^N(1+py^2)\}. \end{aligned} \tag{5.132}$$

Both verifiers check this identity or independently reconstruct its quotient.

For a polynomial $P(x,y) = \sum c_{kl}x^k y^l$ of bidegree at most (r,s), its tensor Bernstein coefficients on the unit square are

$$B_{ij} = \sum_{k\le i,\, l\le j} c_{kl} \frac{\binom{i}{k}}{\binom{r}{k}} \frac{\binom{j}{l}}{\binom{s}{l}}. \tag{5.133}$$

The Bernstein basis functions are nonnegative and sum to one. Hence a strictly positive minimum of these coefficients proves strict positivity of P on the whole square, including its boundary.

For a mesh of denominator d, first substitute $p = (u+i)/d$, $y = (v+j)/d$ on each cell. Formula (5.133) then gives an exact certificate. The data are summarized by

Polynomial	Bidegree	Mesh	Coefficients	Smallest coefficient
P_2	$(3,3)$	8×8	1024	631/12288
P_3	$(4,5)$	4×4	480	11337/40960

Every coefficient is stored as a rational number in the accompanying JSON file. More importantly, the standard-library verifier regenerates all of them from (5.130)–(5.133); it does not trust a stored minimum. These finitely many exact additions and multiplications prove the two base cases.

5.21.2 An envelope for binomial differences

Introduce

$$J_m(p,y) = \frac{(1-py^2)^m - (1-p)^m}{1-y}, \qquad m \ge 2.$$

For fixed $0 < y < 1$, differentiation in p gives a unique interior maximum at

$$p_m(y) = \frac{1-y^{2/(m-1)}}{1-y^{2m/(m-1)}}. \tag{5.134}$$

Indeed the derivative changes sign exactly when $1-p = y^{2/(m-1)}(1-py^2)$. Its sign is positive at $p=0$ and negative at $p=1$. Substitution yields the exact envelope

$$J_m(p,y) \le B_m(y) := (1+y)\left(\frac{1-y^2}{1-y^{2m/(m-1)}}\right)^{m-1}. \tag{5.135}$$

The endpoint values follow by continuity.

Lemma 5.66 (The binomial envelope decreases in its index)**.** *For every* $0 < y < 1$, $B_{m+1}(y) < B_m(y)$ *for* $m \geq 2$.

Proof. Put $k = m - 1$, $t = -2 \log y > 0$ and $h(v) = \log(1 - e^{-(1+v)t})$. Direct differentiation gives $h''(v) < 0$ for $v \geq 0$. Apart from $\log(1 + y)$, the logarithm of (5.135) is

$$k\{h(0) - h(1/k)\} = -\frac{h(1/k) - h(0)}{1/k}.$$

The secant slope of a strictly concave function decreases as the right endpoint increases. Thus the displayed negative secant slope decreases strictly when k increases. Exponentiation proves the claim. □

5.21.3 The fourth envelope is below nine eighths

Set $y = t^3$, $0 \leq t \leq 1$. The inequality $B_4(y) < 9/8$ is equivalent away from $t = 1$ to

$$9(1 - t^8)^3 - 8(1 + t^3)(1 - t^6)^3 = (1 - t^2)^3 Q(t) > 0, \tag{5.136}$$

where

$$\begin{aligned} Q(t) = {} & 9t^{18} + 27t^{16} - 8t^{15} + 54t^{14} - 24t^{13} + 82t^{12} - 48t^{11} \\ & + 84t^{10} - 56t^9 + 60t^8 - 48t^7 + 34t^6 - 24t^5 + 6t^4 - 8t^3 + 3t^2 + 1. \end{aligned} \tag{5.137}$$

The degree-18 Bernstein coefficients after $t = u/2$ and $t = (1 + u)/2$ have respectively the strictly positive minima

$$\frac{192789}{262144}, \qquad \frac{690189}{1949696}. \tag{5.138}$$

There are only 38 coefficients. Formula (5.133) with one variable regenerates them. At $y = 1$ the envelope limit is $B_4(1) = 2(3/4)^3 = 27/32 < 9/8$, so the endpoint is covered as well.

5.21.4 The infinite family follows without an infinite computation

Since $1/(1 + v) = \sum_{j \geq 0} (1 - v)^j / 2^{j+1}$ on $[0, 1]$,

$$K_N(p, y) = \sum_{j=0}^{\infty} 2^{-j-1} J_{N+j}(p, y). \tag{5.139}$$

For $y < 1$ this follows by subtracting the two convergent geometric series. If $N \geq 4$, (5.135), Lemma 5.66 and (5.136) give

$$J_{N+j}(p, y) \leq B_{N+j}(y) \leq B_4(y) < 9/8.$$

Averaging proves $K_N \leq B_4 < 9/8$ for every $N \geq 4$. Continuity covers $y = 1$. The polynomial base cases cover $N = 2, 3$, completing Theorem 5.65.

Corollary 5.67 (A smaller uniform budget after the first step)**.** *For every* $a \geq 0, b > 0$ *and integer* $N \geq 2$*,*

$$0 < 2^N (E_N - g_{a,b}) < \frac{9}{8}.$$

Proof. Take expectations of the strict pointwise inequality in (5.127), with $p = x^2$ and $y = v$. The measures are probability measures, and their kernel is strictly positive almost everywhere, including the $a = 0$ axis. □

The first-step maximum C_* is strictly larger than $9/8$, already by $C_* > 227/200$ in Theorem 5.64. This does not imply that scaled errors decrease at each fixed pair: the exact incoming replay certifies $R_2(4,1) > R_1(4,1)$, where $R_N = 2^N(E_N - g)$. The unscaled E_N still decrease by the theorem above.

Remark 5.68 (Nature of the finite computation). The inequality is a computer-assisted ordinary proof: three stated polynomial identities, a stated rational basis conversion, and 1542 strictly positive rational numbers close its finite component. The all-index reduction is analytic. The check is reproducible under Python's optimized mode and is not a test of finitely many real sample points. The supplied SymPy replay independently derives the two rational kernels rather than importing their coefficient arrays.

The fresh standard-library replay regenerates all 1,542 coefficients on 82 cells, checks the three defining polynomial identities, and rejects a corrupted polynomial. The same verifier succeeds with optimized Python, whose disabled assertions do not bypass its explicit validations. A separate SymPy program reconstructs the base rational kernels. Receipts are preserved in `verificati on/sixth-replay/sharp/`. The independently replayed golden/Cayley delivery uses another positive-mixture route and its own finite base certificates; it is supporting evidence for the same bound, not a second definition of the sharp all-index constant.

5.22 Strict depth, signed oscillation and minimal positive compensation

The real-order depth-two factorization suggests a structural question at higher strict depth: can the initial zero moments account for every sign change of a representing density? The continuation [106] answers this when the leading indices are positive integers and the last index is positive real. Its construction gives a minimal polynomial compensator, a complete upper-arc count and exact integral identities. The ordinary positive transform supplies the geometry only after the signed-density zeros and initial moment cancellation have been established. The proof chain follows those dependencies here.

5.22.1 Strict single-variable notation

For an index of depth d, define

$$L_{\boldsymbol{s}}(z) = \mathrm{Li}_{s_1,\dots,s_d}(z,1,\dots,1) = \sum_{n_1>\cdots>n_d\geq 1} \frac{z^{n_1}}{n_1^{s_1}\cdots n_d^{s_d}} = \sum_{n\geq 1} c_{\boldsymbol{s}}(n) z^n, \qquad |z|<1. \tag{5.140}$$

Our parameter domain is

$$s_1,\dots,s_{d-1} \in \mathbb{N}_{\geq 1}, \qquad s_d = b > 0. \tag{5.141}$$

At depth one, this means any $b > 0$. Indices are strictly decreasing, not weakly decreasing. Only the outer summation variable receives z^{n_1}. This convention agrees with the single-variable convention in the manuscript and in the multiple-polylogarithm literature. The principal branch conventions are those fixed at the start of the book.

Write

$$C_{\boldsymbol{s}} = c_{\boldsymbol{s}}(d) = \frac{1}{d^{s_1}(d-1)^{s_2}\cdots 2^{s_{d-1}}} > 0. \tag{5.142}$$

The empty product at depth one is one. The value of $C_{\boldsymbol{s}}$ is independent of b, because the innermost index of the first admissible chain is one. Also $c_{\boldsymbol{s}}(1) = \cdots = c_{\boldsymbol{s}}(d-1) = 0$. Throughout, principal continuation means continuation to

$$\Omega = \mathbb{C} \setminus [1, \infty).$$

The word "zero" refers to a zero of a holomorphic function, unless the imaginary part or a signed density is explicitly specified.

5.22.2 Main result

Theorem 5.69 (All-depth compensation and zero geometry)**.** *Let* (5.141) *hold. There is a smooth real density* $\kappa_{\boldsymbol{s}} \in L^1(0,1)$ *such that*

$$\int_0^1 u^{n-1} \kappa_{\boldsymbol{s}}(u)\, \mathrm{d}u = c_{\boldsymbol{s}}(n), \qquad \|\kappa_{\boldsymbol{s}}\|_1 \le 2^{d-1}.$$

It has exactly $d-1$ *simple zeros* $0 < r_1 < \cdots < r_{d-1} < 1$*, with alternating signs and positive sign near one. Set*

$$P_{\boldsymbol{s}}(u) = \prod_{j=1}^{d-1} (u - r_j), \qquad D_{\boldsymbol{s}}(z) = \prod_{j=1}^{d-1} (1 - r_j z).$$

Then $\mathrm{d}\mu_{\boldsymbol{s}} = P_{\boldsymbol{s}} \kappa_{\boldsymbol{s}}\, \mathrm{d}u$ *is a positive measure of mass* $C_{\boldsymbol{s}}$*, and*

$$L_{\boldsymbol{s}}(z) = \frac{z^d}{D_{\boldsymbol{s}}(z)} H_{\boldsymbol{s}}(z), \qquad H_{\boldsymbol{s}}(z) = \int_0^1 \frac{\mathrm{d}\mu_{\boldsymbol{s}}(u)}{1 - zu}. \tag{5.143}$$

The polynomial $D_{\boldsymbol{s}}$ *is the unique normalized polynomial of smallest degree making* $D_{\boldsymbol{s}} L_{\boldsymbol{s}} / z^d$ *a positive Hausdorff–Stieltjes transform. In particular,* $L_{\boldsymbol{s}}$ *has no zero in* Ω *other than its zero of multiplicity* d *at zero.*

For every $0 < \rho \le 1$*, there are exactly* $d-1$ *simple zeros*

$$0 < \theta_1(\rho) < \cdots < \theta_{d-1}(\rho) < \pi$$

of $\theta \mapsto \operatorname{Im} L_{\boldsymbol{s}}(\rho e^{\mathrm{i}\theta})$*. With* $\alpha_k = k\pi/d$ *they satisfy*

$$\alpha_k - \arcsin(\rho \sin \alpha_k) < \theta_k(\rho) < \alpha_k. \tag{5.144}$$

Every branch satisfies $\theta_k'(\rho) < 0$ *for* $0 < \rho \le 1$ *(with the one-sided derivative at one). The imaginary part starts positive and alternates sign at these zeros; the value of* $L_{\boldsymbol{s}}$ *at the* k*th crossing is real with sign* $(-1)^k$*.*

The proof is divided into operator, oscillation, compensation, and angular-count steps below. None of those steps is inferred from the finite verification suite.

5.23 Two bounded operators on signed moment densities

For $f \in L^1(0,1)$ define its moments by

$$m_n(f) = \int_0^1 u^{n-1} f(u)\, \mathrm{d}u, \qquad n \ge 1.$$

We use the two operators

$$(\mathsf{A}f)(u) = \int_u^1 \frac{f(v)}{v}\,\mathrm{d}v, \tag{5.145}$$

$$(\mathsf{B}f)(u) = \int_0^u \frac{f(v)}{1-v}\,\mathrm{d}v - \int_u^1 \frac{f(v)}{v}\,\mathrm{d}v. \tag{5.146}$$

For every fixed $0 < u < 1$, both integrals in (5.146) are absolutely convergent. In particular, this definition never subtracts two divergent endpoint integrals. The operator A is the outer-index-raising operator already present at depth two in the manuscript [106]. The companion B encodes a strict prefix sum of moments.

Lemma 5.70 (Norm, derivative, and moment identities)**.** *Both operators map $L^1(0,1)$ to itself and satisfy*

$$\|\mathsf{A}f\|_1 \le \|f\|_1, \qquad \|\mathsf{B}f\|_1 \le 2\|f\|_1. \tag{5.147}$$

Their moments are

$$m_n(\mathsf{A}f) = \frac{m_n(f)}{n}, \qquad m_n(\mathsf{B}f) = \frac{1}{n}\sum_{j=1}^{n-1} m_j(f). \tag{5.148}$$

If f is continuous in the open interval, then

$$(\mathsf{A}f)'(u) = -\frac{f(u)}{u}, \qquad (\mathsf{B}f)'(u) = \frac{f(u)}{u(1-u)}. \tag{5.149}$$

Moreover $\mathsf{A}f$ extends continuously to one with value zero, and $m_1(\mathsf{B}f) = 0$.

Proof. Tonelli's theorem applied to absolute values gives

$$\int_0^1 \int_u^1 \frac{|f(v)|}{v}\,\mathrm{d}v\,\mathrm{d}u = \int_0^1 |f(v)|\,\mathrm{d}v.$$

Similarly,

$$\int_0^1 \int_0^u \frac{|f(v)|}{1-v}\,\mathrm{d}v\,\mathrm{d}u = \int_0^1 |f(v)|\,\mathrm{d}v.$$

These identities prove the two operator bounds and justify every following application of Fubini to a signed integrand. For the first operator,

$$m_n(\mathsf{A}f) = \int_0^1 \frac{f(v)}{v}\left(\int_0^v u^{n-1}\,\mathrm{d}u\right)\mathrm{d}v = \frac{m_n(f)}{n}.$$

The moment of the first term of $\mathsf{B}f$ is

$$\frac{1}{n}\int_0^1 \frac{1-v^n}{1-v} f(v)\,\mathrm{d}v = \frac{1}{n}\sum_{j=1}^{n} m_j(f).$$

Subtracting $m_n(f)/n$ yields the strict prefix in (5.148). The derivative formulas follow locally from the fundamental theorem of calculus. Finally, $\int_u^1 f(v)/v\,\mathrm{d}v \to 0$ as $u \uparrow 1$, because $1/v$ is bounded near one and f is integrable there. □

5.23.1 Construction at arbitrary depth

The positive starting density is

$$f_b(u) = \frac{(-\log u)^{b-1}}{\Gamma(b)}, \qquad b > 0. \tag{5.150}$$

Its mass is one and the Gamma integral gives $m_n(f_b) = n^{-b}$. Define, with composition read from right to left,

$$\kappa_{\boldsymbol{s}} = \mathsf{A}^{s_1-1}\mathsf{B}\mathsf{A}^{s_2-1}\mathsf{B}\cdots\mathsf{A}^{s_{d-1}-1}\mathsf{B}f_b. \tag{5.151}$$

There are $d-1$ occurrences of B; all powers of A are nonnegative integer powers. At depth one the right side is simply f_b.

Proposition 5.71 (Moment realization and continuation)**.** *The density* (5.151) *is smooth on* $(0,1)$*, belongs to* $L^1(0,1)$*, and satisfies*

$$m_n(\kappa_{\boldsymbol{s}}) = c_{\boldsymbol{s}}(n), \qquad \|\kappa_{\boldsymbol{s}}\|_1 \le 2^{d-1}. \tag{5.152}$$

The principal continuation is

$$L_{\boldsymbol{s}}(z) = z\int_0^1 \frac{\kappa_{\boldsymbol{s}}(u)}{1-zu}\,\mathrm{d}u, \qquad z \in \Omega. \tag{5.153}$$

The density is the unique finite signed measure realizing these moments.

Proof. The coefficient recurrence for a nonempty tail $\boldsymbol{t}$ is

$$c_{(a,\boldsymbol{t})}(n) = \frac{1}{n^a}\sum_{j=1}^{n-1} c_{\boldsymbol{t}}(j).$$

It is exactly the effect of $\mathsf{A}^{a-1}\mathsf{B}$ in (5.148). Induction proves the moments and the norm bound. The derivative identities prove local smoothness, starting from (5.150). Expanding the denominator in the open disk is justified by the integrable bound $|\kappa|/(1-|z|)$, and gives the series (5.140).

On a compact subset of Ω, $|1-zu|$ is bounded away from zero uniformly in $u \in [0,1]$. The signed integral is consequently holomorphic there and continues the same germ. If two finite signed measures have these moments, their difference integrates every polynomial to zero. Uniform polynomial approximation on $[0,1]$, followed by uniqueness of a finite measure from its continuous-function integrals, proves equality. □

Corollary 5.72 (Ordinary convergence on the nontrivial unit circle)**.** *For* (5.141)*,* $c_{\boldsymbol{s}}(n) \to 0$ *and*

$$\sum_{n\ge 1} |c_{\boldsymbol{s}}(n+1) - c_{\boldsymbol{s}}(n)| \le \|\kappa_{\boldsymbol{s}}\|_1 \le 2^{d-1}. \tag{5.154}$$

Thus the original series (5.140) *converges at every* $|z| = 1$*,* $z \ne 1$*, and agrees there with* (5.153)*.*

Proof. Dominated convergence proves the coefficient limit. Also

$$|c(n+1) - c(n)| \le \int_0^1 u^{n-1}(1-u)|\kappa(u)|\,\mathrm{d}u.$$

Summing the positive majorants gives (5.154). Summation by parts against the bounded partial sums of z^n then proves convergence when $|z| = 1$, $z \ne 1$. Abel's limiting theorem identifies its sum with the radial limit of the disk series, which in turn equals the continuous signed integral at that point. □

This conclusion concerns the parameter domain of this article. It is not a claim that all of the manuscript's more general fractional-order boundary series converge ordinarily.

5.24 Exact oscillation and two interlacing laws

The central mechanism is a balance between moment orthogonality and Rolle's theorem. Vanishing moments force sign changes; the two derivative identities prevent extra zeros. This proof does not assume a variation-diminishing theorem for fractional integration.

Lemma 5.73 (Vanishing moments force sign changes)**.** *Suppose a nonzero continuous real $f \in L^1(0,1)$ has finitely many zeros and*

$$\int_0^1 u^j f(u)\,\mathrm{d}u = 0 \qquad (0 \le j \le m-1).$$

Then f has at least m sign changes in $(0,1)$.

Proof. If its sign changes were $t_1, \dots, t_k$ with $k < m$, the polynomial $Q(u) = \prod_{j=1}^{k}(u - t_j)$ would have degree at most $m-1$ and Qf would have one fixed nonzero sign except at finitely many points. Its integral would be nonzero, contradicting orthogonality. A zero at which no sign change occurs is not included in Q. □

Theorem 5.74 (Exact density zeros)**.** *For (5.141), $\kappa_{\boldsymbol{s}}$ has exactly $d-1$ distinct simple zeros in $(0,1)$. Its sign alternates at those zeros, and it is positive near one.*

Proof. At depth one, f_b is strictly positive and has no zero. Suppose first that a density f at depth $d-1$ has exactly $d-2$ simple zeros. By (5.149), the derivative of $\mathsf{B}f$ has exactly those zeros. Rolle's theorem therefore bounds the number of distinct zeros of $\mathsf{B}f$ by $d-1$. This also rules out an infinite set of zeros: any d distinct zeros would already force too many zeros of the derivative.

The first $d-1$ moments of the new density vanish, by (5.148). Lemma 5.73 supplies at least $d-1$ sign changes, so the upper bound is attained. There is one derivative zero between each pair of consecutive new zeros, using all $d-2$ derivative zeros. None remains at a new zero. Hence every new zero is simple.

Now suppose f is a depth-d density with exactly $d-1$ simple zeros, and put $g = \mathsf{A}f$. We have $g(1) = 0$. If g had k interior zeros, Rolle's theorem between successive interior zeros and between the largest one and the endpoint one would give at least k zeros of $g' = -f/u$. Thus $k \le d-1$. Its first $d-1$ moments still vanish, so the same lower bound gives exactly $d-1$ sign changes. The derivative zeros are now all allocated to those successive intervals, including the last interval ending at one. Consequently the interior zeros of g are simple. Repeating this argument for each occurrence of A and B proves the zero count.

Let $P(u)$ be the monic polynomial whose roots are the resulting $d-1$ zeros. The product $P\kappa_{\boldsymbol{s}}$ has a fixed strict sign off its roots. Orthogonality to degrees at most $d-2$ gives

$$\int_0^1 P(u)\kappa_{\boldsymbol{s}}(u)\,\mathrm{d}u = \int_0^1 u^{d-1}\kappa_{\boldsymbol{s}}(u)\,\mathrm{d}u = C_{\boldsymbol{s}} > 0.$$

Thus $P\kappa_{\boldsymbol{s}}$ is positive, and since P is positive to the right of its largest root, so is $\kappa_{\boldsymbol{s}}$. □

Corollary 5.75 (Index raising and depth raising interlace)**.** *For $d \ge 2$, if $f = \kappa_{\boldsymbol{s}}$ has roots $r_1 < \cdots < r_{d-1}$ and $\mathsf{A}f$ has roots $q_1 < \cdots < q_{d-1}$, then*

$$q_1 < r_1 < q_2 < r_2 < \cdots < q_{d-1} < r_{d-1} < 1. \tag{5.155}$$

Thus raising the first index by one moves every density zero strictly toward zero. For $d \ge 3$, if a depth-$(d-1)$ density has roots $t_1 < \cdots < t_{d-2}$ and $\mathsf{B}f$ has roots $r_1 < \cdots < r_{d-1}$, then

$$r_1 < t_1 < r_2 < \cdots < t_{d-2} < r_{d-1}. \tag{5.156}$$

Proof. At $d = 2$, depth raising creates one interior zero and there is no previous zero to interlace. For the stated ranges, these are precisely the derivative-zero allocations in the preceding proof. In the A case the last interval is $(q_{d-1}, 1)$; in the B case all intervals are between consecutive interior zeros. The number of derivative zeros equals the number of allocated intervals, so the inequalities are strict and exhaustive. □

5.24.1 Positive seeds, terminal mixtures, and a Lerch extension

The preceding proof used only positivity, smoothness, integrability, and the positive mass of the initial density. The explicit form (5.150) was needed to identify its moments as n^{-b}.

Theorem 5.76 (Positive-seed extension)**.** *Let f be a smooth strictly positive density in $L^1(0,1)$, of mass $M_0 > 0$. Replace f_b by f in (5.151). All density-zero, interlacing, compensation, slit-plane nonvanishing, and angular-count conclusions of this article remain valid. The norm bound becomes $2^{d-1}M_0$, and the mass of the positive compensated measure is*

$$\frac{M_0}{d^{s_1}(d-1)^{s_2}\cdots 2^{s_{d-1}}}.$$

Proof. The operator identities still hold. After $d-1$ strict prefix steps, the first $d-1$ moments vanish and the first nonzero moment is the stated positive number. The proof of Theorem 5.74 therefore applies unchanged. The arguments in the next two sections use only these moment and oscillation properties, and so establish the remaining assertions as well. □

In particular, for fixed leading indices, every finite positive linear combination

$$\sum_{j=1}^{N} a_j L_{s_1,\dots,s_{d-1},b_j}(z), \qquad a_j > 0, \quad b_j > 0, \tag{5.157}$$

has the same nonvanishing and exact angular-count properties. Its seed is $\sum a_j f_{b_j}$. This is a special convex-closure statement; sums of arbitrary zero-free holomorphic functions need not be zero-free.

For a second application, take $\lambda > -1$ and

$$f_{b,\lambda}(u) = u^{\lambda}\frac{(-\log u)^{b-1}}{\Gamma(b)}.$$

Its moments are $(n+\lambda)^{-b}$ and its mass is $(1+\lambda)^{-b}$. The resulting strict shifted-terminal family is

$$L^{(\lambda)}_{s_1,\dots,s_{d-1};b}(z) = \sum_{n_1>\cdots>n_d\ge 1} \frac{z^{n_1}}{n_1^{s_1}\cdots n_{d-1}^{s_{d-1}}(n_d+\lambda)^b}. \tag{5.158}$$

It has the same minimal compensator degree $d-1$ and exactly $d-1$ angular zeros for $0 < \rho \le 1$. At depth one it is the usual z-multiple of a Lerch series. No assertion is made here about shifts in all denominators simultaneously.

5.25 Canonical positive compensation and slit-plane nonvanishing

Fix $\boldsymbol{s}$ and abbreviate $\kappa = \kappa_{\boldsymbol{s}}$, $P = P_{\boldsymbol{s}}$, $D = D_{\boldsymbol{s}}$, and $C = C_{\boldsymbol{s}}$. Empty products at depth one are one. The preceding section proves that

$$P(u)\kappa(u) > 0 \quad \text{for all } u \in (0,1) \setminus \{r_1, \dots, r_{d-1}\}. \tag{5.159}$$

Since P is bounded on $[0,1]$, the positive measure $\mathrm{d}\mu = P\kappa\,\mathrm{d}u$ is finite.

Lemma 5.77 (Polynomial cancellation)**.** *Suppose $m_1(\kappa) = \cdots = m_{d-1}(\kappa) = 0$. For any real polynomial E of degree at most $d-1$, the removable quotient at zero satisfies*

$$E(z)\frac{L(z)}{z^d} = \int_0^1 \frac{u^{d-1}E(1/u)\kappa(u)}{1-zu}\,\mathrm{d}u. \tag{5.160}$$

Here $u^{d-1}E(1/u)$ is a polynomial, including at $u = 0$.

Proof. First subtract the initial $d-1$ zero moments in (5.153) to obtain

$$L(z) = z^d \int_0^1 \frac{u^{d-1}\kappa(u)}{1-zu}\,\mathrm{d}u.$$

For $0 \le j \le d-1$, multiplying this last transform by z^j and moving the factor z^j to the density changes it to $u^{d-1-j}\kappa(u)$. The difference is a finite linear combination of the moments of degrees $d-1-j, \ldots, d-2$, all zero. Applying this monomial identity to every coefficient of E proves (5.160) in the disk and hence throughout Ω. □

Theorem 5.78 (Positive factorization and nonvanishing)**.** *The measure μ has mass C, and*

$$D(z)L(z) = z^d \int_0^1 \frac{\mathrm{d}\mu(u)}{1-zu}. \tag{5.161}$$

Its transform H has strictly positive imaginary part when $\operatorname{Im} z > 0$, strictly negative imaginary part when $\operatorname{Im} z < 0$, and $H(x) > 0$ for every real $x < 1$. The only zero of L on Ω is its zero of multiplicity d at zero.

Proof. We have $u^{d-1}D(1/u) = P(u)$, so the preceding lemma gives (5.161). Its value after division by z^d at zero is $\int P\kappa = C$, as already shown in the oscillation proof. For nonreal z,

$$\operatorname{Im} H(z) = (\operatorname{Im} z)\int_0^1 \frac{u\,\mathrm{d}\mu(u)}{|1-zu|^2}.$$

The integral is strictly positive because the measure has positive density except at finitely many interior points. For real $x < 1$, the integrand defining $H(x)$ is positive. Thus H has no zero on Ω. All zeros of D are $1/r_j > 1$, outside Ω, while $D(0) = 1$ and $H(0) = C > 0$. The factorization proves the claimed multiplicity and the global nonvanishing assertion. □

Theorem 5.79 (Minimality and uniqueness of the compensator)**.** *Among real polynomials E with $E(0) > 0$ for which $E(z)L(z)/z^d$ is the transform of a finite positive measure on $[0,1]$, the smallest possible degree is $d-1$. Every such polynomial of degree $d-1$ is $E(0)D(z)$. In particular the normalization $E(0) = 1$ singles out D.*

Proof. Existence in degree $d-1$ follows from Theorem 5.78. Suppose that $\deg E \le d-1$. Lemma 5.77 already realizes the transform as the finite signed measure with density

$$Q_E(u)\kappa(u), \qquad Q_E(u) = u^{d-1}E(1/u).$$

Uniqueness of finite Hausdorff moments implies that any purported positive representing measure must be this signed measure. Its density must therefore be nonnegative almost everywhere. Since κ changes sign at every r_j, continuity forces $Q_E(r_j) = 0$. As $r_j \neq 0$, these are $d-1$ distinct roots $1/r_j$ of E. Consequently $\deg E \ge d-1$. At equality those roots determine E up to its constant factor, which is fixed by $E(0)$. □

This is minimality of *polynomial compensation for an ordinary positive Stieltjes representation.* It is not minimal depth in an algebra of numerical periods and not a minimal generalized Stieltjes order. Those are different questions.

5.25.1 Quantitative consequences

For $x > 0$, positivity and $0 < u < 1$ give

$$\frac{Cx^d}{(1+x)D(-x)} < (-1)^d L(-x) < \frac{Cx^d}{D(-x)}. \tag{5.162}$$

Indeed $C/(1+x) < H(-x) < C$ and $D(-x) > 0$. Without computing the compensator roots one still has the coarser bounds

$$C\left(\frac{x}{1+x}\right)^d < (-1)^d L(-x) < Cx^d, \tag{5.163}$$

because $1 \le D(-x) \le (1+x)^{d-1}$, with the relevant strictness supplied by H even at depth one.

Write $P(u) = \sum_{j=0}^{d-1} p_j u^j$ and

$$h_n = \int_0^1 u^n \, \mathrm{d}\mu(u) = \sum_{j=0}^{d-1} p_j c_{\boldsymbol{s}}(n+j+1), \qquad n \ge 0. \tag{5.164}$$

For all $n, k \ge 0$,

$$(-1)^k \Delta^k h_n = \int_0^1 u^n (1-u)^k \, \mathrm{d}\mu(u) > 0, \qquad \Delta h_n = h_{n+1} - h_n. \tag{5.165}$$

Every finite Hankel matrix $(h_{i+j})_{0 \le i,j \le m}$ is positive definite. The same is true of (h_{i+j+1}) and $(h_{i+j} - h_{i+j+1})$. To see this, pair each matrix with a nonzero real vector and integrate the square of its polynomial against, respectively, μ, $u\mu$, or $(1-u)\mu$. Each integral is strictly positive. In particular,

$$h_n h_{n+2} - h_{n+1}^2 > 0.$$

These are exact nonlinear positivity constraints on the original nested harmonic coefficients and the compensator roots. They should not be mistaken for rational-coefficient identities among special values.

5.26 Exactly $d-1$ angular zeros on every upper semicircle

The signed transform gives, for $0 < \rho \le 1$ and $0 < \theta < \pi$,

$$\operatorname{Im} L(\rho e^{\mathrm{i}\theta}) = \rho \sin\theta \, J_\rho(\cos\theta), \qquad J_\rho(c) = \int_0^1 \frac{\kappa(u)}{1 - 2\rho c u + \rho^2 u^2} \, \mathrm{d}u. \tag{5.166}$$

A bound based only on simple sign-change arguments would leave open the possibility of tangencies. We therefore prove a multiplicity-sensitive upper bound before proving existence.

5.26.1 An elementary confluent Cauchy-kernel bound

Lemma 5.80 (Angular variation bound with multiplicities)**.** *Suppose $\kappa \in L^1(0,1)$ is continuous, has exactly m sign changes at $r_1 < \cdots < r_m$, and has a nonzero fixed sign on each complementary interval. Then J_ρ has at most m zeros in $(-1,1)$, counted with analytic multiplicity, for every $0 < \rho \le 1$.*

Proof. The function

$$t(u) = \frac{1+\rho^2u^2}{2\rho u} \tag{5.167}$$

is strictly decreasing on $(0,1)$, since $t'(u) = (\rho^2u^2-1)/(2\rho u^2) < 0$. Its values are at least one, and it tends to infinity as $u \downarrow 0$. Thus

$$J_\rho(c) = \int_0^1 \frac{\kappa(u)}{2\rho u}\frac{1}{t(u)-c}\,\mathrm{d}u.$$

The factor $\kappa(u)/u$ need not be integrable by itself; this causes no difficulty because the displayed Cauchy factor is $O(u)$ at zero. All the integrals below are interpreted with their complete kernels.

Suppose there are distinct zeros $c_1,\dots,c_q \in (-1,1)$ with positive multiplicities $\ell_1,\dots,\ell_q$ summing to $m+1$. If more zeros are available, retain any subcollection of total multiplicity $m+1$. Form the proper rational function

$$R(t) = \frac{\prod_{j=1}^m (t-t(r_j))}{\prod_{a=1}^q (t-c_a)^{\ell_a}}. \tag{5.168}$$

Its partial fractions are linear combinations of $(t-c_a)^{-h}$ for $1 \le h \le \ell_a$. Differentiation of J_ρ under the integral is valid locally for $c \in (-1,1)$; explicitly,

$$J_\rho^{(h-1)}(c) = (h-1)!\int_0^1 \frac{\kappa(u)}{2\rho u}\frac{1}{(t(u)-c)^h}\,\mathrm{d}u.$$

The selected zero multiplicities therefore imply

$$\int_0^1 \frac{\kappa(u)}{2\rho u} R(t(u))\,\mathrm{d}u = 0. \tag{5.169}$$

This integral is absolutely convergent: near zero $R(t) = O(1/t) = O(u)$, and near one its denominators stay positive and bounded away from zero.

But each factor $t(u)-t(r_j)$ changes sign exactly where κ changes sign. The denominator of $R(t(u))$ is strictly positive, because $t(u) \ge 1 > c_a$. Hence the integrand in (5.169) has one fixed nonzero sign except at the r_j. Its integral cannot vanish. This contradiction proves the bound and also excludes an identically zero J_ρ. □

The proof is a finite partial-fraction argument, including repeated poles. No unproved strict-total-positivity assertion is being used.

5.26.2 Positive compensation supplies the missing zeros

For $0 < \theta < \pi$ and $0 \le u \le 1$, put

$$\beta_u(\theta) = -\operatorname{Arg}(1-\rho u e^{\mathrm{i}\theta}).$$

For $0 < u < 1$ one has

$$0 < \beta_u(\theta) < \beta_1(\theta) < \frac{\pi}{2}. \tag{5.170}$$

The angle is strictly increasing in u, since differentiation with respect to u gives $\rho\sin\theta/|1-\rho u e^{\mathrm{i}\theta}|^2 > 0$. A positive mixture of the resolvents $(1-zu)^{-1}$ stays strictly inside this sector. Consequently, with H as in (5.161),

$$0 < \operatorname{Arg} H(\rho e^{\mathrm{i}\theta}) < \beta_1(\theta). \tag{5.171}$$

For example, the second inequality follows by rotating each resolvent through $-\beta_1$ and observing that its imaginary part is strictly negative for $0 < u < 1$. All rays lie in an angle of width less than $\pi/2$, so no argument ambiguity arises.

Theorem 5.81 (Complete angular count and signs)**.** *For* (5.141) *and every* $0 < \rho \le 1$, $\operatorname{Im} L(\rho e^{\mathrm{i}\theta})$ *has exactly* $d-1$ *simple zeros in* $0 < \theta < \pi$*. They have the signs and bounds stated in Theorem 5.69.*

Proof. A continuous unwrapped argument of L along the upper arc is

$$\Phi(\theta) = d\theta + \sum_{j=1}^{d-1} \beta_{r_j}(\theta) + \operatorname{Arg} H(\rho e^{\mathrm{i}\theta}). \tag{5.172}$$

The factorization proves this formula exactly; Φ is not reduced modulo 2π. The sector bounds imply

$$d\theta < \Phi(\theta) < d\big(\theta + \beta_1(\theta)\big) < d\pi. \tag{5.173}$$

At the other end of the arc, $\Phi(\theta) \to d\pi$ as $\theta \uparrow \pi$, since all resolvents approach positive real values and $H(-\rho) > 0$.

If $\rho < 1$, then $\Phi(\theta) \to 0$ as $\theta \downarrow 0$. The case $\rho = 1$ requires more care: $L(1)$ may be singular, and there is no reason to assign it argument zero. Instead, each fixed $r_j < 1$ gives $\beta_{r_j}(\theta) \to 0$, while (5.171) gives

$$\limsup_{\theta \downarrow 0} \operatorname{Arg} H(e^{\mathrm{i}\theta}) \le \pi/2.$$

It follows that $0 < \Phi(\theta) < \pi$ for sufficiently small positive θ. This weaker endpoint statement is all that is needed.

By continuity, the phase crosses each of $\pi, 2\pi, \dots, (d-1)\pi$. These yield at least $d-1$ distinct zeros of the imaginary part. On the other hand, Lemma 5.80 with $m = d-1$, together with (5.166), gives at most $d-1$ zeros counted with multiplicities. The change of variable $c = \cos\theta$ has nonzero derivative in the open arc, and the prefactor $\rho \sin\theta$ is positive. Thus multiplicities are preserved. Every zero is simple, and there is exactly one crossing of each listed phase level. The levels must occur in their stated order, since a reversal would force a repeated crossing of an earlier level.

The initial phase lies in $(0, \pi)$, so the imaginary part starts positive. Simplicity forces alternating signs thereafter. At the kth zero, the phase is $k\pi$ and the modulus is nonzero, so the real value has sign $(-1)^k$.

For the location bounds, write $\alpha_k = k\pi/d$. At a zero, (5.173) gives

$$\theta_k < \alpha_k < \theta_k + \beta_1(\theta_k).$$

The function $h(\theta) = \theta + \beta_1(\theta)$ is strictly increasing, because

$$h'(\theta) = \frac{1 - \rho\cos\theta}{1 - 2\rho\cos\theta + \rho^2} > 0.$$

For $\rho < 1$, solving $h(\theta) = \alpha$ gives $\theta = \alpha - \arcsin(\rho \sin\alpha)$. At $\rho = 1$ the same expression is the solution when $\alpha > \pi/2$; when $\alpha \le \pi/2$ it is zero, and the strict lower bound follows from $\theta_k > 0$. This proves (5.144). □

The argument proves the needed crossings without asserting that Φ is globally monotone. Such an assertion is unnecessary here.

Corollary 5.82 (Analytic motion and no interior collisions)**.** *For fixed integer leading indices, each angular branch is real analytic in* $(\rho, b) \in (0,1) \times (0,\infty)$*, and extends real analytically across* $\rho = 1$ *locally at each of its nontrivial upper-arc zeros. The density roots* r_j *are real analytic in* $b > 0$*. No two of the angular branches collide for* $0 < \rho \le 1$*.*

Proof. On compact subsets of $\operatorname{Re} b > 0$, the seed f_b is holomorphic as an L^1-valued function; factors $|\log(-\log u)|^j$ arising from parameter differentiation have an integrable common majorant on smaller compact subsets. The bounded operators preserve this holomorphic parameter dependence. The signed transform is locally holomorphic in $z \in \Omega$ and in b. Simplicity of its angular zeros permits the real analytic implicit-function theorem. The unit-radius zeros remain away from the cut, so the same local argument crosses $\rho = 1$. The density derivative formulas give joint real analyticity in u, b in the open interval, and their simple roots obey the same theorem. Distinctness follows from Theorem 5.81. □

The local extension across radius one does not imply that the global count persists at all larger radii. For $L_{1,1} = \log^2(1-z)/2$, an upper-half-plane imaginary zero requires $|1-z| = 1$, so its angle obeys $\cos\theta = \rho/2$. There are no open-upper-arc zeros when $\rho > 2$.

5.26.3 Strict radial motion of every angular branch

The compensation yields more than a zero count. It also turns a monotonicity property of ordinary positive Stieltjes transforms into strict motion of all the strict-depth angular zeros.

Lemma 5.83 (Radial argument monotonicity for positive transforms)**.** *Let* ν *be a nonzero finite positive measure on* $[0,1]$ *and* $Q(z) = \int (1-uz)^{-1}\,\mathrm{d}\nu(u)$*. For* $0 < \theta < \pi$ *and* $0 < \rho \le 1$*,*

$$\frac{\partial}{\partial \rho} \operatorname{Arg} Q(\rho e^{\mathrm{i}\theta}) \ge 0. \tag{5.174}$$

The argument is chosen continuously from $Q(0) > 0$*.*

Proof. First take a finite measure with positive weights at distinct points $0 < u_1 < \cdots < u_n < 1$. The rational function $\sum_j w_j/(t-u_j)$ is strictly decreasing between consecutive poles, with limits $+\infty$ and $-\infty$ at the two ends. Its numerator therefore has exactly one root $v_j \in (u_j, u_{j+1})$ in each such interval. Factoring the numerator gives

$$Q(z) = Q(0)\frac{\prod_{j=1}^{n-1}(1-v_j z)}{\prod_{j=1}^{n}(1-u_j z)}.$$

Consequently the radial derivative of its argument is

$$\sin\theta \left\{ f(u_1) + \sum_{j=1}^{n-1} \big(f(u_{j+1}) - f(v_j)\big) \right\}, \qquad f(t) = \frac{t}{1 - 2\rho t\cos\theta + \rho^2 t^2}.$$

Every summand is nonnegative, because

$$f'(t) = \frac{1-\rho^2 t^2}{(1 - 2\rho t\cos\theta + \rho^2 t^2)^2} \ge 0 \qquad (0 \le t \le 1).$$

Approximate an arbitrary finite positive ν weakly by such finite measures, with their masses converging to $\nu([0,1])$. For each compact subset of Ω, the integrands defining the transform and

its first complex derivative are bounded and continuous uniformly in $u \in [0,1]$. The transforms and derivatives converge locally uniformly. Also $Q(\rho e^{\mathrm{i}\theta}) \neq 0$: it has positive imaginary part unless the measure is supported at zero, in which case it is a positive constant. Passing to the limit in $\mathrm{Im}(e^{\mathrm{i}\theta}Q'(z)/Q(z))$ proves the nonnegative derivative. The same argument works at $\rho = 1$ because θ is strictly between zero and π. □

Theorem 5.84 (Strict inward angular motion)**.** *For $d \geq 2$ and (5.141), every angular branch satisfies*

$$\frac{\mathrm{d}\theta_k}{\mathrm{d}\rho} < 0, \qquad 0 < \rho \leq 1, \quad 1 \leq k \leq d-1. \tag{5.175}$$

The derivative at one is the one-sided derivative, or equivalently the derivative of the local continuation. The same result holds for the positive-seed and shifted-terminal families of Theorem 5.76.

Proof. At fixed θ, differentiate the unwrapped phase (5.172) with respect to the radius:

$$\Phi_\rho = \sum_{j=1}^{d-1} \frac{r_j \sin\theta}{1 - 2\rho r_j \cos\theta + \rho^2 r_j^2} + \partial_\rho \operatorname{Arg} H(\rho e^{\mathrm{i}\theta}) > 0.$$

The first sum is strictly positive since $d \geq 2$ and $r_j > 0$; the last term is nonnegative by Lemma 5.83. At the unique simple crossing of the phase level $k\pi$, one has $\Phi_\theta > 0$. Indeed the phase starts below that level, ends above it, and crosses it exactly once; simplicity excludes a zero phase derivative there. Implicit differentiation of $\Phi(\theta_k(\rho), \rho) = k\pi$ now gives

$$\theta_k'(\rho) = -\frac{\Phi_\rho}{\Phi_\theta} < 0.$$

All ingredients also hold for a general positive seed. □

For depth two, angular decrease alone does *not* settle the higher-leading-order question about $\eta(\rho) = \cos\theta(\rho)/\rho$: the derivative of that quotient has a second term, and its sign is not implied by $\theta'(\rho) < 0$. The two kinds of radial monotonicity must remain separate in the integration ledger.

5.27 Small-radius and large-outer-index expansions

The global count removes an ambiguity from local expansions: the zeros found near the first Fourier-mode zeros are the entire list of angular zeros, not just selected branches.

Theorem 5.85 (Two-term radial expansion)**.** *Fix an index satisfying (5.141) with $d \geq 2$, and put*

$$p = \frac{c_{\boldsymbol{s}}(d+1)}{C_{\boldsymbol{s}}}, \qquad q = \frac{c_{\boldsymbol{s}}(d+2)}{C_{\boldsymbol{s}}}, \qquad \alpha_k = \frac{k\pi}{d}.$$

The kth angular branch extends real analytically to $\rho = 0$, and

$$\begin{aligned}\theta_k(\rho) = \alpha_k &- \frac{p}{d}\sin\alpha_k\,\rho \\ &+ \left(\frac{d+1}{d^2}p^2 - \frac{2q}{d}\right)\sin\alpha_k\cos\alpha_k\,\rho^2 + O_{\boldsymbol{s}}(\rho^3).\end{aligned} \tag{5.176}$$

In particular, every branch initially moves strictly toward smaller angles as the radius increases from zero.

Proof. Divide the imaginary part of the disk series by $C_{\boldsymbol{s}}\rho^d$. The resulting analytic equation is

$$0 = \sin(d\theta) + p\rho\sin((d+1)\theta) + q\rho^2\sin((d+2)\theta) + O(\rho^3).$$

At $(\rho,\theta) = (0,\alpha_k)$ its θ derivative is $d(-1)^k \neq 0$, so the analytic implicit-function theorem applies. There are $d-1$ such branches; for small positive radius they exhaust the global list from Theorem 5.81.

Write $\theta = \alpha_k + A\rho + B\rho^2 + O(\rho^3)$ and divide by $(-1)^k$. The coefficients of ρ and ρ^2 are respectively

$$dA + p\sin\alpha_k, \qquad dB + (d+1)pA\cos\alpha_k + q\sin(2\alpha_k).$$

Solving gives (5.176). The first derivative is negative because $p > 0$ and $0 < \alpha_k < \pi$. □

Higher Taylor coefficients are obtained by the same finite coefficient recursion. At each order the new unknown occurs with coefficient $d(-1)^k$, while all remaining terms depend only on earlier coefficients and finitely many moments. This is an exact algebraic recurrence, not a proof of any global radial sign.

5.27.1 Uniformity as the first index increases

Fix $d \geq 2$ and positive integer middle indices $s_2, \ldots, s_{d-1}$. Let the first index be $a \in \mathbb{N}$ and the last be $b > 0$. Define

$$T_N(b) = \sum_{N \geq n_2 > \cdots > n_d \geq 1} \frac{1}{n_2^{s_2}\cdots n_{d-1}^{s_{d-1}} n_d^b}, \qquad T_* = T_{d-1}(b) > 0.$$

The latter number is a single-chain value and is independent of b. At depth two, $T_N(b) = H_N^{(b)}$ and $T_* = 1$.

Theorem 5.86 (First displacement, uniformly in the final order)**.** *Set $\lambda = d/(d+1)$ and $\mu = d/(d+2)$. As $a \to \infty$ through positive integers, uniformly for $b > 0$, $0 \leq \rho \leq 1$, and $1 \leq k \leq d-1$,*

$$\theta_k(\rho) = \alpha_k - \frac{T_d(b)}{dT_*}\sin\alpha_k\,\rho\lambda^a + O\big(\rho^2\mu^a\big). \tag{5.177}$$

At $\rho = 0$, the branch is interpreted by its analytic extension. The implied constant and the starting threshold for a may depend on the fixed depth and middle indices, but not on b, ρ, or k.

Proof. The normalized imaginary part is

$$\sin(d\theta) + \frac{T_d(b)}{T_*}\rho\lambda^a\sin((d+1)\theta) + R(\theta), \tag{5.178}$$

where

$$R(\theta) = \sum_{n \geq d+2} \frac{T_{n-1}(b)}{T_*}\rho^{n-d}\left(\frac{d}{n}\right)^a \sin(n\theta).$$

All inner denominators are at least one, so $0 \leq T_{n-1}(b) \leq n^{d-1}$ uniformly for $b > 0$. Choose the fixed integer $a_0 = d+3$. For $a \geq a_0$, termwise differentiation is absolutely justified and

$$|R(\theta)| + |R'(\theta)| \leq K\rho^2\mu^a, \quad K = \frac{2}{T_*}\sum_{n \geq d+2} n^d\left(\frac{d+2}{n}\right)^{a_0} < \infty. \tag{5.179}$$

Also $T_d(b)/T_* \leq d^{d-1}/T_*$, independently of b.

On disjoint fixed small neighborhoods of the α_k, the derivative of the first term of (5.178) is bounded away from zero. Its perturbation tends to zero uniformly by (5.179). Evaluating at $\alpha_k \pm K_1\rho\lambda^a$, for a sufficiently large fixed K_1, gives opposite signs for all sufficiently large a. This follows from the linear term $d(\theta - \alpha_k)$ and the uniform bounds on the perturbations; $\mu/\lambda < 1$. Thus each neighborhood contains a unique root with displacement $\delta = O(\rho\lambda^a)$. The global count identifies these roots as all the branches in their established order.

Expand (5.178) at α_k and divide by $(-1)^k$. The root equation becomes

$$0 = d\delta + \frac{T_d(b)}{T_*}\rho\lambda^a \sin\alpha_k + O\big(\delta^2 + \rho\lambda^a|\delta| + \rho^2\mu^a\big).$$

Because $\lambda^2 < \mu$, the previously obtained displacement bound makes the error $O(\rho^2\mu^a)$. Division by d proves (5.177). All constants used above are independent of the final order. □

At depth two and unit radius this recovers the leading scale $(2/3)^a$ already present in the manuscript's more detailed depth-two expansion [106]. The added content is the simultaneous all-depth statement, the exact branch indexing, and uniformity over the entire range $b > 0$ and $0 \le \rho \le 1$.

5.28 Explicit logistic kernels and positive integral identities

The compensator is generally implicit: its roots are the zeros of a signed density. The all-ones family makes every part explicit and gives an infinite collection of exact positive integral identities.

5.28.1 The all-ones density and its roots

Let $\mathbf{1}_d = (1, \dots, 1)$ and define

$$\ell(u) = \log\frac{u}{1-u}, \qquad 0 < u < 1.$$

The familiar identity

$$L_{\mathbf{1}_d}(z) = \frac{(-\log(1-z))^d}{d!} \tag{5.180}$$

follows recursively from $(1-z)L'_{\mathbf{1}_d}(z) = L_{\mathbf{1}_{d-1}}(z)$ and the vanishing initial value at zero. We use it as a classical identity with a short proof, not as a new reduction of multiple polylogarithms.

Theorem 5.87 (Explicit signed kernels and logistic roots)**.** *For every $d \ge 1$,*

$$\begin{aligned}\kappa_{\mathbf{1}_d}(u) &= \frac{\operatorname{Im}(\ell(u) + \mathrm{i}\pi)^d}{\pi d!} \\ &= \sum_{j=0}^{\lfloor (d-1)/2\rfloor} \frac{(-1)^j \pi^{2j} \ell(u)^{d-1-2j}}{(2j+1)!(d-1-2j)!}.\end{aligned} \tag{5.181}$$

Its increasing list of roots is

$$r_j = \frac{1}{1 + \exp(\pi\cot(j\pi/d))}, \qquad 1 \le j \le d-1. \tag{5.182}$$

In particular, $r_j + r_{d-j} = 1$.

Proof. Consider the generating function

$$G(t,u) = e^{t\ell(u)}\frac{\sin \pi t}{\pi t} = \sum_{d\geq 1} K_d(u) t^{d-1}. \tag{5.183}$$

For $|\operatorname{Re} t| < 1$, its integral at moment one is the beta integral

$$\int_0^1 G(t,u)\,\mathrm{d}u = \frac{\sin \pi t}{\pi t}\Gamma(1+t)\Gamma(1-t) = 1.$$

The reflection formula supplies the last equality, with removable values understood at zero. Thus $\int K_1 = 1$ and $\int K_d = 0$ for $d \geq 2$. Coefficient extraction is legitimate on a smaller complex neighborhood of zero, since $u^{\operatorname{Re} t}(1-u)^{-\operatorname{Re} t}$ and its logarithmic parameter derivatives have an integrable majorant.

Now $K_1 = 1$ and

$$\frac{\mathrm{d}}{\mathrm{d}u}G(t,u) = \frac{t}{u(1-u)}G(t,u),$$

so $K_d' = K_{d-1}/(u(1-u))$. This is exactly the derivative of $\mathsf{B}K_{d-1}$. Their difference is constant, and both have zero integral for $d \geq 2$, so $K_d = \mathsf{B}K_{d-1}$. Hence these are the constructed all-ones kernels. Expanding the exponential and sine gives (5.181).

The imaginary part of $(\ell + \mathrm{i}\pi)^d$ vanishes when $\operatorname{Arg}(\ell + \mathrm{i}\pi)$ is one of $j\pi/d$. This gives $\ell = \pi\cot(j\pi/d)$. Reversing the order of these real values and solving $u/(1-u) = e^{\ell}$ yields (5.182). The symmetry follows from $\cot(\pi - x) = -\cot x$. □

Proposition 5.88 (A continuous Mellin identity and Stirling moments)**.** *For real $x > 0$ and complex t satisfying $-x < \operatorname{Re} t < 1$,*

$$\int_0^1 u^{x-1}e^{t\ell(u)}\frac{\sin \pi t}{\pi t}\,\mathrm{d}u = \frac{\Gamma(x+t)}{\Gamma(x+1)\Gamma(1+t)}. \tag{5.184}$$

The values at removable singularities are obtained by continuation. For every integer $n \geq 1$,

$$\int_0^1 u^{n-1}\kappa_{\mathbf{1}_d}(u)\,\mathrm{d}u = \frac{\left[\begin{smallmatrix} n \\ d \end{smallmatrix}\right]}{n!}, \tag{5.185}$$

where $\left[\begin{smallmatrix} n \\ d \end{smallmatrix}\right]$ is the unsigned Stirling number of the first kind, zero for $d > n$.

Proof. The integral equals $\sin(\pi t)B(x+t, 1-t)/(\pi t)$. The beta formula and Gamma reflection simplify it to (5.184). At integer $x = n$, the right side is

$$\frac{(t+1)(t+2)\cdots(t+n-1)}{n!}.$$

Its coefficient of t^{d-1} is $\left[\begin{smallmatrix} n \\ d \end{smallmatrix}\right]/n!$, proving (5.185) by (5.183). □

These beta and Stirling identities provide an independent normalization check on the general signed-moment construction.

5.28.2 An explicit infinite family of compensated identities

For the roots (5.182), define

$$P_d(u) = \prod_{j=1}^{d-1}(u - r_j), \qquad D_d(z) = \prod_{j=1}^{d-1}(1 - r_j z).$$

The integrable function $P_d\kappa_{\mathbf{1}_d}$ is strictly positive away from its finitely many zeros. Combining (5.161) and (5.180) gives the exact identity

$$\boxed{\int_0^1 \frac{P_d(u)\kappa_{\mathbf{1}_d}(u)}{1 - zu}\,\mathrm{d}u = \frac{D_d(z)}{d!}\left(\frac{-\log(1-z)}{z}\right)^d, \qquad z \in \Omega.} \tag{5.186}$$

At $z = 0$ the removable value is $1/d!$. At $z = -1$ this becomes

$$\int_0^1 \frac{P_d(u)\kappa_{\mathbf{1}_d}(u)}{1 + u}\,\mathrm{d}u = \frac{D_d(-1)}{d!}\log^d 2. \tag{5.187}$$

The content is not a new formula for $L_{\mathbf{1}_d}$; it is the explicit positive density, its canonical compensation, and the resulting positive-integrand identities. They remain valid at every depth without a numerical relation search.

5.28.3 Depth three

Put

$$h_3 = \frac{\pi}{\sqrt{3}}, \qquad q_3 = \frac{1}{2 + 2\cosh h_3}.$$

Then

$$P_3(u) = u^2 - u + q_3, \qquad D_3(z) = 1 - z + q_3 z^2, \qquad \kappa_{\mathbf{1}_3}(u) = \frac{\ell(u)^2 - \pi^2/3}{2}.$$

Both factors P_3 and $\ell^2 - \pi^2/3$ are negative between the same two roots and positive outside them. Hence their product is nonnegative throughout the interval. Formula (5.187) gives

$$\boxed{\int_0^1 \frac{(u^2 - u + q_3)(\ell(u)^2 - \pi^2/3)}{1 + u}\,\mathrm{d}u = \frac{2 + q_3}{3}\log^3 2.} \tag{5.188}$$

The companion normalization is

$$\int_0^1 (u^2 - u + q_3)(\ell(u)^2 - \pi^2/3)\,\mathrm{d}u = \frac{1}{3}. \tag{5.189}$$

These identities are proved statements, not high-precision conjectures.

5.28.4 Depth four

Set

$$q_4 = \frac{1}{2 + 2\cosh \pi}, \qquad P_4(u) = \left(u - \frac{1}{2}\right)(u^2 - u + q_4).$$

Now

$$D_4(z) = \left(1 - \frac{z}{2}\right)(1 - z + q_4 z^2), \qquad \kappa_{\mathbf{1}_4}(u) = \frac{\ell(u)^3 - \pi^2\ell(u)}{6}.$$

The roots are the logistic images of $-\pi, 0, \pi$, so again the product $P_4\kappa_{\mathbf{1}_4}$ is nonnegative. The corresponding pair of identities is

$$\int_0^1 \frac{P_4(u)(\ell(u)^3 - \pi^2\ell(u))}{1+u}\,\mathrm{d}u = \frac{3(2+q_4)}{8}\log^4 2, \tag{5.190}$$

$$\int_0^1 P_4(u)(\ell(u)^3 - \pi^2\ell(u))\,\mathrm{d}u = \frac{1}{4}. \tag{5.191}$$

All endpoint singularities are logarithmic powers and are integrable. Thus no principal-value convention is implicit in these formulas.

5.28.5 An explicit density-edge asymptotic

The smallest all-ones density root gives a simple large-depth benchmark:

$$\log r_1 = -d + \frac{\pi^2}{3d} + \frac{\pi^4}{45d^3} + O(d^{-5}) + O(e^{-d}). \tag{5.192}$$

Indeed $\log r_1 = -\pi\cot(\pi/d) - \log(1 + e^{-\pi\cot(\pi/d)})$, and the Taylor expansion of cot at zero gives the displayed terms. This concerns a density root, not an angular zero. Symmetry supplies $1 - r_{d-1} = r_1$.

5.29 A divided-difference identity across depths and colors

The explicit logistic density suggests an identity search that keeps the depth parameter symbolic instead of recognizing one numerical value at a time. Its beta moments already determine the one-resolvent transform. Partial fractions then introduce several independent colors, while coalescing those colors produces finite formulas at every resolvent order. The resulting Gaussian logarithmic moments are exact polynomials in π and $\log 2$, with no integer-relation premise.

For $-1 < \operatorname{Re} t < 1$ put

$$W_t(u) = \left(\frac{u}{1-u}\right)^t \frac{\sin \pi t}{\pi t}, \qquad 0 < u < 1,$$

using the removable value at $t = 0$. This is the generating density (5.183). Its integral is one, even for complex t; for real $-1 < t < 1$ it is positive. Denote divided differences by $[z_1, \dots, z_m]f$, with the continuous confluent interpretation when nodes coincide.

Theorem 5.89 (Several resolvents and their confluent limits)**.** *Let $z_1, \dots, z_m \in \Omega$, where $\Omega = \mathbb{C} \setminus [1, \infty)$. For distinct nodes and $m \geq 2$,*

$$\int_0^1 \frac{W_t(u)}{\prod_{j=1}^m (1 - z_j u)}\,du = \frac{1}{t}[z_1, \dots, z_m]\{z^{m-2}((1-z)^{-t} - 1)\}. \tag{5.193}$$

For $m = 1$ the transform is

$$\int_0^1 \frac{W_t(u)}{1 - zu}\,du = \frac{(1-z)^{-t} - 1}{tz}. \tag{5.194}$$

All powers use the principal logarithm; the apparent singularities at $t = 0$, $z = 0$, and coincident nodes are removable. In particular, for $m \geq 2$,

$$\int_0^1 \frac{W_t(u)}{(1-zu)^m}\,du = (1-z)^{-t-m+1}\mathcal{P}_m(t, z),$$

$$\mathcal{P}_m(t, z) = \sum_{j=0}^{m-2} \binom{m-2}{j} \frac{(t+1)_j}{(j+1)!} z^j (1-z)^{m-2-j}, \tag{5.195}$$

where $(t+1)_j$ *is the rising factorial and* $(t+1)_0 = 1$.

Proof. The beta identity (5.184) gives

$$\int_0^1 u^n W_t(u)\,du = \frac{(1+t)_n}{(n+1)!}.$$

For $|z| < 1$, expand the geometric kernel and sum the binomial series. Its result is (5.194). The integrable endpoint majorants $u^{-1+\epsilon}(1-u)^{-1+\epsilon}$, on smaller parameter strips, justify local holomorphy and continuation to the principal slit plane. For distinct nonzero nodes, the exact partial fractions are

$$\frac{1}{\prod_j (1-z_j u)} = \sum_{j=1}^{m} \frac{z_j^{m-1}}{\prod_{k\neq j}(z_j - z_k)} \frac{1}{1-z_j u}.$$

Integrating and using (5.194) gives the Lagrange formula for (5.193). Continuity allows zero nodes.

When every node tends to z, the divided difference becomes the $(m-1)$st derivative divided by $(m-1)!$. Apply the product rule to $z^{m-2}((1-z)^{-t}-1)/t$. The term differentiating the second factor zero times vanishes, because the polynomial has degree $m-2$. For $j+1 \geq 1$ derivatives of the second factor one has

$$\frac{d^{j+1}}{dz^{j+1}} \frac{(1-z)^{-t}-1}{t} = (t+1)_j (1-z)^{-t-j-1}.$$

Collecting the binomial factors proves (5.195). All confluent limits also follow directly from domination of the original integral on compact subsets of Ω. □

For example, the first three confluent cases are

$$\begin{aligned}
\mathcal{P}_2 &= 1,\\
\mathcal{P}_3 &= 1 + (t-1)z/2,\\
\mathcal{P}_4 &= (1-z)^2 + (t+1)z(1-z) + (t+1)(t+2)z^2/6.
\end{aligned}$$

Thus the squared-resolvent identity has the particularly simple form

$$\int_0^1 \frac{W_t(u)}{(1-zu)^2}\,du = (1-z)^{-t-1}.$$

The several-color identity is symmetric in its nodes; repeated colors are derivatives, rather than a separate guessed formula.

Corollary 5.90 (Every Gaussian logarithmic moment)**.** *For integers* $m \geq 2$ *and* $k \geq 0$*, define*

$$I_{m,k} = \int_0^1 \frac{\log^k(u/(1-u))}{(1-\mathrm{i}u)^m}\,du, \qquad \lambda = \tfrac{1}{2}\log 2 - \tfrac{\mathrm{i}\pi}{4}.$$

These are ordinary absolutely convergent integrals, and

$$I_{m,k} = \frac{k!}{(1-\mathrm{i})^{m-1}}[t^k]\left\{\mathcal{P}_m(t,\mathrm{i})\exp\left(-\lambda t + \sum_{j\geq 1}\frac{\zeta(2j)}{j}t^{2j}\right)\right\}. \tag{5.196}$$

In particular $I_{m,k} \in \mathbb{Q}[\mathrm{i}, \pi, \log 2]$ *for every* m, k. *For the squared resolvent,*

$$I_{2,0} = \frac{1}{1-\mathrm{i}}, \qquad I_{2,1} = -\frac{\lambda}{1-\mathrm{i}},$$
$$I_{2,2} = \frac{\lambda^2 + \pi^2/3}{1-\mathrm{i}}, \qquad I_{2,3} = -\frac{\lambda^3 + \pi^2\lambda}{1-\mathrm{i}},$$
$$I_{2,4} = \frac{\lambda^4 + 2\pi^2\lambda^2 + 7\pi^4/15}{1-\mathrm{i}}.$$

Proof. Multiply (5.195) by $\pi t/\sin \pi t$ and differentiate k times at zero. Powers of the endpoint logarithms are integrable, so this differentiation is legitimate. The reflection formula and its local logarithmic expansion give

$$\frac{\pi t}{\sin \pi t} = \Gamma(1+t)\Gamma(1-t) = \exp\left(\sum_{j \geq 1} \frac{\zeta(2j)}{j} t^{2j}\right), \qquad |t| < 1.$$

At $z = \mathrm{i}$, the principal logarithm is $\log(1-\mathrm{i}) = \lambda$. This proves the coefficient formula and the listed values. Only finitely many terms enter a coefficient of order k, and every even zeta value is a rational multiple of π^{2j}. □

For a concrete weight-two pair, the real and imaginary parts of $I_{2,2}$ give the ordinary integral identities

$$\int_0^1 \frac{(1-u^2)\log^2(u/(1-u))}{(1+u^2)^2}\, du = \frac{\log^2 2}{8} + \frac{13\pi^2}{96} + \frac{\pi \log 2}{8},$$
$$\int_0^1 \frac{2u\log^2(u/(1-u))}{(1+u^2)^2}\, du = \frac{\log^2 2}{8} + \frac{13\pi^2}{96} - \frac{\pi \log 2}{8}. \tag{5.197}$$

Both follow by expanding λ^2 in the proved formula, so their constants are derived rather than recognized numerically.

Corollary 5.91 (Two real Gaussian generating identities)**.** *For real* $-1 < t < 1$,

$$\int_0^1 \frac{(u/(1-u))^t}{1+u^2}\, du = \frac{\pi\, 2^{-t/2}\sin(\pi t/4)}{\sin \pi t},$$
$$\int_0^1 \frac{u(u/(1-u))^t}{1+u^2}\, du = \frac{\pi\{1 - 2^{-t/2}\cos(\pi t/4)\}}{\sin \pi t}. \tag{5.198}$$

At $t = 0$ *the removable values are respectively* $\pi/4$ *and* $\log 2/2$. *Differentiating at zero supplies every real logarithmic moment of these two kernels.*

Proof. Use (5.194) at $z = \mathrm{i}$ and $z = -\mathrm{i}$ and the identities

$$\frac{1}{1+u^2} = \frac{1}{2}\left(\frac{1}{1-\mathrm{i}u} + \frac{1}{1+\mathrm{i}u}\right), \qquad \frac{u}{1+u^2} = \frac{1}{2\mathrm{i}}\left(\frac{1}{1-\mathrm{i}u} - \frac{1}{1+\mathrm{i}u}\right).$$

Multiply by $\pi t/\sin \pi t$ and simplify the conjugate principal powers. Continuity gives the zero-parameter values. □

The finite confluent polynomials and moment recurrence are checked exactly in `verification/certify_logistic_resolvents.py`. Independent logistic-coordinate quadratures are labeled as numerical diagnostics; the arbitrary-order identities follow from the beta moments, partial fractions and analytic parameter differentiation. These formulas add a several-color integral family to the development; they do not settle the separate mixed harmonic sums S_6 or S_8, and no literature-priority claim is made for this classical beta-integral mechanism.

5.30 Exact rational evaluation and finite certificates

The density zeros need not be computed in order to obtain numerical certificates. The unsigned norm bound $\|\kappa\|_1 \le 2^{d-1}$ already gives an explicit geometric remainder for a centered resolvent expansion. The proof works with real final order; the delivered exact implementation accepts positive integer indices so that all required moments are rational.

5.30.1 A centered-moment expansion

Define

$$B_k = \int_0^1 (2u-1)^k \kappa(u)\,\mathrm{d}u = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} 2^j c_{\boldsymbol{s}}(j+1). \tag{5.199}$$

Then $|B_k| \le M$, where $M = 2^{d-1}$. For $\operatorname{Re} z < 1$, put

$$w = \frac{z}{2-z}, \qquad |w| < 1.$$

The inequality is equivalent to $|z| < |2-z|$. The identity $1 - zu = (1 - z/2)(1 - w(2u-1))$ gives

Theorem 5.92 (Geometric exact-evaluation certificate)**.** *For every integer $N \ge 1$,*

$$V_N(z) = \frac{z}{1-z/2} \sum_{k=0}^{N-1} B_k w^k, \tag{5.200}$$

$$|L(z) - V_N(z)| \le \left|\frac{z}{1-z/2}\right| \frac{M|w|^N}{1-|w|}. \tag{5.201}$$

At $z = \mathrm{i}$ and even $N = 2m$, a purely rational bound is

$$|L(\mathrm{i}) - V_{2m}(\mathrm{i})| < \frac{2^d}{5^m}. \tag{5.202}$$

For $|z| \le 1/2$, the uniform bound is

$$|L(z) - V_N(z)| \le \frac{2^{d-1}}{3^N}. \tag{5.203}$$

Proof. The centered geometric series converges uniformly for $u \in [0,1]$, and its tail is bounded by $|w|^N/(1-|w|)$. Integrating against $|\kappa|$ proves (5.201). At $z = \mathrm{i}$, $|w| = 1/\sqrt{5}$ and $|z/(1-z/2)| = 2/\sqrt{5}$. The multiplicative constant is therefore $2M/(\sqrt{5}-1) < 2M = 2^d$ after extracting 5^{-m}. On $|z| \le 1/2$, one has $|w| \le 1/3$ and $|z/(1-z/2)| \le 2/3$, proving (5.203). □

For integer indices, (5.199) and (5.200) use only rational arithmetic at rational complex z. An enclosing complex disk gives enclosing intervals for each coordinate by adding and subtracting its radius. The evaluator never treats a small residual as a proof of an identity.

5.30.2 Implementation choices and replay

The file `code/exact_evaluator.py` first obtains the moments by dynamic strict prefix sums. It computes B_k with a common integer denominator and a repeated-difference transform, rather than a floating-point alternating binomial sum. Horner evaluation then gives the exact rational center. The generic branch can upper-bound square roots by dyadic rationals; the Gaussian and half-disk

cases use the closed rational bounds above. Near $\operatorname{Re} z = 1$, the fixed norm-bound precision in the generic branch can be insufficient, in which case the implementation raises an error rather than returns an uncertified value. It is not advertised as a complete optimal evaluator on the entire slit plane.

From the package root, the principal replay is

```
python code/verify.py
```

It reads frozen rational certificates and independently recomputes them. The optional command

```
python code/verify.py --regenerate
```

uses floating point to *propose* angular brackets, then requires rational arithmetic to certify both endpoint signs before saving them. The proposal stage uses a strict-disk series and does not decide any accepted sign. This discovery/replay separation is deliberate.

5.30.3 Gaussian values

The delivered Gaussian certificate file contains eight mixed-depth examples, all evaluated with $N = 240$ centered terms. Each complex-disk radius is smaller than 10^{-81}. The decimal entries below are readable midpoint diagnostics; the certificates are the rational endpoints in `data/gaussian_enclosures.json`.

Index	$\operatorname{Re} L_{\boldsymbol{s}}(\mathrm{i})$, approximately	$\operatorname{Im} L_{\boldsymbol{s}}(\mathrm{i})$, approximately
$(2,1)$	-0.174715661158	-0.113648917600
$(1,1,1)$	0.099953993665	-0.033577147847
$(2,1,2)$	0.032383901801	-0.026348629072
$(1,2,1,1)$	0.001158016810	0.008343470492
$(2,1,1,2)$	0.001825088603	0.006510815484
$(1,1,1,1,1)$	-0.003396506582	-0.001886779708
$(2,1,2,1,1)$	-0.000326599679	-0.000010475684
$(1,1,1,1,1,1)$	0.000443168800	-0.000335617002

In addition, the eight all-ones Gaussian values at depths one through eight are checked against *independent exact* intervals for

$$L_{\mathbf{1}_d}(\mathrm{i}) = \frac{(-\frac{1}{2}\log 2 + \frac{\mathrm{i}\pi}{4})^d}{d!}.$$

The logarithm and π intervals are derived from different rational series, as detailed in Section 5.31. These eight normalization controls are distinct from the eight stored mixed-depth examples above.

5.30.4 Certified angular brackets

On the semicircle of radius $1/2$, set

$$z(t) = \frac{1-t^2}{2(1+t^2)} + \mathrm{i}\frac{t}{1+t^2}, \qquad \theta = 2\arctan t, \quad t > 0. \tag{5.204}$$

Rational t gives rational real and imaginary coordinates. Every saved root bracket has t-width at most 10^{-12}; the corresponding angle-width is less than $2 \cdot 10^{-12}$. With $N = 104$, the error bound (5.203) proves the two endpoint signs by strict rational comparisons. For illustration:

Index	k	$10^{12}t_{\mathrm{left}}$	$10^{12}t_{\mathrm{right}}$
$(2,1,2)$	1	472099547153	472099547154
$(2,1,2)$	2	1487963538987	1487963538988
$(1,2,1,1)$	1	314972569528	314972569529
$(1,2,1,1)$	2	806524527583	806524527584
$(1,2,1,1)$	3	2019771345242	2019771345243
$(1,1,1,1,1,1)$	3	774596669241	774596669242

The file `data/angular_brackets.json` contains 24 brackets in eight complete index families of depths two through six, together with all exact coordinate intervals. The existence of a zero in each bracket follows from the opposite signs. The theorem, rather than a scan, proves uniqueness in every bracket and excludes all additional angular zeros. For the displayed middle branch at depth six, the all-ones formula independently gives $\theta_3(1/2) = \arccos(1/4)$ and $t = \sqrt{3/5}$.

5.30.5 What the exact suite checks

The fresh replay reports 5,713 finite check groups:

Check family	Count
Strict coefficients against direct finite nested enumeration	1080
Initial zero moments and leading coefficients	120
Centered moments: independent binomial formula and norm bound	2880
Polynomial compensation identities for arbitrary test roots	1200
All-ones moments against unsigned Stirling recurrence	240
Logistic-kernel derivatives through depth twelve	11
Depth-three/four symbolic kernel formulas	2
Exact generalized-kernel counterexample checks	2
Rejected negative controls	2
Rational atomic radial-argument derivative signs	80
Independent exact all-ones Gaussian controls	8
Replayed stored Gaussian enclosures	8
Replayed angular endpoint signs	48
Replayed root brackets and complete families	$24 + 8$
Total	5713

The polynomial compensation tests deliberately use arbitrary rational roots: they test the cancellation identity, *not* positivity for those roots. Positivity requires the actual density zeros and is proved analytically. Similarly, finite atomic derivative checks do not replace the interlacing-and-approximation proof of Lemma 5.83.

5.31 Independent rational logarithm and Gaussian controls

The exact controls use only rational arithmetic and elementary remainder bounds. They do not share the centered-moment evaluator's recurrence. For $0 < x < 1$, let

$$A_N(x) = \sum_{j=0}^{N-1} \frac{(-1)^j x^{2j+1}}{2j+1}.$$

The alternating-series estimate gives

$$\arctan x \in \begin{cases} [A_N(x),\ A_N(x) + x^{2N+1}/(2N+1)], & N \text{ even}, \\ [A_N(x) - x^{2N+1}/(2N+1),\ A_N(x)], & N \text{ odd}. \end{cases}$$

The identity

$$\pi = 16\arctan(1/5) - 4\arctan(1/239) \tag{5.205}$$

is verified by tangent addition: if $a = \arctan(1/5)$, then $\tan(4a) = 120/119$, and $\tan(4a - \arctan(1/239)) = 1$. Elementary bounds put the angle in $(0, \pi/2)$, so it equals $\pi/4$ and not another tangent branch. The implementation uses $N = 120$ and $N = 40$ for the two arctangent series.

For the logarithm, set

$$B_N = 2\sum_{j=0}^{N-1} \frac{(1/3)^{2j+1}}{2j+1}.$$

Integrating the geometric series for $2/(1-x^2)$, or using the power series of $2\operatorname{arctanh} x$, gives

$$B_N < \log 2 < B_N + \frac{2(1/3)^{2N+1}}{(2N+1)(1-1/9)}. \tag{5.206}$$

The code uses $N = 160$. Both interval endpoints are exact rational numbers. Cartesian complex interval multiplication then encloses $(-\frac{1}{2}\log 2 + \frac{\mathrm{i}\pi}{4})^d/d!$.

For each depth $1 \le d \le 8$, the independently generated real and imaginary intervals lie wholly inside the corresponding coordinate intervals returned by the centered evaluator with 240 terms. This is an exact containment check, not a decimal comparison. It is a strong normalization and branch control, while the identity (5.180) still rests on its differential-equation proof.

The exact evaluator also supplies a dyadic upper bound on $\sqrt{x}$ for a nonnegative rational $x = a/b$. With an integer precision p, set

$$q = \frac{\lfloor\sqrt{\lfloor 2^{2p}a/b\rfloor}\rfloor + 1}{2^p}.$$

Then $q^2 > x$. This is computed by integer square root and exact comparison. If its use in the generic geometric-tail formula fails to produce an upper bound smaller than one for $|w|$, the code stops rather than accepting a possibly divergent bound.

The complete original source and delivered rational certificates remain under `reports/angular-zero-continuation/continuations/all-depth-signed-oscillation/`. The isolated replay record in `verification/sixth-replay/all-depth/` checks the finite arithmetic independently of the analytic zero-count proof. The manuscript uses the stronger positive-transform radial theorem proved next, while retaining the incoming interlacing argument as an independent route. Fractional leading indices at higher depth still require a new construction: iterating an integer number of moment operators does not establish a fractional variation theorem.

5.32 A strict radial theorem for every positive Stieltjes transform

The preceding interlacing proof gives a nonnegative radial argument derivative. A direct symmetrization strengthens it to strict positivity, gives a quantitative bound, and identifies the sharp universal radius range. This also removes the integer-leading restriction from angular motion at depth two when combined with the earlier real-order factorization.

Theorem 5.93 (Strict radial phase and its quantitative bound)**.** *Let ν be a nonzero finite positive measure on $[0,1]$, and put*

$$Q(z) = \int_{[0,1]} \frac{d\nu(u)}{1-uz}, \qquad M_j = \int_{[0,1]} u^j\, d\nu(u).$$

For $0 < \theta < \pi$ and $0 < \rho \le 1$, the continuous phase satisfies

$$\partial_\rho \operatorname{Arg} Q(\rho e^{\mathrm{i}\theta}) \ge \frac{4\rho^2 \sin^3\theta (1-\cos\theta)^2}{(1+\rho)^8} \frac{M_1 M_2}{M_0^2}. \tag{5.207}$$

In particular, this derivative is strictly positive unless ν is supported at zero, in which case Q is constant. The range $\rho \le 1$ is sharp as a universal assertion over positive measures on $[0,1]$: at every prescribed radius greater than one a two-atom measure gives a strictly negative radial phase derivative on the entire open upper arc.

Proof. Write $c = \cos\theta$, $h = \sin\theta$, $x = \rho u$, $y = \rho v$ and $D_x = 1 - 2cx + x^2$. These denominators are positive on the arc, and Q is nonzero: its imaginary part is positive unless ν is supported at zero, where it is a positive constant. Local domination justifies differentiation and the double integral for $\operatorname{Im}(e^{\mathrm{i}\theta} Q'(z)\overline{Q(z)})$. Rationalizing and symmetrizing in u, v gives

$$\partial_\rho \operatorname{Arg} Q(\rho e^{\mathrm{i}\theta}) = \frac{h}{\rho |Q(\rho e^{\mathrm{i}\theta})|^2} \iint \frac{\mathcal{P}(x,y,c)}{D_x^2 D_y^2}\, d\nu(u)\, d\nu(v), \tag{5.208}$$

where

$$\mathcal{P} = \frac{1}{2}\{x D_y^2 + y D_x^2 - (x-y)^2 (x D_y + y D_x)\}.$$

The exact polynomial identity that determines the sign is

$$\begin{aligned} \mathcal{P} = {} & \frac{(1-x)(1-y)}{2}\{(x-y)^2 + x(1-y)^2 + y(1-x)^2\} \\ & + 4xy(1-x)(1-y)(1-c) + 2xy(x+y)(1-c)^2. \end{aligned} \tag{5.209}$$

Every term is nonnegative for $0 \le x, y \le 1$ and $-1 < c < 1$. The last term is strictly positive when $x, y > 0$, including $x = y = 1$. Thus a positive mass on $(0,1]$ gives a strictly positive integral in (5.208); no approximation argument is needed for strictness.

For the explicit lower bound, retain only the last term and use $D_x, D_y \le (1+\rho)^2$. Then

$$\partial_\rho \operatorname{Arg} Q \ge \frac{4\rho^2 h (1-c)^2 M_1 M_2}{(1+\rho)^8 |Q|^2},$$

since $\iint uv(u+v)\, d\nu(u)\, d\nu(v) = 2M_1 M_2$. Also $|1 - te^{\mathrm{i}\theta}|^2 = (t-c)^2 + h^2 \ge h^2$, so $|Q| \le M_0/h$. Substitution proves (5.207). All operations also apply at $\rho = 1$, because the chosen arc point is away from the cut.

For sharpness, let $R > 1$, choose $R^{-2} < v < 1$, and take $\nu = v\delta_0 + (1-v)\delta_1$. Its transform is

$$Q_v(z) = \frac{1 - vz}{1 - z}.$$

Direct differentiation gives

$$\partial_\rho \operatorname{Arg} Q_v(\rho e^{\mathrm{i}\theta}) = \frac{(1-v)(1-\rho^2 v)\sin\theta}{(1 - 2\rho\cos\theta + \rho^2)(1 - 2\rho v\cos\theta + \rho^2 v^2)}. \tag{5.210}$$

It is strictly negative at $\rho = R$ for every $0 < \theta < \pi$. This proves sharpness of the uniform disk range, without asserting that every individual transform loses monotonicity just outside it. □

Corollary 5.94 (All-real-order inward angular motion at depth two)**.** *For every $a \geq 0, b > 0$, let $\theta_{a,b}(\rho)$ be the unique upper-semicircle imaginary zero of the principal continuation of $F_{a,b}$. Then*

$$\theta'_{a,b}(\rho) < 0, \qquad 0 < \rho \leq 1.$$

The derivative at one is the derivative of the local analytic branch, equivalently its left derivative. For $0 < \rho < \sigma \leq 1$ one also has

$$\frac{\sigma \cos \theta_{a,b}(\sigma)}{\rho \cos \theta_{a,b}(\rho)} > \frac{\sigma}{\rho},$$

so the Cartesian abscissa grows faster than the radius in this precise ratio sense. This is distinct from the sign of $\partial_\rho(\cos\theta/\rho)$.

Proof. Theorems 5.18 and 5.26 supply, throughout the parameter domain, a factorization

$$F_{a,b}(z) = \frac{z^2 Q_{a,b}(z)}{1 - rz}, \qquad 0 < r \leq 1,$$

with $Q_{a,b}$ a finite positive Stieltjes transform. In the supercritical region with $a > 0$, r is the interior density crossing; on the critical/subcritical region and the entire $a = 0$ axis one may take $r = 1$. The positive measures have mass 2^{-a} and positive density except possibly at the crossing. For an upper-arc point their continuous unwrapped phase is

$$\Phi = 2\theta - \operatorname{Arg}(1 - r\rho e^{\mathrm{i}\theta}) + \operatorname{Arg} Q_{a,b}(\rho e^{\mathrm{i}\theta}).$$

The resolvent sector bounds give $0 < \Phi < 2\pi$. Hence at the unique imaginary zero, $\Phi = \pi$ and $F_{a,b}$ is strictly negative real. The preceding real-order zero theorems prove $\partial_\theta \operatorname{Im} F_{a,b} < 0$ there. Since $\operatorname{Im} F = |F| \sin \Phi$, this is equivalent to $\Phi_\theta > 0$. At fixed angle,

$$\Phi_\rho = \frac{r \sin\theta}{1 - 2r\rho\cos\theta + r^2\rho^2} + \partial_\rho \operatorname{Arg} Q_{a,b} > 0.$$

Implicit differentiation of $\Phi(\theta_{a,b}(\rho), \rho) = \pi$ therefore gives $\theta' = -\Phi_\rho/\Phi_\theta < 0$. The analytic continuation and angular simplicity justify this argument at the endpoint and on parameter regions whose boundary series fail the ordinary term test. Finally the unique angle lies in $(0, \pi/2)$, so its strict decrease makes $\cos\theta$ strictly increase. The displayed Cartesian ratio follows immediately. □

The direct pair identity and positive decomposition are independently re-expanded symbolically and tested by exact rational arithmetic in `verification/certify_stieltjes_radial.py`. Separate controls check one-atom normalization and two-atom examples outside the disk, including a prescribed-radius counterexample approaching radius one. These finite checks catch algebraic mistakes; the measure theorem follows from the sign decomposition and dominated integral argument, and the all-real-order consequence uses the previously established positive factorization.

5.33 A geometric evaluation formula that survives $xy = 1$

The preceding sections concern finite identities. We now give an evaluation method for the full double polylogarithm, including components for which no finite classical reduction has been established.

Lemma 5.95 (One-dimensional integral)**.** *For $a, b \geq 1$, $|x| \leq 1$, $x \neq 1$, and $|y| \leq 1$,*

$$\operatorname{Li}_{a,b}(x, y) = \frac{x}{(a-1)!} \int_0^1 \frac{(-\log t)^{a-1} \operatorname{Li}_b(xyt)}{1 - xt}\, dt. \tag{5.211}$$

Proof. For $|x|<1$, the Mellin identity $n^{-a}=\int_0^1 t^{n-1}(-\log t)^{a-1}\,dt/(a-1)!$ and absolute convergence give

$$\sum_{n>m\ge 1}(xt)^n y^m/m^b=\frac{xt}{1-xt}\operatorname{Li}_b(xyt).$$

This proves (5.211). For a nontrivial unit outer argument, $1-rxt$ stays uniformly away from zero for r sufficiently close to one. Also $|\operatorname{Li}_b(rxyt)|\le \operatorname{Li}_b(t)$, and

$$\frac{1}{(a-1)!}\int_0^1(-\log t)^{a-1}\operatorname{Li}_b(t)\,dt=\sum_{m\ge 1}\frac{1}{m^b(m+1)^a}\le\zeta(a+b). \tag{5.212}$$

Dominated convergence and summation by parts in the nested series therefore give the boundary case. Notice that $xy=1$ causes no problem: the logarithmic endpoint at $b=1$ is integrable in (5.212). □

Let T_n denote the Chebyshev polynomial defined by $T_n(\cos u)=\cos(nu)$. Put

$$d=\sqrt{1-x},\qquad r=\frac{1-d}{1+d},\qquad R=|r|, \tag{5.213}$$

where the square root has positive real part. For $|x|\le 1$, $x\ne 1$, we have $\operatorname{Re} d>0$ and therefore $R<1$.

Lemma 5.96 (The kernel expansion)**.** *Uniformly for $0\le t\le 1$,*

$$\frac{1}{1-xt}=\frac{1}{d}\left(1+2\sum_{n\ge 1}r^n T_n(2t-1)\right). \tag{5.214}$$

Proof. For $s\in[-1,1]$, summing the two geometric series associated with $s=\cos u$ gives the classical identity

$$1+2\sum_{n\ge 1}r^n T_n(s)=\frac{1-r^2}{1-2rs+r^2}.$$

This is also a direct consequence of the generating function in [49]. Substituting (5.213) shows

$$1-xt=d\frac{1-2r(2t-1)+r^2}{1-r^2}.$$

The series is uniform because $|T_n(s)|\le 1$ and $R<1$. □

Definition 5.97. *For $v=xy$ and $n\ge 0$, define the Chebyshev moment*

$$\mathcal{M}_n^{a,b}(v)=\frac{1}{(a-1)!}\int_0^1(-\log t)^{a-1}T_n(2t-1)\operatorname{Li}_b(vt)\,dt. \tag{5.215}$$

Theorem 5.98 (Certified Chebyshev–moment expansion)**.** *Under the hypotheses of Lemma 5.95, with $x\ne 0$,*

$$\operatorname{Li}_{a,b}(x,y)=\frac{x}{d}\left(\mathcal{M}_0^{a,b}(v)+2\sum_{n\ge 1}r^n\mathcal{M}_n^{a,b}(v)\right). \tag{5.216}$$

For $K \geq 1$, let E_K be the difference between the left side and the truncation with indices $0 \leq n < K$. Then

$$|E_K| \leq \frac{2|x|\zeta(a+b)}{|d|(1-R)} R^K. \tag{5.217}$$

This estimate is uniform over $|y| \leq 1$, including $xy = 1$. For nontrivial level-three or level-four outer roots,

$$|E_K| \leq \frac{16}{3\,4^K}. \tag{5.218}$$

Proof. Insert (5.214) into (5.211). The integrable majorant in (5.212) justifies termwise integration and gives $|\mathcal{M}_n^{a,b}(v)| \leq \zeta(a+b)$ for all n. The tail of the resulting geometric series is (5.217). For the specified roots, direct substitution in (5.213) gives $R < 1/4$ and $|d| > 1$; the code also proves these inequalities by interval arithmetic before using them. Since $\zeta(a+b) \leq \zeta(2) < 2$, the bound becomes $4(1/4)^K/(1-1/4) = 16/(3\,4^K)$. The zero outer argument, omitted only to avoid unnecessary notation, gives $\mathrm{Li}_{a,b}(0,y) = 0$ directly. □

The proof does not appeal to cancellation in a sampled sequence, a fitted asymptotic expansion, or an empirical convergence diagnostic. In particular, when the product of two roots is one, the same argument and the same bound apply. The only new feature is the elementary cancellation in the exact moment formula, to which we now turn.

5.34 Every moment reduces to classical polylogarithms

Write the shifted Chebyshev polynomial as

$$Q_n(t) := T_n(2t-1) = \sum_{j=0}^{n} c_{n,j} t^j. \tag{5.219}$$

Its coefficients are integers and can be generated without symbolic integration:

$$Q_0 = 1, \quad Q_1 = 2t-1, \quad Q_{n+1} = 2(2t-1)Q_n - Q_{n-1}. \tag{5.220}$$

Thus

$$\mathcal{M}_n^{a,b}(v) = \sum_{j=0}^{n} c_{n,j} J_{a,b}(j+1; v), \tag{5.221}$$

where

$$J_{a,b}(c; v) = \sum_{m \geq 1} \frac{v^m}{m^b (m+c)^a}, \qquad c \in \mathbb{Z}_{>0}. \tag{5.222}$$

The power series in (5.222) is absolutely convergent for $|v| \leq 1$.

Lemma 5.99 (Two-pole partial fractions)**.** *Put*

$$\begin{aligned} A_s &= (-1)^{b-s} \binom{a+b-s-1}{a-1} c^{-(a+b-s)} && (1 \leq s \leq b), \\ C_s &= (-1)^{b} \binom{a+b-s-1}{b-1} c^{-(a+b-s)} && (1 \leq s \leq a). \end{aligned} \tag{5.223}$$

Then, for $u \neq 0, -c$,

$$\frac{1}{u^b (u+c)^a} = \sum_{s=1}^{b} \frac{A_s}{u^s} + \sum_{s=1}^{a} \frac{C_s}{(u+c)^s}. \tag{5.224}$$

In particular, $A_1 + C_1 = 0$.

Proof. The principal part at $u = 0$ follows by expanding $(u + c)^{-a} = c^{-a}(1 + u/c)^{-a}$. Its coefficient of u^{-s} after multiplication by u^{-b} is exactly A_s. At $u = -c$, write $w = u + c$ and expand $u^{-b} = (-1)^b c^{-b}(1 - w/c)^{-b}$; this gives C_s. The difference between the two sides has no finite poles and tends to zero at infinity, so it is identically zero. The equality $A_1 = -C_1$ follows from $\binom{a+b-2}{a-1} = \binom{a+b-2}{b-1}$. □

For $s \geq 1$, let $H_c^{(s)}(v) = \sum_{k=1}^{c} v^k/k^s$ and $H_c^{(s)} = H_c^{(s)}(1)$. A coefficient outside the ranges in (5.223) is understood to be zero in the following theorem.

Theorem 5.100 (Finite classical moment formula)**.** *For $0 < |v| \leq 1$, $v \neq 1$,*

$$J_{a,b}(c; v) = \sum_{s=1}^{b} A_s \operatorname{Li}_s(v) + v^{-c} \sum_{s=1}^{a} C_s\big(\operatorname{Li}_s(v) - H_c^{(s)}(v)\big). \tag{5.225}$$

At the resonant endpoint $v = 1$ the finite expression is instead

$$J_{a,b}(c; 1) = -C_1 H_c^{(1)} + \sum_{s=2}^{\max(a,b)} (A_s + C_s)\zeta(s) - \sum_{s=2}^{a} C_s H_c^{(s)}. \tag{5.226}$$

No regularized zeta value is needed. At $v = 0$, $J_{a,b}(c; 0) = 0$.

Proof. Multiply (5.224) by v^m and sum over $m \geq 1$. For $|v| < 1$, all rearrangements are absolute, and

$$\sum_{m\geq 1} \frac{v^m}{(m + c)^s} = v^{-c}\big(\operatorname{Li}_s(v) - H_c^{(s)}(v)\big).$$

Radial limits give (5.225) at $|v| = 1$, $v \neq 1$. At $v = 1$, combine the two $s = 1$ pieces at the level of partial sums:

$$A_1 \sum_{m=1}^{N} \frac{1}{m} + C_1 \sum_{m=1}^{N} \frac{1}{m + c} = -C_1 H_c^{(1)} + C_1 \sum_{m=N+1}^{N+c} \frac{1}{m}.$$

The last term tends to zero. Every $s \geq 2$ term converges separately, which gives (5.226). □

For example, when $a = b = 1$, (5.226) reduces to

$$J_{1,1}(c; 1) = \frac{H_c}{c}. \tag{5.227}$$

Consequently the resonant depth-two function has the explicit expansion

$$\operatorname{Li}_{1,1}(x, x^{-1}) = \frac{x}{\sqrt{1 - x}} \left[1 + 2 \sum_{n\geq 1} r^n \sum_{j=0}^{n} c_{n,j} \frac{H_{j+1}}{j + 1}\right], \tag{5.228}$$

valid for a nontrivial unit outer argument. Its coefficients are rational. The error after K terms is bounded by (5.217), and the uniform level-three/four bound (5.218) applies. This is an exact infinite identity for a conditionally convergent depth-two case; it is not a finite rational expression in classical constants.

5.34.1 A resonant example beyond the first weight

For $(a,b)=(2,1)$, the coefficients are $A_1=c^{-2}$, $C_1=-c^{-2}$, and $C_2=-c^{-1}$, giving

$$J_{2,1}(c;1)=\frac{H_c}{c^2}+\frac{H_c^{(2)}-\zeta(2)}{c}. \tag{5.229}$$

Substitution in (5.221) and (5.216) yields a geometrically convergent formula for $\mathrm{Li}_{2,1}(\rho,\rho^{-1})$, $\rho=e^{2\pi\mathrm{i}/3}$, whose coefficient data involve only rational numbers and $\zeta(2)$. The location of the singular product $v=1$ has become an explicit, finite harmonic-sum correction rather than a numerical failure mode.

5.35 Optimal geometric rate and the conductor dependence

The rate in Theorem 5.98 is not merely an artifact of an arbitrary expansion point. It is the best exponential rate obtainable by uniformly approximating its outer rational kernel with polynomials. This statement concerns the kernel, not the minimal complexity of all possible algorithms for evaluating a special value.

Theorem 5.101 (Sharp exponential rate for polynomial kernels)**.** *Fix $0<|x|\le 1$, $x\neq 1$, and let d,r,R be as in (5.213). For $K\ge 1$, define*

$$e_{K-1}(x)=\inf_{\deg p<K}\max_{0\le t\le 1}\left|\frac{1}{1-xt}-p(t)\right|,$$

where the infimum allows complex polynomial coefficients. Then

$$\frac{R^K}{|d|}\le e_{K-1}(x)\le\frac{2R^K}{|d|(1-R)}. \tag{5.230}$$

In particular, $\lim_{K\to\infty}e_{K-1}(x)^{1/K}=R$.

Proof. The upper bound is the uniform tail bound in (5.214). For the lower bound, use the Kth Chebyshev coefficient of $f(s)=(1-x(s+1)/2)^{-1}$. By (5.214), it is $2r^K/d$. The corresponding coefficient of a polynomial of degree less than K vanishes. For any continuous complex function g on $[-1,1]$,

$$\left|\frac{2}{\pi}\int_0^\pi g(\cos u)\cos(Ku)\,du\right|\le 2\|g\|_\infty.$$

Apply this with $g=f-p$ to obtain $2R^K/|d|\le 2\|f-p\|_\infty$ and take the infimum. Taking Kth roots proves the final statement. □

Corollary 5.102 (Adjacent roots)**.** *For $x_q=e^{2\pi\mathrm{i}/q}$, as $q\to\infty$,*

$$-\log R_q=2\sqrt{\frac{\pi}{q}}+O(q^{-3/2}). \tag{5.231}$$

For fixed a,b, the bound (5.217) guarantees an absolute error at most 10^{-P} using

$$K=O\big(\sqrt{q}(P+\log q)\big) \tag{5.232}$$

outer-kernel terms, uniformly in $|y|\le 1$.

Proof. With $\theta = 2\pi/q$, $d = \sqrt{1-e^{\mathrm{i}\theta}} = e^{-\mathrm{i}\pi/4}\sqrt{\theta}\,(1+O(\theta))$. Hence

$$-\log|r| = \operatorname{Re}\left(\log(1+d) - \log(1-d)\right) = 2\operatorname{Re}(d + d^3/3 + \cdots) = \sqrt{2\theta} + O(\theta^{3/2}),$$

which is (5.231). Also $|d| \asymp q^{-1/2}$ and $1 - R \asymp q^{-1/2}$, so the prefactor in (5.217) is $O(q)$. Solving the resulting geometric inequality gives (5.232). □

For comparison, expansion around $t = 1/2$ gives

$$\frac{1}{1-xt} = \frac{1}{1-x/2}\sum_{n\geq 0}\left(\frac{x(t-1/2)}{1-x/2}\right)^n.$$

At a unit outer argument its uniform ratio is $R_{\mathrm{mid}} = 1/|2-x|$. For $x = x_q$,

$$-\log R_{\mathrm{mid}} = \tfrac{1}{2}\log(5 - 4\cos(2\pi/q)) = \frac{4\pi^2}{q^2} + O(q^{-4}). \tag{5.233}$$

Thus its corresponding sufficient degree is $O(q^2(P + \log q))$. This is a comparison between two proved expansions, not a claim to improve every previously published polylogarithm algorithm.

Arithmetic cost versus approximation degree. With weights fixed, the monomial moments for $1 \leq c \leq K$ can be assembled from finitely many single values and incrementally updated $H_c^{(s)}(v)$ in $O((a+b)K)$ arithmetic operations. The triangular transform (5.221) costs $O(K^2)$ operations in the direct implementation. Thus, after the necessary classical values are available, $O(K^2 + (a+b)K)$ arithmetic operations suffice for the spectral truncation. These are unit-cost arithmetic counts, not bit complexity bounds. At general large conductor, computing the inner single values and representing the algebraic arguments have their own costs. The implementation supplied here certifies levels three and four; the analytic theorem has the wider stated domain.

5.36 Outward-rounded evaluation and the treatment of cancellation

A theorem about a truncation error is not by itself a certificate for a floating-point implementation. This section specifies how finite arithmetic errors are enclosed in the accompanying executable code.

5.36.1 An independent geometric expansion for the single values

To avoid using the desired parity identities inside their own numerical checks, the certified evaluator computes its classical constants from a separate Euler expansion.

Lemma 5.103 (Euler expansion with a simple rigorous tail)**.** *Let $s \geq 1$ be an integer and let $z \neq 1$, with $\tau = z/(z-1)$ satisfying $|\tau| < 1$. For the arguments used below,*

$$\mathrm{Li}_s(z) = -\sum_{N\geq 1}\tau^N\frac{h^\star_{s-1}(N)}{N}, \tag{5.234}$$

where $h^\star_d(N)$ is the complete homogeneous symmetric polynomial of degree d in $1, 1/2, \ldots, 1/N$, and $h^\star_0(N) = 1$. If $|\tau| \leq r_0 < 1$, then the error after $N \leq M$ is at most

$$\frac{r_0^{M+1}}{1-r_0}. \tag{5.235}$$

Proof. In the standard Mellin integral for $\mathrm{Li}_s(z)$, expand

$$\frac{z}{1-zt} = -\sum_{N\ge 1} \tau^N (1-t)^{N-1}.$$

The coefficient integral is

$$\frac{1}{(s-1)!}\int_0^1 (-\log t)^{s-1}(1-t)^{N-1}\,dt = \frac{h^\star_{s-1}(N)}{N}.$$

Indeed, summing its generating series in u^{s-1} gives

$$\int_0^1 t^{-u}(1-t)^{N-1}\,dt = \frac{1}{N}\prod_{j=1}^{N}(1-u/j)^{-1},$$

first for u close to zero. This can be proved by the beta integral or by $N-1$ integrations by parts. Each coefficient integral is between zero and one, since $(1-t)^{N-1} \le 1$. The geometric tail then proves (5.235). □

For the nontrivial roots of levels three and four, and for $z=-1$, $|\tau| \le 1/\sqrt{2} < 3/4$. The implementation uses the rational choice $r_0 = 3/4$ and computes the coefficients incrementally from

$$h^\star_d(N) = h^\star_d(N-1) + \frac{h^\star_{d-1}(N)}{N}. \tag{5.236}$$

It obtains $\zeta(s)$ for $s \ge 2$ from $-\mathrm{Li}_s(-1)/(1-2^{1-s})$, obtains π from $4\,\mathrm{Im}\,\mathrm{Li}_1(\mathrm{i})$, and obtains L from $-\mathrm{Li}_1(-1)$. No conjectural reduction enters these evaluations.

5.36.2 The interval arithmetic contract

Fix a decimal scale $Q = 10^D$. A real interval is stored as integers $[l,u]/Q$. Addition is exact at this scale. For multiplication, the four integer endpoint products are enclosed by a floor of their minimum and a ceiling of their maximum after division by Q. A reciprocal is permitted only when zero is excluded, and is again rounded outward. A nonnegative square root is enclosed with integer square roots of lQ and uQ. Complex values are rectangular pairs of such real intervals.

All these rules are elementary rational inequalities. By induction over the expression tree, the computed rectangle contains the exact finite expression. The analytic disk radius from (5.217) or (5.235) is then added to both real and imaginary coordinate intervals. Enclosing a disk by a square is conservative but valid. The square-root branch in (5.213) is fixed by its positive real part and the sign of the imaginary part of $1-x$.

The executable checks $R < 1/4$ and $|d| \ge 1$ by interval inequalities before using (5.218). It rejects unsupported outer arguments instead of silently using an unproved bound. In particular, the supplied certified implementation does not accept outer argument one.

5.36.3 Why extra working precision is necessary

The coefficients in (5.219) alternate in sign and satisfy

$$\sum_{j=0}^{n} |c_{n,j}| = T_n(3) = \frac{(3+2\sqrt{2})^n + (3-2\sqrt{2})^n}{2}. \tag{5.237}$$

The alternating signs follow, for example, from the explicit formula

$$c_{n,j} = (-1)^{n-j} \frac{n}{n+j} \binom{n+j}{2j} 4^j \quad (n \geq 1),$$

and evaluating Q_n at $t = -1$ gives (5.237). Thus computing the moments by monomial conversion can lose a number of digits proportional to K, even though the moments themselves are bounded.

The interval code does not assume that this cancellation is harmless. For target $P = 35$ it uses $K = 73$ spectral terms and $D = 246$ working decimal places. Each single value uses $M = 1997$ Euler terms and its own rigorous tail. These are conservative choices rather than tuned speed records. The actual final widths, not a heuristic guard-digit count, determine whether a certificate is accepted. A future stable recurrence or validated Clenshaw-style evaluation could reduce the guard overhead; that is not needed for the present certificates.

5.36.4 A positive-measure midpoint alternative

For inverse colors, the same integral defines the positive measure

$$d\mu_{a,b}(t) = \frac{(-\log t)^{a-1} \operatorname{Li}_b(t)}{(a-1)!}\, dt.$$

Its total mass is $J_{a,b}(1;1) \leq J_{1,1}(1;1) = 1$. Set $\alpha = z/(1-z/2)$ and $m_r = \int_0^1 (t-1/2)^r \, d\mu_{a,b}(t)$. The kernel's geometric series gives

$$\operatorname{Li}_{a,b}(z, z^{-1}) = \alpha \sum_{r \geq 0} \alpha^r m_r, \qquad q = \frac{|z|}{|2-z|} < 1.$$

The tail after $r < N$ is at most $|\alpha| J_{a,b}(1;1) q^N/(1-q)$. Each m_r is an exact binomial combination of $J_{a,b}(j+1;1)$ from Theorem 5.100. This is useful at Gaussian and Eisenstein points; near the outer root one, the Chebyshev expansion has the stronger kernel rate proved above. Neither rate statement is a lower bound for every evaluation algorithm.

5.37 Exact rational certificates for Gaussian values

We use the outermost-letter-first convention

$$G(a_1, \ldots, a_w; z) = \int_0^z \frac{G(a_2, \ldots, a_w; t)}{t - a_1}\, dt, \qquad G(;z) = 1.$$

Define the permitted alphabet region

$$\mathcal{A} = \{0, 1\} \cup \{a \in \mathbb{Q}(i) : |a| \geq 1,\ |1-a| \geq 1\}.$$

The two inequalities are exact rational comparisons after squaring the moduli. The Gaussian alphabet and the letter $1 - i$ both belong to this region.

Theorem 5.104 (Uniform exact certificate)**.** *Let $w \geq 1$, let $a_1, \ldots, a_w \in \mathcal{A}$, and assume $a_1 \neq 1$ and $a_w \neq 0$. Then $G(a_1, \ldots, a_w; 1)$ exists as an ordinary convergent iterated integral. It has the Hölder decomposition*

$$G(a_1, \ldots, a_w; 1) = \sum_{j=0}^{w} (-1)^j G(1-a_j, \ldots, 1-a_1; 1/2) G(a_{j+1}, \ldots, a_w; 1/2), \tag{5.238}$$

where a missing word denotes 1. *Replace each factor by its Taylor polynomial through degree* N, *evaluated at* $1/2$, *and call the resulting Gaussian rational number* C_N. *Then*

$$|G(a_1,\dots,a_w;1)-C_N|\le 2(w+1)\,T_w(N),\qquad T_d(N)=2^{-N}\sum_{k=0}^{d-1}\binom{N}{k}. \tag{5.239}$$

Here $\binom{N}{k}=0$ *for* $k>N$ *and* $T_0(N)=0$. *All the Taylor polynomials required by one word are computed using* $O(wN)$ *exact rational arithmetic operations. For an integer* $P\ge 0$, *the explicit choice*

$$N=3\bigl(P+w+\lceil\log_2(w+1)\rceil\bigr) \tag{5.240}$$

guarantees an error at most 2^{-P}. *In particular,* $N=O(P+w)$.

Proof. The hypotheses exclude singularities in the open integration segment. They also ensure integrability at its endpoints: the last letter is nonzero, and any logarithmic singularities at 1 are integrated against an outermost kernel regular there. Partition the ordered integration simplex according to the number j of variables above $1/2$. Replacing those variables by their complements reverses their word and contributes $(-1)^j$. This gives (5.238); this is the standard Hölder convolution, with its endpoint hypotheses made explicit.

Every nonzero letter in either factor of (5.238) has modulus at least 1. Moreover, every nonempty right factor ends at $a_w\neq 0$, and every nonempty left factor ends at $1-a_1\neq 0$. Thus these factors have Taylor series with zero constant coefficient.

Consider a word v of this kind, with d nonzero letters, and write $G(v;z)=\sum_{n\ge 0}c_nz^n$. Coefficientwise absolute values are bounded by replacing each nonzero kernel with $1/(1-z)$. Inserting a zero kernel divides the coefficient of z^n by $n\ge 1$, so deleting all zero letters can only increase this majorant. Consequently

$$|c_n|\le [z^n]\frac{(-\log(1-z))^d}{d!}\le\binom{n-1}{d-1}\quad(n\ge d),\qquad c_n=0\quad(n<d).$$

For the second inequality, expand the d logarithms; each composition of n into d positive parts contributes at most 1, and the additional factor $1/d!$ can be discarded. The empty word is handled separately. It follows that

$$|G(v;1/2)|\le 1,\qquad \left|\sum_{n=0}^{N}c_n2^{-n}\right|\le 1,$$

because $\sum_{n\ge d}\binom{n-1}{d-1}2^{-n}=1$. The tail of this majorant is exactly

$$\sum_{n>N}\binom{n-1}{d-1}2^{-n}=2^{-N}\sum_{k=0}^{d-1}\binom{N}{k}=T_d(N).$$

One proof interprets the left side as the probability that the dth success in fair independent Bernoulli trials occurs after trial N; the right side counts fewer than d successes among the first N trials. Equivalently, the equality follows by Pascal's identity.

For each product AB in the Hölder sum, with truncations A_N,B_N,

$$|AB-A_NB_N|\le|A-A_N|\,|B|+|A_N|\,|B-B_N|\le T_{d_A}(N)+T_{d_B}(N)\le 2T_w(N).$$

Summing over the $w+1$ cuts proves (5.239).

For the complexity claim, let b_n be the Taylor coefficients of a tail and c_n those of the word obtained by prepending a. For $n \geq 1$,

$$c_n = \begin{cases} b_n/n, & a = 0, \\ \big((n-1)c_{n-1} - b_{n-1}\big)/(an), & a \neq 0. \end{cases} \tag{5.241}$$

In both cases $c_0 = 0$, and the empty tail has $b_0 = 1$, $b_n = 0$ for $n \geq 1$. The second recurrence is coefficient comparison in $(z-a)G'(a, v; z) = G(v; z)$. It takes $O(N)$ operations per prepended letter. Compute the original suffix chain starting from the last letter; this gives every right factor. Compute the reflected reversed-prefix chain by successively prepending $1 - a_1, 1 - a_2, \ldots, 1 - a_w$; this gives every left factor. Each chain requires $O(wN)$ operations in total. Summing the $w + 1$ products adds $O(w)$ operations.

Finally, the binomial theorem gives the useful bound

$$T_w(N) \leq 2^{w-1}(3/4)^N.$$

Indeed, for $k \leq w - 1$ one has $2^{-k} \geq 2^{-(w-1)}$, so

$$\sum_{k=0}^{w-1} \binom{N}{k} \leq 2^{w-1} \sum_{k=0}^{N} \binom{N}{k} 2^{-k} = 2^{w-1}(3/2)^N.$$

With $M = P + w + \lceil \log_2(w+1) \rceil$ and $N = 3M$,

$$2(w+1)T_w(N) \leq (w+1)2^w(3/4)^{3M} \leq (w+1)2^{w-M} \leq 2^{-P},$$

since $27/64 < 1/2$. This proves (5.240). □

Corollary 5.105 (Fixed imaginary quadratic fields)**.** *Let $K \subset \mathbb{C}$ be an imaginary quadratic field with its fixed complex embedding. Replace $\mathbb{Q}(i)$ in the alphabet definition by K. The same Hölder sum, tail bound, precision prescription, and $O(wN)$ arithmetic-operation count hold, with the center now in K.*

Proof. The proof of Theorem 5.104 uses the coefficient field only for exact additions, multiplications, and reciprocals. Each Taylor coefficient therefore lies in K, and the analytic majorant is unchanged. The tests $|a|^2 \geq 1$ and $|1 - a|^2 \geq 1$ are exact rational comparisons because these squared moduli are field norms. Arithmetic in a fixed quadratic field requires a bounded number of rational operations per field operation. □

In particular, every sixth root of unity satisfies the alphabet condition: apart from 1, its distance from 1 is at least 1. Thus the theorem also supplies an exact-certificate route for the Eisenstein values. The supplied core implementation specializes to $\mathbb{Q}(i)$; the four mixed Eisenstein formulas are established analytically and checked numerically in the independent scripts.

A safe bit-complexity statement. The count $O(wN)$ is an arithmetic-operation count; it does not assign unit cost to arbitrarily large integers. Let Q be a common positive denominator for the real and imaginary components of the reciprocals of all nonzero letters used in both Hölder chains. Let $L_N = \operatorname{lcm}(1, \ldots, N)$. For a word of length $\ell \leq w$, each component of its degree-n Taylor coefficient has denominator dividing $Q^n L_n^\ell$. This follows from the convolution form of (5.241): a

term uses total reciprocal-letter degree n and one integer divisor at most n for each integration. Thus all evaluated partial sums have denominators dividing

$$D_N = (2Q)^N L_N^w.$$

The coefficient majorant bounds their numerator bit lengths by

$$B = O\big(N \log(2Q) + wN \log(N+1) + N\big),$$

using only the elementary estimate $L_N \le N!$. All product sums have denominator dividing D_N^2 and comparable operand lengths. Therefore standard exact rational arithmetic yields polynomial bit complexity in $w, N, \log Q$. For an entirely conservative explicit estimate, naive multiplication and division use $O(B^2)$ bit operations, the Euclidean algorithm uses at most $O(B)$ such divisions, and normalization after every operation yields $O(wNB^3)$ bit operations. This deliberately loose bound requires no fast-arithmetic assumptions. Standard fast multiplication and gcd algorithms improve it.

What is and is not new here. Hölder convolution and its use for numerical multiple-polylogarithm evaluation are established; see Borwein–Bradley–Broadhurst–Lisoněk [67] and Vollinga–Weinzierl [68]. The contribution of this specialization is the explicit rational tail certificate, the permitted rational Gaussian letter region, the reuse of both chains, and the uniform linear bound $N = O(P + w)$. This is an error bound for computing periods, not a proof of an identity inferred from a small residual.

Application to the manuscript's constants. The letters needed for π, $\log 2$, $\zeta(n)$ for $n \ge 2$, $\beta(n)$, and $\lambda_n = \Im \operatorname{Li}_n((1+i)/2)$ all satisfy the hypotheses:

$$\pi = 4\Im \operatorname{Li}_1(i), \quad \log 2 = -2\Re \operatorname{Li}_1(i), \quad \operatorname{Li}_n(z) = -G(0^{n-1}, z^{-1}; 1).$$

For the half point the nonzero letter is $1 - i$, whose reflected letter is i. Hence both sides of the Gaussian conjectures can be enclosed using one exact Gaussian-rational arithmetic framework.

5.37.1 Agreement with ordinary boundary series

For completeness, suppose $s_1, \dots, s_d$ are positive integers and $|z_j| = 1$. The outer-first colored multiple-polylogarithm series converges absolutely if $s_1 \ge 2$, and converges ordinarily if $s_1 = 1$ and $z_1 \ne 1$. These sufficient conditions include every colored value used in the Gaussian tables.

For $d \ge 2$, put

$$B_n = \sum_{n > n_2 > \cdots > n_d \ge 1} \prod_{j=2}^{d} \frac{z_j^{n_j}}{n_j^{s_j}}.$$

The elementary symmetric harmonic bound gives $|B_n| = O((1 + \log n)^{d-1})$ and

$$|B_{n+1} - B_n| = O\big((1 + \log n)^{d-2}/n\big).$$

The first estimate proves absolute convergence for $s_1 \ge 2$. For $s_1 = 1$ set $b_n = B_n/n$. Then $b_n \to 0$ and

$$|b_{n+1} - b_n| = O\big((1 + \log n)^{d-1}/n^2\big),$$

which is summable. The geometric partial sums of z_1^n are bounded when $z_1 \ne 1$, so summation by parts proves convergence of $\sum z_1^n b_n$. At $d = 1$ this is the usual Dirichlet test.

Introduce $0 < r < 1$ in the outer color, $z_1 \mapsto r z_1$. The absolutely convergent series/integral dictionary gives the GPL with the original word and endpoint r. Abel's theorem identifies the limit of the series with its ordinary boundary sum. On the integral side, the last letter is nonzero. Near zero every nonempty suffix is analytic and $O(t)$. Near one, recursive integration gives at worst powers of $|\log(1-t)|$ in an interior suffix. The outer letter is either 0 (when $s_1 \geq 2$) or $z_1^{-1} \neq 1$ (when $s_1 = 1$), so its kernel is bounded near one and these logarithmic powers are integrable. Thus the GPL has a continuous limit as $r \uparrow 1$ and agrees with the ordinary boundary series. No regularization is being silently used in these cases.

Chapter 6

Gamma grids and exact certificates

An experiment proposes a relation; a certificate explains why it holds. For gamma values at rational points, taking logarithms turns reflection and multiplication into rational rows. Their row combinations yield an exact, terminating reduction procedure relative to a specified law matrix. This computational completeness must be distinguished from the open arithmetic completeness conjecture.

6.1 The lattice model

Work on the denominator-N grid, in coordinates $x_k = \log\Gamma(k/N)$, $k = 1, \dots, N-1$. The two classical law families become exact $\mathbb{Q}$-linear rows with symbolic right-hand sides over the atoms $\log\pi$, $\log n$, $\log\sin(\pi k/N)$:

- reflection, for $k < N/2$:

$$x_k + x_{N-k} = \log\pi - \log\sin\Big(\frac{\pi k}{N}\Big);$$

- Gauss multiplication, for each divisor $d \geq 2$ of N and $k = 1, \dots, N/d$:

$$\sum_{j=0}^{d-1} x_{k+jN/d} \;-\; [dk < N]\, x_{dk} = \frac{d-1}{2}\big(\log 2 + \log\pi\big) + \Big(\frac{1}{2} - \frac{dk}{N}\Big)\log d,$$

  the instance at $z = k/N$ of the Gauss multiplication theorem [11],

$$\prod_{j=0}^{d-1} \Gamma\Big(z + \frac{j}{d}\Big) = (2\pi)^{(d-1)/2}\, d^{\,1/2-dz}\, \Gamma(dz),$$

  with the convention $x_N := \log\Gamma(1) = 0$ in the boundary case $dk = N$.

The raw matrix has $\lfloor (N-1)/2 \rfloor + \sigma(N) - N$ rows and $(N-1)$ columns, before redundant rows are removed. One exact row reduction of the augmented block $[M \mid I]$ over $\mathbb{Q}$ (columns ordered numerator-descending, so pivots land on large numerators) yields simultaneously:

1. a reduction $\Gamma(k/N) = \exp(\text{elementary}) \cdot \prod \Gamma(\text{basis})^{\mathbb{Q}}$ for *every* non-basis k at once, the elementary side computed exactly as (combination matrix) $\cdot$ (right-hand-side vector) — no `FullSimplify` anywhere;

2. the grid's *basis* — its free formal gamma coordinates — and with it a termination guarantee: the lattice rank says the number of free coordinates in this presented system.

Proposition 6.1 (Completeness relative to the law matrix)**.** *For a fixed denominator grid and a fixed finite family of reflection and multiplication rows, exact row reduction computes every linear consequence of those rows. The free columns give a basis of the formal quotient by that row space, and the augmented matrix records a certificate for each reduction.*

Proof. Elementary row operations preserve the row space and are invertible. The reduced echelon matrix therefore spans exactly the same consequences. Each pivot equation is a recorded rational combination of the original rows. The nonpivot columns are free in the presented quotient. □

Unrestricted completeness of the standard relations for gamma monomials at rational arguments is Rohrlich's conjecture. Koblitz–Ogus prove a sufficient algebraicity criterion, not its converse. Thus irreducibility of the numerical gamma basis is conditional, and a finite-grid computation also needs care if it is compared with relations using larger grids. See the discussion of standard relations in Rivoal and Roques, *Hadamard products of algebraic functions* (Section 3) [34].

Result 6.2 (benchmarks and coverage). Implementations: `tools/gamma-reducer.wl` (Wolfram) and `tools/gamma_reducer.py` (pure Python: exact $\mathbb{Q}$ arithmetic via `fractions.Fraction`, sparse atom-coefficient right-hand sides, mpmath verification). Both produce *identical* output (same rules, same bases).

workload (generation only)	Wolfram	Python
denominator-120 grid (103 rules, 16-element basis)	0.42 s	0.056 s
all 151 legacy-database grids, denominators ≤ 308 (15 588 rules)	75.3 s	7.44 s
mpmath verification of all 15 588 rules	—	8.32 s

Coverage of the 550 recovered `GammaRules.wdx` arguments: 550/550. The original prover needed multi-hour sessions for 550 rules with no terminus; either implementation beats it by orders of magnitude *with* a completeness guarantee. Sample output (the sine values collapse to radicals on their own):

$$\Gamma\Big(\frac{5}{12}\Big) = \frac{2^{2/3}\sqrt{\pi}\,\Gamma\big(\frac{1}{12}\big)}{(1+\sqrt{3})\,\Gamma\big(\frac{1}{6}\big)}.$$

Remark 6.3. The row-combination matrix produced by the augmented reduction is, at the same time, a *multiplier certificate* in the sense of Section 6.5: the reducer's output is born proved, not merely verified.

6.2 Trig-root Γ identities: mechanism and census

The prompt for this investigation was the identity

$$\frac{\Gamma\big(\frac{2}{7}\big)\Gamma\big(\frac{11}{42}\big)}{\Gamma\big(\frac{1}{21}\big)} = \frac{8\sin\big(\frac{\pi}{7}\big)\sqrt{\pi\,\sin\big(\frac{\pi}{21}\big)\sin\big(\frac{4\pi}{21}\big)\sin\big(\frac{5\pi}{21}\big)}}{2^{1/42}\,7^{1/3}\,3^{9/28}},$$

of the kind that has surprised mathematicians before the mechanism was recognized [historical experiment: verified RootReduce-exactly against the legacy table].

Result 6.4 (the mechanism, in lattice terms). When the lattice solve expresses a target $\log\Gamma(k/N)$, the elimination may pass through a combination in which the target appears with multiplicity $D > 1$ — i.e. the relation natively proves $\Gamma(k/N)^D = (\text{elementary}) \cdot (\Gamma\text{-monomial})$. Dividing by D puts coefficient $1/D$ on the elementary $\log\sin$ atoms, and exponentiating yields a D-th root of a sine product; positivity of all factors fixes the branch. The root is a property of the *derivation path*, not of the value: the same $\Gamma(11/42)$ carries a square root in the harvested spelling above and cube roots in the lattice's own reduction — both exactly correct.

The surprise dissolves further at grids whose sines are radical-expressible: for $N = 30, 60$ the renderer collapses $\sqrt{\sin\cdots}$ into golden-ratio radicals and nothing looks trig-rooted at all. The phenomenon is only *visible* when the sine arguments have large totient (21, 39, 78, …), where nested square-root expressions need not exist; general radical expressibility is a different question.

Result 6.5 (census, denominator grids $N \le 200$, primitive arguments). 856 lattice reductions carry fractional $\log\sin$ coefficients. By root degree D (the lcm of the coefficient denominators):

D	2	3	4	6	8	16
rules	616	46	106	26	42	20

Demanding *integral* Γ-exponents (so only the trigonometric part is rooted, as in the prompt identity) leaves exactly 10 rules, all $D = 4$, at $\Gamma(71/78)$, $\Gamma(85/156)$, $\Gamma(65/114)$, $\Gamma(167/198)$ and conjugate-numerator siblings. The $D = 16$ cases live on the $N = 168$ grid with 26 sines under the radical.

Result 6.6 (showpieces). [historical experiment: verified to 30+ digits in both implementations] The fourth-root champion ($N = 78$; $\varphi(78) = 24$, so square-root-only expressions are unavailable):

$$\Gamma\Big(\frac{71}{78}\Big) = 2\cdot 2^{55/312}\,\pi\,\sqrt{\cos\tfrac{\pi}{78}}\;\cos^{1/8}\tfrac{\pi}{39}\;\cos^{3/4}\tfrac{\pi}{26}\;\cos^{5/8}\tfrac{2\pi}{39}\;\cos^{11/8}\tfrac{4\pi}{39}\;\big(\cos\tfrac{\pi}{13}\sec\tfrac{7\pi}{78}\big)^{3/8}\,\Gamma\Big(\frac{1}{39}\Big)$$
$$\Big/\Big[\,13^{1/48}\,\cos^{1/4}\tfrac{5\pi}{78}\;\cos^{3/2}\tfrac{3\pi}{26}\;\cos^{7/8}\tfrac{5\pi}{39}\;\cos^{5/8}\tfrac{11\pi}{78}\;\Gamma\Big(\frac{2}{39}\Big)\Gamma\Big(\frac{5}{78}\Big)\Big],$$

and the lattice's cube-root reduction of the prompt's own $\Gamma(11/42)$:

$$\Gamma\Big(\frac{11}{42}\Big) = \frac{7^{1/6}\sqrt{\pi}\;\Gamma\big(\frac{1}{42}\big)\Gamma\big(\frac{1}{21}\big)\big(\sec\frac{\pi}{42}\sec\frac{\pi}{21}\big)^{2/3}\sec^2\frac{\pi}{14}\,\big(\cos\frac{2\pi}{21}\sec\frac{5\pi}{42}\big)^{1/3}}{8\cdot 2^{5/21}\,3^{3/7}\,\Gamma\big(\frac{2}{21}\big)\Gamma\big(\frac{3}{14}\big)}.$$

6.3 The trigamma analogue

The same construction applies one weight up. On the grid $y_k = \psi^{(1)}(k/N)$ the lattice rows are the reflection law $\psi^{(1)}(k/N) + \psi^{(1)}(1 - k/N) = \pi^2\csc^2(\pi k/N)$ and the multiplication theorem $\psi^{(1)}(dk/N) = d^{-2}\sum_{j=0}^{d-1}\psi^{(1)}(k/N + j/d)$, with the pin $\psi^{(1)}(1) = \pi^2/6$ wherever an argument hits 1; the elementary atoms are π^2 and the $\pi^2\csc^2(\pi k/N)$ values.

Result 6.7. `tools/psi_reducer.py` derives the complete trigamma reduction tables for grids $N = 3, \ldots, 100$: 3428 rules, all mpmath-verified, in 7.7 s total, with basis dimensions equal to $\varphi(N)/2$ throughout — the same dimension count as the cyclotomic-polylogarithm lattice, which the exact DFT bridge between the two worlds explains (the DFT provides a change of algebraic coordinates between the value systems).

This subsumes, by a factor of ∼50, a legacy 70-entry hand/PSLQ-era workbook of $\psi^{(1)}(k/n)$ closed forms (denominators 2–16), which is preserved separately as a harvested store.

6.4 The harvest: legacy notebooks, verified and taxonomized

A kernel-free pipeline (notebook box parser, custom `TraditionalForm`/`InputForm` interpreter, mpmath 60-digit verification) mined the owner's OneDrive notebook corpus into verified identity stores:

- `harvested-gammaprover-rules.wl` — the recovered 550-rule saturation-prover database, every rule verified in-kernel to $|\,\mathrm{N}[\Gamma(\mathrm{arg}) - \text{closed form}, 40]\,| < 10^{-30}$, zero failures;
- `harvested-gamma-values.wl` — 508 verified Γ reductions from four further notebook families (including a 451-rule table recovered from a single 700 KB output cell, and 637 sine-product reductions from a working-table pair identified during the sweep);
- `harvested-trigamma-values.wl` — all 70 identities of the trigamma workbook, mpmath-verified at 60 digits and re-verified in-kernel.

Result 6.8 (novelty taxonomy). Measured against the 670-rule `FunctionTransformations` Γ table [12] (denominators ≤ 120) that anchors the corpus: 314 distinct harvested arguments lie beyond the table — 280 ceiling-breakers from the prover database (denominators 123–308) plus 34 arguments that only the notebook harvest reaches: one further ceiling-breaker, $\Gamma(67/160)$, and 33 upper-half reflection twins of table arguments. (Eight more harvest ceiling-breakers duplicate prover-database arguments; an earlier recorded decomposition "$280 + 8 + 34$" double-counted them.)

6.5 Multiplier certificates

Before this program, every identity in the project's stores carried one badge: *numerically verified* ($\gtrsim 25$ digits). The certificate program adds a qualitatively stronger badge: *derived* — a recorded, machine-checkable certificate exhibiting the value as an exact linear combination of verified functional-equation instances, replayed with no search and no numerics in the algebraic replay path of the historical checker. Its numerical branch guards remain outside the exact certificate guarantee.

6.5.1 The model

A certificate stores law *names* plus concrete substitution strings (parsed through a restricted substitution grammar; strings alone do not guarantee safe or valid instantiation) and two exact multiplier vectors over a canonical real/imaginary splitting of the instances. Validity means the claim lhs − rhs equals the recorded linear combination *identically*: every inert Li/ψ atom coefficient cancels `RootReduce`-exactly, and the elementary residual is killed by a dedicated sound zero test (certified principal-branch log normalization, certified multiplicative log-basis reduction, formal coefficient comparison). Numerics appear in exactly two proposal/guard roles: proposing root-of-unity decompositions that exact algebra then certifies, and a numerical per-instance branch guard. That guard can reject bad instances, but is not an exact domain proof; branch validity is a separate obligation. Linear systems subsume orbit words, fixed-point arguments, and multi-value systems in one format.

A `derived` certificate proves its claim *relative to* a declared axiom layer: (a) the classical laws of the project's functional-equation store [1] — Euler reflection, inversion, Landen, duplication/triplication/multiplication, the five-term family, the Kummer-class relations, Schwarz conjugation, the cyclotomic polygamma sum, and the polygamma reflection/multiplication laws, each numerically verified at recorded test points; (b) definitional pins ($\mathrm{Li}_n(1) = \zeta(n)$, $\psi^{(m)}(1) = (-1)^{m+1} m!\,\zeta(m+1)$,

and the two Catalan pins below); (c) Euler's $\zeta(2k) \in \mathbb{Q}\pi^{2k}$ and trigonometric values at rational multiples of π; (d) `RootReduce` as the exact algebraic-number oracle.

6.5.2 Findings the linear algebra produced

Finding 6.9 (Schwarz conjugation is a necessary axiom). The anharmonic orbit of i is the six-point family $\{i, -i, 1 \pm i, (1 \pm i)/2\}$, and inversion and duplication at i produce the same sum relation. The full system of reflection/inversion/Landen/duplication instances over the orbit is rank-deficient by *exactly one* dimension, and the missing direction is precisely $\mathrm{Li}_2(\bar{z}) = \overline{\mathrm{Li}_2(z)}$. No combination of the single-variable transformation laws separates $\mathrm{Li}_2(i)$ from $\mathrm{Li}_2(-i)$ — the transformation laws are Möbius/algebraic in the argument and blind to conjugation. Obvious in hindsight; the linear algebra *measured* it.

Finding 6.10 (Catalan is definitional; the rest is not). $\beta(2)$ admits no derivation from the stored laws: it enters every certificate that needs it through one of two recorded definitional pins, both termwise series identities,

$$\operatorname{Im}\mathrm{Li}_2(i) = G \qquad \text{or} \qquad \psi^{(1)}\!\left(\tfrac{1}{4}\right) - \psi^{(1)}\!\left(\tfrac{3}{4}\right) = 16\,G.$$

By contrast $\mathrm{Li}_2(-1) = -\pi^2/12$ (duplication at $z = 1$), the four golden-ratio dilogarithm values, $\mathrm{Li}_3(\frac{1}{2})$, the golden trilogarithm $\mathrm{Li}_3(1/\varphi^2)$, and $\mathrm{Li}_2(2)$ are all honestly derived; the flagship compaction is $\mathrm{Li}_2(\frac{1}{2})$ as a *one-instance* certificate (Euler reflection at its own fixed point), and each golden-ratio dilogarithm carries exactly the classical four-equation linear system, the coincidence $1 - 1/\varphi = 1/\varphi^2 = (1/\varphi)^2$ doing all the work. Within this law vocabulary, the tested imaginary orbit uses Catalan as its single additional named constant. The whole imaginary line also carries $\mathrm{Ti}_2(y)$; this finite orbit calculation is not a universal reduction theorem.

Finding 6.11 (prime denominators need the full multiplication row). For the cyclotomic values $\mathrm{Li}_n(e^{2\pi i p/q})$ the polygamma vocabulary pins the ψ-grid via proper-divisor multiplication instances when q is composite; for *prime* q the only ψ-sum relation is the full row $d = q$ — the multiplication theorem at $z = 1/q$,

$$\psi^{(m)}(1) = q^{-m-1} \sum_{k=1}^{q} \psi^{(m)}\!\left(\frac{k}{q}\right),$$

whose $k = q$ term is $\psi^{(m)}(1)$ itself — and the DFT coefficients are the source of the algebraic irrational coefficients — e.g. the $\sqrt{5}\,\zeta(3)$ terms in the $q = 5$ trilogarithm closed forms. Omitting $d = q$ cost precisely the prime-q certificates and nothing else.

Finding 6.12 (a trusted layer must be applied, not assumed). In the historical checker, a trigonometric normalization step was omitted: the $q = 8$ certificates initially failed because $\csc^2(\pi/8)$ survived as an opaque atom in the formal zero test while being numerically zero all along. The fix is policy, not code volume: trig at rational multiples of π is part of the declared trusted base and is applied explicitly during law instantiation and residual checking. A trusted constant layer is only trusted if the checker actually *invokes* it; silent reliance on auto-evaluation makes the trust boundary version-dependent.

6.5.3 Coverage ledger

At the repository snapshot, the certificate ledger replayed by the independent checker under the regression guard is 212 **derived** + 4 **definitional, zero failures**:

store	derived / attempted	notes
curated values	$45 + 4$ definitional	every concrete curated value
legacy polylog table, order-2 real	82/83	one honest holdout (below)
legacy Γ table, denominators ≤ 40	85/100	complete-lattice law pool

The Γ certificates use the complete denominator-q relation lattice of Section 6.1 as the law pool (20/20 on the pilot slice, versus 3/20 for the earlier heuristic pool). The polylog order-2 sweep was driven by three cumulative generator upgrades — atom-closure five-term pairs with structured one-parameter pair families $(x, 1-x)$, $(x, x/(x-1))$, $(x, -x)$; square-root argument seeding for duplication ladders; and a height-weighted cluster retry that pools orbit vocabulary across related rules without ever using a related *claim* (no circularity) — the largest certificate being 80 instances.

Search observation 6.13 (the honest holdout). $\mathrm{Li}_2(-7/10)$, the deepest reduction ladder in its slice (10 residual dilogarithms), remains rank-deficient even at deep search caps: the orbit pairs demand an internal 17-flavoured cancellation that no implemented pair family reaches. The natural next lever is second-order stray matching or ladder relations beyond the stored law set. On the Γ side, the frontier is 15 no-certs of $q = 40$ grade (sine-product exact algebra blow-ups) and the $q > 40$ tail, awaiting cyclotomic-unit arithmetic or certificate transport through the exact DFT bridge to the lighter ψ-side atoms.

Chapter 7

Polygamma row sums at complex multiplication points

Rational gamma grids account for periods that reappear in modular forms. At imaginary quadratic arguments the polygamma series becomes a row of a lattice Eisenstein sum. Symmetry isolates elementary components; modular periods and class-field coordinates account for the non-elementary ones. Individual values and full row sums require different evidence.

7.1 Lattice normalization and row sums

For even $w \geq 4$ and $\Im\tau > 0$, define the absolutely convergent

$$G_w(\tau) = \sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}} \frac{1}{(m+n\tau)^w} = 2\zeta(w)E_w(\tau).$$

Pair positive and negative n, and use the defining polygamma series:

$$G_w(\tau) = 2\zeta(w) + \frac{2}{(w-1)!}\sum_{n\geq 1}\left[\psi^{(w-1)}(n\tau) + \psi^{(w-1)}(1-n\tau)\right]. \tag{7.1}$$

The hexagonal point in this chapter is $\rho = e^{i\pi/3}$. The modular invariant $\mathfrak{g}_2 = E_4/\eta^8$ below is unrelated to the second Stieltjes constant. Algebraic fits and historical numerical residuals are labeled as such; class polynomials are formed from normalized modular functions, rather than untreated modular-form values.

7.2 The elementary layer: reflection composed with conjugation

Before any CM input, one family of components is fixed by symmetry alone. The Schwarz reflection $\psi^{(m)}(\bar{z}) = \overline{\psi^{(m)}(z)}$ turns the reflection law into a statement about a single real component, leaving the other component untouched.

Result 7.1 (vertical-line trigamma law). On the line $\operatorname{Re} z = \frac{1}{2}$ the reflection law alone evaluates the real part of the trigamma function at every height: for all real t,

$$\operatorname{Re}\psi^{(1)}\Big(\tfrac{1}{2} + it\Big) = \frac{\pi^2}{2\cosh^2(\pi t)}.$$

In particular $\operatorname{Re}\psi^{(1)}\big(\frac{1+i}{2}\big) = \pi^2/(2\cosh^2(\pi/2))$ [historical experiment: residual 0 at 50 dps]. Off the symmetric line, combining reflection, conjugation and the recurrence gives e.g.

$$\operatorname{Re}\psi^{(1)}(i) = -\frac{1}{2} - \frac{\pi^2}{2\sinh^2\pi} \qquad \text{[historical experiment: residual 0 at 50 dps].}$$

The proof of the first identity is one line: on $\operatorname{Re} z = \frac{1}{2}$ one has $1 - z = \bar{z}$, so the reflection law $\psi^{(1)}(z) + \psi^{(1)}(1-z) = \pi^2/\sin^2(\pi z)$ reads $2\operatorname{Re}\psi^{(1)}(z) = \pi^2/\sin^2(\pi z)$ there.

The imaginary parts on this line stay opaque — they are odd-row lattice data in the sense of Section 7.12.

7.2.1 The FoxTrot identity demystified

MathWorld's *Digamma Function* page [17] calls the following identity "surprising"; it arises from the FoxTrot series $\sum(-1)^{n-1}n^2/(n^3+1)$, whose partial fractions hit the roots of $n^2 - n + 1$:

$$-\psi\big(\tfrac{\omega}{2}\big) - \psi\big(-\tfrac{\omega^2}{2}\big) + \psi\big(\tfrac{1+\omega}{2}\big) + \psi\big(\tfrac{1-\omega^2}{2}\big) = 2\pi\ \operatorname{sech}\Big(\tfrac{\sqrt{3}\,\pi}{2}\Big), \qquad \omega = e^{i\pi/3}. \tag{7.2}$$

Law 7.2 (line law for digamma differences)**.** *For every shift $c \in (0,1)$, on the line $\operatorname{Re} z = (1-c)/2$,*

$$\operatorname{Re}\big[\psi(z+c) - \psi(z)\big] = \pi\ \operatorname{Re}\cot(\pi\bar{z}).$$

[historical experiment: verified for $c = \frac{1}{2}, \frac{1}{3}, \frac{2}{3}$ at generic heights t]

The mechanism is the same one-liner: on that line the shift satisfies $z + c = 1 - \bar{z}$, so reflection composed with conjugation eliminates the digamma values, leaving an elementary $\sinh$ / $\cos$ closed form. The FoxTrot identity (7.2) is the instance $c = \frac{1}{2}$, $z = \frac{1}{4} + i\frac{\sqrt{3}}{4}$: its four arguments pair into $2\operatorname{Re}[\psi(z+\frac{1}{2}) - \psi(z)]$, and on $\operatorname{Re} z = \frac{1}{4}$ one has $\pi\operatorname{Re}\cot(\pi\bar{z}) = \pi\operatorname{sech}(2\pi t)$. The identity is thus the point $t = \sqrt{3}/4$ on a line of identities, itself one member of a two-parameter family — no complex multiplication required.

Differentiating in t ladders the law up the polygamma tower [historical experiment: weights $2, 3$ verified at 10^{-50}]:

$$\operatorname{Im}\big[\psi^{(1)}(\tfrac{3}{4}+it) - \psi^{(1)}(\tfrac{1}{4}+it)\big] = 2\pi^2\operatorname{sech}(2\pi t)\tanh(2\pi t),$$
$$\operatorname{Re}\big[\psi^{(2)}(\tfrac{3}{4}+it) - \psi^{(2)}(\tfrac{1}{4}+it)\big] = 8\pi^3\operatorname{sech}^3(2\pi t) - 4\pi^3\operatorname{sech}(2\pi t).$$

More generally, for $0 < c < 1$, let $z = (1-c)/2 + \mathrm{i}t$ and $R_c(t) = \pi\sin(\pi c)/(\cosh(2\pi t) + \cos(\pi c))$. Reflection and conjugation give $\operatorname{Re}(\psi(z+c) - \psi(z)) = R_c(t)$. Differentiating m times yields

$$\operatorname{Re}\Big(\mathrm{i}^m[\psi^{(m)}(z+c) - \psi^{(m)}(z)]\Big) = R_c^{(m)}(t).$$

Thus the real part is determined for even m and the imaginary part for odd m. With weight $w = m+1$, these are the real component at odd weight and imaginary component at even weight. Their complementary components are not determined by this reflection identity; that leaves open reductions by other methods. This shifted-difference statement is distinct from individual polygamma values on $\operatorname{Re} z = 1/2$.

7.3 Maximal symmetry: vanishing identities and Γ-periods

At the two CM points with extra symmetry — $\tau = i$ (square lattice, $j = 1728$) and $\tau = \rho = e^{i\pi/3}$ (hexagonal lattice, $j = 0$) — classical theory pins every G_{2k} exactly. The square lattice admits no weight-6 form ($i \cdot (\mathbb{Z} + \mathbb{Z}i) = \mathbb{Z} + \mathbb{Z}i$ forces $G_6(i) = i^{-6}G_6(i) = -G_6(i)$), and the hexagonal lattice no weight-4 form; $G_4(i)$ and $G_6(\rho)$ are the lemniscatic and equianharmonic period powers — monomials in $\Gamma(1/4)$, resp. $\Gamma(1/3)$, by Chowla–Selberg [13].

7.3.1 Vanishing identities

Result 7.3 (the CM gems).

$$\sum_{n\geq 1}\Big[\psi^{(5)}(ni) + \psi^{(5)}(1-ni)\Big] = -120\,\zeta(6) \qquad \text{[historical experiment: residual } 1.7\times 10^{-49}\text{]},$$

$$\sum_{n\geq 1}\Big[\psi^{(3)}(n\rho) + \psi^{(3)}(1-n\rho)\Big] = -6\,\zeta(4), \qquad \rho = e^{i\pi/3} \qquad \text{[historical experiment: residual } 6.8\times 10^{-50}\text{]}.$$

Each individual bracket is a transcendental-looking complex number (elementary in coth / csch of $\pi n\sqrt{3}/2$ and the like after the bilateral cotangent expansion, but nobody would guess the totals); the sums collapse to rational multiples of $\zeta(6)$, $\zeta(4)$ *because lattice rotation multiplies the corresponding value by a nontrivial root of unity.* Via (7.1) these are exactly the statements $G_6(i) = 0$ and $G_4(\rho) = 0$.

7.3.2 Chowla–Selberg made polygamma-visible

Result 7.4 (period identities).

$$2\zeta(4) + \frac{1}{3}\sum_{n\geq 1}\Big[\psi^{(3)}(ni) + \psi^{(3)}(1-ni)\Big] = \frac{\Gamma(1/4)^8}{960\,\pi^2} \qquad \text{[historical experiment: residual } 2.1\times 10^{-50}\text{]},$$

$$2\zeta(6) + \frac{1}{60}\sum_{n\geq 1}\Big[\psi^{(5)}(n\rho) + \psi^{(5)}(1-n\rho)\Big] = \frac{\Gamma(1/3)^{18}}{8960\,\pi^6} \qquad \text{[historical experiment: residual } 2.0\times 10^{-60}\text{]}.$$

Polygamma values at purely imaginary (resp. hexagonal) points, summed over the integer multiples of the point, know the lemniscate constant and its equianharmonic sibling. The atoms $\Gamma(1/4)^2/\pi$ and $\Gamma(1/3)$ are precisely the CM-period currency of the repository's hypergeometric reducer (`cmPeriodReduce` in `src/Hypergeometric`), so these identities tie the polygamma world directly to that corpus.

7.3.3 The complete low-weight table

The symmetry argument and the classical Hurwitz recursion [15]

$$(2n+1)(2n-1)(n-3)\,G_{2n} = 3\sum_{m=2}^{n-2}(2m-1)(2n-2m-1)\,G_{2m}G_{2n-2m}$$

extend both subsections through weight 12 with no new constants [historical experiment: all entries verified, residuals 10^{-60}–10^{-263}]:

weight $2k$	square lattice $\tau = i$	hexagonal lattice $\tau = \rho$
4	$\Gamma(1/4)^8/(960\pi^2)$	$\mathbf{0}$
6	$\mathbf{0}$	$\Gamma(1/3)^{18}/(8960\pi^6)$
8	$\frac{3}{7}G_4(i)^2 = 3\,\Gamma(1/4)^{16}/(7\cdot 960^2\pi^4)$	$\mathbf{0}$
10	$\mathbf{0}$	$\mathbf{0}$
12	$\frac{1}{143}\big(18\,G_4(i)^3 + 25\,G_6(i)^2\big)$ type	$\frac{25}{143}G_6(\rho)^2 \propto \Gamma(1/3)^{36}$

Every bold zero is a vanishing identity $\sum_{n\geq 1}[\psi^{(w-1)}(n\tau) + \psi^{(w-1)}(1-n\tau)] = -(w-1)!\,\zeta(w)$, and every non-zero entry is a $\psi^{(w-1)}$-sum evaluation. The pattern is exhaustive: $G_{2k}(i) = 0$ iff $2k \not\equiv 0 \pmod 4$, $G_{2k}(\rho) = 0$ iff $2k \not\equiv 0 \pmod 6$, and all non-zero values are $\mathbb{Q}$-monomials in the two period powers — so the *entire* Eisenstein tower over both lattices is polygamma-explicit.

7.4 Beyond maximal symmetry: $2i$ and $i\sqrt{2}$

The same row-sum at the next two CM points, with closed forms found by PSLQ [historical experiment: 120-dps canaries]:

Theorem 7.5.

$$G_4(2i) = \frac{11}{15360}\,\frac{\Gamma(1/4)^8}{\pi^2} \quad \textit{[historical experiment: canary } 0 \textit{ at } 120 \textit{ dps]},$$

$$G_4(i\sqrt{2}) = \frac{\Gamma(1/8)^4\,\Gamma(3/8)^4}{4608\,\pi^2} \quad \textit{[historical experiment: canary } 3.9\times 10^{-121}\textit{]}.$$

The weight-6 row at the same points [historical experiment: 160-dps canaries]:

$$G_6(2i) = \frac{\Gamma(1/4)^{12}}{2^{14}\cdot 5\cdot \pi^3}, \qquad G_6(i\sqrt{2}) = \frac{\Gamma(1/8)^6\,\Gamma(3/8)^6}{2^{12}\cdot 3^3\cdot 5\cdot \pi^3}.$$

The first value confirms the isogeny picture: $2i$ sits over the same field $\mathbb{Q}(i)$, so only the rational factor changes ($G_4(i)/G_4(2i) = 16/11$). The second uses the period coordinate associated with discriminant -8, field $\mathbb{Q}(\sqrt{-2})$ — the polygamma sum $\sum_n[\psi^{(3)}(ni\sqrt{2}) + \psi^{(3)}(1-ni\sqrt{2})]$ evaluates in $\Gamma(1/8)\Gamma(3/8)$. The fundamental unit $1+\sqrt{2}$ was offered in the PSLQ basis and declined (exponent 0). Together with Section 7.3 this makes three distinct Chowla–Selberg periods polygamma-visible.

7.5 Class number 2: discriminants -20 and -15

At $\mathbb{Q}(\sqrt{-5})$ (class number 2) a monomial-only logarithmic basket can miss the value: the algebraic factor in G_4/Ω^4 lives in the Hilbert class field and is not a monomial. The structure that works is the Hecke/Damerell pair [14] over the two reduced forms x^2+5y^2 ($\tau_1 = i\sqrt{5}$) and $2x^2+2xy+3y^2$ ($\tau_2 = (-1+i\sqrt{5})/2$).

Theorem 7.6 (disc -20)**.**

$$\frac{G_4(\tau_1)}{G_4(\tau_2)} = \frac{29+12\sqrt{5}}{44} = \frac{(3+2\sqrt{5})^2}{44} \quad \textit{[historical experiment: canary } 0 \textit{ at } 160 \textit{ dps]},$$

$$G_4(\tau_1)\,G_4(\tau_2) = \frac{11\,\varphi^4\,\Gamma(3/20)^8\,\Gamma(7/20)^8}{2^{14}\cdot 3^4\cdot 5^4\cdot \pi^4} \quad \textit{[historical experiment: canary } 5.3\times 10^{-160}\textit{]},$$

where φ is the golden ratio. The ratio is algebraic of degree 2 with norm $N(29+12\sqrt{5}) = 121 = 11^2$; the class product is a period monomial carrying the prime 11 and a golden unit.

Theorem 7.7 (disc -15)**.** *With $\tau_1 = (1+i\sqrt{15})/2$ for $x^2+xy+4y^2$ and the conjugate pair $\tau_{2,3} = (\pm 1+i\sqrt{15})/4$ representing $2x^2+xy+2y^2$, write $\widehat{G}_4 = \frac{1}{2}\big[G_4(\tau_2)+G_4(\tau_3)\big] = \operatorname{Re} G_4(\tau_2)$ for the class-pair average. Then [historical experiment: 160-dps canaries]:*

$$\frac{G_4(\tau_1)}{\widehat{G}_4} = \frac{2\,(4+\sqrt{5})^2}{77}, \qquad G_4(\tau_1)\,\widehat{G}_4 = \frac{77\left(\Gamma(\frac{1}{15})\Gamma(\frac{2}{15})\Gamma(\frac{4}{15})\Gamma(\frac{8}{15})\right)^4}{2^{13}\cdot 3^6\cdot 5^5\cdot \pi^4}.$$

Unlike disc -20, the non-principal value here is genuinely complex (the form $(2,1,2)$ has $|\tau_2|=1$, where conjugation acts nontrivially on G_4); the real-part projection matches the representative convention used in these formulas; its period factor is fixed in the proof below.

The Γ-block in Result 7.7 is exactly the set of residues $\{1,2,4,8\}$ with $\chi_{-15}=+1$; the golden unit is declined by PSLQ (exponent 0); and the prime content $77 = 7\cdot 11$ of the product matches the ratio's denominator ($N(4+\sqrt{5}) = 11$). The heuristic that emerges, used at every later discriminant: *run the ratio first; its norm names the primes the class-product basis needs.*

7.6 Class number 3: discriminant -23

$\mathbb{Q}(\sqrt{-23})$ has $h=3$, with Hilbert class field generated over $\mathbb{Q}(\sqrt{-23})$ by a root of x^3-x-1. The three reduced forms give $\tau_1 = (1+i\sqrt{23})/2$ and $\tau_{2,3} = (\pm 1+i\sqrt{23})/4$; $G_4(\tau_1)$ is real and $G_4(\tau_{2,3})$ form a conjugate pair.

Theorem 7.8 (disc -23 class product)**.** *[historical experiment: canary 1.4×10^{-160}]*

$$G_4(\tau_1)\,G_4(\tau_2)\,G_4(\tau_3) = \frac{11\cdot 17\,P^4}{2^{24}\cdot 3^6\cdot 23^5\cdot \pi^{16}}, \qquad P = \prod_{\substack{k \bmod 23 \\ (\frac{k}{23})=+1}} \Gamma\Big(\frac{k}{23}\Big) \quad \textit{(11 factors).}$$

Theorem 7.9 (the $\mathfrak{g}_2$ class polynomial from row-sums)**.** *Let $P(X) = X^3+155X^2+650X+23375$. For each of the three singular moduli j_Q, define ξ_Q as the unique root of P satisfying $\xi_Q^3 = j_Q$. Thus ξ_Q equals $E_4(\tau_Q)/\eta(\tau_Q)^8$ times a cube root of unity, with its phase fixed by this algebraic condition. The three ξ_Q have the exact class polynomial*

$$P(X) = X^3+155X^2+650X+23375.$$

Its constant term $23375 = 5^3\cdot 11\cdot 17$ identifies the same prime support as the proved class product. The cube-image identity and uniqueness of the phase choice are proved below in Proposition 7.21.

Search observation 7.10 (weight-4 sums are not Galois-clean). The dimensionless symmetric functions e_1^3/e_3 and e_2^3/e_3^2 of the raw G_4 values were not recognized as rational in the source search (height 10^{30}, 300 digits). Unlike the value *product* (a norm) and the weight-0 $\mathfrak{g}_2$ invariants, sums of weight-4 CM values across ideal classes are not Galois-stable — the period correction under $\mathrm{Gal}(H/K)$ is nontrivial at weight 4. Class polynomials must be formed from $\mathfrak{g}_2 = E_4/\eta^8$ (or j), not from the G_4 values themselves.

7.7 Class number 4: discriminant -39 and genus structure

The four reduced forms of discriminant -39, namely $(1,1,10)$, $(2,\pm 1,5)$, $(3,3,4)$, give $\tau \in \{(1+i\sqrt{39})/2,\ (\pm 1+i\sqrt{39})/4,\ (3+i\sqrt{39})/6\}$.

Theorem 7.11 (disc -39 class product). *[historical experiment: 160-dps canary]*

$$\prod_{i=1}^{4} G_4(\tau_i) = \frac{17 \cdot 23 \cdot 29\, P^4}{2^{28} \cdot 3^9 \cdot 5^4 \cdot 13^8 \cdot \pi^{16}}, \qquad P = \prod_{\substack{k \bmod 39 \\ \chi_{-39}(k)=+1}} \Gamma\Big(\frac{k}{39}\Big) \quad \textit{(12 factors).}$$

At this discriminant no cube root of a singular modulus can belong to the Hilbert class field: Proposition 7.21 proves its degree is twelve, while that class field has degree eight over $\mathbb{Q}$. This supplies an exact obstruction to every cube-root phase choice. The weight-zero j-invariant still gives the Hilbert class polynomial directly.

Theorem 7.12 (Hilbert class polynomial of disc -39).

$$H_{-39}(x) = x^4 + 331531596\,x^3 - 429878960946\,x^2 + 109873509788637459\,x + 20919104368024767633$$

[historical experiment: $e_1, \ldots, e_3$ PSLQ-exact integers; e_4 integral to 10^{-137}].

Worth dwelling on: a quartic with 20-digit integer coefficients recovered from sums of polygamma values at quadratic-irrational complex points. The polygamma row-sum is a perfectly serviceable special-function front door to the classical CM tower.

Theorem 7.13 (genus structure inside the quartet). *The class group is $\mathbb{Z}/4$, with the ambiguous form $(3,3,4)$ as the order-2 class. The genus-wise ratio — principal genus $\{1, A^2\}$ over $\{A, A^3\}$ — is exactly quadratic, in the real part of the genus field $\mathbb{Q}(\sqrt{-3}, \sqrt{13})$; its precise period cancellation and positive-root extraction are proved by Theorem 7.23 [historical experiment: 160-dps canary; $\sqrt{3}$, $\sqrt{39}$ declined by PSLQ]:*

$$\frac{G_4(\tau_1)\,G_4(\tau_4)}{G_4(\tau_2)\,G_4(\tau_3)} = \frac{103299 + 4440\sqrt{13}}{181424}, \qquad N = \Big(\frac{9}{16}\Big)^2, \qquad 181424 = 16 \cdot 17 \cdot 23 \cdot 29.$$

The denominator reuses the class-product primes 17, 23, 29.

The raw cross-class ratio $G_4(\tau_1)/G_4(\tau_4)$ was separately reported as a quartic candidate. No degree-two polynomial was found up to height 10^{12} in that search. Neither its quartic candidate nor that unsuccessful search establishes its minimal polynomial or degree.

7.8 Class number 5: discriminant -47

The five reduced forms $(1,1,12)$, $(2,\pm1,6)$, $(3,\pm1,4)$ give the quintet $\tau \in \{(1+i\sqrt{47})/2,\ (\pm1+i\sqrt{47})/4,\ (\pm1+i\sqrt{47})/6\}$.

Theorem 7.14 (disc -47 class product). *[historical experiment: 200-dps canary]*

$$\prod_{i=1}^{5} G_4(\tau_i) = \frac{11^2 \cdot 23 \cdot 29\, P^4}{2^{52} \cdot 3^6 \cdot 47^9 \cdot \pi^{36}}, \qquad P = \prod_{\substack{k \bmod 47 \\ (\frac{k}{47})=+1}} \Gamma\Big(\frac{k}{47}\Big) \quad \textit{(23 factors).}$$

Theorem 7.15 (Hilbert class polynomial of disc -47). *The row-sum j-values deliver the quintic Hilbert class polynomial [historical experiment: all symmetric functions integral to 10^{-129} or better]:*

$$\begin{aligned} H_{-47}(x) = x^5 &+ 2257834125\,x^4 - 9987963828125\,x^3 + 5115161850595703125\,x^2 \\ &- 14982472850828613281250\,x + 16042929600623870849609375. \end{aligned}$$

The CM tower is now polygamma-explicit through class number 5 with one uniform recipe: *row-sum → class-product log-PSLQ (norm-forecast primes) → j-symmetric-function integrality.*

7.9 Exact CM row products from class polynomials

The manuscript's complex-multiplication chapter records striking products of weight-four polygamma row sums at discriminants $-15, -20, -23, -39, -47$. Its displayed numerical residuals establish useful evidence, but do not explain the prime factors or provide proofs of the gamma products. The following argument supplies that missing step. It also evaluates the corresponding products at every even weight by a finite computation with a Hilbert class polynomial.

The analytic ingredients are classical: the Kronecker limit formula, the Lerch–Chowla–Selberg formula, and the algebra of modular forms. The contribution here is an explicit, uniformly normalized consequence for the manuscript's row sums, accompanied by certified class-polynomial data and new displayed weight-six evaluations. We do not claim that the underlying CM period theory is new.

7.9.1 Conventions and the main theorem

Let $D = -d < -4$ be a fundamental discriminant, let $\chi_D(r) = (D/r)$ be its primitive quadratic Dirichlet character, and let $h = h(D)$. Take the standard reduced primitive positive definite forms

$$\mathcal{Q}_D = \{(a,b,c) : b^2 - 4ac = D,\ |b| \le a \le c,\ b \ge 0 \text{ if } |b| = a \text{ or } a = c\}.$$

There is one representative for each proper ideal class. Write

$$\tau_Q = \frac{-b + i\sqrt{d}}{2a}, \qquad A_D = \prod_{Q \in \mathcal{Q}_D} a, \qquad P_D = \prod_{\substack{1 \le r < d \\ \chi_D(r) = 1}} \Gamma(r/d), \tag{7.3}$$

$$H_D(X) = \prod_{Q \in \mathcal{Q}_D} (X - j(\tau_Q)), \qquad C_d = \Phi_d(1), \qquad \varphi_d = \varphi(d).$$

Thus $H_D \in \mathbb{Z}[X]$ is monic; $C_d = p$ if d is a positive power of a prime p, and $C_d = 1$ otherwise. All fractional powers of positive real numbers below denote their positive values.

For even $w \ge 4$, use exactly the lattice normalization of the manuscript:

$$G_w(\tau) = \sum_{(m,n) \ne (0,0)} (m + n\tau)^{-w} = 2\zeta(w) E_w(\tau). \tag{7.4}$$

The modular discriminant is $\Delta(\tau) = \eta(\tau)^{24}$; in particular $j = E_4^3/\Delta$ and $j - 1728 = E_6^2/\Delta$.

Lemma 7.16 (The polynomial attached to an even weight)**.** *For every even integer $w \ge 4$, there is a unique monic polynomial $F_w \in \mathbb{Q}[X]$ of degree w such that*

$$\frac{E_w(\tau)^{12}}{\Delta(\tau)^w} = F_w(j(\tau)). \tag{7.5}$$

It is effectively computable using exact rational Fourier coefficients.

Proof. The left side is a weight-zero modular function, holomorphic on the upper half-plane since Δ has no zeros there, with a pole of order w at the cusp and no other poles. The field of modular functions is $\mathbb{C}(j)$, and a modular function with no finite poles is a polynomial in j. Since $E_w = 1 + O(q)$, $\Delta = q + O(q^2)$, and $j = q^{-1} + 744 + O(q)$, this polynomial is monic of degree w. Successively removing its Laurent coefficients at $q^{-w}, \dots, q^0$ computes all coefficients. The input coefficients are rational, hence so are the polynomial coefficients. Equivalently, one can work in the exact graded algebra $\mathbb{Q}[E_4, E_6]$. □

Set

$$\mathcal{R}_{D,w} = \mathrm{Res}_X(H_D(X), F_w(X)).$$

Our resultant convention is $\mathrm{Res}(H_D, F_w) = \prod_Q F_w(j(\tau_Q))$, since H_D is monic. No unidentified sign or leading coefficient is implicit.

7.9.2 An exact generator for the modular weight polynomial

Normalize the Eisenstein series to have constant coefficient one:

$$E_w(\tau) = 1 - \frac{2w}{B_w} \sum_{m \geq 1} \sigma_{w-1}(m) q^m, \qquad q = e^{2\pi i \tau},$$

and set

$$\Delta = q \prod_{m \geq 1} (1 - q^m)^{24} = \frac{E_4^3 - E_6^2}{1728}, \qquad j = \frac{E_4^3}{\Delta}.$$

For each even $w \geq 4$, let $a \in \{0, 1, 2\}$, $b \in \{0, 1\}$, and $c \geq 0$ be the unique integers satisfying

$$w = 4a + 6b + 12c. \tag{7.6}$$

The excluded weight two is precisely the exceptional small weight in this residue prescription.

The level-one graded-ring structure [88] gives a unique monic polynomial $P_w \in \mathbb{Q}[X]$ of degree c such that

$$E_w = E_4^a E_6^b \Delta^c P_w(j). \tag{7.7}$$

This can be computed using only rational arithmetic. Write $P_w(X) = \sum_{r=0}^{c} p_r X^r$ and expand the basis

$$E_4^{a+3r} E_6^b \Delta^{c-r}, \qquad 0 \leq r \leq c.$$

The basis element indexed by r has first term q^{c-r} with coefficient one. Successive coefficient comparison at $q^0, q^1, \ldots, q^c$ therefore determines $p_c, p_{c-1}, \ldots, p_0$ by unit triangular elimination. Since E_w has constant term one, $p_c = 1$.

Consequently the polynomial in the CM norm formula is explicitly

$$\boxed{F_w(X) = X^{4a}(X - 1728)^{6b} P_w(X)^{12}.} \tag{7.8}$$

Indeed, raising (7.7) to the twelfth power and using $E_4^3 = j\Delta$, $E_6^2 = (j - 1728)\Delta$, and (7.6) gives $E_w^{12}/\Delta^w = F_w(j)$. The degree of F_w is exactly w and its leading coefficient is one.

Finite certificates. The companion generator computes Δ independently from its Euler product and verifies its normalization against $E_4^3 - E_6^2$. For every even w from four through twenty-four, it verifies exactly the coefficients of $q^0, \ldots, q^w$ in

$$E_w^{12} = \sum_{r=0}^{w} [X^r] F_w(X)\, E_4^{3r} \Delta^{w-r}. \tag{7.9}$$

Both sides are holomorphic modular forms of weight $12w$ on $\mathrm{SL}_2(\mathbb{Z})$. The level-one Sturm bound is w, so these finitely many rational equalities certify the identity. In characteristic zero this bound follows directly from the valence formula [88, Theorem 2.11]: a nonzero weight-$12w$ holomorphic modular form has order at infinity at most w.

The JSON artifact stores the exact rational coefficients, the compact factorization, the degree and monicity checks, and the finite coefficient certificates. It contains the full range, including

$$F_{14}(X) = X^8(X-1728)^6, \tag{7.10}$$

$$F_{16}(X) = X^4\left(X - \frac{3456000}{3617}\right)^{12}, \tag{7.11}$$

$$F_{18}(X) = (X-1728)^6\left(X - \frac{9504000}{43867}\right)^{12}, \tag{7.12}$$

$$F_{20}(X) = X^8\left(X - \frac{209520000}{174611}\right)^{12}, \tag{7.13}$$

$$F_{22}(X) = X^4(X-1728)^6\left(X - \frac{35424000}{77683}\right)^{12}, \tag{7.14}$$

$$F_{24}(X) = \left(X^2 - \frac{340364160000}{236364091}X + \frac{30710845440000}{236364091}\right)^{12}. \tag{7.15}$$

For example,

$$E_{14} = E_4^2E_6, \qquad E_{16} = \frac{1617E_4^4 + 2000E_4E_6^2}{3617} = E_4\Delta\left(j - \frac{3456000}{3617}\right),$$

which directly explain (7.10)–(7.11). The generator implements a standard modular-form construction, supplying explicit reusable coefficients and proof certificates for the CM recipe.

7.9.3 CM nonvanishing with a rational principal-lattice bound

The possible zero denominators in the product and genus formulas can be excluded uniformly. Non-elliptic CM nonvanishing is classical, through Kohnen's conjugation argument [90]; see the introduction of Chiriac–Jorza [91] for its place in zero theory. The following proof supplies an explicit principal-point bound, certified using rational arithmetic. It applies to nonmaximal orders as well.

Theorem 7.17 (Nonvanishing at every non-elliptic CM point)**.** *Let $D < -4$ be any quadratic-order discriminant. For every CM point τ of discriminant D and every even integer $w \geq 4$,*

$$G_w(\tau) \neq 0.$$

For the principal reduced point

$$\tau_D = \begin{cases} i\sqrt{|D|}/2, & D \equiv 0 \pmod 4, \\ (-1+i\sqrt{|D|})/2, & D \equiv 1 \pmod 4, \end{cases}$$

one has the weight- and discriminant-uniform bound $|G_w(\tau_D)| > 7/20$.

Proof. The principal lattice has, for $n \neq 0$, squared lengths bounded below by one of the two integer quadratic forms

$$q_7(m,n) = m^2 - mn + 2n^2, \qquad q_8(m,n) = m^2 + 2n^2.$$

Indeed the odd principal discriminant has $|D| \geq 7$, and the even one has $|D| \geq 8$. Both forms are at least two when $n \neq 0$. Hence for every even $w \geq 4$ the off-axis absolute sum is bounded by

$$T_d = \sum_{n\neq 0}\sum_{m\in\mathbb{Z}} q_d(m,n)^{-2}, \qquad d = 7, 8.$$

We prove $T_7 < 33/20$ and $T_8 < 11/8$ by finite rational sums and explicit infinite-tail bounds.

For any $A > 0$ and real shift s, layer-cake integration gives

$$\sum_{m\in\mathbb{Z}} \frac{1}{((m-s)^2+A^2)^2} \le A^{-4} + \frac{\pi}{2A^3}.$$

To see this, each superlevel set of the even decreasing function is an interval. Its number of translated lattice points is at most its length plus one. Integrating that inequality bounds the lattice sum by the integral of the function plus its maximum. The integral is $\pi/(2A^3)$.

Use $N = 10$, $M = 20$ in the finite rectangle. The imaginary coordinate in both lattices is greater than one. Thus the part with $|n| > N$, summed over every m, is less than

$$\frac{2}{3N^3} + \frac{2}{N^2},$$

by $\pi < 4$, $\sum_{n>N} n^{-4} \le 1/(3N^3)$ and $\sum_{n>N} n^{-3} \le 1/(2N^2)$. For $1 \le |n| \le N$ and $|m| > M$, the squared length is at least $(|m| - N/2)^2$. Integral comparison bounds this part by $4N/[3(M - N/2)^3]$. The finite certificate therefore verifies

$$2\sum_{n=1}^{10}\sum_{m=-20}^{20} q_7(m,n)^{-2} + \frac{31}{1500} + \frac{8}{2025} < \frac{33}{20},$$

and the same expression with q_8 is less than $11/8$. Every term and comparison in these two finite inequalities is rational; the full fractions are recorded in `verification/CM-nonvanishing.json`.

The $n = 0$ axis contributes exactly $2\zeta(w) > 2$. The triangle inequality now gives $|G_w(\tau_D)| > 2 - 33/20 = 7/20$ in the odd case, and the stronger $5/8$ in the even case.

Suppose a CM value at the same discriminant vanished. By (7.5), its singular modulus would be a zero of the rational polynomial F_w. The class polynomial H_D is the minimal polynomial of that singular modulus, including for a nonmaximal order [86, 16]. Thus H_D divides F_w, and the principal singular modulus is also a zero. Since Δ never vanishes on the upper half-plane, this would make $G_w(\tau_D) = 0$, contradicting the bound. Modular transformations multiply a positive-weight value by a nonzero factor, so the conclusion covers every representative. □

Consequently all class and genus resultants in the following formulas are nonzero at $D < -4$. Their positive roots and quotients are defined at every even weight. The excluded discriminants $-3, -4$ retain their familiar elliptic vanishing patterns.

7.9.4 Class products in every even weight

Theorem 7.18 (CM product formula at every even weight)**.** *With the preceding definitions, for every even $w \ge 4$,*

$$\prod_{Q\in\mathcal{Q}_D} |G_w(\tau_Q)| = \left(\frac{|B_w|}{w!}\right)^h \left(\frac{A_D}{d^h}\right)^{w/2} C_d^{w/4} (2\pi)^{w(2h-\varphi_d)/4} P_D^w \, |\mathcal{R}_{D,w}|^{1/12}. \tag{7.16}$$

Here the Bernoulli numbers satisfy $B_2 = 1/6$. Every factor and the resultant are nonzero by the preceding CM nonvanishing theorem. The identity itself also applies to a zero product when extended to an elliptic discriminant in the order formula below.

The absolute values are essential in a uniform statement. Selecting one standard representative for each class need not make the product of positive-weight modular forms real. Section 7.9.7 gives the phase information needed for the manuscript's examples.

7.9.5 Derivation of the period factor

Put

$$R_D = \prod_{r=1}^{d-1} \Gamma(r/d)^{\chi_D(r)}.$$

We first establish the precise normalization

$$\prod_{Q \in \mathcal{Q}_D} |\Delta(\tau_Q)| = \frac{A_D^6 R_D^6}{(2\pi d)^{6h}}. \tag{7.17}$$

This is the fundamental-discriminant case of the classical Lerch–Chowla–Selberg formula. For a modern primary treatment with all normalization factors, see Barquero-Sanchez et al. [84] and Cohen[85]. Here is a derivation that makes the powers of 2, π, and d checkable.

For $\operatorname{Re} s > 1$, define the full Epstein sum

$$Z(\tau, s) = \sum_{(m,n) \neq (0,0)} \frac{(\operatorname{Im} \tau)^s}{|m + n\tau|^{2s}}.$$

The class decomposition of the Dedekind zeta function gives

$$\sum_{Q \in \mathcal{Q}_D} Z(\tau_Q, s) = 2 \left(\frac{\sqrt{d}}{2} \right)^s \zeta(s) L(s, \chi_D). \tag{7.18}$$

Indeed $Q(m, -n) = a|m + n\tau_Q|^2$ (the sign change in n does not affect the full lattice sum), while $a \operatorname{Im} \tau_Q = \sqrt{d}/2$; the factor 2 is the number of units, since $D < -4$.

The Kronecker limit formula at zero reads

$$Z(\tau, 0) = -1, \qquad Z'(\tau, 0) = -2 \log\bigl(2\pi \sqrt{\operatorname{Im} \tau} |\eta(\tau)|^2\bigr).$$

Also $L(0, \chi_D) = h$, and the finite Hurwitz-zeta expression for L and Lerch's identity give

$$L'(0, \chi_D) = -h \log d + \log R_D.$$

Differentiate (7.18) at zero, using $\zeta(0) = -1/2$ and $\zeta'(0) = -\frac{1}{2} \log(2\pi)$. After collecting terms, one obtains

$$\prod_{Q} \sqrt{\operatorname{Im} \tau_Q} |\eta(\tau_Q)|^2 = (4\pi \sqrt{d})^{-h/2} R_D^{1/2}. \tag{7.19}$$

Finally, $\prod_Q \operatorname{Im} \tau_Q = (\sqrt{d}/2)^h / A_D$. Dividing (7.19) by the square root of this product and raising to the twelfth power proves (7.17).

Because χ_D is odd, the positive and negative character residues are interchanged by $r \mapsto d - r$. Euler reflection therefore gives

$$R_D = \frac{P_D^2}{\pi^{\varphi_d/2}} \prod_{\chi_D(r)=1} \sin(\pi r/d) = \frac{P_D^2 \sqrt{C_d}}{(2\pi)^{\varphi_d/2}}. \tag{7.20}$$

To verify the last equality, square the sine product: it becomes the product over all reduced residues, because paired sines are equal. The identity $|1 - e^{2\pi i r/d}| = 2 \sin(\pi r/d)$ and the cyclotomic polynomial evaluated at 1 give its square as $C_d / 2^{\varphi_d}$. Every sine is positive, fixing the square-root sign.

Proof of Theorem 7.18. Multiply (7.5) over the class representatives and take absolute values:

$$\prod_Q |E_w(\tau_Q)| = \left(\prod_Q |\Delta(\tau_Q)|\right)^{w/12} |\mathcal{R}_{D,w}|^{1/12}.$$

Substitute (7.17), then $2\zeta(w) = |B_w|(2\pi)^w/w!$ and (7.20). Directly collecting exponents gives (7.16). No logarithm is taken of E_w, so the same argument applies when a factor vanishes. □

7.9.6 A finite arithmetic recipe and useful low weights

The general formula is particularly simple in the first few weights:

w	$F_w(X)$	$\lvert\mathcal{R}_{D,w}\rvert^{1/12}$
4	X^4	$\lvert H_D(0)\rvert^{1/3}$
6	$(X-1728)^6$	$\lvert H_D(1728)\rvert^{1/2}$
8	X^8	$\lvert H_D(0)\rvert^{2/3}$
10	$X^4(X-1728)^6$	$\lvert H_D(0)\rvert^{1/3}\lvert H_D(1728)\rvert^{1/2}$
12	$(X-432000/691)^{12}$	$\lvert H_D(432000/691)\rvert$

For the last row, use $E_{12} = (441E_4^3 + 250E_6^2)/691 = \Delta(j - 432000/691)$. This table is an exact algebraic calculation. In particular, the prime factors of a class product are no longer guessed from a ratio: at weights four and six they are determined by the factorizations of $H_D(0)$ and $H_D(1728)$, together with the explicitly displayed elementary factors.

Specializing the theorem to weight four gives

$$\prod_Q |G_4(\tau_Q)| = \frac{|H_D(0)|^{1/3} A_D^2 C_d}{180^h d^{2h} 2^{\varphi_d} \pi^{\varphi_d - 2h}} P_D^4. \tag{7.21}$$

At weight six it gives

$$\prod_Q |G_6(\tau_Q)| = \frac{A_D^3 C_d^{3/2} |H_D(1728)|^{1/2}}{30240^h d^{3h}} (2\pi)^{3h - 3\varphi_d/2} P_D^6. \tag{7.22}$$

7.9.7 The phase and the exceptional class-pair average

For a form strictly inside the reduced domain with $b \neq 0$, the form with $-b$ also occurs. Their G_w values are complex conjugates, so their product is positive unless a factor is zero. Boundary forms with $b = 0$ or $b = a < c$ give real G_4 and G_6, and these values are positive. For G_4 this follows from its positive value high on each boundary ray, continuity, and the fact that E_4 has no zero there except the hexagonal endpoint; for G_6 the same argument uses the unique elliptic zero at i. The excluded discriminants $-3, -4$ are exactly the exceptional endpoints. These are the usual zero-divisor consequences of the valence formula.

The remaining boundary type is $a = c$ and $0 < b < a$. Then $|\tau_Q| = 1$. Modularity and conjugation imply $\overline{E_4(\tau_Q)} = \tau_Q^4 E_4(\tau_Q)$. Following the unit arc from i and using the absence of an interior zero fixes the sign and yields

$$G_4(\tau_Q) = -\tau_Q^{-2} |G_4(\tau_Q)| = \left(\frac{d - b^2}{4a^2} - \frac{ib\sqrt{d}}{2a^2}\right) |G_4(\tau_Q)|. \tag{7.23}$$

Thus an unqualified equality between a raw class product and its positive gamma norm is not valid for arbitrary discriminants. Recent detailed work on the analogous eta-phase issue is Cohen[85]; the elementary modular calculation above suffices for the present cases.

For $D=-15$ the nonprincipal form is $(2,1,2)$. Consequently

$$G_4((-1+i\sqrt{15})/4)=\frac{7-i\sqrt{15}}{8}|G_4((-1+i\sqrt{15})/4)|.$$

The manuscript uses its real part, which is $7/8$ of the absolute value. This explains its factor 7 exactly; averaging a conjugate pair must not be confused with taking one class representative in a norm.

7.9.8 Proofs of the manuscript's five class products

The needed class-polynomial constants and leading-coefficient products are as follows. The supplied interval certificate proves these are the exact Hilbert class polynomials, independently of numerical recognition.

D	h	A_D	$H_D(0)$	$H_D(1728)$
-15	2	2	$-3^6 5^3 11^3$	$3^6 7^4 11^2$
-20	2	2	$-2^{12} 5^3 11^3$	$-2^{16} 11^2 19^2$
-23	3	4	$5^9 11^3 17^3$	$7^6 11^4 19^2 23$
-39	4	12	$3^{15} 17^3 23^3 29^3$	$3^{12} 7^8 19^4 23^2$
-47	5	36	$5^{15} 11^6 23^3 29^3$	$11^6 19^4 23^4 31^2 43^2 47$

The signs and factorizations in this table are exact integer arithmetic; they are also recorded in the machine receipt.

Corollary 7.19 (The five recorded class products are proved)**.** *The product formulas in Theorems 7.7, 7.6, 7.8, 7.11 and 7.14 hold exactly, with the real-part projection specified at discriminant* -15.

Proof. For $D=-23,-39,-47$ there are no unit-arc boundary representatives. Interior pairs are conjugates and all remaining weight-four boundary values are positive, so the absolute norm is the raw product. Substitute the factorizations in the preceding table into (7.21); these give exactly the three displayed class products. For $D=-15$, the real-part factor $7/8$ in (7.23) gives the recorded factor 7.

For $D=-20$, the norm first gives

$$\prod_Q G_4(\tau_Q)=\frac{11P_{-20}^4}{2^{14}3^4 5^5\pi^4}.$$

The multiplication and reflection formulas give

$$\frac{\Gamma(1/20)\Gamma(9/20)}{\Gamma(3/20)\Gamma(7/20)}=4\,5^{1/4}\sin(3\pi/20)\sin(7\pi/20)=5^{1/4}\varphi,$$

where $\varphi=(1+\sqrt{5})/2$. Indeed apply the fivefold multiplication formula at $1/20$ and reflect the factors at $13/20$ and $17/20$. Thus $P_{-20}=5^{1/4}\varphi\Gamma(3/20)^2\Gamma(7/20)^2$; substitution proves the remaining displayed formula. No period independence or numerical recognition is used in this deduction. □

7.9.9 Three explicit weight-six extensions

The same exact class data now produce formulas that were not displayed in the baseline manuscript:

$$\prod_{Q\in\mathcal{Q}_{-23}} G_6(\tau_Q) = \frac{11^2\cdot 19\,P_{-23}^6}{2^{33}3^9 5^3 23^7\pi^{24}}, \tag{7.24}$$

$$\prod_{Q\in\mathcal{Q}_{-39}} G_6(\tau_Q) = \frac{19^2\cdot 23\,P_{-39}^6}{2^{38}3^{15}5^4 13^{12}\pi^{24}}, \tag{7.25}$$

$$\prod_{Q\in\mathcal{Q}_{-47}} G_6(\tau_Q) = \frac{11^3 19^2 23^2\cdot 31\cdot 43\,P_{-47}^6}{2^{73}3^9 5^5 7^5 47^{13}\pi^{54}}. \tag{7.26}$$

They follow by inserting the factorizations of $H_D(1728)$ in (7.22); positivity was already established. For example, at $D=-23$ the square root is $7^3 11^2 19\sqrt{23}$, and $C_{23}^{3/2}=23^{3/2}$. Their product contributes 23^2, which changes 23^9 in the denominator to 23^7. This calculation is a useful check on the normalization.

Every factor here is a polygamma row sum. For example,

$$G_6(\tau) = 2\zeta(6) + \frac{1}{60}\sum_{n\ge 1}\left[\psi^{(5)}(n\tau)+\psi^{(5)}(1-n\tau)\right].$$

Consequently (7.24)–(7.26) are exact identities for finite products of exponentially convergent polygamma-row combinations. Weight twelve gives a further family whose entire arithmetic factor is the rational number $|H_D(432000/691)|$.

7.9.10 How the class-polynomial data are certified

The program `cm_norms.py` enumerates reduced forms by exact integer arithmetic. It then evaluates all singular moduli in rectangular complex intervals using 120 decimal digits of directed interval arithmetic and 70 terms of the Lambert series

$$E_4 = 1+240\sum_{n\ge1}\frac{n^3q^n}{1-q^n}, \qquad E_6 = 1-504\sum_{n\ge1}\frac{n^5q^n}{1-q^n}, \qquad j=\frac{1728E_4^3}{E_4^3-E_6^2}.$$

The interval engine encloses π, $\sqrt{d}$, and the complex exponential. Its output endpoints are finite binary rationals; the program converts them to exact Python fractions before every final containment comparison.

Reduced points satisfy $\operatorname{Im}\tau\ge\sqrt{3}/2$, hence $|q|\le e^{-\pi\sqrt{3}}<1/200$. This last strict inequality needs no floating-point assumption: use $\pi>314/100$, $\sqrt{3}>173/100$, their product greater than $27/5$, and the rational partial sum $\sum_{k=0}^{15}(27/5)^k/k!>200$. For $r=1/200$ and $p=3,5$, the omitted tail satisfies

$$\left|\sum_{n>N}\frac{n^pq^n}{1-q^n}\right| \le \frac{(N+1)^p r^{N+1}\mathcal{A}_p(r)}{(1-r)^{p+2}}, \tag{7.27}$$

where

$$\mathcal{A}_3(r)=1+4r+r^2, \qquad \mathcal{A}_5(r)=1+26r+66r^2+26r^3+r^4.$$

Indeed $n=N+1+j\le(N+1)(j+1)$, and $\sum_{j\ge0}(j+1)^pr^j=\mathcal{A}_p(r)/(1-r)^{p+1}$. The code adds the resulting error bound in both real and imaginary coordinates before forming j or multiplying the class polynomial.

Classical CM theory proves in advance that each coefficient of H_D is an integer. The interval product encloses every coefficient in an interval strictly contained in its asserted integer plus $(-1/2, 1/2)$; its imaginary interval contains zero. Thus the unique possible integer is proved, rather than merely recognized to many digits. The maximum real interval widths were

$D = -15$	$D = -20$	$D = -23$	$D = -39$	$D = -47$
$1.36 \cdot 10^{-109}$	$3.95 \cdot 10^{-108}$	$1.83 \cdot 10^{-103}$	$2.80 \cdot 10^{-95}$	$1.51 \cdot 10^{-88}$

The displayed bounds have been rounded upward. Exact interval endpoints, all polynomial coefficients, and the integer factorizations are saved in `cm_norms_receipt.json`.

An additional 25 numerical comparisons test the gamma norm formula at weights $4, 6, 8, 10, 12$ for each of the five discriminants. They have relative residuals below $1.1 \cdot 10^{-119}$ at 120 digits. These comparisons are regression checks: they are not interval proofs of the gamma identities, whose proof is Theorem 7.18. Similarly, the certified-polynomial computation is a numerical proof conditional on the standard CM integrality theorem and on the interval library's directed-rounding implementation; it is not a Lean formalization. This evidence boundary is explicit and reproducible.

7.9.11 Extension to arbitrary quadratic orders

There is also a direct version for $D = D_0 f^2$, where D_0 is fundamental, including the unit exceptions $D_0 = -3, -4$. Let $h = h(D)$ and $d = |D|$, and let $\mathrm{CS}(D_0)$ be Cohen's normalized gamma quotient

$$\mathrm{CS}(D_0) = \left(\prod_{r=1}^{|D_0|-1} \Gamma(r/|D_0|)^{\chi_{D_0}(r)} \right)^{w(D_0)/(2h(D_0))}.$$

Define the explicit conductor factor

$$c(D_0, f) = \prod_{p \mid f} p^{e(p)}, \qquad e(p) = \frac{(1 - p^{-v_p(f)})(1 - \chi_{D_0}(p))}{(1 - 1/p)(p - \chi_{D_0}(p))}.$$

The order form of Lerch–Chowla–Selberg[84, 85] gives the following immediate extension of the proof:

$$\prod_{Q \in \mathcal{Q}_D} |G_w(\tau_Q)| = (2\zeta(w))^h A_D^{w/2} \left(\frac{c(D_0, f)\mathrm{CS}(D_0)}{2\pi d} \right)^{hw/2} |\operatorname{Res}(H_D, F_w)|^{1/12}. \tag{7.28}$$

Thus nonmaximal orders require an explicit conductor correction, not a new unidentified period basket. Setting $f = 1$ and $D < -4$ recovers the main theorem.

Corollary 7.20 (The four class-number-one row evaluations)**.** *The four formulas at $2i$ and $i\sqrt{2}$ in Theorem 7.5 hold exactly.*

Proof. The additional coefficient certificates give $H_{-8}(X) = X - 8000$ and $H_{-16}(X) = X - 287496$, each with one reduced class. At $D = -8$ insert $H_{-8}(0)$ and $H_{-8}(1728)$ into the fundamental formula; $C_8 = 2$ and $P_{-8} = \Gamma(1/8)\Gamma(3/8)$. This gives the two recorded coefficients. At $D = -16$ use (7.28) with $D_0 = -4$, $f = 2$, $c(-4, 2) = \sqrt{2}$ and

$$\mathrm{CS}(-4) = \left(\frac{\Gamma(1/4)}{\Gamma(3/4)} \right)^2 = \frac{\Gamma(1/4)^4}{2\pi^2}.$$

The weight-four and weight-six resultant roots are $287496^{1/3} = 66$ and $(287496 - 1728)^{1/2} = 378\sqrt{2}$. Substitution gives the stated factors $11/15360$ and $1/(2^{14}5)$. These principal imaginary-axis row sums are positive by the principal-point proof, so absolute values may be removed. The two new class polynomials are certified by directed intervals and CM integrality at 120 digits and 70 Lambert terms, with exact recorded endpoints. □

Proposition 7.21 (Two exact Weber degree statements)**.** *At discriminant -23, the phase-corrected cubic coordinate defined in Theorem 7.9 is unique and belongs to the same cubic field as its singular modulus. At discriminant -39, every cube root of a singular modulus has degree twelve over $\mathbb{Q}$, and none belongs to the Hilbert class field of degree eight over $\mathbb{Q}$.*

Proof. Exact polynomial arithmetic gives

$$\mathrm{Res}_Y(Y^3 + 155Y^2 + 650Y + 23375,\, Y^3 - X) = -H_{-23}(X).$$

The three cube images are therefore the three distinct singular moduli. For each one exactly one root of the cubic has that image. Since $[\mathbb{Q}(j_Q) : \mathbb{Q}] = 3$ and $j_Q = \xi_Q^3$, the cubic equation forces $\mathbb{Q}(\xi_Q) = \mathbb{Q}(j_Q)$. The quotient of ξ_Q by E_4/η^8 is necessarily a cube root of unity, giving the precise phase correction.

For $D = -39$, the integer polynomial $H_{-39}(Y^3)$ reduces modulo five to

$$f(Y) = Y^{12} + Y^9 + 4Y^6 + 4Y^3 + 3.$$

The independent finite-field certificate verifies

$$Y^{5^{12}} \equiv Y \pmod{f}, \qquad \gcd(f, Y^{5^6} - Y) = \gcd(f, Y^{5^4} - Y) = 1.$$

Rabin's criterion proves f irreducible over $\mathbf{F}_5$; its monic integer lift is thus irreducible over $\mathbb{Q}$. Every possible phase has cube j_Q, so it is a root of this degree-twelve polynomial. Such a field cannot be a subfield of the degree-eight Hilbert class field. The resultant, the modular witness and its independent standard-library Frobenius/gcd replay are retained in the CM verification receipts. □

7.9.12 Research directions and scope

The theorem suggests several concrete continuations.

1. Extend the exact polynomial F_w generator supplied here and combine it with certified class-polynomial software such as Sutherland's CRT methods [86]. This extends the product database to large weights and class numbers without integer-relation searches.

2. Determine signed, rather than absolute, products for all weights by tracking the phases of boundary representatives and the zero signs of E_w along the unit arc. The present phase proof covers the displayed weight-four and weight-six examples; a general norm alone does not settle this problem.

3. Use factorizations of $H_D(0)$ and $H_D(1728)$ to compare the prime support with arithmetic formulas for norms of singular-modulus differences. This gives an explanation for the prime patterns observed in the manuscript, beyond using those patterns to build a PSLQ basis.

4. Extend the certified computations to class-by-class algebraic factors and genus-wise products. A full class norm does not by itself prove class ratios. Section 7.10 supplies the additional genus-character argument for the recorded ratios at $D = -15, -20, -39$; extending it beyond these examples remains a natural direction.

5. Build a uniform certificate format containing the discriminant, reduced forms, exact H_D, exact F_w, resultant, conductor factor, phase convention, gamma residue set, and analytic tail bound. Each claimed identity then has a finite, reviewable provenance chain.

None of these norm computations proves linear or algebraic independence of individual gamma values or individual polygamma rows. Nor does it imply that arbitrary sums of raw positive-weight CM values are class invariants. The result is a precise product theorem with completely specified representative dependence.

7.10 Genus characters prove the recorded CM ratios

A class product and a class ratio contain different information. The preceding norm formula controls the former. A nontrivial genus character supplies the missing signed product and proves all three recorded weight-four ratios at discriminants $-15, -20, -39$. The same calculation gives a theorem for every even weight and three explicit weight-six ratios.

The analytic mechanism is classical: twist the Kronecker limit formula by a genus character and use Kronecker's factorization of its Hecke L-function. The character factorization and unit normalization are stated explicitly in Masri[89, eqs. (1.3)–(1.5)]. We give the normalization argument because confusing the ordinary and narrow real-quadratic class numbers, or overlooking the units at discriminants -3 and -4, changes the answer.

7.10.1 A genus ratio theorem in arbitrary even weight

Suppose that the fundamental discriminant $D < -4$ has a factorization

$$D = eD_-, \qquad e > 1, \qquad D_- < 0,$$

where e, D_- are coprime fundamental discriminants. Let $\varepsilon : \mathrm{Cl}(D) \longrightarrow \{1, -1\}$ be the corresponding nontrivial genus character. Concretely, if a form Q represents an integer n coprime to D, then $\varepsilon(Q) = \chi_e(n)$. This value is independent of the represented integer chosen. Retain the reduced representatives τ_Q and the normalizations of Section 7.9, and put

$$A_\varepsilon = \prod_{Q \in \mathcal{Q}_D} a_Q^{\varepsilon(Q)}, \qquad H_D^{\pm}(X) = \prod_{\varepsilon(Q)=\pm 1} (X - j(\tau_Q)), \tag{7.29}$$

$$\lambda_- = L(0, \chi_{D_-}) = \frac{2h(D_-)}{\omega(D_-)}, \qquad \omega(D_-) = |\mathcal{O}_{D_-}^\times|.$$

Here $h(D_-)$ is the imaginary-quadratic class number. Let $h(e)$ be the *ordinary* real-quadratic class number, and let $u_e > 1$ be the least unit greater than one in the real embedding of $\mathbb{Q}(\sqrt{e})$. Its norm is allowed to be -1. In particular,

$$\lambda_{-3} = \frac{1}{3}, \qquad \lambda_{-4} = \frac{1}{2};$$

replacing either of these values by its imaginary class number would give an incorrect result.

Both genus factors are monic of degree $h(D)/2$, with $H_D = H_D^+ H_D^-$. The two sign sets are unions of genera; in the three examples below they are the two genera themselves. Their coefficients lie in the ring of integers of $\mathbb{Q}(\sqrt{e})$. Indeed, the kernel of ε fixes the quadratic genus extension $\mathbb{Q}(\sqrt{D}, \sqrt{e})$ of $\mathbb{Q}(\sqrt{D})$; the two products are invariant under this kernel. Complex conjugation inverts ideal classes, and inversion preserves ε, so both products are real. Their coefficients therefore belong to the real quadratic subfield. The coefficients are algebraic integers because the singular moduli are algebraic integers.

Theorem 7.22 (Genus ratios at every even weight)**.** *For every even $w \geq 4$, all factors are nonzero by Theorem 7.17, and*

$$\prod_{Q \in \mathcal{Q}_D} |G_w(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^{w/2}\, u_e^{-wh(e)\lambda_-} \left| \frac{\mathrm{Res}(H_D^+, F_w)}{\mathrm{Res}(H_D^-, F_w)} \right|^{1/12}. \tag{7.30}$$

The polynomials F_w and the resultant convention are those of Lemma 7.16. All real roots on the right are positive roots. Every numerator and denominator is nonzero.

Proof. Write $d = |D|$ and

$$Z(\tau, s) = \sum_{(m,n)\neq(0,0)} \frac{(\operatorname{Im}\tau)^s}{|m + n\tau|^{2s}}.$$

The quadratic-form and ideal-class zeta correspondence, with two units because $D < -4$, gives

$$\sum_Q \varepsilon(Q) Z(\tau_Q, s) = 2\left(\frac{\sqrt{d}}{2}\right)^s L(s, \chi_e) L(s, \chi_{D_-}). \tag{7.31}$$

Here we used Kronecker's factorization $L_{\mathbb{Q}(\sqrt{D})}(s, \varepsilon) = L(s, \chi_e)L(s, \chi_{D_-})$. For completeness, at an unramified split prime the two quadratic characters have the same sign and give the two equal prime-ideal factors; at an inert prime their signs are opposite and their product is $(1 - p^{-2s})^{-1}$, the factor for the principal ideal (p). At a ramified prime, coprimality of e, D_- leaves precisely the factor of the unramified Dirichlet character. Thus the Euler products agree in $\operatorname{Re} s > 1$, and continuation gives the identity.

Since χ_e is nontrivial and even, $L(0, \chi_e) = 0$. Differentiate (7.31) at zero and use

$$Z'(\tau, 0) = -2\log\bigl(2\pi\sqrt{y}\,|\eta(\tau)|^2\bigr), \qquad y = \operatorname{Im}\tau.$$

The constant disappears because $\sum_Q \varepsilon(Q) = 0$. We obtain

$$\sum_Q \varepsilon(Q) \log\bigl(\sqrt{y_Q}|\eta(\tau_Q)|^2\bigr) = -\lambda_- L'(0, \chi_e). \tag{7.32}$$

Since $y_Q = \sqrt{d}/(2a_Q)$, its signed logarithmic sum is $\sum_Q \varepsilon(Q) \log y_Q = -\log A_\varepsilon$. Consequently

$$\prod_Q |\Delta(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^6 \exp\bigl(-12\lambda_- L'(0, \chi_e)\bigr). \tag{7.33}$$

To fix the real-quadratic normalization, the completed function

$$\Lambda_e(s) = (e/\pi)^{s/2}\Gamma(s/2)L(s, \chi_e)$$

satisfies $\Lambda_e(s) = \Lambda_e(1 - s)$. Its values at zero and one give $2L'(0, \chi_e) = \sqrt{e}\, L(1, \chi_e)$. The real-quadratic class number formula, in the ordinary-class-number and least-unit convention specified above, is

$$L(1, \chi_e) = \frac{2h(e)\log u_e}{\sqrt{e}}.$$

Thus $L'(0, \chi_e) = h(e)\log u_e$, and (7.33) becomes

$$\boxed{\prod_Q |\Delta(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^6 u_e^{-12h(e)\lambda_-}.} \tag{7.34}$$

Finally, apply $|E_w(\tau_Q)|^{12} = |\Delta(\tau_Q)|^w |F_w(j(\tau_Q))|$ to the signed product. The factors $2\zeta(w)$ cancel because the plus and minus character fibers have the same cardinality. The remaining products of $F_w(j)$ are exactly the stated resultants. Taking the positive twelfth root proves the formula. □

In particular, write $R_w(D) = \prod_Q |G_w(\tau_Q)|^{\varepsilon(Q)}$. The two smallest weights give the especially convenient identities

$$R_4(D)^3 = A_\varepsilon^6 u_e^{-12h(e)\lambda_-} \left| \frac{H_D^+(0)}{H_D^-(0)} \right|, \tag{7.35}$$

$$R_6(D)^2 = A_\varepsilon^6 u_e^{-12h(e)\lambda_-} \left| \frac{H_D^+(1728)}{H_D^-(1728)} \right|. \tag{7.36}$$

Genus theory identifies the field of the factors $H_D^\pm$; these equations additionally account for the periods of the raw modular forms. Claiming a quadratic raw ratio requires verifying the root extraction. We now perform that step exactly.

7.10.2 The three recorded ratios, now theorems

The required character and unit data are as follows. A plus or minus sign attached to a form denotes its genus-character value.

D	(e, D_-)	plus forms	A_ε	u_e	λ_-
-15	$(5,-3)$	$(1,1,4)$	$1/2$	$(1+\sqrt{5})/2$	$1/3$
-20	$(5,-4)$	$(1,0,5)$	$1/2$	$(1+\sqrt{5})/2$	$1/2$
-39	$(13,-3)$	$(1,1,10),(3,3,4)$	$3/4$	$(3+\sqrt{13})/2$	$1/3$

All the other reduced forms have minus sign. These signs can be checked by represented integers: at $D=-15$, the nonprincipal form represents 2 and $\chi_5(2) = -1$; at $D=-20$, the nonprincipal form represents 7 and $\chi_5(7) = -1$; at $D=-39$, the forms with $a=2$ have $\chi_{13}(2) = -1$, whereas $(3,3,4)$ represents 10 and $\chi_{13}(10) = 1$. The class numbers $h(5) = h(13) = 1$ follow from the real-quadratic Minkowski bound $\sqrt{e}/2 < 2$. The displayed units are fundamental: checking $x^2 - ey^2 = \pm 4$ below their sizes leaves no smaller positive unit.

The genus factors are explicitly

$$H_{-15}^\pm(X) = X + \frac{191025 \pm 85995\sqrt{5}}{2}, \tag{7.37}$$

$$H_{-20}^\pm(X) = X - 632000 \mp 282880\sqrt{5}, \tag{7.38}$$

$$H_{-39}^\pm(X) = X^2 + (165765798 \pm 45975573\sqrt{13})X + \frac{63399280527 \pm 17399806263\sqrt{13}}{2}. \tag{7.39}$$

Multiplication gives the certified H_D polynomials of Section 7.9. The correspondence between these quadratic-field factors and the displayed genera is also certified: for each CM root, evaluate the other factor in a directed interval enclosure and exclude zero. The attached receipt records the exact rational interval endpoints, so the assignment is not a numerical guess about which radical sign to choose.

Theorem 7.23 (Exact weight-four genus ratios)**.** *With the character data in the table,*

$$R_4(-15) = \frac{(4+\sqrt{5})^2}{44}, \tag{7.40}$$

$$R_4(-20) = \frac{29+12\sqrt{5}}{44}, \tag{7.41}$$

$$R_4(-39) = \frac{103299+4440\sqrt{13}}{181424}. \tag{7.42}$$

The first equation is an absolute-value ratio; the precise real-part version in the manuscript is

$$\frac{G_4((1+i\sqrt{15})/2)}{\operatorname{Re} G_4((1+i\sqrt{15})/4)} = \frac{2(4+\sqrt{5})^2}{77}. \tag{7.43}$$

For $D=-20$ and $D=-39$, the positive products represented by $R_4(D)$ are exactly the products and ratios in the manuscript, without an additional phase factor.

Proof. Substitute (7.37)–(7.39) and the table into (7.35). In each of the two quadratic fields, direct multiplication verifies that the cube of the claimed positive number is the right-hand side. For example, for $D=-20$ this is the explicit identity

$$\left(\frac{29+12\sqrt{5}}{44}\right)^3 = \frac{1}{64}\left(\frac{1+\sqrt{5}}{2}\right)^{-6}\left|\frac{-632000-282880\sqrt{5}}{-632000+282880\sqrt{5}}\right|.$$

The identical finite verification with exponents -4 and -4, and leading-coefficient ratios $1/2$ and $3/4$, proves the other two cases. The attached exact-arithmetic script verifies all three equalities in $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\sqrt{13})$ before any decimal evaluation. Positivity makes the cube root unique.

For $D=-15$, the phase calculation of Section 7.9.7 gives $\operatorname{Re} G_4((1+i\sqrt{15})/4) = \frac{7}{8}|G_4((1+i\sqrt{15})/4)|$. Multiplying (7.40) by $8/7$ proves (7.43); translation by one and reflection in the imaginary axis account for the manuscript's positive-real-part representatives. The phase discussion in the preceding section also shows that the numerator and denominator genus products at -20 and -39 are positive: boundary representatives contribute positive real G_4, and the two interior representatives at -39 form a conjugate pair. □

This supplies the missing proofs for the ratios originally displayed in the manuscript's Results `cm:res:d15`, `cm:res:d20`, and `cm:res:genus`. Combined with the proved total products, it also determines the individual absolute values at class number two, and the absolute products over each genus at class number four: if T_w is the full positive product, the two genus products are $\sqrt{T_w R_w}$ and $\sqrt{T_w/R_w}$.

7.10.3 Three explicit weight-six identities

The same genus polynomials have further consequences beyond the recorded weight-four data.

Theorem 7.24 (Weight-six genus ratios)**.** *With the same genera and representatives,*

$$R_6(-15) = \frac{29+12\sqrt{5}}{88}, \tag{7.44}$$

$$R_6(-20) = \frac{601+252\sqrt{5}}{1672}, \tag{7.45}$$

$$R_6(-39) = \frac{1323+324\sqrt{13}}{1472}. \tag{7.46}$$

The last two again hold for the corresponding raw genus products. At discriminant -15 *an explicit real-part identity is*

$$\frac{G_6((1+i\sqrt{15})/2)}{\operatorname{Re} G_6((1+i\sqrt{15})/4)} = \frac{2(29+12\sqrt{5})}{121}. \tag{7.47}$$

Proof. Use (7.36). Exact arithmetic in the relevant quadratic field shows that its right-hand sides are, respectively,

$$\frac{1561+696\sqrt{5}}{7744} = \left(\frac{29+12\sqrt{5}}{88}\right)^2,$$
$$\frac{678721+302904\sqrt{5}}{2795584} = \left(\frac{601+252\sqrt{5}}{1672}\right)^2,$$
$$\frac{3115017+857304\sqrt{13}}{2166784} = \left(\frac{1323+324\sqrt{13}}{1472}\right)^2.$$

The roots on the right are positive. The valence formula gives the only zero of E_6 in the standard fundamental domain at i; none of the CM representatives under consideration is this point. Along the boundary rays $\operatorname{Re}\tau = 0$ and $\operatorname{Re}\tau = -1/2$ above the relevant representatives, E_6 is real, has no zero, and tends to one at infinity, so its values are positive. Conjugate interior points again contribute a positive product. This proves the raw-product assertion at $-20, -39$.

On the unit arc $|\tau| = 1$ with $\pi/2 < \arg\tau \le 2\pi/3$, modularity and conjugation imply that $\tau^3 G_6(\tau)$ is real. It is positive at the hexagonal endpoint, where $\tau^3 = 1$ and $G_6 > 0$, and cannot change sign before i. Thus $G_6(\tau) = \tau^{-3}|G_6(\tau)|$ on this open arc. At $\tau = (-1+i\sqrt{15})/4$,

$$\tau^{-3} = \frac{11+3i\sqrt{15}}{16}, \qquad \operatorname{Re} G_6(\tau) = \frac{11}{16}|G_6(\tau)|.$$

Reflection preserves the real part. Multiplying (7.44) by $16/11$ proves (7.47). □

The quadratic norms provide useful exact checks:

$$N(R_4(-15)) = N(R_4(-20)) = \frac{1}{16}, \qquad N(R_4(-39)) = \frac{81}{256},$$

and

$$N(R_6(-15)) = N(R_6(-20)) = \frac{1}{64}, \qquad N(R_6(-39)) = \frac{729}{4096}.$$

These follow by conjugating $\sqrt{e} \mapsto -\sqrt{e}$ in the displayed exact formulas. The identities are explicit consequences of the classical genus-period mechanism; no assertion of historical priority for individual numerical evaluations is intended.

7.10.4 A fixed field contains all even-weight genus ratios

The low-weight identities imply a uniform field-degree theorem. For a fixed genus character, increasing the weight introduces no additional algebraic field extension beyond the two seed ratios.

Theorem 7.25 (Uniform algebraic degree of genus ratios)**.** *Under the assumptions of Theorem 7.22, put* $K = \mathbb{Q}(\sqrt{e})$ *and*

$$\mathcal{K}_{D,\varepsilon} = K(R_4(D), R_6(D)).$$

Then

$$[\mathcal{K}_{D,\varepsilon} : K] \le 6, \qquad [\mathcal{K}_{D,\varepsilon} : \mathbb{Q}] \le 12,$$

and every ratio $R_w(D)$ with even $w \ge 4$ belongs to this same field. The degree bound is independent of w and of the imaginary class number $h(D)$.

Proof. The genus period ratio

$$B_{D,\varepsilon} = A_\varepsilon^6 u_e^{-12h(e)\lambda_-}$$

belongs to K. Indeed,

$$12h(e)\lambda_- = \frac{24h(e)h(D_-)}{\omega(D_-)} \in \mathbb{Z},$$

because $\omega(D_-)$ is 2, 4, or 6. Equations (7.35) and (7.36) therefore give $R_4(D)^3 \in K$ and $R_6(D)^2 \in K$. The seed ratios are nonzero: E_4 vanishes only at the elliptic orbit of discriminant -3, and E_6 only at that of discriminant -4, whereas $D < -4$ is fundamental. Adjoining the two positive seed ratios has degree at most $3 \cdot 2 = 6$ over K.

To prove that this one field suffices, use the reduced modular polynomial of (7.7):

$$E_w = E_4^a E_6^b \Delta^m P_w(j), \qquad a \in \{0,1,2\}, \quad b \in \{0,1\}, \quad w = 4a + 6b + 12m, \tag{7.48}$$

where $m \ge 0$ and $P_w \in \mathbb{Q}[X]$ is monic of degree m. The algebraic reason is short: the weight condition forces every monomial of $E_w \in \mathbb{Q}[E_4, E_6]$ to have exponents $a+3r, b+2s$ with $r+s = m$. Substituting $E_4^3 = \Delta j$ and $E_6^2 = \Delta(j - 1728)$ gives the displayed form; the constant term $E_w = 1 + O(q)$ makes P_w monic.

The signed absolute product gives the formula without fractional root extraction

$$R_w(D) = R_4(D)^a R_6(D)^b B_{D,\varepsilon}^m \left| \frac{\mathrm{Res}(H_D^+, P_w)}{\mathrm{Res}(H_D^-, P_w)} \right|. \tag{7.49}$$

The resultant ratio is in K, because both genus factors have coefficients there. Its absolute value is also in K, which is a real field. The CM nonvanishing theorem makes every resultant at these points nonzero, so the quotient is defined at every even weight. This proves the field assertion. □

Corollary 7.26 (Quadratic closure for the recorded discriminants)**.** *For $D = -15, -20, -39$, every genus ratio $R_w(D)$ with even $w \ge 4$ lies in $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{5})$, or $\mathbb{Q}(\sqrt{13})$, respectively. Each is computable in that quadratic field using the reduced polynomial P_w, whose degree is at most $\lfloor w/12 \rfloor$.*

Proof. Theorems 7.23 and 7.24 place both seed ratios in the indicated quadratic field. Formula (7.49) completes the proof. □

For example, $E_{14} = E_4^2 E_6$ gives $R_{14}(D) = R_4(D)^2 R_6(D)$. At weight 16, $a = m = 1, b = 0$ and $P_{16}(X) = X - 3456000/3617$, so

$$R_{16}(D) = R_4(D) B_{D,\varepsilon} \left| \frac{H_D^+(3456000/3617)}{H_D^-(3456000/3617)} \right|.$$

The equal degrees of the two genus factors cancel the signs in the linear-polynomial resultants. This formula, together with the three explicit pairs $H_D^\pm$, generates further exact quadratic identities using rational arithmetic alone.

7.10.5 Certificates and further directions

The companion `cm_genus.py` imports the directed-interval Lambert-series evaluator from `cm_norms.py`. It first certifies the integral Hilbert polynomials, then verifies the displayed genus factorizations by exact symbolic multiplication. For every reduced form it records a represented integer witnessing the genus character, and proves that the opposite genus factor excludes zero at its CM j-value. It next checks the cubic and square identities exactly. The receipt `cm_genus_receipt.json` separates these proof steps from six ordinary 120-digit Fourier-series regressions. A second checker, `verify_genus_arithmetic.py`, independently verifies all six radical equalities and their positive-root choices using only exact standard-library rational arithmetic in the quadratic fields. The numerical regressions have relative discrepancies below 3.1×10^{-120} and are independent consistency checks, not the justification for replacing the manuscript's numerical evidence by theorems.

The genus theorem answers the three recorded ratio questions while leaving several precise extensions open for this project.

1. Extend the genus-ratio certificate to nonmaximal quadratic orders, including the local conductor factors in the twisted Kronecker factorization. The untwisted conductor correction proved earlier cannot simply be copied into a signed genus product.

2. Beyond the three discriminants of Corollary 7.26, classify the discriminants and weights for which the raw ratio $R_w(D)$ itself belongs to $\mathbb{Q}(\sqrt{e})$, rather than merely having a controlled algebraic power in that field. Exact root extraction and representative phases must be part of the statement.

3. For class groups with several independent genus characters, combine their signed formulas by finite Fourier inversion to recover the product over every genus. This determines genus products but does not distinguish classes inside one genus.

4. Investigate twists by nonreal class characters. Their Hecke L-functions no longer reduce to a pair of quadratic Dirichlet L-functions; the resulting period data measure exactly the information absent from genus theory.

7.11 Norms, exact quadratic degree and oscillating genus ratios

The fixed-field formula gives sharper arithmetic information at multiples of twelve. Together with the lattice bounds used for CM nonvanishing, it also proves that the recorded quadratic fields are attained at every even weight. These are further consequences developed from the incoming genus and modular-polynomial results.

Theorem 7.27 (Quadratic norm at every weight divisible by twelve)**.** *Under the hypotheses of the genus theorem, put $K = \mathbb{Q}(\sqrt{e})$ and let σ be its nontrivial automorphism. For every integer $m \geq 1$,*

$$R_{12m}(D) \in K_{>0}, \qquad \sigma(R_{12m}(D)) = \frac{A_\varepsilon^{12m}}{R_{12m}(D)} > 0, \qquad N_{K/\mathbb{Q}}(R_{12m}(D)) = A_\varepsilon^{12m}.$$

Thus $A_\varepsilon^{-6m} R_{12m}(D)$ is a totally positive norm-one element of K. No integrality or unit assertion is implicit.

Proof. In (7.49), weight $12m$ has $a = b = 0$. Write $B = A_\varepsilon^6 u_e^{-t}$, where $t = 12h(e)\lambda_- = 24h(e)h(D_-)/\omega(D_-)$. This is an even integer: the unit counts $\omega(D_-) = 2, 4, 6$ give respectively factors $12, 6, 4$. Since $N(u_e) = \pm 1$,

$$\sigma(B) = A_\varepsilon^6 u_e^t, \qquad N(B) = A_\varepsilon^{12}.$$

The two genus factors are exchanged by σ. Since P_{12m} has rational coefficients, their resultant quotient $x \in K^\times$ satisfies $\sigma(x) = 1/x$. Both real embeddings of x have the same sign, so $\sigma(|x|) = 1/|x|$. This last conclusion uses the reciprocal identity; absolute value does not generally commute with arbitrary field embeddings. Now $R_{12m} = B^m|x|$, and multiplication proves all statements. The CM nonvanishing theorem makes every quotient nonzero. □

Theorem 7.28 (Exact quadratic degree at all even weights)**.** *For $D = -15, -20, -39$ and every even $w \geq 4$, respectively,*

$$R_w(D) \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}, \quad R_w(D) \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}, \quad R_w(D) \in \mathbb{Q}(\sqrt{13}) \setminus \mathbb{Q}.$$

Each ratio is totally positive in its quadratic field, and

$$N(R_w(-15)) = N(R_w(-20)) = 2^{-w}, \qquad N(R_w(-39)) = (3/4)^w.$$

Proof. Quadratic closure follows from the preceding seed theorem. The explicit seed ratios are totally positive and have norms A_ε^4 and A_ε^6. Formula (7.49) and the reciprocal resultant argument just used therefore give total positivity and norm A_ε^w at every even weight. If a ratio were rational, its positivity and norm would force $R_w = A_\varepsilon^{w/2}$.

We exclude that value by uniform rational lattice bounds. At $D = -15$, the absolute fourth-power lattice sum for $(2, 1, 2)$ is less than 8; at $D = -20$, that for $(2, 2, 3)$ is less than 5. Every nonzero vector in a reduced lattice has length at least one, so these are upper bounds for the absolute sums at all even $w \geq 4$. Combine them with the principal lower bound $7/20$ to get

$$R_w(-15) > 7/160 > 2^{-5}, \qquad R_w(-20) > 7/100 > 2^{-4}.$$

They exclude rationality for $w \geq 10$ and $w \geq 8$, respectively. At $D = -39$, the weight-twelve tails after removing the two axis unit vectors are less than $1/2, 1/2, 3/4$ at forms $(1, 1, 10), (2, 1, 5), (3, 3, 4)$. All remaining vectors have length greater than one. The same bounds hold at every $w \geq 12$, and give

$$R_w(-39) > \frac{(3/2)(5/4)}{(5/2)^2} = 3/10 > (3/4)^6 \geq (3/4)^{w/2}.$$

The conjugate pair of forms with $a = 2$ has equal absolute values, which explains the squared denominator.

These five lattice caps are certified by the finite rectangle $|n| \leq 10$, $|m| \leq 20$ and the tail estimates from the nonvanishing proof. For a twelfth-power sum the layer-cake integral is $63\pi/(256A^{11})$; using $\pi < 4$ gives the n-tail $2/(11N^{11}) + 63/(320N^{10})$. The m-tail is $4N/[11(M - N/2)^{11}]$, with the additional axis tail $2/(11M^{11})$. The fourth-power cap at $(2, 1, 2)$ uses the rational imaginary lower bound $9/10$; all other caps use one. The exact fractions and comparisons are retained in `verification/CM-genus-extensions.json`.

The only smaller weights left are $4, 6, 8$, and where needed 10. The seed formulas at $4, 6$ have nonzero radical coefficients, and $R_8 = R_4^2$, $R_{10} = R_4 R_6$ have positive radical coefficients as well. Thus none is rational. This proves exact degree two at every stated weight. □

7.11.1 New explicit quadratic evaluations

The reduced polynomials give nine further formulas at weights $12, 16, 18$. For instance,

$$R_{12}(-15) = \frac{34829 + 15576\sqrt{5}}{1216}, \tag{7.50}$$

$$R_{16}(-15) = \frac{307761 + 137632\sqrt{5}}{511744}, \tag{7.51}$$

$$R_{18}(-15) = \frac{64494289 + 28842228\sqrt{5}}{193434112}. \tag{7.52}$$

For the other two discriminants the complete coefficients are recorded as rational pairs in `verification/CM-genus-extensions.json`. These are exact evaluations by the resultant recipe, rather than decimal fits. In particular the first has norm $1/4096$ even though its positive embedding is approximately 57.28; the other embedding is very small.

7.11.2 A quadratic sequence with a full interval of limit points

The large first value is part of a structural phenomenon. Let $\tau = (-1 + i\sqrt{15})/4 = e^{i\theta}$, and write $r_m = R_{12m}(-15)$.

Theorem 7.29 (The cluster set at discriminant -15)**.** *The finite limit points of r_m, as $m \to \infty$, are precisely $[1/2, \infty)$. Also*

$$\liminf_{m\to\infty} r_m = 1/2, \qquad \limsup_{m\to\infty} r_m = \infty, \qquad \sigma(r_m) = 2^{-12m}/r_m \longrightarrow 0.$$

Thus exact quadratic degree at every weight coexists with dense limiting behavior in one embedding and exponential decay in the other.

Proof. The principal lattice has only the unit vectors ± 1, and all other vectors have length greater than one. Its row sum tends to two. In the second lattice the unit vectors are exactly $\pm 1, \pm\tau$; the next length is $\sqrt{3/2}$. Absolute convergence at weight four therefore gives the additive estimate

$$G_{12m}(\tau) = 2 + 2\tau^{-12m} + O((3/2)^{-6m}), \qquad |G_{12m}(\tau)| = 4|\cos(6m\theta)| + O((3/2)^{-6m}).$$

The error is additive; it is not divided by a small cosine.

The angle θ/π is irrational. Otherwise τ would be a root of unity, making $\tau + \tau^{-1} = -1/2$ an algebraic integer, which is impossible for that rational number. Thus $6m\theta$ is dense modulo π. Along a subsequence with $|\cos(6m\theta)| \to c > 0$, $r_m \to 1/(2c)$, giving every finite limit in $[1/2, \infty)$. Along a subsequence with the cosine tending to zero, the denominator tends to zero while the numerator tends to two; CM nonvanishing makes each term defined, so $r_m \to \infty$. The upper bound $|G_{12m}(\tau)| \le 4 + o(1)$ excludes finite limit points below $1/2$. Finally the norm theorem gives the conjugate formula, and the positive lower bound on r_m proves its convergence to zero. □

7.12 The opaque components

The components *not* fixed by reflection-conjugation or by the even-row mechanism — $\operatorname{Im}\psi^{(1)}(i)$, $\operatorname{Re}\psi(i)$ beyond γ, $\operatorname{Im}\psi^{(1)}\left(\frac{1+i}{2}\right)$ — are *odd-row* lattice data: for example

$$\operatorname{Im}\psi^{(1)}(i) = -2\sum_{k\ge 1} \frac{k}{(k^2+1)^2}$$

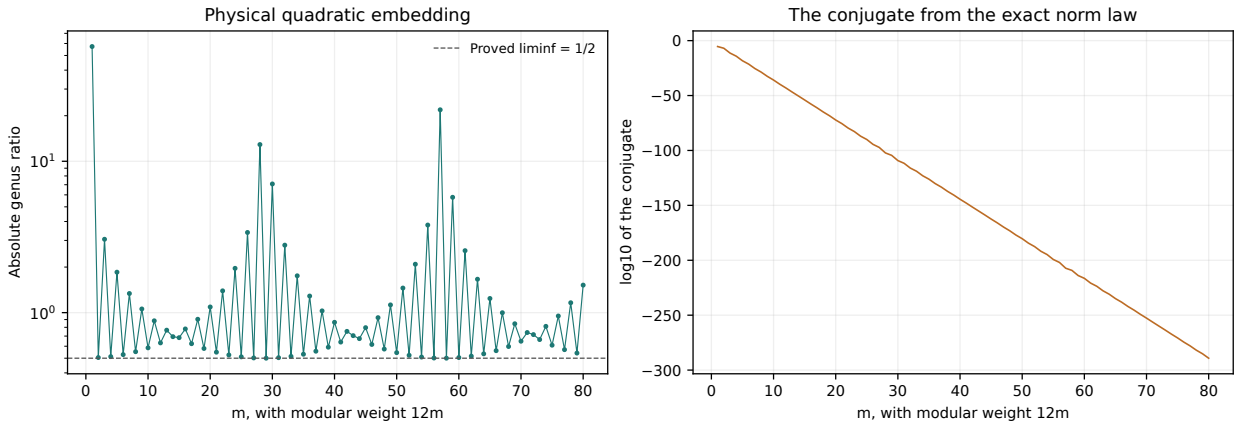


Figure 7.1: Numerical illustrations of the proved genus-ratio behavior at $D = -15$. The physical embedding has large oscillations; its quadratic conjugate is computed from the exact norm law. Finite lattice sums supply the plotted values and explicit analytic tail budgets. The cluster theorem, not these finitely many points, proves the full interval of limit points.

is an odd-power analogue of the coth family, which the bilateral (even) cotangent expansion cannot reach. These are Eichler-integral/quasi-modular territory — the E_2-flavoured layer; compare $\sum_{k\geq 1} k/(e^{2\pi k}-1) = \frac{1}{24} - \frac{1}{8\pi}$, which is $E_2(i) = 3/\pi$ in Lambert-series clothing.

Search observation 7.30. A PSLQ basket for these constants against products drawn from

$$\{1,\ \pi,\ \coth\pi,\ \pi^2/\sinh^2\pi,\ \pi\coth\pi,\ \dots\}$$

found nothing at small coefficient heights — consistent with genuinely new constants. Recorded as an honest negative; the Lambert-series face ($\sum \sigma_5(n)q^n$ at $q = e^{-2\pi}$ and $q = -e^{-\pi\sqrt{3}}$) is the natural next probe surface.

7.13 Summation order, differentiation and certified tails

Absolute convergence made the preceding weights straightforward. Weight two needs a specified order of summation. Once that boundary is handled, Ramanujan's differential algebra turns the existing CM period seeds into all derivative evaluations, with explicit exponential tail estimates.

7.14 Regularized modular rows and all derivative evaluations

The modular draft obtains even-weight Eisenstein series by pairing polygamma values along lattice rows. Its weight-two question requires care: the paired rows converge rapidly, although the ungrouped lattice sum is not absolutely convergent. We first prove an explicit bound for every differentiated row. We then determine the change of summation order directly, before using any modular transformation law. Finally, Ramanujan's differential system evaluates all the resulting derivative rows at the square and hexagonal CM points.

Throughout this section, let

$$\tau \in \mathbb{H} := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}, \qquad q = e^{2\pi i\tau}, \qquad r = |q| = e^{-2\pi \operatorname{Im}\tau} < 1, \qquad \rho = \tau_{\mathrm{hex}} = \frac{1+i\sqrt{3}}{2}.$$

For even $w \geq 2$ and an integer $j \geq 0$, put

$$\mathcal{T}_{w,j}(\tau) = \sum_{n\geq 1} n^j \left[\psi^{(w-1+j)}(n\tau) + (-1)^j \psi^{(w-1+j)}(1-n\tau)\right]. \tag{7.53}$$

Convergence will follow from the next theorem. The sign on the reflected term is part of the definition; differentiating that term changes its sign at every step.

7.14.1 Rational row formulas and explicit tail bounds

The negative-index polylogarithms appearing below are elementary rational functions. For $d \geq 0$, define

$$R_d(t) = \mathrm{Li}_{-d}(t), \qquad R_0(t) = \frac{t}{1-t}, \qquad R_{d+1}(t) = t\frac{d}{dt}R_d(t).$$

These relations are identities of rational functions, since on $|t| < 1$ they follow by differentiating $R_d(t) = \sum_{\ell\geq 1} \ell^d t^\ell$. In particular,

$$R_1(t) = \frac{t}{(1-t)^2}, \qquad R_2(t) = \frac{t(1+t)}{(1-t)^3}, \qquad R_3(t) = \frac{t(1+4t+t^2)}{(1-t)^4}.$$

Theorem 7.31 (Rational rows and an explicit remainder)**.** *For w, j, τ as above,*

$$\psi^{(w-1+j)}(n\tau) + (-1)^j \psi^{(w-1+j)}(1-n\tau) = (2\pi\mathrm{i})^{w+j}\,\mathrm{Li}_{1-w-j}(q^n), \tag{7.54}$$

$$\mathcal{T}_{w,j}(\tau) = (2\pi\mathrm{i})^{w+j} \sum_{n\geq 1} n^j\,\mathrm{Li}_{1-w-j}(q^n). \tag{7.55}$$

The series converges normally on every compact subset of $\mathbb{H}$. If $\mathcal{T}_{w,j}^{[N]}$ is its sum through $n = N$, where $N \geq 0$, then

$$\left|\mathcal{T}_{w,j}(\tau) - \mathcal{T}_{w,j}^{[N]}(\tau)\right| \leq (2\pi)^{w+j}(N+1)^j \frac{\mathrm{Li}_{-j}(r)}{r}\,\mathrm{Li}_{1-w-j}\left(r^{N+1}\right). \tag{7.56}$$

Thus the bound contains only rational functions of r and r^{N+1}, together with an explicit power of π.

Proof. For $\mathrm{Im}\, z > 0$, splitting the absolutely convergent bilateral power sum into its nonnegative and negative indices gives

$$\sum_{m\in\mathbb{Z}} (m+z)^{-w} = \frac{\psi^{(w-1)}(z) + \psi^{(w-1)}(1-z)}{(w-1)!}. \tag{7.57}$$

Here evenness of w is essential. The classical cotangent expansion and its partial-fraction interpretation [49] read

$$\pi\cot(\pi z) = -\pi\mathrm{i} - 2\pi\mathrm{i}\sum_{\ell\geq 1} e^{2\pi\mathrm{i}\ell z}, \qquad \pi\cot(\pi z) = \lim_{M\to\infty} \sum_{m=-M}^{M} \frac{1}{m+z}.$$

Differentiate $w-1$ times. On compact subsets of the upper half-plane the exponential series and its derivatives converge uniformly. After the first derivative, the bilateral series and all its subsequent derivatives also converge normally. Since $(-1)^{w-1} = -1$, the two minus signs cancel and yield

$$\psi^{(w-1)}(z) + \psi^{(w-1)}(1-z) = (2\pi\mathrm{i})^w \sum_{\ell\geq 1} \ell^{w-1} e^{2\pi\mathrm{i}\ell z}.$$

Differentiating another j times with respect to z proves (7.54). Multiplication by n^j gives the summand in (7.55).

To prove the remainder bound, expand the rational polylogarithm in its nonnegative majorizing series. The absolute value of the omitted rows is at most

$$(2\pi)^{w+j}\sum_{\ell\geq 1}\ell^{w+j-1}\sum_{n>N}n^j r^{n\ell}. \tag{7.58}$$

Write $n = N+1+a$. For $a\geq 0$ and $\ell\geq 1$,

$$n^j \leq (N+1)^j(a+1)^j, \qquad r^{a\ell}\leq r^a.$$

Consequently

$$\sum_{n>N}n^j r^{n\ell} \leq (N+1)^j r^{(N+1)\ell}\sum_{a\geq 0}(a+1)^j r^a = (N+1)^j r^{(N+1)\ell}\frac{\mathrm{Li}_{-j}(r)}{r}.$$

Substitution in (7.58) proves (7.56). In particular, the majorant is finite. On a compact subset of $\mathbb{H}$, one has $r\leq r_0<1$, and each of its power series with nonnegative coefficients is bounded by its value at r_0. This proves normal convergence, for every fixed w, j, and justifies all subsequent termwise differentiations. □

For $j=0$, the tail formula simplifies to

$$\left|\mathcal{T}_{w,0}-\mathcal{T}_{w,0}^{[N]}\right| \leq \frac{(2\pi)^w}{1-r}\,\mathrm{Li}_{1-w}\left(r^{N+1}\right).$$

The case $w=2$, which will be particularly useful, is

$$\left|\mathcal{T}_{2,0}-\mathcal{T}_{2,0}^{[N]}\right| \leq \frac{4\pi^2 r^{N+1}}{(1-r)(1-r^{N+1})^2}. \tag{7.59}$$

Theorem 7.31 also proves

$$\frac{d^j}{d\tau^j}\mathcal{T}_{w,0}(\tau) = \mathcal{T}_{w,j}(\tau). \tag{7.60}$$

Computationally, (7.54) is preferable to evaluating its two polygamma terms separately. Their algebraic asymptotic contributions cancel, leaving the exponentially small paired row. The rational formula performs that cancellation exactly before numerical evaluation.

7.14.2 The weight-two ordering correction from telescoping

For $w\geq 4$, the ordinary absolutely convergent lattice series is

$$G_w(\tau) = \sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}}(m+n\tau)^{-w} = 2\zeta(w)+\frac{2}{(w-1)!}\mathcal{T}_{w,0}(\tau).$$

For $w=2$, specify the order explicitly:

$$G_2(\tau) = \sum_{n\in\mathbb{Z}}\left(\sum_{m\in\mathbb{Z}}{}'(m+n\tau)^{-2}\right) = \frac{\pi^2}{3}+2\mathcal{T}_{2,0}(\tau), \tag{7.61}$$

$$C_2(\tau) = \sum_{m\in\mathbb{Z}}\left(\sum_{n\in\mathbb{Z}}{}'(m+n\tau)^{-2}\right). \tag{7.62}$$

The prime omits only $(m,n)=(0,0)$. In the first expression, each inner row and the outer series of completed rows converge absolutely. The reverse expression exists as well: changing the names and signs of the one-dimensional indices gives

$$C_2(\tau)=\tau^{-2}G_2(-1/\tau). \tag{7.63}$$

This is a coordinate identity for the prescribed iterated sums, not an assumption of modularity. The series of absolute values over all lattice points diverges, so an interchange of the two orders still requires a calculation.

Theorem 7.32 (Summation-order anomaly)**.** *For every $\tau\in\mathbb{H}$,*

$$C_2(\tau)=G_2(\tau)-\frac{2\pi\mathrm{i}}{\tau}. \tag{7.64}$$

Consequently

$$G_2(-1/\tau)=\tau^2G_2(\tau)-2\pi\mathrm{i}\tau,\qquad G_2(\tau+1)=G_2(\tau). \tag{7.65}$$

Proof. We prove the difference of the two orders by replacing their common part with an absolutely convergent series. Put

$$h(z)=\frac{1}{z}-\frac{1}{z+1},\qquad f(z)=\frac{1}{z^2}-h(z)=\frac{1}{z^2(z+1)},$$

and define

$$F(\tau)=\sum_{\substack{n\in\mathbb{Z}\setminus\{0\}\\ m\in\mathbb{Z}}} f(m+n\tau).$$

No pole occurs in this sum. The real-linear map $(m,n)\mapsto m+n\tau$ is invertible, so for some $c_\tau>0$, $|m+n\tau|\ge c_\tau\sqrt{m^2+n^2}$. Outside a finite set, $|f(z)|\le 2|z|^{-3}$; hence F converges absolutely. It can therefore be summed in either order.

For each fixed $n\neq 0$, the series $\sum_m h(m+n\tau)$ is absolutely convergent and its symmetric partial sum telescopes:

$$\sum_{m=-M}^{M} h(m+n\tau)=\frac{1}{-M+n\tau}-\frac{1}{M+1+n\tau}\longrightarrow 0.$$

Separating the row $n=0$ proves

$$G_2(\tau)=\frac{\pi^2}{3}+F(\tau). \tag{7.66}$$

For the reverse order, define for integer m

$$A(m)=\lim_{K\to\infty}\sum_{0<|n|\le K}\frac{1}{m+n\tau}.$$

This symmetric limit exists: pairing n and $-n$ gives an $O(n^{-2})$ summand. The cotangent partial-fraction formula gives

$$A(m)=\frac{\pi}{\tau}\cot\Big(\frac{\pi m}{\tau}\Big)-\frac{1}{m}\quad(m\neq 0),\qquad A(0)=0. \tag{7.67}$$

The value at zero is also the removable limit of the expression on the right. For each fixed m, the n-series of $h(m+n\tau)$ is absolutely convergent. Taking its symmetric partial sums shows

$$\sum_{n\neq 0}h(m+n\tau)=A(m)-A(m+1).$$

It follows that

$$C_2(\tau) - \frac{\pi^2}{3} - F(\tau) = \lim_{M\to\infty} \sum_{m=-M}^{M} \sum_{n\neq 0} h(m+n\tau)$$
$$= \lim_{M\to\infty} \big(A(-M) - A(M+1)\big).$$

The extraction of F here is legitimate by its absolute convergence; no rearrangement of the original double power sum has been made.

Since $\operatorname{Im}(1/\tau) < 0$,

$$\cot(\pi m/\tau) \longrightarrow \mathrm{i} \quad (m \to +\infty), \qquad \cot(\pi m/\tau) \longrightarrow -\mathrm{i} \quad (m \to -\infty).$$

Equation (7.67) therefore gives $A(-M) - A(M+1) \to -2\pi\mathrm{i}/\tau$. Combining this with (7.66) proves (7.64). Equation (7.63) now gives the first transformation in (7.65). Periodicity follows directly from the q-series in (7.55). □

The anomaly is classical; Romik and Scherer discuss it together with more general shape-dependent summation procedures [57]. The proof above fixes both its sign and the normalization needed for the draft's polygamma sums.

Corollary 7.33 (The missing weight-two CM rows)**.** *One has*

$$G_2(\mathrm{i}) = \pi, \qquad G_2(\rho) = \frac{2\pi}{\sqrt{3}}, \tag{7.68}$$

$$\mathcal{T}_{2,0}(\mathrm{i}) = \frac{\pi}{2} - \frac{\pi^2}{6}, \qquad \mathcal{T}_{2,0}(\rho) = \frac{\pi}{\sqrt{3}} - \frac{\pi^2}{6}. \tag{7.69}$$

Proof. At $\tau = \mathrm{i}$, the first transformation in (7.65) gives $G_2(\mathrm{i}) = -G_2(\mathrm{i}) + 2\pi$. For ρ, use $-1/\rho = \rho - 1$ and periodicity. The same transformation becomes

$$(\rho^2 - 1)G_2(\rho) = 2\pi\mathrm{i}\rho.$$

Since $\rho^2 - 1 = \mathrm{i}\sqrt{3}\rho$, this proves the second value in (7.68). Substitution into (7.61) proves (7.69). □

7.14.3 CM seeds and the complete differential tower

Normalize the Eisenstein series by

$$E_2 = \frac{3G_2}{\pi^2}, \qquad E_4 = \frac{G_4}{2\zeta(4)}, \qquad E_6 = \frac{G_6}{2\zeta(6)}.$$

The seed values, in precisely this normalization, are

$$(E_2(\mathrm{i}), E_4(\mathrm{i}), E_6(\mathrm{i})) = \left(\frac{3}{\pi}, \frac{3\Gamma(1/4)^8}{64\pi^6}, 0\right), \tag{7.70}$$

$$(E_2(\rho), E_4(\rho), E_6(\rho)) = \left(\frac{2\sqrt{3}}{\pi}, 0, \frac{27\Gamma(1/3)^{18}}{512\pi^{12}}\right). \tag{7.71}$$

We have already proved the weight-two entries. The zero entries follow by rotating the absolutely convergent lattices. Multiplication by i fixes the square lattice and sends its weight-six sum to

its negative; multiplication by ρ fixes the hexagonal lattice and multiplies its weight-four sum by $\rho^{-4} \neq 1$.

For completeness, the classical period integrals fix the nonzero entries without numerical identification. The standard inverse Weierstrass integral [49], with $g_2 = 60G_4$ and $g_3 = 140G_6$, gives positive real periods

$$p_{\mathrm{sq}} = 2\int_1^\infty \frac{dx}{\sqrt{4x^3-4x}} = 2\int_0^1 \frac{du}{\sqrt{1-u^4}} = \frac{1}{2}B\Big(\frac{1}{4},\frac{1}{2}\Big) = \frac{\Gamma(1/4)^2}{2\sqrt{2\pi}},$$
$$p_{\mathrm{hex}} = 2\int_1^\infty \frac{dx}{\sqrt{4x^3-4}} = \frac{1}{3}B\Big(\frac{1}{6},\frac{1}{2}\Big) = \frac{\Gamma(1/3)^3}{2^{4/3}\pi}.$$

The corresponding lattices are $p_{\mathrm{sq}}(\mathbb{Z}+\mathrm{i}\mathbb{Z})$ for $(g_2,g_3)=(4,0)$, and $p_{\mathrm{hex}}(\mathbb{Z}+\rho\mathbb{Z})$ for $(g_2,g_3)=(0,4)$. Their square and equianharmonic shapes follow from the rotations of the respective cubics; the real period integral fixes the scale and orientation. The first substitution above is $x = u^{-2}$; in the second beta integral use $t = x^{-3}$. Euler's reflection and duplication formulas give the displayed gamma evaluations. Scaling the lattice invariants consequently yields

$$G_4(\mathrm{i}) = \frac{p_{\mathrm{sq}}^4}{15} = \frac{\Gamma(1/4)^8}{960\pi^2}, \qquad G_6(\rho) = \frac{p_{\mathrm{hex}}^6}{35} = \frac{\Gamma(1/3)^{18}}{8960\pi^6}.$$

Using $2\zeta(4) = \pi^4/45$ and $2\zeta(6) = 2\pi^6/945$ proves the remaining entries of (7.70)–(7.71).

Ramanujan's classical differential identities [58] are

$$\theta E_2 = \frac{E_2^2 - E_4}{12}, \qquad \theta E_4 = \frac{E_2E_4 - E_6}{3}, \qquad \theta E_6 = \frac{E_2E_6 - E_4^2}{2}, \qquad \theta = \frac{1}{2\pi\mathrm{i}}\frac{d}{d\tau}. \tag{7.72}$$

Encode these identities by the rational derivation

$$\mathcal{V} = \frac{X^2-Y}{12}\frac{\partial}{\partial X} + \frac{XY-Z}{3}\frac{\partial}{\partial Y} + \frac{XZ-Y^2}{2}\frac{\partial}{\partial Z}, \qquad P_0 = X, \quad P_{j+1} = \mathcal{V}P_j. \tag{7.73}$$

For example,

$$P_1 = \frac{X^2-Y}{12},$$
$$P_2 = \frac{X^3-3XY+2Z}{72},$$
$$P_3 = \frac{X^4-6X^2Y+8XZ-3Y^2}{288}.$$

Theorem 7.34 (All differentiated weight-two rows)**.** *Set $\mathcal{S}_j(\tau) = \mathcal{T}_{2,j}(\tau)$. For every $\tau \in \mathbb{H}$,*

$$\mathcal{S}_0(\tau) = \frac{\pi^2}{6}\big(E_2(\tau) - 1\big), \tag{7.74}$$
$$\mathcal{S}_j(\tau) = \frac{\pi^2}{6}(2\pi\mathrm{i})^j P_j(E_2(\tau), E_4(\tau), E_6(\tau)), \qquad j \geq 1. \tag{7.75}$$

In particular, substitution of (7.70)–(7.71) evaluates every $\mathcal{S}_j(\mathrm{i})$ and $\mathcal{S}_j(\rho)$ by a finite polynomial calculation over $\mathbb{Q}$ and the indicated gamma periods.

Proof. Equation (7.61) and the definition of E_2 give (7.74). By the chain rule and (7.72), every polynomial $P \in \mathbb{Q}[X,Y,Z]$ satisfies

$$\theta\big(P(E_2,E_4,E_6)\big) = (\mathcal{V}P)(E_2,E_4,E_6).$$

Induction from $P_0 = X$ therefore gives $\theta^j E_2 = P_j(E_2, E_4, E_6)$. Differentiate (7.74) $j \geq 1$ times and use (7.60). The constant term disappears, and $d/d\tau = 2\pi\mathrm{i}\theta$ contributes the factor $(2\pi\mathrm{i})^j$. This proves (7.75); the CM specializations use the established seed values. □

For clarity, the first two differentiated evaluations are

$$\sum_{n\geq 1} n\left[\psi''(n\mathrm{i}) - \psi''(1-n\mathrm{i})\right] = \frac{\mathrm{i}\pi}{4} - \frac{\mathrm{i}\Gamma(1/4)^8}{768\pi^3}, \tag{7.76}$$

$$\sum_{n\geq 1} n\left[\psi''(n\rho) - \psi''(1-n\rho)\right] = \frac{\mathrm{i}\pi}{3}, \tag{7.77}$$

$$\sum_{n\geq 1} n^2\left[\psi'''(n\mathrm{i}) + \psi'''(1-n\mathrm{i})\right] = -\frac{\pi}{4} + \frac{\Gamma(1/4)^8}{256\pi^3}, \tag{7.78}$$

$$\sum_{n\geq 1} n^2\left[\psi'''(n\rho) + \psi'''(1-n\rho)\right] = -\frac{2\sqrt{3}\pi}{9} - \frac{\Gamma(1/3)^{18}}{1024\pi^8}. \tag{7.79}$$

These follow by substituting the seed values into P_1, P_2, so their derivation contains no integer-relation search.

Corollary 7.35 (Polynomial degree bounds for the CM row formulas)**.** *Put $A = \Gamma(1/4)^8/\pi^4$ and $B = \Gamma(1/3)^{18}/\pi^9$. For each $j \geq 1$, there are polynomials $U_j \in \mathbb{Q}[t]$ and $V_j \in \mathbb{Q}(\sqrt{3})[t]$ such that*

$$\frac{\mathcal{S}_j(\mathrm{i})}{\mathrm{i}^j\pi} = U_j(A), \qquad \frac{\mathcal{S}_j(\rho)}{\mathrm{i}^j\pi} = V_j(B),$$

with

$$\deg U_j \leq \left\lfloor \frac{j+1}{2} \right\rfloor, \qquad \deg V_j \leq \left\lfloor \frac{j+1}{3} \right\rfloor.$$

Proof. Give X, Y, Z weights $2, 4, 6$. Each term of $\mathcal{V}$ raises weight by two, so P_j is homogeneous of weight $2j+2$. At the square point, the surviving monomials $X^a Y^b$ have $a + 2b = j + 1$. On inserting the seeds into (7.75) and dividing by $\mathrm{i}^j\pi$, each becomes a rational multiple of A^b. At the hexagonal point, the surviving monomials $X^a Z^c$ satisfy $a + 3c = j + 1$, and the same substitution produces an element of $\mathbb{Q}(\sqrt{3})B^c$. The stated degree bounds follow. □

7.14.4 What this resolves and how it is verified

Corollary 7.33 resolves the weight-two row-sum question posed in the modular draft. Theorem 7.34 extends it to all polynomially weighted derivative rows, with the explicit error estimate of Theorem 7.31. These are evaluations of specified infinite aggregates. They do not evaluate an isolated component such as $\operatorname{Im}\psi'(\mathrm{i})$, nor do they prove that such a component is an Eichler integral. Any proposed identification of that isolated constant needs its own exact transform or functional equation.

The accompanying `code/modular_rows.py` constructs the rational functions R_d and the polynomials P_j by exact arithmetic. It checks individual rows against a separate complex-polygamma implementation, compares the CM formulas for $j = 0, \ldots, 4$ with their rapidly convergent row sums, and checks representative tails. The output `data/modular_rows.json` records the exact coefficients, analytic truncation bounds, and numerical residuals. Its decimal calculations use ordinary high-precision arithmetic; they are numerical checks, not outward-rounded interval certificates. A machine enclosure can be obtained by evaluating the rational formulas and (7.56) with directed interval arithmetic. The identities themselves are proved above and do not depend on the numerical checks.

Chapter 8

Zeta jets, Stieltjes antiderivatives and negative polygamma

The gamma function is the first spectral derivative of Hurwitz zeta at zero. Further derivatives give a natural language for integrating Stieltjes functions. Integration in the Hurwitz parameter moves the spectral argument downward, creating the negative-order polygamma ladder. This chapter develops that common mechanism before tabulating its moments and rational values.

8.1 Below the digamma: $\psi^{(-2)}$ at rationals

One integration below the digamma sits $\psi^{(-2)}(z) = \int_0^z \log\Gamma(t)\,dt$. At denominators $3, 4, 6$, reflection and distribution reduce the integral to Clausen values, $\zeta'(-1)$ and logarithms. The odd character part is Clausen-explicit at general rational points, but the even part can retain Dirichlet derivatives. The three original examples below are correct; their earlier extrapolation to every denominator was too broad.

Result 8.1 (log-gamma integrals in the Clausen lattice).

$$\int_0^{1/4} \log\Gamma(t)\,dt = \frac{3}{32} + \frac{\log 2}{8} + \frac{\log\pi}{8} + \frac{G}{4\pi} - \frac{9}{8}\,\zeta'(-1),$$

$$\int_0^{1/3} \log\Gamma(t)\,dt = \frac{1}{9} + \frac{\log 2}{6} - \frac{\log 3}{72} + \frac{\log\pi}{6} + \frac{\mathrm{Cl}_2(2\pi/3)}{4\pi} - \frac{4}{3}\,\zeta'(-1),$$

$$\int_0^{1/6} \log\Gamma(t)\,dt = \frac{5}{72} + \frac{7\log 2}{72} + \frac{\log 3}{144} + \frac{\log\pi}{12} + \frac{\mathrm{Cl}_2(\pi/3)}{4\pi} - \frac{5}{6}\,\zeta'(-1).$$

The structural punchline: the quarter-point integral carries Catalan ($\mathrm{Cl}_2(\pi/2) = G$) and the third-point integral carries $\mathrm{Cl}_2(2\pi/3) = \frac{2}{3}\,\mathrm{Cl}_2(\pi/3)$, two-thirds of Gieseking's constant — the "level-3 Catalan" that had surfaced independently in the repository's Eisenstein-ladder work — now arriving from a completely different direction.

Search observation 8.2 (recorded on purpose). (i) The half-point probe $\int_0^{1/2} \log\Gamma$ initially found no relation — because $\log\Gamma(1/2) = \frac{1}{2}\log\pi$ made the PSLQ basis $\mathbb{Q}$-dependent, and PSLQ returned that degeneracy instead of the target. Remove known dependencies or search modulo their known relation space, and require a nonzero target coefficient. Unknown arithmetic independence is not a prerequisite for searching or proving a target identity. (ii) A $\psi^{(-3)}(1/4)$ probe via Cauchy's repeated-integration kernel produced only a large-coefficient candidate (heights $\sim 10^6$) with a weak

canary (10^{-46} at 100 dps) — *rejected* as a PSLQ false positive. The weight-3 level needs a properly prepared basis (presumably $\zeta'(-2)$-equivalents plus Barnes-G-layer logarithms).

8.2 The general log-gamma integral and the fifth-point obstruction

Proposition 8.3 (An exact denominator-five correction)**.** *Let χ_5 be the quadratic character modulo 5, with values $(1,-1,-1,1)$ on residues $1,2,3,4$. Then*

$$\int_0^{1/5} \log\Gamma(t)\,dt = \frac{2}{25} + \frac{\log(2\pi)}{10} + \frac{\mathrm{Cl}_2(2\pi/5)}{4\pi} - \frac{6}{5}\zeta'(-1) + \frac{1}{20}L'(-1,\chi_5) - \frac{29}{1200}\log 5. \tag{8.1}$$

More generally, for $a > 0$,

$$\int_0^a \log\Gamma(t)\,dt = \zeta'(-1,a) - \zeta'(-1) + \frac{a-a^2}{2} + \frac{a}{2}\log(2\pi). \tag{8.2}$$

Proof. Differentiate $\partial_a\zeta(s,a) = -s\zeta(s+1,a)$ with respect to s at $s=-1$. Lerch's identity and $\zeta(0,a) = \frac{1}{2} - a$ give

$$\partial_a\zeta'(-1,a) = \zeta'(0,a) - \zeta(0,a) = \log\Gamma(a) - \tfrac{1}{2}\log(2\pi) - \tfrac{1}{2} + a.$$

The shift identity for Hurwitz zeta shows that $\zeta'(-1,a) \to \zeta'(-1)$ as $a \downarrow 0$, since the additional term is $-a\log a$. Integration proves (8.2). These are classical Hurwitz-zeta identities [49]; the integrated theory is also developed by Espinosa and Moll [54].

Put $Z_r = \zeta'(-1,r/5)$ and $Z = \zeta'(-1)$. The Fourier reflection formula at this weight is

$$\zeta'(-1,a) - \zeta'(-1,1-a) = \frac{\mathrm{Cl}_2(2\pi a)}{2\pi}, \qquad 0 < a < 1. \tag{8.3}$$

For completeness, subtract the two Hurwitz Fourier expansions

$$\zeta(1-s,a) - \zeta(1-s,1-a) = \frac{4\Gamma(s)}{(2\pi)^s}\sin\frac{\pi s}{2}\sum_{m\ge 1}\frac{\sin(2\pi m a)}{m^s}$$

and differentiate at $s=2$. Absolute local convergence of the series and its logarithmic derivative justifies the differentiation; the zero of the sine factor leaves precisely (8.3).

Differentiating distribution and the character decomposition at $s=-1$ gives

$$Z_1 + Z_2 + Z_3 + Z_4 = -\frac{4}{5}Z - \frac{\log 5}{60},$$
$$Z_1 - Z_2 - Z_3 + Z_4 = \frac{1}{5}\big(L'(-1,\chi_5) + \log 5\, L(-1,\chi_5)\big).$$

Here $L(-1,\chi_5) = -2/5$, obtained directly from $\zeta(-1,a) = -B_2(a)/2$. Combining these equations with $Z_1 - Z_4 = \mathrm{Cl}_2(2\pi/5)/(2\pi)$ yields

$$Z_1 = -\frac{1}{5}Z + \frac{1}{20}L'(-1,\chi_5) - \frac{29}{1200}\log 5 + \frac{\mathrm{Cl}_2(2\pi/5)}{4\pi}.$$

Insert this into (8.2) at $a = 1/5$. □

The correction concerns the scope of the available reduction laws. It does not prove that $L'(-1,\chi_5)$ is transcendental or independent of the other constants in (8.1). A precise replacement for the earlier assertion is: the odd reflection component is Clausen-explicit, while the even component requires the appropriate Dirichlet-L derivatives unless further relations have been proved. The general Hurwitz-jet ladder below agrees with this corrected scope.

8.3 Binet's second formula and Malmsten's integrals: the Lambert-kernel face

Binet's second formula

$$\log\Gamma(z) = \Big(z - \tfrac{1}{2}\Big)\log z - z + \tfrac{1}{2}\log(2\pi) + 2\int_0^\infty \frac{\arctan(t/z)}{e^{2\pi t} - 1}\, dt$$

is the integrated Abel–Plana formula with the *Lambert kernel* $1/(e^{2\pi t} - 1)$ — the same kernel whose moment sums are the Eisenstein q-series of the companion CM article. Malmsten/Vardi log-log integrals $\int_0^1 \ln\ln(1/x)\, R(x)\, dx$ are the *Mellin avatars* of Kummer's Fourier series: expanding the rational kernel R in a geometric series produces $L'(\chi, 1)$-type sums, and Kummer's formula converts odd-character $L'(1)$ into $\log\Gamma$ at rationals.

Result 8.4. Binet's formula replays exactly at $z = 1, \frac{1}{2}, \frac{1}{4}$ [historical experiment: residuals $\le 10^{-52}$], turning every Γ-store atom into a closed-form arctan–Lambert integral; e.g.

$$\int_0^\infty \frac{\arctan(4t)}{e^{2\pi t} - 1}\, dt = \frac{1}{2}\Big[\log\Gamma\big(\tfrac{1}{4}\big) - \tfrac{1}{2}\log 2 + \tfrac{1}{4} - \tfrac{1}{2}\log(2\pi)\Big].$$

Result 8.5 (Vardi's integral [5], re-derived blind). PSLQ over $\{\pi\gamma,\ \pi\log 2,\ \pi\log\pi,\ \pi\log\Gamma(1/4)\}$, with the γ atom *declined* (coefficient 0) [historical experiment: canary exact at 80 dps]:

$$\int_0^1 \frac{\ln\ln(1/x)}{1 + x^2}\, dx = \frac{\pi}{4}\,\ln\Big(\frac{4\pi^3}{\Gamma(1/4)^4}\Big).$$

The $q = 3$ Malmsten variant [historical experiment: canary 3.4×10^{-81}; γ declined again]:

$$\int_0^1 \frac{\ln\ln(1/x)}{1 + x + x^2}\, dx = \frac{\pi}{6\sqrt{3}}\,\ln\Big(\frac{2^8\pi^8}{3^3\,\Gamma(1/3)^{12}}\Big).$$

Both integrals carry exactly the harvested-Γ-store atoms: the log-log family is a systematic *integral-representation generator* for the identity stores — every odd Dirichlet character (each denominator q) yields one such evaluation through Kummer's formula, with the Γ-lattice reducer supplying the closed form mechanically.

8.4 The log-gamma moment ladder

Result 8.6 (Hurwitz ladder; classical (Whittaker–Watson [8])). For $\operatorname{Re} z > 0$, use the holomorphic branch of $\log\Gamma(z)$ that is real on the positive axis, and the principal logarithm of z. Then

$$\log\Gamma(z) = \Big(z - \tfrac{1}{2}\Big)\log z - z + \tfrac{1}{2}\log(2\pi) + \frac{1}{2}\sum_{n\ge 2}\frac{n-1}{n(n+1)}\,\zeta(n, z+1).$$

The series is absolutely and locally uniformly convergent in this half-plane. On $\operatorname{Re} z \ge \sigma > 0$, its Hurwitz terms are bounded by $\zeta(n, 1+\sigma)$, a geometrically decaying majorant. The domain matters: at $z = -1/2$, $\zeta(n, z+1) = (2^n - 1)\zeta(n)$ and the series terms do not tend to zero. Since $\zeta(n, z+1)$ is $\psi^{(n-1)}(z+1)$ up to factorials, this is a $\mathbb{Q}$-linear ladder between $\log\Gamma$ and the *entire polygamma tower at the same point* — a candidate certificate law binding the Γ and trigamma stores into one linear system.

Result 8.7 (second moment; classical (Espinosa–Moll [7])). With $L = \log(2\pi)$,

$$\int_0^1 \big[\log\Gamma(x)\big]^2 dx = \frac{\gamma^2}{12} + \frac{\pi^2}{48} + \frac{\gamma L}{6} + \frac{L^2}{3} - (\gamma + L)\frac{\zeta'(2)}{\pi^2} + \frac{\zeta''(2)}{2\pi^2}.$$

The quadratic-moment layer adds $\zeta''(2)$ to the atom inventory.

The numerical cubic moment originally suggested enlarging the constant basket. Its known Tornheim representation and the convergent certificate below settle that representation question; reduction to a specified smaller algebra remains separate.

8.5 Log-gamma moments: poles, late coefficients, and effective remainders

Let

$$M_n = \int_0^1 \log^n \Gamma(x)\, dx, \qquad L = \log(2\pi), \qquad A = \gamma + L.$$

The cubic moment is a test of whether an experiment has selected the right coordinate system. Bailey, D. Borwein and J. M. Borwein evaluated it in Tornheim derivatives [65, Theorem 6]. A smaller single-zeta basket can miss that structure; failure of such a search gives no independence theorem. We give the explicit representation and its convergence proof.

Similarly, the full fixed-order expansion of $M_n/n!$ into the exponential scales k^{-n} is Theorem 8.1 of Amdeberhan et al. [64]. The results proved below add an explicit late-coefficient asymptotic, identify the divergent character of that expansion, and give a uniform remainder bound when the number of retained terms grows linearly with n. These are refinements of a known expansion, not a claim to have discovered that expansion or the cubic Tornheim representation.

8.5.1 A compact form of the cubic identity

Define the holomorphic germ near $(1,1,1)$

$$T(r,s,t) = \sum_{m,n\ge 1} \frac{1}{m^r n^s (m+n)^t}.$$

A subscript j denotes partial differentiation in its jth argument; all unmarked Tornheim terms in the next formula are evaluated at $(1,1,1)$. The double sums and all the derivatives used here converge absolutely.

Theorem 8.8 (Cubic moment in one Tornheim differential operator)**.** *With $A = \gamma + L$,*

$$\begin{aligned} M_3 = {} & \frac{L^3}{8} + \frac{\pi^2 L}{32} + \frac{3\zeta(3)}{16} + \frac{3L}{4\pi^2}\big(A^2\zeta(2) - 2A\zeta'(2) + \zeta''(2)\big) \\ & + \frac{3}{8\pi^2}\left(A^2 T - 2AT_3 + 2T_{13} - T_{12}\right). \end{aligned} \tag{8.4}$$

This is an equivalent form of the known cubic evaluation in [65, Theorem 6].

Proof. Write the Kummer Fourier series as

$$\log\Gamma(x) = \frac{L}{2} + \frac{1}{2}C(x) + \frac{1}{\pi}S(x), \quad C(x) = \sum_{j\ge 1}\frac{\cos(2\pi j x)}{j}, \quad S(x) = \sum_{j\ge 1}\frac{A + \log j}{j}\sin(2\pi j x).$$

Reflection $x \mapsto 1-x$ makes C even and S odd. Parseval gives

$$\int_0^1 C^2 = \frac{1}{2}\zeta(2), \qquad \int_0^1 S^2 = \frac{1}{2}\bigl(A^2\zeta(2) - 2A\zeta'(2) + \zeta''(2)\bigr).$$

The cosine resonance gives $\int C^3 = 3\zeta(3)/2$. Equivalently, one can differentiate the beta integral for $\int_0^1 (2\sin \pi x)^u dx$ three times at zero and use $C = -\log(2\sin \pi x)$. For the only remaining cubic term, the elementary product rule for two sines and one cosine gives

$$\begin{aligned}\int_0^1 CS^2 &= \frac{1}{2}\sum_{m,n\ge 1} \frac{(A+\log m)(A+\log(m+n))}{mn(m+n)} \\ &\quad - \frac{1}{4}\sum_{m,n\ge 1} \frac{(A+\log m)(A+\log n)}{mn(m+n)} \\ &= \frac{1}{4}\left(A^2T - 2AT_3 + 2T_{13} - T_{12}\right).\end{aligned}$$

Expansion of the cube now proves the formula.

For the convergence justification, first put an Abel factor r^j in every Fourier series, $0 < r < 1$. All products can then be integrated termwise. The functions C and S belong to every finite $L^p(0,1)$, since the identities with $\log\Gamma(x)$ and $\log\Gamma(1-x)$ bound them by endpoint logarithms. Poisson convolution therefore converges to them in L^3, so the integrated cubic products converge. In the resulting resonance sums the logarithmic numerator is bounded by a constant times $(1+\log(m+n))^2$. Summing first over $m+n=q$ uses

$$\sum_{m=1}^{q-1} \frac{1}{m(q-m)q} = \frac{2H_{q-1}}{q^2},$$

which proves absolute convergence and legitimizes the Abel limit. □

The double-coordinate derivative in (8.4) can also be written as a positive sum. Put

$$\mathcal{R}_A = \sum_{m,n\ge 1} \frac{(A+\log(m+n))^2 - \log((m+n)/m)\log((m+n)/n)}{mn(m+n)}.$$

Symmetry in m,n proves $\mathcal{R}_A = A^2T - 2AT_3 + 2T_{13} - T_{12}$. The numerator is positive: after expansion it is $A^2 + 2A\log(m+n) + (\log m + \log n)\log(m+n) - \log m \log n$. It is bounded above by $(A+\log(m+n))^2$. Consequently, if $\mathcal{R}_{A,N}$ restricts $m+n \le N$ and $N \ge 3$ is an integer, then

$$0 < \mathcal{R}_A - \mathcal{R}_{A,N} \le \frac{2}{N}\bigl(P(\log N) + P'(\log N) + P''(\log N) + P'''(\log N)\bigr), \quad P(u) = (1+u)(A+u)^2. \tag{8.5}$$

Indeed $H_{q-1} \le 1 + \log q$, and $(1+\log x)(A+\log x)^2/x^2$ decreases for $x \ge 3$. Integration of the resulting tail gives (8.5). This bound is elementary and rigorous but converges only as $O(\log^3 N/N)$; it is useful as an independent sign and normalization check, not as the preferred high-precision algorithm.

8.5.2 A positive summand form and an explicit tail bound

The combination in (8.4) can be computed without cancellation between separate Tornheim constants. For $k = m + n$, define

$$R_A(m,n) = A^2 + 2A\log k + \log m \log k + \log n \log k - \log m \log n.$$

Since $A > 0$, $\log k \geq \log m, \log n \geq 0$, we have $R_A(m,n) > 0$. Writing $p = \log m, q = \log n, r = \log k$, one also has $r(p+q) - pq \leq r^2$: for fixed r, the expression is increasing separately in $p, q \in [0, r]$. Consequently

$$0 < R_A(m,n) \leq (A + \log k)^2.$$

Therefore the diagonal partial sums

$$C_N = \frac{3}{32} \sum_{m+n \leq N} \frac{1}{mn(m+n)} + \frac{3}{8\pi^2} \sum_{m+n \leq N} \frac{R_A(m,n)}{mn(m+n)}$$

increase to the centered cubic moment $M_3 - a^3 - 3aV$. Moreover,

$$0 < (M_3 - a^3 - 3aV) - C_N \leq \sum_{k>N} \frac{2(1+\log k)}{k^2} \left(\frac{3}{32} + \frac{3(A+\log k)^2}{8\pi^2} \right).$$

This gives a transparent computable enclosure, although its convergence is only $O(\log^3 N/N)$. An integral bound may be used once the explicit summand is decreasing; direct summation can handle the finite initial range. This positivity observation is an immediate corollary of the Fourier calculation, not a claim of a new classical-constant evaluation.

8.5.3 Exponentially convergent numerical reproduction

For $t > 0$, put

$$C(t) = \mathrm{Li}_1(e^{-t}) = -\log(1 - e^{-t}), \qquad F(t) = \partial_s \mathrm{Li}_s(e^{-t})\big|_{s=1}.$$

The usual Mellin representation of the Tornheim sum gives

$$T_{12} = \int_0^\infty F(t)^2\, dt, \quad T_{13} = \int_0^\infty F(t)C(t)(\log t + \gamma)\, dt, \quad T_3 = \int_0^\infty C(t)^2(\log t + \gamma)\, dt. \tag{8.6}$$

For $0 < t < 2\pi$, cancellation of the two poles in Jonquière's expansion produces

$$C(t) = -\log t + \sum_{k \geq 1} c_k t^k, \qquad c_k = \frac{(-1)^k \zeta(1-k)}{k!},$$

$$F(t) = -\alpha - \gamma \log t - \tfrac{1}{2}\log^2 t + \sum_{k \geq 1} b_k t^k, \qquad b_k = \frac{(-1)^k \zeta'(1-k)}{k!},$$

$$\alpha = \gamma_1 + \frac{\gamma^2}{2} + \frac{\pi^2}{12}.$$

Here the Stieltjes convention is $\zeta(1+z) = z^{-1} + \sum_{j \geq 0} (-1)^j \gamma_j z^j / j!$. For computational stability the coefficient formulas can be rewritten as

$$c_1 = \tfrac{1}{2}, \qquad b_1 = \tfrac{1}{2}\log(2\pi),$$

$$c_{2j} = \frac{(-1)^j \zeta(2j)}{j(2\pi)^{2j}}, \qquad b_{2j} = c_{2j}\left(\log(2\pi) - \psi(2j) - \frac{\zeta'(2j)}{\zeta(2j)} \right),$$

$$c_{2j+1} = 0 \quad (j \geq 1), \qquad b_{2j+1} = \frac{(-1)^{j+1}\pi\zeta(2j+1)}{(2j+1)(2\pi)^{2j+1}} \quad (j \geq 1).$$

The functional equation of the Riemann zeta function proves these identities. Elementary bounds $\zeta(k) < 2$, $|\zeta'(k)| < 2$, $H_{k-1} \le k-1$, $\gamma < 1$, $\log(2\pi) < 2$, and $2\pi > 6$ give, for $k \ge 2$,

$$|b_k| \le 12\,6^{-k}, \qquad |c_k| \le 2\,6^{-k}.$$

Thus after degree $K \ge 1$ their remainders on $0 < t \le 1$ are at most

$$\epsilon_b = \frac{12}{5}6^{-K}, \qquad \epsilon_c = \frac{2}{5}6^{-K}.$$

All integrals of the resulting polynomials in t and $\log t$ are evaluated using $\int_0^1 t^k \log^j t\,dt = (-1)^j j!/(k+1)^{j+1}$.

On $t \ge 1$, insert $C(t) = \sum_{n\ge1} e^{-nt}/n$ and $F(t) = -\sum_{n\ge1}(\log n)e^{-nt}/n$. The integral with the logarithmic factor uses

$$\int_1^\infty e^{-kt}(\log t + \gamma)\,dt = \frac{E_1(k) + \gamma e^{-k}}{k}.$$

Truncate this sum at $m + n \le N$. A common upper bound for the three omitted tails is

$$B_N = 4\sum_{k>N} ke^{-k} = \frac{4e^{-(N+1)}((N+1) - Ne^{-1})}{(1-e^{-1})^2}.$$

For the three integrals in the order T_{12}, T_{13}, T_3, safe total series-truncation bounds are respectively

$$\epsilon_b(12+\epsilon_b) + B_N, \qquad 5\epsilon_b + 15\epsilon_c + 2\epsilon_b\epsilon_c + B_N, \qquad 10\epsilon_c + 2\epsilon_c^2 + B_N.$$

For example the elementary estimates $C(t) \le -\log t + 1$ and $|F(t)| \le 4 - \log t + \frac{1}{2}\log^2 t$ on $(0,1]$ imply all the constants above. The sharper bound $\int_0^1 |F(t)|dt < 6$ follows directly from its coefficient expansion (one may use $0 < \alpha < 2$). An elementary verification of the latter constant bound is available: the defining limit for γ_1 and the total variation of $x \mapsto \log x/x$ on $[1,\infty)$ give $|\gamma_1| \le 2/e < 3/4$; combine this with $0 < \gamma < 3/5$ and $3 < \pi < \sqrt{12}$. This is an analytic truncation certificate; floating-point rounding must be handled separately to turn an implementation into interval arithmetic.

The independent implementation `verify_tornheim.py` evaluates the three partials and compares (8.4) against direct log-gamma quadrature. Its output explicitly labels the rounding qualification.

8.5.4 A sign check for an auxiliary identity

If the June 2012 preprint is used, an auxiliary sign should be checked before copying its Remark 3. Put

$$S_\gamma = \sum_{m,n\ge1} \frac{(\gamma + \log m)(\gamma + \log n)}{mn(m+n)}.$$

Directly expanding the numerator, with absolutely convergent sums, gives

$$T_{12} = S_\gamma + 2\gamma\big(\zeta'(2,1) + \zeta'(3)\big) - 2\gamma^2\zeta(3).$$

The coefficient of S_γ is positive. The displayed minus sign in equations (63) and (65) of the author-hosted June 2012 preprint would make their right side negative, although $T_{12} > 0$. This issue does not affect the cubic formula proved above. This note does not assert that the same typographical error occurs in the final journal typesetting.

8.5.5 Meromorphic moment generating function

For $\Re z < 1$, let

$$F(z) = \int_0^1 \Gamma(x)^z \, dx = \sum_{n \geq 0} \frac{M_n}{n!} z^n \quad (|z| < 1).$$

The power uses the real logarithm of $\Gamma(x) > 0$. Introduce the positive numbers

$$c_k = \frac{1}{k}[u^{k-1}]\Gamma(1-u)^k, \qquad k \geq 1. \tag{8.7}$$

Their first values are

$$c_1 = 1, \quad c_2 = \gamma, \quad c_3 = \frac{3\gamma^2 + \zeta(2)}{2}, \quad c_4 = \frac{8\gamma^3 + 6\gamma\zeta(2) + \zeta(3)}{3}.$$

These agree with the coefficients in [64, Theorem 8.1].

Proposition 8.9 (All moment poles and their residues)**.** *The function F extends meromorphically to the whole z-plane. Its only poles are simple poles at every positive integer, with*

$$\operatorname*{Res}_{z=k} F(z) = (-1)^k k c_k \neq 0.$$

Proof. Fix $0 < a < 1$ and expand

$$\Gamma(x)^z = x^{-z}\Gamma(1+x)^z = x^{-z} \sum_{j=0}^{J-1} b_j(z) x^j + x^{J-z} E_J(x, z).$$

The b_j are entire functions (in fact polynomials) of z, and E_J is uniformly bounded for $0 \leq x \leq a$ and z in compact subsets of the plane. Integration from zero to a gives

$$\sum_{j=0}^{J-1} \frac{b_j(z) a^{j+1-z}}{j+1-z} + \int_0^a x^{J-z} E_J(x, z) \, dx.$$

The second term is holomorphic for $\Re z < J + 1$; the integral from a to one is entire. Since J is arbitrary, these formulas give the claimed continuation. At $z = k$ the residue is $-b_{k-1}(k) = (-1)^k [u^{k-1}]\Gamma(1-u)^k$. Finally

$$\log \Gamma(1-u) = \gamma u + \sum_{j \geq 2} \frac{\zeta(j)}{j} u^j$$

has strictly positive coefficients, so (8.7) is positive. □

The pole at k contributes $(-1)^{k-1} c_k / k^n$ to the Taylor coefficient of F at zero. This explains the exponential scales in the known moment asymptotics, but an infinite partial-fraction sum cannot be inferred without controlling the growth of c_k. That growth is decisive.

8.5.6 Late coefficients and the inverse reciprocal gamma function

Set $\phi(u) = \Gamma(1-u)$ and let τ be the unique point in $(0,1)$ satisfying

$$-\tau\psi(1-\tau) = 1. \tag{8.8}$$

Equivalently $\psi(-\tau) = 0$. Define

$$\rho = \frac{\tau}{\Gamma(1-\tau)}, \qquad \alpha = -\log\rho, \qquad v = 1 + \tau^2\psi'(1-\tau), \qquad C = \frac{\tau}{\sqrt{2\pi v}}. \tag{8.9}$$

Numerically,

$$\begin{aligned}
\tau &= 0.50408300826445540925826930453330249895538518236858\ldots,\\
\rho &= 0.28211580900629804215379660062046372823230190757940\ldots,\\
\alpha &= 1.26543762211086561338203591116547399218544556554213\ldots,\\
v &= 2.27159994608893009911993485126846348661898954658806\ldots,\\
C &= 0.13342776130944528489602057053941782432703493711204\ldots.
\end{aligned}$$

These displayed approximations are numerical evaluations; the theorem uses the exact definitions (8.8)–(8.9).

Theorem 8.10 (Late-coefficient asymptotic)**.** *The coefficients* (8.7) *satisfy*

$$c_k = C\rho^{-k}k^{-3/2}\bigl(1 + O(k^{-1})\bigr). \tag{8.10}$$

In fact they have a full expansion in inverse powers of k. The power series $h(w) = \sum_{k\ge1} c_k w^k$ has radius of convergence ρ, converges absolutely on $|w| \le \rho$, and satisfies

$$h(w) = w\Gamma(1 - h(w)), \qquad h(\rho) = \tau.$$

For every fixed real n, the formal moment series $\sum_{k\ge1}(-1)^{k-1}c_k/k^n$ diverges, since its terms do not tend to zero.

Proof. Let $D = u\,d/du$. The function

$$\mu(u) = D\log\phi(u) = -u\psi(1-u) = \gamma u + \sum_{j\ge2}\zeta(j)u^j$$

is strictly increasing from zero to infinity on $(0,1)$. Hence (8.8) has exactly one solution. Moreover $\phi(u) > 1$ there, so $0 < \rho < 1$ and $\alpha > 0$.

Write $\phi(u) = \sum_{j\ge0} a_j u^j$ and introduce an integer-valued random variable X by

$$\Pr(X = j) = \frac{a_j\tau^j}{\phi(\tau)}.$$

All its probabilities are positive, in particular those at 0 and 1; it has lattice span one. Analyticity of ϕ on $|u| < 1$ gives an exponential moment. Its mean is $\mu(\tau) = 1$ and its variance is $D^2\log\phi(\tau) = 1 + \tau^2\psi'(1-\tau) = v > 0$. For independent copies and $S_k = X_1 + \cdots + X_k$, coefficient extraction gives

$$c_k = \frac{\tau\rho^{-k}}{k}\Pr(S_k = k-1). \tag{8.11}$$

Fourier inversion gives

$$\Pr(S_k = k-1) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\frac{\phi(\tau e^{it})}{\phi(\tau)}e^{-it}\right)^k e^{it}\,dt.$$

For $t \neq 0$ modulo 2π, the modulus of the bracket is strictly less than one, because the distribution has positive mass at 0 and 1. On a small interval around zero its logarithm is $-vt^2/2 + O(t^3)$. Split off the complement of that interval, whose contribution is exponentially small, and put $t = y/\sqrt{k}$ in the remaining integral. Taylor expansion, dominated by a Gaussian on a shrinking central interval, gives

$$\Pr(S_k = k-1) = \frac{1}{\sqrt{2\pi v k}}\bigl(1 + O(k^{-1})\bigr).$$

For a fully quantitative split, use $|t| \le k^{-2/5}$, the band $k^{-2/5} < |t| \le \delta$, and the fixed outer band. On the middle band the modulus is at most $e^{-ck^{1/5}}$; on the outer band it is $O(q^k)$ for some $q < 1$. The nominal $k^{-1/2}$ correction on the central band is odd and integrates to zero. Taylor expansion to arbitrary even order with its analytic remainder gives the full inverse-power expansion. This elementary saddle argument proves (8.10) and avoids any assumption about unexamined complex critical values of the reciprocal gamma function.

Lagrange inversion gives the stated functional equation for h near zero. Equation (8.10) proves that its radius is ρ and that it converges absolutely at the boundary. On $[0, \tau]$, the function $f(u) = u/\phi(u)$ increases from zero to ρ, with positive derivative before τ, since $f'(u) = (1 - \mu(u))/\phi(u)$. Its inverse is the positive real branch of h, hence continuity at the boundary gives $h(\rho) = \tau$. Finally (8.10) proves the stated divergence. □

For completeness, the first correction in (8.10) is

$$c_k = C\rho^{-k}k^{-3/2}\left(1 + \frac{\beta_1}{k} + O(k^{-2})\right),$$

where, with $\kappa_j = D^j \log\phi(\tau)$,

$$\begin{aligned}\beta_1 &= -\frac{1}{2v} + \frac{\kappa_3}{2v^2} + \frac{\kappa_4}{8v^2} - \frac{5\kappa_3^2}{24v^3}\\ &= 0.30196282878120524112458668271647811378573305248468\ldots.\end{aligned}$$

This follows by retaining the quartic and squared cubic terms, together with the factor e^{it}, in the Fourier integral.

8.5.7 An effective remainder with a growing number of terms

Theorem 8.11 (Explicit truncation bound)**.** *For integers $n \ge 1$ and $K \ge 1$ with $K\alpha < n$, put*

$$R_{n,K} = \frac{M_n}{n!} - \sum_{k=1}^{K}\frac{(-1)^{k-1}c_k}{k^n}.$$

Then, writing $\Gamma(n,z) = \int_z^\infty t^{n-1}e^{-t}\,dt$,

$$|R_{n,K}| \le \frac{\alpha^n}{n!}\left(1 + \frac{n\tau}{n - K\alpha}\right) + \frac{\tau\rho^{-(K+1)}}{(K+1)^n}\frac{\Gamma(n,(K+1)\alpha)}{\Gamma(n)}. \tag{8.12}$$

In particular, whenever $n \ge \max\{2, 4\alpha^2\}$, the choice $K = \lfloor n/\alpha - \sqrt{n} \rfloor$ is positive and gives

$$|R_{n,K}| \le \left(\frac{1}{\sqrt{2\pi n}} + \frac{\tau}{\alpha\sqrt{2\pi}} + \tau e^{2\alpha^2} \right) \left(\frac{e\alpha}{n} \right)^n. \tag{8.13}$$

In particular every integer $n \ge 7$ meets the displayed hypothesis. Indeed $\rho \ge 1/(2\sqrt{\pi})$ by evaluating $u/\Gamma(1-u)$ at $u = 1/2$, and therefore $\alpha \le \log(2\sqrt{\pi}) < 13/10$, so $4\alpha^2 < 169/25 < 7$.

Proof. The function $\log\Gamma(x)$ decreases strictly from infinity to zero on $(0, 1]$, because $\psi'(x) > 0$ and $\psi(1) = -\gamma < 0$. Let $x(y) \in (0, 1]$ be its inverse. The layer-cake formula, or integration by parts after this monotone change of variables, gives

$$\frac{M_n}{n!} = \frac{1}{\Gamma(n)} \int_0^\infty y^{n-1} x(y)\, dy. \tag{8.14}$$

For $y \ge \alpha$, Lagrange inversion and continuity on the convergence circle give

$$x(y) = -h(-e^{-y}) = \sum_{k \ge 1} (-1)^{k-1} c_k e^{-ky}.$$

The identity initially holds for sufficiently large y and extends along the negative real inverse branch. This branch cannot fail before $w = -\rho$: for $0 < x \le 1$, the derivative of $1/\Gamma(x)$ is $-\psi(x)/\Gamma(x) > 0$.

For $y \ge \alpha$, positivity and $\sum c_k \rho^k = \tau$ give

$$\left| x(y) - \sum_{k=1}^{K} (-1)^{k-1} c_k e^{-ky} \right| \le \tau \rho^{-(K+1)} e^{-(K+1)y}.$$

Integrating this on $[\alpha, \infty)$ produces the second term of (8.12). For the interval $[0, \alpha]$ use $0 < x(y) \le 1$ and, since $n > k\alpha$,

$$\int_0^\alpha y^{n-1} e^{-ky}\, dy \le \frac{\alpha^n e^{-k\alpha}}{n - k\alpha}.$$

To see this inequality, differentiate $y^n e^{-ky}$ and use $n - ky \ge n - k\alpha$. Sum over $1 \le k \le K$ and bound $\sum c_k e^{-k\alpha} \le \tau$ to obtain the first term.

For the stated choice of K, $n - K\alpha \ge \alpha\sqrt{n}$. Let $\theta = (K+1)\alpha/n$. Then $1 - \alpha/\sqrt{n} < \theta \le 1$ and the hypothesis ensures $\theta \ge 1/2$. Since

$$\theta - 1 - \log\theta \le 2(1-\theta)^2 \qquad (1/2 \le \theta \le 1),$$

we have

$$\rho^{-(K+1)} (K+1)^{-n} = \left(\frac{e\alpha}{n} \right)^n e^{n(\theta - 1 - \log\theta)} \le e^{2\alpha^2} \left(\frac{e\alpha}{n} \right)^n.$$

Finally use $\Gamma(n, z)/\Gamma(n) \le 1$ and the lower Stirling bound $n! \ge \sqrt{2\pi n}(n/e)^n$. □

For each fixed K, retaining one additional term in (8.12) recovers the usual asymptotic interpretation. The stronger content of (8.13) is that K may grow with n. The bound has the scale $(e\alpha/n)^n$. The late-coefficient theorem puts the least term near $(n + 3/2)/\alpha$ and gives its magnitude

$$C \left(\frac{e\alpha}{n + 3/2} \right)^{n+3/2} (1 + O(n^{-1})).$$

The rigorously proved remainder bound is within a polynomial factor of this scale. A least-term estimate alone would not prove a remainder bound; the inverse-function integral is what supplies that missing control.

The following values are numerical diagnostics from an independent integral in the variable $t = -\log x$, computed at 180 decimal digits. The rightmost column evaluates the explicit bound in (8.13); the inequality itself is proved above.

n	K	Observed $\lvert R_{n,K}\rvert$	Explicit upper bound
8	3	$3.1162 \cdot 10^{-5}$	$1.4837 \cdot 10^{-2}$
20	11	$2.1101 \cdot 10^{-18}$	$6.4881 \cdot 10^{-15}$
40	25	$2.8961 \cdot 10^{-46}$	$3.0208 \cdot 10^{-42}$
80	54	$1.8102 \cdot 10^{-113}$	$5.9717 \cdot 10^{-109}$

The displayed bound decimals have been rounded upward. They serve as a scale comparison; exact theorem constants are defined by (8.9). Full numerical data are in `moment_diagnostics.json`.

8.5.8 A 112-digit rational certificate for the cubic moment

The file `certify_cubic.py` implements a separate high-precision calculation using integer outward intervals. No floating-point special function is trusted. Put $a = 1/2$, $\ell = \log 2$ and

$$b_1 = \gamma, \qquad b_j = \frac{\zeta(j)}{j} \ (j \geq 2), \qquad B_M(x) = \sum_{j=1}^{M} b_j x^j.$$

On $0 < x \leq a$ we approximate $\log\Gamma(x)$ by $-\log x + B_M(-x)$; on the upper half of the interval, put $t = 1 - x$ and approximate $\log\Gamma(1-t)$ by $B_M(t)$. Both errors are at most

$$Dx^{M+1}, \qquad D = \frac{2\zeta(2)}{M+1},$$

with the corresponding variable x or t. This follows from the Taylor series and $\zeta(j) \leq \zeta(2)$ for $j \geq 2$. Define the exact elementary integrals

$$J_{j,p} = \int_0^{1/2} x^j (-\log x)^p dx = \frac{2^{-j-1} p!}{(j+1)^{p+1}} \sum_{r=0}^{p} \frac{((j+1)\ell)^r}{r!}.$$

If b_j^{*r} are the coefficients of B_M^r, the integrated polynomial approximation is

$$V_M = J_{0,3} + 3\sum_{j=1}^{M} (-1)^j b_j J_{j,2} + 3\sum_{j=2}^{2M} (-1)^j b_j^{*2} J_{j,1} + 2 \sum_{\substack{j \leq 3M \\ j \text{ even}}} b_j^{*3} J_{j,0}.$$

The last sum starts at $j = 4$. Since $0 \leq \log\Gamma(x) \leq -\log x$ on the lower half and $0 \leq \log\Gamma(1-t) \leq 1$ on the upper half,

$$\begin{aligned} |M_3 - V_M| \leq E_M := {} & 3D(J_{M+1,2} + J_{M+1,0}) \\ & + 3D^2(J_{2M+2,1} + J_{2M+2,0}) + 2D^3 J_{3M+3,0}. \end{aligned} \tag{8.15}$$

Here $\Gamma(1+x) \leq 1$ follows from log-convexity between 1 and 2; the upper-half bound follows from $\gamma < 1$, $\zeta(j) < 2$, and $\sum_{j \geq 2} (1/2)^j / j \leq 1/4$.

For reproducibility, the constant intervals come from the positive atanh series for $\log 2$, the Euler–Maclaurin formula for $\zeta(s)$ at $N = 128$, and the harmonic-number formula for γ. With $m = 96$, their respective analytic error bounds are

$$\frac{|B_{2m}|}{(2m)!}(s)_{2m-1}N^{-s-2m+1}, \qquad \frac{|B_{2m}|}{2mN^{2m}}.$$

The sharper first-omitted-term zeta bound uses the Stieltjes remainder property, not just an absolute periodic-Bernoulli estimate. Namely,

$$\frac{1}{1-e^{-t}} = \frac{1}{t} + \frac{1}{2} + 2t\sum_{r\geq 1}\frac{1}{t^2+4\pi^2r^2}.$$

Expanding each denominator geometrically through the term preceding B_{2m} gives a signed remainder bounded in absolute value by $|B_{2m}|t^{2m-1}/(2m)!$. Integrate this against $t^{s-1}e^{-Nt}/\Gamma(s)$ to obtain the stated bound. This argument also specifies exactly which Euler–Maclaurin terms are retained: $j = 1, \ldots, m-1$. The gamma bound follows by expanding the geometric factor in Binet's integral for the digamma function. For $s \geq 64$ the code instead brackets the tail of the direct positive sum by the two adjacent integrals. All rational operations round outwards at a fixed denominator 10^{130}. Exact Bernoulli numbers are provided by SymPy; no numerical Bernoulli approximation is involved.

At $M = 360$, the total interval has width less than $2.387 \cdot 10^{-113}$. In particular it certifies the displayed digits

$$M_3 = 5.74038880722947428001957168810246146296101300745487\ldots.$$

The full lower and upper rational endpoints, and the analytic error bound, are stored in `cubic_certificate.json`. An independent 150-digit mpmath quadrature lies inside this proved interval; that numerical check is a diagnostic, whereas (8.15) and the outward arithmetic supply the certificate.

8.6 Sharp truncation of the log-gamma moment expansion

Use the constants τ, ρ, C of the preceding late-coefficient theorem, and put $\Lambda = -\log\rho$. In this section write $b_k = c_k > 0$ for those positive coefficients, and $\widetilde{c}_k = (-1)^{k-1}b_k$. The real inverse $g(t)$ of $1/\Gamma(x)$ on $0 \leq x \leq 1$ satisfies $g(t) = -h(-t)$, where $h(w) = w\Gamma(1-h(w))$ is the earlier inverse germ. Thus $g(t) = \sum_{k\geq 1}\widetilde{c}_k t^k$ near zero. The moment integral is

$$m_n := M_n/n! = \frac{1}{\Gamma(n)}\int_0^\infty y^{n-1}g(e^{-y})\,dy.$$

The first relative coefficient d_1 below is the earlier β_1. Set $P = \psi'(-\tau)$, $Q = \psi''(-\tau)$, and $R = \psi^{(3)}(-\tau)$; equivalently $d_1 = R/(8P^2) - 5Q^2/(24P^3)$.

Lemma 8.12 (The dominant inverse singularity)**.** *The germ g continues to a slit disk of radius greater than ρ, with its only singularity on $|t| = \rho$ at $t = -\rho$, a square-root branch point.*

Proof. The positive inverse h has $h(\rho) = \tau$ and is absolutely convergent on that circle. Its positive coefficients of degrees one and two imply $|h(t)| < \tau$ at every other point of the circle. With $\Phi(u) = \Gamma(1-u)$, one then has $|t\Phi'(h(t))| \leq \rho\Phi'(|h(t)|) < \rho\Phi'(\tau) = 1$. The implicit-function theorem continues the inverse across those points. At $t = \rho$, local inversion at the simple critical point gives a convergent square-root expansion. Compactness supplies a slightly larger slit disk; replace t by $-t$ for g. This verifies the usual smooth implicit-function hypotheses directly.[79] □

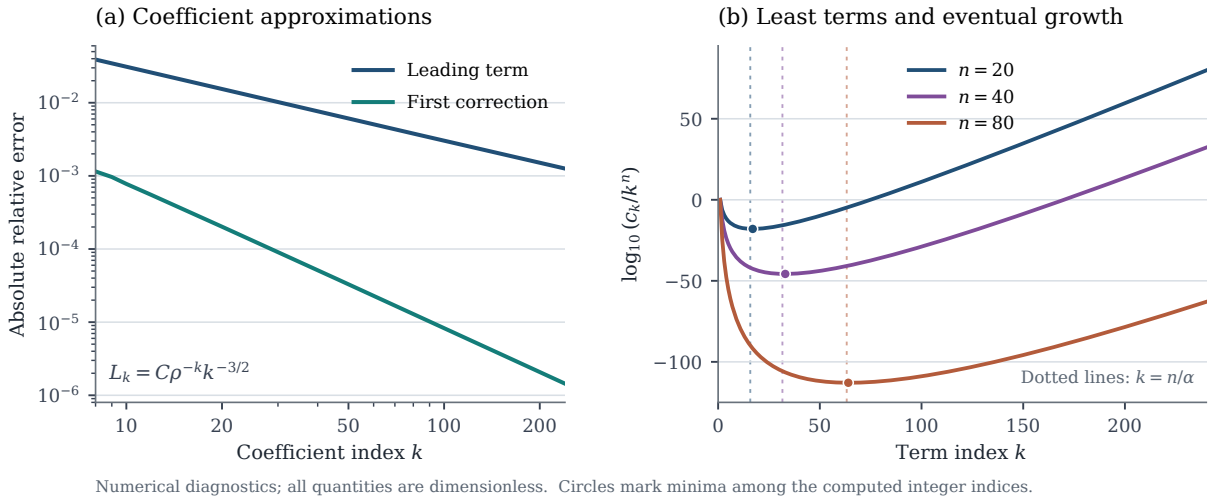


Figure 8.1: Numerical diagnostics for the proved coefficient and moment asymptotics. Left: absolute relative errors of $C\rho^{-k}k^{-3/2}$ and $C\rho^{-k}k^{-3/2}(1+\beta_1/k)$, using coefficients computed from their positive defining recurrence. Right: $\log_{10}(c_k/k^n)$ for $n = 20, 40, 80$. Dotted vertical lines mark $k = n/\alpha$; circles mark the sampled minima. All plotted quantities are dimensionless. The figure illustrates the theorems and supplies no independent error certificate.

8.6.1 A uniform Taylor-remainder lemma

The next statement is the key passage from fixed-order asymptotics to optimal truncation. It explicitly crosses the radius of convergence on the positive real axis; merely summing an infinite Taylor tail there would be invalid.

Lemma 8.13 (Uniform inverse remainder)**.** *For some $\delta > 0$, uniformly for $t \in [\rho e^{-\delta}, \rho e^{\delta}]$, put $r = t/\rho$. Then*

$$\frac{g(t) - \sum_{k=1}^{N-1} \widetilde{c}_k t^k}{\widetilde{c}_N t^N} = \frac{1}{1+r} + \frac{3r}{2N(1+r)^2} + O(N^{-2}). \tag{8.16}$$

On each of the remaining real intervals $0 < t \le \rho e^{-\delta}$ and $\rho e^{\delta} \le t \le 1$, the same normalized remainder is bounded uniformly in N sufficiently large.

Proof. The preceding theorem gives a slit disk of continuation for g, with its only singularity at $-\rho$. Choose its outer radius $R > \rho$, and then $\delta > 0$ so small that $\rho e^{\delta} < R$ and $\rho e^{\delta} < 1$. Cauchy's integral formula for $g(t)$, less its first $N-1$ Taylor terms, has kernel $t^N/(z^N(z-t))$. Deform its contour to the two banks of the negative cut and an outer contour. The coefficient $\widetilde{c}_N$ uses kernel z^{-N-1}. On the cut $z = -s$, the ratio of these kernels, after removing t^N, is

$$\frac{z}{z-t} = \frac{s}{s+t}.$$

The jump of g at $s = \rho$ is a nonzero constant times $(s-\rho)^{1/2}(1 + O(s-\rho))$, with a complete expansion. The coefficient integral is thus a Watson-type endpoint integral whose normalized variable $u = (s-\rho)/\rho$ has

$$\langle u \rangle = \frac{3}{2N} + O(N^{-2}), \qquad \langle u^2 \rangle = O(N^{-2}).$$

These formulas follow directly by comparing $\int_0^\epsilon u^{j+1/2}(1+u)^{-N-1}\,\mathrm{d}u$ with its beta integral extended to infinity. Terms outside a fixed small neighborhood of zero are exponentially smaller. Expanding

$$\frac{s}{s+t} = \frac{1}{1+r} + \frac{r}{(1+r)^2}u + O(u^2)$$

proves (8.16). The outer contour contributes $O(R^{-N})$ to the coefficient and $O((t/R)^N)$ to the remainder; after normalization these errors are exponentially small uniformly on the stated compact t-interval. This contour argument defines and estimates the actual analytic remainder on both sides of $t = \rho$.

It remains to bound the two outer real intervals. The estimate $b_k \asymp \rho^{-k}k^{-3/2}$ implies that, below $\rho e^{-\delta}$, the absolutely convergent tail is bounded by a constant times $b_N t^N$. Above ρe^{δ}, use

$$\sum_{k<N} \frac{b_k t^k}{b_N t^N} \ll \sum_{k<N} (N/k)^{3/2}(\rho/t)^{N-k}.$$

Splitting at $k = N/2$ bounds this uniformly by a geometric series plus an exponentially small term times a polynomial in N. Furthermore $0 \le g(t) \le 1$, whereas $b_N t^N$ grows exponentially uniformly on this interval. This proves the last claim. □

8.6.2 Sharp optimal truncation

Theorem 8.14 (Half the first omitted term)**.** *Let n, N tend to infinity through positive integers with $\eta = N\Lambda - n$ in a fixed bounded interval. Then*

$$\boxed{m_n - \sum_{k=1}^{N-1} \frac{(-1)^{k-1}b_k}{k^n} = \frac{(-1)^{N-1}b_N}{N^n}\left(\frac{1}{2} + \frac{3-2\eta}{8N} + O(N^{-2})\right).} \tag{8.17}$$

The O-constant is uniform for bounded η. The ratio in parentheses has a complete asymptotic expansion in $1/N$, with coefficients computable from the local inverse expansion and gamma moments. In particular, taking the nearest integer to $(n+3/2)/\Lambda$ gives a rigorously justified least-term approximation. The error eventually has the sign of the first omitted term and is asymptotic to one half of that term.

Proof. Subtract the finite sum under (8.14), using

$$\frac{1}{(n-1)!}\int_0^\infty y^{n-1}e^{-ky}\,\mathrm{d}y = k^{-n}.$$

With Y having the gamma density $N^n y^{n-1}e^{-Ny}/(n-1)!$, the error divided by $\widetilde{c}_N/N^n$ is the expectation of the normalized inverse remainder at $t = e^{-Y}$.

The gamma distribution concentrates at Λ:

$$\mathbb{E}Y = \frac{n}{N} = \Lambda - \frac{\eta}{N}, \qquad \operatorname{Var} Y = \frac{n}{N^2}.$$

For every fixed small $\delta > 0$, its probability outside $[\Lambda - \delta, \Lambda + \delta]$ is $O(e^{-a_\delta N})$. This follows from its moment-generating function $(1 - u/N)^{-n}$ and Chernoff's inequality. The uniform outer bounds in Lemma 8.13 therefore make the outer contribution exponentially small.

On the middle interval, the leading kernel is

$$L(Y-\Lambda)=\frac{1}{1+e^{\Lambda-Y}},\qquad L(v)=\frac{1}{2}+\frac{v}{4}-\frac{v^3}{48}+O(v^4).$$

Writing $V=Y-\Lambda$, its centered gamma moments give

$$\mathbb{E}V=-\frac{\eta}{N},\qquad \mathbb{E}V^3=O(N^{-2}),\qquad \mathbb{E}V^4=O(N^{-2}),$$

uniformly for bounded η. Thus $\mathbb{E}L(V)=1/2-\eta/(4N)+O(N^{-2})$. It is essential here to use the signed third moment; an absolute cubic bound alone would lose half a power. The correction kernel in (8.16) is

$$\frac{3e^{-V}}{2N(1+e^{-V})^2}.$$

Its value at zero is $3/(8N)$, and its first derivative vanishes there, so its expectation is $3/(8N)+O(N^{-2})$. Adding these terms proves (8.17). The contour calculation and the Taylor/gamma-moment calculation can both be continued to arbitrary fixed order. Gamma central moments are polynomials in n and N^{-1}; their parity and degree give integer powers of N^{-1}. □

The theorem strengthens the classical expansion in two ways. It allows N to grow linearly with n, and it estimates the actual error after truncation. The least-term estimate alone would not justify either conclusion. No global sign assertion for every n, N is needed.

Corollary 8.15 (Second correction)**.** *Under the hypotheses of Theorem 8.14, the normalized error has the more precise expansion*

$$\frac{m_n-\sum_{k<N}\widetilde{c}_k/k^n}{\widetilde{c}_N/N^n}=\frac{1}{2}+\frac{3-2\eta}{8N}+\frac{1}{N^2}\left(\frac{d_1}{4}+\frac{\Lambda(6\eta-13)}{96}\right)+O(N^{-3}).\tag{8.18}$$

Proof. Here are the additional local coefficients, so that the assertion is independently reproducible. Let $v=B-\tau$, and use P,Q,R from (8.12). The characteristic inverse satisfies

$$\log\frac{B/\Gamma(1-B)}{\rho}=-\frac{P}{2}v^2+\frac{Q}{6}v^3-\frac{R}{24}v^4+O(v^5).$$

Putting $w=\sqrt{1-t/\rho}$ and $\alpha_*=\sqrt{2/P}$, inversion gives

$$B(t)=\tau-\alpha_* w+\frac{Q}{3P^2}w^2+\mu w^3+O(w^4),\qquad D:=\frac{\mu}{\alpha_*}=-\frac{1}{4}+\frac{R}{12P^2}-\frac{5Q^2}{36P^3}.$$

Thus the positive cut density for B is $(\alpha_*/\pi)u^{1/2}(1+Du+O(u^2))$, where $u=t/\rho-1$. Transfer gives $d_1=3/8+3D/2$, proving (8.12) as well. The normalized endpoint moments are

$$\langle u\rangle=\frac{3}{2N}+\frac{9/4+3D/2}{N^2}+O(N^{-3}),\qquad \langle u^2\rangle=\frac{15}{4N^2}+O(N^{-3}),\qquad \langle u^3\rangle=O(N^{-3}).$$

They follow from beta integrals and the displayed cut density. The coefficient of N^{-2} in the inverse remainder at $r=t/\rho$ is therefore

$$K_2(r)=\frac{r}{(1+r)^2}\left(\frac{9}{4}+\frac{3D}{2}\right)-\frac{15r}{4(1+r)^3},\qquad K_2(1)=d_1/4.$$

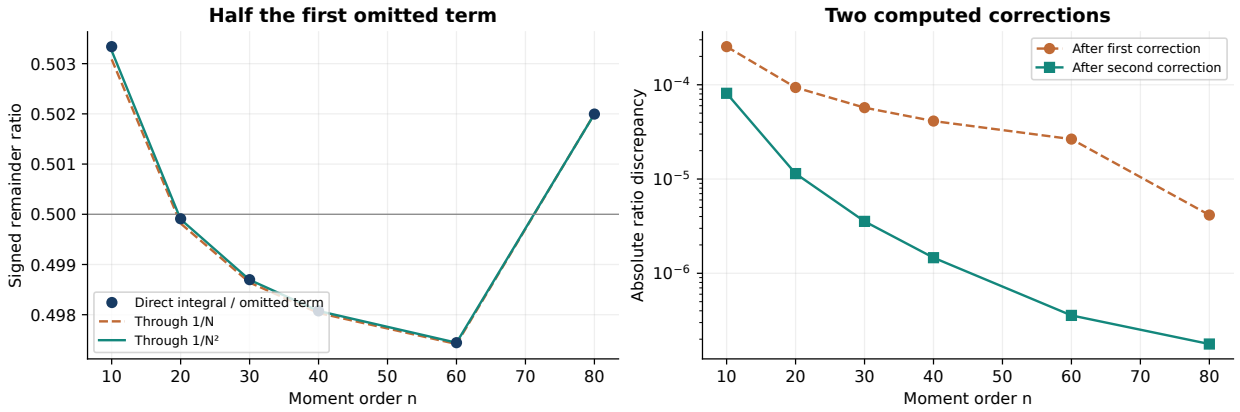


Figure 8.2: The signed error divided by the first omitted term, at a truncation near $(n+3/2)/\Lambda$. The first-correction formula tracks the small changes caused by rounding the truncation index. The plot is numerical corroboration; Theorem 8.14 supplies the proof.

For the gamma average, put $K_1(v) = 3e^{-v}/[2(1+e^{-v})^2]$ and $V = Y - \Lambda$. Taylor expansion through degree five, using $\mathbb{E}|V|^6 = O(N^{-3})$, gives

$$\mathbb{E}L(V) = \frac{1}{2} - \frac{\eta}{4N} + \frac{\Lambda(3\eta-2)}{48N^2} + O(N^{-3}), \qquad \mathbb{E}K_1(V) = \frac{3}{8} - \frac{3\Lambda}{32N} + O(N^{-2}).$$

For example $\mathbb{E}V^3 = \Lambda(2-3\eta)N^{-2} + O(N^{-3})$; the signed fifth moment is $O(N^{-3})$. Also $\mathbb{E}K_2(e^{-V}) = K_2(1) + O(N^{-1})$. The uniform outer bound again makes all outer contributions exponentially small. Combining the three terms proves (8.18). □

8.6.3 A convergent identity with an explicit tail bound

An asymptotic series should not be used as an infinite exact sum. The same inverse coordinate gives an exact convergent alternative.

Proposition 8.16 (Cutoff identity)**.** *For $n \geq 1$ and $Y > \Lambda$,*

$$m_n = \frac{1}{(n-1)!}\int_0^Y y^{n-1} g(e^{-y})\,dy + \sum_{k\geq 1}\frac{\widetilde{c}_k}{k^n}Q(n,kY), \qquad Q(n,z) = e^{-z}\sum_{j=0}^{n-1}\frac{z^j}{j!}. \tag{8.19}$$

The sum is absolutely convergent. If $Y > \log 4$, the absolute tail after $k = K$ is at most

$$\frac{(4e^{-Y})^{K+1}}{2(1-4e^{-Y})}\sum_{j=0}^{n-1}\frac{Y^j}{j!(K+1)^{n-j}}. \tag{8.20}$$

Proof. Split (8.14) at Y, substitute the convergent inverse series on the tail, and integrate term by term. The coefficient growth theorem proves absolute convergence. For the elementary explicit bound, note that

$$\frac{1/2}{\Phi(1/2)} = \frac{1}{2\sqrt{\pi}} > \frac{1}{4}.$$

The smaller solution to $B = t\Phi(B)$ at $t = 1/4$ is therefore less than $1/2$; equivalently $B(1/4) < 1/2$. Positivity gives $b_k \leq 4^k/2$. For $k > K$ and $j \leq n-1$, $k^{j-n} \leq (K+1)^{j-n}$. The geometric tail now gives (8.20). □

The numerical implementation uses ordinary high-precision arithmetic and explicitly distinguishes analytic truncation bounds from interval rounding bounds. Formula (8.19) is also suitable for a future ball-arithmetic evaluator: the finite inverse integral and all constant evaluations can be enclosed separately.

8.7 Reflected log-gamma moments and their late coefficients

Set $f(x) = \log\Gamma(x)$ for $0 < x < 1$, and define

$$M_{n,m} = \int_0^1 f(x)^n f(1-x)^m \, dx, \qquad n, m \in \mathbb{Z}_{\geq 0}.$$

These integrals already occur as $T_{n+m,n}$ in Amdeberhan et al. [64, Section 5]. Their symmetry $M_{n,m} = M_{m,n}$ and the identities obtained by expanding Euler's reflection formula are therefore classical. The asymptotic question here is different: keep the reflected exponent m fixed and let n tend to infinity. The factor $f(1-x)^m$ vanishes to order m at the endpoint that produces the factorial growth. This shifts the first exponential scale from 1 to $m+1$ without removing the eventual divergence of the full expansion. Theorem 8.18 identifies its late terms, including their signs and the first relative correction.

8.7.1 The meromorphic transform and all finite asymptotic orders

Put

$$B(x) = \log\Gamma(1-x) = \gamma x + \sum_{j\geq 2} \frac{\zeta(j)}{j} x^j, \qquad |x| < 1,$$

where γ is Euler's constant. For $k \geq 1$ define

$$A_{k,m} = \frac{1}{k}[x^{k-1}] \exp\{kB(-x)\} B(x)^m. \tag{8.21}$$

The convention $B(x)^0 = 1$ is in force. In particular $A_{k,m} = 0$ when $k \leq m$. Every coefficient is a finite polynomial over $\mathbb{Q}$ in $\gamma, \zeta(2), \ldots, \zeta(k-1)$.

Theorem 8.17 (Reflected moment expansion)**.** *For each fixed integer $m \geq 0$, the function*

$$F_m(z) = \int_0^1 \Gamma(x)^z f(1-x)^m \, dx$$

is holomorphic for $\operatorname{Re} z < m+1$, has Taylor expansion

$$F_m(z) = \sum_{n\geq 0} \frac{M_{n,m}}{n!} z^n \qquad (|z| < m+1),$$

and extends meromorphically to $\mathbb{C}$. Its only possible poles are simple poles at integers $k \geq m+1$, with

$$\operatorname*{Res}_{z=k} F_m(z) = -kA_{k,m}. \tag{8.22}$$

The first pole is present. For every fixed integer $K \geq m+1$,

$$\frac{M_{n,m}}{n!} = \sum_{k=m+1}^{K} \frac{A_{k,m}}{k^n} + O_{m,K}\big((K+1)^{-n}\big), \qquad n \longrightarrow \infty. \tag{8.23}$$

Consequently

$$M_{n,m} \sim \frac{\gamma^m n!}{(m+1)^{n+1}}. \tag{8.24}$$

Proof. As $x \downarrow 0$, $f(x) = -\log x + O(x)$ and $B(x) = \gamma x + O(x^2)$. At the other endpoint, $f(x) = O(1-x)$ and $f(1-x) = O(1 + |\log(1-x)|)$. These estimates prove absolute convergence and compact domination in the asserted half-plane, including after any number of differentiations in z. They also justify the Taylor expansion near zero.

Fix $0 < a < 1$ and an integer $J \geq 1$. Taylor expansion in x gives

$$\Gamma(1+x)^z B(x)^m = \sum_{j=0}^{J-1} b_{j,m}(z) x^j + x^J E_{J,m}(x,z),$$

where each $b_{j,m}$ is an entire function of z, and the remainder is uniformly bounded for $0 \leq x \leq a$ and z in compact sets. The integral over $[0,a]$ is therefore

$$\sum_{j=0}^{J-1} \frac{b_{j,m}(z) a^{j+1-z}}{j+1-z} + \int_0^a x^{J-z} E_{J,m}(x,z)\, dx.$$

The last integral is holomorphic for $\operatorname{Re} z < J+1$. The integral over $[a,1]$ is entire, since its logarithmic singularity at 1 is integrable uniformly for z in compact sets. Letting J increase gives the continuation. At $z = k$ its residue is $-b_{k-1,m}(k) = -kA_{k,m}$. Since $B(x)^m$ starts with $\gamma^m x^m$, the first nonzero coefficient is $A_{m+1,m} = \gamma^m/(m+1) > 0$.

For the asymptotic assertion, subtract the polar functions

$$\sum_{k=m+1}^{K+1} \frac{kA_{k,m}}{k-z}$$

from $F_m(z)$. The difference is holomorphic on $|z| < K+2$. Cauchy's estimate on any circle $K+1 < R < K+2$ bounds its nth Taylor coefficient by $O_{m,K,R}(R^{-n})$. Restoring the term at $K+1$ proves (8.23), even if that particular coefficient happens to vanish. Taking $K = m+1$ proves (8.24). □

The first three coefficients are explicit. Write $k = m+3$ in the last line:

$$\begin{aligned} A_{m+1,m} &= \frac{\gamma^m}{m+1}, \\ A_{m+2,m} &= \frac{\gamma^{m-1}}{m+2}\left(\frac{m\zeta(2)}{2} - (m+2)\gamma^2\right), \\ A_{m+3,m} &= \frac{\gamma^m}{k}\left\{\frac{m\zeta(3)}{3\gamma} + \frac{m(m-1)\zeta(2)^2}{8\gamma^2} + \frac{k(1-m)\zeta(2)}{2} + \frac{k^2\gamma^2}{2}\right\}. \end{aligned} \tag{8.25}$$

These displays also apply to $m = 0$, since $\gamma > 0$ and factors of m remove the superfluous terms. For example,

$$\begin{aligned} \frac{M_{n,1}}{n!} &= \frac{\gamma}{2^{n+1}} + \frac{\zeta(2)/2 - 3\gamma^2}{3^{n+1}} + \frac{\zeta(3)/3 + 8\gamma^3}{4^{n+1}} + O(5^{-n}), \\ \frac{M_{n,2}}{n!} &= \frac{\gamma^2}{3^{n+1}} + \frac{\gamma\zeta(2) - 4\gamma^3}{4^{n+1}} + O(5^{-n}). \end{aligned}$$

The second displayed correction is positive, whereas the corresponding correction for $m = 1$ is negative. Thus strict alternation from the first term cannot be imported from the unreflected family. The two signs can, for instance, be checked using the elementary enclosures $0.577 < \gamma < 0.578$ and $1.644 < \zeta(2) < 1.646$. Eventual alternation has a different, precise rule.

8.7.2 Late coefficients and a universal inverse-gamma scale

Let $\phi(u) = \Gamma(1-u)$ and $D = u\,d/du$. Define $\tau \in (0,1)$ by

$$D \log \phi(\tau) = -\tau\psi(1-\tau) = 1, \tag{8.26}$$

and put

$$\rho = \frac{\tau}{\phi(\tau)}, \quad v = D^2 \log \phi(\tau) = 1 + \tau^2\psi'(1-\tau), \quad C = \frac{\tau}{\sqrt{2\pi v}}, \quad b = -\log\Gamma(1+\tau). \tag{8.27}$$

The equation for τ has a unique solution: its left side equals $\gamma u + \sum_{j\ge 2}\zeta(j)u^j$, which is strictly increasing from zero to infinity. Also $0 < \rho < 1$ and $b > 0$, the latter by strict log-convexity of Γ between 1 and 2. Numerically, $\tau = 0.5040830082\ldots$ and $\rho = 0.2821158090\ldots$. Only the exact definitions are used below.

Theorem 8.18 (Late reflected coefficients)**.** *For every fixed nonnegative integer m,*

$$A_{k,m} = (-1)^{k+m-1} C b^m \rho^{-k} k^{-3/2} \left(1 + \frac{\beta_m}{k} + O_m(k^{-2})\right). \tag{8.28}$$

There is a full asymptotic expansion in integer powers of $1/k$. To specify the first correction, set $G(u) = \log\Gamma(1+u)$, $H_m(u) = G(u)^m$, and $\kappa_j = D^j \log\phi(\tau)$. All quantities on the right below are evaluated at $u = \tau$:

$$\beta_m = -\frac{(D+1)^2 H_m}{2vH_m} + \frac{\kappa_3(D+1)H_m}{2v^2 H_m} + \frac{\kappa_4}{8v^2} - \frac{5\kappa_3^2}{24v^3}. \tag{8.29}$$

In particular, every sufficiently large possible pole in Theorem 8.17 is present. More explicitly,

$$\operatorname{Res}_{z=k} F_m(z) = (-1)^{k+m} C b^m \rho^{-k} k^{-1/2} \left(1 + \frac{\beta_m}{k} + O_m(k^{-2})\right),$$

so the eventual residue sign is $(-1)^{k+m}$. For every fixed real n, the formal series $\sum_{k\ge m+1} A_{k,m}/k^n$ diverges because its terms do not tend to zero.

Proof. Replacing x by $-u$ in (8.21) gives

$$A_{k,m} = \frac{(-1)^{k-1}}{k}[u^{k-1}]\phi(u)^k H_m(u).$$

The function H_m is holomorphic for $|u| < 1$. If $\phi(u) = \sum_{j\ge 0} a_j u^j$, all the a_j are strictly positive, since $\log\phi$ has positive coefficients. Thus

$$p_j = \frac{a_j \tau^j}{\phi(\tau)}$$

is a probability distribution with exponential moments, mean one, variance $v > 0$, and positive mass at both 0 and 1. It follows that the function

$$Q(t) = \frac{\phi(\tau e^{it})}{\phi(\tau)} e^{-it}$$

has modulus strictly less than one for $0 < |t| \le \pi$, while

$$\log Q(t) = -\frac{v}{2}t^2 - \frac{i\kappa_3}{6}t^3 + \frac{\kappa_4}{24}t^4 + O(t^5)$$

near zero. Cauchy's coefficient formula becomes

$$A_{k,m} = \frac{(-1)^{k-1}\tau\rho^{-k}}{2\pi k} \int_{-\pi}^{\pi} Q(t)^k e^{it} H_m(\tau e^{it})\, dt. \tag{8.30}$$

Fix a sufficiently small $\delta > 0$. On $\delta \le |t| \le \pi$ the integral is $O(q^k)$ for some $q < 1$. On $k^{-2/5} \le |t| \le \delta$ it is $O(e^{-ck^{1/5}})$, because $|Q(t)| \le e^{-ct^2}$. The analytic amplitude is bounded on the whole circle. On the remaining interval set $t = y/\sqrt{k}$. Gaussian domination and Taylor expansion to any fixed order, with an analytic remainder, give a full expansion. Odd terms integrate to zero, so only integer powers of $1/k$ occur. The leading integral is

$$H_m(\tau)\sqrt{\frac{2\pi}{vk}} = (-1)^m b^m \sqrt{\frac{2\pi}{vk}},$$

which proves the leading term in (8.28).

For the first correction, the amplitude derivatives at $t = 0$ are H_m, $i(D{+}1)H_m$, and $-(D{+}1)^2 H_m$. The Gaussian moments of degrees 2, 4, and 6 contribute, respectively, the amplitude's quadratic term, the amplitude–cubic cross term together with the quartic phase term, and the square of the cubic phase term. Their normalized sum is exactly (8.29). Expanding two further even orders gives a relative $O_m(k^{-2})$ remainder after the displayed correction. Since $Cb^m > 0$ and $\rho < 1$, the assertions about nonvanishing, signs, and divergence follow at once. □

This proof uses the circle $|u| = \tau$ and a strict modulus inequality; it does not presume a classification of all complex critical values of the reciprocal gamma function. The same circle governs every fixed reflected exponent. With $\alpha = -\log\rho$, the continuous minimum of the leading term magnitude $\rho^{-k}k^{-n-3/2}$ occurs at $k = (n + 3/2)/\alpha$. A least-term calculation alone does not bound the remainder. The next result supplies an explicit bound with a growing number of terms.

8.7.3 An effective remainder despite the initial sign changes

Theorem 8.19 (Explicit reflected truncation bound)**.** *Let $m \ge 0$, $n \ge 1$, and $K \ge 1$ be integers, put $\alpha = -\log\rho$ and $T_m = \tau B(\tau)^m$, and suppose $K\alpha < n$. With*

$$R_{n,K}^{(m)} = \frac{M_{n,m}}{n!} - \sum_{k=1}^{K} \frac{A_{k,m}}{k^n}, \qquad \Gamma(n,s) = \int_s^{\infty} t^{n-1}e^{-t}\, dt,$$

one has the entirely explicit estimate

$$|R_{n,K}^{(m)}| \le \frac{\alpha^n}{n!}\left(m! + \frac{nT_m}{n - K\alpha}\right) + \frac{T_m\rho^{-K-1}}{(K+1)^n}\frac{\Gamma(n,(K+1)\alpha)}{\Gamma(n)}. \tag{8.31}$$

If $n \ge \max\{2, 4\alpha^2\}$ and $K = \lfloor n/\alpha - \sqrt{n}\rfloor$, then $K \ge 1$ and

$$|R_{n,K}^{(m)}| \le \left(\frac{m!}{\sqrt{2\pi n}} + \frac{T_m}{\alpha\sqrt{2\pi}} + T_m e^{2\alpha^2}\right)\left(\frac{e\alpha}{n}\right)^n. \tag{8.32}$$

This bound holds for every m, but no useful uniform relative error is asserted when m grows with n.

Proof. The function f decreases from infinity to zero on $(0,1]$: $\psi'(x) > 0$ and $\psi(1) = -\gamma < 0$. Let $x(y)$ denote its inverse, and set $W_m(x) = \int_0^x B(t)^m\,dt$. Tonelli's theorem applied to $f(x)^n = n\int_0^{f(x)} y^{n-1}\,dy$ gives

$$\frac{M_{n,m}}{n!} = \frac{1}{\Gamma(n)}\int_0^\infty y^{n-1} W_m(x(y))\,dy. \tag{8.33}$$

Let $h(q)$ be the local solution of $h = q\phi(h)$ with $h(0) = 0$. Lagrange inversion gives positive coefficients

$$h(q) = \sum_{k\ge 1} c_k q^k, \qquad c_k = \frac{1}{k}[u^{k-1}]\phi(u)^k.$$

The $m = 0$ case of Theorem 8.18 proves that its radius is ρ and that the series converges absolutely on $|q| \le \rho$. The map $u \mapsto u/\phi(u)$ increases strictly from 0 to ρ on $[0,\tau]$, so inversion along this interval and continuity imply $h(\rho) = \tau$. No other inverse branch is used.

Lagrange–Bürmann inversion now gives

$$W_m(x(y)) = \sum_{k\ge 1} A_{k,m} e^{-ky} \qquad (y \ge \alpha).$$

Indeed $x(y) = -h(-e^{-y})$ initially for large y, and then throughout $y > \alpha$ by analytic continuation along the strictly monotone real inverse. The endpoint follows by absolute convergence. To prove that convergence and obtain its quantitative majorant, define

$$d_{k,m} = \frac{1}{k}[u^{k-1}]\phi(u)^k B(u)^m \ge 0.$$

Coefficientwise absolute values in $B(-u)^m$ are $B(u)^m$. Consequently $|A_{k,m}| \le d_{k,m}$, while another application of Lagrange–Bürmann gives

$$\sum_{k\ge 1} d_{k,m}\rho^k = W_m(h(\rho)) = W_m(\tau) \le \tau B(\tau)^m = T_m.$$

Here B has positive Taylor coefficients and is increasing on $[0,\tau]$. Thus on $y \ge \alpha$,

$$\left| W_m(x(y)) - \sum_{k=1}^{K} A_{k,m} e^{-ky} \right| \le T_m \rho^{-K-1} e^{-(K+1)y}.$$

Its integral in (8.33) is the second term of (8.31).

On $0 \le y \le \alpha$, use $0 \le W_m(x(y)) \le M_{m,0} \le m!$. The last inequality follows from $0 \le \log\Gamma(x) \le -\log x$, since log-convexity gives $\Gamma(1+x) \le 1$ for $0 \le x \le 1$. Also, for $k \le K$,

$$\int_0^\alpha y^{n-1} e^{-ky}\,dy \le \frac{\alpha^n e^{-k\alpha}}{n - k\alpha}.$$

This follows by differentiating $y^n e^{-ky}$ and using $n - ky \ge n - k\alpha > 0$. Summing with $\sum |A_{k,m}|\rho^k \le T_m$ proves the first term of (8.31).

For the stated choice of K, $n - K\alpha \ge \alpha\sqrt{n}$ and $\theta = (K+1)\alpha/n$ lies in $[1/2, 1]$. The inequality $\theta - 1 - \log\theta \le 2(1-\theta)^2$ gives

$$\rho^{-K-1}(K+1)^{-n} \le e^{2\alpha^2}(e\alpha/n)^n.$$

Finally, $\Gamma(n,s) \le \Gamma(n)$ and $n! \ge \sqrt{2\pi n}(n/e)^n$ yield (8.32). □

8.7.4 Sharp truncation for every fixed reflected exponent

The reflected coefficient expansion and the sharp inverse remainder now combine. This statement uses the two incoming developments together; it is stronger than their separate absolute-bound and unreflected results. Write $\Lambda = -\log \rho$, and retain β_m from (8.29).

Corollary 8.20 (Half the first omitted reflected term)**.** *Fix an integer $m \geq 0$. Let $n, N \to \infty$ through positive integers, with $\eta = N\Lambda - n$ in a fixed bounded interval. Then*

$$\frac{M_{n,m}/n! - \sum_{k=1}^{N-1} A_{k,m}/k^n}{A_{N,m}/N^n} = \frac{1}{2} + \frac{3-2\eta}{8N} + \frac{1}{N^2}\left[\frac{\beta_m}{4} + \frac{\Lambda(6\eta-13)}{96}\right] + O_m(N^{-3}), \tag{8.34}$$

uniformly for bounded η. In particular the error eventually has sign $(-1)^{N+m-1}$ and is asymptotic to half the first omitted term. The denominator is nonzero for every sufficiently large N.

Proof. Set $W_m(x) = \int_0^x B(u)^m\, du$ and $G_m(t) = W_m(g(t))$, where g is the real inverse used in the preceding sharp-moment section. Lagrange–Bürmann inversion gives $G_m(t) = \sum_{k\geq 1} A_{k,m} t^k$ near zero. The layer-cake formula (8.33) becomes

$$M_{n,m}/n! = \frac{1}{\Gamma(n)} \int_0^\infty y^{n-1} G_m(e^{-y})\, dy.$$

Since W_m is holomorphic on $|x| < 1$, it can be composed with the slit-disk inverse of Lemma 8.12. At the dominant point $g(-\rho) = -\tau$ its derivative is

$$W_m'(-\tau) = B(-\tau)^m = \big(\log \Gamma(1+\tau)\big)^m = (-b)^m \neq 0.$$

Thus G_m has the same sole dominant square-root singularity, with a nonzero leading amplitude. Its late coefficients have the relative coefficient β_m established in Theorem 8.18. The Cauchy cut-contour argument in Lemma 8.13 therefore applies to G_m with the same leading kernel $1/(1+t/\rho)$ and the same $3/(2N)$ endpoint-moment correction. The sign of the leading jump cancels against $A_{N,m}$ in this ratio. The second normalized cut coefficient at $t = \rho$ is $\beta_m/4$, by the same square-root transfer and beta-moment calculation used in Corollary 8.15.

The outer real-interval bounds also persist. Above the convergence radius, the finite coefficient sum is bounded using $|A_{k,m}| \asymp_m \rho^{-k} k^{-3/2}$, while $0 \leq G_m(t) \leq m!$ for $0 \leq t \leq 1$. The finitely many exceptional early coefficients contribute only an exponentially small term after normalization. Below the radius, the absolute Taylor tail gives the corresponding uniform bound. Average the normalized remainder against the gamma density with shape n and rate N. Its signed moments are exactly those used in the unreflected calculation. The first two corrections are consequently (8.34), with d_1 replaced by β_m; the analytic cut expansion and moments through degree six give the stated uniform $O_m(N^{-3})$ remainder. Eventual nonvanishing and the sign follow from (8.28). □

This fixed-m corollary does not supply uniform relative estimates when m grows with n. That joint regime remains a separate problem.

8.7.5 An exact identity coupling all reflected residues

The coefficients across different m satisfy a finite identity in which Euler's constant and all odd zeta values cancel.

Proposition 8.21 (Reflection sum for the residues)**.** *For every integer $k \geq 1$,*

$$\sum_{m=0}^{k-1} \frac{k^m}{m!} A_{k,m} = \begin{cases} 0, & k \text{ even}, \\ \dfrac{\pi^{k-1}}{k\,2^{k-1}} \dbinom{k-1}{(k-1)/2}, & k \text{ odd}. \end{cases} \tag{8.35}$$

Proof. Because $B(x)^m$ has order m, extending the sum over m to infinity does not affect $[x^{k-1}]$. Euler's reflection formula gives

$$\sum_{m=0}^{k-1} \frac{k^m}{m!} A_{k,m} = \frac{1}{k}[x^{k-1}]\big(\Gamma(1+x)\Gamma(1-x)\big)^k = \frac{1}{k}[x^{k-1}]\left(\frac{\pi x}{\sin \pi x}\right)^k.$$

The coefficient is zero for even k. For odd k, compute it as the residue at $x = 0$ of $(\pi/\sin \pi x)^k\,dx$. The local change of variable $t = \sin \pi x$ gives

$$[x^{k-1}]\left(\frac{\pi x}{\sin \pi x}\right)^k = \pi^{k-1}[t^{k-1}](1-t^2)^{-1/2},$$

and the binomial expansion proves (8.35). □

For example, the cases $k = 2, 3$ read $A_{2,0} + 2A_{2,1} = 0$ and $A_{3,0} + 3A_{3,1} + \frac{9}{2}A_{3,2} = \pi^2/6$. They are exact identities among the asymptotic coefficients, rather than convergent sums of the moment expansions. No summation over the late index k is used.

8.7.6 A recorded log-sine conjecture has a formal proof

There is also a small but useful correction to the logical status of a claim in the literature. The author preprint of [64] labels a coefficient identity as Conjecture 5.5 and motivates it using transcendence assumptions on $\log 2$ and $\log \pi$. For the canonical formal polynomials supplied by the beta integral, the coefficient identity needs none of those assumptions. Under the preprint's assumptions these formal coefficients coincide with its uniquely specified numerical coefficients, so the result also proves that stated implication. We do not assert uniqueness of arbitrary numerical expressions in zeta values and logarithms, nor priority over any later treatment.

Define polynomials $\mathcal{P}_N(X)$ by

$$\sum_{N \geq 0} \mathcal{P}_N(X) \frac{z^N}{N!} = \exp\left\{ Xz + \sum_{j \geq 2} \frac{(1 - 2^{1-j})\zeta(j)}{j} z^j \right\}. \tag{8.36}$$

Proposition 8.22 (Unconditional coefficient and translation identities)**.** *For $N \geq 0$,*

$$\int_0^1 [-\log \sin(\pi x)]^N\,dx = \mathcal{P}_N(\log 2), \tag{8.37}$$

$$[X^j]\mathcal{P}_N(X) = \binom{N}{j} \mathcal{P}_{N-j}(0), \qquad 0 \leq j \leq N,$$

$$\sum_{j=0}^{N} \binom{N}{j} M_{j,N-j} = \mathcal{P}_N(\log(2\pi)).$$

Moreover, the coefficients admit the finite partition formula

$$\mathcal{P}_N(X) = N! \sum_{r_1 + 2r_2 + \cdots + N r_N = N} \frac{X^{r_1}}{r_1!} \prod_{j=2}^{N} \frac{1}{r_j!} \left(\frac{(1-2^{1-j})\zeta(j)}{j}\right)^{r_j}. \tag{8.38}$$

Proof. For $\operatorname{Re} z < 1$, the beta integral gives

$$\int_0^1 \sin(\pi x)^{-z}\,dx = \frac{\Gamma((1-z)/2)}{\sqrt{\pi}\,\Gamma(1-z/2)}.$$

Expanding its logarithm at zero gives the right side of (8.36) with $X = \log 2$; its degree-j coefficient for $j \geq 2$ is $(1-2^{1-j})\zeta(j)/j$. Extracting Taylor coefficients proves (8.37). The factor e^{Xz} gives $[X^j]\mathcal{P}_N = \binom{N}{j}\mathcal{P}_{N-j}(0)$ by formal coefficient extraction. Euler's reflection formula turns the sum of mixed moments into the integral of $(\log\pi - \log\sin\pi x)^N$, which translates X by $\log\pi$. Expanding each exponential factor separately proves (8.38). All formal operations in a fixed degree involve only finitely many coefficients. □

The constant-column generating function has the additional factorization

$$\sum_{N\geq 0} \mathcal{P}_N(0)\frac{z^N}{N!} = \sqrt{\frac{\tan(\pi z/2)}{\pi z/2}}\exp\left\{\sum_{\substack{j\geq 3\\ j\text{ odd}}} \frac{(1-2^{1-j})\zeta(j)}{j} z^j\right\}, \tag{8.39}$$

where the square root is the formal branch with constant term one. Indeed, multiplying the left side by its value at $-z$ and applying the reflection formula yields $\tan(\pi z/2)/(\pi z/2)$. The even logarithmic part is therefore the displayed square root. This makes the multiplicative factorization of its formal odd-zeta monomial coefficients explicit, while the even-zeta contribution has an elementary trigonometric generating function.

Verification and remaining questions. The accompanying script `code/verify_reflected_moments.py` checks the three coefficient formulas by exact symbolic arithmetic, checks finite reflection sums and the formal Appell formulas, and compares independent quadrature with the fixed-order expansion. It also compares the late coefficients with (8.28) and (8.29). The numerical calculations are diagnostics; the preceding analytic and formal arguments are the proofs. The fixed-reflected-exponent signed remainder is now supplied by Corollary 8.20. Uniformity when m increases with n remains a substantive next question. In the former regime the endpoint approximation $B(x)^m \approx \gamma^m x^m$ need not be uniform, so the fixed-m theorem does not settle the simultaneous large-exponent problem.

8.8 The Stieltjes layer

Differentiating $L(s,\chi) = q^{-s}\sum_k \chi(k)\,\zeta(s,k/q)$ at $s=1$ gives the exact ladder

$$L'(\chi,1) = -\ln q\; L(1,\chi) - \frac{1}{q}\sum_k \chi(k)\,\gamma_1\Big(\frac{k}{q}\Big),$$

so every Malmsten log-log integral is a statement about *first generalized Stieltjes constants* $\gamma_1(k/q)$ — Blagouchine's program, equivalent to Deninger's $\zeta''(0,a)$ layer via the functional equation.

Result 8.23 (even-character Malmsten integral). [historical experiment: verified to 10^{-51}; even character mod 5; φ the golden ratio]

$$\int_0^1 \ln\ln\frac{1}{x}\cdot\frac{(1-x)(1-x^2)}{1-x^5}\,dx = -(\gamma+\ln 5)\,\frac{2\ln\varphi}{\sqrt{5}} - \frac{\gamma_1(\frac{1}{5}) - \gamma_1(\frac{2}{5}) - \gamma_1(\frac{3}{5}) + \gamma_1(\frac{4}{5})}{5}.$$

Result 8.24 (the parity dichotomy). The dichotomy is empirically sharp. The *odd* γ_1-combination reduces exactly as differentiated-Kummer predicts [historical experiment: verified blind]:

$$\gamma_1\tfrac{1}{3} - \gamma_1\tfrac{2}{3} = -\frac{\pi}{\sqrt{3}}\Big(\gamma + 4\ln 2 - \tfrac{1}{2}\ln 3 + 4\ln\pi - 6\ln\Gamma\tfrac{1}{3}\Big),$$

which is *Malmsten's 1846 reflection identity* [4] (long misattributed to Almkvist–Meurman). The even combination did not reduce in the tested standard basket [historical experiment: PSLQ junk at coefficient height 10^8 — honest negative]; pushing Blagouchine's rational-arguments theorem [6] through it collapses everything except a ζ''-block [historical experiment: derived and verified to 10^{-50}]:

$$\gamma_1\tfrac{1}{5} - \gamma_1\tfrac{2}{5} - \gamma_1\tfrac{3}{5} + \gamma_1\tfrac{4}{5} = \sqrt{5}\left[\zeta''\tfrac{1}{5} - \zeta''\tfrac{2}{5} - \zeta''\tfrac{3}{5} + \zeta''\tfrac{4}{5}\right] - 2\sqrt{5}\,\ln\varphi\,(\gamma + \ln 10\pi).$$

The canonical atom for the even layer is thus the even-character $\zeta''(0, l/q)$ combination (Deninger's layer [9]); Malmsten integrals, even γ_1-combinations, and even $L'(\chi, 1)$ are all elementary-equivalent to it. Two practical records: Blagouchine's particular value $\gamma_1(1/4)$ was re-verified exactly against `mpmath`'s generalized Stieltjes constants, and the reduction coefficients live in $\mathbb{Q}(\sqrt{q})$ — rational-coefficient PSLQ baskets *miss* the relation, so a $\sqrt{q}$-weighted basket or direct derivation is required.

8.9 The common Hurwitz-jet calculus

The generalized Stieltjes constants $\gamma_n(x)$ are defined by the Laurent expansion of the Hurwitz zeta function at its pole,

$$\zeta(s, x) = \frac{1}{s-1} + \sum_{n\geq 0} \frac{(-1)^n}{n!}\,\gamma_n(x)\,(s-1)^n, \tag{8.40}$$

so that $\gamma_0(x) = -\psi(x)$ and $\gamma_n(1) = \gamma_n$ are the classical Stieltjes constants, $\gamma_0 = \gamma$. Although universally called *constants*, the $\gamma_n(x)$ are real-analytic functions of the argument on $(0, \infty)$, and it is this function-theoretic point of view we adopt. Following Blagouchine [22, 6] we write

$$\Gamma_n(p) \;:=\; \int_1^p \gamma_n(x)\,dx, \qquad \text{so that}\;\; \Gamma_n(1) = 0,$$

for the antiderivative normalized at 1. (The notation is Blagouchine's; the clash with the Euler Γ-function is resolved by the subscript, and indeed $\Gamma_0(p) = -\ln\Gamma(p)$, as Theorem 8.25 makes plain.) Throughout, $\zeta^{(k)}(s_0, a) := \partial_s^k \zeta(s, a)\big|_{s=s_0}$ and $\zeta^{(k)}(s_0) := \zeta^{(k)}(s_0, 1)$ denote derivative jets with respect to the *order*; $\Lambda_q := \gamma + \ln(2\pi q)$; $H_n = \sum_{i\leq n} 1/i$ and $H_n^{(2)} = \sum_{i\leq n} 1/i^2$ are harmonic numbers; A is the Glaisher–Kinkelin constant, $\ln\mathsf{A} = \frac{1}{12} - \zeta'(-1)$; and

$$S_j\Big(\frac{p}{q}\Big) := \sum_{k=1}^{q-1} \sin\frac{2\pi kp}{q}\,\zeta^{(j)}\Big(0, \frac{k}{q}\Big), \qquad S_0\Big(\frac{p}{q}\Big) = \tfrac{1}{2}\cot\frac{\pi p}{q}, \quad S_1\Big(\frac{p}{q}\Big) = \sum_{k=1}^{q-1} \sin\frac{2\pi kp}{q}\,\ln\Gamma\Big(\frac{k}{q}\Big)$$

(the closed forms for S_0, S_1 use $\zeta(0, a) = \frac{1}{2} - a$, $\zeta'(0, a) = \ln\Gamma(a) - \frac{1}{2}\ln(2\pi)$, and $\sum_k \sin(2\pi kp/q) = 0$).

The starting observation is elementary but organizing: differentiating (8.40) in x against $\partial_x \zeta(s, x) = -s\,\zeta(s+1, x)$ identifies every derivative and antiderivative of every γ_n with a derivative jet of the Hurwitz zeta function at an *integer* value of s. Antiderivatives live at $s = 0$; the functions themselves at $s = 1$; their x-derivatives at $s = 2$ [25] — a single ladder.

The preceding gamma-integral examples are the first rung of this calculus. We now prove the integration identity, build its spectral jets, and use reflection and distribution to compute rational arguments. All source residuals in this chapter are historical; locally rerun checks are recorded separately in the verification directory.

8.10 Underived Stieltjes grids and character coordinates

Expanding the Hurwitz distribution identity at $s = 1$ gives

$$\sum_{j=0}^{d-1} \gamma_1((x+j)/d) = d\gamma_1(x) + d\log d\,\psi(x) - \frac{d}{2}\log^2 d,$$
$$\sum_{j=0}^{d-1} \gamma_2((x+j)/d) = d\left[\gamma_2(x) - 2\log d\,\gamma_1(x) - \log^2 d\,\psi(x) + \frac{1}{3}\log^3 d\right].$$

For a nonprincipal character,

$$L'(1,\chi) = -\log q\,L(1,\chi) - \frac{1}{q}\sum_{p=1}^{q}\chi(p)\gamma_1(p/q).$$

For a real quadratic character χ_D, $\zeta_{\mathbb{Q}(\sqrt{D})}(s) = \zeta(s)L(s,\chi_D)$ shows $L'(1,\chi_D)/L(1,\chi_D) = \gamma_{\mathbb{Q}(\sqrt{D})} - \gamma$, where the first term is the field's Euler–Kronecker constant. The source's exact finite matrices for $3 \le q \le 30$ left $\varphi(q)/2 - 1$ coordinates after distribution and first-constant reflection. Their negative searches constrain the selected baskets, without proving independence of Euler–Kronecker constants. The reflection formulas below retain the conductor factor $\gamma + \log(2\pi q)$: the earlier printed $q = 5$ specialization without $\log 5$ was corrected in the source reports.

8.11 The ladder

Theorem 8.25 (antiderivative ladder)**.** *For $n \ge 0$ and $p > 0$,*

$$\frac{d}{dp}\,\zeta^{(n+1)}(0,p) = (-1)^{n+1}(n+1)\,\gamma_n(p), \tag{8.41}$$

and consequently

$$\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1}\Big[\zeta^{(n+1)}(0,p) - \zeta^{(n+1)}(0)\Big]. \tag{8.42}$$

Proof. Differentiating $\partial_x\zeta(s,x) = -s\,\zeta(s+1,x)$ is legitimate termwise for $\operatorname{Re} s > 1$ and extends by analytic continuation. Expand both sides around $s = 0$: on the right, $-s\,\zeta(s+1,x) = -s\big[\frac{1}{s} + \sum_k \frac{(-1)^k}{k!}\gamma_k(x)s^k\big] = -1 - \sum_k \frac{(-1)^k}{k!}\gamma_k(x)s^{k+1}$, whose s^{n+1}-coefficient is $(-1)^{n+1}\gamma_n(x)/n!$. On the left the s^{n+1}-coefficient is $\partial_x\zeta^{(n+1)}(0,x)/(n+1)!$. Equating gives (8.41); integrating from 1 to p gives (8.42). □

The differential form (8.41) is Lemma 2 of Chakraborty–Kanemitsu–Kuzumaki [23] (for $n = 1$ it is Deninger's R'-law [9]); the integrated form (8.42) is eq. (2.16) of Connon [26]. For $n = 0$, $\zeta'(0,p) = \ln\Gamma(p) - \frac{1}{2}\ln(2\pi)$ recovers $\Gamma_0(p) = -\ln\Gamma(p)$.

Theorem 8.26 (interpolation of power sums of logarithms)**.** *For $k \ge 1$ and integer $N \ge 0$,*

$$\Gamma_{k-1}(N+1) \;=\; -\frac{1}{k}\sum_{j=1}^{N}\ln^k j.$$

Proof. From $\zeta(s,N+1) = \zeta(s) - \sum_{j=1}^{N} j^{-s}$, $\zeta^{(k)}(0,N+1) = \zeta^{(k)}(0) - \sum_j(-\ln j)^k = \zeta^{(k)}(0) - (-1)^k\sum_j \ln^k j$. Insert into (8.42) with $n = k-1$. □

Thus Γ_{k-1} is an analytic interpolant of the power sums $\sum \ln^k j$ (up to the factor $-1/k$), exactly as $\ln\Gamma$ interpolates $\ln N!$: the heuristic "differentiate $1^r+\cdots+N^r=\zeta(-r)-\zeta(-r,N+1)$ k times at $r=0$" becomes precise through Theorem 8.25.

Corollary 8.27 (unit translates)**.** *For $k\ge 0$ and integer $n\ge 1$,*

$$\int_n^{n+1}\gamma_k(x)\,dx=-\frac{\ln^{k+1}n}{k+1};\qquad \textit{in particular}\quad \int_1^2\gamma_k(x)\,dx=0.$$

(The case $n=1$ is eqs. (2.18)–(2.19) of [26].)

8.12 Zeta jets and the conversion dictionary

Making (8.42) explicit requires the constants $\zeta^{(k)}(0)$ and, for the moment evaluations of Section 8.14, the jets at negative integers. All of them follow mechanically from the functional equation $\zeta(s)=2^s\pi^{s-1}\sin(\pi s/2)\,\Gamma(1-s)\,\zeta(1-s)$ by expanding $\Gamma(1-s)=\exp\bigl(\gamma s+\sum_{k\ge2}\zeta(k)s^k/k\bigr)$ and the Laurent series of $\zeta(1-s)$ at the pole.

Proposition 8.28 (jets at 0)**.** *With $L:=\ln(2\pi)$,*

$$\begin{aligned}
\zeta'(0)&=-\tfrac{L}{2},\\
\zeta''(0)&=\tfrac{\gamma^2}{2}+\gamma_1-\tfrac{\pi^2}{24}-\tfrac{L^2}{2},\\
\zeta'''(0)&=\gamma^3+3\gamma\gamma_1+\tfrac{3}{2}\gamma_2-\zeta(3)+\tfrac{3}{2}\gamma^2L+3\gamma_1L-\tfrac{\pi^2}{8}L-\tfrac{L^3}{2},\\
\zeta^{(4)}(0)&=\tfrac{3}{2}\gamma^4+4\gamma^3L+6\gamma^2\gamma_1+3\gamma^2L^2+\tfrac{\pi^2}{4}\gamma^2+12\gamma\gamma_1L+6\gamma\gamma_2+6\gamma_1L^2+\tfrac{\pi^2}{2}\gamma_1\\
&\quad+6\gamma_2L+2\gamma_3-\tfrac{L^4}{2}-\tfrac{\pi^2}{4}L^2-4\zeta(3)L-\tfrac{19\pi^4}{480}.
\end{aligned}$$

Proposition 8.29 (fifth jet at 0)**.** *With $L:=\ln(2\pi)$,*

$$\begin{aligned}
\zeta^{(5)}(0)&=2\gamma^5+\tfrac{15}{2}\,\gamma^4L+\gamma^3\Bigl(10L^2+\tfrac{5\pi^2}{6}\Bigr)+\gamma^2\Bigl(5L^3+\tfrac{5\pi^2}{4}\,L+10\,\zeta(3)\Bigr)\\
&\quad+\gamma\gamma_1\Bigl(30L^2+\tfrac{5\pi^2}{2}\Bigr)+\gamma_1\Bigl(10\gamma^3+30\gamma^2L+10L^3+\tfrac{5\pi^2}{2}\,L+20\,\zeta(3)\Bigr)\\
&\quad+\gamma_2\Bigl(15\gamma^2+30\gamma L+15L^2+\tfrac{5\pi^2}{4}\Bigr)+\gamma_3\,(10\gamma+10L)+\tfrac{5}{2}\,\gamma_4\\
&\quad-\zeta(3)\Bigl(10L^2+\tfrac{5\pi^2}{6}\Bigr)-12\,\zeta(5)-\tfrac{L^5}{2}-\tfrac{5\pi^2}{12}\,L^3-\tfrac{19\pi^4}{96}\,L.
\end{aligned}$$

The values $\zeta''(0)$ and $\zeta'''(0)$ are classical (Ramanujan [21]; Apostol [19]; Choudhury [24]). Apostol's Theorem 3 [19] in fact gives a closed form of $\zeta^{(n)}(0)$ at every order, in terms of the Laurent coefficients of $\Gamma(s)\zeta(s)$ at $s=1$, and prints the example $n=4$ in that vocabulary; we have not located the fully expanded forms above — nor any explicit form of $\zeta^{(5)}(0)$ — in print. The pattern continues to every order by the same expansion, which packages into the following closed recurrence.

Remark 8.30 (explicit recurrence for the jets at 0). In $\zeta(s)=\chi(s)\,\zeta(1-s)$ with $\chi(s)=2^s\pi^{s-1}\sin(\pi s/2)\,\Gamma(1-s)$, the simple zero of χ at $s=0$ cancels the pole of $\zeta(1-s)$; factor it out as $\chi(s)=\frac{s}{2}\,e^{g(s)}$ with $g(s):=\ln\bigl(2\chi(s)/s\bigr)=\ln\bigl[(2\pi)^s\,\frac{\sin(\pi s/2)}{\pi s/2}\,\Gamma(1-s)\bigr]$, analytic on $|s|<1$ and real on $(-1,1)$ (the bracket is positive there), $g(0)=0$. From $\ln\Gamma(1-s)=\gamma s+\sum_{k\ge2}\zeta(k)s^k/k$ and $\ln\bigl[\sin(\pi s/2)/(\pi s/2)\bigr]=-\sum_{m\ge1}\zeta(2m)(s/2)^{2m}/m$,

$$g(s)=(\gamma+L)\,s+\sum_{k\ge2}c_k s^k,\qquad c_k=\frac{\zeta(k)}{k}\cdot\begin{cases}1, & k\text{ odd},\\ 1-2^{1-k}, & k\text{ even},\end{cases}$$

so $c_2 = \frac{\pi^2}{24}$, $c_3 = \frac{\zeta(3)}{3}$, $c_4 = \frac{7\pi^4}{2880}$, $c_5 = \frac{\zeta(5)}{5}$. Since $s\,\zeta(1-s) = -1 + \sum_{m\ge1} \gamma_{m-1}\, s^m/(m-1)!$ is entire, the Leibniz rule gives

$$\zeta^{(n)}(0) = \frac{1}{2}\sum_{k=0}^{n} \binom{n}{k} B_k\big(1!\,c_1, 2!\,c_2, \dots, k!\,c_k\big)\, h_{n-k}, \qquad h_0 = -1, \quad h_m = m\,\gamma_{m-1}\ (m \ge 1),$$

where $c_1 = \gamma + L$ and B_k is the complete Bell polynomial, $B_k\big(g'(0), \dots, g^{(k)}(0)\big) = \big(e^g\big)^{(k)}(0)$: $B_0 = 1$, $B_1 = c_1$, $B_2 = c_1^2 + 2c_2$, $B_3 = c_1^3 + 6c_1c_2 + 6c_3$. The cases $n = 0, 1$ recover $\zeta(0) = -\frac{1}{2}$ and $\zeta'(0) = -\frac{L}{2}$; $n = 2, \dots, 5$ reproduce the two propositions verbatim.

Proposition 8.31 (jets at -1 and -2)**.** *Expanding the functional equation around $s = -1$ (where $\zeta(2-\varepsilon)$ is regular) and $s = -2$ (a trivial zero) gives, with $L = \ln(2\pi)$:*

$$\zeta(-1) = -\tfrac{1}{12}, \qquad \zeta'(-1) = \frac{\zeta'(2)}{2\pi^2} + \frac{1-\gamma-L}{12}, \tag{8.43}$$

$$\zeta''(-1) = -\frac{\zeta''(2)}{2\pi^2} + \frac{\zeta'(2)}{\pi^2}(\gamma + L - 1) - \frac{(\gamma+L)^2}{12} + \frac{\gamma+L}{6} + \frac{\pi^2}{144}, \tag{8.44}$$

$$\zeta(-2) = 0, \qquad \zeta'(-2) = -\frac{\zeta(3)}{4\pi^2}, \qquad \zeta''(-2) = \frac{\zeta(3)}{4\pi^2}\,(3 - 2\gamma - 2L) + \frac{\zeta'(3)}{2\pi^2}. \tag{8.45}$$

Remark 8.32 (Glaisher spelling). Since $\ln \mathsf{A} = \frac{1}{12} - \zeta'(-1)$, eq. (8.43) is equivalent to

$$\zeta'(2) = \zeta(2)\big(\gamma + \ln 2 + \ln \pi - 12 \ln \mathsf{A}\big),$$

the form returned by Mathematica's `FunctionExpand[Zeta'[2]]`. Either spelling may be used in all the moment formulas below; we keep $\zeta'(2), \zeta''(2), \zeta'(3)$ as the canonical second-layer constants.

8.13 Reflection formulas

Hurwitz's formula (see, e.g., [20, Ch. 12]), for $0 < p/q < 1$,

$$\zeta\Big(1-s, \frac{p}{q}\Big) = \frac{2\,\Gamma(s)}{(2\pi q)^s} \sum_{k=1}^{q} \Big[\cos\Big(\frac{\pi s}{2} - \frac{2\pi k p}{q}\Big)\Big] \zeta\Big(s, \frac{k}{q}\Big),$$

expanded around $s = 0$, expresses every $\gamma_n(p/q)$ through the jets $\zeta^{(j)}(0, k/q)$, $j \le n+1$. Taking the part odd under $p \mapsto q - p$ isolates the sine sums S_j and yields reflection formulas. Write $E(s) := \exp\big(-\Lambda_q s + \sum_{k\ge2} (-1)^k \zeta(k) s^k/k\big) = \sum_i e_i s^i$ for the prefactor expansion $(2\Gamma(s)(2\pi q)^{-s} = \frac{2}{s}\,E(s))$.

Theorem 8.33 (reflection, general mechanism)**.** *For $n \ge 1$ and $0 < p < q$,*

$$\gamma_n\Big(\frac{p}{q}\Big) - \gamma_n\Big(1 - \frac{p}{q}\Big) \;=\; 4\,n! \sum_{i+2l+1+j\,=\,n+1} e_i\, \frac{(-1)^l (\pi/2)^{2l+1}}{(2l+1)!}\, \frac{S_j(p/q)}{j!}\,.$$

Explicitly, with $\Lambda = \Lambda_q = \gamma + \ln 2\pi q$ (the letter distinguishes it from the Glaisher constant A):

$$\gamma_2\Big(\tfrac{p}{q}\Big) - \gamma_2\Big(1 - \tfrac{p}{q}\Big) = 2\pi S_2 - 4\pi\Lambda\, S_1 + \Big(2\Lambda^2 + \tfrac{\pi^2}{6}\Big)\pi S_0, \tag{8.46}$$

$$\gamma_3\Big(\tfrac{p}{q}\Big) - \gamma_3\Big(1 - \tfrac{p}{q}\Big) = 2\pi S_3 - 6\pi\Lambda\, S_2 + \Big(6\Lambda^2 + \tfrac{\pi^2}{2}\Big)\pi S_1 - \Big(2\Lambda^3 + \tfrac{\pi^2}{2}\Lambda + 4\zeta(3)\Big)\pi S_0, \tag{8.47}$$

$$\begin{aligned}\gamma_4\Big(\tfrac{p}{q}\Big) - \gamma_4\Big(1 - \tfrac{p}{q}\Big) &= 2\pi S_4 - 8\pi\Lambda\, S_3 + \big(12\Lambda^2 + \pi^2\big)\pi S_2 - \big(8\Lambda^3 + 2\pi^2\Lambda + 16\zeta(3)\big)\pi S_1 \\ &\quad + \Big(2\Lambda^4 + \pi^2\Lambda^2 + 16\Lambda\,\zeta(3) + \tfrac{19\pi^4}{120}\Big)\pi S_0.\end{aligned} \tag{8.48}$$

Proof. Split the cosine, $\cos(\pi s/2-\theta)=\cos(\pi s/2)\cos\theta+\sin(\pi s/2)\sin\theta$. Under $p\mapsto q-p$ the cosine sums are even and the sine sums odd, so the odd part of $\zeta(1-s,p/q)$ equals $\frac{2}{s}\,E(s)\sin(\pi s/2)\sum_j S_j\,s^j/j!$. By (8.40), $\zeta(1-s,p/q)=-\frac{1}{s}+\sum_n\gamma_n(p/q)\,s^n/n!$, whose odd part has s^n-coefficient $\frac{1}{2\,n!}\big[\gamma_n(p/q)-\gamma_n(1-p/q)\big]$. Matching coefficients of s^n (i.e. s^{n+1} of $E\cdot\sin\cdot\sum S_j s^j/j!$) gives the general formula; instantiating $e_0=1$, $e_1=-\Lambda$, $e_2=\frac{\Lambda^2}{2}+\frac{\pi^2}{12}$, $e_3=-\frac{\Lambda^3}{6}-\frac{\pi^2}{12}\Lambda-\frac{\zeta(3)}{3}$, $e_4=\frac{\Lambda^4}{24}+\frac{\pi^2}{24}\Lambda^2+\frac{\zeta(3)}{3}\Lambda+\frac{\pi^4}{288}+\frac{\pi^4}{360}$ yields (8.46)–(8.48). □

Formula (8.46) is eq. (89) of Blagouchine [6], where a footnote records that it also appears in unpublished work of D. Connon; (8.47) and (8.48) appear to be new. The same expansion, one order down, recovers Malmsten's reflection formula for γ_1 [6]; an elementary alternative proof of that $n=1$ case, via Kummer's Fourier expansion of $\ln\Gamma$ rather than Hurwitz's functional equation, is given by Kouba [30]. For $q=3$ and $q=4$ the sums collapse ($S_j(\frac{1}{3})=\frac{\sqrt{3}}{2}\big[\zeta^{(j)}(0,\frac{1}{3})-\zeta^{(j)}(0,\frac{2}{3})\big]$, $S_j(\frac{1}{4})=\zeta^{(j)}(0,\frac{1}{4})-\zeta^{(j)}(0,\frac{3}{4})$), so (8.46)–(8.48) can be *inverted* to express the odd jet differences through the Stieltjes differences $\gamma_n(p)-\gamma_n(1-p)$ — the key to Section 8.15.

8.14 Moments

Theorem 8.34 (Hurwitz transform and variants)**.** *For $m\ge 1$ and $\operatorname{Re}s<1$,*

$$\int_0^1 x^m\,\zeta(s,x)\,dx=-\sum_{j=1}^{m}\frac{m!}{(m-j+1)!}\,\frac{\zeta(s-j)}{(s-1)(s-2)\cdots(s-j)}, \tag{8.49}$$

and $\int_0^1\zeta(s,x)\,dx=0$. The same recursion with boundary terms $\zeta(\sigma,\frac{1}{2})=(2^\sigma-1)\zeta(\sigma)$ and $\zeta(\sigma,2)=\zeta(\sigma)-1$ gives the analogous closed forms for $\int_{1/2}^1$ and $\int_1^2$.

Proof. Since $\zeta(s,x)=-\partial_x\zeta(s-1,x)/(s-1)$, integration by parts gives $I_m(s)=-\zeta(s-1)/(s-1)+\frac{m}{s-1}I_{m-1}(s-1)$ for $m\ge1$. In the stated half-plane the factor x^m kills the lower boundary. For the base case $m=0$, the shift relation gives equal limits of $\zeta(s-1,x)$ at $x=0+$ and $x=1$ when $\operatorname{Re}s<1$; these generally nonzero values cancel, giving $I_0(s)=0$. Unrolling yields (8.49), which then extends meromorphically in s. □

For the unit interval, (8.49) is Theorem 3.7 of Espinosa–Moll [28]. Expanding (8.49) at $s=1+t$ against $\int_0^1 x^m\zeta(1+t,x)\,dx=\frac{1}{(m+1)t}+M_0-M_1t+\frac{M_2}{2}t^2-\cdots$, where $M_k=\int_0^1 x^m\gamma_k(x)\,dx$, reads off every Stieltjes moment. The expansion $1/[(t-1)\cdots(t-j+1)]=\frac{(-1)^{j-1}}{(j-1)!}\big(1+H_{j-1}t+\frac{H_{j-1}^2+H_{j-1}^{(2)}}{2}t^2+\cdots\big)$ gives the coefficients in closed form:

Theorem 8.35 (general first moment)**.** *For every integer $m\ge1$,*

$$\int_0^1 x^m\,\gamma_1(x)\,dx=\sum_{j=1}^{m}(-1)^{j-1}\binom{m}{j-1}\left[\frac{\zeta''(1-j)}{2}+H_{j-1}\,\zeta'(1-j)+\frac{H_{j-1}^2+H_{j-1}^{(2)}}{2}\,\zeta(1-j)\right].$$

The γ_n-moment carries the analogous formula with the order-$(n+1)$ jet leading and complete homogeneous symmetric functions of $1,\frac{1}{2},\dots,\frac{1}{j-1}$ as coefficients.

Combining Theorems 8.34 and 8.35 with the dictionary of Proposition 8.31 produces the following table. Items (8.50)–(8.53) are restated from the corpus that motivated this work; the remaining

evaluations appear to be new.

$$\int_{1/2}^{1} \gamma_1(x)\,dx = \frac{\gamma_1}{2} + \frac{\gamma^2}{4} - \frac{\pi^2}{48} + \frac{\ln^2 2}{2} - \frac{\ln^2 \pi}{4}, \tag{8.50}$$

$$\int_{1}^{2} x\,\gamma_1(x)\,dx = 1 + \frac{\gamma_1}{2} + \frac{\gamma^2}{4} - \frac{\pi^2}{48} - \frac{\ln^2 2\pi}{4}, \tag{8.51}$$

$$\int_{0}^{1} x^2\,\gamma_1(x)\,dx = \frac{\gamma_1}{2} + \frac{\gamma^2}{3} - \frac{\pi^2}{36} - \frac{\ln^2 2\pi}{6} + \frac{\gamma}{6}\ln 2\pi - \frac{\zeta'(2)}{\pi^2}\,(\gamma + \ln 2\pi) + \frac{\zeta''(2)}{2\pi^2}, \tag{8.52}$$

$$\begin{aligned}\int_{1/2}^{1} x\,\gamma_1(x)\,dx = {} & \frac{\gamma_1}{2} + \frac{5\gamma^2}{16} - \frac{5\pi^2}{192} + \frac{\ln^2 2}{8} - \frac{\ln 2\,\ln \pi}{6} - \frac{3\ln^2 \pi}{16} + \gamma\Big(\frac{\ln 2}{12} + \frac{\ln \pi}{8}\Big) \\ & - \frac{\zeta'(2)}{4\pi^2}\,(3\gamma + 2\ln 2 + 3\ln \pi) + \frac{3\,\zeta''(2)}{8\pi^2},\end{aligned} \tag{8.53}$$

$$\int_{0}^{1} x\,\gamma_1(x)\,dx = \frac{\zeta''(0)}{2} = \frac{\gamma^2}{4} + \frac{\gamma_1}{2} - \frac{\pi^2}{48} - \frac{\ln^2 2\pi}{4}, \tag{8.54}$$

$$\begin{aligned}\int_{0}^{1} x^3\,\gamma_1(x)\,dx = {} & \frac{3\gamma^2}{8} + \frac{\gamma_1}{2} - \frac{\pi^2}{32} + \frac{\gamma\ln 2\pi}{4} - \frac{\ln^2 2\pi}{8} - \frac{3\,\big(\zeta(3) + 2\zeta'(2)\big)}{4\pi^2}\,(\gamma + \ln 2\pi) \\ & + \frac{3\,\big(\zeta'(3) + \zeta''(2)\big)}{4\pi^2},\end{aligned} \tag{8.55}$$

$$\int_{1}^{2} x^2\,\gamma_1(x)\,dx = \frac{9}{4} + \frac{3\gamma_1}{2} + \frac{5\gamma^2}{6} - \frac{5\pi^2}{72} + \frac{\gamma\ln 2\pi}{6} - \frac{2\ln^2 2\pi}{3} - \frac{\zeta'(2)}{\pi^2}(\gamma + \ln 2\pi) + \frac{\zeta''(2)}{2\pi^2}, \tag{8.56}$$

$$\int_{0}^{1} x\,\gamma_2(x)\,dx = -\frac{\zeta'''(0)}{3} = \frac{\zeta(3)}{3} - \frac{\gamma^3}{3} - \gamma\gamma_1 - \frac{\gamma_2}{2} - \frac{\gamma^2}{2}\ln 2\pi - \gamma_1 \ln 2\pi + \frac{\pi^2}{24}\ln 2\pi + \frac{\ln^3 2\pi}{6}, \tag{8.57}$$

$$\int_{1}^{2} x\,\gamma_2(x)\,dx = \int_{0}^{1} x\,\gamma_2(x)\,dx - 2, \qquad \int_{1/2}^{1} \gamma_2(x)\,dx = -\Gamma_2\big(\tfrac{1}{2}\big). \tag{8.58}$$

At $m = 3$ the constants $\zeta(3)$ and $\zeta'(3)$ enter through (8.45), and each further unit of m opens one more column of jets at the next negative integer; the second moment of γ_1 over $[1/2, 1]$ and the higher γ_2-moments are equally mechanical but bulkier, and we omit them.

8.15 Explicit values of Γ_n

8.15.1 The first antiderivative

Theorem 8.36 (Γ_1 in closed form)**.** *For all $p > 0$,*

$$\Gamma_1(p) = c + \tfrac{1}{2}\,\zeta''(0, p), \qquad c := -\tfrac{1}{2}\zeta''(0) = \frac{\pi^2}{48} + \frac{\ln^2 2\pi}{4} - \frac{\gamma^2}{4} - \frac{\gamma_1}{2}.$$

Moreover $\int_0^1 \Gamma_1(x)\,dx = c$, equivalently (8.54) (since $\int_0^1 \zeta''(0, x)\,dx = 0$ [9, 25]).

Theorem 8.37 (half-integral and even-sum values)**.**

$$\Gamma_1\tfrac{1}{2} = \frac{\pi^2}{48} - \frac{\gamma^2}{4} - \frac{\gamma_1}{2} - \frac{\ln^2 2}{4} - \frac{\ln 2\,\ln 2\pi}{2} + \frac{\ln^2 2\pi}{4},$$

$$\Gamma_1\tfrac{1}{3} + \Gamma_1\tfrac{2}{3} = 2c - \frac{\ln^2 3}{4} - \frac{\ln 3\,\ln 2\pi}{2}, \qquad \Gamma_1\tfrac{1}{4} + \Gamma_1\tfrac{3}{4} = 2c - \frac{3\ln^2 2}{4} - \frac{\ln 2\,\ln 2\pi}{2},$$

$$\Gamma_1\tfrac{1}{6} + \Gamma_1\tfrac{5}{6} = 2c - \frac{\ln 2\,\ln 3}{2}.$$

Proof. $\zeta(s,\frac12) = (2^s-1)\zeta(s)$ gives $\zeta''(0,\frac12) = -\frac12\ln^2 2 - \ln 2\,\ln(2\pi)$; the multiplication identities $\zeta(s,\frac13)+\zeta(s,\frac23) = (3^s-1)\zeta(s)$, $\zeta(s,\frac14)+\zeta(s,\frac34) = (4^s-2^s)\zeta(s)$, $\zeta(s,\frac16)+\zeta(s,\frac56) = (6^s-3^s-2^s+1)\zeta(s)$, twice differentiated at $s=0$, give the even sums (for $q=6$ the $\zeta(0)$-coefficient $\ln^2 6 - \ln^2 3 - \ln^2 2 = 2\ln 2\ln 3$ and the $\zeta'(0)$-coefficient $\ln 6 - \ln 3 - \ln 2 = 0$ produce the strikingly short third identity). Apply Theorem 8.36. □

Theorem 8.38 (third and quarter values)**.**

$$\Gamma_1\tfrac{1}{3} = \frac{\pi^2}{72} - \frac{\ln^2 2\pi}{3} + \frac{\ln^2 3}{24} - \frac{2}{3}\ln 3\,\ln 2\pi - \frac{\gamma}{12}\left(4\gamma + 8\ln 2\pi - \ln 3\right) + \left(\gamma + \ln 6\pi\right)\ln\Gamma\tfrac{1}{3}$$
$$- \frac{\gamma_1}{2} + \frac{\sqrt3}{12\pi}\left(\gamma_2\tfrac{1}{3} - \gamma_2\tfrac{2}{3}\right),$$

$$\Gamma_1\tfrac{1}{4} = \frac{\pi^2}{96} - \frac{3}{8}\ln^2\pi - \frac{9}{4}\ln 2\,\ln 2\pi - \frac{\gamma}{8}\left(3\gamma + 8\ln 2 + 6\ln\pi\right) + \left(\gamma + \ln 8\pi\right)\ln\Gamma\tfrac{1}{4}$$
$$- \frac{\gamma_1}{2} + \frac{1}{8\pi}\left(\gamma_2\tfrac{1}{4} - \gamma_2\tfrac{3}{4}\right),$$

and $\Gamma_1(\frac23)$, $\Gamma_1(\frac34)$ *follow from Theorem 8.37. Furthermore, with the abbreviation* $D_j := \zeta^{(j)}(0,\frac13) - \zeta^{(j)}(0,\frac23)$,

$$\Gamma_1\tfrac{1}{6} - \Gamma_1\tfrac{5}{6} = \frac{\ln 2\,\ln 18}{6} + \ln 2\left(2\ln\Gamma\tfrac{1}{3} + \frac{\ln 3}{6} - \ln 2 - \ln\pi\right) + D_2,$$

which together with the even sum of Theorem 8.37 gives $\Gamma_1(\frac16)$ *and* $\Gamma_1(\frac56)$ *explicitly — in the* same *obstruction atom* $\gamma_2(\frac13) - \gamma_2(\frac23)$*, since inverting (8.46) at* $q=3$ *expresses* $D_2 = \frac{2}{\sqrt3}S_2(\frac13)$ *through it.*

Proof. By Theorem 8.36 everything reduces to the single jets $\zeta''(0,p)$ at $p = \frac13, \frac14, \frac16$. Their even parts are in Theorem 8.37; their odd parts are the inversions of (8.46) at $q = 3, 4$:

$$S_2 = \frac{1}{2\pi}\Big[\gamma_2\tfrac{p}{q} - \gamma_2\big(1-\tfrac{p}{q}\big) + 4\pi\Lambda\,S_1 - \Big(2\Lambda^2 + \tfrac{\pi^2}{6}\Big)\pi S_0\Big],$$

with $S_1(\frac13) = \frac{\sqrt3}{2}\big(2\ln\Gamma(\frac13) - \ln\frac{2\pi}{\sqrt3}\big)$, $S_0(\frac13) = \frac{1}{2\sqrt3}$, $S_1(\frac14) = 2\ln\Gamma(\frac14) - \ln(\pi\sqrt2)$, $S_0(\frac14) = \frac12$. For $q=6$, the Dirichlet character χ_{-3} gives $\zeta(s,\frac16) - \zeta(s,\frac56) = (6^s+3^s)\,L(s,\chi_{-3})$ while $\zeta(s,\frac13) - \zeta(s,\frac23) = 3^s\,L(s,\chi_{-3})$; eliminating the jets of $L(s,\chi_{-3})$ at $s=0$ ($L(0,\chi_{-3}) = \frac13$, $L'(0,\chi_{-3}) = 2\ln\Gamma(\frac13) + \frac{\ln 3}{6} - \ln 2 - \ln\pi$) yields the displayed transport identity. □

8.15.2 The second and third antiderivatives

Theorem 8.39 (Γ_2 values)**.**

$$\Gamma_2\tfrac{1}{2} = \frac{\gamma_2}{2} + \frac{\gamma^3}{3} - \frac{\zeta(3)}{3} + \ln^3 2 - \frac{\ln^3\pi}{6} + \frac{\ln\pi}{24}\left(12\gamma^2 - \pi^2 + 24\ln^2 2\right) + (\gamma + \ln\pi)\,\gamma_1,$$

and at the third and quarter points, with $Z_3(\frac{1}{q}) := \zeta'''(0,\frac{1}{q}) + \zeta'''(0,1-\frac{1}{q})$ *and* $D_3(\frac{1}{q}) := \zeta'''(0,\frac{1}{q}) - \zeta'''(0,1-\frac{1}{q})$,

$$\Gamma_2\big(\tfrac{1}{q}\big) = -\frac{1}{3}\Big[\frac{Z_3 + D_3}{2} - \zeta'''(0)\Big], \qquad q \in \{3,4\},$$

where the even parts are elementary,

$$Z_3\big(\tfrac{1}{3}\big) = -\tfrac{\ln^3 3}{2} - \tfrac{3\ln^2 3\,\ln 2\pi}{2} + 3\ln 3\,\zeta''(0), \qquad Z_3\big(\tfrac{1}{4}\big) = \partial_s^3\big[(4^s - 2^s)\zeta(s)\big]_{s=0},$$

and the odd parts are the inversions of (8.47)*:* $D_3 = \frac{2}{\sqrt{3}} S_3$ *at* $q=3$, $D_3 = S_3$ *at* $q=4$, *with*

$$S_3 = \frac{1}{2\pi}\Big[\gamma_3\big(\tfrac{1}{q}\big) - \gamma_3\big(1-\tfrac{1}{q}\big) + 6\pi\Lambda\, S_2 - \Big(6\Lambda^2 + \tfrac{\pi^2}{2}\Big)\pi S_1 + \Big(2\Lambda^3 + \tfrac{\pi^2}{2}\Lambda + 4\zeta(3)\Big)\pi S_0\Big]$$

and S_2 *as in the proof of Theorem 8.38. Thus* $\Gamma_2(\frac{1}{3})$ *and* $\Gamma_2(\frac{1}{4})$ *are explicit in the constants* $\big\{\gamma, \gamma_1, \gamma_2, \zeta(3), \ln 2, \ln 3, \ln\pi, \ln\Gamma(\frac{1}{q}),\ \gamma_2(\frac{1}{q}) - \gamma_2(1-\frac{1}{q}),\ \gamma_3(\frac{1}{q}) - \gamma_3(1-\frac{1}{q})\big\}$, *the two Stieltjes differences being the only non-classical ingredients.*

Proof. $\Gamma_2(p) = -\frac{1}{3}[\zeta'''(0,p) - \zeta'''(0)]$ by Theorem 8.25. At $p = \frac{1}{2}$ use $\zeta'''(0,\frac{1}{2}) = \partial_s^3[(2^s - 1)\zeta(s)]_{s=0} = \ln^3 2\,\zeta(0) + 3\ln^2 2\,\zeta'(0) + 3\ln 2\,\zeta''(0)$ and Proposition 8.28; at $p = \frac{1}{3}, \frac{1}{4}$ split into even and odd parts as displayed. □

Corollary 8.40 (Γ_2 even sums)**.** *With* $L = \ln 2\pi$,

$$\begin{aligned}
\Gamma_2\big(\tfrac{1}{3}\big) + \Gamma_2\big(\tfrac{2}{3}\big) &= \tfrac{2}{3}\,\zeta'''(0) - \ln 3\,\zeta''(0) + \tfrac{L}{2}\ln^2 3 + \tfrac{\ln^3 3}{6},\\
\Gamma_2\big(\tfrac{1}{4}\big) + \Gamma_2\big(\tfrac{3}{4}\big) &= \tfrac{2}{3}\,\zeta'''(0) - \ln 2\,\zeta''(0) + \tfrac{3L}{2}\ln^2 2 + \tfrac{7\ln^3 2}{6},\\
\Gamma_2\big(\tfrac{1}{6}\big) + \Gamma_2\big(\tfrac{5}{6}\big) &= \tfrac{2}{3}\,\zeta'''(0) + L\,\ln 2\,\ln 3 + \tfrac{\ln 2\,\ln 3\,\ln 6}{2},
\end{aligned}$$

so that $\Gamma_2(\frac{2}{3})$, $\Gamma_2(\frac{3}{4})$ *follow from Theorem 8.39; as in Theorem 8.37, the* $\zeta''(0)$*-term vanishes at* $q=6$ *because* $\ln 6 - \ln 3 - \ln 2 = 0$. *(Generally, with the inclusion–exclusion combination* $\zeta(s,\frac{1}{q}) + \zeta(s,1-\frac{1}{q}) = \big(\sum_i c_i m_i^s\big)\zeta(s)$, *the even sum equals* $\frac{2}{3}\zeta'''(0) + \frac{1}{6}\sum_i c_i \ln^3 m_i + \frac{L}{2}\sum_i c_i \ln^2 m_i - \zeta''(0)\sum_i c_i \ln m_i$.*)*

Theorem 8.41 (Γ_3 values)**.**

$$\begin{aligned}
\Gamma_3\big(\tfrac{1}{2}\big) = {}& -\frac{7\ln^4 2}{4} + \frac{\ln^4\pi}{8} - \frac{3\gamma^4}{8} - \frac{\gamma_3}{2} + \frac{19\pi^4}{1920} + \gamma_1\Big(3\ln^2 2 - \frac{3\gamma^2}{2} - \frac{3\ln^2\pi}{2} - \frac{\pi^2}{8}\Big)\\
& - \frac{3\gamma_2}{2}(\gamma + \ln\pi) + \zeta(3)\ln\pi + \gamma^2\Big(\frac{3\ln^2 2}{2} - \frac{3\ln^2\pi}{4} - \frac{\pi^2}{16}\Big) - \gamma^3\ln\pi - 3\gamma\gamma_1\ln\pi\\
& - 3\ln\pi\,\ln^3 2 - \frac{3\ln^2 2\,\ln^2\pi}{2} - \frac{\pi^2\ln^2 2}{8} + \frac{\pi^2\ln^2\pi}{16},
\end{aligned}$$

and $\Gamma_3(\frac{1}{3})$, $\Gamma_3(\frac{1}{4})$ *are given by* $\Gamma_3(\frac{1}{q}) = \frac{1}{4}[\zeta^{(4)}(0,\frac{1}{q}) - \zeta^{(4)}(0)]$ *with the even part* $\partial_s^4[(q\text{-orbit})\,\zeta(s)]_{s=0}$ *as in Theorem 8.39 and the odd part inverted from* (8.48)*; the resulting expressions are explicit in the same vocabulary augmented by* $\gamma_4(\frac{1}{q}) - \gamma_4(1-\frac{1}{q})$.

8.15.3 The fourth antiderivative at $1/2$

Theorem 8.42 ($\Gamma_4(1/2)$)**.**

$$\begin{aligned}\Gamma_4\tfrac{1}{2} = {}& \frac{\gamma_4}{2} + 3\ln^5 2 - \frac{12\,\zeta(5)}{5} - \frac{\ln^5\pi}{10} + \frac{2\gamma^5}{5} + \frac{3\gamma^4\ln\pi}{2} \\ & + \gamma_1\Big(2\gamma^3 - 12\ln^3 2 + 2\ln^3\pi + 4\zeta(3) + \frac{\pi^2\ln\pi}{2} - 12\ln\pi\,\ln^2 2 + 6\gamma^2\ln\pi\Big) \\ & + \gamma\gamma_1\Big(\frac{\pi^2}{2} - 12\ln^2 2 + 6\ln^2\pi\Big) + \gamma_2\Big(3\gamma^2 - 6\ln^2 2 + 3\ln^2\pi + \frac{\pi^2}{4} + 6\gamma\ln\pi\Big) \\ & + 2\gamma_3\,(\gamma + \ln\pi) + \zeta(3)\Big(4\ln^2 2 - 2\ln^2\pi - \frac{\pi^2}{6}\Big) \\ & + \gamma^2\Big(\ln^3\pi - 6\ln^3 2 + 2\zeta(3) - 6\ln\pi\,\ln^2 2 + \frac{\pi^2\ln\pi}{4}\Big) + \gamma^3\Big(2\ln^2\pi - 4\ln^2 2 + \frac{\pi^2}{6}\Big) \\ & + \frac{\pi^2\ln^3 2}{2} + 2\ln^2 2\,\ln^3\pi + 6\ln^3 2\,\ln^2\pi + 7\ln\pi\,\ln^4 2 \\ & - \frac{19\pi^4\ln\pi}{480} - \frac{\pi^2\ln^3\pi}{12} + \frac{\pi^2\ln\pi\,\ln^2 2}{2}.\end{aligned}$$

Proof. $\Gamma_4(\frac{1}{2}) = -\frac{1}{5}[\zeta^{(5)}(0,\frac{1}{2}) - \zeta^{(5)}(0)]$, with

$$\zeta^{(5)}\big(0,\tfrac{1}{2}\big) = \partial_s^5\big[(2^s-1)\zeta(s)\big]_{s=0} = \sum_{k=0}^{4}\binom{5}{k}\ln^{5-k}2\;\zeta^{(k)}(0)$$

(the $k=5$ term carries the factor $2^0 - 1 = 0$) and Proposition 8.29. □

The pattern of Theorems 8.39–8.42 is general: each rung Γ_n adds the next Stieltjes constant γ_n and the functional-equation zeta terms, and one degree to the logarithmic monomials at $p = \frac{1}{2}$; at $p = \frac{1}{3}, \frac{1}{4}$ it adds the single new obstruction atom $\gamma_{n+1}(p) - \gamma_{n+1}(1-p)$, delivered by the order-$(n+1)$ reflection formula of Theorem 8.33. At denominators such as $5, 8, 12$ the even character decomposition introduces additional coordinates (the multiplication identities no longer close over the principal character), and the values $\Gamma_n(p/q)$ additionally involve the even-character combinations of $\zeta^{(n+1)}(0,\cdot)$ — the obstruction layer studied, at $n=1$, in [6].

8.16 The negative-order polygamma floor

The ladder of Section 8.11 integrates the coefficient functions of the Laurent expansion at $s=1$ in their argument. The same Hurwitz-jet calculus has a second, equally natural direction, and this section — a second main act rather than an appendix — follows it downward. Integrating in the *argument* at the s-jet floor produces the polygamma functions of negative integer order ("negapolygamma", a term Adamchik [18] credits to Gosper): for real $z > 0$,

$$\psi^{(-1)}(z) = \ln\Gamma(z), \qquad \psi^{(-n)}(z) = \int_0^z \psi^{(-(n-1))}(t)\,dt = \frac{1}{(n-2)!}\int_0^z (z-t)^{n-2}\ln\Gamma(t)\,dt,$$

which is precisely Mathematica's `PolyGamma[-n, z]` (all identities below are stated on $0 < z < 1$, where the rational arguments live, and extend to $z > 0$ by the same formulas wherever both sides are defined). Everything built for the Stieltjes side transfers wholesale: the ladder identity has a one-line analogue, the reflection mechanism of Section 8.13 reappears at the jet floor with bare

Clausen values in place of the sine sums S_j, the obstruction constants again organize along Dirichlet characters — and the bridge identities for those constants close a circle the Stieltjes side leaves open. The tools: a one-line ladder, reflection and multiplication laws, bridge identities, and explicit rational-argument values.

8.16.1 The ladder

Theorem 8.43. *For $n \geq 2$ and $z > 0$,*

$$\psi^{(-n)}(z) = \frac{\zeta'(1-n,z) - \zeta'(1-n)}{(n-1)!} + Q_n(z),$$

where Q_n is an explicit polynomial with coefficients in $\mathbb{Q}\big[\ln 2\pi,\, \zeta'(-1), \ldots, \zeta'(2-n)\big]$:

$$\begin{aligned}
Q_2(z) &= \tfrac{z}{2}\ln 2\pi + \tfrac{z(1-z)}{2},\\
Q_3(z) &= \tfrac{z^2}{4}\ln 2\pi - \tfrac{z^3}{4} + \tfrac{3z^2}{8} - \tfrac{z}{24} - z\,\zeta'(-1),\\
Q_4(z) &= \tfrac{z^3}{12}\ln 2\pi - \tfrac{11z^4}{144} + \tfrac{11z^3}{72} - \tfrac{5z^2}{144} - \tfrac{z^2}{2}\zeta'(-1) - \tfrac{z}{2}\zeta'(-2),
\end{aligned}$$

and in general Q_n has leading term $-H_{n-1}z^n/n!$ and $\ln 2\pi$-coefficient $z^{n-1}/(2\,(n-1)!)$.

Proof. $\int_0^z \zeta(s,t)\,dt = -[\zeta(s-1,z) - \zeta(s-1)]/(s-1)$ for $\operatorname{Re} s < 1$; differentiating at $s = -k$,

$$\int_0^z \zeta'(-k,t)\,dt = \frac{\zeta'(-k-1,z) - \zeta'(-k-1)}{k+1} + \frac{\zeta(-k-1,z) - \zeta(-k-1)}{(k+1)^2},$$

with $\zeta(-m,z) = -B_{m+1}(z)/(m+1)$. Iterate from $\psi^{(-1)} = \zeta'(0,z) + \frac{1}{2}\ln 2\pi$: each step shifts the single ζ'-atom by one and deposits Bernoulli and constant terms into the polynomial. □

An equivalent expansion is Adamchik's Proposition 2 [18]; the spelling above isolates the one z-dependent transcendental.

8.16.2 Reflection

Theorem 8.44 (reflection at the jet floor)**.** *For every integer $k \geq 0$ and $0 < z < 1$,*

$$\zeta'(-k,z) + (-1)^k\,\zeta'(-k,1-z) = (-1)^{\lfloor k/2\rfloor}\,\frac{k!}{(2\pi)^k}\,\operatorname{Cl}_{k+1}(2\pi z),$$

where $\operatorname{Cl}_{2m}(\theta) = \operatorname{Im}\operatorname{Li}_{2m}(e^{i\theta})$, $\operatorname{Cl}_{2m+1}(\theta) = \operatorname{Re}\operatorname{Li}_{2m+1}(e^{i\theta})$.

Proof. In Hurwitz's formula, written as $\zeta(1-s,z) = \frac{2\Gamma(s)}{(2\pi)^s}[\cos(\frac{\pi s}{2})C_s(z) + \sin(\frac{\pi s}{2})S_s(z)]$ with C_s, S_s the cosine/sine Dirichlet series, the combination even (odd) under $z \mapsto 1-z$ keeps only the C (S) term. At the integer $s = k+1$, exactly one of $\cos\frac{\pi s}{2}$, $\sin\frac{\pi s}{2}$ vanishes, and it is the one attached to the parity-matching combination; differentiating in s there, the product rule retains only the term in which d/ds falls on the vanishing trigonometric factor, so a single *bare* Clausen value survives and no derivative of the Dirichlet series enters. Collecting constants gives the display. □

For $k \geq 1$ this is the real part of Adamchik's Proposition 1 [18], stated there for every positive integer order with the Bernoulli term as the imaginary part (equivalently Miller–Adamchik [31], eq. (21)); the bare-Clausen real spelling, the $k = 0$ inclusion, and the mechanism note are the reformulation here. $k = 0$ is Kummer-flavoured ($\operatorname{Cl}_1(2\pi z) = -\ln(2\sin\pi z)$). The parity-*opposite*

combination does not close: it produces the series $\sum \ln n\,\{\cos\,|\,\sin\}(2\pi n z)/n^{k+1}$ — the generalized-Glaisher atoms of §8.16.4.

Propagating Theorem 8.44 through the ladder yields reflection formulas at the $\psi^{(-n)}$-level — a relation which Espinosa and Moll [27] record as not known to them ("*We have been unable to find a generalization of the other well-known functional relation for the polygamma function*"):

Theorem 8.45 (negapolygamma reflection)**.**

$$\begin{aligned}
\psi^{(-2)}(z)-\psi^{(-2)}(1-z) &= \frac{\mathrm{Cl}_2(2\pi z)}{2\pi}+\Big(z-\tfrac12\Big)\ln 2\pi,\\
\psi^{(-3)}(z)+\psi^{(-3)}(1-z) &= -\frac{\mathrm{Cl}_3(2\pi z)}{4\pi^2}+\Big(\tfrac{z^2}{2}-\tfrac{z}{2}+\tfrac14\Big)\ln 2\pi-\zeta'(-1)-\zeta'(-2)+\tfrac{1}{12},\\
\psi^{(-4)}(z)-\psi^{(-4)}(1-z) &= -\frac{\mathrm{Cl}_4(2\pi z)}{8\pi^3}+\Big(\tfrac{z^3}{6}-\tfrac{z^2}{4}+\tfrac{z}{4}-\tfrac{1}{12}\Big)\ln 2\pi\\
&\quad+\Big(\tfrac12-z\Big)\big(\zeta'(-1)+\zeta'(-2)\big)+\tfrac{z}{12}-\tfrac{1}{24},\\
\psi^{(-5)}(z)+\psi^{(-5)}(1-z) &= \frac{\mathrm{Cl}_5(2\pi z)}{16\pi^4}+\Big(\tfrac{z^4}{24}-\tfrac{z^3}{12}+\tfrac{z^2}{8}-\tfrac{z}{12}+\tfrac{1}{48}\Big)\ln 2\pi\\
&\quad+\Big(\tfrac{z-z^2}{2}\Big)\big(\zeta'(-1)+\zeta'(-2)\big)\\
&\quad-\tfrac{\zeta'(-1)}{6}-\tfrac{\zeta'(-2)}{4}-\tfrac{\zeta'(-3)}{6}-\tfrac{\zeta'(-4)}{12}+\tfrac{z^2-z}{24}+\tfrac{49}{4320}.
\end{aligned}$$

8.16.3 Multiplication

Differentiating the Hurwitz distribution relation at $s=-k$ gives, for all $k\ge 0$, $m\ge 2$,

$$\sum_{j=0}^{m-1}\zeta'\Big(-k,\,z+\frac{j}{m}\Big)=m^{-k}\Big[\zeta'(-k,mz)-\ln m\,\frac{B_{k+1}(mz)}{k+1}\Big] \tag{8.59}$$

($k=0$ is Gauss's multiplication theorem for $\ln\Gamma$), and through the ladder,

$$\sum_{j=0}^{m-1}\psi^{(-n)}\Big(z+\frac{j}{m}\Big)=m^{1-n}\Big[\psi^{(-n)}(mz)-\ln m\,\frac{B_n(mz)}{n!}\Big]+\rho_{n,m}(z)$$

with elementary $\rho_{n,m}$; for $n=2$,

$$\rho_{2,m}(z)=\frac{(m-1)(2z+1)}{4}\ln 2\pi+\frac{m^2-1}{12m}-\frac{m^2-1}{m}\,\zeta'(-1),$$

so only the *rational* part of the remainder is $(m^2-1)/(12m)$. The ψ-level law is, in the "balanced" convention of their negapolygamma [29], Theorem 3.2 of Espinosa–Moll [27]; the form above is its transcription to the Wolfram convention.

8.16.4 Bridges

At rational arguments the layers $\zeta'(-k,p/q)$ decompose along Dirichlet characters, and two laws organize the resulting constants.

Proposition 8.46 (trivial-zero bridges)**.** *Let χ be primitive or principal, $k \geq 1$. If $\chi(-1) = (-1)^k$ (so that $L(-k,\chi) = 0$, since $B_{k+1}(\chi) \neq 0$ exactly when $\chi(-1) = (-1)^{k+1}$), then $L'(-k,\chi)$ is an elementary multiple of $L(k+1,\bar{\chi})$. For the principal character and χ_{-4}:*

$$\zeta'(-2k) = \frac{(-1)^k(2k)!}{2(2\pi)^{2k}}\,\zeta(2k+1), \qquad \beta'(1-2k) = (-1)^{k+1}(2k-1)!\,\frac{2^{2k-1}}{\pi^{2k-1}}\,\beta(2k),$$

so in particular $\beta'(-1) = 2G/\pi$ (G Catalan's constant) and $\beta'(-3) = -48\,\beta(4)/\pi^3$.

Theorem 8.47 (harmonic-number bridge)**.** *Let χ be a primitive character mod q with $\chi(-1) = (-1)^{k+1}$ (so that $L(-k,\chi) \neq 0$; for χ principal read ζ and require k odd). Then*

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)} + \overline{\left(\frac{L'(-k,\chi)}{L(-k,\chi)}\right)} = \gamma + \ln\frac{2\pi}{q} - H_k.$$

For real χ the conjugation bar is invisible; for complex χ it is essential — equivalently, the bridge pairs χ at $s = k+1$ with $\bar{\chi}$ at $s = -k$, and the un-conjugated sum has the purely imaginary defect $2i\,\mathrm{Im}[L'(-k,\chi)/L(-k,\chi)]$.

Proof. Write the completed function $\Lambda(s) = (q/\pi)^{(s+\delta)/2}\,\Gamma\big(\frac{s+\delta}{2}\big)L(s,\chi)$ with $\delta = (1-\chi(-1))/2$, so $\Lambda(s) = \varepsilon\overline{\Lambda}(1-\bar{s})$, whose logarithmic derivative at real s gives $\Lambda'/\Lambda(s) = -\overline{\Lambda'/\Lambda(1-s)}$; at $s = k+1$,

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)} + \overline{\left(\frac{L'(-k,\chi)}{L(-k,\chi)}\right)} = -\ln\frac{q}{\pi} - \frac{1}{2}\,\psi\Big(\frac{k+1+\delta}{2}\Big) - \frac{1}{2}\,\psi\Big(\frac{\delta-k}{2}\Big).$$

Since $\chi(-1) = (-1)^{k+1}$ means $\delta \equiv k+1 \pmod 2$, the argument $(\delta-k)/2$ is a (possibly negative) half-integer, where $\cot \pi x = 0$; the reflection formula thus gives $\psi\big(\frac{\delta-k}{2}\big) = \psi\big(\frac{k+2-\delta}{2}\big)$, and $\big\{\frac{k+1+\delta}{2}, \frac{k+2-\delta}{2}\big\}$ is a duplication pair $\{x, x+\frac{1}{2}\}$ with $2x = k+1$. Hence the two digammas sum to $2\psi(k+1) - 2\ln 2 = 2(H_k - \gamma) - 2\ln 2$, and the display follows. □

Consequently, at each level $-k$ every character direction of the grid $\{\zeta'(-k, j/q)\}$ either reduces (trivial-zero bridge: to the $\zeta(k+1)$/Catalan-family values) or carries exactly one new constant (the parity-opposite L'), the harmonic-number bridge converting freely between its $s = k+1$ and $s = -k$ spellings. Exact rational row-reduction of the full known-relation lattice (all reflection rows and all-divisor multiplication rows) makes this quantitative:

Corollary 8.48 (Uniform negative-jet ranks)**.** *For all $q \geq 3$ and $k \geq 1$, the formal negative-jet grid modulo the specified reflection, distribution and endpoint rows has residual dimension $\varphi(q)/2 - 1$ for odd k, and $\varphi(q)/2$ for even k.*

The exact character proof is given in Corollary 9.17. The original 112-cell finite experiment supported the result; the polynomial normal form now proves it for every denominator and order. This does not assert arithmetic independence of the remaining L-derivative coordinates.

8.16.5 Rational values

Theorem 8.49 (the half-argument tower)**.** *For every $n \geq 2$,*

$$\psi^{(-n)}\big(\tfrac{1}{2}\big) = \frac{(2^{1-n}-2)\,\zeta'(1-n) - 2^{1-n}\ln 2 \cdot B_n/n}{(n-1)!} + Q_n\big(\tfrac{1}{2}\big);$$

in lowest cases

$$\psi^{(-2)}\big(\tfrac{1}{2}\big) = \frac{3}{2}\ln \mathsf{A} + \frac{5\ln 2}{24} + \frac{\ln \pi}{4} \quad \textit{(classical)}, \qquad \psi^{(-3)}\big(\tfrac{1}{2}\big) = \frac{7\,\zeta(3)}{32\pi^2} + \frac{\ln \mathsf{A}}{2} + \frac{\ln 2\pi}{16}.$$

Theorem 8.50 (third- and quarter-argument values at $n = 3, 4$)**.**

$$\psi^{(-3)}\big(\tfrac14\big) = \frac{35\,\zeta(3)}{256\pi^2} + \frac{\beta'(3)}{4\pi^3} + \frac{\ln \mathsf{A}}{4} + \frac{\ln 2\pi}{128} - \frac{\gamma}{128},$$
$$\psi^{(-3)}\big(\tfrac13\big) = \frac{13\,\zeta(3)}{72\pi^2} + \frac{\sqrt{3}\,L'(3,\chi_{-3})}{8\pi^3} + \frac{\ln \mathsf{A}}{3} + \frac{7\ln 2\pi}{324} - \frac{\gamma}{162},$$

via the exact spellings $\beta'(-2) = \frac14\big(3 - 2\gamma - 2\ln\frac{\pi}{2}\big) + 16\beta'(3)/\pi^3$ *and* $L'(-2,\chi_{-3}) = \frac29\big(\ln\frac{3}{2\pi} + \frac32 - \gamma\big) + \frac{9\sqrt3}{2\pi^3}L'(3,\chi_{-3})$*; and at the next level the parity alternates, so that* $\psi^{(-4)}(\frac14)$ *is explicit with* no *new constant:*

$$\psi^{(-4)}\big(\tfrac14\big) = \frac{\zeta'(-3,\frac14) - \zeta'(-3)}{6} + Q_4\big(\tfrac14\big), \qquad \zeta'\big(-3,\tfrac14\big) = \frac{Z + D}{2}$$

with the even part $Z = (4^{-3} - 2^{-3})\zeta'(-3) + (4^{-3}\ln 4 - 2^{-3}\ln 2)\zeta(-3)$ *and the odd part* $D = -48\,\beta(4)/(64\pi^3)$ *from Proposition 8.46.*

Remark 8.51. $\psi^{(-3)}(\frac12)$ appears in Adamchik [18]; the flattened quarter- and third-values appear to be new — notably, Mathematica's `FunctionExpand` reduces `PolyGamma[-3, 1/2]` fully but leaves the jet $\zeta'(-2,\frac14)$ unevaluated inside `PolyGamma[-3, 1/4]`: the atom $\beta'(3)$ is exactly what is missing. The same atom resolves a naive integer-relation search over $\{\zeta(3)/\pi^2, \ln \mathsf{A}, \text{logs}\}$, which must fail for this value.

Corollary 8.52 (reflection and transport companions)**.** *With the same atoms as Theorem 8.50,*

$$\psi^{(-3)}\big(\tfrac23\big) = \frac{13\,\zeta(3)}{72\pi^2} - \frac{\sqrt{3}\,L'(3,\chi_{-3})}{8\pi^3} + \frac{2\ln \mathsf{A}}{3} + \frac{19\ln 2\pi}{162} + \frac{\gamma}{162},$$
$$\psi^{(-3)}\big(\tfrac34\big) = \frac{35\,\zeta(3)}{256\pi^2} - \frac{\beta'(3)}{4\pi^3} + \frac{3\ln \mathsf{A}}{4} + \frac{19\ln 2\pi}{128} + \frac{\gamma}{128},$$
$$\psi^{(-3)}\big(\tfrac16\big) = \frac{\zeta(3)}{12\pi^2} + \frac{5\sqrt{3}\,L'(3,\chi_{-3})}{32\pi^3} + \frac{\ln \mathsf{A}}{6} - \frac{\ln 2\pi}{1296} - \frac{\ln 2}{648} - \frac{5\gamma}{648},$$
$$\psi^{(-3)}\big(\tfrac56\big) = \frac{\zeta(3)}{12\pi^2} - \frac{5\sqrt{3}\,L'(3,\chi_{-3})}{32\pi^3} + \frac{5\ln \mathsf{A}}{6} + \frac{235\ln 2\pi}{1296} + \frac{\ln 2}{648} + \frac{5\gamma}{648}.$$

Proof. The first two from Theorem 8.45 with $\mathrm{Cl}_3(2\pi/3) = -\frac49\zeta(3)$ and $\mathrm{Cl}_3(\pi/2) = -\frac{3}{32}\zeta(3)$; the sixths from the transport $\zeta(s,\frac16) - \zeta(s,\frac56) = (6^s + 3^s)\,L(s,\chi_{-3})$ differentiated at $s = -2$, the even part being $\zeta'(-2,\frac16) + \zeta'(-2,\frac56) = \frac23\zeta'(-2)$ by inclusion–exclusion. The entire sixth family thus rides on the one atom $L'(3,\chi_{-3})$. □

8.16.6 Denominators 5, 8, 12: quadratic and quartic atoms

The denominators 5, 8, 12 are the first where Proposition 8.48 forces genuinely new constants beyond the χ_{-3}/χ_{-4} families: at $k = 1$ each grid carries one new atom (the even real quadratic direction), and at $k = 2$, $q = 5$ the residual space is two-dimensional (a conjugate quartic pair).

Theorem 8.53 (the $k = 1$ grids)**.** *Write* $Z_1 = \zeta'(-1)$ *and let* χ_q *be the even real quadratic primitive character mod* $q \in \{5, 8, 12\}$ *(Legendre mod* 5*; Kronecker* $(8|\cdot)$*,* $(12|\cdot)$*), with the exact*

values $L(-1,\chi_5)=-\frac{2}{5}$, $L(-1,\chi_8)=-1$, $L(-1,\chi_{12})=-2$. *Then for* $\gcd(p,q)=1$,

$$\zeta'\big(-1,\tfrac{p}{5}\big)=-\frac{Z_1}{5}+\frac{\mathrm{Cl}_2(2\pi p/5)}{4\pi}+\chi_5(p)\Big[\frac{L'(-1,\chi_5)}{20}-\frac{\ln 5}{50}\Big]-\frac{\ln 5}{240},$$
$$\zeta'\big(-1,\tfrac{p}{8}\big)=-\frac{Z_1}{32}+\frac{\mathrm{Cl}_2(\pi p/4)}{4\pi}+\chi_8(p)\Big[\frac{L'(-1,\chi_8)}{32}-\frac{3\ln 2}{32}\Big]+\frac{\ln 2}{384},$$
$$\zeta'\big(-1,\tfrac{p}{12}\big)=\frac{Z_1}{24}+\frac{\mathrm{Cl}_2(\pi p/6)}{4\pi}+\chi_{12}(p)\Big[\frac{L'(-1,\chi_{12})}{48}-\frac{\ln 2}{12}-\frac{\ln 3}{24}\Big]+\frac{\ln 3}{576},$$

whence $\psi^{(-2)}(p/q)=\zeta'(-1,p/q)-Z_1+Q_2(p/q)$ *through the ladder; for instance*

$$\psi^{(-2)}\big(\tfrac{1}{5}\big)=-\frac{6Z_1}{5}+\frac{\mathrm{Cl}_2(2\pi/5)}{4\pi}+\frac{L'(-1,\chi_5)}{20}-\frac{29\ln 5}{1200}+\frac{\ln 2\pi}{10}+\frac{2}{25}.$$

Proof. On each grid the two reflection rows (Theorem 8.44), the multiplication rows (8.59) for every divisor (non-coprime entries reduced through the sub-grids $q\in\{2,3,4,6\}$), and the character row $\sum_p \chi_q(p)\,\zeta'(-1,\frac{p}{q})=\frac{1}{q}\big[L'(-1,\chi_q)+\ln q\,L(-1,\chi_q)\big]$ form a system of full rank $\varphi(q)=4$; solving it and reducing the composite-argument Clausen values ($\mathrm{Cl}_2(\frac{3\pi}{4})=\mathrm{Cl}_2(\frac{\pi}{4})-\frac{G}{2}$, $\mathrm{Cl}_2(\frac{\pi}{6})=\frac{2}{3}G+\frac{3}{8}\,\mathrm{Cl}_2(\frac{2\pi}{3})$) gives the display. For the composite q the known rows are one short of independent — the single dependency is itself a Clausen duplication law in disguise (at $q=8$: $\mathrm{refl}(\frac{1}{8})-\mathrm{refl}(\frac{3}{8})=m_2(\frac{1}{8})-m_2(\frac{3}{8})$ is equivalent to $\mathrm{Cl}_2(\frac{3\pi}{4})=\mathrm{Cl}_2(\frac{\pi}{4})-\frac{G}{2}$) — and the quadratic character row supplies the missing dimension, exactly as Proposition 8.48 predicts ($r_{q,1}=1$ for $q\in\{5,8,12\}$). □

Theorem 8.54 (the quartic floor: $k=2$, $q=5$)**.** *Let* χ *be the odd quartic character mod* 5 *with* $\chi(2)=i$, *for which* $L(-2,\chi)=-\frac{4+2i}{5}$ *exactly, and write* $L'(-2,\chi)=R+iI$. *Then*

$$\zeta'\big(-2,\tfrac{1}{5}\big),\ \zeta'\big(-2,\tfrac{4}{5}\big)=-\frac{\mathrm{Cl}_3(2\pi/5)}{4\pi^2}\pm\Big[\frac{R}{50}-\frac{2\ln 5}{125}\Big],$$
$$\zeta'\big(-2,\tfrac{2}{5}\big),\ \zeta'\big(-2,\tfrac{3}{5}\big)=-\frac{\mathrm{Cl}_3(4\pi/5)}{4\pi^2}\pm\Big[\frac{I}{50}-\frac{\ln 5}{125}\Big],$$

and, flattening through the ladder with $\mathrm{Cl}_3(\frac{2\pi}{5}),\mathrm{Cl}_3(\frac{4\pi}{5})=-\frac{6}{25}\zeta(3)\pm\frac{\sqrt{5}}{4}L(3,\chi_5)$,

$$\psi^{(-3)}\big(\tfrac{1}{5}\big)=\frac{31\,\zeta(3)}{200\pi^2}-\frac{\sqrt{5}\,L(3,\chi_5)}{32\pi^2}+\frac{R}{100}-\frac{\ln 5}{125}+\frac{\ln 2\pi}{100}+\frac{7}{1500}-\frac{Z_1}{5},$$
$$\psi^{(-3)}\big(\tfrac{2}{5}\big)=\frac{31\,\zeta(3)}{200\pi^2}+\frac{\sqrt{5}\,L(3,\chi_5)}{32\pi^2}+\frac{I}{100}-\frac{\ln 5}{250}+\frac{\ln 2\pi}{25}+\frac{41}{1500}-\frac{2Z_1}{5},$$
$$\psi^{(-3)}\big(\tfrac{3}{5}\big)=\frac{31\,\zeta(3)}{200\pi^2}+\frac{\sqrt{5}\,L(3,\chi_5)}{32\pi^2}-\frac{I}{100}+\frac{\ln 5}{250}+\frac{9\ln 2\pi}{100}+\frac{7}{125}-\frac{3Z_1}{5},$$
$$\psi^{(-3)}\big(\tfrac{4}{5}\big)=\frac{31\,\zeta(3)}{200\pi^2}-\frac{\sqrt{5}\,L(3,\chi_5)}{32\pi^2}-\frac{R}{100}+\frac{\ln 5}{125}+\frac{4\ln 2\pi}{25}+\frac{59}{750}-\frac{4Z_1}{5}.$$

Remark 8.55. The remaining derivative coordinates in these formulas are the quartic pair $\{R,I\}$, matching $r_{5,2}=\varphi(5)/2=2$ of Proposition 8.48; the multiplication row $\sum_p \zeta'(-2,\frac{p}{5})=\frac{6}{25}\zeta(3)/\pi^2$ is *dependent* on the reflections — it is the Clausen multiplication law $\mathrm{Cl}_3(\frac{2\pi}{5})+\mathrm{Cl}_3(\frac{4\pi}{5})=-\frac{12}{25}\zeta(3)$ in disguise. Through the conjugated harmonic-number bridge (Theorem 8.47 with $k=2$, $H_2=\frac{3}{2}$) the pair $\{R,I\}$ is interchangeable with $\{\mathrm{Re},\mathrm{Im}\}$ of $L'(3,\chi)$.

The single-function jet calculus extends to products once the endpoint normalization is fixed. The four independent continuations [108, 109, 110, 111] agree on the scale-one finite parts used below. Their common proof is organized through Fourier normalization, singular Laurent coefficients, convolution closure, translated products and normalized primitive lifting. The contact terms belong to the differentiated extension in the next chapter.

8.17 Fourier conventions and the Hurwitz semigroup

We work on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with normalized Lebesgue measure. For a periodic distribution T, use the convention

$$\widehat{T}(k) = \langle T, e^{-2\pi i k x} \rangle, \qquad k \in \mathbb{Z}.$$

The constant function 1 and the point mass δ_0 both have mean one, but their nonzero Fourier coefficients are respectively 0 and 1. Thus

$$\mathsf{I} = \delta_0 - 1$$

is the convolution identity on the space of mean-zero distributions. In particular, $\mathsf{I} * T = T$ whenever $\widehat{T}(0) = 0$.

The logarithm used throughout is

$$\lambda_k = \log(2\pi i k) = \log(2\pi|k|) + \frac{\pi i}{2} \operatorname{sgn} k, \qquad k \neq 0. \tag{8.60}$$

All powers below refer to this fixed branch.

Definition 8.56. *For $\alpha \in \mathbb{C}$, define R_α by*

$$\widehat{R}_\alpha(0) = 0, \qquad \widehat{R}_\alpha(k) = e^{-\alpha \lambda_k} \quad (k \neq 0). \tag{8.61}$$

Equivalently, its Fourier series is $\sum_{k \neq 0} (2\pi i k)^{-\alpha} e^{2\pi i k x}$ in the distribution sense.

Lemma 8.57 (Entire distribution family)**.** *The family $\alpha \mapsto R_\alpha$ is entire as a periodic-distribution family. It is represented by an absolutely convergent Fourier series for $\operatorname{Re}\alpha > 1$. For $\operatorname{Re}\alpha > 0$, it is the locally integrable periodic extension of*

$$R_\alpha(x) = \frac{\zeta(1-\alpha, x)}{\Gamma(\alpha)}, \qquad 0 < x < 1. \tag{8.62}$$

Proof. On each compact set of parameters, the coefficients in (8.61), together with any fixed number of parameter derivatives, have at most polynomial growth in $|k|$. Fourier coefficients of a smooth test function decay faster than every polynomial. Hence the paired Fourier series and all its parameter derivatives converge locally uniformly. This proves the first assertion; the absolute-convergence assertion follows immediately from $\sum k^{-\operatorname{Re}\alpha}$.

For $\operatorname{Re}\alpha > 1$, equation (8.62) is precisely Hurwitz's Fourier formula. To continue to $\operatorname{Re}\alpha > 0$, use

$$\zeta(1-\alpha, x) = x^{\alpha-1} + \zeta(1-\alpha, 1+x).$$

On compact parameter subsets of that half-plane, the first term and its parameter derivatives are dominated by integrable powers of x times powers of $|\log x|$. The second term is regular in $x \in [0,1]$; any apparent parameter singularity at $\alpha = 0$ is outside the stated open half-plane. The two holomorphic distribution families agree in an open subregion, and hence agree throughout the half-plane. □

Theorem 8.58 (Semigroup and differential ladder)**.** *For all $\alpha, \beta \in \mathbb{C}$,*

$$R_\alpha * R_\beta = R_{\alpha+\beta}, \qquad D R_\alpha = R_{\alpha-1}, \qquad R_0 = \mathsf{I}. \tag{8.63}$$

If $J_{\alpha,r} = \partial_\alpha^r R_\alpha$, then

$$J_{\alpha,r} * J_{\beta,s} = J_{\alpha+\beta,r+s}, \qquad D^k J_{\alpha,r} = J_{\alpha-k,r}. \tag{8.64}$$

Proof. Periodic distributions can be convolved: equivalently, their polynomially growing Fourier coefficients can be multiplied. For $k \neq 0$,

$$e^{-\alpha\lambda_k} e^{-\beta\lambda_k} = e^{-(\alpha+\beta)\lambda_k}, \qquad (2\pi i k) e^{-\alpha\lambda_k} = e^{-(\alpha-1)\lambda_k}.$$

Both sides have zero zeroth coefficient. Fourier uniqueness proves (8.63). Differentiating the locally uniformly paired Fourier series gives (8.64). □

For real $\alpha, \beta > 1$, the first identity, expressed using (8.62), is the nonsingular law in Sun [107, Remark 3.2]. The new use here is at its singular integer parameters and their spectral jets. At positive integers,

$$R_m(x) = -\frac{B_m(\{x\})}{m!} \quad (m \geq 1),$$

up to immaterial point values of the sawtooth for $m = 1$. At negative integers,

$$R_{-k} = \delta_0^{(k)} \quad (k \geq 1). \tag{8.65}$$

The quotient in (8.62) vanishes pointwise at $\alpha = -k$ away from the integers, while the distribution is nonzero. This is the basic reason contact terms must be retained.

8.18 Canonical finite parts of the Stieltjes functions

The generalized Stieltjes functions are defined for $x > 0$ by

$$\zeta(1 - z, x) = -\frac{1}{z} + \sum_{n=0}^{\infty} \gamma_n(x) \frac{z^n}{n!}. \tag{8.66}$$

Thus $\gamma_0(x) = -\psi(x)$. The Hurwitz shift relation gives

$$\gamma_n(x) = \frac{\log^n x}{x} + \gamma_n(1 + x). \tag{8.67}$$

Consequently, periodic extension of γ_n is not locally integrable at 0; a convention is necessary.

Definition 8.59 (Canonical periodic finite part)**.** *For a smooth periodic test function ϕ, define*

$$\langle G_n, \phi \rangle = \int_0^1 \left(\gamma_n(x)\phi(x) - \frac{\log^n x}{x} \phi(0) \right) \mathrm{d}x \tag{8.68}$$

$$= \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 \gamma_n(x)\phi(x)\,\mathrm{d}x + \frac{\log^{n+1} \epsilon}{n+1} \phi(0) \right\}. \tag{8.69}$$

The integrand in (8.68) is integrable: its singular part is $\log^n x\,[\phi(x) - \phi(0)]/x$, which is $O(|\log x|^n)$. No arbitrary finite multiple of δ_0 is added.

Lemma 8.60 (The distributional Laurent expansion)**.** *Let*

$$Z(s) = \Gamma(1 - s) R_{1-s}.$$

It is a meromorphic periodic-distribution family, agreeing with the periodic locally integrable Hurwitz zeta for $\operatorname{Re} s < 1$*. Near $z = 0$,*

$$Z(1 - z) = \frac{\mathsf{I}}{z} + \sum_{n=0}^{\infty} G_n \frac{z^n}{n!}. \tag{8.70}$$

Every G_n has mean zero.

Proof. The meromorphic continuation follows from Lemma 8.57. For $\operatorname{Re} z > 0$, split the pairing using $\zeta(1-z,x) = x^{z-1} + \zeta(1-z,1+x)$. The first term has expansion

$$\int_0^1 x^{z-1}\phi(x)\,\mathrm{d}x = \frac{\phi(0)}{z} + \sum_{n\geq 0}\frac{z^n}{n!}\int_0^1 \frac{\log^n x}{x}[\phi(x)-\phi(0)]\,\mathrm{d}x.$$

The second term contributes $-z^{-1}\int_0^1 \phi$ and the coefficients $\int_0^1 \gamma_n(1+x)\phi(x)\,\mathrm{d}x$. These expansions converge locally near zero after pairing, by the same endpoint domination used above. Combining them proves (8.70) with the exact subtraction in Definition 8.59. Since $\widehat{Z}(s)(0) = 0$, every coefficient has mean zero. Equivalently, direct integration over $[1,2]$ gives $\int_1^2 \zeta(1-z,x)\,\mathrm{d}x = -1/z$, whose regular coefficients vanish. □

Remark 8.61 (Two different residues). The scalar function in (8.66) has residue -1 as a function of z, whereas the periodic-distribution family has residue $\delta_0 - 1$. Analytic continuation across a nonintegrable endpoint adds a point-supported residue. Treating the scalar residue as the whole periodic residue gives wrong convolution constants.

8.19 All-index convolution closure and exact normal forms

Write $\mathcal{G}(z) = \sum_{n\geq 0} G_n z^n/n!$, holomorphic as a distribution family for $|z| < 1$. By (8.70),

$$\mathsf{I} + z\mathcal{G}(z) = \Gamma(1+z)R_z. \tag{8.71}$$

Let $X = G_0$. For a complete exponential Bell polynomial we use

$$\exp\left(\sum_{j\geq 1} b_j \frac{z^j}{j!}\right) = \sum_{r\geq 0}\mathcal{B}_r(b_1,\dots,b_r)\frac{z^r}{r!}. \tag{8.72}$$

When its arguments are distributions, all products in this definition are convolution products and its scalar unit is I.

Theorem 8.62 (One-generator convolution algebra)**.** *For every $n \geq 0$,*

$$(n+1)G_n = \mathcal{B}_{n+1}(b_1,\dots,b_{n+1}), \qquad b_1 = X, \quad b_j = (-1)^j(j-1)!\zeta(j)\mathsf{I}\ (j\geq 2). \tag{8.73}$$

The span of $\mathsf{I}, G_0, G_1, \dots$ is a convolution algebra isomorphic to the polynomial algebra $\mathbb{C}[X]$. These distributions form a basis of that algebra, with the unit listed first.

Proof. For $k \neq 0$, the Fourier transform of (8.71) is

$$1 + z\widehat{\mathcal{G}}(z)(k) = \Gamma(1+z)e^{-z\lambda_k} = \exp\left[(-\gamma-\lambda_k)z + \sum_{j\geq 2}\frac{(-1)^j\zeta(j)}{j}z^j\right].$$

The linear coefficient is $\widehat{X}(k) = -\gamma - \lambda_k$. Comparing with (8.72) proves (8.73); the zero Fourier coefficient of each distributional expression vanishes. The leading term in G_n is $X^{*(n+1)}/(n+1)$, so the listed distributions span exactly the polynomial algebra generated by X.

There are no polynomial relations among its powers. Indeed, if $P(X) = 0$ distributionally, then $P(-\gamma - \log(2\pi k) - i\pi/2) = 0$ for every positive integer k. These are infinitely many distinct complex numbers, and hence P is the zero polynomial. This also proves the basis assertion. □

This algebraic freeness concerns distributions as functions of the variable x. It is not an independence theorem about their values at a fixed rational point, nor about zeta or Stieltjes constants as numbers.

The first Bell forms are

$$\begin{aligned} 2G_1 &= X^{*2} + \zeta(2)\mathsf{I}, \\ 3G_2 &= X^{*3} + 3\zeta(2)X - 2\zeta(3)\mathsf{I}, \\ 4G_3 &= X^{*4} + 6\zeta(2)X^{*2} - 8\zeta(3)X + [3\zeta(2)^2 + 6\zeta(4)]\mathsf{I}. \end{aligned} \tag{8.74}$$

They already imply many exact integrals. An efficient all-index multiplication rule is supplied by the following Gamma quotient.

Theorem 8.63 (Bivariate multiplication law)**.** *Define, near* $(u, v) = (0, 0)$,

$$\begin{aligned} B(u,v) &= \frac{\Gamma(1+u)\Gamma(1+v)}{\Gamma(1+u+v)} \\ &= \exp\left(\sum_{j\geq 2} \frac{(-1)^j \zeta(j)}{j}[u^j + v^j - (u+v)^j]\right). \end{aligned} \tag{8.75}$$

Then

$$\begin{aligned} uv\,\mathcal{G}(u) * \mathcal{G}(v) = [B(u,v) - 1]\mathsf{I} \\ + B(u,v)(u+v)\mathcal{G}(u+v) - u\mathcal{G}(u) - v\mathcal{G}(v). \end{aligned} \tag{8.76}$$

The right side is divisible by uv *as a convergent distribution-valued power series. In particular,*

$$G_m * G_n - \frac{m+n+2}{(m+1)(n+1)} G_{m+n+1} \in \operatorname{span}_{\mathbb{C}}(\{\mathsf{I}\} \cup \{G_j : 0 \leq j < m+n\}), \tag{8.77}$$

where the lower Stieltjes span is empty when $m + n = 0$*. The index* $m + n$ *is absent because the Gamma quotient has no linear term. Every coefficient is a polynomial over* $\mathbb{Q}$ *in zeta values at positive integers. Its complete value is obtained by multiplying the coefficient of* $u^{m+1}v^{n+1}$ *in* (8.76) *by* $m!n!$*.*

Proof. Convolve (8.71) at parameters u and v and apply Theorem 8.58. The result is

$$[\mathsf{I} + u\mathcal{G}(u)] * [\mathsf{I} + v\mathcal{G}(v)] = B(u,v)[\mathsf{I} + (u+v)\mathcal{G}(u+v)].$$

Subtract the unit and the two linear summands. Both restrictions $u = 0$ and $v = 0$ vanish identically, proving divisibility. Coefficient extraction is legitimate in a sufficiently small bidisc. The largest-index term comes from $(u+v)\mathcal{G}(u+v)$ with the constant coefficient of B; its coefficient is $\frac{m!n!}{(m+n+1)!}\binom{m+n+2}{m+1}$, which equals the number in (8.77). The exponential expansion of B proves the zeta-polynomial assertion. □

Four useful multiplication rules are

$$G_0 * G_0 = 2G_1 - \zeta(2)\mathsf{I}, \tag{8.78}$$

$$G_0 * G_1 = \frac{3}{2}G_2 - \zeta(2)G_0 + \zeta(3)\mathsf{I}, \tag{8.79}$$

$$G_0 * G_2 = \frac{4}{3}G_3 - 2\zeta(2)G_1 + 2\zeta(3)G_0 - 2\zeta(4)\mathsf{I}, \tag{8.80}$$

$$G_1 * G_1 = G_3 - 2\zeta(2)G_1 + 2\zeta(3)G_0 - \frac{1}{4}\zeta(4)\mathsf{I}. \tag{8.81}$$

The last simplification uses $\zeta(2)^2 = 5\zeta(4)/2$. The verification catalog retains independent formal symbols for zeta values until such a specific relation is explicitly applied.

8.20 Ordinary integral realizations and explicit Stieltjes identities

The singular support of G_n is contained in $\{0\}$. Thus its convolution with another G_m is smooth away from 0: in a local convolution integral at $0 < x < 1$, the two singular factors occur at the distinct integration points $t = 0$ and $t = x$. We now give a direct formula that also proves this local realization without requiring a product of distributions at the same point.

Theorem 8.64 (Absolutely convergent subtracted integrals)**.** *Put $a_n(t) = \log^n(t)/t$. For $m, n \geq 0$ and $0 < x < 1$, define*

$$\begin{aligned} C_{m,n}(x) = &\int_0^x \big[\gamma_m(t)\gamma_n(x-t) - \gamma_n(x)a_m(t) - \gamma_m(x)a_n(x-t)\big]\,\mathrm{d}t \\ &+ \int_x^1 \gamma_m(t)\gamma_n(1+x-t)\,\mathrm{d}t \\ &+ \gamma_n(x)\frac{\log^{m+1} x}{m+1} + \gamma_m(x)\frac{\log^{n+1} x}{n+1}. \end{aligned} \tag{8.82}$$

Both displayed integrals are absolutely convergent, and $C_{m,n}(x) = (G_m * G_n)(x)$. In particular, every expression (8.82) has a finite exact reduction using Theorem 8.63.

Proof. Near $t = 0$, the shift relation (8.67) leaves a possible singularity

$$a_m(t)[\gamma_n(x-t) - \gamma_n(x)] = O(|\log t|^m).$$

The other endpoint is the same argument with m, n interchanged. The second integral is nonsingular. These observations prove absolute convergence, locally uniformly with derivatives in x after using endpoint-fixed local coordinates.

For the normalization, approximate G_m by the cutoff function $\mathbf{1}_{[\epsilon,1)}\gamma_m$ plus $\log^{m+1}\epsilon\,\delta_0/(m+1)$, and similarly for G_n. On a compact subinterval of $0 < x < 1$, their convolution consists of

$$\int_\epsilon^{x-\epsilon} \gamma_m(t)\gamma_n(x-t)\,\mathrm{d}t + \int_x^1 \gamma_m(t)\gamma_n(1+x-t)\,\mathrm{d}t$$

plus the two logarithmic counterterms multiplying $\gamma_n(x)$ and $\gamma_m(x)$. The product of the two delta counterterms is supported at 0, outside that compact subinterval. Integrating the two subtracted a-terms explicitly and taking the limit gives (8.82). The locally uniform endpoint argument identifies this limit with the restriction of the distributional convolution. □

Corollary 8.65 (Four scalar identity families)**.** *For $0 < x < 1$,*

$$C_{0,0}(x) = 2\gamma_1(x) + \zeta(2), \tag{8.83}$$

$$C_{0,1}(x) = \frac{3}{2}\gamma_2(x) - \zeta(2)\gamma_0(x) - \zeta(3), \tag{8.84}$$

$$C_{0,2}(x) = \frac{4}{3}\gamma_3(x) - 2\zeta(2)\gamma_1(x) + 2\zeta(3)\gamma_0(x) + 2\zeta(4), \tag{8.85}$$

$$C_{1,1}(x) = \gamma_3(x) - 2\zeta(2)\gamma_1(x) + 2\zeta(3)\gamma_0(x) + \frac{1}{4}\zeta(4). \tag{8.86}$$

Proof. Restrict (8.78)–(8.81) to $0 < x < 1$, where G_n agrees with γ_n and I agrees with the constant -1. □

8.20.1 A concrete digamma integral

Since $\gamma_0 = -\psi$, the first case reads

$$\int_0^x \left[\psi(t)\psi(x-t) + \psi(x)\left(\frac{1}{t} + \frac{1}{x-t}\right)\right] \mathrm{d}t \tag{8.87}$$
$$+ \int_x^1 \psi(t)\psi(1+x-t)\,\mathrm{d}t - 2\psi(x)\log x = 2\gamma_1(x) + \frac{\pi^2}{6}.$$

The two divergent-looking summands inside the first bracket must be integrated together; their sum is integrable. Equation (8.87) is not an assertion that either unregularized convolution integral exists.

At $x = 1/2$, the Hurwitz identity $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$ implies

$$\gamma_1(1/2) = \gamma_1 - 2\gamma\log 2 - \log^2 2.$$

Thus the left side of (8.87) at $1/2$ equals

$$2\gamma_1 - 4\gamma\log 2 - 2\log^2 2 + \frac{\pi^2}{6}. \tag{8.88}$$

The same method at any rational argument turns the full multiplication table into exact relations among rational-argument Stieltjes functions, with no integer-relation search.

8.21 Conventions and endpoint normalization

Throughout,

$$0 < a < 1, \qquad b = 1 - a, \qquad \ell = \log(2\pi), \qquad \kappa = \gamma + \ell, \qquad B_2(a) = a^2 - a + \frac{1}{6}.$$

The fractional part $\{x + a\}$ is interpreted periodically; its value at the single jump point does not affect ordinary integrals. The notation

$$\zeta^{[j]}(s, x) = \partial_s^j \zeta(s, x), \qquad \mathrm{Li}_s^{[j]}(z) = \partial_s^j \mathrm{Li}_s(z)$$

always means *spectral* differentiation. In contrast, $\gamma_m^{(r)}(x)$ and $\psi^{(r)}(x)$ mean derivatives in the real argument. Euler's constant is $\gamma = \gamma_0(1)$, not a new parameter.

The generalized Stieltjes constants are normalized by

$$\zeta(1+u, x) = \frac{1}{u} + R(u, x), \qquad R(u, x) = \sum_{m \geq 0} \frac{(-1)^m}{m!}\gamma_m(x)u^m. \tag{8.89}$$

Thus $\gamma_0(x) = -\psi(x)$. The Hurwitz shift identity gives the exact local decomposition

$$\gamma_m(x) = \frac{\log^m x}{x} + \gamma_m(1+x), \qquad x > 0. \tag{8.90}$$

Every logarithm of a positive real argument is the real logarithm. For $z = e^{2\pi i a} \neq 1$, $\mathrm{Li}_s(z)$ and $\mathrm{Li}_s(z^{-1})$ denote their usual boundary values from the unit disk, continued in s; these are entire in s. For real s they are conjugate, but we retain them separately when s is complex.

Definition 8.66 (Canonical shifted Stieltjes finite part)**.** *For nonnegative integers m, n, define*

$$J_{mn}(a) = \lim_{\epsilon\downarrow 0}\Bigg\{ \int_\epsilon^b \gamma_m(x)\gamma_n(x+a)\,\mathrm{d}x + \int_\epsilon^a \gamma_m(b+t)\gamma_n(t)\,\mathrm{d}t + \frac{\gamma_n(a)}{m+1}\log^{m+1}\epsilon + \frac{\gamma_m(b)}{n+1}\log^{n+1}\epsilon\Bigg\}. \tag{8.91}$$

Equivalently, this is denoted $\mathrm{FP}\int_0^1 \gamma_m(x)\gamma_n(\{x+a\})\,\mathrm{d}x$.

The two cutoffs can tend to zero independently: each counterterm belongs to its own local endpoint. The coordinates are exactly x and $t = x - b$, with no rescaling or additional subtraction length. A different finite-part convention can change local constants and therefore change the identities.

Lemma 8.67 (Existence and an ordinary-integral version)**.** *The limit* (8.91) *exists and is equivalent to*

$$\begin{aligned} J_{mn}(a) = {} & \int_0^b \left[\gamma_m(x)\gamma_n(x+a) - \gamma_n(a)\frac{\log^m x}{x}\right]\mathrm{d}x \\ & + \int_0^a \left[\gamma_m(b+t)\gamma_n(t) - \gamma_m(b)\frac{\log^n t}{t}\right]\mathrm{d}t \\ & + \frac{\gamma_n(a)}{m+1}\log^{m+1} b + \frac{\gamma_m(b)}{n+1}\log^{n+1} a. \end{aligned} \tag{8.92}$$

Both integrals on the right are ordinary improper integrals. Moreover, $J_{mn}(a) = J_{nm}(1-a)$.

Proof. By (8.90), subtracting the singular term in the first integral leaves $O(1+|\log x|^m)$ near zero: the other factor differs from $\gamma_n(a)$ by $O(x)$. The second integral has the same property. Integrating $\log^m x/x$ explicitly yields (8.92). Interchanging the two pieces and replacing a by b proves the symmetry. □

We use analogous finite parts for polygamma and derivative products: expand each smooth factor at the distinct singular point, integrate the resulting powers and logarithms, and retain the constant term after deleting all divergent cutoff terms. Section 9.2 makes this prescription precise through periodic distributions.

8.22 The shifted Hurwitz correlation

Proposition 8.68 (Classical Fourier method, shifted form)**.** *For $\Re s < 1$, $\Re t < 1$, define the ordinary integral*

$$C(s,t;a) = \int_0^1 \zeta(s,x)\zeta(t,\{x+a\})\,\mathrm{d}x. \tag{8.93}$$

It is jointly holomorphic in this domain. Put $w = 2-s-t$ and $z = e^{2\pi\mathrm{i}a}$. Then

$$C(s,t;a) = \frac{\Gamma(1-s)\Gamma(1-t)}{(2\pi)^w}\Big[e^{-\pi\mathrm{i}(s-t)/2}\,\mathrm{Li}_w(z) + e^{\pi\mathrm{i}(s-t)/2}\,\mathrm{Li}_w(z^{-1})\Big]. \tag{8.94}$$

It also has the meromorphic representation

$$C(s,t;a) = B(1-s,s+t-1)\zeta(s+t-1,a) + B(1-t,s+t-1)\zeta(s+t-1,b). \tag{8.95}$$

At exceptional arguments of the beta factors, the sum *in* (8.95) *is continued before evaluation. Apparent poles of separate summands need not be poles of* C*.*

Proof. Split the integral at b. At $x = 0$ only the first factor is singular and has the form x^{-s} plus a smooth function. At $x = b+$ only the second factor is singular. Compact subsets of $\Re s < 1$, $\Re t < 1$ therefore admit integrable dominating functions, including arbitrary spectral derivatives, because powers of $|\log x|$ remain integrable against $x^{-\sigma}$ for $\sigma < 1$. This proves convergence and holomorphy. There is no additional condition $\Re(s+t) < 1$ for a nonzero shift.

For $\Re s < 0$, the Hurwitz Fourier expansion [93, Eq. 25.11.9] is absolutely convergent and its nonzero coefficients are

$$\widehat{\zeta}_s(k) = \Gamma(1-s)(2\pi|k|)^{s-1}\exp\left(-\frac{\pi\mathrm{i}}{2}(1-s)\operatorname{sgn} k\right), \qquad k \neq 0. \tag{8.96}$$

The zero coefficient vanishes. For $\Re s, \Re t < 0$, multiply the two absolutely convergent series and integrate. Orthogonality gives

$$C(s,t;a) = \sum_{k\neq 0} \widehat{\zeta}_s(k)\widehat{\zeta}_t(-k)e^{-2\pi\mathrm{i}ka}.$$

The positive and negative frequencies give exactly (8.94). Both sides are holomorphic on the larger connected domain stated in the proposition; the identity theorem extends the equality there.

To obtain (8.95), apply the same Fourier formula to $\zeta(1-w,a)$ and $\zeta(1-w,b)$, initially away from gamma poles. Euler reflection gives $\Gamma(w)\Gamma(1-w) = \pi/\sin\pi w$ and $1/\Gamma(t) = \Gamma(1-t)\sin\pi t/\pi$. The coefficient of $\mathrm{Li}_w(z)$ reduces using

$$\frac{\sin(\pi t)e^{-\pi\mathrm{i}w/2} + \sin(\pi s)e^{\pi\mathrm{i}w/2}}{\sin(\pi w)} = e^{-\pi\mathrm{i}(s-t)/2}, \qquad w = 2-s-t.$$

The other coefficient is obtained by reversing the phase. This proves equality on an open set and hence meromorphically everywhere. □

Corollary 8.69 (All ordinary spectral-jet correlations)**.** *For integers* $p, q \geq 0$ *and* $\Re s, \Re t < 1$*,*

$$\int_0^1 \zeta^{[p]}(s,x)\zeta^{[q]}(t,\{x+a\})\,\mathrm{d}x = \partial_s^p\partial_t^q C(s,t;a).$$

Thus (8.94) *generates ordinary identities at nonpositive spectral integers, while continuation of* (8.95) *generates finite-part identities at positive integers.*

Proof. Differentiate under the integral using the compact domination established above. The formulas on the right are finite applications of the product and chain rules. □

The distinction between nonzero and zero shifts is essential. At $a = 0$, the singularities coincide and the original convergence region changes to include a constraint on $s+t$. One must not substitute $a = 0$ directly into a separated-singularity finite-part formula. The continuous ordinary log-gamma correlation will have a legitimate limit at zero, but that is a different assertion.

Proposition 8.70 (Coefficient extraction at the double pole)**.** *For $a \in (0,1)$,*

$$J_{mn}(a) = (-1)^{m+n} m!n![u^m v^n] C(1+u, 1+v; a). \tag{8.97}$$

The brackets mean the regular Laurent coefficient at the specified bidegree. In particular, the double-pole coefficient is $-1/(uv)$, not $1/(uv)$.

Proof. Write $Z_s = Z(s)$ for the periodic family of Lemma 8.60. Its raising-parameter expansion is

$$Z_{1+u} = -\mathsf{I}/u + \sum_{m \geq 0} (-1)^m G_m u^m / m!. \tag{8.98}$$

The singular supports of $Z_{1+u}(x)$ and $Z_{1+v}(x+a)$ are 0 and b, which are distinct. Choose disjoint neighborhoods of these points and a smooth partition of unity. On each neighborhood only one factor is a distribution and the other is smooth, so their product and integral are defined without any product of distributions at a common singularity. This gives the meromorphic continuation of the original integral.

The product of the regular coefficients in (8.98), integrated with this partition, is exactly (8.91). The pole terms give

$$\begin{aligned} C(1+u, 1+v; a) = &-\frac{1}{uv} - \frac{R(v,a)}{u} - \frac{R(u,b)}{v} \\ &+ \sum_{m,n \geq 0} \frac{(-1)^{m+n}}{m!n!} J_{mn}(a) u^m v^n. \end{aligned} \tag{8.99}$$

The integrated product of $1 - \delta_0$ and $1 - \delta_b$ is $1 - 1 - 1 = -1$ away from $a = 0$, and each G_m has mean zero. This proves the statement. □

Corollary 8.71 (Polylogarithmic generator)**.** *Set $E(u) = \Gamma(1-u)(2\pi)^u$ and*

$$\begin{aligned} \mathcal{H}(u,v;a) = E(u)E(v)\big[&e^{-\pi \mathrm{i}(u-v)/2} \operatorname{Li}_{-u-v}(z) \\ &+ e^{\pi \mathrm{i}(u-v)/2} \operatorname{Li}_{-u-v}(z^{-1})\big]. \end{aligned} \tag{8.100}$$

Then

$$J_{mn}(a) = (-1)^{m+n} m!n![u^{m+1} v^{n+1}] \mathcal{H}(u,v;a). \tag{8.101}$$

Proof. In (8.94), set $s = 1+u$, $t = 1+v$ and use $\Gamma(-u) = -\Gamma(1-u)/u$. Thus $C = \mathcal{H}/(uv)$, and (8.97) applies. □

This representation explains how polylogarithm *order derivatives*, rather than only argument derivatives, enter the exact Stieltjes product calculus.

8.23 An all-order linear Stieltjes closure theorem

The beta representation is particularly effective near $(s,t) = (1,1)$. Define

$$A(u,v) = \frac{\Gamma(1-u)\Gamma(1+u+v)}{\Gamma(1+v)}, \qquad T(u,v) = \frac{A(u,v) - 1}{u(u+v)}. \tag{8.102}$$

The apparent singularities in T are removable: $A(0,v)=1$ and $A(u,-u)=1$. Its Taylor coefficients belong to the polynomial algebra generated by positive integer zeta values. More explicitly,

$$\log A(u,v)=\sum_{k\geq 2}\frac{\zeta(k)}{k}\left[u^k+(-1)^k\big((u+v)^k-v^k\big)\right]. \tag{8.103}$$

Euler's constant cancels identically. This series converges in a neighborhood of the origin; it can also be used as an exact formal power series.

Theorem 8.72 (All-index finite-part closure)**.** *Let $m,n\geq 0$ and $d=m+n$. Define the analytic germ*

$$\begin{aligned}\mathcal{F}(u,v;a)&=T(u,v)+T(v,u)\\&\quad+(u+v)\big[T(u,v)R(u+v,a)+T(v,u)R(u+v,b)\big].\end{aligned} \tag{8.104}$$

Then

$$\boxed{J_{mn}(a)=\frac{\gamma_{d+1}(a)}{m+1}+\frac{\gamma_{d+1}(b)}{n+1}-(-1)^d m!n![u^m v^n]\mathcal{F}(u,v;a).} \tag{8.105}$$

Every coefficient is obtained by a finite calculation. Apart from the two leading terms displayed explicitly, the right side contains only $\gamma_j(a)$ and $\gamma_j(b)$ with $0\leq j\leq d-1$, and a constant term. Its coefficients lie in $\mathbb{Q}[\zeta(2),\dots,\zeta(d+2)]$. There are no products of generalized Stieltjes functions and no term involving γ_d.

Proof. Since $B(-u,1+u+v)=-A(u,v)/u$, equation (8.95) becomes

$$C(1+u,1+v;a)=-\frac{A(u,v)}{u}\zeta(1+u+v,a)-\frac{A(v,u)}{v}\zeta(1+u+v,b). \tag{8.106}$$

Substitute $A(u,v)=1+u(u+v)T(u,v)$ and the Laurent expansion (8.89). The result is

$$C(1+u,1+v;a)=-\frac{1}{uv}-\frac{R(u+v,a)}{u}-\frac{R(u+v,b)}{v}-\mathcal{F}(u,v;a). \tag{8.107}$$

Here the apparent diagonal pole at $u+v=0$ has canceled exactly.

The regular coefficient of $-R(u+v,a)/u$ at $u^m v^n$ is

$$-\frac{(-1)^{d+1}\gamma_{d+1}(a)}{(d+1)!}\binom{d+1}{m+1}=\frac{(-1)^d\gamma_{d+1}(a)}{(m+1)!n!}.$$

The other term similarly contributes $(-1)^d\gamma_{d+1}(b)/(m!(n+1)!)$. Now apply (8.97) to get (8.105).

To obtain a coefficient of total degree d in T, only terms of total degree at most $d+2$ in A are required. Formula (8.103) proves the asserted zeta coefficient algebra. In the product $(u+v)TR$, a Stieltjes index j occupies degree $j+1$ before multiplication by T, so $j\leq d-1$. All appearances of R are linear. These observations prove the remaining claims. □

Corollary 8.73 (Formal weight conservation)**.** *Assign weight $j+1$ to $\gamma_j(a)$ and $\gamma_j(b)$ and weight k to $\zeta(k)$. Every monomial in (8.105) has weight $m+n+2$.*

Proof. The degree-k component of $\log A$ has coefficient weight k. Thus a coefficient of total degree r in T has weight $r+2$. A contribution from $(u+v)TR$ to total degree d has $r+j+1=d$ and weight $(r+2)+(j+1)=d+2$. The leading Stieltjes terms have the same weight. □

This is a grading of the displayed symbolic algebra, not a theorem that the evaluated constants have no additional relations.

8.23.1 The first four nontrivial identities

The beginning of the universal kernel is

$$T(u,v)=\zeta(2)-\zeta(3)v+\frac{\zeta(4)+\zeta(2)^2}{2}(u^2+uv)+\zeta(4)v^2+O((|u|+|v|)^3). \tag{8.108}$$

Write $S_j(a)=\gamma_j(a)+\gamma_j(b)$. Coefficient extraction gives

$$J_{00}(a)=S_1(a)-2\zeta(2), \tag{8.109}$$
$$J_{10}(a)=\tfrac{1}{2}\gamma_2(a)+\gamma_2(b)+\zeta(2)S_0(a)-\zeta(3), \tag{8.110}$$
$$J_{11}(a)=\tfrac{1}{2}S_3(a)+2\zeta(2)S_1(a)+\zeta(3)S_0(a)-\zeta(4)-\zeta(2)^2, \tag{8.111}$$
$$J_{20}(a)=\tfrac{1}{3}\gamma_3(a)+\gamma_3(b)+2\zeta(2)S_1(a)+2\zeta(3)\gamma_0(b) - 3\zeta(4)-\zeta(2)^2. \tag{8.112}$$

These are exact analytic identities under Definition 8.66, not conjectured PSLQ relations. The asymmetric coefficients in J_{10} and J_{20} are necessary; exchanging a and b exchanges the two Laurent indices.

Because $\gamma_0=-\psi$, the first identity takes the particularly simple form

$$\operatorname{FP}\int_0^1 \psi(x)\psi(\{x+a\})\,\mathrm{d}x=\gamma_1(a)+\gamma_1(b)-\frac{\pi^2}{3}. \tag{8.113}$$

An equivalent identity containing only ordinary integrals is

$$\int_0^b\left[\psi(x)\psi(x+a)+\frac{\psi(a)}{x}\right]\mathrm{d}x+\int_0^a\left[\psi(b+t)\psi(t)+\frac{\psi(b)}{t}\right]\mathrm{d}t = \gamma_1(a)+\gamma_1(b)-\frac{\pi^2}{3}+\psi(a)\log b+\psi(b)\log a. \tag{8.114}$$

The endpoint singularities cancel inside each bracket. This version is useful when a conventional improper integral is preferable to distribution notation.

8.23.2 A finite exact coefficient algorithm

For target total index d, truncate (8.103) at degree $d+2$. Exponentiate by

$$A=\sum_{j=0}^{\lfloor (d+2)/2\rfloor}\frac{(\log A)^j}{j!}\pmod{\text{total degree}>d+2}.$$

Exact polynomial division of $A-1$ by $u(u+v)$ has zero remainder. Let $T(u,v)=\sum t_{ij}u^iv^j$. The correction coefficient needed in (8.105) is

$$[u^mv^n]\mathcal{F}=t_{mn}+t_{nm} + \sum_{\substack{i\le m,\ j\le n\\ k=d-i-j-1\ge 0}} t_{ij}\frac{(-1)^k}{k!}\binom{k+1}{m-i}\gamma_k(a) + \sum_{\substack{j\le m,\ i\le n\\ k=d-i-j-1\ge 0}} t_{ij}\frac{(-1)^k}{k!}\binom{k+1}{m-j}\gamma_k(b). \tag{8.115}$$

Only polynomial arithmetic over rational numbers and formal zeta symbols is used. The implementation does not perform floating-point relation recognition. Exact divisibility, reflection symmetry, linearity, and weight conservation provide independently checkable invariants of the generated formulas.

For the collision comparison, write the already proved linear closure as

$$J_{mn}(a) = e_{mn} + \sum_j (c_{mn,j}\gamma_j(a) + d_{mn,j}\gamma_j(b)).$$

Use the local germs $\mathcal{A}(u,v) = A(v,u)$ and $\mathcal{B}(u,v) = A(u,v)$ from (8.102), and put $w = u+v$. The beta formula then reads

$$K(u,v;a) := uvC(1+u,1+v;a) = -u\mathcal{A}(u,v)\zeta(1+w,b) - v\mathcal{B}(u,v)\zeta(1+w,a). \tag{8.116}$$

Put $Q(u,v) = (u\mathcal{A}+v\mathcal{B})/w$. Its numerator is divisible by w: both germs equal one at $v=-u$. This removes the apparent diagonal pole and supplies the finite coefficient construction used below.

8.24 Coincident endpoints and canonical collision subtraction

A translated correlation has two separated logarithmic endpoint divergences. At zero translation they merge into an additional x^{-2} singularity. We now show that the same coefficient table still applies, but only after the merged singularity is subtracted explicitly.

For an integer $r \geq 0$, put

$$E_r(L) = \sum_{k=0}^{r} \frac{r!}{k!} L^k. \tag{8.117}$$

The identity $E_r - E_r' = L^r$ implies

$$\int_\epsilon^1 \frac{\log^r x}{x^2}\,\mathrm{d}x = \frac{E_r(\log\epsilon)}{\epsilon} - r!. \tag{8.118}$$

Definition 8.74 (Coincident-endpoint finite part)**.** *Write $\gamma_j = \gamma_j(1)$ and define*

$$J_{mn}^{\mathrm{coll}} = \lim_{\epsilon\downarrow 0}\Big\{ \int_\epsilon^1 \gamma_m(x)\gamma_n(x)\,\mathrm{d}x - \frac{E_{m+n}(\log\epsilon)}{\epsilon} + \frac{\gamma_n}{m+1}\log^{m+1}\epsilon + \frac{\gamma_m}{n+1}\log^{n+1}\epsilon \Big\}. \tag{8.119}$$

Theorem 8.75 (Coincident closure)**.** *The finite part (8.119) exists. With exactly the coefficients of Theorem 8.72,*

$$J_{mn}^{\mathrm{coll}} = e_{mn} + \sum_{j=0}^{m+n+1} (c_{mn,j} + d_{mn,j})\gamma_j. \tag{8.120}$$

Equivalently it is the same mixed coefficient of

$$K_0(u,v) = -(u\mathcal{A}(u,v) + v\mathcal{B}(u,v))\zeta(1+u+v). \tag{8.121}$$

In particular,

$$J_{00}^{\mathrm{coll}} = 2\gamma_1 - \frac{\pi^2}{3}. \tag{8.122}$$

Proof. By (8.90), the cutoff in Definition 8.74 converges to the absolutely convergent expression

$$\begin{aligned} J^{\mathrm{coll}}_{mn} = & -(m+n)! \\ & + \int_0^1 \left\{ \frac{\log^m x}{x}[\gamma_n(1+x)-\gamma_n] + \frac{\log^n x}{x}[\gamma_m(1+x)-\gamma_m] \right. \\ & \qquad \left. + \gamma_m(1+x)\gamma_n(1+x) \right\} \mathrm{d}x. \end{aligned} \tag{8.123}$$

The two bracketed differences vanish linearly at zero, and (8.118) supplies the constant $-(m+n)!$.

To identify the coefficients, start where the ordinary coincident Hurwitz integral converges, for example $\operatorname{Re} s, \operatorname{Re} t < 0$. Fourier orthogonality gives

$$C_0(s,t) = 2\Gamma(1-s)\Gamma(1-t)(2\pi)^{s+t-2}\cos\left(\frac{\pi(s-t)}{2}\right)\zeta(2-s-t).$$

The Riemann functional equation, equivalently the specialization of the Fourier calculation in Theorem 8.68, gives $uvC_0(1+u,1+v) = K_0(u,v)$. Meromorphic continuation is justified by subtracting the local terms as above.

More explicitly, the holomorphic generating expression near zero is

$$\begin{aligned} R_0(u,v) = & -\frac{1}{1+u+v} \\ & + \int_0^1 \left\{ x^{-1-u}[R(v,1+x)-R(v,1)] + x^{-1-v}[R(u,1+x)-R(u,1)] \right. \\ & \qquad \left. + R(u,1+x)R(v,1+x) \right\} \mathrm{d}x. \end{aligned}$$

The first term is the analytic continuation of $\int_0^1 x^{-2-u-v}\,\mathrm{d}x$, not a discarded divergence. The remaining integrals converge locally uniformly near zero. On a starting convergence region, and hence by continuation,

$$R_0 = \frac{K_0 + 1 + uR(u,1) + vR(v,1)}{uv}.$$

Its Taylor coefficients are (8.123). Decomposing (8.121) as $-Q - (u\mathcal{A} + v\mathcal{B})R(u+v,1)$ now gives (8.120) by the identical finite algebra used earlier. □

For the first case, an ordinary integral version is worth recording:

$$\int_0^1 \left\{ \psi(1+x)^2 - \frac{2[\psi(1+x)+\gamma]}{x} \right\} \mathrm{d}x = 1 + 2\gamma_1 - \frac{\pi^2}{3}. \tag{8.124}$$

The apparently singular numerator in the second term vanishes at zero. There is no positivity contradiction in the negative value of J^{coll}_{00}: it is a finite part of a positive divergent square, not its ordinary nonnegative integral.

8.24.1 An exact convergent expansion at collision

The relationship between the separated and coincident finite parts can be expressed by an equality of convergent series, not merely an asymptotic prescription.

Theorem 8.76 (Collision polynomial and convergent regular part)**.** *For $m, n \geq 0$ define a polynomial*

$$\mathcal{P}_{mn}(L) = (-1)^{m+n+1} m! n! [u^{m+1} v^n] \mathcal{B}(u, v) e^{-(u+v)L}. \tag{8.125}$$

It has degree $m+n+1$, leading coefficient $1/(m+1)$, and coefficients in positive integer zeta values. There is an exact expansion

$$J_{mn}(a) = \frac{\mathcal{P}_{mn}(\log a)}{a} + J_{mn}^{\mathrm{coll}} + \sum_{k \geq 1} h_{mn,k} a^k, \qquad 0 < a < 1, \tag{8.126}$$

whose regular power series converges locally uniformly for $|a| < 1$. Its coefficients are

$$\begin{aligned} h_{mn,k} = \frac{(-1)^{m+n+1} m! n!}{k!} &[u^{m+1} v^{n+1}] \\ &\big(u\mathcal{A}(u, v) + (-1)^k v \mathcal{B}(u, v)\big)(1 + u + v)_k \zeta(1 + u + v + k). \end{aligned} \tag{8.127}$$

For $m = n$, all odd-indexed $h_{mm,k}$ vanish. In particular,

$$\lim_{a \downarrow 0} \left\{ J_{mn}(a) - \frac{\mathcal{P}_{mn}(\log a)}{a} \right\} = J_{mn}^{\mathrm{coll}}. \tag{8.128}$$

Proof. The classical parameter Taylor series at one gives, locally uniformly in the spectral variable,

$$\zeta(1 + w, 1 - a) = \sum_{k \geq 0} \frac{(1 + w)_k}{k!} \zeta(1 + w + k) a^k,$$

while the shift law gives

$$\zeta(1 + w, a) = a^{-1-w} + \sum_{k \geq 0} \frac{(1 + w)_k}{k!} \zeta(1 + w + k)(-a)^k.$$

These are meromorphic in w at zero, with the $k = 0$ pole explicitly visible. Substitution in (8.116) yields

$$\begin{aligned} K(u, v; a) = &- v\mathcal{B}(u, v) a^{-1-w} \\ &- \sum_{k \geq 0} \frac{u\mathcal{A}(u, v) + (-1)^k v \mathcal{B}(u, v)}{k!} (1 + w)_k \zeta(1 + w + k) a^k. \end{aligned} \tag{8.129}$$

The $k = 0$ summand is exactly K_0; its apparent pole at $w = 0$ is removable. The remaining series is holomorphic for small u, v and $|a| < 1$, uniformly on compact subsets. For example, on a fixed small spectral polydisc its terms and spectral derivatives grow at most polynomially in k times powers of $\log k$, whereas $|a|^k$ decays geometrically on compact subdiscs. Coefficient extraction is therefore legitimate. The singular first term gives (8.125), the constant gives Theorem 8.75, and the other terms give (8.127).

The highest power of L comes from $B_0 = 1$ and has degree $m + n + 1$; a binomial coefficient gives its stated leading coefficient. For $m = n$ and odd k, the factor $u\mathcal{A} - v\mathcal{B}$ is antisymmetric under $u \leftrightarrow v$, while every other factor in (8.127) is symmetric. Its diagonal mixed coefficient is zero. Finally the regular series has zero limit at $a = 0$, proving (8.128). □

The first collision polynomials are

$$\mathcal{P}_{00}(L) = L, \qquad \mathcal{P}_{10}(L) = \frac{L^2}{2} + \zeta(2), \qquad \mathcal{P}_{01}(L) = L^2 + \zeta(2). \tag{8.130}$$

The asymmetry of the last two is required: the factor that becomes singular at $a = 0$ has changed its index. In particular,

$$J_{00}(a) - \frac{\log a}{a} \longrightarrow 2\gamma_1 - \frac{\pi^2}{3}.$$

There is no arbitrary additional finite constant in this limit. It is fixed by the initial endpoint subtraction and agrees with the coincident prescription.

8.25 Ordinary integrals of all antiderivative orders

The finite-part identities have a convergent counterpart. We first choose mean-zero periodic primitives, rather than implicitly choosing constants of integration at a singular endpoint.

Definition 8.77 (Normalized primitive tower)**.** *For integers $m \geq 0$ and $r \geq 1$, put*

$$\mathfrak{h}_j^{(r-1)} = [u^j] \prod_{\nu=1}^{r-1} (1 - u/\nu)^{-1},$$

$$U_{m,r}(x) = \frac{(-1)^{m+1} m!}{(r-1)!} \sum_{j=0}^{m+1} \mathfrak{h}_{m+1-j}^{(r-1)} \frac{\zeta^{[j]}(1-r, x)}{j!}. \tag{8.131}$$

The empty product convention gives $\mathfrak{h}_0^{(0)} = 1$ and $\mathfrak{h}_j^{(0)} = 0$ for $j > 0$. Write $P_m = U_{m,1}$ and $Q_m = U_{m,2}$. Explicitly,

$$P_m(x) = \frac{(-1)^{m+1}}{m+1} \zeta^{[m+1]}(0, x), \tag{8.132}$$

$$Q_m(x) = (-1)^{m+1} m! \sum_{j=0}^{m+1} \frac{\zeta^{[j]}(-1, x)}{j!}. \tag{8.133}$$

These are normalized Hurwitz-jet primitives. The repository's primitive based at 1 is $P_m(x) - P_m(1)$; it is not $P_m(x)$ itself. Likewise, zero-based negative polygamma conventions must be converted explicitly before using the following correlations.

Lemma 8.78 (Primitive identities and regularity)**.** *Each $U_{m,r}$ has mean zero on $(0, 1)$. One has*

$$DP_m = G_m, \qquad DU_{m,r} = U_{m,r-1} \quad (r \geq 2), \qquad D^r U_{m,r} = G_m$$

as periodic distributions. The functions P_m lie in $L^2(\mathbb{T})$, and for $r \geq 2$, $U_{m,r}$ is a periodic C^{r-2} function.

Proof. Consider the spectral primitive

$$V_r(u, x) = \frac{(-1)^r \zeta(1 + u - r, x)}{(1 + u - r)_r} = -\frac{\zeta(1 + u - r, x)}{(r-1)! u} \prod_{\nu=1}^{r-1} (1 - u/\nu)^{-1}. \tag{8.134}$$

Its regular coefficient $(-1)^m m![u^m]V_r$ is (8.131). Pointwise differentiation gives $\partial_x V_r = V_{r-1}$ for $r \geq 2$, and $\partial_x V_1 = \zeta(1+u,x)$. Comparing regular coefficients proves the pointwise primitive relations.

For $r = 1$, the shift identity gives

$$P_m(x) = \frac{\log^{m+1} x}{m+1} + P_m(1+x).$$

This proves square integrability. Integration by parts against a periodic test function, with a lower cutoff, yields the boundary term $\phi(0)\log^{m+1}\epsilon/(m+1)$ and hence $DP_m = G_m$ by (8.59). For $r \geq 2$, the singular term in $\zeta^{[j]}(1-r,x)$ is $x^{r-1}(-\log x)^j$, which tends to zero at $x = 0$. The endpoint values at $0+$ and 1 consequently match. Iterating the pointwise derivative relations shows periodic C^{r-2} regularity and no extra jump terms. This proves the distributional relations. Finally, $\int_0^1 \zeta(s,x)\,\mathrm{d}x = 0$ for $\Re s < 1$, and its spectral derivatives also integrate to zero. Apply this to each term in (8.131). □

Theorem 8.79 (All-order convergent lifting)**.** *Suppose the closure formula for fixed m, n is written as*

$$J_{mn}(a) = \sum_j c_j \gamma_j(a) + \sum_j d_j \gamma_j(b) + c, \tag{8.135}$$

where all sums are finite and the coefficients are independent of a. For $r, k \geq 1$, $N = r + k$, define the ordinary integral

$$W_{m,n}^{r,k}(a) = \int_0^1 U_{m,r}(x) U_{n,k}(\{x+a\})\,\mathrm{d}x.$$

Then

$$\begin{aligned} W_{m,n}^{r,k}(a) = (-1)^r \sum_j c_j U_{j,N}(a) + (-1)^k \sum_j d_j U_{j,N}(b) \\ + \frac{(-1)^r c}{N!} B_N(a), \end{aligned} \tag{8.136}$$

where B_N is the ordinary Bernoulli polynomial. Thus every such integral reduces to finitely many Hurwitz-zeta jets at the single spectral integer $1 - r - k$ and zeta-valued coefficients. The highest required spectral derivative is $m + n + 2$.

Proof. The integral exists by the L^2 property and Cauchy–Schwarz. Its mean in a is zero by Fubini and the mean-zero normalization of both factors. In the distributional sense, periodic integration by parts gives

$$\frac{d^N}{da^N} W_{m,n}^{r,k}(a) = (-1)^r J_{mn}(a), \qquad 0 < a < 1. \tag{8.137}$$

The right side is smooth there.

We need enough endpoint regularity to fix all integration constants. From (8.96) and (8.98), for $q \neq 0$,

$$\begin{aligned} \widehat{G}_m(q) = (-1)^{m+1} m! [u^{m+1}] \exp\Bigg[\left(\kappa + \log|q| + \frac{\pi \mathrm{i}}{2} \operatorname{sgn} q \right) u \\ + \sum_{j \geq 2} \frac{\zeta(j)}{j} u^j \Bigg]. \end{aligned} \tag{8.138}$$

In particular $\widehat{G}_m(q) = O_m((1+\log|q|)^{m+1})$. The primitive relations imply $\widehat{U}_{m,r}(q) = \widehat{G}_m(q)/(2\pi i q)^r$. The Fourier coefficients of W are therefore $O((1+\log|q|)^{m+n+2}/|q|^N)$. After $N-2$ derivatives the series is still absolutely convergent. Hence W is periodic C^{N-2}.

Let $F(a)$ be the proposed right side of (8.136). It also has mean zero and is periodic C^{N-2}, using Lemma 8.78 and the endpoint identities for Bernoulli polynomials. Its Nth derivative on $(0,1)$ is $(-1)^r J_{mn}$: the reflected primitive contributes $(-1)^N$, and $B_N^{(N)} = N!$. Thus $W - F$ is a polynomial of degree at most $N-1$ on $(0,1)$. Matching its endpoint derivatives of orders $0,\dots,N-2$ forces every nonconstant coefficient to vanish, starting with the leading one. Its zero mean then forces the remaining constant to vanish.

The closure uses indices at most $m+n+1$, and $U_{j,N}$ uses spectral derivatives at most $j+1$. This gives the final derivative-order assertion. □

8.25.1 The first-primitive correlation

For $K_{mn}(a) = W^{1,1}_{m,n}(a)$, the theorem simplifies to

$$K_{mn}(a) = -\sum_j c_j Q_j(a) - \sum_j d_j Q_j(b) - \frac{c}{2}B_2(a), \qquad K''_{mn}(a) = -J_{mn}(a). \tag{8.139}$$

Equivalently, ordinary differentiation of the master integral at $(s,t) = (0,0)$ gives

$$K_{mn}(a) = (-1)^{m+n} m!n![s^{m+1}t^{n+1}]C(s,t;a). \tag{8.140}$$

The signs and factorials follow from (8.132); no finite-part operation occurs in this integral.

For example, (8.111) gives the fully explicit convergent identity

$$\begin{aligned} K_{11}(a) = {} & -\tfrac{1}{2}\big(Q_3(a)+Q_3(b)\big) - 2\zeta(2)\big(Q_1(a)+Q_1(b)\big) \\ & -\zeta(3)\big(Q_0(a)+Q_0(b)\big) + \tfrac{1}{2}\big(\zeta(4)+\zeta(2)^2\big)B_2(a). \end{aligned} \tag{8.141}$$

This evaluates a product of two second spectral derivatives at zero using derivatives at -1 of order at most four.

An example with unequal integration orders is

$$\int_0^1 Q_0(x)P_0(\{x+a\})\,\mathrm{d}x = U_{1,3}(a) - U_{1,3}(b) - \frac{\zeta(2)}{3}B_3(a), \tag{8.142}$$

where

$$U_{1,3}(x) = \frac{7}{8}\zeta(-2,x) + \frac{3}{4}\zeta^{[1]}(-2,x) + \frac{1}{4}\zeta^{[2]}(-2,x).$$

Thus the same finite closure gives exact identities involving successive antiderivatives rather than merely a formal recurrence for integration constants.

8.26 Explicit log-gamma and mixed identities

Let

$$f(x) = \log\Gamma(x) - \frac{\ell}{2}, \qquad \mathcal{K}(a) = \int_0^1 f(x)f(\{x+a\})\,\mathrm{d}x.$$

Lerch's identity gives $P_0 = -f$, so $\mathcal{K} = K_{00}$. No regularization is used in this definition: logarithmic singularities are square integrable.

Theorem 8.80 (Translated log-gamma correlation)**.** *For* $0 < a < 1$,

$$\mathcal{K}(a) = (1+\zeta(2))B_2(a) - \zeta^{[1]}(-1,a) - \zeta^{[1]}(-1,b) - \frac{1}{2}\big[\zeta^{[2]}(-1,a) + \zeta^{[2]}(-1,b)\big]. \tag{8.143}$$

Equivalently, for $z = e^{2\pi i a}$,

$$\mathcal{K}(a) = \frac{1}{2\pi^2}\Re\left[\mathrm{Li}_2^{[2]}(z) - 2\kappa\,\mathrm{Li}_2^{[1]}(z) + \left(\kappa^2 + \frac{\pi^2}{4}\right)\mathrm{Li}_2(z)\right]. \tag{8.144}$$

For the uncentered product,

$$\int_0^1 \log\Gamma(x)\log\Gamma(\{x+a\})\,\mathrm{d}x = \frac{\ell^2}{4} + \mathcal{K}(a). \tag{8.145}$$

Finally,

$$\mathcal{K}''(a) = \frac{\pi^2}{3} - \gamma_1(a) - \gamma_1(b). \tag{8.146}$$

Proof. Apply (8.139) to $J_{00} = \gamma_1(a) + \gamma_1(b) - 2\zeta(2)$. Since

$$Q_1(x) = \zeta(-1,x) + \zeta^{[1]}(-1,x) + \tfrac{1}{2}\zeta^{[2]}(-1,x)$$

and $\zeta(-1,x) = -B_2(x)/2$, the result is (8.143).

For an independent Fourier derivation of (8.144), differentiate (8.96) at $s = 0$. The positive-frequency coefficient of f is

$$\widehat{f}(n) = \frac{1}{4n} - \frac{\mathrm{i}(\kappa + \log n)}{2\pi n}, \qquad n \geq 1.$$

The negative-frequency coefficient is its conjugate. These coefficients also give the classical Kummer Fourier expansion. The L^2 correlation therefore equals the absolutely convergent series

$$\mathcal{K}(a) = \frac{1}{2\pi^2}\sum_{n\geq 1}\frac{\cos(2\pi n a)}{n^2}\left[(\kappa+\log n)^2 + \frac{\pi^2}{4}\right]. \tag{8.147}$$

Taking the first two spectral derivatives of the absolutely convergent polylogarithm series at order 2 proves (8.144). The mean of f is zero, so expanding the uncentered product proves (8.145). Equation (8.146) is the $m = n = 0$ case of (8.139). □

Corollary 8.81 (A mixed log-gamma–digamma identity)**.** *The same finite-part convention gives*

$$\mathrm{FP}\int_0^1 \log\Gamma(x)\psi(\{x+a\})\,\mathrm{d}x = \frac{1}{2}\big[\zeta^{[2]}(0,b) - \zeta^{[2]}(0,a)\big] + \zeta(2)(2a-1). \tag{8.148}$$

It is unchanged when $\log\Gamma(x)$ *is replaced by* $f(x)$.

Proof. Since $DP_0 = G_0$ with no contact term at this first primitive stage, $\mathcal{K}'(a) = \mathrm{FP}\int f(x)\psi(\{x+a\})\,\mathrm{d}x$. Differentiate (8.143) using

$$\partial_x \zeta^{[j]}(-1,x) = \zeta^{[j]}(0,x) - j\zeta^{[j-1]}(0,x).$$

The first spectral derivatives at zero cancel, and the terms from $\zeta(0,x) = 1/2 - x$ simplify to $\zeta(2)(2a-1)$. This gives (8.148). The canonical periodic finite part of ψ has mean zero, so subtracting $\ell/2$ from the log-gamma factor does not change the answer. □

For example, its left side with the centered function equals the entirely convergent expression

$$\int_0^b f(x)\psi(x+a)\,\mathrm{d}x + \int_0^a \left[(f(b+t)-f(b))\psi(t)+f(b)\psi(1+t)\right]\,\mathrm{d}t - f(b)\log a.$$

The recurrence $\psi(1+t)=\psi(t)+1/t$ removes the endpoint pole explicitly. This form also provides a direct numerical check independent of differentiating a finite-part integral.

8.26.1 Half shift and the zero-shift limit

At $a=1/2$, use $\mathrm{Li}_s(-1)=(2^{1-s}-1)\zeta(s)$. With $L_2=\log 2$,

$$\mathcal{K}(1/2) = -\frac{\zeta^{[2]}(2)}{4\pi^2}+\frac{\kappa-L_2}{2\pi^2}\zeta^{[1]}(2) + \frac{L_2^2+2\kappa L_2-\kappa^2-\pi^2/4}{24}. \tag{8.149}$$

This formula concerns a *translated* correlation, not the reflected product $f(x)f(1-x)$.

The absolutely convergent series (8.147) is continuous at $a=0$. Its value there recovers the classical second log-gamma moment [28]:

$$\int_0^1 \log^2\Gamma(x)\,\mathrm{d}x = \frac{\ell^2}{4}+\frac{\zeta^{[2]}(2)}{2\pi^2}-\frac{\kappa\zeta^{[1]}(2)}{\pi^2}+\frac{\kappa^2}{12}+\frac{\pi^2}{48}. \tag{8.150}$$

Equivalently, taking the continuous limit in (8.143) gives

$$\mathcal{K}(0)=\frac{1+\zeta(2)}{6}-2\zeta^{[1]}(-1)-\zeta^{[2]}(-1).$$

These are useful consistency checks, not priority claims. In contrast, $J_{00}(a)=\log a/a+O(1)$ as $a\downarrow 0$, by (8.90). The finite-part product has no continuous zero-shift specialization. The ordinary correlation remains continuous but its second derivative develops the expected singularity.

8.27 Rational shifts and exact arithmetic coordinates

For integers $q\ge 2$ and $1\le p<q$, residue-class decomposition gives the classical finite Fourier identity

$$\mathrm{Li}_s(e^{2\pi\mathrm{i}p/q})=q^{-s}\sum_{r=1}^{q}e^{2\pi\mathrm{i}pr/q}\zeta(s,r/q). \tag{8.151}$$

It is proved by absolutely convergent series for $\Re s>1$ and then by analytic continuation. Spectral differentiation yields

$$\mathrm{Li}_s^{[j]}(e^{2\pi\mathrm{i}p/q})=q^{-s}\sum_{r=1}^{q}e^{2\pi\mathrm{i}pr/q}\sum_{h=0}^{j}\binom{j}{h}(-\log q)^{j-h}\zeta^{[h]}(s,r/q). \tag{8.152}$$

At $s=1$ the poles of the finite sum cancel because the roots sum to zero. One must combine the pole terms before evaluating; separately substituting $\zeta(1,r/q)$ is invalid.

At a rational shift, Theorem 8.72 reduces all Stieltjes correlations to rational-argument Stieltjes constants and ordinary zeta values. Theorem 8.79 places every primitive correlation at a negative integer spectral argument. Equation (8.152) instead evaluates the polylogarithmic side using positive-order Hurwitz jets. The two routes give independent choices of arithmetic coordinates for numerical replay and exact symbolic comparison.

For a Dirichlet character modulo q, including an imprimitive one with its actual modulus, the identity

$$\sum_{r=1}^{q} \chi(r)\zeta(s, r/q) = q^s L(s, \chi)$$

and its spectral derivatives transform the character components into Dirichlet-L jets. This gives a reduction framework, not a claim of numerical independence or minimality of the resulting coordinates.

Corollary 8.82 (Small-denominator digamma correlations)**.** *Write* $\gamma_1 = \gamma_1(1)$*. Then*

$$\Pi_{0,0}(1/2) = 2\gamma_1 - 4\gamma \log 2 - 2\log^2 2 - \frac{\pi^2}{3}, \tag{8.153}$$

$$\Pi_{0,0}(1/3) = 2\gamma_1 - 3\gamma \log 3 - \frac{3}{2}\log^2 3 - \frac{\pi^2}{3}, \tag{8.154}$$

$$\Pi_{0,0}(1/4) = 2\gamma_1 - 6\gamma \log 2 - 7\log^2 2 - \frac{\pi^2}{3}. \tag{8.155}$$

Proof. Expand at $s = 1$ the exact identities

$$\begin{aligned} \zeta(s, 1/2) &= (2^s - 1)\zeta(s), \\ \zeta(s, 1/3) + \zeta(s, 2/3) &= (3^s - 1)\zeta(s), \\ \zeta(s, 1/4) + \zeta(s, 3/4) &= (4^s - 2^s)\zeta(s). \end{aligned}$$

Compare the coefficient of $s - 1$ with (8.89) and insert the resulting sums into (8.113). In the first line the correlation contains two copies of $\gamma_1(1/2)$, which accounts for its different coefficients. □

Higher indices require no new search: expand these finite exponential factors to the needed order and use (8.105). At larger denominators, retaining a nontrivial character block is legitimate; failing to reduce that block does not establish transcendence or independence.

The canonical finite-part and primitive identities above are audited against the independent coefficient engines and their separately labeled numerical replays under `verification/eighth-replay/`. The full contact law is proved in the next chapter before derivatives of these finite-part identities are used. The singular same-point finite part and the continuous ordinary log-Gamma autocorrelation have different zero-shift behavior.

8.28 All circular convolution powers of log Gamma

At the integrated end no finite part is needed. Put

$$L = \log(2\pi), \qquad Z_j(x) = \zeta^{[j]}(0, x), \qquad g(x) = Z_1(x) = \log \Gamma(x) - \frac{L}{2}. \tag{8.156}$$

Each Z_j has mean zero. It is locally a polynomial in $\log x$ plus a smooth function near zero, and hence belongs to every finite $L^p(0, 1)$.

Theorem 8.83 (All mixed integrated jets)**.** *Let $d \geq 2$ and $r_1, \dots, r_d \geq 0$ be integers. Write $S = s_1 + \cdots + s_d$. Then, as an identity of continuous functions after circular convolution,*

$$Z_{r_1} * \cdots * Z_{r_d}(x) = \partial_{s_1}^{r_1} \cdots \partial_{s_d}^{r_d} \left[\frac{\prod_{i=1}^{d} \Gamma(1-s_i)}{\Gamma(d-S)} \zeta(1-d+S, x) \right] \Bigg|_{s_1 = \cdots = s_d = 0}. \tag{8.157}$$

It holds for $0 < x < 1$ and at the integer point by continuous extension. Thus any mixed convolution of the centered primitives A_n reduces to finitely many jets at the single integer $1-d$.

Proof. Before differentiation, write each Hurwitz factor as $\zeta(s_i, x) = \Gamma(1-s_i) R_{1-s_i}(x)$ and apply Theorem 8.58 repeatedly. For parameters sufficiently close to zero, each factor is holomorphic as an L^2-valued family: the endpoint singularities are bounded by $x^{-\eta}(1 + |\log x|^M)$ with $2\eta < 1$ on each fixed derivative order and compact parameter set. Convolution of two L^2 functions is continuous, by Cauchy–Schwarz and translation continuity; further convolution with L^1 factors preserves continuity. These same estimates justify differentiating all the convolutions. The right side is continuous at the integer point since $\mathrm{Re}(1-d+S) < 0$ in a small parameter neighborhood. This proves the formula and its endpoint extension. □

The case with every $r_i = 1$ admits a compact all-d formula. Define $H_m^{(j)} = \sum_{a=1}^{m} a^{-j}$, and, for fixed $d \geq 2$, set

$$b_1 = H_{d-1}, \qquad b_j = (j-1)!\big[H_{d-1}^{(j)} - \zeta(j)\big] \quad (j \geq 2). \tag{8.158}$$

Theorem 8.84 (All convolution powers of centered log-gamma)**.** *For every integer $d \geq 2$,*

$$g^{*d}(x) = \frac{1}{(d-1)!} \sum_{j=0}^{d} \binom{d}{j} \mathcal{B}_{d-j}(b_1, \dots, b_{d-j}) \zeta^{[j]}(1-d, x), \tag{8.159}$$

where $\mathcal{B}_0 = 1$ and the b_j are scalar quantities in (8.158).

Proof. In (8.157), differentiate once in each s_i. Only the squarefree coefficient $s_1 \cdots s_d$ matters. Modulo $s_i^2 = 0$, one has

$$\prod_{i=1}^{d} \Gamma(1-s_i) = \prod_{i=1}^{d} (1 + \gamma s_i) = \exp(\gamma S).$$

Consequently the required mixed derivative equals the ordinary dth derivative at $S = 0$ of $e^{\gamma S} \zeta(1-d+S, x)/\Gamma(d-S)$. Now

$$\log \frac{e^{\gamma S} \Gamma(d)}{\Gamma(d-S)} = H_{d-1} S + \sum_{j \geq 2} \frac{H_{d-1}^{(j)} - \zeta(j)}{j} S^j.$$

This follows directly from $\Gamma(d-S) = \Gamma(1-S) \prod_{a=1}^{d-1} (a-S)$ and the Taylor series of $\log \Gamma(1-S)$. Apply (8.72) and the Leibniz rule; $\Gamma(d) = (d-1)!$ supplies the prefactor. □

The first two nontrivial cases are

$$g * g = \zeta''(-1, x) + 2\zeta'(-1, x) + [2 - \zeta(2)] \zeta(-1, x), \tag{8.160}$$

$$\begin{aligned} g^{*3} = {} & \frac{1}{2} \zeta'''(-2, x) + \frac{9}{4} \zeta''(-2, x) + \left[\frac{21}{4} - \frac{3}{2} \zeta(2) \right] \zeta'(-2, x) \\ & + \left[\frac{45}{8} - \frac{9}{4} \zeta(2) - \zeta(3) \right] \zeta(-2, x). \end{aligned} \tag{8.161}$$

The first left side is explicitly

$$\int_0^x g(t)g(x-t)\,\mathrm{d}t + \int_x^1 g(t)g(1+x-t)\,\mathrm{d}t.$$

For uncentered $f = \log\Gamma$, mean-zero convolution annihilates all mixed constant/nonconstant terms, so

$$f^{*d}(x) = \left(\frac{L}{2}\right)^d + g^{*d}(x). \tag{8.162}$$

Remark 8.85 (Convolution powers are not pointwise moments). The value $g^{*3}(0)$ is a two-dimensional circular-convolution integral. It is not $\int_0^1 g(t)^3\,\mathrm{d}t$. Thus (8.161) does not eliminate the Tornheim derivative coordinates of the ordinary cubic log-gamma moment discussed in the repository. Likewise, the mixed reflected pointwise moments in `07-reflected-moments.tex` are different objects. This distinction prevents a spurious claim of resolving that separate reduction problem.

8.29 Nonintegral twists: Lerch jets, contact terms, and the half-twist

The convolution-algebra continuation explicitly asks for a finite-part Stieltjes normal form for nonintegral frequency shifts, with an all-index half-twist multiplication table that includes its point-supported counterterms [108, "Twists, characters, and alternate endpoint normalizations"]. The twisted continuation [119] supplies the full proof below. The underlying Lerch function is classical [115, §25.14]; at the half-twist its order jets are the modified Stieltjes constants of Hu and Kim [117]. An ordinary, nonsingular alternating Hurwitz convolution is already given by Zhu, Hu, and Kim [118, Theorem 1.13]. The additional content below is the continuation through the singular orders, including the exact contact terms, covariant primitives, and degeneration to the untwisted normalization. In particular, twisting changes the special-function jets and the convolution identity; the positive-integer zeta values in the Gamma correction remain the same.

8.29.1 The meromorphic family and its endpoint normalization

Fix $0 < \theta < 1$, put $\omega = 2\pi\theta$ and $z = e^{-i\omega}$, and retain the Fourier and convolution conventions of Section 8.17. Any other real nonintegral twist $\theta = m + \vartheta$, $m \in \mathbb{Z}$, $0 < \vartheta < 1$, is obtained by the periodic modulation $U(x) \mapsto e^{-2\pi i m x}U(x)$. Thus the chosen interval loses no real nonintegral shifts. Define

$$\lambda_{n,\theta} = 2\pi i(n+\theta), \qquad L_{n,\theta} = \log(2\pi|n+\theta|) + \frac{i\pi}{2}\operatorname{sgn}(n+\theta), \qquad n \in \mathbb{Z}. \tag{8.163}$$

Every power of $\lambda_{n,\theta}$ uses this logarithm. There is no zero frequency to remove. We write

$$D_\theta = D + 2\pi i\theta, \qquad \widehat{D_\theta U}(n) = \lambda_{n,\theta}\widehat{U}(n).$$

Here D_θ acts on ordinary periodic distributions, rather than on functions satisfying a different boundary condition. The nonperiodic factor $e^{-i\omega x}$ below is used only for ordinary representatives on $0 < x < 1$ and for local endpoint Taylor coefficients. It is not used as a global multiplier on periodic distributions; the Fourier or cutoff definitions fix those extensions. Explicitly, for $m \in \mathbb{Z}$ the extension to shifted intervals is $W_s^{\theta+m}(x) = e^{-2\pi i m x}W_s^\theta(x)$, with the same identity for Z_s, U_n, and P_k after their definitions below.

Theorem 8.86 (Twisted Lerch family)**.** *The Fourier coefficients*

$$\widehat{W_s^\theta}(n) = \lambda_{n,\theta}^{s-1}, \qquad n \in \mathbb{Z}, \tag{8.164}$$

define an entire distribution-valued function of s. The family

$$Z_s^\theta = \Gamma(1-s) W_s^\theta \tag{8.165}$$

is meromorphic, with simple poles precisely at $s = k+1$, $k \geq 0$, and

$$\operatorname{Res}_{s=k+1} Z_s^\theta = \frac{(-1)^{k+1}}{k!} D_\theta^k \delta_0. \tag{8.166}$$

For $\operatorname{Re} s < 1$, it is the locally integrable periodic extension of

$$f_s^\theta(x) = e^{-i\omega x} \Phi(z, s, x), \qquad 0 < x < 1. \tag{8.167}$$

For every s, the restriction of the meromorphic distribution to $(0,1)$ is the entire function of s on the right of (8.167). The convolution and differential identities are

$$W_s^\theta * W_t^\theta = W_{s+t-1}^\theta, \qquad W_1^\theta = \delta_0, \tag{8.168}$$

$$D_\theta W_s^\theta = W_{s+1}^\theta, \qquad D_\theta Z_s^\theta = -s Z_{s+1}^\theta. \tag{8.169}$$

Consequently,

$$Z_s^\theta * Z_t^\theta = \frac{\Gamma(1-s)\Gamma(1-t)}{\Gamma(2-s-t)} Z_{s+t-1}^\theta \tag{8.170}$$

as a meromorphic identity, interpreted through (8.168) at intersections of polar divisors.

Proof. On compact sets of s, the coefficients in (8.164), and each fixed number of their order derivatives, have a common polynomial growth bound in $|n|$. Pairing their Fourier series with the rapidly decreasing Fourier coefficients of a smooth test function therefore gives normal convergence. This proves entire dependence. The Fourier multiplier of $D_\theta^k \delta_0$ is $\lambda_{n,\theta}^k$; the residues of $\Gamma(1-s)$ prove (8.166). These distributions are nonzero, so every displayed pole is present.

For completeness, the Lerch identification can be proved without assuming a boundary-value Fourier formula. Initially let $0 < \operatorname{Re} s < 1$ and $\epsilon > 0$. Abel damping gives

$$f_{s,\epsilon}^\theta(x) = \sum_{m \geq 0} e^{-(\epsilon + i\omega)(m+x)} (m+x)^{-s}.$$

Absolute convergence permits unfolding its nth Fourier coefficient:

$$\begin{aligned} \int_0^1 f_{s,\epsilon}^\theta(x) e^{-2\pi i n x}\, dx &= \int_0^\infty t^{-s} e^{-(\epsilon + \lambda_{n,\theta})t}\, dt \\ &= \Gamma(1-s)(\epsilon + \lambda_{n,\theta})^{s-1}. \end{aligned}$$

Since $z \neq 1$, the partial sums of z^m are bounded. Summation by parts shows that the tail starting at $m = 1$ converges uniformly in $x \in [0,1]$, and locally uniformly for s in this strip, as $\epsilon \downarrow 0$. The remaining term is bounded by an integrable multiple of $x^{-\operatorname{Re} s}$. Thus the limit can be taken in $L^1(0,1)$ and in its Fourier coefficients, giving (8.167). Repeated summation by parts, or finite differences of $(m+x)^{-s}$, continues the Lerch tail holomorphically to every s, locally uniformly when x stays away from zero. Near zero its only nonsmooth term is

$$f_s^\theta(x) = e^{-i\omega x} x^{-s} + e^{-i\omega x} z \Phi(z, s, 1+x). \tag{8.171}$$

This proves L^1 holomorphy throughout $\operatorname{Re} s < 1$ and the asserted continuation away from the endpoint. The remaining identities follow by multiplying the Fourier coefficients; the Gamma recurrence gives the last identity of (8.169). □

The pointwise function in (8.167) has no pole in s. Every pole in (8.165) is an endpoint distribution. This is different from the untwisted Hurwitz family, whose pole at $s = 1$ also has an ordinary constant contribution away from the endpoint.

For $n \geq 0$ define ordinary Lerch jets and their periodic finite parts by

$$c_n^\theta(x) = (-1)^n \, \partial_s^n f_s^\theta(x)\Big|_{s=1}, \qquad 0 < x < 1, \tag{8.172}$$

$$Z_{1+u}^\theta = -\frac{\delta_0}{u} + \sum_{n\geq 0} \frac{(-1)^n}{n!} U_n^\theta u^n. \tag{8.173}$$

Set, also,

$$P_k^\theta = (-1)^{k+1} k! \, \mathrm{FP}_{s=k+1} \, Z_s^\theta, \qquad k \geq 0. \tag{8.174}$$

In particular $P_0^\theta = -U_0^\theta$. Off the endpoint it equals $p_k^\theta(x) = (-1)^{k+1} k! f_{k+1}^\theta(x)$.

Proposition 8.87 (Explicit singular normalization)**.** *For a smooth periodic test function ϕ, put $h(x) = e^{-i\omega x}\phi(x)$ near zero. Then*

$$\langle U_n^\theta, \phi \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 c_n^\theta(x)\phi(x)\, dx + \frac{\phi(0)}{n+1} \log^{n+1} \epsilon \right\}. \tag{8.175}$$

If $F_k^\theta = \mathrm{FP}_{s=k+1} \, Z_s^\theta$ and $k \geq 1$, then

$$\langle F_k^\theta, \phi \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_\epsilon^1 f_{k+1}^\theta(x)\phi(x)\, dx - \sum_{j=0}^{k-1} \frac{h^{(j)}(0)}{j!} \frac{\epsilon^{j-k}}{k-j} + \frac{h^{(k)}(0)}{k!} \log \epsilon \right\}. \tag{8.176}$$

These normalizations retain the zero Fourier coefficient; no mean-zero condition is imposed.

Proof. Write the second term in (8.171) as $r_s^\theta(x)$. After subtracting the degree-N Taylor polynomial of h, the continuation of the test-function pairing is

$$\begin{aligned} \langle Z_s^\theta, \phi \rangle = {} & \int_0^1 x^{-s} \left(h(x) - \sum_{j=0}^{N} \frac{h^{(j)}(0)}{j!} x^j \right) dx + \sum_{j=0}^{N} \frac{h^{(j)}(0)}{j!(j+1-s)} \\ & + \int_0^1 r_s^\theta(x)\phi(x)\, dx. \end{aligned} \tag{8.177}$$

The first integral is holomorphic for $\mathrm{Re}\, s < N + 2$, and the last integral is entire. At $s = 1 + u$, expand $x^{-1-u} = x^{-1} \sum_n (-1)^n u^n \log^n x/n!$ after the constant subtraction. This gives (8.175). At $s = k + 1$, take the constant Laurent coefficient and integrate each subtracted monomial explicitly; this gives (8.176). The residue computed this way agrees with (8.166), because

$$\langle D_\theta^k \delta_0, \phi \rangle = (-1)^k h^{(k)}(0).$$

This also fixes the signs of all the lower derivatives of δ_0 inside $D_\theta^k \delta_0$. □

8.29.2 All-index multiplication and covariant contact terms

Let

$$B_\theta(u) = -u Z_{1+u}^\theta = \Gamma(1-u) W_{1+u}^\theta, \qquad G_\theta(u) = \sum_{n\geq 0} \frac{(-1)^n}{n!} U_n^\theta u^n. \tag{8.178}$$

Thus $B_\theta(u) = \delta_0 - u G_\theta(u)$. All the following equalities are analytic germs at zero in the algebra of periodic distributions with its usual convolution identity δ_0.

Theorem 8.88 (Twisted Stieltjes normal form)**.** *Put* $C(u,v)=\Gamma(1-u)\Gamma(1-v)/\Gamma(1-u-v)$. *Then*

$$B_\theta(u)*B_\theta(v)=C(u,v)B_\theta(u+v), \tag{8.179}$$

$$B_\theta(u)=\exp_*\left(uP_0^\theta+\sum_{r\ge2}\frac{\zeta(r)}{r}u^r\delta_0\right). \tag{8.180}$$

In particular,

$$U_n^\theta=\frac{(-1)^{n+1}}{n+1}\mathcal{B}^*_{n+1}\big(P_0^\theta,\zeta(2)\delta_0,2!\zeta(3)\delta_0,\dots,n!\zeta(n+1)\delta_0\big). \tag{8.181}$$

An explicit all-index multiplication table is furnished by the removable coefficient formula

$$\begin{aligned}G_\theta(u)*G_\theta(v)=\frac{1}{uv}\big\{&uG_\theta(u)+vG_\theta(v)-(u+v)C(u,v)G_\theta(u+v)\\&+(C(u,v)-1)\delta_0\big\}.\end{aligned} \tag{8.182}$$

More precisely, $U_m^\theta*U_n^\theta$ *is* $(-1)^{m+n}m!n!$ *times the coefficient of* u^mv^n *on the right. This is a finite combination of* $\delta_0,U_0^\theta,\dots,U_{m+n+1}^\theta$*, with coefficients polynomial in ordinary zeta values.*

Proof. The first identity follows from (8.168). At every Fourier mode, including $n=0$,

$$\widehat{P_0^\theta}(n)=\gamma+L_{n,\theta},\qquad \widehat{B_\theta(u)}(n)=\exp\left((\gamma+L_{n,\theta})u+\sum_{r\ge2}\frac{\zeta(r)}{r}u^r\right).$$

This proves (8.180) and the Bell formula, with normal convergence after test-function pairing. Substitute $B_\theta(u)=\delta_0-uG_\theta(u)$ into (8.179) and solve for $G_\theta(u)*G_\theta(v)$ to obtain (8.182). Its numerator vanishes when either variable is zero, so the quotient is removable. Extracting a fixed coefficient uses only finitely many coefficients of the Gamma quotient and of G_θ. □

For example, the full identities, including their contact masses, are

$$U_0^\theta*U_0^\theta=2U_1^\theta-\zeta(2)\delta_0, \tag{8.183}$$

$$U_0^\theta*U_1^\theta=\tfrac{3}{2}U_2^\theta-\zeta(2)U_0^\theta+\zeta(3)\delta_0, \tag{8.184}$$

$$U_0^\theta*U_2^\theta=\tfrac{4}{3}U_3^\theta-2\zeta(2)U_1^\theta+2\zeta(3)U_0^\theta-2\zeta(4)\delta_0, \tag{8.185}$$

$$U_1^\theta*U_1^\theta=U_3^\theta-2\zeta(2)U_1^\theta+2\zeta(3)U_0^\theta-\tfrac{1}{4}\zeta(4)\delta_0. \tag{8.186}$$

The presence of δ_0, rather than δ_0-1, matters even when these formulas are eventually evaluated away from zero.

Theorem 8.89 (All covariant contact corrections)**.** *For* $k\ge0$*, with* $H_0=0$*,*

$$\widehat{P_k^\theta}(n)=\lambda_{n,\theta}^k(\gamma+L_{n,\theta}-H_k), \tag{8.187}$$

$$D_\theta P_k^\theta=P_{k+1}^\theta+\frac{1}{k+1}D_\theta^{k+1}\delta_0, \tag{8.188}$$

$$Q_k^\theta:=D_\theta^kP_0^\theta=P_k^\theta+H_kD_\theta^k\delta_0. \tag{8.189}$$

Hence the all-index twisted polygamma product is

$$P_k^\theta*P_\ell^\theta=D_\theta^{k+\ell}\big\{2U_1^\theta+(H_k+H_\ell)U_0^\theta+(H_kH_\ell-\zeta(2))\delta_0\big\}. \tag{8.190}$$

More generally, let $R^{\theta}_{n,k}$ be the raw Hadamard finite part of the ordinary function $D^k_\theta c^\theta_n$ on $(0,1)$, using Taylor subtraction of its powers of x and $\log x$ at zero and no additional endpoint mass. Then

$$D^k_\theta U^\theta_n = R^\theta_{n,k} - h_{k,n} D^k_\theta \delta_0, \qquad h_{k,n} = (-1)^n n! [u^{n+1}] \prod_{j=1}^{k} \left(1 + \frac{u}{j}\right). \tag{8.191}$$

In particular $h_{k,0} = H_k$, and $h_{k,n} = 0$ for $n \geq k$.

Proof. At $s = k + 1 + u$,

$$\Gamma(-k-u) = \frac{(-1)^{k+1}}{k!u} + \frac{(-1)^k}{k!}(H_k - \gamma) + O(u).$$

Multiply this by $\lambda^k_{n,\theta} e^{uL_{n,\theta}}$ and take the finite part. This proves (8.187). The two derivative formulas follow by multiplication by $\lambda_{n,\theta}$ and $H_{k+1} - H_k = 1/(k+1)$. Multiplying the two affine polynomials $\gamma + L - H_k$ and $\gamma + L - H_\ell$ and using (8.183) gives (8.190).

For clarity, the raw finite part in (8.191) can equivalently be defined by the coefficientwise constant cutoff term of $D^k_\theta f^\theta_{1+u}$. The Mellin subtraction (8.177) identifies the regular Laurent coefficients of Z^θ_{k+1+u} with those cutoff finite parts of its order derivatives. Repeated application of (8.169) gives

$$D^k_\theta Z^\theta_{1+u} = (-1)^k (1+u)_k Z^\theta_{k+1+u}.$$

The contribution from the pole on the right is exactly

$$-\frac{1}{u} \prod_{j=1}^{k} \left(1 + \frac{u}{j}\right) D^k_\theta \delta_0.$$

Its constant and higher coefficients supply all the difference between derivative-compatible and raw finite parts. Equating the coefficients of $(-1)^n u^n/n!$ proves (8.191). □

For example, $D^2_\theta \delta_0 = \delta_0'' + 4\pi i\theta \delta_0' - (2\pi\theta)^2 \delta_0$. Thus (8.189) specifies the lower point-supported terms as well as the highest derivative. Writing only $H_k \delta_0^{(k)}$ would give the wrong normalization for a nonzero twist.

8.29.3 Rational twists and ordinary special functions

Proposition 8.90 (Finite Hurwitz reduction)**.** *Let $\theta = p/q$, where $1 \leq p < q$ are relatively prime, and put $z = e^{-2\pi i p/q}$. Then*

$$f^{p/q}_s(x) = \frac{e^{-2\pi i p x/q}}{q^s} \sum_{r=0}^{q-1} z^r \zeta\left(s, \frac{x+r}{q}\right), \qquad 0 < x < 1. \tag{8.192}$$

The sum is entire in s: its Hurwitz residues cancel. Consequently,

$$c^{p/q}_n(x) = \frac{e^{-2\pi i p x/q}}{q} \sum_{r=0}^{q-1} z^r \sum_{j=0}^{n} \binom{n}{j} (\log q)^{n-j} \gamma_j \left(\frac{x+r}{q}\right), \tag{8.193}$$

$$p^{p/q}_k(x) = \frac{e^{-2\pi i p x/q}}{q^{k+1}} \sum_{r=0}^{q-1} z^r \psi^{(k)}\left(\frac{x+r}{q}\right), \qquad k \geq 0. \tag{8.194}$$

These ordinary functions, extended by (8.175) and (8.176), give the finite-part distributions in Theorems 8.88 and 8.89.

Proof. For $\operatorname{Re} s > 1$, split the defining Lerch series into its residue classes modulo q. This gives (8.192); analytic continuation proves it everywhere. Since $\sum_r z^r = 0$, the common Hurwitz pole vanishes before the factor q^{-s} is expanded. Differentiating the resulting Taylor expansion at $s = 1$ gives (8.193). At positive integers use the Hurwitz–polygamma identity; its limiting case at $s = 1$ uses $\gamma_0 = -\psi$ and the same pole cancellation. This proves (8.194) also for $k = 0$. □

Thus additive rational twists require finite colored combinations of Hurwitz jets, while the coefficients governing their convolution remain the ordinary $\zeta(r)$ from $\log\Gamma(1-u)$. No replacement by Dirichlet eta values is needed in (8.181) or (8.182).

8.29.4 The half-twist and convergent alternating identities

At $\theta = \frac{1}{2}$, set

$$\zeta_E(s,x) = \Phi(-1,s,x) = 2^{-s}\left\{\zeta\left(s,\frac{x}{2}\right) - \zeta\left(s,\frac{x+1}{2}\right)\right\}, \tag{8.195}$$

$$\widetilde{\gamma}_n(x) = (-1)^n\,\partial_s^n\zeta_E(s,x)|_{s=1}\,. \tag{8.196}$$

These modified Stieltjes constants are the established functions of Hu and Kim [117]. The explicit conversion is

$$\widetilde{\gamma}_n(x) = \frac{1}{2}\sum_{j=0}^{n}\binom{n}{j}(\log 2)^{n-j}\left\{\gamma_j\left(\frac{x}{2}\right) - \gamma_j\left(\frac{x+1}{2}\right)\right\}, \qquad c_n^{1/2}(x) = e^{-\pi i x}\widetilde{\gamma}_n(x). \tag{8.197}$$

In particular, the Nielsen beta function is

$$\beta(x) = \widetilde{\gamma}_0(x) = \frac{1}{2}\left\{\psi\left(\frac{x+1}{2}\right) - \psi\left(\frac{x}{2}\right)\right\}. \tag{8.198}$$

The elementary shift relation

$$\widetilde{\gamma}_n(x) + \widetilde{\gamma}_n(x+1) = \frac{\log^n x}{x} \tag{8.199}$$

fixes its endpoint singularity, including every logarithmic order.

Theorem 8.91 (All-index alternating finite-part convolution)**.** *For $0 < a < 1$ and $m, n \geq 0$, the following expression consists of absolutely convergent integrals:*

$$\begin{aligned}\mathcal{A}_{mn}(a) = &\int_0^a \left\{\widetilde{\gamma}_m(x)\widetilde{\gamma}_n(a-x) - \widetilde{\gamma}_n(a)\frac{\log^m x}{x} - \widetilde{\gamma}_m(a)\frac{\log^n(a-x)}{a-x}\right\}dx\\ &- \int_a^1 \widetilde{\gamma}_m(x)\widetilde{\gamma}_n(1+a-x)\,dx\\ &+ \frac{\widetilde{\gamma}_n(a)}{m+1}\log^{m+1} a + \frac{\widetilde{\gamma}_m(a)}{n+1}\log^{n+1} a.\end{aligned} \tag{8.200}$$

It equals $e^{\pi i a}(U_m^{1/2} * U_n^{1/2})(a)$, where the convolution is smooth on $(0,1)$. Every $\mathcal{A}_{mn}$ is therefore the finite linear combination of $\widetilde{\gamma}_0, \ldots, \widetilde{\gamma}_{m+n+1}$ obtained from (8.182) by discarding its δ_0 term. In particular,

$$\mathcal{A}_{00}(a) = 2\widetilde{\gamma}_1(a), \tag{8.201}$$

$$\mathcal{A}_{01}(a) = \tfrac{3}{2}\widetilde{\gamma}_2(a) - \zeta(2)\widetilde{\gamma}_0(a), \tag{8.202}$$

$$\mathcal{A}_{11}(a) = \widetilde{\gamma}_3(a) - 2\zeta(2)\widetilde{\gamma}_1(a) + 2\zeta(3)\widetilde{\gamma}_0(a). \tag{8.203}$$

Proof. For integrable orders, convolution on the circle splits into $0 < x < a$ and $a < x < 1$. The phases of the two factors have product $e^{-\pi i a}$ in the first interval and $e^{-\pi i(1+a)} = -e^{-\pi i a}$ in the second. This is the source of the alternating sign in (8.200), and agrees with the nonsingular alternating convolution of [118, Theorem 1.13].

To continue to the jets, use the normalized cutoff distributions in (8.175). On a compact subinterval of $0 < a < 1$, the two endpoint singularities in the first integral occur at distinct points, $x = 0$ and $x = a$. Their only divergent terms are the two explicit subtractions in (8.200). For instance, at zero, (8.199) writes the first factor as $\log^m x/x$ plus a smooth function, while $\widetilde{\gamma}_n(a-x) - \widetilde{\gamma}_n(a) = O(x)$. The subtracted integrand is therefore $O(1 + |\log x|^m)$ there, and has the analogous integrable behavior at a. The wrapped integral is smooth. Integration of the two subtracted singular monomials gives exactly the two finite $\log a$ terms shown. This verifies the cutoff expression and permits all parameter coefficients to be taken under the subtracted integrals. Equivalently it is the restriction of the meromorphic convolution identity (8.170). The identities (8.201)–(8.203) now follow from (8.183), (8.184), and (8.186). □

For ordinary digamma functions this gives the concrete identity

$$\begin{aligned}\int_0^a \left\{\beta(x)\beta(a-x) - \beta(a)\left(\frac{1}{x} + \frac{1}{a-x}\right)\right\} dx - \int_a^1 \beta(x)\beta(1+a-x)\,dx + 2\beta(a)\log a \\ = \gamma_1\left(\frac{a}{2}\right) - \gamma_1\left(\frac{a+1}{2}\right) + 2\log 2\,\beta(a).\end{aligned} \tag{8.204}$$

The full distributional identity behind it still contains $-\zeta(2)\delta_0$. Omitting that mass would preserve (8.204) on $0 < a < 1$ but would destroy the Fourier coefficients and every identity involving endpoint tests.

Corollary 8.92 (A quarter-Gamma evaluation)**.** *At $a = \frac{1}{2}$, the absolutely convergent integral in* (8.204) *has the exact value*

$$\mathcal{A}_{00}\left(\tfrac{1}{2}\right) = \gamma_1\left(\tfrac{1}{4}\right) - \gamma_1\left(\tfrac{3}{4}\right) + \pi\log 2 = \pi\left\{4\log\Gamma\left(\tfrac{1}{4}\right) - \gamma - 3\log(2\pi)\right\}. \tag{8.205}$$

Proof. Distinguish the Dirichlet beta function $\beta_{\mathrm{D}}(s) = \sum_{r\ge 0}(-1)^r(2r+1)^{-s}$ from the Nielsen function $\beta(x)$ in (8.198). The identity $\zeta_E(s, \frac{1}{2}) = 2^s\beta_{\mathrm{D}}(s)$ gives

$$2\widetilde{\gamma}_1\left(\tfrac{1}{2}\right) = -\pi\log 2 - 4\beta_{\mathrm{D}}'(1),$$

since $\beta_{\mathrm{D}}(1) = \pi/4$. The completed Dirichlet functional equation [116, §25.15(i)] is, for this character,

$$\Lambda(s) = \left(\frac{4}{\pi}\right)^{(s+1)/2}\Gamma\left(\frac{s+1}{2}\right)\beta_{\mathrm{D}}(s), \qquad \Lambda(s) = \Lambda(1-s).$$

Also

$$\beta_{\mathrm{D}}(s) = 4^{-s}\left\{\zeta\left(s, \tfrac{1}{4}\right) - \zeta\left(s, \tfrac{3}{4}\right)\right\}, \quad \beta_{\mathrm{D}}(0) = \tfrac{1}{2}, \quad \beta_{\mathrm{D}}'(0) = \log\Gamma\left(\tfrac{1}{4}\right) - \log\Gamma\left(\tfrac{3}{4}\right) - \log 2.$$

Logarithmic differentiation of the completed functional equation at 0 and 1, followed by the Gamma reflection formula, yields

$$\beta_{\mathrm{D}}'(1) = \frac{\pi}{4}\left\{\gamma + 2\log 2 + 3\log\pi - 4\log\Gamma\left(\tfrac{1}{4}\right)\right\}.$$

Substitution proves (8.205); the first equality also uses $\beta(\frac{1}{2}) = \pi/2$ in (8.204). □

8.29.5 Reflection at the half-twist

For the half-twist, the map $n \mapsto -n-1$ reverses the shifted frequency $n+\frac{1}{2}$. Its periodic realization is

$$\mathcal{J}U(x) = e^{-2\pi i x}U(-x), \qquad \widehat{\mathcal{J}U}(n) = \widehat{U}(-n-1). \tag{8.206}$$

This is an involutive convolution automorphism with $\mathcal{J}\delta_0 = \delta_0$ and $D_{1/2}\mathcal{J} = -\mathcal{J}D_{1/2}$. Ordinary reflection alone does not reverse the half-shifted frequencies. Also define

$$\widehat{\mathcal{H}_{1/2}U}(n) = -i\,\operatorname{sgn}\left(n+\tfrac{1}{2}\right)\widehat{U}(n), \qquad n \in \mathbb{Z}.$$

Here $\mathcal{H}_{1/2}^2 = -\mathrm{Id}$ on all periodic distributions, because every shifted frequency is nonzero.

Corollary 8.93 (Reflected half-twist table)**.** *Writing $U_n = U_n^{1/2}$ and $B = B_{1/2}$ in this corollary,*

$$B(u) * \mathcal{J}B(v) = C(u,v)\big(\cos(\pi v)\,\mathrm{Id} + \sin(\pi v)\,\mathcal{H}_{1/2}\big)B(u+v). \tag{8.207}$$

Thus the coefficient prescription gives an all-index reflected table as well. Two explicit instances are

$$U_0 * \mathcal{J}U_1 = \tfrac{1}{2}U_2 + \mathcal{J}U_2 + \zeta(2)(U_0 + \mathcal{J}U_0) + \zeta(3)\delta_0, \tag{8.208}$$

$$U_1 * \mathcal{J}U_1 = \tfrac{1}{2}(U_3 + \mathcal{J}U_3) + 2\zeta(2)(U_1 + \mathcal{J}U_1) + \zeta(3)(U_0 + \mathcal{J}U_0) + \tfrac{7}{2}\zeta(4)\delta_0. \tag{8.209}$$

For the polygamma finite parts,

$$P_k^{1/2} * \mathcal{J}P_\ell^{1/2} = (-1)^\ell D_{1/2}^{k+\ell}\big\{U_1 + \mathcal{J}U_1 + H_\ell U_0 + H_k\mathcal{J}U_0 + (H_kH_\ell + 2\zeta(2))\delta_0\big\}. \tag{8.210}$$

Proof. At mode n put $\sigma = \operatorname{sgn}(n+\frac{1}{2})$. The logarithms of $\lambda_{n,1/2}$ and $-\lambda_{n,1/2}$ differ by $i\pi\sigma$. Dividing the Fourier coefficient of the left side of (8.207) by that of $B(u+v)$ gives $C(u,v)e^{-i\pi v\sigma}$, which is exactly the multiplier on the right. Coefficient extraction, with $\zeta(2)^2 = \frac{5}{2}\zeta(4)$, proves the two displayed instances. Multiplication of $\gamma + L_{n,1/2} - H_k$ and $\gamma + L_{-n-1,1/2} - H_\ell$ proves (8.210); the factor $(-1)^\ell$ comes from $\lambda_{-n-1,1/2}^\ell = (-\lambda_{n,1/2})^\ell$. □

The modulation in (8.206) is periodic specifically because $2\theta = 1$. For a general twist, frequency reversal connects θ with $1-\theta$; the same fixed-twist involution should not be asserted without changing the space or the twist.

8.29.6 Unique covariant primitives

Because $\lambda_{n,\theta} \neq 0$ for every n, the operator D_θ has a unique inverse on periodic distributions:

$$\widehat{D_\theta^{-q}U}(n) = \lambda_{n,\theta}^{-q}\widehat{U}(n), \qquad q \geq 1.$$

This removes the choice of an integration constant that occurs for ordinary periodic primitives.

Theorem 8.94 (All-order covariant antiderivatives)**.** *Define the complete homogeneous reciprocal-harmonic coefficients by*

$$\sum_{j\geq 0} h_j^{(q-1)}w^j = \prod_{r=1}^{q-1}\left(1-\frac{w}{r}\right)^{-1}; \qquad h_0^{(0)} = 1, \quad h_j^{(0)} = 0 \ (j>0). \tag{8.211}$$

Then for $q \geq 1$ and $n \geq 0$,

$$D_\theta^{-q} U_n^\theta = \frac{(-1)^{n+1} n!}{(q-1)!} \sum_{j=0}^{n+1} \frac{h_{n+1-j}^{(q-1)}}{j!} \, \partial_s^j Z_s^\theta \Big|_{s=1-q}. \tag{8.212}$$

Its ordinary value on $(0,1)$ is obtained by replacing Z_s^θ by f_s^θ. For $q \geq 2$ this is a periodic C^{q-2} function. For $q > k \geq 0$, setting $p = q - k$ gives

$$D_\theta^{-q} P_k^\theta = \frac{\partial_s Z_s^\theta\big|_{s=1-p} + (H_{p-1} - H_k) Z_{1-p}^\theta}{(p-1)!}, \tag{8.213}$$

$$D_\theta^{-q} Q_k^\theta = \frac{\partial_s Z_s^\theta\big|_{s=1-p} + H_{p-1} Z_{1-p}^\theta}{(p-1)!}. \tag{8.214}$$

At the half-twist, the first primitive has the explicit Gamma form

$$D_{1/2}^{-1} U_0^{1/2}(x) = e^{-\pi i x} \left\{ \log \Gamma\left(\frac{x+1}{2}\right) - \log \Gamma\left(\frac{x}{2}\right) + \frac{1}{2} \log 2 \right\}, \qquad 0 < x < 1. \tag{8.215}$$

The displayed additive term is fixed by the periodic distributional normalization.

Proof. The spectral definition and the Gamma recurrence give

$$D_\theta^{-q} Z_{1+u}^\theta = \frac{Z_{1-q+u}^\theta}{(-u)_q}, \qquad \frac{1}{(-u)_q} = -\frac{1}{(q-1)! u} \sum_{j \geq 0} h_j^{(q-1)} u^j.$$

The family in the numerator is holomorphic at $u = 0$. Comparing the coefficients of $(-1)^n u^n / n!$ proves (8.212). Its Fourier coefficients are $O(|n|^{-q}(1 + \log |n|)^{n_0+1})$ when the fixed Stieltjes index is denoted by n_0. Therefore the Fourier series and its first $q-2$ ordinary derivatives converge absolutely for $q \geq 2$. The derivative formulas for P_k^θ and Q_k^θ follow directly from (8.187) and $\psi(p) = H_{p-1} - \gamma$.

For (8.215), the case $q = 1, n = 0$ is $-\partial_s Z_s^{1/2}|_{s=0}$. Use (8.195), $\zeta(0,a) = \frac{1}{2} - a$, and $\zeta'(0,a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$. The difference of the two zeta values at zero is $\frac{1}{2}$, which supplies the term $\frac{1}{2} \log 2$ after differentiation. □

8.29.7 Degeneration to the untwisted finite parts

The zero Fourier coefficient becomes singular as $\theta \downarrow 0$. Deleting it exactly gives a continuous transition to the untwisted normalization, including all the Stieltjes jets.

Theorem 8.95 (Exact zero-mode renormalization)**.** *For every $\rho < 1$,*

$$B_\theta(u) - \Gamma(1-u)(2\pi i \theta)^u \, 1 \longrightarrow B(u) \qquad (\theta \downarrow 0) \tag{8.216}$$

locally uniformly for $|u| < \rho$ after pairing with any smooth periodic test function, and with every fixed number of derivatives in u. Here $B(u) = \Gamma(1-u) R_{-u} = \mathsf{I} - u \sum_{n \geq 0} (-1)^n G_n u^n / n!$ is the already proved untwisted family. The symbol 1 is the constant distribution, which has only the zero Fourier coefficient. Put

$$\mu_n(\theta) = \frac{(-1)^{n+1}}{n+1} \mathcal{B}_{n+1}\big(\gamma + \log(2\pi i \theta), \zeta(2), 2!\zeta(3), \ldots, n!\zeta(n+1)\big). \tag{8.217}$$

Then $\mu_n(\theta) = \widehat{U_n^\theta}(0)$ *and*

$$U_n^\theta - \mu_n(\theta)\,1 \longrightarrow G_n \qquad \text{in } \mathcal{D}'(\mathbb{R}/\mathbb{Z}). \tag{8.218}$$

In particular,

$$\mu_0(\theta) = -\gamma - \log(2\pi i\theta), \qquad \mu_1(\theta) = \frac{(\gamma + \log(2\pi i\theta))^2 + \zeta(2)}{2}.$$

Proof. The subtracted quantity in (8.216) is exactly the $n = 0$ Fourier coefficient. For each $n \neq 0$, the remaining coefficient tends to $\Gamma(1-u)(2\pi in)^u = \widehat{B(u)}(n)$ with the previously fixed branch. Uniformly for $0 < \theta \leq \frac{1}{2}$ and u on a compact subdisk, these coefficients and their fixed order derivatives are bounded by a common polynomial in $|n|$, because $|n+\theta|$ is comparable with $|n|$. Rapid decrease of the test-function Fourier coefficients proves normal convergence and allows termwise order differentiation. Taking the coefficient of $(-1)^{n+1}u^{n+1}/n!$ gives (8.218); the scalar Gamma expansion gives (8.217). □

At $u = 0$, the left side of (8.216) is already $\delta_0 - 1 = \mathsf{I}$. Thus the change of convolution identity is caused exactly by the disappearing zero mode. The same observation explains why the interior alternating identity (8.201) has no added constant, whereas the untwisted digamma convolution has the constant contributed by $-\zeta(2)\mathsf{I}$.

8.30 Unequal twists through beta mixtures and finite resolvent identities

The fixed-twist calculus suggests a next question: how do kernels with different frequency shifts convolve? The Fourier multipliers give an explicit answer. Fractional orders admit a Dirichlet beta mixture of intermediate twists, while positive integer orders reduce to finitely many elementary kernels. These statements use the classical beta integral and partial fractions; they make no literature-priority or arithmetic-independence claim.

Write $K_\alpha^\theta = W_{1-\alpha}^\theta$, so that

$$\widehat{K}_\alpha^\theta(n) = (2\pi i(n+\theta))^{-\alpha}.$$

The twists in the fractional-mixture theorem lie in $(0,1)$, with the logarithms of (8.163). More generally one may put them in a common open interval between consecutive integers. This condition keeps the frequencies at each fixed mode on the same imaginary ray.

Theorem 8.96 (Any finite number of fractional twist kernels)**.** *Let* $d \geq 2$, $\theta_j \in (0,1)$ *and* $\operatorname{Re}\alpha_j > 0$. *Put* $A = \sum_j \alpha_j$ *and let* $\Sigma_{d-1} = \{t_j \geq 0 : \sum_j t_j = 1\}$, *with measure* $dt_1 \cdots dt_{d-1}$. *Then, as periodic distributions,*

$$K_{\alpha_1}^{\theta_1} * \cdots * K_{\alpha_d}^{\theta_d} = \frac{\Gamma(A)}{\prod_j \Gamma(\alpha_j)} \int_{\Sigma_{d-1}} \left(\prod_j t_j^{\alpha_j - 1} \right) K_A^{\sum_j t_j \theta_j}\, dt. \tag{8.219}$$

The mixture includes the zero Fourier coefficient. Its normalized right side, continued as a whole, is entire in all α_j. *For two twists the simplex is the interval* $0 \leq t \leq 1$ *and the intermediate twist is* $t\theta_1 + (1-t)\theta_2$.

Proof. For positive real a_j and $\operatorname{Re}\alpha_j > 0$, change variables $y_j = rt_j$ in the product of Gamma integrals

$$\prod_j a_j^{-\alpha_j} = \frac{1}{\prod_j \Gamma(\alpha_j)} \int_{(0,\infty)^d} \prod_j y_j^{\alpha_j - 1} e^{-\sum_j a_j y_j}\, dy.$$

The Jacobian is r^{d-1}. Integrating r gives the scalar Dirichlet formula with $(\sum_j t_j a_j)^{-A}$. At Fourier mode n, all $n+\theta_j$ have the same sign. Apply the scalar formula to $a_j = |2\pi(n+\theta_j)|$ and restore their common phase $\exp(-i\pi A \operatorname{sgn}(n+\theta_j)/2)$. The resulting coefficient is exactly that of (8.219), including $n = 0$.

The intermediate twists range in a compact subinterval of $(0,1)$. On compact parameter subsets, the kernel coefficients have a common polynomial growth bound, and the simplex weight is integrable. Rapid decrease of the Fourier coefficients of each smooth test function therefore justifies integration and summation. Fourier uniqueness proves the distribution identity. For continuation outside the initial half-planes, subtract finite Taylor polynomials at the simplex faces before integrating; the added monomials have meromorphic beta integrals. Their combined normalized continuation has the same Fourier coefficients as the left side, which are entire in every α_j. Test-function pairing gives normal convergence on compact parameter sets. Thus all apparent poles of the combined continuation are removable. It is this continuation, not a divergent unsubtracted simplex integral, that is asserted to be entire. □

Theorem 8.97 (Finite reduction at every positive integer order)**.** *For distinct real nonintegral θ, ϕ, set $\Delta = 2\pi i(\phi - \theta)$. For positive integers r, s,*

$$K_r^\theta * K_s^\phi = \sum_{j=1}^{r} \frac{(-1)^{r-j}\binom{r+s-j-1}{r-j}}{\Delta^{r+s-j}} K_j^\theta + \sum_{k=1}^{s} \frac{(-1)^{r}\binom{r+s-k-1}{s-k}}{\Delta^{r+s-k}} K_k^\phi. \tag{8.220}$$

Every ordinary representative on $(0,1)$ is elementary, since

$$K_1^\theta(x) = \frac{e^{-2\pi i\theta x}}{1-e^{-2\pi i\theta}}, \qquad K_r^\theta(x) = \frac{(-1)^{r-1}}{(2\pi i)^{r-1}(r-1)!}\partial_\theta^{r-1} K_1^\theta(x). \tag{8.221}$$

If two twists coincide, their orders add: $K_r^\theta * K_s^\theta = K_{r+s}^\theta$.

Proof. At a Fourier mode put $X = 2\pi i(n+\theta)$; the second multiplier has denominator $X + \Delta$. The principal part at $X = 0$ of $X^{-r}(X+\Delta)^{-s}$ is obtained by expanding $(X+\Delta)^{-s}$ through degree $r-1$; this gives the first sum. At $X = -\Delta$, expand X^{-r} in $X + \Delta$ through degree $s-1$; this gives the second sum. Their difference from the rational function has no finite poles and tends to zero at infinity, so it is zero. Apply this identity at every Fourier mode and use uniqueness. The representative for K_1 is (8.167) at $s = 0$. Differentiation of its multiplier gives $\partial_\theta K_r^\theta = -2\pi i r K_{r+1}^\theta$; distributional differentiation in the parameter is legitimate by the same compact Fourier estimates. This proves (8.221). The coincident identity is (8.168). □

For instance, the first unequal-twist convolution is the explicit ordinary integral

$$(K_1^\theta * K_1^\phi)(x) = \frac{K_1^\theta(x) - K_1^\phi(x)}{2\pi i(\phi - \theta)}.$$

Its coincident limit is $-\partial_\theta K_1^\theta/(2\pi i) = K_2^\theta$. Any finite number of distinct twists and positive integer orders is handled by the same principal-part argument; repeated twists are combined first. These integer-order reductions require no common frequency interval. Fractional orders have the precise mixture (8.219). Finite reduction of every mixed Stieltjes order jet to a prescribed list of single Hurwitz values remains a separate question; the mixture does not assert such a reduction.

8.31 Gamma coefficients prove every harmonic-order identity

The continuous Gamma correlations above have a discrete counterpart. Finite harmonic products are coefficients of a Gamma ratio; a normally convergent Gauss sum therefore evaluates an entire family at once. This gives corrected all-order forms of the fifth- and sixth-order conjectures in Choi [113], rather than isolated numerical relations. The harmonic-resonance continuation supplies this route and its broader mixed-jet context [114].

For $n, r \geq 0$, define

$$\prod_{j=1}^{n}\left(1+\frac{u}{j}\right)=\sum_{r=0}^{n}\frac{\mathcal{E}_r(n)}{r!}u^r, \qquad \mathcal{E}_0(n)=1. \tag{8.222}$$

Thus $\mathcal{E}_r(n)$ is $r!$ times the elementary symmetric polynomial in $1, 1/2, \ldots, 1/n$. In particular $\mathcal{E}_1(n) = H_n$ and $\mathcal{E}_2(n) = H_n^2 - H_n^{(2)}$. Equivalently it is the complete exponential Bell polynomial with arguments $H_n, -H_n^{(2)}, 2H_n^{(3)}, \ldots, (-1)^{r-1}(r-1)!H_n^{(r)}$.

Theorem 8.98 (Every pair of harmonic coefficient orders)**.** *For u, v in a neighborhood of zero,*

$$\sum_{n=0}^{\infty}\frac{(1+u)_n(1+v)_n}{(n!)^2(n+1)(n+2)}=\frac{\Gamma(1-u-v)}{\Gamma(2-u)\Gamma(2-v)}. \tag{8.223}$$

Every fixed Taylor coefficient can be taken termwise. Consequently, for all $r, t \geq 0$,

$$\sum_{n=1}^{\infty}\frac{\mathcal{E}_r(n)\mathcal{E}_t(n)}{(n+1)(n+2)}=r!t![u^r v^t]\frac{\Gamma(1-u-v)}{\Gamma(2-u)\Gamma(2-v)}-\frac{1}{2}\mathbf{1}_{r=t=0}. \tag{8.224}$$

These are ordinary absolutely convergent series.

Proof. Since $(3)_n = n!(n+1)(n+2)/2$, the series in (8.223) is $\frac{1}{2}{}_2F_1(1+u, 1+v; 3; 1)$. Gauss summation [112] evaluates it as the displayed Gamma quotient. On a sufficiently small closed polydisc, its summands are uniformly $O(n^{-2+\mathrm{Re}(u+v)})$ by the Gamma-ratio asymptotic. Enlarging that polydisc slightly and applying Cauchy's coefficient formula justifies each derivative or coefficient extraction. Formula (8.222) now proves (8.224). The $n = 0$ summand is $1/2$ and contributes only at $r = t = 0$. □

Corollary 8.99 (Two explicit all-order families)**.** *For every $r \geq 1$ and every $t \geq 0$, respectively,*

$$\sum_{n=1}^{\infty}\frac{\mathcal{E}_r(n)}{(n+1)(n+2)}=r!, \tag{8.225}$$

$$\sum_{n=1}^{\infty}\frac{H_n\mathcal{E}_t(n)}{(n+1)(n+2)}=t!\left(1+\sum_{j=2}^{t+1}\zeta(j)\right). \tag{8.226}$$

The sum on the second right side is empty at $t = 0$.

Proof. Set $v = 0$ in (8.223). Its right side is $1/(1-u)$, proving (8.225). Differentiate first in v and then put $v = 0$. The result is

$$\frac{\psi(2)-\psi(1-u)}{1-u}=\frac{1+\sum_{j\geq 1}\zeta(j+1)u^j}{1-u}, \qquad |u|<1.$$

The normally convergent digamma Taylor expansion at one proves this last equality. Taking the coefficient of u^t proves (8.226). Its $n = 0$ derivative is zero, so no additional constant occurs. □

The cases $r = 5, 6$ prove the intended pure harmonic identities in Choi's Conjecture 4; the cases $t = 5, 6$ prove corrected versions of Conjecture 6. His printed formulas (41)–(46) carry twice the multiplier in (8.226). The finite-product definition also fixes the harmonic orders in every numerator, including the sixth-order term $-120H_n^{(6)}$ and, after multiplication by H_n, the partition term $45H_n^3(H_n^{(2)})^2$. The unambiguous coefficient formula is the corrected statement at all orders. The source comparison and its text-versus-visual evidence boundary are recorded in `verification/ninth-analytic-audit.md`.

For a short independent check of the simplest normalization, summation by parts gives

$$\begin{aligned}\sum_{n\geq 1}\frac{H_n^2}{(n+1)(n+2)} &= \sum_{k\geq 1}\frac{H_k^2 - H_{k-1}^2}{k+1}\\ &= 2\sum_{k\geq 1}\frac{H_{k-1}}{k(k+1)} + \sum_{k\geq 1}\frac{1}{k^2(k+1)}\\ &= 2 + \zeta(2) - 1 = 1 + \zeta(2).\end{aligned}$$

Indeed Tonelli's theorem gives $\sum_{k\geq 1} H_{k-1}/(k(k+1)) = \sum_{j\geq 1} 1/(j(j+1)) = 1$. This isolates the factor of two without any special-function differentiation. Higher mixed orders are finite Gamma Taylor coefficients; no arithmetic-independence assumption is needed.

8.32 Mixed Gamma derivatives and exact harmonic tail identities

The harmonic coefficient calculus has a useful companion at a double zero of a hypergeometric residue. The mixed spectral continuation [123] identifies a normally convergent two-parameter Gamma identity and a weighted square of trigamma differences. We prove the needed first resonance directly from the classical Rogers–Dougall sum [124], rather than assuming the entire pending all-resonance programme. Its mixed derivatives produce ordinary harmonic identities. Differentiating the resulting square identity also suggests an all-orders zeta-tail family; an independent Euler partial-fraction proof of that family closes the loop between experiment and exact algebra.

8.32.1 A normally convergent double-zero germ

Put $\eta = a/2$, $x_n = n + \eta$, and $A = a - 1$. Define

$$Q_n(a;p,q) = \frac{\Gamma(n+a)\Gamma(n+1+p)\Gamma(n+1+q)\Gamma(n+a-p-q)}{\Gamma(n+1)\Gamma(n+a-p)\Gamma(n+a-q)\Gamma(n+1+p+q)}, \tag{8.227}$$

$$B_a(p,q) = \psi(a/2) + \tfrac{1}{2}\{\psi(2) - \psi(1+p) - \psi(1+q) - \psi(a-p-q)\}. \tag{8.228}$$

These are holomorphic for $\Re a > 0$ and p, q sufficiently small on each compact a-set. Both $Q_n(a;p,0)$ and $Q_n(a;0,q)$ are one.

Theorem 8.100 (The first mixed resonance)**.** *For every real $a > 0$, and sufficiently small complex p, q,*

$$\sum_{n\geq 0}\left\{x_n\big(Q_n(a;p,q) - 1\big) + \frac{(A-p-q)pq}{x_n}\right\} = -(A-p-q)pq\,B_a(p,q). \tag{8.229}$$

The bracketed series converges locally normally, including after division by pq at either axis. Every fixed parameter derivative can be taken termwise. The identity extends holomorphically to the corresponding germs over $\Re a > 0$.

Proof. First take $a > 1$, $b = 1 + p$, $c = 1 + q$ and $d_0 = 1 + a - b - c > 0$. Introduce a separate spectral variable u and the centered summand

$$r_n(u) = x_n \frac{\Gamma(n+a)\Gamma(n+b)\Gamma(n+c)\Gamma(n+d_0-u)}{\Gamma(n+1)\Gamma(n+1+a-b)\Gamma(n+1+a-c)\Gamma(n+b+c+u)}.$$

For $0 < \Re u < d_0$, the Rogers–Dougall formula with upper parameters $a, 1 + a/2, b, c, d_0 - u$ and lower parameters $a/2, 1 + a - b, 1 + a - c, b + c + u$ gives

$$\sum_{n\geq 0} r_n(u) = G(u) := \frac{\Gamma(b)\Gamma(c)\Gamma(d_0-u)\Gamma(u)}{2\Gamma(d_0)\Gamma(b+u)\Gamma(c+u)}. \tag{8.230}$$

The classical excess is $2u$; the factor $1/2$ follows from the normalization $x_n = n + a/2$, not the usual Pochhammer summand.

Write

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (\eta, b - \eta, c - \eta, d_0 - u - \eta).$$

The denominator shifts relative to x_n are $1 - \alpha_i$. The shifted Stirling expansion [49] and Bernoulli reflection therefore give uniformly on compact parameter sets

$$r_n(u) = x_n^{-1-2u}\{1 + C_1(u)x_n^{-2} + O(n^{-4})\}, \qquad C_1(u) = -\tfrac{1}{3}\sum_{i=1}^{4} B_3(\alpha_i).$$

Centering cancels the odd inverse powers. Thus

$$R_2(u) = \sum_{n\geq 0}\{r_n(u) - x_n^{-1-2u} - C_1(u)x_n^{-3-2u}\} = G(u) - \zeta(1+2u,\eta) - C_1(u)\zeta(3+2u,\eta) \tag{8.231}$$

extends holomorphically to $-2 < \Re u < d_0$. The series bound is $O(n^{-5-2\Re u})$ on compact subsets; Cauchy's formula gives normal convergence for each fixed spectral or parameter derivative. The equality first follows by summing the counterterms in the ordinary convergence region, and then by continuation.

At $u = -1$, direct Bernoulli algebra gives

$$\begin{aligned} C_1(-1) &= \rho := -(A - p - q)pq, \\ \tfrac{1}{2}C_1'(-1) + \zeta(-1,\eta) &= \tfrac{1}{2}\{B_2(\eta - p - q) - B_2(\eta)\} = -\tfrac{1}{2}(A - p - q)(p + q). \end{aligned} \tag{8.232}$$

For generic nonzero p, q, G has residue $\rho/2$ and constant term

$$\frac{\rho}{2}\{\psi(2) - \psi(d_0 + 1) - \psi(p) - \psi(q)\}.$$

The resonant Hurwitz term has constant coefficient $C_1'(-1)/2 - \rho\psi(\eta)$. Combining these constants with (8.232), and using $\psi(1 + p) = \psi(p) + 1/p$ and its q counterpart, yields $R_2(-1) = \rho B_a(p, q)$. Since $r_n(-1) = x_n Q_n$, this is (8.229). Holomorphy extends it to the spectral axes.

Finally all Gamma arguments in Q_n and all digamma arguments in B_a stay pole-free for $\Re a > 0$ and small p, q. The centered bound remains $O(n^{-3})$ uniformly on every compact a-set. Each Taylor coefficient is consequently holomorphic in that half-plane and agrees with the asserted coefficient for $\Re a > 1$; the identity theorem extends it throughout. The bracket vanishes identically on both axes. Cauchy's formula on a slightly larger polydisc transfers its $O(n^{-3})$ bound to its holomorphic quotient by pq and to every fixed derivative, proving the divided-series assertion. □

8.32.2 Finite mixed coefficient tables

For $k \geq 2$ set $\Lambda_k(n;a) = (-1)^k \psi^{(k-1)}(n+a) - \psi^{(k-1)}(n+1)$. Logarithmic differentiation of the eight Gamma factors gives

$$\log Q_n = \sum_{r,s \geq 1} \Lambda_{r+s}(n;a) \frac{p^r q^s}{r!s!}. \tag{8.233}$$

Define the finite bivariate Bell coefficient

$$\mathcal{B}_{rs}(n;a) = r!s![p^r q^s] \exp\left(\sum_{i,j \geq 1} \Lambda_{i+j}(n;a) \frac{p^i q^j}{i!j!} \right).$$

Only $i \leq r$, $j \leq s$ and powers at most $\min(r,s)$ are needed. For example

$$\mathcal{B}_{11} = \Lambda_2, \quad \mathcal{B}_{31} = \Lambda_4, \quad \mathcal{B}_{22} = \Lambda_4 + 2\Lambda_2^2, \quad \mathcal{B}_{33} = \Lambda_6 + 9\Lambda_2\Lambda_4 + 9\Lambda_3^2 + 6\Lambda_2^3.$$

Write $B_{rs} = \partial_p^r \partial_q^s B_a(0,0)$. Then

$$B_{00} = \psi(a/2) + \tfrac{1}{2}(1 + \gamma - \psi(a)),$$

and, for $r + s = k \geq 1$,

$$B_{rs} = -\tfrac{1}{2}\{\mathbf{1}_{s=0}\psi^{(r)}(1) + \mathbf{1}_{r=0}\psi^{(s)}(1) + (-1)^k \psi^{(k)}(a)\}. \tag{8.234}$$

Terms with a negative subscript are omitted below.

Corollary 8.101 (All fixed mixed orders)**.** *For $r, s \geq 1$ the absolutely convergent single bracket*

$$S_{rs}(a) = \sum_{n \geq 0} \left\{ x_n \mathcal{B}_{rs}(n;a) + \frac{A\mathbf{1}_{r=s=1} - 2\mathbf{1}_{(r,s)=(2,1)} - 2\mathbf{1}_{(r,s)=(1,2)}}{x_n} \right\} \tag{8.235}$$

equals

$$-rs\{AB_{r-1,s-1} - (r-1)B_{r-2,s-1} - (s-1)B_{r-1,s-2}\}. \tag{8.236}$$

When $r + s \geq 4$ the counterterm is absent.

Proof. Differentiate the normally convergent identity (8.229). Its counterterm is a polynomial of total degree three, while Leibniz's rule on the right gives (8.236). The quotient and normal-convergence statements were proved before differentiation. □

8.32.3 A trigamma square and its arithmetic specializations

Theorem 8.102 (Weighted trigamma square)**.** *For every real $a > 0$,*

$$\sum_{n \geq 0} \left(n + \frac{a}{2}\right) \left(\psi'(n+a) - \psi'(n+1)\right)^2 = \frac{a-1}{4}\psi''(a) + \frac{\psi'(a) - \zeta(2)}{2} + \frac{3(a-1)}{2}\zeta(3). \tag{8.237}$$

The ordinary series is locally normally convergent over $\Re a > 0$, where the square is algebraic.

Proof. The $(2,2)$ and $(3,1)$ rows of (8.236) give

$$\sum_n x_n(\Lambda_4+2\Lambda_2^2)=2A\psi''(a)+4\psi'(a)-4\zeta(2),$$

$$\sum_n x_n\Lambda_4=\tfrac{3A}{2}\{\psi''(1)+\psi''(a)\}+3\{\psi'(a)-\zeta(2)\}.$$

Subtract and divide by two, using $\psi''(1)=-2\zeta(3)$. All series in that calculation converge absolutely. Independently, the polygamma expansion gives $\psi'(n+a)-\psi'(n+1)=O(n^{-2})$ locally uniformly, so the weighted square is $O(n^{-3})$, including at $a=1$. □

The half-integer recurrences reduce the cases $a=3/2$ and $a=1/2$ to

$$\sum_{n\ge0}(4n+3)\{2H^{(2)}_{2n+1}-H^{(2)}_n-\zeta(2)\}^2=\zeta(2)-\zeta(3), \tag{8.238}$$

$$\sum_{n\ge0}(4n+1)\{\zeta(2)-2H^{(2)}_{2n}+H^{(2)}_n\}^2=\zeta(2)+\zeta(3). \tag{8.239}$$

Indeed $\psi'(3/2)=3\zeta(2)-4$, $\psi''(3/2)=-14\zeta(3)+16$, $\psi'(1/2)=3\zeta(2)$ and $\psi''(1/2)=-14\zeta(3)$. The polygamma series express the respective differences as twice the bracketed harmonic tails. At $a=2$, (8.237) reduces to $\sum_{k\ge1}k^{-3}=\zeta(3)$.

8.32.4 An all-orders zeta-tail identity with an independent proof

Put $T_s(n)=\zeta(s)-H_n^{(s)}=\sum_{j>n}j^{-s}$ for $s\ge2$. Differentiating (8.237) at $a=1$ suggests the following finite combination at every order. We prove it independently by Euler partial fractions and positive double sums, so its validity does not depend on the hypergeometric continuation.

Theorem 8.103 (All-order quadratic tail reduction)**.** *For every integer $k\ge2$,*

$$\sum_{n\ge0}\Bigg[\big(n+\tfrac12\big)\sum_{j=1}^{k-1}(j+1)(k-j+1)T_{j+2}(n)T_{k-j+2}(n) - \tfrac12\sum_{j=1}^{k-2}(j+1)(k-j)T_{j+2}(n)T_{k-j+1}(n)\Bigg]=\frac{(k+1)(k+2)}{4}\zeta(k+2). \tag{8.240}$$

Every series displayed on the left is ordinary and absolutely convergent.

Proof. For $a,b\ge3$, summing first over $0\le n<\min(l,m)$ gives

$$\sum_{n\ge0}(n+\tfrac12)T_a(n)T_b(n)=\tfrac12\{\zeta(a+b-2)+\zeta(a,b-2)+\zeta(b,a-2)\}, \tag{8.241}$$

$$\sum_{n\ge0}T_a(n)T_b(n)=\zeta(a+b-1)+\zeta(a,b-1)+\zeta(b,a-1). \tag{8.242}$$

The finite sums of the weights are $\min(l,m)^2/2$ and $\min(l,m)$, respectively. Positivity justifies Tonelli, and the resulting double zeta sums converge. Here $\zeta(r,s)=\sum_{l>m\ge1}l^{-r}m^{-s}$. Substituting these two formulas, using the symmetry in j and $k-j$ or $k-1-j$, reduces (8.240) to

$$\sum_{j=1}^{k-2}(j+1)\zeta(j+2,k-j)+2k\zeta(k+1,1)=\zeta(k+2). \tag{8.243}$$

The diagonal single-zeta coefficient before this last reduction is $(k^2+3k-2)/4$; the proposed value exceeds it by one.

For completeness the classical relation (8.243) has an elementary proof. Iterating

$$\frac{1}{x^r y^s} = \frac{x^{-(r-1)}y^{-s} + x^{-r}y^{-(s-1)}}{x+y}$$

gives, for $x, y > 0$,

$$\frac{1}{x^2y^k} = \sum_{i=0}^{1} \frac{\binom{k+i-1}{i}}{(x+y)^{k+i}x^{2-i}} + \sum_{i=0}^{k-1} \frac{i+1}{(x+y)^{2+i}y^{k-i}}. \tag{8.244}$$

Summing over positive integer x, y is justified by positivity. The first sum becomes $\zeta(k,2) + k\zeta(k+1,1)$; the second becomes $\sum_{i=0}^{k-1}(i+1)\zeta(i+2,k-i)$. On the other hand, separating equal and unequal indices in the product gives the stuffle identity $\zeta(2)\zeta(k) = \zeta(2,k) + \zeta(k,2) + \zeta(k+2)$. Subtracting the two expressions proves (8.243), including $k=2$ where the empty sum is zero. This proves the theorem. □

The first member is the simple ordinary identity

$$\boxed{\sum_{n\geq 0}(2n+1)\{\zeta(3) - H_n^{(3)}\}^2 = \tfrac{3}{2}\zeta(4).} \tag{8.245}$$

The next member is

$$\sum_{n\geq 0}\{12(n+\tfrac{1}{2})T_3(n)T_4(n) - 2T_3(n)^2\} = 5\zeta(5).$$

Corollary 8.104 (Centered tail squares)**.** *For every integer $s \geq 3$,*

$$\sum_{k\geq 1} k\{2T_s(k-1) - k^{-s}\}^2 = 2\zeta(2s-2) + 4\zeta(s,s-2) - \zeta(2s-1). \tag{8.246}$$

At $s=3$ this is the mixed-report evaluation

$$\boxed{\sum_{k\geq 1} k\{2(\zeta(3) - H_{k-1}^{(3)}) - k^{-3}\}^2 = 3\zeta(4) - \zeta(5).} \tag{8.247}$$

Proof. Expand the square and use (8.241), (8.242), and $\sum_{k\geq 1} k^{1-s}T_s(k-1) = \zeta(s,s-1) + \zeta(2s-1)$. The two $\zeta(s,s-1)$ terms cancel. For $s=3$, (8.243) at $k=2$ gives $4\zeta(3,1) = \zeta(4)$, yielding (8.247). All rearranged sums converge absolutely. □

These identities show how a Gamma mixed-derivative experiment can expose an exact Euler relation. The additional all-order tail and centered-tail formulas are collective deductions from classical partial fractions; global priority or independence of their period coordinates is not asserted.

8.33 All centered trigamma-square moments and a polylogarithmic generator

The invariant numerator $(\zeta(2) - H_n^{(2)})^2$ provides an explicit family beyond raw powers of H_n. Define

$$A_n = \zeta(2) - H_n^{(2)} = \psi'(n+1), \qquad \mathcal{T}(s) = \sum_{n=0}^{\infty} \frac{A_n^2}{(n+1/2)^s}. \tag{8.248}$$

The harmonic parity continuation [130] gives an ordinary finite-sum route to these moments, independent of its broader difference-equation classification. The defining series already converges for $\Re s > -1$. Its centered asymptotic expansion contains only even inverse powers. Subtracting finitely many terms continues the Dirichlet series normally to successively larger half-planes, with possible poles only at odd integers $1-2k$. Hence it is regular at every nonpositive even integer. All its values there reduce to a rational number plus a rational multiple of $\zeta(3)$.

Theorem 8.105 (Every nonpositive even trigamma-square value)**.** *For every integer $r \geq 0$,*

$$\begin{aligned}\mathcal{T}(-2r) &= 3B_{2r}(1/2)\zeta(3) \\ &\quad - \frac{1}{2}\sum_{j=1}^{r}\binom{2r}{2j}\frac{2j-1}{2j+1}B_{2r-2j}(1/2)B_{2j-2}.\end{aligned} \tag{8.249}$$

The Bernoulli number convention is $B_1 = -1/2$, and the sum is empty when $r = 0$.

For example,

$$\begin{aligned}\mathcal{T}(0) &= 3\zeta(3), \\ \mathcal{T}(-2) &= -\frac{1}{4}\zeta(3) - \frac{1}{6}, \\ \mathcal{T}(-4) &= \frac{7}{80}\zeta(3) + \frac{1}{30}, \\ \mathcal{T}(-6) &= -\frac{31}{448}\zeta(3) + \frac{1}{672}, \\ \mathcal{T}(-8) &= \frac{127}{1280}\zeta(3) - \frac{17}{540}.\end{aligned} \tag{8.250}$$

The first value is the classical Euler sum. The theorem supplies a single finite formula for all subsequent centered moments.

8.33.1 Equivalent identities of absolutely convergent ordinary sums

Let

$$c_k = \sum_{i=0}^{k-1} B_{2i}(1/2)B_{2k-2-2i}(1/2), \qquad k \geq 1. \tag{8.251}$$

Thus $c_1 = 1$, $c_2 = -1/6$, $c_3 = 47/720$, and $c_4 = -257/5040$. The shifted Stirling expansion gives $A_n \sim \sum_{i\geq 0} B_{2i}(1/2)(n+1/2)^{-2i-1}$. Squaring this expansion gives $A_n^2 \sim \sum_{k\geq 1} c_k(n+1/2)^{-2k}$. For every $r \geq 0$ the exact ordinary convergent identity is

$$\sum_{n=0}^{\infty}\left\{(n+1/2)^{2r}\psi'(n+1)^2 - \sum_{k=1}^{r} c_k(n+1/2)^{2r-2k}\right\} = \mathcal{T}(-2r), \tag{8.252}$$

where the right side is explicitly (8.249). The summands after subtraction are $O(n^{-2})$.

For instance,

$$\sum_{n=0}^{\infty}\left((n+1/2)^2\psi'(n+1)^2 - 1\right) = -\frac{1}{4}\zeta(3) - \frac{1}{6},$$

and

$$\sum_{n=0}^{\infty}\left((n+1/2)^4\psi'(n+1)^2 - (n+1/2)^2 + \frac{1}{6}\right) = \frac{7}{80}\zeta(3) + \frac{1}{30}.$$

These statements need no prescription for summing a divergent series. To prove (8.252), subtract the displayed asymptotic terms in the half-plane of convergence and continue. Their Dirichlet sums at $s = -2r$ are $c_k\zeta(2k - 2r, 1/2) = 0$, since $\zeta(-2m, 1/2) = 0$ for $m \geq 0$.

8.33.2 Proof of the Bernoulli formula by finite summation

For a sequence admitting an expansion in powers of N and $\log N$, write CT_N for its coefficient of $N^0(\log N)^0$. Only finite exact identities followed by known asymptotic expansions will be used here. Put

$$S_r(k) = \sum_{n=0}^{k-1}(n + 1/2)^{2r} = \frac{B_{2r+1}(k + 1/2)}{2r + 1} = \sum_{j=0}^{r} b_{r,j}k^{2j+1}, \tag{8.253}$$

where

$$b_{r,j} = \frac{\binom{2r}{2j}}{2j + 1}B_{2r-2j}(1/2). \tag{8.254}$$

The Bernoulli polynomial $B_{2r+1}(1/2)$ is zero, so this is an odd polynomial in k and has zero constant term.

Proof of Theorem 8.105. First, (8.252) and (8.253) show that

$$\mathcal{T}(-2r) = \mathrm{CT}_N \sum_{n=0}^{N-1}(n + 1/2)^{2r}A_n^2. \tag{8.255}$$

Indeed every subtracted even power has partial sum with zero constant term in N, and the convergent remainder tends to the stated value.

The identity $A_{k-1} = A_k + k^{-2}$ gives finite summation by parts:

$$\sum_{n=0}^{N-1}(n + 1/2)^{2r}A_n^2 = \sum_{k=1}^{N} S_r(k)\left(\frac{2A_k}{k^2} + \frac{1}{k^4}\right) + S_r(N)A_N^2. \tag{8.256}$$

The coefficient $b_{r,0}$ of k contributes

$$\mathrm{CT}_N\left(2\sum_{k=1}^{N}\frac{A_k}{k} + H_N^{(3)} + NA_N^2\right) = 3\zeta(3).$$

Here $\sum_{k\geq 1} A_k/k = \zeta(2, 1) = \zeta(3)$ is the classical Euler identity and $NA_N^2 \to 0$. For completeness, the Euler identity follows from

$$\zeta(2, 1) = \frac{1}{2}\int_0^1 \frac{\log^2(1 - t)}{t}\,dt = \frac{1}{2}\int_0^\infty \frac{u^2}{e^u - 1}\,du = \zeta(3).$$

For $j \geq 1$, it remains to compute the constant of

$$E_j(N) = 2\sum_{k=1}^{N} k^{2j-1}A_k + \sum_{k=1}^{N} k^{2j-3} + N^{2j+1}A_N^2. \tag{8.257}$$

Let

$$F_j(N) = \sum_{k=1}^{N} k^{2j-1} = \frac{B_{2j}(N + 1) - B_{2j}}{2j}.$$

Changing the order in a finite sum gives

$$\sum_{k=1}^{N} k^{2j-1} A_k = A_N F_j(N) + \sum_{\ell=1}^{N} \frac{F_j(\ell-1)}{\ell^2}. \tag{8.258}$$

For $j \geq 2$, the second summand is a sum of polynomial powers of ℓ and has zero constant term as a polynomial in N; $B_{2j-1} = 0$ removes a possible $1/\ell$ term. The other power sum in (8.257) also has zero constant. For $j = 1$ the second sum in (8.258) equals $(N - H_N)/2$; its harmonic term cancels exactly against $\sum k^{-1}$ in (8.257). In all cases,

$$\mathrm{CT}_N\, E_j(N) = 2\,\mathrm{CT}_N(A_N F_j(N)) + \mathrm{CT}_N(N^{2j+1} A_N^2). \tag{8.259}$$

Now use the uncentered tail expansion

$$A_N = \frac{1}{N} - \frac{1}{2N^2} + \sum_{k\geq 1} \frac{B_{2k}}{N^{2k+1}}.$$

The finite coefficient calculation gives

$$\mathrm{CT}_N(A_N F_j(N)) = \frac{3-2j}{4} B_{2j-2}, \qquad \mathrm{CT}_N(N^{2j+1} A_N^2) = -B_{2j-2}. \tag{8.260}$$

To check the first identity, the coefficient of N^ℓ in F_j is $\binom{2j}{\ell} B_{2j-\ell}(1)/(2j)$. Only $\ell = 2$ and $\ell = 2j-1$ can pair with nonzero inverse coefficients of A_N when $j \geq 2$; their contributions are $-(2j-1)B_{2j-2}/4$ and $B_{2j-2}/2$. The case $j = 1$ is the direct product with $F_1(N) = (N^2+N)/2$ and gives $1/4$. For the second identity, an odd inverse exponent in A_N^2 arises only by multiplying $-1/(2N^2)$ by the appropriate odd inverse power; this gives $-B_{2j-2}$, also at $j = 1$.

Combining (8.259) and (8.260) yields

$$\mathrm{CT}_N\, E_j(N) = -\frac{2j-1}{2} B_{2j-2}.$$

Multiplication by $b_{r,j}$, followed by the $j = 0$ contribution, proves (8.249). □

8.33.3 A convergent generator involving classical polylogarithms

The complete sequence has an analytic exponential generating function. Let

$$\mathcal{G}(t) = \sum_{r=0}^{\infty} \mathcal{T}(-2r) \frac{t^{2r}}{(2r)!}.$$

Theorem 8.106 (Polylogarithmic moment generator)**.** *The series $\mathcal{G}$ converges for $|t| < 2\pi$ and equals*

$$\mathcal{G}(t) = \frac{t}{2\sinh(t/2)} \left[3\zeta(3) - \frac{1}{4t} \int_0^t (t-u) u^2 \coth(u/2)\, du \right]. \tag{8.261}$$

The singularities at zero in this expression are removable. For real $0 < t < 2\pi$, an equivalent formula in classical polylogarithms is

$$\mathcal{G}(t) = \frac{t}{2\sinh(t/2)} \left[2\zeta(3) - \frac{t^3}{48} - \frac{t}{2} \mathrm{Li}_2(e^{-t}) - 2\,\mathrm{Li}_3(e^{-t}) + \frac{3\big(\zeta(4) - \mathrm{Li}_4(e^{-t})\big)}{t} \right]. \tag{8.262}$$

Its continuation specified by (8.261) is holomorphic and even throughout $|t| < 2\pi$.

Proof. The Bernoulli generating functions give

$$\frac{t}{2\sinh(t/2)} = \sum_{r\ge 0} B_{2r}(1/2)\frac{t^{2r}}{(2r)!}, \qquad \frac{u}{2}\coth(u/2) = \sum_{k\ge 0} B_{2k}\frac{u^{2k}}{(2k)!}.$$

Both converge in the disk of radius 2π. Integrating the second series twice gives

$$\frac{1}{2t}\int_0^t (t-u)u^2\coth(u/2)\,du = \sum_{j\ge 1}\frac{2j-1}{2j+1}B_{2j-2}\frac{t^{2j}}{(2j)!}.$$

Thus (8.261) is exactly the binomial convolution in (8.249). This proves convergence, analyticity and the integral formula. In particular all apparent odd or logarithmic terms in any branch representation must cancel.

For positive t, expand $(e^u-1)^{-1}$ in its convergent positive exponential series. Direct integration, justified for the integrable factor $(t-u)u^2$, gives

$$\int_0^t \frac{(t-u)u^2}{e^u-1}\,du = 2t\zeta(3) - 6\zeta(4) + t^2\operatorname{Li}_2(e^{-t}) + 4t\operatorname{Li}_3(e^{-t}) + 6\operatorname{Li}_4(e^{-t}).$$

Finally $\coth(u/2) = 1 + 2/(e^u-1)$ and $\int_0^t (t-u)u^2\,du = t^4/12$ give (8.262). □

The finite-sum proof uses no transcendence assumption and no formal difference Galois theorem. The subtractions in the ordinary sums are part of each identity; without them the higher moments would diverge. The broader reflection-invariant numerator classification, resolvent transforms and independently marked higher-tail moments retain separate audit scopes.

8.34 Digamma powers, harmonic spectral data and exact finite parts

From harmonic Gamma identities to nonlinear finite parts. The three independent fifth-power continuations [131] converge on one all-power mechanism: specify each pole, determine the entire remainder, and only then compare the cutoff constant with spectral Laurent data. We retain the complete polynomial subtraction and every finite Stieltjes correction. This extends the preceding harmonic arithmetic, rather than assuming closure of all nonlinear periods in ordinary constants.

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$ and, for a positive integer word $w = (w_1,\dots,w_d)$, put

$$H_n(w) = \sum_{n\ge j_1>\cdots>j_d\ge 1} j_1^{-w_1}\cdots j_d^{-w_d}, \qquad Z(s,w) = \sum_{n\ge 1} H_{n-1}(w)n^{-s}. \tag{8.263}$$

The empty word has value one; a nonempty word has $H_0(w) = 0$. All Z coefficients below mean meromorphic Laurent coefficients, including at a pole. The Stieltjes convention is

$$\zeta(1+u,a) = u^{-1} + \sum_{j\ge 0}(-1)^j\gamma_j(a)u^j/j!, \qquad \gamma = \gamma_0(1). \tag{8.264}$$

If $f(x) = \sum_{j=1}^k a_j x^{-j} + g(x)$ with g continuous on $[0,1]$, the original-coordinate scalar finite part is

$$\operatorname{FP}_x\int_0^1 f(x)\,\mathrm{d}x = \lim_{\epsilon\downarrow 0}\left(\int_\epsilon^1 f(x)\,\mathrm{d}x - \sum_{j=2}^k \frac{a_j\epsilon^{1-j}}{j-1} + a_1\log\epsilon\right). \tag{8.265}$$

Set $Q_k = \mathrm{FP}_x \int_0^1 \psi(x)^k \, \mathrm{d}x$, $k \geq 1$. This scalar problem uses the pole at zero and the upper endpoint one; it is distinct from the distributional shift-contact comparison. For a sequence $F_N = P(\log N) + o(1)$, $\mathrm{CT}_{\log N} F_N = P(0)$. The cutoff comparison below proves its relation with Laurent coefficients.

8.34.1 Pole coefficients and the normalized global expansion

The standard Taylor expansion of $\psi(1+\epsilon)$ and the recurrence $\psi(z+1) = \psi(z) + 1/z$ give [132, §§5.5,5.7]

$$\psi(-n+\epsilon) = -\frac{1}{\epsilon} + b_n + \sum_{r\geq 2} c_{r,n}\epsilon^{r-1}, \tag{8.266}$$

$$b_n = H_n - \gamma, \qquad c_{r,n} = H_n^{(r)} + (-1)^r \zeta(r), \qquad n \geq 0.$$

This is a convergent local expansion for $0 < |\epsilon| < 1$.

For each fixed positive integer k, define the finite polynomial data

$$p_{k,j}(n) = [\epsilon^{k-j}] \left(-1 + b_n\epsilon + \sum_{r=2}^{k} c_{r,n}\epsilon^r \right)^k, \qquad 1 \leq j \leq k, \tag{8.267}$$

$$r_k(n) = p_{k,1}(n), \qquad P_{k,n}(z) = \sum_{j=1}^{k} \frac{p_{k,j}(n)}{(z+n)^j}, \qquad \kappa_k = [\epsilon^0]\psi(\epsilon)^k. \tag{8.268}$$

The symbol κ_k is a local Laurent constant; it will not be confused with the universal contour constant C_k introduced later. The pole coefficients satisfy

$$p_{k,j}(n) = O_k\big((1+\log(n+1))^{k-j}\big). \tag{8.269}$$

Indeed $b_n = O(1+\log(n+1))$ and every $c_{r,n}$ with $r \geq 2$ is bounded. In a term of degree $k-j$ in ϵ, at most $k-j$ factors of b_n can occur.

Theorem 8.107 (Normalized Mittag–Leffler formula for every power)**.** *For $z \notin \{0, -1, -2, \ldots\}$,*

$$\boxed{\psi(z)^k = P_{k,0}(z) + \kappa_k + \sum_{n\geq 1} \big(P_{k,n}(z) - P_{k,n}(0)\big).} \tag{8.270}$$

The series converges normally on compact sets avoiding the poles. In particular the entire remainder is exactly the displayed constant, with no uncomputed entire function.

Proof. Normal convergence of the candidate series follows from (8.269): on a fixed compact set, for sufficiently large n,

$$(z+n)^{-j} - n^{-j} = O(n^{-j-1}),$$

uniformly in z. It remains to determine the possible entire remainder. Put $f(w) = \psi(w)^k - P_{k,0}(w)$, so f is analytic at zero and $f(0) = \kappa_k$. Integrate

$$\frac{z f(w)}{w(w-z)}$$

around $|w| = R_N = N + \frac{1}{2}$, with positive orientation and N large.

We need a uniform estimate on the full circle, including near the negative real axis. On its right half, the usual sectorial digamma asymptotic gives $\psi(w) = O(1 + \log R_N)$. On its left half, use

$$\psi(w) = \psi(1 - w) - \pi \cot(\pi w). \tag{8.271}$$

Here $1 - w$ lies in the right half-plane and has size comparable to R_N. If $|\Im w| \geq 1$, the cotangent is uniformly bounded. If $|\Im w| < 1$, the points of this circle are a fixed positive distance from the integers for all sufficiently large N, because its real intersections are half-integers. The cotangent is again uniformly bounded. Thus $f(w) = O_k((1 + \log R_N)^k)$ on the whole contour, and the integral is $O_k((1 + \log R_N)^k / R_N) = o(1)$.

The residues at $w = z$ and $w = 0$ are $f(z)$ and $-f(0)$. At $w = -n$, the residue is $-P_{k,n}(z) + P_{k,n}(0)$, as follows by expanding $z/[w(w - z)] = 1/(w - z) - 1/w$ there. The residue theorem and passage to the limit give (8.270). This proves the normalization, not just agreement of the principal parts. □

8.34.2 A finite recurrence for the coefficient differences

Write $\Delta p_{k,j}(n) = p_{k,j}(n) - p_{k,j}(n - 1)$ for $n \geq 1$, and $\Delta r_k(n) = \Delta p_{k,1}(n)$.

Proposition 8.108 (Exact difference recurrence)**.** *For $1 \leq j \leq k$,*

$$\Delta p_{k,j}(n) = \sum_{\ell=1}^{k-j} \binom{k}{\ell} \sum_{h=j}^{k-\ell} \binom{\ell + h - j - 1}{h - j} \frac{p_{k-\ell,h}(n-1)}{n^{\ell+h-j}}. \tag{8.272}$$

An empty sum is zero; in particular $\Delta p_{k,k}(n) = 0$.

Proof. The recurrence for ψ gives the local identity

$$\psi(-n + \epsilon) = \psi(-(n-1) + \epsilon) + \frac{1}{n - \epsilon}.$$

Expand its kth power by the binomial theorem. In the term with ℓ copies of $(n - \epsilon)^{-1}$, multiply the pole $p_{k-\ell,h}(n-1)\epsilon^{-h}$ by

$$(n - \epsilon)^{-\ell} = \sum_{q \geq 0} \binom{\ell + q - 1}{q} \frac{\epsilon^q}{n^{\ell+q}}.$$

The coefficient of ϵ^{-j} has $q = h - j$, which gives (8.272). Only finitely many coefficients contribute. □

The recurrence proves, in particular, $\Delta p_{k,j}(n) = O_k((1 + \log n)^k / n)$ for $j < k$. Sharper logarithmic powers are immediate, but are not needed for the exact identities. Formula (8.272) is also a convenient implementation specification: it involves only lower powers and exact rational binomial coefficients.

8.35 The universal contour polynomial

The subtraction constants in (8.270) are individually awkward harmonic expressions. Their combined cutoff asymptotic is much simpler than their separate expansions suggest.

Theorem 8.109 (All coefficients of the large-circle boundary term)**.** *For every fixed integer $k \geq 1$,*

$$\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (\log N + i\theta)^k \, \mathrm{d}\theta + o(1)$$
$$= \sum_{m=0}^{\lfloor k/2 \rfloor} \binom{k}{2m} \frac{(-1)^m \pi^{2m}}{2m+1} (\log N)^{k-2m} + o(1). \tag{8.273}$$

Consequently

$$C_k := \mathrm{CT}_{\log N}\left(\kappa_k - \sum_{n=1}^{N} P_{k,n}(0)\right) = \begin{cases} (-1)^m \pi^{2m}/(2m+1), & k = 2m, \\ 0, & k \text{ odd.} \end{cases} \tag{8.274}$$

Proof. The residue at zero of $\psi(w)^k/w$ is κ_k. At $-n$ it is $-P_{k,n}(0)$. Thus, on $R = N + \frac{1}{2}$, the residue theorem gives the exact identity

$$\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \psi(Re^{i\theta})^k \, \mathrm{d}\theta. \tag{8.275}$$

We compare this with the angular average of the principal logarithm. On the right half-circle, $\psi(w) - \operatorname{Log} w = O(1/R)$ uniformly. On the left half-circle with $|\Im w| \geq \log R$, reflection (8.271) and the exponential expansion of cotangent give

$$\psi(w) - \operatorname{Log} w = O(1/R) + O(e^{-2\pi|\Im w|}).$$

For example, in the upper left quadrant, $\operatorname{Log}(1-w) = \operatorname{Log} w - i\pi + O(1/R)$ and $\pi \cot \pi w = -i\pi + O(e^{-2\pi \Im w})$; the lower quadrant is the conjugate branch calculation.

The remaining arc near the negative real axis has angular length $O(\log R/R)$. On that arc, the bounded-cotangent argument in Theorem 8.107 gives $\psi(w) - \operatorname{Log} w = O(1)$, while both terms are $O(1 + \log R)$. Factoring the difference of the kth powers shows that its contribution to the angular average is $O_k((1+\log R)^k/R)$. The rest of the circle contributes a quantity of the same or smaller order. The logarithm has a jump at the one endpoint of the angular parametrization, which does not affect the integral. Therefore

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \left(\psi(Re^{i\theta})^k - (\log R + i\theta)^k\right) \mathrm{d}\theta = o(1).$$

Replacing $\log R$ by $\log N$ changes this by $o(1)$. Odd powers of θ integrate to zero and the even angular moments are $(-1)^m \pi^{2m}/(2m+1)$, proving both assertions. □

For orientation, $C_1 = C_3 = C_5 = 0$, $C_2 = -2\zeta(2)$, and $C_4 = 18\zeta(4)$. These are combined boundary constants. They are not the local Laurent constants κ_k.

8.36 A finite-part formula for every positive integer power

Define

$$J_k = \sum_{n \geq 1} \sum_{j=2}^{k} \frac{\Delta p_{k,j}(n)}{(j-1)n^{j-1}}, \tag{8.276}$$

$$D_k(s) = \sum_{n \geq 1} \frac{\Delta r_k(n)}{n^s}, \qquad \Re s > 0. \tag{8.277}$$

The first series is ordinary and absolutely convergent, by Proposition 8.108. The second converges absolutely in its initial half-plane. Its meromorphic continuation to a neighborhood of zero will follow from the proof below.

Theorem 8.110 (The all-power harmonic spectral formula)**.** *For every integer $k \geq 1$,*

$$\boxed{Q_k = C_k + J_k + [s]D_k(s).} \tag{8.278}$$

All data on the right are specified by the finite polynomial calculation (8.267), the exact difference recurrence (8.272), and the coefficient of a meromorphic germ. The identity uses the original-coordinate finite part (8.265).

We first record the exact cutoff comparison that determines the sign and the order of the spectral coefficient.

Lemma 8.111 (Logarithmic cutoffs and Laurent coefficients)**.** *Suppose $a_n = P(\log n) + O(n^{-\delta}(1+\log n)^M)$, where P is a polynomial and $\delta > 0$. Let $A(s) = \sum_{n\geq 1} a_n n^{-s}$ in its right half-plane of absolute convergence. Then $A(1+u)$ continues meromorphically near $u = 0$, and*

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \frac{a_n}{n} = [u^0]A(1+u), \tag{8.279}$$

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \frac{a_n \log n}{n} = -[u]A(1+u). \tag{8.280}$$

Proof. For $P(L) = \sum_j c_j L^j$, write

$$A(1+u) = \sum_j c_j (-1)^j \zeta^{(j)}(1+u) + \sum_{n\geq 1} \frac{a_n - P(\log n)}{n^{1+u}}.$$

The last series and all its derivatives converge normally for $\Re u > -\delta$. The elementary Stieltjes limit formula gives

$$\sum_{n=1}^{N} \frac{(\log n)^j}{n} = \frac{(\log N)^{j+1}}{j+1} + \gamma_j + O((1+\log N)^j/N).$$

The stated error follows by one sum–integral comparison applied to the derivative of $(\log x)^j/x$ on the tail. On the Laurent side, $[u^0](-1)^j\zeta^{(j)}(1+u) = \gamma_j$ and $[u](-1)^j\zeta^{(j)}(1+u) = -\gamma_{j+1}$. Termwise comparison proves the first formula and, after inserting one further factor of $\log n$, the second. Absolute convergence handles the subtracted remainder in both cases. □

Proof of Theorem 8.110. The harmonic-number asymptotic expansions, or reflection followed by the positive-axis digamma expansion, show that

$$r_k(n) = R_k(\log n) + O_k((1+\log n)^M/n), \tag{8.281}$$

$$\Delta r_k(n) = \frac{R_k'(\log n)}{n} + O_k((1+\log n)^M/n^2), \tag{8.282}$$

$$R_k(L) = \operatorname*{Res}_{\epsilon=0} (L - \pi\cot\pi\epsilon)^k. \tag{8.283}$$

Here and below M may depend on k. To justify the second estimate, use the complete harmonic asymptotic expansions through the $1/n$ term: $r_k(n) = R_k(\log n) + n^{-1}U_k(\log n) + O((1+\log n)^M/n^2)$

with a polynomial U_k. The difference of the explicit $n^{-1}U_k$ terms is $O((1+\log n)^M/n^2)$, and the leading difference is $R_k'(\log n)/n$ to that same accuracy. Thus the second formula does not follow from differencing an unspecified first-order error. It also follows directly from (8.272) and the first two harmonic asymptotic terms.

Apply Lemma 8.111 to $a_n = n\Delta r_k(n)$. It proves the meromorphic continuation of D_k and gives

$$\mathrm{CT}_{\log N} \sum_{n=1}^{N} \Delta r_k(n) \log n = -[s]D_k(s). \tag{8.284}$$

Now integrate the normally convergent series (8.270). After the pole $P_{k,0}$ is removed, the same $O(n^{-j-1})$ estimate proves uniform convergence also on the closed interval $[0,1]$, so integration across its regular endpoint is justified. The finite part of the pole at zero contributes $-\sum_{j=2}^{k} p_{k,j}(0)/(j-1)$; its simple pole has finite part zero. For a pole at $-n$, the integrated simple-pole term is $r_k(n)\log(1+1/n)$, and the higher-pole terms have numerators $p_{k,j}(n)(n^{1-j}-(n+1)^{1-j})/(j-1)$. The constant subtraction integrates to $-P_{k,n}(0)$.

Use the following *finite* summation identities:

$$\begin{aligned}
&\sum_{n=1}^{N} p_{k,j}(n)\big(n^{1-j}-(n+1)^{1-j}\big) \\
&\qquad = p_{k,j}(0) + \sum_{n=1}^{N} \frac{\Delta p_{k,j}(n)}{n^{j-1}} - \frac{p_{k,j}(N)}{(N+1)^{j-1}}, \qquad j \ge 2, \qquad (8.285)\\
&\sum_{n=1}^{N} r_k(n)\log(1+1/n) = r_k(N)\log(N+1) - \sum_{n=1}^{N} \Delta r_k(n)\log n. \qquad (8.286)
\end{aligned}$$

The first terms in (8.285) cancel the finite part of the pole at zero exactly, while its final terms tend to zero. If $J_{k,N}$ denotes the cutoff in (8.276), we obtain

$$\begin{aligned}
Q_k = \lim_{N\to\infty} \Bigg\{ &\kappa_k - \sum_{n=1}^{N} P_{k,n}(0) + J_{k,N} + r_k(N)\log(N+1) \\
&- \sum_{n=1}^{N} \Delta r_k(n)\log n \Bigg\}. \qquad (8.287)
\end{aligned}$$

All terms here arose from finite sums or normally convergent series. By (8.281), the polynomial asymptotic of the boundary $r_k(N)\log(N+1)$ is $(\log N)R_k(\log N)$, which has no constant term. Theorem 8.109 supplies the constant C_k for the first two terms of (8.287), and (8.284) supplies $+[s]D_k$. Finally $J_{k,N} \to J_k$. Taking the constant coefficient of the complete cutoff expression proves (8.278). □

8.36.1 A closed ordinary-series form

The residue polynomial in (8.283) itself has a closed form. This makes every Stieltjes correction explicit, without naming any multiple-zeta Laurent coefficient.

Lemma 8.112 (Closed asymptotic residue polynomial)**.** *For an integer $k \ge 0$, define*

$$R_k(L) = \operatorname*{Res}_{t=0}\big(L - \pi\cot(\pi t)\big)^k.$$

Then

$$\boxed{R_k(L) = -\frac{(L+i\pi)^k - (L-i\pi)^k}{2i\pi}.} \tag{8.288}$$

For $k \geq 1$, if $R_k'(L) = \sum_{j\geq 0} a_{k,j} L^j$, the ordinary Stieltjes contribution is the explicit finite expression

$$\boxed{-\sum_{j\geq 0} a_{k,j}\gamma_{j+1} = k \sum_{m=0}^{\lfloor (k-2)/2 \rfloor} (-1)^m \binom{k-1}{2m+1} \pi^{2m} \gamma_{k-2m-1}.} \tag{8.289}$$

The right side is understood as zero for $k = 1$. Here $\zeta(1+s) = s^{-1} + \sum_{j\geq 0}(-1)^j \gamma_j s^j/j!$.

Proof. Put $f(t) = L - \pi\cot(\pi t)$. The elementary differential identity

$$f'(t) = (f(t) - L)^2 + \pi^2$$

and the zero residue of a meromorphic derivative give, for $k \geq 2$,

$$0 = \frac{1}{k-1} \operatorname*{Res}_{t=0} \frac{\mathrm{d}}{\mathrm{d}t} f(t)^{k-1} = R_k(L) - 2LR_{k-1}(L) + (L^2+\pi^2)R_{k-2}(L).$$

Since $R_0 = 0$ and $R_1 = -1$, solving this recurrence with characteristic roots $L \pm i\pi$ proves (8.288). Expanding its numerator yields

$$R_k(L) = -\sum_{m=0}^{\lfloor (k-1)/2 \rfloor} (-1)^m \binom{k}{2m+1} \pi^{2m} L^{k-2m-1}.$$

On differentiating in L, the possible term with $k - 2m - 1 = 0$ vanishes. Use $(k-2m-1)\binom{k}{2m+1} = k\binom{k-1}{2m+1}$. This gives (8.289). □

To see directly why this finite expression is the Stieltjes term, suppose the residue difference satisfies the previously established asymptotic formula

$$r_k(n) - r_k(n-1) = \frac{R_k'(\log n)}{n} + O\left(\frac{(1+\log n)^M}{n^2}\right).$$

For $D_k(s) = \sum_{n\geq 1}(r_k(n) - r_k(n-1))n^{-s}$, subtraction therefore gives

$$D_k(s) = \sum_{j\geq 0} a_{k,j}(-1)^j \zeta^{(j)}(1+s) + \sum_{n\geq 1}\left(r_k(n) - r_k(n-1) - \frac{R_k'(\log n)}{n}\right) n^{-s}.$$

The second series and its first spectral derivative converge normally near $s = 0$. Since $[s](-1)^j\zeta^{(j)}(1+s) = -\gamma_{j+1}$, it follows that

$$\begin{aligned}[s]D_k(s) = k &\sum_{m=0}^{\lfloor (k-2)/2 \rfloor} (-1)^m \binom{k-1}{2m+1} \pi^{2m}\gamma_{k-2m-1} \\ &- \sum_{n\geq 1}\left(r_k(n) - r_k(n-1) - \frac{R_k'(\log n)}{n}\right)\log n.\end{aligned} \tag{8.290}$$

The final sum is absolutely convergent. For example, the finite Stieltjes terms at powers four and five are respectively $12\gamma_3 - 24\zeta(2)\gamma_1$ and $20\gamma_4 - 120\zeta(2)\gamma_2$.

Corollary 8.113 (Ordinary subtracted series at every power)**.** *With the definitions above,*

$$Q_k = C_k + J_k + k\sum_{m=0}^{\lfloor (k-2)/2\rfloor}(-1)^m\binom{k-1}{2m+1}\pi^{2m}\gamma_{k-2m-1} - \sum_{n\ge 1}\left(\Delta r_k(n) - \frac{R_k'(\log n)}{n}\right)\log n. \tag{8.291}$$

The finite Stieltjes sum is zero at $k = 1$. Both infinite sums are ordinary absolutely convergent sums. The last summand is $O_k((1+\log n)^{M+1}/n^2)$.

This is useful both conceptually and computationally. The right side contains no unspecified regularization and no uncomputed entire remainder. Further terms of the same harmonic asymptotic expansions give convergent Hurwitz-zeta tail accelerations if numerical evaluation is wanted. Such acceleration is optional for the identity itself.

8.37 A finite algorithm in strict harmonic words

The spectral formula becomes finite algebra once its target class is specified. Work over $\mathbb{Q}[\gamma, \zeta(2), \dots, \zeta(k)]$ and put $h_r = H_{n-1}^{(r)}$, $v = 1/n$. Compute the polynomial $p_{k,j}$ from (8.267), and replace $H_n^{(r)}$ by $h_r + v^r$ to obtain its value at n. Subtract its value at $n-1$. This produces every $\Delta p_{k,j}$ as a finite polynomial in $v, h_1, \dots, h_k$.

The finite stuffle recursion, with (a, A) denoting a leading letter followed by a word, is

$$(a, A) * (b, B) = (a, A * (b, B)) + (b, (a, A) * B) + (a + b, A * B). \tag{8.292}$$

It expresses products of the h_r as a finite integer combination of strict words $H_{n-1}(w)$. The coefficients count equal-index possibilities as well as the two strict orderings. Thus

$$h_1^2 = 2H_{n-1}(1,1) + H_{n-1}^{(2)},$$

and a further multiplication gives

$$h_1^3 = 6H_{n-1}(1,1,1) + 3H_{n-1}(2,1) + 3H_{n-1}(1,2) + H_{n-1}^{(3)}.$$

These are finite identities before any infinite sum is taken.

Every positive-integer word has a polynomial harmonic asymptotic

$$H_{n-1}(w) = B_w(\log n) + O((1+\log n)^M/n) \tag{8.293}$$

for some polynomial B_w and integer M. Here is a direct induction that also supplies the continuation needed term by term. For the empty word the assertion is immediate. Given it for w, use $H_N(a, w) = \sum_{m=1}^{N} H_{m-1}(w)m^{-a}$. If $a = 1$, the Stieltjes cutoff formula applied to each power of $\log m$ and the absolutely convergent subtracted remainder give a polynomial in $\log N$ with an $O((1+\log N)^{M'}/N)$ error. If $a \ge 2$, the sum converges and its tail has that same error form, so the polynomial is a constant. Replacing N by $n-1$ preserves the claimed form. Consequently each $Z(1+s, w)$ has a meromorphic germ at zero by Lemma 8.111; for first entry at least two its defining series is already holomorphic near zero.

Theorem 8.114 (Terminating harmonic-word reduction)**.** *At a fixed positive integer power k, the following procedure terminates and yields the exact formula* (8.278)*:*

1. *Compute $p_{k,j}$ and their finite differences by (8.267) or (8.272).*

2. *Expand the summand of J_k and the polynomial $\Delta r_k(n)$ in $v, h_1, \dots, h_k$.*

3. *Apply the finite stuffle recursion to every harmonic monomial.*

4. *Replace $v^q H_{n-1}(w)$ by $Z(q,w)$ for J_k, and by $Z(q+s,w)$ for $D_k(s)$.*

Every first entry in J_k is at least two. Every first entry in $D_k(0)$ is at least one. The resulting word weights are at most $k-1$; coefficient factors in γ and ordinary zeta values account for the remaining formal weight. The prescription introduces only a finite number of specified linear Laurent coefficients.

Proof. Coefficient extraction and finite differences are finite polynomial operations. Each application of the stuffle recursion reduces the combined remaining word length, so it terminates. By (8.272), every monomial in $\Delta p_{k,j}$ contains at least one v, and division by n^{j-1} gives $q \geq j \geq 2$ in J_k. For Δr_k, it gives $q \geq 1$. Therefore all replacements in the initial half-planes are identities of absolutely convergent series. Their continuations at zero agree by the meromorphic subtraction justified by (8.293). Finally assign weight 1 to h_1, γ, v, weight r to $h_r, \zeta(r)$, and use (8.267). Both relevant summand polynomials have weight $k-1$, and stuffle preserves it. □

This is a reduction to a precise finite class of multiple-zeta values and their spectral coefficients. It does not prove that all those values reduce further to products of ordinary zeta values. For higher powers it is appropriate to retain admissible multiple-zeta words unless an exact additional relation is supplied.

8.37.1 Recovery of the lower-power formulas

Define the height-one coefficients

$$\eta = [u]Z(1+u,1), \qquad \theta = [u]Z(1+u,1,1), \qquad \chi = [u]Z(1+u,1,1,1). \tag{8.294}$$

The first instances of Theorem 8.110 are

$$\begin{aligned}
Q_1 &= 0, \\
Q_2 &= 2\gamma_1 - 2\zeta(2), && (8.295)\\
Q_3 &= 3\zeta(2) - 6\eta - 6\gamma\gamma_1, && (8.296)\\
Q_4 &= 12\zeta(3) - 12\gamma\zeta(2) + 18\zeta(4) + 4\zeta'(3) - 24\theta + 24\gamma\eta + 12(\gamma^2 - \zeta(2))\gamma_1. && (8.297)
\end{aligned}$$

The cubic and quartic identities are consistency checks against the incoming reports [131]; they are not presented as new evaluations here. For example,

$$\begin{aligned}
D_2(s) &= -2\zeta(1+s), && J_2 = 0,\\
D_3(s) &= -6Z(1+s,1) + 6\gamma\zeta(1+s), && J_3 = 3\zeta(2),\\
D_4(s) &= -24Z(1+s,1,1) + 24\gamma Z(1+s,1) \\
&\quad + 12(\zeta(2) - \gamma^2)\zeta(1+s) + 4\zeta(3+s), && J_4 = 12\zeta(3) - 12\gamma\zeta(2).
\end{aligned}$$

At $k=1$ the residue is the constant -1, hence $J_1 = D_1 = 0$; the finite part also follows directly from $\log\Gamma(1) = 0$ and $\log\Gamma(\epsilon) = -\log\epsilon + O(\epsilon)$.

8.38 The fifth power with every finite correction

For compactness write $\zeta_r = \zeta(r)$ in this section. At a fixed pole use $b = b_n$, $c = c_{2,n}$, $d = c_{3,n}$, and $e = c_{4,n}$. Direct coefficient extraction gives

$$\begin{aligned} p_{5,5} &= -1, \qquad p_{5,4} = 5b, \qquad p_{5,3} = 5c - 10b^2, \\ p_{5,2} &= 10b^3 - 20bc + 5d, \\ r_5 &= -5b^4 + 30b^2c - 10c^2 - 20bd + 5e. \end{aligned} \tag{8.298}$$

For reference, the local constant is

$$\kappa_5 = -\gamma^5 + 20\gamma^3\zeta_2 - 55\gamma\zeta_4 - 30\gamma^2\zeta_3 + 20\zeta_2\zeta_3 - 5\zeta_5. \tag{8.299}$$

The universal constant is $C_5 = 0$, a different quantity.

For the finite differences, let b, c, d instead denote the preceding index $n-1$. Then

$$\begin{aligned} \Delta p_{5,4}(n) &= \frac{5}{n}, \\ \Delta p_{5,3}(n) &= -\frac{5}{n^2} - \frac{20b}{n}, \\ \Delta p_{5,2}(n) &= -\frac{5}{n^3} + \frac{10b}{n^2} + \frac{30b^2 - 20c}{n}, \\ \Delta r_5(n) &= \frac{-20b^3 + 60bc - 20d}{n} + \frac{10c}{n^2} + \frac{20b}{n^3}. \end{aligned} \tag{8.300}$$

Equivalently the first numerator in the last line is $5r_4(n-1)$. These identities can be checked simply by replacing (b, c, d, e) by $(b + 1/n, c + 1/n^2, d + 1/n^3, e + 1/n^4)$ in (8.298); Proposition 8.108 gives an independent binomial derivation.

The higher poles therefore contribute

$$\begin{aligned} J_5 &= \sum_{n\geq 1} \left\{ \frac{30(H_{n-1} - \gamma)^2 - 20(H^{(2)}_{n-1} + \zeta_2)}{n^2} - \frac{35}{6n^4} \right\} \\ &= 30\gamma^2\zeta_2 - 60\gamma\zeta_3 + \frac{35}{3}\zeta_4. \end{aligned} \tag{8.301}$$

For completeness, the low-weight evaluations needed in this last step are

$$Z(2,1) = \zeta_3, \quad Z(2,1,1) = \zeta_4, \quad Z(2,2) = \frac{3}{4}\zeta_4, \quad \zeta_2^2 = \frac{5}{2}\zeta_4.$$

They follow from the standard duality, sum, and stuffle identities [132, 133]. In particular,

$$\begin{aligned} \sum_{n\geq 1} \frac{(H_{n-1} - \gamma)^2}{n^2} &= \frac{11}{4}\zeta_4 - 2\gamma\zeta_3 + \gamma^2\zeta_2, \\ \sum_{n\geq 1} \frac{H^{(2)}_{n-1} + \zeta_2}{n^2} &= \frac{13}{4}\zeta_4, \end{aligned}$$

which proves every scalar term in (8.301).

Applying finite stuffle to the last line of (8.300) yields the complete spectral identity

$$\begin{aligned} D_5(s) = &-120Z(1+s,1,1,1)+120\gamma Z(1+s,1,1)\\ &+60(\zeta_2-\gamma^2)Z(1+s,1)+20(\gamma^3-3\gamma\zeta_2+\zeta_3)\zeta(1+s)\\ &+20Z(1+s,3)+10Z(2+s,2)+20Z(3+s,1)\\ &+10\zeta_2\zeta(2+s)-20\gamma\zeta(3+s). \end{aligned} \tag{8.302}$$

All equalities here hold first for $\Re s>0$, and then as meromorphic germs. In the first numerator of (8.300), the words $(2,1)$ and $(1,2)$ from H_{n-1}^3 cancel against the corresponding words in $H_{n-1}H_{n-1}^{(2)}$. The residual (3) word does not cancel. It is responsible for $Z(1+s,3)$.

Set

$$D_{p,q}=[u]Z(p+u,q),\qquad \Omega_4=2D_{1,3}+D_{2,2}+2D_{3,1}. \tag{8.303}$$

For $p\geq 2$ these coefficients are ordinary outer-order derivatives, $D_{p,q}=\partial_1 Z(p,q)$. For $p=1,q=3$, coefficient extraction is required at a simple pole.

Theorem 8.115 (Complete fifth-power finite part)**.** *In the conventions* (8.265)*,* (8.264)*,* (8.294)*, and* (8.303)*,*

$$\begin{aligned} Q_5 = &\,30\gamma^2\zeta_2-60\gamma\zeta_3+\frac{35}{3}\zeta_4-120\chi+120\gamma\theta+60(\zeta_2-\gamma^2)\eta\\ &-20(\gamma^3-3\gamma\zeta_2+\zeta_3)\gamma_1+20D_{1,3}+10D_{2,2}+20D_{3,1}\\ &+10\zeta_2\zeta'(2)-20\gamma\zeta'(3). \end{aligned} \tag{8.304}$$

Equivalently the three mixed terms are $10\Omega_4$*.*

Proof. Use $C_5=0$, (8.301), and the linear coefficient of (8.302) in Theorem 8.110. In particular $[s]\zeta(1+s)=-\gamma_1$, which fixes the sign of its correction. The other coefficients are exactly those named in (8.294) and (8.303). Every step rests on a finite algebraic identity or the normalized analytic results already proved. □

This answers the fifth-power research question in [131]: the fourfold height-one coefficient enters with multiplier -120, and the mixed twofold coefficients and both ordinary zeta derivatives are retained explicitly. The result is not a reduction of χ or Ω_4 to ordinary zeta and Stieltjes constants.

8.38.1 An ordinary convergent formula for the same finite part

The residue polynomial and its derivative are

$$R_5(L)=-5L^4+60\zeta_2L^2-90\zeta_4,\qquad R_5'(L)=-20L^3+120\zeta_2L.$$

Consequently Corollary 8.113 gives the exact alternative

$$\begin{aligned} Q_5 = &\,30\gamma^2\zeta_2-60\gamma\zeta_3+\frac{35}{3}\zeta_4+20\gamma_4-120\zeta_2\gamma_2\\ &-\sum_{n\geq 1}\left\{\Delta r_5(n)+\frac{20\log^3 n-120\zeta_2\log n}{n}\right\}\log n. \end{aligned} \tag{8.305}$$

Here Δr_5 is the explicit harmonic expression (8.300), with b,c,d at $n-1$. It can equally be written in polygamma values at positive integers:

$$b=\psi(n),\qquad c=2\zeta_2-\psi'(n),\qquad d=\tfrac{1}{2}\psi''(n). \tag{8.306}$$

The final series in (8.305) converges absolutely; its summand is $O((1+\log n)^4/n^2)$. This gives a definition-free numerical route to Q_5 using only ordinary Stieltjes constants and ordinary polygamma evaluations, with no multivariate analytic continuation routine.

8.38.2 Explicit kernels for the height-one coefficients

The retained height-one coefficient has a concrete normalized integral representation. We give it at every depth so that the fourfold coefficient is specified independently of the digamma moment. Height-one Laurent constants and related normalized integrals have antecedents in Young's work [134, Section 4]. The proposition below records a direct Mellin subtraction with the coefficient and Stieltjes conventions fixed explicitly for this article. Let

$$\frac{1}{\Gamma(1+u)} = \sum_{j\ge 0} \alpha_j u^j, \qquad K_d(t) = \frac{[-\log(1-e^{-t})]^{d-1}}{(d-1)!(e^t-1)}, \quad d \ge 1. \tag{8.307}$$

Proposition 8.116 (Normalized height-one Mellin coefficient)**.** *For $d \ge 1$,*

$$Z(1+u, 1^{d-1}) = \sum_{j=0}^{d} \alpha_j u^{j-d} + L_d u + O(u^2), \tag{8.308}$$

where

$$\begin{aligned} L_d = \alpha_{d+1} &+ \int_0^1 \log t \left\{ K_d(t) - \frac{(-\log t)^{d-1}}{(d-1)!t} \right\} \, \mathrm{d}t \\ &+ \int_1^\infty \log t \, K_d(t) \, \mathrm{d}t. \end{aligned} \tag{8.309}$$

Both integrals are absolutely convergent. In particular $L_2 = \eta$, $L_3 = \theta$, and $L_4 = \chi$.

Proof. The generating series of $H_{n-1}(1^{d-1})$ gives, for $\Re u > 0$,

$$\Gamma(1+u) Z(1+u, 1^{d-1}) = \int_0^\infty t^u K_d(t) \, \mathrm{d}t.$$

For example it follows by dividing $\mathrm{Li}_{1^{d-1}}(z) = (-\log(1-z))^{d-1}/(d-1)!$ by $1-z$, multiplying by z, and setting $z = e^{-t}$; for $d = 1$ the kernel is simply $1/(e^t - 1)$. Subtract the leading kernel on $(0, 1)$ to write this integral as $u^{-d} + B_d(u)$, with B_d holomorphic near zero. The subtraction is integrable with any fixed power of $\log t$, because its remainder is at worst a polynomial in $|\log t|$ near zero; the kernel decays exponentially at infinity.

The derivative identity

$$K_d(t) = -\frac{1}{d!} \frac{\mathrm{d}}{\mathrm{d}t} [-\log(1-e^{-t})]^d$$

shows that $B_d(0) = 0$: the integrals from ϵ to infinity and of the subtracted kernel from ϵ to 1 have difference

$$\frac{[-\log(1-e^{-\epsilon})]^d - (-\log \epsilon)^d}{d!} \longrightarrow 0.$$

Multiplication by $1/\Gamma(1+u)$ now proves (8.308), with $L_d = \alpha_{d+1} + B_d'(0)$ exactly as in (8.309). □

For the fifth moment the coefficient used in this kernel is explicitly

$$\alpha_5 = \frac{\gamma^5 - 10\gamma^3\zeta_2 + 20\gamma^2\zeta_3 + 15\gamma\zeta_2^2 - 30\gamma\zeta_4 - 20\zeta_2\zeta_3 + 24\zeta\alpha_5}{120}. \tag{8.310}$$

This follows by exponentiating $\log(1/\Gamma(1+u)) = \gamma u + \sum_{r\ge 2} (-1)^{r+1} \zeta(r) u^r / r$. The kernel is an exact analytic representation, not by itself a reduction of the arithmetic content of χ.

8.38.3 Positive Cauchy series and the height-one sign convention

The height-one coefficient formula has a second, ordinary-series form [134]. Put

$$\rho_n = \frac{1}{(n-1)!}\int_0^1 (r)_n \, \mathrm{d}r, \qquad n \geq 1.$$

These are positive rational numbers, with $\rho_n \leq n$.

Proposition 8.117 (An ordinary series for every height-one coefficient)**.** *With $\alpha_j = [u^j]\Gamma(1+u)^{-1}$ and $L_d = [u]Z(1+u, 1^{d-1})$, for every integer $d \geq 1$,*

$$L_d = \alpha_{d+1} + \sum_{n\geq 1} \frac{\rho_n}{n^{d+2}}. \tag{8.311}$$

The sum is ordinary and absolutely convergent. In particular $L_1 = -\gamma_1$; L_2, L_3, L_4 are η, θ, χ in the fifth-power formula.

Proof. Let $\ell(t) = -\log(1-e^{-t})$. It satisfies $\ell(\ell(t)) = t$ and $K_d(t) = -\ell(t)^{d-1}\ell'(t)/(d-1)!$. Substitution into the height-one Mellin integral gives, initially for $\Re u > 0$,

$$\Gamma(1+u)Z(1+u, 1^{d-1}) = \frac{1}{(d-1)!}\int_0^\infty y^{d-1}\ell(y)^u \, \mathrm{d}y.$$

Subtract the elementary integral of $y^{d-1}e^{-uy}/(d-1)!$, which is u^{-d}. The difference is holomorphic near zero: at infinity its integrand is $O(e^{-(1+\Re u)y}y^{d-1})$ locally uniformly in u, and at zero fixed powers of $\log \ell(y)$ are integrable against y^{d-1}. Consequently its derivative at zero is

$$\frac{1}{(d-1)!}\int_0^\infty y^{d-1}\{\log \ell(y) + y\} \, \mathrm{d}y.$$

For $0 < z < 1$, ordinary binomial integration gives

$$\int_0^1 (1-z)^{-r} \, \mathrm{d}r = -\frac{z}{(1-z)\log(1-z)} = 1 + \sum_{n\geq 1}\frac{\rho_n}{n}z^n.$$

Differentiating $\log \ell(y) + y$ and using its zero limit at infinity therefore gives

$$\log \ell(y) + y = \sum_{n\geq 1}\frac{\rho_n}{n^2}e^{-ny}.$$

All terms are positive; Tonelli and the elementary Gamma integral yield $\sum \rho_n/n^{d+2}$. Multiplication by $\Gamma(1+u)^{-1}$ adds α_{d+1}, since the subtracted holomorphic remainder vanishes at zero. This proves the claimed coefficient. Finally $\rho_n \leq n$ follows from $(r)_n \leq n!$ on $0 \leq r \leq 1$, proving convergence. □

This representation is established literature in Cauchy-number notation; the direct involution proof here makes its coefficient and sign explicit. Appendix A.5.2 checks the published description of that sign.

8.38.4 The mixed derivative and its polylogarithmic representations

The stuffle identity in a right half-plane gives

$$Z(1+u,3)=\zeta(1+u)\zeta_3-Z(3,1+u)-\zeta(4+u).$$

Consequently

$$D_{1,3}=-\gamma_1\zeta_3-\partial_2 Z(3,1)-\zeta'(4). \tag{8.312}$$

The inner-order derivative has an absolutely convergent defining series. In particular this transformation replaces a singular outer-order coefficient by a convergent inner-order derivative; it does not remove the nonsymmetric spectral information.

Define the Tornheim function

$$T(a,b,c)=\sum_{m,n\geq 1}\frac{1}{m^a n^b(m+n)^c}$$

in its absolute-convergence domain. A finite partial fraction calculation gives, for u near zero,

$$T(2,2,u)=2Z(2+u,2)+4Z(3+u,1). \tag{8.313}$$

Indeed, for $l=m+n$, summation over $1\leq m<l$ uses

$$\frac{1}{m^2(l-m)^2}=\frac{1}{l^2}\left(\frac{1}{m^2}+\frac{1}{(l-m)^2}+\frac{2}{m(l-m)}\right)$$

and gives $2H^{(2)}_{l-1}/l^2+4H_{l-1}/l^3$. Thus

$$\Omega_4=2D_{1,3}+\tfrac{1}{2}\partial_3 T(2,2,0) \tag{8.314}$$

$$=2(\partial_1-\partial_2)Z(3,1)+\partial_1 Z(2,2)-2\gamma_1\zeta_3-2\zeta'(4). \tag{8.315}$$

Spectral derivatives of Tornheim sums have substantial prior literature, including [32]. These equalities are useful coordinate changes; no arithmetic independence is asserted for the displayed coordinates.

There are also completely convergent one-dimensional representations:

$$D_{1,3}=-\gamma_1\zeta_3+\sum_{n\geq 1}\frac{\zeta(3,n)\log n}{n}, \tag{8.316}$$

$$\partial_3 T(2,2,0)=\int_0^\infty\frac{\mathrm{Li}_2(e^{-t})^2-e^{-t}\zeta_2^2}{t}\,\mathrm{d}t. \tag{8.317}$$

For (8.316), use $H^{(3)}_{n-1}=\zeta_3-\zeta(3,n)$ and extract the linear coefficient. For (8.317), insert the Frullani identity $-\log(m+n)=\int_0^\infty(e^{-(m+n)t}-e^{-t})\,\mathrm{d}t/t$ and sum. Absolute integrability follows either from $\sum_{m,n\geq 1}\log(m+n)/(m^2n^2)<\infty$, or directly from the $O(\log t)$ integrand at zero and exponential decay at infinity. Equations (8.314), (8.316), and (8.317) connect the first mixed correction in a digamma power to a dilogarithm integral and a Hurwitz-zeta tail sum.

8.38.5 What scalar polynomial cancellation proves

Corollary 8.118 (A normalized fifth-degree cancellation)**.** *Use the fixed-coordinate finite parts* $Q_k = \mathrm{FP}_x \int_0^1 \psi(x)^k \, \mathrm{d}x$*, and put*

$$\Omega_4 = 2D_{1,3} + D_{2,2} + 2D_{3,1}, \qquad D_{p,q} = [s]\zeta(p+s,q).$$

Here $[s]$ *denotes the coefficient of* s *in the Laurent expansion, including when a pole is present. Define*

$$\begin{aligned} P_5(T) = T^5 + 5\gamma T^4 + 10\big(\gamma^2 + \zeta(2)\big)T^3 \\ + 10\big(\gamma^3 + 3\gamma\zeta(2) + \zeta(3)\big)T^2. \end{aligned} \tag{8.318}$$

Then

$$\begin{aligned} \mathrm{FP}_x \int_0^1 P_5\big(\psi(x)\big) \, \mathrm{d}x = & -120\chi + 10\Omega_4 + 10\zeta(2)\zeta'(2) \\ & + \frac{260}{3}\zeta(4) - 60\gamma\zeta(4) - 20\gamma^3\zeta(2) - 20\zeta(2)\zeta(3). \end{aligned} \tag{8.319}$$

Among monic polynomials $T^5 + a_4T^4 + a_3T^3 + a_2T^2$ *with coefficients in* $\mathbb{Q}[\gamma, \zeta(2), \zeta(3)]$*, the displayed polynomial is the unique one whose coefficients of* θ, η, γ_1 *vanish in the stated formulas for* Q_2, Q_3, Q_4, Q_5*, when these symbols are treated as formal coordinates.*

Proof. The finite-part operation in one fixed coordinate is linear. In $Q_5 + a_4Q_4 + a_3Q_3 + a_2Q_2$, cancellation of the displayed θ, η, γ_1 coefficients gives, successively,

$$\begin{aligned} 120\gamma - 24a_4 &= 0, \\ 60\big(\zeta(2) - \gamma^2\big) + 24\gamma a_4 - 6a_3 &= 0, \\ -20\big(\gamma^3 - 3\gamma\zeta(2) + \zeta(3)\big) + 12\big(\gamma^2 - \zeta(2)\big)a_4 - 6\gamma a_3 + 2a_2 &= 0. \end{aligned}$$

This triangular system has exactly the coefficients in (8.318). The $\zeta'(3)$ coefficient simultaneously becomes $-20\gamma + 4a_4 = 0$, while the coefficients of χ, Ω_4, and $\zeta(2)\zeta'(2)$ remain respectively $-120, 10, 10$. Before using any even-zeta reduction, the remaining scalar is

$$\frac{35}{3}\zeta(4) + 90\gamma\zeta(4) + 30\zeta(2)^2 - 20\gamma^3\zeta(2) - 60\gamma\zeta(2)^2 - 20\zeta(2)\zeta(3).$$

Substitute $\zeta(2)^2 = \frac{5}{2}\zeta(4)$ to obtain (8.319). □

The uniqueness assertion is deliberately relative to the displayed coefficient presentation and to the normalization excluding constant and linear terms. It does not assert arithmetic independence of $\chi, \Omega_4, \theta, \eta, \gamma_1$, or the impossibility of rewriting Ω_4 in another family of constants. In particular, "cancellation of γ_1" means cancellation of its explicit coefficient while Ω_4 is retained as a coordinate; a different representation of Ω_4 may contain γ_1. Further arithmetic reduction of the mixed-depth term remains a separate question.

8.38.6 Reconciling the regular and singular mixed coordinates

The other two fifth-moment routes [131] package the mixed term as a convergent derivative. Define

$$R(s) = Z(s,3) - \zeta(3)\zeta(s), \qquad J(s) = 2R(s) + 2Z(s+2,1) + Z(s+1,2) - 2\gamma\zeta(s+2) + \zeta(2)\zeta(s+1).$$

Here R is holomorphic for $\Re s > -1$ and J for $\Re s > 0$. Indeed $R(s) = -\sum_{n\ge1} \zeta(3,n)n^{-s}$, whose tail is $O(n^{-\Re s-2})$; the other defining series have the stated normal convergence. Thus $J'(1)$ is an ordinary derivative, and

$$J'(1) = \Omega_4 + 2\zeta(3)\gamma_1 - 2\gamma\zeta'(3) + \zeta(2)\zeta'(2). \tag{8.320}$$

To prove this, first extract $[u]$ in $R(1+u) = Z(1+u,3) - \zeta(3)\zeta(1+u)$; the result is $R'(1) = D_{1,3} + \zeta(3)\gamma_1$. The remaining terms are ordinary derivatives, giving the display. This also proves exact agreement of the three fifth-power formulas. Dropping the $\zeta(3)\gamma_1$ term would break that agreement.

The polynomial P_5 just displayed cancels explicit lower coordinates in the Ω_4 presentation. In the $J'(1)$ presentation, the corresponding nonlinear part is $\widetilde{P}_5(T) = P_5(T) + 10\zeta(3)T^2$. Linearity and $Q_2 = 2\gamma_1 - 2\zeta(2)$ give the exact conversion

$$\mathrm{FP}_x \int_0^1 \big(\widetilde{P}_5 - P_5\big)(\psi(x))\,\mathrm{d}x = 20\zeta(3)\gamma_1 - 20\zeta(2)\zeta(3).$$

The different coefficients therefore reflect different retained coordinates and normalization choices. Neither formal cancellation proves that the remaining mixed period has no alternative arithmetic reduction.

8.39 Normalized primitives and polygamma identities

The same normalization gives a primitive, not only a finite moment. On $\mathbb{C}\setminus(-\infty,0]$ use the principal logarithm.

Theorem 8.119 (An all-power primitive with fixed normalization)**.** *For $k \ge 1$, put*

$$\begin{aligned}\mathcal{A}_k(z) = \kappa_k z + p_{k,1}(0)\operatorname{Log} z + \sum_{j=2}^{k} \frac{p_{k,j}(0)z^{1-j}}{1-j} \\ + \sum_{n\ge1}\Bigg\{ p_{k,1}(n)\log(1+z/n) - zP_{k,n}(0) \\ + \sum_{j=2}^{k} p_{k,j}(n)\frac{(z+n)^{1-j} - n^{1-j}}{1-j}\Bigg\}.\end{aligned} \tag{8.321}$$

The series converges normally on compact subsets of that slit plane, and

$$\mathcal{A}_k'(z) = \psi(z)^k, \qquad \mathcal{A}_k(1) = Q_k. \tag{8.322}$$

Its local Laurent–log expansion at zero has constant coefficient zero, which fixes the otherwise arbitrary additive constant.

Proof. Each brace in (8.321) is the integral from zero to z of $P_{k,n}(w) - P_{k,n}(0)$. For large n and z in a fixed compact set its magnitude is $O_k((1+\log n)^{k-1}/n^2)$, as follows by cancellation of its linear terms or from the bound in the proof of Theorem 8.107. The normally convergent series may therefore be differentiated, and (8.270) gives the first identity. Every term in the braces vanishes at zero and the analytic part begins with $\kappa_k z$. The Laurent and logarithmic terms outside the braces also have no constant coefficient. The definition of the original-coordinate finite part now gives $\mathcal{A}_k(1) = Q_k$. □

For $k = 1$, this is the normalized branch of $\log\Gamma(z)$, whose local expansion begins $-\operatorname{Log} z - \gamma z + \cdots$. For every higher power it gives an explicitly normalized antiderivative through harmonic pole coefficients. Ordinary integrals along paths contained in the slit plane follow by endpoint subtraction of $\mathcal{A}_k$. On other pole-free intervals the same conclusion holds after choosing a consistent analytic continuation.

Corollary 8.120 (Differentiated polygamma identities)**.** *For $m \geq 1$ and z away from the nonpositive integers,*

$$\frac{\mathrm{d}^m}{\mathrm{d}z^m}\psi(z)^k = (-1)^m \sum_{n\geq 0}\sum_{j=1}^{k} \frac{(j)_m p_{k,j}(n)}{(z+n)^{j+m}}. \tag{8.323}$$

The series is normally convergent. Its left side is explicitly

$$\sum_{m_1+\cdots+m_k=m} \frac{m!}{m_1!\cdots m_k!}\prod_{i=1}^{k}\psi^{(m_i)}(z), \tag{8.324}$$

where $\psi^{(0)} = \psi$ and $(j)_m$ is the rising factorial.

Proof. Differentiate (8.270) normally on compact subsets. The constant subtractions disappear after the first derivative and each pole gives the displayed rising factorial. The denominator exponent is at least two, so (8.269) proves normal convergence directly as well. The multinomial identity is the ordinary product rule. □

For example, with $b_n = H_n - \gamma$ the case $k = 2, m = 1$ gives

$$\psi(z)\psi'(z) = -\sum_{n\geq 0}\left\{\frac{1}{(z+n)^3} - \frac{b_n}{(z+n)^2}\right\}.$$

This agrees with the classical polygamma pole series at $k = 1$ and extends it to polynomial products without an undetermined analytic term. The full formulas apply at arbitrary complex z in their specified domains, not only at integer arguments.

The all-power primitive has zero local Laurent–log constant at zero. For a primitive anchored at one, use $\widehat{\mathcal{A}}_k(z) = \mathcal{A}_k(z) - Q_k$; then $\widehat{\mathcal{A}}_k(1) = 0$ and its derivative is unchanged. The numerical value of Q_5 is approximately -8.6903039626245382663, confirmed by separately replayed arithmetic and local-subtraction quadrature routes, and by independent native Wolfram quadrature. These diagnostics carry no interval error certificate. The analytic proofs establish the identities; finite algebra and numerical checks test their implementation.

Appendix: fully expanded values at the third and quarter points

The reduction recipes of Theorems 8.39 and 8.41, carried out to the end, give the following flattened closed forms. The Γ_2 values live in the atom set announced in Theorem 8.39; the Γ_3 values pick up, exactly as Theorem 8.41 predicts, the three further atoms γ_3, π^4 and $\gamma_4(\frac{1}{q}) - \gamma_4(1-\frac{1}{q})$. The combinations $\gamma + \ln 6\pi$ and $\gamma + \ln 8\pi$ are the constants Λ_3, Λ_4 of Theorem 8.33. Each display has been verified at 90-digit working precision (residuals below 10^{-88}) against $\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1}\left[\zeta^{(n+1)}(0,p) - \zeta^{(n+1)}(0)\right]$ evaluated from Hurwitz-zeta derivative jets.

$$
\begin{aligned}
\Gamma_2\left(\tfrac{1}{3}\right) = {} & \frac{1}{2}\,\gamma_2 + \gamma\,\gamma_1 + \gamma_1\,\ln 2\pi - \frac{1}{2}\,\gamma_1\,\ln 3 \\
& + \frac{4}{9}\,\gamma^3 + \frac{4}{3}\,\gamma^2\,\ln 2\pi - \frac{1}{6}\,\gamma^2\,\ln 3 + \frac{4}{3}\,\gamma\,\ln^2 2\pi \\
& + \frac{7}{6}\,\gamma\,\ln 2\pi\,\ln 3 - \frac{1}{6}\,\gamma\,\ln^2 3 - \frac{4}{9}\,\zeta(3) - \frac{\pi^2}{12}\,\ln 2\pi \\
& + \frac{\pi^2}{24}\,\ln 3 + \frac{4}{9}\,\ln^3 2\pi + \frac{4}{3}\,\ln^2 2\pi\,\ln 3 + \frac{7}{12}\,\ln 2\pi\,\ln^2 3 \\
& - \frac{1}{18}\,\ln^3 3 \\
& - \left(\left(\gamma + \ln 6\pi\right)^2 - \frac{\pi^2}{12}\right)\ln\Gamma\left(\tfrac{1}{3}\right) \\
& - \frac{\sqrt{3}}{6\pi}\left(\gamma + \ln 6\pi\right)\left(\gamma_2(\tfrac{1}{3}) - \gamma_2(\tfrac{2}{3})\right) \\
& - \frac{\sqrt{3}}{18\pi}\left(\gamma_3(\tfrac{1}{3}) - \gamma_3(\tfrac{2}{3})\right)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_2\left(\tfrac{1}{4}\right) = {} & \frac{1}{2}\,\gamma_2 + \gamma\,\gamma_1 + \frac{1}{2}\,\gamma_1\,\ln 2 + \gamma_1\,\ln\pi \\
& + \frac{1}{2}\,\gamma^3 + 2\,\gamma^2\,\ln 2 + \frac{3}{2}\,\gamma^2\,\ln\pi + 6\,\gamma\,\ln^2 2 \\
& + \frac{13}{2}\,\gamma\,\ln 2\,\ln\pi + \frac{3}{2}\,\gamma\,\ln^2\pi - \frac{1}{2}\,\zeta(3) - \frac{\pi^2}{24}\,\ln 2 \\
& - \frac{\pi^2}{12}\,\ln\pi + \frac{49}{6}\,\ln^3 2 + \frac{45}{4}\,\ln^2 2\,\ln\pi + \frac{9}{2}\,\ln 2\,\ln^2\pi \\
& + \frac{1}{2}\,\ln^3\pi \\
& - \left(\left(\gamma + \ln 8\pi\right)^2 - \frac{\pi^2}{12}\right)\ln\Gamma\left(\tfrac{1}{4}\right) \\
& - \frac{1}{4\pi}\left(\gamma + \ln 8\pi\right)\left(\gamma_2(\tfrac{1}{4}) - \gamma_2(\tfrac{3}{4})\right) \\
& - \frac{1}{12\pi}\left(\gamma_3(\tfrac{1}{4}) - \gamma_3(\tfrac{3}{4})\right)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_3\left(\tfrac{1}{3}\right) = &-\frac{1}{2}\,\gamma_3 - \frac{3}{2}\,\gamma\,\gamma_2 - \frac{3}{2}\,\gamma_2\,\ln 2\pi + \frac{3}{4}\,\gamma_2\,\ln 3 \\
&- \frac{3}{2}\,\gamma^2\,\gamma_1 - 3\,\gamma\,\gamma_1\,\ln 2\pi + \frac{3}{2}\,\gamma\,\gamma_1\,\ln 3 - \frac{3}{2}\,\gamma_1\,\ln^2 2\pi \\
&+ \frac{3}{2}\,\gamma_1\,\ln 2\pi\,\ln 3 + \frac{3}{4}\,\gamma_1\,\ln^2 3 - \frac{\pi^2}{8}\,\gamma_1 - \frac{1}{2}\,\gamma^4 \\
&- 2\,\gamma^3\,\ln 2\pi + \frac{1}{4}\,\gamma^3\,\ln 3 - 3\,\gamma^2\,\ln^2 2\pi - \frac{3}{2}\,\gamma^2\,\ln 2\pi\,\ln 3 \\
&+ \frac{3}{8}\,\gamma^2\,\ln^2 3 - \frac{\pi^2}{24}\,\gamma^2 - 2\,\gamma\,\ln^3 2\pi - \frac{15}{4}\,\gamma\,\ln^2 2\pi\,\ln 3 \\
&- \frac{3}{2}\,\gamma\,\ln 2\pi\,\ln^2 3 + \frac{1}{4}\,\gamma\,\ln^3 3 + \frac{\pi^2}{6}\,\gamma\,\ln 2\pi - \frac{\pi^2}{48}\,\gamma\,\ln 3 \\
&+ \frac{5\pi^2}{24}\,\ln^2 2\pi + \frac{\pi^2}{24}\,\ln 2\pi\,\ln 3 - \frac{7\pi^2}{96}\,\ln^2 3 + \frac{\pi^4}{120} \\
&- \frac{1}{2}\,\ln^4 2\pi - 2\,\ln^3 2\pi\,\ln 3 - \frac{15}{8}\,\ln^2 2\pi\,\ln^2 3 - \frac{1}{2}\,\ln 2\pi\,\ln^3 3 \\
&+ \frac{1}{16}\,\ln^4 3 \\
&+ \left(\left(\gamma + \ln 6\pi\right)^3 - \frac{\pi^2}{4}\left(\gamma + \ln 6\pi\right) + 2\,\zeta(3)\right)\ln\Gamma\left(\tfrac{1}{3}\right) \\
&+ \frac{\sqrt{3}}{48\pi}\left(12\left(\gamma + \ln 6\pi\right)^2 - \pi^2\right)\left(\gamma_2(\tfrac{1}{3}) - \gamma_2(\tfrac{2}{3})\right) \\
&+ \frac{\sqrt{3}}{6\pi}\left(\gamma + \ln 6\pi\right)\left(\gamma_3(\tfrac{1}{3}) - \gamma_3(\tfrac{2}{3})\right) \\
&+ \frac{\sqrt{3}}{24\pi}\left(\gamma_4(\tfrac{1}{3}) - \gamma_4(\tfrac{2}{3})\right)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma_3\left(\tfrac{1}{4}\right) = &-\frac{1}{2}\,\gamma_3 - \frac{3}{2}\,\gamma\,\gamma_2 - \frac{3}{4}\,\gamma_2\,\ln 2 - \frac{3}{2}\,\gamma_2\,\ln\pi \\
&- \frac{3}{2}\,\gamma^2\,\gamma_1 - \frac{3}{2}\,\gamma\,\gamma_1\,\ln 2 - 3\,\gamma\,\gamma_1\,\ln\pi + \frac{9}{4}\,\gamma_1\,\ln^2 2 \\
&- \frac{3}{2}\,\gamma_1\,\ln 2\,\ln\pi - \frac{3}{2}\,\gamma_1\,\ln^2\pi - \frac{\pi^2}{8}\,\gamma_1 - \frac{9}{16}\,\gamma^4 \\
&- 3\,\gamma^3\,\ln 2 - \frac{9}{4}\,\gamma^3\,\ln\pi - \frac{45}{4}\,\gamma^2\,\ln^2 2 - \frac{51}{4}\,\gamma^2\,\ln 2\,\ln\pi \\
&- \frac{27}{8}\,\gamma^2\,\ln^2\pi - \frac{\pi^2}{32}\,\gamma^2 - 27\,\gamma\,\ln^3 2 - \frac{153}{4}\,\gamma\,\ln^2 2\,\ln\pi \\
&- \frac{33}{2}\,\gamma\,\ln 2\,\ln^2\pi - \frac{9}{4}\,\gamma\,\ln^3\pi + \frac{\pi^2}{4}\,\gamma\,\ln 2 + \frac{3\pi^2}{16}\,\gamma\,\ln\pi \\
&+ \frac{3\pi^2}{8}\,\ln^2 2 + \frac{11\pi^2}{16}\,\ln 2\,\ln\pi + \frac{7\pi^2}{32}\,\ln^2\pi + \frac{29\pi^4}{3840} \\
&- \frac{207}{8}\,\ln^4 2 - \frac{179}{4}\,\ln^3 2\,\ln\pi - 27\,\ln^2 2\,\ln^2\pi - \frac{27}{4}\,\ln 2\,\ln^3\pi \\
&- \frac{9}{16}\,\ln^4\pi \\
&+ \left(\left(\gamma + \ln 8\pi\right)^3 - \frac{\pi^2}{4}\left(\gamma + \ln 8\pi\right) + 2\,\zeta(3)\right)\ln\Gamma\left(\tfrac{1}{4}\right) \\
&+ \frac{1}{32\pi}\left(12\left(\gamma + \ln 8\pi\right)^2 - \pi^2\right)\left(\gamma_2(\tfrac{1}{4}) - \gamma_2(\tfrac{3}{4})\right) \\
&+ \frac{1}{4\pi}\left(\gamma + \ln 8\pi\right)\left(\gamma_3(\tfrac{1}{4}) - \gamma_3(\tfrac{3}{4})\right) \\
&+ \frac{1}{16\pi}\left(\gamma_4(\tfrac{1}{4}) - \gamma_4(\tfrac{3}{4})\right)
\end{aligned}
$$

Chapter 9

The parameter-derivative tower

Differentiating in the Hurwitz parameter moves the spectral argument upward. The resulting coefficients are elementary symmetric functions of reciprocal integers, so harmonic numbers control every rung. The functional equation then pairs this tower with the integrated tower of the preceding chapter. Explicit character tables show how experiments can be replaced by uniform derivations.

9.1 The uniform master identity

Theorem 9.1 (Harmonic-number master identity)**.** *For all $k \geq 1$ and $n \geq 0$,*

$$\gamma_n^{(k)}(a) = (-1)^{n+k}\, k!\, n! \sum_{i+j=n} e_i(k)\, \frac{\zeta^{(j)}(k{+}1, a)}{j!}, \tag{9.1}$$

where $e_i(k)$ is the i-th elementary symmetric function of $\{1, \frac{1}{2}, \ldots, \frac{1}{k}\}$; in particular $e_0 = 1$, $e_1 = H_k$, $e_2 = \frac{1}{2}(H_k^2 - H_k^{(2)})$. The first two rows are

$$\gamma_1^{(k)}(a) = (-1)^{k+1} k! \left[\zeta'(k{+}1, a) + H_k\, \zeta(k{+}1, a)\right], \tag{9.2}$$

$$\gamma_2^{(k)}(a) = (-1)^{k} k! \left[\zeta''(k{+}1, a) + 2H_k\, \zeta'(k{+}1, a) + (H_k^2 - H_k^{(2)})\, \zeta(k{+}1, a)\right]. \tag{9.3}$$

Proof. From $\partial_a \zeta(s, a) = -s\, \zeta(s{+}1, a)$, iteration gives $\partial_a^k \zeta(s, a) = (-1)^k (s)_k\, \zeta(s{+}k, a)$ with the rising factorial $(s)_k = s(s{+}1)\cdots(s{+}k{-}1)$. For $k \geq 1$ this annihilates the pole term in (8.40), so

$$\sum_{n \geq 0} \frac{(-1)^n}{n!}\, \gamma_n^{(k)}(a)\, (s-1)^n = (-1)^k (s)_k\, \zeta(s{+}k, a).$$

Set $s = 1 + \varepsilon$. Then $(s)_k = (1+\varepsilon)_k = \prod_{m=1}^{k}(m + \varepsilon) = k! \prod_{m=1}^{k}(1 + \varepsilon/m) = k! \sum_{i \geq 0} e_i(k)\, \varepsilon^i$, the generating identity for the elementary symmetric functions of $\{1/m\}$, while $\zeta(s{+}k, a) = \zeta(k{+}1{+}\varepsilon, a) = \sum_{j \geq 0} \zeta^{(j)}(k{+}1, a)\, \varepsilon^j / j!$ is analytic at $\varepsilon = 0$ because $k{+}1 \geq 2$. Comparing coefficients of ε^n yields (9.1); $n = 1, 2$ give (9.2)–(9.3). □

Theorem 9.1 is the harmonic-number repackaging of Coffey's all-orders Stirling-number ladder [25]: the change of basis from the unsigned Stirling numbers $\left|s(k{+}1, i{+}1)\right|$ to the symmetric functions $e_i(k)$ turns the coefficients into H_k, $H_k^2 - H_k^{(2)}$, ... The specializations $\gamma_1' = \zeta'(2, a) + \zeta(2, a)$, $\gamma_1'' = -2\zeta'(3, a) - 3\zeta(3, a)$, $\gamma_2' = -[\zeta''(2, a) + 2\zeta'(2, a)]$ and $\gamma_2'' = 2\zeta''(3, a) + 6\zeta'(3, a) + 2\zeta(3, a)$ are the $s = 2, 3$ layers used below. Identity (9.1) was verified against numerical k-th parameter-derivatives of γ_1, γ_2 for $k = 1, \ldots, 5$.

9.2 Differentiation and the contact-term correction

For $k \geq 1$, the ordinary function $\gamma_n^{(k)}(x)$ has a power-log singularity of order $k+1$. Its canonical Hadamard finite part on $\mathbb{T}$ is defined by

$$\langle F_{n,k}, \phi \rangle = \mathrm{CT}_{\epsilon \downarrow 0} \int_\epsilon^1 \gamma_n^{(k)}(x)\phi(x)\,\mathrm{d}x. \tag{9.4}$$

Here CT means the coefficient independent of ϵ after the finite expansion in negative powers of ϵ and nonconstant powers of $\log \epsilon$ is removed. The remaining error tends to zero. This convention uses the coordinate $x \in (0,1)$ and no finite counterterm. It agrees with Definition 8.59 when $k = 0$.

Lemma 9.2 (Higher Hurwitz poles as distributions)**.** *For each $k \geq 1$, as $z \to 0$,*

$$Z(1+k-z) = \frac{(-1)^k}{k!}\frac{\delta_0^{(k)}}{z} + \sum_{j \geq 0} \frac{(-1)^j z^j}{j!}\,\mathrm{Fp}_{\mathbb{T}}[\zeta^{[j]}(k+1,x)]. \tag{9.5}$$

The finite parts in this formula have the cutoff convention (9.4).

Proof. Split $\zeta(1+k-z,x) = x^{-k-1+z} + \zeta(1+k-z,1+x)$. Subtract the Taylor polynomial of ϕ at zero through degree k. Its remainder gives an analytic integral near $z=0$. The subtracted monomials give terms $\phi^{(r)}(0)/(r!(z+r-k))$, $0 \leq r \leq k$; only $r=k$ has a pole at zero. Since $\langle \delta_0^{(k)}, \phi \rangle = (-1)^k \phi^{(k)}(0)$, its residue is the one stated. Expanding each regular term and each analytic remainder is exactly the constant-term prescription for the cutoff integrals. This proves the entire Laurent expansion. □

Theorem 9.3 (All-index differentiation anomaly)**.** *For $n \geq 0$ and $k \geq 1$,*

$$D^k G_n = F_{n,k} + (-1)^{n+1} n! e_{n+1}(k)\delta_0^{(k)}. \tag{9.6}$$

In particular, there is no anomaly when $n \geq k$.

Proof. The meromorphic distribution family satisfies

$$D^k Z(s) = (-1)^k (s)_k Z(s+k). \tag{9.7}$$

This follows directly from its Fourier coefficients and the Gamma recurrence; it also follows by repeated integration by parts in a left half-plane and meromorphic continuation. Put $s = 1-z$ and expand

$$(1-z)_k = k! \prod_{r=1}^{k}(1 - z/r) = k! \sum_{i=0}^{k} (-1)^i e_i(k) z^i.$$

On the left of (9.7), the pole is $\delta_0^{(k)}/z$ and the regular coefficient of z^n is $D^k G_n / n!$. On the right, multiplying the regular part of Lemma 9.2 by the polynomial gives $F_{n,k}/n!$: this is the finite part of the ordinary pointwise parameter-derivative identity. Multiplying its polar part by that same polynomial gives, in addition, the regular coefficient $(-1)^{n+1} e_{n+1}(k)\delta_0^{(k)}$. Comparison proves the formula. □

The first examples are

$$\begin{aligned} DG_0 &= F_{0,1} - \delta_0', \\ D^2G_0 &= F_{0,2} - \tfrac{3}{2}\delta_0'', \\ D^2G_1 &= F_{1,2} + \tfrac{1}{2}\delta_0''. \end{aligned} \tag{9.8}$$

The point-supported terms disappear upon restriction to $(0,1)$, so ordinary differentiation there cannot detect them. But convolving $\delta_0^{(k)}$ with another distribution differentiates that other factor everywhere. Therefore these terms cannot be discarded before convolution, even when the desired final identity is off the singular point.

The pointwise harmonic master identity (9.1) remains valid. The new contact term governs its canonical periodic extension: point-supported derivatives vanish on the open interval but contribute after convolution.

9.2.1 Convolutions of arbitrary differentiated Stieltjes functions

The same argument gives a closed law for all derivative indices, not only the zeroth Stieltjes function. Put $F_{n,0} = G_n$ and define

$$a_{n,k} = (-1)^{n+1} n! e_{n+1}(k) \quad (k \geq 1), \qquad a_{n,0} = 0.$$

Thus $F_{n,k} = D^k G_n - a_{n,k}\delta_0^{(k)}$ when $k \geq 1$.

Corollary 9.4 (The full differentiated convolution calculus)**.** *Suppose $n, m, k, l \geq 0$, $r = k + l \geq 1$, and the exact rule from Theorem 8.63 is*

$$G_n * G_m = c_{-1}\mathsf{I} + \sum_{j=0}^{n+m+1} c_j G_j.$$

Then

$$\begin{aligned} F_{n,k} * F_{m,l} = {} & \sum_{j=0}^{n+m+1} c_j D^r G_j - a_{n,k} D^r G_m - a_{m,l} D^r G_n \\ & + (c_{-1} + a_{n,k} a_{m,l})\delta_0^{(r)}. \end{aligned} \tag{9.9}$$

Away from the integer point this reduces to

$$\sum_{j=0}^{n+m+1} c_j \gamma_j^{(r)}(x) - a_{n,k}\gamma_m^{(r)}(x) - a_{m,l}\gamma_n^{(r)}(x).$$

In particular, every such convolution has a finite reduction to $\zeta^{[j]}(r+1, x)$ with $0 \leq j \leq n + m + 1$.

Proof. Multiply the two identities $F = DG - a\delta$ at their indicated orders, interpreting the zero-order case through $a_{n,0} = 0$. The product of the first terms is $D^r(G_n * G_m)$, and $D^r\mathsf{I} = \delta_0^{(r)}$. Restricting off zero and applying (9.1) proves the last assertion. □

As an explicit higher-index example, the contact term vanishes for $F_{1,1} = DG_1$. Hence

$$\begin{aligned} (F_{1,1} * F_{1,1})(x) = {} & -2\zeta'''(3,x) - 9\zeta''(3,x) \\ & + [4\zeta(2) - 6]\zeta'(3,x) + [6\zeta(2) + 4\zeta(3)]\zeta(3,x). \end{aligned} \tag{9.10}$$

This follows by twice differentiating (8.81) and using (9.1). It illustrates that the full tower still closes at one spectral integer even when higher Stieltjes indices occur.

9.3 A two-jet collapse for every polygamma convolution

Let $H_0 = 0$ and $H_k = \sum_{j=1}^{k} 1/j$ for $k \geq 1$. Define

$$\Psi_k = \mathrm{Fp}_{\mathbb{T}}\, \psi^{(k)} \qquad (k \geq 0)$$

using the same canonical cutoff convention. Since $\gamma_0 = -\psi$, Theorem 9.3 gives

$$\Psi_0 = -G_0, \qquad \Psi_k = -D^k G_0 - H_k \delta_0^{(k)} \quad (k \geq 1). \tag{9.11}$$

Theorem 9.5 (Polygamma convolution reduction)**.** *Let $k, l \geq 0$ and $r = k + l \geq 1$. Then*

$$\begin{aligned} \Psi_k * \Psi_l = {} & 2D^r G_1 + (H_k + H_l) D^r G_0 \\ & + [H_k H_l - \zeta(2)] \delta_0^{(r)}. \end{aligned} \tag{9.12}$$

Consequently, for $0 < x < 1$,

$$(\Psi_k * \Psi_l)(x) = (-1)^{r+1} r! \left[2\zeta'(r+1, x) + (2H_r - H_k - H_l)\zeta(r+1, x) \right]. \tag{9.13}$$

For $k = l = 0$, the appropriate separate formula is $\Psi_0 * \Psi_0 = 2G_1 - \zeta(2)\mathsf{I}$.

Proof. Convolving (9.11) and using that derivatives may be placed on either factor gives

$$\Psi_k * \Psi_l = D^r (G_0 * G_0) + (H_k + H_l) D^r G_0 + H_k H_l \delta_0^{(r)}.$$

The same expression is valid when one index is zero, since its harmonic number vanishes. Apply (8.78) and $D^r \mathsf{I} = \delta_0^{(r)}$. Restriction to $0 < x < 1$ gives $2\gamma_1^{(r)} + (H_k + H_l)\gamma_0^{(r)}$. Now use (9.1) with $n = 0, 1$:

$$\gamma_1^{(r)} = (-1)^{r+1} r! [\zeta'(r+1, x) + H_r \zeta(r+1, x)], \quad \gamma_0^{(r)} = (-1)^r r! \zeta(r+1, x).$$

This proves (9.13). □

For example,

$$\begin{aligned} (\Psi_0 * \Psi_1)(x) &= 2\zeta'(2, x) + \zeta(2, x), \\ (\Psi_1 * \Psi_1)(x) &= -4\zeta'(3, x) - 2\zeta(3, x), \\ (\Psi_0 * \Psi_2)(x) &= -4\zeta'(3, x) - 3\zeta(3, x). \end{aligned} \tag{9.14}$$

Although the last two convolutions have the same total derivative order, they have different coefficients. Confusing Ψ_k with the kth derivative of Ψ_0 would incorrectly erase this distinction.

9.3.1 A convergent trigamma identity with every subtraction visible

For $0 < x < 1$, set $h = x/2$ and $f(t) = \psi'(t)$. Then

$$\begin{aligned} \mathcal{T}(x) = {} & 2 \int_0^h \left[\frac{f(x-t) - f(x) + t f'(x)}{t^2} + \psi'(1+t) f(x-t) \right] \mathrm{d}t \\ & + \int_x^1 f(t) f(1+x-t)\, \mathrm{d}t - \frac{2f(x)}{h} - 2f'(x) \log h \end{aligned} \tag{9.15}$$

satisfies

$$\mathcal{T}(x) = -4\zeta'(3, x) - 2\zeta(3, x). \tag{9.16}$$

Indeed, the shift relation $\psi'(t) = t^{-2} + \psi'(1+t)$ and symmetry about $t = x/2$ reduce the first singular convolution integral to twice its left half. Subtracting the constant and linear Taylor terms of $f(x-t)$ leaves a bounded quotient. The constant terms of $\int_\epsilon^h t^{-2}\, \mathrm{d}t$ and $\int_\epsilon^h t^{-1}\, \mathrm{d}t$ are $-1/h$ and $\log h$, respectively. This proves that (9.15) is exactly the canonical finite-part convolution, so Theorem 9.5 applies. Every integral in (9.15) is ordinary and absolutely convergent.

9.4 Contact terms in translated derivative products

For $r \geq 1$ let $h_{r,m} = (-1)^m m! e_{m+1}(r)$, and put $h_{0,m} = 0$. Thus (9.6) reads $F_{m,r} = D^r G_m + h_{r,m}\delta_0^{(r)}$.

Theorem 9.6 (Translated contact law)**.** *For $r, k, m, n \geq 0$, $N = r+k$, and $b = 1-a$ with $0 < a < 1$,*

$$\begin{aligned} &\mathrm{FP}\int_0^1 \gamma_m^{(r)}(x)\gamma_n^{(k)}(\{x+a\})\,dx \\ &\quad = (-1)^r J_{mn}^{(N)}(a) + (-1)^k h_{k,n}\gamma_m^{(N)}(b) + (-1)^r h_{r,m}\gamma_n^{(N)}(a). \end{aligned} \tag{9.17}$$

Proof. The local finite parts use the constant-term convention (9.4). At nonzero shift the singular supports of the two factors are disjoint, so a partition into disjoint endpoint neighborhoods defines the product unambiguously. Substitute the two contact identities. Periodic integration by parts gives the first term. The shifted delta derivative acts at $x = b$, giving $(-1)^k h_{k,n}\gamma_m^{(N)}(b)$, and the unshifted one acts at zero, giving $(-1)^r h_{r,m}\gamma_n^{(N)}(a)$. The product of the two point-supported terms vanishes because their supports are distinct. This proves the formula with the exact scale-one endpoint normalization. □

9.4.1 Every polygamma product in first Hurwitz derivatives

For $r, k \geq 0$ define

$$\Pi_{r,k}(a) = \mathrm{FP}\int_0^1 \psi^{(r)}(x)\psi^{(k)}(\{x+a\})\,\mathrm{d}x.$$

Corollary 9.7 (Polygamma closure)**.** *For $N = r + k \geq 1$,*

$$\begin{aligned} \Pi_{r,k}(a) = {} & (-1)^r\big[\gamma_1^{(N)}(a) - H_r\psi^{(N)}(a)\big] \\ & + (-1)^k\big[\gamma_1^{(N)}(b) - H_k\psi^{(N)}(b)\big]. \end{aligned} \tag{9.18}$$

Equivalently,

$$\begin{aligned} \Pi_{r,k}(a) = N!\Big\{ & (-1)^{k+1}\big[\zeta^{[1]}(N+1,a) + (H_N - H_r)\zeta(N+1,a)\big] \\ & + (-1)^{r+1}\big[\zeta^{[1]}(N+1,b) + (H_N - H_k)\zeta(N+1,b)\big]\Big\}. \end{aligned} \tag{9.19}$$

The case $N = 0$ is (8.113).

Proof. Take $m = n = 0$ in (9.17), use $\gamma_0^{(r)} = -\psi^{(r)}$, $h_{r,0} = H_r$, and differentiate (8.109). This proves (9.18). The pointwise identities

$$\begin{aligned} \psi^{(N)}(x) &= (-1)^{N+1}N!\zeta(N+1,x), \\ \gamma_1^{(N)}(x) &= (-1)^{N+1}N!\big[\zeta^{[1]}(N+1,x) + H_N\zeta(N+1,x)\big] \end{aligned}$$

then give (9.19). The second follows directly by expanding $\partial_x^N\zeta(1+u,x) = (-1)^N(1+u)_N\zeta(N+1+u,x)$ at $u = 0$; these pointwise ladders are also part of the repository's existing development [109]. □

Two instructive special cases are

$$\Pi_{0,1}(a) = \zeta^{[1]}(2,a) + \zeta(2,a) - \zeta^{[1]}(2,b) = J'_{00}(a) + \psi'(b), \tag{9.20}$$

$$\Pi_{1,1}(a) = 2\zeta^{[1]}(3,a) + 2\zeta^{[1]}(3,b) + \zeta(3,a) + \zeta(3,b). \tag{9.21}$$

The term $+\psi'(b)$ in (9.20) is an explicit counterexample to the unqualified rule "differentiate the finite-part identity under the integral." It is not a counterexample to ordinary differentiation under a genuinely dominated integral.

For reproducibility, let $I_{r,k}(\epsilon)$ be the sum of the two raw cutoff integrals, with arguments split as in (8.91). Then

$$\Pi_{0,1}(a) = \lim_{\epsilon\downarrow 0}\left[I_{0,1}(\epsilon) - \frac{\psi(b)}{\epsilon} - \big(\psi'(a) - \psi'(b)\big)\log\epsilon\right], \tag{9.22}$$

$$\Pi_{1,1}(a) = \lim_{\epsilon\downarrow 0}\left[I_{1,1}(\epsilon) - \frac{\psi'(a) + \psi'(b)}{\epsilon} + \big(\psi''(a) + \psi''(b)\big)\log\epsilon\right]. \tag{9.23}$$

These counterterms can be derived directly from $\psi(x) = -1/x - \gamma + O(x)$ and $\psi'(x) = x^{-2} + O(1)$. They give numerical tests of the contact law that do not implement distributional differentiation.

The full default replays are preserved under `verification/eighth-replay/`. The direct cutoff polygamma calculations check the contact correction without implementing distributional differentiation. The all-index law rests on the Laurent residues and the proof above. The ordinary pointwise master identity has not been retracted or altered.

9.5 Dilation, distribution traces and exact logarithmic contacts

The contact theorem answers an extension question suggested by the correlation reports: classical distribution rows need a normalization correction when interpreted as finite-part identities. Fourier traces make the correction computable at every Stieltjes and derivative order. This also describes how dilation acts on the one-generator convolution algebra, without any arithmetic-independence assumption.

For a positive integer q, define the trace $\mathcal{S}_q$ and covering pullback $\mathcal{P}_q$ on periodic distributions by

$$\widehat{\mathcal{S}_q T}(k) = q\widehat{T}(qk), \qquad \widehat{\mathcal{P}_q T}(k) = \begin{cases} \widehat{T}(k/q), & q \mid k, \\ 0, & q \nmid k. \end{cases}$$

For ordinary periodic functions these are $\mathcal{S}_q f(x) = \sum_{j=0}^{q-1} f((x+j)/q)$ and $\mathcal{P}_q f(x) = f(\{qx\})$. Put $\ell_q = \log q$ and $\mathcal{L}_q = q^{-1}\mathcal{S}_q$.

Theorem 9.8 (The trace translates the convolution generator)**.** *For every $m \geq 0$,*

$$\mathcal{S}_q G_m = q\sum_{j=0}^{m}\binom{m}{j}(-\ell_q)^{m-j}G_j + \frac{q(-\ell_q)^{m+1}}{m+1}\mathsf{I}. \tag{9.24}$$

The normalized trace $\mathcal{L}_q$ is a unital homomorphism of the mean-zero convolution algebra. On its generator $X = G_0$ it acts by $X \mapsto X - \ell_q\mathsf{I}$. These actions compose as $\mathcal{L}_q\mathcal{L}_r = \mathcal{L}_{qr}$.

Proof. The Fourier definition gives $\mathcal{S}_q R_\alpha = q^{1-\alpha}R_\alpha$. Apply this to the Laurent family (8.70):

$$\mathcal{S}_q\left(\frac{\mathsf{I}}{z} + \sum_{m\geq 0} G_m\frac{z^m}{m!}\right) = qe^{-\ell_q z}\left(\frac{\mathsf{I}}{z} + \sum_{m\geq 0} G_m\frac{z^m}{m!}\right).$$

The regular coefficients give (9.24), including the term produced by the pole. On Fourier coefficients, $\mathcal{L}_q(T * S) = (\mathcal{L}_q T) * (\mathcal{L}_q S)$ and $\mathcal{L}_q \mathsf{I} = \mathsf{I}$. The generator formula is the $m = 0$ case, and repeated traces evaluate a Fourier coefficient at qrk. □

Theorem 9.9 (Finite part does not commute with covering pullback)**.** *Let* $\operatorname{Fp}_x \gamma_m(\{qx\})$ *use the constant term in the unscaled local coordinate* $x - j/q$ *at each of the* q *singularities, with right-hand cutoffs tending independently to zero. Then*

$$\mathcal{P}_q G_m = \operatorname{Fp}_x \gamma_m(\{qx\}) + \frac{\ell_q^{m+1}}{q(m+1)} \sum_{j=0}^{q-1} \delta_{j/q}. \tag{9.25}$$

In particular the raw-coordinate finite part has mean $-\ell_q^{m+1}/(m+1)$*, while the covering pullback has mean zero.*

Proof. On the jth segment substitute $y = q(x - j/q)$. Pulling back the cutoff prescription (8.69) gives its local counterterm

$$\frac{\log^{m+1}(q\epsilon)}{q(m+1)} \phi(j/q).$$

The raw-coordinate constant-term prescription keeps only the divergent powers of $\log \epsilon$ in this expression. Their difference is its constant coefficient, $\ell_q^{m+1}\phi(j/q)/(q(m+1))$. All Taylor remainder integrals converge as in the definition of G_m. Summing the segments proves (9.25). The Fourier pullback preserves the zeroth coefficient, and pairing the contact sum with the constant function proves the mean statement. □

For the trace itself, put $c_{m,q} = -q(-\ell_q)^{m+1}/(m+1)$. The classical pointwise multiplication row and its raw-coordinate extension read

$$\operatorname{Fp}_x[\mathcal{S}_q \gamma_m] = q \sum_{j=0}^{m} \binom{m}{j} (-\ell_q)^{m-j} G_j + c_{m,q}\, 1.$$

Comparing with (9.24) gives the separate identity

$$\mathcal{S}_q G_m = \operatorname{Fp}_x[\mathcal{S}_q \gamma_m] - c_{m,q}\delta_0. \tag{9.26}$$

The logarithmic term thus has different signs and support for covering pullback and trace; substituting one convention for the other is invalid.

Theorem 9.10 (Every differentiated trace and its contact)**.** *Use* $F_{m,r}$ *and* $a_{m,r} = (-1)^{m+1} m! e_{m+1}(r)$ *from the contact theorem, and let* $r \geq 1$*. Then*

$$\begin{aligned} \mathcal{S}_q F_{m,r} &= q^{r+1} \sum_{j=0}^{m} \binom{m}{j} (-\ell_q)^{m-j} F_{j,r} + q^{r+1} \kappa_{m,r}(\ell_q) \delta_0^{(r)}, \\ \kappa_{m,r}(\ell) &= \frac{(-\ell)^{m+1}}{m+1} + \sum_{j=0}^{m} \binom{m}{j} (-\ell)^{m-j} a_{j,r} - a_{m,r}. \end{aligned} \tag{9.27}$$

The first sum is exactly $\operatorname{Fp}_x[\mathcal{S}_q \gamma_m^{(r)}]$*. For the canonical polygamma finite parts, the particularly simple all-order specialization is*

$$\mathcal{S}_q \Psi_r = q^{r+1}(\Psi_r + \ell_q \delta_0^{(r)}), \qquad r \geq 1. \tag{9.28}$$

Proof. Fourier coefficients give $\mathcal{S}_q D^r = q^r D^r \mathcal{S}_q$ and $\mathcal{S}_q \delta_0^{(r)} = q^{r+1}\delta_0^{(r)}$. Apply these identities to $F_{m,r} = D^r G_m - a_{m,r}\delta_0^{(r)}$, use (9.24), and then replace each $D^r G_j$ by $F_{j,r} + a_{j,r}\delta_0^{(r)}$. Because $D^r \mathsf{I} = \delta_0^{(r)}$, the contact coefficient is precisely the displayed $\kappa_{m,r}$. Differentiating the classical pointwise multiplication row r times proves the raw-coordinate interpretation of the first sum. Finally $\Psi_r = -F_{0,r}$ and $\kappa_{0,r} = -\ell_q$, proving (9.28). □

These identities answer the normalization question for integer covering maps and their distribution traces at every index. They also show that the polynomial convolution algebra is preserved by the normalized trace. Twisted character projections and products at several independent shifted singularities still require separate calculations. Exact Fourier-polynomial and local logarithmic-counterterm controls, with a corruption check, are recorded in `verification/certify_stieltjes_dilation.py`.

9.6 The uniform rational-argument landscape

Three structural facts hold uniformly in k; all follow from $\sum_p \chi(p)\,\zeta(s,p/q) = q^s L(s,\chi)$ (and its multiplicative cousin) differentiated in s, fed through Theorem 9.1.

Proposition 9.11 (Character coordinate)**.** *For a primitive nonprincipal Dirichlet character χ modulo $q > 1$ and all $k \geq 1$,*

$$\sum_{p=1}^{q-1} \chi(p)\,\gamma_1^{(k)}(p/q) = (-1)^{k+1}k!\,q^{k+1}\big[L'(k{+}1,\chi) + (H_k + \ln q)\,L(k{+}1,\chi)\big]. \tag{9.29}$$

Proposition 9.12 (Distribution row)**.** *For $d \geq 2$ and all $k \geq 1$, with $\psi^{(k)}(a) = (-1)^{k+1}k!\,\zeta(k{+}1,a)$,*

$$\sum_{j=0}^{d-1} \gamma_1^{(k)}\Big(\tfrac{a+j}{d}\Big) = d^{\,k+1}\gamma_1^{(k)}(a) + d^{\,k+1}\ln d\;\psi^{(k)}(a). \tag{9.30}$$

Proof of (9.29) *and* (9.30). Apply the character sum to (9.2) for (9.29), and the divisor sum for (9.30). For the character identity, use $\sum_p \chi(p)\zeta(s,p/q) = q^s L(s,\chi)$ and its s-derivative $q^s \ln q\, L(s,\chi) + q^s L'(s,\chi)$ at $s = k{+}1$; collect. For (9.30), use $\sum_{j<d}\zeta(s,(a{+}j)/d) = d^s\zeta(s,a)$ and its s-derivative, and recognize $(-1)^{k+1}k!\,\zeta(k{+}1,a) = \psi^{(k)}(a)$. □

Corollary 9.13 (Uniform formal distribution ranks)**.** *For every $q \geq 2$ and every parameter derivative order $k \geq 1$, the specified distribution presentation, with the endpoint and lower-index data treated as background, has residual dimension $\varphi(q) - 1$. This is a formal quotient statement, not arithmetic independence of the evaluated constants.*

The original finite rank experiments suggested this law. The following polynomial character normal form proves it without extrapolation, also at arbitrary weights and in finite jet algebras.

9.7 Weighted distribution modules and exact rational-grid ranks

The distribution relations in the Stieltjes drafts have a simple representation-theoretic structure. The relevant dimension is the dimension of a formal quotient by specified relations. The classical theory of universal distributions provides the broader setting [52, 53]. Here we give an explicit character normal form, including its behavior at exceptional weights and in finite Taylor jets. This proves the two all-denominator rank statements suggested by the source computations, without making an independence assertion about evaluated special values.

9.7.1 The relation module and its complete character decomposition

Fix $q \geq 2$, and put

$$G_q = \frac{1}{q}\mathbb{Z}/\mathbb{Z}, \qquad U(q) = (\mathbb{Z}/q\mathbb{Z})^\times.$$

Let $\mathcal{A}$ be a commutative unital $\mathbb{C}$-algebra and choose arbitrary $t_p \in \mathcal{A}$, one for each prime $p \mid q$. The choices may vanish or be nilpotent. Let

$$\mathcal{V}_q = \bigoplus_{x \in G_q} \mathcal{A}[x].$$

For $p \mid q$ and $x \in pG_q$, define the prime distribution row

$$D_p(x) = \sum_{\substack{y \in G_q \\ py = x}} [y] - t_p[x]. \tag{9.31}$$

Write $\mathcal{R}_q$ for the $\mathcal{A}$-span of these rows and $\mathcal{U}_q = \mathcal{V}_q/\mathcal{R}_q$. The unit group acts by $u[x] = [ux]$, and each relation map is equivariant.

For $d \mid q$, set $t_d = \prod_{p \mid d} t_p^{v_p(d)}$. Rows for composite divisors introduce no additional relations. Indeed, if $ab \mid q$ and $x \in abG_q$, then

$$D_{ab}(x) = \sum_{\substack{z \in G_q \\ az = x}} D_b(z) + t_b D_a(x). \tag{9.32}$$

Every z in this sum belongs to bG_q, so the rows on the right are legitimate grid rows. Expanding the two sums proves the identity. Induction on the number of prime factors of d, counted with multiplicity, proves the assertion for every divisor.

Let χ run through all characters of $U(q)$, and let $f = f_\chi$ be the conductor of its primitive inducing character χ^*. For $f \mid d \mid q$, let χ_d denote the induced character modulo d, and define

$$Y_{d,\chi} = \sum_{a \in U(d)} \chi_d(a)[a/d]. \tag{9.33}$$

For the principal character at $d = 1$, this means $Y_{1,\mathbf{1}} = [0]$. These are formal vectors; no numerical L-value is being declared independent.

Lemma 9.14 (All character components of the rows)**.** *The χ^{-1}-eigenspace of $\mathcal{V}_q$ is free over $\mathcal{A}$, with basis*

$$\{Y_{d,\chi} : f_\chi \mid d \mid q\}.$$

Its relation submodule is generated exactly by the edges

$$\begin{aligned} Y_{pd,\chi} - t_p Y_{d,\chi}, &\qquad p \mid d, \\ Y_{pd,\chi} - (t_p - \chi^*(p)) Y_{d,\chi}, &\qquad p \nmid d, \end{aligned} \qquad f_\chi \mid d, \quad pd \mid q. \tag{9.34}$$

In particular, there are no additional character relations arising from denominators smaller than, or not divisible by, f_χ.

Proof. The elements of G_q of exact denominator d form a single $U(q)$-orbit, naturally identified with $U(d)$. Reduction $U(q) \to U(d)$ is surjective. The character χ factors through this map exactly when $f_\chi \mid d$; otherwise its projection on that orbit is zero. More explicitly, the idempotent

$$E_\chi = \frac{1}{\varphi(q)} \sum_{u \in U(q)} \chi(u) u$$

satisfies, for $a \in U(d)$,

$$E_\chi[a/d] = \begin{cases} \dfrac{\chi_d(a)^{-1}}{\varphi(d)} Y_{d,\chi}, & f_\chi \mid d, \\ 0, & f_\chi \nmid d. \end{cases} \tag{9.35}$$

This follows by grouping u according to its residue modulo d and using character orthogonality on the kernel of reduction. The idempotents are mutually orthogonal and sum to the identity. All their scalar denominators are units in $\mathcal{A}$, so the argument applies equally to a nonreduced algebra.

For a fixed $p \mid q$, consider the equivariant map

$$D_p : \mathcal{A}[pG_q] \longrightarrow \mathcal{V}_q, \qquad [x] \longmapsto D_p(x).$$

The exact-denominator orbits in its domain have $pd \mid q$. By (9.35), their only nonzero χ-components have $f_\chi \mid d$, and each such component has the single generator $Y_{d,\chi}$. Consequently, computing $D_p(Y_{d,\chi})$ computes *every* projected relation for that p; an orbit on which the projector vanishes cannot create a hidden relation in the target.

If $p \mid d$, all p preimages of each a/d, $a \in U(d)$, have exact denominator pd. Summing with weights $\chi_d(a)$ therefore gives $Y_{pd,\chi} - t_p Y_{d,\chi}$. If $p \nmid d$, one preimage has denominator d, while the other $p-1$ have denominator pd. In the former preimage, $pc \equiv a \pmod{d}$, so its character coefficient is $\chi_d(a) = \chi^*(p)\chi_d(c)$. Thus the sum is

$$Y_{pd,\chi} + \chi^*(p) Y_{d,\chi} - t_p Y_{d,\chi}.$$

Here $p \nmid f_\chi$, as $f_\chi \mid d$, so $\chi^*(p)$ is a nonzero root of unity. This proves both the edge formulas and their completeness. □

Theorem 9.15 (Polynomial normal form)**.** *For $f = f_\chi \mid d \mid q$, define*

$$A_{d,\chi}(\mathbf{t}) = \prod_{p \mid f} t_p^{v_p(d) - v_p(f)} \prod_{\substack{p \mid d \\ p \nmid f}} t_p^{v_p(d)-1}(t_p - \chi^*(p)). \tag{9.36}$$

Empty products are 1, so $A_{f,\chi} = 1$. Then the full χ-relation submodule is

$$\mathcal{R}_{q,\chi} = \bigoplus_{\substack{f \mid d \mid q \\ d \neq f}} \mathcal{A}\big(Y_{d,\chi} - A_{d,\chi} Y_{f,\chi}\big). \tag{9.37}$$

In particular,

$$\mathcal{U}_q \cong \bigoplus_{\chi \in \widehat{U(q)}} \mathcal{A}\, Y_{f_\chi,\chi} \cong \mathcal{A}^{\varphi(q)}. \tag{9.38}$$

The character normal form of every vector is obtained by replacing $Y_{d,\chi}$ with $A_{d,\chi} Y_{f_\chi,\chi}$.

Proof. The polynomials $A_{d,\chi}$ satisfy precisely the edge recursions (9.34): a prime already dividing d contributes t_p, and a prime introduced for the first time outside f contributes $t_p - \chi^*(p)$. The factors for distinct primes commute.

Follow any chain of prime extensions from f to d. Successively eliminating the intermediate Y's expresses $Y_{d,\chi} - A_{d,\chi}Y_{f,\chi}$ as a combination of the edge relations. Conversely, write an edge as $Y_{pd,\chi} - cY_{d,\chi}$, where $A_{pd,\chi} = cA_{d,\chi}$. It equals

$$\big(Y_{pd,\chi} - A_{pd,\chi}Y_{f,\chi}\big) - c\big(Y_{d,\chi} - A_{d,\chi}Y_{f,\chi}\big).$$

Lemma 9.14 therefore gives equality of the two relation spans.

The displayed generators in (9.37) are independent: the coefficient of $Y_{d,\chi}$, for $d \neq f$, in a linear combination is exactly the coefficient of its own generator. Together with $Y_{f,\chi}$, they form a triangular basis of the whole character block. Its quotient is consequently free of rank one. Summing over all $\varphi(q)$ characters proves (9.38). □

Remark 9.16. The proof divides by neither t_p nor $t_p - \chi^*(p)$. It proves freeness first over $\mathbb{C}[t_p : p \mid q]$ and then over every $\mathbb{C}$-algebra obtained by specialization. Exceptional weights can therefore make a preferred coordinate vanish, but cannot change the rank of the full quotient. This distinction is essential when Euler factors are used to solve a relation system.

9.7.2 Anchors, reflection, and the ranks in the drafts

In this subsection take $\mathcal{A} = \mathbb{C}$. In a Hurwitz-zeta grid, use the representatives $0 < a \leq 1$: the group element 0 represents $a = 1$, not the singular Hurwitz parameter $a = 0$. Treating the endpoint as known means imposing the homogeneous anchor $[0] = 0$. It removes exactly the principal generator $Y_{1,\mathbf{1}}$. Hence

$$\dim_{\mathbb{C}} \mathcal{V}_q/(\mathcal{R}_q + \mathbb{C}[0]) = \varphi(q) - 1. \tag{9.39}$$

There are $q - 1$ remaining grid variables, so their distribution coefficient matrix has rank $q - \varphi(q)$. At rational weights t_p, the matrix has rational entries and this is also its rank over $\mathbb{Q}$.

Let $J[x] = [-x]$. On the χ-block, J acts as $\chi(-1)$. Thus the additional homogeneous reflection rows

$$[x] + (-1)^k[-x] = 0, \qquad x \neq 0, \tag{9.40}$$

retain exactly the characters with $\chi(-1) = (-1)^{k+1}$. After anchoring, the omission of $x = 0$ in (9.40) has no effect.

Corollary 9.17 (All-denominator formal ranks)**.** *For $q \geq 3$ and every integer $k \geq 0$, the formal quotient by distribution, endpoint, and the reflection rows (9.40) has dimension*

$$r_{q,k} = \begin{cases} \varphi(q)/2 - 1, & k \text{ odd}, \\ \varphi(q)/2, & k \text{ even}. \end{cases} \tag{9.41}$$

This conclusion holds for arbitrary prime weights. In particular it proves the all-q, all-k formal rank law for the negative Hurwitz-zeta derivative ladder. The formal distribution-only rank in the Stieltjes parameter derivative tower is $q - \varphi(q)$ for every $q \geq 2$, every parameter derivative order $k \geq 1$, and every Stieltjes index $n \geq 1$, relative to the endpoint and lower-index data.

Proof. For $q \geq 3$, the element -1 is a nonidentity involution in $U(q)$. Character orthogonality gives $\sum_\chi \chi(-1) = 0$, and each summand is 1 or -1. There are consequently $\varphi(q)/2$ characters of each

parity. Reflection retains the parity stated above. The principal character is even, so the endpoint removes one additional dimension exactly when k is odd. This proves (9.41).

For the negative ladder, differentiation of the Hurwitz multiplication formula at $s = -k$ gives

$$\sum_{j=0}^{d-1} \zeta'\left(-k, \frac{a+j}{d}\right) = d^{-k}\big(\zeta'(-k,a) + \log d\,\zeta(-k,a)\big). \tag{9.42}$$

The Bernoulli-polynomial values $\zeta(-k,a)$ are lower data. Thus the homogeneous distribution weight is $t_p = p^{-k}$. The homogeneous reflection is (9.40); an explicit right-hand side is given below. This applies the first part to the ladder in the source article.

For the parameter derivative tower, the Laurent expansion is

$$\zeta(s,a) = \frac{1}{s-1} + \sum_{n\ge 0} \frac{(-1)^n}{n!}\gamma_n(a)(s-1)^n. \tag{9.43}$$

For $k \ge 1$, its pole disappears after differentiation in a:

$$F_k(\epsilon,a) := \partial_a^k \zeta(1+\epsilon,a) = (-1)^k(1+\epsilon)_k\zeta(k+1+\epsilon,a) = \sum_{n\ge 0}\frac{(-1)^n}{n!}\gamma_n^{(k)}(a)\epsilon^n. \tag{9.44}$$

This standard identity is the starting point of the all-order parameter derivative formula of Coffey [25]. Differentiating the Hurwitz multiplication formula k times in a, including its chain-rule factors, gives

$$\sum_{j=0}^{d-1} F_k\left(\epsilon, \frac{a+j}{d}\right) = d^{k+1+\epsilon}F_k(\epsilon,a).$$

Comparing coefficients yields the exact triangular identity

$$\sum_{j=0}^{d-1}\gamma_n^{(k)}\left(\frac{a+j}{d}\right) = d^{k+1}\sum_{r=0}^{n}\binom{n}{r}(-\log d)^{n-r}\gamma_r^{(k)}(a). \tag{9.45}$$

With lower indices $r < n$ treated as specified data, the homogeneous weight is $t_p = p^{k+1}$, independently of n. Equation (9.39) now gives rank $q - \varphi(q)$. □

For completeness, the right-hand side of the negative reflection can be written without introducing another Hurwitz derivative. For $0 < a < 1$, set

$$C_j(a) = \sum_{m\ge 1}\frac{\cos(2\pi m a)}{m^j}, \qquad S_j(a) = \sum_{m\ge 1}\frac{\sin(2\pi m a)}{m^j}.$$

For $k \ge 1$, these series at $j = k+1$ converge absolutely, and the Fourier form of the Hurwitz functional equation [49] gives

$$\zeta'(-k,a) + (-1)^k\zeta'(-k,1-a) = \frac{(-1)^{\lfloor k/2\rfloor}k!}{(2\pi)^k}\begin{cases} C_{k+1}(a), & k \text{ even},\\ S_{k+1}(a), & k \text{ odd}.\end{cases} \tag{9.46}$$

Indeed, the Fourier expansion of $\zeta(s,a)$ has the factor

$$\frac{2\Gamma(1-s)}{(2\pi)^{1-s}}\left(\sin\frac{\pi s}{2}\,C_{1-s}(a) + \cos\frac{\pi s}{2}\,S_{1-s}(a)\right).$$

The indicated reflection selects a sine or cosine prefactor that vanishes at $s = -k$. Only the derivative of that prefactor survives, giving (9.46). This also explains why reflection is an inhomogeneous relation with depth-one data in the inventory.

Corollary 9.18 (The formal gamma grid)**.** *For $q \geq 3$, the $q-1$ formal coordinates $\log\Gamma(a/q)$, $1 \leq a < q$, modulo multiplication and reflection and relative to their stated elementary right-hand sides, have quotient dimension $\varphi(q)/2$.*

Proof. Lerch's identity $\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi)$ translates gamma multiplication into (9.42) with $k=0$, hence $t_p = 1$. The reflection formula

$$\zeta'(0,a) + \zeta'(0,1-a) = -\log(2\sin\pi a)$$

has homogeneous part $[x] + [-x] = 0$. Apply (9.41) with $k=0$. □

Remark 9.19 (What these rank theorems settle). The first Stieltjes source states a finite verification followed by an all-denominator conjecture; the parameter-derivative source states a general rank theorem while reporting only a finite calculation. Corollary 9.17 supplies a proof for the specified relation modules in both cases. The README had already identified the second proof gap. Corollary 9.18 likewise determines the rank of the stated gamma relation matrix. None of these results says that the evaluated numbers admit no further relations. In particular, the claim that multiplication and reflection generate every algebraic multiplicative relation among rational gamma values remains part of the conjectural completeness problem, not a consequence of the matrix rank.

9.7.3 Analytic weights, exceptional coordinates, and finite jets

Specialize $t_p = p^s = \exp(s\log p)$, using the real logarithm of the prime. Formula (9.36) becomes

$$A_{d,\chi}(s) = \left(\frac{d}{f_\chi}\right)^s \prod_{\substack{p\mid d \\ p\nmid f_\chi}} \left(1 - \chi^*(p)p^{-s}\right). \tag{9.47}$$

This is exactly the familiar imprimitive Euler-factor pattern for Dirichlet L-functions [49]. The normal-form proof shows why it appears in the formal relations before any evaluation is made.

The symbols of exact denominator q have character coordinates $Y_{q,\chi}$. In the quotient, their χ-coordinate is

$$Y_{q,\chi} = A_{q,\chi}(s) Y_{f_\chi,\chi}. \tag{9.48}$$

They form a basis of the full quotient precisely when every $A_{q,\chi}(s)$ is nonzero. In particular this holds when $\operatorname{Re} s \neq 0$, since $\chi^*(p)$ has modulus one. All nonzero integer weights in the two Stieltjes articles are therefore free of this coordinate obstruction. On the imaginary axis the top-denominator coordinates can fail, while the conductor coordinates of Theorem 9.15 remain a basis.

Theorem 9.20 (Exact defect of top-denominator jets)**.** *Fix $s_0 \in \mathbb{C}$ and $N \geq 0$, and work over $\mathcal{A}_N = \mathbb{C}[\epsilon]/(\epsilon^{N+1})$ with $t_p = p^{s_0+\epsilon}$, truncated in this ring. The full distribution module is free of rank $\varphi(q)$ over $\mathcal{A}_N$, hence has complex dimension $(N+1)\varphi(q)$. Set*

$$r_\chi(s_0) = \#\{p \mid q : p \nmid f_\chi,\ p^{s_0} = \chi^*(p)\}. \tag{9.49}$$

The image of the top-denominator submodule in its χ-block has complex dimension

$$\max\{N+1-r_\chi(s_0), 0\}, \tag{9.50}$$

and its cokernel has dimension $\min\{r_\chi(s_0), N+1\}$. An endpoint anchor or a fixed-parity reflection restricts these formulas to the retained characters. The resulting full quotient remains free over $\mathcal{A}_N$ with the ranks already computed.

Proof. Freeness is Theorem 9.15, applied to $\mathcal{A}_N$. A resonant factor in (9.47) has a simple zero at s_0:

$$\frac{d}{ds}(1-\chi^*(p)p^{-s})\Big|_{s=s_0} = \log p \neq 0.$$

All other factors, including $(q/f_\chi)^s$, are nonzero there. In the ring of analytic germs,

$$A_{q,\chi}(s_0+\epsilon) = \epsilon^{r_\chi(s_0)}u_\chi(\epsilon), \qquad u_\chi(0)\neq 0.$$

After truncation, u_χ is a unit. The top-denominator image in the free rank-one block is therefore the ideal $\epsilon^{r_\chi(s_0)}\mathcal{A}_N$. Its basis consists of the powers $\epsilon^{r_\chi},\dots,\epsilon^N$ when $r_\chi \le N$, and it is zero otherwise. This proves both dimensions. Anchoring and reflection remove entire character blocks, so the same calculation applies to each retained block. □

Comparing Taylor coefficients of a weighted distribution identity gives the corresponding stacked finite-jet system. If one identifies the coefficient at order j with $\epsilon^{N-j}[x]$, its relation space is the $\mathcal{A}_N$-span above; factorial normalizations merely rescale coordinates. Thus freeness also gives the rank of the complete simultaneous jet system, rather than just its successive highest layers.

Proposition 9.21 (Multiplicity of nonzero resonances)**.** *Resonances occur only on* $\operatorname{Re} s_0 = 0$*. For* $s_0 \neq 0$*, each* $r_\chi(s_0)$ *is at most one. Multiple resonant primes in a single character block can occur only at* $s_0 = 0$*.*

Proof. Taking absolute values in $p^{s_0} = \chi^*(p)$ gives $\operatorname{Re} s_0 = 0$. Write $s_0 = it$, with $t \neq 0$. Because every character value is a root of unity, resonance at p implies $t\log p/(2\pi) \in \mathbb{Q}$. This rational number is nonzero. If distinct primes p and ℓ both resonate, then $\log p/\log \ell \in \mathbb{Q}$. Writing this positive rational as a/b would give $p^b = \ell^a$ for positive integers a, b, contradicting unique factorization. □

For a concrete origin resonance, take $q = 14$ and let χ^* be the quadratic character modulo 7. Its conductor is 7, its parity is odd, and $\chi^*(2) = 1$. Then

$$Y_{14,\chi}(s) = (2^s - 1)Y_{7,\chi}(s). \tag{9.51}$$

At $s = 0$, the top-denominator coordinate vanishes although the conductor direction persists. For any analytic realization of this relation,

$$\begin{aligned} Y'_{14,\chi}(0) &= \log 2\, Y_{7,\chi}(0), \\ Y''_{14,\chi}(0) &= 2\log 2\, Y'_{7,\chi}(0) + (\log 2)^2 Y_{7,\chi}(0). \end{aligned} \tag{9.52}$$

Thus recovering the first conductor derivative from these top-denominator data requires the second top-denominator jet. The actual realization

$$Y_{d,\chi}(s) = \sum_{a\in U(d)} \chi_d(a)\zeta(s, a/d) = d^s L(s,\chi_d)$$

obeys the same identity. This example concerns an imprimitive Euler factor; it is distinct from the trivial-zero regularization of the primitive L-function in the next section.

9.8 Integral distribution, reflection and modular specialization

The character basis above gives efficient coordinates over a splitting field. To settle the integral question, retain the original grid symbols: Fourier transport is exact over its cyclotomic coefficient field, but inverse transforms introduce group-order denominators and need not be unimodular over integral rings. The following division-free construction proves an all-ring basis before imposing reflection. The resulting reflection quotients can have two-torsion even though the unreflected module is free. This develops the two overlapping integral continuations through a single proof chain: point basis, resolution, fixed-point homology, scalar torsion, jets and analytic identities.

The ordinary distribution and its sign cohomology are classical [52, 92]; Ouyang's universal norm distribution and Anderson-type resolution supply broader antecedents [53]. We give the complete proofs for the specified weighted presentation. Neither formal freeness nor a formal torsion calculation asserts independence of evaluated periods.

For a positive integer q, let

$$X_q = (q^{-1}\mathbb{Z})/\mathbb{Z}, \qquad P(q) = \{p : p \text{ is prime and } p \mid q\}, \qquad R_q = \mathbb{Z}[A_p : p \in P(q)].$$

We write $\mathrm{den}(x)$ for the exact order of x in $\mathbb{Q}/\mathbb{Z}$, and set $\mathrm{den}(0) = 1$. A symbol e_x is not itself a special value. It is a generator in a free module. Under polylogarithm evaluation, e_0 will become $\mathrm{Li}_s(1) = \zeta(s)$. Under Hurwitz evaluation, it becomes $\zeta(s, 1)$, not the undefined expression $\zeta(s, 0)$.

For $d \mid q$, put $A_d = \prod_{p|d} A_p^{v_p(d)}$. Weights on composite divisors are thus multiplicative; an unrelated variable for every divisor would define a different module. The universal module and its relation submodule are

$$V_q = \bigoplus_{x \in X_q} R_q e_x, \tag{9.53}$$

$$r_{p,x} = \sum_{\substack{y \in X_q \\ py = x}} e_y - A_p e_x \qquad (p \mid q,\ x \in X_{q/p}), \tag{9.54}$$

$$\mathcal{R}_q = \langle r_{p,x} \rangle_{R_q}, \qquad \mathcal{D}_q = V_q/\mathcal{R}_q. \tag{9.55}$$

Every sum in (9.54) has exactly p terms. For a commutative ring S and weights $a_p \in S$, write $\mathcal{D}_q(S, a)$ for the same presentation after $A_p \mapsto a_p$.

Lemma 9.22 (Prime rows suffice)**.** *The relation $r_{d,x} = \sum_{dy=x} e_y - A_d e_x$, for $d \mid q$ and $x \in X_{q/d}$, follows from the prime rows without division.*

Proof. For $ab \mid q$ and $x \in X_{q/(ab)}$, direct expansion gives

$$r_{ab,x} = \sum_{ay=x} r_{b,y} + A_b r_{a,x}.$$

Here all y belong to $X_{q/b}$, so every displayed prime or divisor row is legitimate. The intermediate symbols cancel. Induction on the number of prime factors of d, counted with multiplicity, proves the claim. □

This elementary observation already prevents a common ambiguity: "distribution rank" is meaningful only after the row vocabulary, endpoint convention, coefficient ring, and divisor-weight convention have all been fixed.

9.8.1 A division-free integral basis and a unit relation minor

Choosing the surviving grid points

Write $U_m = (\mathbb{Z}/m\mathbb{Z})^\times$ and use a one-element set U_1. For every prime p and exponent $a \geq 1$, choose a distinguished lift in each fiber of $U_{p^a} \to U_{p^{a-1}}$. At $a = 1$, this means choosing one member of U_p. Let $E_{p,a}$ be the remaining members. Thus

$$|E_{p,a}| = \varphi(p^a) - \varphi(p^{a-1}), \qquad \varphi(1) = 1. \tag{9.56}$$

The following fixed choice makes the construction executable:

$$\begin{aligned} E_{p,1} &= \{2, 3, \dots, p-1\}, \\ E_{p,a} &= \{b + jp^{a-1} : 1 \leq b < p^{a-1},\ p \nmid b,\ 1 \leq j \leq p-1\} \quad (a \geq 2). \end{aligned} \tag{9.57}$$

For $p = 2$, the first set is empty. This is intentional.

Every point x of exact denominator m has a unique additive primary expansion

$$x = \sum_{p^a \| m} \frac{b_p}{p^a} \pmod{\mathbb{Z}}, \qquad b_p \in U_{p^a}. \tag{9.58}$$

For $x = n/m$ in lowest terms, $b_p \equiv n(m/p^a)^{-1} \pmod{p^a}$. Define $\mathcal{B}_q$ to contain 0 and all points of denominator $m \mid q$ for which $b_p \in E_{p,a}$ at every nonzero primary component. We call a primary component *bad* when its residue is the distinguished, discarded lift. This adjective refers only to a basis choice.

The cardinality telescopes prime by prime:

$$|\mathcal{B}_q| = \prod_{p^e \| q} \left(1 + \sum_{a=1}^{e} \big(\varphi(p^a) - \varphi(p^{a-1})\big)\right) = \varphi(q). \tag{9.59}$$

The point 0 must be retained for this count and for the normal forms.

Theorem 9.23 (Integral normal form)**.** *The images of e_b, $b \in \mathcal{B}_q$, form an R_q-basis of $\mathcal{D}_q$. Consequently, for every commutative ring S and every choice of weights $a_p \in S$, the same points form an S-basis of $\mathcal{D}_q(S, a)$. In particular this module is free of rank $\varphi(q)$ even in positive characteristic, at zero weights, and over nonreduced rings.*

The spanning algorithm is immediate and is proved next. Its independence is established in Section 9.8.2, without dividing by any group order or weight.

The terminating straightening rule

Let $y \notin \mathcal{B}_q$, let $m = \operatorname{den}(y)$, and choose any bad prime p for its primary coordinates. The row with target py gives

$$e_y = A_p e_{py} - \sum_{\substack{pz = py \\ z \neq y}} e_z \quad \text{in } \mathcal{D}_q. \tag{9.60}$$

All terms on the right have denominator dividing m. At exact denominator m, the p-primary components of the other solutions are the allowed lifts in the same fiber; all other primary components are unchanged. When the p-component has order p, there is also one solution with zero p-component, hence lower denominator. The point py itself has denominator m/p.

Order points first by their positive denominator and then by the number of bad primary components. Every right-hand term of (9.60) is strictly earlier in this order. Recursion therefore terminates, and proves that $\mathcal{B}_q$ spans. The implementation always chooses the smallest bad prime, but the theorem permits any choice. Independence will imply that all choices yield the same normal form.

Proposition 9.24 (Support and polynomial degree)**.** *Write the normal form of e_y, with $m = \operatorname{den}(y)$, as*

$$\mathrm{NF}(e_y) = \sum_{b \in \mathcal{B}_q} c_{y,b}(\mathbf{A}) e_b.$$

A nonzero coefficient has $\operatorname{den}(b) = d \mid m$, and

$$\deg_{A_p} c_{y,b} \le v_p(m/d) \qquad (p \mid q). \tag{9.61}$$

All coefficients belong to $\mathbb{Z}[\mathbf{A}]$.

Proof. The recursion never increases a primary denominator. Multiplication by A_p occurs only in the term e_{py}, where that denominator has lost one power of p. Every path from denominator m to denominator d can therefore contain at most $v_p(m/d)$ such multiplications. The other coefficients in the recursion are -1. Summing the finitely many paths proves the assertions, including possible cancellations. □

There is no assertion here that the coefficient of e_0 factors as a product of $A_p - 1$. At level 15 the coefficient in one normal form is $1 - A_3A_5$; this distinction will matter at the spectral pole.

An explicit maximal minor equal to one

For each nonbasis point y, select the single row $r_{p,py}$ used in its first straightening step. Select the nonbasis points as columns, and order rows by their selected pivot. The restricted matrix is triangular, with every diagonal entry equal to 1: all other nonbasis columns are earlier in the terminating order.

The selected rows are distinct. Indeed a selected row has its chosen point as its unique latest nonbasis point, with coefficient 1. Two identical rows could not have different such latest points. We have proved the following stronger integrality statement.

Theorem 9.25 (Universal unit minor and relation basis)**.** *The prime-relation matrix contains an explicitly selected $(q - \varphi(q)) \times (q - \varphi(q))$ minor equal to 1 in R_q. The selected rows form an R_q-basis of the full relation module $\mathcal{R}_q$. After specializing the weights to integers, all nonzero Smith invariants of the complete prime-relation matrix are 1; there are exactly $q - \varphi(q)$ of them.*

Proof. Triangularity proves the minor statement. Selected-row elimination reduces every vector to a combination of basis-point symbols. By Theorem 9.23, a vector in $\mathcal{R}_q$ has zero such remainder, so every raw row is a combination of the selected ones. The minor proves their independence. The same argument specializes without division. Equivalently, after an integer specialization the image lattice is saturated and has the indicated rank; the unit minor also proves directly that all Smith invariants are units. □

The proof order is not circular: the resolution below proves Theorem 9.23 independently of the assertion that the selected rows span all relations.

Certificates rather than opaque elimination

The straightener can retain the exact row combination at every step. If C_y denotes a certificate for $e_y - \mathrm{NF}(e_y)$, then

$$C_y = r_{p,py} + A_p C_{py} - \sum_{\substack{pz=py \\ z\neq y}} C_z, \qquad C_b = 0 \ (b \in \mathcal{B}_q). \tag{9.62}$$

A verifier need not trust the recursion. It can expand the listed raw rows in the original q-generator free module, multiply their explicit integer polynomials, and compare coefficients. The accompanying verifier does exactly this for the three frozen identities.

Example 9.26 (Small levels). The specified basis choices give

$$\mathcal{B}_4 = \{0, 3/4\}, \qquad \mathcal{B}_{12} = \{0, 5/12, 2/3, 3/4\},$$

$$\mathcal{B}_{15} = \{0, 1/15, 4/15, 2/5, 7/15, 3/5, 2/3, 4/5\}.$$

Also $\mathcal{B}_{30} = \mathcal{B}_{15}$ as sets of rational points. An exact primary component of order two has no allowed residue, which explains this last equality. It does not make the weight A_2 irrelevant to all normal-form coefficients.

The procedure terminates with polynomial coefficients, but we do not assert a polynomial-time bound in the bit length of q. There are q raw grid points to begin with, and certificate coefficients can grow. The finite support and degree bounds are rigorous; sharper complexity and coefficient-height estimates remain separate questions.

A uniform coefficient-height bound

For $q > 1$, let $s = |P(q)|$, let $P_{\max}$ be its largest prime divisor, and put $H(x) = \sum_p v_p(\mathrm{den}(x))$, $b(x) = \#\{\text{bad primary components of } x\}$ and $W(x) = (s+1)H(x) + b(x)$. Summing absolute values over all normal-form coordinates and all polynomial monomials gives

$$\| \mathrm{NF}(e_x)\|_1 \leq P_{\max}^{W(x)}.$$

Indeed every straightening edge lowers W by at least one: at equal denominator a bad component is removed, while a decrease in H absorbs any possible increase of at most s in b. There are at most $P_{\max}$ terms in a step, and multiplication by A_p preserves the coefficient norm. Induction proves the bound. The sharper coordinate multidegrees remain those in (9.61). The estimate is a uniform height bound, not an optimal complexity assertion.

The complete level-twelve normal-form table

Let $E_a = [e_{a/12}]$, $\alpha = A_2$, $\beta = A_3$. The retained basis is E_0, E_5, E_8, E_9. Each row gives the four coefficients of E_a in that basis.

a	E_0	E_5	E_8	E_9
0	1	0	0	0
1	$\alpha\beta(\alpha-1)$	-1	0	$-\beta-1$
2	$\alpha(\beta-1)$	0	$-\alpha-1$	0
3	$\alpha(\alpha-1)$	0	0	-1
4	$\beta-1$	0	-1	0
5	0	1	0	0
6	$\alpha-1$	0	0	0
7	$\alpha(\beta-\alpha)$	1	$-\alpha(\alpha+1)$	$\beta+1$
8	0	0	1	0
9	0	0	0	1
10	$1-\beta$	0	$\alpha+1$	0
11	$\alpha(1-\beta)$	-1	$\alpha(\alpha+1)$	0

The companion file `normal_forms_q12.json` contains the unfactored integer monomial data. At arbitrary complex order, substitute $\alpha = 2^{1-s}$, $\beta = 3^{1-s}$ and $E_a = \mathrm{Li}_s(e^{2\pi i a/12})$. At formal integral jet weights, substitute polynomials modulo u^L instead. The same table supports both uses.

9.8.2 A finite weighted resolution and a proof of independence

The complex

Order the primes dividing q. For $S \subseteq P(q)$ put $p_S = \prod_{p\in S} p$, and define

$$C_j = \bigoplus_{\substack{S\subseteq P(q)\\ |S|=j}} R_q[X_{q/p_S}]. \tag{9.63}$$

A generator is written $[S, x]$. Define

$$d[S,x] = \sum_{p\in S} (-1)^{\#\{\ell\in S:\ell<p\}} \left(\sum_{py=x} [S\setminus\{p\}, y] - A_p[S\setminus\{p\}, x] \right). \tag{9.64}$$

The roots in each inner sum belong to $X_{q/p_{S\setminus\{p\}}}$. The augmentation $C_0 \to \mathcal{D}_q$ is the defining quotient.

Lemma 9.27. *The operators in distinct prime directions commute before the exterior signs are applied. Hence $d^2 = 0$, and the image of $C_1 \to C_0$ is exactly $\mathcal{R}_q$.*

Proof. Both successive root sums in prime directions p and ℓ enumerate once all solutions of $p\ell z = x$. The two terms consisting of one root sum and one multiplication by a weight agree in either order, and so do the terms with coefficient A_pA_ℓ. All intermediate points lie in the required grids. The exterior signs make each pair cancel in d^2. The assertion about C_1 is its definition. □

A filtration that removes the weights

Associate to a generator the integer

$$\nu(S,x) = p_S \operatorname{den}(x), \tag{9.65}$$

which divides q. Filter the complex by increasing ν. If $p \in S$, a root of $py = x$ has denominator either $\operatorname{den}(x)$ or $p\operatorname{den}(x)$. Thus the differential does not increase ν. Its top-filtration part retains exactly the roots of denominator $p\operatorname{den}(x)$; the term with coefficient A_p lies strictly lower. The associated graded differential is consequently independent of all weights.

Fix a divisor $m \mid q$. In its associated graded summand, a generator with subset S has exact denominator m/p_S, so necessarily $S \subseteq P(m)$. For $p^a \parallel m$, the local coordinate belongs to $U_{p^{a-1}}$ when $p \in S$, and to U_{p^a} otherwise. At exponent zero the unique local coordinate is zero, represented by the one-element set U_1.

There is a harmless CRT twist: division by a prime changes the other primary coordinates by a unit. We spell out its removal because omitting it would leave a gap in a claimed tensor-product argument. If x_ℓ is the ℓ-primary component of x, replace it in the S-summand by

$$z_\ell = \left(\prod_{p \in S \setminus \{\ell\}} p \right)^{-1} x_\ell. \tag{9.66}$$

The inverse exists on the ℓ-primary group. Under a differential edge removing p, the coordinates z_ℓ for $\ell \neq p$ are unchanged: the factor p^{-1} in a root coordinate cancels the removal of p from the product. The p-coordinate is carried to all its lifts under multiplication by p.

It follows that the associated graded summand at m is the tensor product, with its usual exterior signs, of the two-term complexes

$$R_q[U_{p^{a-1}}] \xrightarrow{N_{p,a}} R_q[U_{p^a}], \qquad N_{p,a}(b) = \sum_{\substack{u \in U_{p^a} \\ u \mapsto b}} u, \quad p^a \parallel m. \tag{9.67}$$

For $a = 1$, the image of the single generator is the sum of all $p-1$ nonzero residues. For $a \geq 2$, each fiber has p elements.

Lemma 9.28 (Local splitting)**.** *Each $N_{p,a}$ is a split injection. Its cokernel is free with basis the images of $E_{p,a}$.*

Proof. Different fibers have disjoint supports. In one fiber, the sum of its basis vectors has a coefficient equal to 1 on the distinguished lift. Replace that one vector by the fiber sum; this is a unimodular basis change. The remaining vectors are exactly the allowed lifts. Doing this in all fibers proves both claims over any coefficient ring. □

Theorem 9.29 (Exact weighted resolution)**.** *The augmented complex*

$$0 \longrightarrow C_{|P(q)|} \longrightarrow \cdots \longrightarrow C_1 \longrightarrow C_0 \longrightarrow \mathcal{D}_q \longrightarrow 0$$

is exact. The images of $\mathcal{B}_q$ are a basis of $\mathcal{D}_q$, proving Theorem 9.23.

Proof. By Lemma 9.28, each local two-term complex is a direct sum of a contractible identity complex and its free cokernel in degree zero. Its finite tensor product is therefore acyclic in positive

degrees. Its degree-zero homology has the tensor-product basis of the allowed local residues. In the graded summand at m, these are exactly the basis points of denominator m.

Now lift through the finite filtration. If a positive-degree cycle has highest nonzero filtration m, its highest component is a boundary in the associated graded complex. Subtract a lift of that boundary to obtain a cycle of lower filtration. Repetition makes the cycle a boundary in the original complex. This proves positive-degree exactness without any convergence issue.

For degree zero, suppose a linear combination of allowed points is dw with $w \in C_1$, and let M be the highest filtration present in that linear combination. If w has higher filtration, its leading component is a cycle in degree one of the associated graded complex. Positive-degree exactness there allows us to subtract a lifted boundary dv, $v \in C_2$, from w and lower its filtration without changing dw. Repetition gives a preimage of filtration at most M. Its leading boundary would now be a relation among the allowed points in the graded summand at M. Those points form a free basis of its zero-degree homology, so all their coefficients must vanish. Descending M proves independence. Spanning was proved by (9.60). Thus the claimed points form a basis of $H_0(C_\bullet) = \mathcal{D}_q$. □

Base change and the rank identity

The proof works verbatim over an arbitrary commutative ring with arbitrary weights. Alternatively, the augmented complex over R_q is split exact as an underlying R_q-module complex: its degree-zero homology is free, and successive surjections onto projective kernels split. Tensoring with any R_q-algebra therefore preserves exactness. No flatness hypothesis on the target algebra is needed.

Taking the alternating ranks of the chain groups gives a useful check:

$$\operatorname{rank} \mathcal{D}_q = \sum_{S \subseteq P(q)} (-1)^{|S|} \frac{q}{p_S} = q \prod_{p \mid q} (1 - p^{-1}) = \varphi(q). \tag{9.68}$$

This rank computation alone would not prove integral freeness. The local unimodular splittings and the lifted basis are the missing integral information.

Remark 9.30 (Relation to character normal forms). Over a characteristic-zero splitting field, both the conductor-character basis in the source and the present raw-point basis describe the same free module. Their transition matrix is a legitimate field-valued change of coordinates. Its construction can involve divisions by character-group orders. The raw-point basis is designed to avoid those divisions, not to replace the arithmetic convenience of character coordinates where those coordinates are available.

9.8.3 Reflection and the exact integral torsion formula

Reflection is a separate quotient operation

The map $c(e_x) = e_{-x}$ preserves the relations and satisfies $c^2 = 1$. It acts on the entire resolution by $c[S, x] = [S, -x]$. For integer weights a_p, let $D = \mathcal{D}_q(\mathbb{Z}, a)$ and define

$$D^+ = D/(c-1)D, \qquad D^- = D/(c+1)D.$$

These are *coinvariants with a prescribed sign*, not the submodules of invariant and anti-invariant elements. Over a field of characteristic not two, the corresponding eigenspaces and coinvariants agree. Over $\mathbb{Z}$ this distinction is essential.

For any q, put

$$r(q) = \#\{p > 2 : p \mid q\} + \mathbf{1}_{4|q}. \tag{9.69}$$

For $q > 2$, we have $r(q) \geq 1$ and set $n(q) = \varphi(q)/2$, $h(q) = 2^{r(q)-1}$.

Theorem 9.31 (Weighted integral reflection theorem)**.** *Let $q > 2$ and specialize every A_p to an integer a_p. For each $\varepsilon \in \{1, -1\}$,*

$$D/(c-\varepsilon)D \simeq \mathbb{Z}^{n(q)} \oplus (\mathbb{Z}/2\mathbb{Z})^{t(q,a)}, \tag{9.70}$$

where

$$t(q,a) = \begin{cases} h(q), & a_p \text{ is odd for every odd prime } p \mid q, \\ 0, & \text{some odd prime } p \mid q \text{ has } a_p \text{ even.} \end{cases} \tag{9.71}$$

In particular the answer is independent of the integer a_2. At $q = 1$ or $q = 2$, the separate answers are $D^+ = \mathbb{Z}$ and $D^- = \mathbb{Z}/2\mathbb{Z}$.

The absence of odd torsion is an elementary property of an involution on a free lattice. The exact multiplicity is the substantive calculation. We prove it through the fixed points of the resolution.

Two elementary facts about involutions

For a module M with an involution c, define its two periodic cohomology groups by

$$\widehat{H}^0(M) = \ker(c-1)/(1+c)M, \qquad \widehat{H}^1(M) = \ker(1+c)/(c-1)M, \tag{9.72}$$

and repeat with period two. These are the cohomology groups of the complex whose differentials alternate between $c-1$ and $1+c$, with $c-1$ leaving even degree. This definition, rather than any unmentioned cohomology convention, fixes all signs and parity labels.

Lemma 9.32 (Torsion in sign coinvariants)**.** *If M is a finite-rank free abelian group, every torsion class in $M/(c-\varepsilon)M$ is killed by 2. Its torsion subgroup is $\widehat{H}^1(M)$ for $\varepsilon = 1$, and $\widehat{H}^0(M)$ for $\varepsilon = -1$.*

Proof. Suppose $kv = (c-\varepsilon)w$ with $k \neq 0$. Applying $c+\varepsilon$ gives $k(c+\varepsilon)v = 0$, hence $(c+\varepsilon)v = 0$ because M is torsion-free. Then $(c-\varepsilon)v = -2\varepsilon v$, so the class of v is killed by 2. Conversely a vector in $\ker(c+\varepsilon)$ represents a class killed by 2. Taking its quotient by $(c-\varepsilon)M$ gives exactly the two groups in (9.72). □

Lemma 9.33 (Balanced free ranks)**.** *For every field of characteristic different from two, and arbitrary weights in that field, the two reflection coinvariants of $\mathcal{D}_q$ have dimension $\varphi(q)/2$ when $q > 2$.*

Proof. On a permutation module with u basis points and f fixed points, the two eigenspace dimensions are the integers $(u+f)/2$ and $(u-f)/2$. The sign-eigenspace functors are exact when 2 is invertible, so apply this count to the resolution. The alternating sum of the u's is $\varphi(q)$ by (9.68). The fixed-point count for X_m is one if m is odd and two if m is even. Thus its alternating sum is

$$\sum_{S \subseteq P(q)} (-1)^{|S|} \gcd(2, q/p_S) = 0 \qquad (q > 2).$$

For odd q, this is $(1-1)^{|P(q)|}$. If $v_2(q) = 1$, summing first over whether $2 \in S$ leaves $(2-1)(1-1)^k$, where k is the number of odd prime divisors and is positive. If $v_2(q) \geq 2$, that first sum is $2-2$. Hence both dimensions are $\varphi(q)/2$.

This uses integer dimension counts in the chain modules, not merely a trace equality in the coefficient field. The latter alone would not determine dimensions in positive characteristic. □

Removing every nonfixed orbit from periodic cohomology

This step explains why the calculation is small despite the q original grid points. Let C_j^{mov} be the span of the nonfixed points in each summand of C_j. It is a subcomplex: if $x \neq -x$ and $py = x$, then y cannot be fixed; the term $-a_p[S \setminus \{p\}, x]$ is also nonfixed. Each term of this subcomplex is a direct sum of copies of the regular permutation module $S[\mathbb{Z}/2\mathbb{Z}]$ over the chosen coefficient ring S.

The periodic complex of a regular two-point orbit is exact over *every* commutative ring. Indeed the kernel of the difference map is the diagonal, which is the image of the sum map; the kernel of the sum map consists of $(u, -u)$, which is the image of the difference map. This assertion requires neither division by 2 nor absence of 2-torsion.

Let $K_\bullet = C_\bullet / C_\bullet^{\mathrm{mov}}$. Its generators are only the fixed symbols $[S, 0]$ and, when available, $[S, 1/2]$. The action on $K_\bullet$ is trivial. Forming the double complex of the periodic cohomology complexes gives

$$\widehat{H}^\bullet(\mathcal{D}_q(S, a)) \simeq H^\bullet \operatorname{Tot}\big(T(K_\bullet)\big). \tag{9.73}$$

Here T denotes the alternating difference/sum complex just defined, and the chain degree of K_j contributes $-j$ to total degree.

For completeness, this comparison can be justified without assuming a spectral sequence degenerates. The augmented resolution is exact after base change, so taking horizontal homology in the double complex replaces $C_\bullet$ by D in degree zero. Removing the moving subcomplex does not change total cohomology because each of its vertical columns is exact. In both statements the horizontal width is bounded by $|P(q)|$, so in each total degree only finitely many columns occur; successive elimination along these columns justifies the two comparisons. In particular, no convergence of an infinite horizontal filtration is being used.

The fixed-point complex is a Koszul complex

For a sequence $b_1, \dots, b_r$ in a ring S, let $K_S(b_1, \dots, b_r)$ denote the tensor product of the two-term complexes $S \xrightarrow{b_i} S$, with source in chain degree one. This is the Koszul complex; its definition is all that is needed below.

For an odd prime p, the fixed-point part of its distribution map is multiplication by $1 - a_p$ on each available fixed symbol. Of the p roots of a fixed point, exactly one is fixed, and is the original point. The other roots disappear in the quotient by moving orbits. The factor at 2 has two different forms.

When $v_2(q) = 1$, its source has only 0 and its target has $0, 1/2$. Its column is

$$\begin{pmatrix} 1 - a_2 \\ 1 \end{pmatrix}. \tag{9.74}$$

The unit entry splits off an identity complex, leaving one free copy of S in degree zero. Therefore this prime contributes no Koszul generator to the fixed-point complex.

When $v_2(q) \geq 2$, both grids have two fixed points. In the order $(0, 1/2)$, the matrix is

$$B(a_2) = \begin{pmatrix} 1 - a_2 & 0 \\ 1 & -a_2 \end{pmatrix}. \tag{9.75}$$

The two roots of $1/2$ are $1/4, 3/4$, which are moving points; this explains the second column. Unimodular row and column operations transform this matrix to $\operatorname{diag}(1, a_2(a_2 - 1))$, up to a unit sign. These operations are valid over $\mathbb{Z}[a_2]$ and over every specialization, because they pivot on the

displayed 1. Odd-prime maps are scalar, so the operations commute with all other directions of the complex.

We have proved the following ring-uniform reduction.

Proposition 9.34 (Fixed-point reduction)**.** *Over any commutative ring, $K_\bullet$ is chain-homotopy equivalent to*

$$K_S(\boldsymbol{\delta}), \qquad \boldsymbol{\delta} = \big(a_p - 1 : \ p > 2, \ p \mid q\big) \text{ together with } a_2(a_2-1) \text{ if } 4 \mid q. \tag{9.76}$$

It has $r(q)$ active scalar entries. At $q = 1, 2$, this is the empty Koszul complex S in degree zero.

Completing the integer calculation

For $S = \mathbb{Z}$, the periodic complex on a trivial module alternates between 0 and multiplication by 2. It separates into adjacent two-term blocks. Tensoring one such block with the free complex $K_{\mathbb{Z}}(\boldsymbol{\delta})$ computes $K_{\mathbb{Z}}(\boldsymbol{\delta}) \otimes \mathbb{F}_2$: multiplication by 2 is a free resolution of reduction modulo two. Consequently

$$\widehat{H}^{\varepsilon}(D) \simeq \bigoplus_{\substack{0 \le j \le r(q) \\ j \equiv \varepsilon \ (2)}} H_j\big(K_{\mathbb{F}_2}(\overline{\boldsymbol{\delta}})\big), \qquad \varepsilon = 0, 1. \tag{9.77}$$

Only an isomorphism of abelian groups is asserted here; no extra multiplicative structure is required.

If some odd a_p is even, its active scalar reduces to 1, so its two-term complex is contractible and all the homology vanishes. Otherwise every odd-prime scalar is zero modulo two. The remaining scalar $a_2(a_2 - 1)$, when present, is also zero modulo two for every integer a_2. All differentials in the reduced active complex therefore vanish, and

$$\dim_{\mathbb{F}_2} H_j = \binom{r(q)}{j}.$$

For $r(q) \ge 1$, the sums over even and odd j are both $2^{r(q)-1}$. Lemmas 9.32 and 9.33 now prove Theorem 9.31. When $r(q) = 0$, the only homology is in degree zero, giving the stated distinct answers for $q = 1, 2$.

Examples and the classical specialization

At level four, put $a = a_2$. The basis is $(e_0, e_{3/4})$, and

$$e_{1/2} = (a-1)e_0, \qquad e_{1/4} = a(a-1)e_0 - e_{3/4}, \qquad c = \begin{pmatrix} 1 & a(a-1) \\ 0 & -1 \end{pmatrix}.$$

For integer a, the nonzero Smith factor of either $c - 1$ or $c + 1$ is 2, because $a(a-1)$ is even. This two-dimensional example exhibits both the unavoidable parity and the irrelevance of the integer value of a_2.

For all odd-prime weights odd, the following values result:

q	4	12	15	30	60	105
$n(q)$	1	2	4	4	8	24
$r(q)$	1	2	2	2	3	3
$h(q)$	1	2	2	2	4	4

For example, either sign at level 60 gives $\mathbb{Z}^8 \oplus (\mathbb{Z}/2)^4$ in the exceptional parity case, but only $\mathbb{Z}^8$ as soon as a_3 or a_5 is even.

When all $a_p = 1$, the count agrees with the classical ordinary sign-cohomology theorem of Kubert [92]; twice an odd level has the same effective prime count as that odd level. The weighted parity criterion, the fixed-point matrices, and the later jet calculation are the present explicit continuation of the manuscript's integral question.

9.8.4 Characteristic two, the universal support, and all jet lengths

A characteristic-two homology theorem over arbitrary algebras

Assume now that the coefficient ring S has characteristic two. Write $N = c - 1 = c + 1$. Then $N^2 = 0$, and both periodic cohomology groups are the same module $\ker N / \operatorname{im} N$. On a fixed-point module both periodic differentials vanish. The comparison (9.73) therefore has a particularly simple consequence.

Theorem 9.35 (Koszul model in characteristic two)**.** *For every commutative characteristic-two ring S and every choice of its weights,*

$$\frac{\ker(N : \mathcal{D}_q(S,a) \to \mathcal{D}_q(S,a))}{\operatorname{im} N} \simeq \bigoplus_{j=0}^{r(q)} H_j(K_S(\boldsymbol{\delta})), \tag{9.78}$$

where the active sequence is (9.76)*.*

Proof. Remove the moving-orbit subcomplex as in Section 9.8.3. On its fixed quotient the vertical periodic differential is zero. Total cohomology is therefore the direct sum of the horizontal homology groups, one from each of the finitely many chain degrees. Apply the chain-homotopy reduction of Proposition 9.34. □

The universal obstruction is a reduced complete intersection

Let $R_2 = \mathbb{F}_2[A_p : p \mid q]$ and let $I_q \subseteq R_2$ be the ideal

$$I_q = (A_p - 1 : \ p > 2,\ p \mid q) + (A_2(A_2 - 1) : \ 4 \mid q), \tag{9.79}$$

where the second summand is omitted unless indicated. Its generators form a regular sequence. Indeed the odd-prime entries are independent linear variables shifted by one; after removing them, the possible last entry is a nonzero monic polynomial in the still-free variable A_2. At each step multiplication by that entry is injective in the quotient by the earlier ones.

The Koszul complex on a regular sequence resolves its quotient. This can be seen inductively here without a general regular-sequence criterion: tensor a free resolution of the previous quotient with $[R_2 \xrightarrow{\delta} R_2]$. Its homology consists of the kernel and cokernel of multiplication by δ on that quotient. The kernel is zero, and the cokernel is the next quotient.

Corollary 9.36 (Universal characteristic-two support)**.** *Over R_2, the reflection homology is a cyclic module:*

$$\ker N / \operatorname{im} N \simeq R_2 / I_q. \tag{9.80}$$

Its zeroth Fitting ideal is exactly I_q. The closed locus defined by I_q is reduced. For odd $q > 1$ it is the single point with every weight equal to one. For $4 \mid q$ it has two points, with all odd-prime weights equal to one and $A_2 = 0$ or 1. For q twice an odd number, A_2 is free and the locus is an affine line.

Proof. The positive Koszul homology vanishes by the preceding argument; apply Theorem 9.35. A cyclic module R_2/I_q has zeroth Fitting ideal I_q by its one-generator presentation. After the odd linear equations, the only possible further equation is $A_2(A_2-1)$, whose two factors are relatively prime. Its quotient is a product of two fields, or of two copies of the remaining coefficient ring if one extends scalars. The stated descriptions and reducedness follow. □

The cyclic universal answer does *not* imply that every fiber of the reflection homology is one-dimensional. Homology need not commute with a nonflat specialization. At an exceptional field-valued point the differentials of the specialized Koszul complex all vanish, so its homology has total dimension $2^{r(q)}$. This is the precise source of the larger fiber defect.

There is also an integral universal formulation. With the periodic convention (9.72), the same fixed-complex calculation, followed by reduction modulo two, gives

$$\widehat{H}^{2j}(\mathcal{D}_q) \simeq R_q/(2,\boldsymbol{\delta}), \qquad \widehat{H}^{2j+1}(\mathcal{D}_q) = 0. \tag{9.81}$$

Here $\mathcal{D}_q$ is still over the polynomial ring, not an integer weight fiber. The regularity argument is in $R_q/2R_q$. Odd periodic cohomology appearing after integer specialization is another nonflat-base-change effect, not a contradiction to (9.81).

The exact rank locus over every field

For a field k of characteristic two, a nonzero active scalar is a unit and makes the Koszul complex contractible. If all active scalars vanish, its differentials vanish and its total homology dimension is $2^{r(q)}$. To pass from homology to the desired coinvariants, note that an endomorphism N with $N^2=0$ on a d-dimensional vector space satisfies

$$\dim(\ker N/\operatorname{im} N) = d - 2\operatorname{rank} N, \qquad \dim \operatorname{coker} N = \frac{d+\dim(\ker N/\operatorname{im} N)}{2}. \tag{9.82}$$

Theorem 9.37 (All-field reflection dimension)**.** *Let $q>2$, let k be a field, and use arbitrary weights $a_p \in k$. If $\operatorname{char} k \neq 2$, both sign coinvariants have dimension $n(q)$. If $\operatorname{char} k = 2$, the signs coincide and*

$$\dim_k \mathcal{D}_q(k,a)/(c-1)\mathcal{D}_q(k,a) = n(q) + h(q)\mathbf{1}_{\mathcal{E}_q(a)}, \tag{9.83}$$

where $\mathcal{E}_q(a)$ means

$$a_p = 1 \quad \textit{for every odd } p \mid q, \qquad a_2 \in \{0,1\} \quad \textit{if } 4 \mid q.$$

There is no condition on a_2 when $v_2(q)=1$.

Proof. For characteristic different from two use Lemma 9.33. In characteristic two use the Koszul calculation and (9.82) with $d=\varphi(q)$. Over a field, $a_2(a_2-1)=0$ is equivalent to $a_2=0$ or 1. □

Over $\mathbb{F}_2$, the condition at 2 is automatic. Over $\mathbb{F}_4$ it is not: a weight satisfying $a_2^2+a_2+1=0$ is neither zero nor one, and the reflection defect disappears when $4 \mid q$. Thus an experiment restricted to binary weights would miss part of the geometry. The exact regression suite includes all four field weights at selected levels to test this distinction.

A complete one-parameter Smith law

Let k have characteristic two, let $O = k[[t]]$, and choose power series weights $a_p(t)$. The module $\mathcal{D}_q(O, a)$ is free of rank $2n = \varphi(q)$. Form the active sequence $\boldsymbol{\delta}(t)$ as before, and define its contact order by

$$d = \min_i v_t(\delta_i(t)), \qquad v_t(0) = \infty. \tag{9.84}$$

The word "contact" refers to the pullback of the explicit ideal I_q along the chosen formal curve. It is not an approximation to a root of a transcendental function.

Each active prime contributes a factor at least two to $\varphi(q)$, so $\varphi(q) \geq 2^{r(q)}$ and $n \geq h$. Thus all block sizes in the following statement are nonnegative.

Theorem 9.38 (Contact-order Smith law)**.** *Let $q > 2$, $n = n(q)$, $h = h(q)$, and use the above characteristic-two power series weights. If $d < \infty$, the Smith diagonal of $N = c - 1$ over O is, up to units,*

$$\operatorname{diag}\bigl(\underbrace{1, \dots, 1}_{n-h}, \underbrace{t^d, \dots, t^d}_{h}, \underbrace{0, \dots, 0}_{n}\bigr). \tag{9.85}$$

When $d = 0$, the first two blocks together are n unit entries. When $d = \infty$, replace the h entries t^d by zero; there are then $n - h$ unit entries and $n + h$ zeros.

For every integer $L \geq 1$, put $O_L = k[t]/(t^L)$. The complete jet cokernel formula is

$$\dim_k \frac{\mathcal{D}_q(O_L, a)}{(c-1)\mathcal{D}_q(O_L, a)} = nL + h \min(d, L), \tag{9.86}$$

with $\min(\infty, L) = L$.

Proof. First work over O_L and put $d_L = \min(d, L)$. Elementary invertible operations on the active sequence identify its Koszul complex with that on $(t^{d_L}, 0, \dots, 0)$, with the convention that $t^L = 0$. To see this explicitly when $d_L < L$, choose an entry with least valuation, multiply it by a unit to make it t^{d_L}, and subtract its multiples from all other entries. Every other entry is divisible by it in O_L. If $d_L = L$, all entries already vanish. Invertible changes of the sequence induce invertible changes on its exterior algebra, hence isomorphisms of Koszul complexes.

The kernel and cokernel of multiplication by t^{d_L} on O_L each have k-dimension d_L. Tensoring with $r - 1$ zero-differential two-term complexes multiplies total homology dimension by 2^{r-1}. The right-hand side of (9.78) therefore has k-dimension $2^r d_L$. The underlying vector space of $\mathcal{D}_q(O_L, a)$ has dimension $2nL$, by the integral basis theorem. Applying (9.82) proves (9.86).

It remains to show that these length formulas determine all Smith exponents, rather than just their sum. A matrix over the discrete valuation ring O has nonzero invariant factors t^{e_i} up to units, together with some zero factors. If there are z zero factors, its cokernel modulo t^L has length

$$zL + \sum_i \min(e_i, L).$$

The first difference in L is $z + \#\{i : e_i \geq L\}$ for $L \geq 1$. For finite d, the formula already proved has first difference $n + h$ for $1 \leq L \leq d$, and n for $L > d$. Thus exactly n factors are zero and exactly h nonunit factors have exponent d; the remaining $n - h$ factors are units. For $d = 0$ there are no positive exponents. For $d = \infty$, the first difference is always $n + h$, so there are $n + h$ zero factors and no positive finite exponents. This proves every case of (9.85). □

Examples of the jet law

At level 15, take

$$a_3(t) = 1 + t^2, \qquad a_5(t) = 1 + t^5.$$

Here $(n, r, h) = (4, 2, 2)$ and $d = 2$. The reflection matrix has Smith entries $1, 1, t^2, t^2, 0, 0, 0, 0$, and its jet cokernel dimensions at $L = 1, 2, 3, 4$ are $6, 12, 16, 20$. A rank calculation at $t = 0$ alone would detect the first number, but not the exponent two.

At level 12, take $a_2(t) = t^3$ and $a_3(t) = 1 + t^5$. Then $a_2(a_2 - 1) = t^3(1 + t^3)$, so $d = 3$. Since $(n, h) = (2, 2)$, there are no unit Smith factors and the diagonal is $t^3, t^3, 0, 0$. Every jet of length at most three has trivial reflection matrix up to equivalence; later lengths detect the nonzero factors.

If the entire curve is contained in the exceptional locus, then $d = \infty$ and the dimension is $(n + h)L$. If any active scalar has a nonzero constant term, $d = 0$ and the dimension is nL. These are exact boundary cases of the same formula. In contrast, the *unreflected* distribution module always has dimension $\varphi(q)L$ at length L and never exhibits this jump.

9.8.5 Integral jets and why abelian freeness is insufficient

For a finite free $\mathbb{Z}$-algebra B, put $C_\pm(\mathcal{D}_B) = \mathcal{D}_q(B, a)/(c \mp 1)\mathcal{D}_q(B, a)$. The scalar Tate calculation extends to this coefficient ring.

Proposition 9.39. *With $\overline{B} = B/2B$ and the reduced active sequence $\overline{\boldsymbol{\delta}}$, the two integer-torsion submodules are*

$$\operatorname{Tor}_{\mathbb{Z}} C_+(\mathcal{D}_B) \simeq \bigoplus_{j \text{ odd}} H_j(K_{\overline{B}}(\overline{\boldsymbol{\delta}})), \qquad \operatorname{Tor}_{\mathbb{Z}} C_-(\mathcal{D}_B) \simeq \bigoplus_{j \text{ even}} H_j(K_{\overline{B}}(\overline{\boldsymbol{\delta}})).$$

These are isomorphisms of B-modules. Every torsion class is killed by two. For $q > 2$ and $T = \operatorname{rank}_{\mathbb{Z}} B$, the free abelian rank of both quotients is $T\varphi(q)/2$.

Proof. The underlying distribution lattice is free over $\mathbb{Z}$ by the all-ring basis theorem. Thus the torsion lemma still applies. In the fixed-point periodic complex, multiplication by two is injective on B and its free chain modules. Its adjacent two-term blocks resolve reduction modulo two, exactly as in (9.77); the bounded chain width gives the same parity sums as B-modules. Over $B \otimes \mathbb{Q}$, reflection eigenspaces are exact direct summands. The fixed-point Euler count in the balanced-rank proof is multiplied by T and remains zero, giving the asserted free ranks. This argument does not assume that the reflected torsion-free part is a free B-module. □

9.8.6 An exact Smith formula for arbitrary finite integral jets

Let

$$B_L = \mathbb{Z}[u]/(u^L), \quad L \geq 1, \qquad a_p(u) \in B_L,$$

and reduce coefficients modulo two. In $\overline{B}_L = \mathbb{F}_2[u]/(u^L)$, form the parameters (9.76). For an element f of this ring, define

$$\nu(f) = \min\{j : [u^j]f \neq 0\}, \qquad \nu(0) = L, \quad v = \min_{f \in \boldsymbol{f}} \nu(f).$$

Here $\boldsymbol{f} = \overline{\boldsymbol{\delta}}$ and $r(q)$ is the active-prime count in (9.69). For $q > 2$, the list is nonempty.

Theorem 9.40 (Minimum-valuation jet formula)**.** *For $q > 2$, the two reflected jet quotients have the following Smith decomposition as abelian groups:*

$$C_{\pm}(\mathcal{D}_{B_L}) \simeq \mathbb{Z}^{L\varphi(q)/2} \oplus (\mathbb{Z}/2\mathbb{Z})^{v\,2^{r(q)-1}}. \tag{9.87}$$

More precisely, their integer-torsion submodules are each isomorphic, as B_L-modules, to

$$\big(B_L/(2,u^v)\big)^{2^{r(q)-1}}. \tag{9.88}$$

The direct-sum decomposition in (9.87) is a statement over $\mathbb{Z}$, not a claim that the torsion-free part is free over B_L.

Proof. In the principal ideal ring $\overline{B}_L$, choose a parameter of least valuation. It is a unit times u^v unless every parameter is zero, which is the same argument with $v = L$. A change of basis in the free module defining the Koszul complex transforms the parameter list into $(u^v, 0, \dots, 0)$: first divide by its unit factor, then subtract its multiples from the other entries. All these are invertible operations over $\overline{B}_L$.

The complex is now the tensor product of the two-term complex given by multiplication by u^v and $r(q) - 1$ zero-differential exterior factors. Both the kernel and the cokernel of multiplication by u^v are isomorphic to $\overline{B}_L/(u^v)$; the kernel is the ideal $u^{L-v}\overline{B}_L$. Consequently

$$H_k\big(K_{\overline{B}_L}(\boldsymbol{f})\big) \simeq \big(\overline{B}_L/(u^v)\big)^{\binom{r(q)}{k}}.$$

Summing either degree parity in Theorem 9.39 proves (9.88). Each copy has $\mathbb{F}_2$-dimension v, so Lemma 9.32 gives the torsion multiplicity in (9.87).

The unreflected underlying lattice has rank $L\varphi(q)$. The alternating fixed-point trace used in Theorem 9.31 is multiplied by L, and still vanishes. Thus each rational eigenspace has dimension $L\varphi(q)/2$. This proves the free rank and completes the Smith calculation. □

Example 9.41 (Five jet coefficients at level fifteen). Take $L = 5$, $a_3(u) = 1+u^2$, and $a_5(u) = 1+2u+u^3$. The binary parameters are u^2, u^3, so $v = 2$ and $r(q) = 2$. Therefore

$$C_{\pm}(\mathcal{D}_{B_5}) \simeq \mathbb{Z}^{20} \oplus (\mathbb{Z}/2\mathbb{Z})^4.$$

The constant jets $a_3 = a_5 = 1$ would instead give ten two-torsion factors. The constant terms of the weights agree; their higher binary coefficients change the integral reflection structure.

Example 9.42 (The two-adic jet parameter matters). At $q = 12$, $L = 4$, let $a_2(u) = 1 + u$ and $a_3(u) = 1 + u^3$. The two parameters are u^3 and $u + u^2$, not simply $1 - a_3$ and $1 - a_2$. Thus $v = 1$, $r(q) = 2$, and

$$C_{\pm}(\mathcal{D}_{B_4}) \simeq \mathbb{Z}^8 \oplus (\mathbb{Z}/2\mathbb{Z})^2.$$

Although the parity of a *scalar* two-prime weight does not affect Theorem 9.31, a nonconstant two-prime jet can affect Theorem 9.40.

Why the free abelian part need not be jet-free

Take $q = 3$, $L = 2$, and $a_3 = 1 + u$. In the unreflected basis $e = e_0$, $f = e_{2/3}$, reflection satisfies

$$ce = e, \qquad cf = ue - f.$$

Hence

$$C_{+}(\mathcal{D}_{B_2}) = B_2 e \oplus B_2 f \big/ \langle ue - 2f \rangle. \tag{9.89}$$

As an abelian group this has free generators e, f and a torsion generator uf of order two. In its torsion-free quotient, multiplication by u sends e to $2f$ and f to zero. This is not a free rank-one B_2-module: the image of u in a free rank-one module is a primitive rank-one sublattice, whereas here the image is $2\mathbb{Z}f$. Thus a claim of B_L-freeness for the reflected torsion-free part would be false. By contrast, $C_- = B_2 f \oplus (B_2/(2,u))e$ in this example. Both signs have the same abelian Smith factors, but different module extensions over the jet ring.

Multivariate jets and arithmetic weights

For a finite free $\mathbb{Z}$-algebra B of rank L, the Koszul theorem remains an exact answer. If $q > 2$ and $h_k = \dim_{\mathbb{F}_2} H_k(K_{B/2B}(\boldsymbol{f}))$, then

$$\operatorname{Tor}_{\mathbb{Z}} C_+ \simeq (\mathbb{Z}/2\mathbb{Z})^{\sum_{k \text{ odd}} h_k}, \qquad \operatorname{Tor}_{\mathbb{Z}} C_- \simeq (\mathbb{Z}/2\mathbb{Z})^{\sum_{k \text{ even}} h_k}.$$

The two sums are equal: the Euler characteristic of the Koszul complex is $L(1-1)^{r(q)} = 0$. A closed minimum-valuation description is special to the one-variable principal ideal ring; it must not be transferred unchanged to a multivariate Artinian ring.

For actual positive integral polylogarithm orders, $A_p = p^{1-n}$ usually belongs to $\mathbb{Z}[1/q]$, rather than $\mathbb{Z}$. The corresponding coefficient ring must be stated. If q is even, inverting q also inverts two and removes the reflected two-torsion. For odd q, every p^{1-n} is one modulo two, and the ordinary maximal multiplicity persists over $\mathbb{Z}[1/q]$. These are statements about the chosen formal coefficient lattice, not about independence of the evaluated periods.

9.8.7 Separable multivariable jets: an exact product law

The one-variable minimum-valuation formula has a useful multivariable extension when the active parameters separate into coordinate powers. The general Koszul answer remains necessary for parameters that do not have this form. In particular, replacing several coordinate valuations by their minimum would give the wrong answer here.

Let $q > 2$, put $r = r(q)$, $n = \varphi(q)/2$, $h = 2^{r-1}$, and set

$$S = k[t_1, \dots, t_r]/(t_1^{L_1}, \dots, t_r^{L_r}), \qquad L_i \geq 1, \qquad \operatorname{char} k = 2.$$

Suppose the active sequence, after permutation and multiplication of individual entries by units, is $(t_1^{d_1}, \dots, t_r^{d_r})$, where $d_i \geq 1$; a zero entry is allowed by taking $d_i = \infty$. Write $v_i = \min(d_i, L_i)$ and $T = \prod_i L_i$.

Theorem 9.43 (Multivariable reflection product law)**.** *Under these hypotheses,*

$$\dim_k H_j(K_S(\boldsymbol{\delta})) = \binom{r}{j} \prod_{i=1}^{r} v_i, \qquad 0 \leq j \leq r, \tag{9.90}$$

$$\dim_k \mathcal{D}_q(S,a)/(c-1)\mathcal{D}_q(S,a) = nT + h \prod_{i=1}^{r} v_i. \tag{9.91}$$

The unreflected dimension is $\varphi(q)T$. For integral weights in $B = \mathbb{Z}[t_1, \dots, t_r]/(t_1^{L_1}, \dots, t_r^{L_r})$ whose reduced active sequence has the above form, both reflected coinvariants have the abelian-group decomposition

$$C_\pm(\mathcal{D}_B) \simeq \mathbb{Z}^{nT} \oplus (\mathbb{Z}/2\mathbb{Z})^{h \prod_i v_i}.$$

Their integer-torsion submodules are, more precisely, isomorphic as B-modules to

$$(B/(2, t_1^{v_1}, \dots, t_r^{v_r}))^h .$$

The integral assertion uses $k = \mathbb{F}_2$.

Proof. Multiplying an active entry by a unit and permuting entries induces an invertible change of the exterior generators of its Koszul complex. After that change, the complex is the tensor product over k of the two-term complexes

$$k[t_i]/(t_i^{L_i}) \xrightarrow{t_i^{d_i}} k[t_i]/(t_i^{L_i}).$$

Each has kernel and cokernel of dimension v_i, in degrees one and zero respectively. As a module over its truncated polynomial ring, each homology group is isomorphic to $k[t_i]/(t_i^{v_i})$; multiplication by $t_i^{L_i - v_i}$ gives the kernel isomorphism. This includes the zero-map case $v_i = L_i$.

The Künneth formula over the field k has no Tor corrections. In total degree j there is one tensor summand for each j-element subset of the r factors. Each summand has dimension $\prod_i v_i$ and is isomorphic, as an S-module, to $S/(t_1^{v_1}, \dots, t_r^{v_r})$. This proves (9.90) and its module refinement. The characteristic-two fixed-point theorem gives total reflection homology of dimension $2^r \prod_i v_i$. Since $\mathcal{D}_q(S, a)$ has dimension $2nT$, the square-zero rank identity (9.82) gives (9.91). For the integral statement apply the finite-free-algebra Tate formula of Proposition 9.39. The even and odd binomial sums are both 2^{r-1}; the balanced-rank calculation gives free rank nT. Finally a finitely generated abelian group splits into its free part and its torsion subgroup. No splitting as B-modules or freeness of its torsion-free part is claimed. □

Example 9.44 (Two independent deformation directions at level fifteen). Take $B = \mathbb{Z}[t, u]/(t^3, u^4)$, $a_3 = 1 + t^2$ and $a_5 = 1 + u^3$. Then $n = 4$, $r = 2$, $h = 2$, $T = 12$ and $(v_1, v_2) = (2, 3)$. Both integral signs give

$$C_{\pm}(\mathcal{D}_B) \simeq \mathbb{Z}^{48} \oplus (\mathbb{Z}/2\mathbb{Z})^{12},$$

and the characteristic-two reflected dimension is 60. The defect depends on the product $2 \cdot 3$, not on a single minimum. At an active two-prime coordinate one may take $a_2 = t_i^{d_i}$ or $a_2 = 1 + t_i^{d_i}$: in either case $a_2(a_2 - 1) = t_i^{d_i}(1 + t_i^{d_i})$, and the second factor is a unit.

The independent raw-matrix verifier expands the original prime rows and reflection rows in point/monomial coordinates over $\mathbb{F}_2$. It imports neither the normal-form generator nor the Koszul calculation. It checks 210 finite cases, including both two-prime branches, vanished coordinate powers, unequal jet lengths, and a three-prime level. A corruption control rejects omission of reflection rows. These computations corroborate the formula; the proof above establishes its full scope.

9.8.8 Exact cyclotomic identities and pole-cancelled spectral jets

Evaluating an integral certificate

For $x \in \mathbb{Q}/\mathbb{Z}$, write

$$\mathcal{L}_s(x) = \mathrm{Li}_s(e^{2\pi i x}).$$

At $x = 0$, this means $\zeta(s)$. For $x \neq 0$, the spectral function is entire. One way to see this, and to fix its continuation, is the finite residue-class formula

$$\mathcal{L}_s(a/q) = q^{-s} \sum_{u=1}^{q} e^{2\pi i a u/q} \zeta(s, u/q). \tag{9.92}$$

For $a \not\equiv 0 \pmod{q}$, the residues of the simple poles at $s = 1$ sum to zero. The standard meromorphic continuation of Hurwitz zeta has no other spectral poles [93].

For $\Re s > 1$, absolute convergence gives

$$\sum_{py=x} \mathcal{L}_s(y) = p^{1-s}\mathcal{L}_s(x). \tag{9.93}$$

Indeed the root sum cancels every series index not divisible by p; for index pk it contributes $pe^{2\pi ikx}$. Continuation then proves the meromorphic identity for every complex s. Thus an integral polynomial certificate becomes an exact analytic identity under

$$e_x \longmapsto \mathcal{L}_s(x), \qquad A_p \longmapsto p^{1-s}, \tag{9.94}$$

where real positive p fixes the exponential unambiguously. No integer relation search or floating-point acceptance threshold occurs in this step.

Proposition 9.45 (A removable endpoint coefficient)**.** *For $y \neq 0$, the coefficient $c_{y,0}(\mathbf{A})$ in its integral normal form vanishes at $A_p = 1$ for all p. Hence its polylogarithmic specialization cancels the simple pole of $\zeta(s)$ at $s = 1$.*

Proof. At ordinary weights, the functional taking e_0 to 1 and all other grid symbols to 0 annihilates every distribution row. For $x = 0$, both the root sum and e_x contain exactly one copy of e_0; for $x \neq 0$, neither does. Apply the functional to the normal form of e_y. The second assertion follows because $p^{1-s} = 1$ at $s = 1$ and the coefficient is analytic in s. □

The vanishing need only be first order. It is not legitimate to infer one vanishing factor for each prime merely from the number of prime variables in a raw-point normal form.

A three-row level-twelve identity

In the level-12 free module, define

$$F_{12} = e_{1/12} + e_{5/12} + (A_3 + 1)e_{3/4} - A_2A_3(A_2 - 1)e_0.$$

The following is an equality before taking the quotient:

$$F_{12} = r_{3,1/4} + A_3r_{2,1/2} + A_2A_3r_{2,0}. \tag{9.95}$$

For example, the first row contains $e_{1/12} + e_{5/12} + e_{3/4} - A_3e_{1/4}$; the next two rows cancel $e_{1/4}$ and then $e_{1/2}$. This verifies the certificate by direct inspection.

Corollary 9.46. *For every complex s, with the removable value used at $s = 1$,*

$$\mathcal{L}_s(1/12) + \mathcal{L}_s(5/12) + (3^{1-s} + 1)\mathcal{L}_s(3/4) = 6^{1-s}(2^{1-s} - 1)\zeta(s). \tag{9.96}$$

At $s = 1$, the right-hand side is $-\log 2$.

Proof. Evaluate (9.95) using (9.94). The value at one follows from $2^{1-s} - 1 = -(s-1)\log 2 + O((s-1)^2)$. □

A four-row level-fifteen identity

Set

$$U = \{1/15, 4/15, 7/15, 2/5, 4/5\}. \tag{9.97}$$

In the level-15 free module define

$$F_{15} = e_{8/15} - \sum_{u\in U} e_u - (A_3+1)e_{3/5} - (A_5+1)e_{2/3} - (1 - A_3A_5)e_0. \tag{9.98}$$

A short exact certificate is

$$F_{15} = r_{3,3/5} - r_{5,1/3} - A_5 r_{3,0} - r_{5,0}. \tag{9.99}$$

The roots in the first two rows are respectively

$$\{1/5, 8/15, 13/15\}, \qquad \{1/15, 4/15, 7/15, 2/3, 13/15\}.$$

The 13/15 terms cancel. The last two rows remove 1/3 and 1/5, and leave precisely (9.98). In particular the coefficient of e_0 in the normal form of $e_{8/15}$ is $1 - A_3A_5$, not $(A_3 - 1)(A_5 - 1)$.

Theorem 9.47 (All-order level-fifteen identity)**.** *For every complex spectral order s,*

$$\begin{aligned}\mathcal{L}_s(8/15) - \sum_{u\in U} \mathcal{L}_s(u) - (3^{1-s}+1)\mathcal{L}_s(3/5) - (5^{1-s}+1)\mathcal{L}_s(2/3)& \\ = (1 - 15^{1-s})\zeta(s).&\end{aligned} \tag{9.100}$$

Both sides extend holomorphically through $s = 1$, where the right-hand side is $\log 15$.

Proof. Apply (9.94) to the four-row identity. The endpoint value follows from $1 - 15^{1-s} = (s - 1)\log 15 + O((s-1)^2)$ and the residue one of $\zeta(s)$. □

The theorem supplies infinitely many exact identities, including ordinary positive integer weights, negative orders, and derivatives in spectral order. It is generated by classical distribution rather than by a new numerical relation conjecture.

A compatible level-30 normal form follows from the extra row $r_{2,1/15}$:

$$\begin{aligned}e_{1/30} = (A_2 - 1)e_{1/15} - (A_3+1)e_{3/5} - (A_5+1)e_{2/3}& \\ - e_{4/15} - e_{7/15} - e_{2/5} - e_{4/5} + (A_3A_5 - 1)e_0.&\end{aligned} \tag{9.101}$$

Its residual has the five-row certificate

$$r_{2,1/15} - r_{3,3/5} + r_{5,1/3} + A_5 r_{3,0} + r_{5,0}.$$

All rows can be read in the level-30 grid. This illustrates compatibility of the straightener with an enlarged level and a weight whose order-one specialization vanishes in a particular coefficient.

Every spectral derivative, with the pole term retained

Let

$$\mathcal{L}^{[j]}(x) = \frac{\partial^j}{\partial s^j}\mathcal{L}_s(x)\Big|_{s=1}, \qquad x \neq 0,$$

and write the Laurent expansion

$$\zeta(1+t)=\frac{1}{t}+\sum_{m\ge 0}\frac{(-1)^m\gamma_m}{m!}t^m. \tag{9.102}$$

For a positive prime p and $N\ge 0$, define the explicit finite operator

$$\mathcal{T}_{p,N}(x)=\mathcal{L}^{[N]}(x)+\sum_{j=0}^{N}\binom{N}{j}(-\log p)^j\mathcal{L}^{[N-j]}(x). \tag{9.103}$$

It is the Nth derivative of $(p^{1-s}+1)\mathcal{L}_s(x)$ at $s=1$.

Theorem 9.48 (A level-fifteen Stieltjes–polylogarithm jet family)**.** *For every integer $N\ge 0$, put $\ell=\log 15$. Then*

$$\begin{aligned}\mathcal{L}^{[N]}(8/15)-\sum_{u\in U}\mathcal{L}^{[N]}(u)-\mathcal{T}_{3,N}(3/5)-\mathcal{T}_{5,N}(2/3)\\ =(-1)^N\left\{\frac{\ell^{N+1}}{N+1}-\sum_{j=1}^{N}\binom{N}{j}\ell^j\gamma_{N-j}\right\}.\end{aligned} \tag{9.104}$$

The sum on the right is empty at $N=0$.

Proof. All nonzero-root terms of (9.100) are entire in s, so differentiate the holomorphic continuation N times. The weighted terms give (9.103) by Leibniz's rule. For the right side use

$$1-e^{-\ell t}=\sum_{j\ge 1}\frac{(-1)^{j+1}\ell^j}{j!}t^j$$

and multiply by (9.102). The pole term contributes $(-1)^N\ell^{N+1}/(N+1)$ to the Nth derivative. The regular Laurent coefficients contribute $(-1)^{N+1}\sum_{j=1}^{N}\binom{N}{j}\ell^j\gamma_{N-j}$. This is the stated formula. □

The first derivative gives the especially compact identity

$$\begin{aligned}\mathcal{L}^{[1]}(8/15)-\sum_{u\in U}\mathcal{L}^{[1]}(u)-2\mathcal{L}^{[1]}(3/5)-2\mathcal{L}^{[1]}(2/3)+\log 3\,\mathcal{L}^{[0]}(3/5)+\log 5\,\mathcal{L}^{[0]}(2/3)\\ =\gamma_0\log 15-\tfrac{1}{2}\log^2 15.\end{aligned} \tag{9.105}$$

For $N=2$, the scalar right-hand side is

$$\frac{\log^3 15}{3}-\gamma_0\log^2 15-2\gamma_1\log 15.$$

These terms would be wrong if one substituted $s=1$ into the coefficient of $\zeta(s)$ before cancelling its pole.

More generally, if an analytic coefficient $c(t)$ satisfies $c(0)=0$, then

$$\frac{d^N}{dt^N}\{c(t)\zeta(1+t)\}\Big|_{t=0}=\frac{c^{(N+1)}(0)}{N+1}+\sum_{j=0}^{N}\binom{N}{j}c^{(j)}(0)(-1)^{N-j}\gamma_{N-j}. \tag{9.106}$$

This coefficient identity gives a general implementation rule for every raw-point certificate, not just the displayed level.

An elementary endpoint and branch check

For $0 < x < 1$, the chosen branch gives

$$\mathcal{L}^{[0]}(x) = -\log(2\sin \pi x) + i\pi(1/2 - x).$$

Taking real parts of (9.100) at $s = 1$ gives

$$\frac{\prod_{u\in U}(2\sin \pi u)(2\sin(3\pi/5))^2(2\sin(2\pi/3))^2}{2\sin(8\pi/15)} = 15.$$

Since $\sin(8\pi/15) = \sin(7\pi/15)$, it simplifies to

$$(2\sin(\pi/15))(2\sin(4\pi/15))(2\sin(2\pi/5))^3(2\sin(\pi/5)) = 5. \tag{9.107}$$

All factors are positive, so exponentiating the real logarithmic identity introduces no ambiguity. The imaginary parts cancel by the displayed linear argument formula. This is an elementary specialization and branch check, not a claim of novelty for the trigonometric product itself.

A division-free Hurwitz–Stieltjes translation

The same polynomial certificates evaluate under

$$A_p = p^s, \qquad e_x \mapsto Z_x(s), \qquad Z_x(s) = \begin{cases} \zeta(s,x), & 0 < x < 1, \\ \zeta(s,1), & x = 0. \end{cases}$$

The distribution identity here is $\sum_{py=x} Z_y(s) = p^s Z_x(s)$; it follows by splitting the Hurwitz series into residue classes, including the endpoint convention. Let $c_b(t)$ be the normal-form coefficient of e_b for e_x after $A_p = p^{1+t}$, and put $C(t) = \sum_b c_b(t)$. Residue comparison gives $C(0) = 1$. If $\gamma_j(b^*)$ denotes the generalized Stieltjes coefficient with $b^* = b$ for $b \neq 0$ and $b^* = 1$ for $b = 0$, expansion of the exact normal form gives, for all $N \geq 0$,

$$\gamma_N(x^*) = \frac{(-1)^N C^{(N+1)}(0)}{N+1} + \sum_{b\in\mathcal{B}_q}\sum_{j=0}^{N}(-1)^j\binom{N}{j}c_b^{(j)}(0)\gamma_{N-j}(b^*). \tag{9.108}$$

To verify the sign, multiply each Laurent series by $c_b(t)$, compare t^N, and multiply by $(-1)^N N!$. The pole part is $C(t)/t$ and contributes the first term. Thus even the underived constants require a coefficient derivative. This is the raw-basis version of the manuscript's pole-corrected jet philosophy, now with division-free integral certificates for the coefficients.

Every coefficient derivative in (9.108) is a finite integer combination of products of logarithms of primes and integer prime powers. There is no need to differentiate a solved matrix inverse or to redo a relation search separately at each derivative order.

Level-twelve derivatives and cyclotomic products

For the coefficient $c(t) = 12^{-t} - 6^{-t}$ in (9.96), set $D_j = (-\log 12)^j - (-\log 6)^j$. The finite-part rule (9.106) yields, for every $N \geq 0$,

$$\mathcal{L}^{[N]}(1/12) + \mathcal{L}^{[N]}(5/12) + \mathcal{T}_{3,N}(3/4) = \frac{D_{N+1}}{N+1} + \sum_{j=1}^{N}\binom{N}{j}D_j(-1)^{N-j}\gamma_{N-j}. \tag{9.109}$$

For example its first derivative has right side $(\log^2 12 - \log^2 6)/2 - \gamma_0 \log 2$. At order one of the polylogarithm, exponentiating the exact logarithmic identities gives, with $\zeta_q = e^{2\pi i/q}$,

$$(1-\zeta_{12})(1-\zeta_{12}^5)(1-\zeta_{12}^9)^2 = 2, \tag{9.110}$$

$$(1-\zeta_{30})(1-\zeta_{30}^8)(1-\zeta_{30}^{12})(1-\zeta_{30}^{14})(1-\zeta_{30}^{18})^2(1-\zeta_{30}^{20})^2(1-\zeta_{30}^{24}) = 15. \tag{9.111}$$

The level-thirty identity is the analytic specialization of (9.101); its endpoint coefficient has removable value $-\log 15$. Exponentiation proves the product without a branch ambiguity, while the converse would lose an additive multiple of $2\pi i$. The preserved independent polynomial verifier also proves that each product polynomial minus its displayed value is divisible by the corresponding cyclotomic polynomial.

9.9 Primitive-grid determinants, local defects and finite descent

The polynomial character normal form above remains valid over arbitrary complex coefficient algebras. Here identify the prime weights A_p with t_p, write $\mathcal{D}_q$ for that quotient, and put $v_{m,\chi} = Y_{m,\chi}$, $b_\chi = Y_{f,\chi}$ for $f = \mathrm{cond}(\chi)$. The coefficient $E_{m,\chi}$ is $A_{m,\chi}$ of Theorem 9.15. Thus

$$[v_{m,\chi}] = E_{m,\chi} b_\chi. \tag{9.112}$$

No Euler-factor division is used. Let $U_q = \{u \bmod q : (u,q) = 1\}$, and use $r = 0$ for the Hurwitz endpoint $x_r = 1$. Distinguish the rank of the full quotient from the image of its primitive-grid coordinate map.

Conductor coordinates are polynomial and globally valid, but it is often more convenient to use the $\varphi(q)$ primitive-grid points $e_{u/q}$, $u \in U_q$. Fourier-transforming those points gives $v_{q,\chi}$. By Theorem 9.15, the map from their formal span into $\mathcal{D}_q$ is diagonal in character coordinates:

$$v_{q,\chi} \longmapsto E_{q,\chi} b_\chi. \tag{9.113}$$

Its determinant in these specified Fourier and conductor bases is $\Delta_q = \prod_{\chi \bmod q} E_{q,\chi}$.

For $p^e \| q$, write

$$M_p = q/p^e, \qquad h_p = \mathrm{ord}_{M_p}(p),$$

with $h_p = 1$ when $M_p = 1$.

Theorem 9.49 (Determinant factorization)**.** *One has the polynomial identity*

$$\boxed{\Delta_q(\mathbf{A}) = \prod_{p^e \| q} A_p^{\varphi(M_p)(p^{e-1}-1)} \big(A_p^{h_p} - 1\big)^{\varphi(M_p)/h_p}.} \tag{9.114}$$

Thus primitive-grid coordinates form a basis precisely when all the factors in (9.114) are invertible in a field specialization. Their failure does not change the rank of $\mathcal{D}_q$.

Proof. Fix $p^e \| q$ and inspect the p factor of $\prod_\chi E_{q,\chi}$. Characters whose conductor is not divisible by p are exactly the characters of U_{M_p}. Their additional factors are $A_p - \chi(p)$. In the character group of U_{M_p}, the values $\chi(p)$ run through every h_pth root of unity with multiplicity $\varphi(M_p)/h_p$. This follows by duality from the fact that p has order h_p in U_{M_p}. Their product is therefore $(A_p^{h_p} - 1)^{\varphi(M_p)/h_p}$.

It remains to count the pure powers of A_p. If $a = v_p(f)$, the exponent contributed by one character is $e - \max(1, a)$. Writing this as a sum of indicators gives

$$\sum_{\chi \bmod q} \big(e - \max(1, v_p(f_\chi))\big) = \sum_{j=1}^{e-1} \#\{\chi : v_p(f_\chi) \le j\}.$$

The count in the last sum is $\varphi(p^j)\varphi(M_p)$, since these are exactly the characters induced from level $p^j M_p$. Hence the exponent is $\varphi(M_p)\sum_{j=1}^{e-1}\varphi(p^j) = \varphi(M_p)(p^{e-1}-1)$. Multiply over the primes. □

Example 9.50 (Level twelve). Let χ_3 and χ_4 be the nonprincipal quadratic characters of conductors three and four, and let $\chi_{12} = \chi_3\chi_4$. Their conductor generators are

$$b_3 = e_{1/3} - e_{2/3}, \qquad b_4 = e_{1/4} - e_{3/4},$$
$$b_{12} = e_{1/12} - e_{5/12} - e_{7/12} + e_{11/12}.$$

For $r \in U_{12}$, the normal form reads

$$[e_{r/12}] = \frac{1}{4}\big[A_2(A_2-1)(A_3-1)e_0 + \chi_3(r)A_2(A_2+1)b_3 + \chi_4(r)(A_3+1)b_4 + \chi_{12}(r)b_{12}\big]. \tag{9.115}$$

Consequently

$$\Delta_{12} = A_2^2(A_2^2-1)(A_3^2-1).$$

At $A_2 = A_3 = 1$, the primitive-grid span misses the principal conductor coordinate, even though the full quotient still has rank four.

9.9.1 Local jet defects of the primitive-grid map

Specialize to $A_p(s) = p^s$, and let $s = s_0 + t$. All exponential factors are nonzero. A factor $p^s - \chi(p)$ vanishes precisely when $p^{s_0} = \chi(p)$, and any such zero is simple because its derivative is $p^{s_0}\log p \neq 0$.

Theorem 9.51 (Local Smith factors and primitive-jet ranks)**.** *Define*

$$\rho_\chi(s_0) = \#\{p \mid q : p \nmid f_\chi,\ p^{s_0} = \chi(p)\}. \tag{9.116}$$

Over $\mathbb{C}[[t]]$, the diagonal factors of the primitive-grid map are units times $t^{\rho_\chi(s_0)}$. Consequently its rank on order-N truncated jets, as a complex linear map, is

$$\sum_{\chi \bmod q} \max\{N+1-\rho_\chi(s_0), 0\}. \tag{9.117}$$

At $s_0 = 0$, this is determined by $\rho_\chi(0) = \#\{p \mid q : p \nmid f_\chi,\ \chi(p) = 1\}$.

Proof. Factor $E_{q,\chi}(s_0+t)$ into its simple vanishing factors and a unit. The primitive Fourier matrix is constant and invertible, so these are also the local invariant factors, up to permutation and units. Multiplication by t^r on $\mathbb{C}[t]/(t^{N+1})$ has rank $\max(N+1-r, 0)$. Summing over the diagonal coordinates proves the formula. This is the rank of a *coordinate inclusion map*, not the rank of the distribution relation matrix. □

Corollary 9.52 (Multiple-prime resonances are central)**.** *If $s_0 \neq 0$, at most one prime can satisfy $p^{s_0 h_p} = 1$. At a nonzero resonant point for that prime p, the primitive-grid rank defect is $\varphi(M_p)/h_p$, and every affected local factor has order one. The order-N inclusion rank is then $(N+1)\varphi(q) - \varphi(M_p)/h_p$ for all $N \geq 0$.*

Proof. A resonance forces s_0 to be purely imaginary. Resonances at distinct primes would make $\log p/\log \ell$ rational, because each phase is a rational multiple of 2π. Unique prime factorization rules this out unless $s_0 = 0$. For the one resonant prime, the multiplicity count in Theorem 9.49 gives the number of affected characters. Apply Theorem 9.51. □

Example 9.53 (Two exact jet-rank profiles). At level 12 and $s_0 = 0$, the principal character has $\rho = 2$ and the other three characters have $\rho = 0$. For $N = 0, 1, 2$, the primitive-jet ranks are $3, 6, 10$, while the full quotient dimensions are $4, 8, 12$. At level 21, the principal character has $\rho = 2$, the character induced from conductor three has $\rho = 1$, and the other ten characters have $\rho = 0$. The inclusion ranks are $10, 21, 33$, compared with full quotient dimensions $12, 24, 36$. Indeed,

$$\Delta_{21} = (A_3^6 - 1)(A_7 - 1)^2.$$

Exact character-phase certificates for these examples are supplied.

9.10 Explicit primitive-grid reduction coefficients

Away from the exceptional locus, the conductor normal form can be inverted to express all grid points through the primitive ones. For Hurwitz weights there is a particularly concrete answer that avoids character-field arithmetic.

Let $\mathcal{S}_q$ be the multiplicative semigroup of positive integers whose prime factors divide q, including 1. For $\operatorname{Re} s > 0$, define

$$C_{r,u}(s) = \sum_{\substack{d \in \mathcal{S}_q \\ du \equiv r \ (q)}} d^{-s}, \qquad 0 \le r < q, \quad u \in U_q. \tag{9.118}$$

The series is absolutely convergent, since $\sum_{d \in \mathcal{S}_q} d^{-\sigma} = \prod_{p|q} (1 - p^{-\sigma})^{-1}$ for $\sigma > 0$.

Theorem 9.54 (Primitive-grid extension). *For $\operatorname{Re} s > 0$, the $q \times \varphi(q)$ matrix $C(s)$ satisfies every prime and divisor distribution relation with weights p^s, and its primitive rows form the identity matrix. It is the unique extension matrix from primitive-grid data. In particular,*

$$\zeta(s, r/q) = \sum_{u \in U_q} C_{r,u}(s) \zeta(s, u/q) \tag{9.119}$$

with the endpoint convention for $r = 0$, initially for $\operatorname{Re} s > 1$ and then meromorphically wherever the coefficients are defined.

Proof. For a primitive row r, the congruence $du \equiv r \pmod q$ forces d to be prime to q, hence $d = 1$. This gives the identity submatrix. For a prime distribution row indexed by $a \equiv 0 \pmod p$,

$$\sum_{pr \equiv a \ (q)} C_{r,u}(s) = \sum_{\substack{d \in \mathcal{S}_q \\ pdu \equiv a \ (q)}} d^{-s} = p^s \sum_{\substack{d' \in \mathcal{S}_q \\ d'u \equiv a \ (q)}} (d')^{-s}.$$

In the last step $d' = pd$; conversely every d' in the last sum is divisible by p, because u is a unit and a is divisible by p. The last expression is $p^s C_{a,u}(s)$. Prime sufficiency gives all divisor rows. Theorem 9.15 and the identity submatrix show that these $\varphi(q)$ independent columns exhaust the solution space. Hurwitz zeta satisfies the same distribution equations by regrouping its absolutely convergent defining series, proving (9.119). □

9.10.1 A finite formula, including negative spectral parameters

For $0 \le r < q$, let

$$g = \gcd(r, q), \quad m = q/g, \quad a = r/g, \quad P_r = \{p \mid q : p \nmid m\}.$$

When $r = 0$, this means $g = q$, $m = 1$, $a = 0$. For $p \in P_r$, put $h_{p,m} = \operatorname{ord}_m(p)$, with order one modulo one.

Theorem 9.55 (Finite residue formula)**.** *The meromorphic continuation of* (9.118) *is*

$$C_{r,u}(s) = g^{-s} \frac{\displaystyle\sum_{0\le b_p<h_{p,m}\ (p\in P_r)} \ \prod_{p\in P_r} p^{-sb_p}}{\displaystyle\prod_{p\in P_r}(1-p^{-sh_{p,m}})}, \qquad u\prod_{p\in P_r} p^{b_p}\equiv a\ (m) \tag{9.120}$$

Empty products have their usual value one. In particular,

$$C_{0,u}(s) = q^{-s}\prod_{p\mid q}(1-p^{-s})^{-1}, \tag{9.121}$$

independently of u.

Proof. The congruence in (9.118) forces the valuation of d at an unsaturated prime, one dividing m, to be exactly $v_p(g)$. At a saturated prime, one not dividing m, it is at least $v_p(g)$. Thus $d = g\prod_{p\in P_r} p^{c_p}$ with $c_p \ge 0$, and the remaining congruence is $u\prod p^{c_p} \equiv a \pmod m$. Write $c_p = b_p + h_{p,m}j_p$ and sum the independent geometric series in j_p. This proves the formula in $\operatorname{Re} s > 0$ and hence its meromorphic continuation. The endpoint has trivial congruence modulo one. □

Formula (9.120), *not* the divergent smooth-number series, is the definition used at negative real s. It has no pole at any nonzero real s, and the continued distribution identities remain valid there. At a zero of a displayed denominator, one should use the polynomial conductor normal form rather than cancel poles numerically. Some individual coefficients may have removable singularities, but Theorem 9.49 gives the criterion for the entire primitive-grid coordinate system.

Corollary 9.56 (Exact support and signs)**.** *Let H_r be the subgroup of U_m generated by the classes of the primes in P_r. For every nonzero real s,*

$$C_{r,u}(s) \ne 0 \quad\Longleftrightarrow\quad au^{-1} \in H_r.$$

The number of supported primitive coordinates is

$$\frac{\varphi(q)}{\varphi(m)}|H_r|. \tag{9.122}$$

All supported coefficients are positive for $s > 0$. For $s < 0$, they all have sign $(-1)^{|P_r|}$.

Proof. The numerator of (9.120) is a sum of positive real terms and is nonempty exactly under the subgroup condition. The reduction map $U_q \to U_m$ has fibers of cardinality $\varphi(q)/\varphi(m)$, giving (9.122). For $s > 0$ every denominator factor is positive. For $s < 0$ each of the $|P_r|$ denominator factors is negative, while the numerator and g^{-s} are positive. □

Corollary 9.57 (Complete monotonicity and normalization)**.** *For $s > 0$ and $j \ge 0$, put*

$$W^{[j]}_{r,u}(s) = (-1)^j \frac{d^j}{ds^j} C_{r,u}(s) = \sum_{\substack{d\in\mathcal{S}_q \\ du\equiv r\ (q)}} \frac{(\log d)^j}{d^s}. \tag{9.123}$$

These quantities are nonnegative. Moreover, $\sum_u C_{r,u}(1) = 1$ for every r, and $0 \le C_{r,u}(k) \le 1$ for integer $k \ge 1$.

Proof. The logarithmically weighted semigroup series converges locally uniformly in $\operatorname{Re} s > 0$, so termwise differentiation is valid. The constant grid vector satisfies the distribution equations at $s = 1$, and its primitive entries are all one. Uniqueness of extension proves the row-sum identity. Monotonicity gives $C_{r,u}(k) \le C_{r,u}(1) \le 1$. □

Example 9.58 (Two reductions at denominator six). The nontrivial coefficients for $r = 2$ are

$$C_{2,1}(s) = \frac{2^s}{2^{2s} - 1}, \qquad C_{2,5}(s) = \frac{1}{2^{2s} - 1}.$$

Thus, meromorphically,

$$\zeta(s, 1/3) = \frac{2^s \zeta(s, 1/6) + \zeta(s, 5/6)}{2^{2s} - 1}, \tag{9.124}$$

$$\zeta(s, 1/2) = \frac{\zeta(s, 1/6) + \zeta(s, 5/6)}{3^s - 1}. \tag{9.125}$$

At $s = 1$, the coefficients in (9.124) are $2/3$ and $1/3$. At $s = -1$ they are $-2/3$ and $-4/3$, illustrating the sign theorem without using a divergent series. The poles at $s = 0$ reflect a singular primitive basis, not a failure of the underlying distribution relations.

9.10.2 Denominator certificates and construction cost

Let M_p also denote, in this paragraph only, the pushforward matrix on grid vectors defined by

$$(M_p f)_r = \sum_{pu \equiv r \ (q)} f_u.$$

To avoid confusion with the integer complement in Section 9.9, write that complement as $q_p = q/p^e$ here and put $h_p = \operatorname{ord}_{q_p}(p)$. Multiplication of residues gives $M_p^{e+h_p} = M_p^e$. For $z = p^{-s}$,

$$(I - zM_p)^{-1} = \sum_{j=0}^{e-1} z^j M_p^j + \frac{z^e}{1 - z^{h_p}} \sum_{j=0}^{h_p - 1} z^j M_p^{e+j}. \tag{9.126}$$

This follows first from a geometric series for $|z| < 1$, then as a rational matrix identity. The commuting product of these inverses, applied to vectors supported on U_q, is the extension $C(s)$.

Proposition 9.59 (A uniform rational denominator)**.** *For integer $k \ge 1$, every reduced denominator of $C_{r,u}(k)$ divides*

$$D_q(k) = \prod_{p^e \| q} p^{k(e-1)} \bigl(p^{kh_p} - 1\bigr). \tag{9.127}$$

Its bit length is $O(kq \log q)$. The coefficients of the raw derivative $C_{r,u}^{(j)}(k)$, viewed as a polynomial in the symbols $\log p$, have denominators dividing $D_q(k)^{j+1}$.

Proof. Every scalar coefficient on the right of (9.126) has denominator dividing $p^{k(e-1)}(p^{kh_p} - 1)$. Multiplication over primes proves the first claim. Since $h_p \le q$ and $\sum_{p|q} \log p \le \log q$, its logarithm is at most a constant times $kq \log q$. The matrix product can be written as a ratio of integer polynomials in p^s, with denominator obtained from (9.127) by replacing k with s. Repeated differentiation introduces polynomial factors in $\log p$ and raises the denominator power by at most one each time. For normalized Taylor coefficients, an additional factor $j!$ may be needed. □

The implementation computes (9.120) by convolution on residue classes modulo m, separately for each $m \mid q$. At a fixed m, each prime update takes at most $\varphi(m)^2$ rational arithmetic operations. A complete matrix can therefore be constructed in

$$O\left(\omega(q)\sum_{m|q}\varphi(m)^2 + q\varphi(q)\right) = O\big(\omega(q)q^2\big)$$

rational arithmetic operations, using $\sum_{m|q}\varphi(m) = q$. This is an arithmetic-operation bound, not a constant-cost bit model or a claim of optimality. The denominator estimate supplies a conservative size control at positive integer orders.

9.11 All-orders Hurwitz–Stieltjes identities

Write the Laurent expansion as

$$\zeta(1+t,x) = \frac{1}{t} + \sum_{n\ge 0}\frac{(-1)^n}{n!}\gamma_n(x)t^n, \qquad x > 0. \tag{9.128}$$

Here $\gamma_n^{(k)}$ always means k derivatives in x; a derivative on $C(s)$ is in the spectral variable. The coefficient identities in this section are analytic theorems, distinct from formal rank claims.

Theorem 9.60 (Derived Stieltjes reduction)**.** *For $k \ge 1$, $n \ge 0$, and $x_r = r/q$ with the endpoint convention,*

$$\boxed{\gamma_n^{(k)}(x_r) = \sum_{u\in U_q}\sum_{j=0}^{n}\binom{n}{j}W_{r,u}^{[j]}(k+1)\,\gamma_{n-j}^{(k)}(u/q).} \tag{9.129}$$

All the coefficient functions are given by the finite formula (9.120) and its derivatives.

Proof. Differentiating Hurwitz zeta in its parameter yields

$$\partial_x^k\zeta(1+t,x) = (-1)^k(1+t)_k\zeta(k+1+t,x).$$

The pole in (9.128) is independent of x, so for $k \ge 1$ the left side is $\sum_n(-1)^n\gamma_n^{(k)}(x)t^n/n!$. Apply (9.119) at spectral parameter $k+1+t$ and multiply by the common scalar $(-1)^k(1+t)_k$. Taylor multiplication gives (9.129), because $(-1)^jC_{r,u}^{(j)} = W_{r,u}^{[j]}$. □

This proof also explains why the harmonic-number master identity in Chapter 8 is compatible with the same reduction matrix at every Stieltjes order: the rising factorial is common to every rational argument. It does not create new independent distribution coordinates.

Theorem 9.61 (Underived Stieltjes reduction, with the pole term)**.** *For $n \ge 0$,*

$$\boxed{\gamma_n(x_r) = \sum_{u\in U_q}\sum_{j=0}^{n}\binom{n}{j}W_{r,u}^{[j]}(1)\gamma_{n-j}(u/q) - \frac{1}{n+1}\sum_{u\in U_q}W_{r,u}^{[n+1]}(1).} \tag{9.130}$$

Proof. Apply (9.119) at $s = 1+t$. The coefficients are analytic there. Multiplying their Taylor expansions by (9.128), the coefficient of t^n receives the usual product terms and also $\sum_u C_{r,u}^{(n+1)}(1)/(n+1)!$ from the pole $1/t$. Multiplication by $(-1)^n n!$ converts that additional term to the negative last term of (9.130). The coefficient of $1/t$ agrees because $\sum_u C_{r,u}(1) = 1$. □

Omitting the last term would already give an incorrect formula for γ_0. The point is not a rank correction: a known scalar right side does not change a homogeneous rank. It is a correction to the actual analytic value identity.

Example 9.62 (An explicit first-order identity). From (9.124), direct differentiation at $s = 1$ gives

$$W_{2,1}^{[1]}(1) = \frac{10}{9}\log 2, \qquad W_{2,5}^{[1]}(1) = \frac{8}{9}\log 2.$$

Consequently

$$\gamma_0(1/3) = \frac{2}{3}\gamma_0(1/6) + \frac{1}{3}\gamma_0(5/6) - 2\log 2. \tag{9.131}$$

At $s = 2$, the analogous coefficients give

$$\begin{aligned}\gamma_1'(1/3) = \frac{4}{15}\gamma_1'(1/6) + \frac{1}{15}\gamma_1'(5/6) \\ + \frac{68\log 2}{225}\gamma_0'(1/6) + \frac{32\log 2}{225}\gamma_0'(5/6).\end{aligned} \tag{9.132}$$

Since $\gamma_0' = -\psi'$, the signs can also be checked against the parameter-differentiated digamma multiplication identity. The verification program checks (9.130) through $n = 2$ and (9.132) by numerical parameter differentiation of Stieltjes functions.

9.12 Finite conductor descent for all Stieltjes layers

For a primitive character χ of conductor $f \mid d$, define

$$G_{d,\chi}^{(n,k)} = \sum_{a \in \mathcal{U}_d} \chi(a)\gamma_n^{(k)}(a/d). \tag{9.133}$$

Set

$$R(d,f) = \prod_{\substack{p \mid d \\ p \nmid f}} p, \qquad m_r = \frac{d}{fr} \quad (r \mid R(d,f)). \tag{9.134}$$

Each m_r is a positive integer. Upon substituting $t_p = p^s$, (9.36) becomes the entire exponential polynomial

$$\begin{aligned}A_{d,f,\chi}(s) &= \left(\frac{d}{f}\right)^s \prod_{\substack{p \mid d \\ p \nmid f}} (1 - \chi(p)p^{-s}) \\ &= \sum_{r \mid R(d,f)} \mu(r)\chi(r)m_r^s.\end{aligned} \tag{9.135}$$

In particular,

$$A_{d,f,\chi}^{(j)}(s) = \sum_{r \mid R(d,f)} \mu(r)\chi(r)m_r^s \log^j m_r. \tag{9.136}$$

Theorem 9.63 (All-index, all-parameter-order descent)**.** *For $n \geq 0$, $k \geq 1$ and $f \mid d$,*

$$\begin{aligned}G_{d,\chi}^{(n,k)} = \sum_{r \mid R(d,f)} \mu(r)\chi(r)m_r^{k+1} \\ \cdot \sum_{j=0}^{n} (-1)^j \binom{n}{j} \log^j m_r \, G_{f,\chi}^{(n-j,k)}.\end{aligned} \tag{9.137}$$

No Euler-factor denominator occurs. Equivalently,

$$G_{d,\chi}^{(n,k)} = \sum_{j=0}^{n} (-1)^j \binom{n}{j} A_{d,f,\chi}^{(j)}(k+1) G_{f,\chi}^{(n-j,k)}. \tag{9.138}$$

Proof. For $\operatorname{Re} s > 1$, sorting a Dirichlet series by residue classes gives

$$\sum_{a \in \mathcal{U}_d} \chi(a)\zeta(s, a/d) = d^s L(s, \chi_d),$$

where χ_d is the induced character modulo d. Removing the extra Euler factors from its primitive L-series yields

$$L(s, \chi_d) = L(s, \chi) \prod_{\substack{p \mid d \\ p \nmid f}} (1 - \chi(p)p^{-s}).$$

Thus the character sum at d is $A_{d,f,\chi}(s)$ times the character sum at f. This is also the analytic realization of Theorem 9.15.

Insert $s = k + 1 + \varepsilon$, multiply by the common factor $(-1)^k(1+\varepsilon)_k$, and use (9.1). Comparison of $(-1)^n \varepsilon^n / n!$ gives (9.138); the sign $(-1)^j$ comes from the Stieltjes Laurent convention. Finally apply (9.136). □

9.12.1 Concrete second-index identities at level twelve

The multipliers are

$$A_{12,3,\chi_3}(s) = 4^s + 2^s, \qquad A_{12,4,\chi_4}(s) = 3^s + 1.$$

Therefore, at $n = 2$, $k = 1$,

$$G_{12,\chi_3}^{(2,1)} = 20 G_{3,\chi_3}^{(2,1)} - 72 \log 2 \, G_{3,\chi_3}^{(1,1)} + 68 \log^2 2 \, G_{3,\chi_3}^{(0,1)}, \tag{9.139}$$

$$G_{12,\chi_4}^{(2,1)} = 10 G_{4,\chi_4}^{(2,1)} - 18 \log 3 \, G_{4,\chi_4}^{(1,1)} + 9 \log^2 3 \, G_{4,\chi_4}^{(0,1)}. \tag{9.140}$$

For instance,

$$G_{12,\chi_3}^{(2,1)} = \gamma_2'(1/12) - \gamma_2'(5/12) + \gamma_2'(7/12) - \gamma_2'(11/12),$$

and $G_{3,\chi_3}^{(2,1)} = \gamma_2'(1/3) - \gamma_2'(2/3)$. These are exact identities between explicitly specified signed sums; they require no integer-relation search. The two identities were also checked numerically using independent finite Hurwitz character sums.

9.12.2 The undifferentiated layer and its pole correction

For $k = 0$, a principal character sum still contains the Hurwitz pole. The same formula without a correction would be wrong.

Proposition 9.64 (The $k = 0$ correction)**.** *For nonprincipal primitive χ the formula* (9.138) *remains valid with $k = 0$. For $f = 1$, put $A_d(s) = A_{d,1,1}(s)$. Then*

$$G_{d,1}^{(n,0)} = \sum_{j=0}^{n} (-1)^j \binom{n}{j} A_d^{(j)}(1) \gamma_{n-j}(1) + \frac{(-1)^n}{n+1} A_d^{(n+1)}(1). \tag{9.141}$$

Proof. For a nonprincipal character, the sum of its values over units is zero, so the $1/\varepsilon$ residues cancel. Coefficient comparison is unchanged. For the principal conductor, multiply

$$A_d(1+\varepsilon)\left(\frac{1}{\varepsilon}+\sum_{n\ge 0}\frac{(-1)^n\gamma_n(1)}{n!}\varepsilon^n\right).$$

The pole term contributes $A_d^{(n+1)}(1)/(n+1)!$ to the coefficient of ε^n. Multiplying by $(-1)^n n!$ gives the correction in (9.141). The residue itself is $A_d(1) = d\prod_{p\mid d}(1-p^{-1}) = \varphi(d)$, as required. □

For the non-character multiplication formula this specializes to

$$\sum_{j=0}^{d-1}\gamma_n\left(\frac{a+j}{d}\right) = d\sum_{r=0}^{n}\binom{n}{r}(-\log d)^r\gamma_{n-r}(a) + \frac{(-1)^n d\log^{n+1} d}{n+1}. \tag{9.142}$$

The inhomogeneous term is the residue contribution, not an optional normalization.

9.13 Cyclotomic polylogarithm traces

The same finite module applies to a second evaluation. Put $\zeta_m = \exp(2\pi\mathrm{i}/m)$ and evaluate $e_x \mapsto \mathrm{Li}_s(\mathrm{e}^{2\pi\mathrm{i}x})$. For $\operatorname{Re} s > 1$, grouping the defining series by multiples of p gives

$$\sum_{py=x}\mathrm{Li}_s(\mathrm{e}^{2\pi\mathrm{i}y}) = p^{1-s}\,\mathrm{Li}_s(\mathrm{e}^{2\pi\mathrm{i}x}).$$

Thus the weights are now $A_p = p^{1-s}$, and analytic continuation extends the identity wherever the individual expressions are defined.

For a primitive character χ of conductor $f \mid q$, define

$$\mathcal{P}_{q,\chi}(s) = \sum_{u\in U_q}\chi(u)\,\mathrm{Li}_s(\zeta_q^u), \qquad \tau(\chi) = \sum_{a\in U_f}\chi(a)\zeta_f^a. \tag{9.143}$$

At conductor one take $\tau(\mathbf{1}) = 1$ and $L(s,\mathbf{1}) = \zeta(s)$.

Theorem 9.65 (Conductor-lifted trace identity)**.** *As a meromorphic identity, and with removable singularities interpreted by continuation,*

$$\boxed{\mathcal{P}_{q,\chi}(s) = (q/f)^{1-s}\prod_{\substack{p\mid q\\ p\nmid f}}\bigl(1-\chi(p)p^{s-1}\bigr)\tau(\chi)L(s,\overline{\chi}).} \tag{9.144}$$

For every $q > 1$, the left side is an entire function of the spectral variable s.

Proof. At primitive conductor $f > 1$, absolute convergence and the primitive Gauss-sum identity give

$$\sum_{a\in U_f}\chi(a)\,\mathrm{Li}_s(\zeta_f^a) = \sum_{n\ge 1}\frac{\sum_a \chi(a)\zeta_f^{an}}{n^s} = \tau(\chi)L(s,\overline{\chi}).$$

The Gauss-sum convention, including the complex conjugate on χ, is the one in [80]. Applying (9.112) with $A_p = p^{1-s}$ multiplies this primitive trace by $E_{q,\chi}$. Extracting p^{1-s} from each first-appearance factor gives precisely the factor in (9.144). The conductor-one case follows directly from $\mathrm{Li}_s(1) = \zeta(s)$ and the same polynomial identity.

For completeness, when $z = \zeta_q^u \neq 1$, regrouping by residues gives

$$\mathrm{Li}_s(z) = q^{-s} \sum_{a=1}^{q} z^a \zeta(s, a/q). \tag{9.145}$$

Hurwitz zeta has only its simple pole at $s = 1$, whose coefficient in this finite sum is $\sum_a z^a = 0$. Hence $\mathrm{Li}_s(z)$ is entire in s. All arguments in the primitive level-q trace differ from one. □

Remark 9.66 (The character phase is consequential). For the order-four character modulo five with $\chi(2) = \mathrm{i}$, the level-ten lifting factor is $2^{1-s} - \mathrm{i}$, not $2^{1-s} + \mathrm{i}$. A complex-character numerical check is included to catch an accidental interchange of χ and $\overline{\chi}$. Real quadratic examples alone would not detect that error.

9.13.1 Exact nonprincipal vanishing orders

For a nonprincipal primitive χ, set

$$\rho_\chi = \#\{p \mid q : p \nmid f,\ \chi(p) = 1\}.$$

These are the same conductor statistics that controlled the primitive-grid jet defects at Hurwitz order zero.

Theorem 9.67 (Character trace zeros at order one)**.** *The entire function $\mathcal{P}_{q,\chi}(s)$ has a zero of exact order ρ_χ at $s = 1$, where order zero means a nonzero value. All lower derivatives vanish, and*

$$\begin{aligned}\mathcal{P}_{q,\chi}^{(\rho_\chi)}(1) = (-1)^{\rho_\chi} \rho_\chi!\, \tau(\chi) L(1, \overline{\chi}) \\ \times \prod_{\substack{p \mid q,\ p \nmid f \\ \chi(p) = 1}} \log p \prod_{\substack{p \mid q,\ p \nmid f \\ \chi(p) \neq 1}} (1 - \chi(p)).\end{aligned} \tag{9.146}$$

Proof. Each factor with $\chi(p) = 1$ in (9.144) is $1 - p^{s-1} = -(s-1)\log p + O((s-1)^2)$. All other displayed factors are nonzero at $s = 1$. A primitive Gauss sum is nonzero, and Dirichlet's nonvanishing theorem states $L(1, \overline{\chi}) \neq 0$ for nonprincipal characters [80, Eq. 25.15.9]. Thus there is no additional zero. Taking the first nonzero Taylor coefficient proves the formula. □

Increasing the exponents of primes already present in q changes higher Taylor coefficients but not this first nonzero derivative: $(q/f)^{1-s}$ equals one at $s = 1$. This invariance provides an additional family of identities between traces at different levels.

9.13.2 The principal trace and its pole cancellation

Write $\mathcal{P}_q = \mathcal{P}_{q,\mathbf{1}}$ and define

$$r = \omega(q), \quad P = \prod_{p \mid q} \log p, \qquad B = \log q - \tfrac{1}{2} \log \mathrm{rad}(q).$$

The conductor-one factor contains a pole, so the vanishing order is one less than the number of primes, not equal to that number.

Theorem 9.68 (Principal trace jet generator)**.** *In a neighborhood of $t = 0$,*

$$\boxed{\mathcal{P}_q(1+t) = (-1)^r P\, t^{r-1}\, \mathrm{e}^{-Bt}\, [t\zeta(1+t)] \prod_{p|q} \frac{\sinh(t\log p/2)}{t\log p/2}.} \tag{9.147}$$

In particular its zero at $s = 1$ has exact order $r - 1$, and

$$\mathcal{P}_q^{(j)}(1) = 0 \qquad (0 \le j < r-1), \tag{9.148}$$

$$\mathcal{P}_q^{(r-1)}(1) = (-1)^r (r-1)! P, \tag{9.149}$$

$$\mathcal{P}_q^{(r)}(1) = (-1)^r r! P(\gamma - B), \tag{9.150}$$

$$\mathcal{P}_q^{(r+1)}(1) = (-1)^r (r+1)! P \left[-\gamma_1 - B\gamma + \frac{B^2}{2} + \frac{1}{24} \sum_{p|q} (\log p)^2 \right]. \tag{9.151}$$

Here γ_1 is the ordinary first Stieltjes constant in (9.128)*.*

Proof. The principal case of (9.144) is

$$\mathcal{P}_q(1+t) = q^{-t} \prod_{p|q} (1 - p^t)\zeta(1+t).$$

Use $1 - \mathrm{e}^x = -x\mathrm{e}^{x/2}\sinh(x/2)/(x/2)$ in every factor. This proves (9.147). The factors in square brackets and in the hyperbolic-sine product are analytic and equal to one at zero. Expanding them gives

$$t\zeta(1+t) = 1 + \gamma t - \gamma_1 t^2 + O(t^3), \qquad \prod_p \frac{\sinh(t\log p/2)}{t\log p/2} = 1 + \frac{t^2}{24} \sum_p (\log p)^2 + O(t^4).$$

Multiplication by e^{-Bt} gives all the displayed derivatives. □

9.13.3 Concrete exact identities

The following examples are proved consequences of the preceding theorems, not integer-relation conjectures. The derivatives are in the *order* of the polylogarithm, not its argument.

At level thirty,

$$\sum_{u \in U_{30}} \mathrm{Li}_1(\zeta_{30}^u) = 0, \tag{9.152}$$

$$\sum_{u \in U_{30}} \partial_s^1 \mathrm{Li}_s(\zeta_{30}^u)\big|_{s=1} = 0, \tag{9.153}$$

$$\sum_{u \in U_{30}} \partial_s^2 \mathrm{Li}_s(\zeta_{30}^u)\big|_{s=1} = \boxed{-2\log 2\log 3\log 5}. \tag{9.154}$$

The third derivative of this trace is $-6\log 2\log 3\log 5\,[\gamma - \frac{1}{2}\log 30]$. At level 210, derivatives of orders zero, one, and two vanish, while

$$\sum_{u \in U_{210}} \partial_s^3 \mathrm{Li}_s(\zeta_{210}^u)\big|_{s=1} = 6\log 2\log 3\log 5\log 7. \tag{9.155}$$

For the quadratic character χ_3 with $\chi_3(1) = 1$ and $\chi_3(2) = -1$, the primitive trace at order one is $\tau(\chi_3)L(1,\chi_3) = \mathrm{i}\pi/3$. Since $\chi_3(7) = \chi_3(13) = 1$, one obtains

$$\sum_{u\in U_{21}} \chi_3(u)\,\partial_s^1 \operatorname{Li}_s(\zeta_{21}^u)\big|_{s=1} = -\frac{\mathrm{i}\pi}{3}\log 7, \tag{9.156}$$

$$\sum_{u\in U_{273}} \chi_3(u)\,\partial_s^2 \operatorname{Li}_s(\zeta_{273}^u)\big|_{s=1} = \frac{2\mathrm{i}\pi}{3}\log 7\log 13. \tag{9.157}$$

In the first equation the order-zero trace vanishes; in the second, both order-zero and first-order traces vanish. Likewise the character χ_4 gives

$$\sum_{u\in U_{20}} \chi_4(u)\,\partial_s^1 \operatorname{Li}_s(\zeta_{20}^u)\big|_{s=1} = -\frac{\mathrm{i}\pi}{2}\log 5. \tag{9.158}$$

These primitive trace constants follow directly from $\operatorname{Li}_1(z) = -\log(1-z)$ with the values continued from inside the unit disk, so the signs do not depend on an unspecified logarithm branch.

For the real quadratic character χ_5, positive on residues $1, 4$ and negative on $2, 3$, let $\phi = (1+\sqrt{5})/2$. Pairing conjugates in the primitive trace gives

$$\mathcal{P}_{5,\chi_5}(1) = 2\log\frac{\sin(2\pi/5)}{\sin(\pi/5)} = 2\log\phi.$$

As $\chi_5(11) = 1$,

$$\sum_{u\in U_{55}} \chi_5(u)\,\partial_s^1 \operatorname{Li}_s(\zeta_{55}^u)\big|_{s=1} = -2\log\phi\log 11. \tag{9.159}$$

This derivation needs no numerical integer-relation search or unproved independence statement about logarithms.

9.13.4 A stable independent formula for numerical checks

Differentiation of (9.145) at its cancelled pole gives, for $z = \zeta_q^u \neq 1$,

$$\partial_s^k \operatorname{Li}_s(z)\Big|_{s=1} = \frac{(-1)^k}{q}\sum_{a=1}^{q} z^a \sum_{j=0}^{k}\binom{k}{j}(\log q)^{k-j}\gamma_j(a/q). \tag{9.160}$$

There is no pole term, because its complete Fourier coefficient vanishes identically. This supplies a finite Stieltjes-coordinate evaluator for the trace identities, independently of their Euler-factor right sides. In development, direct tiny-step numerical differentiation of a polylogarithm routine near integral order gave an unreliable trace derivative; the pole-cancelled formula avoids that cancellation problem. This observation is an implementation diagnostic, not a claimed theorem about all versions of a library.

9.13.5 A level-260 specialization and convention reconciliation

We use the unbarred weight $\chi(a)$ in $\mathcal{P}_{q,\chi}$. An alternative convention weights the trace by $\overline{\chi}(a)$; it conjugates the character factors and interchanges the primitive L-character. All formulas here use the first convention. The following exact specialization adds two primes with character value one.

Let χ be primitive of conductor $f > 1$, and $f \mid q$. Among the added primes, define

$$\mathcal{S} = \{p \mid q : p \nmid f,\ \chi(p) = 1\}, \qquad r_\chi = |\mathcal{S}|.$$

The remaining added primes form $\mathcal{N}$.

Theorem 9.69 (Twisted leading jet)**.** *The trace $\mathcal{P}_{q,\chi}$ vanishes to order at least r_χ at one, and*

$$\mathcal{P}_{q,\chi}^{(r_\chi)}(1) = (-1)^{r_\chi} r_\chi!\, \mathcal{P}_{f,\chi}(1) \prod_{p\in\mathcal{S}} \log p \prod_{p\in\mathcal{N}} (1-\chi(p)). \tag{9.161}$$

Its order is exactly r_χ whenever $\mathcal{P}_{f,\chi}(1) \neq 0$. More generally its exact order is $r_\chi + \operatorname{ord}_{s=1} \mathcal{P}_{f,\chi}(s)$.

Proof. The multiplier is

$$B_{q,f,\chi}(1+\varepsilon) = (q/f)^{-\varepsilon} \prod_{\substack{p\mid q\\ p\nmid f}} (1-\chi(p)p^{\varepsilon}).$$

The factors indexed by $\mathcal{S}$ have simple zeros with leading coefficient $-\log p$. All other factors are nonzero at zero and have value $1-\chi(p)$. Multiply their leading terms by the Taylor series of $\mathcal{P}_{f,\chi}(1+\varepsilon)$. □

For primitive nonprincipal characters, the classical nonvanishing $L(1,\overline{\chi}) \neq 0$, together with the nonzero primitive Gauss sum in (9.144), supplies the stated nonvanishing hypothesis [80]. The explicit example below does not need that theorem; its primitive trace is computed directly from logarithms.

For $\chi_4(a) = (-1)^{(a-1)/2}$ on odd integers,

$$\mathcal{P}_{4,\chi_4}(1) = \operatorname{Li}_1(\mathrm{i}) - \operatorname{Li}_1(-\mathrm{i}) = \frac{\mathrm{i}\pi}{2}.$$

Take $q = 260 = 4\cdot 5\cdot 13$. Both added primes have $\chi_4(p) = 1$. Consequently

$$\sum_{a\in\mathcal{U}_{260}} \chi_4(a) \operatorname{Li}_1(\zeta_{260}^a) = 0, \tag{9.162}$$

$$\sum_{a\in\mathcal{U}_{260}} \chi_4(a)\, \partial_s^1 \operatorname{Li}_s(\zeta_{260}^a)\big|_{s=1} = 0, \tag{9.163}$$

$$\sum_{a\in\mathcal{U}_{260}} \chi_4(a)\, \partial_s^2 \operatorname{Li}_s(\zeta_{260}^a)\big|_{s=1} = \mathrm{i}\pi \log 5 \log 13. \tag{9.164}$$

This is a proved identity, not an integer-relation conjecture. More generally, for distinct primes $p_1,\dots,p_r \equiv 1 \pmod 4$ and $q = 4p_1\cdots p_r$,

$$\sum_{a\in\mathcal{U}_q} \chi_4(a)\, \partial_s^r \operatorname{Li}_s(\zeta_q^a)\big|_{s=1} = (-1)^r r! \frac{\mathrm{i}\pi}{2} \prod_{j=1}^{r} \log p_j, \tag{9.165}$$

with all lower order traces vanishing. The same leading identity persists if the added prime factors occur with higher exponents.

9.14 Explicit tables for the second constant

We record the closed forms of $\gamma_2' = -[\zeta''(2,a) + 2\zeta'(2,a)]$ (the $s=2$ layer) and $\gamma_2'' = 2\zeta''(3,a) + 6\zeta'(3,a) + 2\zeta(3,a)$ (the $s=3$ layer) at the elementary denominators. The new constants entering are the *second* s-derivatives $\zeta''(2), \zeta''(3)$, $\beta''(s) = L''(s,\chi_{-4})$ and $L''(s,\chi_{-3})$; the values $L(3,\chi_{-4}) = \pi^3/32$ and $L(3,\chi_{-3}) = 4\pi^3/(81\sqrt{3})$ are elementary. As at the first-derivative layers, the odd-character block flips sign as a single unit under $p \mapsto q-p$ (including any elementary π-power value-part it carries). All entries are exact and were verified to $< 10^{-70}$.

Second constant, first parameter-derivative.

$$\begin{aligned}
\gamma_2'(1) &= -\zeta''(2) - 2\zeta'(2),\\
\gamma_2'(\tfrac12) &= -3\zeta''(2) - (6+8\ln 2)\zeta'(2) - \tfrac43\pi^2\ln 2 - \tfrac23\pi^2\ln^2 2,\\
\gamma_2'(\tfrac13) &= -4\zeta''(2) - (8+9\ln 3)\zeta'(2) - \tfrac32\pi^2\ln 3 - \tfrac34\pi^2\ln^2 3\\
&\quad - \Big[\tfrac92 L''(2,\chi_{-3}) + 9(1+\ln 3)L'(2,\chi_{-3}) + \big(9\ln 3 + \tfrac92\ln^2 3\big)L(2,\chi_{-3})\Big],\\
\gamma_2'(\tfrac14) &= -6\zeta''(2) - (12+28\ln 2)\zeta'(2) - \tfrac{14}{3}\pi^2\ln 2 - 5\pi^2\ln^2 2\\
&\quad - \Big[8\beta''(2) + 16(1+2\ln 2)\beta'(2) + \big(32\ln 2 + 32\ln^2 2\big)G\Big],
\end{aligned}$$

with $\gamma_2'(2/3), \gamma_2'(3/4)$ obtained by negating the bracketed odd-character block, and the $q=6$ entries given in the source report.

Second constant, second parameter-derivative.

$$\begin{aligned}
\gamma_2''(1) &= 2\zeta''(3) + 6\zeta'(3) + 2\zeta(3),\\
\gamma_2''(\tfrac12) &= 14\zeta''(3) + (42+32\ln 2)\zeta'(3) + (14+48\ln 2+16\ln^2 2)\zeta(3),\\
\gamma_2''(\tfrac13) &= 26\zeta''(3) + (78+54\ln 3)\zeta'(3) + (26+81\ln 3+27\ln^2 3)\zeta(3)\\
&\quad + \Big[27L''(3,\chi_{-3}) + (81+54\ln 3)L'(3,\chi_{-3}) + (27+81\ln 3+27\ln^2 3)L(3,\chi_{-3})\Big],\\
\gamma_2''(\tfrac14) &= 56\zeta''(3) + (168+240\ln 2)\zeta'(3) + (56+360\ln 2+248\ln^2 2)\zeta(3)\\
&\quad + \Big[64\beta''(3) + (192+256\ln 2)\beta'(3) + (2+12\ln 2+8\ln^2 2)\pi^3\Big],
\end{aligned}$$

again with $\gamma_2''(2/3), \gamma_2''(3/4)$ the odd-block negations. The reducible fractions collapse correctly ($\frac24=\frac36=\frac12$, $\frac26=\frac13$), an internal consistency check.

9.15 The derivative $\leftrightarrow$ negapolygamma duality

The functional equation pairs the layer $s=k+1$ of $\gamma_1^{(k)}$ with the layer $s=-k$, which is exactly where the negative-order polygamma $\psi^{(-(k+1))}$ lives (its ladder uses $\zeta'(-k,\cdot)$). The seam is a single uniform identity.

Theorem 9.70 (Harmonic-number bridge law)**.** *Let χ be a primitive character mod q for which $L(-k,\chi)\neq 0$. Then*

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)} + \overline{\left(\frac{L'(-k,\chi)}{L(-k,\chi)}\right)} = \gamma + \ln\frac{2\pi}{q} - H_k. \tag{9.166}$$

The same H_k that sets the master coefficient in Theorem 9.1 sets the bridge constant; the case $k=1$ recovers the generalized-Glaisher constant $C_q = \gamma + \ln(2\pi/q) - 1$.

Proof of the bridge. For a primitive nonprincipal character of parity $\chi(-1)=(-1)^\epsilon$, the completed function is $\Lambda(s,\chi) = (q/\pi)^{(s+\epsilon)/2}\Gamma((s+\epsilon)/2)L(s,\chi)$. Its functional equation gives $\Lambda'/\Lambda(s,\chi) = -\Lambda'/\Lambda(1-s,\bar\chi)$. At $s=k+1$ with no trivial zero at $-k$, the two digamma terms, using reflection and recurrence, sum to $2(H_k-\gamma-\log 2)$ before their factor $1/2$ is applied. Collecting the conductor factor proves (9.166). The conductor-one principal character requires the meromorphic completed zeta function; the same nonzero-value calculation is valid for $k\geq 1$ where neither point is a pole. □

Remark 9.71 (Trivial-zero bridge). For $k \geq 1$, $L(-k,\chi) = 0$ when $\epsilon \equiv k \pmod 2$. At a simple trivial zero, cancellation of the pole in the completed gamma factor gives instead

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)} + \overline{\frac{L''(-k,\chi)}{2L'(-k,\chi)}} = \gamma + \log(2\pi/q) - H_k.$$

The factor 2 in the denominator is essential. This follows by writing $L(-k+u) = L'(-k)u + \frac{1}{2}L''(-k)u^2 + O(u^3)$ in the same logarithmic functional equation and taking its finite part.

At $z = \frac{1}{4}$ the governing character is χ_{-4} (odd). For k even there is no trivial zero, and $\psi^{(-(k+1))}(\frac{1}{4})$ carries the unreduced derivative coordinate $\beta'(k+1) = L'(k+1,\chi_{-4})$, so $\gamma_1^{(k)}(\frac{1}{4})$ is an exact finite combination of $\psi^{(-(k+1))}(\frac{1}{4})$ and the remaining zeta-value and zeta-derivative terms. The representative instance $k = 2$ is, after eliminating $\beta'(3)$ entirely,

$$\begin{aligned}\gamma_1''(\tfrac{1}{4}) = \; & -256\,\pi^3\,\psi^{(-3)}(\tfrac{1}{4}) - 56\,\zeta'(3) - 84\,\zeta(3) - 120\ln 2\,\zeta(3) + 35\pi\,\zeta(3) \\ & - 3\pi^3 - 4\ln 2\,\pi^3 + 64\ln \mathsf{A}\,\pi^3 - 2\gamma\,\pi^3 + 2\ln(2\pi)\,\pi^3,\end{aligned} \tag{9.167}$$

verified to $< 10^{-118}$. (The irrational coefficient $-256\pi^3$ on $\psi^{(-3)}(\frac{1}{4})$ is why a rational integer-relation search for this identity must take $\pi^3\psi^{(-3)}$, not $\psi^{(-3)}$, as the search atom.) For k odd the value-level duality *degenerates*: by Remark 9.71, $\psi^{(-(k+1))}(\frac{1}{4})$ then carries only the lower *value* atom (for $k = 1$, $\beta'(-1) = 2G/\pi$, a value rather than a derivative at the positive point; for $k = 3$, $\beta'(-3) \sim \beta(4)/\pi^3$), while the genuine derivative atom $\beta'(k+1)$ of $\gamma_1^{(k)}(\frac{1}{4})$ is tied instead to the second derivative $\beta''(-k)$, absent from $\psi^{(-(k+1))}$. Numerically the coefficient of $\beta'(k+1)$ in $\psi^{(-(k+1))}(\frac{1}{4})$ is zero to $\sim 10^{-163}$ for $k = 1, 3$. The clean odd-k value-level duality therefore lives at even-character points (e.g. $q = 5$), where the parity is reversed.

9.16 One normalized functional equation for all derivative bridges

The Stieltjes drafts connect derivatives of $L(s,\chi)$ at a positive integer with derivatives at its reflected negative integer. A single identity of analytic germs organizes all these formulas and treats trivial zeros without differentiating singular logarithms. It also makes the conjugation required for complex characters unavoidable. The input is the classical primitive Dirichlet functional equation and the reflection and duplication formulas for Γ [49]; the all-order normalized form below is a systematic continuation of the bridges used in the corpus.

9.16.1 Regularization and the normalized germ

Let χ be primitive of conductor q, allowing the trivial character of conductor 1, for which $L(s,\chi) = \zeta(s)$. Fix $k \geq 1$, and write

$$\delta = \frac{1-\chi(-1)}{2} \in \{0,1\}, \qquad \nu = \begin{cases} 1, & \chi(-1) = (-1)^k, \\ 0, & \chi(-1) \neq (-1)^k. \end{cases}$$

Define

$$B(t) = \frac{L(-k+t,\chi)}{t^\nu}, \qquad B^{\#}(z) = \overline{B(\overline{z})}, \tag{9.168}$$

with the removable value filled in when $\nu = 1$. The function $B^{\#}$ is holomorphic near zero; it is coefficientwise conjugation of the Taylor series of B.

Theorem 9.72 (Normalized positive–negative germ)**.** *The quantities $L(k+1,\chi)$ and $B(0)$ are nonzero. On a sufficiently small disk about $z=0$,*

$$\frac{L(k+1+z,\chi)}{L(k+1,\chi)} = K_\nu(z)\frac{B^{\#}(-z)}{B^{\#}(0)}, \tag{9.169}$$

where

$$\begin{aligned} K_\nu(z) &= \left(\frac{2\pi}{q}\right)^{z}\frac{\Gamma(k+1)}{\Gamma(k+1+z)}\,T_\nu(z),\\ T_0(z) &= \sec\frac{\pi z}{2}, \qquad T_1(z) = \frac{\pi z/2}{\sin(\pi z/2)}, \quad T_1(0)=1. \end{aligned} \tag{9.170}$$

All three normalized factors in (9.169) equal 1 at the origin.

Proof. The Euler product gives $L(k+1,\chi)\neq 0$, since $k+1>1$. Use the completed function

$$\Lambda(s,\chi) = \left(\frac{q}{\pi}\right)^{(s+\delta)/2}\Gamma\!\left(\frac{s+\delta}{2}\right)L(s,\chi)$$

and its functional equation

$$\Lambda(s,\chi) = \varepsilon_\chi\Lambda(1-s,\overline{\chi}), \qquad |\varepsilon_\chi| = 1. \tag{9.171}$$

At $s=-k$, the gamma factor is finite and nonzero if $\nu=0$, and has a simple pole if $\nu=1$. The reflected completed value at $k+1$ is finite and nonzero. Consequently $L(-k,\chi)\neq 0$ in the first case, and $L(s,\chi)$ has a simple zero at $-k$ in the second. In either case B is holomorphic and $B(0)\neq 0$.

Put

$$a = \frac{\delta-k}{2}, \qquad b = \frac{k+1+\delta}{2}.$$

Solving (9.171) at $s=k+1+z$ gives

$$L(k+1+z,\chi) = \varepsilon_\chi\left(\frac{q}{\pi}\right)^{-k-\frac{1}{2}-z}\frac{\Gamma(a-z/2)}{\Gamma(b+z/2)}L(-k-z,\overline{\chi}). \tag{9.172}$$

Since $L(-k-z,\overline{\chi}) = (-z)^\nu B^{\#}(-z)$, the numerator

$$F_\nu(z) = (-z)^\nu\Gamma(a-z/2)$$

has a nonzero removable value at the origin. Dividing (9.172) by its value at $z=0$ cancels the root number and all constant factors:

$$\frac{L(k+1+z,\chi)}{L(k+1,\chi)} = \left(\frac{q}{\pi}\right)^{-z}\frac{F_\nu(z)}{F_\nu(0)}\frac{\Gamma(b)}{\Gamma(b+z/2)}\frac{B^{\#}(-z)}{B^{\#}(0)}. \tag{9.173}$$

If $\nu=0$, then a is a half-integer. The gamma reflection formula and $\sin(\pi(a-z/2)) = \sin(\pi a)\cos(\pi z/2)$ give

$$\frac{F_0(z)}{F_0(0)} = \frac{\Gamma(1-a)}{\Gamma(1-a+z/2)}\sec\frac{\pi z}{2}. \tag{9.174}$$

If $\nu=1$, write $a=-m$, $m\geq 0$. Then $F_1(0) = 2(-1)^m/m!$, and the same reflection formula gives

$$\frac{F_1(z)}{F_1(0)} = \frac{\Gamma(1-a)}{\Gamma(1-a+z/2)}\frac{\pi z/2}{\sin(\pi z/2)}. \tag{9.175}$$

For example, the pole in $\Gamma(-m-z/2)$ is canceled by the explicit factor $-z$, so no logarithm of a function vanishing at zero has been taken.

The unordered pair $\{b, 1-a\}$ is $\{(k+1)/2, (k+2)/2\}$, for either value of δ. Duplication therefore yields

$$\frac{\Gamma(b)\Gamma(1-a)}{\Gamma(b+z/2)\Gamma(1-a+z/2)} = 2^z \frac{\Gamma(k+1)}{\Gamma(k+1+z)}. \tag{9.176}$$

Substituting (9.174) or (9.175), followed by (9.176), into (9.173) proves the theorem. □

Remark 9.73. The statement concerns analytic germs and local logarithms. It requires no claim about a global zero-free region. For a complex character the functional equation involves $\overline{\chi}$, and therefore the reflected Taylor coefficients are conjugated. Replacing $B^{\#}$ by B is justified for a real character but generally false for a complex one.

9.16.2 Closed logarithmic coefficients at every order

Choose the normalized local logarithms, each zero at the origin, of $L(k+1+z,\chi)/L(k+1,\chi)$ and $B(t)/B(0)$. For $r \geq 1$, set

$$\mathcal{L}_r = \frac{d^r}{ds^r} \log L(s,\chi)\Big|_{s=k+1}, \qquad b_r = \frac{d^r}{dt^r} \log B(t)\Big|_{t=0}.$$

These derivatives do not depend on a choice of logarithm branch. Write

$$H_k^{(r)} = \sum_{j=1}^{k} j^{-r}, \qquad H_k = H_k^{(1)}, \qquad \eta(r) = (1-2^{1-r})\zeta(r) \quad (r \geq 2).$$

Theorem 9.74 (All-order logarithmic bridges)**.** *For every $r \geq 1$,*

$$\mathcal{L}_r - (-1)^r \overline{b_r} = \kappa_r, \tag{9.177}$$

where

$$\kappa_r = \begin{cases} \gamma + \log\dfrac{2\pi}{q} - H_k, & r = 1, \\ (r-1)!\big(\zeta(r) - H_k^{(r)}\big), & r \geq 3 \text{ odd}, \\ (r-1)!\big(H_k^{(r)} + (-1)^{\nu}\eta(r)\big), & r \geq 2 \text{ even}. \end{cases} \tag{9.178}$$

Proof. Taking normalized logarithms of (9.169) gives

$$\log \frac{L(k+1+z,\chi)}{L(k+1,\chi)} = \log K_\nu(z) + \overline{\log \frac{B(-\overline{z})}{B(0)}}.$$

The r-th derivative of the last term at zero is $(-1)^r\overline{b_r}$. It remains to compute the derivatives of $\log K_\nu$.

The convergent gamma expansion, on $|z| < 1$, is

$$\log \frac{\Gamma(k+1+z)}{\Gamma(k+1)} = (H_k - \gamma)z + \sum_{r\geq 2} \frac{(-1)^r}{r}\big(\zeta(r) - H_k^{(r)}\big)z^r. \tag{9.179}$$

The sine and cosine products give, in the same disk,

$$\log\sec\frac{\pi z}{2}=\sum_{m\ge1}\frac{(1-2^{-2m})\zeta(2m)}{m}z^{2m},\tag{9.180}$$

$$\log\frac{\pi z/2}{\sin(\pi z/2)}=\sum_{m\ge1}\frac{2^{-2m}\zeta(2m)}{m}z^{2m}.\tag{9.181}$$

These identities also follow directly by taking logarithms of the absolutely locally convergent products $\cos(\pi z/2)=\prod_{j\ge0}(1-z^2/(2j+1)^2)$ and $\sin(\pi z/2)/(\pi z/2)=\prod_{j\ge1}(1-z^2/(2j)^2)$.

The linear term of $\log K_\nu$ is the stated κ_1. Odd coefficients of order at least three come only from the negative of (9.179), giving the second case. For $r=2m$, the secant coefficient combines with the gamma coefficient to give

$$(r-1)!\left[H_k^{(r)}-\zeta(r)+2(1-2^{-r})\zeta(r)\right]=(r-1)!\big(H_k^{(r)}+\eta(r)\big).$$

Replacing the secant factor by the sine quotient instead gives

$$(r-1)!\left[H_k^{(r)}-\zeta(r)+2^{1-r}\zeta(r)\right]=(r-1)!\big(H_k^{(r)}-\eta(r)\big).$$

These are exactly the two even cases in (9.178). □

At a nonzero reflected value, $\nu=0$, the first two identities read

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)}+\overline{\frac{L'(-k,\chi)}{L(-k,\chi)}}=\gamma+\log\frac{2\pi}{q}-H_k,\tag{9.182}$$

$$(\log L)''(k+1,\chi)-\overline{(\log L)''(-k,\chi)}=H_k^{(2)}+\frac{\pi^2}{12}.\tag{9.183}$$

At a trivial zero, $\nu=1$, the Taylor expansion

$$B(t)=L'(-k,\chi)+\frac{L''(-k,\chi)}{2}t+\frac{L'''(-k,\chi)}{6}t^2+\cdots$$

instead gives

$$\frac{L'(k+1,\chi)}{L(k+1,\chi)}+\frac12\overline{\frac{L''(-k,\chi)}{L'(-k,\chi)}}=\gamma+\log\frac{2\pi}{q}-H_k,\tag{9.184}$$

$$(\log L)''(k+1,\chi)-\overline{\left(\frac{L'''(-k,\chi)}{3L'(-k,\chi)}-\frac{L''(-k,\chi)^2}{4L'(-k,\chi)^2}\right)}=H_k^{(2)}-\frac{\pi^2}{12}.\tag{9.185}$$

These formulas preserve the familiar first-order harmonic bridge and supply a uniform extension to every derivative order and both character parities.

9.16.3 Raw jets and an exact triangular implementation

The normalized germ also gives direct formulas for ordinary derivatives, without first converting each order to logarithmic derivatives. Define K_m by

$$K_\nu(z)=\sum_{m\ge0}K_m\frac{z^m}{m!},\qquad K_0=1.$$

Since $\log K_\nu(z) = \sum_{r\ge1} \kappa_r z^r/r!$, these coefficients are the complete exponential Bell polynomials:

$$K_m = m! \sum_{\substack{m_1,\dots,m_m\ge0\\ m_1+2m_2+\cdots+mm_m=m}} \prod_{r=1}^{m} \frac{1}{m_r!}\left(\frac{\kappa_r}{r!}\right)^{m_r} \qquad (m\ge1). \tag{9.186}$$

Leibniz's rule in (9.169) proves

$$\frac{L^{(m)}(k+1,\chi)}{L(k+1,\chi)} = \sum_{j=0}^{m} \binom{m}{j} K_{m-j}(-1)^j \frac{\overline{B^{(j)}(0)}}{B(0)}. \tag{9.187}$$

The required regularized derivatives are explicitly

$$\frac{B^{(j)}(0)}{B(0)} = \frac{j!\,\nu!}{(j+\nu)!}\frac{L^{(j+\nu)}(-k,\chi)}{L^{(\nu)}(-k,\chi)}. \tag{9.188}$$

The diagonal coefficient in (9.187) is $(-1)^m$, so the transformation is triangular and invertible. Its coefficients belong to the ring generated by the explicit κ_r's. This supplies exact relation certificates at arbitrary order and avoids repeated symbolic differentiation across gamma poles.

Remark 9.75 (Correction of the complex-character formulas). The older derivative-relations report omits the conjugation in versions of the trivial-zero first bridge and the nonzero second bridge. Equations (9.184) and (9.183) give the required replacements. The two main Stieltjes articles already include conjugation in their displayed first bridges; those correct formulas should be retained. Exact source locators and numerical counterexamples to the omitted-conjugation versions appear in the correction register.

9.17 Experimental independence tests

The tables introduce $\zeta''(2), \zeta''(3), \beta''(2), \beta''(3)$ and $L''(2,\chi_{-3}), L''(3,\chi_{-3})$. Historical searches at 200 digits with planted controls recovered the known value reductions and found no accepted relation for the six derivative coordinates in the lower-order baskets. This supports using them as named coordinates in a reduction table. Their mutual independence, and the impossibility of reductions in larger baskets, remain open.

9.18 Elementary grids and the sixth-point completion

For $q = 3, 4, 6$ define

$$A_3(s) = \frac{3^s-1}{2}\zeta(s), \quad A_4(s) = \frac{4^s-2^s}{2}\zeta(s), \quad A_6(s) = \frac{(2^s-1)(3^s-1)}{2}\zeta(s),$$

$$B_3(s) = \frac{3^s}{2}L(s,\chi_{-3}), \quad B_4(s) = \frac{4^s}{2}\beta(s), \quad B_6(s) = \frac{6^s+3^s}{2}L(s,\chi_{-3}).$$

Residue-class decomposition proves $\zeta(s,1/q) = A_q(s) + B_q(s)$ and $\zeta(s,1-1/q) = A_q(s) - B_q(s)$ for these three q. Consequently all four derivative tables, including the omitted sixth-point rows in the later article, follow from the differential operators

$$\gamma_1'(a) = (\partial_s+1)\zeta(s,a)|_{s=2}, \qquad \gamma_1''(a) = -(2\partial_s+3)\zeta(s,a)|_{s=3},$$
$$\gamma_2'(a) = -(\partial_s^2+2\partial_s)\zeta(s,a)|_{s=2}, \qquad \gamma_2''(a) = (2\partial_s^2+6\partial_s+2)\zeta(s,a)|_{s=3}.$$

This supplies an exact symbolic recipe for every entry, without new fitted constants. Write $a = \log 2$, $b = \log 3$, and $L_j(s) = L^{(j)}(s, \chi_{-3})$. At the sixth points the first-constant rows are

$$\begin{aligned}
\gamma_1'(1/6),\ \gamma_1'(5/6) &= 12\zeta'(2) + 2\pi^2 + \frac{8}{3}\pi^2 a + \frac{9}{4}\pi^2 b \\
&\quad \pm \left[\frac{45}{2}L_1(2) + \left(\frac{45}{2} + 18a + \frac{45}{2}b\right) L_0(2)\right], \\
\gamma_1''(1/6),\ \gamma_1''(5/6) &= -182\zeta'(3) - (273 + 208a + 189b)\zeta(3) \\
&\quad \mp \left[243L_1(3) + \left(\frac{729}{2} + 216a + 243b\right) L_0(3)\right].
\end{aligned}$$

The second-constant rows are

$$\begin{aligned}
\gamma_2'(1/6),\ \gamma_2'(5/6) &= S_2 \mp O_2, \\
S_2 &= -12\zeta''(2) - (24 + 32a + 27b)\zeta'(2) \\
&\quad - \pi^2\left(\frac{16}{3}a + \frac{9}{2}b + \frac{8}{3}a^2 + \frac{9}{4}b^2 + 6ab\right), \\
O_2 &= \frac{45}{2}L_2(2) + (45 + 36a + 45b)L_1(2) \\
&\quad + \left(36a + 45b + 18a^2 + \frac{45}{2}b^2 + 36ab\right) L_0(2), \\
\gamma_2''(1/6),\ \gamma_2''(5/6) &= S_3 \pm O_3, \\
S_3 &= 182\zeta''(3) + (546 + 416a + 378b)\zeta'(3) \\
&\quad + (182 + 624a + 567b + 208a^2 + 189b^2 + 432ab)\zeta(3), \\
O_3 &= 243L_2(3) + (729 + 432a + 486b)L_1(3) \\
&\quad + (243 + 648a + 729b + 216a^2 + 243b^2 + 432ab)L_0(3).
\end{aligned}$$

The upper sign is the 1/6 entry. These formulas consolidate the tables from the first-derivative, second-derivative and general-order reports.

9.19 Fixed-coordinate periodic collision contacts

The differentiation anomaly of Theorem 9.3 already fixes the individual periodic factors. A second question remains: does convolution of those factors equal the coordinate finite-part extension of their separated correlation? The independent continuations [121] give the same answer at every index and derivative order. The comparison below keeps the coordinate x for each original factor and the independent coordinates a and $1 - a$ for the shift. In this convention there is exactly one contact term. Its coefficient is distinct from the scalar constant in an off-point collision expansion.

Write $T_{m,p} = F_{m,p}$, with $T_{m,0} = G_m$, in the notation of (9.4). Reflection is denoted by a check, so $\widehat{\check{U}}(k) = \widehat{U}(-k)$. Set

$$P_p(u) = \frac{(1+u)_p}{p!} = \prod_{j=1}^{p}(1 + u/j), \qquad P_0 = 1, \qquad H_p^{(j)} = \sum_{l=1}^{p} l^{-j}.$$

All indices m, n, p, q are nonnegative integers; put $r = p + q$ and $w = u + v$. The scalar Stieltjes

germ and its argument derivatives are

$$g(u,x)=\zeta(1+u,x)-1/u=\sum_{m\ge0}(-1)^m\gamma_m(x)u^m/m!, \tag{9.189}$$

$$G_p(u,x)=\partial_x^p g(u,x)=(-1)^p(1+u)_p\zeta(1+p+u,x)-\delta_{p0}/u. \tag{9.190}$$

The apparent pole in g is removable for each $x>0$.

9.19.1 Whole resonant numerators and Fourier input

Let $\mathcal{Z}_s=Z(s)$ be the meromorphic family of Lemma 8.60. Its Fourier coefficients are

$$\widehat{\mathcal{Z}_s}(0)=0,\qquad \widehat{\mathcal{Z}_s}(k)=\Gamma(1-s)(2\pi\mathrm{i}k)^{s-1}\quad(k\neq0), \tag{9.191}$$

on the branch (8.60).

Lemma 9.76 (Full resonant completion)**.** *The germ generating the unit-coordinate finite parts is*

$$T_p(u):=\sum_{m\ge0}\frac{(-1)^m}{m!}T_{m,p}u^m=\mathrm{D}^p\mathcal{Z}_{1+u}-\frac{\delta_{p0}}{u}\,1+\frac{P_p(u)}{u}\delta_0^{(p)}. \tag{9.192}$$

It is holomorphic near zero, and every $T_{m,p}$ has mean zero.

Proof. The local singular part of G_p is $c_p(u)x^{-p-1-u}$, with $c_p(u)=(-1)^p p!P_p(u)$; its remaining shifted Hurwitz part is jointly holomorphic. Subtract the test-function Taylor polynomial through degree p. Its indices $j<p$ contribute $c_p(u)\varphi^{(j)}(0)/(j!(j-p-u))$, holomorphic near zero. The index p contributes

$$-\frac{c_p(u)\varphi^{(p)}(0)}{p!u}=-\frac{P_p(u)}{u}\langle\delta_0^{(p)},\varphi\rangle.$$

Every coefficient of this resonant monomial is proportional to $x^{-1}\log^j x$, whose unit-coordinate finite-part integral on $(0,1)$ is zero. Thus its whole numerator must be removed, including its regular Taylor coefficients. This gives (9.192), using $\mathrm{D}^p\mathcal{Z}_{1+u}=(-1)^p(1+u)_p\mathcal{Z}_{1+p+u}$. After subtraction the local remainder, with any fixed number of spectral derivatives, is dominated by $Cx^{-\eta}(1+|\log x|^M)$, $\eta<1$. Consequently pairing and coefficient extraction are holomorphic and give exactly the cutoff finite parts. The nonlocal zero Fourier mode is zero; for $p=0$ the last two terms cancel in mean, and for $p>0$ both have mean zero. □

The existing differentiation anomaly becomes

$$T_{m,p}=\mathrm{D}^pG_m+\kappa_{m,p}\delta_0^{(p)},\qquad \kappa_{m,p}=(-1)^m m![u^{m+1}]P_p(u). \tag{9.193}$$

This is the earlier elementary-symmetric formula, since $[u^{m+1}]P_p=e_{m+1}(p)$. It also follows immediately by subtracting $\mathrm{D}^pT_0(u)$ from (9.192). In particular $\kappa_{0,p}=H_p$ and $\kappa_{m,p}=0$ when $m\ge p$.

For a scalar complete exponential Bell polynomial B_j, define

$$b_m(z)=\frac{(-1)^{m+1}}{m+1}B_{m+1}(z,1!\zeta(2),2!\zeta(3),\dots,m!\zeta(m+1)). \tag{9.194}$$

Equivalently, B_j is defined by $\exp(\sum_{l\ge1}x_lu^l/l!)=\sum_{j\ge0}B_ju^j/j!$. For $k\neq0$, put $L_k=\gamma+\log(2\pi\mathrm{i}k)$. Then

$$\widehat{T}_{m,p}(k)=(2\pi\mathrm{i}k)^p\big(b_m(L_k)+\kappa_{m,p}\big),\qquad \widehat{T}_{m,p}(0)=0. \tag{9.195}$$

Indeed (9.192) gives the nonzero-mode germ $(2\pi\mathrm{i}k)^p\{\Gamma(-u)(2\pi\mathrm{i}k)^u + P_p(u)/u\}$. Insert $\Gamma(-u) = -\Gamma(1-u)/u$ and

$$\Gamma(1-u)(2\pi\mathrm{i}k)^u = \exp\left(L_k u + \sum_{j\geq 2}\zeta(j)u^j/j\right),$$

the classical gamma expansion [122]. This proves (9.195); for example $b_0(z) = -z$ and $b_1(z) = (z^2+\zeta(2))/2$.

The lowest-index comparison, proved in the general theorem below, is

$$\check{G}_0 * G_0 = \mathrm{FP}_{\mathbb{T},a}\left[\gamma_1(a) + \gamma_1(1-a) - \pi^2/3\right] + \frac{\pi^2}{3}\delta_0. \tag{9.196}$$

Both factors have zero mean. The atom restores that same mean in the convolution; it is invisible in the punctured-circle function.

9.20 The off-collision kernel and its second finite part

For $0 < a < 1$, set $b = 1-a$ and define, in the original integration variable,

$$J^{p,q}_{mn}(a) = \mathrm{FP}\int_0^1 \gamma_m^{(p)}(x)\gamma_n^{(q)}(\{x+a\})\,\mathrm{d}x. \tag{9.197}$$

There are two independent right-hand singularities, at $x = 0+$ and $x = b+$; the second coordinate is $x-b$. This is the convention of [121]. Define the distinct distribution

$$\mathsf{J}^{p,q}_{mn} = \mathrm{FP}_{\mathbb{T},a}\, J^{p,q}_{mn}(a). \tag{9.198}$$

The second finite part is taken in the shift variable, at both $a = 0+$ and $a = 1-$.

We recall the off-point generating kernel, deriving the ingredients needed for the distributional comparison. Put

$$\mathcal{E}(u,v) = \frac{\Gamma(1-u)\Gamma(1+u+v)}{\Gamma(1+v)}, \tag{9.199}$$

$$\mathcal{A}(u,v) = \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)}\frac{\cos(\pi(u-v)/2)}{\cos(\pi(u+v)/2)}. \tag{9.200}$$

These are holomorphic near the origin and satisfy

$$v\mathcal{E}(u,v) + u\mathcal{E}(v,u) = (u+v)\mathcal{A}(u,v), \qquad \mathcal{A}(u,0) = \mathcal{A}(0,v) = 1. \tag{9.201}$$

The identity follows by inserting the gamma reflection formula. Equivalently, it follows by comparing the two Hurwitz forms of the classical spectral correlation.

For $r \geq 1$, write

$$W_r(t,x) = (1+t)_r\zeta(1+r+t,x) = (-1)^r G_r(t,x).$$

The off-point germ whose coefficients are $(-1)^{m+n}J^{p,q}_{mn}/(m!n!)$ is

$$\begin{aligned}\mathcal{J}_{pq}(u,v;a) = (-1)^q&\frac{P_p(u)W_r(v,a) - \mathcal{E}(u,v)W_r(w,a)}{u}\\ &+ (-1)^p\frac{P_q(v)W_r(u,b) - \mathcal{E}(v,u)W_r(w,b)}{v}.\end{aligned} \tag{9.202}$$

For $p = q = 0$, the corresponding expression is

$$\mathcal{J}_{00}(u, v; a) = \frac{g(v, a) - \mathcal{E}(u, v)g(w, a)}{u} + \frac{g(u, b) - \mathcal{E}(v, u)g(w, b)}{v} + \frac{1 - \mathcal{A}(u, v)}{uv}. \tag{9.203}$$

Each divided difference is holomorphic: its numerator vanishes on the relevant spectral axis.

To verify these formulas, start where all ordinary integrals converge and use the Hurwitz Fourier series. The shifted spectral kernel is

$$C(s, t; a) = B(1 - s, s + t - 1)\zeta(s + t - 1, a) + B(1 - t, s + t - 1)\zeta(s + t - 1, 1 - a). \tag{9.204}$$

It follows either by gamma reflection from the product of Fourier coefficients, or from the classical polylogarithmic expression. This product-kernel method is classical [28]; the shifted specialization and its Stieltjes completion are already used in [121]. Multiply by $(-1)^{p+q}(1 + u)_p(1 + v)_q$, set $s = 1 + p + u, t = 1 + q + v$, and remove the *full* resonant Taylor contribution at each of the two original singularities. The gamma identity

$$(-1)^p(1 + u)_p\Gamma(-p - u) = \Gamma(-u)$$

then gives (9.202); the subtractions of $1/u$ and $1/v$ give (9.203). The same finite Taylor-subtraction argument as in Lemma 9.76 justifies holomorphy and coefficient extraction.

The exact Hurwitz shift shows that, near $a = 0+$,

$$J^{p,q}_{mn}(a) = a^{-r-1}\mathcal{Q}^{p,q}_{mn}(\log a) + \text{an analytic function of } a,$$

where $\deg \mathcal{Q}^{p,q}_{mn} \le m + n + 1$. The other endpoint is obtained by interchanging (m, p) and (n, q) and reflecting. Consequently (9.198) is well defined, with only finitely many test-function Taylor subtractions at each endpoint.

Taking the shift-variable finite part in (9.202) yields, for $r \ge 1$,

$$\mathsf{J}_{pq}(u, v) = (-1)^p \frac{P_p(u)T_r(v) - \mathcal{E}(u, v)T_r(w)}{u} + (-1)^q \frac{P_q(v)\check{T}_r(u) - \mathcal{E}(v, u)\check{T}_r(w)}{v}. \tag{9.205}$$

For $r = 0$, use the same formula with $P_0 = 1$ and add $(1 - \mathcal{A})/(uv)$ times the constant distribution 1. This expression specifies the full second finite part; it is not merely a choice of extension having the correct off-point values.

9.21 The all-index collision-contact theorem

Define the convolution distribution

$$\mathsf{C}^{p,q}_{mn} = \check{T}_{m,p} * T_{n,q}. \tag{9.206}$$

Its restriction to the punctured circle is (9.197). Indeed, for a nonzero shift the singular supports of the two factors are disjoint. Multiplication by the other, smooth factor near each singular point gives exactly the two original coordinate finite parts. A partition of unity in x makes this assertion a direct consequence of the distribution definitions. It does not involve defining a pointwise product of two distributions at a coincident singularity.

Theorem 9.77 (Complete periodic contact correction)**.** *For all nonnegative m, n, p, q, with $r = p+q$,*

$$\boxed{\mathsf{C}^{p,q}_{mn} = \mathsf{J}^{p,q}_{mn} + \Delta^{p,q}_{mn}\delta_0^{(r)}.} \tag{9.207}$$

The coefficient is

$$\Delta^{p,q}_{mn} = (-1)^{m+n+p} m!n![u^{m+1}v^{n+1}]\mathcal{N}_{pq}(u,v), \tag{9.208}$$

$$\begin{aligned}\mathcal{N}_{pq}(u,v) = {}& \mathcal{A}(u,v)P_r(u+v) - P_p(u)P_r(v) \\ & - P_q(v)P_r(u) + P_p(u)P_q(v).\end{aligned} \tag{9.209}$$

Equivalently, the spectral generating coefficient is the holomorphic germ

$$\Delta_{pq}(u,v) = \frac{(-1)^p}{uv}\mathcal{N}_{pq}(u,v). \tag{9.210}$$

There are no additional lower-order delta derivatives in this fixed convention.

Proof. First let $r \geq 1$. Work with meromorphic distribution germs before taking finite parts. Set $U_p(u) = \mathrm{D}^p \mathcal{Z}_{1+u}$. By (9.204) and the gamma cancellation preceding (9.205), the raw correlation is

$$\check{U}_p(u) * U_q(v) = -\frac{(-1)^p \mathcal{E}(u,v)}{u}\mathrm{D}^r \mathcal{Z}_{1+w} - \frac{(-1)^q \mathcal{E}(v,u)}{v}\mathrm{D}^r \check{\mathcal{Z}}_{1+w}.$$

This identity can also be checked at each nonzero Fourier mode from (9.191). It holds as a meromorphic distribution identity because it holds in an ordinary convergence region and all its Fourier coefficients continue meromorphically.

Insert (9.192) in $\check{T}_p(u) * T_q(v)$. Constant-distribution terms vanish when $r \geq 1$, since at least one factor has zero mean and a positive derivative. The two cross terms and the delta–delta term give

$$\begin{aligned}\mathsf{C}_{pq}(u,v) = {}& (-1)^p \frac{P_p(u)\mathrm{D}^r \mathcal{Z}_{1+v} - \mathcal{E}(u,v)\mathrm{D}^r \mathcal{Z}_{1+w}}{u} \\ & + (-1)^q \frac{P_q(v)\mathrm{D}^r \check{\mathcal{Z}}_{1+u} - \mathcal{E}(v,u)\mathrm{D}^r \check{\mathcal{Z}}_{1+w}}{v} + \frac{(-1)^p P_p(u)P_q(v)}{uv}\delta_0^{(r)}.\end{aligned}$$

Replace $\mathrm{D}^r \mathcal{Z}_{1+t}$ by $T_r(t) - P_r(t)\delta_0^{(r)}/t$ and use $\check{\delta}_0^{(r)} = (-1)^r \delta_0^{(r)}$. The nonlocal terms are exactly (9.205). The remaining coefficient of $\delta_0^{(r)}$, after factoring $(-1)^p$, is

$$-\frac{P_p(u)P_r(v)}{uv} - \frac{P_q(v)P_r(u)}{uv} + \frac{P_p(u)P_q(v)}{uv} + \frac{P_r(w)}{w}\left(\frac{\mathcal{E}(u,v)}{u} + \frac{\mathcal{E}(v,u)}{v}\right).$$

Equation (9.201) gives (9.210).

For $p = q = 0$, keep the zero-mode terms explicitly. Since $T_0(u) = \mathcal{Z}_{1+u} + (\delta_0 - 1)/u$, their convolution is

$$\mathsf{C}_{00}(u,v) = \frac{\mathcal{Z}_{1+v} - \mathcal{E}(u,v)\mathcal{Z}_{1+w}}{u} + \frac{\check{\mathcal{Z}}_{1+u} - \mathcal{E}(v,u)\check{\mathcal{Z}}_{1+w}}{v} + \frac{\delta_0 - 1}{uv}.$$

Replace $\mathcal{Z}_{1+t} = T_0(t) + (1-\delta_0)/t$ and apply (9.201). The result is

$$\mathsf{C}_{00}(u,v) = \mathsf{J}_{00}(u,v) + \frac{\mathcal{A}(u,v) - 1}{uv}\delta_0,$$

which is exactly (9.210) with all P's equal to one.

The numerator (9.209) vanishes identically on $u = 0$ and on $v = 0$, because $P_j(0) = 1$ and $\mathcal{A}$ equals one on both axes. Therefore it is divisible by uv as a holomorphic germ. Taking coefficients proves (9.207). The displayed calculation exhausts the difference, and produces only $\delta_0^{(r)}$; hence there are no omitted lower contacts. □

9.21.1 Why the correction has no Stieltjes constants

The logarithm of the gamma–trigonometric factor is

$$\log \mathcal{A}(u,v) = \sum_{k\geq 2} \frac{\zeta(k)}{k}\big(u^k + v^k - (u+v)^k\big) - \sum_{j\geq 1} \frac{(1-2^{-2j})\zeta(2j)}{j}\big((u-v)^{2j} - (u+v)^{2j}\big). \tag{9.211}$$

Euler's constant cancels from the gamma ratio. It follows that

$$\Delta_{mn}^{p,q} \in \mathbb{Q}[\zeta(2), \zeta(3), \ldots],$$

with the finite harmonic data encoded in rational coefficients of P_p, P_q, P_r. In particular there are no ordinary or generalized Stieltjes constants and no spectral derivatives of zeta in the contact coefficient. This is an assertion about the explicit expression, not an arithmetic independence theorem.

The reflection law is

$$\Delta_{nm}^{q,p} = (-1)^r \Delta_{mn}^{p,q}. \tag{9.212}$$

It follows both from (9.209) and from reflecting the convolution and its coordinate extension. Reflection changes $\delta_0^{(r)}$ by precisely the necessary sign.

9.21.2 All derivative orders at Stieltjes index zero

Corollary 9.78 (Polygamma contact formula)**.** *For $r = p + q$, including $r = 0$,*

$$\boxed{\Delta_{00}^{p,q} = (-1)^p \left[\frac{\pi^2}{3} - H_r^{(2)} + (H_r - H_p)(H_r - H_q)\right].} \tag{9.213}$$

Proof. The coefficient of uv in $\mathcal{A}$ is $2\zeta(2)$, and its linear part is zero. The coefficient of uv in $P_r(u+v)$ is $H_r^2 - H_r^{(2)}$. Extracting the remaining products in (9.209) gives

$$2\zeta(2) + H_r^2 - H_r^{(2)} - (H_p + H_q)H_r + H_p H_q,$$

which is (9.213). □

Some low-order values are

(p,q)	r	$\Delta_{00}^{p,q}$
$(0,0)$	0	$\pi^2/3$
$(0,1)$	1	$\pi^2/3 - 1$
$(1,0)$	1	$1 - \pi^2/3$
$(1,1)$	2	$1 - \pi^2/3$
$(0,2)$	2	$\pi^2/3 - 5/4$
$(1,2)$	3	$13/12 - \pi^2/3$

These are corrections to a *distribution in the shift*, not the values of the common-point product integrals studied in [121].

9.21.3 A closed parity law and derivative-split stabilization

Corollary 9.79 (The complete first row)**.** *For every $m \geq 0$,*

$$\Delta_{m0}^{0,0} = m!\zeta(m+2)\begin{cases} 3-2^{-m}, & m \text{ even}, \\ 1, & m \text{ odd}. \end{cases} \tag{9.214}$$

Proof. Differentiating (9.200) in v and setting $v = 0$ gives

$$\partial_v \mathcal{A}(u,0) = \psi(1-u) - \psi(1) + \pi \tan\frac{\pi u}{2}. \tag{9.215}$$

The coefficient of u^k is $-\zeta(k+1)$ for even $k \geq 2$, and $(3-2^{1-k})\zeta(k+1)$ for odd $k \geq 1$. This follows from the gamma series and the cosine product, or by differentiating (9.211). Substitute $k = m+1$ in (9.208). □

For example,

$$\begin{aligned} &\Delta_{10}^{0,0} = \zeta(3), && \Delta_{20}^{0,0} = \tfrac{11}{2}\zeta(4), && \Delta_{30}^{0,0} = 6\zeta(5), \\ &\Delta_{11}^{0,0} = \tfrac{7}{2}\zeta(4), && \Delta_{21}^{0,0} = 4\zeta(5) + 4\zeta(2)\zeta(3), && \Delta_{22}^{0,0} = 4\zeta(3)^2 + \tfrac{83}{2}\zeta(6). \end{aligned}$$

Corollary 9.80 (Stabilization of the derivative split)**.** *If $m \geq r$ or $n \geq r$, then*

$$\Delta_{mn}^{p,q} = (-1)^p \Delta_{mn}^{0,r}. \tag{9.216}$$

Proof. Every product in (9.209) other than $\mathcal{A}P_r(u+v)$ has degree at most r in each variable. The indicated coefficient therefore vanishes in those products as soon as either $m+1 > r$ or $n+1 > r$. The remaining expression depends only on r, apart from the factor $(-1)^p$. □

An explicit first-row generator for arbitrary derivative orders is also useful:

$$\begin{aligned} \Delta_{m0}^{p,q} = (-1)^{m+p} m![u^{m+1}]\big[&P_r(u)\partial_v \mathcal{A}(u,0) + P_r'(u) \\ &- H_q P_r(u) + (H_q - H_r)P_p(u)\big]. \end{aligned} \tag{9.217}$$

For $m \geq r$, the last three polynomial terms disappear. Formula (9.215) then gives a finite sum of zeta values with indices from $m+2-r$ through $m+2$, weighted by elementary symmetric polynomials in reciprocal integers.

9.22 Exact convergent integral identities from finite Fourier data

The distributional result produces ordinary integrals once the test function vanishes to sufficient order at the collision. Put

$$V_{M,k}(a) = (1-\cos(2\pi k a))^M, \qquad M, k \in \mathbb{N}_{>0}. \tag{9.218}$$

Its finite Fourier expansion is

$$V_{M,k}(a) = 2^{-M} \sum_{j=-M}^{M} (-1)^j \binom{2M}{M+j} e^{2\pi \mathrm{i} j k a}. \tag{9.219}$$

This follows by expanding $2^{-M}(2 - e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta})^M$. It vanishes to order $2M$ at both identified endpoints.

Theorem 9.81 (Ordinary all-index correlation integrals)**.** *If $2M > r = p + q$, then*

$$\int_0^1 V_{M,k}(a) J_{mn}^{p,q}(a)\,\mathrm{d}a = 2^{1-M} \sum_{j=1}^{M} (-1)^j \binom{2M}{M+j} \operatorname{Re}\Big(\widehat{T}_{m,p}(-jk)\widehat{T}_{n,q}(jk)\Big). \tag{9.220}$$

The integral is ordinary and absolutely convergent. The right-hand side is a finite exact expression given by (9.195)*.*

Proof. The endpoint expansion in Section 9.20 bounds the weighted integrand by a constant times $a^{2M-r-1}(1 + |\log a|^{m+n+1})$ near zero, and similarly at one. Since $2M > r$, the integral converges. All derivatives of $V_{M,k}$ through degree r vanish at zero, so the contact in (9.207) pairs to zero. Pair the convolution distribution with (9.219); its zero Fourier coefficient is zero. Realness of the original distributions combines the positive and negative modes into the displayed real part. □

At Stieltjes index zero, put $\lambda_j = \gamma + \log(2\pi jk)$. Then the Fourier product in (9.220) is

$$(-1)^p (2\pi \mathrm{i} jk)^r \left[(\lambda_j - H_p)(\lambda_j - H_q) + \frac{\pi^2}{4} + \frac{\pi \mathrm{i}}{2}(H_q - H_p)\right]. \tag{9.221}$$

For even $r = 2d$ its real part is

$$(-1)^{p+d} (2\pi jk)^r \left[(\lambda_j - H_p)(\lambda_j - H_q) + \frac{\pi^2}{4}\right],$$

whereas for odd $r = 2d + 1$ it is

$$(-1)^{p+d+1} (2\pi jk)^r \frac{\pi}{2}(H_q - H_p).$$

Thus all logarithms disappear from these symmetric weighted polygamma correlations when the total derivative degree is odd.

At the first index, with $\lambda_k = \gamma + \log(2\pi k)$,

$$\int_0^1 (1 - \cos(2\pi ka)) \left[\gamma_1(a) + \gamma_1(1-a) - \frac{\pi^2}{3}\right] \mathrm{d}a = -\lambda_k^2 - \frac{\pi^2}{4}. \tag{9.222}$$

The corresponding single-factor consequences of (9.195) are

$$\int_0^1 (1 - \cos(2\pi kx))[-\psi(x)]\,\mathrm{d}x = \lambda_k, \tag{9.223}$$

$$\int_0^1 (1 - \cos(2\pi kx))\gamma_1(x)\,\mathrm{d}x = \frac{\pi^2}{24} - \frac{1}{2}\lambda_k^2. \tag{9.224}$$

These elementary Fourier specializations are included as checks and applications of the conventions, not as claims of priority over the classical Fourier theory.

9.23 A Hurwitz kernel for all positive-integer spectral jets

A second route makes the spectral zeta derivatives in the weighted identities explicit. Define the finite exponential sum

$$S_M(t) = \sum_{j=1}^{M} (-1)^j \binom{2M}{M+j} j^t, \qquad j^t = e^{t \log j}. \tag{9.225}$$

It is entire, and $S_M^{(h)}(t)$ is the same finite sum with a factor $\log^h j$.

Theorem 9.82 (Weighted Hurwitz kernel)**.** *For* $\operatorname{Re} s < 2M+1$, $s \neq 1$,

$$\int_0^1 V_{M,k}(x)\zeta(s,x)\,\mathrm{d}x = 2^{1-M}\Gamma(1-s)(2\pi k)^{s-1}\sin\frac{\pi s}{2}\,S_M(s-1). \tag{9.226}$$

At $s = 2,\dots,2M$*, the right-hand side is interpreted by its removable limit. The identity can be spectrally differentiated to every finite order in this domain, except at the genuine pole* $s = 1$*.*

Proof. For $\operatorname{Re} s < 0$, pair the absolutely convergent Hurwitz Fourier expansion with (9.219). Its zero mode vanishes. The two signs of each nonzero frequency combine to $2\Gamma(1-s)(2\pi jk)^{s-1}\cos(\pi(s-1)/2)$, and the cosine equals $\sin(\pi s/2)$. This gives (9.226).

For the larger domain, use $\zeta(s,x) = x^{-s} + \zeta(s,1+x)$. The weight is $O(x^{2M})$, so the first term and each of its spectral derivatives are integrable when $\operatorname{Re} s < 2M+1$. The shifted factor is holomorphic in s except for its usual pole at one and locally bounded in x. Analytic continuation proves the formula on the stated connected domain and shows that the apparent poles at the other displayed integers are removable. □

The cancellations are finite identities, not numerical limiting phenomena:

$$S_M(2d) = 0 \qquad (1 \le d < M). \tag{9.227}$$

Indeed, differentiate (9.219) $2d$ times at zero and use the order-$2M$ zero of the weight. At even spectral integers the sine factor instead cancels the gamma pole.

9.23.1 Complete holomorphic resonance germs

The next formulas give every derivative, not just the value at a resonance. Let h be a small spectral perturbation. For an odd integer $s_0 = 2d+1$, with $1 \le d < M$,

$$\int_0^1 V_{M,k}(x)\zeta(2d+1+h,x)\,\mathrm{d}x = \frac{(-1)^{d+1}2^{1-M}(2\pi k)^{2d}}{(2d)!} \times \frac{\Gamma(1-h)(2\pi k)^h\cos(\pi h/2)}{P_{2d}(h)}\,\frac{S_M(2d+h)}{h}. \tag{9.228}$$

For an even integer $s_0 = 2d$, with $1 \le d \le M$,

$$\int_0^1 V_{M,k}(x)\zeta(2d+h,x)\,\mathrm{d}x = \frac{(-1)^d 2^{1-M}(2\pi k)^{2d-1}}{(2d-1)!} \times \frac{\Gamma(1-h)(2\pi k)^h}{P_{2d-1}(h)}\,\frac{\sin(\pi h/2)}{h}\,S_M(2d-1+h). \tag{9.229}$$

Both right-hand sides are holomorphic. These formulas follow directly from

$$\Gamma(-n-h) = \frac{(-1)^{n+1}\Gamma(1-h)}{n!\,hP_n(h)}.$$

Consequently, the integral with $\zeta^{(j)}(s_0,x)$ is exactly $j!$ times the coefficient of h^j in the corresponding right-hand side. The logarithm of its common gamma–harmonic factor is

$$\log\frac{\Gamma(1-h)(2\pi k)^h}{P_n(h)} = (\gamma + \log(2\pi k) - H_n)h + \sum_{j\ge 2}\frac{\zeta(j)+(-1)^j H_n^{(j)}}{j}h^j. \tag{9.230}$$

This is a finite coefficient algorithm in gamma, harmonic numbers, ordinary zeta values, and logarithms of the integers $1,\dots,M$. No derivative of an infinite special-function series has to be fitted numerically.

At the pole $s=1$, put $c_M = 2^{-M}\binom{2M}{M}$, the mean of the weight. The Stieltjes counterpart is the holomorphic identity

$$\int_0^1 V_{M,k}(x)g(h,x)\,\mathrm{d}x = \frac{-2^{1-M}\Gamma(1-h)(2\pi k)^h\cos(\pi h/2)S_M(h) - c_M}{h}. \tag{9.231}$$

Its numerator vanishes at zero because $S_M(0) = -\binom{2M}{M}/2$. Extracting $(-1)^m m![h^m]$ evaluates the ordinary weighted integral of $\gamma_m(x)$ at every index. For $M=1$, this recovers (9.223) and (9.224).

9.23.2 Explicit logarithmic identities at order three

For every positive integer k, put $\lambda_k = \gamma + \log(2\pi k)$. Taking $M=2, d=1$ in (9.228), where $S_2(t) = -4 + 2^t$, gives

$$\boxed{\int_0^1 (1-\cos(2\pi kx))^2\zeta(3,x)\,\mathrm{d}x = 4\pi^2k^2\log 2.} \tag{9.232}$$

$$\boxed{\int_0^1 (1-\cos(2\pi kx))^2\zeta'(3,x)\,\mathrm{d}x = 4\pi^2k^2\log 2\left(\lambda_k - \frac{3}{2} + \frac{\log 2}{2}\right).} \tag{9.233}$$

Here ζ' denotes the spectral derivative. To see the derivative constant explicitly, the common factor in (9.228) has logarithmic derivative $\lambda_k - H_2 = \lambda_k - 3/2$ at zero, while

$$\frac{S_2(2+h)}{h} = 4\log 2 + 2\log^2 2\,h + O(h^2).$$

The local integrands behave like x and $x\log x$, respectively, so these are ordinary absolutely convergent integrals. Their polygamma form follows from $\psi^{(2)}(x) = -2\zeta(3,x)$.

The correlation identity in Theorem 9.81, at $(p,q)=(1,1)$, gives the compatible combination

$$\int_0^1 (1-\cos(2\pi kx))^2\big[\zeta(3,x) + 2\zeta'(3,x)\big]\,\mathrm{d}x = 4\pi^2k^2\big[2\log 2(\lambda_k - 1) + \log^2 2\big]. \tag{9.234}$$

This agreement checks the derivative signs and the harmonic constants by two exact routes: the shifted-contact calculus and the single-Hurwitz spectral kernel.

9.24 Polylogarithm boundary values and invisible local terms

For $\sigma \in \{+1,-1\}$, let

$$\mathfrak{L}_s^\sigma(a) = \sum_{k\ge 1}\frac{e^{\sigma 2\pi i k a}}{k^s} \tag{9.235}$$

be interpreted as a periodic distribution. Its Fourier coefficients have polynomial growth on every compact spectral set, so this is an entire distribution-valued function of s. It is also the Abel limit of $\mathrm{Li}_s(\rho e^{\sigma 2\pi i a})$ as $\rho\uparrow 1$. Spectral derivatives multiply the coefficients by powers of $-\log k$. Off the integer points these distributions agree with the usual analytic boundary values of polylogarithms [96].

Put $c=\gamma+\log(2\pi)$ and expand the finite polynomial

$$(-1)^p(\sigma 2\pi\mathrm{i})^r\big[b_m(c+X-\sigma\mathrm{i}\pi/2)+\kappa_{m,p}\big]\big[b_n(c+X+\sigma\mathrm{i}\pi/2)+\kappa_{n,q}\big]=\sum_{j=0}^{m+n+2}a_{j,\sigma}X^j. \tag{9.236}$$

Equation (9.195) proves the all-index polylogarithmic identity

$$\mathsf{C}^{p,q}_{mn}=\sum_{\sigma=\pm1}\sum_{j=0}^{m+n+2}a_{j,\sigma}(-1)^j\,\partial_s^j\mathfrak{L}^\sigma_s\big|_{s=-r}. \tag{9.237}$$

This is a finite formula; its equality is checked at every Fourier mode, including the zero mode, which vanishes on both sides. Together with Theorem 9.77, it identifies the exact delta derivative that must be added to the coordinate extension of the corresponding off-point polylogarithm identity.

The elementary cautionary example is

$$\mathfrak{L}^+_0+\mathfrak{L}^-_0=\delta_0-1. \tag{9.238}$$

For $a\notin\mathbb{Z}$, the ordinary rational values satisfy $\mathrm{Li}_0(e^{2\pi\mathrm{i}a})+\mathrm{Li}_0(e^{-2\pi\mathrm{i}a})=-1$. The missing delta cannot be read from that pointwise identity. Similarly, for $r\geq1$,

$$\mathfrak{L}^+_{-r}+(-1)^r\mathfrak{L}^-_{-r}=\frac{\delta_0^{(r)}}{(2\pi\mathrm{i})^r}. \tag{9.239}$$

These identities follow immediately by adding positive and negative Fourier modes, or by differentiating (9.238). They explain why substituting rational negative-order polylogarithms before fixing distributional boundary values can erase precisely the terms computed by the contact theorem.

9.25 Exact primitives and the log-gamma bridge

Define the mean-zero periodic inverse derivative by

$$\widehat{\mathcal{I}U}(k)=\frac{\widehat{U}(k)}{2\pi\mathrm{i}k}\quad(k\neq0),\qquad\widehat{\mathcal{I}U}(0)=0.$$

It satisfies $\mathrm{D}\mathcal{I}U=U-\widehat{U}(0)1$. Let B_h be the Bernoulli polynomial, periodically interpreted when necessary. Its standard Fourier coefficients, obtainable also from the negative-integer Hurwitz values [93], give

$$\mathcal{I}^h\delta_0^{(r)}=\begin{cases}\delta_0^{(r-h)}, & h<r,\\ \delta_0-1, & h=r\geq1,\\ -B_{h-r}(\{a\})/(h-r)!, & h>r.\end{cases} \tag{9.240}$$

For $h=r=0$, the left side is simply δ_0 and the middle rule is not used.

Corollary 9.83 (All normalized contact primitives)**.** *Apply $\mathcal{I}^h$ to Theorem 9.77. For $h>r$ the contact correction is exactly*

$$\mathcal{I}^h\mathsf{C}^{p,q}_{mn}-\mathcal{I}^h\mathsf{J}^{p,q}_{mn}=-\Delta^{p,q}_{mn}\frac{B_{h-r}(\{a\})}{(h-r)!}. \tag{9.241}$$

There is no undetermined polynomial or integration constant: the zero Fourier mode is fixed by definition.

For $h \geq r+2$, the integrated convolution has an absolutely convergent Fourier series because its coefficients are $O(|k|^{r-h}(1+\log^{m+n+2}|k|))$. Thus sufficiently integrated identities are equalities of continuous ordinary functions, even though their derivatives contain contact distributions.

For a concrete gamma example, set

$$\ell(x) = \log\Gamma(x) - \frac{1}{2}\log(2\pi).$$

Raabe's mean and the Hurwitz identity for $\zeta'(0,x)$ give $\int_0^1 \ell(x)\,\mathrm{d}x = 0$. Distributionally $\mathrm{D}\ell = -T_{0,0}$; no extra delta occurs because the constant term in the endpoint expansion of $\log\Gamma(x)$ is zero and $\log\Gamma(1) = 0$. Hence $\ell = -\mathcal{I}T_{0,0}$ and

$$K_\Gamma(a) := \int_0^1 \ell(x)\ell(\{x+a\})\,\mathrm{d}x = -\mathcal{I}^2\mathsf{C}_{00}^{0,0}. \tag{9.242}$$

The integral exists also at $a = 0$ because logarithmic endpoint singularities are square integrable. The contact theorem gives the normalized form

$$K_\Gamma(a) = -\mathcal{I}^2\mathsf{J}_{00}^{0,0}(a) + \frac{\pi^2}{6}B_2(\{a\}). \tag{9.243}$$

The Bernoulli term is the integrated image of the $\pi^2\delta_0/3$ term in (9.196).

The ordinary Fourier representation is already (8.144), after subtracting the Raabe mean. The additional information here is its fixed-coordinate Bernoulli correction: it is the integrated image of the collision contact, which a punctured-circle polylogarithm identity cannot determine.

Evidence and remaining scope. The two independent contact generators agree after aligning their derivative signs and unit-coordinate conventions. Their finite exact tables and independent subtracted Fourier-mode quadratures test those normalizations; the analytic comparison above proves the all-index law. This completes the specified periodic extension problem. Cubic geometric collision limits, ordered higher-depth germs and canonical changes of subtraction coordinates retain their separate audit status. The contact coefficient does not assert independence of its displayed zeta coordinates.

Chapter 10

Positive kernels and the geometry of Stieltjes zeros

The derivative tower gives exact spectral coordinates. Its Mellin form does more: a reciprocal-gamma polynomial becomes a sign-changing kernel inside a positive Laplace transform. Root preservation bounds the total number of positive zeros, gamma concentration explains their motion, and outward rational computation certifies concrete base cases. This chapter unites the fixed-index zero geometry with the complementary spectral regime in which the Laurent index grows with the derivative order.

10.1 A unified Stieltjes–Lerch derivative family

For $a > 0$, write

$$\zeta(s,a) = \frac{1}{s-1} + \sum_{n\geq 0} \frac{(-1)^n}{n!}\gamma_n(a)(s-1)^n. \tag{10.1}$$

The superscript in $\gamma_n^{(k)}(a)$ means differentiation k times in a. In contrast, $\zeta^{(j)}(s,a)$ denotes differentiation j times in s. Set

$$P_k(t) = \prod_{j=1}^{k}\left(1+\frac{t}{j}\right) = \sum_{i=0}^{k} e_i(k)t^i, \qquad H_k^{(r)} = \sum_{j=1}^{k} j^{-r}, \quad H_k = H_k^{(1)}. \tag{10.2}$$

We take $H_0^{(r)} = 0$.

For $0 \leq \rho < 1$, define the convergent Lerch coefficients

$$\ell_n(\rho,a) = \sum_{m\geq 0} \frac{\rho^m \log^n(a+m)}{a+m} = (-1)^n\, \partial_s^n \Phi(\rho,s,a)|_{s=1}\,, \qquad \Phi(\rho,s,a) = \sum_{m\geq 0}\frac{\rho^m}{(a+m)^s}. \tag{10.3}$$

For $k \geq 1$ put

$$\mathcal{D}_{n,k}^{\rho}(a) = \begin{cases} \partial_a^k \ell_n(\rho,a), & 0 \leq \rho < 1, \\ \gamma_n^{(k)}(a), & \rho = 1. \end{cases} \tag{10.4}$$

Only the derivative family is unified at $\rho = 1$: the series defining ℓ_n generally diverges there. For $0 < \rho < 1$ the specialization $a = 1$ is directly polylogarithmic,

$$\ell_n(\rho,1) = \frac{(-1)^n}{\rho}\, \partial_s^n \operatorname{Li}_s(\rho)|_{s=1}\,. \tag{10.5}$$

At $\rho = 0$ it is simply $\log^n a/a$.

Proposition 10.1 (The classical master identity)**.** *For $k \geq 1$, $n \geq 0$, and $a > 0$,*

$$\gamma_n^{(k)}(a) = (-1)^{n+k} k! n! \sum_{i+j=n} e_i(k) \frac{\zeta^{(j)}(k+1, a)}{j!}. \tag{10.6}$$

The same formula with $\Phi(\rho, \cdot, a)$ in place of $\zeta(\cdot, a)$ holds for $\mathcal{D}_{n,k}^{\rho}$ when $\rho < 1$.

Proof. The normally convergent defining series gives $\partial_a^k \zeta(s, a) = (-1)^k (s)_k \zeta(s+k, a)$ first in a right half-plane, and then by meromorphic continuation. Apply it to (10.1), noting that the pole term is independent of a. Substituting $s = 1 + t$ and using $(1+t)_k = k! P_k(t)$ gives

$$\sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n^{(k)}(a) t^n = (-1)^k k! P_k(t) \zeta(k + 1 + t, a). \tag{10.7}$$

Coefficient comparison proves (10.6). For $\rho < 1$ the same calculation is justified directly by locally uniform convergence of the Lerch series. This is Coffey's Stirling-number formula in the harmonic/symmetric-function coordinates used by the source draft. □

Definition 10.2 (Reciprocal-gamma Appell polynomials)**.** *Define the monic polynomials Q_n by*

$$\frac{e^{xt}}{\Gamma(1+t)} = \sum_{n \geq 0} Q_n(x) \frac{t^n}{n!}. \tag{10.8}$$

Set $W_\rho(x) = (1 - \rho e^{-x})^{-1}$ for $x > 0$ and $0 \leq \rho \leq 1$.

Writing $y = x + \gamma$ and $\zeta_r = \zeta(r)$, the first polynomials are

$$\begin{aligned} Q_0(x) &= 1, & Q_1(x) &= y, \\ Q_2(x) &= y^2 - \zeta_2, & Q_3(x) &= y^3 - 3\zeta_2 y + 2\zeta_3, \\ Q_4(x) &= y^4 - 6\zeta_2 y^2 + 8\zeta_3 y + 3\zeta_2^2 - 6\zeta_4. \end{aligned} \tag{10.9}$$

They satisfy $Q_n' = n Q_{n-1}$ and the explicit recurrence

$$Q_{n+1}(x) = (x + \gamma) Q_n(x) + \sum_{r=1}^{n} (-1)^r \binom{n}{r} r! \zeta(r+1) Q_{n-r}(x). \tag{10.10}$$

This follows by differentiating (10.8) in t and using $-\psi(1+t) = \gamma + \sum_{r \geq 1} (-1)^r \zeta(r+1) t^r$ near zero.

Theorem 10.3 (Appell–Laplace representation)**.** *For all $n \geq 0$, $k \geq 1$, $a > 0$, and $0 \leq \rho \leq 1$,*

$$F_{n,k}^{\rho}(a) := (-1)^{n+k} \mathcal{D}_{n,k}^{\rho}(a) = \int_0^\infty x^k e^{-ax} W_\rho(x) Q_n(\log x) \, \mathrm{d}x. \tag{10.11}$$

The integral and every fixed number of its a-derivatives converge locally uniformly for $a > 0$, uniformly in $\rho \in [0, 1]$.

Proof. The Hurwitz and Lerch Mellin integrals give, for $|t|$ sufficiently small,

$$k! P_k(t) \Phi(\rho, k + 1 + t, a) = \frac{1}{\Gamma(1+t)} \int_0^\infty x^{k+t} e^{-ax} W_\rho(x) \, \mathrm{d}x,$$

where $\Phi(1, s, a)$ means $\zeta(s, a)$ in its convergent half-plane. The cancellation of $\Gamma(k + 1 + t)$ is the reason for introducing Q_n. Extract the coefficient of t^n and apply (10.7).

For justification, $W_\rho(x) \leq W_1(x) = O(x^{-1})$ at zero and $W_\rho(x) = O(1)$ at infinity, uniformly in ρ. On $|t| \leq \eta < k$, the integrand near zero is bounded by a constant times $x^{k-1-\eta}$, and its coefficient derivatives introduce only powers of $|\log x|$. At infinity, a fixed positive lower bound on a supplies exponential domination. Multiplication by any fixed power of x handles the a-derivatives. These estimates also prove continuity at $\rho = 1$. □

10.2 Real-rootedness and a global zero bound

Lemma 10.4 (A real-root preserving operator)**.** *If a real polynomial f of degree n has only real roots and $c \neq 0$ is real, then $f + cf'$ has only real roots. An inherited root of multiplicity m has multiplicity $m-1$ if $m \geq 2$, and every other root of $f + cf'$ is simple. Consequently, after $n-1$ such operators, any real-rooted polynomial of degree n becomes strictly real-rooted.*

Proof. Away from the distinct roots r_i of f, of multiplicities m_i, the new roots solve

$$\frac{f'(x)}{f(x)} = \sum_i \frac{m_i}{x - r_i} = -\frac{1}{c}.$$

The left side is strictly decreasing on each complementary interval because its derivative is $-\sum_i m_i/(x-r_i)^2 < 0$. It has one solution between each consecutive pair of distinct roots and one further solution in the appropriate exterior interval. All these solutions are simple. Factoring at r_i proves the inherited multiplicity assertion, and counting degrees accounts for all roots. Repeating reduces every possible multiple root at each step; new roots never introduce multiplicity. □

Theorem 10.5 (Strictly real-rooted reciprocal-gamma polynomials)**.** *For every $n \geq 1$, Q_n has exactly n distinct real roots. The roots of Q_{n-1} strictly interlace those of Q_n.*

Proof. The classical Weierstrass product [44] is

$$\frac{1}{\Gamma(1+t)} = e^{\gamma t} \prod_{m=1}^{\infty} (1 + t/m) e^{-t/m}, \tag{10.12}$$

with locally uniform convergence. For a finite product, extracting $n![t^n]$ after multiplication by e^{xt} is equivalent to applying operators $1 + \partial_x/m$ and real translations to x^n. Lemma 10.4 therefore proves real-rootedness of each finite-product polynomial. Coefficient convergence and closure of the set of monic real-rooted polynomials prove real-rootedness in the limit.

A limit of simple-rooted polynomials alone would not prove simplicity. To obtain it, remove the first $n-1$ factors $(1+t/m)e^{-t/m}$ from (10.12). The Appell polynomial associated with the remaining tail is still real-rooted by the same finite-product argument. Restore the removed factors. The translations preserve multiplicity and commute with the differential operators; the $n-1$ operators $1 + \partial_x/m$ make all roots simple by Lemma 10.4. Finally $Q_n' = nQ_{n-1}$ and Rolle's theorem give strict interlacing. □

Lemma 10.6 (Laplace variation bound, with multiplicity)**.** *Let $g : (0, \infty) \to \mathbb{R}$ have exactly N sign changes, with constant nonzero sign between the sign-change points, up to null sets. Suppose all exponentially weighted polynomial moments needed below are absolutely integrable. Its Laplace transform $G(a) = \int_0^\infty e^{-ax} g(x)\,\mathrm{d}x$ is not identically zero and has at most N zeros on $(0, \infty)$, counted with multiplicity.*

Proof. For $N = 0$ the transform has a fixed strict sign. For $N > 0$, choose a sign-change point x_0. Multiplying g by $x_0 - x$ removes that sign change and introduces no others. The transform of the new function is

$$(\partial_a + x_0)G(a) = e^{-x_0 a} \frac{\mathrm{d}}{\mathrm{d}a}\big(e^{x_0 a} G(a)\big).$$

By induction it has at most $N-1$ zeros. If G has any finite collection of zeros of total multiplicity M, Rolle's theorem with multiplicity gives at least $M-1$ zeros of the displayed derivative between

the first and last zero, including inherited multiplicities at the endpoints when appropriate. Thus $M \le N$. This bounds every finite collection, and hence all zeros. The induction hypothesis also says that the transform of the new kernel is not identically zero. If G were identically zero, its displayed transformed derivative would be identically zero as well, a contradiction. This completes the induction including nonvanishing. □

Theorem 10.7 (Global bound on positive zeros)**.** *For every $n \ge 0$, $k \ge 1$, and $0 \le \rho \le 1$, the function $\mathcal{D}^{\rho}_{n,k}(a)$ has at most n positive zeros, counted with multiplicity. In particular, this holds for $\gamma^{(k)}_n(a)$.*

Proof. The factor $x^k W_\rho(x)$ in (10.11) is strictly positive. By Theorem 10.5, $Q_n(\log x)$ has precisely n simple sign changes on $(0,\infty)$. The estimates in Theorem 10.3 justify every step of Lemma 10.6. The transform is not identically zero: as $a \downarrow 0$, scaling $x = u/a$ gives

$$F^{\rho}_{n,k}(a) \sim k! a^{-k-1} \log^n(1/a), \tag{10.13}$$

uniformly in ρ, with the convention $\log^0(1/a) = 1$. This follows from monicity of Q_n and $W_\rho(u/a) \to 1$, using the same integrable bounds after splitting at $u = a$. The variation bound now applies. □

Remark 10.8 (The elementary endpoint explains possible multiplicity). At $\rho = 0$ one has

$$F^{0}_{n,k}(a) = k! a^{-k-1}\, n! [t^n] P_k(t) e^{-t \log a}.$$

The polynomial in $x = -\log a$ is $\prod_{j=1}^{k}(1 + \partial_x/j)x^n$. It has n real roots counted with multiplicity by Lemma 10.4. When $n > k$, the root $x = 0$ has multiplicity exactly $n - k$. Thus all zeros cannot be declared simple uniformly in ρ at small k. Any uniform simplicity threshold for $n \ge 2$ must be at least $n - 1$.

10.3 An all-order concentration expansion

The scale on which the zeros move is $a = ck$, with c bounded away from zero and infinity. Set

$$\mathcal{T}^{\rho}_{n,k}(c) = \frac{(-1)^{n+k}(ck)^{k+1}}{k!} \mathcal{D}^{\rho}_{n,k}(ck), \qquad h_{n,\rho}(x) = W_\rho(x) Q_n(\log x). \tag{10.14}$$

Let Y_k have gamma density

$$\frac{k^{k+1}}{k!} y^k e^{-ky}, \qquad y > 0.$$

Then $\mathbb{E}Y_k = 1 + 1/k$, and (10.11) becomes the exact identity

$$\mathcal{T}^{\rho}_{n,k}(c) = \mathbb{E}\, h_{n,\rho}(Y_k/c). \tag{10.15}$$

The shape is $k+1$, not k; the resulting $1/k$ shift matters for the constant term of each zero.

Define polynomials C_j by the formal generating function

$$\sum_{j \ge 0} C_j(z) w^j = \exp\left\{ \sum_{\ell \ge 1} w^\ell \left(\frac{z^\ell}{\ell} + \frac{z^{\ell+1}}{\ell+1} \right) \right\}. \tag{10.16}$$

For $C_j(z) = \sum_r c_{j,r} z^r$, use the *normal-ordered* differential operator

$$(A_j h)(x) = \sum_r c_{j,r} x^r h^{(r)}(x). \tag{10.17}$$

This notation does not mean $C_j(x\partial_x)$: the powers of x stand to the left of the derivatives. For example,

$$\begin{aligned} A_0h &= h, \\ A_1h &= xh' + \tfrac{1}{2}x^2h'', \\ A_2h &= x^2h'' + \tfrac{5}{6}x^3h''' + \tfrac{1}{8}x^4h^{(4)}. \end{aligned} \tag{10.18}$$

Theorem 10.9 (Uniform complete expansion)**.** *Fix $n, J, r \geq 0$ and a compact interval $I \subset (0, \infty)$. As $k \to \infty$,*

$$\mathcal{T}^{\rho}_{n,k}(c) = \sum_{j=0}^{J} \frac{(A_j h_{n,\rho})(1/c)}{k^j} + O_{n,J,r,I}(k^{-J-1}), \tag{10.19}$$

uniformly for $c \in I$ and $0 \leq \rho \leq 1$, also after any number of c-derivatives up to r.

Proof. The centered moment generating function is

$$\mathbb{E}e^{z(Y_k-1)} = e^{-z}(1 - z/k)^{-k-1}.$$

For fixed z near zero its logarithm is exactly the convergent expansion

$$\log \mathbb{E}e^{z(Y_k-1)} = \sum_{\ell \geq 1} k^{-\ell} \left(\frac{z^\ell}{\ell} + \frac{z^{\ell+1}}{\ell+1} \right). \tag{10.20}$$

Consequently the coefficient of k^{-j} in $\mathbb{E}(Y_k - 1)^m/m!$ is $[z^m]C_j(z)$. For fixed m, these moments are finite polynomials in $1/k$, since only finitely many terms of the exponential can contribute to z^m. The degree of C_j is at most $2j$.

Here are the estimates needed to turn coefficient comparison into a uniform asymptotic statement. On $1/2 \leq y \leq 3/2$, all derivatives of $h_{n,\rho}(y/c)$ of any fixed order in y and c are uniformly bounded for $c \in I$ and $\rho \in [0, 1]$. Indeed y/c stays in a compact subinterval of $(0, \infty)$ and $1 - \rho e^{-y/c}$ is uniformly positive there. Taylor's theorem at $y = 1$, through degree $2J + 1$, has remainder bounded by $C|y - 1|^{2J+2}$. Formula (10.20) gives

$$\mathbb{E}|Y_k - 1|^{2J+2} = O(k^{-J-1}).$$

Replacing truncated central moments by full central moments changes the answer only exponentially, as we now verify.

For each fixed derivative order, the functions to be averaged admit bounds

$$|\partial_c^u h_{n,\rho}(y/c)| \leq C\big(y^{-B} + y^B\big)(1 + |\log y|^n)$$

for some fixed B and C, uniformly in the stated parameters. These follow from $W_1(x) = O(1 + x^{-1})$ and its differentiated forms; enlarging B absorbs the logarithm if convenient. For a fixed real exponent v,

$$\mathbb{E}\big[Y_k^v \mathbf{1}_{Y_k \notin [1/2,3/2]}\big] = \frac{\Gamma(k+1+v)}{\Gamma(k+1)k^v} \Pr\{Z_{k,v} \notin [1/2, 3/2]\},$$

where $Z_{k,v}$ has shape $k + 1 + v$ and rate k, for all sufficiently large k. The prefactor is bounded, and a Chernoff bound obtained from the gamma moment generating function makes the probability $O(e^{-\eta k})$ for some $\eta > 0$. The same holds for the finitely many polynomial and inverse-polynomial bounds just used. This proves exponentially small tails for the Taylor remainder and all retained moments.

Taylor expansion in (10.15), followed by (10.20), therefore gives exactly (10.19) and the coefficients (10.17). Applying the same argument to $\partial_c^u h_{n,\rho}(y/c)$ proves the stated derivative uniformity. No differentiation of an uncontrolled big-O term is used. □

Remark 10.10 (What uniformity does not say). The constants in Theorem 10.9 may depend on n and on the compact c-interval. The theorem does not supply estimates uniform when n grows with k, nor when $c \downarrow 0$ or $c \to \infty$. These are separate problems, especially because the extreme roots of Q_n then control very different scales.

10.4 Exactly n eventual zeros and their locations

Order the Appell roots and define the corresponding slopes by

$$x_{n,1} > x_{n,2} > \cdots > x_{n,n}, \qquad c_{n,j} = e^{-x_{n,j}}, \quad 0 < c_{n,1} < \cdots < c_{n,n}. \tag{10.21}$$

Theorem 10.11 (Uniform eventual zero theorem)**.** *For every fixed $n \geq 1$ there is an integer K_n such that, for every $k \geq K_n$ and every $\rho \in [0,1]$, $\mathcal{D}^{\rho}_{n,k}$ has exactly n positive zeros, all simple. Number them increasingly as $a^{\rho}_{n,j,k}$. Uniformly in ρ,*

$$a^{\rho}_{n,j,k} = kc_{n,j} + d_{n,j}(\rho) + O_n(k^{-1}), \tag{10.22}$$

where

$$d_{n,j}(\rho) = \frac{c_{n,j}}{2}\left(1 + \frac{Q_n''(x_{n,j})}{Q_n'(x_{n,j})}\right) - \frac{\rho}{\exp(1/c_{n,j}) - \rho}. \tag{10.23}$$

More generally, for every fixed $J \geq 0$ there are coefficients $b_{n,j,\ell}(\rho)$ such that

$$a^{\rho}_{n,j,k} = kc_{n,j} + d_{n,j}(\rho) + \sum_{\ell=1}^{J} b_{n,j,\ell}(\rho)k^{-\ell} + O_{n,J}(k^{-J-1}), \tag{10.24}$$

uniformly for $\rho \in [0,1]$. These coefficients are recursively computable from (10.16).

Proof. The leading term in Theorem 10.9 is

$$G_0(c,\rho) = W_\rho(1/c)Q_n(-\log c).$$

Its zeros are exactly the $c_{n,j}$, and its derivative at such a zero is

$$\partial_c G_0(c_{n,j},\rho) = -\frac{W_\rho(1/c_{n,j})}{c_{n,j}}Q_n'(x_{n,j}) \neq 0. \tag{10.25}$$

Choose disjoint closed intervals around these slopes on whose endpoints $Q_n(-\log c)$ has opposite signs. Since $1 \leq W_\rho(1/c) \leq W_1(1/c)$ on their union, the endpoint absolute values are bounded away from zero uniformly in ρ. The uniform $J = 0$ approximation gives one zero in each interval for all sufficiently large k, uniformly in ρ. The global bound of Theorem 10.7 excludes any further zero and, because it counts multiplicity, makes every one of the n zeros simple.

Put $c = c_{n,j} + d/k + O(k^{-2})$. The k^{-1} coefficient of (10.19) gives

$$d = -\frac{G_1(c_{n,j},\rho)}{\partial_c G_0(c_{n,j},\rho)}, \qquad G_1(c,\rho) = (A_1 h_{n,\rho})(1/c). \tag{10.26}$$

At $x = 1/c_{n,j}$, write $Q' = Q_n'(x_{n,j})$ and $Q'' = Q_n''(x_{n,j})$. Because $Q_n(\log x) = 0$,

$$xh' + \tfrac{1}{2}x^2h'' = \tfrac{1}{2}W_\rho(x)(Q' + Q'') + xW_\rho'(x)Q'.$$

Also

$$\frac{W'_\rho(x)}{W_\rho(x)} = -\frac{\rho}{e^x - \rho}.$$

Substitution in (10.26) proves (10.23).

For completeness, the all-order assertion does not require a convergent power series in $1/k$. Apply Theorem 10.9 to one additional order at each step. Substitute a finite ansatz for c into the finite expansion. At every new order the unknown coefficient occurs linearly with the nonzero multiplier (10.25); hence it is uniquely determined. This derivative is bounded away from zero uniformly in ρ. The residual of a truncation, divided by a uniform lower derivative bound on a smaller root interval, bounds the error in the true zero by the mean value theorem. Expanding one order farther than the desired expansion of $a = kc$ gives (10.24) with its stated uniform remainder. □

10.4.1 Examples and the first zero

For $n = 1$, $x_{1,1} = -\gamma$ and $c_{1,1} = e^\gamma$. Thus the unique zero of $\gamma_1^{(k)}$ satisfies

$$a^1_{1,1,k} = e^\gamma k + \frac{e^\gamma}{2} - \frac{1}{\exp(e^{-\gamma}) - 1} + O(k^{-1}). \tag{10.27}$$

The constant term is approximately -0.4370805068. At $\rho = 0$ the zero is exactly e^{H_k}, whose constant term is $e^\gamma/2$; the Lerch kernel therefore changes the constant term, not the slope. A fully explicit refinement, obtained from Appendix 10.5, is available throughout the deformation. Put $c = e^\gamma$ and $g = \rho/(e^{1/c} - \rho)$. Then, uniformly in $\rho \in [0, 1]$,

$$a_k(\rho) = ck + \frac{c}{2} - g + \frac{1}{k}\left\{\frac{c}{24} + \frac{g(g+1)}{c} - \frac{g(g+1)(2g+1)}{2c^2}\right\} + O(k^{-2}). \tag{10.28}$$

At $\rho = 1$, the coefficient of k^{-1} is approximately 0.0288707254211. At $\rho = 0$ it is $c/24$, agreeing with the expansion of the exact zero e^{H_k}.

For $n = 2$ the roots are $-\gamma \pm \sqrt{\zeta(2)}$, and hence

$$c_{2,1} = e^{\gamma - \pi/\sqrt{6}}, \qquad c_{2,2} = e^{\gamma + \pi/\sqrt{6}}. \tag{10.29}$$

The identity $Q''_n(x_j)/Q'_n(x_j) = 2\sum_{i \neq j}(x_j - x_i)^{-1}$ gives an alternative expression for every offset entirely in terms of the Appell roots.

10.4.2 Nonasymptotic information at $n = 1$

Theorem 10.12 (Harmonic brackets and monotonicity)**.** *For each $k \geq 1$ and $\rho \in [0, 1]$, $\mathcal{D}^\rho_{1,k}$ has exactly one positive zero, denoted $a_k(\rho)$. It is simple, and*

$$e^{H_{k-1}} < a_k(\rho) \leq e^{H_k}. \tag{10.30}$$

Equality at the upper endpoint occurs exactly when $\rho = 0$. The function $\rho \mapsto a_k(\rho)$ is continuous on $[0, 1]$ and strictly decreasing. In particular, $a_k(1) < a_k(\rho) < e^{H_k}$ for $0 < \rho < 1$.

Proof. Normalize the positive density proportional to $x^k e^{-ax} W_\rho(x)$ and write $m(a, \rho)$ for its expectation of $\log X + \gamma$. The sign of $F^\rho_{1,k}(a)$ is the sign of m. Under an unweighted gamma density of shape $k + 1$ and rate a, this expectation equals $H_k - \log a$. Weighting by the decreasing function

n	j	slope $c_{n,j}$	offset $d_{n,j}(1)$
1	1	1.781072417990	−0.437080506782
2	1	0.493943487920	0.287385687369
2	2	6.422230550060	−5.227782104747
3	1	0.261424326196	0.354674627668
3	2	1.064376550729	−0.506464646434
3	3	20.305020815096	−21.208385701520
4	1	0.172347335379	0.355378064541
4	2	0.465599872096	0.040510911109
4	3	2.062627326998	−2.178710104429
4	4	60.797835737559	−70.712693097880

Table 10.1: Numerical evaluations of the exact slope and offset formulas. These decimals are not isolating intervals. The accompanying CSV records zero computations at $k = 10, 40, 160$.

W_ρ lowers the expectation of the increasing function $\log X$; the decrease is strict when $\rho > 0$. The elementary covariance identity

$$2\operatorname{Cov}(f(X), g(X)) = \mathbb{E}\big[(f(X) - f(X'))(g(X) - g(X'))\big]$$

for independent identically distributed X, X' justifies this comparison without an external stochastic-order theorem. Hence $m(e^{H_k}, \rho) \le 0$, with equality exactly at $\rho = 0$.

Alternatively use a gamma density of shape k and rate a and weight it by $xW_\rho(x)$. This weight is strictly increasing, since

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{x}{1-\rho e^{-x}} = \frac{1-\rho(1+x)e^{-x}}{(1-\rho e^{-x})^2} > 0 \qquad (x > 0).$$

The unweighted expectation is $H_{k-1} - \log a$, so $m(e^{H_{k-1}}, \rho) > 0$. A zero exists between the brackets, and Theorem 10.7 proves uniqueness and simplicity.

For fixed ρ, exponential tilting gives $\partial_a m = -\operatorname{Cov}(\log X, X) < 0$. For $\rho_2 > \rho_1$, the ratio

$$\frac{W_{\rho_2}(x)}{W_{\rho_1}(x)} = \frac{1-\rho_1 e^{-x}}{1-\rho_2 e^{-x}}$$

is strictly decreasing in x. The same covariance comparison therefore gives $m(a, \rho_2) < m(a, \rho_1)$. Together with strict decrease in a, this implies $a_k(\rho_2) < a_k(\rho_1)$. Continuity follows from locally uniform dominated convergence in (10.11), uniqueness of the root, and the common compact bracket. No differentiability in ρ at the endpoint $\rho = 1$ is assumed. □

10.4.3 Propagation and interlacing in the derivative order

Corollary 10.13 (Persistence and rightward interlacing)**.** *Fix $n \ge 1$ and $\rho \in [0, 1]$. Once $F^\rho_{n,k}$ has n distinct positive zeros, the same is true at every larger derivative order. At consecutive such orders,*

$$a^\rho_{n,j,k} < a^\rho_{n,j,k+1} < a^\rho_{n,j+1,k} \quad (1 \le j < n), \qquad a^\rho_{n,n,k} < a^\rho_{n,n,k+1}. \tag{10.31}$$

Thus the unique $n = 1$ zero moves strictly rightward for every $k \ge 1$.

Proof. Differentiating the integral gives $F^\rho_{n,k+1} = -\partial_a F^\rho_{n,k}$. Rolle's theorem supplies a zero between each pair of consecutive simple zeros. Also $F^\rho_{n,k}(a) \to 0$ as $a \to \infty$, and it has a constant nonzero

sign after its last zero. There must be a stationary point after that zero: otherwise its nonzero derivative would have constant sign and could not both depart from zero and tend back to zero. Hence there are at least n distinct zeros at the next order. Theorem 10.7 gives exactly those zeros, all simple, and excludes any zero before the first interval or a coincident zero at an old simple root. Induction proves persistence. □

10.5 A recurrence for the next zero coefficient

For a particular root, abbreviate $c_0 = c_{n,j}$ and $G_m(c,\rho) = (A_m h_{n,\rho})(1/c)$. Suppose

$$a^{\rho}_{n,j,k} = kc_0 + d + \frac{b_1}{k} + O(k^{-2}).$$

Taylor expansion of (10.19) gives

$$b_1 = -\frac{\frac{1}{2}\,\partial_c^2 G_0(c_0,\rho)d^2 + \partial_c G_1(c_0,\rho)d + G_2(c_0,\rho)}{\partial_c G_0(c_0,\rho)}. \tag{10.32}$$

Here d is (10.23), G_2 is explicit from (10.18), and the denominator is nonzero by (10.25). This is often more usable than a long expanded expression involving several root separations. It also explains why the quantity $k(a - kc_0 - d)$ tends to a finite limit. Higher coefficients follow by the same triangular recursion.

10.6 The classical half-unit bracket

For this specialization abbreviate $h_n = h_{n,1}$, and put $t_0 = e^{-\gamma}$.

Theorem 10.14 (Unique zero and an explicit bracket)**.** *For every integer $k \geq 1$, $\gamma_1^{(k)}(a)$ has a unique positive zero a_k, which is simple. With $H_0 = 0$,*

$$e^{H_{k-1}} < a_k < e^{H_{k-1}} + \tfrac{1}{2}. \tag{10.33}$$

The zeros are strictly increasing in k. Their asymptotic expansion starts with

$$a_k = e^{\gamma}\left(k + \tfrac{1}{2}\right) - \frac{1}{e^{e^{-\gamma}} - 1} + O(k^{-1}). \tag{10.34}$$

Formula (10.32) supplies the next coefficient and an $O(k^{-2})$ remainder.

Proof. The endpoint signs, the Laplace representation above, and the bound of one zero prove existence, uniqueness, and simplicity. To obtain the bracket, normalize the positive density $t^k e^{-at}/(1 - e^{-t})$. The sign of $F_{1,k}(a)$ is the sign of $\mathbb{E}(\log T + \gamma)$ under this density.

Relative to a gamma density of shape k and rate a, the tilting factor is $r(t) = t/(1 - e^{-t})$, which is strictly increasing: its derivative has numerator $1 - e^{-t}(1 + t) > 0$. A strictly increasing tilt strictly increases the expectation of the strictly increasing function $\log t$. This elementary fact follows from

$$2\,\mathrm{Cov}(u(X), v(X)) = \mathbb{E}[(u(X) - u(Y))(v(X) - v(Y))] > 0$$

for independent X, Y with the same continuous distribution and strictly increasing u, v. Consequently

$$\mathbb{E}\log T > \psi(k) - \log a = H_{k-1} - \gamma - \log a.$$

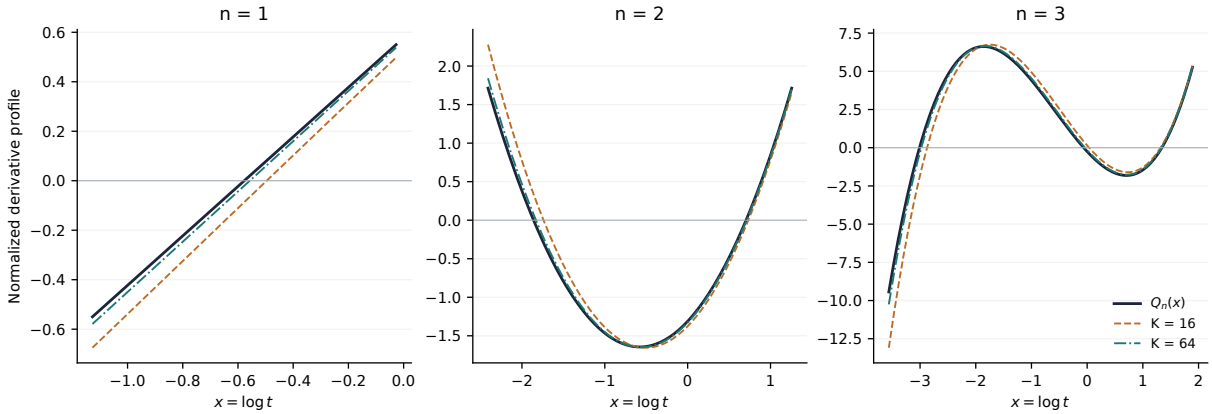


Figure 10.1: Normalized derivative profiles $(1 - e^{-e^x})\mathcal{H}_{n,K}(e^x)$ approach $Q_n(x)$ as $K = k + 1$ increases. Here $\mathcal{H}_{n,K}(t) = \mathbb{E}[Q_n(\log(tY/K))/(1 - e^{-tY/K})]$ for $Y \sim \text{Gamma}(K, 1)$. The plots illustrate $n = 1, 2, 3$. Quadrature produces the curves; Theorems 10.7 and 10.11, rather than the plots, prove the zero counts and asymptotic completeness.

At the zero this gives $a_k > e^{H_{k-1}} \geq 1$.

For $a > 1/2$, compare instead with a gamma density of shape k and rate $a - 1/2$. The tilting factor is $s(t) = t/(2\sinh(t/2))$, which is strictly decreasing because $(t/2)\cosh(t/2) > \sinh(t/2)$. The covariance inequality reverses:

$$\mathbb{E}\log T < \psi(k) - \log(a - 1/2).$$

Applying it at $a_k > 1$ gives $a_k < e^{H_{k-1}} + 1/2$.

Finally, at a zero the signed kernel $f(t)$ in (10.11) satisfies $\int e^{-a_k t} f(t)\, dt = 0$ and $(t - t_0)f(t) > 0$ except at t_0. Thus

$$F_{1,k+1}(a_k) = \int_0^\infty (t - t_0)e^{-a_k t} f(t)\, dt > 0.$$

Since $F_{1,k+1}$ crosses from positive to negative at its unique zero, $a_{k+1} > a_k$. The root of Q_1 is $-\gamma$, so (10.34) is Theorem 10.11. □

The interval in (10.33) is an analytic certificate for bracketing a unique root, valid even at $k = 1$. Its width is uniformly $1/2$; no assertion that this is the best possible width is made. The accompanying computation uses this bracket rather than guessing an initial zero from a graph.

10.7 Exact low-index zero counts

The general theorem proves eventual saturation. At the classical endpoint $\rho = 1$, a finite exact certificate supplies a stronger statement for $1 \leq n \leq 4$ at every $k \geq 1$. Define the sign-preserving normalization

$$\mathcal{V}_{n,k}(a) = \frac{a^{k+1}(-1)^{n+k}}{k!}\gamma_n^{(k)}(a).$$

10.7.1 A reproducible Euler–Maclaurin sign certificate

The finite certificates above can be checked without floating-point special functions. Put $p = k + 1$, and define

$$R_{n,\ell}(v) = n! \sum_{j=0}^{n} e_{n-j}(\ell) \frac{(-v)^j}{j!}.$$

The master identity gives the absolutely convergent spectral representation

$$\mathcal{V}_{n,k}(a) = \sum_{m \geq 0} \left(\frac{a}{a+m} \right)^p R_{n,k}(\log(a+m)).$$

For positive rational a, integers $M, R \geq 1$, $A = a + M > 1$ and $L = k + 2R$, let

$$\begin{aligned} \mathcal{E}_{M,R} = & \sum_{m=0}^{M-1} \left(\frac{a}{a+m} \right)^p R_{n,k}(\log(a+m)) \\ & + \left(\frac{a}{A} \right)^p \left[\frac{A}{k} R_{n,k-1}(\log A) + \frac{1}{2} R_{n,k}(\log A) \right] \\ & + \left(\frac{a}{A} \right)^p \sum_{r=1}^{R} \frac{B_{2r}}{(2r)!} \left(\prod_{j=1}^{2r-1} \frac{k+j}{A} \right) R_{n,k+2r-1}(\log A). \end{aligned}$$

Proposition 10.15 (An explicit rational-certifiable remainder)**.** *One has*

$$|\mathcal{V}_{n,k}(a) - \mathcal{E}_{M,R}| \leq \left(\frac{a}{A} \right)^p \frac{|B_{2R}|}{(2R)!} \left(\prod_{j=1}^{2R-1} \frac{k+j}{A} \right) \mathcal{B}_{n,L}(\log A),$$

$$\mathcal{B}_{n,L}(v) = \sum_{i=0}^{\min(n,L)} \sum_{j=0}^{n-i} \frac{n! e_i(L)}{(n-i-j)! L^j} v^{n-i-j}.$$

Every factor in the bound is nonnegative.

Proof. Apply Euler–Maclaurin through the Rth Bernoulli term to the spectral sum, or equivalently apply it to the Hurwitz series before extracting the nth spectral coefficient. The integral and half-endpoint terms give the second line of $\mathcal{E}_{M,R}$. The identity $(1+u)_k (k+1+u)_{2r-1} = (1+u)_{k+2r-1}$ gives its correction terms.

The periodic Bernoulli remainder has absolute kernel bound $|B_{2R}|$. After differentiation, bound the absolute value of $R_{n,L}(\log t)$ by $n! \sum_i e_i(L) (\log t)^{n-i}/(n-i)!$ for $t \geq A > 1$. For each $d \geq 0$,

$$\int_A^\infty t^{-L-1} \log^d t \, dt = \frac{A^{-L}}{L} \sum_{j=0}^{d} \frac{d!}{(d-j)! L^j} \log^{d-j} A.$$

The last factor in $(k+1)_{2R}$ cancels the leading L in this integral. Substitution yields precisely the displayed bound. □

Logarithms of positive rationals have elementary rational enclosures. Reduce their argument to $1 \leq y < 2$ by a power of two, put $u = (y-1)/(y+1)$, and use

$$0 \leq \log y - 2 \sum_{j=0}^{T-1} \frac{u^{2j+1}}{2j+1} \leq \frac{2u^{2T+1}}{(2T+1)(1-u^2)}.$$

The interval verifier uses integer endpoints scaled by 10^{90} and rounds every operation outward. The Bernoulli numbers and $e_i(\ell)$ are exact rationals. A final enclosure lying strictly on one side of zero is therefore a sign certificate; the proof of the analytic remainder and the rational arithmetic play separate roles.

The fresh replay verified fourteen base signs for $n \le 4$ and sixteen endpoint signs isolating the unique $n = 1$ roots for $k = 1, 2, 5, 10, 20, 50, 100, 200$. The endpoint pairs have width 10^{-20}. These are global root certificates because the zero-count theorem proves uniqueness, rather than merely sampled sign changes near a candidate root.

The following enclosures use integer interval arithmetic with outward rounding, rational logarithm bounds and a proved Euler–Maclaurin remainder. They have been replayed in this consolidation. The factor $a^{k+1}/k!$ is part of the tabulated value.

n	a	Certified enclosure for $\mathcal{V}_{n,1}(a)$	Sign
1	1	$[0.70738581,\ 0.70738582]$	$+$
1	1.5	$[-0.20183770,\ -0.20183769]$	$-$
2	1	$[0.11418372,\ 0.11418373]$	$+$
2	1.3	$[-0.12008911,\ -0.12008910]$	$-$
2	2	$[0.45673490,\ 0.45673491]$	$+$
3	0.8	$[0.16385907,\ 0.16385908]$	$+$
3	1.1	$[-0.04095264,\ -0.04095263]$	$-$
3	1.5	$[0.05603756,\ 0.05603757]$	$+$
3	2.1	$[-0.24542536,\ -0.24542535]$	$-$
4	0.8	$[0.03897530,\ 0.03897531]$	$+$
4	0.94	$[-0.00338878,\ -0.00338877]$	$-$
4	1.2	$[0.02176370,\ 0.02176371]$	$+$
4	1.7	$[-0.03656822,\ -0.03656821]$	$-$
4	2.2	$[0.15091738,\ 0.15091739]$	$+$

Corollary 10.16 (Saturation at every derivative order for $n \le 4$)**.** *For $1 \le n \le 4$ and every $k \ge 1$, $\gamma_n^{(k)}$ has exactly n positive zeros, all simple. They interlace to the right as k increases.*

Proof. At $k = 1$ each row group gives n sign-changing disjoint intervals, so there are at least n distinct zeros. The global bound with multiplicity makes them the only zeros and makes each simple. Persistence and interlacing follow from Corollary 10.13. □

This finite certification does not settle every small-k case for $n \ge 5$, and it does not assert saturation at every small order uniformly in the Lerch deformation.

10.8 High parameter derivatives of Stieltjes constants

The parameter-derivative tower admits a useful separation into a finite polynomial coefficient and an exponentially small spectral tail. This gives a different asymptotic regime from the frequently studied limit of a large Stieltjes index with derivative order fixed. Here the derivative order k tends to infinity, while the index n may grow proportionally to $\log k$. The resulting profile is the reciprocal gamma function.

10.8.1 An exact spectral expansion

Fix the Laurent convention

$$\zeta(s,a) = \frac{1}{s-1} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!}\gamma_n(a)(s-1)^n, \qquad a > 0. \tag{10.35}$$

The superscript $\gamma_n^{(k)}(a)$ always denotes differentiation with respect to a, not to the index or the zeta variable. Define

$$P_{n,k}(x) = n![z^n]\left(e^{xz}\prod_{j=1}^{k}\left(1+\frac{z}{j}\right)\right). \tag{10.36}$$

This is a degree-n polynomial in x; its coefficients are rational. In particular $P_{0,k} = 1$ and $P_{1,k}(x) = x + H_k$.

Theorem 10.17 (Spectral expansion and a finite-tail certificate)**.** *For $a > 0$, integers $k \geq 1$, and $n \geq 0$,*

$$\frac{(-1)^{n+k}}{k!}\gamma_n^{(k)}(a) = \sum_{m=0}^{\infty} \frac{P_{n,k}(-\log(a+m))}{(a+m)^{k+1}}. \tag{10.37}$$

The series converges absolutely, locally uniformly for $a > 0$. If the sum is truncated before $m = M$, where $b = a + M \geq 1$, then for every real $0 < R < k$ the magnitude of the omitted part is at most

$$\frac{n!}{R^n}\prod_{j=1}^{k}\left(1+\frac{R}{j}\right)\left(b^{-k-1+R} + \frac{b^{-k+R}}{k-R}\right). \tag{10.38}$$

Proof. The Hurwitz-zeta parameter identity

$$\partial_a^k \zeta(s,a) = (-1)^k (s)_k \zeta(s+k,a)$$

follows initially by differentiating its absolutely convergent Dirichlet series and then by analytic continuation; here $(s)_k = s(s+1)\cdots(s+k-1)$. The pole in (10.35) is independent of a. Put $s = 1+z$, use $(1+z)_k = k!\prod_{j=1}^{k}(1+z/j)$, and compare coefficients. For $|z| < k$, the series for $\zeta(k+1+z,a)$ converges normally. This gives (10.37) and justifies the coefficient interchange. Alternatively, absolute convergence follows directly because each summand is a linear combination of $(a+m)^{-k-1}\log^{\ell}(a+m)$, $0 \leq \ell \leq n$. The parameter identity is also the starting point of Coffey's all-order formulas [25].

On the circle $|z| = R$, and for $x \geq b \geq 1$,

$$\left|e^{-z\log x}\prod_{j=1}^{k}(1+z/j)\right| \leq x^R \prod_{j=1}^{k}(1+R/j).$$

Cauchy's coefficient estimate bounds the m-th omitted summand by $n!R^{-n}\prod_j(1+R/j)(a+m)^{-k-1+R}$. For the positive decreasing power, comparison with its integral gives

$$\sum_{m=M}^{\infty}(a+m)^{-k-1+R} \leq b^{-k-1+R} + \int_b^{\infty} x^{-k-1+R}\,dx.$$

This is (10.38). □

The condition $b \geq 1$ belongs to the stated majorant, rather than to the exact expansion. It always holds for the tail after the first term, because then $b = a + 1 > 1$. The choice of R can be optimized for a given n, k, M; there is no need to differentiate a highly oscillatory numerical representation of γ_n.

10.8.2 Exact rational enclosures at parameter one

Write $\begin{bmatrix} v \\ u \end{bmatrix}$ for the unsigned Stirling number of the first kind. The elementary identity

$$\prod_{j=1}^{k}(1+z/j) = \frac{1}{k!}\sum_{n=0}^{k}\begin{bmatrix} k+1 \\ n+1 \end{bmatrix} z^n \tag{10.39}$$

turns the first spectral term at $a = 1$ into an exact rational number.

Corollary 10.18 (Rational certificates)**.** *Let $k \geq 2$, $n \geq 0$, and*

$$e_n(k) = \begin{cases} \begin{bmatrix} k+1 \\ n+1 \end{bmatrix}/k!, & 0 \leq n \leq k, \\ 0, & n > k. \end{cases}$$

For any integer $1 \leq h < k$, set

$$B_{k,n,h} = h^{-n}\binom{k+h}{h}\left(2^{-k-1+h} + \frac{2^{-k+h}}{k-h}\right). \tag{10.40}$$

Then the following is an exact inclusion between real numbers:

$$\frac{(-1)^{n+k}\gamma_n^{(k)}(1)}{k!\,n!} \in [e_n(k) - B_{k,n,h},\, e_n(k) + B_{k,n,h}]. \tag{10.41}$$

One may take the minimum radius over all $h = 1, \ldots, k-1$. In particular, the simple choice $h = 1$ gives

$$B_k = (k+1)\left(2^{-k} + \frac{2^{-k+1}}{k-1}\right), \tag{10.42}$$

independently of n.

Proof. Apply Theorem 10.17 with $a = M = 1$, $b = 2$, and $R = h$, and divide its bound by $n!$. The product in that bound is $\prod_{j=1}^{k}(1+h/j) = \binom{k+h}{h}$. Equation (10.39) supplies the center. All quantities in the resulting endpoints are rational. □

Thus the term *certificate* has its literal meaning here: integer arithmetic constructs the endpoints, and the proved tail bound supplies the inclusion. Floating-point agreement is not being renamed a proof. For example, positivity of an endpoint $e_n(k) - B_{k,n,h}$ rigorously proves $(-1)^{n+k}\gamma_n^{(k)}(1) > 0$. The accompanying script emits centers, radii, and both endpoints as numerator–denominator pairs, and checks their arithmetic exactly.

10.8.3 A uniform coefficient lemma

The main asymptotic step is an elementary coefficient estimate. It is convenient to set $n^{\underline{j}} = n(n-1)\cdots(n-j+1)$ for $j \leq n$, and $n^{\underline{j}} = 0$ for $j > n$.

Lemma 10.19 (Two uniform coefficient terms)**.** *Suppose f is analytic on a neighborhood of the closed disk $|z| \leq R$, with $|f(z)| \leq M$ there. Let $L > 0$, $n \geq 1$, and $r = n/L < R$. Put*

$$Q_n(L; f) = \frac{n!}{L^n}[z^n]e^{Lz}f(z).$$

Then

$$|Q_n(L;f) - f(r)| \le \frac{Mn}{L^2R^2(1-r/R)^3}, \tag{10.43}$$

$$\left|Q_n(L;f) - f(r) + \frac{n}{2L^2}f''(r)\right| \le \frac{3Mn}{L^3R^3(1-r/R)^5}. \tag{10.44}$$

For $n = 0$, $Q_0(L;f) = f(0)$ exactly.

Proof. Write $f(z) = \sum_{j\ge0} c_j z^j$. Since $|c_j| \le MR^{-j}$,

$$Q_n(L;f) = \sum_{j\ge0} c_j \frac{n^{\underline{j}}}{L^j}$$

is a finite sum, which we compare with the absolutely convergent Taylor series at r.

For completeness, the needed inequalities remain valid when $j > n$. Choose j independent uniform elements of a set of n elements. The probability of no repeated choice is $n^{\underline{j}}/n^j$, including its value zero when $j > n$. For the $K = \binom{j}{2}$ pair-collision events, every event has probability $1/n$, and every intersection of two distinct events has probability $1/n^2$, even if the pairs share a position. The union bound and the first two Bonferroni inequalities give

$$0 \le n^j - n^{\underline{j}} \le \binom{j}{2} n^{j-1},$$

$$0 \le n^{\underline{j}} - n^j + \binom{j}{2} n^{j-1} \le \binom{\binom{j}{2}}{2} n^{j-2}.$$

At $j = 0, 1, 2$, the second remainder is zero, so no negative-power exception is needed. Now use

$$\sum_{j\ge2} \binom{j}{2} x^{j-2} = (1-x)^{-3}, \qquad \sum_{j\ge3} \binom{\binom{j}{2}}{2} x^{j-3} = 3(1-x)^{-5}.$$

The first inequality gives (10.43). In the second comparison,

$$\sum_{j\ge2} c_j \binom{j}{2} \frac{n^{j-1}}{L^j} = \frac{n}{2L^2} f''(r).$$

This proves (10.44). □

10.8.4 The reciprocal-gamma profile

Theorem 10.20 (A joint logarithmic-range asymptotic)**.** *Fix $0 < a_0 \le a_1 < \infty$ and $C > 0$. For $a \in [a_0, a_1]$, put*

$$L = \log(k/a), \qquad r = \frac{n}{L}, \qquad g(r) = \frac{1}{\Gamma(1+r)}, \qquad \rho = \frac{a_1}{a_1+1} < 1.$$

Uniformly over integers $0 \le n \le CL$, as $k \to \infty$,

$$\frac{(-1)^{n+k} a^{k+1} \gamma_n^{(k)}(a)}{k!\, L^n} = g(r) - \frac{n}{2L^2} g''(r) + O_{a_0,a_1,C}\left(\frac{n}{L^3} + k^{C+1}\rho^k\right). \tag{10.45}$$

Equivalently, the two displayed terms are

$$\frac{1}{\Gamma(1+r)}\left[1+\frac{n}{2L^2}\left\{\psi^{(1)}(1+r)-\psi(1+r)^2\right\}\right]. \tag{10.46}$$

The formula includes $n=0$, for which the first spectral term is exactly one. In particular, if $n/\log(k/a)\to c\ge 0$ in a bounded range, then the left side of (10.45) tends to $1/\Gamma(1+c)$.

Proof. Separate the $m=0$ term of (10.37). Its generating function factors as

$$e^{-z\log a}\prod_{j=1}^{k}(1+z/j)=e^{Lz}f_k(z),\qquad f_k(z)=k^{-z}\prod_{j=1}^{k}(1+z/j).$$

The gamma product gives

$$f_k(z)=\frac{\Gamma(k+1+z)}{\Gamma(k+1)k^z\Gamma(1+z)}=g(z)e^{D_k(z)}.$$

On $|z|<k+1$, define the analytic function

$$D_k(z)=z(H_k-\log k-\gamma)+\sum_{\ell=2}^{\infty}\frac{(-1)^\ell z^\ell}{\ell}\big(\zeta(\ell)-H_k^{(\ell)}\big). \tag{10.47}$$

The factorization follows first on a small disk from the Weierstrass product and then throughout $|z|<k+1$ by analytic continuation. Defining D_k by its series avoids taking the logarithm of g at its zeros.

The elementary bounds

$$0<H_k-\log k-\gamma<\frac{1}{2k},\qquad \sum_{j>k}j^{-2}\le\frac{1}{k}$$

imply, for $|z|\le R<k+1$,

$$|D_k(z)|\le\frac{R}{2k}+\frac{R^2}{2k(1-R/(k+1))}. \tag{10.48}$$

Indeed, for $\ell\ge 2$, bound the tail of $\zeta(\ell)$ by $(k+1)^{2-\ell}/k$, and use $1/\ell\le 1/2$. Consequently f_k is uniformly bounded on every fixed disk, $f_k-g=O_R(k^{-1})$, and Cauchy's estimate gives $f_k''-g''=O_C(k^{-1})$ on $[0,C]$. The same estimate with $R=r$, or the real product, also gives the slightly sharper

$$f_k(r)-g(r)=O_C(r/k)\qquad(0\le r\le C).$$

It is this factor r that preserves the exact $n=0$ first term.

Choose $R=C+1$ in Lemma 10.19. For $n\ge 1$ the normalized first spectral term is

$$Q_n(L;f_k)=f_k(r)-\frac{n}{2L^2}f_k''(r)+O_C(n/L^3).$$

Replacing f_k by g costs $O_C(n/(kL)+n/(kL^2))$, which is absorbed in $O(n/L^3)$ because $L^2\le k$ for all sufficiently large k, uniformly on the compact parameter interval.

It remains to control the other spectral terms. Apply (10.38) with $M=1$, and multiply the bound by a^{k+1}/L^n. The result is

$$\frac{n!}{(RL)^n}\prod_{j=1}^{k}(1+R/j)\left(\frac{a}{a+1}\right)^{k+1}(a+1)^R\left(1+\frac{a+1}{k-R}\right).$$

The first factor is at most one: for $n \geq 1$, use $n! \leq n^n$ and $n/L \leq C < R$; for $n = 0$, it is one by convention. The product is $O_R(k^R)$, as follows already from (10.48) at $z = R$. The remaining factors are uniformly bounded apart from $(a/(a+1))^{k+1} \leq \rho^{k+1}$. Thus the normalized tail is $O_{a_0,a_1,C}(k^{C+1}\rho^k)$. The case $n = 0$ follows directly from $P_{0,k} = 1$. Finally, $g''(r) = g(r)\{\psi(1+r)^2 - \psi^{(1)}(1+r)\}$. □

Corollary 10.21 (A uniform eventual sign)**.** *For every compact interval $[a_0, a_1] \subset (0, \infty)$ and every $C > 0$, there is k_0 such that*

$$(-1)^{n+k}\gamma_n^{(k)}(a) > 0$$

for all $k \geq k_0$, $a \in [a_0, a_1]$, and integers $0 \leq n \leq C \log(k/a)$.

Proof. The continuous function g is strictly positive on $[0, C]$, so its minimum there is positive. Both the displayed correction and the remainder in Theorem 10.20 tend uniformly to zero. The other factors in the normalization are positive. □

The corollary is restricted to its stated range. It gives no sign theorem when n grows arbitrarily quickly compared with $\log k$. For a concrete finite pair n, k at $a = 1$, the rational interval in Corollary 10.18 is a separate, effective certificate.

10.8.5 Fixed-index expansions and the classical mechanism

For a fixed n, it is also useful to retain the polynomial in $L = \log(k/a)$. Define Appell polynomials

$$A_n(L) = n![z^n]\frac{e^{Lz}}{\Gamma(1+z)}.$$

The first few are

$$\begin{aligned} A_0(L) &= 1, \\ A_1(L) &= L + \gamma, \\ A_2(L) &= (L+\gamma)^2 - \zeta(2), \\ A_3(L) &= (L+\gamma)^3 - 3\zeta(2)(L+\gamma) + 2\zeta(3). \end{aligned}$$

The standard gamma-ratio expansion [49] gives, uniformly on fixed z-disks,

$$e^{D_k(z)} = 1 + \frac{z+z^2}{2k} + \frac{3z^4 + 2z^3 - 3z^2 - 2z}{24k^2} + O(k^{-3}).$$

Consequently, for fixed $n \geq 1$ and a in a compact positive interval,

$$\begin{aligned} \frac{(-1)^{n+k}a^{k+1}\gamma_n^{(k)}(a)}{k!} &= A_n(L) + \frac{nA_{n-1}(L) + n(n-1)A_{n-2}(L)}{2k} \\ &\quad + O_n\left(\frac{(1+L)^{n-1}}{k^2}\right) + O_{a_0,a_1,n}(k^2\rho^k). \end{aligned} \tag{10.49}$$

Terms with a negative Appell index are omitted. To see the stated polynomial error, note that the analytic remainder after the first correction vanishes at $z = 0$; its coefficient convolution with $e^{Lz}g(z)$ therefore has degree at most $n-1$ in L. For $n = 0$ the first term is exact and only the

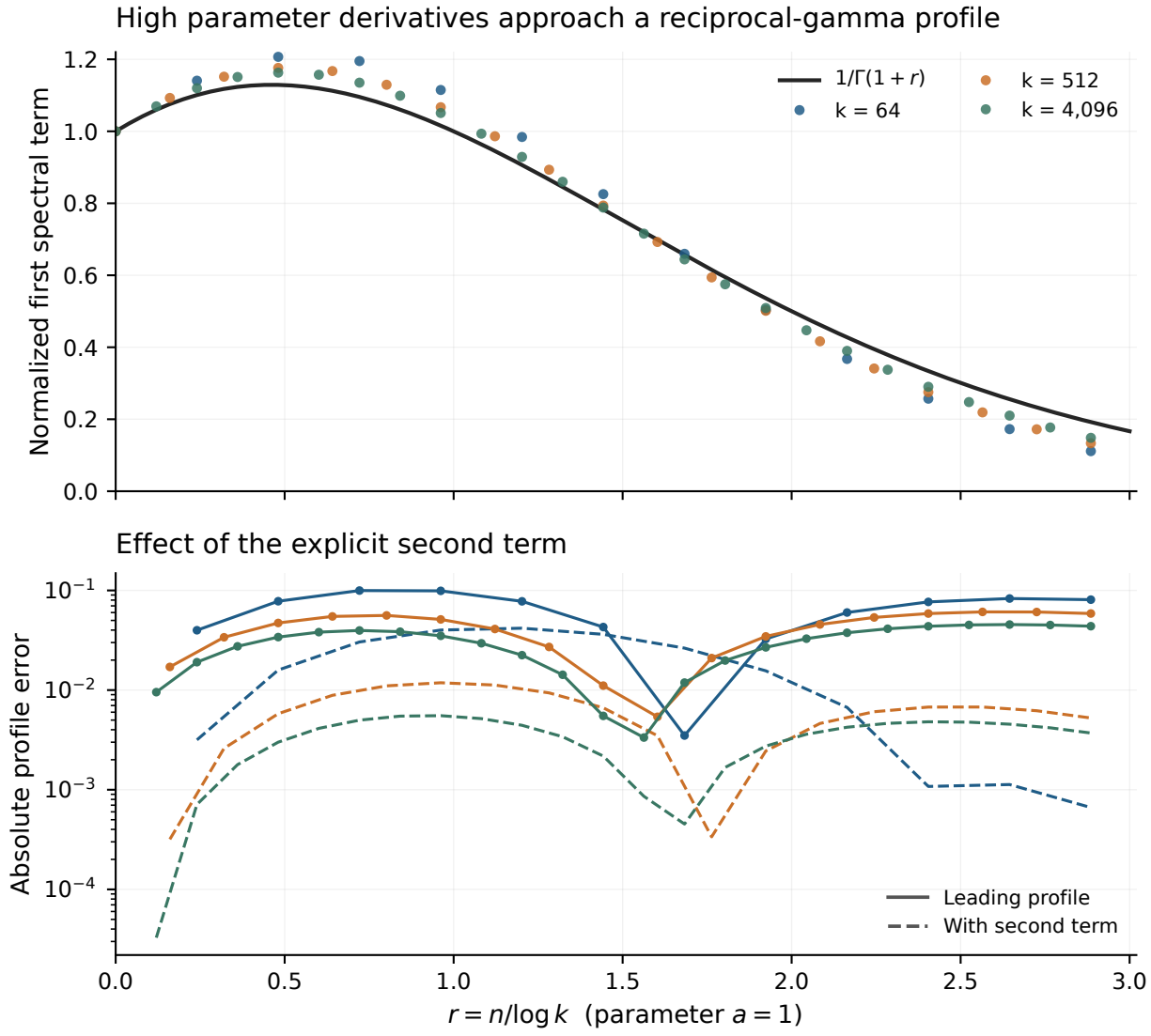


Figure 10.2: The reciprocal-gamma profile at $a = 1$. The first spectral term is computed from the elementary-symmetric coefficient recurrence; its rigorously bounded spectral error is smaller than the plotted resolution. The lower panel compares its deviation from the limiting profile with its deviation after the explicit second term is included. These plotted arithmetic evaluations are floating point; the separate rational certificates do not depend on them.

spectral error remains. Higher powers of k^{-1} follow by the same coefficient procedure. For example the k^{-2} numerator is

$$\frac{1}{24}\left(3n^{\underline{4}}A_{n-4}+2n^{\underline{3}}A_{n-3}-3n^{\underline{2}}A_{n-2}-2nA_{n-1}\right).$$

At $a=1$, (10.39) identifies the first spectral term with first-kind Stirling numbers. These are also the cycle-count coefficients of permutations. The reciprocal-gamma factor is thus a classical mechanism, closely related to mod-Poisson asymptotics [56]. The contribution to this programme is the explicit spectral transfer, the uniform second coefficient estimate, and the exact rational enclosure, not a claim that the reciprocal-gamma phenomenon itself is new.

10.9 The first classical failure of saturation: the fifth index

The first four Stieltjes indices in the preceding discussion have the maximum possible number of positive parameter-derivative zeros at every derivative order. That pattern fails at the next index. The failure is not restricted to a Lerch deformation: it occurs at the classical Hurwitz endpoint itself. We prove the exact transition

$$\#\{a>0:\gamma_5^{(k)}(a)=0\}=\begin{cases}3, & k=1,2,\\ 5, & k\geq 3,\end{cases} \tag{10.50}$$

and every zero in this display is simple. Thus the least derivative order after which fifth-index saturation persists is exactly three.

Two ingredients distinguish this conclusion from a numerical zero scan. A general upper bound of five controls the entire positive axis at order three. Certified signs at its stationary points then count the zeros at orders two and one by monotonicity. The large positive zeros near 148.9122 and 319.6184 are essential to this argument; a scan confined to a small neighborhood of $a=1$ misses them.

10.9.1 Endpoint signs, parity, and monotonicity of zero counts

Retain the Laurent convention

$$\zeta(s,a)=\frac{1}{s-1}+\sum_{n\geq 0}\frac{(-1)^n}{n!}\gamma_n(a)(s-1)^n, \qquad a>0.$$

For the classical family write

$$U_{n,k}(a)=(-1)^{n+k}\gamma_n^{(k)}(a), \qquad k\geq 1.$$

Then $U'_{n,k}=-U_{n,k+1}$. The established Appell–Laplace formula is

$$U_{n,k}(a)=\int_0^\infty \frac{t^k e^{-at}}{1-e^{-t}}\,Q_n(\log t)\,dt, \qquad \frac{e^{xz}}{\Gamma(1+z)}=\sum_{n\geq 0}Q_n(x)\frac{z^n}{n!}. \tag{10.51}$$

The reciprocal-gamma product makes Q_n strictly real-rooted, and Laplace variation diminution therefore bounds the positive zero count by n, including multiplicities. These are the global zero theorem and its proof mechanism, reproduced in Section 10.2 for completeness. The classical parameter-derivative identity underlying (10.51) is also contained in Coffey's all-order formulas [25].

For fixed $n \geq 0$ and $k \geq 1$ the endpoint expansions are

$$U_{n,k}(a) \sim k! a^{-k-1} \log^n(1/a), \qquad a \downarrow 0, \tag{10.52}$$

$$U_{n,k}(a) \sim (-1)^n (k-1)! a^{-k} \log^n a, \qquad a \to \infty. \tag{10.53}$$

When $n = 0$, the logarithmic powers mean one. The first formula follows either by scaling the integral or by differentiating the shift term $\log^n a/a$ in $\gamma_n(a) = \log^n a/a + \gamma_n(a+1)$. For the second, set $t = u/a$ in (10.51) and use $(1-e^{-u/a})^{-1} = a/u + O(1+u/a)$ together with monicity of Q_n. The bound $(1-e^{-v})^{-1} \leq 1 + v^{-1}$ supplies an integrable majorant after expanding the fixed polynomial in $\log u - \log a$. In particular, every $U_{n,k}$ tends to zero at infinity and has fixed strict signs near both endpoints.

Proposition 10.22 (Parity and persistence before saturation)**.** *Let $N_{n,k}$ be the total multiplicity of the positive zeros of $\gamma_n^{(k)}$. Then*

$$0 \leq N_{n,k} \leq n, \qquad N_{n,k} \equiv n \pmod 2, \qquad N_{n,k+1} \geq N_{n,k}. \tag{10.54}$$

Consequently every strict increase in the total zero count is by an even integer. The same assertions hold, at each fixed $0 \leq \rho \leq 1$, for the Stieltjes–Lerch family $D_{n,k}^{\rho}$.

Proof. The upper bound is the global variation theorem. The endpoint signs in (10.52)–(10.53) show that the total multiplicity has parity n: a real analytic zero reverses sign exactly when its multiplicity is odd.

Suppose the distinct zeros of $U_{n,k}$ are $r_1 < \cdots < r_q$, with multiplicities $m_1, \ldots, m_q$ and total $N = \sum_i m_i > 0$. Its derivative has inherited multiplicity $m_i - 1$ at each r_i, and Rolle's theorem supplies a zero between each successive pair. These account for at least $(N-q) + (q-1) = N - 1$ derivative zeros counted with multiplicity. There is one more stationary point strictly after r_q: a nonzero function leaving zero at r_q, with constant sign afterwards and limit zero at infinity, must attain an interior extremum. Thus $U_{n,k+1} = -U'_{n,k}$ has at least N zeros counted with multiplicity. The case $N = 0$ is immediate.

For fixed $\rho < 1$, the analogous Mellin weight $(1-\rho e^{-t})^{-1}$ is positive and tends to $(1-\rho)^{-1}$ at zero. The large-a leading term is then $(-1)^n k!(1-\rho)^{-1} a^{-k-1} \log^n a$; the small-$a$ leading term is unchanged. The same endpoint and Rolle arguments apply, including the possible multiple zeros at $\rho = 0$. □

Proposition 10.23 (A classical lower bound from matching endpoint data)**.** *For every $k \geq 1$, $\gamma_n^{(k)}$ has at least two distinct positive zeros if $n \geq 2$ is even, and at least three distinct positive zeros if $n \geq 3$ is odd. In particular, saturation for $n = 2, 3$ at every derivative order follows analytically, without a finite sign certificate.*

Proof. For $n \geq 2$, the shift identity at $a = 1$ and its first derivative give

$$\gamma_n(1) = \gamma_n(2), \qquad \gamma_n'(1) = \gamma_n'(2), \qquad \int_1^2 \gamma_n'(a)\, da = 0.$$

Put $g = \gamma_n'$. This is a nonzero real analytic function. If its common endpoint value is positive, its integral forces a negative interior value; hence there are at least two sign-changing zeros in $(1, 2)$. The same conclusion holds with signs reversed when the common endpoint value is negative. If the common endpoint value is zero, the integral forces both signs in the interior, and there are at least three distinct zeros in $[1, 2]$, including the endpoints.

For odd n in the nonzero-endpoint case, the two interior sign-changing zeros have odd multiplicities. The odd total parity from Proposition 10.22 forces an additional odd-multiplicity zero, distinct from them. This proves the asserted lower bounds at $k = 1$. Rolle's theorem between distinct zeros, together with the stationary point after the last zero used above, preserves a lower bound on the number of distinct zeros at every subsequent derivative order. The global upper bound makes the two or three zeros simple when $n = 2$ or $n = 3$. □

Thus the count three in (10.50) is the smallest possible classical fifth-index count. Passing from order two to order three raises the positive zero count by exactly two.

10.9.2 A finite certificate that controls the whole positive axis

For the remainder of the proof abbreviate

$$U_k = U_{5,k}, \qquad V_k(a) = \frac{a^{k+1}}{k!} U_k(a).$$

The normalization V_k preserves signs, but its stationary points are not the zeros of U_{k+1}. All monotonicity arguments below are applied to U_k, while V_k is used only for stable sign enclosures.

Let $\varepsilon = 10^{-16}$. Each row of Table 10.2 specifies a rational interval $[\ell, \ell + \varepsilon]$. The stated endpoint signs of V_k are certified by the rational Euler–Maclaurin enclosure already derived above; the displayed error bound is valid for every fixed nonnegative index.

k	j	ℓ	sign at ℓ	sign at $\ell + 10^{-16}$
3	1	0.9409528529131887	+	−
3	2	1.1292935180347063	−	+
3	3	1.6357795548293846	+	−
3	4	6.1361342793896981	−	+
3	5	319.6184025568026870	+	−
2	1	1.3713298168772550	+	−
2	2	1.9113619446041762	−	+
2	3	148.9122045692035618	+	−
1	1	1.1378900542449414	+	−
1	2	1.6389372768448503	−	+
1	3	2.1249028498579805	+	−

Table 10.2: Rational isolating intervals of width 10^{-16}. These endpoint signs and the critical-value signs in the next table are finite certificates. The proof establishes uniqueness and completeness.

Denote the five third-order roots by $c_1 < \cdots < c_5$ and the three second-order roots by $d_1 < d_2 < d_3$. The interval computation also encloses the preceding-order function uniformly on every indicated root interval. Table 10.3 lists rational enclosures with outward rounding; they apply in particular at the true roots.

Theorem 10.24 (The exact fifth-index transition)**.** *The positive zero counts are exactly those in* (10.50)*. All zeros are simple. The eleven intervals in Table 10.2 isolate all zeros at derivative orders one, two, and three. For every $k \geq 3$, the five zeros at order $k + 1$ strictly interlace to the right of those at order k.*

Quantity	Certified rational enclosure
$V_2(c_1)$	$[0.006154632340,\ 0.006154632341]$
$V_2(c_2)$	$[0.041073722584,\ 0.041073722585]$
$V_2(c_3)$	$[-0.097090480335,\ -0.097090480334]$
$V_2(c_4)$	$[106.545043234228,\ 106.545043234229]$
$V_2(c_5)$	$[-135615.425337743697,\ -135615.425337743695]$
$V_1(d_1)$	$[-0.014754703693,\ -0.014754703692]$
$V_1(d_2)$	$[0.028710652775,\ 0.028710652776]$
$V_1(d_3)$	$[-466915.576951780840,\ -466915.576951780837]$

Table 10.3: Signs of the lower-order function at every critical point. The verifier encloses the function on the entire rational root bracket, so these signs do not presume a floating-point approximation to a root.

Proof. The five third-order interval signs give at least five distinct zeros of U_3. The global upper bound of five, counting multiplicity, shows that they are all its positive zeros and that each is simple. Starting with the positive left endpoint sign, the signs of U_3 on the six complementary intervals are

$$+,\ -,\ +,\ -,\ +,\ -.$$

Consequently $U_2' = -U_3$ has constant nonzero sign on each such interval. Its values at the five critical points have signs

$$\operatorname{sgn}(U_2(c_1), \dots, U_2(c_5)) = (+, +, -, +, -).$$

Therefore there is exactly one second-order zero in each of (c_2, c_3), (c_3, c_4), and (c_4, c_5), and no zero in $(0, c_1)$ or (c_1, c_2). After c_5, the function U_2 is strictly increasing from a negative value to its limit zero, so there is no zero there either. The three zeros lie away from the critical points, so each is simple. Their brackets in Table 10.2 consequently identify them as d_1, d_2, d_3.

The signs of U_2 are now $+, -, +, -$ on the four intervals cut out by the d_j, and $U_1' = -U_2$. The critical-value signs are $-, +, -$. Since U_1 is positive near zero, strict monotonicity gives exactly one zero in each of $(0, d_1)$, (d_1, d_2), and (d_2, d_3). After d_3, U_1 increases from a negative value to zero, so it has no further zero. Again all three crossings are simple.

Finally, five simple zeros at order three force at least five distinct zeros at the next order: Rolle supplies four between adjacent roots and the vanishing tail supplies one after the last root. The global upper bound gives exactly five simple zeros, with no additional zero before the first interval. Repeating proves saturation and strict right interlacing for every $k \geq 3$. □

This proof uses no claim that a dense set of sampled values accounts for every zero. Completeness at the highest checked order comes from a global analytic theorem; completeness at lower orders comes from the complete list of stationary points and their certified signs.

The fresh replay supplies twenty-two root endpoint signs and eight signs on whole critical-point intervals, using denominator 10^{80}, 120 logarithm-series terms, 24 initial Hurwitz summands and 16 Bernoulli corrections. The generic remainder proof in the preceding certificate subsection applies with $n = 5$. No floating-point sign enters acceptance. The data and executable verifier remain with the full incoming package.

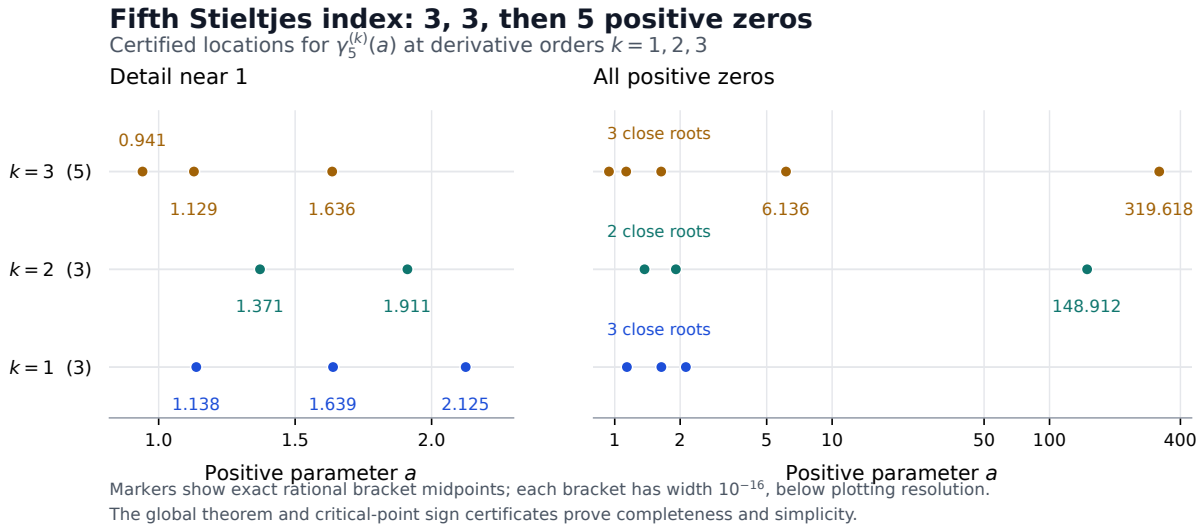


Figure 10.3: The eleven certified zero locations. The local panel separates the near-unit roots; the logarithmic panel includes the large roots. The brackets have width 10^{-16}, much smaller than a plotted marker.

10.10 Exact sixth- and seventh-index saturation thresholds

The incoming fifth-index method suggests a reproducible extension of the classical zero classification. Its next two first-derivative counts were reported only as numerical conjectures. Exact critical-value descent now settles both indices at every derivative order, not only the sampled first order. The argument retains the existing global bound and adds finite rational signs; it does not infer completeness from an approximate scan.

Theorem 10.25 (The next two classical transitions)**.** *Let $N_{n,k}$ count the positive zeros of $\gamma_n^{(k)}$, with multiplicity. For every integer $k \geq 1$,*

$$N_{6,k} = \begin{cases} 4, & 1 \leq k \leq 3, \\ 6, & k \geq 4, \end{cases} \qquad N_{7,k} = \begin{cases} 5, & 1 \leq k \leq 4, \\ 7, & k \geq 5. \end{cases}$$

Every zero is simple. Thus the least classical saturation thresholds are $K_6^{\mathrm{cl}} = 4$ and $K_7^{\mathrm{cl}} = 5$. At and above each threshold, the next-order zeros strictly interlace to the right of the current ones.

Retain $U_{n,k} = (-1)^{n+k}\gamma_n^{(k)}$ and the sign-preserving normalization $V_{n,k} = a^{k+1}U_{n,k}/k!$ of the previous section. The endpoint expansions give $U_{n,k} > 0$ near zero, $\operatorname{sgn} U_{n,k} = (-1)^n$ near infinity, and $U_{n,k} \to 0$ there. Also $U'_{n,k} = -U_{n,k+1}$.

Lemma 10.26 (Complete descent from critical-value signs)**.** *Suppose all positive zeros $c_1 < \cdots < c_r$ of $U_{n,k+1}$ are simple, and $U_{n,k}(c_j)$ has a nonzero certified sign s_j at each one. Put $s_0 = 1$. Then every positive zero of $U_{n,k}$ is simple and*

$$N_{n,k} = \#\{0 \leq j < r : s_j \neq s_{j+1}\}.$$

There is no zero on the final interval (c_r, ∞).

Proof. On each open interval cut out by the complete critical-point list, $U'_{n,k} = -U_{n,k+1}$ is nonzero and has constant sign. Hence $U_{n,k}$ is strictly monotone there. Its limits or endpoint values show one crossing exactly when the stated signs differ. It cannot vanish at a critical point, since each critical value is nonzero. On the final interval it is strictly monotone with limit zero; a nonzero strictly monotone function tending to zero cannot cross zero there. Every crossing lies where the derivative is nonzero, so it is simple. These intervals cover the whole positive axis. □

n	k	j	Lower endpoint ℓ	$V_{n,k}(\ell)$	$V_{n,k}(\ell + 10^{-14})$
6	4	1	0.95661907907155	+	−
6	4	2	1.07179934147775	−	+
6	4	3	1.51367699674930	+	−
6	4	4	3.14848332199473	−	+
6	4	5	16.34597601190147	+	−
6	4	6	1389.12290266995208	−	+
7	5	1	0.97564797285268	+	−
7	5	2	1.03011037496236	−	+
7	5	3	1.39376738084727	+	−
7	5	4	2.31266214479726	−	+
7	5	5	5.93513467415295	+	−
7	5	6	38.92782595180997	−	+
7	5	7	5297.39431175614964	+	−

Table 10.4: All saturated-order root brackets. Each displayed decimal is an exact rational number, and each bracket has width 10^{-14}. The signs are determined by outward integer intervals, rather than floating-point root approximations.

n	Known critical order	Predecessor order	Signs at all critical points	Count
6	4	3	$+,+,-,+,-,+$	4
6	3	2	$-,+,-,+$	4
6	2	1	$-,+,-,+$	4
7	5	4	$+,+,-,+,-,+,-$	5
7	4	3	$-,+,-,+,-$	5
7	3	2	$-,+,-,+,-$	5
7	2	1	$-,+,-,+,-$	5

Table 10.5: The complete critical-value descent. Each predecessor sign is certified uniformly over the entire rational root bracket of its derivative.

Proof of Theorem 10.25. The terminal brackets in Table 10.4 have opposite strict endpoint signs, so give six distinct zeros at $(n, k) = (6, 4)$ and seven at $(7, 5)$. The global bound with multiplicity, Theorem 10.7, makes these complete lists and all their zeros simple. On every terminal bracket the preceding derivative has the sign recorded in the first applicable row of Table 10.5. Thus Lemma 10.26 gives exactly four sixth-index roots at order three and five seventh-index roots at order four.

The same exact certificate file supplies disjoint brackets with opposite endpoint signs for every one of these predecessor roots. Since the preceding descent already proves the total count, these brackets form the complete critical-point lists for the next step. Their whole-interval predecessor

signs are the next rows of Table 10.5. Iterating the lemma reaches order one, with the counts displayed in the theorem and simple zeros throughout. This use of lower-order brackets has no circular completeness assumption: their completeness is established before the next descent step.

At a saturated order there are n distinct simple zeros. Rolle's theorem supplies $n-1$ next-order zeros between adjacent ones and one after the last, because the function has a nonzero sign there and tends back to zero. The same global bound makes these exactly n simple zeros, with strict right interlacing. Induction covers every larger order. The lower counts prove minimality of the two thresholds. □

The exact evidence and its analytic contract. The finite proof data contain 45 rational root brackets: 18 for the sixth index and 27 for the seventh. They require **90** endpoint enclosures and **36** whole critical-interval enclosures. The first implementation uses the generic Euler–Maclaurin bound already proved in this chapter, with fixed denominator 10^{80}, 120 logarithm-series terms, 24 initial Hurwitz terms and 16 Bernoulli corrections. Its elementary symmetric coefficient table is rebuilt through degree seven; this changes no analytic hypothesis in the index-uniform remainder formula.

A second implementation independently extracts rising-factorial coefficients and encloses all 126 signs. It uses denominator 10^{100}, 128 logarithm-series terms, 32 initial Hurwitz terms and 16 Bernoulli corrections. Its precision was increased because the delivered 55-digit default could not resolve some new narrow brackets after very small powers were multiplied by large factorial coefficients. Unresolved intervals were never accepted as signs. Both replays use exact integer and fraction arithmetic throughout acceptance. The complete endpoints, analytic error budgets, source digests and descent signs are retained in `verification/higher-zero-certificates.json` and `verification/higher-zero-independent.json`.

The resulting first thresholds are

$$(K_1^{\mathrm{cl}}, \dots, K_7^{\mathrm{cl}}) = (1, 1, 1, 1, 3, 4, 5).$$

This finite sequence suggests further threshold questions; it does not prove a formula in the index. In particular no claim that $K_n^{\mathrm{cl}} = n - 2$ for every $n \geq 5$ is made. Extending the terminating critical-point method with effective saturation estimates is a concrete next research problem.

Chapter 11

Herglotz arithmetic and quadratic regulators

The Herglotz function joins the digamma series to finite dilogarithm sums. At rational arguments this supplies exact evaluations; at quadratic units it connects continued-fraction cycles with real-quadratic Kronecker limits. The same experimental strategy thus leads from a finite cyclotomic sum to regulator candidates in class fields.

11.1 The function and the exact dilogarithm sum

For $x > 0$ define

$$F(x) = \sum_{n\geq 1} \frac{\psi(nx) - \log(nx)}{n}, \qquad J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t}\, dt.$$

The series is locally uniformly convergent on the positive line and extends analytically to the cut plane $\mathbb{C} \setminus (-\infty, 0]$. Radchenko–Zagier [35] relate the two functions by

$$J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t}\, dt, \qquad J(x) = F(2x) - 2F(x) + F(x/2) + \frac{\pi^2}{12x}, \tag{11.1}$$

which relates the companion J to the Herglotz function F. For the rational-argument results the engine is Radchenko–Zagier's Theorem 3, an *exact, finite* evaluation obtained from the second integral representation of F via the distribution relations:

$$F\Big(\tfrac{P}{Q}\Big) - F(1) = \sum_{\alpha^P=1} \Big(\sum_{\substack{\beta^Q=1 \\ \beta\neq 1}} f(\alpha,\beta) - \sum_{\substack{\beta^P=1 \\ \beta\neq 1}} f(\alpha,\beta)\Big), \quad f(\alpha,\beta) = \mathrm{Li}_2\Big(\tfrac{\beta}{\beta-1}\Big) - \mathrm{Li}_2\Big(\tfrac{\alpha-\beta}{1-\beta}\Big). \tag{11.2}$$

The finite sum is an exact analytic identity, with compatible principal branches at its arguments. Floating-point evaluation still has rounding and cancellation errors. It is a useful independent comparator for the defining integral, not a zero-error numerical oracle.

11.2 Quadratic-unit values: Tables 1 & 2, and two errata

For real quadratic units the relevant object is the companion $\mathcal{J}(x) = J(x) - \frac{\log^2 2}{2} + \frac{\pi^2}{24}(x - \frac{1}{x})$, and the arithmetic atoms are $S_{p,q} = \log \varepsilon_p \, \log \varepsilon_q$ where ε_d is the fundamental unit of $\mathbb{Q}(\sqrt{d})$ for a fundamental

discriminant $d > 1$ and $\varepsilon_1 := 2$. Tables 1 and 2 of the source paper express, for the 45 known one-class-per-genus cases, the quantities $4\big(\mathcal{J}(n+\sqrt{n^2-1}) - n\frac{\pi^2}{24}\big)$ and $2\big(\mathcal{J}(n+\sqrt{n^2+1}) - n\frac{\pi^2}{24}\big)$ as integer combinations of the $S_{p,q}$.

We re-implemented $\mathcal{J}$ and $S_{p,q}$ in PARI/GP (`intnum` integrals, `bnfinit` regulators) and evaluated both sides of all 45 rows at 200-digit precision. All 16 rows of Table 2 and 27 of the 29 rows of Table 1 agree to better than 10^{-80}. The two failures are $O(1)$:

$$n = 43\text{: LHS} - \text{RHS} = -13.0160\ldots, \qquad n = 61\text{: LHS} - \text{RHS} = -2.9011\ldots.$$

Each resolves to a single exact correction, confirmed against the high-DPI scan to be a typo in the printed paper rather than an OCR artifact.

Proposition 11.1 (Two corrections to Table 1)**.**

- $n = 43$ *($n^2 - 1 = 2^3 \cdot 3 \cdot 7 \cdot 11$): the printed right-hand side overshoots by exactly* $S_{33,56}$ *$\big((LHS - RHS)/S_{33,56} = -1.000\ldots$ to 110 digits$\big)$; the spurious* $+S_{33,56}$ *term should be deleted.*
- $n = 61$ *($n^2 - 1 = 2^3 \cdot 3 \cdot 5 \cdot 31$): the printed* $S_{60,1032}$ *should read* $S_{60,248}$*. The printed subscript* $1032 = 2^3{\cdot}3{\cdot}43$ *carries the prime* 43*, foreign to* $n^2{-}1 = 3720$*, whereas* $60{\cdot}248 = 14880 = 4{\cdot}3720$ *is the correct genus splitting.*

A brute-force search over all single edits (delete a term, change a coefficient in $[-4, 4]$*, or replace a subscript by any admissible fundamental-discriminant pair) returns these as the* unique *fixes. With both applied, all* 45 *rows verify to better than* 10^{-80}*.*

The quadratic-unit tables concern specified arithmetic families. They do not imply that every even-trace unit has a pure-logarithm evaluation: outside genus characters the Kronecker-limit formula introduces additional class-group coordinates. Those coordinates are developed below.

11.3 Rational arguments: a family of theorems

The paper writes out only two rational-argument cases: $F(n)$ (pure logarithms, its eq. 23) and the lone fractional example $F(2/5)$ (its eq. 24). We now give the general laws.

11.3.1 Two reductions on the line $\mathrm{Re} = \frac{1}{2}$

Everything rests on one observation: for a unit $\beta = e^{i\theta}$,

$$\frac{\beta}{\beta - 1} \text{ has real part } \tfrac{1}{2}, \qquad \arg\frac{\beta}{\beta-1} = \frac{\theta}{2} - \frac{\pi}{2}, \qquad \Big|\frac{\beta}{\beta-1}\Big| = \frac{1}{2\sin(\theta/2)}. \tag{11.3}$$

By the reflection $\mathrm{Li}_2(z) + \mathrm{Li}_2(1-z) = \frac{\pi^2}{6} - \log z \log(1-z)$ with $1 - z = \bar{z}$ on $\mathrm{Re}\, z = \frac{1}{2}$,

$$2\,\mathrm{Re}\,\mathrm{Li}_2\Big(\tfrac{1}{2} + iy\Big) = \frac{\pi^2}{6} - \log^2|z| - (\arg z)^2. \tag{11.4}$$

The second reduction handles the $\alpha = -1$ part of (11.2): $\frac{-1-\beta}{1-\beta} = -i\cot\frac{\theta}{2}$ is purely imaginary, and for real y

$$\mathrm{Re}\,\mathrm{Li}_2(iy) = \tfrac{1}{4}\,\mathrm{Li}_2(-y^2). \tag{11.5}$$

11.3.2 $F(1/q)$: pure logarithms

Theorem 11.2. *For all $q \geq 2$, with $m = \lfloor (q-1)/2 \rfloor$,*

$$F\Big(\tfrac{1}{q}\Big) - F(1) = -\frac{(q-1)(3q-2)}{24q}\,\pi^2 - \sum_{k=1}^{m} \log^2\Big(2\sin\tfrac{\pi k}{q}\Big) \;-\; [\,q \text{ even}\,]\,\tfrac{1}{2}\log^2 2.$$

Proof. With $P = 1$ only $\alpha = 1$ occurs and the inner β^P-sum is empty, so (11.2) reads $F(1/q) - F(1) = \sum_{\beta^q=1,\beta\neq 1}(\mathrm{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6})$. Pairing β with $\bar\beta$ and applying (11.4) gives the π^2 coefficient $-\big[\frac{q-1}{12} + \sum_{k=1}^{m}(\frac{k}{q} - \frac{1}{2})^2\big] = -\frac{(q-1)(3q-2)}{24q}$ and the log-sine sum. For even q the self-conjugate root $\beta = -1$ contributes $\mathrm{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2}\log^2 2$. □

11.3.3 $F(2/q)$: the general theorem

Theorem 11.3. *For every $q \geq 3$,*

$$F\Big(\tfrac{2}{q}\Big) - F(1) = -\frac{(q-1)(q-2)}{12q}\pi^2 + \log^2 2 - 2\sum_{k=1}^{\lfloor (q-1)/2\rfloor} \log^2\Big(2\sin\tfrac{\pi k}{q}\Big) - \tfrac{1}{2}\sum_{k=1}^{\lfloor (q-1)/2\rfloor} \mathrm{Li}_2\Big(-\cot^2\tfrac{\pi k}{q}\Big),$$

where for even q the two sums run to $k = \frac{q}{2} - 1$ and a further $-\log^2 2$ is added.

Proof (odd q). Here $\alpha \in \{1, -1\}$ and the inner β^P-sum is the single term $\beta = -1$. For the first part, $f(1,\beta) + f(-1,\beta) = 2\,\mathrm{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6} - \mathrm{Li}_2(\frac{-1-\beta}{1-\beta})$. Summing over $\beta \neq 1$ and using (11.4) for the first pieces and (11.5) (with $\frac{-1-\beta}{1-\beta} = -i\cot\frac{\pi k}{q}$, pairing $k \leftrightarrow q-k$) for the last gives $2S_1 - (q-1)\frac{\pi^2}{6} - \frac{1}{2}\sum_k \mathrm{Li}_2(-\cot^2\frac{\pi k}{q})$, where $S_1 = \sum_{k=1}^{(q-1)/2}[\frac{\pi^2}{6} - \log^2(2\sin\frac{\pi k}{q}) - \pi^2(\frac{k}{q} - \frac{1}{2})^2]$. The boundary gives $-\sum_\alpha f(\alpha,-1) = +\log^2 2$ from $\mathrm{Li}_2(\frac{1}{2})$. Collecting, the $\frac{\pi^2}{6}$ terms cancel and the π^2 coefficient is $-2\sum_{k=1}^{(q-1)/2}(\frac{k}{q} - \frac{1}{2})^2$, which equals $-\frac{(q-1)(q-2)}{12q}$ by an elementary identity (verified symbolically). The even-q case differs only in that $\beta = -1$ is now a genuine q-th root, contributing the extra $-\log^2 2$. □

The algebraic dilogarithm arguments make the field structure explicit. At $q = 5$ the symmetric sum can be reduced further as described below. Unsuccessful searches at larger conductors do not prove that no other closed form exists; the finite-sum theorem remains valid at every q.

11.3.4 The $F(n/(n+1))$ family and the $F(p/2)$ reciprocity

Theorem 11.4. *For all $n \geq 2$, with $S(q) = \sum_{k=1}^{q-1}\log^2(2\sin\frac{\pi k}{q})$,*

$$F\Big(\tfrac{n}{n+1}\Big) - F(1) = -\frac{3n^2+3n+2}{24n(n+1)}\pi^2 + \mathrm{Li}_2\Big(\tfrac{n}{n+1}\Big) + \tfrac{1}{2}\log^2\tfrac{n+1}{n} - \tfrac{1}{2}S(n+1) + \tfrac{1}{2}S(n),$$

the single dilogarithm $\mathrm{Li}_2(n/(n+1))$ appearing with coefficient exactly $+1$ (the source paper's "bilinear-log" Example 2, made explicit). The π^2 coefficient equals $-\frac{1}{8} - \frac{1}{12n(n+1)}$.

Proof. The two functional equations of F are

$$F(x) - F(x+1) - F(x/(x+1)) = -F(1) + \mathrm{Li}_2(1/(x+1)),$$
$$F(x) + F(1/x) - 2F(1) = \tfrac{1}{2}\log^2 x - \frac{\pi^2}{6x}(x-1)^2.$$

The second equation and Theorem 11.2 give

$$F(n) - F(1) = -\frac{(n-1)(n-2)}{24n}\pi^2 + \frac{1}{2}\log^2 n + \frac{1}{2}S(n).$$

Subtract the cases n and $n+1$ in the first equation, and apply Euler reflection to $\mathrm{Li}_2(1/(n+1))$. Collecting the logarithms and rational π^2 coefficients yields the displayed formula. □

Theorem 11.5 (Reciprocity)**.** *For all odd $p \geq 3$,*

$$F\Big(\tfrac{p}{2}\Big) + F\Big(\tfrac{2}{p}\Big) - 2F(1) = -\frac{(p-2)^2}{12p}\pi^2 + \tfrac{1}{2}\log^2 \tfrac{p}{2}.$$

The $\mathrm{Li}_2(-\cot^2 \frac{\pi k}{p})$ terms of $F(p/2)$ are exactly minus those of $F(2/p)$ and cancel in the sum; combined with Theorem 11.3 this yields $F(p/2)$ in closed form.

Proof. Substitute $x = p/2$ into the second functional equation above. □

11.3.5 The derivative law

Theorem 11.6 (Radchenko–Zagier Proposition 4, $P = 2$)**.** *For all $q \geq 3$, with $T(q) := \frac{2}{q}F'(2/q) - (1 + \log 2)$,*

$$T(q) = \frac{q^2-4}{24q}\pi^2 + W(q), \qquad W(q) = \sum_{k=1}^{q-1} C(2k/q)\ \mathrm{Cl}_2\Big(\tfrac{2\pi k}{q}\Big),$$

where $C(t) = \cot(\pi t)$ for $t \notin \mathbb{Z}$ and $C(t) = 0$ for $t \in \mathbb{Z}$, with an extra $-\log 2$ for even q. This assigned value at integers is part of the formula; a literal cotangent there would be undefined. The sum provides a Clausen representation, without an independence assertion.

11.3.6 Rational-argument samples and the companion integral

The source's proposed criterion $\mathrm{rad}(q) \mid 30$ is false as a necessary condition for a reduction to rational dilogarithms: Theorem 11.2 gives a pure-logarithm answer for $F(1/7)$. Numerator, denominator and the available functional equations must all be considered. The formal five-term classification is proved in the following section. The following experimentally recovered samples are useful:

$$\begin{aligned}
F(3/4) - F(1) &= -\frac{19}{144}\pi^2 - \frac{3}{4}\log^2 2 - 2\log 2\log 3 + \frac{7}{4}\log^2 3 + 2\,\mathrm{Li}_2(1/3),\\
F(4/5) - F(1) &= \frac{3}{80}\pi^2 + \frac{11}{4}\log^2 2 - \frac{1}{2}\log^2 5 - \mathrm{Li}_2(1/5) - \log^2(2\sin(\pi/5)) - \log^2(2\sin(2\pi/5)),\\
J(2/3) &= -\frac{\pi^2}{144} + \frac{3}{4}\log^2 2,\\
J(2/5) &= \frac{11\pi^2}{240} + \frac{3}{4}\log^2 2 - 2\log^2 \phi.
\end{aligned}$$

The last two values follow by combining the F evaluations through (11.1). Their existence does not prove that these are the only q for which $J(2/q)$ simplifies.

11.4 From rational examples to an exact formal criterion

The denominator-only condition was false, but the newer cyclotomic argument supplies a precise replacement. Its equivalences concern the specified five-term calculus. Successful cases give actual analytic evaluations; an unsuccessful formal test is not a proof of numerical period independence.

11.5 Cyclotomic symbols and formal reduction

We use additive notation for a multiplicative group after rationalization:

$$\mathcal{A}(K) = K^\times \otimes_{\mathbb{Z}} \mathbb{Q}, \qquad \langle uv\rangle = \langle u\rangle + \langle v\rangle.$$

Roots of unity map to zero. Exterior powers below are over $\mathbb{Q}$ unless otherwise specified.

Definition 11.7 (Pre-Bloch group and formal reduction)**.** *Let $\mathcal{P}(K)$ be the $\mathbb{Q}$-vector space on symbols $[z]$, $z \in K$, with $[0] = [1] = 0$, modulo the five-term relations. Its differential is*

$$\partial[z] = \langle z\rangle \wedge \langle 1-z\rangle \quad (z \neq 0, 1), \qquad \partial[0] = \partial[1] = 0, \qquad \mathcal{B}(K) = \ker \partial.$$

For an invariant dilogarithm element $\xi \in \mathcal{P}(K)$, a formal logarithmic reduction *means $\xi = 0$ in $\mathcal{P}(K)$. A* formal rational-dilogarithm reduction *means that ξ belongs to the image of $\mathcal{P}(\mathbb{Q})$. Upon evaluation, five-term relations carry products of algebraic logarithms and rational multiples of π^2; these are the terms suppressed by this terminology.*

For a finite Galois number field $K/\mathbb{Q}$, two standard rationalized facts will be used:

$$\mathcal{B}(K)^{\mathrm{Gal}(K/\mathbb{Q})} = \mathcal{B}(\mathbb{Q}) = 0, \qquad \partial\mathcal{P}(\mathbb{Q}) = \Lambda^2\mathcal{A}(\mathbb{Q}). \tag{11.6}$$

The first is Bloch-group descent and the torsion of the Bloch group of $\mathbb{Q}$; the second is equivalent to the torsion of $K_2(\mathbb{Q})$. These are established algebraic K-theory inputs, not new results of this article; see [46, 45]. The same inputs underpin the symbol argument in [35, §7.2].

Let $\zeta_q = e^{2\pi i/q}$ and, for $(a, q) = 1$, set

$$\beta_q(a) = \sum_{k=1}^{q-1} \langle 1-\zeta_q^{ak}\rangle \wedge \langle 1-\zeta_q^{k}\rangle \in \Lambda^2\mathcal{A}(\mathbb{Q}(\zeta_q)). \tag{11.7}$$

We put $\beta_1 = \beta_2 = 0$. The term $k = 0$ is excluded: there is no multiplicative-group element $\langle 0\rangle$. For coprime positive p, q, the finite cyclotomic dilogarithm sum for $F(p/q) - F(1)$ defines a Galois-invariant element $\xi_{p/q}$ with

$$\partial\xi_{p/q} = \beta_q(p) - \beta_p(q) - \langle p\rangle \wedge \langle q\rangle. \tag{11.8}$$

One can take $K = \mathbb{Q}(\zeta_{pq})$. We retain this classical normalization throughout; the rational wedge in (11.8) matters for the distinction between logarithms and rational dilogarithms.

Lemma 11.8 (Elementary symmetries)**.** *For every conductor and every invertible residue a,*

$$\beta_q(-a) = \beta_q(a), \qquad \beta_q(a^{-1}) = -\beta_q(a).$$

Each $\beta_q(a)$ is Galois invariant.

Proof. The identity $1 - \zeta_q^{-ak} = -\zeta_q^{-ak}(1-\zeta_q^{ak})$ proves the first claim after roots of unity are killed. In the sum for $\beta_q(a^{-1})$, replace k by ak and reverse the exterior factors. Finally, every Galois automorphism multiplies k by a unit modulo q, permuting the summation set. □

11.6 The all-conductor relation theorem

The following is the main algebraic theorem.

Theorem 11.9 (Complete kernel, every conductor)**.** *Let $q \geq 3$, $\mathcal{G}_q = (\mathbb{Z}/q\mathbb{Z})^\times/\{\pm 1\}$, and $(c_a)_{a\in\mathcal{G}_q} \in \mathbb{Q}^{\mathcal{G}}_q$. Then*

$$\sum_{a\in\mathcal{G}_q} c_a\beta_q(a) = 0 \quad\Longleftrightarrow\quad c_a = c_{a^{-1}} \quad \text{for every } a \in \mathcal{G}_q. \tag{11.9}$$

In particular,

$$\beta_q(a) = 0 \quad\Longleftrightarrow\quad a^2 \equiv \pm 1 \pmod q, \tag{11.10}$$

$$\dim_{\mathbb{Q}} \operatorname{span}\{\beta_q(a) : a \in \mathcal{G}_q\} = \frac{|\mathcal{G}_q| - |\mathcal{G}_q[2]|}{2}. \tag{11.11}$$

Choosing one a from each inverse pair $\{a, a^{-1}\}$ with $a \neq a^{-1}$ gives a basis of the symbol space.

The proof is short after the correct positive matrix is identified. Its use of *all* nontrivial roots, including those of smaller order, is essential at composite conductor.

11.6.1 Cyclotomic logarithms and a positive Gram matrix

For $1 \leq k < q$, define the real vector

$$v_k = \big(\log|1-\zeta_q^{ak}|\big)_{a\in\mathcal{G}_q} \in \mathbb{R}^{\mathcal{G}}_q, \qquad A_q = \sum_{k=1}^{q-1} v_k v_k^{\mathsf{T}}. \tag{11.12}$$

The entries are independent of the representative of a modulo sign. Let P_b be the permutation matrix $(P_b v)_a = v_{ab}$. Then

$$P_b v_k = v_{bk}, \qquad P_b A_q P_b^{\mathsf{T}} = A_q, \qquad P_b A_q = A_q P_b. \tag{11.13}$$

Lemma 11.10 (Positive definiteness)**.** *The vectors v_k span $\mathbb{R}^{\mathcal{G}}_q$. Hence A_q is positive definite.*

Proof. Complexify the regular representation of the finite abelian group $\mathcal{G}_q$. Each character occurs once. The span of the v_k is $\mathcal{G}_q$-invariant by (11.13), so it suffices to show a nonzero Fourier component for every character.

Let χ be nontrivial and let $f \mid q$ be its primitive Dirichlet conductor. It is an even character, so $f > 2$. Write χ^* for the primitive character modulo f. Reduction gives a surjection $\mathcal{G}_q \longrightarrow G_f$. For the vector with $k = q/f$, its Fourier coefficient is a nonzero positive integer multiple of

$$\sum_{a\in G_f} \chi^*(a)\log|1-\zeta_f^a| = -\frac{1}{2}\tau(\chi^*)L(1,\overline{\chi^*}) \neq 0. \tag{11.14}$$

The equality follows by Abel-regularizing $\log|1-re^{i\theta}| = -\sum_{n\geq 1} r^n\cos(n\theta)/n$, summing over a, using the primitive Gauss-sum identity, and letting $r \uparrow 1$. The Gauss sum is nonzero and Dirichlet's nonvanishing theorem gives $L(1,\overline{\chi^*}) \neq 0$. This is a standard cyclotomic-logarithm identity; cf. [47, 48].

For the trivial character, choose a prime $\ell \mid q$. If ℓ is odd, use $k = q/\ell$ and

$$\sum_{a\in G_\ell} \log|1-\zeta_\ell^a| = \tfrac{1}{2}\log\ell \neq 0.$$

If $\ell = 2$, the vector $v_{q/2}$ is the nonzero constant vector $(\log 2)_{a\in\mathcal{G}_q}$. Thus every character component occurs. This proves the spanning assertion, and $w^{\mathsf{T}}A_q w = \sum_k (w^{\mathsf{T}}v_k)^2 > 0$ for $w \neq 0$. □

11.6.2 From exterior symbols to skew matrices

There is a $\mathbb{Q}$-linear logarithm map

$$\ell : \mathcal{A}(\mathbb{Q}(\zeta_q)) \longrightarrow \mathbb{R}_q^{\mathcal{G}}, \qquad \ell(\langle u \rangle) = (\log |\sigma_a(u)|)_{a \in \mathcal{G}_q}.$$

Composing its exterior square with the passage to the real exterior power and then identifying $v \wedge w$ with $vw^{\mathsf{T}} - wv^{\mathsf{T}}$ gives a $\mathbb{Q}$-linear map to real skew matrices. We do *not* assume this map is injective on the whole exterior power. Its value on (11.7) is explicitly

$$\ell_\wedge(\beta_q(b)) = P_b A_q - A_q P_b^{\mathsf{T}} = A_q(P_b - P_{b^{-1}}). \tag{11.15}$$

Proof of Theorem 11.9. If the left side of (11.9) vanishes, apply (11.15) and multiply by the invertible matrix A_q^{-1}. We get

$$0 = \sum_{b \in \mathcal{G}_q} c_b (P_b - P_{b^{-1}}) = \sum_{b \in \mathcal{G}_q} (c_b - c_{b^{-1}}) P_b.$$

The regular permutation matrices P_b are linearly independent, even over $\mathbb{R}$: their values on a single coordinate vector are the distinct coordinate vectors. Therefore $c_b = c_{b^{-1}}$.

Conversely, if these equalities hold, group the sum into inverse pairs and use Lemma 11.8; self-inverse classes have $2\beta_q(b) = 0$ and hence contribute zero. This proves (11.9). Applying it to a coefficient vector supported at one class proves (11.10), and counting nontrivial inverse pairs proves (11.11) and the basis assertion. □

Remark 11.11 (Why the composite case is not a numerical rank claim). All matrix entries involving logarithms are used only in the proof of positive definiteness. The conclusion follows from exact character nonvanishing, not from a numerical determinant. In particular, no lower bound for a numerically computed singular value is required. The resulting coordinate model $\beta_q(a) \mapsto P_a - P_{a^{-1}}$ is exact on the symbol span, even though its derivation used a real logarithm map.

11.6.3 Explicit rank formula and zero conductors

Corollary 11.12. *For $q \geq 3$,*

$$d(q) := \dim_{\mathbb{Q}} \operatorname{span}\{\beta_q(a)\} = \frac{\varphi(q) - r_+(q) - r_-(q)}{4}.$$

Write $q = 2^e \prod_{j=1}^{s} p_j^{e_j}$ with distinct odd primes p_j. Then

$$r_+(q) = 2^s \begin{cases} 1 & e = 0, 1, \\ 2 & e = 2, \\ 4 & e \geq 3, \end{cases}$$

and $r_-(q) = 0$ if $e \geq 2$ or some $p_j \equiv 3 \pmod 4$; otherwise $r_-(q) = 2^s$.

Proof. The self-inverse elements of $\mathcal{G}_q$ are the sign classes of roots of $u^2 = 1$ or $u^2 = -1$. These two sets are disjoint for $q > 2$, and every sign class has two representatives. Thus $|\mathcal{G}_q[2]| = (r_+ + r_-)/2$. The stated root counts follow from the Chinese remainder theorem and the elementary square-root counts at prime powers. □

Corollary 11.13 (Complete list of identically zero conductors)**.** *With the conventions at $q = 1, 2$, the function β_q is identically zero exactly for*

$$q \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}. \tag{11.16}$$

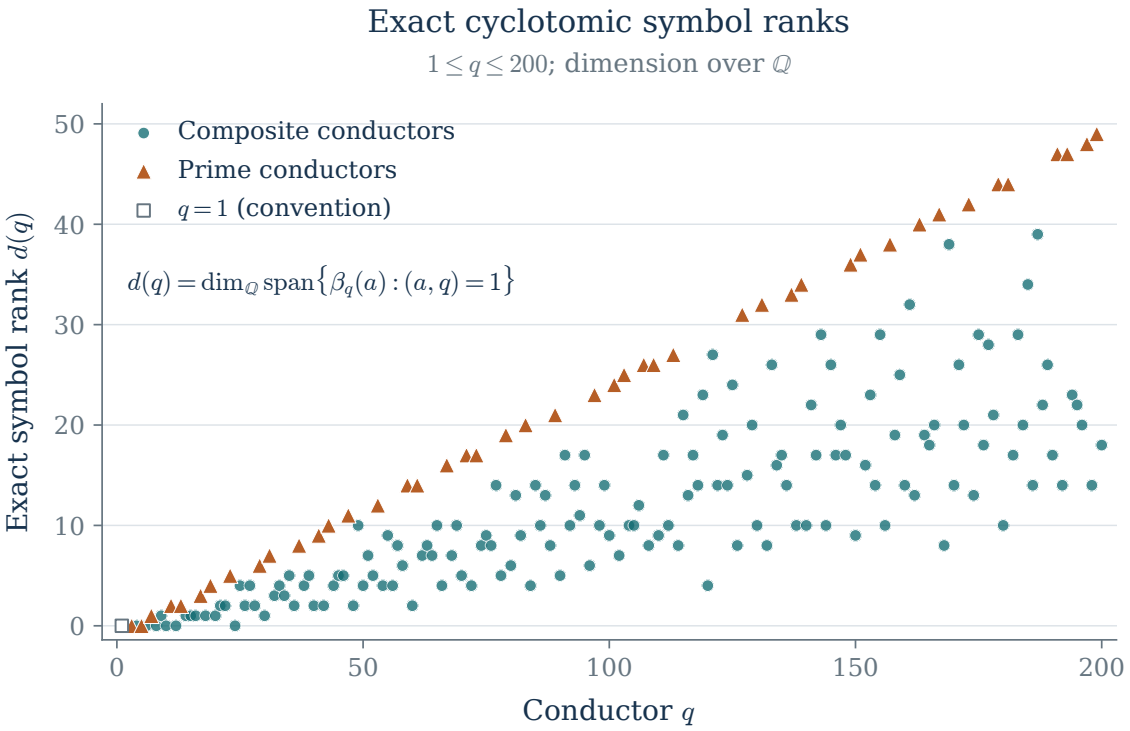


Figure 11.1: Exact ranks of the cyclotomic symbol spaces for conductors $1 \le q \le 200$. The plotted values come from integer residue counts and the proved rank formula. They are not numerical estimates of linear independence among evaluated periods.

Proof. The condition is that $\mathcal{G}_q$ have exponent at most 2. Projecting to each odd prime-power factor forces that factor to be 3 or 5 to the first power, since its unit group is cyclic and has order dividing 4. The 2-part is at most 8: the square of 3 modulo 16 is neither 1 nor -1. If 5 occurs with 3 or with 4, the Chinese remainder theorem produces a unit whose square is -1 at 5 and 1 at the other factor; its square is not a global sign. Hence the cases containing 5 are only $5, 10$. Without 5, all divisors of 24 work. This is exactly (11.16). □

q	7	9	13	15	16	24	25	60	105
$d(q)$	1	1	2	1	1	0	4	2	10

The rank concerns symbols, not the dimension of the numerical span of Herglotz values. Formula (11.11) makes the distinction computable without a height-bounded integer-relation search.

11.7 Exact reduction criteria and constructive rational cores

The two cyclotomic pieces in (11.8) cannot cancel across coprime conductors. This needs a norm on the *individual exterior factors*; averaging an exterior symbol under Galois action would not work, because the symbols themselves are invariant.

Lemma 11.14 (Separation by normalized field norms)**.** *Let* $(p, q) = 1$ *and regard both symbols in* $\Lambda^2\mathcal{A}(K)$*,* $K = \mathbb{Q}(\zeta_{pq})$*. Then*

$$\beta_q(p) - \beta_p(q) \in \Lambda^2\mathcal{A}(\mathbb{Q}) \quad \Longleftrightarrow \quad \beta_q(p) = \beta_p(q) = 0. \tag{11.17}$$

Proof. Let $E = \mathbb{Q}(\zeta_p)$ and define the linear normalized norm

$$N_p : \mathcal{A}(K) \longrightarrow \mathcal{A}(E), \qquad \langle u \rangle \longmapsto \frac{\langle \mathrm{Norm}_{K/E}(u) \rangle}{[K : E]}.$$

It fixes the vectors from E. The fields $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\zeta_q)$ are linearly disjoint over $\mathbb{Q}$. In each summand of $\beta_q(p)$, the two factors are Galois conjugate in $\mathbb{Q}(\zeta_q)$, so N_p sends them to the same rational vector. Hence

$$(\Lambda^2 N_p)\beta_q(p) = 0, \qquad (\Lambda^2 N_p)\beta_p(q) = \beta_p(q).$$

The analogous statements hold with p, q exchanged. If the difference equals a rational wedge W, applying $\Lambda^2 N_p$ gives $W = -\beta_p(q)$. Applying $\Lambda^2 N_q$ to this equality gives $W = 0$, and both symbols vanish. The converse is immediate. The conventions for conductor 1 or 2 make the same argument valid in the degenerate field cases. □

Theorem 11.15 (Complete formal reduction criterion)**.** *For coprime positive integers p, q, the following are equivalent:*

1. *$\xi_{p/q}$ has a formal rational-dilogarithm reduction.*
2. *$\beta_q(p) = \beta_p(q) = 0$.*
3. *$p^2 \equiv \pm 1 \pmod{q}$ and $q^2 \equiv \pm 1 \pmod{p}$.*

Moreover, $\xi_{p/q}$ has a formal logarithmic reduction if and only if $p = 1$ or $q = 1$.

Proof. If $\xi_{p/q}$ is represented by rational dilogarithms, its differential is a rational wedge. Equations (11.8) and (11.17) imply (2). Conversely, if (2) holds, the differential is $-\langle p \rangle \wedge \langle q \rangle$. By (11.6) choose $\eta \in \mathcal{P}(\mathbb{Q})$ with this differential. Then $\xi_{p/q} - \eta$ is a Galois-invariant Bloch-group element and hence zero. The equivalence with (3) is Theorem 11.9.

For the last assertion, a normalized norm to $\mathbb{Q}$ sends every β-summand to zero on the exterior square. Thus $\partial\xi_{p/q} = 0$ implies $\langle p \rangle \wedge \langle q \rangle = 0$. If $p, q > 1$ are coprime, their prime-exponent vectors have disjoint nonempty supports, and their wedge is nonzero. If $p = 1$ or $q = 1$, all terms in (11.8) vanish; invariant Bloch descent gives the reduction. □

Remark 11.16 (Interpretation of a failed test). The theorem gives a necessary and sufficient condition for a certificate in a specified algebraic calculus. A nonzero symbol is not a proof that the corresponding real number satisfies no other numerical identity. Such a statement would require additional information about the kernel of period evaluation. In particular, the theorem should not be restated as an unconditional transcendence theorem.

11.7.1 Checking the boundary formula directly

Because the reduction criterion depends on (11.8), it is useful to audit that formula algebraically. The finite evaluation uses the formal summand

$$f(\alpha, \beta) = \left[\frac{\beta}{\beta - 1}\right] - \left[\frac{\alpha - \beta}{1 - \beta}\right],$$

summed over $\alpha^p = 1$, first over nontrivial qth roots β, and then with the analogous sum over nontrivial pth roots subtracted. Terms with $\alpha = 1$ or $\alpha = \beta$ have zero differential and are set aside before taking multiplicative logarithms. For the remaining roots of unity α, β, put $A = \langle 1 - \alpha \rangle$,

$B = \langle 1-\beta\rangle$, $C = \langle \alpha-\beta\rangle$. Away from the terms $[0], [1]$, which contribute zero, direct expansion gives

$$\partial f(\alpha,\beta) = -C\wedge A + C\wedge B + B\wedge A.$$

The differential of the subtracted summand is antisymmetric under $\alpha \leftrightarrow \beta$, so that sum has zero differential. In the first sum use

$$\prod_{\substack{\beta^q=1\\ \beta\neq 1}} (\alpha-\beta) = \frac{\alpha^q-1}{\alpha-1}, \qquad \prod_{\substack{\alpha^p=1\\ \alpha\neq 1}} (\beta-\alpha) = \frac{\beta^p-1}{\beta-1}.$$

The first two wedge terms then give $-\beta_p(q)$ and $\beta_q(p)$. The last gives $\langle q\rangle\wedge\langle p\rangle$. This proves (11.8) independently of any finite-field cocycle claim.

11.7.2 A recurrence parametrization of the reducible pairs

The congruence criterion has an elementary arithmetic description that connects it to the golden-ratio ladder thread of the repository.

Theorem 11.17 (Quadratic recurrence families)**.** *Apart from* $(1,1)$*, all coprime pairs* $1 \le p < q$ *satisfying* (11.10) *and Theorem 11.15 are exactly the adjacent pairs in the following three families. Their union need not be disjoint.*

1. *For each integer* $h \ge 2$*,* $u_0 = 1$*,* $u_1 = h$*,* $u_{j+1} = hu_j - u_{j-1}$*.*
2. $u_0 = 1$*,* $u_1 = 2$*,* $u_{j+1} = 3u_j - u_{j-1}$*.*
3. *For each integer* $h \ge 1$*,* $u_0 = 1$*,* $u_1 = h$*,* $u_{j+1} = hu_j + u_{j-1}$*, retaining pairs in increasing order.*

The first family has both square congruences $+1$*, the second both* -1*, and the third permits opposite signs, with the usual coincidences of signs modulo* 1 *or* 2*.*

Proof. Suppose $1 < a < b$ and write $a^2 \equiv s \pmod b$, $b^2 \equiv t \pmod a$ with $s,t \in \{\pm 1\}$. Set

$$c = \frac{a^2 - s}{b}.$$

Then $1 \le c < a$, $(c,a) = 1$, and the new pair (c,a) satisfies the same conditions with signs (t,s). Indeed $bc \equiv -s \pmod a$ implies $c^2 \equiv b^{-2} \equiv t \pmod a$, while $a^2 \equiv s \pmod c$. This gives a strictly descending process.

If $s = t$, the positive integer $h = (a^2+b^2-s)/(ab)$ is preserved under descent and satisfies $b = ha - c$. For $s = 1$, the terminal pair is $(1,h)$ with $h \ge 2$. For $s = -1$, a terminal pair $(1,b)$ has $b \mid 2$, so it is $(1,1)$ or $(1,2)$ and the invariant is $h = 3$.

If $s = -t$, the positive integer $h = (b^2-a^2-t)/(ab)$ is preserved and satisfies $b = ha + c$. The terminal pairs are seeds $(1,h)$, with the terminal $(1,2)$ also lying in the $h = 1$ sequence $1,1,2,3,5,\ldots$. Reversing descent gives the three displayed recurrences. Conversely their preserved quadratic identities, or the same recurrence step in reverse, verify both congruences and coprimality at every stage. □

For example, the second family is $1,2,5,13,34,\ldots$, the odd-index Fibonacci numbers. The first family with $h = 3$ is $1,3,8,21,55,\ldots$, and the third with $h = 3$ is $1,3,10,33,109,\ldots$. This is an arithmetic parametrization of *which* rational arguments admit formal reduction, not an assertion that all their logarithmic correction terms are already tabulated.

11.7.3 An explicit rational-dilogarithm core

Theorem 11.18 (A short core with a boundary certificate)**.** *For every pair satisfying Theorem 11.15, one can produce a rational combination $C_{p/q}$ of symbols $[z]$, $0 < z < 1$, such that*

$$\partial C_{p/q} = -\langle p\rangle \wedge \langle q\rangle, \qquad \xi_{p/q} - C_{p/q} = 0 \quad \text{in } \mathcal{P}(K).$$

The following recursion suffices for $1 \le p < q$:

1. *If $p = 1$, use $C_{p/q} = 0$.*
2. *If $q = p + 1$, use $C_{p/q} = [p/q]$.*
3. *Otherwise choose $s \in \{\pm 1\}$ with $q \mid p^2 - s$, put $c = (p^2 - s)/q$, and use*

$$C_{p/q} = C_{c/p} + \begin{cases} -\frac{1}{2}[1 - p^{-2}], & s = 1, \\ \frac{1}{2}[p^2/(p^2+1)], & s = -1. \end{cases} \tag{11.18}$$

Put $C_{1/1} = 0$. For $p > q$, negate the core for q/p. After the consecutive-pair shortcut, the number of terms is $O(\log \max(p, q))$.

Proof. The first two cases have the required differential. In the last case $cq = p^2 - s$. Direct calculation gives

$$\begin{aligned} -\tfrac{1}{2}\partial[1 - p^{-2}] &= -\langle p\rangle \wedge \langle q\rangle - \langle p\rangle \wedge \langle c\rangle && (s = 1), \\ \tfrac{1}{2}\partial[p^2/(p^2+1)] &= -\langle p\rangle \wedge \langle q\rangle - \langle p\rangle \wedge \langle c\rangle && (s = -1). \end{aligned}$$

Adding $\partial C_{c/p} = -\langle c\rangle \wedge \langle p\rangle$ leaves the required $-\langle p\rangle \wedge \langle q\rangle$. Descent terminates by the previous theorem. Both $C_{p/q}$ and $\xi_{p/q}$ are invariant; their difference lies in the Bloch group and vanishes by (11.6).

For the length bound, the $h = 2$ minus recurrence consists of consecutive integers and is handled in one step. Every remaining minus recurrence has $h \ge 3$ and grows by a factor greater than 2 after its seed. Each plus recurrence grows at least as fast as the Fibonacci recurrence. Thus the number of descent steps is logarithmic in the larger input. □

Argument	A rational-dilogarithm core for $F(p/q) - F(1)$
2/5	$\frac{1}{2}\operatorname{Li}_2(4/5)$
3/8	$-\frac{1}{2}\operatorname{Li}_2(8/9)$
4/17	$\frac{1}{2}\operatorname{Li}_2(16/17)$
5/13	$\frac{1}{2}\operatorname{Li}_2(4/5) + \frac{1}{2}\operatorname{Li}_2(25/26)$
10/33	$\frac{1}{2}\operatorname{Li}_2(9/10) - \frac{1}{2}\operatorname{Li}_2(99/100)$

In this table the difference between $F(p/q) - F(1)$ and the displayed core is a combination of logarithm products and π^2. The core algorithm certifies the dilogarithm part; it does not yet print the full logarithmic correction. Constructing that correction efficiently and with a complete branch certificate is a further research task.

Corollary 11.19 (A lower bound for a family of formal values)**.** *Fix $q \ge 3$ and choose one positive numerator representing each nontrivial inverse pair in $\mathcal{G}_q$. The corresponding $d(q)$ elements $\xi_{a/q}$ are linearly independent modulo rational dilogarithms.*

Proof. Take a common cyclotomic field for all numerators and q, and apply the exterior square of its normalized norm to $\mathbb{Q}(\zeta_q)$. The numerator-conductor symbols disappear and the denominator symbols $\beta_q(a)$ remain. If their combination were a rational wedge, the normalized norm to $\mathbb{Q}$ would kill it, so it would be zero. The chosen $\beta_q(a)$ are independent by Theorem 11.9. □

11.8 Complete classification and evaluations of $J(2/q)$

Define

$$S(n) = \sum_{k=1}^{n-1} \log^2\left(2\sin\frac{\pi k}{n}\right), \qquad S(1) = 0, \qquad \phi = \frac{1+\sqrt{5}}{2}.$$

The reciprocal-integer identity, obtainable by pairing conjugate roots in the finite dilogarithm formula, is

$$F(1/n) - F(1) = -\frac{(n-1)(3n-2)}{24n}\pi^2 - \frac{1}{2}S(n), \qquad n \geq 1. \tag{11.19}$$

All logarithms here are real logarithms of positive algebraic numbers.

Theorem 11.20 (Complete formal classification)**.** *For integers $q \geq 2$:*

1. *$J(2/q)$ has a formal logarithmic reduction exactly when $q \in \{2,3,5\}$ or $4 \mid q$.*
2. *It has a formal rational-dilogarithm reduction exactly when $q \in \{2,3,5,6,10\}$ or $4 \mid q$.*

In every positive case the evaluations below give the corresponding analytic identity.

Proof. First suppose q is odd. Applying (11.8) to (11.1) at $x = 2/q$, the numerator symbols at conductors $1, 2, 4$ vanish and the rational wedges cancel. Thus the boundary is

$$\beta_q(4) - 2\beta_q(2).$$

Put $g = 2 \in \mathcal{G}_q$. By Theorem 11.9, this vanishes exactly when g has order at most 2. To check the assertion directly, an element of order 3 gives $\beta(g^2) - 2\beta(g) = -3\beta(g)$; order 4 gives $-2\beta(g)$; for order at least 5, the two relevant inverse pairs are distinct. Hence vanishing is equivalent to $4 \equiv \pm 1 \pmod{q}$, or $q = 3, 5$. A rational wedge cannot remove a nonzero combination of these symbols, since the normalized norm to $\mathbb{Q}$ kills the combination and fixes rational wedges. This proves both classifications for odd q.

If $4 \mid q$, all three arguments $4/q, 2/q, 1/q$ in (11.1) are reciprocal integers. Equation (11.19) gives a logarithmic reduction.

The remaining case is $q = 2m$ with m odd. Its boundary is

$$\beta_m(2) - \langle 2\rangle \wedge \langle m\rangle. \tag{11.20}$$

For $m = 1$ it is zero. For $m > 1$, applying the normalized norm to $\mathbb{Q}$ gives the nonzero rational wedge $-\langle 2\rangle \wedge \langle m\rangle$; there is no formal logarithmic reduction. Rational dilogarithms are allowed exactly when $\beta_m(2) = 0$, equivalently $m = 3, 5$. Invariant Bloch descent supplies the stated formal reductions. □

11.8.1 All positive evaluations

The isolated cases are

$$J(1) = \tfrac{1}{2}\log^2 2, \tag{11.21}$$

$$J(2/3) = -\frac{\pi^2}{144} + \frac{3}{4}\log^2 2, \tag{11.22}$$

$$J(2/5) = \frac{11\pi^2}{240} + \frac{3}{4}\log^2 2 - 2\log^2\phi, \tag{11.23}$$

$$J(1/3) = \frac{\pi^2}{9} + \frac{1}{2}\log^2 2 - \frac{1}{2}\log^2 3 - \mathrm{Li}_2(1/3), \tag{11.24}$$

$$J(1/5) = \frac{7\pi^2}{60} + \frac{1}{2}\log^2 2 - \frac{1}{4}\log^2 5 - \log^2\phi - \frac{1}{2}\mathrm{Li}_2(1/5). \tag{11.25}$$

For all even $m \geq 2$, the infinite family is

$$\boxed{J(1/m) = \frac{\pi^2(m^2-2)}{48m} + S(m) - \frac{1}{2}S(m/2) - \frac{1}{2}S(2m).} \tag{11.26}$$

Thus (11.26) evaluates $J(2/q)$ for every multiple q of 4, without a dilogarithm term.

Derivation of the evaluations. Equation (11.21) follows by integrating $\log(1+t)/(1+t)$. For (11.26), substitute (11.19) at $n = m/2, m, 2m$ in (11.1). The constant $F(1)$ cancels; the polynomial coefficient simplifies to $m/48 - 1/(24m)$, yielding the formula.

The small odd cases are obtained from the finite root-of-unity dilogarithm formula and the reflection, duplication, and Landen relations. For clarity, the two fractional F evaluations needed in the calculation can be taken as

$$F(2/3) - F(1) = \frac{\pi^2}{36} + \log^2 2 - \frac{3}{4}\log^2 3 - \mathrm{Li}_2(1/3),$$
$$F(2/5) - F(1) = \tfrac{1}{2}\,\mathrm{Li}_2(4/5) - \frac{\pi^2}{5} + \log^2\!\left(2\sin\frac{\pi}{5}\right) + \log^2\!\left(2\sin\frac{2\pi}{5}\right) + \log 2\log(2/5).$$

The second is the previously known evaluation [35]; the first follows by using $\cot^2(\pi/3) = 1/3$ in the $F(2/q)$ sum and $\mathrm{Li}_2(-1/3) = 2\,\mathrm{Li}_2(1/3) - \pi^2/6 + \frac{1}{2}\log^2 3$. Substitution in (11.1), together with (11.19) and the two standard functional equations

$$F(x) + F(1/x) = 2F(1) + \tfrac{1}{2}\log^2 x - \frac{\pi^2(x-1)^2}{6x},$$
$$F(x) - F(x+1) - F(x/(x+1)) = -F(1) + \mathrm{Li}_2(1/(1+x)),$$

gives (11.22)–(11.25). Reflection of $\mathrm{Li}_2(4/5)$ supplies the $\mathrm{Li}_2(1/5)$ in (11.25). All manipulations use arguments in $(0,1)$ or the real negative axis, so no unstated complex logarithm branches are involved. □

The first member of (11.26) is particularly useful for auditing the earlier claim:

$$J(1/2) = \frac{\pi^2}{48} + \frac{1}{4}\log^2 2. \tag{11.27}$$

The complete classification proves a precise replacement for the unsupported numerical "if and only if" in the draft. It also shows that the distinction between logarithms and rational dilogarithms is real at the level of the boundary: the cases $q = 6, 10$ have a nonzero rational wedge, even though their cyclotomic obstruction vanishes.

11.9 Bridges to the golden ladders and the Clausen lattice

The new constants are not isolated; they touch two corpora we have already mapped.

The golden bridge ($q = 5$). The arguments $-\cot^2\frac{\pi}{5} = -\frac{5+2\sqrt{5}}{5}$ and $-\cot^2\frac{2\pi}{5} = -\frac{5-2\sqrt{5}}{5}$ lie in $\mathbb{Q}(\sqrt{5})$, the field of the golden polylogarithm ladders. Comparing Theorem 11.3 ($q = 5$) with eq. 24 gives the functional equation

$$\mathrm{Li}_2\!\left(\tfrac{4}{5}\right) + \mathrm{Li}_2\!\left(-\cot^2\tfrac{\pi}{5}\right) + \mathrm{Li}_2\!\left(-\cot^2\tfrac{2\pi}{5}\right) = 2\log^2 2 - 2\log 2\log\tfrac{2}{5} - 6\big(\log^2(2\sin\tfrac{\pi}{5}) + \log^2(2\sin\tfrac{2\pi}{5})\big),$$

hence the golden-sum closed form (with $\varphi = \frac{1+\sqrt{5}}{2}$)

$$\mathrm{Li}_2\Big(-\cot^2\tfrac{\pi}{5}\Big) + \mathrm{Li}_2\Big(-\cot^2\tfrac{2\pi}{5}\Big) = \mathrm{Li}_2\Big(\tfrac{1}{5}\Big) - \frac{\pi^2}{6} - 3\log^2\varphi + \tfrac{1}{4}\log^2 5.$$

The symmetric sum has the displayed reduction. The antisymmetric combination $X = \frac{1}{2}[\mathrm{Li}_2(-\cot^2(\pi/5)) - \mathrm{Li}_2(-\cot^2(2\pi/5))]$ did not reduce in the source's tested basket. Its independence is open. A formal symbol with nonzero Bloch boundary is not subject to a vanishing-rank conclusion for the Bloch group.

The Clausen bridge. The coefficients of $W(q)$ are real algebraic numbers contained in $\mathbb{Q}(\zeta_{\mathrm{lcm}(4,q)})^+$; at odd conductors the factor i implicit in the cotangent can require an extension of $\mathbb{Q}(\zeta_q)^+$. This is an algebraic coefficient issue, not a proof of numerical period independence. Applying the distribution identities of Chapter 2 gives

$$W(8) = 2\,\mathrm{Cl}_2(\pi/4) - 2\,\mathrm{Cl}_2(3\pi/4) = G, \qquad W(12) = \frac{11\sqrt{3}}{9}\,\mathrm{Cl}_2(\pi/3).$$

For the second formula, pair k with $12-k$:

$$W(12) = 2\sqrt{3}\,[\mathrm{Cl}_2(\pi/6) - \mathrm{Cl}_2(5\pi/6)] + \frac{2}{\sqrt{3}}\,[\mathrm{Cl}_2(\pi/3) - \mathrm{Cl}_2(2\pi/3)]\,.$$

Use $\mathrm{Cl}_2(\pi/6) - \mathrm{Cl}_2(5\pi/6) = V/2$ and $\mathrm{Cl}_2(2\pi/3) = 2V/3$. This proves the original report's coefficient $22\sqrt{3}/9$ multiplying the first difference. It is also checked against direct derivative quadrature in the fresh verification battery.

11.10 Positive measures and sharp approximation

Binet's kernel, already encountered in the gamma-integral chapter, gives the Herglotz function a positive-measure representation. The same representation certifies truncation errors and explains the optimal cutoff's arithmetic fluctuations. These analytic results complement the exact symbol calculus without relying on period conjectures.

11.11 A positive measure and certified remainder bounds

The analytic part begins with a representation already obtained by Radchenko and Zagier [35, §7.3]. We derive its consequences explicitly, keeping signs and normalizations visible. Put

$$G(x) = -F(x) - \frac{\zeta(2)}{2x}, \qquad \mathcal{L}(u) = \sum_{m=1}^{\infty} \sigma_{-1}(m) e^{-2\pi m u}, \qquad \sigma_{-1}(m) = \sum_{d\mid m} \frac{1}{d}. \tag{11.28}$$

Expanding a geometric series gives

$$\mathcal{L}(u) = \sum_{n=1}^{\infty} \frac{1}{n(e^{2\pi n u} - 1)}.$$

Binet's digamma integral, followed by a positive change of variables and Tonelli's theorem, yields

$$G(x) = 2\int_0^{\infty} \frac{u\mathcal{L}(u)}{x^2+u^2}\,du = \int_0^{\infty} \frac{d\mu(s)}{x^2+s}, \qquad d\mu(s) = \mathcal{L}(\sqrt{s})\,ds. \tag{11.29}$$

Thus $y \mapsto G(\sqrt{y})$ is a Stieltjes transform of a positive measure with a positive density on $(0,\infty)$.

Lemma 11.21 (Explicit moments)**.** *Every nonnegative moment exists, and*

$$\mu_j := \int_0^\infty s^j \, d\mu(s) = \frac{2\Gamma(2j+2)\zeta(2j+2)\zeta(2j+3)}{(2\pi)^{2j+2}} = \frac{|B_{2j+2}|}{2j+2}\zeta(2j+3). \tag{11.30}$$

Proof. Substitute $s = u^2$, integrate each positive exponential term, and use $\sum_m \sigma_{-1}(m)m^{-w} = \zeta(w)\zeta(w+1)$ for $w > 1$. Euler's formula for the even zeta values gives the second equality. □

Define the approximation and its first omitted term by

$$F_N(x) = -\frac{\zeta(2)}{2x} - \sum_{j=0}^{N-1} \frac{(-1)^j \mu_j}{x^{2j+2}}, \qquad T_N(x) = \frac{\mu_N}{x^{2N+2}}, \qquad N \geq 0. \tag{11.31}$$

The empty sum for $N = 0$ is zero. The series is asymptotic and divergent; we will work with its exact finite remainder.

Theorem 11.22 (Signed remainder and three-moment bounds)**.** *For $x > 0$ and $N \geq 0$,*

$$F(x) - F_N(x) = (-1)^{N+1} R_N(x), \tag{11.32}$$

$$R_N(x) = x^{-2N} \int_0^\infty \frac{s^N}{x^2+s} \, d\mu(s) > 0, \tag{11.33}$$

$$R_N(x) + R_{N+1}(x) = T_N(x). \tag{11.34}$$

Set

$$A = \frac{\mu_{N+1}}{\mu_N x^2}, \qquad B = \frac{\mu_{N+2}}{\mu_N x^4}.$$

Then the strict bounds

$$\frac{1}{1+A} < \frac{R_N(x)}{T_N(x)} < 1 - \frac{A^2}{A+B} < 1 \tag{11.35}$$

hold. In particular, two consecutive truncations bracket $F(x)$.

Proof. The finite geometric identity

$$\frac{1}{x^2+s} = \sum_{j=0}^{N-1} \frac{(-s)^j}{x^{2j+2}} + \frac{(-1)^N s^N}{x^{2N}(x^2+s)}$$

gives (11.32)–(11.33), and multiplying the remainder integrand by $1 + s/x^2$ gives (11.34).

Under the probability measure $s^N \, d\mu(s)/\mu_N$ let $U = s/x^2$. Then $R_N/T_N = \mathbb{E}(1+U)^{-1}$, $\mathbb{E}U = A$, and $\mathbb{E}U^2 = B$. Strict Jensen convexity gives the lower bound. Cauchy–Schwarz gives

$$\mathbb{E}\frac{U}{1+U} \; \mathbb{E}\big(U(1+U)\big) \geq (\mathbb{E}U)^2,$$

so $\mathbb{E}(1+U)^{-1} \leq 1 - A^2/(A+B)$. Both inequalities are strict because the measure has a nonconstant positive-density support. □

For interval computation, an equivalent and useful form is

$$L_N(x) = \frac{x^{-2N}\mu_N^2}{x^2\mu_N + \mu_{N+1}},$$

$$L_N(x) < R_N(x) < T_N(x) - L_{N+1}(x). \tag{11.36}$$

It uses only three adjacent moments, and hence only rational Bernoulli numbers and odd zeta values. No oscillatory integral is needed.

11.11.1 An exact rational certification scheme

For integer $s \geq 2$, integer $M \geq 2$, and $J \geq 1$, Euler–Maclaurin gives

$$Z_{M,J}(s) = \sum_{n=1}^{M-1} n^{-s} + \frac{M^{1-s}}{s-1} + \frac{1}{2M^s} + \sum_{j=1}^{J-1} \frac{B_{2j}}{(2j)!}(s)_{2j-1} M^{-s-2j+1}, \tag{11.37}$$

$$|\zeta(s) - Z_{M,J}(s)| \leq \frac{|B_{2J}|}{(2J)!}(s)_{2J-1} M^{-s-2J+1}. \tag{11.38}$$

Here $(s)_k$ is a rising factorial. A direct positive-kernel proof is useful because it fixes the endpoint convention in this bound. The partial fraction expansion of the hyperbolic cotangent gives

$$k(t) := \frac{1}{1-e^{-t}} - \frac{1}{t} - \frac{1}{2} = 2t \sum_{n \geq 1} \frac{1}{t^2 + (2\pi n)^2}, \qquad t > 0.$$

Expand each denominator through $J-1$ terms by the finite geometric identity. After summing, the result is

$$k(t) = \sum_{j=1}^{J-1} \frac{B_{2j}}{(2j)!} t^{2j-1} + \varepsilon_J(t), \qquad |\varepsilon_J(t)| \leq \frac{|B_{2J}|}{(2J)!} t^{2J-1}.$$

Indeed the remainder before summing is $(-1)^{J-1} 2t^{2J-1}/((2\pi n)^{2J-2}((2\pi n)^2 + t^2))$, whose magnitude is bounded by replacing the last denominator by $(2\pi n)^2$. Integrate against $t^{s-1}e^{-Mt}/\Gamma(s)$, using

$$\sum_{n=M}^{\infty} n^{-s} = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{t^{s-1}e^{-Mt}}{1-e^{-t}}\, dt.$$

This proves (11.37)–(11.38), including the first-omitted-term bound, rather than assuming a truncation-sign rule. Related sharper Hurwitz-zeta remainder bounds are discussed in [50].

All quantities in (11.37)–(11.38) are rational. Combining their enclosing intervals with (11.30), (11.31), and (11.36) yields exact rational endpoints for $F(x)$ when x is rational. The code uses interval operations with integer numerators and denominators throughout; decimal endpoints are rounded outward only for display.

x	N	Certified width of the interval for $F(x)$, at most
2	6	$4.18751510340 \times 10^{-7}$
4	12	$4.88272699850 \times 10^{-13}$
8	25	$2.15471915709 \times 10^{-24}$
16	50	$1.10232929493 \times 10^{-46}$

These are reproducible certificates, not estimates based on the number of floating-point digits. The file `data/rational_intervals.json` contains the complete exact endpoints. The parameters here are $M = J = 48$; no optimality of that computational choice is asserted.

11.11.2 Classical Stieltjes consequences

The representation also settles the qualitative approximation question without any new special-function identity.

Proposition 11.23 (Determinacy and positive rational approximation)**.** *The measure μ is uniquely determined by its moments. Gaussian quadrature for μ produces lower bounds that are rational functions of y for $G(\sqrt{y})$, $y > 0$; Gauss–Radau quadrature with a prescribed node at 0 produces rational upper bounds. These bounds converge pointwise to $G(\sqrt{y})$ as their orders tend to infinity.*

Proof. Stirling's formula in (11.30) gives $\mu_j^{-1/(2j)} \sim \pi e/j$. Thus the Stieltjes Carleman sum diverges, which implies determinacy. The quadrature error signs follow from the alternating derivatives of $(y+s)^{-1}$: Gaussian quadrature has a positive even-order remainder, while the Radau rule with left endpoint 0 has a negative odd-order remainder. For convergence, the discrete positive quadrature measures have the same total mass and, eventually, every fixed finite set of moments as μ. A first-moment bound gives tightness; higher moment bounds give uniform integrability of each fixed power. Every subsequential weak limit therefore has all the moments of μ and must equal μ by determinacy. Integration of the bounded continuous function $(y+s)^{-1}$ proves the claim. These are standard Stieltjes and Padé consequences; see [51]. □

The value of the proposition here is practical: it explains why positive rational approximations are available even in parameter regimes where an asymptotic series is not an efficient evaluator. An effective convergence rate for this particular measure remains a separate problem.

11.12 Uniform sharp asymptotics at the optimal scale

The scale $N \sim \pi x$ can be located from the ratio of consecutive moments. To obtain the error constant and its corrections, it is more efficient to convert the exact remainder into gamma expectations. Write

$$a = 2\pi x, \quad K = 2N + 2, \quad r = K - a = 2\delta + 2, \quad \delta = N - \pi x. \tag{11.39}$$

Let T have gamma density $t^{K-1}e^{-t}/\Gamma(K)$ on $(0,\infty)$, and define

$$E_K(c) = \mathbb{E}\frac{1}{1+(T/c)^2}, \qquad w_m = \sigma_{-1}(m)m^{-K}, \qquad Z_K = \sum_{m\geq 1} w_m = \zeta(K)\zeta(K+1).$$

Lemma 11.24 (Exact gamma mixture)**.** *For all $x > 0$ and $N \geq 0$,*

$$R_N(x) = \frac{2\Gamma(K)}{a^K}\sum_{m=1}^{\infty} w_m E_K(ma), \qquad T_N(x) = \frac{2\Gamma(K)}{a^K}Z_K. \tag{11.40}$$

For $K \geq 4$,

$$0 < Z_K - 1 \leq 4\,2^{-K}. \tag{11.41}$$

Proof. In each positive exponential summand of (11.33), written in the variable $u = \sqrt{s}$, substitute $t = 2\pi m u$. Collecting powers gives (11.40). For the tail, use $\sigma_{-1}(m) \leq m$ and

$$\sum_{m\geq 2} m^{1-K} \leq 2^{1-K} + \frac{2^{2-K}}{K-2} \leq 4\,2^{-K}.$$

□

The arithmetic terms $m \geq 2$ therefore do not affect any algebraic coefficient in the optimal-scale expansion. They do, however, move the exact transition point by an exponentially small amount; this effect is treated separately in Section 11.14.

11.12.1 Gamma moments and a uniform remainder mechanism

Let $f(v) = (1+v^2)^{-1}$ and $H = (T-a)/a$. The local Taylor expansion is

$$f(1+H) = \frac{1}{2} - \frac{H}{2} + \frac{H^2}{4} - \frac{H^4}{8} + \frac{H^5}{8} - \frac{H^6}{16} + \cdots,$$

and the moments are exactly

$$\mathbb{E}H^\ell = a^{-\ell} \sum_{j=0}^{\ell} \binom{\ell}{j} (-a)^{\ell-j} (a+r)_j. \tag{11.42}$$

This is an algebraic coefficient algorithm, not a fit to numerical data.

For rigor at any fixed order J, Taylor-expand f through degree $2J+1$. Every derivative of f is bounded on $[0,\infty)$, so the absolute remainder is bounded by a constant times $|H|^{2J+2}$. The centered gamma cumulants are $K(j-1)!$ in orders $j \geq 2$. Consequently each centered even moment of order $2J+2$ is a polynomial in K of degree at most $J+1$. Translating the center from K to $a = K - r$, with r bounded, gives

$$\mathbb{E}|H|^{2J+2} = O_J(a^{-J-1})$$

uniformly on each compact interval of r. This proves the required uniform remainder after taking expectations. In particular,

$$\begin{aligned} 2E_K(a) = 1 &+ \frac{1/2 - r}{a} + \frac{(r^2+r)/2 - 3/4}{a^2} + \frac{-3r^2/2 + r/4 + 13/8}{a^3} \\ &- \frac{4r^4 - 16r^3 - 66r^2 + 74r + 59}{16a^4} + O_D(a^{-5}). \end{aligned} \tag{11.43}$$

The subscript D signifies that the constant may depend on a fixed bound for $|r|$; the same convention will be used for δ.

The classical shifted Stirling expansion [49, §5.11.8] is

$$\frac{\Gamma(a+r)}{a^{a+r}} \sim \sqrt{2\pi}\, a^{-1/2} e^{-a} \exp\left(\sum_{j=1}^{\infty} \frac{(-1)^{j+1} B_{j+1}(r)}{j(j+1)a^j} \right). \tag{11.44}$$

It is uniform for r in compact sets. Combining its rigorous finite forms with the gamma-moment argument proves the following theorem.

Theorem 11.25 (Complete optimal-scale expansion)**.** *For each fixed $D > 0$ and each integer $J \geq 0$, there are explicitly computable rational polynomials $C_j(\delta)$ such that, uniformly for integers $N \geq 0$ with $|N - \pi x| \leq D$,*

$$R_N(x) = \frac{e^{-2\pi x}}{\sqrt{x}} \left(\sum_{j=0}^{J} \frac{C_j(\delta)}{(2\pi x)^j} + O_{D,J}(x^{-J-1}) \right). \tag{11.45}$$

The first three are

$$C_0(\delta) = 1,$$

$$C_1(\delta) = 2\delta^2 + \delta - \frac{5}{12}, \tag{11.46}$$

$$C_2(\delta) = 2\delta^4 + \frac{2}{3}\delta^3 - \frac{10}{3}\delta^2 - \frac{7}{12}\delta + \frac{205}{288}. \tag{11.47}$$

Moreover, $R_N(x)/T_N(x) \longrightarrow 1/2$ *uniformly in that regime.*

Proof. Use (11.42) to expand $2E_K(a)$, multiply by (11.44), and put $r = 2\delta + 2$. The frequency tail (11.41) is smaller than every power of a^{-1}. The leading factor is $\sqrt{2\pi}\, a^{-1/2}e^{-a} = x^{-1/2}e^{-2\pi x}$. For an explicit check of the first two coefficients, put $g_1 = B_2(r)/2$ and $g_2 = -B_3(r)/6$. Before substituting for r they are

$$g_1 + \tfrac{1}{2} - r, \qquad \tfrac{1}{2}g_1^2 + g_2 + g_1(\tfrac{1}{2} - r) + \tfrac{1}{2}(r + r^2) - \tfrac{3}{4}.$$

These simplify to (11.46)–(11.47). The ratio assertion follows directly from (11.40), since $E_K(a) \to 1/2$ and $Z_K \to 1$. □

The leading constant is therefore exactly 1. Replacing the true remainder by the first omitted term loses a factor of 2 at the optimal scale. The three-moment bounds also approach this factor: there $A \to 1$ and $B \to 1$, so both sides of (11.35) tend to $1/2$.

The program `code/truncation_coefficients.py` generates C_j symbolically using rational polynomial operations and Bernoulli polynomials. The theorem's bounded-δ condition is essential to the stated uniformity. A growing window, for example $N - \pi x$ of order $\sqrt{x}$, requires a different scaling.

11.13 The exact best cutoff and its transitions

An asymptotic nearest-integer rule is useful, but it cannot determine the best cutoff arbitrarily close to a transition. The positive measure supplies an exact answer.

Theorem 11.26 (Unique transitions and global minimizers)**.** *For every* $N \geq 0$*, there is a unique* $x_N > 0$ *such that* $R_N(x_N) = R_{N+1}(x_N)$*. The sequence* x_N *is strictly increasing and tends to infinity. With* $x_{-1} = 0$*, the index* N *is a global minimizer of* $R_j(x)$*,* $j \geq 0$*, exactly when*

$$x_{N-1} \leq x \leq x_N. \tag{11.48}$$

There are exactly two adjacent minimizing indices at an endpoint and exactly one elsewhere on $x > 0$*. An exact transition equation is*

$$x_N^2 \int_0^\infty \frac{s^N}{x_N^2 + s}\, d\mu(s) = \frac{\mu_N}{2}. \tag{11.49}$$

Proof. Put $y = x^2$ and $M_N(y) = \int s^N (y+s)^{-1}\, d\mu(s)$. Strict Cauchy–Schwarz gives

$$M_{N+1}(y)^2 < M_N(y) M_{N+2}(y).$$

Hence $R_N = y^{-N} M_N(y)$ is strictly log-convex as a sequence in N, and its consecutive ratios are strictly increasing. The recurrence $M_{N+1} = \mu_N - yM_N$ gives

$$\frac{R_{N+1}(x)}{R_N(x)} = \frac{\mu_N}{yM_N(y)} - 1. \tag{11.50}$$

By dominated convergence, $yM_N(y) = \int s^N y/(y+s)\,d\mu(s)$ increases continuously and strictly from 0 to μ_N as y increases from 0 to infinity. The ratio in (11.50) thus decreases continuously from infinity to 0, proving the unique transition and (11.49).

Strict increase of the ratios in N forces $x_{N+1} > x_N$. For fixed y, the moment ratios M_{N+1}/M_N tend to infinity. Indeed, if their increasing sequence were bounded by B, the moments would be at most $M_0 B^N$, contradicting positive mass on an interval strictly to the right of B. This also proves $x_N \to \infty$ and existence of a global minimum. The strict ratio monotonicity now gives (11.48) and the assertions about ties. □

To expand the transition, let α_K be the unique positive root of

$$E_K(\alpha_K) = \frac{1}{2}, \tag{11.51}$$

and write $\beta_K = 2\pi x_N$ for $K = 2N + 2$. The full transition is the unique root of

$$\Phi_K(a) := \sum_{m \geq 1} w_m \left(E_K(ma) - \frac{1}{2} \right) = 0. \tag{11.52}$$

The symbols α_K, β_K in this analytic section are real numbers; the subscript and argument distinguish β_K from the cyclotomic symbol $\beta_q(a)$.

Theorem 11.27 (Algebraic transition expansion)**.** *Put $b = K - \frac{1}{2} = 2N + \frac{3}{2}$. As $N \to \infty$,*

$$\alpha_K = b + \frac{3}{8b} - \frac{1}{b^2} + \frac{407}{128b^3} + O(b^{-4}). \tag{11.53}$$

The same expansion holds for $\beta_K = 2\pi x_N$, and the two quantities have identical expansions to every algebraic order.

Proof. The gamma law gives $T/K \to 1$ in probability. Evaluating at $a = cK$ shows that both roots have $a/K \to 1$. More precisely, substitution of $a = K + C$ for fixed constants on either side of $C = -1/2$ in (11.43) brackets both roots for large K; the frequency tail does not change these opposite signs. Therefore $r = K - a$ is bounded at the roots, and the uniform expansion applies.

Solve $2E_K(a) = 1$ from (11.43) to obtain

$$r = \frac{1}{2} - \frac{3}{8a} + \frac{1}{a^2} - \frac{425}{128a^3} + O(a^{-4}).$$

For example, putting $r = 1/2 + q_1/a + q_2/a^2 + q_3/a^3$ gives successively $q_1 = -3/8$, $q_2 = 1$, $q_3 = -425/128$. Since $\partial_a[2E_K(a)]$, with K fixed, is $1/K + O(K^{-2})$ near the root (see (11.56)), a residual of order a^{-j-1} locates the root to order a^{-j}. Thus the formal inversion has the asserted error control. Reexpressing it in $b = K - 1/2$ yields (11.53). Finally, the perturbing tail in (11.52) is $O(2^{-K})$, while its derivative is of order $1/K$; it changes the root by $O(K2^{-K})$. This proves the last assertion. □

To leading order, choose the integer nearest to $\pi x - 1/4$. The exact prescription remains (11.48); no rounding formula with an unspecified remainder resolves every near-tie case.

11.14 An arithmetic displacement beyond all algebraic orders

The comparison in this section is between two exactly defined transition points. Subtracting a finite approximation such as $b + 3/(8b) - 1/b^2$ would leave an algebraic error much larger than the effect we want to measure.

Theorem 11.28 (First arithmetic shift)**.** *For every $K = 2N + 2 \geq 2$, the full transition satisfies $0 < \beta_K < \alpha_K$. As $K \to \infty$,*

$$\boxed{\beta_K - \alpha_K = -\frac{9}{10} K 2^{-K} \left(1 + O(K^{-1}) + O((2/3)^K)\right).} \tag{11.54}$$

Equivalently, with $x_N^{(1)} = \alpha_K/(2\pi)$, $x_N - x_N^{(1)}$ has leading term $-9K2^{-K}/(20\pi)$.

Proof. The function $E_K(c)$ is strictly increasing in c. At $a = \alpha_K$, the $m = 1$ term of (11.52) vanishes, while every $m \geq 2$ term is positive. Hence $\Phi_K(\alpha_K) > 0$. Since Φ_K is strictly increasing, its root lies below α_K.

For the coefficient, $\alpha_K = K + O(1)$ and gamma concentration with its second-moment bound gives

$$E_K(2\alpha_K) = \frac{4}{5} + O(K^{-1}).$$

Also $w_2 = (3/2)2^{-K}$ and $\sum_{m \geq 3} w_m = O(3^{-K})$. Therefore

$$\Phi_K(\alpha_K) = \frac{9}{20} 2^{-K} \left(1 + O(K^{-1}) + O((2/3)^K)\right). \tag{11.55}$$

Differentiation under the expectation gives

$$E_K'(a) = \frac{1}{a} \mathbb{E}h(T/a), \qquad h(v) = \frac{2v^2}{(1+v^2)^2}.$$

On a bounded neighborhood of K, $h(1) = 1/2$ and a second-moment expansion yield $E_K'(a) = (2K)^{-1}(1 + O(K^{-1}))$. For any m, the derivative of $E_K(ma)$ with respect to a is $a^{-1}\mathbb{E}h(T/(ma))$, and $0 < h \leq 1/2$. Thus

$$\Phi_K'(a) = \frac{1}{2K}\big(1 + O(K^{-1}) + O(2^{-K})\big) \tag{11.56}$$

uniformly in that neighborhood. In particular it has a positive lower bound of order K^{-1}. The mean value theorem first gives $\beta_K - \alpha_K = O(K2^{-K})$ and then, using (11.55)–(11.56), gives (11.54). □

The constant has a transparent decomposition:

$$\frac{9}{10} = \underbrace{\frac{3}{2}}_{\text{divisor weight at } m=2} \underbrace{\left(\frac{4}{5} - \frac{1}{2}\right)}_{\text{kernel gap}} \Big/ \underbrace{\frac{1}{2}}_{\text{scaled derivative}}.$$

It records the second Lambert frequency, rather than a correction to the Stirling expansion.

Remark 11.29 (A transferable positive-mixture calculation). The same argument applies to a positive exponential mixture with frequencies $1, \rho, \ldots$, $\rho > 1$, and coefficients $c_1, c_\rho > 0$. If its higher-frequency Dirichlet tail is $o(\rho^{-K})$, the corresponding normalized transition displacement is

$$-\frac{c_\rho}{c_1}\frac{\rho^2 - 1}{\rho^2 + 1} K \rho^{-K}(1 + o(1)).$$

This follows by replacing the kernel limit $4/5$ by $\rho^2/(1+\rho^2)$ in the proof. It is a useful extension of the calculation, rather than a claim to a new general method of exponentially improved asymptotics; see [49, §2.11] for the broader established theory.

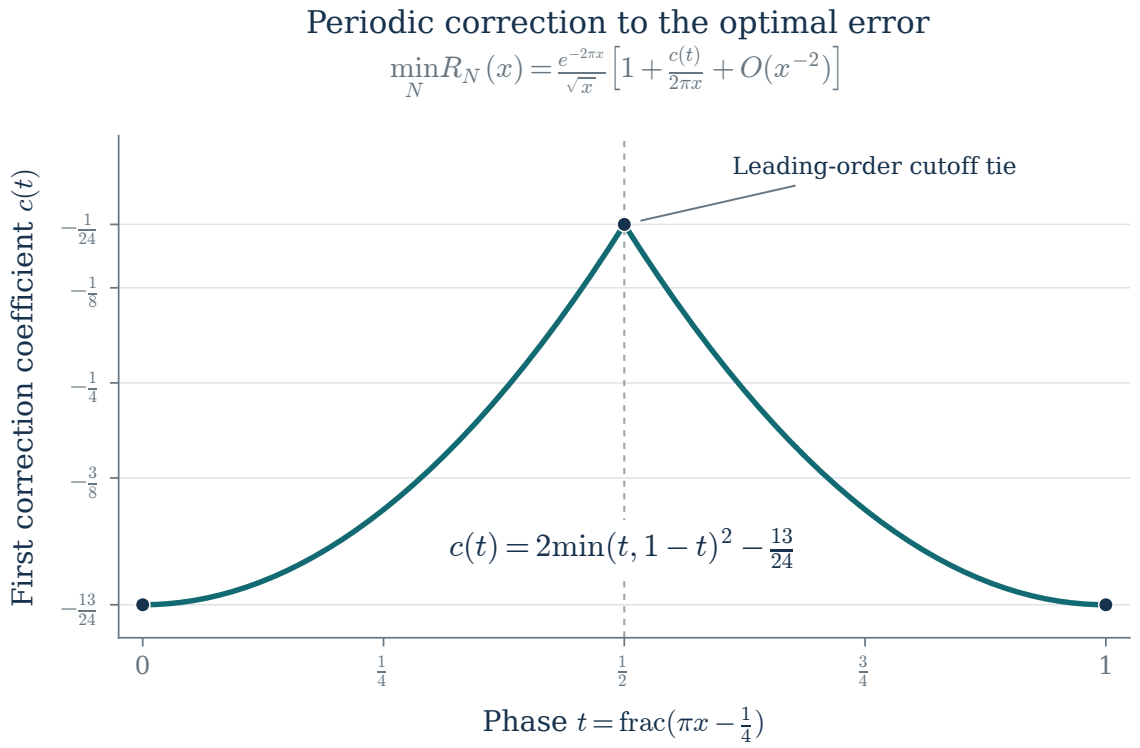


Figure 11.2: The explicit first correction in (11.57), as a function of the fractional part of $\pi x - 1/4$. This plots a proved coefficient function, not fitted numerical data. The two endpoint phases represent the same point of the unit circle.

11.14.1 The periodic correction to the best attainable error

Corollary 11.30 (Integer-cutoff oscillation)**.** *Uniformly for real $x \to +\infty$,*

$$\min_{N\geq 0} R_N(x) = \frac{e^{-2\pi x}}{\sqrt{x}}\left[1 + \frac{2\,\mathrm{dist}(\pi x - \frac{1}{4}, \mathbb{Z})^2 - \frac{13}{24}}{2\pi x} + O(x^{-2})\right]. \tag{11.57}$$

The numerator of the first correction ranges from $-13/24$ to $-1/24$.

Proof. Complete the square in (11.46): $C_1(\delta) = 2(\delta + 1/4)^2 - 13/24$. The exact transition expansion shows that the optimizing index differs from a nearest integer to $\pi x - 1/4$ only within $O(x^{-1})$ of a tie phase. There the squared distances for the two competing integers differ by $O(x^{-1})$; after multiplication by $(2\pi x)^{-1}$ this contributes only $O(x^{-2})$. The bounded-δ uniformity of Theorem 11.25 completes the proof, including arbitrarily close approaches to transition points. □

11.15 Rational Herglotz derivatives and an exponentially small oscillation

Let

$$F(x) = \sum_{n\geq 1} \frac{\psi(nx) - \log(nx)}{n}, \qquad D_r(x) = \frac{(-1)^{r+1}x^{r+1}}{r!}F^{(r)}(x) \quad (r \geq 1, x > 0).$$

The all-orders assertion that rational jets reduce to depth-one cyclotomic polylogarithms is already stated by Radchenko–Zagier [35], immediately after Proposition 4 of their preprint, p. 9. It must not be presented as a new closure theorem. The explicit Bernoulli coefficient formula below gives a constructive version; the large-order estimate and its effective remainder bound are derived independently here. The limit considered below is derivative order tending to infinity at a fixed positive argument.

11.15.1 An explicit finite rational-jet formula

Write $B_j(X)$ for the Bernoulli polynomials with $B_1(X) = X - 1/2$. For positive integers p, q (coprimality is unnecessary for this formula), put

$$c_{a,b} = \frac{a}{q} + \frac{b}{p}, \qquad d_{r,j}(a,b) = \binom{r}{j} B_{r-j}(1 - b/p) - \mathbf{1}_{j=0} B_r(a/q),$$

where $1 \le a \le q$, $0 \le b < p$, and $0 \le j < r$.

Theorem 11.31 (Finite Bernoulli–Hurwitz evaluator)**.** *For $r \ge 1$,*

$$F^{(r)}(p/q) = \frac{(-1)^{r+1}(r-1)!\,q^{r-1}}{p^{r+1}} \left\{ \sum_{a=1}^{q} \sum_{b=0}^{p-1} \left[-\psi(c_{a,b}) + \sum_{j=0}^{r-1} d_{r,j}(a,b)\,\zeta(r+1-j, c_{a,b}) \right] + pq \left(\frac{1}{r} - \gamma - \log q \right) \right\}.$$

In particular every Hurwitz zeta value in the formula has integer order between 2 and $r+1$; no spectral derivatives occur.

Proof. Use the classical integral representation (Radchenko–Zagier, eq. (6))

$$F^{(r)}(x) = (-1)^{r+1} \int_0^\infty t^r \operatorname{Li}_{1-r}(e^{-xt}) \left(\frac{1}{1-e^{-t}} - \frac{1}{t} \right) dt.$$

After $t = qu$ introduce, initially for $\Re s > 0$,

$$I(s) = \int_0^\infty u^{r+s} \operatorname{Li}_{1-r}(e^{-pu}) \left(\frac{1}{1-e^{-qu}} - \frac{1}{qu} \right) du.$$

The combined integral is holomorphic on $\Re s > -1$ since its integrand is $O(u^{\Re s})$ at zero. In $\Re s > 0$ the two terms can be integrated separately, giving

$$I(s) = \Gamma(r+1+s) \sum_{n \ge 1, m \ge 0} \frac{n^{r-1}}{(pn+qm)^{r+1+s}} - \frac{\Gamma(r+s)}{qp^{r+s}} \zeta(1+s).$$

Set $n = qk + a$, $m = p\ell + b$, and $h = k + \ell$. Faulhaber's identity gives

$$\sum_{k=0}^{h} (qk+a)^{r-1} = \frac{q^{r-1}}{r} \left[B_r(h+1+a/q) - B_r(a/q) \right].$$

Writing $v = h + c_{a,b}$ expands the bracket as $\sum_{j=0}^{r} d_{r,j}(a,b) v^j$, with $d_{r,r} = 1$. Consequently

$$\sum_{n \ge 1, m \ge 0} \frac{n^{r-1}}{(pn+qm)^{r+1+s}} = \frac{q^{r-1}}{r(pq)^{r+1+s}} \sum_{a,b} \sum_{j=0}^{r} d_{r,j}(a,b)\,\zeta(r+1+s-j, c_{a,b}).$$

The $j = r$ terms have Laurent expansion $\zeta(1+s, c) = s^{-1} - \psi(c) + O(s)$. Their combined residue cancels that of the second term of $I(s)$. The finite part of the two gamma factors contributes $pq[1/r - \gamma - \log q]$ after factoring out $(r-1)!/(p^{r+1}q^2)$. Finally $F^{(r)}(p/q) = (-1)^{r+1}q^{r+1}I(0)$. All rearrangements took place in an absolutely convergent half-plane; holomorphy of the combined integral justifies taking the finite part. □

Cyclotomic interpretation. For coprime p, q, the rational function

$$R_{r,p,q}(z) = \frac{\mathrm{Li}_{1-r}(z^p)}{1 - z^q}$$

has poles only at pth and qth roots of unity. Its coefficient sequence has a unique finite decomposition

$$[z^N]R_{r,p,q}(z) = \sum_{\omega^p=1 \text{ or } \omega^q=1} \sum_{j=0}^{r} a_{\omega,j} N^j \omega^N \quad (N \geq 1).$$

The coefficients lie in $\mathbb{Q}(\mu_{pq})$, $a_{1,r} = 1/(rp^r q)$, and $a_{\omega,r} = 0$ for $\omega \neq 1$. Partial fractions therefore give the equivalent formula

$$F^{(r)}(p/q) = (-1)^{r+1}(r-1)!(q/p)^r \left(\frac{1}{r} + \log p\right) + (-1)^{r+1} r! q^{r+1} \sum_{\omega} \sum_{j=0}^{r-1} a_{\omega,j} \,\mathrm{Li}_{r+1-j}(\omega).$$

This proves the already known closure statement and gives all its coefficients. The special $a_{1,r}\zeta(1+s)$ term is the only pole: its finite-part contribution is exactly the displayed elementary term.

For example,

$$F''(1) = -\frac{1}{2} - \zeta(2), \qquad F'''(1) = \frac{2}{3} + 3\zeta(2) + \zeta(3),$$
$$F''(1/2) = -2 - 8\zeta(2) - \frac{7}{2}\zeta(3), \qquad F'''(1/2) = \frac{16}{3} + 48\zeta(2) + 32\zeta(3).$$

At $p = q = 1$ the complete formula reduces to

$$F^{(r)}(1) = (-1)^{r+1}(r-1)! \left[\frac{1}{r} + \sum_{j=1}^{r-1} \binom{r}{j} B_{r-j}(1)\zeta(r+1-j)\right].$$

For high r this exact expression can suffer substantial floating-point cancellation; exact coefficients do not make low-precision evaluation reliable.

11.15.2 The complete algebraic large-order expansion

Define

$$E_r(x) = D_r(x) - \zeta(2) - \frac{x}{r}(\log x - H_{r-1}).$$

The following result concerns $r \to \infty$ with x fixed, not the large-x Bernoulli expansion of F.

Theorem 11.32 (Large derivative order with effective remainder)**.** *For $x > 0$, integers $r \geq 2$, real $1 < A < r$, and $\pi/4 < \theta < \pi/2$,*

$$|E_r(x)| \leq \frac{2x\zeta(A)\zeta(A+1)}{\pi(2\theta - \pi/2)\cos\theta} \left(\frac{x}{2\pi\cos^2\theta}\right)^A \frac{\Gamma(r-A)\Gamma(A)\Gamma(A+1)}{\Gamma(r+1)}.$$

Consequently, uniformly for x in each compact subinterval of $(0,\infty)$,

$$E_r(x) = O\left(r^{-1/2}\exp[-2\sqrt{\pi r/x}]\right).$$

In particular, the exact harmonic-number expression $\zeta(2) + (x/r)(\log x - H_{r-1})$ contains every algebraic order in $1/r$.

Proof. Binet's digamma integral and Mellin inversion give, for $1 < c < 2$,

$$F(x) + \frac{\zeta(2)}{2x} = -\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{\pi\Gamma(s)\zeta(s)\zeta(s+1)}{(2\pi x)^s \sin(\pi s/2)}\,ds.$$

One derivation is to Mellin-transform $\psi(y) - \log y + 1/(2y)$ in $1 < \Re s < 2$ using $\int_0^\infty v^{s-1}/(1+v^2)\,dv = \pi/[2\sin(\pi s/2)]$; the sum over n supplies $\zeta(s+1)$. Absolute convergence of Binet's integral and exponential decay on vertical lines justify inversion and fixed-order differentiation. After r differentiations,

$$D_r(x) = \frac{\zeta(2)}{2} + \frac{1}{2\pi i}\int_{(c)} \frac{\pi\Gamma(r+s)\zeta(s)\zeta(s+1)x^{1-s}}{\Gamma(r+1)(2\pi)^s \sin(\pi s/2)}\,ds.$$

Move the contour to $\Re s = -A$. The only crossed poles are $s = 1$ and $s = 0$. At the negative even integers the zero of $\zeta(s)$ cancels the sine denominator; gamma poles lie to the left since $A < r$. The residue at 1 is $\zeta(2)/2$. Using $\zeta'(0) = -\frac{1}{2}\log(2\pi)$ and $\psi(r) = H_{r-1} - \gamma$, the residue at 0 is $x(\log x - H_{r-1})/r$. Thus the integral on the new line equals $E_r(x)$.

Apply the zeta functional equation twice to its kernel:

$$\frac{\pi\zeta(s)\zeta(s+1)}{(2\pi)^s\sin(\pi s/2)} = 2(2\pi)^s\cos(\pi s/2)\Gamma(1-s)\Gamma(-s)\zeta(1-s)\zeta(-s).$$

On $s = -A + it$, use $|\zeta(1-s)\zeta(-s)| \le \zeta(A+1)\zeta(A)$ and $|\Gamma(r-A+it)| \le \Gamma(r-A)$. Rotating the Euler gamma integral through angle $\theta\,\mathrm{sgn}(t)$ gives the elementary inequality

$$|\Gamma(a+it)| \le \Gamma(a)(\cos\theta)^{-a}e^{-\theta|t|} \qquad (a > 0, 0 < \theta < \pi/2).$$

Use it for the other two gamma factors, with parameters A and $A+1$, and use $|\cos(\pi s/2)| \le e^{\pi|t|/2}$. The remaining integral is $\int_{\mathbb{R}} e^{-(2\theta-\pi/2)|t|}dt = 2/(2\theta - \pi/2)$, yielding the stated effective bound. The same estimates justify the horizontal-contour limits.

For the stretched-exponential estimate take an integer $m = \sqrt{\pi r/x} + O(1)$ and $\theta = \pi/4 + 1/(2m)$. For r sufficiently large, $1 < m < r$ and the bound applies. Uniformly on compact positive x-intervals,

$$\frac{\Gamma(r-m)}{\Gamma(r+1)} = r^{-m-1}\exp\left[\frac{m(m+1)}{2r} + O(r^{-1/2})\right], \qquad \Gamma(m)\Gamma(m+1) = 2\pi m^{2m}e^{-2m}(1+O(m^{-1})).$$

Also $(2\pi\cos^2\theta)^{-m} = \pi^{-m}e^{1+O(m^{-1})}$, and the prefactor is $O(xm)$. The power and exponential factors reduce to $O(r^{-1/2}e^{-2\sqrt{\pi r/x}})$, as required. □

An elementary factorial form. At $\theta = \pi/3$ and integer $2 \le m < r$ the estimate becomes

$$|E_r(x)| \le \frac{24x}{\pi^2}\zeta(m)\zeta(m+1)\left(\frac{2x}{\pi}\right)^m \frac{(r-m-1)!(m-1)!m!}{r!}.$$

It is useful as a directly computable error certificate even without asymptotic optimization.

Positivity and a convergent Taylor certificate. The derivative integral has a positive kernel with $1/2 < 1/(1-e^{-t}) - 1/t < 1$, so

$$\frac{\zeta(2)}{2} < D_r(x) < \zeta(2).$$

Therefore the Taylor expansion of $F(x+h)$ about $x > 0$ has, for $\rho = |h|/x < 1$, the explicit remainder bound

$$\left| F(x+h) - \sum_{j=0}^{N} \frac{F^{(j)}(x)}{j!} h^j \right| \le \frac{\zeta(2)}{x} \frac{\rho^{N+1}}{1-\rho}.$$

Its radius is exactly x, since $D_r(x) \to \zeta(2)$.

11.15.3 Exact arithmetic sectors and the leading oscillation

The divisor-weighted eta/Binet representation in Radchenko–Zagier [35, Section 7.3] is classical context for this calculation. The following identity is a differentiated and reciprocity-transformed form useful for large derivative order. The quantitative sum estimates are needed in addition to a one-sector expansion.

Write $\sigma_{-1}(n) = \sum_{d|n} d^{-1}$. Let U denote Tricomi's confluent hypergeometric function with its principal branch. Its integral representation is DLMF 13.4.4 [82]. The large-first-parameter asymptotics used below are classical (DLMF 13.8.11–16; Temme [83]); a direct saddle calculation is included to fix the branch and the first correction in this application. The arithmetic sum over all sectors and its aggregate error bound must be justified in addition to that classical one-sector asymptotic.

Theorem 11.33 (Exact divisor-weighted expansion)**.** *For every integer $r \ge 2$ and $x > 0$,*

$$\begin{aligned} E_r(x) &= 2x\Gamma(r) \sum_{n\ge1} \sigma_{-1}(n) \operatorname{Re} U\left(r, 0, \frac{2\pi i n}{x}\right) \\ &= 2x \sum_{n\ge1} \sigma_{-1}(n) \int_0^\infty \frac{u^{r-1}}{(1+u)^{r+1}} \cos\left(\frac{2\pi n u}{x}\right) du. \end{aligned}$$

The series of the integrals is absolutely convergent. The assertion does not require absolute convergence after moving the infinite sum inside the oscillatory u-integral.

More precisely, for $1 < A < r$ and $\pi/4 < \theta < \pi/2$, the contribution of an arbitrary subset $\mathcal{N} \subset \mathbb{N}$ is bounded by the previous effective estimate with $\zeta(A)\zeta(A+1)$ replaced by $\sum_{n\in\mathcal{N}} \sigma_{-1}(n) n^{-A}$.

Proof. Set $w = -s$ in the post-functional-equation contour representation:

$$E_r(x) = \frac{2x}{2\pi i \Gamma(r+1)} \int_{(A)} \Gamma(r-w)\Gamma(w)\Gamma(w+1)\zeta(w)\zeta(w+1)\cos(\pi w/2) \left(\frac{x}{2\pi}\right)^w dw.$$

The Dirichlet series $\zeta(w)\zeta(w+1) = \sum_{n\ge1} \sigma_{-1}(n) n^{-w}$ converges absolutely on this line. The effective gamma bound proves absolute convergence of the resulting series of contour integrals.

For $\Re z > 0$ put

$$M_r(z) = \int_0^\infty e^{-zu} \frac{u^{r-1}}{(1+u)^{r+1}} du = \Gamma(r) U(r, 0, z).$$

For $0 < \Re w < r$, taking its Mellin transform in z gives

$$\int_0^\infty z^{w-1} M_r(z)\, dz = \frac{\Gamma(w)\Gamma(r-w)\Gamma(w+1)}{\Gamma(r+1)}.$$

Mellin inversion is valid in that strip. Both the defining integral and the Barnes integral extend continuously to $z = \pm i\tau$, $\tau > 0$: the first by dominated convergence, since its rational weight is integrable, and the second by its exponential vertical decay. Averaging the two boundary values replaces z^{-w} by $\tau^{-w}\cos(\pi w/2)$. Substitution proves the formula and the termwise bounds. □

A finite-sum tail bound. For an integer cutoff $N \geq 1$ and $A > 1$,

$$\sum_{n>N} \sigma_{-1}(n) n^{-A} \leq N^{1-A}\left[\frac{1+\log N}{A-1} + \frac{1}{(A-1)^2}\right],$$

because $\sigma_{-1}(n) \leq H_n \leq 1 + \log n$ and $(1+\log t)t^{-A}$ is decreasing for $t \geq 1$. Thus the exact series comes with a computable truncation bound.

An alternative bound avoids the auxiliary contour parameters. Rotate the defining u-integral for $M_r(i\tau)$ to the negative imaginary ray. There are no poles in that quadrant; the small and large arcs vanish. Then

$$|M_r(i\tau)| \leq \int_0^\infty e^{-\tau y} \frac{y^{r-1}}{|1-iy|^{r+1}}\, \mathrm{d}y \leq \Gamma(r)\tau^{-r}.$$

It follows that the arithmetic tail after N terms is at most

$$2x\Gamma(r)\left(\frac{x}{2\pi}\right)^r N^{1-r}\left(\frac{1+\log N}{r-1} + \frac{1}{(r-1)^2}\right), \qquad r \geq 2. \tag{11.58}$$

This is often the simplest finite-sum certificate; the earlier gamma-factor bound is more useful for the sharp large-r exponent.

Theorem 11.34 (Two-term oscillatory asymptotic)**.** *For fixed $x > 0$ define*

$$\begin{aligned}
\mathcal{A}_r(x) &= 2^{5/4}\pi^{3/4}x^{3/4}r^{-3/4}e^{-2\sqrt{\pi r/x}},\\
\Phi_r(x) &= 2\sqrt{\pi r/x} - \frac{\pi}{x} - \frac{\pi}{8},\\
\kappa(x) &= \frac{3}{16}\sqrt{\frac{x}{2\pi}} + \frac{1}{12}\left(\frac{2\pi}{x}\right)^{3/2}.
\end{aligned}$$

Then, uniformly for x in compact subintervals of $(0,\infty)$,

$$E_r(x) = \mathcal{A}_r(x)\left[\cos\Phi_r(x) + \frac{\kappa(x)}{\sqrt{r}}\cos\left(\Phi_r(x) + \frac{\pi}{4}\right) + O(r^{-1})\right].$$

This is an additive oscillatory expansion; a relative-error claim after division by $\cos\Phi_r(x)$ would be invalid near its zeros.

Proof. We first prove, uniformly for $z = i\tau$ with τ in any compact positive interval,

$$M_r(z) = \sqrt{\pi}\, z^{1/4} r^{-3/4} e^{z/2 - 2\sqrt{rz}}\left[1 + \frac{1}{\sqrt{r}}\left(\frac{3}{16\sqrt{z}} - \frac{z^{3/2}}{12}\right) + O(r^{-1})\right]. \tag{11.59}$$

All powers use their principal values.

Rotate the u-contour in the integral defining $M_r(i\tau)$ to argument $-\pi/4$. There are no poles in that quadrant. On the large arc the rational factor is $O(R^{-2})$, while $|e^{-i\tau u}| \le 1$; a small arc of radius ϵ contributes $O(\epsilon^r)$. Hence the rotation is justified directly. Put

$$u = \sqrt{r/z}\, v, \qquad \lambda = \sqrt{rz}, \qquad \epsilon = \sqrt{z/r}.$$

This gives the exact representation

$$M_r(z) = \sqrt{z/r} \int_0^\infty v^{-2} \exp\left[-\lambda v - (r+1)\log(1+\epsilon/v)\right] dv.$$

On any fixed compact positive v-interval,

$$\begin{aligned} -\lambda v - (r+1)\log(1+\epsilon/v) = & -\lambda(v+v^{-1}) + \frac{z}{2v^2} \\ & + r^{-1/2}\left(-\frac{\sqrt{z}}{v} - \frac{z^{3/2}}{3v^3}\right) + O(r^{-1}), \end{aligned}$$

with the same estimates after any fixed number of v-derivatives.

For completeness, localization at $v = 1$ is uniform. Write $a = \sqrt{\tau/(2r)}$, so $\epsilon = a(1+i)$ and $\Re\lambda = ra$. On a compact v-interval the real part of the exponent is $-ra(v+v^{-1}) + O(1)$, and $v + v^{-1}$ has its unique minimum 2 at 1. For the small tail $0 < v < \delta < 1/2$, use $|1+\epsilon/v| \ge 1 + a/v$ and substitute $y = a/v$:

$$\int_0^\delta v^{-2}|1+\epsilon/v|^{-r-1} dv \le \frac{1}{ar}(1+a/\delta)^{-r} = O\left(r^{-1/2} e^{-ra/\delta + O(1)}\right).$$

This is exponentially smaller than e^{-2ra}. For $v > M > 2$, the factor $e^{-\lambda v}$ gives the analogous bound $O(e^{-Mra})$. The remaining compact interval outside any fixed neighborhood of 1 also has a strict exponential gap.

Apply complex Laplace expansion at $v = 1$ to $\phi(v) = v + v^{-1}$ and $g(v) = v^{-2} e^{z/(2v^2)}$. It is legitimate since $\arg\lambda = \pi/4$ lies strictly inside the right half-plane. The Gaussian integral along the real v-axis gives $e^{-2\lambda}\sqrt{\pi/\lambda}$. The coefficient of $1/\lambda$, divided by $g(1)$, is

$$\frac{g''(1)}{4g(1)} + \frac{3g'(1)}{4g(1)} + \frac{3}{16} = z + \frac{z^2}{4} + \frac{3}{16}.$$

Adding the explicit $r^{-1/2}$ correction in the exponent at $v = 1$ gives

$$-\sqrt{z} - \frac{z^{3/2}}{3} + \frac{1}{\sqrt{z}}\left(z + \frac{z^2}{4} + \frac{3}{16}\right) = \frac{3}{16\sqrt{z}} - \frac{z^{3/2}}{12}.$$

This proves (11.59); further terms follow by the same finite Taylor and Gaussian-moment calculation, with standard local remainder estimates. As an independent classical check, in DLMF 13.8.11 with $b = 0$ one has $p_0(z) = 1$ and $q_0(z) = -z/12$; the standard large-argument expansions of K_1 and K_0 give exactly the same first correction.

At $z = 2\pi i/x$, taking $2x$ times the real part of (11.59) gives the stated main term and correction. Indeed $z^{1/4}$ has argument $\pi/8$, and the correction equals $\kappa(x)e^{-\pi i/4}$.

It remains to control all the other arithmetic sectors uniformly. For $A \ge 2$,

$$\zeta(A)\zeta(A+1) - 1 = \sum_{n\ge 2} \sigma_{-1}(n) n^{-A} \le 4\big(\zeta(2)\zeta(3) - 1\big)2^{-A}.$$

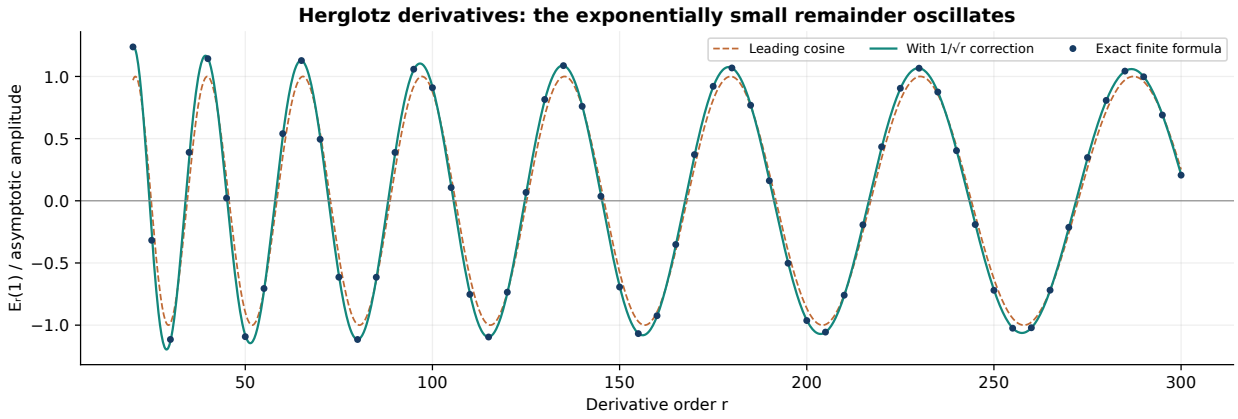


Figure 11.3: The Herglotz remainder divided by its asymptotic amplitude at $x = 1$. The leading cosine and its first correction track the exact finite-formula values. The asymptotic is additive, so no division by a small cosine is used.

Use the effective sector bound with $A = \sqrt{2\pi r/x} + O(1)$ an integer and $\theta = \pi/4 + 1/(2A)$. Exactly the previous optimization yields

$$\sum_{n\geq 2} 2x\sigma_{-1}(n)\,\mathrm{Re}\,M_r(2\pi i n/x) = O\left(r^{-1/2}e^{-2\sqrt{2\pi r/x}}\right).$$

This is absorbed into $\mathcal{A}_r(x)O(r^{-1})$, uniformly on compact positive x-intervals. □

Corollary 11.35 (Sharp exponential rate and infinitely many signs)**.** *For each fixed $x > 0$ the real sequence $E_r(x)$ takes positive and negative values infinitely often. More precisely,*

$$\limsup_{r\to\infty} \frac{E_r(x)}{\mathcal{A}_r(x)} = 1, \qquad \liminf_{r\to\infty} \frac{E_r(x)}{\mathcal{A}_r(x)} = -1,$$

and

$$\limsup_{r\to\infty} \frac{\log|E_r(x)|}{\sqrt{r}} = -2\sqrt{\pi/x}.$$

Here $\log 0 = -\infty$ if an exact zero occurs.

Proof. The phase $\Phi_r(x)$ increases to infinity, with successive differences tending to zero. Consequently it has subsequences approaching any fixed phase modulo 2π. Apply the additive expansion and choose phases 0 and π. The same subsequences establish the lower bound for the limsup of the logarithmic rate; the asymptotic envelope gives the upper bound. □

Further research specific to this section. Compute and certify several further half-integer corrections in (11.59); build an interval implementation of the exact sector sum; locate the sign-transition indices beyond the leading $\sqrt{r}$ phase law; and obtain a uniform transition theory when x also varies with r. The fixed-x theorem above does not justify such a double-scaling limit automatically.

11.16 Quadratic units and Kronecker-limit coordinates

Define the normalized functions

$$\mathcal{F}(x) = F(x) - F(1) + \frac{\pi^2}{12}(x - 2 + x^{-1}) - \frac{1}{4}\log^2 x, \quad \mathcal{J}(x) = J(x) - \frac{1}{2}\log^2 2 + \frac{\pi^2}{24}(x - x^{-1}).$$

For a real quadratic order of discriminant D, a narrow ideal class $\mathcal{B}$ has a partial zeta function. Its Kronecker constant is

$$\varrho(\mathcal{B}) = \lim_{s\to 1}\left[D^{s/2}\zeta(\mathcal{B}, s) - \frac{\log \varepsilon_D}{s-1}\right],$$

where $\varepsilon_D > 1$ is the least norm-one unit. In the reduced-form cycle representation, $w_j > 1 > w_j' > 0$, Radchenko–Zagier give

$$\varrho(\mathcal{B}) = 2\gamma \log \varepsilon_D - \ell(\mathcal{B})\frac{\pi^2}{6} + \sum_{j \bmod \ell}\left[\mathcal{F}(w_j) - \mathcal{F}(w_j') + L(w_j'/w_j)\right], \tag{11.60}$$

where L is the unnormalized Rogers dilogarithm. Class characters turn the ϱ coordinates into Hecke L-values. Their Theorem 5 expresses $\mathcal{J}$ at specified even-trace units in these coordinates. The source's independent cycle engine reported checks at $n = 2, 4, 6, 8$ with residuals about 5×10^{-38}; the engine was not imported into this checkout.

11.17 Regulator candidates and field degrees

The published Stark examples and the source's follow-up fits are conjectural or numerical evaluations. A cubic polynomial over $\mathbb{Q}$ defines a degree-three field E. It cannot itself be a Hilbert or ring class field containing a quadratic field K. In the examples at issue, its degree-six splitting field can be a cyclic cubic extension of K; E is a cubic subfield. The distinction corrects a repeated field-degree error in the report.

For $u = 16 + \sqrt{257}$ the source reports

$$\mathcal{J}(u) \overset{\text{num}}{=} 4\zeta(2) + \log 2 \log u + 2\,\mathrm{reg}(E), \quad E = \mathbb{Q}[x]/(x^3 - 2x^2 - 3x + 1). \tag{11.61}$$

The determinant of the two selected unit-logarithm columns is signed; with the source's ordering it is $-\mathrm{reg}(E)$. Vertical bars around a matrix denote its determinant, not an absolute value. Replacing this negative determinant by its absolute value without changing its external coefficient reverses the regulator term.

The other published example concerns $u = 10 + 3\sqrt{11}$ and the order of discriminant 396. Its unit-logarithm expression is a Stark minor rather than a full field regulator. We retain that structural distinction; the unimported algebraic-unit definitions do not provide a locally replayable formula for the minor.

The source reports that $n = 6, 10, 14, 16, 26$ are the narrow-class-number-three cases in its scan of $2 \le n \le 5000$. This is a bounded census, not a proof that the family is finite. Negative fits at class numbers five and seven likewise do not establish a universal "single regulator if and only if cyclic cubic" theorem.

There is a precise conditional route to a leading-L interpretation. Suppose a non-Galois totally real cubic field E has quadratic resolvent K, and its normal closure supplies a nontrivial order-three class character χ of K. Artin induction gives $\zeta_E(s) = \zeta(s)L_K(s, \chi)$. The analytic class-number formula then gives

$$L_K(s, \chi) = h_E\,\mathrm{reg}(E)s^2 + O(s^3),$$

n	D_0	cubic field polynomial	c_1	c_2	c_3
6	37	$x^3 - x^2 - 3x + 1$	3/2	2/3	2/3
10	101	$x^3 - x^2 - 5x - 1$	5/2	2/3	2/3
14	197	$x^3 - x^2 - 7x - 3$	7/2	2/3	2/3
16	257	$x^3 - x^2 - 4x + 3$	4	1	2
26	677	$x^3 - x^2 - 11x - 7$	13/2	2/3	2/3

Table 11.1: Historical numerical candidates $\mathcal{J}(n + \sqrt{n^2+1}) = c_1\zeta(2) + c_2 \log 2 \log u + c_3 \operatorname{reg}(E)$. The source reports residuals below 1.2×10^{-109}. These are not unconditional theorems.

because $\zeta_E(s) = -h_E \operatorname{reg}(E) s^2/2 + O(s^3)$ and $\zeta(0) = -1/2$. Thus the coefficient equals $\operatorname{reg}(E)$ only after $h_E = 1$ and the field-character identification have been established. This argument does not convert the numerical Herglotz fits into proofs of Stark's conjecture.

11.18 Exact class-number-one certificates

Proposition 11.36 (Class-number certificates for the five cubic fields)**.** *Each of the five totally real cubic fields in Table 11.2 has class number one. In the table, α denotes a root of the polynomial in its row, d_H is the field discriminant, and $B = \lfloor 2\sqrt{d_H}/9 \rfloor$ is the integer Minkowski bound. The final column lists a generator of a principal ideal of norm p for each indicated rational prime p; the field norm of the generator may be $-p$.*

Polynomial for α	d_H	B	p : generator
$X^3 - X^2 - 3X + 1$	148	2	$2 : \alpha - 1$
$X^3 - X^2 - 5X - 1$	404	4	$2 : \alpha + 1$; $3 : \alpha^2 - 2\alpha - 2$
$X^3 - X^2 - 7X - 3$	788	6	$2 : \alpha + 1$; $3 : \alpha$; $5 : \alpha^2 - 2\alpha - 4$
$X^3 - X^2 - 4X + 3$	257	3	$3 : \alpha$
$X^3 - X^2 - 11X - 7$	2708	11	$2 : \alpha + 1$; $3 : \alpha + 2$; $7 : \alpha$
			$5 : -\alpha^2 + 2\alpha + 10$; $11 : 2\alpha^2 - 4\alpha - 17$

Table 11.2: Exact generators used in the class-number-one proof.

Proof. The rational-root test proves irreducibility of each polynomial f. Their polynomial discriminants are, in the displayed order, $148, 404, 788, 257, 2708$, all positive. Thus all five fields are totally real. At discriminant 257, squarefreeness immediately gives $\mathcal{O}_{H_0} = \mathbb{Z}[\alpha]$. In each other row only 2 could divide the index. In all four cases $\overline{f} = (X+1)^3$ modulo 2, while

$$\left.\frac{f(X) - (X+1)^3}{2}\right|_{X=1} \in \{-5, -7, -9, -13\}.$$

These values are odd. Dedekind's index criterion therefore excludes 2 from the index. Equivalently, each shifted polynomial $f(X+1)$ is Eisenstein at 2, so its root generates the local ring of integers [55, Proposition 7.55 and Remark 7.56]. This proves maximality and the asserted field discriminants. The use of Dedekind factorization below is consequently valid also at 2.

Minkowski's theorem gives an integral ideal in every ideal class with norm at most $2\sqrt{d_H}/9$ for a totally real cubic field [55]. It suffices to prove principalness for every prime ideal whose norm is at most B. In the four even-discriminant rows, 2 is totally ramified; the displayed norm-2 element

generates its unique prime ideal. At discriminant 257, the polynomial is irreducible modulo 2, so the prime above 2 has norm $8 > B$.

For each relevant odd rational prime $p \leq B$, direct factorization of f modulo p has type $1 + 2$. There is therefore exactly one prime ideal of norm p. The corresponding element in the table has norm $\pm p$, so this degree-one prime is principal. Its degree-two partner is principal as well, since their product is (p). This last observation covers in particular the prime of norm 9 in the field of discriminant 2708. The table accounts for all rational primes $p \leq B$, hence for all prime ideals within the Minkowski bound. Every ideal class is therefore trivial.

All finite calculations in this proof—discriminants, index tests, modular splitting types, and generator norms—are reproduced with integer arithmetic by `../reports/corpus-corrections/code/12-relation-spaces-verify_cubic_class_numbers.py`. □

For the other polynomial used in the 257 example, $x^3 - 2x^2 - 3x + 1$, the same bound is three: it is irreducible modulo two, and its unique norm-three prime is generated by $\alpha - 1$. Its squarefree discriminant is 257, so its order is maximal. Thus that cubic also has class number one. The leading primitive Artin coefficient consequently specializes to the regulator in these cases. Inflating a primitive character to a ray modulus can remove Euler factors; at the 257 modulus-two example the factor at zero is three. The primitive and ray-modulus Dirichlet series must be identified before comparing their leading coefficients. This still does not turn the fitted Herglotz evaluations into proved Stark identities.

Chapter 12

Experimental discovery and the route to proof

The mathematical progression of the manuscript suggests a practical method. One begins with a value that is easy to compute but hard to recognize, finds an exact transformation that organizes its arguments, and then asks which coordinates remain after known laws are imposed. Numerical experiments are especially effective at proposing those coordinates. Their usefulness increases when the proposed identity is subsequently derived, and when a negative result is recorded with enough detail to explain its limits.

12.1 Constructing a meaningful integer-relation experiment

An experiment consists of a target, a finite ordered basket of constants, their coefficient field, a working precision, a coefficient bound and an acceptance rule. Weight and character parity give strong guidance for the basket. For instance a rational-angle Fourier transform can require $\sqrt{q}$-weighted atoms; searching only with rational coefficients in the unweighted values misses that structure. The $\pi^3\psi^{(-3)}(1/4)$ coordinate in Chapter 9 is another example: omitting the π^3 factor changes the question.

Before searching, remove known basket dependencies. A search containing both $\log\Gamma(1/2)$ and $\frac{1}{2}\log\pi$ may return their tautology instead of a relation involving the target. Reflecting arguments, distributing roots of unity and replacing elementary odd beta values reduce this problem before numerical linear algebra starts.

Working precision must exceed the cancellation loss as well as the digits needed to discriminate the proposed coefficient height. The heuristic $10^{P/d}$, for P available digits and d basket coordinates, estimates a scale at which accidental short numerical relations can appear. It is neither a theorem about a specific basket nor a rigorous exclusion bound. Coefficient sizes, conditioning and the number of searches all matter. The earlier false negative for S_5 illustrates this: a 51-coordinate basket at 150 digits obscured the height-3840 relation that a smaller, properly structured basket exposed.

12.2 Controls and independent recomputation

A planted positive control checks that the search can recover a known relation of comparable scale. A perturbation of one coefficient tests the resolution of the residual check. A negative control and an auxiliary “canary” constant can expose numerical coincidences. These controls are diagnostics.

In particular no independence theorem is known between the Champernowne constant and the periods in these baskets, and a zero canary coefficient cannot certify an identity.

A successful candidate should be frozen as exact rational or algebraic coefficients, then reevaluated at higher precision using an independent representation. For a nested sum that may be an iterated integral; for a CM row sum it may be a modular Fourier series; for Herglotz values it may be a defining integral compared with the finite dilogarithm sum. Agreement between two engines using the same incorrect formula is less informative than agreement between different mathematical representations. Error-controlled interval evaluation can establish a finite residual bound, but even an arbitrarily small nonzero numerical bound alone does not establish equality of general transcendental constants.

12.3 What exact linear algebra proves

An exact relation matrix proves an upper bound on the number of necessary coordinates in its presented quotient. Its row-combination matrix is a replayable finite certificate relative to the input identities. A free column is independent in that formal quotient, not automatically among the evaluated numerical periods. Additional functional equations or regularization can reduce the quotient further.

Likewise a motivic dimension theorem concerns motivic objects. To infer numerical independence one needs a period-injectivity statement, generally conjectural. A dimension for a chosen word-length grading need not be the dimension of the classical depth filtration. The Gaussian triples are a natural test case for this distinction.

Certificates require both exact atom cancellation and sound law instantiation. Substitution strings must be parsed within a restricted grammar; branches and domains must be established mathematically. A numerical branch guard is useful for rejecting errors but leaves a separate domain obligation. These issues motivate a future formal development rather than a current claim of proof-assistant verification.

12.4 Computational conventions

The manuscript uses decreasing nested indices. In the tested Wolfram Language 15 interface,

$$\mathrm{Li}_{a,b}(z_1,z_2) = \texttt{MultiplePolyLog[\{a,b\},\{z1,z2\}]}, \qquad \zeta(a,b) = \texttt{MultipleZeta[\{a,b\}]}.$$

Finite checks such as $\zeta(2,1,1) = \zeta(4)$ and $H(0,1;z) = \mathrm{Li}_2(z)$ calibrate the interface. A divergent call does not resolve ordering ambiguities on its own. The same calibration is needed for signed harmonic indices and generalized-polylogarithm letter order.

Negative-order polygamma means the integral based at zero: $\psi^{(-1)} = \log\Gamma$ and $\psi^{(-n)}(z) = \int_0^z \psi^{(-(n-1))}(t)\,dt$ for $n \geq 2$. Balanced conventions can differ by explicit polynomials. In Hurwitz jets, $\zeta^{(j)}$ always differentiates the spectral variable; the superscript (k) on γ_n differentiates its parameter. The CM hexagonal point $e^{i\pi/3}$ and the Eisenstein cube root $e^{2\pi i/3}$ are different arguments even when a local section denotes each by ρ.

12.5 The surviving research programme

Several concrete proof problems now fit into one framework. The odd-weight S_p directions need a precise choice of comparison algebra and a rigorous reduction or obstruction; the even-weight

direction has the generating-integral proof of Chapter 4. The Gaussian and Eisenstein inverse-color points now have an all-weight partial-fraction reduction, and the distribution ranks have a complete character normal form. A full regularized relation computation remains useful for the other colored points. Numerical independence of character derivative coordinates remains a separate arithmetic problem.

The gamma programme can transport its exact row certificates across the DFT and prove branch validity for each law instance. Its finite-matrix termination should remain separate from Rohrlich's conjecture. The CM programme can replace fitted coefficients by modular equations and class-field computations, while taking account of periods under Galois conjugation. The Herglotz programme needs complete explicit class-character and unit definitions for the regulator candidates before they become locally replayable, followed by the relevant Stark argument.

The cubic log-gamma moment now has the explicit Tornheim representation and certified enclosure of Section 8.5. Reduction of those derivative coordinates into a stated smaller algebra remains a precise question, alongside the opaque individual CM components. The Gaussian weight-five relation, weight-six doubles, weight-four triples and cubic/quartic ladders likewise now have exact proofs. The S_4 relation now has the exact certificate of Theorem 4.8. Higher golden ladder certificates and numerical period independence remain open; the S_6 candidate supplies a concrete next mixed-reduction problem. A richer representation is often more productive than a larger unstructured search. The scientific value of the experiments is that they tell us which representation to seek next.

12.6 Questions sharpened by the exact continuations

12.6.1 Real-order geometry after the positive-difference theorem

The finite signed-moment threshold no longer limits the zero-free theorem. The positive consecutive-difference measure of Theorem 5.24 supplies the missing subcritical representation, and Theorem 5.26 proves nonvanishing away from the double origin for every $a \geq 0, b > 0$ on the principal slit plane. Angular uniqueness holds throughout this domain. On and below $a + b = 1$, and on the entire $a = 0$ axis, the normalized angular radius strictly increases. The full-radius sign conjecture for integer $a \geq 2$ remains open; the fractional turning points show why local coefficient signs alone cannot settle a general global motion law.

The Gaussian experiments have also led to proofs rather than just narrower searches. Fixed-total monotonicity is proved in Theorem 5.31; the unique global Gaussian maximum has the exact stationary-point bracket $1 < b_* < 2$ in Theorem 5.32. Its location and height now have sharp rational enclosures in Theorem 5.36. Corollary 5.33 settles the optimal uniform Euler constant $\pi/4 + \log(2)/2$ for positive orders with $a + b \leq 1$. The contraction argument of Theorem 5.64 now settles the universal real-order Euler constant as the unique Gaussian axis maximum $C(b_*)$. The number of quartic transition points and the Bessel and uniform Lerch continuations remain further proof problems. The universal bound uses the separately proved tail-kernel contraction, rather than Gaussian magnitude alone.

12.6.2 Zero geometry beyond a fixed Laurent index

The complete classical counts are now known through Laurent index seven. The fifth-index transition has threshold three, and Theorem 10.25 proves sharp thresholds four and five at indices six and seven. Critical-point descent determines the deficits below saturation with exact signs, rather than extrapolating from a root scan. Small-order classification for higher indices remains open. Effective

thresholds uniform in the positive Lerch parameter require quantitative separation of the Appell roots and explicit concentration remainders. The spectral coefficient expansion for $n = O(\log k)$ does not by itself supply moving-zero asymptotics throughout that range, because the extreme root slopes can leave a compact scale. For higher zero branches, monotonicity in the deformation parameter is suggested by the first correction but is proved globally only at $n = 1$.

12.6.3 Certificates, integral structure and arithmetic evaluation

The complex distribution module is free even at resonant weights. What changes there is the usefulness of selected coordinates, not its rank. The division-free point basis and unit minor now settle integral unreflected Smith factors and arbitrary-ring base change. Reflection torsion, all-field rank loci and one-parameter Smith exponents are proved in Section 9.8; Theorem 9.43 also gives a closed product law for separable multivariable jets. General multivariable module extensions, sharper certificate-height bounds and arithmetic evaluation kernels remain separate questions. The same distinction applies to the exterior-symbol calculus: a nonzero formal obstruction rules out a certificate in that calculus, while the kernel of numerical period evaluation is a deeper arithmetic problem. Constructive reduction can now be benchmarked against an exact admissibility criterion rather than a denominator heuristic.

12.6.4 Further positive kernels and approximation regimes

The harmonic-index transition and Herglotz optimal cutoff illustrate two ways that positive kernels convert experiments into uniform asymptotics. Other roots of unity introduce complex weights and can destroy positivity; their extensions need a replacement argument rather than a formal reuse of the signed bounds. The general $J(p/q)$ symbol problem extends the proved $J(2/q)$ classification. At the optimal approximation scale, algebraic corrections and exponentially smaller arithmetic shifts require separately defined reference transitions. Their numerical diagnostics should retain that separation of scales.

12.6.5 A focused path toward formal verification

The finite algebraic interfaces are suitable first targets: character projectors, prime distribution edges, polynomial normal forms, exact partial fractions and the rational interval operations. Analytical formalization can then isolate the root-preserving operator, Laplace variation argument, positive-kernel remainder theorems and local normalized functional equation. These are explicit future tasks, not claims that the present ordinary proofs have already been checked by a proof assistant.

12.7 From decimal fits to short executable proofs

The Gaussian continuation illustrates a complete experimental cycle. A weight-five relation first found with large coefficients is the combination of two simple shuffle rows; the parity theorem then explains all five weight-six double reductions. The one-two integral supplies the three weight-four triples, while algebraic argument substitutions and rational row operations certify the cubic and quartic ladders. The cubic log-gamma moment requires a Tornheim derivative coordinate rather than a larger unstructured single-value basket.

The accompanying fresh replay re-expands all 511 ending-in-one binary words through weight nine and checks 45 bigraded sectors. It verifies 35 exact algebraic ladder substitutions and reproduces the

integer cubic moment certificate, whose endpoints share 112 decimal places. Independent defining-integral comparisons check 64 double and one-two formulas. The Chebyshev implementation supplies 89 interval residual enclosures, including one for S_4, which was conjectural at that checkpoint. That enclosure containing zero did not prove equality; the later octahedral certificate provides the exact argument. Six separate midpoint rectangles enclose the Gaussian and Eisenstein inverse-color values to widths below 10^{-70}. These finite replays test the implementations. The all-weight and asymptotic statements rest on the analytic and algebraic arguments in the chapters.

12.8 What the new reports change in the research programme

The exact S_4 proof adds a convergent octahedral word transformation and higher-depth lifted rows to a vocabulary that the finite obstructions had shown insufficient. The weight-three companion in Theorem 4.11 supplies a second exact mixed reduction. The product-matrix rank conjecture is now proved by Theorem 5.15. The next task is to compress the larger certificate and investigate the frozen S_6 relation with the same analytic control. Pure complementary seeds now have a complete collision classification and an algebraic-degree cutoff; arbitrary multiplicative unit relations and maximal ladder weights remain broader questions.

Primitive-grid determinants and Smith factors quantify coordinate failures without changing the free distribution quotient. Explicit finite descent and pole-cancelled Fourier jets give constructive all-index identities and trace zeros; integral lattices and certified high-order implementations remain distinct developments. The new sharp and reflected moment theories separate fixed-exponent asymptotics, optimal truncation and simultaneous large-exponent limits. Likewise, the Herglotz oscillation is proved for fixed positive argument and large derivative order; joint scaling requires further uniform estimates.

12.9 A certified rejection beyond the discovery precision

A new Cayley continuation supplies a useful counterexample to premature promotion of integer-relation output. At weight nine, define the ordered basket

$$\mathbf{b}_8 = (S_8, g_{8,1}, g_{7,2}, g_{6,3}, g_{5,4}, \pi^9, \beta(8)\log 2, \beta(2)\zeta(7), \beta(4)\zeta(5), \beta(6)\zeta(3)).$$

A search with 110 working digits, tolerance 10^{-90} and coefficient ceiling 10^{12} returned the integer vector

$$\begin{aligned}\mathbf{c}_8 = (&-6451250972, -4072849461, -710107844, -10984374763,\\ &-6393592639, 31875, -1955138863, 3459066102, -1034180101, -1477786143).\end{aligned}$$

Proposition 12.1 (The frozen weight-nine vector is false)**.** *With* $R_8 = \mathbf{c}_8 \cdot \mathbf{b}_8$*,*

$$-1.3274179020580459 \times 10^{-86} < R_8 < -1.3274179020580457 \times 10^{-86} < 0.$$

Proof. The delivered rational evaluator encloses each harmonic double and each single atom with 1000 Euler terms, using the explicit signed tail budgets. It forms all products and the frozen linear combination in exact interval arithmetic. The resulting rational residual interval has width less than 2.080×10^{-290} and lies strictly inside the displayed negative interval. The fresh isolated replay recomputes every rational atom endpoint and the residual, comparing exact fractions with the preserved record `reports/cayley-s4-continuation/data/S8_exclusion_N1000.json`. Both the negative sign and the containment are exact rational comparisons. This computation proves

the displayed exclusion; it uses no floating-point sign or integer-relation search as an acceptance criterion. □

This rejects precisely the frozen vector, not every possible relation in the basket. A later unsuccessful higher-precision search likewise proves no arithmetic independence. In contrast the new alternative S_6 vector still has an enclosure containing zero; it remains conjectural, and Proposition 4.41 proves by two convergent shuffles that it is the earlier S_6 conjecture in a different basket.

12.10 The complete Cayley ideal does not close the bounded weight-seven search

The pinned manuscript leaves the mixed Gaussian identities for S_6 and S_8 conjectural. Two recent source contributions sharpened the algebraic problem in complementary ways. One supplied an exact integer functional separating the S_6 target from a specified collection of 5131 relations of depth at most four [125]. The other constructed a depth-preserving retraction for the *full* shuffle ideal generated by convergent Cayley identities [126]. It explicitly proposed projecting the bounded rows by that retraction as a next proof attempt.

The exact resonant continuation [129] completes that proposed calculation. The inherited integer functional also annihilates the intersection of the *entire* Cayley shuffle ideal with the weight-seven space of depth at most four. This includes all Cayley rows and all their shuffle multiples, even when their construction uses higher-depth intermediate words. Thus completing the Cayley side alone cannot prove S_6 from the stated bounded additional relations. The numerical S_6 identity remains open. This is a result about an explicit formal presentation; it is not nonvanishing of its evaluated period residual, and it gives no period-independence assertion.

12.10.1 The target and the exact formal presentation

Use the convention

$$\mathrm{Li}_{r,s}(u,v)=\sum_{n>m\ge 1}\frac{u^n v^m}{n^r m^s},\qquad g_{r,s}=\operatorname{Im}\mathrm{Li}_{r,s}(i,1),\qquad S_p=\sum_{n\ge 0}\frac{(-1)^n H_n}{(2n+1)^p}.$$

Let $L=\log 2$, and write the conjectural residual in its already reconciled short coordinate form:

$$\begin{aligned}R_6 = S_6 &+ \frac{12334}{527}g_{6,1}+\frac{384}{31}g_{5,2}+\frac{2136}{527}g_{4,3}-\frac{3179\pi^7}{5713920}\\ &+\frac{801}{2108}\beta(4)\zeta(3)+2\beta(6)L.\end{aligned}\tag{12.1}$$

The conjecture is $R_6=0$.

For formal words take the positive forms

$$z=\frac{dt}{t},\quad c=\frac{dt}{1-t},\quad a=\frac{i\,dt}{1-it},\quad b=-\frac{dt}{1+t},\quad \bar a=-\frac{i\,dt}{1+it}.$$

Their iterated integrals $H(w)$ are written outer letter first on $[0,1]$. An admissible nonempty word begins with a letter other than c and ends with a letter other than z. Let $\mathcal{A}$ be the graded rational shuffle algebra on the admissible words and the empty word. Conjugation K interchanges $a,\bar a$ and

fixes z, c, b. The superscript $-$ denotes its negative eigenspace, and $F_D\mathcal{A}$ is the span of words with at most D nonzero letters. This is the usual series-depth filtration.

The map induced by $t \mapsto (1-t)/(1+t)$ is

$$C(v_1 \cdots v_n) = \tau(v_n) \cdots \tau(v_1), \qquad \begin{array}{c|ccccc} v & z & c & a & b & \bar{a} \\ \hline \tau v & c-b & z+b & b-\bar{a} & b & b-a. \end{array}$$

Here $\tau = -\phi^*$ incorporates the reversal of orientation. Every transformed admissible word is admissible, and change of variables gives $H(Cw) = H(w)$. Consequently the period map annihilates the graded, K-stable shuffle ideal

$$\mathcal{J} = \langle f - Cf : f \in \mathcal{A} \rangle_{\Pi}. \tag{12.2}$$

No restriction on weight, depth, or multiplier is made in this definition.

In the following polynomial, juxtaposition is concatenation and $\star$ is the usual series stuffle product, transferred to words by cumulative colors. Define

$$\begin{aligned} P_6 = {} & 128588 z^5 a\bar{a} + 3138084 z^5 aa + 1592832 z^4 aza + 521184 z^3 az^2 a - 216172 z^6 a \\ & + 48861\big((z^3 a) \star (z^2 c)\big) - 257176\big((z^5 a) \star b\big), \qquad T = \frac{P_6 - K P_6}{2}. \end{aligned} \tag{12.3}$$

This is the exact target used by the earlier bounded certificate. It belongs to $F_2\mathcal{A}_7^-$. The elementary harmonic split gives $S_6 = g_{6,1} + \operatorname{Im} \operatorname{Li}_{6,1}(i, -1)$, while $\beta(7) = 61\pi^7/184320$. Using also $H(b) = -L$ gives

$$H(T) = 128588\, i R_6. \tag{12.4}$$

Both products in (12.3) use convergent inputs, so their stuffle expansion requires no regularization.

For precision, $\mathcal{R}_4 \subset F_4\mathcal{A}_7^-$ denotes the *same* inherited finite row family as in [125]. Its construction is retained in the delivered independent enumerator:

1. All weight-seven shuffle-minus-stuffle products with sum of input depths at most four. Inputs are admissible except for the permitted single factor $\operatorname{Li}_1(1)$; the standard one-divergence comparison cancels its divergent term.

2. All convergent level-two-to-level-four distribution rows and their products with admissible complementary-weight words, provided the sum of depths is at most four. Products are expanded by stuffle.

3. The already proved imaginary S_2 and S_4 rows multiplied by real parts of all admissible complementary-weight words of depth at most two, again expanded by stuffle.

4. The inherited balanced Cayley rows and their specified products. In the adapted alphabet below, the seed u begins with a letter other than x, ends with a letter other than z, and satisfies $\min(n_z(u), n_x(u)) \geq 1$. A multiplier v is allowed when $|u| - \min(n_z(u), n_x(u)) + \operatorname{dep}(v) \leq 4$ and $|u| + |v| = 7$. Its product is expanded by stuffle.

Imaginary projection, rational normalization, and duplicate removal give, in this order, 3460, 1394, 40, and 237 new nonzero rows, respectively: 5131 rows in total. These counts specify a formal enumeration, not dimensions of periods. The fourth family may be retained even though the new assertion explicitly imposes the complete ideal (12.2).

Theorem 12.2 (Full-Cayley extension of the weight-seven obstruction)**.** *With the formal target and row family just specified,*

$$\boxed{T \notin \mathcal{J}_7^- + \operatorname{span}_{\mathbb{Q}} \mathcal{R}_4.} \tag{12.5}$$

In particular the inherited finite proof search remains unsuccessful even after all *convergent Cayley relations and arbitrary shuffle products of them have been included.*

The distinction between the full ideal and its bounded intersection is essential. We verify a finite basis in $F_4\mathcal{A}_7^-$, but the theorem allows $\mathcal{J}$ in the entire admissible word algebra. The next two subsections justify why those finite checks imply that stronger assertion.

12.10.2 Why the depth-preserving projector controls the entire ideal

We state and justify the precise prerequisite from [126], so the logic of the extension does not depend on a finite-rank extrapolation. Introduce

$$x = c - b, \qquad y = a - \bar{a}, \qquad h = a + \bar{a} - b, \qquad \mathcal{E} = \{z, y, h, b, x\}.$$

Then

$$c = x + b, \qquad a = (y + h + b)/2, \qquad \bar{a} = (-y + h + b)/2.$$

In this basis admissibility means first letter different from x and last letter different from z. Depth is $|w| - n_z(w)$, conjugation acts by $(-1)^{n_y(w)}$, and C reverses a word, interchanges z, x, and multiplies by $(-1)^{n_h(w)}$.

Let $\widetilde{\mathcal{A}}$ be the full shuffle algebra, including nonadmissible formal words. It is used here only algebraically. With the order $z < y < h < b < x$, every Lyndon word of length at least two is admissible. The shuffle-polynomial theorem [127] consequently gives

$$\widetilde{\mathcal{A}} = \mathcal{A}[z, x].$$

Let R_0 be the algebra retraction setting the two *shuffle polynomial* variables z, x to zero. It preserves both letter counts, commutes with C, K, and is explicitly

$$R_0(w) = \sum_{u=0}^{l} \sum_{v=0}^{r} (-1)^{u+v} x^u \sqcup w[u : |w| - v] \sqcup z^v, \tag{12.6}$$

where l, r are the initial x run and terminal z run. Deleting an initial x and deleting a terminal z are commuting shuffle derivations; their finite Taylor substitutions prove (12.6). The factorials cancel because shuffle powers of a single letter are factorial multiples of its concatenation powers. In particular this is not literal erasure of letters inside a word, and assigns no analytic value to divergent integrals.

For nonempty w put

$$e(w) = \sum_{k=1}^{|w|} \frac{(-1)^{k-1}}{k} \sum_{w = w_1 \cdots w_k} w_1 \sqcup \cdots \sqcup w_k, \qquad E = R_0 e, \tag{12.7}$$

where the blocks are consecutive and nonempty. The map e is the first Eulerian logarithm. It annihilates products of positive-degree factors and equals the identity modulo such products [128]. These facts also follow from the convolution logarithm of the identity character into the commutative shuffle algebra: its logarithm is an infinitesimal character, and the term $k = 1$ gives the congruence.

Thus $e^2 = e$. Reversal permutes consecutive cuts, so e commutes with C; it also commutes with K and preserves both counts.

The subspace $U = E(\mathcal{A})$ maps isomorphically onto the indecomposables of $\mathcal{A}$, and multiplication identifies $\mathcal{A}$ with $\mathrm{Sym}(U)$. To see this directly, E kills positive-degree products and is the identity modulo them. Any homogeneous lift of an indecomposable basis therefore gives polynomial generators, by a triangular change from the Lyndon generators. The lifts are bigraded by (n_z, n_x), and C exchanges these two degrees.

For a homogeneous $u \in U$ of degree (p, q) choose the orientation

$$Q_\uparrow u = \begin{cases} u, & p > q, \\ Cu, & p < q, \\ (u + Cu)/2, & p = q. \end{cases}$$

On each pair of different bidegrees this identifies the two transported generators. On the diagonal it retains the positive eigenspace of C and kills its negative eigenspace. Extending multiplicatively gives an algebra retraction whose kernel is exactly $\mathcal{J}$: on the quotient all $u - Cu$ vanish, hence C acts identically on every polynomial, and conversely $u - Cu$ are among the ideal generators. With $E_\uparrow = Q_\uparrow E$, the retraction is the finite formula

$$\Pi_\uparrow(w) = \sum_{k=1}^{|w|} \frac{1}{k!} \sum_{w = w_1 \cdots w_k} E_\uparrow(w_1) \,\Pi\, \cdots \,\Pi\, E_\uparrow(w_k), \qquad \Pi_\uparrow(1) = 1. \tag{12.8}$$

Convolution exponentiation of (12.7), followed by R_0 and the multiplicative orientation, proves this expression. For completeness, $E(w) \in U$ even if a block w is not admissible: in degree at least two, the polynomial decomposition $\mathcal{A}[z, x]$ makes $w - R_0 w$ a sum of positive-degree products, killed by e; in degree one, $R_0 e$ kills the two extra generators directly.

The orientation never decreases the number of z's in any nonzero output. Shuffle multiplication adds their counts. Consequently, on $\mathcal{A}$,

$$\Pi_\uparrow^2 = \Pi_\uparrow, \quad \ker \Pi_\uparrow = \mathcal{J}, \quad \Pi_\uparrow K = K \Pi_\uparrow, \quad \Pi_\uparrow(F_D \mathcal{A}) \subseteq F_D \mathcal{A}. \tag{12.9}$$

These are all-weight algebraic statements. Notice that $C\Pi_\uparrow = \Pi_\uparrow$ is neither required nor generally true.

The verifier evaluates (12.7) using a permutation formula; its equivalence with consecutive cuts is also elementary in every weight. For a permutation of the n positions let d be the number of descents of its inverse. A shuffle of consecutive blocks produces that permutation exactly when cuts are placed at those d required positions, with any additional cuts chosen freely. Its total coefficient in (12.7) is therefore

$$\sum_{k=d+1}^{n} \frac{(-1)^{k-1}}{k} \binom{n-1-d}{k-1-d} = (-1)^d \int_0^1 t^d (1-t)^{n-1-d}\, dt = \frac{(-1)^d}{n\binom{n-1}{d}}.$$

This is precisely the implemented coefficient. Repeated letters simply combine the integer multiplicities of equal output words. Thus the finite audits of the two formulas are convention checks; their equality at weight seven does not depend on extrapolating those audits.

Lemma 12.3 (The full ideal intersection has a finite spanning certificate)**.** *If $\mathcal{B}_4$ is any rational basis of $F_4\mathcal{A}_7^-$, then*

$$\mathcal{J}_7^- \cap F_4 \mathcal{A}_7 = \mathrm{span}_{\mathbb{Q}}\{w - \Pi_\uparrow(w) : w \in \mathcal{B}_4\}.$$

Proof. Every displayed difference belongs to the full ideal by the kernel property and belongs to the bounded space by depth preservation. Conversely, for f in the intersection, $\Pi_{\uparrow} f = 0$, so $f = (1 - \Pi_{\uparrow}) f$. Expanding f in the finite basis proves the opposite inclusion. No restriction on the construction of f inside $\mathcal{J}$ has been imposed. □

12.10.3 The integer certificate and the proof

Use the adapted-letter encoding $Z = 0, Y = 1, H = 2, B = 3, X = 4$. Enumerate all length-seven words lexicographically, retaining exactly those with first letter different from X, last letter different from Z, at least three Z's, and an odd number of Y's. This gives a basis $\mathcal{B}_4$ of $F_4\mathcal{A}_7^-$ with 2546 elements. The exact-depth counts are $1, 35, 390, 2120$. They can be checked directly from the enumeration, or by choosing the nonzero positions and projecting to odd Y parity.

The certificate's `separator.json` supplies an integer functional λ on this basis. It is obtained by expressing the inherited functional of [125] in adapted coordinates; the functional itself is not being claimed as a newly discovered vector. The new fact is the second row of the following exact certificate:

$$\lambda(R) = 0 \qquad (R \in \mathcal{R}_4), \tag{12.10}$$

$$\lambda(w - \Pi_{\uparrow} w) = 0 \qquad (w \in \mathcal{B}_4), \tag{12.11}$$

$$\lambda(T) = -\mathcal{N} \neq 0, \tag{12.12}$$

where the positive integer is

$$\begin{aligned}\mathcal{N} = {} & 18666062728923681262058369113049668 2 \cdot 10^{36} \\ & + 78843750753489297279472069700354208.\end{aligned}$$

The integer is an exact functional value, not a numerical residual of the polylogarithm conjecture. Its decimal expansion is also recorded in the certificate JSON.

Proof of Theorem 12.2. The standalone verifier, freshly replayed on an isolated copy, regenerates the complete inherited row family, checks (12.10), and independently reconstructs each $\Pi_{\uparrow}(w)$ by (12.6)–(12.8). It verifies (12.11) for all 2546 basis words, and reconstructs (12.3) before checking (12.12). All arithmetic is integral or rational. Thus Lemma 12.3 shows that λ annihilates $\mathcal{J}_7^- \cap F_4\mathcal{A}_7$.

If, contrary to the theorem, $T = j + r$ with $j \in \mathcal{J}_7^-$ and $r \in \operatorname{span} \mathcal{R}_4$, then $T, r \in F_4\mathcal{A}_7^-$ implies $j = T - r \in F_4\mathcal{A}_7^-$. Consequently $\lambda(j) = 0$; also $\lambda(r) = 0$ by (12.10). This contradicts $\lambda(T) \neq 0$ and proves (12.5). □

For reproducibility, execute

```
py certificates/cayley/code/verify_complete_cayley.py
```

from the package root. It requires only Python's standard library. The verifier checks SHA-256 digests of the two retained source modules and the functional data before use. It regenerates 5131 rows, verifies 2546 complete-ideal basis differences, and checks the nonzero target pairing. There are 2300 nonzero projector differences; no claim of their linear independence is made. Additional finite audits compare the Eulerian permutation formula against consecutive cuts on all 780 words through weight four and check 2150 shuffle products, as well as retraction, conjugation, and depth conventions. These audits support the implementation; the preceding algebraic proof supplies the all-weight projector properties and the passage to the full ideal.

12.10.4 Consequences for further research

The immediate computational recommendation is sharper than in the inspected sources. Completing Cayley symmetry while retaining precisely $\mathcal{R}_4$ does not settle S_6. A successful continuation must change the additional relation input: for example by admitting suitable higher-depth double-shuffle or distribution rows, by adding another functional equation, or by deriving a uniform generating identity that implies the target. A larger numerical-precision calculation does not address this missing algebraic information.

The theorem does not prove that every conceivable derivation must pass through depth five. It excludes exactly the full ideal plus the specified finite additional span; another depth-four family could carry information outside that span. Likewise, no conclusion for the revised S_8 candidate follows merely by analogy. Useful next questions are to identify the lowest-weight additional row whose projected class pairs nontrivially with λ, to test whether selected depth-five double-shuffle rows suffice, and to find a conceptual functional equation explaining the whole even-index family. These are exact proof-search questions, while $R_6 = 0$ itself remains the separate numerical-period conjecture.

Appendix A

Literature problems and source provenance

The two source catalogs organize questions that motivated or could extend the programme. Their categories and answer status describe the June 2026 survey, not a current audit of the websites. A catalog entry is a research lead, not an accepted identity. Closely overlapping questions are retained under a single thread identifier. The external reports numbered one through four contributed leads to these catalogs; their unverified prose and claims are not used as mathematical authorities.

One erroneous external formula is corrected here with a proof. Euler reflection at $1/3$, together with Landen at $-1/3$ and duplication, gives

$$6\,\mathrm{Li}_2(1/3) - \mathrm{Li}_2(1/9) = \frac{\pi^2}{3} - \log^2 3.$$

For completeness, put $A = \mathrm{Li}_2(1/3)$ and $B = \mathrm{Li}_2(-1/3)$. The five-term relation at $x = y = 1/2$ and Landen at $-1/3$ give

$$2\,\mathrm{Li}_2(1/2) = \mathrm{Li}_2(1/4) + 2A + \log^2(2/3), \quad \mathrm{Li}_2(1/4) = -B - \frac{1}{2}\log^2(4/3).$$

With $2\,\mathrm{Li}_2(1/2) = \pi^2/6 - \log^2 2$, collecting the logarithms gives $2A - B = \pi^2/6 - \frac{1}{2}\log^2 3$. Duplication says $\mathrm{Li}_2(1/9) = 2A + 2B$, so subtracting it from $6A$ proves the corrected formula. This derivation fixes the factor-two error in the external report without relying on numerical recognition.

A.1 Polylogarithm questions

Thread	Research question	Historical category
MSE 1424600	Real part of $\mathrm{Li}_2(1/2 + i/6)$	Seed question
MSE 1430169	Dilogarithm values at algebraic complex arguments	Seed question
MSE 1373123	An identity for six rational tetralogarithm values	Seed question
MSE 945972	Relations among rational polylogarithm values	Seed question
MSE 1579355	Trilogarithms and the imaginary golden ratio	Seed question
MSE 918680	Closed Form for the Imaginary Part of Li_3((1+i)/2)	Primary answered matches on Math Stack Exchange
MSE 3290798	A conjectured value for Re Li_4(1+i)	Primary answered matches on Math Stack Exchange

Thread	Research question	Historical category
MSE 3447529	On the relationship between Re Li_n(1+i) and Li_n(1/2) when n >= 5	Primary answered matches on Math Stack Exchange
MSE 676124	Proving a Li_3(-1/3) - 2 Li_3(1/3) identity	Primary answered matches on Math Stack Exchange
MSE 1450917	A special value polylogarithm identity involving Li_3(-1/2), Li_3(-1/3), Li_3(2/3), Li_2(-1/3), Li_2(2/3)	Primary answered matches on Math Stack Exchange
MSE 935366	Simplification of an expression containing Li_3(x) terms	Primary answered matches on Math Stack Exchange
MSE 5092144	How to prove this zeta(3) form with Li_3 and Li_2	Primary answered matches on Math Stack Exchange
MSE 463682	Closed form of Li_2(phi) and Li_2(phi-1)	Primary answered matches on Math Stack Exchange
MSE 1410276	Closed-form of Li_2(1 +- i sqrt(3))	Primary answered matches on Math Stack Exchange
MSE 967398	Extract real and imaginary parts of Li_2(i(2 +- sqrt(3)))	Primary answered matches on Math Stack Exchange
MSE 1602647	Extract imaginary part of Li_3(2/3 - i 2 sqrt(2)/3) in closed form	Primary answered matches on Math Stack Exchange
MSE 980179	Closed-form of a special value dilogarithm identity	Primary answered matches on Math Stack Exchange
MSE 989704	Closed-forms of real parts of special value dilogarithm identities from inverse tangent integral function	Primary answered matches on Math Stack Exchange
MSE 1436493	Dilogarithm identity containing the tribonacci constant	Primary answered matches on Math Stack Exchange
MSE 481682	Polylogarithm ladders for the tribonacci and n-nacci constants	Primary answered matches on Math Stack Exchange
MSE 1064100	Why does the tribonacci constant have a trilogarithm ladder?	Primary answered matches on Math Stack Exchange
MSE 1959503	Proof of a dilogarithm identity	Primary answered matches on Math Stack Exchange
MSE 932932	Known exact values of the Li_3 function	Primary answered matches on Math Stack Exchange
MSE 1411312	Closed form of a series (dilogarithm)	Primary answered matches on Math Stack Exchange
MSE 2389425	A very odd-looking statement about zeta(3) and Li_2(1/phi^2)	Primary answered matches on Math Stack Exchange
MO 492746	Proving identity involving dilogarithms and pi/9	Primary answered matches on MathOverflow
MO 85986	A dilogarithm identity: known or new?	Primary answered matches on MathOverflow
MO 144322	The relationship between the dilogarithm and the golden ratio	Primary answered matches on MathOverflow
MO 261408	Several conjectured identities for polylogarithms	Primary answered matches on MathOverflow
MO 450252	An identity involving polylogarithms	Primary answered matches on MathOverflow
MO 511718	Two-term dilogarithmic identity simplification	Primary answered matches on MathOverflow
MO 511765	Daniel Shanks "simplest cubic fields" and dilogarithm identities	Primary answered matches on MathOverflow
MO 131111	Is this combination of generalized polygamma and dilogarithm actually zero?	Primary answered matches on MathOverflow

Thread	Research question	Historical category
MSE 5080786	Ramanujan type formulas for polylogarithm	Open prompts with usable identity examples but no meaningful answers
MSE 5023055	How to find closed form for a complex expression involving special functions and polylogarithms?	Open prompts with usable identity examples but no meaningful answers
MO 173492	Dilogarithm of -1/2?	Open prompts with usable identity examples but no meaningful answers
MO 141185	A second polylogarithm ladder for the tribonacci and n-nacci constants	Open prompts with usable identity examples but no meaningful answers
MO 490573	The imaginary part of two-term complex dilogarithms	Open prompts with usable identity examples but no meaningful answers
MO 496682	Deriving a 4-term functional dilogarithm identity by Kirillov	Open prompts with usable identity examples but no meaningful answers
MSE 3938336	Evaluate a series equal to Li_3((3-sqrt(5))/2)	Duplicate-like or low-new-information threads
MSE 5110303	Does the polylogarithm Li_k(x) solve a first or second order ODE/ADE?	Rejected or parked: no answer and no special-value identity example
MSE 5127863	Completing proof of formula for a hard polylogarithmic integral	Rejected or parked: no answer and no special-value identity example
MSE 4986171	Understanding polylogarithm branch points, monodromy and multivaluedness	Rejected or parked: no answer and no special-value identity example
MO 85774	Is the quantum dilogarithm related in any way to cohomology of quantum groups?	Rejected or parked: no answer and no special-value identity example
MO 160646	Inverse of polylogarithm	Rejected or parked: no answer and no special-value identity example
MO 333610	Limiting property of polylogarithm ratio	Rejected or parked: no answer and no special-value identity example
MO 279588	Approximations of polylogarithm and Lerch transcendent?	Rejected or parked: no answer and no special-value identity example
MO 218158	Root polylogarithm dominance questions	Rejected or parked: no answer and no special-value identity example
MSE 1210644	Proving a sum cos(n)/n^4 evaluation	Near misses and background threads
MSE 376180	Integral identity involving Li_2(-1-e^{ix})	Near misses and background threads
MSE 4284361	Closed form evaluation of a trigonometric integral in terms of polylogarithms	Near misses and background threads
MSE 5125161	Verify a polylogarithmic evaluation of a 5D integral over a positive orthant	Near misses and background threads

Thread	Research question	Historical category
MO 347747	Simplify the difference of two dilogarithms	Near misses and background threads
MO 2402	Abel's equation for the dilog	Near misses and background threads
MO 25428	What is special about polylogarithms that leads to so many interesting identities and applications?	Near misses and background threads
MO 242920	Polylogarithm sheaves	Near misses and background threads
MO 309945	A remarkable almost-identity	Near misses and background threads

A.2 Polygamma and Hurwitz questions

Thread	Research question	Historical category
MSE 897967	Trigamma identity 4 psi_1(1/5)+psi_1(2/5)-psi_1(1/10)=4 pi^2/(phi sqrt(5))	Primary answered matches on Math Stack Exchange
MSE 4290297	New trigamma identity for Psi_1(3/20)+6 Psi_1(1/5)+10 Psi_1(2/5)-Psi_1(1/20)	Primary answered matches on Math Stack Exchange
MSE 953976	Prove psi^(1)(5/6)-psi^(1)(1/6)=5(psi^(1)(2/3)-psi^(1)(1/3))	Primary answered matches on Math Stack Exchange
MSE 2802872	Trigamma identity psi_1(11/12)-psi_1(5/12)=4 sqrt(3) pi^2-80G	Primary answered matches on Math Stack Exchange
MSE 3063097	Alternative proof for zeta(2,1/4)=psi^(1)(1/4)=pi^2+8G	Primary answered matches on Math Stack Exchange
MSE 2952474	Is it possible to simplify psi^(2)(1/8) or psi^(2)(p/q)?	Primary answered matches on Math Stack Exchange
MSE 4040435	General summation formula for psi^(s)(1/2), including negative orders	Primary answered matches on Math Stack Exchange
MSE 4784920	Closed form for psi^(-2)(1/4)	Primary answered matches on Math Stack Exchange
MSE 893486	Identity connecting polylogarithm and polygamma functions of arguments 1/4 and 3/4	Primary answered matches on Math Stack Exchange
MSE 4504218	Identity relating the polygamma function and the Lerch transcendent	Primary answered matches on Math Stack Exchange
MSE 3653746	Decomposition of psi^(n)(1) in terms of psi^(n)(k)	Primary answered matches on Math Stack Exchange
MSE 1154373	An identity on Polygamma	Primary answered matches on Math Stack Exchange
MSE 597867	Multiplication formula for the Hurwitz zeta function	Primary answered matches on Math Stack Exchange
MSE 1335002	An infinite series in polygamma function	Primary answered matches on Math Stack Exchange
MSE 4441095	Closed form for a derivative limit involving psi^(0) and higher polygamma values at 1	Primary answered matches on Math Stack Exchange
MO 312479	pi in terms of polygamma	Primary answered matches on MathOverflow
MO 312550	psi(2,1/6), psi(4,1/6) in terms of zeta and pi only	Primary answered matches on MathOverflow

Thread	Research question	Historical category
MO 42696	Connection between Bernoulli polynomials and polygamma function	Primary answered matches on MathOverflow
MO 347520	Reference request: sums of rational functions and polygamma functions	Primary answered matches on MathOverflow
MO 251727	Closed form for sum involving digamma?	Primary answered matches on MathOverflow
MSE 1909595	Interesting connection between polylogarithms and polygamma functions	Open or no-meaningful-answer leads
MSE 4319642	Simplification of a difficult identity involving the digamma function	Open or no-meaningful-answer leads
MSE 3555737	Sum involving 2nd antiderivative of the digamma function	Open or no-meaningful-answer leads
MSE 7183	Connection between Bernoulli polynomials and polygamma function	Open or no-meaningful-answer leads
MSE 521203	Recurrence relation for polygamma reflection polynomials	Open or no-meaningful-answer leads
MSE 4845364	Closed form for psi^(1/k)(1), where k is an integer	Open or no-meaningful-answer leads
MO 152299	Recurrence formula for digamma function with rational number	Open or no-meaningful-answer leads
MO 372726	Can a digamma generating-function expression be simplified?	Open or no-meaningful-answer leads
MO 455136	Negativity of a determinant involving trigamma, tetragamma, and pentagamma	Open or no-meaningful-answer leads
MSE 1427206	Conjecture for an integral involving psi^(1)(1/3)	Adjacent answered leads
MSE 1346396	Logarithmic Integral I	Adjacent answered leads
MSE 3075324	Improper integral with trigamma identity in the solution path	Adjacent answered leads
MSE 4634827	Closed form for a fractional-part integral	Adjacent answered leads
MSE 2683542	Meaning and definition of psi^(-2)(x)	Adjacent answered leads
MSE 5080998	Integral of a trigamma difference at conjugate half-line arguments	Adjacent answered leads
MO 220759	Closed form for sum 1/(1+n+...+n^a)	Adjacent answered leads
MO 378665	Closed form of sum_{r>=2} zeta(r)/r^2	Adjacent answered leads

A.3 The provenance boundary

The original drafts retain their snapshot identifiers. The accompanying source inventory records the SHA-256 digest of all thirty-nine textual sources. The editorial ledger maps each source to the merged mathematical development, records which results later drafts supersede, and identifies claims that have been withdrawn or weakened. The old PDFs are source evidence and are superseded as reading artifacts by this manuscript.

The original identity stores, reduction tools, notebook parsers, certificate checker and many high-precision search scripts were not included in the import. A historical receipt in a source report is therefore not a fresh replay here. The new verification scripts exercise representative corrected formulas from independent definitions. Their receipt specifies engine, precision, tolerance and actual residuals; it does not assert a complete replay of every historical experiment.

A.4 A version-specific external consistency check

The final paragraph of §7.2, printed page 16 of the inspected 18-page 2020 preprint [35], displays the untwisted identity

$$\beta_p(n) = \beta_p(n+1) + \beta_p\left(\frac{n}{n+1}\right). \tag{A.1}$$

For its stated exterior-square symbol, this identity is false. Take $p = 7$, $n = 2$. Since $2/3 = 3$ in $\mathbb{F}_7$ and the class of 3 is inverse to that of 2 modulo sign,

$$\beta_7(3) = -\beta_7(2).$$

Equation (A.1) would therefore imply $3\beta_7(2) = 0$. But $2^2 = 4$ is neither 1 nor -1 modulo 7, so Theorem 11.9 proves $\beta_7(2) \neq 0$.

There is also a small exact matrix certificate. In the regular basis of sign classes $1, 2, 3$, the image of the defect under the coordinate model $\beta_7(a) \mapsto P_a - P_{a^{-1}}$ is

$$\begin{pmatrix} 0 & 3 & -3 \\ -3 & 0 & 3 \\ 3 & -3 & 0 \end{pmatrix} \neq 0. \tag{A.2}$$

Its entries are integers; no numerical logarithm independence test enters this counterexample.

Proposed correction and version scope. Withdraw the untwisted identity as printed, or supply a corrected coefficient action or quotient together with a proof. This article does not guess what a missing action should be. The definition of β and the boundary formula (11.8) remain valid; the latter was independently derived in Section 11.7. The statement was verified against the 18-page author preprint and its 2020 version; the separately published 2023 journal version is not covered by this particular check. The correction is therefore specifically to the inspected 2020 preprint, and must not be attributed to an unchecked version.

A.5 Two Stieltjes index and coefficient corrections

These checks concern the identified displays in the inspected source PDFs. They also illustrate why the coefficient convention and the initial summation index must be fixed before comparing harmonic spectral identities.

A.5.1 The initial term in the Hurwitz–Stieltjes limit

On printed page 1 of *On certain harmonic zeta functions*, arXiv:2403.07123v1, dated 11 March 2024 [135], the Hurwitz series begins at $n = 0$, but the following unnumbered Stieltjes-limit display begins at $n = 1$. For real $a > 0$ and integer $m \geq 0$, the correct unit-coordinate limit is

$$\gamma_m(a) = \lim_{N\to\infty}\left[\sum_{n=0}^{N} \frac{\log^m(n+a)}{n+a} - \frac{\log^{m+1}(N+a)}{m+1}\right]. \tag{A.3}$$

Equivalently, retain the lower index one and add $\log^m(a)/a$.

Proof. The standard sum–integral subtraction for the Laurent expansion of $\zeta(s,a)$ gives (A.3); changing the upper endpoint by one contributes $O((1+\log^m N)/N)$ and does not change the limit. The recurrence $\zeta(1+u,a+1)=\zeta(1+u,a)-a^{-1-u}$ gives, coefficient by coefficient,

$$\gamma_m(a+1)=\gamma_m(a)-\frac{\log^m a}{a}.$$

The sum starting at one is the sum for $a+1$, hence its printed limit is $\gamma_m(a+1)$, not $\gamma_m(a)$. □

At $m=0, a=1$, the exact counterexample is

$$\lim_{N\to\infty}\{H_{N+1}-1-\log(N+1)\}=\gamma-1.$$

At $a=1, m>0$ the omitted term vanishes; testing only those cases would miss the index defect. This correction is specifically to the inspected version-one introductory display.

A.5.2 The sign of the height-one linear coefficient

Young's published *Series of height one multiple zeta functions*, *Integers* 24 (2024), A43, printed page 12 [134], describes the symbol in Corollary 3 as the linear coefficient. Its displayed integral expression instead gives the negative of the literal coefficient $L_d=[u]Z(1+u,1^{d-1})$ in our convention. The ordinary Stieltjes value in Example 5 is consistent with the conventional $\gamma_1=-L_1$.

The sign can be checked at every depth without a decimal fit. For $0\le r\le 1$ and $s>1$, define the classical Barnes series

$$\mathcal{B}_r(s)=\sum_{n\ge 0}\frac{(r)_n}{n!}(n+1)^{-s}.$$

At $r=0$ only the initial term remains. Put

$$U_d=\int_0^1(1-r)\mathcal{B}_r(d+2)\,\mathrm{d}r,\qquad V_d=\int_0^1\mathcal{B}_r(d+1)\,\mathrm{d}r.$$

The right side of the inspected Corollary 3 is $U_d-V_d-\alpha_{d+1}$. Using $(r)_{n+1}=(r+n)(r)_n$ and termwise integration gives

$$U_d-V_d=-\sum_{n\ge 1}\frac{\rho_n}{n^{d+2}},$$

where ρ_n is defined in Proposition 8.117. All relevant series converge absolutely because $(r)_n/n!\le 1$ on $0\le r\le 1$ and $d\ge 1$. The proved Cauchy coefficient formula therefore shows exactly that

$$U_d-V_d-\alpha_{d+1}=-L_d. \tag{A.4}$$

Thus one may keep the displayed expression and describe the negative linear coefficient, or negate it when naming the literal $[u]$ coefficient. At depth one it gives γ_1, whereas the literal linear coefficient of $\zeta(1+u)$ is $-\gamma_1$. The depth-one discrepancy is nonzero rigorously: the exact rational certificate in `verification/digamma-coordinate-source-audit.json` gives

$$-\frac{3}{40}<\gamma_1<-\frac{7}{100}<0.$$

Here is its analytic error budget. For $f(x)=\log(x)/x$ and $N=10$, one Euler–Maclaurin correction gives

$$\gamma_1=\sum_{n=1}^{N}\frac{\log n}{n}-\frac{\log^2 N}{2}-\frac{\log N}{2N}+\frac{\log N-1}{12N^2}+E_N,\qquad |E_N|\le\frac{\log N-1}{12N^2}.$$

The bound follows from $|\widetilde{B}_2|/2! \le 1/12$ and $f''(x) = (2\log x - 3)/x^3 \ge 0$ for $x \ge 10$; thus $\int_N^\infty |f''| = -f'(N)$. Each logarithm is enclosed rationally by 128 terms of its atanh series, with the geometric positive-tail bound. The resulting endpoint comparisons are exact fractions, not floating sign tests. The separately replayed 100-digit Cauchy/Barnes diagnostic also confirms the sign; (A.4) proves the all-depth coefficient identity. The correction concerns that coefficient description, with Example 5's ordinary Stieltjes formula retained.

Bibliography

[1] L. Lewin, *Polylogarithms and Associated Functions*, North-Holland, 1981; and (ed.) *Structural Properties of Polylogarithms*, AMS, 1991. (The trigamma Clausen-DFT law; cited via MathWorld, *Trigamma Function*, eq. (14).)

[2] Wolfram Research, *The Wolfram Functions Site*, `functions.wolfram.com`, PolyLog identity 10.08.03.0007.01. (The forward cyclotomic formula.)

[3] E. E. Kummer, *Beitrag zur Theorie der Function* $\Gamma(x)$, J. reine angew. Math. **35** (1847), 1–4. (The Fourier series of $\log\Gamma$.)

[4] C. J. Malmsten, *De integralibus quibusdam definitis, seriebusque infinitis*, J. reine angew. Math. **38** (1849), 1–39 (dated 1846). (The log-log integrals and the reflection identity for γ_1.)

[5] I. Vardi, *Integrals, an introduction to analytic number theory*, Amer. Math. Monthly **95** (1988), 308–315.

[6] I. V. Blagouchine, *A theorem for the closed-form evaluation of the first generalized Stieltjes constant at rational arguments and some related summations*, J. Number Theory **148** (2015), 537–592. (Vendored in the repository at `src/PolyLog/docs/pdfs/1401.3724v3/`.)

[7] O. Espinosa and V. Moll, *On some integrals involving the Hurwitz zeta function, Part 1*, Ramanujan J. **6** (2002), 159–188. (The second log-gamma moment.)

[8] E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis*, 4th ed., Cambridge University Press, 1927. (Binet's formulas; the Hurwitz-series ladder.)

[9] C. Deninger, *On the analogue of the formula of Chowla and Selberg for real quadratic fields*, J. reine angew. Math. **351** (1984), 171–191. (The $\zeta''(0,a)$ layer.)

[10] N. Koblitz and A. Ogus, *Algebraicity of some products of values of the* Γ *function*, appendix (pp. 343–346) to P. Deligne, *Valeurs de fonctions L et périodes d'intégrales*, in *Automorphic Forms, Representations and L-Functions* (Corvallis, 1977), Proc. Sympos. Pure Math. **33**, part 2, Amer. Math. Soc., 1979, 313–346.

[11] C. F. Gauss, *Disquisitiones generales circa seriem infinitam*, Comment. Soc. Regiae Sci. Gottingensis Rec. **2** (1813), 3–46; Werke **3**, 265–327. (The multiplication theorem.)

[12] V. Reshetnikov, *FunctionTransformations* (Wolfram Language package), 2016, https://github.com/VladimirReshetnikov/FunctionTransformations. (The 437-rule polylog and 670-rule Γ reduction tables anchoring the corpus.)

[13] A. Selberg and S. Chowla, *On Epstein's zeta-function*, J. reine angew. Math. **227** (1967), 86–110.

[14] R. M. Damerell, *L-functions of elliptic curves with complex multiplication, I*, Acta Arith. **17** (1970), 287–301.

[15] A. Hurwitz, *Über die Entwicklungskoeffizienten der lemniskatischen Funktionen*, Math. Ann. **51** (1899), 196–226.

[16] D. A. Cox, *Primes of the Form $x^2 + ny^2$: Fermat, Class Field Theory, and Complex Multiplication*, Wiley, 1989. (Genus theory; class invariants; the $3 \mid D$ obstruction for γ_2.)

[17] E. W. Weisstein, *Digamma Function*, MathWorld — A Wolfram Web Resource. (Source of the FoxTrot identity.)

[18] V. S. Adamchik, *Polygamma functions of negative order*, J. Comput. Appl. Math. **100** (1998), 191–199.

[19] T. M. Apostol, *Formulas for higher derivatives of the Riemann zeta function*, Math. Comp. **44** (1985), 223–232.

[20] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976.

[21] B. C. Berndt, *Ramanujan's Notebooks, Part I*, Springer, 1985.

[22] I. V. Blagouchine, *Rediscovery of Malmsten's integrals, their evaluation by contour integration methods and some related results*, Ramanujan J. **35** (2014), 21–110; Erratum–Addendum: **42** (2017), 777–781.

[23] K. Chakraborty, S. Kanemitsu and T. Kuzumaki, *Finite expressions for higher derivatives of the Dirichlet L-function and the Deninger R-function*, Hardy–Ramanujan J. **32** (2009), 38–53.

[24] B. K. Choudhury, *The Riemann zeta-function and its derivatives*, Proc. R. Soc. Lond. A **450** (1995), 477–499.

[25] M. W. Coffey, *Series representations for the Stieltjes constants*, Rocky Mountain J. Math. **44** (2014), 443–477 (arXiv:0905.1111).

[26] D. F. Connon, *On some new formulae involving the Stieltjes constants*, arXiv:1902.00510 (2019).

[27] O. Espinosa and V. H. Moll, *A generalized polygamma function*, Integral Transforms Spec. Funct. **15** (2004), 101–115.

[28] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: Part 1*, Ramanujan J. **6** (2002), 159–188.

[29] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: Part 2*, Ramanujan J. **6** (2002), 449–468.

[30] O. Kouba, *Generalized Stieltjes constants and integrals involving the log–log function: Kummer's theorem in action*, J. Classical Anal. **9** (2016), no. 2, 79–88 (arXiv:1702.05439).

[31] J. Miller and V. S. Adamchik, *Derivatives of the Hurwitz zeta function for rational arguments*, J. Comput. Appl. Math. **100** (1998), 201–206.

[32] D. H. Bailey and J. M. Borwein, *Computation and structure of character polylogarithms with applications to character Mordell–Tornheim–Witten sums*, Math. Comp. **85** (2016), 295–324.

[33] E. Panzer, *The parity theorem for multiple polylogarithms*, J. Number Theory **172** (2017), 93–113. https://arxiv.org/abs/1512.04482.

[34] T. Rivoal and J. Roques, *Hadamard products of algebraic functions*, Section 3 (gamma relations and Rohrlich's conjecture). https://rivoal.perso.math.cnrs.fr/articles/algcoef.pdf.

[35] D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*. https://arxiv.org/abs/2012.15805. Author version: https://people.mpim-bonn.mpg.de/zagier/files/preprints/HerglotzFunction.pdf.

[36] M. Abouzahra and L. Lewin, *The polylogarithm in algebraic number fields*, Journal of Number Theory **21** (1985), no. 2, 214–244. doi:10.1016/0022-314X(85)90052-6.

[37] M. Abouzahra, L. Lewin and H. Xiao, *Polylogarithms in the field of omega (a root of a given cubic): Functional equations and ladders*, Aequationes Mathematicae **33** (1987), 23–45. doi:10.1007/BF01836149.

[38] D. H. Bailey and D. J. Broadhurst, *A seventeenth-order polylogarithm ladder*, preprint, 1999. https://arxiv.org/abs/math/9906134.

[39] H. Gangl, *Functional equations and ladders for polylogarithms*, Communications in Number Theory and Physics **7** (2013), no. 3, 397–410. doi:10.4310/CNTP.2013.v7.n3.a1.

[40] H. Cohen, L. Lewin and D. Zagier, *A Sixteenth-order Polylogarithm Ladder*, Experimental Mathematics **1** (1992), no. 1. https://people.mpim-bonn.mpg.de/zagier/files/exp-math-1/fulltext.pdf. (The higher-order ladders in this paper are explicitly conjectural numerical identities.)

[41] C. Hakimoglu-Brown, *Analytic Dilogarithm Identities* (2025). https://arxiv.org/abs/2506.10206.

[42] E. Remiddi and J. A. M. Vermaseren, *Harmonic polylogarithms*, Int. J. Mod. Phys. A **15** (2000), 725–754. https://arxiv.org/abs/hep-ph/9905237.

[43] F. Brown, *Mixed Tate motives over* $\mathbb{Z}$, Annals of Mathematics **175** (2012), 949–976. https://arxiv.org/abs/1102.1312.

[44] NIST Digital Library of Mathematical Functions, §5.8, *Infinite Products*; reciprocal-gamma Weierstrass product. https://dlmf.nist.gov/5.8. Accessed 7 October 2026.

[45] D. Zagier, *The dilogarithm function*, in P. Cartier, B. Julia, P. Moussa, and P. Vanhove (eds.), *Frontiers in Number Theory, Physics, and Geometry II*, Springer, 2007, pp. 3–65. https://doi.org/10.1007/978-3-540-30308-4_1.

[46] J. Milnor, *Introduction to Algebraic K-Theory*, Annals of Mathematics Studies **72**, Princeton University Press, 1971. https://doi.org/10.1515/9781400881796.

[47] L. C. Washington, *Introduction to Cyclotomic Fields*, second edition, Graduate Texts in Mathematics **83**, Springer, 1997. https://doi.org/10.1007/978-1-4612-1934-7.

[48] R. Sharifi, *Iwasawa theory*, online lecture notes, Chapter 4 (cyclotomic units and Dirichlet L-values). https://www.math.ucla.edu/~sharifi/notes/iwasawa-ch04.html. Accessed 8 October 2026.

[49] NIST, *Digital Library of Mathematical Functions*, §§2.11, 5.11, and 25.15; in particular the shifted Stirling expansion and Dirichlet nonvanishing. https://dlmf.nist.gov/. Accessed 8 October 2026.

[50] G. Nemes, *Error bounds for the asymptotic expansion of the Hurwitz zeta function*, *Proc. R. Soc. A* **473** (2017), 20170363. https://doi.org/10.1098/rspa.2017.0363; https://arxiv.org/abs/1702.05316.

[51] B. Simon, *The classical moment problem as a self-adjoint finite difference operator*, *Adv. Math.* **137** (1998), 82–203. https://doi.org/10.1006/aima.1998.1728; https://arxiv.org/abs/math-ph/9906008.

[52] D. S. Kubert, *The universal ordinary distribution*, Bulletin de la Société Mathématique de France **107** (1979), 179–202. https://doi.org/10.24033/bsmf.1891.

[53] Y. Ouyang, *On the universal norm distribution*, Journal of the Ramanujan Mathematical Society **17** (2002), no. 4, 287–311. https://arxiv.org/abs/math/0210297.

[54] O. Espinosa and V. H. Moll, "On Some Integrals Involving the Hurwitz Zeta Function: Part 2," *Ramanujan Journal* **6** (2002), 449–468. https://www.math.tulane.edu/~vhm/papers_html/hurwitz2.pdf.

[55] J. S. Milne, *Algebraic Number Theory*, version 3.08, 19 July 2020, especially Theorem 4.3 and Proposition 7.55 with Remark 7.56. https://www.jmilne.org/math/CourseNotes/ANT.pdf.

[56] E. Kowalski and A. Nikeghbali, *Mod-Poisson convergence in probability and number theory*, author manuscript. https://people.math.ethz.ch/~kowalski/mod-poisson.pdf. The permutation-cycle example explains the classical reciprocal-gamma factor underlying the coefficient mechanism.

[57] D. Romik and R. Scherer, *Alternative summation orders for the Eisenstein series G_2 and Weierstrass $\wp$-function*, arXiv:1811.01523, version 2 (2020). https://arxiv.org/abs/1811.01523.

[58] S. Ramanujan, *On certain arithmetical functions*, Transactions of the Cambridge Philosophical Society **22** (1916), no. 9, 159–184. Transcription of the original paper: https://ramanujan.sirinudi.org/Volumes/published/ram18.html.

[59] C. Xu and W. Wang, *Dirichlet type extensions of Euler sums*, Comptes Rendus Mathématique **361** (2023), 979–1010. https://doi.org/10.5802/crmath.453. The specialization used here is equation (20), section 3.4, printed page 994.

[60] D. E. Radford, A natural ring basis for the shuffle algebra and an application to group schemes, *Journal of Algebra* **58** (1979), no. 2, 432–454. doi:10.1016/0021-8693(79)90171-6.

[61] R. Umezawa, *An explicit parity theorem for multiple polylogarithms*, arXiv:2508.02040, 2025. https://arxiv.org/abs/2508.02040.

[62] D. Zagier, Special values and functional equations of polylogarithms, in L. Lewin (ed.), *Structural Properties of Polylogarithms*, Mathematical Surveys and Monographs **37**, American Mathematical Society, Providence, RI, 1991, 377–400. Author copy: https://people.mpim-bonn.mpg.de/zagier/files/tex/LewinPolylogarithms/fulltext.pdf.

[63] H. Nakamura and D. Shiraishi, Landen's trilogarithm functional equation and ℓ-adic Galois multiple polylogarithms, in M. Morishita, H. Nakamura, and J. Ueki (eds.), *Low Dimensional Topology and Number Theory*, Springer Proceedings in Mathematics & Statistics **456**, Springer, Singapore, 2025, 237–262. doi:10.1007/978-981-97-3778-9_8. Author preprint: https://arxiv.org/abs/2210.17182.

[64] T. Amdeberhan, M. W. Coffey, O. Espinosa, C. Koutschan, D. V. Manna, and V. H. Moll, Integrals of powers of loggamma, *Proceedings of the American Mathematical Society* **139** (2011), no. 2, 535–545. doi:10.1090/S0002-9939-2010-10589-0. Author preprint: https://www.math.tulane.edu/~vhm/papers_html/lg-subm.pdf.

[65] D. H. Bailey, D. Borwein, and J. M. Borwein, On Eulerian log-gamma integrals and Tornheim–Witten zeta functions, *The Ramanujan Journal* **36** (2015), nos. 1–2, 43–68. doi:10.1007/s11139-012-9427-1. Author preprint: https://www.davidhbailey.com/dhbpapers/log-gamma.pdf. The version-specific auxiliary sign check concerns the author copy dated 27 June 2012: https://carmamaths.org/resources/jon/log-gamma.pdf, Remark 3, equations (63)–(65). The separately linked July copy has different equation numbering; the sign observation is not a journal-version audit.

[66] J. M. Borwein and A. Straub, *Relations for Nielsen polylogarithms*, Journal of Approximation Theory **193** (2015), 74–88. doi:10.1016/j.jat.2013.07.003. https://arminstraub.com/downloads/pub/nielsenrelations.pdf.

[67] J. M. Borwein, D. M. Bradley, D. J. Broadhurst, and P. Lisoněk, *Special values of multiple polylogarithms*, Transactions of the American Mathematical Society **353** (2001), 907–941. doi:10.1090/S0002-9947-00-02616-7. arXiv:math/9910045.

[68] J. Vollinga and S. Weinzierl, *Numerical evaluation of multiple polylogarithms*, Computer Physics Communications **167** (2005), 177–194. doi:10.1016/j.cpc.2004.12.009. arXiv:hep-ph/0410259.

[69] J. Zhao, *Standard relations of multiple polylogarithm values at roots of unity*, arXiv:0707.1459. https://arxiv.org/abs/0707.1459.

[70] W. Hintze, *Closed form of the sum* $s_4 = \sum_{n=1}^{\infty}(-1)^n H_n/(2n+1)^4$, Mathematics Stack Exchange question 4160026 and answer by pisco, 3 June 2021. https://math.stackexchange.com/questions/4160026/.

[71] M. H. Zhao, *MZV evaluation of hypergeometric series*, arXiv:2007.02508, version 10, 27 September 2020. https://arxiv.org/abs/2007.02508.

[72] J. Zhao, *Multiple polylogarithm values at roots of unity*, C. R. Math. Acad. Sci. Paris **346** (2008), 1029–1032; expanded version arXiv:0810.1064. https://arxiv.org/abs/0810.1064.

[73] G. Racinet, *Doubles mélanges des polylogarithmes multiples aux racines de l'unité*, Publications Mathématiques de l'IHÉS **95** (2002), 185–231. https://arxiv.org/abs/math/0202142.

[74] J. Aarts, R. Fokkink, and G. Kruijtzer, Morphic numbers, *Nieuw Archief voor Wiskunde*, series 5, **2**, no. 1 (2001), 56–58. https://www.nieuwarchief.nl/serie5/pdf/naw5-2001-02-1-056.pdf.

[75] W. Ljunggren, On the irreducibility of certain trinomials and quadrinomials, *Mathematica Scandinavica* **8** (1960), 65–70. https://doi.org/10.7146/math.scand.a-10593.

[76] H. Tverberg, On the irreducibility of the trinomials $x^n \pm x^m \pm 1$, *Mathematica Scandinavica* **8** (1960), 121–126. https://doi.org/10.7146/math.scand.a-10599. Original scanned text: https://journals.msp.org/mscand/article/download/2744/2743/2775.

[77] D. W. Boyd, G. Martin, and M. Thom, Squarefree values of trinomial discriminants, *LMS Journal of Computation and Mathematics* **18** (2015), 148–169, especially Lemma 4.1(a),(b). https://doi.org/10.1112/S1461157014000436.

[78] NIST Digital Library of Mathematical Functions, §2.3(iii), Laplace's method. https://dlmf.nist.gov/2.3#iii.

[79] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. The smooth implicit-function schema and singularity transfer are the classical background for the late-coefficient analysis.

[80] NIST Digital Library of Mathematical Functions, §25.15, *Dirichlet L-functions*, including (25.15.4), (25.15.6), and the nonvanishing statement (25.15.9). Accessed 9 October 2026. https://dlmf.nist.gov/25.15.

[81] J. Zhao, *Multiple polylogarithm values at roots of unity*, C. R. Acad. Sci. Paris, Ser. I **346** (2008), 1029–1032. doi:10.1016/j.crma.2008.09.011. https://www.numdam.org/item/10.1016/j.crma.2008.09.011.pdf. Its standard-relation framework includes more relations and depths than the restricted presentation computed in this article.

[82] National Institute of Standards and Technology, *NIST Digital Library of Mathematical Functions*, Section 13.4, equation 13.4.4, and Section 13.8(iii), equations 13.8.11–16. https://dlmf.nist.gov/13.4.E4; https://dlmf.nist.gov/13.8.iii. Consulted 9–10 October 2026.

[83] N. M. Temme, *Uniform asymptotic expansions of a class of integrals in terms of modified Bessel functions, with application to confluent hypergeometric functions*, SIAM J. Math. Anal. **21** (1990), no. 1, 241–261. doi:10.1137/0521013.

[84] A. Barquero-Sanchez, L. Cadwallader, O. Cannon, T. Genao, and R. Masri, *Faltings heights of CM elliptic curves and special Gamma values*, Research in Number Theory **3** (2017), article 13. doi:10.1007/s40993-017-0077-7. Sections 3–5 give the analytic normalization of the Kronecker and Chowla–Selberg formulas, including nonmaximal quadratic orders.

[85] H. Cohen, *On the Argument of the Lerch, Chowla–Selberg Formula and CM Values of* $\eta(\tau)$, arXiv:2606.02904v1 (2026). https://arxiv.org/html/2606.02904v1. Theorem 1.3 states the period-product formula with conductor correction; Theorem 2.1 addresses the choice of representatives and eta phases.

[86] A. V. Sutherland, *Computing Hilbert class polynomials with the Chinese Remainder Theorem*, Mathematics of Computation **80** (2011), 501–538. Author's implementation and references.

[87] N. Yui and D. Zagier, *On the singular values of Weber modular functions*, Mathematics of Computation **66** (1997), 1645–1662. Author-hosted full text.

[88] W. Stein, *Modular Forms: A Computational Approach*, Graduate Studies in Mathematics, vol. 79, American Mathematical Society, 2007. https://wstein.org/books/modform/modform/level_one.html.

[89] R. Masri, *Kronecker's solution of Pell's equation for CM fields*, Annales de l'Institut Fourier **63** (2013), 2287–2306. https://aif.centre-mersenne.org/item/10.5802/aif.2830.pdf.

[90] W. Kohnen, *Transcendence of zeros of Eisenstein series and other modular functions*, Commentarii Mathematici Universitatis Sancti Pauli **52** (2003), no. 1, 55–57. https://doi.org/10.14992/00008713.

[91] L. Chiriac and A. Jorza, *The variation of zeros of the Miller basis*, arXiv:2605.09731v1 (2026), introduction and Section 7. https://arxiv.org/abs/2605.09731.

[92] D. S. Kubert, *The $\mathbb{Z}/2\mathbb{Z}$ cohomology of the universal ordinary distribution*, Bulletin de la Société Mathématique de France **107** (1979), 203–224. doi:10.24033/bsmf.1892.

[93] NIST Digital Library of Mathematical Functions, *Section 25.11: Hurwitz Zeta Function*, especially the defining series, meromorphic continuation and multiplication formulas. https://dlmf.nist.gov/25.11.

[94] ProveIt research contribution, *Gaussian Polylogarithms Beyond Parity: Exact relation modules, an octahedral proof of the mixed harmonic sum, and global angular-zero geometry*, October 2026. Preserved source: `reports/angular-zero-continuation/article.tex`.

[95] NIST Digital Library of Mathematical Functions, §5.9, *Integral Representations*, especially formula 5.9.13. https://dlmf.nist.gov/5.9.

[96] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*. https://dlmf.nist.gov/25.12.

[97] NIST Digital Library of Mathematical Functions, §4.36, *Infinite Products and Partial Fractions*, especially formula 4.36.3. https://dlmf.nist.gov/4.36.

[98] J.-G. Liu and R. L. Pego, *On generating functions of Hausdorff moment sequences*, Transactions of the American Mathematical Society **368** (2016), 8499–8518. The author's arXiv version 4 includes a corrigendum. https://arxiv.org/abs/1401.8052v4.

[99] ProveIt Contributors, *Sharp Gaussian Bounds, Cyclic Distribution Torsion, and Extremal Lerch Zeros*, research continuation, 10 October 2026. Prepared against repository revision `a0a90ef318`. Original source and certificates are preserved under `reports/formal-reductions-continuation/continuations/extremal-cyclic-descent/`.

[100] V. Reshetnikov, *A conjectured identity for tetralogarithms* Li_4, Math.StackExchange question 1373123 (2015). https://math.stackexchange.com/questions/1373123/.

[101] V. Reshetnikov, *Several conjectured identities for polylogarithms*, MathOverflow question 261408 (2017). https://mathoverflow.net/questions/261408/.

[102] ProveIt Contributors, *Polylogarithm Certificates and Sharp Euler Bounds*, research continuation, 10 October 2026, prepared against `4c173c06cc`. Preserved under `reports/uniform-transition-continuation/continuations/tetralogarithm-euler-polynomial-zeros/`.

[103] ProveIt Contributors, *Global Radial Monotonicity for Leading-Index-One Double Polylogarithms*, research continuation, 10 October 2026, prepared against `1539f353af`. Preserved under `reports/angular-zero-continuation/continuations/leading-one-global-radius/`.

[104] ProveIt Contributors, *Sharp Universal Euler Bounds*, research continuation, 10 October 2026. Preserved under `reports/real-order-threshold/continuations/sharp-universal-euler/`.

[105] ProveIt Contributors, *Polylogarithm Research: Euler Bounds, Golden Ladders and Cayley Reductions*, research continuation, 10 October 2026. Preserved under `reports/fractional-cayley-scaling/continuations/golden-cayley-double-turning/`.

[106] ProveIt Contributors, *Signed-Kernel Oscillation at Every Depth: Minimal Pick Compensation, Zero Geometry, and Certified Polylogarithmic Identities*, research continuation, 10 October 2026, prepared against `3447bc59b78d138a7f0d536b66ac6e5cc5d5e49f`. Original source and certificates are preserved under `reports/angular-zero-continuation/continuations/all-depth-signed-oscillation/`.

[107] Z.-H. Sun, *On the properties of invariant functions*, arXiv:2209.14625v1 (2022), Remark 3.2, printed p. 15. https://arxiv.org/abs/2209.14625.

[108] ProveIt Contributors, *A Convolution Algebra of Stieltjes Functions*, research continuation, 10 October 2026, prepared against `0efdbaa694`. Preserved under `reports/stieltjes-correlation-calculus/continuations/convolution-algebra/`.

[109] ProveIt Contributors, *Shifted Hurwitz-Zeta Correlations: Exact Stieltjes Closure, Polygamma Contact Terms, and Log-Gamma Antiderivatives*, research continuation, 10 October 2026, observed source `0e5ab2f60f`. Preserved under `reports/stieltjes-correlation-calculus/continuations/shifted-hurwitz-jets/`.

[110] ProveIt Contributors, *Bilinear Closure of Generalized Stieltjes Correlations*, research continuation, 10 October 2026. Individual source observations and scope are retained in the original provenance note. Preserved under `reports/stieltjes-correlation-calculus/continuations/bilinear-correlation-closure/`.

[111] ProveIt Contributors, *A Convolution Calculus for Stieltjes Functions*, research continuation, 10 October 2026, prepared against `570b0567f3`. Preserved under `reports/stieltjes-correlation-calculus/continuations/convolution-calculus/`.

[112] NIST Digital Library of Mathematical Functions, Eq. 15.4.20, Gauss summation of the hypergeometric function at one. https://dlmf.nist.gov/15.4.E20.

[113] J. Choi, Summation formulas involving binomial coefficients, harmonic numbers, and generalized harmonic numbers, *Abstract and Applied Analysis* **2014** (2014), Article 501906, printed p. 6, Conjectures 4 and 6 and Eqs. (40)–(46). https://doi.org/10.1155/2014/501906.

[114] ProveIt Contributors, *Exact Stieltjes and Harmonic Identities*, research continuation, 10 October 2026, prepared against `fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed`. Preserved under `reports/stieltjes-harmonic-resonance/continuations/stieltjes-harmonic-identities/`.

[115] NIST Digital Library of Mathematical Functions, §25.14, Lerch's transcendent. https://dlmf.nist.gov/25.14.

[116] NIST Digital Library of Mathematical Functions, §25.15, Dirichlet L-functions. https://dlmf.nist.gov/25.15.

[117] S. Hu and M.-S. Kim, On the Stieltjes constants and gamma functions with respect to alternating Hurwitz zeta functions, *Journal of Mathematical Analysis and Applications* **509**(1) (2022), 125930. https://arxiv.org/abs/2106.14674.

[118] H. Zhu, S. Hu and M.-S. Kim, On the properties of alternating invariant functions, arXiv:2505.10985v3 (2025), Theorem 1.13. https://arxiv.org/abs/2505.10985.

[119] ProveIt Contributors, *Twisted Stieltjes Convolutions and Harmonic Laurent Identities*, research continuation, 10 October 2026, prepared against `e0d9463bdee9685dfb1dddb819059cc738540c57`. Preserved under `reports/stieltjes-correlation-calculus/continuations/twisted-harmonic-laurent/`.

[120] ProveIt external research continuation, *Exact Identities: Collision Calculus, Shifted Harmonic Jets and Weighted Arctangent Families*, 10 October 2026. Original package retained at `reports/stieltjes-correlation-calculus/continuations/collision-and-weighted-identities/`. The weighted real evaluations are proved in Chapter 3; the remaining claims retain separate intake and audit status. Global priority is not established.

[121] ProveIt research continuations, *Periodic Stieltjes Collisions: Exact Contact Terms, Finite-Part Fubini Laws, and Gamma–Polylogarithm Primitives* and *Periodic Stieltjes Contact Calculus: All-Index Collision Corrections, Harmonic Derivative Anomalies, and Exact Zeta-Jet Integrals*, 10 October 2026. Independent routes retained in the correlation spine as `continuations/periodic-collision-contacts/` and `continuations/periodic-stieltjes-contact/`; their immediate coincident source is preserved as `continuations/coincident-stieltjes/`. The fixed-coordinate periodic comparison is reconciled in Chapter 9; the derivative anomaly was already established there.

[122] NIST Digital Library of Mathematical Functions, §5.7, *Series Expansions of the Gamma Function*, especially formula 5.7.3. https://dlmf.nist.gov/5.7.E3. Inspected 10 October 2026.

[123] ProveIt research continuation, *Mixed Spectral Identities*, 10 October 2026, prepared against `dcd95baeb7c0e914b138bddef6829c5b1d52b726`. Preserved at `reports/stieltjes-harmonic-resonance/continuations/mixed-spectral-identities/`. The mixed Gamma identity is derived from the classical Rogers–Dougall sum in Chapter 8. Its further resonance, harmonic and collision claims retain separate audit scopes. No global priority is asserted.

[124] NIST Digital Library of Mathematical Functions, §16.4, *Rogers–Dougall Very Well-Poised Sum*, formula 16.4.9. https://dlmf.nist.gov/16.4.E9. Inspected 10 October 2026.

[125] ProveIt research contribution, *A certified obstruction in the weight-seven reduction search*, in the extremal cyclic-descent continuation, `article/s6_obstruction.tex`. Pinned repository commit `5a790187c8e186e41e2b990b4941cb7a1a3c7b6b`. Source article.

[126] ProveIt research contribution, *A Cayley normal form that preserves series depth*, in the polylogarithm–Stieltjes critical-transition continuation, `article/05_cayley_depth.tex`. Pinned repository commit `5a790187c8e186e41e2b990b4941cb7a1a3c7b6b`. Source article.

[127] D. E. Radford, A natural ring basis for the shuffle algebra and an application to group schemes, *Journal of Algebra* **58** (1979), no. 2, 432–454. doi:10.1016/0021-8693(79)90171-6.

[128] F. Patras, La décomposition en poids des algèbres de Hopf, *Annales de l'Institut Fourier* **43** (1993), no. 4, 1067–1087. doi:10.5802/aif.1365.

[129] ProveIt research continuation, *Exact Resonant Identities Beyond the Diagonal*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/exact-resonant-identities/`. The complete-Cayley formal obstruction is audited in Chapter 12; the numerical S_6 conjecture remains open.

[130] ProveIt research continuation, *Harmonic Parity and Resolvent Identities*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/harmonic-parity-resolvents/`. The centered trigamma-square moments and their polylogarithmic generator are audited in Chapter 8; the broader classification and resolvent claims retain separate scopes.

[131] ProveIt research continuations, *Higher Digamma Powers and Ordered Gamma Sums*, *Canonical Digamma Polynomials and the Fifth Harmonic Bridge*, and *Mixed Resolvent Products, Higher Digamma Powers, and Stieltjes Collisions*, 11 October 2026. Original packages are preserved as `reports/stieltjes-harmonic-resonance/continuations/higher-digamma-powers/`, `reports/stieltjes-harmonic-resonance/continuations/canonical-digamma-fifth/`, and `reports/stieltjes-correlation-calculus/continuations/mixed-resolvents-harmonic-powers/`. Their all-power and fifth-moment cores are reconciled in Chapter 8; other Gamma, resolvent, collision and ordered branches retain separate audit scopes.

[132] NIST Digital Library of Mathematical Functions, Sections 5.5 (digamma recurrence and reflection), 5.7 (Taylor and reciprocal-Gamma series), and 5.11 (sectorial asymptotics). In particular formulas 5.5.2, 5.5.4, 5.7.4 and 5.11.2. https://dlmf.nist.gov/5. Inspected 11 October 2026 UTC.

[133] K. Ihara, M. Kaneko and D. Zagier, Derivation and double shuffle relations for multiple zeta values, *Compositio Mathematica* **142** (2006), 307–338. https://doi.org/10.1112/S0010437X0500182X.

[134] P. T. Young, Series of height one multiple zeta functions, *Integers* **24** (2024), A43, published 20 May 2024. https://math.colgate.edu/~integers/y43/y43.pdf. https://doi.org/10.5281/zenodo.11221606. The coefficient-description check in Appendix A concerns the inspected publisher PDF, printed page 12; its ordinary Stieltjes formula is retained.

[135] M. Can, L. Kargın, M. Cenkci and A. Dil, *On certain harmonic zeta functions*, arXiv:2403.07123v1, 11 March 2024, printed page 1. https://arxiv.org/pdf/2403.07123v1. The initial-index correction in Appendix A is scoped to that version's introductory Stieltjes-limit display. Inspected 11 October 2026 UTC.